Mathematics

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Notation List

Collision Index and Scope Rules

bare $A, B, \ldots$; $\mathcal{A}, \mathcal{B}, \ldots$	unadorned Latin and calligraphic letters denote local objects unless a decorated form is listed below; local declarations in a theorem, proof, or subsection take precedence
$B,\ B_r(x),\ \overline{B}_r(x),\ \mathrm{B}(x,y),\ \mathcal{B}(\mathbb{R}^n)$	matrix or set variable / open ball / closed ball / beta function / Borel σ-algebra; font, arguments, and ambient subject determine the referent
$I_{\mathcal{V}},\ I_n,\ I^k,\ I_R$	identity operator, identity matrix, k-cell, and power-series interval
$P,\ \mathcal{P}(X),\ \mathscr{P},\ P_{\mathcal{U}},\ P_j$	locally declared projection or map / power set / umbral polynomial algebra / orthogonal projection / spectral projection
$\mathcal{F},\ \mathscr{F},\ \mathcal{F}(K),\ \mathcal{F}(\mathcal{V})$	local family of functions or operators / umbral formal-series algebra / finite subsets of K / finite-rank operators
$\mathcal{L}(\mathcal{V},\mathcal{W}),\ \mathcal{L}^n,\ \mathscr{L}^1,\ \mathrm{LS}(A,\mathbf{y}),\ \mathcal{L}(\theta,\alpha)$	linear maps / Lebesgue measurable sets / integrable functions / least-squares solution set / local Lagrangian or loss notation
$R(f,\Pi,\{c_i\}),\ R,\ R_k(A),\ A=QR$	Riemann sum, locally declared radius or scalar, row-sum quantity, and QR factorization; arity and argument type disambiguate the notation
$\Delta x_i,\ \Delta A,\ \Delta(A),\ \Delta(\mathcal{V}),\ \Delta_{k-1}$	increment, matrix perturbation, defect from normality, metric discriminant, and coefficient simplex
$\Omega,\ \Omega_T,\ \Omega_n,\ \mathrm{vol}(\Omega)$	local domain or set / finite spectrum of T / primitive roots of unity or DFT diagonal, as specified by context / volume domain
$Q^k,\ Q_n,\ \Phi_n$	standard k-simplex / normalized Fourier matrix / cyclotomic polynomial; cyclotomic polynomials use Φ_n
$\overline{z},\ \overline{E},\ \overline{\mathcal{A}},\ A^*,\ T^*\omega,\ f',\ \mathcal{V}'$	conjugation, closure, uniform closure, adjoint or conjugate transpose, pullback, derivative, and dual space; the decorated object determines the operation

Global Conventions and Basic Notation

Convention. Notation is scoped by subject, argument type, font, decoration, and any local definition in force. Unadorned letters denote local objects by default; persistent meanings are carried by subscripts, superscripts, operators, fonts, or declared argument patterns. When a glyph is reused, the distinguishing feature is recorded below.

Set, Function, and Logical Notation

$x \in E,\ x \notin E$	membership / non-membership
$E \subseteq F,\ \emptyset$	subset inclusion, not necessarily proper / empty set
$X \times Y,\ f : X \to Y$	Cartesian product / function with domain X and codomain Y
$\Longrightarrow,\ \Longleftrightarrow$	logical implication / logical equivalence
$:=$	definition by equality

Number, Size, and Scalar Notation

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	natural numbers, integers, rationals, reals, and complexes
$\mathbb{F},\ \mathbb{F}_0$	ambient scalar field unless fixed as $\mathbb{R}$ or $\mathbb{C}$ / prime subfield of $\mathbb{F}$; probabilities use explicit Pr or locally declared probability spaces
$\lvert x\rvert,\ \lvert E\rvert,\ \lVert x\rVert$	absolute value or complex modulus for scalars; interval length or cardinality according to argument type; norm of a vector, function, or operator
a, b, c	locally declared scalars or interval endpoints

Linear and Operator Structures Notation

Vector Spaces and Lists

$\mathbb{F},\ \mathbb{F}_0$	ambient scalar field, typically $\mathbb{R}$ or $\mathbb{C}$ / prime subfield of $\mathbb{F}$
$V(n,q),\ \mathbb{F}_q^n$	n-dimensional coordinate space over the finite field with q elements
$\mathbb{F}^{(\mathbb{N})}$	vector space of finitely supported sequences over $\mathbb{F}$
$\mathbf{u},\mathbf{v},\mathbf{w},\mathbf{0}$	vectors / zero vector
$\mathcal{U},\mathcal{V},\mathcal{W}$	vector spaces or subspaces; other calligraphic letters such as $\mathcal{A},\mathcal{B},\mathcal{F},\mathcal{M}$ are local unless listed below; umbral algebras use script $\mathscr{P},\mathscr{F}$
$I_{\mathcal{V}},\ I_n,\ 0$	identity operator on $\mathcal{V}$ / identity matrix / zero map or zero matrix; distinct from the cell I^k and interval I_R
$\beta=(\mathbf{v}_1,\dots,\mathbf{v}_m)$	ordered vector list; a basis iff it is linearly independent and spans
$(),\ \beta_{\widehat{j}}$	empty list / list β with the jth entry omitted
$(\beta,\mathbf{v}),\ \beta_{j\leftarrow\mathbf{v}}$	list with $\mathbf{v}$ appended / list with the jth entry replaced by $\mathbf{v}$
$\mathcal{B}$	locally declared basis set, such as a Hamel, Hilbert, module, or affine basis; $\mathcal{B}(\mathbb{R}^n)$ denotes the Borel σ-algebra in analysis
$\operatorname{span}(E),\ \dim(\mathcal{V})$	linear span of E / dimension of $\mathcal{V}$
$\operatorname{codim}_{\mathcal{V}}(\mathcal{U})$	codimension of $\mathcal{U}$ in $\mathcal{V}$; equal to $\dim(\mathcal{V}/\mathcal{U})$ in finite dimension
$\bigwedge_{i\in I}\mathcal{U}_i,\ \bigvee_{i\in I}\mathcal{U}_i$	meet / join of a family of subspaces
$\sum_{i\in I}\mathcal{U}_i$	sum of an indexed family of subspaces; equivalently the join
$\mathcal{U}+\mathcal{W},\ \mathcal{U}\oplus\mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$\mathcal{V}_1\times\cdots\times\mathcal{V}_m$	product vector space with componentwise operations
$\prod_{i\in I}\mathcal{V}_i,\ \bigoplus_{i\in I}^{\text{ext}}\mathcal{V}_i$	external direct product / external direct sum of finite-support families
$\mathcal{V}\cong\mathcal{W},\ \mathbf{e}_{j,k}$	isomorphic vector spaces / basis vector supported in one product component
$\mathbf{v}+\mathcal{U},\ \pi(\mathbf{v})$	coset modulo $\mathcal{U}$ / locally declared quotient map $\mathcal{V}\to\mathcal{V}/\mathcal{U}$

Linear Maps and Duality

$T:\mathcal{V}\to\mathcal{W},\ \mathcal{L}(\mathcal{V},\mathcal{W})$	linear map / vector space of linear maps; distinct from measure-theoretic $\mathcal{L}^n$, least-squares $\operatorname{LS}(A,\mathbf{y})$, and local loss notation $\mathcal{L}(\theta)$
$ST,\ T\vert_{\mathcal{U}},\ T/\mathcal{U}$	composition $S\circ T$ / restriction to $\mathcal{U}$ / induced quotient operator
$T^{-1},\ T^m,\ p(T)$	inverse of an invertible map or operator / operator power / polynomial functional calculus
$\ker(T),\ \operatorname{range}(T)$	kernel and range of T
$\mathcal{V}_{\mathbb{C}},\ \operatorname{cpx},\ T_{\mathbb{C}}$	complexification of a real vector space / canonical embedding $\mathbf{v}\mapsto\mathbf{v}+i\mathbf{0}$ / complexification of a real operator
$\operatorname{graph}(T),\ [S,T]$	graph of a map / commutator $ST-TS$
$\operatorname{Comm}(\mathcal{F}),\ \operatorname{Comm}(A)$	commutant of a family / commutant of a single matrix or operator
$A\sim_{\text{row}}B,\ S\sim T$	row-equivalent matrices / operator similarity when declared; distinct from cardinal, a.e., and coefficient equivalence
$\mathcal{V}',\ \beta',\ T'$	dual space, dual basis, and dual map; prime denotes duality in this context
$\varphi,\psi,\ \mathcal{U}^0$	linear functionals / annihilator
$\mathcal{V}'',\ \Lambda_{\mathcal{V}}$	double dual / natural evaluation map $\mathcal{V}\to\mathcal{V}''$
$[\mathbf{v}]_\beta,\ [T]_{\gamma\leftarrow\beta}$	coordinate vector in basis β / matrix of T from β-coordinates to γ-coordinates
$C_{\gamma\leftarrow\beta}$	change-of-basis matrix sending β-coordinate columns to γ-coordinate columns

Matrices and Entrywise Structure

$A, B, C,\ \mathrm{Mat}(m, n, \mathbb{F})$	matrix variables / space of $m \times n$ matrices over $\mathbb{F}$; decorated forms such as C_f and C_{ij} have separate meanings
$A_{ij},\ A_{j,\cdot},\ A_{\cdot,k}$	entry in row i and column j / jth row / kth column
$[a_{ij}],\ E_{j,k},\ A\mathbf{x}$	matrix with entries a_{ij} / matrix unit / matrix-vector product
$A^{-1},\ A^t,\ A^T,\ A^*$	inverse matrix when it exists / transpose / alternate transpose notation / conjugate transpose or adjoint
$\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$	diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$
$\overline{z},\ \mathrm{Re}\, z,\ \mathrm{Im}\, z$	complex conjugate / real part / imaginary part; for sets, overline denotes closure
$\mathrm{tr}(A), \det(A)$	trace and determinant
$\mathfrak{S}_n,\ \mathrm{sign}(\sigma),\ \mathrm{adj}(A)$	permutation group / permutation sign / adjugate matrix
$M_{ij},\ C_{ij},\ A_j(\mathbf{b})$	minor matrix / cofactor / matrix obtained from A by replacing column j with $\mathbf{b}$
$\mathrm{col}(A),\ \mathrm{row}(A)$	column space and row space of A
$\mathrm{rank}(A),\ \mathrm{nullity}(A)$	rank and nullity of A
$A \oplus B,\ A \otimes B$	matrix direct sum / Kronecker product; vector-space direct sums use subspace notation
$A \otimes I_q + I_n \otimes B$	Kronecker sum of square matrices A and B in compatible dimensions
$A \circ B,\ \mathrm{vec}(A)$	Hadamard product / column-stacking vectorisation
$M/A,\ M/D$	Schur complement of the block A or D in a compatible block matrix M; distinct from quotient-space notation
$B,\ L = B^t B$	directed incidence matrix / associated incidence Laplacian in a graph context

Inner Product Geometry

$\|\mathbf{v}\|,\ \|T\|_{\mathrm{op}}$	vector norm / operator norm
$\|\cdot\|_1,\ \|\cdot\|_2,\ \|\cdot\|_\infty$	coordinate one-norm / Euclidean or spectral two-norm according to argument type / sup norm
$\|\cdot\|_F$	Frobenius norm
$\langle \mathbf{u}, \mathbf{v} \rangle,\ \langle A, B \rangle_F$	inner product / Frobenius inner product
$\mathbf{u} \perp \mathbf{v},\ \mathcal{U}^\perp,\ \delta_{ij}$	orthogonal vectors, orthogonal complement, Kronecker delta
$\mathrm{proj}_{\mathbf{v}}(\mathbf{u}),\ P_{\mathcal{U}}$	orthogonal projection of $\mathbf{u}$ onto $\mathrm{span}(\mathbf{v})$ / orthogonal projection onto $\mathcal{U}$
$\mathrm{LS}(A, \mathbf{y})$	least-squares solution set $\arg\min_{\mathbf{x} \in \mathbb{F}^n} \|A\mathbf{x} - \mathbf{y}\|_2$ for $A \in \mathrm{Mat}(m, n, \mathbb{F})$; roman notation avoids collision with $\mathcal{L}$ and $\mathcal{L}^n$
$d(\mathbf{v}, \mathcal{U})$	distance $\inf_{\mathbf{u} \in \mathcal{U}} \|\mathbf{v} - \mathbf{u}\|$ from $\mathbf{v}$ to $\mathcal{U}$
$\mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram matrix of $\mathbf{u}_1, \ldots, \mathbf{u}_n$
$\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$	Gram determinant
$P_{\mathbf{u}},\ U_{\mathbf{w}}$	rank-one projection matrix / Householder matrix, with defining vector indicated by the subscript
$\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$	ordered basis; orthonormal when declared in an inner-product context
$\mathbf{e} = (1, \ldots, 1)^t$	all-ones vector
$\mathrm{Ray}_A(\mathbf{x}),\ \mathbf{x}^* A \mathbf{x}$	Rayleigh quotient notation; for unit $\mathbf{x}$, $\mathrm{Ray}_A(\mathbf{x}) = \mathbf{x}^* A \mathbf{x}$ / Hermitian quadratic form
$\mathrm{Sym}(B),\ \mathrm{Skew}(B),\ \mathrm{Alt}(B)$	predicates that the bilinear form B is symmetric, skew-symmetric, or alternating
$M_\beta,\ \Delta(\mathcal{V}),\ \mathrm{rad}(\mathcal{V})$	matrix of a metric bilinear form in basis β / discriminant set or square class / radical of a metric vector space
$\mathcal{V} = \mathcal{S} \odot \mathcal{T},\ \mathcal{H},\ w(\mathcal{V})$	orthogonal direct sum / hyperbolic space or plane in metric-vector-space contexts / Witt index; affine hyperplanes use $\mathcal{H}(\mathbf{N}, b)$
$\mathcal{O}(\mathcal{V}),\ \mathrm{Sp}(\mathcal{V}),\ \tau_{\mathbf{v},a},\ \sigma_{\mathbf{u}}$	orthogonal group / symplectic group / symplectic transvection / orthogonal symmetry

Affine, Projective, and Convex Geometry

$\mathcal{A}(\mathcal{V}),\ \mathcal{A}_0(\mathcal{V}),\ \mathrm{Sub}(\mathcal{V})$	flats in $\mathcal{V}$ / $\mathcal{A}(\mathcal{V}) \cup \{\emptyset\}$ / subspaces of $\mathcal{V}$; $\mathcal{A}$ without arguments remains local
$\mathrm{affspan}(E),\ \mathrm{affhull}(E)$	affine span / least flat containing E
$X \parallel Y,\ X \vee Y$	parallel flats / join of flats or projective subspaces; meet is intersection in the relevant lattice
$\mathbb{R}^n_+,\ \mathbb{R}^n_{++},\ \mathbf{x} \geq \mathbf{0},\ \mathbf{x} > \mathbf{0},\ \mathbf{x} \gg \mathbf{0}$	nonzero nonnegative orthant / coordinatewise positive orthant / coordinatewise nonnegative, nonzero nonnegative, and coordinatewise positive inequalities
$\mathrm{conv}(S),\ \Delta_{k-1},\ C_A$	convex hull / coefficient simplex $\{(t_i) : t_i \geq 0,\ \sum_i t_i = 1\}$ / nonnegative cone generated by the columns of A; distinct from cofactor and companion notation
$\mathcal{H}(\mathbf{N}, b),\ \mathcal{H}_\pm(\mathbf{N}, b),\ \langle \mathbf{N}, X \rangle < b$	affine hyperplane / closed half-spaces / setwise inequality, meaning $\langle \mathbf{N}, \mathbf{x} \rangle < b$ for all $\mathbf{x} \in X$

Spectral and Polynomial Structure

$E_\lambda(T),\ \mathrm{spec}(T)$	eigenspace $\ker(T - \lambda I_\mathcal{V})$ / spectrum
$\mathrm{Inv}(T)$	lattice or set of T-invariant subspaces
$\lambda_i(A),\ m_A(\lambda)$	eigenvalues in the locally declared order / algebraic multiplicity of λ as a root of p_A; distinct from measure notation $m(E)$
$\mathcal{G}_\lambda(T),\ J_k(\lambda)$	generalized eigenspace for λ / Jordan block of size k and eigenvalue λ
$p_A,\ q_T,\ \chi_T$	matrix characteristic polynomial / operator characteristic polynomial / characteristic polynomial determined by elementary divisors
$\mu_A,\ \mu_T,\ w_p(\lambda, A)$	minimal polynomial of a matrix or operator / Weyr count of λ-Jordan blocks of size at least p; distinct from measure notation $\mu(E)$
$P_j,\ T = \sum_j \mu_j P_j$	spectral projections / spectral decomposition with distinct eigenvalues μ_j
$\sqrt{T},\ A^{1/2},\ \rho(A)$	positive square root of a positive operator or matrix / spectral radius of A; distinct from the interval parameter in K_ρ
$\mathbb{F}[z]$	polynomials in z over $\mathbb{F}$
$\mathscr{P} = \mathbb{C}[x],\ \mathscr{P}',\ \mathscr{F} = \mathbb{C}[[t]]$	umbral polynomial algebra / algebraic dual / formal power-series algebra; script notation distinguishes these from $\mathcal{P}(X)$ and local families $\mathcal{F}$
$\mathcal{V}_T,\ \mathcal{C}_T(\mathbf{v}),\ \mathrm{ann}_T(\mathbf{v})$	$\mathbb{F}[z]$-module induced by T / T-cyclic subspace generated by $\mathbf{v}$ / annihilator ideal of $\mathbf{v}$
$m_\mathbf{v},\ C_f,\ \mathrm{InvFact}(T),\ \mathrm{ElemDiv}(T)$	order polynomial of $\mathbf{v}$ under T / companion matrix of f / invariant factors / elementary divisors
e^{tA}	matrix exponential $\sum_{k \geq 0} (tA)^k / k!$

Matrix Analysis and Numerical Structure

$A \succeq 0,\ A \succ 0$	Hermitian positive semidefinite / Hermitian positive definite
$\mathrm{in}(A),\ n_+(A), n_-(A), n_0(A)$	inertia of a Hermitian matrix / positive, negative, and zero eigenvalue counts
$A = QR,\ A = LL^*,\ A = LDL^t$	QR factorization / Cholesky factorization / symmetric or Hermitian LDL^t factorization
$A = V\Sigma W^*,\ A = X\Sigma_r Y^*$	singular value decomposition / compact SVD with rank r
$\sigma_j(A),\ \sigma_{\max}(A),\ \sigma_{\min}(A)$	singular values and extreme singular values
$\lvert T\rvert,\ \lvert A\rvert,\ \mathrm{vol}(\Omega)$	operator or matrix modulus / volume of a set; scalar absolute value remains $\lvert x\rvert$
$A^\dagger,\ A_k$	Moore–Penrose pseudoinverse / rank-k truncated SVD approximation in an SVD context
$\kappa_2(A),\ \Delta A,\ \Delta\mathbf{x}$	spectral condition number of an invertible matrix / matrix and solution perturbations
$\Delta(A)$	defect from normality; distinct from perturbation notation ΔA
$R_k(A),\ R_k^{\mathrm{off}}(A),\ R_{\max}(A)$	absolute row sum $\sum_j \lvert a_{kj}\rvert$ / deleted row sum $\sum_{j\neq k} \lvert a_{kj}\rvert$ / maximum absolute row sum
$\mathrm{Ger}_k(A),\ \mathrm{Ger}(A)$	kth Gersgorin disk / union of all Gersgorin disks
$\mathrm{tridiag}(a, b, c)$	tridiagonal matrix with subdiagonal a, diagonal b, and superdiagonal c
$w,\ F_n,\ Q_n,\ \omega_n$	half-bandwidth / unnormalised Fourier matrix / normalised Fourier matrix / primitive nth root used for the DFT
$\mathcal{K}_m(A, \mathbf{b})$	Krylov subspace $\mathrm{span}(\mathbf{b}, A\mathbf{b}, \ldots, A^{m-1}\mathbf{b})$
$Q_m,\ H_m$	Krylov basis matrix / projected Hessenberg or tridiagonal matrix
$A = S - T,\ B = S^{-1}T,\ \mathbf{e}_k$	matrix splitting / stationary iteration matrix / kth error vector
$A = D + L + U,\ B_J,\ B_{\mathrm{GS}}$	diagonal, lower, and upper splitting / Jacobi and Gauss–Seidel iteration matrices
$\omega,\ S_\omega,\ T_\omega,\ B_\omega$	SOR parameter / SOR splitting matrices / SOR iteration matrix; distinct from the differential form ω
$M,\ M^{-1}A,\ AM^{-1}$	preconditioner / left and right preconditioned matrices

Algebraic Structures Notation

Algebraic Language and Categories

$\mathcal{C},\ A \xrightarrow{f} B,\ \mathrm{id}_A$	ambient category / morphism $f : A \to B$ / identity morphism of A; $\mathcal{C}$ remains local unless declared
$\mathrm{Hom}_{\mathcal{C}}(A, B),\ g \circ f$	set of morphisms $A \to B$ in $\mathcal{C}$ / composition for compatible morphisms
$A \simeq B,\ F : \mathcal{C} \to \mathcal{D}$	isomorphic objects / functor with domain $\mathcal{C}$ and codomain $\mathcal{D}$

Groups, Actions, Rings, and Ideals

$G, H, K,\ e_G,\ \langle S\rangle$	groups or subgroups / identity element of G / substructure generated by S in the ambient algebraic category
$H \leq G,\ H \trianglelefteq G,\ G/H,\ [G : H]$	subgroup / normal subgroup / quotient group / subgroup index
$\mathrm{ord}(g),\ \mathrm{Aut}(G)$	order of an element / automorphism group
$R, S,\ 0_R,\ 1_R,\ R^\times$	rings / zero and identity elements of R / unit group of R
$Z(R),\ \mathrm{char}(R)$	center of R / characteristic of R
$I, J,\ \mathfrak{a}, \mathfrak{p},\ R/I$	ideals; fraktur letters denote distinguished ideals, such as prime ideals, when declared / quotient ring
$R[x],\ \mathbb{F}[z],\ \deg f$	polynomial rings in the displayed indeterminate / degree of a polynomial
$U_n,\ \Omega_n,\ \Phi_n(x),\ \varphi(n)$	complex nth roots of unity / primitive nth roots / nth cyclotomic polynomial / Euler totient; Ω_n is algebraic root notation, distinct from domain notation

Modules and Associative Algebras

$M, N,\ S \le M,\ M/N$	modules / submodule S of M / quotient module
$\mathrm{Hom}_R(M,N),\ \mathrm{End}_R(M)$	set of R-module homomorphisms $M \to N$ / endomorphism ring of M
$\ker\tau,\ \mathrm{im}\,\tau,\ \overline{\tau}$	kernel / image / quotient-induced map $M/\ker\tau \to \mathrm{im}\,\tau$ for a homomorphism $\tau : M \to N$
$\bigoplus_i M_i,\ M = N \oplus H$	external direct sum of modules / internal direct-sum decomposition
$\mathcal{C}_S(N),\ \mathcal{L}_\sigma,\ \mathcal{R}_\sigma$	set $\{H \le N : N = S \oplus H\}$ of complements / left inverses of σ / right inverses of σ; subscript distinguishes these from $\mathcal{L}$ and $\mathcal{L}^n$
$\mathcal{F}(\mathcal{V})$	finite-rank operators $\{T \in \mathcal{L}(\mathcal{V}) : \dim \mathrm{im}\, T < \infty\}$ in associative-algebra contexts; distinct from local function-family notation

Algebraic Geometry and Commutative Algebra Notation

Status. This part of the notes is still mostly scaffolded. The entries below reserve the notation promised by the section headings and keep it distinct from already-active algebra and analysis notation.

Commutative Algebra

$\mathrm{Spec}\, R,\ \mathfrak{p} \in \mathrm{Spec}\, R$	set of prime ideals of a commutative ring R / prime ideal $\mathfrak{p} \subseteq R$
$D(f),\ V(I)$	standard open set $\{\mathfrak{p} \in \mathrm{Spec}\, R : f \notin \mathfrak{p}\}$ / Zariski closed set $\{\mathfrak{p} : I \subseteq \mathfrak{p}\}$
$S^{-1}R,\ R_{\mathfrak{p}},\ k(\mathfrak{p})$	localization at a multiplicative set S / localization $(R \setminus \mathfrak{p})^{-1}R$ / residue field $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$
$\mathrm{rad}(I),\ \sqrt{I}$	radical ideal $\{r \in R : r^n \in I \text{ for some } n \ge 1\}$; use $\mathrm{rad}(\mathcal{V})$ for metric-vector-space radicals

Affine and Projective Geometry

$\mathbb{A}^n_k,\ \mathbb{P}^n_k$	affine n-space over a field k / projective n-space over k; geometric notation distinct from probability notation
$I(X),\ Z(I),\ k[X]$	vanishing ideal $\{f \in k[x_1,\dots,x_n] : f|_X = 0\}$ / zero locus of an ideal / coordinate ring, typically $k[x_1,\dots,x_n]/I(X)$
$f^{\#},\ \varphi^*$	coordinate-ring pullback, e.g. $f^{\#} : k[Y] \to k[X]$ for $f : X \to Y$; distinct from adjoints and differential-form pullbacks by context

Analysis and Continuous Structure Notation

Sets and Functions in Analysis

$\mathcal{P}(X)$	power set $\{A : A \subseteq X\}$; projection symbols use $P_{\mathcal{U}}$ or P_j
$f(A),\ f^{-1}(B)$	image $\{f(a) : a \in A\}$ / preimage $\{x : f(x) \in B\}$ under f
$\mathrm{range}(f),\ f\|_A$	range $f(\mathrm{dom}\, f)$ / restriction of f to A
$\{A_i\}_{i\in I},\ \bigcup_{i\in I} A_i,\ \bigcap_{i\in I} A_i$	indexed family of sets / set of points lying in some A_i / set of points lying in every A_i
$g \circ f,\ \mathrm{id}_X$	composition of functions with compatible domain and codomain / identity function on X
$J_n,\ \#E,\ A \sim_{\#} B,\ [x]$	finite initial segment $J_0 = \emptyset$, $J_n = \{1,\dots,n\}$ / cardinality of a finite set / existence of a bijection $A \to B$ / equivalence class of x for the specified relation

Order and Completeness

$\sup E,\ \inf E$	least upper bound / greatest lower bound in the ambient ordered set, with values in $\overline{\mathbb{R}}$ when extended values are allowed
$\|x\|,\ \|a-b\|$	real absolute value or complex modulus / usual real-line distance $d(a,b) = \|a-b\|$
$\overline{\mathbb{R}}$	extended real line $\mathbb{R} \cup \{-\infty, +\infty\}$ ordered by $-\infty < x < +\infty$ for $x \in \mathbb{R}$; the bar is conventional, not topological closure
$x < y,\ x \le y$	strict order / non-strict order in the ambient ordered set

Sequences and Series

$(a_n),\ a_n \to a$	sequence indexed by $n \in \mathbb{N}$ / convergence of the sequence to a
$(a_{n_k}),\ n_1 < n_2 < \cdots$	subsequence indexed by a strictly increasing sequence (n_k) of positive integers
$s_n = \sum_{k=1}^{n} a_k$	partial-sum sequence associated to $\sum a_k$
$\limsup a_n = s^*,\ \liminf a_n = s_*$	upper and lower limits in $\overline{\mathbb{R}}$, equivalently $\inf_N \sup_{n\geq N} a_n$ and $\sup_N \inf_{n\geq N} a_n$ for real sequences
$\sum_{n=1}^{\infty} a_n,\ \prod_{n=1}^{\infty} a_n$	limit of partial sums / limit of finite partial products when defined

Metric and Topological Notation

$d(x,y),\ B_r(x)$	metric distance / open ball $\{y : d(x,y) < r\}$
$B_\varepsilon(a) = V_\varepsilon(a)$	real-line open ball $(a-\varepsilon, a+\varepsilon)$; $V_\varepsilon(a)$ is Abbott's notation
$\overline{B}_r(x),\ \{G_\alpha\}_{\alpha\in A}$	closed ball $\{y : d(x,y) \leq r\}$ / indexed open cover
$d_E,\ d_{\max},\ d_{\text{sum}}$	Euclidean, maximum, and sum product metrics on a product space
$E^\circ,\ \overline{E},\ E^{\text{acc}}$	interior / topological closure / set of limit points of E
$\text{bd}(E),\ E^c$	boundary $\overline{E} \setminus E^\circ$ / complement in the ambient space
$\operatorname{diam} E$	diameter $\sup\{d(p,q) : p,q \in E\}$ of a nonempty metric set
$X \simeq Y$	homeomorphic metric spaces, i.e. related by a bijection with continuous inverse
$A \cap \overline{B} = \emptyset,\ \overline{A} \cap B = \emptyset$	definition of separated subsets A and B
$F_\sigma,\ G_\delta,$ meager/thin/first category	countable union of closed sets / countable intersection of open sets / subset of a countable union of nowhere-dense sets
residual/thick, generic	set containing a countable intersection of dense open sets / property whose truth set is residual

Functions, Differentiation, and Integration

$\lim_{x\to p} f(x),\ f(x+),\ f(x-)$	limit of f at p / right and left limits at x
$C(X),\ C([a,b]),\ C_b(X),\ \|f\|_\infty$	continuous functions on X and $[a,b]$ / bounded continuous functions / supremum norm
$f_n \to f,\ f_n \rightrightarrows f$	pointwise convergence / uniform convergence on the ambient domain
$f'(x),\ Df(a),\ D_j f_i,\ \partial_i f,\ J_f(x)$	one-variable derivative, total derivative at a, component partial $D_j f_i$, scalar partial, and Jacobian determinant; J_f is distinct from J_n and $J_k(\lambda)$
$\nabla f,\ D_{\mathbf{u}} f,\ C^r,\ C^\infty,\ C^\omega$	gradient, directional derivative, and classes C^r, C^∞, C^ω
$\Pi = \{x_0, \ldots, x_n\},\ c_i,\ \Delta x_i,\ \Delta\alpha_i$	partition of $[a,b]$, tags $c_i \in [x_{i-1}, x_i]$, and increments $\Delta x_i = x_i - x_{i-1}$, $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$
$R(f,\Pi,\{c_i\}),\ \delta,\ \delta(x)$	$R(f,\Pi,\{c_i\}) = \sum_{i=1}^{n} f(c_i)\Delta x_i$ / constant mesh bound / gauge; uppercase $R(\cdots)$ is local to Riemann sums
$U(\Pi,f,\alpha),\ L(\Pi,f,\alpha)$	Riemann–Stieltjes upper and lower sums with integrator α
$f \in \mathcal{R},\ f \in \mathcal{R}(\alpha)$	ordinary Riemann integrability / Riemann–Stieltjes integrability with respect to α
$\int_a^b f,\ \int_a^b f\,d\alpha,\ \int_E f\,d\mu$	ordinary Riemann integral / Riemann–Stieltjes integral / measure integral
$\int_a^\infty f,\ \int_{-\infty}^\infty f$	one-sided and two-sided improper integrals as limits over finite truncations
$\ell(\Pi,\gamma),\ \ell(\gamma)$	polygonal length of γ along Π / curve length
$I^k,\ Q^k,\ \operatorname{supp}(f)$	k-cell, standard k-simplex, and closed support $\overline{\{x : f(x) \neq 0\}}$; I^k is not an identity operator
$dx_I,\ dx_I \wedge dx_J$	basic form indexed by I / wedge product of basic forms
$\omega,\ d\omega,\ T^*\omega,\ \partial_{\text{ch}}\Psi$	differential form, exterior derivative, pullback by T, and chain boundary
$\int_\Psi \omega,\ \partial_{\text{ch}}[\mathbf{p}_0,\ldots,\mathbf{p}_k]$	integral of ω over the chain Ψ / oriented boundary $\sum_{j=0}^{k}(-1)^j[\mathbf{p}_0,\ldots,\widehat{\mathbf{p}_j},\ldots,\mathbf{p}_k]$
$dV,\ dA,\ ds$	volume, area, and arc-length elements

Power Series, Fourier, and Special Functions

$\sum_{n\geq 0} c_n(x-a)^n,\ R$	power series centered at a / radius of convergence in this context, distinct from $R(\cdots)$ and the R in QR
$I_R,\ K_\rho$	centered interval $I_R = (-R, R)$ / compact subinterval $K_\rho = [-\rho, \rho] \subset I_R$ for $0 < \rho < R$
$\widehat{f}(m),\ s_N(f;x),\ s_N(f)$	$\widehat{f}(m) = (2\pi)^{-1}\int_{-\pi}^{\pi} f(x)e^{-imx}\,dx$ / $s_N(f;x) = \sum_{\lvert m\rvert\leq N}\widehat{f}(m)e^{imx}$, viewed pointwise or as a function
$f \sim_{\text{coef}} \sum \widehat{f}(n)e^{inx},\ f \sim_{\text{coef}} \sum c_n\phi_n$	formal Fourier or orthonormal coefficient expansion; no convergence assertion is implied
$D_N(t)$	Dirichlet kernel $D_N(t) = \sum_{m=-N}^{N} e^{imt}$; uppercase D_N is not derivative notation
$\mathsf{E},\ \mathsf{L},\ \mathsf{C},\ \mathsf{S}$	power-series exponential, inverse logarithm, cosine, and sine
$\Gamma(x),\ \mathrm{B}(x,y)$	gamma function / beta function in roman type, distinct from balls $B_r(x)$ and variables B

Measure and Function Spaces

$E \in \mathcal{M},\ E \in \mathcal{M}(\mu)$	E measurable in the ambient σ-algebra / Rudin measurable set generated by μ
$\mathcal{B}(\mathbb{R}^n),\ \mathcal{L}^n$	Borel σ-algebra on $\mathbb{R}^n$ / Lebesgue measurable subsets of $\mathbb{R}^n$; not a basis $\mathcal{B}$, a linear-map space $\mathcal{L}$, or a least-squares set LS
$\mathcal{E},\ \mathcal{M}_F(\mu),\ A\triangle B,\ d_\mu(A,B)$	elementary sets, finitely μ-measurable sets, symmetric difference, and $d_\mu(A,B) = \mu^*(A\triangle B)$; $\mathcal{M}$ is measure-theoretic here
$\mu^*(E),\ m^*(E),\ \mathbf{1}_E$	μ-outer measure / Lebesgue outer measure / indicator $\mathbf{1}_E(x) = 1$ for $x \in E$ and 0 otherwise
$\mathrm{vol}_n(B),\ m_n^*(\Omega)$	n-dimensional box volume / box-cover outer measure $\inf\{\sum_{j\in J}\mathrm{vol}_n(B_j) : (B_j)_{j\in J} \text{ covers } \Omega,\ J \text{ at most countable}\}$
$m(E),\ \mu(E)$	Lebesgue or elementary volume of E / measure of E; not algebraic multiplicity $m_A(\lambda)$ or minimal polynomial μ_A
$\lambda^+,\ \lambda^-,\ \lvert\lambda\rvert,\ \lambda \ll \mu,\ \lambda \perp \mu,\ \frac{d\lambda}{d\mu}$	positive variation, negative variation, total variation, absolute continuity, mutual singularity, and Radon–Nikodym derivative
$\bowtie,\ \{f \bowtie a\}$	relation in $\{>, \geq, <, \leq\}$ / level set $\{x : f(x) \bowtie a\}$
$f^+,\ f^-,\ \mathcal{S}^+(f),\ \mathcal{I}_E(s)$	$f^+ = \max(f, 0),\ f^- = -\min(f, 0)$ / simple minorants of f / simple-function integral on E
$f \sim_{\text{a.e.}} g \text{ on } E,\ \text{a.e.}$	$\mu(\{x \in E : f(x) \neq g(x)\}) = 0$ / outside a null set
$E_x,\ E^y,\ f_x,\ f^y$	product slices $E_x = \{y : (x,y) \in E\}$, $E^y = \{x : (x,y) \in E\}$, and $f_x(y) = f^y(x) = f(x,y)$
$\mathcal{U}f,\ F_a$	undergraph $\{(x,t) : 0 < t < f(x)\}$ for $f \geq 0$ / superlevel set $F_a = \{x : f(x) > a\}$; $\mathcal{U}f$ is not a vector subspace
$[f : A],\ [B : Q],\ \mathrm{dp}(E)$	$[f : A] = m(A)^{-1}\int_A f$ / $[B : Q] = m(B \cap Q)/m(Q)$ / density-point set of E
$\mathscr{L}^1(\mu),\ \mathscr{L}^1_{\text{loc}}(\mathbb{R}^n)$	measurable f with $\int \lvert f\rvert\,d\mu < +\infty$ / locally integrable functions; script $\mathscr{L}$ is distinct from $\mathcal{L}^n$ and $\mathcal{L}$
$L^p(\mu),\ L^p(E),\ \lVert f\rVert_p$	Lebesgue space modulo a.e. equality with respect to μ or the ambient measure on E / $\lVert f\rVert_p = (\int \lvert f\rvert^p\,d\mu)^{1/p}$

Probability and Stochastic Structure Notation

Probability Spaces and Distributions

$(\Omega, \mathcal{F}, \mathbb{P})$, $\Pr(A)$	probability space / event probability $\Pr(A) = \mathbb{P}(A)$ for $A \in \mathcal{F}$; explicit probability notation keeps $\mathbb{P}_k^n$ and $\mathbb{F}_0$ unambiguous
X, Y, $X_*\mathbb{P}$, $\mathcal{L}(X)$	random variables / pushforward law $X_*\mathbb{P}(B) = \mathbb{P}(X^{-1}(B))$ / law of X when locally declared; not Lebesgue $\mathcal{L}^n$
$X \sim F$, $X \overset{\text{dist}}{=} Y$, $X_1, \dots, X_n$ iid	X has distribution F / X and Y have the same distribution / independent and identically distributed sample
$F(t)$, $g(t)$, $\Phi(t)$	distribution function $F(t) = \Pr(X \le t)$ / density g / standard normal distribution function in probabilistic contexts
$\mathbb{E}[X]$, $\operatorname{Var}(X)$, $\operatorname{Cov}(X, Y)$	expectation / $\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$ / $\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, when finite
$\operatorname{Corr}(X, Y)$, ρ, σ_{XY}	correlation coefficient / local correlation parameter / covariance shorthand $\sigma_{XY} = \operatorname{Cov}(X, Y)$ when declared
$\psi_X(t)$	moment-generating function $\psi_X(t) = \mathbb{E}e^{tX}$ on its domain of finiteness
$X \perp\!\!\!\perp Y$	independence of random variables, equivalently independence of $\sigma(X)$ and $\sigma(Y)$ when defined
$X \perp\!\!\!\perp Y \mid Z$	conditional independence of X and Y given Z
$X_n \xrightarrow{\mathbb{P}} X$, $X_n \rightsquigarrow X$, $X_n \xrightarrow{L^p} X$, $X_n \xrightarrow{\text{a.s.}} X$, $X_n \xrightarrow{qm} X$	convergence in probability / convergence in distribution / convergence in L^p / almost-sure convergence / convergence in quadratic mean

Stochastic Processes and Markov Structure

$(X_t)_{t \in T}$, $\mathcal{F}_t$	stochastic process indexed by T / filtration $(\mathcal{F}_t)$ when locally declared; this $\mathcal{F}_t$ is not the umbral algebra $\mathscr{F}$
$\mathbf{x}$, $\mathbf{e}$, $A\mathbf{e} = \mathbf{e}$	probability vector $\mathbf{x} \ge 0$, $\sum_i x_i = 1$ / all-ones vector $\mathbf{e} = (1, \dots, 1)^t$ / row-stochastic Markov convention $A = [a_{ij}] \ge 0$, $A\mathbf{e} = \mathbf{e}$
π, $\pi A = \pi$, A^p	stationary distribution written as a row probability vector / stationarity equation / p-step Markov matrix
P_t, $p_t(x, y)$	transition semigroup / transition probability, kernel, or density from x to y, when introduced

Numerical and Computational Mathematics Notation

Status. Apart from numerical linear algebra, already covered under Linear and Operator Structures, the numerical and computational chapters remain placeholders. The entries below reserve notation only; they do not introduce PDE or machine-learning conventions.

Floating Point, Approximation, and Error

$\operatorname{fl}(x)$, u, $\varepsilon_{\text{mach}}$	floating-point rounded value, unit roundoff, and machine epsilon
x_h, e_h, $\mathcal{O}(h^p)$	approximation at discretization scale h, corresponding error, and Landau order as $h \to 0$
$\widehat{x}$, Δx, $r = b - A\widehat{x}$	computed approximation, perturbation or error in x, and residual for $Ax = b$

Monte Carlo and Stochastic Computation

$X_1, \dots, X_N$, $\widehat{\mu}_N$	samples / sample-mean estimator $\widehat{\mu}_N = N^{-1} \sum_{i=1}^N X_i$ when estimating $\mu = \mathbb{E}X$
$\operatorname{Var}(\widehat{\mu}_N)$, $\operatorname{SE}(\widehat{\mu}_N)$	estimator variance / standard error $\operatorname{SE}(\widehat{\mu}_N) = \sqrt{\operatorname{Var}(\widehat{\mu}_N)}$
w_i, $\widehat{I}_N$	importance weights / Monte Carlo estimate of an integral or expectation, when locally introduced

Part I
Set Theory, Logic, and Foundations

Foundational prerequisites collect the logical, set-theoretic, functional, order-theoretic, and cardinal language used before the dedicated algebraic and analytic foundations sections.

1 Sets

Sets provide the membership language used throughout the notes: elements, subsets, products, power sets, operations, indexed families, and the first warnings about unrestricted comprehension.

1.1 Introduction to Sets

Definition 1.1.1. *Sets and Operations*
A *set* is a collection. Write $x \in E$, $x \notin E$, $\emptyset$, and $A \subseteq B$; $A = B$ iff $A \subseteq B$ and $B \subseteq A$. For $A, B \subseteq X$,

$$\begin{aligned} A \cup B &:= \{x \in X : x \in A \text{ or } x \in B\}, \\ A \cap B &:= \{x \in X : x \in A \text{ and } x \in B\}, \\ A \setminus B &:= \{x \in X : x \in A \text{ and } x \notin B\}, \\ A^c &:= X \setminus A. \end{aligned}$$

1.2 Cartesian Products

Definition 1.2.1. *Cartesian Product*
For sets X and Y, the *Cartesian product* is $X \times Y := \{(x, y) : x \in X,\ y \in Y\}$.
The order matters: the first coordinate comes from X, and the second comes from Y.

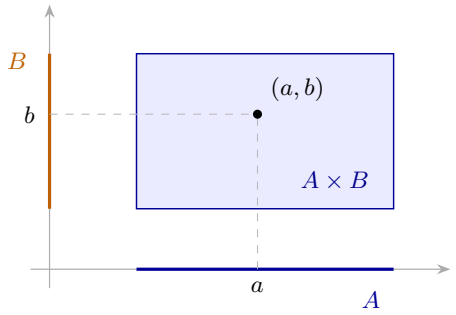


Figure 1: $A \times B$: first coordinate in A, second in B.

1.3 Subsets and Power Sets

Definition 1.3.1. *Power Set*
Let X be a set. The *power set* is $\mathcal{P}(X) := \{A : A \subseteq X\}$.

Method 1.3.2. *Proof by Two Containments*
To prove $A = B$, prove both inclusions.

(i) To prove $A \subseteq B$, let $x \in A$ be arbitrary and use the definitions and hypotheses to show $x \in B$.
(ii) To prove $B \subseteq A$, let $x \in B$ be arbitrary and use the definitions and hypotheses to show $x \in A$.
(iii) Once both containments are proved, conclude $A = B$ by extensionality of sets.

If both sides are described by membership conditions, this is a chain, $x \in A \iff \cdots \iff x \in B$.

Placeholder: Add extensionality examples, proper subset notation, and power-set examples.

1.4 Union, Intersection, Difference, and Complement

Remark 1.4.1. *Set-Builder Notation*
$\{x \in X : P(x)\}$ means all $x \in X$ satisfying $P(x)$.
For sets (A_n), $\bigcup_n A_n$ means membership in some A_n; $\bigcap_n A_n$, membership in every A_n.
Key Insight: Expand membership conditions.

Proposition 1.4.2. *Basic Set Algebra*
Let $A, B, C \subseteq X$. Then

(i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
(ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
(iii) $(A \cup B)^c = A^c \cap B^c$;
(iv) $(A \cap B)^c = A^c \cup B^c$.

Proof. It suffices to prove (i) and (iii); the others are similar. Let $x \in X$. Then

(i) $x \in A \cap (B \cup C) \iff x \in A$ and $(x \in B$ or $x \in C) \iff (x \in A \cap B)$ or $(x \in A \cap C) \iff x \in (A \cap B) \cup (A \cap C)$
(iii) $x \in (A \cup B)^c \iff x \notin A \cup B \iff x \notin A$ and $x \notin B \iff x \in A^c \cap B^c$

Structural Commentary: Set algebra is membership logic.
Key Insight: Translate sets into conditions on x. □

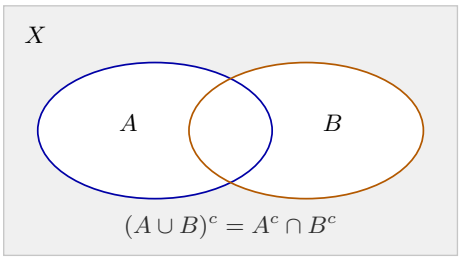


Figure 2: De Morgan's law says that being outside a union is the same as being outside each set.

Theorem 1.4.3. *Arbitrary De Morgan Laws*
Let X be a set. For any collection $\{E_\alpha\}$ of subsets of X, $\left(\bigcup_\alpha E_\alpha\right)^c = \bigcap_\alpha E_\alpha^c$ *and* $\left(\bigcap_\alpha E_\alpha\right)^c = \bigcup_\alpha E_\alpha^c$.

Proof. The first identity follows from membership: $x \in \left(\bigcup_\alpha E_\alpha\right)^c \iff x \notin E_\alpha \ \forall \alpha \iff x \in \bigcap_\alpha E_\alpha^c$.
The second follows by applying the first identity to the complements E_α^c, or by the same membership argument.
Structural Commentary: Complements exchange unions and intersections.
Key Insight: Negating "for some" gives "for every", and conversely. □

1.5 Indexed Sets and Number Systems

Definition 1.5.1. *Indexed Families of Sets*
An *indexed family* $\{A_i\}_{i \in I}$ is one set $A_i \subseteq X$ for each index $i \in I$. Define

$$\bigcup_{i \in I} A_i := \{x \in X : x \in A_i \text{ for some } i \in I\},$$
$$\bigcap_{i \in I} A_i := \{x \in X : x \in A_i \text{ for every } i \in I\}.$$

Key Insight: Union means "some index"; intersection means "every index".

Placeholder: Add the standard number-system inclusions and the convention that indexed notation binds its index variable.

1.6 Russell's Paradox and Axioms

Placeholder: Add Russell's paradox as a motivation for restricted comprehension, then state the informal role of extensionality, separation, pairing, union, power set, infinity, replacement, foundation, and choice.

2 Logic and Proofs

Logic fixes the grammar of mathematical statements before the notes use that grammar for definitions, theorems, and proofs.

2.1 Statements and Well-Formed Language

Principle 2.1.1. *Well-Formed Statements and Bound Variables*
A mathematical statement is well formed only after every symbol has a declared role and every expression has a defined meaning. A variable introduced by a quantifier, summation sign, integral, set-builder condition, or similar construction is *bound* inside that construction; changing its name does not change the statement as long as its scope is unchanged and no free variable is captured.

2.2 Connectives, Conditionals, and Truth Tables

Placeholder: Add the truth-functional meanings of $\wedge, \vee, \neg, \Rightarrow, \iff$, inclusive "or", truth tables, logical equivalence, converse, inverse, and contrapositive.

2.3 Quantifiers, Bound Variables, and Negation

Principle 2.3.1. *Language Carries Structure*
Objects, hypotheses, conclusions, and quantifier order matter. For instance,

$$\forall x \in X\ \exists y \in Y\ P(x,y), \qquad \exists y \in Y\ \forall x \in X\ P(x,y)$$

are different: in the first y may depend on x; in the second one y works for all x.

Method 2.3.2. *Proof Grammar and Negation*
In theorem statements, put the hypotheses before the conclusion and keep quantifiers in the order meant.
In particular, switching $\forall$ and $\exists$ usually changes the statement; see Principle 2.3.1.
Before applying a rule, also check that the displayed expressions are well formed; an undefined expression makes the sentence ill formed, not merely false.
The word "or" is inclusive in mathematics, and the basic negation rules are

$$\neg(a \wedge b) \iff \neg a \vee \neg b, \qquad \neg(a \vee b) \iff \neg a \wedge \neg b, \qquad \neg(a \Rightarrow b) \iff a \wedge \neg b$$

Thus a proof by contradiction of $a \Rightarrow b$ assumes both a and $\neg b$ and derives an impossibility.
Key Insight: Before proving or negating a statement, parse its quantifiers and logical connectives.

Placeholder: Add negation rules for quantified statements and examples translating English into symbolic logic.

2.4 Proof Grammar and Inference

Placeholder: Add inference rules such as modus ponens, universal instantiation, existential introduction, and proof templates for definitions.

2.5 Direct, Contrapositive, and Contradiction Proofs

Placeholder: Add direct proof, proof by cases, contrapositive proof, contradiction, biconditional proof, existence proof, uniqueness proof, and disproof by counterexample.

2.6 Induction, Recursion, and Counterexamples

Principle 2.6.1. *Mathematical Induction*
Let $P_1, P_2, \ldots$ be statements indexed by positive integers. To prove P_n for every $n \in \mathbb{N}$, it is enough to prove:

(i) *basis*: P_1 is true;
(ii) *induction step*: for every $n \in \mathbb{N}$, P_n implies P_{n+1}.

The induction step is an implication; it does not prove P_{n+1} in isolation.
The basis starts the chain, and the implication carries truth from one statement to the next.

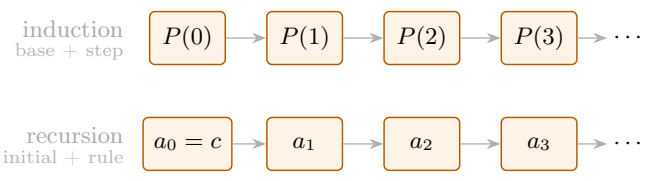


Figure 3: Induction propagates truth, while recursion propagates values along the same index chain.

Principle 2.6.2. *Well-Ordering of $\mathbb{N}$*
Every nonempty subset of $\mathbb{N}$ has a least element, justifying first unused indices, first repeated values, and increasing listings of infinite subsets of $\mathbb{N}$.

Placeholder: Add strong induction, smallest-counterexample arguments, recursion, and the relation between induction and well-ordering of $\mathbb{N}$.

3 Counting and Finite Sums

Counting records finite constructions before cardinality and infinite sets are treated abstractly.

3.1 Lists and Counting Principles

Counting arguments turn structured finite choices into cardinalities. In probability, this is the bridge from a uniform finite sample space to numerical probabilities.

Definition 3.1.1. *Lists, Words, and Product Sets* Let $S_1, \dots, S_p$ be finite sets.

(i) A *list* of length p with ith entry in S_i is an element $(s_1, \dots, s_p) \in S_1 \times \cdots \times S_p$.
(ii) If $S_1 = \cdots = S_p = S$, a list in S^p may be called a *word* of length p over the alphabet S.
(iii) A counting model is *ordered* when changing the order of the selected objects changes the outcome.

Theorem 3.1.2. *Multiplication Principle*
Let $p \geq 1$ and let $S_1, \dots, S_p$ be finite sets with $|S_i| = n_i$. Then $|S_1 \times \cdots \times S_p| = \prod_{i=1}^{p} n_i$.
Equivalently, a p-stage experiment whose ith stage has n_i possible choices after every history has $\prod_{i=1}^{p} n_i$ ordered outcomes; in particular, two stages of sizes m, n give mn ordered outcomes.

Proof. The case $p = 1$ is immediate.
For $p = 2$, write $|S_1| = m$ and $|S_2| = n$, and choose enumerations $S_1 = \{a_1, \dots, a_m\}$ and $S_2 = \{b_1, \dots, b_n\}$.
The map $\{1, \dots, m\} \times \{1, \dots, n\} \to S_1 \times S_2$, $(i, j) \mapsto (a_i, b_j)$, is a bijection, so $|S_1 \times S_2| = mn$.
For $p > 2$, use induction: $|S_1 \times \cdots \times S_p| = |(S_1 \times \cdots \times S_{p-1}) \times S_p| = (\prod_{i=1}^{p-1} n_i) n_p = \prod_{i=1}^{p} n_i$.
Structural Commentary: Outcomes form a finite product, or a constant-branching tree.
Key Insight: Counts multiply because every history has equally many extensions. □

Example 3.1.3. *Finite Words by the Multiplication Principle*

(i) A playing card can be described by a suit and a face value, so there are $4 \cdot 13 = 52$ suit-value combinations.
(ii) There are $2^8 = 256$ binary words of length 8.
(iii) A DNA word of length N over the alphabet $\{A, C, G, T\}$ has 4^N possible sequences.
A codon of length 3 therefore has $4^3 = 64$ possible words.

Key Insight: Independent slots multiply; word counts grow exponentially with length.

Proposition 3.1.4. *Addition, Subtraction, and Division Principles*

(i) Addition: if finite sets $A_1, \dots, A_r$ are pairwise disjoint, then $\left|\bigcup_{i=1}^{r} A_i\right| = \sum_{i=1}^{r} |A_i|$.
(ii) Subtraction: if $B \subseteq A$ and A is finite, then $|A \setminus B| = |A| - |B|$.
(iii) Division: if $f : Y \to X$ is surjective, X, Y are finite, $d \geq 1$, and $\left|f^{-1}(\{x\})\right| = d$ for every $x \in X$. Then $|Y| = d\,|X|$ and $|X| = |Y|/d$.

Proof.

(i) Let $n_i = |A_i|$ and enumerate $A_i = \{a_{i,1}, \dots, a_{i,n_i}\}$. Since the A_i are pairwise disjoint, $(i, k) \mapsto a_{i,k}$ is a bijection $\{(i, k) : 1 \leq i \leq r,\ 1 \leq k \leq n_i\} \to \bigcup_{i=1}^{r} A_i$, so $\left|\bigcup_{i=1}^{r} A_i\right| = \sum_{i=1}^{r} n_i = \sum_{i=1}^{r} |A_i|$.
(ii) Since A is the disjoint union of B and $A \setminus B$, part (i) gives $|A| = |B| + |A \setminus B|$, so $|A \setminus B| = |A| - |B|$.
(iii) For $x \neq x'$, $f^{-1}(\{x\}) \cap f^{-1}(\{x'\}) = \varnothing$, and $Y = \bigcup_{x \in X} f^{-1}(\{x\})$. Thus the fibers partition Y, so by (i), $|Y| = \sum_{x \in X} \left|f^{-1}(\{x\})\right| = \sum_{x \in X} d = d\,|X|$; since $d \geq 1$, $|X| = |Y|/d$.

Structural Commentary: Count parts; quotient equal fibers.
Key Insight: Make representations disjoint, then add their fiber sizes. □

3.2 Permutations, Subsets, and Binomial Coefficients

Definition 3.2.1. *Permutations and Sign*
Let $n \geq 1$. Let $\mathfrak{S}_n$ denote the set of permutations of $\{1, \ldots, n\}$.
Sign of $\sigma \in \mathfrak{S}_n$ is $\operatorname{sign}(\sigma) = 1$ if number of pairs $j < k$ with $\sigma(j) > \sigma(k)$ is even, and $\operatorname{sign}(\sigma) = -1$ otherwise.

Definition 3.2.2. *Factorials and Binomial Coefficients*

(i) The *factorial* is defined by $0! := 1$ and, for $n \in \mathbb{N}$ with $n \geq 1$, $n! := n(n-1)\cdots 3 \cdot 2 \cdot 1$.
(ii) For integers $0 \leq r \leq n$, the *falling factorial* is $(n)_r := n(n-1)\cdots(n-r+1)$, with $(n)_0 := 1$.
(iii) For integers $0 \leq k \leq n$, the *binomial coefficient* is $\binom{n}{k} := \frac{n!}{k!(n-k)!}$.
(iv) The symbol $\binom{n}{k}$ is read *n choose k* and counts the k-element subsets of an n-element set.

Proposition 3.2.3. *Ordered Sampling With and Without Replacement*
Let S be finite with $|S| = n$, let $0 \leq r \leq n$, and set $D_r(S) := \{(s_1, \ldots, s_r) \in S^r : s_i \neq s_j \text{ for all } 1 \leq i < j \leq r\}$. Use $S^0 = \{()\}$ and empty products equal 1.

(i) With replacement: the ordered samples are S^r, and $|S^r| = n^r$.
(ii) Without replacement: the ordered samples are $D_r(S)$, and $|D_r(S)| = (n)_r$, where $(n)_0 = 1$ and $(n)_r = n(n-1)\cdots(n-r+1)$ for $r \geq 1$.
(iii) Full orderings: $|D_n(S)| = n!$.

Proof.

(i) If $r = 0$, then $|S^0| = 1 = n^0$; if $r \geq 1$, Theorem 3.1.2 gives $|S^r| = \prod_{k=1}^{r} n = n^r$.
(ii) For $k = 1, \ldots, r$, after choosing distinct $s_1, \ldots, s_{k-1}$, the entry s_k has $n - k + 1$ choices, so $|D_r(S)| = \prod_{k=1}^{r}(n-k+1) = (n)_r$.
(iii) Taking $r = n$ in (ii) gives $|D_n(S)| = (n)_n = n!$.

Structural Commentary: Replacement counts all tuples; no replacement counts injective tuples.
Key Insight: Order gives slots; replacement determines whether choices stay fixed or decrease. □

Example 3.2.4. *Committees*
The number of ways to choose a committee of 3 people from a class of 20 people is $\binom{20}{3} = \frac{20!}{3!17!} = \frac{20 \cdot 19 \cdot 18}{3 \cdot 2 \cdot 1} = 1140$.

Proposition 3.2.5. *Unordered Samples Without Replacement*
Let S be finite with $|S| = n$, let $0 \leq r \leq n$, and set $\mathcal{P}_r(S) := \{A \subseteq S : |A| = r\}$. Then $|\mathcal{P}_r(S)| = \binom{n}{r} = \frac{(n)_r}{r!} = \frac{n!}{(n-r)!r!}$.

Proof. Let $\pi : D_r(S) \to \mathcal{P}_r(S)$ be $\pi(s_1, \ldots, s_r) = \{s_1, \ldots, s_r\}$.
This map is surjective, and for each $A \in \mathcal{P}_r(S)$, $\pi^{-1}(\{A\})$ consists of the $r!$ orderings of A.
By Proposition 3.2.3, $|D_r(S)| = (n)_r$; by the division principle, $|\mathcal{P}_r(S)| = (n)_r / r!$.
Since $(n)_r = n!/(n-r)!$, the claimed identity follows.
Structural Commentary: Unordered samples are ordered injective samples modulo reordering.
Key Insight: Count ordered first, then divide by the $r!$ orders of each subset. □

Proposition 3.2.6. *Edge Values and Symmetry*
For $n \in \mathbb{N}$ and $k \in \mathbb{Z}$ with $0 \leq k \leq n$:

(i) Edge values: $\binom{n}{0} = \binom{n}{n} = 1$.
(ii) Symmetry: $\binom{n}{k} = \binom{n}{n-k}$.

Proof. Let $n \in \mathbb{N}$ and $k \in \mathbb{Z}$ with $0 \leq k \leq n$.

(i) Since $0! := 1$, $\binom{n}{0} = \frac{n!}{0!n!} = 1$ and $\binom{n}{n} = \frac{n!}{n!0!} = 1$.
(ii) Since $0 \leq n - k \leq n$, $\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{(n-k)!k!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}$.

Structural Commentary: Choosing the selected k objects is equivalent to choosing the omitted $n - k$ objects.
Key Insight: Complements turn k-subsets into $(n-k)$-subsets. □

Proposition 3.2.7. *Binomial Theorem and Subset Count*
For $n \in \mathbb{N}$ and commuting a, b, $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.
If S is finite with $|S| = n$, then $|\mathcal{P}(S)| = 2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Proof. Write $(a+b)^n = \prod_{j=1}^{n}(a+b)$.
For each $I \subseteq \{1, \ldots, n\}$, choose a from factors in I and b from the rest, giving $a^{|I|}b^{n-|I|}$.
Grouping by $k = |I|$ gives $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.
For $|S| = n$, the sets $\mathcal{P}_k(S) := \{A \subseteq S : |A| = k\}$ partition $\mathcal{P}(S)$, so $|\mathcal{P}(S)| = \sum_{k=0}^{n} |\mathcal{P}_k(S)| = \sum_{k=0}^{n} \binom{n}{k}$.

Also $|\mathcal{P}(S)| = 2^n$ by choosing in/out for each element.
Structural Commentary: Expansion and subset counting both group binary choices by size.
Key Insight: A term is determined by which positions choose a. □

Definition 3.2.8. *Multinomial Coefficients*
For nonnegative integers $n_1, \dots, n_r$ with $n_1 + \cdots + n_r = n$, define $\binom{n}{n_1\, n_2 \cdots n_r} := \frac{n!}{n_1! n_2! \cdots n_r!}$.
This counts the ways to divide n labeled positions into r labeled classes of sizes $n_1, \dots, n_r$.

Proposition 3.2.9. *Multinomial Count and Expansion*
Let $r \geq 1$, let $\mathbf{n} = (n_1, \dots, n_r) \in \mathbb{N}_0^r$ with $n = \sum_{i=1}^r n_i$, and let $x_1, \dots, x_r$ commute.
If $\mathcal{M}_{\mathbf{n}}$ is the set of labeled partitions $(A_1, \dots, A_r)$ of $\{1, \dots, n\}$ with $|A_i| = n_i$,
then $|\mathcal{M}_{\mathbf{n}}| = \binom{n}{n_1\, n_2 \cdots n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}$, and $(x_1 + \cdots + x_r)^n = \sum_{m_1 + \cdots + m_r = n} \binom{n}{m_1\, m_2 \cdots m_r} x_1^{m_1} \cdots x_r^{m_r}$.

Proof. Choose the parts successively. After $A_1, \dots, A_{j-1}$ are chosen, there are $n - \sum_{i=1}^{j-1} n_i$ positions left, so A_j has $\binom{n - \sum_{i=1}^{j-1} n_i}{n_j}$ choices. Thus $|\mathcal{M}_{\mathbf{n}}| = \prod_{j=1}^r \binom{n - \sum_{i=1}^{j-1} n_i}{n_j} = \frac{n!}{n_1! \cdots n_r!}$. For the expansion, write each term in $\prod_{k=1}^n (x_1 + \cdots + x_r)$ as a function $\phi : \{1, \dots, n\} \to \{1, \dots, r\}$, giving $\prod_{k=1}^n x_{\phi(k)}$. If $m_i = \left|\phi^{-1}(\{i\})\right|$, then the number of such ϕ is $\binom{n}{m_1\, m_2 \cdots m_r}$; grouping by $(m_1, \dots, m_r)$ gives the identity.
Structural Commentary: Multinomial coefficients count labeled partitions of positions.
Key Insight: Group product terms by how many times each label is chosen. □

Example 3.2.10. *Repeated Letters, Committees, and Pairings*

(i) The number of arrangements of the multiset $\{A, A, G, G, G, G, C, C, C\}$ is $\binom{9}{2\,4\,3} = 9!/(2!4!3!)$.
(ii) A committee of seven split into labeled subcommittees of sizes $3, 2, 2$ can be done in $\binom{7}{3\,2\,2} = 210$ ways.
(iii) If $2m$ people are paired without assigning the pairs to labeled positions, the count is $(2m)!/(m!2^m)$.

3.3 Inclusion-Exclusion, Pigeonhole, and Combinatorial Proof

Definition 3.3.1. *Multisets*
Let S be a set. A *multiset on S* is a pair (S_0, m) such that

(i) $S_0 \subseteq S$ is the *underlying set*;
(ii) $m : S_0 \to \mathbb{Z}_{>0}$ is the *multiplicity function*;
(iii) $m(s)$ is the *multiplicity* of $s \in S_0$;
(iv) if S_0 is finite, the *size* is $|m| := \sum_{s \in S_0} m(s)$.

Equivalently, record (S_0, m) as $\{(s, m(s)) : s \in S_0\}$.

Example 3.3.2. *Multiset Bookkeeping*
The list $\{a, a, b, b, b, c\}$ is the multiset on $\{a, b, c\}$ with multiplicities $2, 3, 1$. This language is useful whenever order is irrelevant but repetition matters, such as eigenvalues counted with algebraic multiplicity.

Proposition 3.3.3. *Finite Inclusion-Exclusion for Two and Three Sets*
Let A, B, C be finite sets, and write $\mathbf{1}_E(x)$ for membership in E.

(i) $|A \cup B| = |A| + |B| - |A \cap B|$.
(ii) $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

Proof.

(i) Let $U = A \cup B$. For $x \in U$, $\mathbf{1}_U(x) = \mathbf{1}_A(x) + \mathbf{1}_B(x) - \mathbf{1}_{A \cap B}(x)$. Sum over U; $|U| = |A| + |B| - |A \cap B|$.
(ii) Let $U = A \cup B \cup C$ and $m(x) := \mathbf{1}_A(x) + \mathbf{1}_B(x) + \mathbf{1}_C(x)$.
For $x \in U$, $m(x) \in \{1, 2, 3\}$ and $m(x) - \binom{m(x)}{2} + \binom{m(x)}{3} = 1$.
Thus each $x \in U$ contributes once to $|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$, QED.

Structural Commentary: Inclusion-exclusion weights each membership pattern to total weight 1.
Key Insight: Add single memberships, subtract overcounts, then restore triple overlaps. □

Definition 3.3.4. *Symmetric Difference*
For sets A and B, the *symmetric difference* is $A \triangle B := (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.
It records the elements belonging to exactly one of A and B.

Method 3.3.5. *Combinatorial Proof by Classification*
To prove $N_L = N_R$ for finite counts:

(i) choose a finite object set Ω and a pattern map $p : \Omega \to P$;
(ii) write both sides as sums over fibers, $N_L = \sum_{\alpha \in P} w_L(\alpha) \left|p^{-1}(\{\alpha\})\right|$ and $N_R = \sum_{\alpha \in P} w_R(\alpha) \left|p^{-1}(\{\alpha\})\right|$;
(iii) verify $w_L(\alpha) = w_R(\alpha)$ for every $\alpha \in P$.

3.4 Finite Sums

Definition 3.4.1. *Finite Sums Indexed by Integers*

Let $a_i \in \mathbb{C}$ be assigned for integers $m \le i \le n$. Define $\sum_{i=m}^{n} a_i := a_m + a_{m+1} + \cdots + a_n$.

If $m > n$, the sum over the empty index set is 0.

Proposition 3.4.2. *Basic Laws for Finite Sums*

Let $a_i, b_i \in \mathbb{C}$ and $c \in \mathbb{C}$ be assigned for $m \le i \le n$.

(i) If $m \le r < n$, then $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) $\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i$ and $\sum_{i=m}^{n} ca_i = c \sum_{i=m}^{n} a_i$.

(iii) $\left| \sum_{i=m}^{n} a_i \right| \le \sum_{i=m}^{n} |a_i|$.

(iv) If $a_i, b_i \in \mathbb{R}$ and $a_i \le b_i$ for every i, then $\sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i$.

Proof.

(i) By Definition 3.4.1 and associativity of addition in $\mathbb{C}$, $\sum_{i=m}^{n} a_i = \sum_{i=m}^{r} a_i + \sum_{i=r+1}^{n} a_i$.

(ii) This follows from associativity, commutativity, and distributivity in the field $\mathbb{C}$.

(iii) Induct on $n - m$. The case $n = m$ is immediate.
If the claim holds up to n, then the triangle inequality gives

$$\left| \sum_{i=m}^{n+1} a_i \right| \le \left| \sum_{i=m}^{n} a_i \right| + |a_{n+1}| \le \sum_{i=m}^{n} |a_i| + |a_{n+1}| = \sum_{i=m}^{n+1} |a_i| .$$

(iv) Induct on $n - m$. The case $n = m$ is the hypothesis $a_m \le b_m$.
If the claim holds up to n, then compatibility of the order on $\mathbb{R}$ with addition gives

$$\sum_{i=m}^{n+1} a_i = \sum_{i=m}^{n} a_i + a_{n+1} \le \sum_{i=m}^{n} b_i + b_{n+1} = \sum_{i=m}^{n+1} b_i.$$

Structural Commentary: Finite sums are iterated addition, so their laws are inherited from the scalar field.
Key Insight: No convergence hypothesis is needed for a finite sum. □

Definition 3.4.3. *Summation over a Finite Set*

Let X be a finite set with $\#X = n$, and let $f : X \to \mathbb{C}$.
Choose any bijection $g : J_n \to X$, where $J_n = \{1, \ldots, n\}$, and set $\sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i))$.

When $X = \emptyset$, this is the empty sum 0.
Key Insight: Proposition 3.4.4 shows that the value is independent of the chosen bijection.

Proposition 3.4.4. *Finite Set Summation Laws*

Let X, Y be finite sets.

(i) Well-definedness: different bijections $J_{\#X} \to X$ give the same value in Definition 3.4.3.

(ii) Substitution: if $h : Y \to X$ is a bijection and $f : X \to \mathbb{C}$, then $\sum_{x \in X} f(x) = \sum_{y \in Y} f(h(y))$.

(iii) Linearity: for $f, g : X \to \mathbb{C}$ and $c \in \mathbb{C}$, $\sum_{x \in X} (f(x)+g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x)$ and $\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x)$.

(iv) Disjoint unions: if $X \cap Y = \emptyset$ and $f : X \cup Y \to \mathbb{C}$, then $\sum_{z \in X \cup Y} f(z) = \sum_{x \in X} f(x) + \sum_{y \in Y} f(y)$.

(v) Finite Fubini: if $f : X \times Y \to \mathbb{C}$, then $\sum_{x \in X} \sum_{y \in Y} f(x, y) = \sum_{(x,y) \in X \times Y} f(x, y) = \sum_{y \in Y} \sum_{x \in X} f(x, y)$.

Proof.

(i) For $f : X \to \mathbb{C}$, let $n = \#X$ and let $g, k : J_n \to X$ be bijections. Induct on n. The case $n = 0$ is empty. If $n \ge 1$, let $j \in J_n$ satisfy $k(j) = g(n)$. Removing n from g and j from k gives bijections onto $X \setminus \{g(n)\}$; after relabeling by J_{n-1}, the induction hypothesis makes the remaining sums equal. Adding the common last term $f(g(n)) = f(k(j))$ gives equality of the two sums.

(ii) Let $g : J_{\#Y} \to Y$ be a bijection. Then $h \circ g : J_{\#Y} \to X$ is a bijection, so $\#Y = \#X$ by Proposition 6.1.3.
Hence, by (i) and Definition 3.4.3, $\sum_{x \in X} f(x) = \sum_{i=1}^{\#Y} f(h(g(i))) = \sum_{y \in Y} f(h(y))$.

(iii) Choose a bijection $\gamma : J_{\#X} \to X$.
Proposition 3.4.2 gives $\sum_{x\in X}(f(x)+g(x)) = \sum_{x\in X} f(x) + \sum_{x\in X} g(x)$ and $\sum_{x\in X} cf(x) = c\sum_{x\in X} f(x)$

(iv) Let $\alpha : J_{\#X} \to X$ and $\beta : J_{\#Y} \to Y$ be bijections, and enumerate $X \cup Y$ by listing α first and then β.
Since $X \cap Y = \emptyset$, this is a bijection, and Proposition 3.4.2 gives $\sum_{z\in X\cup Y} f(z) = \sum_{x\in X} f(x) + \sum_{y\in Y} f(y)$.

(v) Let $\alpha : J_{\#X} \to X$ and $\beta : J_{\#Y} \to Y$ be bijections.
The maps $\ell((i-1)\#Y + j) = (\alpha(i), \beta(j))$ and $r((j-1)\#X + i) = (\alpha(i), \beta(j))$ for $i \in J_{\#X}$ and $j \in J_{\#Y}$ give the two lexicographic bijections $J_{(\#X)(\#Y)} \to X \times Y$.
Applying Definition 3.4.3, Proposition 3.4.2, and (i) yields $\sum_{x\in X}\sum_{y\in Y} f(x,y) = \sum_{(x,y)\in X\times Y} f(x,y) = \sum_{y\in Y}\sum_{x\in X} f(x,y)$.

Structural Commentary: A finite set sum is an order-free finite sum.
Key Insight: Bijections are bookkeeping devices; the value does not depend on the bookkeeping order. □

4 Relations, Orders, and Lattices

Relations package comparisons, identifications, order structures, and quotient constructions.

4.1 Relations and Equivalence

Definition 4.1.1. *Relations and Equivalence*
A *relation* from X to Y is a subset $R \subseteq X \times Y$. If $(x, y) \in R$, one often writes xRy.
A relation $\sim$ on X is an *equivalence relation* if, for all $x, y, z \in X$,

i. $x \sim x$;
ii. $x \sim y$ implies $y \sim x$;
iii. $x \sim y$ and $y \sim z$ imply $x \sim z$.

The equivalence class of x is $[x] := \{y \in X : y \sim x\}$.

4.2 Partitions, Congruence, and Quotients

Proposition 4.2.1. *Equivalence Classes Partition the Set*
Let $\sim$ be an equivalence relation on X. Then the equivalence classes form a partition of X.

Proof. Every $x \in X$ lies in $[x]$ since $x \sim x$. If $[x] \cap [y] \neq \emptyset$, choose z in the intersection.
Then $x \sim z \sim y$, so $u \in [x]$ implies $u \sim y$ and $u \in [y]$. Hence $[x] \subseteq [y]$, and symmetry gives equality.
Structural Commentary: Equivalence means partition.
Key Insight: Touching classes coincide. □

Placeholder: Add congruence modulo n, quotient sets, and well-defined operations on equivalence classes.

4.3 Ordered Sets, Well-Orders, and Lattices

Definition 4.3.1. *Ordered Set Vocabulary*
Let S be a set, let $E, C \subseteq S$, and let $\leq_S$ be a relation on S. Write $x <_S y$ iff $x \leq_S y$ and $x \neq y$.

(i) $\leq_S$ is a *partial order* if it is reflexive, antisymmetric, and transitive; then $(S, \leq_S)$ is a *poset*.
(ii) $\leq_S$ is an *order* if it is a partial order and any $x, y \in S$ are comparable; then $(S, \leq_S)$ is an *ordered set*.
(iii) C is a *chain* if $\leq_S |_C$ is an order.
(iv) S is *well-ordered* if every nonempty $Y \subseteq S$ has $\ell \in Y$ with $\ell \leq_S y$ for all $y \in Y$.
(v) $\beta \in S$ is an *upper bound* of E if $x \leq_S \beta$ for all $x \in E$; lower bounds reverse the inequality.
(vi) E is *bounded above* if it has an upper bound; bounded below is dual.
(vii) $\sup E$ is the least upper bound of E, and $\inf E$ is the greatest lower bound of E, when these exist.
(viii) $m \in E$ is *maximal* if no $x \in E$ satisfies $m <_S x$; minimal is dual.
(ix) S has the *least-upper-bound property* if every nonempty bounded-above $E \subseteq S$ has $\sup E \in S$.

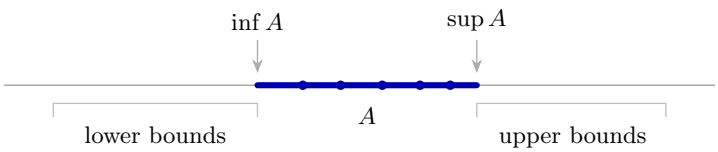


Figure 4: The supremum is the least upper bound, while the infimum is the greatest lower bound.

Example 4.3.2. *Suprema Need Not Be Elements*
In $\mathbb{Q}$, the set $E_1 = \{r \in \mathbb{Q} : r < 0\}$ and the set $E_2 = \{r \in \mathbb{Q} : r \leq 0\}$ both have supremum 0, but $0 \notin E_1$ and $0 \in E_2$. If $E = \{1/n : n = 1, 2, 3, \ldots\}$, then $\sup E = 1$ while $\inf E = 0$, and the infimum is not a member of E.

Theorem 4.3.3. *Least Upper Bounds Give Greatest Lower Bounds*
If S has the least-upper-bound property, then every nonempty bounded-below set $B \subseteq S$ has an infimum.

Proof. Let L be the set of lower bounds of B. Then L is nonempty and every $x \in B$ is an upper bound of L, so $\alpha = \sup L$ exists. Since every $x \in B$ bounds L above, $\alpha \leq x$ for all $x \in B$; hence α is a lower bound of B. If $\beta > \alpha$, then $\beta \notin L$, so β is not a lower bound. Thus $\alpha = \inf B$.
Structural Commentary: Infima are suprema of lower bounds.
Key Insight: Completeness above gives completeness below. □

Definition 4.3.4. *Lattice Vocabulary*
Let $(L, \leq)$ be a partially ordered set.

(i) A *meet* of $x, y \in L$ is $\inf\{x, y\}$ when it exists, often written $x \wedge y$.
(ii) A *join* of $x, y \in L$ is $\sup\{x, y\}$ when it exists, often written $x \vee y$.
(iii) A *lattice* is a partially ordered set in which every two elements have a meet and a join.
(iv) A lattice is *complete* if every subset $E \subseteq L$ has an infimum and a supremum.
(v) A *bottom* element is an element $\bot \in L$ with $\bot \leq x$ for every $x \in L$; a *top* element is an element $\top \in L$ with $x \leq \top$ for every $x \in L$.
(vi) A lattice is *modular* if $x \leq z$ implies $z \wedge (y \vee x) = (z \wedge y) \vee x$ for all $x, y, z \in L$.

4.4 Relations Between Sets and Closures

Placeholder: Add relations between different sets, inverse relations, relation composition, reflexive/transitive closure, and order-isomorphism examples.

5 Functions

Functions are relations with unique outputs; their images, preimages, restrictions, compositions, and inverses organize nearly every later construction.

5.1 Functions, Images, and Preimages

Definition 5.1.1. *Functions, Images, and Preimages*
Let $f : X \to Y$ be a *function*, assigning each $x \in X$ a unique $f(x) \in Y$.

(i) $X = \mathrm{dom}(f)$ is the *domain* and Y is the *codomain*;
(ii) $\mathrm{range}(f) := f(X) := \{f(x) : x \in X\}$ is the *range* of f;
(iii) for $A \subseteq X$, $f(A) := \{f(x) : x \in A\}$ is the *image* of A;
(iv) for $B \subseteq Y$, $f^{-1}(B) := \{x \in X : f(x) \in B\}$ is the *preimage* of B; no inverse function is required;
(v) if Y has distinguished 0, $\mathrm{supp}(f) := \{x \in X : f(x) \neq 0\}$;

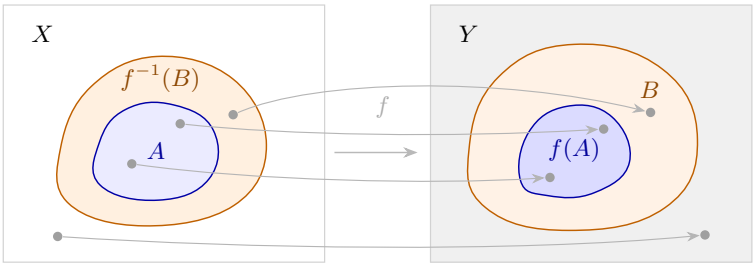


Figure 5: Image–preimage containment: $f(A) \subseteq B \iff A \subseteq f^{-1}(B)$.

Proposition 5.1.2. *Image and Preimage Calculus*
Let $f : X \to Y$, let $\{A_i\}_{i \in I}$ be a nonempty family of subsets of X, and let $\{B_i\}_{i \in I}$ be a nonempty family of subsets of Y. Then

(i) $f\left(\bigcup_{i \in I} A_i\right) = \bigcup_{i \in I} f(A_i)$;
(ii) $f\left(\bigcap_{i \in I} A_i\right) \subseteq \bigcap_{i \in I} f(A_i)$;

(iii) $f^{-1}\left(\bigcup_{i\in I} B_i\right) = \bigcup_{i\in I} f^{-1}(B_i)$;
(iv) $f^{-1}\left(\bigcap_{i\in I} B_i\right) = \bigcap_{i\in I} f^{-1}(B_i)$;
(v) $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$ *for every* $B \subseteq Y$;
(vi) $f(f^{-1}(B)) = B \cap \operatorname{range}(f)$ *for every* $B \subseteq Y$.

Proof.

(i) $y \in f(\bigcup_i A_i)$ iff $y = f(x)$ for some $x \in A_i$ and some i, iff $y \in \bigcup_i f(A_i)$.
(ii) If $y = f(x)$ with $x \in \bigcap_i A_i$, then $y \in f(A_i)$ for every i.
(iii) $x \in f^{-1}(\bigcup_i B_i)$ iff $f(x) \in B_i$ for some i, iff $x \in \bigcup_i f^{-1}(B_i)$.
(iv) $x \in f^{-1}(\bigcap_i B_i)$ iff $f(x) \in B_i$ for every i, iff $x \in \bigcap_i f^{-1}(B_i)$.
(v) For $x \in X$, $x \in f^{-1}(Y \setminus B) \iff f(x) \notin B \iff x \notin f^{-1}(B) \iff x \in X \setminus f^{-1}(B)$
(vi) $y \in f(f^{-1}(B))$ iff $y = f(x)$ for some x with $f(x) \in B$, iff $y \in B \cap \operatorname{range}(f)$.

Structural Commentary: Preimages are Boolean-exact; images need not be.
Key Insight: Pull back conditions through f. □

5.2 Injective, Surjective, and Bijective Maps

Definition 5.2.1. *Injective, Surjective, and Bijective Maps*
Let $f : X \to Y$.

i. f is *injective* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
ii. f is *surjective* if $\operatorname{range}(f) = Y$.
iii. f is *bijective* if it is both injective and surjective.

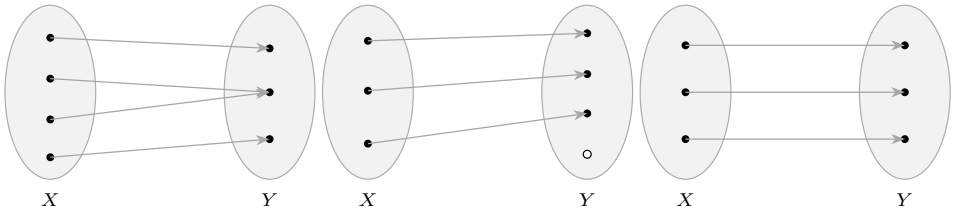


Figure 6: Surjective: covers codomain; injective: no collisions; bijective: both.

Example 5.2.2. *Image and Preimage for the Square Map*
Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$. Then $f([-2,1]) = [0,4]$, $f^{-1}([1,4]) = [-2,-1] \cup [1,2]$. The map is not injective because $f(1) = f(-1)$, and it is not surjective as a map $\mathbb{R} \to \mathbb{R}$ because no real number maps to -1.

5.3 Composition, Restriction, and Inverse Functions

Definition 5.3.1. *Composition, Restriction, Extension, and Identity*
Let $f : X \to Y$.

(i) for $A \subseteq X$, $f|_A : A \to Y$ is the *restriction* of f to A;
(ii) if $X \subseteq U$, an *extension* of f to U is $\widetilde{f} : U \to Y$ with $\widetilde{f}|_X = f$;
(iii) if $g : Y \to Z$, then $g \circ f : X \to Z$ is given by $(g \circ f)(x) = g(f(x))$;
(iv) $\mathrm{id}_X : X \to X$ denotes the *identity function*.

Theorem 5.3.2. *Bijective Means Invertible*
A function $f : X \to Y$ is bijective iff there exists a function $g : Y \to X$ s.t. $g \circ f = \mathrm{id}_X$, $f \circ g = \mathrm{id}_Y$. When such a function exists, it is unique and is denoted $f^{-1} : Y \to X$.

Proof.

($\Rightarrow$) If f is bijective, define $g(y)$ as the unique x with $f(x) = y$. Then $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$.
($\Leftarrow$) If such a g exists, applying g to $f(x_1) = f(x_2)$ gives $x_1 = x_2$, and $y = f(g(y))$ gives surjectivity.

If g and h are two-sided inverses, then $g(y) = g(f(h(y))) = h(y)$, so the inverse is unique.
Structural Commentary: A true inverse undoes f on both sides.
Key Insight: Bijective means exactly one preimage per output. □

Placeholder: Add algebra of composition and image/preimage laws under composition.

6 Cardinality, Choice, and Infinite Sets

Cardinality compares sets by functions rather than by informal size.

6.1 Same Cardinality, Finite, Countable, and Uncountable

Definition 6.1.1. *Same Cardinality*
Two sets A and B have *same cardinality*, or are *equivalent*, if $\exists$ a bijection $f : A \to B$. Then $A \sim_{\#} B$.
The relation $\sim_{\#}$ is an equivalence relation: $A \sim_{\#} A$ by identity map, symmetry follows from Theorem 5.3.2, and transitivity follows by composing bijections.

Definition 6.1.2. *Finite, Countable, and Uncountable*
Set $J_0 := \emptyset$, and for $n \geq 1$ set $J_n := \{1, 2, \ldots, n\}$. A set A is

(i) *finite* if $A \sim_{\#} J_n$ for some $n \in \mathbb{N}$;
(ii) *infinite* if it is not finite;
(iii) *countable* if $A \sim_{\#} \mathbb{N}$;
(iv) *uncountable* if it is neither finite nor countable;
(v) *at most countable* if it is finite or countable.

Countable sets are also called enumerable or denumerable.

Proposition 6.1.3. *Finite Cardinality is Well Defined*
If $A \sim_{\#} J_m$ and $A \sim_{\#} J_n$, then $m = n$. Thus $\#A = n$ is well defined for finite sets.

Proof. By transitivity of $\sim_{\#}$, it suffices to prove $J_m \sim_{\#} J_n \Rightarrow m = n$. Induct on m.
For $m = 0$, only $J_0 = \emptyset$ is bijective to J_0.
If $m, n \geq 1$ and $f : J_m \to J_n$ is bijective, remove m and its partner $f(m)$. The remaining sets are bijective and relabel as J_{m-1} and J_{n-1}, so induction gives $m - 1 = n - 1$, hence $m = n$.
Structural Commentary: Finite counting is invariant under bijection.
Key Insight: Remove one matched element and count down. □

Definition 6.1.4. *Sequences as Enumerations*
A *sequence* in set A is $x : \mathbb{N} \to A$, written (x_n), where $x_n := x(n)$. Terms of sequence need not be distinct.
If A is countable, then its elements can be arranged as a sequence with distinct terms. Conversely, a sequence whose terms are distinct and whose range is A gives a bijection $\mathbb{N} \to A$.

Example 6.1.5. *The Integers Are Countable*
The set $\mathbb{Z}$ is countable. One explicit enumeration is given by

$$f(n) = \begin{cases} \frac{n}{2}, & n \text{ even}, \\ -\frac{n-1}{2}, & n \text{ odd}, \end{cases} \qquad n \in \mathbb{N}.$$

This lists $\mathbb{Z}$ as $0, 1, -1, 2, -2, 3, -3, \ldots$.

Theorem 6.1.6. *Infinite Subsets of Countable Sets*
Every infinite subset of a countable set is countable.

Proof. Let $E \subseteq A$ be infinite, and enumerate A by distinct terms (x_n). Let $n_1 < n_2 < \cdots$ be exactly the indices for which $x_{n_k} \in E$, listed in increasing order. Then $k \mapsto x_{n_k}$ is a bijection from $\mathbb{N}$ onto E.
Structural Commentary: The proof keeps only the entries of the original enumeration that land in E.
Key Insight: An infinite subset of an enumeration inherits an enumeration by first appearances. □

Proposition 6.1.7. *Subsets and Images of At Most Countable Sets*
Let A be at most countable.

(i) If $B \subseteq A$, then B is at most countable.
(ii) If $f : A \to Y$ is a function, then $f(A)$ is at most countable.

Proof. If A is finite, both claims are immediate. Otherwise write $A = \{a_n : n \in \mathbb{N}\}$ with the a_n distinct.

(i) If B is infinite, let $I_B := \{n : a_n \in B\} = \{n_1 < n_2 < \cdots\}$.
Then $k \mapsto a_{n_k}$ bijects $\mathbb{N}$ onto B; the finite case is trivial.
(ii) Let $I := \{n : f(a_n) \notin \{f(a_m) : m < n\}\}$. By (i), I is at most countable, $n \mapsto f(a_n)$ bijects I onto $f(A)$.

Structural Commentary: Subsets select indices; images select first appearances.
Key Insight: Subsets and images preserve at-most-countability. □

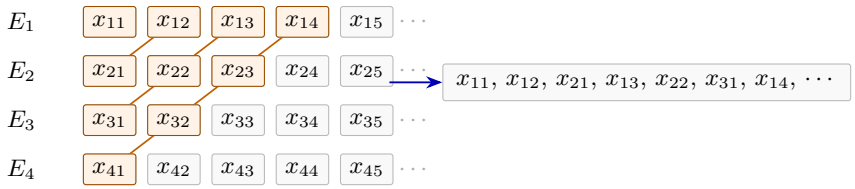


Figure 7: Diagonal scan of countable rows into one sequence, skipping repeats.

Theorem 6.1.8. *Countable Union of Countable Sets*
Let (E_n) be a sequence of countable sets and put $S := \bigcup_{n=1}^{\infty} E_n$. Then S is countable.

Proof. Enumerate each E_n as $(x_{nk})_{k\in\mathbb{N}}$.
Reading the array (x_{nk}) by diagonals, as in Figure 7, gives a sequence whose image is S.
Thus S is at most countable by Proposition 6.1.7.
Since $E_1 \subseteq S$ and E_1 is countable, S is infinite; hence S is countable.
Structural Commentary: The diagonal order turns a two-indexed list into a one-indexed list.
Key Insight: Countably many countable rows can still be scanned by one sequence. □

Corollary 6.1.9. *At Most Countable Unions*
If A is at most countable and B_α is at most countable for each $\alpha \in A$, then $\bigcup_{\alpha\in A} B_\alpha$ is at most countable.

Proof. Let $T := \bigcup_{\alpha\in A} B_\alpha$. If $T = \emptyset$, we are done. Otherwise enumerate the nonempty sets among the B_α as $B_{\alpha_1}, B_{\alpha_2}, \ldots$, repeating indices if needed, and list each B_{α_j} as $(b_{jk})_{k\in\mathbb{N}}$.
A diagonal scan of (b_{jk}) has image T, so Proposition 6.1.7 makes T at most countable.
Structural Commentary: Finite pieces cause no difficulty; they can be embedded into countable rows.
Key Insight: At most countable indexing + at most countable indexed sets gives at most countable total set. □

Theorem 6.1.10. *Finite Tuples over a Countable Set*
Let A be countable. For each fixed $n \in \mathbb{N}$, the set of all n-tuples $(a_1, \ldots, a_n)$ with $a_k \in A$ is countable.

Proof. Let $B_n := \{(a_1, \ldots, a_n) : a_k \in A\}$, $B_1 = A$. If B_{n-1} is countable, $B_n = \bigcup_{b\in B_{n-1}} \{(b, a) : a \in A\}$, each set $\{(b, a) : a \in A\}$ is equivalent to A. Hence B_n is countable by Theorem 6.1.8.
Structural Commentary: Build an n-tuple by appending one coordinate to an $(n-1)$-tuple.
Key Insight: Induction plus countable unions controls finite products. □

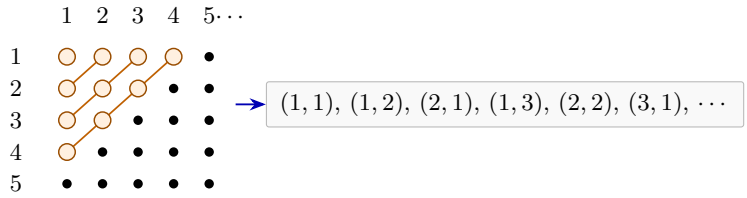


Figure 8: Diagonal enumeration of $\mathbb{N} \times \mathbb{N}$ by finite $i + j$ diagonals.

Corollary 6.1.11. *Countable Cartesian Products*
If A, B are at most countable, then $A \times B$ is at most countable.
If A, B are countable and nonempty, then $A \times B$ is countable. Hence $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. For $a \in A$, $\{a\} \times B \sim B$, and $A \times B = \bigcup_{a\in A} \big(\{a\} \times B\big)$, thus $A \times B$ at most countable (Corollary 6.1.9). If A, B are countable and $b_0 \in B$, then $a \mapsto (a, b_0)$ injects A into $A \times B$, so $A \times B$ is infinite; hence it is countable.
Structural Commentary: A product is a countable union of countable slices.
Key Insight: Two countable coordinates can still be scanned by one sequence. □

Corollary 6.1.12. *The Rationals Are Countable*
The set $\mathbb{Q}$ is countable.

Proof. Let $D := \{(a,b) \in \mathbb{Z} \times \mathbb{Z} : a \neq 0\}$ and $\pi : D \to \mathbb{Q}$ by $\pi(a,b) = b/a$.
Since $\mathbb{Z}$ is countable, Corollary 6.1.11 and Proposition 6.1.7 make D at most countable. Since $\pi(D) = \mathbb{Q}$, Proposition 6.1.7 makes $\mathbb{Q}$ at most countable. As $\mathbb{Z} \subseteq \mathbb{Q}$, it is countable.
Structural Commentary: Repeated fractions only add repetitions to the list.
Key Insight: Countable pairs produce countably many rational values. □

6.2 Diagonalization and Power Sets

Theorem 6.2.1. *Cantor's Theorem*
For every set X, no function $f : X \to \mathcal{P}(X)$ is surjective. Hence X and $\mathcal{P}(X)$ cannot have the same cardinality.

Proof. Suppose $f : X \to \mathcal{P}(X)$ were surjective, and set $D := \{x \in X : x \notin f(x)\}$.
Since $D \subseteq X$, we have $D \in \mathcal{P}(X)$, so surjectivity gives $D = f(a)$ for some $a \in X$.
But then $a \in D \iff a \notin f(a) \iff a \notin D$, a contradiction.
Structural Commentary: The diagonal set is defined to disagree with each proposed row at its own index.
Key Insight: Power sets are too large to be exhausted by their base set. □

$$\begin{array}{lcl} s_1 & = & \boxed{a_{11}}\ a_{12}\ a_{13}\ a_{14}\ a_{15} \\ s_2 & = & a_{21}\ \boxed{a_{22}}\ a_{23}\ a_{24}\ a_{25} \\ s_3 & = & a_{31}\ a_{32}\ \boxed{a_{33}}\ a_{34}\ a_{35}\ \cdots \\ s_4 & = & a_{41}\ a_{42}\ a_{43}\ \boxed{a_{44}}\ a_{45} \\ s_5 & = & a_{51}\ a_{52}\ a_{53}\ a_{54}\ \boxed{a_{55}} \end{array}$$

Figure 9: Cantor diagonalization: $s(n) = 1 - a_{nn}$ differs from row s_n at position n.

Theorem 6.2.2. *Binary Sequences Are Uncountable*
Let A be the set of all sequences whose terms are the digits 0 and 1. Then A is uncountable.

Proof. Suppose A were countable, $A = \{s_1, s_2, \ldots\}$ with $s_n = (s_n(k))_{k \in \mathbb{N}}$. Define $s \in A$ by $s(k) := 1 - s_k(k)$.
Then $s(n) \neq s_n(n)$ for every n, so $s \neq s_n$ for all n. Hence $s \notin A$, a contradiction. Thus A is uncountable.
Structural Commentary: The construction attacks the diagonal of a proposed enumeration.
Key Insight: A sequence can be built to escape every row of a countable list. □

Definition 6.2.3. *Continuum Cardinal*
The *continuum cardinal* is $\mathfrak{c} := 2^{\aleph_0} := \left|\{0,1\}^{\mathbb{N}}\right|$. Equivalently, $\mathfrak{c} = |\mathcal{P}(\mathbb{N})|$ via $s \mapsto \{n \in \mathbb{N} : s(n) = 1\}$.

Perspective 6.2.4. *Continuum Cardinality*
Theorem 6.2.2 and Theorem 6.2.1 give $\aleph_0 < \mathfrak{c}$.
Thus the continuum is the first cardinal jump produced by binary choice over $\mathbb{N}$.

6.3 Schroeder–Bernstein and Cardinal Comparison

Theorem 6.3.1. *Schroeder–Bernstein Theorem*
If there are injections $f : A \to B$ and $g : B \to A$, then there is a bijection $h : A \to B$.

Proof. Let $A_0 := A \setminus g(B)$, and recursively define $B_n := f(A_n)$ and $A_{n+1} := g(B_n)$.
The sets A_n are pairwise disjoint: if $A_i \cap A_j \neq \emptyset$ with $i < j$, then $i = 0$ already contradicts $A_j \subseteq g(B)$; if $i > 0$, injectivity of g and then f steps backward to $A_{i-1} \cap A_{j-1} \neq \emptyset$, and iteration gives the same contradiction.
Since f is injective, the sets $B_n = f(A_n)$ are pairwise disjoint as well.
Let $A_* := A \setminus \bigcup_{n \geq 0} A_n$ and $B_* := B \setminus \bigcup_{n \geq 0} B_n$. The map g restricts to a bijection $B_* \to A_*$.
Indeed, $g(B_*)$ avoids every A_n, while if $a \in A_*$ then $a \notin A_0$, so $a = g(b)$ for a unique $b \in B$; this b cannot lie in any B_n, otherwise $a \in A_{n+1}$.
Define $h(x) := \begin{cases} f(x), & x \in \bigcup_{n \geq 0} A_n, \\ g^{-1}(x), & x \in A_*, \end{cases}$ where g^{-1} denotes the inverse of $g : B_* \to A_*$.
This maps each A_n bijectively onto B_n and maps A_* bijectively onto B_*. Hence h is a bijection $A \to B$.
Structural Commentary: Two injections carve both sets into matched layers plus a common remainder.
Key Insight: Use f on the escaping layers and g^{-1} on the stable remainder. □

Definition 6.3.2. *Cardinal Comparison and Arithmetic*
Let A, B be sets, and let $\kappa = |A|$, $\lambda = |B|$.

(i) *Cardinal comparison*: $|A| \leq |B|$ iff $\exists$ an injection $A \to B$.
(ii) *Strict cardinal comparison*: $|A| < |B|$ iff $|A| \leq |B|$ and $A \not\sim_{\#} B$.
(iii) If $A \cap B = \emptyset$, the *cardinal sum* is $\kappa + \lambda := |A \cup B|$.
(iv) The *cardinal product* is $\kappa\lambda := |A \times B|$.
(v) The *function set* is $A^B := \{f : B \to A\}$, and the *cardinal power* is $\kappa^\lambda := \left|A^B\right|$.

6.4 Cardinal Arithmetic

Proposition 6.4.1. *Cardinal Arithmetic Reference Laws*
Let κ, λ, μ be cardinal numbers.

(i) $\kappa + \lambda = \lambda + \kappa$ *and* $\kappa\lambda = \lambda\kappa$;
(ii) $\kappa + (\lambda + \mu) = (\kappa + \lambda) + \mu$ *and* $\kappa(\lambda\mu) = (\kappa\lambda)\mu$;
(iii) $\kappa(\lambda + \mu) = \kappa\lambda + \kappa\mu$;
(iv) $\kappa^{\lambda+\mu} = \kappa^\lambda \kappa^\mu$, $(\kappa^\lambda)^\mu = \kappa^{\lambda\mu}$, *and* $(\kappa\lambda)^\mu = \kappa^\mu \lambda^\mu$.

Proof. Choose pairwise disjoint sets A, B, C with $|A| = \kappa$, $|B| = \lambda$, and $|C| = \mu$.

(i) Use $A \cup B \cong B \cup A$ and $A \times B \cong B \times A$.
(ii) Use $A \cup (B \cup C) \cong (A \cup B) \cup C$ and $A \times (B \times C) \cong (A \times B) \times C$.
(iii) Use $A \times (B \cup C) \cong (A \times B) \cup (A \times C)$.
(iv) Use $A^{B \cup C} \cong A^B \times A^C$, $(A^B)^C \cong A^{B \times C}$, and $(A \times B)^C \cong A^C \times B^C$.

Taking cardinalities gives (i)–(iv).
Structural Commentary: The laws are canonical bijections.
Key Insight: Cardinal arithmetic is set construction up to bijection. □

Theorem 6.4.2. *Infinite Cardinal Absorption*
Let κ, λ be cardinals and set $\mu := \max\{\kappa, \lambda\}$. If μ is infinite, then $\kappa + \lambda = \mu$.
If also $\kappa, \lambda > 0$, then $\kappa\lambda = \mu$. In particular, $\aleph_0 + \aleph_0 = \aleph_0 \cdot \aleph_0 = \aleph_0$.

Proof. For every infinite cardinal ν, the standard square law gives $\nu + \nu = \nu^2 = \nu$.
Since $\kappa, \lambda \leq \mu$, $\mu \leq \kappa + \lambda \leq \mu + \mu = \mu$, so $\kappa + \lambda = \mu$.
If $\kappa, \lambda > 0$, choose nonempty representatives $|A| = \kappa$ and $|B| = \lambda$; fixing one point in the smaller factor injects the larger factor into $A \times B$, so $\mu \leq \kappa\lambda$. Also $\kappa\lambda \leq \mu^2 = \mu$, so $\kappa\lambda = \mu$.
Structural Commentary: Infinite cardinals absorb smaller summands and nonzero factors.
Key Insight: Squeeze between μ and $\mu^2 = \mu$. □

6.5 Choice, Zorn's Lemma, and Well-Ordering

Principle 6.5.1. *Choice Principle*
For nonempty sets $\{X_i\}_{i \in I}$, the axiom of choice asserts $\exists c : I \to \bigcup_{i \in I} X_i$ such that $c(i) \in X_i$ $\forall i \in I$.
Use direct construction for finite or explicit choices; use choice only for arbitrary simultaneous choices.

Principle 6.5.2. *Zorn's Lemma*
If a nonempty partially ordered set X has the property that every chain in X has an upper bound in X, then X contains a maximal element. This is equivalent in strength to the axiom of choice, Principle 6.5.1.

Method 6.5.3. *Cardinality Proof Patterns*

(i) Same cardinality: construct a bijection.
(ii) Countable: construct a sequence whose range contains every element.
(iii) Uncountable: show every proposed sequence misses some element.
(iv) Choice/maximality: cite Principle 6.5.1 for arbitrary simultaneous choices and Principle 6.5.2 for maximal elements in partially ordered sets.

Key Idea: Use maps: bijections prove equal cardinality, enumerations $\mathbb{N} \to A$ prove countability, and diagonal arguments show every proposed enumeration misses an element.

Placeholder: Add the well-ordering theorem, maximality methods, and examples of where choice is avoidable versus structurally necessary.

7 Axiomatic and Advanced Set Theory

This section reserves space for the deeper set-theoretic material suggested by the set-theory and metatheory contents: axioms, ordinals, models, forcing, large cardinals, and sets of reals.

7.1 Extensionality, Specification, and Replacement

Placeholder: Add ZF-style axiom statements, informal motivations, and the separation/replacement distinction.

7.2 Ordinals, Transfinite Recursion, and Well-Foundedness

Placeholder: Add well-founded relations, transitive sets, ordinal numbers, ordinal arithmetic, transfinite induction, and transfinite recursion.

7.3 Models, Constructibility, and Independence

Placeholder: Add models of set theory, the constructible universe, forcing previews, consistency strength, and independence statements.

7.4 Filters, Ultrafilters, and Sets of Reals

Placeholder: Add filters, ideals, ultrafilters, stationary/closed-unbounded sets, Borel and projective sets, descriptive set theory, and large-cardinal gateways.

Part II
Structural Linear Algebra

1 Vector Spaces and Linear Structure

1.1 Vector Spaces

Vector spaces encode addition and scalar multiplication. Subspaces, spans, bases, dimension, and coordinates describe how linear structure is generated and measured.

Definition 1.1.1. *Vector Space*
Let $\mathbb{F}$ be a field. A vector space over $\mathbb{F}$ is a set $\mathcal{V}$ with vector addition and scalar multiplication such that, $\forall\, \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $\forall\, a, b \in \mathbb{F}$, the following axioms hold:

i. vector addition is commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
ii. vector addition is associative: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
iii. there exists an additive identity $\mathbf{0} \in \mathcal{V}$ such that $\mathbf{0} + \mathbf{v} = \mathbf{v}$;
iv. for each $\mathbf{v} \in \mathcal{V}$ there exists an additive inverse $-\mathbf{v} \in \mathcal{V}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$;
v. scalar multiplication has identity: $1\mathbf{v} = \mathbf{v}$;
vi. scalar multiplication is compatible with multiplication in $\mathbb{F}$: $(ab)\mathbf{v} = a(b\mathbf{v})$;
vii. scalar multiplication distributes over vector addition: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$;
viii. scalar multiplication distributes over addition in $\mathbb{F}$: $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

If $\mathcal{V} = \{\mathbf{0}\}$, then $\mathcal{V}$ is a *zero vector space*; otherwise $\mathcal{V}$ is a *nonzero vector space*.

Definition 1.1.2. *Finitely Nonzero Sequence Space*
Let $\mathbb{F}^{(\mathbb{N})}$ denote the set of sequences $(x_1, x_2, \ldots)$ in $\mathbb{F}$ with finite support, i.e. $x_j = 0$ for all but finitely many $j \in \mathbb{N}$. Addition and scalar multiplication are componentwise.

Proposition 1.1.3. *Finitely Nonzero Sequences Form a Vector Space*
The set $\mathbb{F}^{(\mathbb{N})}$ is a vector space over $\mathbb{F}$ under componentwise addition and scalar multiplication.

Proof. The zero sequence has finite support.
If $\mathbf{x}, \mathbf{y} \in \mathbb{F}^{(\mathbb{N})}$ and $a \in \mathbb{F}$, then $a\mathbf{x} + \mathbf{y}$ can be nonzero only where $\mathbf{x}$ or $\mathbf{y}$ is nonzero, a finite set of indices.
Hence $a\mathbf{x} + \mathbf{y} \in \mathbb{F}^{(\mathbb{N})}$, and the vector space axioms are inherited componentwise from $\mathbb{F}$.
Structural Commentary: Finite support is preserved by finite linear combinations.
Key Insight: Componentwise operations cannot create infinitely many new nonzero entries. □

Example 1.1.4. *Standard Vector Space Models*
Let $\mathbb{F}$ be a field and let X be a set. The following are $\mathbb{F}$-vector spaces.

(i) $\mathbb{F}^X = \{f : X \to \mathbb{F}\}$, with pointwise operations $(f + g)(x) = f(x) + g(x)$ and $(af)(x) = af(x)$.
(ii) $\mathrm{Mat}(m, n, \mathbb{F})$ and $\mathbb{F}^n$, with entrywise operations. For $\mathbb{F} = \mathbb{F}_q$, set $V(n, q) := \mathbb{F}_q^n$.
(iii) $\mathrm{Seq}(\mathbb{F}) = \mathbb{F}^{\mathbb{N}}$, with componentwise operations; $\mathbb{F}^{(\mathbb{N})}$ is its finite-support subspace.
(iv) Over $\mathbb{C}$, the sequence spaces $c_0 = \{(z_n) : z_n \to 0\}$, $\ell^\infty = \{(z_n) : \sup_n |z_n| < \infty\}$, and, for $1 \le p < \infty$, $\ell^p = \left\{(z_n) : \sum_{n=1}^{\infty} |z_n|^p < \infty\right\}$ are vector spaces under componentwise operations; for ℓ^p, closure under addition follows from Minkowski's inequality.

Structural Commentary: The vector-space axioms are independent of finite-dimensional coordinates; function and sequence spaces give canonical infinite-dimensional models.
Key Insight: Pointwise or componentwise operations inherit linear structure from the scalar field.

Theorem 1.1.5. *Basic Consequences of Vector Space Axioms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then:

(i) the additive identity is unique;
(ii) for each $\mathbf{v} \in \mathcal{V}$, the additive inverse of $\mathbf{v}$ is unique;
(iii) $\forall \mathbf{v} \in \mathcal{V}$, $0\mathbf{v} = \mathbf{0}$;
(iv) $\forall a \in \mathbb{F}$, $a\mathbf{0} = \mathbf{0}$;
(v) $\forall a \in \mathbb{F}$ and $\mathbf{v} \in \mathcal{V}$, $a\mathbf{v} = \mathbf{0} \iff a = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof.

(i) If $\mathbf{0}$ and $\mathbf{0}'$ are additive identities, then $\mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}'$.
(ii) If $\mathbf{w}, \mathbf{w}'$ are additive inverses of $\mathbf{v}$, then $\mathbf{w} = \mathbf{w} + \mathbf{0} = \mathbf{w} + (\mathbf{v} + \mathbf{w}') = (\mathbf{w} + \mathbf{v}) + \mathbf{w}' = \mathbf{0} + \mathbf{w}' = \mathbf{w}'$.

(iii) $0\mathbf{v} = (0+0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v}$; adding $-(0\mathbf{v})$ gives $0\mathbf{v} = \mathbf{0}$.
(iv) $a\mathbf{0} = a(\mathbf{0}+\mathbf{0}) = a\mathbf{0} + a\mathbf{0}$; adding $-(a\mathbf{0})$ gives $a\mathbf{0} = \mathbf{0}$.
(v) ($\Rightarrow$) Suppose $a\mathbf{v} = \mathbf{0}$. If $a = 0$, the first alternative holds.
If $a \neq 0$, then $\mathbf{v} = 1\mathbf{v} = (a^{-1}a)\mathbf{v} = a^{-1}(a\mathbf{v}) = a^{-1}\mathbf{0} = \mathbf{0}$.
(v) ($\Leftarrow$) If $a = 0$ or $\mathbf{v} = \mathbf{0}$, then (iii) or (iv) gives $a\mathbf{v} = \mathbf{0}$.

Structural Commentary: These identities make cancellation, zeros, and inverses available.
Key Insight: The vector space axioms force the zero and inverse behaviour to be rigid, not optional structure. □

Corollary 1.1.6. *Scalar Multiplication by -1 Is Negation*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. Then $\forall\, \mathbf{v} \in \mathcal{V}$, $(-1)\mathbf{v} = -\mathbf{v}$.

Proof. For $\mathbf{v} \in \mathcal{V}$, $(-1)\mathbf{v} + \mathbf{v} = (-1)\mathbf{v} + 1\mathbf{v} = (-1+1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$.
Hence $(-1)\mathbf{v}$ is an additive inverse of $\mathbf{v}$; by uniqueness, $(-1)\mathbf{v} = -\mathbf{v}$.
Structural Commentary: Scalar multiplication by -1 is exactly the same operation as taking an additive inverse.
Key Insight: Negation is not an extra operation; it is already encoded by the field action. □

1.2 Subspaces and Spans

Definition 1.2.1. *Subspace*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. A subset $\mathcal{U} \subseteq \mathcal{V}$ is a *subspace*, written $\mathcal{U} \leq \mathcal{V}$, if $\mathcal{U}$ is a vector space over $\mathbb{F}$ under the operations inherited from $\mathcal{V}$.

Theorem 1.2.2. *Subspace Test*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $\mathcal{U}$ be a nonempty subset of $\mathcal{V}$.
Then $\mathcal{U}$ is a subspace of $\mathcal{V}$ iff $\forall\, \mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $\forall\, c \in \mathbb{F}$, one has $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$.

Proof.
($\Rightarrow$) If $\mathcal{U} \leq \mathcal{V}$, then closure under scalar multiplication and addition gives $c\mathbf{u} + \mathbf{v} \in \mathcal{U}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$.
($\Leftarrow$) Assume the stated closure condition. Choose $\mathbf{u}_0 \in \mathcal{U}$. Then $\mathbf{0} = (-1)\mathbf{u}_0 + \mathbf{u}_0 \in \mathcal{U}$. For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ and $c \in \mathbb{F}$, the identities $-\mathbf{u} = (-1)\mathbf{u} + \mathbf{0}$, $c\mathbf{u} = c\mathbf{u} + \mathbf{0}$, and $\mathbf{u} + \mathbf{v} = 1\mathbf{u} + \mathbf{v}$ give closure under additive inverses, scalar multiplication, and addition. The remaining axioms are inherited from $\mathcal{V}$, so $\mathcal{U} \leq \mathcal{V}$.

Structural Commentary: One closure condition captures the subspace axioms.
Key Insight: To prove a subset is a subspace, prove it survives one affine-looking linear combination. □

Definition 1.2.3. *Linear Code and Hamming Weight*
(i) The *ambient code space* is $V(n,q) := \mathbb{F}_q^n$.
(ii) A *linear code* of length n over $\mathbb{F}_q$ is a *subspace* $C \leq V(n,q)$.
(iii) For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{F}_q^n$, its *Hamming weight* is $\mathrm{wt}(\mathbf{x}) := \#\{j : 1 \leq j \leq n,\ x_j \neq 0\}$.

Example 1.2.4. *Even-Weight Code*
In $V(n,2) = \mathbb{F}_2^n$, let $E_n = \left\{ \mathbf{x} \in \mathbb{F}_2^n : \sum_{j=1}^{n} x_j = 0 \right\}$. The condition $\sum_{j=1}^{n} x_j = 0$ is equivalent to $\mathrm{wt}(\mathbf{x})$ being even.
If $\mathbf{x}, \mathbf{y} \in E_n$ and $c \in \mathbb{F}_2$, then $\sum_{j=1}^{n}(cx_j + y_j) = c\sum_{j=1}^{n} x_j + \sum_{j=1}^{n} y_j = 0$, so $c\mathbf{x} + \mathbf{y} \in E_n$. Hence $E_n \leq V(n,2)$ by Theorem 1.2.2, and E_n is a linear code.
Structural Commentary: The even-weight code is the parity-zero subspace in $V(n,2)$.
Key Insight: Parity is linear over $\mathbb{F}_2$.

Proposition 1.2.5. *Intersection of Subspaces*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$. Then $\mathcal{U} \cap \mathcal{W} = \{\mathbf{v} : \mathbf{v} \in \mathcal{U}$ and $\mathbf{v} \in \mathcal{W}\}$ is a subspace of $\mathcal{V}$.

Proof. Since $\mathbf{0} \in \mathcal{U}$ and $\mathbf{0} \in \mathcal{W}$, we have $\mathbf{0} \in \mathcal{U} \cap \mathcal{W}$, so the intersection is nonempty.
If $\mathbf{u}, \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$ and $c \in \mathbb{F}$, then $c\mathbf{u} + \mathbf{v}$ lies in both $\mathcal{U}$ and $\mathcal{W}$ because each is a subspace.
Thus $c\mathbf{u} + \mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, and Theorem 1.2.2 gives $\mathcal{U} \cap \mathcal{W} \leq \mathcal{V}$.
Structural Commentary: Intersections inherit shared closure.
Key Insight: Common vectors remain closed under linear combinations. □

Definition 1.2.6. *Lists, Linear Combinations, and Span*
(i) A *list* in $\mathcal{V}$ is $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$; the empty list is $()$.
(ii) For $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ and $a_1, \ldots, a_m \in \mathbb{F}$, a *linear combination* of β is $\sum_{j=1}^{m} a_j \mathbf{v}_j$.

(iii) For $E \subseteq \mathcal{V}$, the *span* of E is $\operatorname{span}(E) := \left\{ \sum_{j=1}^{m} a_j \mathbf{v}_j : m \in \mathbb{N},\ a_j \in \mathbb{F},\ \mathbf{v}_j \in E \right\}$, with $\operatorname{span}(\emptyset) = \{\mathbf{0}\}$.
(iv) For $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$, the *span* of β is $\operatorname{span}(\beta) := \operatorname{span}(\{\mathbf{v}_1, \ldots, \mathbf{v}_m\})$.
(v) $E \subseteq \mathcal{V}$ *spans* $\mathcal{V}$ iff $\operatorname{span}(E) = \mathcal{V}$; a list β *spans* $\mathcal{V}$ iff $\operatorname{span}(\beta) = \mathcal{V}$.

Theorem 1.2.7. *Span is the Smallest Containing Subspace*
Let $E \subseteq \mathcal{V}$. Then $\operatorname{span}(E)$ is the smallest subspace of $\mathcal{V}$ containing E.

Proof. If $E = \emptyset$, then $\operatorname{span}(E) = \{\mathbf{0}\}$, which is a subspace. Suppose $E \neq \emptyset$. For $\mathbf{x}, \mathbf{y} \in \operatorname{span}(E)$ and $c \in \mathbb{F}$, both $\mathbf{x}$ and $\mathbf{y}$ are finite linear combinations of vectors in E, so $c\mathbf{x} + \mathbf{y}$ is also such a combination. Hence $c\mathbf{x} + \mathbf{y} \in \operatorname{span}(E)$, and Theorem 1.2.2 makes $\operatorname{span}(E)$ a subspace.
Each $\mathbf{v} \in E$ satisfies $\mathbf{v} = 1\mathbf{v} \in \operatorname{span}(E)$, so $E \subseteq \operatorname{span}(E)$. If $\mathcal{U} \leq \mathcal{V}$ and $E \subseteq \mathcal{U}$, then closure under finite linear combinations gives $\operatorname{span}(E) \subseteq \mathcal{U}$. Thus $\operatorname{span}(E)$ is the smallest subspace containing E.
Structural Commentary: Span closes a set of vectors under finite linear combinations, giving exactly the subspace forced by those vectors.
Key Insight: Nothing beyond linear closure is added. Every containing subspace must already contain span. □

Method 1.2.8. *Subspace Diagnostic*
Let $S \subseteq \mathcal{V}$.
(i) To prove $S \leq \mathcal{V}$, show $S \neq \varnothing$ and $\forall\, \mathbf{u}, \mathbf{v} \in S,\ \forall\, c \in \mathbb{F},\ c\mathbf{u} + \mathbf{v} \in S$.
(ii) To disprove $S \leq \mathcal{V}$, exhibit $\mathbf{u}, \mathbf{v} \in S$ and $c \in \mathbb{F}$ with $c\mathbf{u} + \mathbf{v} \notin S$; $\mathbf{0} \in S$ alone is insufficient.
(iii) The generated subspace is $\langle S \rangle = \operatorname{span}(S) = \left\{ \sum_{i=1}^{n} a_i \mathbf{s}_i : n \in \mathbb{N},\ a_i \in \mathbb{F},\ \mathbf{s}_i \in S \right\}$.

Proposition 1.2.9. *Monotonicity of Span*
Let $\mathcal{U}$ and $\mathcal{W}$ be subsets of $\mathcal{V}$. If $\mathcal{U} \subseteq \mathcal{W}$, then $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.

Proof. If $\mathcal{U} = \emptyset$, then $\operatorname{span}(\mathcal{U}) = \{\mathbf{0}\} \subseteq \operatorname{span}(\mathcal{W})$.
Suppose $\mathcal{U} \neq \emptyset$. Every vector in $\operatorname{span}(\mathcal{U})$ is a finite linear combination of vectors in $\mathcal{U}$; since $\mathcal{U} \subseteq \mathcal{W}$, the same expression is a finite linear combination of vectors in $\mathcal{W}$. Hence $\operatorname{span}(\mathcal{U}) \subseteq \operatorname{span}(\mathcal{W})$.
Structural Commentary: Enlarging the generating set can only enlarge the collection of linear combinations.
Key Insight: Span is monotone with respect to inclusion of generating sets. □

1.3 Linear Independence and Bases

Definition 1.3.1. *Linear Independence, Bases, and Dimension*

(i) A list $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is *linearly independent* if $\sum_{j=1}^{m} a_j \mathbf{v}_j = \mathbf{0} \Rightarrow a_1 = \cdots = a_m = 0$.
Otherwise it is *linearly dependent*. The empty list is linearly independent.
(ii) A list β is a *basis* of $\mathcal{V}$ if β is linearly independent and $\operatorname{span}(\beta) = \mathcal{V}$.
(iii) $\mathcal{V}$ is *finite-dimensional* if some finite list β satisfies $\operatorname{span}(\beta) = \mathcal{V}$; otherwise $\mathcal{V}$ is *infinite-dimensional*.

Proposition 1.3.2. *Unique Coefficients for Independent Lists*
Let $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a linearly independent list in $\mathcal{V}$.
Then $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = b_1\mathbf{v}_1 + \cdots + b_m\mathbf{v}_m$ iff $a_j = b_j$ for each $j = 1, \ldots, m$.

Proof.
(⇒) Subtracting the two sides gives $(a_1 - b_1)\mathbf{v}_1 + \cdots + (a_m - b_m)\mathbf{v}_m = \mathbf{0}$.
Linear independence forces $a_j - b_j = 0$ for each j, hence $a_j = b_j$.
(⇐) If $a_j = b_j$ for each j, then the two linear combinations have identical coefficients and are equal.
Structural Commentary: Linear independence makes coefficients in a linear combination unique.
Key Insight: Independent vectors allow no hidden coefficient cancellation. □

Proposition 1.3.3. *Appending to a Dependent List*
Let $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a linearly dependent list in $\mathcal{V}$.
Then for each $\mathbf{v} \in \mathcal{V}$, the list $(\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{v})$ is linearly dependent.

Proof. Since $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly dependent, $\exists$ scalars $a_1, \ldots, a_m$, not all zero, with $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
Appending coefficient 0 gives $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m + 0\mathbf{v} = \mathbf{0}$, and the scalars $a_1, \ldots, a_m, 0$ are not all zero.
Structural Commentary: Longer lists inherit dependence; set new coefficients to 0.
Key Insight: Once a list has a redundancy, adding more vectors cannot remove it. □

Definition 1.3.4. *Omitting a Vector from a List*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a list in $\mathcal{V}$ with $m \geq 2$.
For $j \in \{1, \ldots, m\}$, the list obtained from β by omitting $\mathbf{v}_j$ is denoted $(\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.

Proposition 1.3.5. *Omitting from an Independent List*
If $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent and $m \geq 2$, then $\beta_{\widehat{j}} := (\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$ is linearly independent for each $j \in \{1, \ldots, m\}$.

Proof. Fix j. Any relation $\sum_{i \neq j} a_i \mathbf{v}_i = \mathbf{0}$ on $\beta_{\widehat{j}}$ becomes a relation on β after setting $a_j = 0$. Independence of β gives $a_i = 0$ for all i, so the sublist is independent.
Structural Commentary: Deletion cannot create cancellation.
Key Insight: Every sublist of an independent list remains independent. □

Proposition 1.3.6. *Adjoining and Removing Vectors from Lists*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be a list in $\mathcal{V}$. Write $(\beta, \mathbf{v}) := (\mathbf{v}_1, \ldots, \mathbf{v}_m, \mathbf{v})$ and $\beta_{\widehat{j}} := (\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.

(i) If β is linearly independent and $\mathbf{v} \notin \operatorname{span}(\beta)$, then $(\beta, \mathbf{v})$ is linearly independent.

(ii) If $\sum_{i=1}^{m} a_i \mathbf{v}_i = \mathbf{0}$ is nontrivial and $a_j \neq 0$, then $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$.
Hence $\operatorname{span}(\beta) = \mathcal{V}$ implies $\operatorname{span}(\beta_{\widehat{j}}) = \mathcal{V}$.

Proof.

(i) Suppose $\sum_{i=1}^{m} a_i \mathbf{v}_i + a\mathbf{v} = \mathbf{0}$. If $a \neq 0$, then $\mathbf{v} = -a^{-1} \sum_{i=1}^{m} a_i \mathbf{v}_i \in \operatorname{span}(\beta)$, a contradiction.
Hence $a = 0$, and independence of β gives $a_1 = \cdots = a_m = 0$.

(ii) Solving the relation for $\mathbf{v}_j$ gives $\mathbf{v}_j = -a_j^{-1} \sum_{i \neq j} a_i \mathbf{v}_i \in \operatorname{span}(\beta_{\widehat{j}})$. Thus every vector of β lies in $\operatorname{span}(\beta_{\widehat{j}})$, so $\operatorname{span}(\beta) \subseteq \operatorname{span}(\beta_{\widehat{j}})$. The reverse inclusion is immediate because $\beta_{\widehat{j}}$ is a sublist of β.

Structural Commentary: New directions preserve independence; redundant vectors preserve span.
Key Insight: Add new directions; remove redundant ones. □

Lemma 1.3.7. *Replacement Lemma*
Let $\mathcal{V} \neq \{\mathbf{0}\}$, let $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_r)$ span $\mathcal{V}$, and let $\mathbf{0} \neq \mathbf{v} = \sum_{i=1}^{r} c_i \mathbf{u}_i$.
For $j \in \{1, \ldots, r\}$, set $\beta_{j \leftarrow \mathbf{v}} := (\mathbf{u}_1, \ldots, \mathbf{u}_{j-1}, \mathbf{v}, \mathbf{u}_{j+1}, \ldots, \mathbf{u}_r)$.

(i) $c_j \neq 0$ for some j.
(ii) If $c_j \neq 0$, then $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$; if β is a basis, then $\beta_{j \leftarrow \mathbf{v}}$ is a basis.
(iii) If $r \geq 2$, β is a basis, and $\mathbf{v} \notin \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$ for some $1 \leq k < r$, then $\beta_{j \leftarrow \mathbf{v}}$ is a basis for some $j > k$.

Proof.

(i) If all $c_j = 0$, then $\mathbf{v} = \mathbf{0}$.

(ii) Since $\mathbf{u}_j = c_j^{-1} \left(\mathbf{v} - \sum_{i \neq j} c_i \mathbf{u}_i \right)$, every vector of β lies in $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}})$.
Conversely, $\mathbf{v} \in \operatorname{span}(\beta)$, so every vector of $\beta_{j \leftarrow \mathbf{v}}$ lies in $\operatorname{span}(\beta)$. Hence $\operatorname{span}(\beta_{j \leftarrow \mathbf{v}}) = \mathcal{V}$.
If β is a basis and $\sum_{i<j} b_i \mathbf{u}_i + a\mathbf{v} + \sum_{i>j} b_i \mathbf{u}_i = \mathbf{0}$, substitution gives $\sum_{i \neq j} (b_i + ac_i)\mathbf{u}_i + ac_j \mathbf{u}_j = \mathbf{0}$. Independence of β gives $ac_j = 0$ and $b_i + ac_i = 0$ for $i \neq j$.
Since $c_j \neq 0$, $a = 0$, and then every $b_i = 0$. Thus $\beta_{j \leftarrow \mathbf{v}}$ is linearly independent.

(iii) Since $\mathbf{v} \notin \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, some coefficient among $c_{k+1}, \ldots, c_r$ is nonzero.
Choose $j > k$ with $c_j \neq 0$; by (ii), $\beta_{j \leftarrow \mathbf{v}}$ is a basis.

Structural Commentary: Replacement preserves span.
Key Insight: Replace where the coefficient is nonzero. □

Lemma 1.3.8. *Linear Dependence Lemma*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly dependent with $\mathbf{v}_1 \neq \mathbf{0}$, and set $\beta_{\widehat{j}} := (\mathbf{v}_1, \ldots, \widehat{\mathbf{v}_j}, \ldots, \mathbf{v}_m)$.
Then some $j \in \{2, \ldots, m\}$ satisfies $\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$ and $\operatorname{span}(\beta_{\widehat{j}}) = \operatorname{span}(\beta)$.

Proof. Choose a nontrivial relation $\sum_{i=1}^{m} a_i \mathbf{v}_i = \mathbf{0}$ and let $j = \max\{i : a_i \neq 0\}$.
If $j = 1$, then $a_1 \mathbf{v}_1 = \mathbf{0}$ with $a_1 \neq 0$, so $\mathbf{v}_1 = \mathbf{0}$, contrary to hypothesis. Hence $j \geq 2$.
Since $a_i = 0$ for $i > j$, we have $\mathbf{v}_j = -a_j^{-1} \sum_{i=1}^{j-1} a_i \mathbf{v}_i$, so $\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{j-1})$.

Substituting this expression for $\mathbf{v}_j$ rewrites every linear combination of β without $\mathbf{v}_j$, so $\text{span}(\beta) \subseteq \text{span}(\beta_{\widehat{j}})$. The reverse inclusion follows because $\beta_{\widehat{j}}$ is a sublist of β. Hence the spans are equal.
Structural Commentary: The last nonzero coefficient gives the removable vector.
Key Insight: Dependence gives a later vector removable without losing span. □

Theorem 1.3.9. *Independent Lists Are No Longer Than Spanning Lists*
If $\mathcal{V}$ is finite-dimensional, $\alpha = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is linearly independent, and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ spans $\mathcal{V}$, then $m \leq n$.

Proof. Suppose $m > n$.
If $n = 0$, then $\mathcal{V} = \text{span}(\gamma) = \{\mathbf{0}\}$, contradicting independence of the nonempty list α; hence $n \geq 1$.
Construct spanning lists γ_k of length n that contain $\mathbf{u}_1, \ldots, \mathbf{u}_k$ for $0 \leq k \leq n$. Set $\gamma_0 = \gamma$. Given γ_k with $k < n$, write $\mathbf{u}_{k+1}$ in $\text{span}(\gamma_k)$. Since $\mathbf{u}_{k+1} \notin \text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k)$, some coefficient on an entry of γ_k outside $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is nonzero. Replacing that entry by $\mathbf{u}_{k+1}$ preserves span by Lemma 1.3.7(ii); call the result γ_{k+1}.
Then γ_n spans $\mathcal{V}$ and contains $\mathbf{u}_1, \ldots, \mathbf{u}_n$, so $\text{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n) = \mathcal{V}$.
Since $m > n$, this puts $\mathbf{u}_{n+1}$ in the span of earlier vectors, contradicting independence of α.
Structural Commentary: Independent length is bounded by spanning length.
Key Insight: Independent lists cannot be longer than spanning lists in finite dimensions. □

Proposition 1.3.10. *Unique Coordinate Representation*
$\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ iff each $\mathbf{v} \in \mathcal{V}$ has a unique coefficient vector $(a_1, \ldots, a_n) \in \mathbb{F}^n$ with $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j$.

Proof.

($\Rightarrow$) Spanning gives a representation of each $\mathbf{v} \in \mathcal{V}$. If $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{v}_j = \sum_{j=1}^{n} b_j \mathbf{v}_j$, then Proposition 1.3.2 gives $a_j = b_j$ for each j.

($\Leftarrow$) Existence gives $\text{span}(\beta) = \mathcal{V}$. If $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}$, compare with $\mathbf{0} = \sum_{j=1}^{n} 0\mathbf{v}_j$; uniqueness gives $a_j = 0$ for each j, so β is independent.

Structural Commentary: A basis is exactly existence and uniqueness of coordinates.
Key Insight: Coordinates are basis coefficients. □

Theorem 1.3.11. *Reducing Spanning Lists*
Every spanning list in a finite-dimensional vector space contains a sublist that is a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a spanning list of $\mathcal{V}$. If β is independent, it is a basis. Otherwise remove one vector without changing span: omit $\mathbf{v}_1$ if $\mathbf{v}_1 = \mathbf{0}$, and otherwise use Lemma 1.3.8 to remove a suitable $\mathbf{v}_j$. The resulting sublist still spans $\mathcal{V}$ and has length one less. Repeat this step. The length strictly decreases, so the process stops with a spanning independent sublist, hence a basis.
Structural Commentary: Delete redundant vectors while preserving span.
Key Insight: Spanning gives enough vectors; reduction deletes exactly the unnecessary ones. □

Theorem 1.3.12. *Extending Independent Lists*
Every linearly independent list in a finite-dimensional vector space can be extended to a basis.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$ be linearly independent in $\mathcal{V}$, and choose a spanning list $(\mathbf{w}_1, \ldots, \mathbf{w}_n)$ of $\mathcal{V}$.
Initialize $\mathcal{B} = \beta$. Inspect $\mathbf{w}_1, \ldots, \mathbf{w}_n$ in order, appending $\mathbf{w}_j$ exactly when $\mathbf{w}_j \notin \text{span}(\mathcal{B})$. Each append preserves independence by Proposition 1.3.6.
After the inspection, each $\mathbf{w}_j$ lies in $\text{span}(\mathcal{B})$: it was either already there or was appended. Since the $\mathbf{w}_j$ span $\mathcal{V}$ and $\text{span}(\mathcal{B})$ is a subspace, every vector of $\mathcal{V}$ lies in $\text{span}(\mathcal{B})$.
Thus $\mathcal{B}$ is a spanning independent list, hence a basis.
Structural Commentary: An independent list has no redundancy, but may miss directions. Appending only vectors outside its span fills those directions while preserving independence.
Key Insight: A finite-dimensional independent list becomes a basis once enough new directions are added. □

Corollary 1.3.13. *Bases and Subspaces in Finite Dimension*
If $\mathcal{V}$ is finite-dimensional, then $\mathcal{V}$ has a basis. Moreover, $\mathcal{U} \leq \mathcal{V}$ implies $\mathcal{U}$ is finite-dimensional.

Proof. Choose a finite spanning list γ of $\mathcal{V}$. By Theorem 1.3.11, some sublist of γ is a basis of $\mathcal{V}$.
Let $\mathcal{U} \leq \mathcal{V}$. If $\mathcal{U} = \{\mathbf{0}\}$, then $()$ is a basis. Otherwise start with $\beta = ()$ and append vectors from $\mathcal{U} \setminus \text{span}(\beta)$ while possible. Proposition 1.3.6 preserves independence, while Theorem 1.3.9 bounds the length by the length of γ; hence the process terminates. At termination, $\mathcal{U} \subseteq \text{span}(\beta)$, so β is a finite basis of $\mathcal{U}$.
Structural Commentary: Finite-dimensional spaces have finite coordinate systems, and subspaces cannot contain

longer independent lists than the ambient space permits.
Key Insight: Finite-dimensionality passes downward to subspaces. □

Theorem 1.3.14. *Basis Length Is Well-Defined*
Any two bases of a finite-dimensional vector space have the same length.

Proof. Let $\beta_1 = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\beta_2 = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ be bases of $\mathcal{V}$. Theorem 1.3.9 applied to β_1 against β_2 gives $m \leq n$, and reversing the roles gives $n \leq m$. Hence $m = n$.
Structural Commentary: Bases are simultaneously independent and spanning, so the comparison theorem traps their lengths in both directions. This makes dimension well-defined.
Key Insight: Dimension is an intrinsic number, not an artefact of a particular basis. □

1.4 Dimension and Coordinates

Definition 1.4.1. *Dimension*

(i) If $\mathcal{V} = \{\mathbf{0}\}$, then $\dim(\mathcal{V}) := 0$.
(ii) If $\mathcal{V}$ has a basis of length $n \geq 1$, then $\mathcal{V}$ is *n-dimensional* and $\dim(\mathcal{V}) := n$.

Theorem 1.3.14 makes $\dim(\mathcal{V})$ well-defined.

Proposition 1.4.2. *Subspace Dimension Criterion*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U} \leq \mathcal{V}$. Then $\dim(\mathcal{U}) \leq \dim(\mathcal{V})$, and equality holds iff $\mathcal{U} = \mathcal{V}$.

Proof. Let $\delta = (\mathbf{u}_1, \dots, \mathbf{u}_m)$ be a basis of $\mathcal{U}$ and let β be a basis of $\mathcal{V}$, where $n = \dim(\mathcal{V})$.
Since δ is independent in $\mathcal{V}$ and β spans $\mathcal{V}$, Theorem 1.3.9 gives $m \leq n$.

($\Rightarrow$) If $\mathcal{U} = \mathcal{V}$, then any basis of $\mathcal{U}$ is a basis of $\mathcal{V}$, so $\dim(\mathcal{U}) = \dim(\mathcal{V})$.
($\Leftarrow$) If $\dim(\mathcal{U}) = \dim(\mathcal{V})$, then δ is independent in $\mathcal{V}$ of length n. By Theorem 1.3.12, extending δ to a basis of $\mathcal{V}$ adds no vector, since every basis has length n. Hence $\mathcal{U} = \operatorname{span}(\delta) = \mathcal{V}$.

Structural Commentary: Dimension bounds subspaces; equality forces fullness.
Key Insight: Equal dimension subspaces fill the ambient space. □

Proposition 1.4.3. *Full-Dimension Basis Criteria*
Let $\mathcal{V}$ be finite-dimensional and set $n := \dim(\mathcal{V})$. Then:

(i) if γ spans $\mathcal{V}$ and has length n, then γ is a basis;
(ii) if α is linearly independent in $\mathcal{V}$ and has length n, then α is a basis.

Proof.

(*i*) By Theorem 1.3.11, γ contains a basis sublist. Since every basis has length n, no proper sublist can have length n; hence γ itself is a basis.
(*ii*) Extend α to a basis of $\mathcal{V}$ by Theorem 1.3.12. Since every basis has length n and α already has length n, no vector is added; hence α is a basis.

Structural Commentary: Full-length spanning or independent lists cannot be shortened or extended.
Key Insight: In dimension n, n independent or n spanning vectors already form a basis. □

Theorem 1.4.4. *Basis Characterizations*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a finite list in $\mathcal{V}$, and write $\beta^\frown\mathbf{x} = (\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{x})$. Then the following are equivalent:

(i) β is a basis of $\mathcal{V}$;
(ii) each $\mathbf{x} \in \mathcal{V}$ has a unique coefficient vector $(a_1, \dots, a_n) \in \mathbb{F}^n$ with $\mathbf{x} = \sum_{j=1}^{n} a_j \mathbf{v}_j$;
(iii) $\operatorname{span}(\beta) = \mathcal{V}$, and no proper sublist of β spans $\mathcal{V}$;
(iv) β is linearly independent, and $\beta^\frown\mathbf{x}$ is linearly dependent for each $\mathbf{x} \in \mathcal{V}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) This is Proposition 1.3.10.
(*ii*) $\Rightarrow$ (*iii*) Existence gives $\operatorname{span}(\beta) = \mathcal{V}$.
If a proper sublist $\gamma \subsetneq \beta$ spanned $\mathcal{V}$ and omitted the occurrence $\mathbf{v}_j$, then $\mathbf{v}_j$ would have both its standard β-coordinate representation and one using only γ, contradicting uniqueness.
(*iii*) $\Rightarrow$ (*iv*) If β were dependent, Proposition 1.3.6 would remove some $\mathbf{v}_j$ without changing $\operatorname{span}(\beta)$, contradicting minimality. Thus β is independent.
Since $\operatorname{span}(\beta) = \mathcal{V}$, writing $\mathbf{x}$ as a linear combination of β gives a nontrivial relation on $\beta^\frown\mathbf{x}$.
(*iv*) $\Rightarrow$ (*i*) If $\operatorname{span}(\beta) \neq \mathcal{V}$, choose $\mathbf{x} \notin \operatorname{span}(\beta)$. Then Proposition 1.3.6 makes $\beta^\frown\mathbf{x}$ independent, contradicting (iv). Hence β spans $\mathcal{V}$, and with independence from (iv), it is a basis.

Structural Commentary: A basis is simultaneously just enough to span and too large to extend independently.
Key Insight: Spanning and independence are dual minimality and maximality conditions. □

Definition 1.4.5. *Hamel Basis*
A subset $\mathcal{B} \subseteq \mathcal{V}$ is a *Hamel basis* of $\mathcal{V}$ if it is linearly independent and $\operatorname{span}(\mathcal{B}) = \mathcal{V}$. Equivalently, each $\mathbf{v} \in \mathcal{V}$ has a unique finitely supported coefficient function $a : \mathcal{B} \to \mathbb{F}$ with $\mathbf{v} = \sum_{\mathbf{b}\in\mathcal{B}} a(\mathbf{b})\mathbf{b}$.

Finite ordered bases are orderings of finite Hamel bases.

Theorem 1.4.6. *Hamel Basis Extension*
Let $I \subseteq S \subseteq \mathcal{V}$, with I linearly independent and $\operatorname{span}(S) = \mathcal{V}$.
Assuming Zorn's lemma, there exists a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ such that $I \subseteq \mathcal{B} \subseteq S$. Consequently, every nonzero vector space has a basis; every independent set extends to a basis; every spanning set contains a basis.

Proof. Let $\mathscr{P} = \{E : I \subseteq E \subseteq S,\ E \text{ linearly independent}\}$, ordered by inclusion. Then $I \in \mathscr{P}$.
If $\mathcal{C} \subseteq \mathscr{P}$ is a nonempty chain, put $U = \bigcup_{E\in\mathcal{C}} E$. In any finite relation $\sum_{k=1}^{m} a_k\mathbf{u}_k = \mathbf{0}$ in U, the finitely many vectors $\mathbf{u}_k$ lie in one member of the chain; independence there gives $a_k = 0$ for all k.
Thus $U \in \mathscr{P}$ and is an upper bound for $\mathcal{C}$. The empty chain is bounded by I.
By Zorn's lemma, Principle 6.5.2, choose maximal $\mathcal{B} \in \mathscr{P}$.
If $s \in S \setminus \operatorname{span}(\mathcal{B})$, then $\mathcal{B} \cup \{s\}$ is linearly independent, contradicting maximality. Hence $S \subseteq \operatorname{span}(\mathcal{B})$, so $\operatorname{span}(\mathcal{B}) = \mathcal{V}$ and $\mathcal{B}$ is a Hamel basis. The consequences follow by taking $S = \mathcal{V}$ or $I = \emptyset$.
Structural Commentary: Finite extension uses exchange; arbitrary extension uses maximality.
Key Insight: Maximal independence inside a spanning set forces spanning. □

Theorem 1.4.7. *Dimension as Basis Cardinality*
If $\mathcal{B}$ and $\mathcal{C}$ are Hamel bases of $\mathcal{V}$, then $|\mathcal{B}| = |\mathcal{C}|$. Hence $\dim(\mathcal{V}) := |\mathcal{B}|$ is well-defined for any Hamel basis $\mathcal{B}$.

Proof. If one basis is finite, say $|\mathcal{B}| = n$, then $\mathcal{C}$ cannot contain $n+1$ distinct vectors, since such a finite sublist would be independent while $\mathcal{B}$ spans $\mathcal{V}$, contradicting Theorem 1.3.9. Hence $\mathcal{C}$ is finite, and Theorem 1.3.14 gives $|\mathcal{C}| = n$. Thus assume both bases are infinite. For $\mathbf{c} \in \mathcal{C}$, let $\operatorname{supp}_{\mathcal{B}}(\mathbf{c}) \subseteq \mathcal{B}$ be its finite $\mathcal{B}$-support.
Every $\mathbf{b} \in \mathcal{B}$ appears in some $\operatorname{supp}_{\mathcal{B}}(\mathbf{c})$: otherwise all vectors in $\operatorname{span}(\mathcal{C})$ would have zero $\mathbf{b}$-coordinate, contradicting $\operatorname{span}(\mathcal{C}) = \mathcal{V}$. Hence $\mathcal{B} \subseteq \bigcup_{\mathbf{c}\in\mathcal{C}} \operatorname{supp}_{\mathcal{B}}(\mathbf{c})$. Thus $|\mathcal{B}| \le \sum_{\mathbf{c}\in\mathcal{C}} |\operatorname{supp}_{\mathcal{B}}(\mathbf{c})| \le |\mathcal{C}|\,\aleph_0 = |\mathcal{C}|$ by Theorem 6.4.2.
Interchanging $\mathcal{B}$ and $\mathcal{C}$ gives $|\mathcal{C}| \le |\mathcal{B}|$; Theorem 6.3.1 gives equality.
Structural Commentary: Even in infinite dimensions, a basis size is intrinsic.
Key Insight: Each basis vector is detected inside finite supports relative to the other basis. □

2 Direct Sums, Quotients, and Decomposition

2.1 Sums and Direct Sums

Direct sums split spaces into complementary pieces; quotients collapse selected directions. Dimension formulas keep both constructions controlled.

Definition 2.1.1. *Sums of Subspaces*
Let $\mathcal{U}_1, \dots, \mathcal{U}_m \le \mathcal{V}$.

(i) The *sum* of $\mathcal{U}_1, \dots, \mathcal{U}_m$ is $\mathcal{U}_1 + \cdots + \mathcal{U}_m := \{\mathbf{u}_1 + \cdots + \mathbf{u}_m : \mathbf{u}_j \in \mathcal{U}_j\}$.
(ii) For subspaces $\mathcal{U}, \mathcal{W} \le \mathcal{V}$, $\mathcal{U} + \mathcal{W} = \operatorname{span}(\mathcal{U} \cup \mathcal{W}) = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in \mathcal{U},\ \mathbf{w} \in \mathcal{W}\}$.
(iii) The sum is *direct* if every vector in it has a unique decomposition as $\mathbf{u}_1 + \cdots + \mathbf{u}_m$ with $\mathbf{u}_j \in \mathcal{U}_j$; in this case it is denoted $\mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Proposition 2.1.2. *Sum of Subspaces Is the Smallest Containing Subspace*
Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \dots, \mathcal{U}_m$.

Proof. Since $\mathbf{0} \in \mathcal{U}_j$ for each j, we have $\mathbf{0} \in \mathcal{U}_1 + \cdots + \mathcal{U}_m$. If $\mathbf{x} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$ and $\mathbf{y} = \mathbf{v}_1 + \cdots + \mathbf{v}_m$ lie in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, then for any $c \in \mathbb{F}$, $c\mathbf{x} + \mathbf{y} = (c\mathbf{u}_1 + \mathbf{v}_1) + \cdots + (c\mathbf{u}_m + \mathbf{v}_m) \in \mathcal{U}_1 + \cdots + \mathcal{U}_m$, because $c\mathbf{u}_j + \mathbf{v}_j \in \mathcal{U}_j$ for each j. By Theorem 1.2.2, $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a subspace of $\mathcal{V}$.
Each $\mathcal{U}_j$ is contained in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ by taking all other summands to be $\mathbf{0}$. If $\mathcal{W}$ is any subspace of $\mathcal{V}$ containing $\mathcal{U}_1, \dots, \mathcal{U}_m$, then closure under addition gives $\mathbf{u}_1 + \cdots + \mathbf{u}_m \in \mathcal{W}$ whenever $\mathbf{u}_j \in \mathcal{U}_j$.
Hence $\mathcal{U}_1 + \cdots + \mathcal{U}_m \subseteq \mathcal{W}$, so the sum is the smallest such subspace.
Structural Commentary: Sums of subspaces are obtained by closing several subspaces under finite addition. Minimality follows because every containing subspace is already closed under those sums.
Key Insight: The sum adds no extra structure beyond the linear combinations forced by the given subspaces. □

Perspective 2.1.3. *Sum Is Not Union*
The union $\mathcal{U} \cup \mathcal{W}$ keeps vectors from either subspace; the sum $\mathcal{U} + \mathcal{W}$ adds vectors from both. Unless one subspace contains the other, $\mathcal{U} \cup \mathcal{W}$ is not closed under addition.

Theorem 2.1.4. *Finite Unions of Proper Subspaces*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be over an infinite field $\mathbb{F}$. If $\mathcal{U}_1, \ldots, \mathcal{U}_n \subsetneq \mathcal{V}$ are subspaces, then $\mathcal{V} \neq \bigcup_{j=1}^{n} \mathcal{U}_j$.

Proof. Assume for contradiction that $\mathcal{V} = \bigcup_{j=1}^{n} \mathcal{U}_j$. After deleting redundant subspaces and reindexing, assume the cover is minimal, so $\mathcal{U}_i \not\subseteq \bigcup_{j \neq i} \mathcal{U}_j$. If $n = 1$, then $\mathcal{V} = \mathcal{U}_1$, contradicting $\mathcal{U}_1 \subsetneq \mathcal{V}$; hence $n \geq 2$.

Choose $\mathbf{w} \in \mathcal{U}_1 \setminus \bigcup_{j=2}^{n} \mathcal{U}_j$ and $\mathbf{v} \in \mathcal{V} \setminus \mathcal{U}_1$. Since $\mathbf{0} \in \mathcal{U}_j$ for every j, we have $\mathbf{w} \neq \mathbf{0}$.

Let $A = \{\mathbf{v} + r\mathbf{w} : r \in \mathbb{F}\}$. Then $|A| = |\mathbb{F}| = \infty$, $A \cap \mathcal{U}_1 = \emptyset$, and for $j \geq 2$, $|A \cap \mathcal{U}_j| \leq 1$, since two points $\mathbf{v} + r_1\mathbf{w}, \mathbf{v} + r_2\mathbf{w} \in \mathcal{U}_j$ with $r_1 \neq r_2$ imply $\mathbf{w} \in \mathcal{U}_j$.

Hence $\left| A \cap \bigcup_{j=1}^{n} \mathcal{U}_j \right| \leq n - 1 < \infty$, contradicting $A \subseteq \mathcal{V} = \bigcup_{j=1}^{n} \mathcal{U}_j$.

Structural Commentary: Over an infinite field, finite proper subspaces cannot cover a well-chosen affine line.
Key Insight: A single one-parameter translate has too many points for finitely many proper intersections. □

Perspective 2.1.5. *Sums Always Exist; Directness Is Extra*
For subspaces $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$, the sum $\mathcal{U} + \mathcal{W}$ is always a subspace by Proposition 2.1.2. Writing $\mathcal{U} \oplus \mathcal{W}$ asserts the extra condition $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, equivalently uniqueness of the decomposition $\mathbf{u} + \mathbf{w}$.

Proposition 2.1.6. *Two-Subspace Directness Criterion*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$. Then $\mathcal{U} + \mathcal{W}$ is direct if and only if $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$.

Proof.
(⇒) Suppose $\mathcal{U} + \mathcal{W}$ is direct. If $\mathbf{x} \in \mathcal{U} \cap \mathcal{W}$, then $\mathbf{x} = \mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x}$ gives two decompositions of $\mathbf{x}$ with one term in each subspace. Uniqueness gives $\mathbf{x} = \mathbf{0}$, so $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$.
(⇐) Suppose $\mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$. Every vector in $\mathcal{U} + \mathcal{W}$ has at least one representation $\mathbf{u} + \mathbf{w}$. If $\mathbf{u}_1 + \mathbf{w}_1 = \mathbf{u}_2 + \mathbf{w}_2$, with $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$ and $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, then $\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{w}_2 - \mathbf{w}_1 \in \mathcal{U} \cap \mathcal{W}$, hence equals $\mathbf{0}$.
Thus $\mathbf{u}_1 = \mathbf{u}_2$ and $\mathbf{w}_1 = \mathbf{w}_2$, so the sum is direct.

Structural Commentary: Trivial intersection prevents cancellation between summands.
Key Insight: Directness means unique decomposition. □

Definition 2.1.7. *Subspace Poset, Meet, and Join*
(i) The *subspace poset* of $\mathcal{V}$ is $\mathrm{Sub}(\mathcal{V}) := \{\mathcal{U} \subseteq \mathcal{V} : \mathcal{U} \leq \mathcal{V}\}$, ordered by inclusion.
(ii) For $\{\mathcal{U}_i : i \in I\} \subseteq \mathrm{Sub}(\mathcal{V})$, the *meet* is $\bigwedge_{i \in I} \mathcal{U}_i := \bigcap_{i \in I} \mathcal{U}_i$.
(iii) For $\{\mathcal{U}_i : i \in I\} \subseteq \mathrm{Sub}(\mathcal{V})$, the *join* is $\bigvee_{i \in I} \mathcal{U}_i := \sum_{i \in I} \mathcal{U}_i := \mathrm{span}\left(\bigcup_{i \in I} \mathcal{U}_i\right)$.
(iv) The empty meet is $\mathcal{V}$, and the empty join is $\{\mathbf{0}\}$.

Proposition 2.1.8. *Subspace Meet and Join*
For subspaces $\{\mathcal{U}_i : i \in I\}$ of $\mathcal{V}$, ordered by inclusion, $\bigwedge_{i \in I} \mathcal{U}_i = \inf\{\mathcal{U}_i : i \in I\}$; and $\bigvee_{i \in I} \mathcal{U}_i = \sup\{\mathcal{U}_i : i \in I\}$.

Proof. If $I = \emptyset$, conventions give vacuous bounds $\mathcal{V}$ and $\{\mathbf{0}\}$. Assume $I \neq \emptyset$.
The meet $M = \bigcap_{i \in I} \mathcal{U}_i$ is a subspace by Proposition 1.2.5; moreover $M \subseteq \mathcal{U}_i$ for all i, and any subspace $\mathcal{W}$ with $\mathcal{W} \subseteq \mathcal{U}_i$ for all i satisfies $\mathcal{W} \subseteq M$. The join $J = \mathrm{span}\left(\bigcup_{i \in I} \mathcal{U}_i\right)$ is a subspace by Theorem 1.2.7; moreover $\mathcal{U}_i \subseteq J$ for all i, and any subspace $\mathcal{W}$ with $\mathcal{U}_i \subseteq \mathcal{W}$ for all i contains $\bigcup_{i \in I} \mathcal{U}_i$, hence contains J.
Structural Commentary: Meet enforces common constraints; join closes the combined directions under span.
Key Insight: Meet is intersection, while join is span of the union. □

Theorem 2.1.9. *Subspace Lattice*
The subspace poset $\mathrm{Sub}(\mathcal{V})$ is a complete modular lattice.
Explicitly, for every $\{\mathcal{U}_i : i \in I\} \subseteq \mathrm{Sub}(\mathcal{V})$, $\bot = \{\mathbf{0}\}$, $\top = \mathcal{V}$, $\bigwedge_{i \in I} \mathcal{U}_i = \bigcap_{i \in I} \mathcal{U}_i$, $\bigvee_{i \in I} \mathcal{U}_i = \sum_{i \in I} \mathcal{U}_i = \mathrm{span}\left(\bigcup_{i \in I} \mathcal{U}_i\right)$.
Moreover, if $\mathcal{U} \subseteq \mathcal{S}$, then $\mathcal{S} \cap (\mathcal{T} + \mathcal{U}) = (\mathcal{S} \cap \mathcal{T}) + \mathcal{U}$.

Proof. Proposition 2.1.8 gives arbitrary meets and joins with the displayed formulas; the empty meet and join give $\top = \mathcal{V}$ and $\bot = \{\mathbf{0}\}$. It remains to prove modularity. Assume $\mathcal{U} \subseteq \mathcal{S}$. If $\mathbf{x} \in (\mathcal{S} \cap \mathcal{T}) + \mathcal{U}$, then $\mathbf{x} = \mathbf{y} + \mathbf{u}$ with $\mathbf{y} \in \mathcal{S} \cap \mathcal{T}$ and $\mathbf{u} \in \mathcal{U}$, hence $\mathbf{x} \in \mathcal{S} \cap (\mathcal{T} + \mathcal{U})$. Conversely, if $\mathbf{x} \in \mathcal{S} \cap (\mathcal{T} + \mathcal{U})$, write $\mathbf{x} = \mathbf{t} + \mathbf{u}$ with $\mathbf{t} \in \mathcal{T}$ and $\mathbf{u} \in \mathcal{U}$. Since $\mathbf{x} \in \mathcal{S}$ and $\mathbf{u} \in \mathcal{U} \subseteq \mathcal{S}$, $\mathbf{t} = \mathbf{x} - \mathbf{u} \in \mathcal{S} \cap \mathcal{T}$, so $\mathbf{x} \in (\mathcal{S} \cap \mathcal{T}) + \mathcal{U}$.
Structural Commentary: The subspace lattice is complete and modular, not generally distributive.
Key Insight: If one summand already lies inside $\mathcal{S}$, intersecting with $\mathcal{S}$ can be pushed through the sum. □

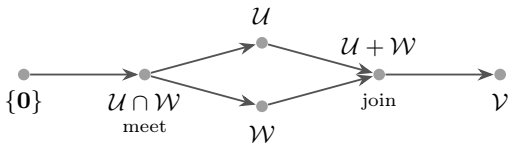


Figure 10: For two subspaces, intersection is the meet and sum is the join under inclusion.

Proposition 2.1.10. *Equivalent Conditions for a Direct Sum*
For subspaces $\mathcal{U}_1, \ldots, \mathcal{U}_m \subseteq \mathcal{V}$, the following are equivalent:

(i) $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$;
(ii) *every* $\mathbf{v} \in \mathcal{V}$ *has a unique representation* $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m$ *with* $\mathbf{u}_j \in \mathcal{U}_j$;
(iii) $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, *and if* $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ *with* $\mathbf{u}_j \in \mathcal{U}_j$, *then* $\mathbf{u}_1 = \cdots = \mathbf{u}_m = \mathbf{0}$.

Proof.

(i) $\Rightarrow$ *(ii)* This follows from the definition of direct sum.
(ii) $\Rightarrow$ *(iii)* Every vector lies in $\mathcal{U}_1 + \cdots + \mathcal{U}_m$, so $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$. If $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$, then this representation of $\mathbf{0}$ must equal $\mathbf{0} + \cdots + \mathbf{0}$ by uniqueness, so each $\mathbf{u}_j = \mathbf{0}$.
(iii) $\Rightarrow$ *(i)* Since $\mathcal{V} = \mathcal{U}_1 + \cdots + \mathcal{U}_m$, every $\mathbf{v} \in \mathcal{V}$ has at least one representation. If $\mathbf{v} = \mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{w}_1 + \cdots + \mathbf{w}_m$, then $(\mathbf{u}_1 - \mathbf{w}_1) + \cdots + (\mathbf{u}_m - \mathbf{w}_m) = \mathbf{0}$. By (iii), $\mathbf{u}_j - \mathbf{w}_j = \mathbf{0}$ for every j, so the representation is unique. Hence $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$.

Structural Commentary: Direct sums require both spanning by the summands and no cancellation among them. The zero-representation condition isolates the no-cancellation part.
Key Insight: Uniqueness is equivalent to saying the only way the summands add to zero is trivially. □

Definition 2.1.11. *Support and Internal Direct Sum of a Family*
Let I be an index set. For each $i \in I$, let $\mathcal{X}_i$ be a vector space with zero vector $\mathbf{0}_i$.

(i) *Support.* For $\mathbf{x} = (\mathbf{x}_i)_{i \in I} \in \prod_{i \in I} \mathcal{X}_i$, define $\operatorname{supp}(\mathbf{x}) := \{i \in I : \mathbf{x}_i \neq \mathbf{0}_i\}$.

(ii) The indexed family $\mathbf{x}$ is *finitely supported* iff $|\operatorname{supp}(\mathbf{x})| < \infty$.
(iii) *Finite sum.* If all $\mathcal{X}_i$ lie in a common vector space and $\mathbf{x}$ is finitely supported, define $\sum_{i \in I} \mathbf{x}_i := \sum_{i \in \operatorname{supp}(\mathbf{x})} \mathbf{x}_i$.
(iv) *Internal direct sum.* For $\{\mathcal{U}_i : i \in I\} \subseteq \operatorname{Sub}(\mathcal{V})$, $\mathcal{V} = \bigoplus_{i \in I} \mathcal{U}_i$ means every $\mathbf{v} \in \mathcal{V}$ has a unique finitely supported $\mathbf{u} = (\mathbf{u}_i)_{i \in I}$ with $\mathbf{u}_i \in \mathcal{U}_i$ and $\mathbf{v} = \sum_{i \in I} \mathbf{u}_i$.

Equivalently, $\mathcal{V} = \sum_{i \in I} \mathcal{U}_i$ and every finite relation $\sum_{i \in F} \mathbf{u}_i = \mathbf{0}$, with $F \subseteq I$ finite and $\mathbf{u}_i \in \mathcal{U}_i$, has $\mathbf{u}_i = \mathbf{0}$ $\forall i \in F$.

Proposition 2.1.12. *Family Direct-Sum Criterion*
For $\{\mathcal{U}_i\}_{i \in I} \subseteq \operatorname{Sub}(\mathcal{V})$, $\sum_{i \in I} \mathcal{U}_i$ is direct $\iff \forall i \in I$, $\mathcal{U}_i \cap \sum_{j \in I \setminus \{i\}} \mathcal{U}_j = \{\mathbf{0}\}$.

Proof.

($\Rightarrow$) If $\mathbf{x} \in \mathcal{U}_i \cap \sum_{j \in I \setminus \{i\}} \mathcal{U}_j$, then $\mathbf{x} = \sum_{j \in F} \mathbf{u}_j$ for some finite $F \subseteq I \setminus \{i\}$ and $\mathbf{u}_j \in \mathcal{U}_j$. Directness applied to $\mathbf{x} - \sum_{j \in F} \mathbf{u}_j = \mathbf{0}$ gives $\mathbf{x} = \mathbf{0}$.

($\Leftarrow$) If $\sum_{k=1}^{m} \mathbf{u}_{i_k} = \mathbf{0}$ with distinct i_k and $\mathbf{u}_{i_k} \in \mathcal{U}_{i_k}$, then for each k, $\mathbf{u}_{i_k} = -\sum_{\ell \neq k} \mathbf{u}_{i_\ell} \in \mathcal{U}_{i_k} \cap \sum_{j \in I \setminus \{i_k\}} \mathcal{U}_j = \{\mathbf{0}\}$. Thus all finite zero-relations are trivial, so the sum is direct.

Structural Commentary: Family directness requires each summand to miss all others together.
Key Insight: Directness is subspace independence. □

Definition 2.1.13. *Product of Vector Spaces*
Let $\mathcal{V}_1, \dots, \mathcal{V}_m$ be vector spaces over $\mathbb{F}$. Their *product* is $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m := \{(\mathbf{v}_1, \dots, \mathbf{v}_m) : \mathbf{v}_j \in \mathcal{V}_j\}$.
Addition and scalar multiplication are defined componentwise:
$(\mathbf{u}_1, \dots, \mathbf{u}_m) + (\mathbf{v}_1, \dots, \mathbf{v}_m) := (\mathbf{u}_1 + \mathbf{v}_1, \dots, \mathbf{u}_m + \mathbf{v}_m)$ and $a(\mathbf{v}_1, \dots, \mathbf{v}_m) := (a\mathbf{v}_1, \dots, a\mathbf{v}_m)$.

Proposition 2.1.14. *Products Are Vector Spaces*
If $\mathcal{V}_1, \dots, \mathcal{V}_m$ are vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is a vector space under componentwise operations.

Proof. The additive identity is $(\mathbf{0}_1, \dots, \mathbf{0}_m)$, where $\mathbf{0}_j$ is the zero vector of $\mathcal{V}_j$, and additive inverse of $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is $(-\mathbf{v}_1, \dots, -\mathbf{v}_m)$. All remaining vector space axioms hold componentwise because they hold in each $\mathcal{V}_j$.
Structural Commentary: Product spaces collect several vector spaces into one coordinatewise structure.
Key Insight: Every vector space axiom reduces to the same axiom in each component. □

Theorem 2.1.15. *Dimension of a Product*
If $\mathcal{V}_1, \dots, \mathcal{V}_m$ are finite-dimensional vector spaces over $\mathbb{F}$, then $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ is finite-dimensional, and $\dim(\mathcal{V}_1 \times \cdots \times \mathcal{V}_m) = \dim(\mathcal{V}_1) + \cdots + \dim(\mathcal{V}_m)$.

Proof. For each j, let $(\mathbf{v}_{j,1}, \dots, \mathbf{v}_{j,n_j})$ be a basis of $\mathcal{V}_j$. For every j and k, let $\mathbf{e}_{j,k}$ denote the element of $\mathcal{V}_1 \times \cdots \times \mathcal{V}_m$ whose jth component is $\mathbf{v}_{j,k}$ and whose other components are zero.
The list of all $\mathbf{e}_{j,k}$ spans the product because each component can be expanded in its own basis. If a linear combination of the $\mathbf{e}_{j,k}$ is zero, then looking at the jth component gives a linear relation among $(\mathbf{v}_{j,1}, \dots, \mathbf{v}_{j,n_j})$, so all coefficients in that component are zero. Hence the list is linearly independent and therefore is a basis of the product. Its length is $n_1 + \cdots + n_m$.
Structural Commentary: A basis for a product is formed by placing each component basis into its own slot.
Key Insight: Product dimension counts the independent directions contributed by each factor. □

Definition 2.1.16. *External Product and Direct Sum*
Let $\{\mathcal{V}_i : i \in I\}$ be $\mathbb{F}$-vector spaces.

(i) The *external direct product* is $\prod_{i \in I} \mathcal{V}_i = \{\mathbf{v} : I \to \bigcup_{i \in I} \mathcal{V}_i : \mathbf{v}(i) \in \mathcal{V}_i\}$.

(ii) Operations are componentwise: $(\mathbf{u} + \mathbf{v})(i) = \mathbf{u}(i) + \mathbf{v}(i)$ and $(a\mathbf{v})(i) = a\mathbf{v}(i)$.

(iii) The *external direct sum* is $\bigoplus^{\text{ext}}_{i \in I} \mathcal{V}_i = \{\mathbf{v} \in \prod_{i \in I} \mathcal{V}_i : |\operatorname{supp}(\mathbf{v})| < \infty\}$. If $|I| < \infty$, then $\bigoplus^{\text{ext}}_{i \in I} \mathcal{V}_i = \prod_{i \in I} \mathcal{V}_i$.

Proposition 2.1.17. *External Sums Are Finite-Support Products*
With componentwise operations, $\prod_{i \in I} \mathcal{V}_i$ is an $\mathbb{F}$-vector space, $\bigoplus^{\text{ext}}_{i \in I} \mathcal{V}_i = \{\mathbf{v} \in \prod_{i \in I} \mathcal{V}_i : |\operatorname{supp}(\mathbf{v})| < \infty\} \leq \prod_{i \in I} \mathcal{V}_i$.

Proof. Let $P = \prod_{i \in I} \mathcal{V}_i$. For $\mathbf{u}, \mathbf{v} \in P$ and $a, b \in \mathbb{F}$, $(a\mathbf{u} + b\mathbf{v})(i) = a\mathbf{u}(i) + b\mathbf{v}(i) \in \mathcal{V}_i$.
The zero $0_P(i) = 0_i$, inverses $(-\mathbf{v})(i) = -\mathbf{v}(i)$, and all axioms are coordinatewise.
Let $S = \bigoplus^{\text{ext}}_{i \in I} \mathcal{V}_i$. If $\mathbf{u}, \mathbf{v} \in S$ and $a, b \in \mathbb{F}$, then $\operatorname{supp}(a\mathbf{u} + b\mathbf{v}) \subseteq \operatorname{supp}(\mathbf{u}) \cup \operatorname{supp}(\mathbf{v})$, so $a\mathbf{u} + b\mathbf{v} \in S$.
Also $0_P \in S$. Hence $S \leq P$.
Structural Commentary: $\prod_i \mathcal{V}_i$: arbitrary support; $\bigoplus^{\text{ext}}_i \mathcal{V}_i$: finite support.
Key Insight: Finite support is exactly what finite linear combinations preserve. □

Proposition 2.1.18. *Direct Sum Map*
Suppose $\mathcal{U}_1, \dots, \mathcal{U}_m$ are subspaces of $\mathcal{V}$. Define $\Gamma : \mathcal{U}_1 \times \cdots \times \mathcal{U}_m \to \mathcal{U}_1 + \cdots + \mathcal{U}_m$ by $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_m) = \mathbf{u}_1 + \cdots + \mathbf{u}_m$. Then $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is a direct sum if and only if Γ is injective.

Proof.

(⇒) Suppose $\mathcal{U}_1 + \cdots + \mathcal{U}_m$ is direct. If $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_m) = \mathbf{0}$, then $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$, so directness gives $\mathbf{u}_j = \mathbf{0}$ for all j. Hence $\ker \Gamma = \{(\mathbf{0}, \dots, \mathbf{0})\}$, and Γ is injective.

(⇐) Suppose Γ is injective. If $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in \mathcal{U}_j$, then $\Gamma(\mathbf{u}_1, \dots, \mathbf{u}_m) = \Gamma(\mathbf{0}, \dots, \mathbf{0})$. Injectivity gives $\mathbf{u}_j = \mathbf{0}$ for all j, so every zero decomposition is trivial. Hence the sum is direct.

Structural Commentary: The addition map identifies decompositions precisely through its kernel.
Key Insight: Directness is injectivity of Γ. □

2.2 Quotient Spaces

Definition 2.2.1. *Affine Subset and Parallel Subspaces*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. For $\mathbf{v} \in \mathcal{V}$, the subset $\mathbf{v}+\mathcal{U} := \{\mathbf{v}+\mathbf{u} : \mathbf{u} \in \mathcal{U}\}$ is an *affine subset* of $\mathcal{V}$ parallel to $\mathcal{U}$.

Proposition 2.2.2. *Parallel Affine Subsets Are Equal or Disjoint*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$ and let $\mathbf{v}, \mathbf{w} \in \mathcal{V}$. The following are equivalent:

(i) $\mathbf{v}-\mathbf{w} \in \mathcal{U}$;
(ii) $\mathbf{v}+\mathcal{U} = \mathbf{w}+\mathcal{U}$;
(iii) $(\mathbf{v}+\mathcal{U}) \cap (\mathbf{w}+\mathcal{U}) \neq \emptyset$.

Proof.

(i) $\Rightarrow$ (ii) If $\mathbf{v}-\mathbf{w} \in \mathcal{U}$ and $\mathbf{u} \in \mathcal{U}$, then $\mathbf{v}+\mathbf{u} = \mathbf{w}+((\mathbf{v}-\mathbf{w})+\mathbf{u}) \in \mathbf{w}+\mathcal{U}$. Thus $\mathbf{v}+\mathcal{U} \subseteq \mathbf{w}+\mathcal{U}$, and the reverse inclusion is similar.
(ii) $\Rightarrow$ (iii) Equal affine subsets have the same elements.
(iii) $\Rightarrow$ (i) If $\mathbf{x} \in (\mathbf{v}+\mathcal{U}) \cap (\mathbf{w}+\mathcal{U})$, then $\mathbf{x} = \mathbf{v}+\mathbf{u}_1 = \mathbf{w}+\mathbf{u}_2$ for some $\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$. Hence $\mathbf{v}-\mathbf{w} = \mathbf{u}_2-\mathbf{u}_1 \in \mathcal{U}$.

Structural Commentary: Parallel affine subsets either coincide or do not meet.
Key Insight: Intersection is controlled by the difference of base points. □

Definition 2.2.3. *Quotient Space*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$.
The *quotient space* $\mathcal{V}/\mathcal{U}$ is the set of all affine subsets of $\mathcal{V}$ parallel to $\mathcal{U}$: $\mathcal{V}/\mathcal{U} := \{\mathbf{v}+\mathcal{U} : \mathbf{v} \in \mathcal{V}\}$.

Perspective 2.2.4. *Cosets as Equivalence Classes*
The quotient relation is $\mathbf{v} \sim \mathbf{w}$ when $\mathbf{v}-\mathbf{w} \in \mathcal{U}$. Proposition 2.2.2 shows that the equivalence class of $\mathbf{v}$ is exactly the coset $\mathbf{v}+\mathcal{U}$. Thus the quotient space remembers vectors only up to directions in $\mathcal{U}$.

Proposition 2.2.5. *Quotient Relation Is a Linear Congruence*
Let $\mathcal{U} \leq \mathcal{V}$ and define $\mathbf{v} \equiv_{\mathcal{U}} \mathbf{w} \iff \mathbf{v}-\mathbf{w} \in \mathcal{U}$.
Then $\equiv_{\mathcal{U}}$ is an equivalence relation and, for $a, b \in \mathbb{F}$, $\mathbf{v}_1 \equiv_{\mathcal{U}} \mathbf{v}_2$, $\mathbf{w}_1 \equiv_{\mathcal{U}} \mathbf{w}_2 \implies a\mathbf{v}_1+b\mathbf{w}_1 \equiv_{\mathcal{U}} a\mathbf{v}_2+b\mathbf{w}_2$.

Proof. Since $\mathcal{U} \leq \mathcal{V}$, $\mathbf{0} \in \mathcal{U}$, $-\mathcal{U} = \mathcal{U}$, and $\mathcal{U}+\mathcal{U} \subseteq \mathcal{U}$; hence $\equiv_{\mathcal{U}}$ is reflexive, symmetric, and transitive.
If $\mathbf{v}_1 \equiv_{\mathcal{U}} \mathbf{v}_2$ and $\mathbf{w}_1 \equiv_{\mathcal{U}} \mathbf{w}_2$, then $(a\mathbf{v}_1+b\mathbf{w}_1)-(a\mathbf{v}_2+b\mathbf{w}_2) = a(\mathbf{v}_1-\mathbf{v}_2)+b(\mathbf{w}_1-\mathbf{w}_2) \in \mathcal{U}$.
Structural Commentary: Subspace congruence makes quotient operations well-defined.
Key Insight: Subspace closure is exactly the well-definedness mechanism for quotient operations. □

Proposition 2.2.6. *Structure of the Quotient*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Addition and scalar multiplication on $\mathcal{V}/\mathcal{U}$ defined by $(\mathbf{v}+\mathcal{U})+(\mathbf{w}+\mathcal{U}) := (\mathbf{v}+\mathbf{w})+\mathcal{U}$ and $a(\mathbf{v}+\mathcal{U}) := a\mathbf{v}+\mathcal{U}$ are well-defined, and with these operations $\mathcal{V}/\mathcal{U}$ is a vector space.

Proof. If $\mathbf{v}+\mathcal{U} = \mathbf{v}'+\mathcal{U}$ and $\mathbf{w}+\mathcal{U} = \mathbf{w}'+\mathcal{U}$, then $\mathbf{v}-\mathbf{v}' \in \mathcal{U}$ and $\mathbf{w}-\mathbf{w}' \in \mathcal{U}$.
Since $\mathcal{U}$ is a subspace, $(\mathbf{v}+\mathbf{w})-(\mathbf{v}'+\mathbf{w}') \in \mathcal{U}$, and $a\mathbf{v}-a\mathbf{v}' \in \mathcal{U}$ so the operations do not depend on the chosen representatives. The vector space axioms are inherited from $\mathcal{V}$ under these operations.
Structural Commentary: Quotients preserve operations modulo $\mathcal{U}$.
Key Insight: Cosets identify vectors differing by elements of $\mathcal{U}$. □

Theorem 2.2.7. *Dimension of a Quotient*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace, then $\dim(\mathcal{V}/\mathcal{U}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_k)$ of $\mathcal{V}$.
The cosets $\mathbf{w}_1+\mathcal{U}, \ldots, \mathbf{w}_k+\mathcal{U}$ span $\mathcal{V}/\mathcal{U}$ because the $\mathbf{u}_j$-terms vanish in the quotient.
If $a_1(\mathbf{w}_1+\mathcal{U})+\cdots+a_k(\mathbf{w}_k+\mathcal{U}) = \mathcal{U}$, then $a_1\mathbf{w}_1+\cdots+a_k\mathbf{w}_k \in \mathcal{U}$, so a linear relation in the extended basis forces $a_1 = \cdots = a_k = 0$. Hence these cosets form a basis of $\mathcal{V}/\mathcal{U}$, and its length is $k = \dim(\mathcal{V}) - \dim(\mathcal{U})$.
Structural Commentary: Quotiening removes the directions in $\mathcal{U}$ and keeps the added basis directions.
Key Insight: Passing to $\mathcal{V}/\mathcal{U}$ subtracts the collapsed dimension. □

This block develops two complementary ways to organise a vector space: build from smaller pieces by spans and direct sums, or collapse a chosen subspace by passing to a quotient. Bases and dimension measure exactly what is added, split off, or made invisible.

2.3 Decomposition and Complementation

Theorem 2.3.1. *Dimension Formula for a Sum*
If $\mathcal{U}, \mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$, then $\dim(\mathcal{U}+\mathcal{W}) = \dim(\mathcal{U}) + \dim(\mathcal{W}) - \dim(\mathcal{U}\cap\mathcal{W})$.

Proof. Choose a basis $(\mathbf{e}_1,\dots,\mathbf{e}_r)$ of $\mathcal{U}\cap\mathcal{W}$, extend it to a basis $(\mathbf{e}_1,\dots,\mathbf{e}_r,\mathbf{u}_1,\dots,\mathbf{u}_p)$ of $\mathcal{U}$, and to a basis $(\mathbf{e}_1,\dots,\mathbf{e}_r,\mathbf{w}_1,\dots,\mathbf{w}_q)$ of $\mathcal{W}$. Then $(\mathbf{e}_1,\dots,\mathbf{e}_r,\mathbf{u}_1,\dots,\mathbf{u}_p,\mathbf{w}_1,\dots,\mathbf{w}_q)$ spans $\mathcal{U}+\mathcal{W}$ because every vector in the sum is the sum of one vector from $\mathcal{U}$ and one vector from $\mathcal{W}$.
To prove linear independence, suppose $a_1\mathbf{e}_1+\cdots+a_r\mathbf{e}_r+b_1\mathbf{u}_1+\cdots+b_p\mathbf{u}_p+c_1\mathbf{w}_1+\cdots+c_q\mathbf{w}_q=\mathbf{0}$. Then $a_1\mathbf{e}_1+\cdots+a_r\mathbf{e}_r+b_1\mathbf{u}_1+\cdots+b_p\mathbf{u}_p$ lies in $\mathcal{U}\cap\mathcal{W}$. Since $(\mathbf{e}_1,\dots,\mathbf{e}_r)$ is a basis of $\mathcal{U}\cap\mathcal{W}$ and $(\mathbf{e}_1,\dots,\mathbf{e}_r,\mathbf{u}_1,\dots,\mathbf{u}_p)$ is a basis of $\mathcal{U}$, we get $b_1=\cdots=b_p=0$. The remaining relation lies in the basis $(\mathbf{e}_1,\dots,\mathbf{e}_r,\mathbf{w}_1,\dots,\mathbf{w}_q)$ of $\mathcal{W}$, so $a_1=\cdots=a_r=c_1=\cdots=c_q=0$. Thus the displayed list is a basis of $\mathcal{U}+\mathcal{W}$.
Therefore $\dim(\mathcal{U}+\mathcal{W}) = r+p+q = (r+p)+(r+q)-r$.
Structural Commentary: The formula corrects double-counting: vectors in $\mathcal{U}\cap\mathcal{W}$ are counted once in each subspace. Extending an intersection basis separates shared directions from new directions.
Key Insight: The dimension of a sum is the sum of dimensions minus the overlap. □

Corollary 2.3.2. *Dimension of a Direct Sum*
If $\mathcal{U}$ and $\mathcal{W}$ are finite-dimensional subspaces of $\mathcal{V}$ with $\mathcal{U}\cap\mathcal{W}=\{\mathbf{0}\}$, then $\dim(\mathcal{U}\oplus\mathcal{W}) = \dim(\mathcal{U})+\dim(\mathcal{W})$.

Proof. Since $\mathcal{U}\cap\mathcal{W}=\{\mathbf{0}\}$, the sum is direct and $\dim(\mathcal{U}\cap\mathcal{W})=0$. The result follows from Theorem 2.3.1.
Structural Commentary: When the overlap is trivial, there is no double-counting in the dimension formula. Direct sums are exactly the case where the overlap term vanishes.
Key Insight: Dimensions add exactly for direct sums. □

Theorem 2.3.3. *Direct Sums and Dimension*
Suppose $\mathcal{V}$ is finite-dimensional and $\mathcal{U}_1,\dots,\mathcal{U}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{U}_1+\cdots+\mathcal{U}_m$ is a direct sum if and only if $\dim(\mathcal{U}_1+\cdots+\mathcal{U}_m) = \dim(\mathcal{U}_1)+\cdots+\dim(\mathcal{U}_m)$.

Proof. For each j, choose a basis β_j of $\mathcal{U}_j$, and concatenate these bases into a list β.
This list spans $\mathcal{U}_1+\cdots+\mathcal{U}_m$ and has length $\dim(\mathcal{U}_1)+\cdots+\dim(\mathcal{U}_m)$.

(⇒) If the sum is direct, any linear relation among the vectors in β gives a zero decomposition $\mathbf{u}_1+\cdots+\mathbf{u}_m=\mathbf{0}$ with $\mathbf{u}_j\in\mathcal{U}_j$. Directness gives $\mathbf{u}_j=\mathbf{0}$ for every j, and independence of each β_j gives all coefficients zero. Thus β is a basis of the sum, and the dimensions add.

(⇐) If the dimensions add, then β is a spanning list whose length equals the dimension of the sum, so Proposition 1.4.3 makes β a basis. A zero decomposition $\mathbf{u}_1+\cdots+\mathbf{u}_m=\mathbf{0}$ expands to a zero linear relation in this basis, so every $\mathbf{u}_j=\mathbf{0}$. Hence the sum is direct.

Structural Commentary: Direct sums add dimensions without hidden overlap.
Key Insight: Equality of dimensions means the concatenated bases have no linear redundancy. □

Definition 2.3.4. *Basis Adapted to a Direct Sum*
Suppose $\mathcal{V}=\mathcal{U}_1\oplus\cdots\oplus\mathcal{U}_m$ and each β_j is a basis of $\mathcal{U}_j$. The concatenated basis $\beta=(\beta_1,\dots,\beta_m)$ of $\mathcal{V}$ is called a *basis adapted to the direct-sum decomposition*.
Equivalently, the vectors in an adapted basis are grouped by the summands whose direct sum is $\mathcal{V}$.

Corollary 2.3.5. *Large Subspaces Must Intersect*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of a finite-dimensional vector space $\mathcal{V}$, and let k be a positive integer.

(i) If $\dim(\mathcal{U})+\dim(\mathcal{W})>\dim(\mathcal{V})$, then $\mathcal{U}\cap\mathcal{W}$ contains a nonzero vector.
(ii) If $\dim(\mathcal{U})+\dim(\mathcal{W})\geq\dim(\mathcal{V})+k$, then $\mathcal{U}\cap\mathcal{W}$ contains k linearly independent vectors.

Proof. Since $\mathcal{U}+\mathcal{W}\subseteq\mathcal{V}$, Theorem 2.3.1 gives
$\dim(\mathcal{U}\cap\mathcal{W}) = \dim(\mathcal{U})+\dim(\mathcal{W})-\dim(\mathcal{U}+\mathcal{W}) \geq \dim(\mathcal{U})+\dim(\mathcal{W})-\dim(\mathcal{V})$.
If $\dim(\mathcal{U})+\dim(\mathcal{W})>\dim(\mathcal{V})$, the right-hand side is at least 1, so the intersection contains a nonzero vector.
If $\dim(\mathcal{U})+\dim(\mathcal{W})\geq\dim(\mathcal{V})+k$, then $\dim(\mathcal{U}\cap\mathcal{W})\geq k$, so intersection contains k linearly independent vectors.
Structural Commentary: Excess dimension appears as overlap.
Key Insight: Two large subspaces cannot fit inside $\mathcal{V}$ without sharing directions. □

Proposition 2.3.6. *Complements Exist in Finite Dimensions*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}\subseteq\mathcal{V}$ is a subspace, then there exists a subspace $\mathcal{W}$ such that $\mathcal{V}=\mathcal{U}\oplus\mathcal{W}$.

Proof. Choose a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_n)$ of $\mathcal{V}$.
Let $\mathcal{W} = \operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_n)$. Every vector in $\mathcal{V}$ is a linear combination of this extended basis, so $\mathcal{V} = \mathcal{U} + \mathcal{W}$. If $\mathbf{v} \in \mathcal{U} \cap \mathcal{W}$, then $\mathbf{v}$ has one representation using only $\mathbf{u}_1, \dots, \mathbf{u}_m$ and another using only $\mathbf{w}_1, \dots, \mathbf{w}_n$.
Uniqueness of coefficients in the basis of $\mathcal{V}$ forces $\mathbf{v} = \mathbf{0}$. Hence $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.
Structural Commentary: Complements split a space into a subspace and the remaining basis directions.
Key Insight: Extend a basis of $\mathcal{U}$ to split $\mathcal{V}$. □

Theorem 2.3.7. *Complements Exist*
Assuming Zorn's lemma, for every $\mathcal{U} \leq \mathcal{V}$, $\exists\, \mathcal{W} \leq \mathcal{V}$ such that $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proof. By Theorem 1.4.6, choose a Hamel basis $\mathcal{B}_\mathcal{U}$ of $\mathcal{U}$ and a Hamel basis $\mathcal{B} \supseteq \mathcal{B}_\mathcal{U}$ of $\mathcal{V}$.
Put $\mathcal{C} = \mathcal{B} \setminus \mathcal{B}_\mathcal{U}$ and $\mathcal{W} = \operatorname{span}(\mathcal{C})$. For $\mathbf{v} \in \mathcal{V}$, its finite $\mathcal{B}$-expansion splits as $\mathbf{v} = \sum_{b \in F_\mathcal{U}} a_b b + \sum_{c \in F_\mathcal{C}} a_c c \in \mathcal{U} + \mathcal{W}$, with finite $F_\mathcal{U} \subseteq \mathcal{B}_\mathcal{U}$ and $F_\mathcal{C} \subseteq \mathcal{C}$. If $\mathbf{x} \in \mathcal{U} \cap \mathcal{W}$, then its $\mathcal{B}$-coordinates are supported in both $\mathcal{B}_\mathcal{U}$ and $\mathcal{C}$; uniqueness gives $\mathbf{x} = \mathbf{0}$. Hence $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.
Structural Commentary: Extend a basis of $\mathcal{U}$; span the new directions.
Key Insight: The uniqueness of basis expansion becomes the uniqueness of direct-sum decomposition. □

3 Linear Maps and Matrix Representations

3.1 Linear Maps and Operations

Linear maps preserve vector-space structure. Kernels, ranges, rank-nullity, matrices, change of basis, and similarity describe them without making coordinates primary.

Definition 3.1.1. *Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces over the same field $\mathbb{F}$.
A function $T : \mathcal{V} \to \mathcal{W}$ is *linear* if $T(a\mathbf{u} + b\mathbf{v}) = aT\mathbf{u} + bT\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $a, b \in \mathbb{F}$.
The set of all linear maps from $\mathcal{V}$ to $\mathcal{W}$ is denoted $\mathcal{L}(\mathcal{V}, \mathcal{W})$. If $\mathcal{V} = \mathcal{W}$, elements of $\mathcal{L}(\mathcal{V})$ are *operators* on $\mathcal{V}$.

Definition 3.1.2. *Zero and Identity Maps*
For vector spaces $\mathcal{V}$ and $\mathcal{W}$, the *zero map* from $\mathcal{V}$ to $\mathcal{W}$ is $0_{\mathcal{V},\mathcal{W}} : \mathcal{V} \to \mathcal{W}$, where $0_{\mathcal{V},\mathcal{W}}\mathbf{v} = \mathbf{0}_\mathcal{W}$ for all $\mathbf{v} \in \mathcal{V}$.
The *identity map* on $\mathcal{V}$ is $I_\mathcal{V} : \mathcal{V} \to \mathcal{V}$, where $I_\mathcal{V}\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathcal{V}$.

Proposition 3.1.3. *Linear Maps Send Zero to Zero*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T\mathbf{0}_\mathcal{V} = \mathbf{0}_\mathcal{W}$.

Proof. Linearity gives $T\mathbf{0}_\mathcal{V} = T(\mathbf{0}_\mathcal{V} + \mathbf{0}_\mathcal{V}) = T\mathbf{0}_\mathcal{V} + T\mathbf{0}_\mathcal{V}$. Adding $-T\mathbf{0}_\mathcal{V}$ to both sides gives $T\mathbf{0}_\mathcal{V} = \mathbf{0}_\mathcal{W}$.
Structural Commentary: Linearity preserves zero.
Key Insight: A linear map cannot move zero away from zero. □

Definition 3.1.4. *Operations on Linear Maps*
For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, define pointwise $(S + T)\mathbf{v} := S\mathbf{v} + T\mathbf{v}$ and $(aT)\mathbf{v} := a(T\mathbf{v})$ for every $\mathbf{v} \in \mathcal{V}$.

Proposition 3.1.5. *Linear Maps Form a Vector Space*
With the operations above, $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is a vector space over $\mathbb{F}$.

Proof. If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a, c \in \mathbb{F}$, then for $\mathbf{u}, \mathbf{v} \in \mathcal{V}$,

$$(aS + T)(c\mathbf{u} + \mathbf{v}) = a(cS\mathbf{u} + S\mathbf{v}) + (cT\mathbf{u} + T\mathbf{v}) = c(aS + T)\mathbf{u} + (aS + T)\mathbf{v}.$$

Thus $\mathcal{L}(\mathcal{V}, \mathcal{W})$ is closed under linear combinations. The zero map is the additive identity, $-T$ is the additive inverse of T, and the remaining vector-space axioms hold pointwise because they hold in $\mathcal{W}$.
Structural Commentary: Linear map operations are pointwise.
Key Insight: Pointwise sums and scalar multiples preserve linearity. □

Definition 3.1.6. *Product of Linear Maps*
For $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *product* $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$ is defined by $(ST)\mathbf{u} := S(T\mathbf{u})$ for all $\mathbf{u} \in \mathcal{U}$.

Proposition 3.1.7. *Algebra of Linear Map Products*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$. Whenever defined, $(RS)T = R(ST)$, $TI_\mathcal{U} = T = I_\mathcal{V}T$, $(S_1 + S_2)T = S_1T + S_2T$, and $S(T_1 + T_2) = ST_1 + ST_2$.

Proof. For $\mathbf{u}, \mathbf{x} \in \mathcal{U}$ and $a \in \mathbb{F}$, $(ST)(a\mathbf{u} + \mathbf{x}) = S(aT\mathbf{u} + T\mathbf{x}) = aS(T\mathbf{u}) + S(T\mathbf{x}) = a(ST)\mathbf{u} + (ST)\mathbf{x}$, hence $ST \in \mathcal{L}(\mathcal{U}, \mathcal{W})$. For each remaining identity $A = B$ and admissible $\mathbf{z}$, $A\mathbf{z} = B\mathbf{z}$ by associativity of evaluation, identity maps, or distributivity in the target space; hence $A = B$.
Structural Commentary: Composition chains linear actions, while addition distributes through evaluation.
Key Insight: Linear map algebra is function composition plus pointwise vector-space algebra. □

Definition 3.1.8. *Graph of a Function*
For a function $T : \mathcal{V} \to \mathcal{W}$, the *graph* of T is $\text{graph}(T) := \{(\mathbf{v}, T\mathbf{v}) : \mathbf{v} \in \mathcal{V}\} \subseteq \mathcal{V} \times \mathcal{W}$.

Proposition 3.1.9. *Linearity and Graphs*
A function $T : \mathcal{V} \to \mathcal{W}$ is linear if and only if $\text{graph}(T)$ *is a subspace of $\mathcal{V} \times \mathcal{W}$.*

Proof.
($\Rightarrow$) If T is linear, then $(\mathbf{0}, \mathbf{0}) \in \text{graph}(T)$.
If $(\mathbf{u}, T\mathbf{u}), (\mathbf{v}, T\mathbf{v}) \in \text{graph}(T)$ and $a \in \mathbb{F}$, then $a(\mathbf{u}, T\mathbf{u}) + (\mathbf{v}, T\mathbf{v}) = (a\mathbf{u} + \mathbf{v}, T(a\mathbf{u} + \mathbf{v})) \in \text{graph}(T)$.
Thus $\text{graph}(T)$ is a subspace.
($\Leftarrow$) If $\text{graph}(T)$ is a subspace, then for $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $a \in \mathbb{F}$, the vector $a(\mathbf{u}, T\mathbf{u}) + (\mathbf{v}, T\mathbf{v}) = (a\mathbf{u} + \mathbf{v}, aT\mathbf{u} + T\mathbf{v})$ lies in $\text{graph}(T)$. Hence $T(a\mathbf{u} + \mathbf{v}) = aT\mathbf{u} + T\mathbf{v}$, so T is linear.
Structural Commentary: A graph records a function as a subset of a product space.
Key Insight: Closure of the graph is exactly linearity of the function. □

Principle 3.1.10. *Coordinate-Free Principle*
A linear map is completely determined by what it does to a basis. Coordinates do not create the map; they only encode it once bases have been chosen.

Proposition 3.1.11. *Linear Maps Are Determined by a Basis*
If $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$ and $\mathbf{w}_1, \dots, \mathbf{w}_n \in \mathcal{W}$ are arbitrary, then $\exists$ a unique linear map $T : \mathcal{V} \to \mathcal{W}$ such that $T(\mathbf{v}_j) = \mathbf{w}_j$ for $j = 1, \dots, n$.

Proof. Every $\mathbf{v} \in \mathcal{V}$ has a unique expression $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Define $T(\mathbf{v}) := a_1\mathbf{w}_1 + \cdots + a_n\mathbf{w}_n$.
This is well-defined because basis coordinates are unique, and linearity follows from coordinate expansions. Any linear map with the prescribed basis values must agree with this formula, so it is unique.
Structural Commentary: A basis reduces a linear map to finitely many choices.
Key Insight: Once basis images are fixed, linearity determines every other value. □

3.2 Kernels, Images, Rank, and Nullity

Definition 3.2.1. *Kernel and Range*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, define $\ker(T) := \{\mathbf{v} \in \mathcal{V} : T\mathbf{v} = \mathbf{0}_{\mathcal{W}}\}$ and $\text{range}(T) := \{T\mathbf{v} : \mathbf{v} \in \mathcal{V}\}$.
The kernel is also called the *nullspace* of T.

Definition 3.2.2. *Injective and Surjective Linear Maps*
A function $T : \mathcal{V} \to \mathcal{W}$ is *injective* if $T\mathbf{u} = T\mathbf{v}$ implies $\mathbf{u} = \mathbf{v}$. It is *surjective* if $\text{range}(T) = \mathcal{W}$.

Proposition 3.2.3. *Kernel and Range Are Subspaces*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $\ker(T)$ *is a subspace of $\mathcal{V}$ and* $\text{range}(T)$ *is a subspace of $\mathcal{W}$.*

Proof. Since $T\mathbf{0}_{\mathcal{V}} = \mathbf{0}_{\mathcal{W}}$, the kernel is nonempty.
If $\mathbf{u}, \mathbf{v} \in \ker(T)$ and $a \in \mathbb{F}$, then $T(a\mathbf{u} + \mathbf{v}) = aT\mathbf{u} + T\mathbf{v} = \mathbf{0}_{\mathcal{W}}$, so $a\mathbf{u} + \mathbf{v} \in \ker(T)$.
For the range, if $T\mathbf{u}, T\mathbf{v} \in \text{range}(T)$ and $a \in \mathbb{F}$, then $aT\mathbf{u} + T\mathbf{v} = T(a\mathbf{u} + \mathbf{v}) \in \text{range}(T)$.
Structural Commentary: The kernel and range are the two canonical subspaces attached to a linear map.
Key Insight: Linearity makes both preimages of zero and images of vectors closed under linear combinations. □

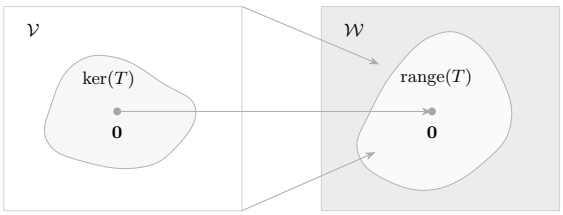


Figure 11: For a linear map $T : \mathcal{V} \to \mathcal{W}$, the range of T may be a proper subspace of $\mathcal{W}$, and the kernel may contain nonzero vectors. The conditions $\text{range}(T) = \mathcal{W}$ and $\ker(T) = \{\mathbf{0}\}$ characterize surjectivity and injectivity, respectively.

Definition 3.2.4. *Projection Along a Complement*
An operator $P \in \mathcal{L}(\mathcal{V})$ is an *algebraic projection*, or *idempotent operator*, if $P^2 = P$.
If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, the *projection onto $\mathcal{U}$ along $\mathcal{W}$* is $\rho_{\mathcal{U},\mathcal{W}} \in \mathcal{L}(\mathcal{V})$ where $\rho_{\mathcal{U},\mathcal{W}}(\mathbf{u} + \mathbf{w}) := \mathbf{u}$ for $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{W}$.

Proposition 3.2.5. *Algebraic Projections and Direct-Sum Splittings*
For $P \in \mathcal{L}(\mathcal{V})$, P is an algebraic projection if and only if $\mathcal{V} = \operatorname{range}(P) \oplus \ker(P)$ and $P = \rho_{\operatorname{range}(P),\ker(P)}$.
In particular, if $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ and $\rho = \rho_{\mathcal{U},\mathcal{W}}$, then ρ is an algebraic projection, $\operatorname{range}(\rho) = \mathcal{U}$, *and* $\ker(\rho) = \mathcal{W}$.

Proof.

($\Rightarrow$) Suppose P is an algebraic projection, so $P^2 = P$. For every $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$, with $P\mathbf{v} \in \operatorname{range}(P)$ and $\mathbf{v} - P\mathbf{v} \in \ker(P)$. If $\mathbf{x} \in \operatorname{range}(P) \cap \ker(P)$, then $\mathbf{x} = P\mathbf{y}$ and $\mathbf{0} = P\mathbf{x} = P^2\mathbf{y} = \mathbf{x}$. Thus the sum is direct. Also P fixes $\operatorname{range}(P)$ and kills $\ker(P)$, so $P = \rho_{\operatorname{range}(P),\ker(P)}$.

($\Leftarrow$) Assume the right-hand side. For $\mathbf{v} = \mathbf{x} + \mathbf{y}$ with $\mathbf{x} \in \operatorname{range}(P)$ and $\mathbf{y} \in \ker(P)$, $P^2\mathbf{v} = P\mathbf{x} = \mathbf{x} = P\mathbf{v}$. Thus P is an algebraic projection. For $\rho_{\mathcal{U},\mathcal{W}}$, the formula $\rho_{\mathcal{U},\mathcal{W}}(\mathbf{u} + \mathbf{w}) = \mathbf{u}$ gives range $\mathcal{U}$ and kernel $\mathcal{W}$, and the equivalence above shows that ρ is an algebraic projection.

Structural Commentary: Idempotents keep one summand and kill its complement.
Key Insight: Direct-sum coordinates turn a projection into $I \oplus 0$. □

Proposition 3.2.6. *Complementary Algebraic Projections*
If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then $\rho_{\mathcal{U},\mathcal{W}} + \rho_{\mathcal{W},\mathcal{U}} = I_{\mathcal{V}}$, and $\rho_{\mathcal{U},\mathcal{W}}\rho_{\mathcal{W},\mathcal{U}} = \rho_{\mathcal{W},\mathcal{U}}\rho_{\mathcal{U},\mathcal{W}} = 0$.

Proof. For $\mathbf{v} = \mathbf{u} + \mathbf{w}$, $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{W}$, $(\rho_{\mathcal{U},\mathcal{W}} + \rho_{\mathcal{W},\mathcal{U}})\mathbf{v} = \mathbf{u} + \mathbf{w} = \mathbf{v}$,
and $\rho_{\mathcal{U},\mathcal{W}}\rho_{\mathcal{W},\mathcal{U}}\mathbf{v} = \rho_{\mathcal{U},\mathcal{W}}\mathbf{w} = \mathbf{0}$; $\rho_{\mathcal{W},\mathcal{U}}\rho_{\mathcal{U},\mathcal{W}}\mathbf{v} = \rho_{\mathcal{W},\mathcal{U}}\mathbf{u} = \mathbf{0}$. Thus the stated operator identities hold.
Structural Commentary: A direct-sum decomposition gives two complementary idempotents.
Key Insight: Projection onto one summand kills the other summand. □

Definition 3.2.7. *Orthogonal Projections in the Algebraic Sense*
Two algebraic projections $\rho, \sigma \in \mathcal{L}(\mathcal{V})$ are *algebraically orthogonal*, written $\rho \perp \sigma$, if $\rho\sigma = \sigma\rho = 0$.

Resolution of identity on $\mathcal{V}$ is a finite family of algebraic projections $(\rho_i)_{i=1}^k$ with $\rho_i \perp \rho_j$ for $i \neq j$ and $\sum_{i=1}^k \rho_i = I_{\mathcal{V}}$.

Theorem 3.2.8. *Resolution Ranges Form a Direct Sum*
If $(\rho_i)_{i=1}^k$ is a resolution of $I_{\mathcal{V}}$ and $R_i = \operatorname{range}(\rho_i)$, then $\mathcal{V} = \bigoplus_{i=1}^k R_i$, and $\ker(\rho_i) = \bigoplus_{j \neq i} R_j$.

Proof. Since $\sum_{i=1}^k \rho_i = I_{\mathcal{V}}$, $\mathbf{v} = \sum_{i=1}^k \rho_i\mathbf{v} \in \sum_{i=1}^k R_i$.
If $\sum_{i=1}^k \mathbf{x}_i = \mathbf{0}$ with $\mathbf{x}_i \in R_i$, then for each $1 \leq m \leq k$, $\mathbf{0} = \rho_m \sum_{i=1}^k \mathbf{x}_i = \mathbf{x}_m$, as ρ_m fixes R_m and kills R_i for $i \neq m$.
Thus $\mathcal{V} = \bigoplus_{i=1}^k R_i$. Also, for $\mathbf{v} = \sum_{j=1}^k \mathbf{x}_j$ with $\mathbf{x}_j \in R_j$, $\rho_i\mathbf{v} = \mathbf{x}_i$, so $\ker(\rho_i) = \bigoplus_{j \neq i} R_j$.
Structural Commentary: A resolution decomposes the identity into independent idempotent pieces.
Key Insight: Pairwise zero products are the operator form of directness. □

Theorem 3.2.9. *Direct Sums Give Resolutions*
If $\mathcal{V} = \bigoplus_{i=1}^k \mathcal{U}_i$ and $\rho_i = \rho_{\mathcal{U}_i,\bigoplus_{j\neq i}\mathcal{U}_j}$, then $(\rho_i)_{i=1}^k$ is a resolution of $I_{\mathcal{V}}$.

Proof. For $\mathbf{v} = \sum_{i=1}^k \mathbf{u}_i$, $\mathbf{u}_i \in \mathcal{U}_i$, $\sum_{i=1}^k \rho_i\mathbf{v} = \sum_{i=1}^k \mathbf{u}_i = \mathbf{v}$. Thus $\sum_{i=1}^k \rho_i = I_{\mathcal{V}}$.
If $i \neq j$, then $\rho_i\rho_j\mathbf{v} = \rho_i\mathbf{u}_j = \mathbf{0}$ and $\rho_j\rho_i\mathbf{v} = \rho_j\mathbf{u}_i = \mathbf{0}$, so $\rho_i \perp \rho_j$.
Structural Commentary: Direct-sum coordinates produce mutually annihilating coordinate projections.
Key Insight: Coordinate projections add to the identity and kill all other summands. □

Theorem 3.2.10. *Algebra of Projections*
Let $\rho, \sigma \in \mathcal{L}(\mathcal{V})$ be algebraic projections and $\operatorname{char}(\mathbb{F}) \neq 2$.

(i) $\rho + \sigma$ *is an algebraic projection* $\iff \rho \perp \sigma$.
Then $\operatorname{range}(\rho + \sigma) = \operatorname{range}(\rho) \oplus \operatorname{range}(\sigma)$; $\ker(\rho + \sigma) = \ker(\rho) \cap \ker(\sigma)$

(ii) $\rho - \sigma$ *is an algebraic projection* $\iff \rho\sigma = \sigma\rho = \sigma$.
Then $\operatorname{range}(\rho - \sigma) = \operatorname{range}(\rho) \cap \ker(\sigma)$; $\ker(\rho - \sigma) = \ker(\rho) \oplus \operatorname{range}(\sigma)$.

(iii) If $\rho\sigma = \sigma\rho$, then $\rho\sigma$ is algebraic projection and $\operatorname{range}(\rho\sigma) = \operatorname{range}(\rho) \cap \operatorname{range}(\sigma)$; $\ker(\rho\sigma) = \ker(\rho) + \ker(\sigma)$.

Proof.

(i) Put $A = \rho\sigma$, $B = \sigma\rho$, and $C = \rho\sigma\rho$. Since $(\rho + \sigma)^2 = \rho + \sigma \iff A + B = 0$,
$A + B = 0$ gives $A + C = \rho(A + B) = 0$ and $C + B = (A + B)\rho = 0$, hence $B = A$ and $2A = 0$.
Thus $A = B = 0$. The converse is immediate. Assume $\rho \perp \sigma$. Then $\operatorname{range}(\rho + \sigma) \subseteq \operatorname{range}(\rho) + \operatorname{range}(\sigma)$,

while $(\rho+\sigma)(\mathbf{x}+\mathbf{y}) = \mathbf{x}+\mathbf{y}$ for $\mathbf{x} \in \text{range}(\rho)$, $\mathbf{y} \in \text{range}(\sigma)$. Also $\text{range}(\rho) \cap \text{range}(\sigma) = \{\mathbf{0}\}$, since $\mathbf{z} \in \text{range}(\rho) \cap \text{range}(\sigma)$ and $\mathbf{z} = \sigma\mathbf{w}$ imply $\mathbf{z} = \rho\mathbf{z} = \rho\sigma\mathbf{w} = \mathbf{0}$. Finally, $(\rho+\sigma)\mathbf{v} = 0 \iff \rho\mathbf{v} = 0 = \sigma\mathbf{v}$.

(ii) Since $P^2 = P \iff (I_\mathcal{V} - P)^2 = I_\mathcal{V} - P$, $\rho - \sigma$ is an algebraic projection $\iff (I_\mathcal{V} - \rho) + \sigma$ is an algebraic projection. By (i), this is equivalent to $(I_\mathcal{V} - \rho)\sigma = \sigma(I_\mathcal{V} - \rho) = 0$, i.e. $\rho\sigma = \sigma\rho = \sigma$.
Assume $\rho\sigma = \sigma\rho = \sigma$ and set $Q = \rho - \sigma$. Then $Q\mathbf{v} \in \text{range}(\rho) \cap \ker(\sigma)$; $Q\mathbf{x} = \mathbf{x}$, $\mathbf{x} \in \text{range}(\rho) \cap \ker(\sigma)$, so $\text{range}(Q) = \text{range}(\rho) \cap \ker(\sigma)$. Also Q kills both $\ker(\rho)$ and $\text{range}(\sigma)$, and $\ker(\rho) \cap \text{range}(\sigma) = \{\mathbf{0}\}$.
If $\mathbf{v} \in \ker(Q)$, then $\rho\mathbf{v} = \sigma\mathbf{v}$ and $\mathbf{v} = (\mathbf{v} - \rho\mathbf{v}) + \sigma\mathbf{v} \in \ker(\rho) \oplus \text{range}(\sigma)$. Hence $\ker(Q) = \ker(\rho) \oplus \text{range}(\sigma)$.

(iii) If $\rho\sigma = \sigma\rho$, then $(\rho\sigma)^2 = \rho^2\sigma^2 = \rho\sigma$. Moreover $\mathbf{x} \in \text{range}(\rho\sigma) \iff \rho\mathbf{x} = \mathbf{x} = \sigma\mathbf{x}$,
so $\text{range}(\rho\sigma) = \text{range}(\rho) \cap \text{range}(\sigma)$. Finally, $\ker(\rho) + \ker(\sigma) \subseteq \ker(\rho\sigma)$, and if $\rho\sigma\mathbf{v} = 0$,
then $\mathbf{v} = (\mathbf{v} - \sigma\mathbf{v}) + \sigma\mathbf{v}$; $\mathbf{v} - \sigma\mathbf{v} \in \ker(\sigma)$, $\sigma\mathbf{v} \in \ker(\rho)$. Thus $\ker(\rho\sigma) = \ker(\rho) + \ker(\sigma)$.

Structural Commentary: Operator identities encode range-kernel geometry.
Key Insight: Idempotence is a rigid equation; expanding it forces the cross terms to encode the geometry. □

Proposition 3.2.11. *Shift Operators on Finitely Nonzero Sequences*
Define $R, S : \mathbb{F}^{(\mathbb{N})} \to \mathbb{F}^{(\mathbb{N})}$ *by* $R(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$, $S(x_1, x_2, \ldots) = (x_2, x_3, \ldots)$. *Then* $R, S \in \mathcal{L}(\mathbb{F}^{(\mathbb{N})})$.
Also, $\ker(R) = \{\mathbf{0}\}$, $\text{range}(R) = \{(x_1, x_2, \ldots) \in \mathbb{F}^{(\mathbb{N})} : x_1 = 0\}$, $\ker(S) = \text{span}((1, 0, 0, \ldots))$, $\text{range}(S) = \mathbb{F}^{(\mathbb{N})}$.

Proof. Both R and S preserve finite support, so they map $\mathbb{F}^{(\mathbb{N})}$ into itself.
Since both maps act componentwise by shifting entries, $R(a\mathbf{x}+\mathbf{y}) = aR\mathbf{x} + R\mathbf{y}$ and $S(a\mathbf{x}+\mathbf{y}) = aS\mathbf{x} + S\mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}^{(\mathbb{N})}$ and $a \in \mathbb{F}$. Thus R and S are linear.
The map R has zero kernel as $R\mathbf{x} = \mathbf{0}$ forces every entry of $\mathbf{x}$ to be zero, and its range is exactly the set of finitely nonzero sequences with first entry zero. For S, $S\mathbf{x} = \mathbf{0}$ forces $x_2 = x_3 = \cdots = 0$, so $\ker(S) = \text{span}((1, 0, 0, \ldots))$.
Finally, for any $(y_1, y_2, \ldots) \in \mathbb{F}^{(\mathbb{N})}$, we have $S(0, y_1, y_2, \ldots) = (y_1, y_2, \ldots)$, so S is surjective.
Structural Commentary: Shift operators expose behavior that finite-dimensional maps cannot have.
Key Insight: The right shift is injective but not surjective, while the left shift is surjective but not injective. □

Example 3.2.12. *Shift Operators Show Noncommutativity*
For the shifts $R, S \in \mathcal{L}(\mathbb{F}^{(\mathbb{N})})$ from Proposition 3.2.11, $SR = I_{\mathbb{F}^{(\mathbb{N})}}$ and $RS(x_1, x_2, \ldots) = (0, x_2, x_3, \ldots)$.
Hence $RS \neq SR$, for instance on $(1, 0, 0, \ldots)$. Therefore $\mathcal{L}(\mathbb{F}^{(\mathbb{N})})$ is not commutative under composition.

Proposition 3.2.13. *Injectivity and the Kernel*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ *is injective if and only if* $\ker(T) = \{\mathbf{0}\}$.

Proof.
($\Rightarrow$) If T is injective and $\mathbf{v} \in \ker(T)$, then $T\mathbf{v} = \mathbf{0} = T\mathbf{0}$, so $\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $\ker(T) = \{\mathbf{0}\}$ and $T\mathbf{v} = T\mathbf{w}$, then $T(\mathbf{v} - \mathbf{w}) = \mathbf{0}$. Hence $\mathbf{v} - \mathbf{w} \in \ker(T)$, so $\mathbf{v} = \mathbf{w}$.
Structural Commentary: The kernel measures exactly where a linear map collapses vectors to zero.
Key Insight: Injectivity is equivalent to collapsing only the zero vector. □

Theorem 3.2.14. *Range from Basis Images*
Let $\mathcal{V}$ *be finite-dimensional with basis* $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$, *and let* $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. *Then* $\text{range}(T) = \text{span}(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$.
In particular, $\text{range}(T)$ *is finite-dimensional and* $\dim(\text{range}(T)) \leq n$.

Proof. If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then $T\mathbf{v} = c_1T\mathbf{v}_1 + \cdots + c_nT\mathbf{v}_n$.
Thus every vector in $\text{range}(T)$ lies in $\text{span}(T\mathbf{v}_1, \ldots, T\mathbf{v}_n)$, and the reverse inclusion is immediate from the definition of range. The dimension bound follows because this spanning list has length n.
Structural Commentary: The range is generated by the images of any basis of the domain.
Key Insight: Linear maps carry a basis-spanning formula directly into a range-spanning formula. □

Definition 3.2.15. *Rank and Nullity*
For a finite-dimensional domain, the rank of T is $\dim(\text{range}(T))$ and the nullity of T is $\dim(\ker(T))$.

Theorem 3.2.16. *Rank-Nullity Theorem*
If $\mathcal{V}$ *is finite-dim and* $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, *then* $\text{range}(T)$ *is finite-dim and* $\dim(\mathcal{V}) = \dim(\ker(T)) + \dim(\text{range}(T))$.

Proof. By Proposition 3.2.3 and Corollary 1.3.13, $\ker(T)$ has a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$.
Extend it to a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}_1, \ldots, \mathbf{w}_r)$ of $\mathcal{V}$ by Theorem 1.3.12.
If $\mathbf{v} = \sum_{i=1}^{m} a_i\mathbf{u}_i + \sum_{j=1}^{r} b_j\mathbf{w}_j$, then $T\mathbf{v} = \sum_{j=1}^{r} b_jT\mathbf{w}_j$, because $T\mathbf{u}_i = \mathbf{0}$. Hence $(T\mathbf{w}_1, \ldots, T\mathbf{w}_r)$ spans $\text{range}(T)$.
If $\sum_{j=1}^{r} c_jT\mathbf{w}_j = \mathbf{0}$, then $\sum_{j=1}^{r} c_j\mathbf{w}_j \in \ker(T)$, so for some scalars d_i, $\sum_{j=1}^{r} c_j\mathbf{w}_j = \sum_{i=1}^{m} d_i\mathbf{u}_i$.
Thus $\sum_{j=1}^{r} c_j\mathbf{w}_j - \sum_{i=1}^{m} d_i\mathbf{u}_i = \mathbf{0}$, and linear independence of extended basis forces every c_j and d_i to be zero.

Hence $(T\mathbf{w}_1, \dots, T\mathbf{w}_r)$ is a basis of range(T). Thus $\dim(\ker(T)) = m$, $\dim(\text{range}(T)) = r$, and $\dim(\mathcal{V}) = m+r$.
Structural Commentary: An adapted basis splits $\mathcal{V}$ into kernel directions and directions detected by T.
Key Insight: Rank-nullity says every domain dimension is either killed by T or appears in its range. □

Corollary 3.2.17. *Maps to Smaller Spaces Are Not Injective*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) > \dim(\mathcal{W})$.
Then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is injective.

Proof. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 3.2.16, $\dim(\ker(T)) = \dim(\mathcal{V}) - \dim(\text{range}(T)) \geq \dim(\mathcal{V}) - \dim(\mathcal{W}) > 0$.
Thus $\ker(T)$ contains a nonzero vector, so T is not injective by Proposition 3.2.13.
Structural Commentary: Excess domain dimension forces collapse.
Key Insight: The dimension deficit must appear inside the kernel. □

Corollary 3.2.18. *Maps to Larger Spaces Are Not Surjective*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) < \dim(\mathcal{W})$.
Then no linear map from $\mathcal{V}$ to $\mathcal{W}$ is surjective.

Proof. Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. By Theorem 3.2.16, $\dim(\text{range}(T)) = \dim(\mathcal{V}) - \dim(\ker(T)) \leq \dim(\mathcal{V}) < \dim(\mathcal{W})$.
Hence range(T) is a proper subspace of $\mathcal{W}$, so T is not surjective.
Structural Commentary: A smaller domain cannot cover a larger codomain.
Key Insight: The range can never have dimension larger than the domain. □

Corollary 3.2.19. *Equal-Dimension Kernel-Range Criterion*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces with $\dim(\mathcal{V}) = \dim(\mathcal{W})$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $\ker(T) = \{\mathbf{0}\}$ if and only if $\text{range}(T) = \mathcal{W}$. Equivalently, T is injective if and only if T is surjective.

Proof.
($\Rightarrow$) If $\ker(T) = \{\mathbf{0}\}$, then rank-nullity gives $\dim(\text{range}(T)) = \dim(\mathcal{V}) = \dim(\mathcal{W})$.
Since range(T) is a subspace of $\mathcal{W}$ with the same dimension, $\text{range}(T) = \mathcal{W}$.
($\Leftarrow$) If $\text{range}(T) = \mathcal{W}$, then rank-nullity gives $\dim(\ker(T)) = \dim(\mathcal{V}) - \dim(\mathcal{W}) = 0$, hence $\ker(T) = \{\mathbf{0}\}$.
Structural Commentary: Equal finite dimensions make injectivity and surjectivity equivalent.
Key Insight: In equal dimensions, having no kernel loss is exactly having full range. □

Definition 3.2.20. *Invertible Linear Map and Isomorphism*
A linear map $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is *invertible* if there exists $S \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ such that $ST = I_{\mathcal{V}}$ and $TS = I_{\mathcal{W}}$.
An invertible linear map is called an *isomorphism*, and if one exists we write $\mathcal{V} \cong \mathcal{W}$.

Proposition 3.2.21. *Inverse Is Unique*
An invertible linear map has a unique inverse. Thus if T is invertible, its inverse is denoted T^{-1}.

Proof. Suppose $S_1, S_2 \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ are inverses of $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $S_1 = S_1 I_{\mathcal{W}} = S_1(TS_2) = (S_1T)S_2 = I_{\mathcal{V}} S_2 = S_2$. Hence the inverse is unique.
Structural Commentary: A two-sided inverse is forced by the composition identities.
Key Insight: Associativity lets the two candidate inverses meet around T and collapse to the same map. □

Proposition 3.2.22. *Inverse of a Product Reverses Order*
Let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ be invertible. Then ST is invertible and $(ST)^{-1} = T^{-1}S^{-1}$.

Proof. Since $T^{-1} \in \mathcal{L}(\mathcal{V},\mathcal{U})$ and $S^{-1} \in \mathcal{L}(\mathcal{W}, \mathcal{V})$, the product $T^{-1}S^{-1} \in \mathcal{L}(\mathcal{W},\mathcal{U})$ is defined.
Moreover, $(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = I_{\mathcal{U}}$ and $(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SS^{-1} = I_{\mathcal{W}}$.
Thus $T^{-1}S^{-1}$ is a two-sided inverse of ST, so $(ST)^{-1} = T^{-1}S^{-1}$.
Structural Commentary: To undo a composition, undo the last map first.
Key Insight: Composition order reverses under inversion. □

Proposition 3.2.23. *Invertibility and Bijectivity*
A linear map is invertible if and only if it is both injective and surjective.

Proof.
($\Rightarrow$) Let S be an inverse of T. If $T\mathbf{v} = T\mathbf{w}$, then $\mathbf{v} = ST\mathbf{v} = ST\mathbf{w} = \mathbf{w}$, so T is injective. Also every $\mathbf{w}$ in the codomain satisfies $\mathbf{w} = T(S\mathbf{w})$, so T is surjective.
($\Leftarrow$) Suppose T is injective and surjective. Define $S\mathbf{w}$ to be the unique vector $\mathbf{v}$ with $T\mathbf{v} = \mathbf{w}$.
Since both sides below map under T to $a\mathbf{w}_1 + \mathbf{w}_2$, $S(a\mathbf{w}_1 + \mathbf{w}_2) = aS\mathbf{w}_1 + S\mathbf{w}_2$. Thus S is linear and satisfies $ST = I$ and $TS = I$, so T is invertible.

Structural Commentary: Bijectivity gives a unique reverse map, and linearity makes it a linear inverse.
Key Insight: A linear map is invertible exactly when it is one-to-one and onto. □

Definition 3.2.24. *Morphism Terminology for Linear Maps*
For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$:

(i) T is a *homomorphism*.
(ii) If $\mathcal{V} = \mathcal{W}$, then T is an *endomorphism*.
(iii) If $\ker(T) = \{\mathbf{0}\}$, then T is a *monomorphism*, or *embedding*.
(iv) If $\text{range}(T) = \mathcal{W}$, then T is an *epimorphism*.
(v) If $\ker(T) = \{\mathbf{0}\}$ and $\text{range}(T) = \mathcal{W}$, then T is an *isomorphism*.
(vi) If $\mathcal{V} = \mathcal{W}$ and T is invertible, then T is an *automorphism*.

Proposition 3.2.25. *Rank-Nullity as a Complement Isomorphism*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and suppose $\mathcal{V} = \ker(T) \oplus \mathcal{M}$. Then $T|_{\mathcal{M}} : \mathcal{M} \to \text{range}(T)$ is an isomorphism.

Proof. Indeed, $\ker(T|_{\mathcal{M}}) = \mathcal{M} \cap \ker(T) = \{\mathbf{0}\}$, and every $T\mathbf{v}$ equals $T\mathbf{m}$ for the decomposition $\mathbf{v} = \mathbf{k} + \mathbf{m}$.
Thus rank-nullity counts $\mathcal{V} = \ker(T) \oplus \mathcal{M}$ with $\mathcal{M} \cong \text{range}(T)$. □

Definition 3.2.26. *Quotient Map*
Let $\mathcal{U}$ be a subspace of $\mathcal{V}$. The *quotient map* is the map $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ defined by $\pi(\mathbf{v}) = \mathbf{v} + \mathcal{U}$.

Proposition 3.2.27. *Properties of the Quotient Map*
The quotient map π is linear, surjective, and $\ker(\pi) = \mathcal{U}$.

Proof. For $\mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $a \in \mathbb{F}$, $\pi(a\mathbf{v}+\mathbf{w}) = (a\mathbf{v}+\mathbf{w})+\mathcal{U} = a(\mathbf{v}+\mathcal{U})+(\mathbf{w}+\mathcal{U}) = a\pi(\mathbf{v})+\pi(\mathbf{w})$ so π is linear. It is surjective because every coset in $\mathcal{V}/\mathcal{U}$ has the form $\mathbf{v}+\mathcal{U} = \pi(\mathbf{v})$. Finally, $\ker(\pi) = \{\mathbf{v} \in \mathcal{V} : \mathbf{v}+\mathcal{U} = \mathcal{U}\} = \mathcal{U}$.
Structural Commentary: The quotient map collapses $\mathcal{U}$ to zero and records cosets.
Key Insight: Quotient maps realise subspaces as kernels. □

Theorem 3.2.28. *Universal Property of the Quotient Map*
Let $\mathcal{U} \leq \mathcal{V}$, $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$, and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ with $\mathcal{U} \leq \ker(T)$.
Then $\exists!\, \overline{T} \in \mathcal{L}(\mathcal{V}/\mathcal{U}, \mathcal{W})$ such that $\overline{T}\pi = T$, $\overline{T}(\mathbf{v}+\mathcal{U}) = T\mathbf{v}$, and $\ker(\overline{T}) = \ker(T)/\mathcal{U}$, $\text{range}(\overline{T}) = \text{range}(T)$.

Proof. Define $\overline{T}(\mathbf{v}+\mathcal{U}) = T\mathbf{v}$. If $\mathbf{v}+\mathcal{U} = \mathbf{w}+\mathcal{U}$, then $\mathbf{v}-\mathbf{w} \in \mathcal{U} \leq \ker(T)$, so $T\mathbf{v} = T\mathbf{w}$; hence $\overline{T}$ is well-defined.
Also $\overline{T}\big(a(\mathbf{v}+\mathcal{U}) + b(\mathbf{w}+\mathcal{U})\big) = T(a\mathbf{v}+b\mathbf{w}) = aT\mathbf{v} + bT\mathbf{w}$.
By definition, $\overline{T}\pi = T$. If $S\pi = T$, then $S(\mathbf{v}+\mathcal{U}) = S\pi(\mathbf{v}) = T\mathbf{v} = \overline{T}(\mathbf{v}+\mathcal{U})$, so $S = \overline{T}$.
Also $\mathbf{v}+\mathcal{U} \in \ker(\overline{T}) \iff T\mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \ker(T)$.
So $\ker(\overline{T}) = \ker(T)/\mathcal{U}$, and $\text{range}(\overline{T}) = \{\overline{T}(\mathbf{v}+\mathcal{U}) : \mathbf{v} \in \mathcal{V}\} = \text{range}(T)$.
Structural Commentary: Maps killing $\mathcal{U}$ factor through $\mathcal{V}/\mathcal{U}$.
Key Insight: Quotients are universal domains for maps with prescribed kernel ambiguity. □

Theorem 3.2.29. *Subspace Correspondence for Quotients*
Let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map. The assignments $\mathcal{T} \longmapsto \mathcal{T}/\mathcal{U}$ and $\mathcal{Y} \longmapsto \pi^{-1}(\mathcal{Y})$ are inverse inclusion-preserving bijections between $\{\mathcal{T}$ subspace of $\mathcal{V} : \mathcal{U} \subseteq \mathcal{T}\}$ and $\{\mathcal{Y}$ subspace of $\mathcal{V}/\mathcal{U}\}$.

Proof. If $\mathcal{U} \subseteq \mathcal{T} \subseteq \mathcal{V}$, then $\mathcal{T}/\mathcal{U}$ is a subspace of $\mathcal{V}/\mathcal{U}$.
If $\mathcal{Y} \subseteq \mathcal{V}/\mathcal{U}$ is a subspace, then $\pi^{-1}(\mathcal{Y})$ is a subspace of $\mathcal{V}$ and contains $\ker(\pi) = \mathcal{U}$.
For $\mathcal{U} \subseteq \mathcal{T} \subseteq \mathcal{V}$, $\pi^{-1}(\mathcal{T}/\mathcal{U}) = \mathcal{T}$, since $\mathbf{v}+\mathcal{U} = \mathbf{t}+\mathcal{U}$ with $\mathbf{t} \in \mathcal{T}$ implies $\mathbf{v}-\mathbf{t} \in \mathcal{U} \subseteq \mathcal{T}$.
For $\mathcal{Y} \subseteq \mathcal{V}/\mathcal{U}$, surjectivity gives $\pi\left(\pi^{-1}(\mathcal{Y})\right) = \mathcal{Y}$, that is, $\pi^{-1}(\mathcal{Y})/\mathcal{U} = \mathcal{Y}$
Both assignments preserve inclusion by monotonicity of images and preimages.
Structural Commentary: Subspaces above $\mathcal{U}$ are exactly the subspaces visible after collapsing $\mathcal{U}$.
Key Insight: The quotient map translates containment above $\mathcal{U}$ into containment inside $\mathcal{V}/\mathcal{U}$. □

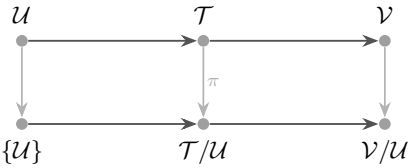


Figure 12: Intermediate subspaces above $\mathcal{U}$ correspond to subspaces of the quotient.

Theorem 3.2.30. *Quotient by the Kernel*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then $\widetilde{T}(\mathbf{v} + \ker(T)) := T\mathbf{v}$ determines a well-defined linear map $\widetilde{T} : \mathcal{V}/\ker(T) \to \mathcal{W}$. Also,

(i) $\widetilde{T}$ is injective;
(ii) $\operatorname{range}(\widetilde{T}) = \operatorname{range}(T)$;
(iii) $\mathcal{V}/\ker(T) \cong \operatorname{range}(T)$.

Proof. If $\mathbf{v} + \ker(T) = \mathbf{v}' + \ker(T)$, then $\mathbf{v} - \mathbf{v}' \in \ker(T)$, so $T\mathbf{v} = T\mathbf{v}'$. Thus $\widetilde{T}$ is well-defined.
For cosets and $a \in \mathbb{F}$, $\widetilde{T}\big(a(\mathbf{v} + \ker(T)) + (\mathbf{v}' + \ker(T))\big) = T(a\mathbf{v} + \mathbf{v}') = aT\mathbf{v} + T\mathbf{v}'$, so $\widetilde{T}$ is linear.

(i) If $\widetilde{T}(\mathbf{v} + \ker(T)) = \mathbf{0}$, then $T\mathbf{v} = \mathbf{0}$, so $\mathbf{v} \in \ker(T)$ and $\mathbf{v} + \ker(T) = \ker(T)$, the zero coset.
Hence $\ker(\widetilde{T}) = \{\ker(T)\}$ and $\widetilde{T}$ is injective.
(ii) For every coset $\mathbf{v} + \ker(T)$, the value $\widetilde{T}(\mathbf{v} + \ker(T)) = T\mathbf{v}$ lies in $\operatorname{range}(T)$.
Conversely, every vector in $\operatorname{range}(T)$ has the form $T\mathbf{v} = \widetilde{T}(\mathbf{v} + \ker(T))$. Hence the ranges are equal.
(iii) Let $\widehat{T} : \mathcal{V}/\ker(T) \to \operatorname{range}(T)$ be the same formula as $\widetilde{T}$ with codomain restricted to $\operatorname{range}(T)$.
The linearity proved above and parts (i)–(ii) show that $\widehat{T}$ is linear, injective, and surjective, so Proposition 3.2.23 gives $\mathcal{V}/\ker(T) \cong \operatorname{range}(T)$.

Structural Commentary: Quotienting by the kernel removes exactly the ambiguity of T.
Key Insight: Every linear map factors as a quotient followed by an isomorphism onto its range. □

Proposition 3.2.31. *Quotient by a Complemented Subspace*
Let $\mathcal{U}$ and $\mathcal{W}$ be subspaces of $\mathcal{V}$ with $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$. Then $J : \mathcal{W} \to \mathcal{V}/\mathcal{U}$, $J(\mathbf{w}) = \mathbf{w} + \mathcal{U}$ is an isomorphism. Hence every complement of $\mathcal{U}$ is isomorphic to $\mathcal{V}/\mathcal{U}$.

Proof. The map is linear by the quotient operations from Proposition 2.2.6.
If $J(\mathbf{w}) = \mathcal{U}$, then $\mathbf{w} \in \mathcal{U} \cap \mathcal{W} = \{\mathbf{0}\}$, so J is injective.
If $\mathbf{v} + \mathcal{U} \in \mathcal{V}/\mathcal{U}$, write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$. Then $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U} = J(\mathbf{w})$, so J is surjective.
Structural Commentary: A complement gives canonical representatives for quotient cosets.
Key Insight: Collapsing $\mathcal{U}$ leaves exactly the complementary directions visible. □

Proposition 3.2.32. *Quotient of a Direct Sum*
If $\mathcal{V} = \mathcal{V}_1 \oplus \mathcal{V}_2$ and $\mathcal{U}_j \le \mathcal{V}_j$ for $j = 1, 2$, then $\frac{\mathcal{V}}{\mathcal{U}_1 \oplus \mathcal{U}_2} \cong \frac{\mathcal{V}_1}{\mathcal{U}_1} \oplus \frac{\mathcal{V}_2}{\mathcal{U}_2}$.

Proof. Define, using the unique decomposition
$\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$, $\Phi : \mathcal{V} \to (\mathcal{V}_1/\mathcal{U}_1) \oplus (\mathcal{V}_2/\mathcal{U}_2)$ and $\Phi(\mathbf{v}_1 + \mathbf{v}_2) = (\mathbf{v}_1 + \mathcal{U}_1, \mathbf{v}_2 + \mathcal{U}_2)$.
For $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$, $\mathbf{y} = \mathbf{y}_1 + \mathbf{y}_2$, and $a, b \in \mathbb{F}$, $\Phi(a\mathbf{x} + b\mathbf{y}) = a\Phi(\mathbf{x}) + b\Phi(\mathbf{y})$, so Φ is linear.
It is surjective since $(\mathbf{v}_1 + \mathcal{U}_1, \mathbf{v}_2 + \mathcal{U}_2) = \Phi(\mathbf{v}_1 + \mathbf{v}_2)$. Finally, $\ker(\Phi) = \{\mathbf{v}_1 + \mathbf{v}_2 : \mathbf{v}_1 \in \mathcal{U}_1,\ \mathbf{v}_2 \in \mathcal{U}_2\} = \mathcal{U}_1 \oplus \mathcal{U}_2$.
The result follows from Theorem 3.2.30.
Structural Commentary: Quotienting respects direct-sum coordinates.
Key Insight: Direct-sum coordinates make quotienting componentwise. □

Perspective 3.2.33. *Quotient and Complement Warnings*
If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, then $\mathcal{W} \cong \mathcal{V}/\mathcal{U}$, but complements are not functorial in $\mathcal{U}$.
For $\mathcal{V}$ countably infinite-dimensional with basis $(\mathbf{e}_n)_{n \ge 1}$, put $\mathcal{S} = \operatorname{span}(\mathbf{e}_n : n \ge 2)$.
Then $\mathcal{S} \cong \mathcal{V}$ and $\mathcal{V} = \mathcal{V} \oplus \{\mathbf{0}\} = \mathcal{S} \oplus \mathbb{F}\mathbf{e}_1$; $\mathcal{V}/\mathcal{V} = \{\mathbf{0}\}$, $\mathcal{V}/\mathcal{S} \cong \mathbb{F}\mathbf{e}_1$.

Definition 3.2.34. *Codimension*
If $\mathcal{U}$ is a subspace of $\mathcal{V}$ and $\dim(\mathcal{V}/\mathcal{U}) < \infty$, define $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) := \dim(\mathcal{V}/\mathcal{U})$.
Thus, if $\mathcal{V}$ is finite-dimensional, Theorem 2.2.7 gives $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.
If $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, Proposition 3.2.31 gives $\operatorname{codim}_{\mathcal{V}}(\mathcal{U}) = \dim(\mathcal{W})$.

Perspective 3.2.35. *Quotients Split Over Fields*
In finite-dimensional vector spaces, choosing a complement $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$ identifies $\mathcal{V}/\mathcal{U}$ with $\mathcal{W}$. The quotient map collapses $\mathcal{U}$ and keeps the complementary directions.

Proposition 3.2.36. *One-Sided Inverses Split in Finite Dimensions*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) If T is injective, then there exists $L \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ such that $LT = I_{\mathcal{V}}$.
(ii) If T is surjective, then there exists $R \in \mathcal{L}(\mathcal{W}, \mathcal{V})$ such that $TR = I_{\mathcal{W}}$.

Proof.

(i) Choose $\mathcal{H} \subseteq \mathcal{W}$ with $\mathcal{W} = \operatorname{range}(T) \oplus \mathcal{H}$. For $\mathbf{w} = T\mathbf{v} + \mathbf{h}$, define $L\mathbf{w} = \mathbf{v}$. This is well-defined since the direct-sum decomposition is unique and T is injective. It is linear, and $LT\mathbf{v} = \mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.

(ii) Choose $\mathcal{H} \subseteq \mathcal{V}$ with $\mathcal{V} = \ker(T) \oplus \mathcal{H}$. Then $T|_{\mathcal{H}} : \mathcal{H} \to \mathcal{W}$ is injective since $\mathcal{H} \cap \ker(T) = \{\mathbf{0}\}$, and surjective since each $\mathbf{v} = \mathbf{k} + \mathbf{h}$ satisfies $T\mathbf{v} = T\mathbf{h}$. Let $R = (T|_{\mathcal{H}})^{-1}$, viewed as a map $\mathcal{W} \to \mathcal{V}$. Then R is linear and $TR = I_{\mathcal{W}}$.

Structural Commentary: Complements supply the missing directions on which a one-sided inverse is chosen.
Key Insight: Finite-dimensional vector spaces split kernels and ranges. □

Proposition 3.2.37. *Isomorphisms Preserve Bases*
Let $T : \mathcal{V} \to \mathcal{W}$ be an isomorphism and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a finite list in $\mathcal{V}$. Set $T\beta = (T\mathbf{v}_1, \dots, T\mathbf{v}_n)$. The following are equivalent:

(i) β is a basis of $\mathcal{V}$.
(ii) $T\beta$ is a basis of $\mathcal{W}$.

Proof.

(i) $\Rightarrow$ (ii) If $\mathbf{w} \in \mathcal{W}$, choose $\mathbf{v} \in \mathcal{V}$ with $\mathbf{w} = T\mathbf{v}$ and write $\mathbf{v} = \sum_j a_j \mathbf{v}_j$. Then $\mathbf{w} = \sum_j a_j T\mathbf{v}_j$, so $T\beta$ spans $\mathcal{W}$. If $\sum_j a_j T\mathbf{v}_j = \mathbf{0}$, applying T^{-1} gives $\sum_j a_j \mathbf{v}_j = \mathbf{0}$; hence all $a_j = 0$. Thus $T\beta$ is a basis.
(ii) $\Rightarrow$ (i) Assume $T\beta$ is a basis of $\mathcal{W}$. If $\mathbf{v} \in \mathcal{V}$, write $T\mathbf{v} = \sum_j a_j T\mathbf{v}_j$; applying T^{-1} gives $\mathbf{v} = \sum_j a_j \mathbf{v}_j$, so β spans $\mathcal{V}$. If $\sum_j a_j \mathbf{v}_j = \mathbf{0}$, applying T gives $\sum_j a_j T\mathbf{v}_j = \mathbf{0}$; hence all $a_j = 0$. Thus β is linearly independent, and therefore a basis of $\mathcal{V}$.

Structural Commentary: Isomorphisms preserve exactly the spanning and independence data that define a basis.
Key Insight: An isomorphism changes the ambient space, not the linear relations among vectors. □

Theorem 3.2.38. *Finite-Dimensional Invertibility Criteria*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then the following are equivalent:

i. T is invertible;
ii. T is injective;
iii. T is surjective.

Proof.

(i) $\Rightarrow$ (ii) follows from Proposition 3.2.23.
(ii) $\Rightarrow$ (iii): If T is injective, then $\ker(T) = \{\mathbf{0}\}$.
Rank-nullity gives $\dim(\text{range}(T)) = \dim(\mathcal{V})$, hence $\text{range}(T) = \mathcal{V}$, so T is surjective.
(iii) $\Rightarrow$ (i): If T is surjective, rank-nullity gives $\dim(\ker(T)) = 0$, so T is injective.
Thus T is bijective, hence invertible by Proposition 3.2.23.

Structural Commentary: For operators on finite-dimensional spaces, kernel and range dimensions trade off exactly. Injectivity and surjectivity therefore force each other.
Key Insight: In equal finite dimensions, losing no information is the same as reaching the whole space. □

Perspective 3.2.39. *Kernel Obstruction to Invertibility*
By Theorem 3.2.38 and Proposition 3.2.13, a finite-dimensional operator is invertible exactly when its kernel is zero. Any nonzero $\mathbf{v} \in \ker T$ therefore blocks invertibility.

Theorem 3.2.40. *Isomorphism and Dimension*
Two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension.

Proof.

($\Rightarrow$) If $T : \mathcal{V} \to \mathcal{W}$ is an isomorphism and $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, then Proposition 3.2.37 shows that $(T\mathbf{v}_1, \dots, T\mathbf{v}_n)$ is a basis of $\mathcal{W}$.
($\Leftarrow$) If both spaces have dimension n, choose bases $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $(\mathbf{w}_1, \dots, \mathbf{w}_n)$.
By Proposition 3.1.11, there is a linear map T with $T\mathbf{v}_j = \mathbf{w}_j$ for every j. Then $T(\sum a_j \mathbf{v}_j) = \sum a_j \mathbf{w}_j$, so $\ker(T) = \{\mathbf{0}\}$ and $\text{range}(T) = \mathcal{W}$. Hence T is an isomorphism.

Structural Commentary: Dimension classifies finite-dimensional spaces up to isomorphism.
Key Insight: Matching bases give an isomorphism; any isomorphism carries bases to bases. □

3.3 Matrix Representations

Definition 3.3.1. *Matrices and Matrix Operations*

(i) *Matrix space*: for $m, n \in \mathbb{Z}_{>0}$, $\text{Mat}(m, n, \mathbb{F}) = \{[A_{j,k}]_{1 \le j \le m, 1 \le k \le n} : A_{j,k} \in \mathbb{F}\}$.
(ii) *Entries, rows, columns*: for $A = [A_{j,k}] \in \text{Mat}(m, n, \mathbb{F})$,
$A_{j,k}$ is the (j,k)-entry, $A_{j,\cdot}$ is row j, and $A_{\cdot,k}$ is column k.
(iii) *Transpose*: for $A \in \text{Mat}(m, n, \mathbb{F})$, $A^t \in \text{Mat}(n, m, \mathbb{F})$ is defined by $(A^t)_{k,j} = A_{j,k}$.

(iv) *Main diagonal*: for $A = [A_{j,k}] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $r = \min(m,n)$, $\mathrm{diag}(A) = (A_{1,1}, \ldots, A_{r,r})$.
(v) *Symmetry*: for $A \in \mathrm{Mat}(n,n,\mathbb{F})$, A is symmetric if $A^t = A$ and skew-symmetric if $A^t = -A$.
(vi) *Triangularity*: for $A \in \mathrm{Mat}(n,n,\mathbb{F})$, A is upper triangular if $A_{j,k} = 0$ for $j > k$, lower triangular if $A_{j,k} = 0$ for $j < k$, and diagonal if $A_{j,k} = 0$ for $j \neq k$.
(vii) *Linear operations*: for $A, C \in \mathrm{Mat}(m,n,\mathbb{F})$ and $a \in \mathbb{F}$, $(A+C)_{j,k} = A_{j,k} + C_{j,k}$ and $(aA)_{j,k} = aA_{j,k}$.
(viii) *Matrix product*: for $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $B \in \mathrm{Mat}(n,p,\mathbb{F})$, $AB \in \mathrm{Mat}(m,p,\mathbb{F})$ is $(AB)_{j,k} := \sum_{r=1}^{n} A_{j,r}B_{r,k}$.
(ix) *Identity and inverse*: $I_n = [\delta_{j,k}] \in \mathrm{Mat}(n,n,\mathbb{F})$; for $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$, B is a two-sided inverse of A if $AB = I_n = BA$, and A is invertible if such B exists.

Lemma 3.3.2. *Uniqueness of Two-Sided Inverses*
Let $A, B, C \in \mathrm{Mat}(n,n,\mathbb{F})$. If $AB = I_n$ and $BC = I_n$, then $A = C$.

Proof. Using $BC = I_n$ and $AB = I_n$, $A = AI_n = A(BC) = (AB)C = I_nC = C$.
Structural Commentary: A common middle inverse forces the left and right inverses to coincide.
Key Insight: Associativity turns $AB = I_n = BC$ into $A = C$. □

Definition 3.3.3. *Inverse Matrix Notation*
If $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, Lemma 3.3.2 shows that its two-sided inverse is unique.
This inverse is denoted A^{-1}, and the set of invertible matrices in $\mathrm{Mat}(n,n,\mathbb{F})$ is denoted $\mathrm{Mat}(n,n,\mathbb{F})^{\times}$.

Proposition 3.3.4. *Matrix Spaces*
For positive integers m, n, $\mathrm{Mat}(m,n,\mathbb{F})$ is a vector space over $\mathbb{F}$ with dimension mn.

Proof. Entrywise addition and scalar multiplication satisfy the vector space axioms because $\mathbb{F}$ does. For $1 \leq j \leq m$ and $1 \leq k \leq n$, let $E_{j,k}$ be the matrix with 1 in entry (j,k) and 0 elsewhere.
Every $A = [A_{j,k}]$ has the unique expansion $A = \sum_{j=1}^{m} \sum_{k=1}^{n} A_{j,k}E_{j,k}$.
Thus the mn matrices $E_{j,k}$ form a basis, so $\dim(\mathrm{Mat}(m,n,\mathbb{F})) = mn$.
Structural Commentary: A matrix is a finite array of independent scalar slots.
Key Insight: The standard entry matrices isolate those slots one at a time. □

Proposition 3.3.5. *Matrix Product Entries Are Row Times Column*
If $A \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(n,p,\mathbb{F})$, then $(AB)_{j,k} = A_{j,\cdot}B_{\cdot,k}$ for $1 \leq j \leq m$ and $1 \leq k \leq p$.

Proof. The row-column product $A_{j,\cdot}B_{\cdot,k}$ is exactly $\sum_{r=1}^{n} A_{j,r}B_{r,k}$, which is the definition of $(AB)_{j,k}$.
Structural Commentary: Matrix multiplication is assembled entry-by-entry from row-column pairings.
Key Insight: Each entry of AB sees one row of A and one column of B. □

Proposition 3.3.6. *Matrix-Vector Products as Column Combinations*
If $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $\mathbf{x} = (x_1, \ldots, x_n)^t \in \mathbb{F}^n$, then $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.

Proof. The jth entry of $A\mathbf{x}$ is $\sum_{k=1}^{n} A_{j,k}x_k$, which is also the jth entry of $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
Structural Commentary: Multiplying by a column selects a linear combination of the columns of A.
Key Insight: The coordinates of $\mathbf{x}$ are precisely the column coefficients. □

Definition 3.3.7. *Column Space, Matrix Rank, and Nullity*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$.
(i) $\mathrm{col}(A) := \mathrm{span}(\mathbf{a}_1, \ldots, \mathbf{a}_n) \subseteq \mathbb{F}^m$, and $\mathrm{row}(A) := \mathrm{col}(A^t) \subseteq \mathbb{F}^n$.
(ii) $\mathrm{rank}(A) := \dim(\mathrm{col}(A))$; column rank is $\dim(\mathrm{col}(A))$, and row rank is $\dim(\mathrm{row}(A))$.
(iii) A has *full rank* if $\mathrm{rank}(A) = \min(m,n)$, *full column rank* if $\mathrm{rank}(A) = n$, and *full row rank* if $\mathrm{rank}(A) = m$.
(iv) $\ker(A) := \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{0}\}$ is the *nullspace*, and $\mathrm{nullity}(A) := \dim(\ker(A))$.

Proposition 3.3.8. *Solvability and Complete Solution Sets*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m,n,\mathbb{F})$ and let $\mathbf{b} \in \mathbb{F}^m$. Then $A\mathbf{x} = \mathbf{b}$ is solvable if and only if $\mathbf{b} \in \mathrm{col}(A)$.
If $\mathbf{x}_p$ is one solution, then every solution is of the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_0$, where $\mathbf{x}_0 \in \ker(A)$.

Proof.
($\Rightarrow$) If $A\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} = (x_1, \ldots, x_n)^T$, then $\mathbf{b} = A\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j \in \mathrm{col}(A)$.
($\Leftarrow$) If $\mathbf{b} \in \mathrm{col}(A)$, then $\mathbf{b} = \sum_{j=1}^{n} x_j\mathbf{a}_j$ for some scalars x_j. Hence, for $\mathbf{x} = (x_1, \ldots, x_n)^T$, we have $A\mathbf{x} = \mathbf{b}$.

Now suppose $A\mathbf{x}_p = \mathbf{b}$. For any $\mathbf{x}$, $A\mathbf{x} = \mathbf{b} \iff A(\mathbf{x} - \mathbf{x}_p) = \mathbf{0} \iff \mathbf{x} - \mathbf{x}_p \in \ker(A)$.
Thus every solution is exactly of the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_0$, where $\mathbf{x}_0 \in \ker(A)$.
Structural Commentary: A consistent system is an affine shift of $\ker(A)$ by any particular solution.
Key Insight: Column space decides existence; kernel decides nonuniqueness. □

Proposition 3.3.9. *Matrix Products by Columns*
If $A \in \mathrm{Mat}(m, r, \mathbb{F})$ and $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \in \mathrm{Mat}(r, n, \mathbb{F})$, then $AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_n]$.

Proof. For each k, the kth column of AB has entries $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k}$, which are exactly the entries of $A\mathbf{b}_k$.
Structural Commentary: Matrix products apply A to columns of B.
Key Insight: The columns of AB are obtained by applying A to the columns of B. □

Proposition 3.3.10. *Matrix Products as Sums of Outer Products*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_r] \in \mathrm{Mat}(m, r, \mathbb{F})$, and write $B \in \mathrm{Mat}(r, n, \mathbb{F})$ by rows as

$$B = \begin{bmatrix} \mathbf{b}_1^t \\ \vdots \\ \mathbf{b}_r^t \end{bmatrix}.$$

Then $AB = \mathbf{a}_1\mathbf{b}_1^t + \cdots + \mathbf{a}_r\mathbf{b}_r^t$.

Proof. The (j,k) entry of $\mathbf{a}_\ell\mathbf{b}_\ell^t$ is $A_{j,\ell}B_{\ell,k}$. Summing over $\ell = 1, \ldots, r$ gives $\sum_{\ell=1}^{r} A_{j,\ell}B_{\ell,k} = (AB)_{j,k}$.
Structural Commentary: Column-row products decompose a matrix product into single column-row pieces.
Key Insight: Each shared index ℓ contributes one outer product. □

Proposition 3.3.11. *Elementary Transpose Rules*
Let $A, C \in \mathrm{Mat}(m, n, \mathbb{F})$ and let $a \in \mathbb{F}$. Then $(A^t)^t = A$; $(A + C)^t = A^t + C^t$; and $(aA)^t = aA^t$.

Proof. For all admissible indices, $((A^t)^t)_{j,k} = (A^t)_{k,j} = A_{j,k}$, so $(A^t)^t = A$.
Also $((A + C)^t)_{j,k} = (A + C)_{k,j} = A_{k,j} + C_{k,j} = (A^t + C^t)_{j,k}$ and $((aA)^t)_{j,k} = (aA)_{k,j} = aA_{k,j} = (aA^t)_{j,k}$.
Structural Commentary: Transpose swaps indices; entrywise algebra is unchanged.
Key Insight: Transpose is linear and involutive. □

Example 3.3.12. *Symmetric and Skew-Symmetric Decomposition*
Assume $\mathrm{char}(\mathbb{F}) \neq 2$, $M = \mathrm{Mat}(n, n, \mathbb{F})$, let $\mathrm{Sym}_n(\mathbb{F}) = \{A \in M : A^t = A\}$, $\mathrm{Skew}_n(\mathbb{F}) = \{A \in M : A^t = -A\}$.
For the map $\tau : M \to M$ defined by $\tau(A) = A^t$, linearity follows from Proposition 3.3.11. Hence $\mathrm{Sym}_n(\mathbb{F}) = \ker(\tau - I)$ and $\mathrm{Skew}_n(\mathbb{F}) = \ker(\tau + I)$, so both are subspaces of M.
For every $A \in M$, $A = \frac{1}{2}(A + A^t) + \frac{1}{2}(A - A^t) \in \mathrm{Sym}_n(\mathbb{F}) + \mathrm{Skew}_n(\mathbb{F})$.
Also $A \in \mathrm{Sym}_n(\mathbb{F}) \cap \mathrm{Skew}_n(\mathbb{F})$ implies $A = A^t = -A$, hence $A = 0$. Thus $M = \mathrm{Sym}_n(\mathbb{F}) \oplus \mathrm{Skew}_n(\mathbb{F})$.
Structural Commentary: Transpose splits a matrix into its even and odd parts.
Key Insight: The directness of the split uses the ability to divide by 2.

Proposition 3.3.13. *Transpose Reverses Products*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(n, p, \mathbb{F})$, then $(AB)^t = B^tA^t$.

Proof. For $1 \leq j \leq p$ and $1 \leq k \leq m$, $((AB)^t)_{j,k} = (AB)_{k,j} = \sum_{\ell=1}^{n} A_{k,\ell}B_{\ell,j} = \sum_{\ell=1}^{n} (B^t)_{j,\ell}(A^t)_{\ell,k} = (B^tA^t)_{j,k}$.
Structural Commentary: Transposition swaps the order of composition encoded by matrix multiplication.
Key Insight: Rows become columns, so the product order reverses. □

Corollary 3.3.14. *Matrix Inverses and Transposes Reverse Products*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ be invertible. Then AB, A^t are invertible, and $(AB)^{-1} = B^{-1}A^{-1}$; $(A^t)^{-1} = (A^{-1})^t$.

Proof. First, $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I_n$; $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I_n$.
Hence $(AB)^{-1} = B^{-1}A^{-1}$. By Proposition 3.3.13, $(A^{-1})^tA^t = (AA^{-1})^t = I_n$; $A^t(A^{-1})^t = (A^{-1}A)^t = I_n$.
Structural Commentary: Inversion and transposition both reverse multiplication order.
Key Insight: Apply the defining inverse equations and reverse products. □

Definition 3.3.15. *Coordinate Vector*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of a finite-dimensional vector space $\mathcal{V}$.
If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then the *$\beta$-coordinate vector* of $\mathbf{v}$ is

$$[\mathbf{v}]_\beta := \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{F}^n.$$

The function $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ is the *β-basis representation function*.

Proposition 3.3.16. *Coordinate Maps Are Linear Bijections*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of a finite-dimensional vector space $\mathcal{V}$.
Then the coordinate map $[\cdot]_\beta : \mathcal{V} \to \mathbb{F}^n$ is linear and bijective.

Proof. If $\mathbf{u} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ and $\mathbf{v} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$, then for $c \in \mathbb{F}$,

$$[c\mathbf{u} + \mathbf{v}]_\beta = \begin{bmatrix} ca_1 + b_1 \\ \vdots \\ ca_n + b_n \end{bmatrix} = c[\mathbf{u}]_\beta + [\mathbf{v}]_\beta$$

Thus $[\cdot]_\beta$ is linear. It is injective because $[\mathbf{v}]_\beta = \mathbf{0}$ forces all basis coefficients of $\mathbf{v}$ to be zero, and it is surjective because every column $(c_1, \dots, c_n)^t \in \mathbb{F}^n$ equals $[c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n]_\beta$.
Structural Commentary: A basis converts abstract vectors into coordinate columns without losing structure.
Key Insight: Coordinates give a structure-preserving copy of the vector space, not extra information. □

Definition 3.3.17. *Matrix Representation*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_m)$ a basis of $\mathcal{W}$. For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the matrix of T relative to (β, γ) is $[T]_{\gamma \leftarrow \beta} := \begin{bmatrix} [T(\mathbf{v}_1)]_\gamma & \cdots & [T(\mathbf{v}_n)]_\gamma \end{bmatrix} \in \mathrm{Mat}(m, n, \mathbb{F})$.

Proposition 3.3.18. *Stacks Intersect Kernels, Concatenations Sum Columns*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$, $B \in \mathrm{Mat}(p, n, \mathbb{F})$, $C \in \mathrm{Mat}(m, r, \mathbb{F})$, and $D \in \mathrm{Mat}(m, s, \mathbb{F})$.
Then $\ker \begin{bmatrix} A \\ B \end{bmatrix} = \ker(A) \cap \ker(B)$*; and* $\mathrm{col} \begin{bmatrix} C & D \end{bmatrix} = \mathrm{col}(C) + \mathrm{col}(D)$.

Proof. For $\mathbf{x} \in \mathbb{F}^n$, $\mathbf{x} \in \ker \begin{bmatrix} A \\ B \end{bmatrix} \iff \begin{bmatrix} A \\ B \end{bmatrix} \mathbf{x} = \mathbf{0} \iff A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0} \iff \mathbf{x} \in \ker(A) \cap \ker(B)$.

For $\mathbf{y} \in \mathbb{F}^m$, $\mathbf{y} \in \mathrm{col} \begin{bmatrix} C & D \end{bmatrix} \iff \exists\, \mathbf{u} \in \mathbb{F}^r,\ \mathbf{v} \in \mathbb{F}^s$ such that $\mathbf{y} = \begin{bmatrix} C & D \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix}$
$\iff \exists\, \mathbf{u}, \mathbf{v}$ such that $\mathbf{y} = C\mathbf{u} + D\mathbf{v} \iff \mathbf{y} \in \mathrm{col}(C) + \mathrm{col}(D)$.
Structural Commentary: Stacking intersects constraints; concatenation adds column directions.
Key Insight: Stacks encode intersections of nullspaces, while concatenations encode sums of column spaces. □

Proposition 3.3.19. *Matrices Respect Linear Combinations of Maps*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and γ be a basis of $\mathcal{W}$.
If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then $[aS + T]_{\gamma \leftarrow \beta} = a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.

Proof. The jth column of $[aS + T]_{\gamma \leftarrow \beta}$ is $[(aS + T)\mathbf{v}_j]_\gamma = [aS\mathbf{v}_j + T\mathbf{v}_j]_\gamma = a[S\mathbf{v}_j]_\gamma + [T\mathbf{v}_j]_\gamma$, which is the jth column of $a[S]_{\gamma \leftarrow \beta} + [T]_{\gamma \leftarrow \beta}$.
Structural Commentary: Matrix representation records basis images column-by-column.
Key Insight: Linear combinations of maps become entrywise linear combinations of their matrices. □

Definition 3.3.20. *Linear Map Induced by a Matrix*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$, the *linear map induced by A* is the map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$.

Proposition 3.3.21. *Matrix-Induced Maps Are Linear*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$, then $T_A : \mathbb{F}^n \to \mathbb{F}^m$ is linear.

Proof. For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and $a \in \mathbb{F}$, $T_A(a\mathbf{x} + \mathbf{y}) = A(a\mathbf{x} + \mathbf{y}) = aA\mathbf{x} + A\mathbf{y} = aT_A(\mathbf{x}) + T_A(\mathbf{y})$.
Structural Commentary: Matrix multiplication preserves addition and scalar multiplication.
Key Insight: Every matrix defines a canonical linear map between coordinate spaces. □

Corollary 3.3.22. *Matrix Rank-Nullity*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$. Then $\dim(\ker(A)) + \dim(\mathrm{col}(A)) = n$ and $\mathrm{nullity}(A) + \mathrm{rank}(A) = n$.
If $m = n$, then $\ker(A) = \{\mathbf{0}\}$ if and only if $\mathrm{col}(A) = \mathbb{F}^n$.

Proof. Apply Theorem 3.2.16 to the induced linear map $T_A : \mathbb{F}^n \to \mathbb{F}^m$. By definition, $\ker(T_A) = \ker(A)$ and $\mathrm{range}(T_A) = \mathrm{col}(A)$, giving the displayed identities. When $m = n$, the remaining equivalence follows as:
($\Rightarrow$) If $\ker(A) = \{\mathbf{0}\}$, then $\dim(\mathrm{col}(A)) = n$. Since $\mathrm{col}(A) \subseteq \mathbb{F}^n$, Proposition 1.4.2 gives $\mathrm{col}(A) = \mathbb{F}^n$.
($\Leftarrow$) If $\mathrm{col}(A) = \mathbb{F}^n$, then $\dim(\mathrm{col}(A)) = n$, so the identity gives $\dim(\ker(A)) = 0$. Hence $\ker(A) = \{\mathbf{0}\}$.
Structural Commentary: Matrix rank-nullity is ordinary rank-nullity written in coordinates.
Key Insight: Kernel loss plus column-space dimension equals the number of columns. □

Theorem 3.3.23. *Row Rank Equals Column Rank*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$, $\dim(\mathrm{row}(A)) = \dim(\mathrm{col}(A))$. Equivalently, $\mathrm{rank}(A^t) = \mathrm{rank}(A)$.

Proof. Let $r = \operatorname{rank}(A)$. If $r = 0$, then $\operatorname{col}(A) = \{\mathbf{0}\}$, so every column of A is zero and $A = 0$; the result follows. Suppose $r > 0$. Choose columns $\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r}$ forming a basis of $\operatorname{col}(A)$, and set $B = [\mathbf{a}_{j_1} \ \cdots \ \mathbf{a}_{j_r}] \in \operatorname{Mat}(m, r, \mathbb{F})$. For each column $\mathbf{a}_k$ of A, there is a unique $\mathbf{x}_k \in \mathbb{F}^r$ such that $\mathbf{a}_k = B\mathbf{x}_k$. With $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n] \in \operatorname{Mat}(r, n, \mathbb{F})$, we have $A = BX$. Therefore $A^t = X^t B^t$, so every column of A^t is a linear combination of the r columns of X^t. Hence $\operatorname{rank}(A^t) \le r = \operatorname{rank}(A)$. Applying the same argument to A^t gives $\operatorname{rank}(A) \le \operatorname{rank}(A^t)$.
Structural Commentary: Column rank controls row rank by factoring A through a basis of its column space.
Key Insight: Rows and columns measure the same intrinsic rank, viewed from transposed coordinates. □

Perspective 3.3.24. *Row and Column Spaces Have Different Hosts*
For $A \in \operatorname{Mat}(m, n, \mathbb{F})$, $\operatorname{row}(A) \subseteq \mathbb{F}^n$ and $\operatorname{col}(A) \subseteq \mathbb{F}^m$.
Hence, when $m \neq n$, Theorem 3.3.23 asserts only $\dim \operatorname{row}(A) = \dim \operatorname{col}(A)$, not equality of subspaces.

Theorem 3.3.25. *Four Fundamental Matrix Subspaces*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ and let $r = \operatorname{rank}(A)$. Then $\operatorname{row}(A), \ker(A) \subseteq \mathbb{F}^n$; $\operatorname{col}(A), \ker(A^t) \subseteq \mathbb{F}^m$, and $\dim(\operatorname{col}(A)) = \dim(\operatorname{row}(A)) = r$; $\dim(\ker(A)) = n - r$; $\dim(\ker(A^t)) = m - r$.

Proof. The inclusions follow from the ambient dimensions of the rows and columns, and from the domains of A and A^t. By definition, $\dim(\operatorname{col}(A)) = r$; by Theorem 3.3.23, $\dim(\operatorname{row}(A)) = r$.
Rank-nullity gives $\dim(\ker(A)) = n - \operatorname{rank}(A) = n - r$. Since $\operatorname{rank}(A^t) = \dim(\operatorname{col}(A^t)) = \dim(\operatorname{row}(A)) = r$, rank-nullity for $A^t : \mathbb{F}^m \to \mathbb{F}^n$ gives $\dim(\ker(A^t)) = m - \operatorname{rank}(A^t) = m - r$.
Structural Commentary: One rank determines all four fundamental dimensions.
Key Insight: Rank counts what survives; nullity counts what is killed. □

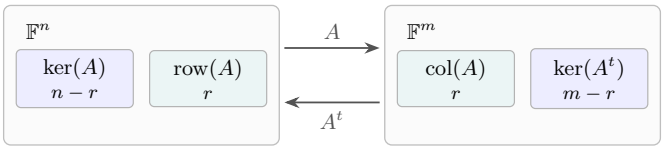


Figure 13: Four fundamental subspaces by host coordinate space.

Corollary 3.3.26. *Homogeneous Systems with More Variables Than Equations*
If $A \in \operatorname{Mat}(m, n, \mathbb{F})$ with $n > m$, then the homogeneous system $A\mathbf{x} = \mathbf{0}$ has a nonzero solution.

Proof. The induced map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ has domain dimension larger than codomain dimension, so Corollary 3.2.17 shows that T_A is not injective.
Hence $\ker(A) = \ker(T_A)$ contains a nonzero vector $\mathbf{x}$, and this vector satisfies $A\mathbf{x} = \mathbf{0}$.
Structural Commentary: Extra variables force a nontrivial kernel.
Key Insight: A nontrivial kernel is exactly a nonzero homogeneous solution. □

Corollary 3.3.27. *Inhomogeneous Systems with More Equations Than Variables*
If $A \in \operatorname{Mat}(m, n, \mathbb{F})$ with $m > n$, then there exists $\mathbf{b} \in \mathbb{F}^m$ such that $A\mathbf{x} = \mathbf{b}$ has no solution.

Proof. The induced map $T_A : \mathbb{F}^n \to \mathbb{F}^m$ has domain dimension smaller than codomain dimension, so Corollary 3.2.18 shows that T_A is not surjective. Choose $\mathbf{b} \in \mathbb{F}^m \setminus \operatorname{range}(T_A)$. Then no $\mathbf{x} \in \mathbb{F}^n$ satisfies $A\mathbf{x} = \mathbf{b}$.
Structural Commentary: Too few variables cannot reach every possible right-hand side.
Key Insight: A missing range vector becomes an inconsistent inhomogeneous system. □

Proposition 3.3.28. *Matrix Representation Acts on Coordinates*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$, let γ be a basis of $\mathcal{W}$, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then for every $\mathbf{v} \in \mathcal{V}$, $[T(\mathbf{v})]_\gamma = [T]_{\gamma \leftarrow \beta} [\mathbf{v}]_\beta$.

Proof. Write $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Then linearity gives $T(\mathbf{v}) = a_1 T(\mathbf{v}_1) + \cdots + a_n T(\mathbf{v}_n)$ and the right-hand side is exactly the linear combination of the columns of $[T]_{\gamma \leftarrow \beta}$ with coefficients $a_1, \dots, a_n$.
Structural Commentary: Matrix representation turns applying T into multiplying coordinate columns.
Key Insight: The columns of $[T]_{\gamma \leftarrow \beta}$ are the coordinate images of the basis vectors. □

Theorem 3.3.29. *Columns of an Invertible Matrix Form a Basis*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible. Then $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$.

Proof. The columns span $\mathbb{F}^n$ because for any $\mathbf{y} \in \mathbb{F}^n$, writing $\mathbf{x} = A^{-1}\mathbf{y}$ gives $\mathbf{y} = A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
They are linearly independent because if $c_1\mathbf{a}_1 + \cdots + c_n\mathbf{a}_n = \mathbf{0}$, then $A\mathbf{c} = \mathbf{0}$, so $\mathbf{c} = A^{-1}\mathbf{0} = \mathbf{0}$.
Structural Commentary: An invertible matrix is a coordinate isomorphism of $\mathbb{F}^n$. Its columns are the images of the standard coordinate directions.
Key Insight: Invertibility forces the columns to span and prevents nontrivial column relations. □

Theorem 3.3.30. *One-Sided Inverses of Square Matrices*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$. If $AB = I_n$, then $BA = I_n$.

Proof. If $B\mathbf{x} = \mathbf{0}$, then $\mathbf{x} = AB\mathbf{x} = \mathbf{0}$, so the induced map $T_B : \mathbb{F}^n \to \mathbb{F}^n$ is injective. By Theorem 3.2.38, T_B is surjective. For any $\mathbf{y} \in \mathbb{F}^n$, choose $\mathbf{x}$ with $\mathbf{y} = B\mathbf{x}$. Then $BA\mathbf{y} = BAB\mathbf{x} = B\mathbf{x} = \mathbf{y}$.
Thus BA acts as the identity on every vector of $\mathbb{F}^n$, so $BA = I_n$.
Structural Commentary: In finite dimensions, injectivity and surjectivity coincide for square linear maps. A left inverse makes B injective, hence onto, which forces the missing inverse relation.
Key Insight: A one-sided inverse of a square matrix is automatically two-sided. □

Proposition 3.3.31. *Composition and Matrix Multiplication*
If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $[ST]_{\delta \leftarrow \alpha} = [S]_{\delta \leftarrow \beta}[T]_{\beta \leftarrow \alpha}$ for any bases α of $\mathcal{U}$, β of $\mathcal{V}$, and δ of $\mathcal{W}$.

Proof. For every $\mathbf{u} \in \mathcal{U}$, $[(ST)\mathbf{u}]_\delta = [S]_{\delta \leftarrow \beta}[T\mathbf{u}]_\beta = [S]_{\delta \leftarrow \beta}[T]_{\beta \leftarrow \alpha}[\mathbf{u}]_\alpha$.
Since coordinate columns $[\mathbf{u}]_\alpha$ range over all of $\mathbb{F}^{\dim(\mathcal{U})}$, the representing matrices are equal.
Structural Commentary: Matrix multiplication is the coordinate shadow of composition of linear maps.
Key Insight: Applying T then S becomes multiplying first by the matrix of T, then by the matrix of S. □

Theorem 3.3.32. *Linear Maps and Matrices Are the Same Data*
If $\dim(\mathcal{V}) = n$ and $\dim(\mathcal{W}) = m$, then choosing bases of $\mathcal{V}$ and $\mathcal{W}$ gives isomorphism $\mathcal{L}(\mathcal{V}, \mathcal{W}) \cong \mathrm{Mat}(m, n, \mathbb{F})$. Consequently, $\dim(\mathcal{L}(\mathcal{V}, \mathcal{W})) = mn$.

Proof. Fix bases β of $\mathcal{V}$ and γ of $\mathcal{W}$. The map $T \mapsto [T]_{\gamma \leftarrow \beta}$ is linear. It is injective because a linear map whose matrix is zero sends every basis vector of $\mathcal{V}$ to $\mathbf{0}$, hence is the zero map. It is surjective because, for any $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, there is a unique linear map satisfying $T(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + \cdots + a_{mj}\mathbf{w}_m$, $j = 1, \ldots, n$.
Thus $\mathcal{L}(\mathcal{V}, \mathcal{W}) \cong \mathrm{Mat}(m, n, \mathbb{F})$, whose standard entry matrices form a basis of length mn.
Structural Commentary: Choosing bases converts every linear map into one and only one array of scalars.
Key Insight: A matrix is exactly the list of coordinate images of the domain basis vectors. □

Motivation 3.3.33. *Coordinate Encoding*
This is the key bridge between abstraction and computation. The linear map is basis-free; the matrix is what the map looks like after coordinates have been chosen.

3.4 Row Reduction and Elementary Matrices

Definition 3.4.1. *Elementary Row Operations and Elementary Matrices*
For row symbols R_i, the *elementary row operations* are:

(i) *Row scaling*: $R_i \leftarrow cR_i$, $c \in \mathbb{F}^\times$;
(ii) *Row swap*: $R_i \leftrightarrow R_j$, $i \neq j$;
(iii) *Row replacement*: $R_i \leftarrow R_i + cR_j$, $i \neq j$, $c \in \mathbb{F}$.

An *elementary matrix* is $E = \epsilon(I_m)$, where ϵ is one elementary row operation.

Proposition 3.4.2. *Elementary Matrices Perform Row Operations*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$.

(i) If ϵ is an elementary row operation and $E_\epsilon = \epsilon(I_m)$, then $E_\epsilon A = \epsilon(A)$.
(ii) $E_\epsilon^{-1} = E_{\epsilon^{-1}}$; hence every elementary matrix is invertible with elementary inverse.
(iii) If $F = E_\epsilon^t \in \mathrm{Mat}(n, n, \mathbb{F})$, then $AF = (\epsilon(A^t))^t$.
Hence right multiplication performs the corresponding column operation.

Proof.

(i) For each row index r, $\mathrm{row}_r(E_\epsilon A) = \mathrm{row}_r(E_\epsilon)A$.
So the rows of $E_\epsilon A$ are exactly the ϵ-linear combinations of the rows of A.
(ii) Since ϵ^{-1} is elementary, $E_{\epsilon^{-1}}E_\epsilon = E_{\epsilon^{-1}}\epsilon(I_m) = \epsilon^{-1}(\epsilon(I_m)) = I_m$, and similarly $E_\epsilon E_{\epsilon^{-1}} = I_m$.
(iii) By (i) and Proposition 3.3.13, $(AF)^t = E_\epsilon A^t = \epsilon(A^t)$, so $AF = (\epsilon(A^t))^t$.

Structural Commentary: Elementary matrices encode row and column operations by multiplication.
Key Insight: Row operations are left multiplication; column operations are right multiplication. □

Definition 3.4.3. *Row Equivalence and Reduced Row Echelon Form*
For $A, B \in \mathrm{Mat}(m, n, \mathbb{F})$, write $A \sim_{\mathrm{row}} B$ if $B = E_k \cdots E_1 A$ for elementary $E_i \in \mathrm{Mat}(m, m, \mathbb{F})$ and some $k \geq 0$.
A matrix $R = [r_{ij}]$ is in *reduced row echelon form* if its nonzero rows have pivot columns $p_1, \ldots, p_s$ such that:

(i) rows $1, \ldots, s$ are nonzero and rows $s+1, \ldots, m$ are zero;
(ii) $r_{i,p_i} = 1$ and $r_{i,j} = 0$ for $j < p_i$;

(iii) $p_1 < \cdots < p_s$;
(iv) $r_{\ell,p_i} = 0$ for all $\ell \neq i$.

Theorem 3.4.4. *Reduced Row Echelon Form*
For every $A \in \mathrm{Mat}(m,n,\mathbb{F})$, there is a unique RREF matrix $\mathrm{rref}(A)$ with $A \sim_{\mathrm{row}} \mathrm{rref}(A)$.
If $R = \mathrm{rref}(A)$, then $R = E_k \cdots E_1 A$ for elementary $E_i \in \mathrm{Mat}(m,m,\mathbb{F})$.

Proof. Gaussian elimination successively chooses a pivot column, swaps/scales to a pivot 1, clears the pivot column, and continues on the remaining submatrix. Proposition 3.4.2 gives $R = E_k \cdots E_1 A$ with R in RREF.
For uniqueness, row operations preserve $\ker(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$. For an RREF matrix, the pivot set and pivot-variable formulas are determined by this kernel.
Hence two row-equivalent RREF matrices have the same pivot columns and rows.
Structural Commentary: RREF is the canonical row-operation representative of a matrix.
Key Insight: Elimination gives existence; the solution space pins down the reduced representative. □

Lemma 3.4.5. *Invertible Matrices Row-Reduce to the Identity*
If $P \in \mathrm{Mat}(m,m,\mathbb{F})$ is invertible, then $\mathrm{rref}(P) = I_m$. Hence there are elementary matrices $E_1, \ldots, E_k \in \mathrm{Mat}(m,m,\mathbb{F})$ such that $E_k \cdots E_1 P = I_m$.

Proof. Let $R = \mathrm{rref}(P)$. By Theorem 3.4.4, $R = E_k \cdots E_1 P$ for elementary matrices E_i. Since each E_i is invertible by Proposition 3.4.2, left multiplication by $E_k \cdots E_1$ preserves the kernel, so $\ker(R) = \ker(P)$. Because P is invertible, $P\mathbf{x} = \mathbf{0}$ implies $\mathbf{x} = P^{-1}P\mathbf{x} = \mathbf{0}$; hence $\ker(R) = \{\mathbf{0}\}$.
Suppose the nonzero rows of R have pivot columns $p_1 < \cdots < p_s$. If $s < m$, choose a non-pivot column j. Set the non-pivot variable $x_j = 1$, set all other non-pivot variables equal to 0, and define the pivot variables recursively from $i = s$ down to 1 so that the ith row equation is satisfied. This gives a nonzero $\mathbf{x}$ with $R\mathbf{x} = \mathbf{0}$, contradicting $\ker(R) = \{\mathbf{0}\}$. Thus $s = m$.
Since $p_1 < \cdots < p_m$ are m columns in an $m \times m$ matrix, $p_i = i$ for every i. The RREF conditions then force the ith pivot column to be the ith standard coordinate column, so $R = I_m$. □

Corollary 3.4.6. *Row Equivalence Characterization*
For $A, B \in \mathrm{Mat}(m,n,\mathbb{F})$, the following are equivalent:

(i) $A \sim_{\mathrm{row}} B$.
(ii) $\mathrm{rref}(A) = \mathrm{rref}(B)$.
(iii) $\exists\, P \in \mathrm{Mat}(m,m,\mathbb{F})^{\times}$ *such that* $B = PA$.

Proof.

(i) $\Rightarrow$ (ii) Row-equivalent matrices have the same unique RREF by Theorem 3.4.4.
(ii) $\Rightarrow$ (iii) If $R = \mathrm{rref}(A) = \mathrm{rref}(B)$, then $R = E_k \cdots E_1 A = F_\ell \cdots F_1 B$.
Since each F_j is invertible, $B = (F_\ell \cdots F_1)^{-1} E_k \cdots E_1 A$.
(iii) $\Rightarrow$ (i) If $B = PA$ and $P \in \mathrm{Mat}(m,m,\mathbb{F})^{\times}$, Lemma 3.4.5 gives elementary matrices with $E_k \cdots E_1 P = I_m$.
Hence $P = E_1^{-1} \cdots E_k^{-1}$, so B is obtained from A by elementary row operations.

Structural Commentary: Row equivalence is invertible left multiplication.
Key Insight: Products of elementary matrices are the coordinate record of row reduction. □

3.5 Change of Basis

Definition 3.5.1. *Change of Basis Matrix*
Let β and γ be two bases of the same vector space $\mathcal{V}$. The change-of-basis matrix from β-coordinates to γ-coordinates is $C_{\gamma\leftarrow\beta} := [I_{\mathcal{V}}]_{\gamma\leftarrow\beta}$. Thus $[\mathbf{v}]_\gamma = C_{\gamma\leftarrow\beta}[\mathbf{v}]_\beta$ for every $\mathbf{v} \in \mathcal{V}$.

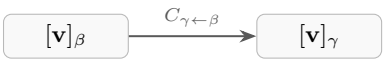


Figure 14: Change of basis: $[\mathbf{v}]_\gamma = C_{\gamma\leftarrow\beta}[\mathbf{v}]_\beta$.

Theorem 3.5.2. *Change-of-Basis Matrices*
Let $\mathcal{V}$ be n-dimensional and let β be a basis of $\mathcal{V}$.

(i) If γ is a basis of $\mathcal{V}$, then $C_{\gamma\leftarrow\beta}$ is invertible and $(C_{\gamma\leftarrow\beta})^{-1} = C_{\beta\leftarrow\gamma}$.
(ii) If $S \in \mathrm{Mat}(n,n,\mathbb{F})$ is invertible, then there is a basis γ of $\mathcal{V}$ such that $S = C_{\gamma\leftarrow\beta}$.

Proof.

(*i*) For any $\mathbf{v} \in \mathcal{V}$, $C_{\beta\leftarrow\gamma}C_{\gamma\leftarrow\beta}[\mathbf{v}]_\beta = [\mathbf{v}]_\beta$, $C_{\gamma\leftarrow\beta}C_{\beta\leftarrow\gamma}[\mathbf{v}]_\gamma = [\mathbf{v}]_\gamma$.
Since $[\cdot]_\beta$ and $[\cdot]_\gamma$ are onto $\mathbb{F}^n$, both products equal I_n. Hence $C_{\gamma\leftarrow\beta}$ is invertible with inverse $C_{\beta\leftarrow\gamma}$.

(*ii*) Write $S^{-1} = [\mathbf{s}_1 \ \cdots \ \mathbf{s}_n]$. For each j, choose the unique $\mathbf{w}_j \in \mathcal{V}$ with $[\mathbf{w}_j]_\beta = \mathbf{s}_j$, and set $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$.
Since columns of S^{-1} form a basis of $\mathbb{F}^n$ and $[\cdot]_\beta$ is an isomorphism, γ is a basis of $\mathcal{V}$.
Moreover $C_{\beta\leftarrow\gamma} = S^{-1}$, so $C_{\gamma\leftarrow\beta} = S$.

Structural Commentary: Change of basis is an invertible coordinate rewrite.
Key Insight: Every basis change is reversible, and every invertible matrix is one. □

Corollary 3.5.3. *Basis Columns Give an Invertible Matrix*
If $(\mathbf{a}_1, \ldots, \mathbf{a}_n)$ is a basis of $\mathbb{F}^n$, then $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible.

Proof. Let $\epsilon = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the standard basis and $\alpha = (\mathbf{a}_1, \ldots, \mathbf{a}_n)$. Then $C_{\epsilon\leftarrow\alpha} = \begin{bmatrix}[\mathbf{a}_1]_\epsilon & \cdots & [\mathbf{a}_n]_\epsilon\end{bmatrix} = A$.
Since α is a basis, Theorem 3.5.2 gives that A is invertible.
Structural Commentary: Basis columns form a change-of-basis matrix.
Key Insight: Changing coordinates is reversible. □

Theorem 3.5.4. *Completing a Full-Column-Rank Matrix*
If $X \in \mathrm{Mat}(m, n, \mathbb{F})$ has rank $n < m$, then $\exists\ X' \in \mathrm{Mat}(m, m-n, \mathbb{F})$ s.t. $\begin{bmatrix}X & X'\end{bmatrix} \in \mathrm{Mat}(m, m, \mathbb{F})$ is invertible.

Proof. Write $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$. Since $\mathrm{rank}(X) = n$, the columns $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are linearly independent in $\mathbb{F}^m$. By Theorem 1.3.12, extend this list to a basis $(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{x}_{n+1}, \ldots, \mathbf{x}_m)$ of $\mathbb{F}^m$.
Set $X' = [\mathbf{x}_{n+1} \ \cdots \ \mathbf{x}_m]$. Corollary 3.5.3 shows that $\begin{bmatrix}X & X'\end{bmatrix}$ is invertible.
Structural Commentary: Full column rank means the existing columns are independent directions. Extending them to a basis turns the rectangular matrix into an invertible square matrix.
Key Insight: Independent columns can always be completed to a basis in finite dimensions. □

Corollary 3.5.5. *Invertibility and Full Rank*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, A is invertible $\iff \mathrm{rank}(A) = n$.

Proof.

($\Rightarrow$) If A is invertible, its columns form a basis of $\mathbb{F}^n$ by Theorem 3.3.29; hence $\mathrm{rank}(A) = n$.

($\Leftarrow$) If $\mathrm{rank}(A) = n$, then $\mathrm{col}(A) = \mathbb{F}^n$.
Thus the columns of A form a basis by Proposition 1.4.3, so Corollary 3.5.3 gives that A is invertible.

Structural Commentary: Full rank is square-matrix invertibility.
Key Insight: Invertibility means the columns fill $\mathbb{F}^n$. □

Corollary 3.5.6. *Transpose Preserves Invertibility*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, the matrix A is invertible if and only if A^t is invertible.

Proof. By Theorem 3.3.23, $\mathrm{rank}(A) = \mathrm{rank}(A^t)$. Thus $\mathrm{rank}(A) = n$ if and only if $\mathrm{rank}(A^t) = n$.
Corollary 3.5.5 gives the result.
Structural Commentary: Transposition exchanges rows and columns without changing rank.
Key Insight: A square matrix is invertible exactly when either its rows or columns have full rank. □

Theorem 3.5.7. *Full-Rank Products Preserve Rank*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$. If $X \in \mathrm{Mat}(p, m, \mathbb{F})$ has full column rank and $Y \in \mathrm{Mat}(n, q, \mathbb{F})$ has full row rank, then $\mathrm{rank}(A) = \mathrm{rank}(XAY)$. This holds when X and Y are square invertible matrices of appropriate sizes.

Proof. Because X has full column rank, the induced map $L_X : \mathbb{F}^m \to \mathbb{F}^p$, $L_X(\mathbf{u}) = X\mathbf{u}$, has kernel $\{\mathbf{0}\}$.
Thus $\mathbf{z} \in \ker(XA) \iff X(A\mathbf{z}) = \mathbf{0} \iff A\mathbf{z} = \mathbf{0} \iff \mathbf{z} \in \ker(A)$.
Since XA and A have the same number of columns, Corollary 3.3.22 gives $\mathrm{rank}(XA) = \mathrm{rank}(A)$.
Because Y has full row rank, Y^t has full column rank. Applying the left-multiplication result to A^t and Y^t, then using Theorem 3.3.23, gives $\mathrm{rank}(AY) = \mathrm{rank}((AY)^t) = \mathrm{rank}(Y^tA^t) = \mathrm{rank}(A^t) = \mathrm{rank}(A)$.
Applying the left-multiplication result once more to AY gives $\mathrm{rank}(XAY) = \mathrm{rank}(AY) = \mathrm{rank}(A)$.
Structural Commentary: Multiplication by a full-column-rank matrix on the left loses no output information, while full row rank on the right loses no input information.
Key Insight: Rank changes only when multiplication introduces a new kernel or collapses the row side. □

Theorem 3.5.8. *Change of Bases for a Linear Map*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Let β, β' be bases of $\mathcal{V}$ and let γ, γ' be bases of $\mathcal{W}$. Then $[T]_{\gamma'\leftarrow\beta'} = C_{\gamma'\leftarrow\gamma}[T]_{\gamma\leftarrow\beta}C_{\beta\leftarrow\beta'}$.

Proof. For every $\mathbf{v} \in \mathcal{V}$, $[T\mathbf{v}]_{\gamma'} = C_{\gamma' \leftarrow \gamma}[T\mathbf{v}]_{\gamma} = C_{\gamma' \leftarrow \gamma}[T]_{\gamma \leftarrow \beta}[\mathbf{v}]_{\beta} = C_{\gamma' \leftarrow \gamma}[T]_{\gamma \leftarrow \beta} C_{\beta \leftarrow \beta'}[\mathbf{v}]_{\beta'}$.
Since $[\cdot]_{\beta'}$ is onto, the representing matrices are equal.
Structural Commentary: Basis changes act by $C_{\gamma' \leftarrow \gamma}$ on the left and $C_{\beta \leftarrow \beta'}$ on the right.
Key Insight: Domain changes act on input coordinates; codomain changes act on output coordinates. □

3.6 Matrix Equivalence and Similarity

Definition 3.6.1. *Matrix Equivalence*
For $A, B \in \mathrm{Mat}(m, n, \mathbb{F})$, write $A \sim_{\mathrm{eq}} B$ if $\exists\ P \in \mathrm{Mat}(m, m, \mathbb{F})^{\times}$ and $Q \in \mathrm{Mat}(n, n, \mathbb{F})^{\times}$ s.t. $B = PAQ$.

Proposition 3.6.2. *Matrix Equivalence and Elementary Operations*
The relation $\sim_{\mathrm{eq}}$ is an equivalence relation on $\mathrm{Mat}(m, n, \mathbb{F})$. Moreover, $A \sim_{\mathrm{eq}} B$ if and only if B can be obtained from A by finitely many elementary row and column operations.

Proof. Reflexivity follows from $A = I_m A I_n$. If $B = PAQ$ with P, Q invertible, then $A = P^{-1}BQ^{-1}$, so symmetry holds. If $B = P_1 A Q_1$ and $C = P_2 B Q_2$, then $C = (P_2 P_1) A (Q_1 Q_2)$, and products of invertible matrices are invertible by Corollary 3.3.14; hence transitivity holds.
Each elementary row operation is left multiplication by an invertible elementary matrix by Proposition 3.4.2. Each elementary column operation is right multiplication by the transpose of an invertible elementary matrix, again by Proposition 3.4.2 and Corollary 3.3.14. Thus any finite sequence of elementary row and column operations has the form $B = PAQ$ with P, Q invertible, so $A \sim_{\mathrm{eq}} B$.
Conversely, suppose $A \sim_{\mathrm{eq}} B$, so $B = PAQ$ with P, Q invertible. Lemma 3.4.5 gives products of elementary matrices expressing P and Q as products of elementary factors. Left multiplication by the factors of P is a sequence of elementary row operations, and right multiplication by the factors of Q is a sequence of elementary column operations. Hence B can be obtained from A by finitely many elementary row and column operations. □

Theorem 3.6.3. *Equivalent Matrices Represent the Same Linear Map*
Let $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$, and $A, B \in \mathrm{Mat}(m, n, \mathbb{F})$.
Then $A \sim_{\mathrm{eq}} B \iff \exists T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ with $A = [T]_{\gamma \leftarrow \beta}$, $B = [T]_{\gamma' \leftarrow \beta'}$ for some bases β, β' of $\mathcal{V}$ and γ, γ' of $\mathcal{W}$.

Proof.
($\Rightarrow$) Let $B = PAQ$ with P, Q invertible. Choose bases β, γ and define $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ by $[T]_{\gamma \leftarrow \beta} = A$.
By Theorem 3.5.2, choose basis γ' with $P = C_{\gamma' \leftarrow \gamma}$, and basis β' with $Q^{-1} = C_{\beta' \leftarrow \beta}$. Then $Q = C_{\beta \leftarrow \beta'}$.
Then Theorem 3.5.8 gives $[T]_{\gamma' \leftarrow \beta'} = PAQ = B$.
($\Leftarrow$) If $A = [T]_{\gamma \leftarrow \beta}$ and $B = [T]_{\gamma' \leftarrow \beta'}$, then $B = C_{\gamma' \leftarrow \gamma} A C_{\beta \leftarrow \beta'}$, so $A \sim_{\mathrm{eq}} B$.

Structural Commentary: Equivalence is two-sided basis change.
Key Insight: Same map; different coordinates. □

Theorem 3.6.4. *Rank Classifies Matrix Equivalence*
For $0 \le r \le \min(m, n)$, set $J_r^{m,n} := \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \in \mathrm{Mat}(m, n, \mathbb{F})$.
For $A, B \in \mathrm{Mat}(m, n, \mathbb{F})$, $A \sim_{\mathrm{eq}} J_{\mathrm{rank}(A)}^{m,n}$ and $A \sim_{\mathrm{eq}} B \iff \mathrm{rank}(A) = \mathrm{rank}(B)$.

Proof.
($\Rightarrow$) If $B = PAQ$ with P, Q invertible, then $\mathrm{rank}(B) = \mathrm{rank}(A)$ by Theorem 3.5.7.
($\Leftarrow$) For any X with $\mathrm{rank}(X) = r$, let $T_X : \mathbb{F}^n \to \mathbb{F}^m$ be $T_X(\mathbf{x}) = X\mathbf{x}$.
Choose $\mathcal{M}_X$ with $\mathbb{F}^n = \ker(T_X) \oplus \mathcal{M}_X$; then $T_X|_{\mathcal{M}_X} : \mathcal{M}_X \to \mathrm{range}(T_X)$ is an isomorphism.
Choose a basis $(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ of $\mathcal{M}_X$, extend it by a basis of $\ker(T_X)$ to β_X, and choose γ_X beginning with $T_X\mathbf{u}_1, \ldots, T_X\mathbf{u}_r$. Then $[T_X]_{\gamma_X \leftarrow \beta_X} = J_r^{m,n}$, so $X \sim_{\mathrm{eq}} J_r^{m,n}$.
If $\mathrm{rank}(A) = \mathrm{rank}(B) = r$, then $A \sim_{\mathrm{eq}} J_r^{m,n}$ and $B \sim_{\mathrm{eq}} J_r^{m,n}$. By Proposition 3.6.2, symmetry gives $J_r^{m,n} \sim_{\mathrm{eq}} B$, and transitivity gives $A \sim_{\mathrm{eq}} B$.

Structural Commentary: Rank is the invariant of two-sided equivalence.
Key Insight: Adapt bases to kernel and range. □

Corollary 3.6.5. *Rank Normal Form for Linear Maps*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $\dim \mathcal{V} = n$, $\dim \mathcal{W} = m$, and $\mathrm{rank}(T) = r$. Then $\exists$ bases β, γ with $[T]_{\gamma \leftarrow \beta} = J_r^{m,n}$.

Proof. For initial bases β_0, γ_0, let $A = [T]_{\gamma_0 \leftarrow \beta_0}$.
By Theorem 3.6.4, $A \sim_{\mathrm{eq}} J_r^{m,n}$; Theorem 3.6.3 gives bases β, γ with $[T]_{\gamma \leftarrow \beta} = J_r^{m,n}$.
Structural Commentary: Rank gives the two-basis normal form.
Key Insight: Rank normalizes the map. □

Corollary 3.6.6. *Linear Maps Classified by Rank*
For $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, write $S \sim_{\text{bc}} T$ if $\exists$ bases β_S, β_T of $\mathcal{V}$ and γ_S, γ_T of $\mathcal{W}$ such that $[S]_{\gamma_S \leftarrow \beta_S} = [T]_{\gamma_T \leftarrow \beta_T}$. Then $S \sim_{\text{bc}} T \iff \operatorname{rank}(S) = \operatorname{rank}(T)$

Proof.

($\Rightarrow$) If $[S]_{\gamma_S \leftarrow \beta_S} = [T]_{\gamma_T \leftarrow \beta_T}$, then the two maps have the same matrix rank, hence the same rank.
($\Leftarrow$) If $\operatorname{rank}(S) = \operatorname{rank}(T) = r$, Corollary 3.6.5 gives bases with $[S]_{\gamma_S \leftarrow \beta_S} = J_r^{m,n} = [T]_{\gamma_T \leftarrow \beta_T}$.

Structural Commentary: Rank is the invariant of basis-change equivalence.
Key Insight: Same rank means same normal form. □

Theorem 3.6.7. *Change of Basis Formula*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let β be a basis of $\mathcal{V}$.

(i) If γ is a basis of $\mathcal{V}$ and $S = C_{\gamma \leftarrow \beta}$, then $[T]_{\gamma \leftarrow \gamma} = S[T]_{\beta \leftarrow \beta} S^{-1}$.
(ii) If $S \in \operatorname{Mat}(n, n, \mathbb{F})$ is invertible, then there is a basis γ of $\mathcal{V}$ such that $[T]_{\gamma \leftarrow \gamma} = S[T]_{\beta \leftarrow \beta} S^{-1}$.

Proof.

(i) Suppose γ is a basis and $S = C_{\gamma \leftarrow \beta}$. By Theorem 3.5.2, $S^{-1} = C_{\beta \leftarrow \gamma}$.
For every $\mathbf{v} \in \mathcal{V}$, $[T\mathbf{v}]_\gamma = S[T\mathbf{v}]_\beta = S[T]_{\beta \leftarrow \beta}[\mathbf{v}]_\beta = S[T]_{\beta \leftarrow \beta} S^{-1}[\mathbf{v}]_\gamma$.
Since $[\cdot]_\gamma$ is onto $\mathbb{F}^n$, the representing matrices are equal.
(ii) If S is invertible, Theorem 3.5.2 supplies a basis γ with $S = C_{\gamma \leftarrow \beta}$, and part (i) applies.

Structural Commentary: Changing basis conjugates operator matrices.
Key Insight: Similarity is a coordinate rewrite. □

Definition 3.6.8. *Similarity*
Matrices $A, B \in \operatorname{Mat}(n, n, \mathbb{F})$ are similar if there exists an invertible matrix P such that $B = PAP^{-1}$.

Proposition 3.6.9. *Similarity Is an Equivalence Relation*
Similarity is an equivalence relation on $\operatorname{Mat}(n, n, \mathbb{F})$.

Proof. For any A, $A = I_n A I_n^{-1}$, so similarity is reflexive. If $B = PAP^{-1}$, then $A = P^{-1}BP$, so it is symmetric. If $B = PAP^{-1}$ and $C = QBQ^{-1}$, then $C = Q(PAP^{-1})Q^{-1} = (QP)A(QP)^{-1}$ so it is transitive.
Structural Commentary: Similarity groups matrices by change of basis.
Key Insight: Identity, inverse, and composition give the three equivalence properties. □

Perspective 3.6.10. *Similarity as Same Operator, New Coordinates*
Similarity does not mean two operators are merely related; it means they are the same operator written in two different bases. This is the structural content behind change of basis.

Remark 3.6.11. *Similarity Convention*
The forms PAP^{-1} and $M^{-1}AM$ are the same convention with $P = M^{-1}$. In Theorem 3.6.7, this means choosing opposite coordinate-change directions.

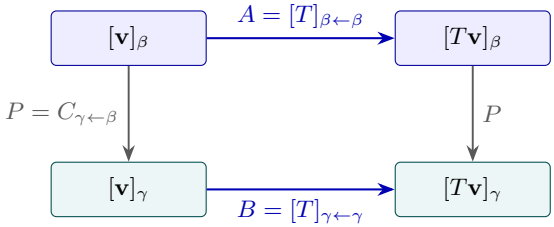


Figure 15: One operator in two bases: $PA = BP$, so $B = PAP^{-1}$.

Corollary 3.6.12. *Similarity as One Operator in Two Bases*
For $A, B \in \operatorname{Mat}(n, n, \mathbb{F})$, the following are equivalent:

(i) A and B are similar.
(ii) There are an n-dim vector space $\mathcal{V}$, bases β, γ of $\mathcal{V}$, and $T \in \mathcal{L}(\mathcal{V})$ such that $A = [T]_{\beta \leftarrow \beta}$ and $B = [T]_{\gamma \leftarrow \gamma}$.

Proof.

(i) $\Rightarrow$ (ii): Suppose $B = PAP^{-1}$ with P invertible.
Let $\mathcal{V} = \mathbb{F}^n$, let β be the standard basis, and let $T_A(\mathbf{x}) = A\mathbf{x}$. By Theorem 3.5.2, choose γ with $P = C_{\gamma \leftarrow \beta}$.
Then $[T_A]_{\beta \leftarrow \beta} = A$, and Theorem 3.6.7 gives $[T_A]_{\gamma \leftarrow \gamma} = B$.

(*ii*) $\Rightarrow$ (*i*): If $A = [T]_{\beta\leftarrow\beta}$ and $B = [T]_{\gamma\leftarrow\gamma}$, then $B = C_{\gamma\leftarrow\beta} A C_{\beta\leftarrow\gamma}$ with $C_{\gamma\leftarrow\beta}$ invertible.

Structural Commentary: Similar matrices are one operator in different bases.
Key Insight: Conjugation records a change of basis. □

Definition 3.6.13. *Similarity of Operators*
For $T \in \mathcal{L}(\mathcal{V})$ and $S \in \mathcal{L}(\mathcal{W})$, $T \sim S \iff \exists\, \Phi : \mathcal{V} \xrightarrow{\sim} \mathcal{W}$ such that $S = \Phi T \Phi^{-1}$.
If $\mathcal{V} = \mathcal{W}$, then $\Phi \in \mathrm{Aut}(\mathcal{V})$ and similarity is conjugacy in $\mathcal{L}(\mathcal{V})$.

Proposition 3.6.14. *Operator Similarity and Common Matrix Representatives*
Let $\dim \mathcal{V} = \dim \mathcal{W} = n$, $T \in \mathcal{L}(\mathcal{V})$, *and* $S \in \mathcal{L}(\mathcal{W})$.
Then $T \sim S \iff \exists\, A \in \mathrm{Mat}(n, n, \mathbb{F})$ *with* $A = [T]_{\beta\leftarrow\beta} = [S]_{\gamma\leftarrow\gamma}$ *for some bases* β *of* $\mathcal{V}$ *and* γ *of* $\mathcal{W}$.

Proof.
($\Rightarrow$) Let $S = \Phi T \Phi^{-1}$ and $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Put $\gamma = (\Phi\mathbf{v}_1, \ldots, \Phi\mathbf{v}_n)$.
Then γ is a basis and, for all j, $S(\Phi\mathbf{v}_j) = \Phi T \mathbf{v}_j$, so $[S]_{\gamma\leftarrow\gamma} = [T]_{\beta\leftarrow\beta}$.
($\Leftarrow$) Suppose $[T]_{\beta\leftarrow\beta} = [S]_{\gamma\leftarrow\gamma}$. Define $\Phi : \mathcal{V} \to \mathcal{W}$ by $\Phi(\beta_j) = \gamma_j$.
Then Φ is an isomorphism and $\Phi T \beta_j = S \Phi \beta_j$ for every j, hence $\Phi T = S \Phi$. Thus $S = \Phi T \Phi^{-1}$.

Structural Commentary: Similarity is equality of matrices after identifying bases.
Key Insight: A common matrix is an operator-level conjugacy written in bases. □

Theorem 3.6.15. *Similarity and Scalar Shifts*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.

(i) If A *and* B *are similar, then* $A - \lambda I_n$ *and* $B - \lambda I_n$ *are similar for every* $\lambda \in \mathbb{F}$.
(ii) If $A - \lambda I_n$ *and* $B - \lambda I_n$ *are similar for some* $\lambda \in \mathbb{F}$, *then* A *and* B *are similar.*

Proof.
(i) If $B = PAP^{-1}$ with P invertible, then $B - \lambda I_n = PAP^{-1} - \lambda PP^{-1} = P(A - \lambda I_n)P^{-1}$.
(ii) If $B - \lambda I_n = P(A - \lambda I_n)P^{-1}$ with P invertible, then adding $\lambda I_n = \lambda PP^{-1}$ gives $B = PAP^{-1}$.

Structural Commentary: Scalar shifts commute with change of basis.
Key Insight: The identity matrix is basis-independent. □

4 Block Matrices and Matrix Operations

4.1 Block Matrix Algebra

Definition 4.1.1. *Block Matrices*
Suppose $\mathcal{V} = \mathcal{U}_1 \oplus \mathcal{U}_2$ and $\mathcal{W} = \mathcal{Z}_1 \oplus \mathcal{Z}_2$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, relative to bases adapted to these decompositions in the sense of Definition 2.3.4, the matrix of T has block form $[T] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$, where each block records the part of T mapping one summand of the domain into one summand of the codomain.

Definition 4.1.2. *Conformal Block Partitions*
For a block matrix $A = [A_{ij}]$, the submatrices A_{ij} are called *blocks*. Two block partitions are *conformal for addition* when corresponding blocks have the same sizes. They are *conformal for multiplication* when every block product appearing in the usual row-column block multiplication is defined.

Proposition 4.1.3. *Block Matrix Arithmetic*
For conformal 2×2 *block matrices,*

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} \\ A_{21} + B_{21} & A_{22} + B_{22} \end{bmatrix}$$

and

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

Proof. Each displayed block consists of the same entries obtained by ordinary entrywise addition or ordinary row-column multiplication, with the common indices grouped according to the partition.
Structural Commentary: This is ordinary matrix arithmetic with indices grouped into rectangular packages.
Key Insight: Conformality is exactly the condition that the grouped row-column products make sense. □

Proposition 4.1.4. *Block Centrosymmetry and Commutation*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \in \mathrm{Mat}(2n, 2n, \mathbb{F})$, $L_{2n} = \begin{bmatrix} 0 & I_n \\ I_n & 0 \end{bmatrix}$.
Then M *has block centrosymmetric form* $A = D$ *and* $B = C$ *if and only if* $L_{2n}M = ML_{2n}$.

Proof. Block multiplication gives $L_{2n}M = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$ and $ML_{2n} = \begin{bmatrix} B & A \\ D & C \end{bmatrix}$.
The two matrices are equal exactly when $A = D$ and $B = C$.
Structural Commentary: The matrix L_{2n} swaps the two block coordinates. Commuting with that swap means M treats the two block positions symmetrically.
Key Insight: Compute both block products; equality forces matching opposite blocks. □

Definition 4.1.5. *Direct Sum of Matrices*
For $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(p, q, \mathbb{F})$, the *direct sum* of A and B is $A \oplus B := \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \in \mathrm{Mat}(m+p, n+q, \mathbb{F})$.

4.2 Matrix Operations and Identities

Definition 4.2.1. *Polynomial Evaluation on a Square Matrix*
If $A \in \mathrm{Mat}(n, n, \mathbb{F})$ and $p(z) = \sum_{j=0}^{d} c_j z^j \in \mathbb{F}[z]$, then $p(A) := \sum_{j=0}^{d} c_j A^j$, $A^0 := I_n$.

Proposition 4.2.2. *Algebra of Matrix Direct Sums*
Whenever the expressions are defined,
(i) $a(A \oplus B) + b(C \oplus D) = (aA + bC) \oplus (aB + bD)$;
(ii) $(A \oplus B)(C \oplus D) = AC \oplus BD$;
(iii) if A *and* B *are square, then* $(A \oplus B)^k = A^k \oplus B^k$ *for every positive integer* k;
(iv) if A *and* B *are square and* p *is a polynomial, then* $p(A \oplus B) = p(A) \oplus p(B)$.

Proof. Block addition gives $a(A \oplus B) + b(C \oplus D) = \begin{bmatrix} aA + bC & 0 \\ 0 & aB + bD \end{bmatrix} = (aA + bC) \oplus (aB + bD)$.
Block multiplication gives $(A \oplus B)(C \oplus D) = \begin{bmatrix} AC & 0 \\ 0 & BD \end{bmatrix} = AC \oplus BD$.
For powers, the case $k = 1$ is the definition. If $(A \oplus B)^k = A^k \oplus B^k$, then part (ii) gives $(A \oplus B)^{k+1} = A^{k+1} \oplus B^{k+1}$. Hence (iii) follows by induction. For the polynomial statement, suppose $A \in \mathrm{Mat}(m, m, \mathbb{F})$ and $B \in \mathrm{Mat}(r, r, \mathbb{F})$. For $j = 0$, Definition 4.2.1 gives $(A \oplus B)^0 = I_{m+r} = I_m \oplus I_r = A^0 \oplus B^0$. If $p(z) = \sum_{j=0}^{d} c_j z^j$,
then $p(A \oplus B) = \sum_{j=0}^{d} c_j (A^j \oplus B^j) = \left(\sum_{j=0}^{d} c_j A^j \right) \oplus \left(\sum_{j=0}^{d} c_j B^j \right) = p(A) \oplus p(B)$.
Structural Commentary: A direct sum matrix keeps the two blocks independent under linear combinations, products, powers, and polynomials.
Key Insight: Block diagonal matrices behave componentwise. □

Proposition 4.2.3. *Diagonal Multiplication Scales Rows and Columns*
Let $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, *let* $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, *and let* $M = \mathrm{diag}(\mu_1, \ldots, \mu_m)$. *Then* $A\Lambda = [\lambda_j a_{ij}]$ *and* $MA = [\mu_i a_{ij}]$. *Thus right multiplication by* Λ *scales columns of* A, *left multiplication by* M *scales rows of* A.

Proof. The (i, j) entry of $A\Lambda$ is $\sum_{r=1}^{n} a_{ir}\Lambda_{rj} = a_{ij}\lambda_j$, and the (i, j) entry of MA is $\sum_{r=1}^{m} M_{ir}a_{rj} = \mu_i a_{ij}$.
Structural Commentary: Diagonal matrices act by independent scalar weights on coordinate directions.
Key Insight: Multiplication on the right scales columns, while multiplication on the left scales rows. □

Proposition 4.2.4. *Inverse of an Upper Block-Triangular Matrix*
Let $Y \in \mathrm{Mat}(m, m, \mathbb{F})$ *and* $Z \in \mathrm{Mat}(n, n, \mathbb{F})$ *be invertible, and let* $X \in \mathrm{Mat}(m, n, \mathbb{F})$.
Then $\begin{bmatrix} Y & X \\ 0 & Z \end{bmatrix}^{-1} = \begin{bmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{bmatrix}$.

Proof. Let $M = \begin{bmatrix} Y & X \\ 0 & Z \end{bmatrix}$ and $N = \begin{bmatrix} Y^{-1} & -Y^{-1}XZ^{-1} \\ 0 & Z^{-1} \end{bmatrix}$. Proposition 4.1.3 gives

$$MN = \begin{bmatrix} I_m & -XZ^{-1} + XZ^{-1} \\ 0 & I_n \end{bmatrix} = I_{m+n}, \qquad NM = \begin{bmatrix} I_m & Y^{-1}X - Y^{-1}X \\ 0 & I_n \end{bmatrix} = I_{m+n}$$

. Thus $N = M^{-1}$.
Structural Commentary: Upper block-triangular inversion preserves the zero lower-left block. The off-diagonal

term is forced by cancelling the original upper-right block.
Key Insight: Inverting a block triangular matrix inverts diagonal blocks and corrects the coupling block. □

Theorem 4.2.5. *Inverse of an Upper Triangular Matrix*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ be upper triangular with nonzero diagonal entries. Then A is invertible, A^{-1} is upper triangular, and the diagonal entries of A^{-1} are $a_{11}^{-1}, \dots, a_{nn}^{-1}$.

Proof. We argue by induction on n. The case $n = 1$ is immediate. For $n > 1$, write $A = \begin{bmatrix} B & \mathbf{x} \\ 0 & a_{nn} \end{bmatrix}$, where B is $(n-1) \times (n-1)$ upper triangular with nonzero diagonal entries.
By induction, B^{-1} is upper triangular with diagonal entries $a_{11}^{-1}, \dots, a_{n-1,n-1}^{-1}$.
Proposition 4.2.4 gives $A^{-1} = \begin{bmatrix} B^{-1} & -B^{-1}\mathbf{x}a_{nn}^{-1} \\ 0 & a_{nn}^{-1} \end{bmatrix}$, which is upper triangular and has the asserted diagonal.
Structural Commentary: An upper triangular matrix splits into a smaller upper triangular block and one final scalar block. The block inverse formula preserves this triangular structure.
Key Insight: Nonzero diagonals make blockwise triangular inversion possible. □

Proposition 4.2.6. *Upper Block-Triangular Similarity Shift*
Let $B \in \mathrm{Mat}(m, m, \mathbb{F})$, let $D \in \mathrm{Mat}(n, n, \mathbb{F})$, and let $C, X \in \mathrm{Mat}(m, n, \mathbb{F})$.
Then $\begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$ is similar to $\begin{bmatrix} B & C + XD - BX \\ 0 & D \end{bmatrix}$.

Proof. The matrices $\begin{bmatrix} I_m & X \\ 0 & I_n \end{bmatrix}$ and $\begin{bmatrix} I_m & -X \\ 0 & I_n \end{bmatrix}$ are inverses in $\mathrm{Mat}(m+n, m+n, \mathbb{F})$, by block multiplication,

$$\begin{bmatrix} I_m & X \\ 0 & I_n \end{bmatrix} \begin{bmatrix} B & C \\ 0 & D \end{bmatrix} \begin{bmatrix} I_m & -X \\ 0 & I_n \end{bmatrix} = \begin{bmatrix} B & C + XD - BX \\ 0 & D \end{bmatrix}.$$

Structural Commentary: Unit upper block-triangular conjugation changes only the coupling block.
Key Insight: Similarity can adjust the connecting block while preserving the diagonal blocks. □

5 Duality and Multilinear Structure

5.1 Dual Spaces and Dual Bases

Duality studies vectors through scalar-valued tests. Functionals, dual maps, annihilators, transposes, bilinear forms, and tensor products extend linear structure into multilinear language.

Definition 5.1.1. *Linear Functional and Dual Space*
A *linear functional* on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$. The *dual space* of $\mathcal{V}$ is $\mathcal{V}' := \mathcal{L}(\mathcal{V}, \mathbb{F})$.

Example 5.1.2. *Evaluation and Integral Functionals*
The evaluation map $\mathrm{ev}_0 : \mathbb{F}[x] \to \mathbb{F}$, $p \mapsto p(0)$ is a linear functional, since $\mathrm{ev}_0(\alpha p + \beta q) = \alpha p(0) + \beta q(0)$ for all $\alpha, \beta \in \mathbb{F}$ and $p, q \in \mathbb{F}[x]$. If $C[a, b]$ is the real vector space of continuous functions on $[a, b]$, then $I : C[a, b] \to \mathbb{R}$, $I(f) := \int_a^b f(x)\,dx$ is a linear functional, since $I(\alpha f + \beta g) = \alpha I(f) + \beta I(g)$ $\forall \alpha, \beta \in \mathbb{R}$ and $f, g \in C[a, b]$.

Definition 5.1.3. *Dual Basis*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$. The *dual basis* of β is the list $\beta' = (\varphi_1, \dots, \varphi_n)$ in $\mathcal{V}'$ defined by

$$\varphi_j(\mathbf{v}_k) = \begin{cases} 1, & k = j, \\ 0, & k \neq j. \end{cases}$$

Existence and uniqueness of each φ_j follow from Proposition 3.1.11.

Theorem 5.1.4. *Dual Basis and Dimension of the Dual Space*
Let $\mathcal{V}$ be finite-dimensional with basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, and let $\beta' = (\varphi_1, \dots, \varphi_n)$ be the dual basis. Then β' is a basis of $\mathcal{V}'$. In particular, $\dim(\mathcal{V}') = \dim(\mathcal{V})$.

Proof. First suppose $a_1\varphi_1 + \cdots + a_n\varphi_n = 0$. Evaluating at $\mathbf{v}_j$ gives $a_j = 0$ for each j, so β' is linearly independent. Now let $\varphi \in \mathcal{V}'$. For $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, $\left(\sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j\right)(\mathbf{v}) = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)c_j = \varphi(\mathbf{v})$.
Thus $\varphi = \sum_{j=1}^{n} \varphi(\mathbf{v}_j)\varphi_j$, so β' spans $\mathcal{V}'$.
Structural Commentary: A basis of $\mathcal{V}$ induces the dual coordinate system.
Key Insight: A linear functional is determined exactly by its values on a basis. □

Proposition 5.1.5. *Coordinate Functionals for a Hamel Basis*
Let $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ be a Hamel basis of $\mathcal{V}$, define $\varepsilon_i \in \mathcal{V}'$ by $\varepsilon_i\left(\sum_{j\in F} a_j\mathbf{b}_j\right) := a_i$, $F \subseteq I$ finite, $a_i = 0$ if $i \notin F$.
Then $\{\varepsilon_i\}_{i\in I}$ is linearly independent in $\mathcal{V}'$. Moreover, the following are equivalent:
(i) $\operatorname{span}\{\varepsilon_i : i \in I\} = \mathcal{V}'$;
(ii) $|I| < \infty$;
(iii) $\dim(\mathcal{V}) < \infty$.

Proof. If $\sum_{r=1}^{n} c_r\varepsilon_{i_r} = 0$ with distinct i_r, then $0 = \left(\sum_{r=1}^{n} c_r\varepsilon_{i_r}\right)(\mathbf{b}_{i_s}) = c_s\ \forall s$. Hence $\{\varepsilon_i\}_{i\in I}$ linearly independent.
(*i*) $\Rightarrow$ (*ii*). Suppose $\operatorname{span}\{\varepsilon_i : i \in I\} = \mathcal{V}'$. Put $\operatorname{supp}_{\mathcal{B}}(\psi) := \{i \in I : \psi(\mathbf{b}_i) \neq 0\}$.
If I were infinite, define $\varphi \in \mathcal{V}'$ by $\varphi(\mathbf{b}_i) = 1$ for all $i \in I$.
Then $\operatorname{supp}_{\mathcal{B}}(\varphi) = I$, while $\operatorname{supp}_{\mathcal{B}}(\psi)$ is finite for every $\psi \in \operatorname{span}\{\varepsilon_i : i \in I\}$, a contradiction.
(*ii*) $\Rightarrow$ (*iii*). Since $\mathcal{B}$ is a basis, $\dim(\mathcal{V}) = |I| < \infty$.
(*iii*) $\Rightarrow$ (*i*). If $\dim(\mathcal{V}) < \infty$, then Theorem 5.1.4 gives $\operatorname{span}\{\varepsilon_i : i \in I\} = \mathcal{V}'$.

Structural Commentary: Coordinate spans are finite-support.
Key Insight: Infinite-dimensional coordinate functionals are independent, not spanning. □

Theorem 5.1.6. *Dimension of the Algebraic Dual*
For every $\mathbb{F}$-vector space $\mathcal{V}$, $\dim(\mathcal{V}) \leq \dim(\mathcal{V}')$. Moreover, the following are equivalent:
(i) $\dim(\mathcal{V}) = \dim(\mathcal{V}')$;
(ii) $\dim(\mathcal{V}) < \infty$.

Proof. Let $\mathcal{B} = \{\mathbf{b}_i\}_{i\in I}$ be a Hamel basis. By Proposition 5.1.5, $\{\varepsilon_i\}_{i\in I} \subseteq \mathcal{V}'$ is linearly independent; hence $\dim(\mathcal{V}) = |I| \leq \dim(\mathcal{V}')$. For the equality criterion:
(*i*) $\Rightarrow$ (*ii*). Suppose $\dim(\mathcal{V}) = \dim(\mathcal{V}')$. If I were infinite, then $\mathcal{V} \cong (\mathbb{F}^I)_0$ and $\mathcal{V}' \cong \mathbb{F}^I$.
Let $\mathbb{F}_0 \subseteq \mathbb{F}$ be the prime subfield. Since $\mathbb{F}_0$ is countable, $\dim_{\mathbb{F}_0}\left((\mathbb{F}_0^I)_0\right) = |I|$ and $\left|\mathbb{F}_0^I\right| > |I|$ by Cantor, so $\dim_{\mathbb{F}_0}(\mathbb{F}_0^I) > |I|$; hence $\dim_{\mathbb{F}_0}\left((\mathbb{F}_0^I)_0\right) = |I| < \dim_{\mathbb{F}_0}(\mathbb{F}_0^I)$.
Every $\mathbb{F}_0$-linearly independent family in $\mathbb{F}_0^I$ is $\mathbb{F}$-linearly independent in $\mathbb{F}^I$, by expanding coefficients in an $\mathbb{F}_0$-basis of $\mathbb{F}$. Thus $\dim_{\mathbb{F}}(\mathcal{V}) = \dim_{\mathbb{F}}\left((\mathbb{F}^I)_0\right) < \dim_{\mathbb{F}}(\mathbb{F}^I) = \dim_{\mathbb{F}}(\mathcal{V}')$, a contradiction. Hence $\dim(\mathcal{V}) < \infty$.
(*ii*) $\Rightarrow$ (*i*). If $\dim(\mathcal{V}) < \infty$, then Theorem 5.1.4 gives $\dim(\mathcal{V}) = \dim(\mathcal{V}')$.

Structural Commentary: Algebraic duals have arbitrary support.
Key Insight: Infinite algebraic duals are bigger than their original spaces. □

Proposition 5.1.7. *Functionals Separate Points*
Let $\mathcal{V}$ be a vector space.
(i) If $\mathbf{v} \neq \mathbf{0}$, then there exists $\varphi \in \mathcal{V}'$ such that $\varphi(\mathbf{v}) \neq 0$.
(ii) A vector $\mathbf{v} \in \mathcal{V}$ is zero if and only if $\varphi(\mathbf{v}) = 0$ for every $\varphi \in \mathcal{V}'$.
(iii) If $\mathcal{U} \leq \mathcal{V}$ and $\mathbf{x} \notin \mathcal{U}$, then there exists $\varphi \in \mathcal{V}'$ such that $\varphi|_{\mathcal{U}} = 0$ and $\varphi(\mathbf{x}) = 1$.
(iv) If $\varphi \in \mathcal{V}'$ and $\varphi(\mathbf{x}) \neq 0$, then $\mathcal{V} = \operatorname{span}(\mathbf{x}) \oplus \ker(\varphi)$.
(v) If nonzero $\varphi, \psi \in \mathcal{V}'$ have the same kernel, then $\varphi = a\psi$ for some nonzero $a \in \mathbb{F}$.

Proof.
(i) Extend $\{\mathbf{v}\}$ to a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.4.6. Define φ on $\mathcal{B}$ by $\varphi(\mathbf{v}) = 1$ and $\varphi(\mathbf{b}) = 0$ for $\mathbf{b} \in \mathcal{B} \setminus \{\mathbf{v}\}$, then extend linearly to finite basis expansions.
(ii) If $\mathbf{v} = \mathbf{0}$, then $\varphi(\mathbf{v}) = 0$ for every $\varphi \in \mathcal{V}'$. Conversely, if $\varphi(\mathbf{v}) = 0$ for every $\varphi \in \mathcal{V}'$ and $\mathbf{v} \neq \mathbf{0}$, then (i) gives some $\psi \in \mathcal{V}'$ with $\psi(\mathbf{v}) \neq 0$, a contradiction.
(iii) Choose a Hamel basis $\mathcal{B}_{\mathcal{U}}$ of $\mathcal{U}$. Since $\mathbf{x} \notin \mathcal{U} = \operatorname{span}(\mathcal{B}_{\mathcal{U}})$, the set $\mathcal{B}_{\mathcal{U}} \cup \{\mathbf{x}\}$ is linearly independent. Extend it to a Hamel basis $\mathcal{B}$ of $\mathcal{V}$ by Theorem 1.4.6. Define φ on $\mathcal{B}$ by $\varphi(\mathbf{x}) = 1$ and $\varphi(\mathbf{b}) = 0$ for every $\mathbf{b} \in \mathcal{B} \setminus \{\mathbf{x}\}$, then extend linearly to finite basis expansions.
(iv) Since $\varphi(\mathbf{x}) \neq 0$, $\operatorname{span}(\mathbf{x}) \cap \ker(\varphi) = \{\mathbf{0}\}$.
For any $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{x})}\mathbf{x} + \left(\mathbf{v} - \frac{\varphi(\mathbf{v})}{\varphi(\mathbf{x})}\mathbf{x}\right)$, and the second summand lies in $\ker(\varphi)$.
(v) Let $\mathcal{K} = \ker(\varphi) = \ker(\psi)$ and choose $\mathbf{x} \notin \mathcal{K}$. By (iv), every $\mathbf{v} \in \mathcal{V}$ can be written $\mathbf{v} = c\mathbf{x} + \mathbf{k}$ with $\mathbf{k} \in \mathcal{K}$. Setting $a = \varphi(\mathbf{x})/\psi(\mathbf{x})$ gives $\varphi(\mathbf{v}) = c\varphi(\mathbf{x}) = a\,c\psi(\mathbf{x}) = a\psi(\mathbf{v})$.

Structural Commentary: Dual vectors separate points and define hyperplanes.
Key Insight: A nonzero functional splits the space into one visible direction plus its kernel. □

Corollary 5.1.8. *Kernels of Nonzero Functionals*
If $0 \neq \varphi \in \mathcal{V}'$, then φ is surjective onto $\mathbb{F}$, $\mathcal{V}/\ker(\varphi) \cong \mathbb{F}$, and $\operatorname{codim}_{\mathcal{V}}(\ker\varphi) = 1$.

Proof. Choose $\mathbf{x}$ with $\varphi(\mathbf{x}) \neq 0$. For each $a \in \mathbb{F}$, $\varphi\big(a\varphi(\mathbf{x})^{-1}\mathbf{x}\big) = a$, so $\operatorname{im}\varphi = \mathbb{F}$.
Hence, by Theorem 3.2.30, $\mathcal{V}/\ker(\varphi) \cong \operatorname{im}\varphi = \mathbb{F}$ and $\operatorname{codim}_{\mathcal{V}}(\ker\varphi) = \dim(\mathcal{V}/\ker\varphi) = 1$.
Structural Commentary: A nonzero functional sees exactly one scalar direction and kills a hyperplane.
Key Insight: Kernels of nonzero functionals have codimension one. □

Definition 5.1.9. *Algebraic Double Dual and Natural Map*

(i) The *algebraic double dual* of $\mathcal{V}$ is $\mathcal{V}'' := (\mathcal{V}')'$.
(ii) The *natural evaluation map* is $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$; $\Lambda_{\mathcal{V}}(\mathbf{v})(\varphi) = \varphi(\mathbf{v})$.
(iii) $\mathcal{V}$ is *algebraically reflexive* if $\Lambda_{\mathcal{V}}$ is surjective.

Theorem 5.1.10. *Algebraic Reflexivity*
Natural map $\Lambda_{\mathcal{V}} \in \mathcal{L}(\mathcal{V}, \mathcal{V}'')$ is linear and injective. It is an isomorphism if and only if $\mathcal{V}$ is finite-dimensional.

Proof. For $\mathbf{v} \in \mathcal{V}$, $\Lambda_{\mathcal{V}}(\mathbf{v}) : \varphi \mapsto \varphi(\mathbf{v})$ is linear on $\mathcal{V}'$.
For $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\Lambda_{\mathcal{V}}(a\mathbf{u} + b\mathbf{v})(\varphi) = a\Lambda_{\mathcal{V}}(\mathbf{u})(\varphi) + b\Lambda_{\mathcal{V}}(\mathbf{v})(\varphi)$, $\varphi \in \mathcal{V}'$, so $\Lambda_{\mathcal{V}} \in \mathcal{L}(\mathcal{V}, \mathcal{V}'')$.
If $\mathbf{v} \in \ker(\Lambda_{\mathcal{V}})$, then $\varphi(\mathbf{v}) = 0$ for all $\varphi \in \mathcal{V}'$, hence $\mathbf{v} = \mathbf{0}$ by Proposition 5.1.7. Thus $\ker(\Lambda_{\mathcal{V}}) = \{\mathbf{0}\}$.
For the isomorphism criterion:

(⇒) If $\Lambda_{\mathcal{V}}$ is an isomorphism and $\dim(\mathcal{V}) = \infty$, then Theorem 5.1.6 gives $\dim(\mathcal{V}'') > \dim(\mathcal{V}') > \dim(\mathcal{V})$, contradicting $\mathcal{V} \cong \mathcal{V}''$. Hence $\dim(\mathcal{V}) < \infty$.
(⇐) If $\dim(\mathcal{V}) < \infty$, then $\dim(\mathcal{V}'') = \dim(\mathcal{V}') = \dim(\mathcal{V})$ by Theorem 5.1.4. Since $\Lambda_{\mathcal{V}}$ is injective, it is an isomorphism.

Structural Commentary: $\Lambda_{\mathcal{V}}$ embeds vectors as evaluation functionals.
Key Insight: Finite-dimensional spaces are exactly the algebraically reflexive vector spaces. □

Definition 5.1.11. *Dual Map*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the *dual map* of T is the map $T' : \mathcal{W}' \to \mathcal{V}'$ defined by $T'\varphi := \varphi T$ for every $\varphi \in \mathcal{W}'$.
Equivalently, $(T'\varphi)(\mathbf{v}) = \varphi(T\mathbf{v})$.

Proposition 5.1.12. *Dual Maps Are Linear*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $T' \in \mathcal{L}(\mathcal{W}', \mathcal{V}')$.

Proof. For $\varphi, \psi \in \mathcal{W}'$, $a \in \mathbb{F}$, and $\mathbf{v} \in \mathcal{V}$, $(T'(a\varphi + \psi))(\mathbf{v}) = (a\varphi+\psi)(T\mathbf{v}) = a\varphi(T\mathbf{v})+\psi(T\mathbf{v}) = (aT'\varphi + T'\psi)(\mathbf{v})$.
Structural Commentary: The dual map pulls functionals back along T and preserves linear structure.
Key Insight: Dual maps reverse direction because a functional can test $T\mathbf{v}$ only after T is applied. □

Theorem 5.1.13. *Algebra of Dual Maps*
The dual operation has the following properties:

(i) If $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $a \in \mathbb{F}$, then $(aS + T)' = aS' + T'$.
(ii) If $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $(ST)' = T'S'$.
(iii) $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.

Proof.

(i) For $\varphi \in \mathcal{W}'$ and $\mathbf{v} \in \mathcal{V}$, $((aS + T)'\varphi)(\mathbf{v}) = \varphi(aS\mathbf{v}+T\mathbf{v}) = a(S'\varphi)(\mathbf{v})+(T'\varphi)(\mathbf{v})$, so $(aS+T)' = aS'+T'$.
(ii) For $\varphi \in \mathcal{W}'$ and $\mathbf{u} \in \mathcal{U}$, $((ST)'\varphi)(\mathbf{u}) = \varphi(S(T\mathbf{u})) = (T'(S'\varphi))(\mathbf{u})$, so $(ST)' = T'S'$.
(iii) For $\varphi \in \mathcal{V}'$ and $\mathbf{v} \in \mathcal{V}$, $(I'_{\mathcal{V}}\varphi)(\mathbf{v}) = \varphi(I_{\mathcal{V}}\mathbf{v}) = \varphi(\mathbf{v})$, so $(I_{\mathcal{V}})' = I_{\mathcal{V}'}$.

Structural Commentary: Duality is linear on maps and contravariant under composition.
Key Insight: Pullbacks compose in the opposite order. □

Corollary 5.1.14. *Double Dual Maps Identify with Original Maps*
Let $\dim\mathcal{V}, \dim\mathcal{W} < \infty$ and $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Under $\Lambda_{\mathcal{V}} : \mathcal{V} \cong \mathcal{V}''$ and $\Lambda_{\mathcal{W}} : \mathcal{W} \cong \mathcal{W}''$, $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}}T$, so T'' identifies with T.

Proof. For $\mathbf{v} \in \mathcal{V}$ and $\psi \in \mathcal{W}'$, $\big(T''\Lambda_{\mathcal{V}}(\mathbf{v})\big)(\psi) = \Lambda_{\mathcal{V}}(\mathbf{v})(T'\psi) = (T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = \Lambda_{\mathcal{W}}(T\mathbf{v})(\psi)$.
Hence $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}}T$.
Structural Commentary: Double dualisation commutes with natural evaluation.
Key Insight: The double dual square commutes because both sides evaluate every functional the same way. □

Corollary 5.1.15. *Dual Map of an Inverse*
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is an isomorphism, then T' is an isomorphism and $(T^{-1})' = (T')^{-1}$

Proof. Since $T^{-1} \in \mathcal{L}(\mathcal{W}, \mathcal{V})$, Theorem 5.1.13 gives $T'(T^{-1})' = (T^{-1}T)' = (I_{\mathcal{V}})' = I_{\mathcal{V}'}$ and $(T^{-1})'T' = (TT^{-1})' = (I_{\mathcal{W}})' = I_{\mathcal{W}'}$. Thus $(T^{-1})'$ is inverse of T', T' is isomorphism and $(T^{-1})' = (T')^{-1}$.
Structural Commentary: Duality preserves inverses contravariantly.
Key Insight: Invertibility survives duality. □

5.2 Annihilators and Transposes

Definition 5.2.1. *Annihilator*
If $\mathcal{U} \subseteq \mathcal{V}$, the *annihilator* of $\mathcal{U}$ is $\mathcal{U}^0 := \{\varphi \in \mathcal{V}' : \varphi(\mathbf{u}) = 0 \text{ for every } \mathbf{u} \in \mathcal{U}\}$.

Proposition 5.2.2. *Annihilators Are Subspaces*
If $\mathcal{U} \subseteq \mathcal{V}$, then $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.

Proof. The zero functional lies in $\mathcal{U}^0$. If $\varphi, \psi \in \mathcal{U}^0$, $a \in \mathbb{F}$, and $\mathbf{u} \in \mathcal{U}$, then $(a\varphi + \psi)(\mathbf{u}) = a\varphi(\mathbf{u}) + \psi(\mathbf{u}) = 0$. Thus $a\varphi + \psi \in \mathcal{U}^0$, and Theorem 1.2.2 shows that $\mathcal{U}^0$ is a subspace of $\mathcal{V}'$.
Structural Commentary: The annihilator collects all linear tests that vanish on a chosen set. Closure follows because vanishing is preserved by linear combinations of tests.
Key Insight: Annihilators are kernels of evaluation constraints. □

Theorem 5.2.3. *Dimension of the Annihilator*
Let $\mathcal{V}$ be finite-dimensional and let $\mathcal{U}$ be a subspace of $\mathcal{V}$. Then $\dim(\mathcal{U}) + \dim(\mathcal{U}^0) = \dim(\mathcal{V})$.

Proof. Choose a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{w}_1, \dots, \mathbf{w}_k)$ of $\mathcal{V}$.
Let $(\varphi_1, \dots, \varphi_m, \psi_1, \dots, \psi_k)$ be the corresponding dual basis.
By Theorem 5.1.4, every $\varphi \in \mathcal{V}'$ has the expansion $\varphi = \sum_{j=1}^{m} \varphi(\mathbf{u}_j)\varphi_j + \sum_{\ell=1}^{k} \varphi(\mathbf{w}_\ell)\psi_\ell$.
If $\varphi \in \mathcal{U}^0$, then $\varphi(\mathbf{u}_j) = 0$ for every j, so $\varphi \in \operatorname{span}(\psi_1, \dots, \psi_k)$.
Conversely, each ψ_ℓ vanishes on every $\mathbf{u}_j$, hence on $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_m)$. Therefore $\mathcal{U}^0 = \operatorname{span}(\psi_1, \dots, \psi_k)$.
The list $(\psi_1, \dots, \psi_k)$ is linearly independent as a sublist of the dual basis, so $\dim(\mathcal{U}^0) = k = \dim(\mathcal{V}) - \dim(\mathcal{U})$.
Structural Commentary: Extending a basis of $\mathcal{U}$ separates directions inside $\mathcal{U}$ from complementary directions. The annihilator is dual to the complementary part.
Key Insight: To vanish on $\mathcal{U}$, a functional can only use dual coordinates attached to directions outside $\mathcal{U}$. □

Theorem 5.2.4. *Annihilator Lattice Identities*
Let $\mathcal{V}$ be finite-dimensional, and let $\mathcal{S}, \mathcal{T}$ be subspaces of $\mathcal{V}$.

(i) If $\mathcal{S} \subseteq \mathcal{T}$, then $\mathcal{T}^0 \subseteq \mathcal{S}^0$.
(ii) $(\mathcal{S} + \mathcal{T})^0 = \mathcal{S}^0 \cap \mathcal{T}^0$.
(iii) $(\mathcal{S} \cap \mathcal{T})^0 = \mathcal{S}^0 + \mathcal{T}^0$.

Proof.

(i) If $\varphi \in \mathcal{T}^0$ and $\mathbf{s} \in \mathcal{S}$, then $\mathbf{s} \in \mathcal{T}$, so $\varphi(\mathbf{s}) = 0$. Hence $\varphi \in \mathcal{S}^0$.
(ii) For $\varphi \in \mathcal{V}'$, $\varphi \in (\mathcal{S} + \mathcal{T})^0$ if and only if $\varphi(\mathbf{s} + \mathbf{t}) = 0$ for all $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$, if and only if $\varphi \in \mathcal{S}^0 \cap \mathcal{T}^0$.
(iii) Since $\mathcal{S} \cap \mathcal{T} \subseteq \mathcal{S}, \mathcal{T}$, part (i) gives $\mathcal{S}^0 + \mathcal{T}^0 \subseteq (\mathcal{S} \cap \mathcal{T})^0$. By part (ii), Theorem 2.3.1, and Theorem 5.2.3, $\dim(\mathcal{S}^0 + \mathcal{T}^0) = \dim(\mathcal{S}^0) + \dim(\mathcal{T}^0) - \dim(\mathcal{S}^0 \cap \mathcal{T}^0) = \dim(\mathcal{V}) - \dim(\mathcal{S} \cap \mathcal{T}) = \dim\big((\mathcal{S} \cap \mathcal{T})^0\big)$. The inclusion is therefore equality.

Structural Commentary: Annihilation is an order-reversing lattice duality.
Key Insight: Vanishing on more vectors gives fewer functionals. □

Theorem 5.2.5. *Double Annihilator*
Let $\dim \mathcal{V} < \infty$ and $\emptyset \neq M \subseteq \mathcal{V}$. Under $\Lambda_{\mathcal{V}} : \mathcal{V} \cong \mathcal{V}''$, $M^{00} = \Lambda_{\mathcal{V}}(\operatorname{span} M)$.
In particular, if $\mathcal{S} \leq \mathcal{V}$, then $\mathcal{S}^{00} = \Lambda_{\mathcal{V}}(\mathcal{S}) \cong \mathcal{S}$.

Proof. If $\mathbf{v} \in \operatorname{span} M$ and $\varphi \in M^0$, then $\varphi(\mathbf{v}) = 0$, so $\Lambda_{\mathcal{V}}(\mathbf{v})(\varphi) = 0$. Hence $\Lambda_{\mathcal{V}}(\operatorname{span} M) \subseteq M^{00}$.
Conversely, let $\eta \in M^{00}$. By Theorem 5.1.10, $\eta = \Lambda_{\mathcal{V}}(\mathbf{v})$ for a unique $\mathbf{v} \in \mathcal{V}$.
If $\mathbf{v} \notin \operatorname{span} M$, Proposition 5.1.7(iii) gives $\varphi \in \mathcal{V}'$ with $\varphi|_{\operatorname{span} M} = 0$ and $\varphi(\mathbf{v}) = 1$.
Then $\varphi \in M^0$ but $\eta(\varphi) = \Lambda_{\mathcal{V}}(\mathbf{v})(\varphi) = 1$, contradicting $\eta \in M^{00}$. Thus $\mathbf{v} \in \operatorname{span} M$, so $M^{00} \subseteq \Lambda_{\mathcal{V}}(\operatorname{span} M)$.
Structural Commentary: Double annihilation returns the span under $\Lambda_{\mathcal{V}}$.
Key Insight: Linear functionals separate any vector outside a finite-dimensional span. □

Theorem 5.2.6. *Kernel and Range of the Dual Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then

(i) $\ker(T') = (\operatorname{range}(T))^0$;

(ii) $\dim(\ker(T')) = \dim(\ker(T)) + \dim(\mathcal{W}) - \dim(\mathcal{V})$;
(iii) $\operatorname{range}(T') = (\ker(T))^0$;
(iv) $\dim(\operatorname{range}(T')) = \dim(\operatorname{range}(T))$.

Proof.

(i) For $\varphi \in \mathcal{W}'$, $T'\varphi = 0$ if and only if $\varphi(T\mathbf{v}) = 0$ for every $\mathbf{v} \in \mathcal{V}$, if and only if φ vanishes on $\operatorname{range}(T)$.
(ii) By (i), Theorem 5.2.3, and rank-nullity,
$\dim(\ker(T')) = \dim(\mathcal{W}) - \dim(\operatorname{range}(T)) = \dim(\ker(T)) + \dim(\mathcal{W}) - \dim(\mathcal{V})$.
(iii) If $\varphi = T'\psi$ and $\mathbf{v} \in \ker(T)$, then $\varphi(\mathbf{v}) = \psi(T\mathbf{v}) = 0$, so $\operatorname{range}(T') \subseteq (\ker(T))^0$. Also, by rank-nullity for T', (ii), Theorem 5.1.4, and Theorem 5.2.3, $\dim(\operatorname{range}(T')) = \dim(\mathcal{V}) - \dim(\ker(T)) = \dim\left((\ker(T))^0\right)$. Thus $\operatorname{range}(T') = (\ker(T))^0$.
(iv) By (iii), Theorem 5.2.3, and rank-nullity, $\dim(\operatorname{range}(T')) = \dim(\mathcal{V}) - \dim(\ker(T)) = \dim(\operatorname{range}(T))$

Structural Commentary: Duality exchanges kernels and ranges through annihilators.
Key Insight: A functional dies after pulling back by T exactly when it vanishes on everything T can reach. □

Corollary 5.2.7. *Surjectivity and Dual Injectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is surjective if and only if T' is injective.

Proof.

($\Rightarrow$) Suppose T is surjective. Then $\operatorname{range}(T) = \mathcal{W}$, so Theorem 5.2.6 gives $\ker(T') = \mathcal{W}^0 = \{0\}$.
($\Leftarrow$) Suppose T' is injective. Then $\ker(T') = \{0\}$, so Theorem 5.2.6 gives $(\operatorname{range}(T))^0 = \{0\}$. By Theorem 5.2.3, $\dim(\operatorname{range}(T)) = \dim(\mathcal{W})$, hence $\operatorname{range}(T) = \mathcal{W}$.

Structural Commentary: A map is onto exactly when no nonzero functional can vanish on its range. Dual injectivity detects this absence of invisible codomain directions.
Key Insight: Surjectivity of T becomes injectivity of the pullback map T'. □

Corollary 5.2.8. *Injectivity and Dual Surjectivity*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T is injective if and only if T' is surjective.

Proof.

($\Rightarrow$) Suppose T is injective. Then $\ker(T) = \{\mathbf{0}\}$, so Theorem 5.2.6 gives $\operatorname{range}(T') = (\ker(T))^0 = \mathcal{V}'$.
($\Leftarrow$) Suppose T' is surjective. Then $\operatorname{range}(T') = \mathcal{V}'$, so Theorem 5.2.6 gives $(\ker(T))^0 = \mathcal{V}'$. By Theorem 5.2.3, this forces $\dim(\ker(T)) = 0$, hence $\ker(T) = \{\mathbf{0}\}$.

Structural Commentary: Injectivity says there are no nonzero domain directions killed by T. The dual statement says every test on the domain comes by pulling back a test from the codomain.
Key Insight: No kernel directions means no obstruction to extending functionals backward through T. □

Corollary 5.2.9. *Dual of a Quotient as an Annihilator*
Let $\mathcal{V}$ be finite-dimensional, let $\mathcal{U} \leq \mathcal{V}$, and let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map. Then $\ker(\pi') = \{0\}$ and $\operatorname{range}(\pi') = \mathcal{U}^0$. Hence π' identifies $(\mathcal{V}/\mathcal{U})'$ with $\mathcal{U}^0$, so $(\mathcal{V}/\mathcal{U})' \cong \mathcal{U}^0$.

Proof. By Theorem 2.2.7, $\mathcal{V}/\mathcal{U}$ is finite-dimensional. Since π is surjective and $\ker(\pi) = \mathcal{U}$, Corollary 5.2.7 gives $\ker(\pi') = \{0\}$, and Theorem 5.2.6 gives $\operatorname{range}(\pi') = (\ker(\pi))^0 = \mathcal{U}^0$.
Structural Commentary: A functional on the quotient is exactly a functional on $\mathcal{V}$ that cannot see $\mathcal{U}$.
Key Insight: Pulling back along the quotient map identifies $(\mathcal{V}/\mathcal{U})'$ with $\mathcal{U}^0$. □

Theorem 5.2.10. *Direct Sums and Annihilators*
Let $\mathcal{V}$ be finite-dimensional, and suppose $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$.

(i) Extension by zero on $\mathcal{S}$ gives an isomorphism $\mathcal{T}' \cong \mathcal{S}^0$.
(ii) $\mathcal{V}' = \mathcal{S}^0 \oplus \mathcal{T}^0$.

Proof.

(i) Define $E : \mathcal{T}' \to \mathcal{S}^0$ by $(E\varphi)(\mathbf{s} + \mathbf{t}) = \varphi(\mathbf{t})$. Since $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$, this is well-defined and linear. Restriction $R : \mathcal{S}^0 \to \mathcal{T}'$, $R\psi = \psi|_{\mathcal{T}}$, satisfies $RE = I_{\mathcal{T}'}$ and $ER = I_{\mathcal{S}^0}$, so E is an isomorphism.
(ii) By Theorem 5.2.4, $\mathcal{S}^0 + \mathcal{T}^0 = (\mathcal{S} \cap \mathcal{T})^0 = \{\mathbf{0}\}^0 = \mathcal{V}'$, and $\mathcal{S}^0 \cap \mathcal{T}^0 = (\mathcal{S} + \mathcal{T})^0 = \mathcal{V}^0 = \{0\}$. Thus $\mathcal{V}' = \mathcal{S}^0 \oplus \mathcal{T}^0$.

Structural Commentary: A direct sum decomposes the dual by complementary annihilators.
Key Insight: Direct sums split the dual space by annihilators. □

Theorem 5.2.11. *Range Membership Dual Alternative*
Let $\mathcal{V}, \mathcal{W}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and $\mathbf{w} \in \mathcal{W}$. Exactly one holds:

(i) $\mathbf{w} \in \operatorname{range}(T)$;
(ii) there exists $\psi \in \mathcal{W}'$ such that $\psi T = 0$ and $\psi(\mathbf{w}) \neq 0$.

Proof.

(i) The alternatives are incompatible: if $\mathbf{w} = T\mathbf{v}$ and $\psi T = 0$, then $\psi(\mathbf{w}) = \psi(T\mathbf{v}) = 0$.
(ii) Suppose $\mathbf{w} \notin \operatorname{range}(T)$. By Proposition 5.1.7(iii), there exists $\psi \in \mathcal{W}'$ such that $\psi|_{\operatorname{range}(T)} = 0$ and $\psi(\mathbf{w}) = 1$.
Then ψ vanishes on $\operatorname{range}(T)$, so $\psi T = 0$, while $\psi(\mathbf{w}) = 1$.

Structural Commentary: A missing range vector is detected by a dual test that vanishes on the whole range.
Key Insight: Either a vector is reached by the map, or a functional separates it from the range. □

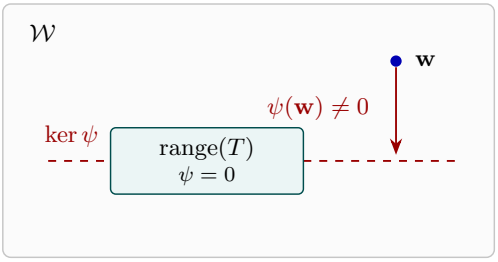


Figure 16: A dual certificate detects a vector outside the range.

Proposition 5.2.12. *Standard Coordinate Dual Identification*
For each positive integer m, the map

$$\Theta : \mathbb{F}^m \to (\mathbb{F}^m)', \qquad \Theta(\mathbf{y})(\mathbf{z}) = \mathbf{y}^t\mathbf{z},$$

is an isomorphism. Hence every $\psi \in (\mathbb{F}^m)'$ has a unique form $\psi(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$ with $\mathbf{y} \in \mathbb{F}^m$.

Proof. Linearity of Θ follows from bilinearity of the coordinate dot product. If $\Theta(\mathbf{y}) = 0$, then evaluating at the standard basis vector $\mathbf{e}_j$ gives $y_j = 0$ for every j, so $\mathbf{y} = \mathbf{0}$. Thus Θ is injective. Since $\dim(\mathbb{F}^m) = m = \dim((\mathbb{F}^m)')$ by Theorem 5.1.4, Corollary 3.2.19 gives that Θ is an isomorphism.
Structural Commentary: Coordinate column vectors represent all coordinate linear functionals by row multiplication.
Key Insight: The standard dual of $\mathbb{F}^m$ is represented by transposed coordinate vectors. □

Corollary 5.2.13. *Matrix Equation Alternative*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ and $\mathbf{b} \in \mathbb{F}^m$. Exactly one holds:

(i) $A\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{F}^n$;
(ii) $\mathbf{y}^t A = \mathbf{0}^t$ and $\mathbf{y}^t\mathbf{b} \neq 0$ for some $\mathbf{y} \in \mathbb{F}^m$.

Proof.

(i) By Theorem 5.2.11, for $T_A : \mathbb{F}^n \to \mathbb{F}^m$ and $\mathbf{w} = \mathbf{b}$, $\mathbf{b} \in \operatorname{range}(T_A)$ is exactly $A\mathbf{x} = \mathbf{b}$ for some $\mathbf{x} \in \mathbb{F}^n$.
(ii) By Proposition 5.2.12, $\psi \in (\mathbb{F}^m)'$ has the form $\psi(\mathbf{z}) = \mathbf{y}^t\mathbf{z}$ for a unique $\mathbf{y} \in \mathbb{F}^m$.
Thus $\psi T_A = 0$ iff $\mathbf{y}^t A = \mathbf{0}^t$, and $\psi(\mathbf{b}) \neq 0$ iff $\mathbf{y}^t\mathbf{b} \neq 0$.

Structural Commentary: Inconsistency is certified by a left-null vector.
Key Insight: Left null vectors are the coordinate form of dual separating functionals. □

Theorem 5.2.14. *Matrix of the Dual Map Is the Transpose*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ and $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be bases of $\mathcal{V}$ and $\mathcal{W}$. If β' and γ' are the corresponding dual bases, then $[T']_{\beta' \leftarrow \gamma'} = ([T]_{\gamma \leftarrow \beta})^t$.

Proof. Let $A = [T]_{\gamma \leftarrow \beta}$ and $C = [T']_{\beta' \leftarrow \gamma'}$. Write $\gamma' = (\psi_1, \ldots, \psi_m)$ and $\beta' = (\varphi_1, \ldots, \varphi_n)$.
For $1 \leq j \leq m$, $T'\psi_j = \sum_{r=1}^{n} C_{r,j}\varphi_r$. Evaluating at $\mathbf{v}_k$ gives $C_{k,j} = (T'\psi_j)(\mathbf{v}_k) = \psi_j(T\mathbf{v}_k)$.
Since $T\mathbf{v}_k = \sum_{r=1}^{m} A_{r,k}\mathbf{w}_r$, we get $\psi_j(T\mathbf{v}_k) = A_{j,k}$. Hence $C_{k,j} = A_{j,k}$ for all j, k, so $C = A^t$.
Structural Commentary: Duality turns the matrix of a map into the matrix of its pullback on coordinate functionals. The row-column indexing is exactly transposition.
Key Insight: Coefficients of $T\mathbf{v}_k$ become values of the dual basis functionals on $T\mathbf{v}_k$. □

5.3 Multilinear Structure

Definition 5.3.1. *Bilinear Forms and Matrices*
A map $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ is *bilinear* if it is linear in each variable separately.
For bases $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ of $\mathcal{V}$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ of $\mathcal{W}$, the *matrix of B in the bases β, γ* is $[B]_{\beta,\gamma} = [B(\mathbf{v}_i, \mathbf{w}_j)]$. If $\mathbf{x} = [\mathbf{v}]_\beta$ and $\mathbf{y} = [\mathbf{w}]_\gamma$, then $B(\mathbf{v}, \mathbf{w}) = \mathbf{x}^t [B]_{\beta,\gamma} \mathbf{y}$.

Proposition 5.3.2. *Bilinear Forms Linearize through Duals*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear, and define $\Phi_B : \mathcal{V} \to \mathcal{W}'$ where $(\Phi_B \mathbf{v})(\mathbf{w}) = B(\mathbf{v}, \mathbf{w})$, and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ where $(\Psi_B \mathbf{w})(\mathbf{v}) = B(\mathbf{v}, \mathbf{w})$. Then Φ_B, Ψ_B are linear. For bases β, γ and $M = [B]_{\beta,\gamma}$, $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$; and $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.

Proof. For fixed $\mathbf{v}$ and $\mathbf{w}$, bilinearity gives $\Phi_B \mathbf{v} \in \mathcal{W}'$ and $\Psi_B \mathbf{w} \in \mathcal{V}'$. For $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$, $a \in \mathbb{F}$, and every $\mathbf{w} \in \mathcal{W}$, $(\Phi_B(a\mathbf{v}_1 + \mathbf{v}_2))(\mathbf{w}) = B(a\mathbf{v}_1 + \mathbf{v}_2, \mathbf{w}) = a(\Phi_B \mathbf{v}_1)(\mathbf{w}) + (\Phi_B \mathbf{v}_2)(\mathbf{w})$, so Φ_B is linear.
For $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$, $a \in \mathbb{F}$, and every $\mathbf{v} \in \mathcal{V}$, $(\Psi_B(a\mathbf{w}_1 + \mathbf{w}_2))(\mathbf{v}) = B(\mathbf{v}, a\mathbf{w}_1 + \mathbf{w}_2) = a(\Psi_B \mathbf{w}_1)(\mathbf{v}) + (\Psi_B \mathbf{w}_2)(\mathbf{v})$, so Ψ_B is linear. Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$.
The jth γ'-coordinate of $\Phi_B(\mathbf{v}_i)$ is $(\Phi_B \mathbf{v}_i)(\mathbf{w}_j) = B(\mathbf{v}_i, \mathbf{w}_j) = M_{i,j}$, so column i of $[\Phi_B]_{\gamma' \leftarrow \beta}$ is row i of M transposed, hence $[\Phi_B]_{\gamma' \leftarrow \beta} = M^t$. Likewise, the ith β'-coordinate of $\Psi_B(\mathbf{w}_j)$ is $M_{i,j}$, so $[\Psi_B]_{\beta' \leftarrow \gamma} = M$.
Structural Commentary: A bilinear form becomes linear once one input is frozen. Tracking both freezes gives two dual-valued linear maps whose matrices are transposes of one another.
Key Insight: Bilinearity is linearity into a dual space. □

Definition 5.3.3. *Radicals and Nonsingularity of a Bilinear Pairing*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear.

(i) The *left* and *right radicals* are
$$\operatorname{rad}_L(B) := \{\mathbf{v} \in \mathcal{V} : B(\mathbf{v}, \mathbf{w}) = 0 \text{ for every } \mathbf{w} \in \mathcal{W}\},$$
$$\operatorname{rad}_R(B) := \{\mathbf{w} \in \mathcal{W} : B(\mathbf{v}, \mathbf{w}) = 0 \text{ for every } \mathbf{v} \in \mathcal{V}\}.$$
(ii) B is *left nondegenerate* if $\operatorname{rad}_L(B) = \{\mathbf{0}\}$ and *right nondegenerate* if $\operatorname{rad}_R(B) = \{\mathbf{0}\}$.
(iii) B is *nonsingular* or *perfect* if $\Phi_B : \mathcal{V} \to \mathcal{W}'$ and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ from Proposition 5.3.2 are isomorphisms.

Proposition 5.3.4. *Radicals, Matrices, and Dual Maps*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear, $\dim \mathcal{V} = m$, $\dim \mathcal{W} = n$, and $M = [B]_{\beta,\gamma}$ in ordered bases β, γ. Then $\{[\mathbf{v}]_\beta : \mathbf{v} \in \operatorname{rad}_L(B)\} = \ker(M^t)$ and $\{[\mathbf{w}]_\gamma : \mathbf{w} \in \operatorname{rad}_R(B)\} = \ker(M)$, therefore $\dim \operatorname{rad}_L(B) = m - \operatorname{rank}(M)$ and $\dim \operatorname{rad}_R(B) = n - \operatorname{rank}(M)$. Moreover, $\ker(\Phi_B) = \operatorname{rad}_L(B)$ and $\ker(\Psi_B) = \operatorname{rad}_R(B)$.

Proof. For $\mathbf{x} = [\mathbf{v}]_\beta$ and $\mathbf{y} = [\mathbf{w}]_\gamma$, $B(\mathbf{v}, \mathbf{w}) = \mathbf{x}^t M \mathbf{y}$.
Thus $\mathbf{v} \in \operatorname{rad}_L(B)$ if and only if $\mathbf{x}^t M \mathbf{y} = 0$ for all $\mathbf{y}$, if and only if $M^t \mathbf{x} = 0$. Similarly, $\mathbf{w} \in \operatorname{rad}_R(B)$ if and only if $M\mathbf{y} = 0$. This proves the coordinate-set identities. Rank-nullity gives the dimension formulas. Finally, $\mathbf{v} \in \ker(\Phi_B) \iff B(\mathbf{v}, \mathbf{w}) = 0$ for every $\mathbf{w} \in \mathcal{W} \iff \mathbf{v} \in \operatorname{rad}_L(B)$; similarly $\ker(\Psi_B) = \operatorname{rad}_R(B)$.
Structural Commentary: Radicals are the kernel obstructions in each slot.
Key Insight: In coordinates, radicals are matrix nullspaces. □

Proposition 5.3.5. *Square Pairing Nonsingularity Criteria*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be bilinear with $\dim \mathcal{V} = \dim \mathcal{W} = n$, and let $M = [B]_{\beta,\gamma}$ in ordered bases β, γ. Then the following are equivalent:

(i) B is left nondegenerate;
(ii) B is right nondegenerate;
(iii) M is invertible;
(iv) B is perfect.

Proof.

(i) $\Rightarrow$ (ii). If $\operatorname{rad}_L(B) = 0$, then by Proposition 5.3.4, $\ker(M^t) = 0 \Rightarrow \operatorname{rank}(M) = n \Rightarrow \ker(M) = 0$.
(ii) $\Rightarrow$ (iii). If $\operatorname{rad}_R(B) = 0$, then $\ker(M) = 0$ by Proposition 5.3.4; since $M \in \operatorname{Mat}(n, n, \mathbb{F})$, M is invertible.
(iii) $\Rightarrow$ (iv). If M is invertible, then $\ker(M) = \ker(M^t) = 0$, so $\operatorname{rad}_L(B) = \operatorname{rad}_R(B) = 0$.
Hence $\ker(\Phi_B) = \ker(\Psi_B) = 0$ by Proposition 5.3.4.
Since $\dim \mathcal{V} = \dim \mathcal{W}' = \dim \mathcal{W} = \dim \mathcal{V}' = n$, Φ_B and Ψ_B are isomorphisms; hence B is perfect.
(iv) $\Rightarrow$ (i). If B is perfect, then Φ_B is an isomorphism. Thus $\operatorname{rad}_L(B) = \ker(\Phi_B) = 0$.

Structural Commentary: In square pairings, one-sided nondegeneracy is invertibility.
Key Insight: Perfect pairings are exactly invertible matrices viewed without coordinates. □

Corollary 5.3.6. *Perfect Pairings Represent Functionals*
Let $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ be perfect.
Then $\forall \psi \in \mathcal{W}'\ \exists! \mathbf{v} \in \mathcal{V} : \psi(\mathbf{w}) = B(\mathbf{v}, \mathbf{w})$, $\mathbf{w} \in \mathcal{W}$; and $\forall \varphi \in \mathcal{V}'\ \exists! \mathbf{w} \in \mathcal{W} : \varphi(\mathbf{v}) = B(\mathbf{v}, \mathbf{w})$, $\mathbf{v} \in \mathcal{V}$.

Proof. Since B is perfect, $\Phi_B : \mathcal{V} \to \mathcal{W}'$ and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ are isomorphisms.
Thus $\psi = \Phi_B(\mathbf{v})$ and $\varphi = \Psi_B(\mathbf{w})$ for unique $\mathbf{v}, \mathbf{w}$, which gives the two formulas.
Structural Commentary: Perfect pairings identify each space with the other's dual.
Key Insight: Functionals are represented by vectors exactly when the appropriate pairing map is onto. □

Definition 5.3.7. *Matrix Congruence*
Matrices $M, N \in \mathrm{Mat}(n, n, \mathbb{F})$ are *congruent* if $\exists\, P \in \mathrm{Mat}(n, n, \mathbb{F})$ invertible such that $N = P^t M P$.

Proposition 5.3.8. *Basis Change for a Square Bilinear Form*
Let $\mathcal{V}$ be finite-dimensional, let $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ be bilinear, and let β, γ be ordered bases of $\mathcal{V}$.
Set $M_\beta := [B]_{\beta,\beta}$; $M_\gamma := [B]_{\gamma,\gamma}$, and $C := C_{\beta \leftarrow \gamma}$.
Then $M_\gamma = C^t M_\beta C$. Hence M_β and M_γ are congruent.

Proof. For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, $[\mathbf{x}]_\beta = C[\mathbf{x}]_\gamma$ and $[\mathbf{y}]_\beta = C[\mathbf{y}]_\gamma$. Therefore $B(\mathbf{x}, \mathbf{y}) = [\mathbf{x}]_\beta^t M_\beta [\mathbf{y}]_\beta = [\mathbf{x}]_\gamma^t C^t M_\beta C [\mathbf{y}]_\gamma$.
By uniqueness of the representing matrix in the basis γ, $M_\gamma = C^t M_\beta C$.
Since C is invertible by Theorem 3.5.2, Definition 5.3.7 gives congruence.
Structural Commentary: Operators change by similarity; square bilinear forms change by congruence because both input slots are rewritten.
Key Insight: A basis change acts on a square bilinear form from both sides. □

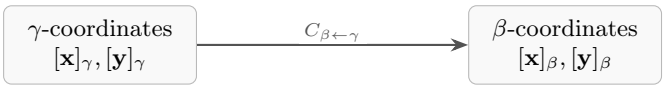


Figure 17: Changing both slots gives $M_\gamma = C_{\beta \leftarrow \gamma}^t M_\beta C_{\beta \leftarrow \gamma}$.

Perspective 5.3.9. *Bilinear Forms, Inner Products, and Tensor Scope*

(i) Bilinear forms: $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$; inner products add $\mathbf{v} \neq \mathbf{0} \Rightarrow B(\mathbf{v}, \mathbf{v}) > 0$ and, over $\mathbb{C}$, $B(\mathbf{v}, \mathbf{w}) = \overline{B(\mathbf{w}, \mathbf{v})}$.
(ii) $\Phi_B : \mathcal{V} \to \mathcal{W}'$ and $\Psi_B : \mathcal{W} \to \mathcal{V}'$ encode B in dual spaces, with $[\Phi_B]_{\gamma' \leftarrow \beta} = [B]_{\beta,\gamma}^t$ and $[\Psi_B]_{\beta' \leftarrow \gamma} = [B]_{\beta,\gamma}$.
(iii) $\mathcal{V} \oplus \mathcal{W}$ and $\mathcal{V} \times \mathcal{W}$ use single-coordinate slots; the finite tensor scaffold uses paired slots (i, j).

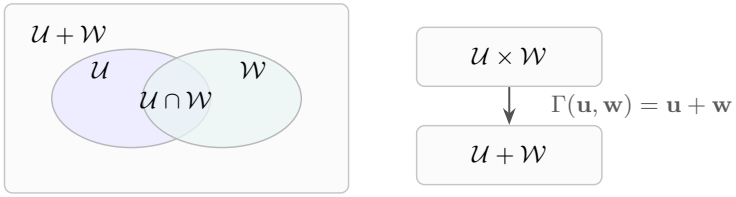


Figure 18: Intersection, sum, and product of subspaces.

6 Finite-Dimensional Topological Vector Spaces

6.1 Natural Topology and Continuity

Finite-dimensional real vector spaces carry a canonical topology: choose coordinates, use the standard Euclidean topology on $\mathbb{R}^n$, and then forget that coordinates were chosen. All coordinate changes are homeomorphisms.

Definition 6.1.1. *Topological Vector Space*
Let $\mathcal{V}$ be a real vector space and let $\mathcal{T}$ be a topology on its underlying set.
The pair $(\mathcal{V}, \mathcal{T})$ is a *topological vector space* if addition $A : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$, $A(\mathbf{u}, \mathbf{v}) = \mathbf{u} + \mathbf{v}$, and scalar multiplication $M : \mathbb{R} \times \mathcal{V} \to \mathcal{V}$, $M(a, \mathbf{v}) = a\mathbf{v}$ are continuous, where products carry the product topology.

Example 6.1.2. *The Standard Topology on $\mathbb{R}^n$*
With the Euclidean topology, $\mathbb{R}^n$ is a topological vector space: the maps $(\mathbf{x}, \mathbf{y}) \longmapsto \mathbf{x} + \mathbf{y}$, $(a, \mathbf{x}) \longmapsto a\mathbf{x}$ have polynomial coordinate functions $x_i + y_i$ and $a x_i$, hence are continuous.

Proposition 6.1.3. *Real Matrix Maps Are Continuous*
Let $A = [a_{j,k}] \in \mathrm{Mat}(m, n, \mathbb{R})$, and let $T_A : \mathbb{R}^n \to \mathbb{R}^m$ be the induced map $T_A(\mathbf{x}) = A\mathbf{x}$. Then T_A is continuous for the Euclidean topologies. If $m = n$ and A is invertible, then T_A is a homeomorphism. In particular, every linear functional on $\mathbb{R}^n$ is continuous.

Proof. For $\mathbf{x} = (x_1, \dots, x_n)^t$, the jth coordinate of $T_A(\mathbf{x})$ is $\sum_{k=1}^{n} a_{j,k}x_k$, hence is continuous. Thus T_A is continuous. If A is invertible, then $(T_A)^{-1} = T_{A^{-1}}$, which is continuous by the first part. Finally, every $f \in (\mathbb{R}^n)'$ equals T_A for a unique $A \in \mathrm{Mat}(1, n, \mathbb{R})$, so f is continuous. *Structural Commentary*: Matrix maps on Euclidean coordinate spaces are continuous coordinate by coordinate.
Key Insight: Invertible real matrices give homeomorphisms of Euclidean space. □

Definition 6.1.4. *Natural Topology*
Let $\mathcal{V}$ be an n-dimensional real vector space, let β be a basis, and let $[\cdot]_\beta : \mathcal{V} \to \mathbb{R}^n$ be the coordinate isomorphism.
The *natural topology* on $\mathcal{V}$ is $\mathcal{T}_\beta := \{[\cdot]_\beta^{-1}(O) : O \subseteq \mathbb{R}^n \text{ is open}\}$.
Equivalently, $U \subseteq \mathcal{V}$ is open if and only if $[\cdot]_\beta(U)$ is open in $\mathbb{R}^n$.

Proposition 6.1.5. *Natural Topology Is Basis-Independent*
If β and γ are bases of $\mathcal{V}$, then $\mathcal{T}_\beta = \mathcal{T}_\gamma$.

Proof. Set $H_{\gamma\leftarrow\beta} := [\cdot]_\gamma \circ [\cdot]_\beta^{-1} : \mathbb{R}^n \to \mathbb{R}^n$.
Then $H_{\gamma\leftarrow\beta}\mathbf{x} = C_{\gamma\leftarrow\beta}\mathbf{x}$. Since $C_{\gamma\leftarrow\beta}$ is invertible with inverse $C_{\beta\leftarrow\gamma}$, Proposition 6.1.3 shows that $H_{\gamma\leftarrow\beta}$ is a homeomorphism with inverse $H_{\beta\leftarrow\gamma}$.
Hence, for $U \subseteq \mathcal{V}$, $U \in \mathcal{T}_\beta \iff [\cdot]_\beta(U)$ is open $\iff H_{\gamma\leftarrow\beta}([\cdot]_\beta(U)) = [\cdot]_\gamma(U)$ is open $\iff U \in \mathcal{T}_\gamma$. *Structural Commentary*: Basis changes are homeomorphisms.
Key Insight: The natural topology is carried by the vector-space structure, not by a preferred basis. □

Theorem 6.1.6. *Characterization of the Natural Topology*
Let $\dim_{\mathbb{R}} \mathcal{V} < \infty$. There is a unique topology $\mathcal{T}$ on $\mathcal{V}$ such that:

(i) $(\mathcal{V}, \mathcal{T})$ is a topological vector space;
(ii) $\mathcal{V}' \subseteq C((\mathcal{V}, \mathcal{T}), \mathbb{R})$.

Moreover, $\mathcal{T} = \mathcal{T}_\beta$ for every basis β.

Proof. Fix an arbitrary basis $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$.

(i) *Existence*: put $\mathcal{T} = \mathcal{T}_\beta$. Since $[\cdot]_\beta : (\mathcal{V}, \mathcal{T}_\beta) \to \mathbb{R}^n$ is a homeomorphism, the vector operations on $\mathcal{V}$ are transported from the continuous vector operations on $\mathbb{R}^n$.
Also, for $\varphi \in \mathcal{V}'$, $\varphi \circ [\cdot]_\beta^{-1} \in (\mathbb{R}^n)'$, hence is continuous by Proposition 6.1.3. Thus (i)–(ii) hold.

(ii) *Uniqueness*: suppose $\mathcal{T}$ satisfies (i)–(ii).
For $1 \le i \le n$, the coordinate functional $\pi_i \circ [\cdot]_\beta \in \mathcal{V}'$ is continuous, so $[\cdot]_\beta : (\mathcal{V}, \mathcal{T}) \to \mathbb{R}^n$ is continuous.
Its inverse $\Psi(a_1, \dots, a_n) = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ is continuous by (i), since it is built from scalar multiplication and finite sums. Hence $[\cdot]_\beta$ is a homeomorphism, so $\mathcal{T} = \mathcal{T}_\beta$.

Structural Commentary: Functionals give coordinates; operations give inverse coordinates.
Key Insight: Functionals and vector operations determine finite-dimensional coordinates topologically. □

Corollary 6.1.7. *Finite-Dimensional Linear Maps Are Continuous*
Let $\dim_{\mathbb{R}} \mathcal{V}, \dim_{\mathbb{R}} \mathcal{W} < \infty$, with natural topologies. Then $\mathcal{L}(\mathcal{V}, \mathcal{W}) \subseteq C(\mathcal{V}, \mathcal{W})$.
If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is an isomorphism, then T is a homeomorphism.

Proof. Choose bases β of $\mathcal{V}$ and γ of $\mathcal{W}$. For $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, $[\cdot]_\gamma \circ T \circ [\cdot]_\beta^{-1}(\mathbf{x}) = [T]_{\gamma\leftarrow\beta}\mathbf{x}$, which is continuous by Proposition 6.1.3. Since $[\cdot]_\beta$ and $[\cdot]_\gamma$ are homeomorphisms, T is continuous.
If T is an isomorphism, apply the same argument to T^{-1}.
Structural Commentary: Finite-dimensional linear maps are just matrix maps in natural coordinates.
Key Insight: A basis turns topological questions about linear maps into continuity of matrix multiplication. □

Perspective 6.1.8. *Finite-Dimensional Topology Is Rigid*
For finite-dimensional real vector spaces, topology is algebraically forced.
Note that $\mathcal{T}_\beta = \mathcal{T}_\gamma$; $\mathcal{L}(\mathcal{V}, \mathcal{W}) \subseteq C(\mathcal{V}, \mathcal{W})$; $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ bijective $\Rightarrow T$ is a homeomorphism.
Infinite-dimensional topological vector spaces need not be rigid in this sense.

Part III
Geometric and Spectral Linear Algebra

1 Inner Product Geometry

1.1 Inner Products and Norms

Throughout this subsection, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Inner products turn linear spaces into geometric spaces with norms, orthogonality, orthonormal bases, and Gram matrices.

Definition 1.1.1. *Preliminary Terminology*

(i) For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the *dot product* is $\mathbf{x} \cdot \mathbf{y} := x_1 y_1 + \cdots + x_n y_n$.
(ii) For $z = a + bi \in \mathbb{C}$, the *complex conjugate* is $\overline{z} := a - bi$, and the *modulus* is $|z| := \sqrt{a^2 + b^2}$.
(iii) A map $f : \mathcal{V} \to \mathbb{F}$ is *conjugate linear* if $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$ and $f(\lambda \mathbf{v}) = \overline{\lambda} f(\mathbf{v})$.
(iv) A function $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is *sesquilinear* if it is linear in one slot and conjugate linear in the other.

Motivation 1.1.2. *From Dot Products to Inner Products*
For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, the Euclidean length is $\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}$.
The expression under the square root is the dot product of $\mathbf{x}$ with itself. Over $\mathbb{C}$, conjugates are needed to keep squared length real and nonnegative: $\|\mathbf{z}\|^2 = |z_1|^2 + \cdots + |z_n|^2 = z_1\overline{z_1} + \cdots + z_n\overline{z_n}$.
Thus the standard inner product on $\mathbb{F}^n$ is $\langle \mathbf{x}, \mathbf{y} \rangle := x_1\overline{y_1} + \cdots + x_n\overline{y_n}$.

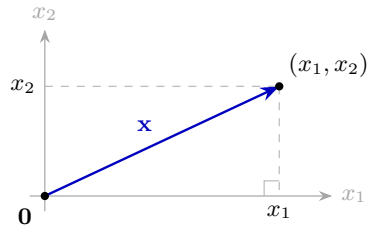


Figure 19: The norm is vector length, with $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$.

Definition 1.1.3. *Inner Product*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. An *inner product* on $\mathcal{V}$ is a function that assigns to each pair $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ a scalar $\langle \mathbf{u}, \mathbf{v} \rangle \in \mathbb{F}$ satisfying, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and all $\lambda \in \mathbb{F}$,

i. *positivity*: $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$;
ii. *definiteness*: $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$;
iii. *additivity in the first slot*: $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$;
iv. *homogeneity in the first slot*: $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle$;
v. *conjugate symmetry*: $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$.

In the positivity axiom, $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ means that $\langle \mathbf{v}, \mathbf{v} \rangle$ is a real nonnegative scalar.

Definition 1.1.4. *Inner-Product Space*
An *inner-product space* is a vector space $\mathcal{V}$ together with an inner product on $\mathcal{V}$.

Example 1.1.5. *Standard Inner Product on $\mathbb{F}^n$*
For $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, the standard inner product is $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{j=1}^{n} x_j \overline{y_j}$. When $\mathbb{F} = \mathbb{R}$, this is the usual dot product.

Example 1.1.6. *Function and Matrix Inner Products*
If $\mathcal{V} = C_{\mathbb{F}}[a, b]$, then $\langle f, g \rangle = \int_a^b f(t)\overline{g(t)}\, dt$ is an inner product.
If $\mathcal{V} = \mathrm{Mat}(m, n, \mathbb{F})$, then the *Frobenius inner product* is $\langle A, B \rangle_F := \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}\overline{b_{ij}}$.

Example 1.1.7. *Square-Summable Sequences*
Set $\ell^2(\mathbb{F}) := \left\{ \mathbf{x} = (x_j)_{j \geq 1} : \sum_{j=1}^{\infty} |x_j|^2 < \infty \right\}$, where $\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{j=1}^{\infty} x_j \overline{y_j}$. Since $|x_j\overline{y_j}| \leq (|x_j|^2 + |y_j|^2)/2$, the series converges absolutely; termwise operations give an inner product on $\ell^2(\mathbb{F})$, with $\mathbb{F}^{(\mathbb{N})} \subseteq \ell^2(\mathbb{F})$.

Proposition 1.1.8. *Basic Inner-Product Identities*
Let $\mathcal{V}$ be an inner-product space. Then for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. for fixed $\mathbf{u}$, the map $\mathbf{v} \mapsto \langle \mathbf{v}, \mathbf{u} \rangle$ is linear;
ii. $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ and $\langle \mathbf{u}, \mathbf{0} \rangle = 0$;
iii. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$;
iv. $\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \overline{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle$.

Proof.

(i) This is exactly additivity and homogeneity in the first slot.
(ii) A linear map sends $\mathbf{0}$ to 0, so $\langle \mathbf{0}, \mathbf{u} \rangle = 0$. Conjugate symmetry gives $\langle \mathbf{u}, \mathbf{0} \rangle = \overline{\langle \mathbf{0}, \mathbf{u} \rangle} = 0$.
(iii) By conjugate symmetry and additivity, $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \overline{\langle \mathbf{v} + \mathbf{w}, \mathbf{u} \rangle} = \overline{\langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle} = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$.
(iv) By conjugate symmetry and homogeneity, $\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \overline{\langle \lambda \mathbf{v}, \mathbf{u} \rangle} = \overline{\lambda} \overline{\langle \mathbf{v}, \mathbf{u} \rangle} = \overline{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle$.

Structural Commentary: First-slot linearity and conjugate symmetry force conjugate linearity in second slot.
Key Insight: Once one slot is linear, the other slot is forced to be conjugate linear. □

Lemma 1.1.9. *Inner Products Separate Vectors*
For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$, $(\forall \mathbf{x} \in \mathcal{V},\ \langle \mathbf{x}, \mathbf{u} \rangle = \langle \mathbf{x}, \mathbf{v} \rangle) \Longrightarrow \mathbf{u} = \mathbf{v}$, and $(\forall \mathbf{x} \in \mathcal{V},\ \langle \mathbf{x}, \mathbf{w} \rangle = 0) \Longleftrightarrow \mathbf{w} = \mathbf{0}$.

Proof. Set $\mathbf{x} = \mathbf{u} - \mathbf{v}$. Then $\|\mathbf{u} - \mathbf{v}\|^2 = \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = 0$, hence $\mathbf{u} = \mathbf{v}$.
Taking $\mathbf{u} = \mathbf{w}$ and $\mathbf{v} = \mathbf{0}$ gives the second assertion; the reverse implication is immediate.
Structural Commentary: Testing against all vectors detects the vector being tested.
Key Insight: A vector orthogonal to every vector, including itself, must be zero. □

Definition 1.1.10. *Norm Induced by an Inner Product*
For $\mathbf{v} \in \mathcal{V}$, the *norm* of $\mathbf{v}$ is $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Example 1.1.11. *Norms from Standard Inner Products*

In $\mathbb{F}^n$, the norm from the standard inner product is the *Euclidean norm* $\|(x_1, \ldots, x_n)\|_2 = \sqrt{|x_1|^2 + \cdots + |x_n|^2}$.

In $C_{\mathbb{F}}[a, b]$, with the integral inner product, the derived norm is the *L^2 norm* $\|f\|_2 = \sqrt{\int_a^b |f(t)|^2 \, dt}$.

In $\mathrm{Mat}(m, n, \mathbb{F})$, with the Frobenius inner product, the *Frobenius norm* is $\|A\|_F = \sqrt{\langle A, A \rangle_F} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$, $A = [a_{ij}]$. The Frobenius norm is also called the *Schur norm* or *Hilbert-Schmidt norm*.

Proposition 1.1.12. *Basic Properties of the Derived Norm*
Let $\mathcal{V}$ be an inner-product space with derived norm $\|\cdot\|$. For all $\mathbf{v} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. $\|\mathbf{v}\|$ is real and nonnegative;
ii. $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$;
iii. $\|\lambda \mathbf{v}\| = |\lambda| \, \|\mathbf{v}\|$.

Proof.

(i) By positivity, $\langle \mathbf{v}, \mathbf{v} \rangle$ is real and nonnegative, so its square root is real and nonnegative.
(ii) By definition, $\|\mathbf{v}\| = 0$ if and only if $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, which holds if and only if $\mathbf{v} = \mathbf{0}$ by definiteness.
(iii) Compute $\|\lambda \mathbf{v}\|^2 = \langle \lambda \mathbf{v}, \lambda \mathbf{v} \rangle = \lambda \overline{\lambda} \langle \mathbf{v}, \mathbf{v} \rangle = |\lambda|^2 \|\mathbf{v}\|^2$. Taking nonnegative square roots, $\|\lambda \mathbf{v}\| = |\lambda| \, \|\mathbf{v}\|$.

Structural Commentary: Positivity gives length; sesquilinearity gives scaling.
Key Insight: Squared length is easier to manipulate than length itself. □

Theorem 1.1.13. *Cauchy-Schwarz Inequality*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$. Equality holds if and only if one of $\mathbf{u}, \mathbf{v}$ is a scalar multiple of the other.

Proof. If $\mathbf{v} = \mathbf{0}$, both sides are 0. Assume $\mathbf{v} \neq \mathbf{0}$.
For any $\lambda \in \mathbb{F}$, positivity gives $0 \leq \|\mathbf{u} - \lambda \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \lambda \langle \mathbf{v}, \mathbf{u} \rangle - \overline{\lambda} \langle \mathbf{u}, \mathbf{v} \rangle + |\lambda|^2 \|\mathbf{v}\|^2$.
Choose $\lambda = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$. Since $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle}$, the inequality becomes $0 \leq \|\mathbf{u}\|^2 - \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^2}$.
Multiplying by $\|\mathbf{v}\|^2$ and taking square roots proves the inequality. Equality in this argument is equivalent to $\|\mathbf{u} - \lambda \mathbf{v}\| = 0$, hence to $\mathbf{u} = \lambda \mathbf{v}$ for the chosen λ.
Conversely, if $\mathbf{u} = \lambda \mathbf{v}$, then $|\langle \mathbf{u}, \mathbf{v} \rangle| = |\lambda| \ \langle \mathbf{v}, \mathbf{v} \rangle = \|\lambda \mathbf{v}\| \, \|\mathbf{v}\| = \|\mathbf{u}\| \, \|\mathbf{v}\|$, so equality holds.
The case $\mathbf{v} = \mathbf{0}$ is also covered because $\mathbf{v} = 0\mathbf{u}$.
Structural Commentary: Positivity of squared length controls the mixed inner-product term.
Key Insight: The best scalar comparison between two vectors bounds their inner product. □

Theorem 1.1.14. *Triangle Inequality*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.
Equality holds if and only if one of $\mathbf{u}, \mathbf{v}$ *is a nonnegative scalar multiple of the other.*

Proof. We compute

$$\|\mathbf{u}+\mathbf{v}\|^2 = \langle \mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v}\rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{u}\rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\operatorname{Re}\langle \mathbf{u}, \mathbf{v}\rangle$$
$$\leq \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\,|\langle \mathbf{u}, \mathbf{v}\rangle| \leq \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\,\|\mathbf{u}\|\,\|\mathbf{v}\| = \big(\|\mathbf{u}\| + \|\mathbf{v}\|\big)^2.$$

Taking square roots gives the inequality. If either vector is $\mathbf{0}$, the equality condition is immediate.
Otherwise, equality in the two displayed inequalities holds exactly when $\operatorname{Re}\langle \mathbf{u}, \mathbf{v}\rangle = |\langle \mathbf{u}, \mathbf{v}\rangle|$ and equality holds in Cauchy-Schwarz. The first condition says $\langle \mathbf{u}, \mathbf{v}\rangle$ is real nonnegative, and the second says $\mathbf{u} = a\mathbf{v}$ for some $a \in \mathbb{F}$. Then $a\,\|\mathbf{v}\|^2 = \langle \mathbf{u}, \mathbf{v}\rangle$ is real nonnegative, so $a \geq 0$.
Conversely, if $\mathbf{u} = a\mathbf{v}$ with $a \geq 0$, then $\|\mathbf{u}+\mathbf{v}\| = (a+1)\,\|\mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$.
Structural Commentary: Square the norm and bound the cross term by Cauchy-Schwarz.
Key Insight: Equality means the two vectors point in the same direction. □

Proposition 1.1.15. *Reverse Triangle Inequality*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $|\|\mathbf{u}\| - \|\mathbf{v}\|| \leq \|\mathbf{u}-\mathbf{v}\|$.

Proof. The triangle inequality gives $\|\mathbf{u}\| \leq \|\mathbf{u}-\mathbf{v}\| + \|\mathbf{v}\|$, hence $\|\mathbf{u}\| - \|\mathbf{v}\| \leq \|\mathbf{u}-\mathbf{v}\|$.
Interchanging $\mathbf{u}$ and $\mathbf{v}$ gives $\|\mathbf{v}\| - \|\mathbf{u}\| \leq \|\mathbf{u}-\mathbf{v}\|$. Combining the two inequalities gives the result.
Structural Commentary: Distances control changes in length.
Key Insight: The ordinary triangle inequality applied twice gives the reverse one. □

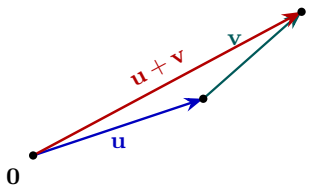


Figure 20: Triangle inequality: $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Definition 1.1.16. *Norm and Normed Vector Space*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$. A *norm* on $\mathcal{V}$ is a function $\|\cdot\| : \mathcal{V} \to [0, \infty)$ such that, $\forall \mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\lambda \in \mathbb{F}$,

i. $\|\mathbf{u}\| \geq 0$;
ii. $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$;
iii. $\|\lambda\mathbf{u}\| = |\lambda|\,\|\mathbf{u}\|$;
iv. $\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

A *normed vector space* is a vector space equipped with a norm. Its *unit ball* is $\{\mathbf{v} \in \mathcal{V} : \|\mathbf{v}\| \leq 1\}$.

Definition 1.1.17. *Distance Induced by an Inner Product*
Let $\mathcal{V}$ be an inner-product space. The *distance* between two vectors in $\mathcal{V}$ is $d(\mathbf{u}, \mathbf{v}) := \|\mathbf{u}-\mathbf{v}\|$.

Proposition 1.1.18. *Inner-Product Distance Is a Metric*
The function $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u}-\mathbf{v}\|$ *is a metric on* $\mathcal{V}$*: for all* $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$,

i. $d(\mathbf{u}, \mathbf{v}) \geq 0$, *and* $d(\mathbf{u}, \mathbf{v}) = 0$ *if and only if* $\mathbf{u} = \mathbf{v}$*;*
ii. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$*;*
iii. $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$.

Proof.

(i) By Proposition 1.1.12, $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u}-\mathbf{v}\| \geq 0$, with equality iff $\mathbf{u}-\mathbf{v} = \mathbf{0}$.
(ii) $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u}-\mathbf{v}\| = \|-(\mathbf{v}-\mathbf{u})\| = \|\mathbf{v}-\mathbf{u}\| = d(\mathbf{v}, \mathbf{u})$.
(iii) Since $\mathbf{u}-\mathbf{v} = (\mathbf{u}-\mathbf{w}) + (\mathbf{w}-\mathbf{v})$, Theorem 1.1.14 gives $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$.

Structural Commentary: The norm turns displacement vectors into distances.
Key Insight: Metric triangle inequalities are norm triangle inequalities applied to differences. □

Perspective 1.1.19. *Metric Viewpoint*
Define $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u}-\mathbf{v}\|$. Then $\mathbf{v}_n \to \mathbf{v}$ iff $\|\mathbf{v}_n - \mathbf{v}\| \to 0$.
In finite-dimensional inner-product spaces, subspaces are closed, linear maps are continuous, and the ambient space is complete. Infinite-dimensional analogues require additional hypotheses; subsequent projection and basis results therefore state finite-dimensional assumptions explicitly.

Theorem 1.1.20. *Parallelogram Identity*
For all $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\big(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\big)$.

Proof. Expanding both squared norms gives

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2, \quad \|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \langle \mathbf{u}, \mathbf{v}\rangle - \langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2.$$

Adding the two identities cancels the cross terms and yields the result.
Structural Commentary: Opposite cross terms cancel when the two diagonal norms are added.
Key Insight: Inner-product norms remember parallelogram geometry through cancellation. □

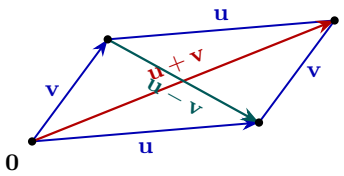


Figure 21: Parallelogram identity: $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$.

Proposition 1.1.21. *Coordinate Norms on* $\mathbb{F}^n$
For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{F}^n$, $\|\mathbf{x}\|_1 = \sum_{j=1}^{n} |x_j|$, $\|\mathbf{x}\|_2 = \left(\sum_{j=1}^{n} |x_j|^2\right)^{1/2}$, $\|\mathbf{x}\|_\infty = \max_{1\le j\le n} |x_j|$ *are norms on* $\mathbb{F}^n$.

Proof.

(i) For $\|\cdot\|_1$, nonnegativity, definiteness, and homogeneity follow from the corresponding facts for modulus. The triangle inequality follows from $\|\mathbf{x}+\mathbf{y}\|_1 = \sum_{j=1}^{n} |x_j + y_j| \le \sum_{j=1}^{n} (|x_j| + |y_j|) = \|\mathbf{x}\|_1 + \|\mathbf{y}\|_1$.

(ii) For $\|\cdot\|_2$, the axioms follow from its standard inner product via Proposition 1.1.12 and Theorem 1.1.14.

(iii) For $\|\cdot\|_\infty$, nonnegativity, definiteness, and homogeneity are immediate from the maximum of finitely many moduli. If k is chosen so that $\|\mathbf{x}+\mathbf{y}\|_\infty = |x_k + y_k|$, then $\|\mathbf{x}+\mathbf{y}\|_\infty \le |x_k| + |y_k| \le \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty$.

Structural Commentary: The ℓ_1 norm sums coordinate magnitudes, ℓ_∞ keeps only the largest coordinate magnitude, and ℓ_2 comes from the standard inner product.
Key Insight: All three obey the same abstract axioms, even though their unit balls have different shapes. □

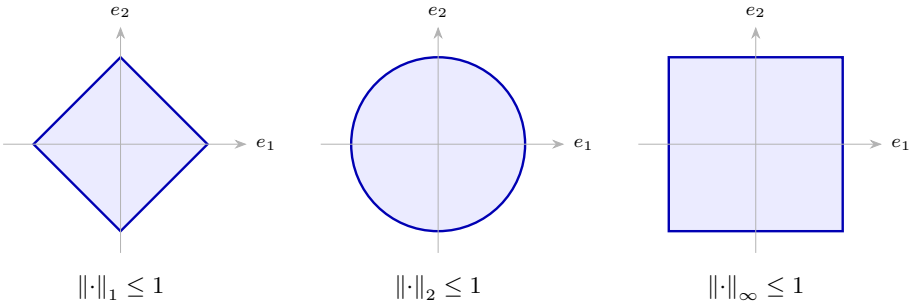


Figure 22: The unit balls of $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$ in $\mathbb{R}^2$.

Corollary 1.1.22. *ℓ_1 and ℓ_∞ Are Not Inner-Product Norms*
If $n \ge 2$, then the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $\mathbb{F}^n$ are not induced by any inner product.

Proof. Let $\mathbf{e}_1, \mathbf{e}_2$ be the first two standard coordinate vectors.
For $\|\cdot\|_1$, $\|\mathbf{e}_1+\mathbf{e}_2\|_1^2 + \|\mathbf{e}_1-\mathbf{e}_2\|_1^2 = 4+4 = 8$, but $2\|\mathbf{e}_1\|_1^2 + 2\|\mathbf{e}_2\|_1^2 = 4$.
For $\|\cdot\|_\infty$, $\|\mathbf{e}_1+\mathbf{e}_2\|_\infty^2 + \|\mathbf{e}_1-\mathbf{e}_2\|_\infty^2 = 1+1 = 2$ but $2\|\mathbf{e}_1\|_\infty^2 + 2\|\mathbf{e}_2\|_\infty^2 = 4$.
Both norms violate Theorem 1.1.20, which every inner-product norm must satisfy.
Structural Commentary: The parallelogram identity detects inner-product norms.
Key Insight: Different unit-ball geometry can rule out hidden inner-product structure. □

Theorem 1.1.23. *Polarization Identities*
Let $\mathcal{V}$ be an inner-product space and let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.

i. If $\mathbb{F} = \mathbb{R}$, then $\langle \mathbf{u}, \mathbf{v}\rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2\right)$.

ii. If $\mathbb{F} = \mathbb{C}$, *then* $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}\left(\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 + i\,\|\mathbf{u}+i\mathbf{v}\|^2 - i\,\|\mathbf{u}-i\mathbf{v}\|^2 \right)$.

Proof. Let $z := \langle \mathbf{u}, \mathbf{v} \rangle$.

(i) If $\mathbb{F} = \mathbb{R}$, then expanding squared norms gives $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\,\mathrm{Re}\, z = 4z$, proving (i).

(ii) If $\mathbb{F} = \mathbb{C}$, the first expansion gives $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\,\mathrm{Re}\, z$. The same expansion with $i\mathbf{v}$ gives $\|\mathbf{u}+i\mathbf{v}\|^2 - \|\mathbf{u}-i\mathbf{v}\|^2 = 4\,\mathrm{Im}\, z$. Combining the real and imaginary parts gives (ii).

Structural Commentary: Two norm differences extract the real and imaginary parts.
Key Insight: The norm determines the inner product because it determines both parts of every cross term. □

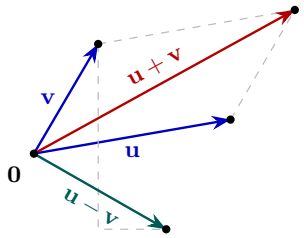


Figure 23: Polarization identity: $\|\mathbf{u}+\mathbf{v}\|^2 - \|\mathbf{u}-\mathbf{v}\|^2 = 4\,\mathrm{Re}\,\langle \mathbf{u}, \mathbf{v} \rangle$.

Proposition 1.1.24. *Operator Zero Tests from Inner Products*
Let $T \in \mathcal{L}(\mathcal{V})$.

i. If $\langle T\mathbf{v}, \mathbf{w} \rangle = 0$ *for every* $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, *then* $T = 0$.
ii. If $\mathbb{F} = \mathbb{C}$ *and* $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$ *for every* $\mathbf{v} \in \mathcal{V}$, *then* $T = 0$.
iii. The preceding diagonal test can fail over $\mathbb{R}$.

Proof.

(i) Taking $\mathbf{w} = T\mathbf{v}$ gives $\|T\mathbf{v}\|^2 = 0$ for every $\mathbf{v}$.

(ii) Put $Q(\mathbf{x}) = \langle T\mathbf{x}, \mathbf{x} \rangle$. For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$ and $r \in \mathbb{C}$, the hypothesis applied to $r\mathbf{x}+\mathbf{y}$ gives

$$0 = Q(r\mathbf{x}+\mathbf{y}) = |r|^2 Q(\mathbf{x}) + Q(\mathbf{y}) + r\,\langle T\mathbf{x}, \mathbf{y} \rangle + \overline{r}\,\langle T\mathbf{y}, \mathbf{x} \rangle = r\,\langle T\mathbf{x}, \mathbf{y} \rangle + \overline{r}\,\langle T\mathbf{y}, \mathbf{x} \rangle .$$

Taking $r = 1, i$ gives $\langle T\mathbf{x}, \mathbf{y} \rangle = 0$ for all $\mathbf{x}, \mathbf{y}$; by (i), $T = 0$.

(iii) On standard $\mathbb{R}^2$, a 90° rotation has $\langle T\mathbf{v}, \mathbf{v} \rangle = 0$ for all $\mathbf{v}$, yet $T \neq 0$.

Structural Commentary: Complex scalar directions recover mixed inner products from diagonal values.
Key Insight: Over $\mathbb{R}$, a skew-symmetric operator can hide on the diagonal; over $\mathbb{C}$, polarization detects it. □

Principle 1.1.25. *Geometry from Algebra*
Most geometric statements in an inner-product space come from expanding $\|\mathbf{u} \pm \mathbf{v}\|^2$ and then interpreting the cross terms $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$.

1.2 Orthogonality and Angles

Definition 1.2.1. *Unit Vector and Normalization*
A vector $\mathbf{u} \in \mathcal{V}$ is a *unit vector* if $\|\mathbf{u}\| = 1$. If $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is called the *normalization* of $\mathbf{v}$.

Proposition 1.2.2. *Normalization Gives a Unit Vector*
If $\mathbf{v} \in \mathcal{V}$ *and* $\mathbf{v} \neq \mathbf{0}$, *then* $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ *is a unit vector in the same direction as* $\mathbf{v}$.

Proof. By Proposition 1.1.12, $\|\mathbf{v}\| > 0$ and $\left\| \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\| = \frac{1}{\|\mathbf{v}\|}\,\|\mathbf{v}\| = 1$.
The vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is a positive scalar multiple of $\mathbf{v}$, so it lies in the same one-dimensional direction. *Structural Commentary*: Normalization changes scale but preserves direction.
Key Insight: Divide by the length to make the length equal to one. □

Definition 1.2.3. *Orthogonal Vectors*
Vectors $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ are *orthogonal*, written $\mathbf{u} \perp \mathbf{v}$, if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Proposition 1.2.4. *Orthogonality and the Zero Vector*
Let $\mathcal{V}$ *be an inner-product space.*

i. $\mathbf{u} \perp \mathbf{v}$ *if and only if* $\mathbf{v} \perp \mathbf{u}$.
ii. $\mathbf{0}$ *is orthogonal to every vector in* $\mathcal{V}$.

iii. If $\mathbf{v}$ *is orthogonal to itself, then* $\mathbf{v} = \mathbf{0}$.

Proof.

(i) If $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, then conjugate symmetry gives $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{0} = 0$, and the converse is identical.
(ii) This follows from Proposition 1.1.8, which gives $\langle \mathbf{0}, \mathbf{u} \rangle = 0$ and $\langle \mathbf{u}, \mathbf{0} \rangle = 0$.
(iii) If $\mathbf{v} \perp \mathbf{v}$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, hence $\mathbf{v} = \mathbf{0}$ by definiteness.

Structural Commentary: Conjugate symmetry makes orthogonality symmetric, and definiteness isolates $\mathbf{0}$.
Key Insight: Conjugate symmetry preserves the equation "equals zero." □

Proposition 1.2.5. *Mutually Orthogonal Nonzero Vectors Are Independent*
Mutually orthogonal nonzero vectors $\mathbf{v}_1, \ldots, \mathbf{v}_m$ *in an inner-product space are linearly independent.*

Proof. If $\sum_{j=1}^{m} c_j \mathbf{v}_j = \mathbf{0}$, then for each k, $0 = \left\langle \sum_{j=1}^{m} c_j \mathbf{v}_j, \mathbf{v}_k \right\rangle = c_k \langle \mathbf{v}_k, \mathbf{v}_k \rangle$.
Since $\langle \mathbf{v}_k, \mathbf{v}_k \rangle > 0$, $c_k = 0$. Thus all coefficients vanish.
Structural Commentary: Orthogonality isolates one coefficient at a time.
Key Insight: Pair a dependence relation with one vector from the list; every cross term disappears. □

1.3 Orthonormal Bases

Definition 1.3.1. *Orthonormal List*
A list $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ of vectors in an inner-product space $\mathcal{V}$ is *orthonormal* if

$$\langle \mathbf{e}_j, \mathbf{e}_k \rangle = \begin{cases} 1, & j = k, \\ 0, & j \neq k. \end{cases}$$

Equivalently, every $\mathbf{e}_j$ has norm 1 and distinct vectors in the list are orthogonal.

Definition 1.3.2. *Orthogonal and Orthonormal Sets*
An indexed family $(\mathbf{u}_i)_{i \in I}$ is *orthogonal* if $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for $i \neq j$, and *orthonormal* if $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$, $i, j \in I$.
A nonzero orthogonal family becomes orthonormal after replacing each $\mathbf{u}_i$ by $\mathbf{u}_i / \|\mathbf{u}_i\|$.

Proposition 1.3.3. *Norm of an Orthonormal Linear Combination*
If $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ *is an orthonormal list in* $\mathcal{V}$, *then* $\|a_1 \mathbf{e}_1 + \cdots + a_m \mathbf{e}_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$ *for all* $a_1, \ldots, a_m \in \mathbb{F}$.

Proof. Using orthonormality, $\left\| \sum_{j=1}^{m} a_j \mathbf{e}_j \right\|^2 = \left\langle \sum_{j=1}^{m} a_j \mathbf{e}_j, \sum_{k=1}^{m} a_k \mathbf{e}_k \right\rangle = \sum_{j=1}^{m} \sum_{k=1}^{m} a_j \overline{a_k} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{m} |a_j|^2$.
Structural Commentary: Orthonormality deletes every cross term in the squared-norm expansion.
Key Insight: In orthonormal coordinates, length is computed coefficient-by-coefficient. □

Corollary 1.3.4. *Orthonormal Lists Are Linearly Independent*
Every orthonormal list of vectors in an inner-product space is linearly independent.

Proof. If $a_1 \mathbf{e}_1 + \cdots + a_m \mathbf{e}_m = \mathbf{0}$, then Proposition 1.3.3 gives $0 = \|\mathbf{0}\|^2 = \|a_1 \mathbf{e}_1 + \cdots + a_m \mathbf{e}_m\|^2 = \sum_{j=1}^{m} |a_j|^2$.
Hence each $a_j = 0$, so the list is linearly independent.
Structural Commentary: The norm identity converts a linear relation into a sum of nonnegative scalars.
Key Insight: A zero orthonormal linear combination can only have zero coefficients. □

Definition 1.3.5. *Orthonormal Basis*
An *orthonormal basis* of $\mathcal{V}$ is a basis of $\mathcal{V}$ that is an orthonormal list.

Proposition 1.3.6. *Right-Length Orthonormal Lists Are Bases*
If $\mathcal{V}$ *is finite-dimensional and* $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *is an orthonormal list in* $\mathcal{V}$ *with* $n = \dim(\mathcal{V})$, *then* $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ *is an orthonormal basis of* $\mathcal{V}$.

Proof. By Corollary 1.3.4, the list is linearly independent.
Since its length is $\dim(\mathcal{V})$, Proposition 1.4.3 implies that it is a basis.
Structural Commentary: Orthonormality gives independence; the correct length forces spanning.
Key Insight: In finite dimensions, n independent vectors in an n-dimensional space already form a basis. □

Theorem 1.3.7. *Gram–Schmidt Process*
Let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ *be a linearly independent list in an inner-product space* $\mathcal{V}$.
Define $\mathbf{u}_1 = \mathbf{v}_1$, $\mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$ *and, for* $2 \le j \le m$, $\mathbf{u}_j = \mathbf{v}_j - \sum_{k=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_k \rangle \, \mathbf{e}_k$, $\mathbf{e}_j = \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}$.
Then $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ *is orthonormal and* $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j) = \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$.

Proof. Proceed by induction on j. The case $j = 1$ is normalization of the nonzero vector $\mathbf{v}_1$.
Assume the result holds through $j-1$. Since $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}) = \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_{j-1})$ and the $\mathbf{v}$-list is linearly independent, $\mathbf{v}_j \notin \operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_{j-1})$. Hence $\mathbf{u}_j \neq \mathbf{0}$, because $\mathbf{u}_j = \mathbf{0}$ would put $\mathbf{v}_j$ in that span.
For $k < j$, $\langle \mathbf{u}_j, \mathbf{e}_k \rangle = \langle \mathbf{v}_j, \mathbf{e}_k \rangle - \sum_{i=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_i \rangle \langle \mathbf{e}_i, \mathbf{e}_k \rangle = \langle \mathbf{v}_j, \mathbf{e}_k \rangle - \langle \mathbf{v}_j, \mathbf{e}_k \rangle = 0$.
Thus $\mathbf{e}_j$ is a unit vector orthogonal to $\mathbf{e}_1, \dots, \mathbf{e}_{j-1}$.
Also $\mathbf{e}_j$ lies in $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ because it is a nonzero scalar multiple of $\mathbf{u}_j$, and $\mathbf{u}_j$ is built from $\mathbf{v}_j$ and the previous $\mathbf{e}_k$, whose span agrees with $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$.
Conversely, $\mathbf{v}_j = \mathbf{u}_j + \sum_{k=1}^{j-1} \langle \mathbf{v}_j, \mathbf{e}_k \rangle \, \mathbf{e}_k$ lies in $\operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$.
Combining this with the induction hypothesis gives equality of the two spans at step j.
Structural Commentary: Subtract the previous-span component, then normalize.
Key Insight: Gram–Schmidt changes the basis vectors but preserves every initial span. □

Theorem 1.3.8. *Gram–Schmidt for Arbitrary Finite Lists*
Let $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ *be a finite list in an inner-product space.*
Define $\mathbf{u}_1 = \mathbf{v}_1$ *and, for* $2 \le k \le m$, $\mathbf{u}_k = \mathbf{v}_k - \sum_{\substack{1 \le j < k \\ \mathbf{u}_j \neq \mathbf{0}}} \frac{\langle \mathbf{v}_k, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \mathbf{u}_j$.
Set $U_k = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ *and* $V_k = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, *with* $U_0 = V_0 = \{\mathbf{0}\}$.
Then $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ *is orthogonal,* $U_k = V_k$ *for* $1 \le k \le m$.

Proof. Induct on k. The case $k = 0$ is trivial. Assume $(\mathbf{u}_1, \dots, \mathbf{u}_{k-1})$ is orthogonal and $U_{k-1} = V_{k-1}$.
For $i < k$ with $\mathbf{u}_i \neq \mathbf{0}$, $\langle \mathbf{u}_k, \mathbf{u}_i \rangle = \langle \mathbf{v}_k, \mathbf{u}_i \rangle - \frac{\langle \mathbf{v}_k, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \langle \mathbf{u}_i, \mathbf{u}_i \rangle = 0$, while $\mathbf{u}_i = \mathbf{0}$ is trivial.
Hence the list remains orthogonal. Also $\mathbf{u}_k \in V_k$ and $\mathbf{v}_k \in U_k$, so $U_k = V_k$.
Structural Commentary: Gram–Schmidt preserves every prefix span.
Key Insight: Each step subtracts previous orthogonal directions. □

Corollary 1.3.9. *Zero Remainders Detect Dependence*
Let $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ *be constructed from* $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ *by Theorem 1.3.8,*
and set $U_k = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$, $V_k = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, *with* $U_0 = V_0 = \{\mathbf{0}\}$.
Then $\mathbf{u}_k = \mathbf{0} \iff \mathbf{v}_k \in V_{k-1}$, $1 \le k \le m$.

Proof. The case $k = 1$ follows from $\mathbf{u}_1 = \mathbf{v}_1$ and $V_0 = \{\mathbf{0}\}$. Let $2 \le k \le m$.
($\Rightarrow$) If $\mathbf{u}_k = \mathbf{0}$, the recursion gives $\mathbf{v}_k \in U_{k-1} = V_{k-1}$.
($\Leftarrow$) If $\mathbf{v}_k \in V_{k-1} = U_{k-1}$, write $\mathbf{v}_k = \sum_{\substack{1 \le j < k \\ \mathbf{u}_j \neq \mathbf{0}}} a_j \mathbf{u}_j$. Orthogonality gives $a_j = \frac{\langle \mathbf{v}_k, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle}$, recursion gives $\mathbf{u}_k = \mathbf{0}$.
Structural Commentary: Zero remainders certify dependence.
Key Insight: A nonzero remainder is a new direction. □

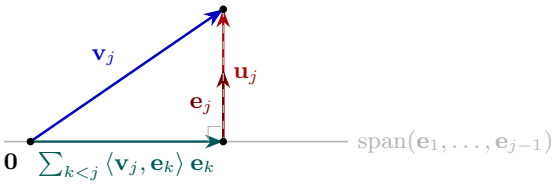


Figure 24: One Gram–Schmidt step: subtract the previous-span component, then normalize.

Example 1.3.10. *Legendre Polynomials from Gram–Schmidt*
Let $\mathbb{R}[x]$ have inner product $\langle p, q \rangle = \int_{-1}^{1} p(x) q(x) \, dx$, and apply Gram–Schmidt to $(1, x, x^2, x^3, \dots)$.
The first unnormalized vectors are $u_1(x) = 1$, $u_2(x) = x$, $u_3(x) = x^2 - \frac{1}{3}$, $u_4(x) = x^3 - \frac{3}{5}x$.
Thus (u_n) is, up to nonzero scalar multiples, the Legendre sequence.
Structural Commentary: Orthogonal polynomials are orthogonalized monomials.
Key Insight: Changing the inner product changes the polynomial family produced by Gram–Schmidt.

Lemma 1.3.11. *Gram–Schmidt on Orthogonal Lists*
If $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is a nonzero mutually orthogonal list, then Gram–Schmidt produces $\mathbf{e}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$, $j = 1, \dots, m$. In particular, if the original list is orthonormal, Gram–Schmidt leaves it unchanged.

Proof. Assume $\mathbf{e}_i = \mathbf{v}_i / \|\mathbf{v}_i\|$ for $i < j$. Since $\mathbf{v}_j \perp \mathbf{v}_i$, we have $\langle \mathbf{v}_j, \mathbf{e}_i \rangle = 0$ for every $i < j$.
Hence the Gram–Schmidt remainder at step j is $\mathbf{u}_j = \mathbf{v}_j$, so $\mathbf{e}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$.
If the original list is orthonormal, then $\|\mathbf{v}_j\| = 1$ for every j.
Structural Commentary: Orthogonal input has no previous-span component to subtract.
Key Insight: Gram–Schmidt only removes nonorthogonal overlap. □

Corollary 1.3.12. *Existence of Orthonormal Bases*
Every finite-dimensional inner-product space has an orthonormal basis.

Proof. Choose any basis of $\mathcal{V}$ and apply Theorem 1.3.7.
The resulting orthonormal list has the same span as the original basis, hence spans $\mathcal{V}$.
Structural Commentary: Gram–Schmidt orthonormalizes a basis without changing its span.
Key Insight: Finite-dimensional inner-product spaces always admit geometry-friendly coordinates. □

Corollary 1.3.13. *Extending an Orthonormal List*
Every orthonormal list in a finite-dimensional inner-product space can be extended to an orthonormal basis.

Proof. Let $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ be orthonormal. It is linearly independent by Corollary 1.3.4, so extend it to a basis $(\mathbf{e}_1, \dots, \mathbf{e}_m, \mathbf{v}_{m+1}, \dots, \mathbf{v}_n)$ by Theorem 1.3.12.
During Gram–Schmidt, the first m remainders are the original $\mathbf{e}_j$ because the list is already orthonormal; after normalization they remain unchanged. The remaining vectors are orthonormalized by Theorem 1.3.7, and the final list spans $\mathcal{V}$ because Gram–Schmidt preserves the total span.
Structural Commentary: Extend to a basis, then orthonormalize the new part.
Key Insight: Existing orthonormal directions are already cleaned up, so Gram–Schmidt does not move them. □

Perspective 1.3.14. *Hamel and Hilbert Basis Caution*
Here an *orthonormal basis* is a Hamel basis that is orthonormal.
In infinite dimensions, a *Hilbert basis* usually means a maximal orthonormal set, not necessarily a Hamel basis. For example, the standard $(\mathbf{e}_j)$ in $\ell^2(\mathbb{F})$ is maximal orthonormal, but $\operatorname{span}\{\mathbf{e}_j : j \geq 1\} = \mathbb{F}^{(\mathbb{N})} \subsetneq \ell^2(\mathbb{F})$.

Definition 1.3.15. *Hilbert Basis*

(i) A *Hilbert basis* is a maximal orthonormal set under inclusion.
(ii) A *Hamel basis* is a vector-space basis.
(iii) An *orthonormal basis* is an orthonormal Hamel basis.

Proposition 1.3.16. *Finite Hilbert Bases Are Orthonormal Bases*
Let $\mathcal{O} = \{\mathbf{e}_1, \dots, \mathbf{e}_m\}$ be orthonormal in $\mathcal{V}$ and set $W = \operatorname{span}(\mathcal{O})$.
Then $\mathcal{O}$ is a Hilbert basis $\iff W = \mathcal{V} \iff \mathcal{O}$ is an orthonormal Hamel basis.

Proof. Since $\mathcal{O}$ is orthonormal, Corollary 1.3.4 gives independence; hence $W = \mathcal{V}$ iff $\mathcal{O}$ is an orthonormal Hamel basis. It remains to prove maximality iff spanning.

($\Rightarrow$) Suppose $\mathcal{O}$ is a Hilbert basis. If $W \neq \mathcal{V}$, choose $\mathbf{v} \notin W$ and set $\mathbf{u} = \mathbf{v} - \sum_j \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$. Then $\langle \mathbf{u}, \mathbf{e}_i \rangle = 0 \ \forall i$, and $\mathbf{u} \neq \mathbf{0}$ since $\mathbf{v} \notin W$. Thus $\mathcal{O} \cup \{\mathbf{u}/\|\mathbf{u}\|\}$ is orthonormal, contradicting maximality. Hence $W = \mathcal{V}$.
($\Leftarrow$) Suppose $W = \mathcal{V}$. If $\mathbf{w} \perp \mathcal{O}$, write $\mathbf{w} = \sum_j a_j \mathbf{e}_j$. Then $a_j = \langle \mathbf{w}, \mathbf{e}_j \rangle = 0$ for all j, so $\mathbf{w} = \mathbf{0}$.
Hence no nonzero vector can be adjoined, and $\mathcal{O}$ is maximal orthonormal.

Structural Commentary: Finite maximal orthonormal sets span.
Key Insight: Missing directions can be orthogonalized and adjoined. □

Example 1.3.17. *A Hilbert Basis Need Not Be a Hamel Basis*
In $\ell^2(\mathbb{F})$, let $\mathbf{e}_j = (0, \dots, 0, 1, 0, \dots)$. Then $\{\mathbf{e}_j : j \geq 1\}$ is orthonormal and maximal, since $\mathbf{x} \perp \mathbf{e}_j$ for all j implies $x_j = \langle \mathbf{x}, \mathbf{e}_j \rangle = 0$ for all j. Its finite span is $\mathbb{F}^{(\mathbb{N})} \subsetneq \ell^2(\mathbb{F})$, so it is a Hilbert basis but not a Hamel basis.

Theorem 1.3.18. *Coordinates in an Orthonormal Basis*
Let $\mathcal{V}$ be finite-dimensional and let $\beta = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$.
Then for every $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$, $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} |\langle \mathbf{v}, \mathbf{e}_j \rangle|^2$ and $[\mathbf{v}]_\beta = \begin{bmatrix} \langle \mathbf{v}, \mathbf{e}_1 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{e}_n \rangle \end{bmatrix}$.

Proof. Since β is a basis, write $\mathbf{v} = \sum_{j=1}^{n} a_j \mathbf{e}_j$. Taking inner product with $\mathbf{e}_k$ gives $\langle \mathbf{v}, \mathbf{e}_k \rangle = \sum_{j=1}^{n} a_j \langle \mathbf{e}_j, \mathbf{e}_k \rangle = a_k$. Thus the displayed expansion and coordinate column follow.

Applying Proposition 1.3.3 to the same expansion gives $\|\mathbf{v}\|^2 = \sum_{j=1}^{n} |a_j|^2 = \sum_{j=1}^{n} |\langle \mathbf{v}, \mathbf{e}_j \rangle|^2$.

Structural Commentary: An orthonormal basis makes coordinate extraction an inner-product computation.
Key Insight: Pairing with $\mathbf{e}_k$ isolates the kth coordinate. □

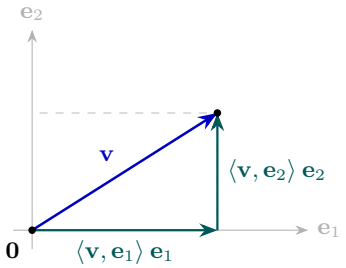


Figure 25: An orthonormal basis resolves a vector into perpendicular coordinate components.

Remark 1.3.19. *Cross Terms*
In an expansion such as $\|\mathbf{u} + \mathbf{v}\|^2$, the terms involving both vectors, $\langle \mathbf{u}, \mathbf{v} \rangle$ and $\langle \mathbf{v}, \mathbf{u} \rangle$, are called *cross terms*. The terms $\|\mathbf{u}\|^2$ and $\|\mathbf{v}\|^2$ are the pure squared-length terms.

Theorem 1.3.20. *Pythagorean Theorem*
If $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\mathbf{u} \perp \mathbf{v}$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Proof. Since $\mathbf{u} \perp \mathbf{v}$, Proposition 1.2.4 gives $\langle \mathbf{v}, \mathbf{u} \rangle = 0$. Thus

$$\|\mathbf{u} + \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 .$$

Structural Commentary: Orthogonality cancels the cross terms in the squared-norm expansion.
Key Insight: Perpendicular components add in squared length, not in length. □

Proposition 1.3.21. *Pythagorean Converse over $\mathbb{R}$*
In a real inner-product space, $\mathbf{u} \perp \mathbf{v} \iff \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Proof. Use $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$.
($\Rightarrow$) If $\mathbf{u} \perp \mathbf{v}$, then $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, so the identity follows.
($\Leftarrow$) If the identity holds, then $2\langle \mathbf{u}, \mathbf{v} \rangle = 0$, hence $\mathbf{u} \perp \mathbf{v}$.

Structural Commentary: In real inner-product spaces, the cross term has no hidden imaginary part.
Key Insight: The Pythagorean identity detects orthogonality exactly over $\mathbb{R}$. □

Theorem 1.3.22. *Bessel's Inequality*
Let $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be an orthonormal list in an inner-product space $\mathcal{V}$.
Then, for every $\mathbf{v} \in \mathcal{V}$, $\sum_{j=1}^{m} |\langle \mathbf{v}, \mathbf{e}_j \rangle|^2 \leq \|\mathbf{v}\|^2$.

Proof. Set $\mathbf{u} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$. For each k, orthonormality gives $\langle \mathbf{v} - \mathbf{u}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$.
Thus $\mathbf{v} - \mathbf{u}$ is orthogonal to $\mathbf{u}$. By Pythagorean Theorem 1.3.20 and Proposition 1.3.3,

$$\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v} - \mathbf{u}\|^2 \geq \|\mathbf{u}\|^2 = \sum_{j=1}^{m} |\langle \mathbf{v}, \mathbf{e}_j \rangle|^2 .$$

Structural Commentary: The chosen directions give an orthogonal part of $\mathbf{v}$; the residual is perpendicular.
Key Insight: Partial orthonormal coordinates cannot contain more energy than the whole vector. □

Corollary 1.3.23. *Coefficients Against an Orthonormal Sequence Vanish*
Let $(\mathbf{e}_1, \mathbf{e}_2, \ldots)$ be an orthonormal sequence in an inner-product space $\mathcal{V}$, meaning every finite sublist is orthonormal. Then, for every $\mathbf{v} \in \mathcal{V}$, $\lim_{n \to \infty} \langle \mathbf{v}, \mathbf{e}_n \rangle = 0$.

Proof. Fix $\epsilon > 0$. By Theorem 1.3.22, every finite set of indices satisfying $\left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right| \geq \epsilon$ has cardinality at most $\|\mathbf{v}\|^2 / \epsilon^2$. Hence only finitely many j have $\left|\langle \mathbf{v}, \mathbf{e}_j \rangle\right| \geq \epsilon$. This holds for every $\epsilon > 0$, so $\langle \mathbf{v}, \mathbf{e}_n \rangle \to 0$.
Structural Commentary: Bessel bounds the number of large orthonormal coefficients.
Key Insight: Infinitely many large coefficients would force infinite energy. □

Theorem 1.3.24. *Parseval's Identity*
Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of an inner-product space $\mathcal{V}$.
Then, for all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle} = \langle [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$.

Proof. By Theorem 1.3.18, $\mathbf{v} = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$ and $\mathbf{w} = \sum_{k=1}^{n} \langle \mathbf{w}, \mathbf{e}_k \rangle \mathbf{e}_k$. By linearity in first slot and conjugate linearity in second, $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{n} \sum_{k=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_k \rangle} \langle \mathbf{e}_j, \mathbf{e}_k \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \overline{\langle \mathbf{w}, \mathbf{e}_j \rangle}$.
The last expression is exactly the standard inner product of the coordinate columns in $\mathbb{F}^n$.
Structural Commentary: Orthonormal coordinates make inner products coordinatewise.
Key Insight: Orthonormal coordinates preserve angles and lengths, not just linear combinations. □

Theorem 1.3.25. *Matrix Entries from an Orthonormal Codomain Basis*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional vector spaces over $\mathbb{F}$, with $\mathcal{W}$ an inner-product space. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$ and let $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_m)$ be an orthonormal basis of $\mathcal{W}$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then

$$[T]_{\gamma \leftarrow \beta} = \begin{bmatrix} \langle T\mathbf{v}_1, \mathbf{w}_1 \rangle & \cdots & \langle T\mathbf{v}_n, \mathbf{w}_1 \rangle \\ \vdots & \ddots & \vdots \\ \langle T\mathbf{v}_1, \mathbf{w}_m \rangle & \cdots & \langle T\mathbf{v}_n, \mathbf{w}_m \rangle \end{bmatrix}.$$

Conversely, for each $A \in \mathrm{Mat}(m, n, \mathbb{F})$, there is a unique $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ such that $[T]_{\gamma \leftarrow \beta} = A$.

Proof. The jth column of $[T]_{\gamma \leftarrow \beta}$ is $[T\mathbf{v}_j]_\gamma$. Since γ is orthonormal, Theorem 1.3.18 gives

$$[T\mathbf{v}_j]_\gamma = \begin{bmatrix} \langle T\mathbf{v}_j, \mathbf{w}_1 \rangle \\ \vdots \\ \langle T\mathbf{v}_j, \mathbf{w}_m \rangle \end{bmatrix}.$$

This proves the displayed matrix formula. The converse is exactly the surjectivity and injectivity of the coordinate encoding in Theorem 3.3.32.
Structural Commentary: Columns are basis-image coordinates, read by inner products.
Key Insight: Each entry measures $T\mathbf{v}_j$ against one output direction. □

Corollary 1.3.26. *Inner Products Through an Operator Matrix*
Let $\beta = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of a finite-dimensional inner-product space $\mathcal{V}$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then, for all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle [T]_{\beta \leftarrow \beta} [\mathbf{v}]_\beta, [\mathbf{w}]_\beta \rangle_{\mathbb{F}^n}$.

Proof. By Proposition 3.3.28, $[T\mathbf{v}]_\beta = [T]_{\beta \leftarrow \beta} [\mathbf{v}]_\beta$. Applying Theorem 1.3.24 to $T\mathbf{v}$ and $\mathbf{w}$ gives the result.
Structural Commentary: Orthonormal coordinates turn $\langle T\mathbf{v}, \mathbf{w} \rangle$ into matrix computation.
Key Insight: Orthonormal coordinates make operator geometry computable by matrices. □

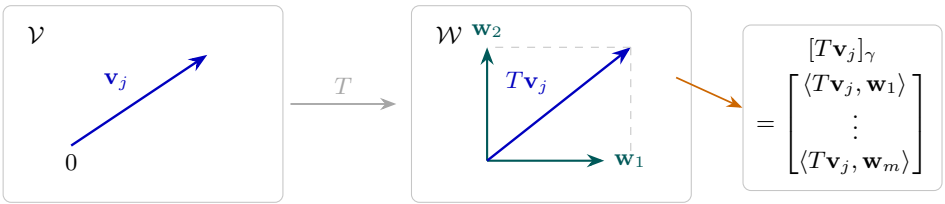


Figure 26: The jth column of $[T]_{\gamma \leftarrow \beta}$ is obtained by resolving $T\mathbf{v}_j$ against the orthonormal basis of $\mathcal{W}$.

Theorem 1.3.27. *Riesz Representation Theorem*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $\varphi : \mathcal{V} \to \mathbb{F}$ be a linear functional. Then there is a unique vector $\mathbf{u} \in \mathcal{V}$ such that $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$.
By Theorem 1.3.18, $\varphi(\mathbf{v}) = \varphi\left(\sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j\right) = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle \varphi(\mathbf{e}_j)$.

Set $\mathbf{u} = \sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)}\,\mathbf{e}_j$. Since inner product is conjugate linear in second slot, $\langle \mathbf{v}, \mathbf{u} \rangle = \sum_{j=1}^{n} \langle \mathbf{v}, \mathbf{e}_j \rangle\, \varphi(\mathbf{e}_j) = \varphi(\mathbf{v})$.
For uniqueness, suppose $\langle \mathbf{v}, \mathbf{u}_1 \rangle = \langle \mathbf{v}, \mathbf{u}_2 \rangle \ \ \forall \mathbf{v} \in \mathcal{V}$. Taking $\mathbf{v} = \mathbf{u}_1 - \mathbf{u}_2$ gives $\|\mathbf{u}_1 - \mathbf{u}_2\|^2 = 0$, hence $\mathbf{u}_1 = \mathbf{u}_2$.
Structural Commentary: An orthonormal basis packages a functional as one representing vector.
Key Insight: Finite-dimensional functionals are inner products with fixed vectors. □

Definition 1.3.28. *Riesz Vector*
Let $\mathcal{V}$ be finite-dimensional and let $\varphi : \mathcal{V} \to \mathbb{F}$ be a linear functional. The unique vector $\mathbf{u} \in \mathcal{V}$ satisfying $\varphi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{u} \rangle$ for every $\mathbf{v} \in \mathcal{V}$ is the *Riesz vector* representing φ.

Theorem 1.3.29. *Riesz Map*
Let $\mathcal{V}$ be finite-dimensional inner-product space, $\tau : \mathcal{V} \to \mathcal{V}'$, $\tau(\mathbf{u}) = \langle \cdot, \mathbf{u} \rangle$. Then τ is bijective, conjugate-linear. Hence $R = \tau^{-1}$ is conjugate-linear, $R(a\varphi + \psi) = \overline{a}\, R(\varphi) + R(\psi)$, $a \in \mathbb{F}$, $\varphi, \psi \in \mathcal{V}'$.
Over $\mathbb{R}$ this is ordinary linearity.

Proof. For $a \in \mathbb{F}$ and $\mathbf{u}, \mathbf{w} \in \mathcal{V}$, $\tau(a\mathbf{u} + \mathbf{w}) = \overline{a}\tau(\mathbf{u}) + \tau(\mathbf{w})$, so τ is conjugate-linear.
If $\tau(\mathbf{u}) = 0$, then $\|\mathbf{u}\|^2 = \tau(\mathbf{u})(\mathbf{u}) = 0$, so τ is injective. Surjectivity is Theorem 1.3.27.
For $a \in \mathbb{F}$, $\varphi, \psi \in \mathcal{V}'$, and $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, \overline{a}R(\varphi) + R(\psi) \rangle = a \langle \mathbf{v}, R(\varphi) \rangle + \langle \mathbf{v}, R(\psi) \rangle = (a\varphi + \psi)(\mathbf{v})$.
Uniqueness gives $R(a\varphi + \psi) = \overline{a}R(\varphi) + R(\psi)$.
Structural Commentary: Riesz identifies $\mathcal{V}$ with $\mathcal{V}'$ conjugate-linearly.
Key Insight: The representing slot conjugates scalars. □

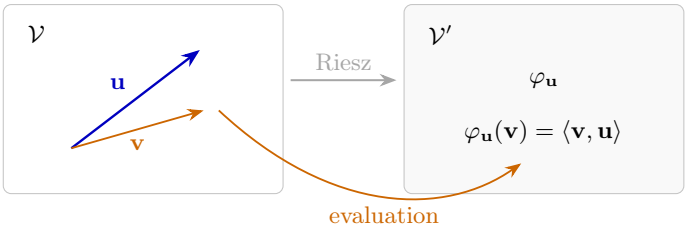


Figure 27: Riesz identifies a functional with its representing vector.

Perspective 1.3.30. *Riesz as Coordinate Duality*
The vector $\sum_{j=1}^{n} \overline{\varphi(\mathbf{e}_j)}\mathbf{e}_j$ seems to depend on the orthonormal basis, but Theorem 1.3.27 shows that the resulting representing vector is intrinsic.

1.4 Gram Matrices

Definition 1.4.1. *Gram Matrix*
For vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ in an inner-product space, the *Gram matrix* is

$$\operatorname{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{u}_n \rangle & \langle \mathbf{u}_2, \mathbf{u}_n \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix}.$$

Proposition 1.4.2. *Gram Matrix Quadratic Form*
For any vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$, Gram matrix $G = \operatorname{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$ is Hermitian and satisfies $\mathbf{c}^* G \mathbf{c} = \left\| \sum_{j=1}^{n} c_j \mathbf{u}_j \right\|^2$ for every $\mathbf{c} = [c_1 \ \cdots \ c_n]^t \in \mathbb{F}^n$. Hence the Gram quadratic form is nonnegative. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, then it is positive for every nonzero $\mathbf{c}$.

Proof. Hermitian symmetry follows from conjugate symmetry of the inner product. Expanding the squared norm gives $\left\| \sum_{j=1}^{n} c_j \mathbf{u}_j \right\|^2 = \sum_{i,j} \overline{c_i} \langle \mathbf{u}_j, \mathbf{u}_i \rangle\, c_j = \mathbf{c}^* G \mathbf{c}$. The right-hand side is a squared norm, so it is nonnegative.
If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, the squared norm is 0 only when all $c_j = 0$.
Structural Commentary: A Gram matrix records every pairwise inner product in a list of vectors.
Key Insight: The quadratic form of a Gram matrix is exactly the squared norm of a linear combination. □

2 Orthogonality, Projections, and Least Squares

2.1 Orthogonal Complements

Orthogonal complements turn orthogonality into a subspace operation; projections turn it into canonical decompositions and best approximations.

Definition 2.1.1. *Orthogonality, Complements, and Direct Sums*
Let $\mathcal{X}, \mathcal{Y}, \mathcal{U} \subseteq \mathcal{V}$ and let $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$.

i. $\mathcal{X}$ and $\mathcal{Y}$ are *orthogonal*, written $\mathcal{X} \perp \mathcal{Y}$, if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$ for all $\mathbf{x} \in \mathcal{X}$ and $\mathbf{y} \in \mathcal{Y}$.
ii. The *orthogonal complement* of $\mathcal{U}$ is $\mathcal{U}^{\perp} := \{\mathbf{v} \in \mathcal{V} : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in \mathcal{U}\}$.
iii. $\mathcal{V}$ is the *orthogonal direct sum* of $\mathcal{S}$ and $\mathcal{T}$, written $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, if $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{S} \perp \mathcal{T}$.

Proposition 2.1.2. *Basic Properties of Orthogonal Complements*
For subsets $\mathcal{U}, \mathcal{W} \subseteq \mathcal{V}$,

(i) $\mathcal{U}^{\perp}$ is a subspace of $\mathcal{V}$;
(ii) $\{\mathbf{0}\}^{\perp} = \mathcal{V}$;
(iii) $\mathcal{V}^{\perp} = \{\mathbf{0}\}$;
(iv) $\mathcal{U} \cap \mathcal{U}^{\perp} \subseteq \{\mathbf{0}\}$;
(v) if $\mathcal{U} \subseteq \mathcal{W}$, then $\mathcal{W}^{\perp} \subseteq \mathcal{U}^{\perp}$.

Proof.

(i) We have $\mathbf{0} \in \mathcal{U}^{\perp}$. If $\mathbf{v}, \mathbf{w} \in \mathcal{U}^{\perp}$ and $a \in \mathbb{F}$, then for every $\mathbf{u} \in \mathcal{U}$, $\langle a\mathbf{v} + \mathbf{w}, \mathbf{u} \rangle = a \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{w}, \mathbf{u} \rangle = 0$.
(ii) For every $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, \mathbf{0} \rangle = 0$, so $\mathbf{v} \in \{\mathbf{0}\}^{\perp}$.
(iii) If $\mathbf{v} \in \mathcal{V}^{\perp}$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$.
(iv) If $\mathbf{v} \in \mathcal{U} \cap \mathcal{U}^{\perp}$, then $\langle \mathbf{v}, \mathbf{v} \rangle = 0$, so $\mathbf{v} = \mathbf{0}$.
(v) If $\mathbf{v} \in \mathcal{W}^{\perp}$, then $\mathbf{v}$ is orthogonal to every vector in $\mathcal{W}$, hence to every vector in $\mathcal{U}$.

Structural Commentary: Orthogonal complements are solution spaces of homogeneous inner-product equations. Larger input sets impose more equations, so their complements become smaller.
Key Insight: Orthogonality constraints are linear constraints. □

Remark 2.1.3. *Definiteness Matters*
For a bilinear form B, $\mathcal{U}^{\perp} = \{\mathbf{v} : B(\mathbf{v}, \mathbf{u}) = 0 \ \forall \mathbf{u} \in \mathcal{U}\}$ is still a subspace.
Without positive definiteness, however, one may have $\mathcal{V}^{\perp} \neq \{\mathbf{0}\}$ or $\mathcal{U} \cap \mathcal{U}^{\perp} \neq \{\mathbf{0}\}$.
Thus $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$ is an inner-product result, not a consequence of bilinearity alone.

Theorem 2.1.4. *Direct Sum with the Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^{\perp}$.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be an orthonormal basis of $\mathcal{U}$. For $\mathbf{v} \in \mathcal{V}$, define $\mathbf{u} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \mathbf{e}_j$ and set $\mathbf{w} = \mathbf{v} - \mathbf{u}$.

Clearly $\mathbf{u} \in \mathcal{U}$. For each k, $\langle \mathbf{w}, \mathbf{e}_k \rangle = \langle \mathbf{v}, \mathbf{e}_k \rangle - \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \langle \mathbf{e}_j, \mathbf{e}_k \rangle = 0$.
If $\mathbf{x} \in \mathcal{U}$, write $\mathbf{x} = \sum_{k=1}^{m} a_k \mathbf{e}_k$. Then $\langle \mathbf{w}, \mathbf{x} \rangle = \sum_{k=1}^{m} \overline{a_k} \langle \mathbf{w}, \mathbf{e}_k \rangle = 0$, so $\mathbf{w} \in \mathcal{U}^{\perp}$ and $\mathbf{v} = \mathbf{u} + \mathbf{w}$.
Hence $\mathcal{V} = \mathcal{U} + \mathcal{U}^{\perp}$. The intersection is $\{\mathbf{0}\}$ by Proposition 2.1.2, so the sum is direct.
Structural Commentary: Each vector splits into a $\mathcal{U}$-component and an orthogonal residual.
Key Insight: Projection plus residual gives the canonical direct-sum decomposition. □

Proposition 2.1.5. *Orthogonal Direct Sum Characterization*
For subspaces $\mathcal{S}, \mathcal{T} \subseteq \mathcal{V}$, the following are equivalent:

i. $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$.
ii. $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$.

Proof.

(i) $\Rightarrow$ *(ii)* If $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, then $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{T} \subseteq \mathcal{S}^{\perp}$. For $\mathbf{v} \in \mathcal{S}^{\perp}$, write $\mathbf{v} = \mathbf{s} + \mathbf{t}$ with $\mathbf{s} \in \mathcal{S}$, $\mathbf{t} \in \mathcal{T}$. Then $\mathbf{s} = \mathbf{v} - \mathbf{t} \in \mathcal{S}^{\perp}$, so $\mathbf{s} \in \mathcal{S} \cap \mathcal{S}^{\perp} = \{\mathbf{0}\}$ and $\mathbf{v} \in \mathcal{T}$. Hence $\mathcal{T} = \mathcal{S}^{\perp}$.
(ii) $\Rightarrow$ *(i)* If $\mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{T} = \mathcal{S}^{\perp}$, then $\mathcal{S} \perp \mathcal{T}$, so $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$.

Structural Commentary: Orthogonal complements are maximal orthogonal partners.
Key Insight: A full orthogonal summand equals the complement. □

Corollary 2.1.6. *Dimension of the Orthogonal Complement*
If $\mathcal{V}$ is finite-dimensional and $\mathcal{U}$ is a subspace of $\mathcal{V}$, then $\dim(\mathcal{U}^{\perp}) = \dim(\mathcal{V}) - \dim(\mathcal{U})$.

Proof. Since $\mathcal{V}$ is finite-dimensional, $\mathcal{U}$ is finite-dimensional by Corollary 1.3.13. Theorem 2.1.4 gives $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$. Thus Corollary 2.3.2 gives $\dim(\mathcal{V}) = \dim(\mathcal{U}) + \dim(\mathcal{U}^\perp)$.
Structural Commentary: Dimensions add across the orthogonal split.
Key Insight: Orthogonal complements contain exactly the dimensions not used by $\mathcal{U}$. □

Corollary 2.1.7. *Double Orthogonal Complement*
If $\mathcal{U}$ is a finite-dimensional subspace of $\mathcal{V}$, then $\mathcal{U} = (\mathcal{U}^\perp)^\perp$.

Proof. If $\mathbf{u} \in \mathcal{U}$, then $\mathbf{u}$ is orthogonal to every vector in $\mathcal{U}^\perp$, so $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$. Conversely, let $\mathbf{v} \in (\mathcal{U}^\perp)^\perp$. By Theorem 2.1.4, write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^\perp$. Since $\mathcal{U} \subseteq (\mathcal{U}^\perp)^\perp$, we have $\mathbf{w} = \mathbf{v} - \mathbf{u} \in (\mathcal{U}^\perp)^\perp$. Hence $\mathbf{w} \in \mathcal{U}^\perp \cap (\mathcal{U}^\perp)^\perp$, so $\mathbf{w} = \mathbf{0}$ and $\mathbf{v} = \mathbf{u} \in \mathcal{U}$.
Structural Commentary: The orthogonal split leaves no extra direction beyond $\mathcal{U}$.
Key Insight: Taking the orthogonal complement twice recovers the original finite-dimensional subspace. □

Corollary 2.1.8. *Double Complement of a Finite Span*
If $\mathcal{X} \subseteq \mathcal{V}$ and $\operatorname{span}(\mathcal{X})$ is finite-dimensional, then $(\mathcal{X}^\perp)^\perp = \operatorname{span}(\mathcal{X})$.

Proof. Let $W = \operatorname{span}(\mathcal{X})$. As $\mathcal{X}^\perp = W^\perp$, W is finite-dimensional, by Corollary 2.1.7, $(\mathcal{X}^\perp)^\perp = (W^\perp)^\perp = W$.
Structural Commentary: Orthogonal complements depend only on the span of the tested set.
Key Insight: Double orthogonality recovers the finite-dimensional span, not merely the original subset. □

Perspective 2.1.9. *Orthogonal Versus Complementary*
Orthogonality gives $\mathcal{W} \subseteq \mathcal{U}^\perp$ and $\mathcal{U} \subseteq \mathcal{W}^\perp$, not necessarily equality. In finite dimensions, equality follows once $\dim(\mathcal{U}) + \dim(\mathcal{W}) = \dim(\mathcal{V})$ by Corollary 2.1.6.

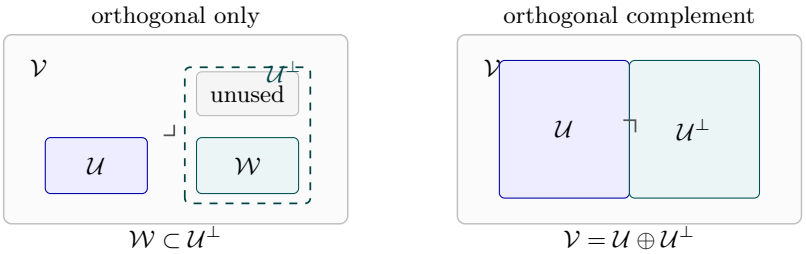


Figure 28: Orthogonal subspaces are complements only when their sum fills $\mathcal{V}$.

2.2 Orthogonal Projections

Proposition 2.2.1. *Orthogonal Projection onto a Line*
Let $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$. Define $c := \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, $\mathbf{w} := \mathbf{u} - c\mathbf{v}$. Then $\mathbf{w} \perp \mathbf{v}$ and $\mathbf{u} = c\mathbf{v} + \mathbf{w}$.
The vector $\operatorname{proj}_{\mathbf{v}}(\mathbf{u}) := c\mathbf{v} = \left\langle \mathbf{u}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle \frac{\mathbf{v}}{\|\mathbf{v}\|}$ is the orthogonal projection of $\mathbf{u}$ onto $\operatorname{span}(\mathbf{v})$.

Proof. The decomposition $\mathbf{u} = c\mathbf{v} + \mathbf{w}$ follows from the definition of $\mathbf{w}$.
Moreover, $\langle \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u} - c\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - c \langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \|\mathbf{v}\|^2 = 0$. Thus $\mathbf{w} \perp \mathbf{v}$.
Structural Commentary: Choose c so the remainder $\mathbf{u} - c\mathbf{v}$ has zero inner product with $\mathbf{v}$.
Key Insight: Projection is the part detected by pairing with the direction vector. □

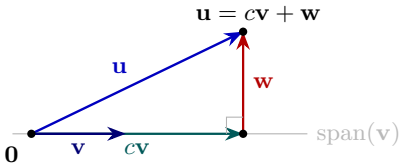


Figure 29: Writing $\mathbf{u}$ as a component parallel to $\mathbf{v}$ plus a component orthogonal to $\mathbf{v}$.

Definition 2.2.2. *Orthogonal Projection onto a Subspace*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$.
For $\mathbf{v} \in \mathcal{V}$, write the unique decomposition $\mathbf{v} = \mathbf{u} + \mathbf{w}$, $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^\perp$, from Theorem 2.1.4.
The *orthogonal projection* of $\mathcal{V}$ onto $\mathcal{U}$ is the map $P_{\mathcal{U}} : \mathcal{V} \to \mathcal{V}$ defined by $P_{\mathcal{U}}\mathbf{v} = \mathbf{u}$.

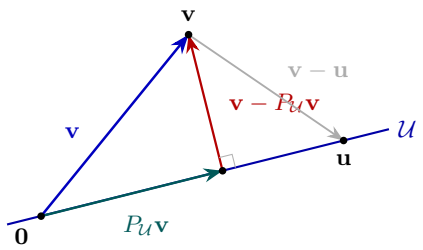


Figure 30: Orthogonal projection decomposes $\mathbf{v}$ into $P_{\mathcal{U}}\mathbf{v}$ plus an orthogonal residual.

Proposition 2.2.3. *Properties of Orthogonal Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. Then:

i. $P_{\mathcal{U}} \in \mathcal{L}(\mathcal{V})$;
ii. $P_{\mathcal{U}}\mathbf{u} = \mathbf{u}$ *for every* $\mathbf{u} \in \mathcal{U}$;
iii. $P_{\mathcal{U}}\mathbf{w} = \mathbf{0}$ *for every* $\mathbf{w} \in \mathcal{U}^{\perp}$;
iv. $\operatorname{range}(P_{\mathcal{U}}) = \mathcal{U}$;
v. $\ker(P_{\mathcal{U}}) = \mathcal{U}^{\perp}$, *i.e. the kernel/nullspace of* $P_{\mathcal{U}}$ *is* $\mathcal{U}^{\perp}$;
vi. $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$ *for every* $\mathbf{v} \in \mathcal{V}$;
vii. $P_{\mathcal{U}}^2 = P_{\mathcal{U}}$;
viii. $\|P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v}\|$ *for every* $\mathbf{v} \in \mathcal{V}$;
ix. if $(\mathbf{e}_1, \dots, \mathbf{e}_m)$ *is an orthonormal basis of* $\mathcal{U}$, *then* $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{m} \langle \mathbf{v}, \mathbf{e}_j \rangle \, \mathbf{e}_j$, $\mathbf{v} \in \mathcal{V}$.

Proof.

(i) For linearity, let $a \in \mathbb{F}$ and write $\mathbf{v}_r = \mathbf{u}_r + \mathbf{w}_r$ with $\mathbf{u}_r \in \mathcal{U}$ and $\mathbf{w}_r \in \mathcal{U}^{\perp}$ for $r = 1, 2$.
Then $a\mathbf{v}_1 + \mathbf{v}_2 = (a\mathbf{u}_1 + \mathbf{u}_2) + (a\mathbf{w}_1 + \mathbf{w}_2)$, where first summand lies in $\mathcal{U}$ and second in $\mathcal{U}^{\perp}$.
By uniqueness of the orthogonal decomposition, $P_{\mathcal{U}}(a\mathbf{v}_1 + \mathbf{v}_2) = aP_{\mathcal{U}}\mathbf{v}_1 + P_{\mathcal{U}}\mathbf{v}_2$.
(ii) This follows from $\mathbf{u} = \mathbf{u} + \mathbf{0}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{0} \in \mathcal{U}^{\perp}$.
(iii) This follows from $\mathbf{w} = \mathbf{0} + \mathbf{w}$ with $\mathbf{0} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{U}^{\perp}$.
(iv) The definition gives $\operatorname{range}(P_{\mathcal{U}}) \subseteq \mathcal{U}$, and (ii) gives $\mathcal{U} \subseteq \operatorname{range}(P_{\mathcal{U}})$.
(v) Part (iii) gives $\mathcal{U}^{\perp} \subseteq \ker(P_{\mathcal{U}})$.
Conversely, if $P_{\mathcal{U}}\mathbf{v} = \mathbf{0}$, then $\mathbf{v} = \mathbf{0} + (\mathbf{v} - P_{\mathcal{U}}\mathbf{v})$ with $\mathbf{v} - P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}^{\perp}$, so $\mathbf{v} \in \mathcal{U}^{\perp}$.
(vi) This is exactly the residual term in the decomposition $\mathbf{v} = P_{\mathcal{U}}\mathbf{v} + (\mathbf{v} - P_{\mathcal{U}}\mathbf{v})$.
(vii) By (ii), $P_{\mathcal{U}}(P_{\mathcal{U}}\mathbf{v}) = P_{\mathcal{U}}\mathbf{v}$ for every $\mathbf{v} \in \mathcal{V}$.
(viii) Write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{U}^{\perp}$. Then $\mathbf{u} \perp \mathbf{w}$ and $\|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{w}\|^2 \ge \|\mathbf{u}\|^2 = \|P_{\mathcal{U}}\mathbf{v}\|^2$.
Since norms are nonnegative, $\|P_{\mathcal{U}}\mathbf{v}\| \le \|\mathbf{v}\|$.
(ix) This is the explicit $\mathcal{U}$-component constructed in the proof of Theorem 2.1.4.

Structural Commentary: Orthogonal projection keeps the $\mathcal{U}$-component and discards the $\mathcal{U}^{\perp}$-component.
Key Insight: $P_{\mathcal{U}}$ is the identity on $\mathcal{U}$ and zero on $\mathcal{U}^{\perp}$. □

Definition 2.2.4. *Rank-One Projection Matrix*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. The *rank-one projection matrix* onto $\operatorname{span}(\mathbf{u})$ is $P_{\mathbf{u}} := \mathbf{u}\mathbf{u}^* \in \operatorname{Mat}(n, n, \mathbb{F})$, where * denotes conjugate transpose. For every $\mathbf{x} \in \mathbb{F}^n$, $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} \mathbf{u}$.

Proposition 2.2.5. *Properties of Rank-One Projection Matrices*
Let $\mathbf{u} \in \mathbb{F}^n$ be a unit vector. Then:

i. $\operatorname{range}(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})$;
ii. $\ker(P_{\mathbf{u}}) = \operatorname{span}(\mathbf{u})^{\perp}$;
iii. $P_{\mathbf{u}}^* = P_{\mathbf{u}}$;
iv. $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.

Proof.

(i) The formula $P_{\mathbf{u}}\mathbf{x} = \langle \mathbf{x}, \mathbf{u} \rangle \mathbf{u}$ gives $\operatorname{range}(P_{\mathbf{u}}) \subseteq \operatorname{span}(\mathbf{u})$, and $P_{\mathbf{u}}\mathbf{u} = \mathbf{u}$ gives the reverse inclusion.
(ii) The same formula gives $P_{\mathbf{u}}\mathbf{x} = 0$ exactly when $\langle \mathbf{x}, \mathbf{u} \rangle \mathbf{u} = \mathbf{0}$.
Since $\mathbf{u} \neq \mathbf{0}$, this is equivalent to $\langle \mathbf{x}, \mathbf{u} \rangle = 0$, i.e. $\mathbf{x} \in \operatorname{span}(\mathbf{u})^{\perp}$.
(iii) Since $P_{\mathbf{u}} = \mathbf{u}\mathbf{u}^*$, $P_{\mathbf{u}}^* = (\mathbf{u}\mathbf{u}^*)^* = \mathbf{u}\mathbf{u}^* = P_{\mathbf{u}}$.
(iv) Because $\mathbf{u}^*\mathbf{u} = \|\mathbf{u}\|_2^2 = 1$, $P_{\mathbf{u}}^2 = \mathbf{u}(\mathbf{u}^*\mathbf{u})\mathbf{u}^* = \mathbf{u}\mathbf{u}^* = P_{\mathbf{u}}$.

Structural Commentary: A rank-one projection keeps the $\mathbf{u}$ direction and kills its orthogonal complement.
Key Insight: The outer product $\mathbf{u}\mathbf{u}^*$ encodes projection onto one unit direction. □

Theorem 2.2.6. *Orthogonal Projection Minimizes Distance*
Let $\mathcal{U}$ be a finite-dimensional subspace of $\mathcal{V}$. If $\mathbf{v} \in \mathcal{V}$ and $\mathbf{u} \in \mathcal{U}$, then $\|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\| \leq \|\mathbf{v} - \mathbf{u}\|$. Equality holds if and only if $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$.

Proof. Set $\mathbf{p} = P_{\mathcal{U}}\mathbf{v}$, $\mathbf{r} = \mathbf{v} - \mathbf{p} \in \mathcal{U}^{\perp}$, and $\mathbf{h} = \mathbf{p} - \mathbf{u} \in \mathcal{U}$.
Then $\mathbf{r} \perp \mathbf{h}$, so $\|\mathbf{v} - \mathbf{u}\|^2 = \|\mathbf{r} + \mathbf{h}\|^2 = \|\mathbf{r}\|^2 + \|\mathbf{h}\|^2 \geq \|\mathbf{r}\|^2 = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|^2$.
($\Rightarrow$) If equality holds, then $\|\mathbf{h}\| = 0$, hence $\mathbf{u} = \mathbf{p} = P_{\mathcal{U}}\mathbf{v}$.
($\Leftarrow$) If $\mathbf{u} = P_{\mathcal{U}}\mathbf{v}$, then equality is immediate.
Structural Commentary: Projection error is orthogonal to every in-subspace correction.
Key Insight: The projection is the unique nearest point. □

Definition 2.2.7. *Fourier Coefficients and Expansion*
Let $\mathcal{O} = (\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be an orthonormal list and $\mathbf{v} \in \mathcal{V}$.
The *Fourier coefficients* of $\mathbf{v}$ relative to $\mathcal{O}$ are $c_j(\mathbf{v}) := \langle \mathbf{v}, \mathbf{e}_j \rangle$, $1 \leq j \leq m$.
The *Fourier expansion* of $\mathbf{v}$ onto $\mathcal{O}$ is $\widehat{\mathbf{v}}_{\mathcal{O}} := \sum_{j=1}^{m} c_j(\mathbf{v})\mathbf{e}_j$.

Theorem 2.2.8. *Fourier Expansion onto a Finite Orthonormal Set*
Let $\mathcal{O} = (\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be orthonormal, let $\mathcal{S} = \operatorname{span}(\mathcal{O})$, and let $\mathbf{v} \in \mathcal{V}$. Then:

i. $\widehat{\mathbf{v}}_{\mathcal{O}}$ *is the unique* $\mathbf{s} \in \mathcal{S}$ *with* $\mathbf{v} - \mathbf{s} \in \mathcal{S}^{\perp}$;
ii. $\widehat{\mathbf{v}}_{\mathcal{O}} = \arg\min_{\mathbf{s} \in \mathcal{S}} \|\mathbf{v} - \mathbf{s}\|$;
iii. $\|\widehat{\mathbf{v}}_{\mathcal{O}}\| \leq \|\mathbf{v}\|$.

Proof.

(i) For $1 \leq k \leq m$, $\langle \mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}}, \mathbf{e}_k \rangle = c_k(\mathbf{v}) - \sum_{j=1}^{m} c_j(\mathbf{v})\delta_{jk} = 0$. Hence $\mathbf{v} - \widehat{\mathbf{v}}_{\mathcal{O}} \in \mathcal{S}^{\perp}$.
(ii) If $\mathbf{s} \in \mathcal{S}$ also satisfies $\mathbf{v} - \mathbf{s} \in \mathcal{S}^{\perp}$, then $\mathbf{s} - \widehat{\mathbf{v}}_{\mathcal{O}} \in \mathcal{S} \cap \mathcal{S}^{\perp} = \{\mathbf{0}\}$.
(iii) By (i), $\widehat{\mathbf{v}}_{\mathcal{O}} = P_{\mathcal{S}}\mathbf{v}$; Theorem 2.2.6 gives the minimizer.
(iv) By Theorem 1.3.22, $\|\widehat{\mathbf{v}}_{\mathcal{O}}\|^2 = \sum_{j=1}^{m} |c_j(\mathbf{v})|^2 \leq \|\mathbf{v}\|^2$, hence $\|\widehat{\mathbf{v}}_{\mathcal{O}}\| \leq \|\mathbf{v}\|$.
Structural Commentary: Fourier expansion is orthogonal projection in orthonormal coordinates.
Key Insight: The coefficients are obtained by pairing with the basis vectors because all cross terms vanish. □

Theorem 2.2.9. *Characterizing Finite Orthonormal Bases*
Let $\mathcal{O} = (\mathbf{e}_1, \ldots, \mathbf{e}_m)$ be orthonormal, let $\mathcal{S} = \operatorname{span}(\mathcal{O})$, and write $c_j(\mathbf{v})$ and $\widehat{\mathbf{v}} := \widehat{\mathbf{v}}_{\mathcal{O}}$ as in Definition 2.2.7. The following are equivalent:

i. $\mathcal{S} = \mathcal{V}$.
ii. $\mathcal{S}^{\perp} = \{\mathbf{0}\}$.
iii. $\widehat{\mathbf{v}} = \mathbf{v}$ *for all* $\mathbf{v} \in \mathcal{V}$.
iv. $\|\widehat{\mathbf{v}}\| = \|\mathbf{v}\|$ *for all* $\mathbf{v} \in \mathcal{V}$.
v. For all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$, $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{m} c_j(\mathbf{v})\overline{c_j(\mathbf{w})}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) If $\mathcal{S} = \mathcal{V}$, then $\mathcal{S}^{\perp} = \mathcal{V}^{\perp} = \{\mathbf{0}\}$.
(*ii*) $\Rightarrow$ (*iii*) By Theorem 2.2.8, $\mathbf{v} - \widehat{\mathbf{v}} \in \mathcal{S}^{\perp} = \{\mathbf{0}\}$.
(*iii*) $\Rightarrow$ (*i*) Since $\widehat{\mathbf{v}} \in \mathcal{S}$ for all $\mathbf{v}$, condition (*iii*) gives $\mathcal{V} \subseteq \mathcal{S}$, hence $\mathcal{S} = \mathcal{V}$.
(*iii*) $\Rightarrow$ (*iv*) If $\widehat{\mathbf{v}} = \mathbf{v}$, then $\|\widehat{\mathbf{v}}\| = \|\mathbf{v}\|$.
(*iv*) $\Rightarrow$ (*iii*) The decomposition $\mathbf{v} = \widehat{\mathbf{v}} + (\mathbf{v} - \widehat{\mathbf{v}})$ is orthogonal, so $\|\mathbf{v}\|^2 = \|\widehat{\mathbf{v}}\|^2 + \|\mathbf{v} - \widehat{\mathbf{v}}\|^2$. Together with (*iv*), this gives $\mathbf{v} = \widehat{\mathbf{v}}$.
(*iii*) $\Rightarrow$ (*v*) If $\widehat{\mathbf{v}} = \mathbf{v}$ and $\widehat{\mathbf{w}} = \mathbf{w}$, then orthonormality gives $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{j=1}^{m} c_j(\mathbf{v})\overline{c_j(\mathbf{w})}$.
(*v*) $\Rightarrow$ (*iv*) Put $\mathbf{w} = \mathbf{v}$ and use $\|\widehat{\mathbf{v}}\|^2 = \sum_{j=1}^{m} |c_j(\mathbf{v})|^2$ to get $\|\mathbf{v}\| = \|\widehat{\mathbf{v}}\|$.
Structural Commentary: Each condition says projection onto $\mathcal{S}$ loses no component.
Key Insight: Basis means lossless projection. □

Definition 2.2.10. *Distance to a Subspace*
Let $\mathcal{U}$ be a nonempty subset of an inner-product space $\mathcal{V}$.
The *distance* from $\mathbf{v} \in \mathcal{V}$ to $\mathcal{U}$ is $d(\mathbf{v}, \mathcal{U}) := \inf_{\mathbf{u} \in \mathcal{U}} \|\mathbf{v} - \mathbf{u}\|$.
If $\mathcal{U}$ is finite-dimensional, Theorem 2.2.6 gives $d(\mathbf{v}, \mathcal{U}) = \|\mathbf{v} - P_{\mathcal{U}}\mathbf{v}\|$.

Perspective 2.2.11. *Why the Finite-Dimensional Hypothesis Matters*
In $\ell^2(\mathbb{F})$, let $\mathcal{S} = \mathbb{F}^{(\mathbb{N})} = \operatorname{span}\{\mathbf{e}_j : j \geq 1\}$.
If $\mathbf{x} \in \mathcal{S}^\perp$, then $x_j = \langle \mathbf{x}, \mathbf{e}_j \rangle = 0 \quad (\forall j)$, so $\mathcal{S}^\perp = \{\mathbf{0}\}$. Since $\mathcal{S} \subsetneq \ell^2(\mathbb{F})$, $\mathcal{S} \oplus \mathcal{S}^\perp \neq \ell^2(\mathbb{F})$.

2.3 Least Squares and Metric Optimization

Least squares is projection theory as optimization: normal equations, full-rank formulas, and minimum-norm solutions.

Principle 2.3.1. *Least Squares as Orthogonal Projection*
In an inner-product space, the best approximation to $\mathbf{v}$ from a finite-dimensional subspace $\mathcal{U}$ is $P_{\mathcal{U}}\mathbf{v}$. The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ is orthogonal to every allowable correction inside $\mathcal{U}$.

Definition 2.3.2. *Matrix Least-Squares Solution*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$.
A vector $\mathbf{x}_0 \in \mathbb{F}^n$ is a *least-squares solution* of $A\mathbf{x} = \mathbf{y}$ if $\|A\mathbf{x}_0 - \mathbf{y}\|_2 = \min\limits_{\mathbf{x} \in \mathbb{F}^n} \|A\mathbf{x} - \mathbf{y}\|_2$.
The vector $A\mathbf{x}_0$ is the fitted vector, and $\mathbf{y} - A\mathbf{x}_0$ is the residual.

Theorem 2.3.3. *Normal Equations for Projection*
Let $\mathcal{U}$ be a finite-dimensional subspace of an inner-product space $\mathcal{V}$, and suppose $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$.
For $\mathbf{v} \in \mathcal{V}$, the projection has the form $P_{\mathcal{U}}\mathbf{v} = \sum\limits_{j=1}^{n} c_j \mathbf{u}_j$, where $(c_1, \ldots, c_n)$ solves the normal equations

$$\begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \langle \mathbf{u}_2, \mathbf{u}_1 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_1 \rangle \\ \langle \mathbf{u}_1, \mathbf{u}_2 \rangle & \langle \mathbf{u}_2, \mathbf{u}_2 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{u}_n \rangle & \langle \mathbf{u}_2, \mathbf{u}_n \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \langle \mathbf{v}, \mathbf{u}_1 \rangle \\ \langle \mathbf{v}, \mathbf{u}_2 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{u}_n \rangle \end{bmatrix}.$$

This system is consistent. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, then the solution is unique.

Proof. Since $P_{\mathcal{U}}\mathbf{v} \in \mathcal{U}$, write $P_{\mathcal{U}}\mathbf{v} = \sum_{j=1}^{n} c_j \mathbf{u}_j$.
The residual $\mathbf{v} - P_{\mathcal{U}}\mathbf{v}$ lies in $\mathcal{U}^\perp$, so for each i, $0 = \left\langle \mathbf{v} - \sum\limits_{j=1}^{n} c_j \mathbf{u}_j, \mathbf{u}_i \right\rangle = \langle \mathbf{v}, \mathbf{u}_i \rangle - \sum\limits_{j=1}^{n} c_j \langle \mathbf{u}_j, \mathbf{u}_i \rangle$.
These equations are exactly the displayed system, so consistency follows from the existence of $P_{\mathcal{U}}\mathbf{v}$.
Conversely, if $(d_1, \ldots, d_n)$ solves displayed system and $\mathbf{q} := \sum_{j=1}^{n} d_j \mathbf{u}_j$, then $\langle \mathbf{v} - \mathbf{q}, \mathbf{u}_i \rangle = 0$ for every i.
Since $\mathbf{u}_i$ span $\mathcal{U}$, it follows that $\mathbf{v} - \mathbf{q} \in \mathcal{U}^\perp$, hence $\mathbf{q} = P_{\mathcal{U}}\mathbf{v}$ by uniqueness of the decomposition. If $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are linearly independent, they form a basis of $\mathcal{U}$, so the coordinates of this projection in that basis are unique.
Structural Commentary: Normal equations express residual orthogonality.
Key Insight: The projection error is perpendicular to $\mathcal{U}$. □

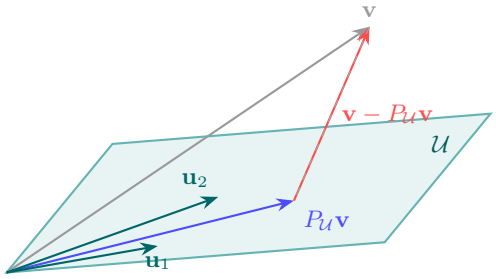


Figure 31: Normal equations impose $\langle \mathbf{v} - P_{\mathcal{U}}\mathbf{v}, \mathbf{u}_i \rangle = 0$ for each spanning vector.

Theorem 2.3.4. *Matrix Least Squares and Normal Equations*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \mathbb{F}^m$, where A^* denotes the conjugate transpose of A.
A vector $\mathbf{x}_0 \in \mathbb{F}^n$ satisfies $\|\mathbf{y} - A\mathbf{x}_0\|_2 = \min_{\mathbf{x} \in \mathbb{F}^n} \|\mathbf{y} - A\mathbf{x}\|_2$ if and only if $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.
The normal equations are always consistent, and their solution is unique if and only if $\operatorname{rank}(A) = n$.

Proof. Let $\mathcal{U} = \operatorname{col}(A)$, and write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Since $\{A\mathbf{x} : \mathbf{x} \in \mathbb{F}^n\} = \mathcal{U}$, Theorem 2.2.6 shows that $\mathbf{x}_0$ is least-squares optimal exactly when $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$, or equivalently when $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$.

($\Rightarrow$) If $\mathbf{x}_0$ is least-squares optimal, then $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$. Hence $\langle \mathbf{y} - A\mathbf{x}_0, \mathbf{a}_j \rangle = 0$, $j = 1, \ldots, n$.
These are the entries of $A^*(\mathbf{y} - A\mathbf{x}_0)$, so $A^*(\mathbf{y} - A\mathbf{x}_0) = \mathbf{0}$ and $A^*A\mathbf{x}_0 = A^*\mathbf{y}$.

($\Leftarrow$) If $A^*A\mathbf{x}_0 = A^*\mathbf{y}$, then $A^*(\mathbf{y} - A\mathbf{x}_0) = \mathbf{0}$. Thus $\langle \mathbf{y} - A\mathbf{x}_0, \mathbf{a}_j \rangle = 0$ for every j, so $\mathbf{y} - A\mathbf{x}_0$ is orthogonal to every linear combination of the columns of A. Hence $\mathbf{y} - A\mathbf{x}_0 \in \mathcal{U}^\perp$.
Hence $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$, so $\mathbf{x}_0$ is least-squares optimal.

Consistency follows by choosing $\mathbf{x}_0$ such that $A\mathbf{x}_0 = P_{\mathcal{U}}\mathbf{y}$.
If $\mathbf{x}_0, \mathbf{x}_1$ solve the normal equations, then $\|A(\mathbf{x}_0 - \mathbf{x}_1)\|_2^2 = \langle \mathbf{x}_0 - \mathbf{x}_1, A^*A(\mathbf{x}_0 - \mathbf{x}_1) \rangle = 0$, so $\mathbf{x}_0 - \mathbf{x}_1 \in \ker(A)$.
Conversely, if $\mathbf{z} \in \ker(A)$ and $\mathbf{x}_0$ solves the normal equations, then $\mathbf{x}_0 + \mathbf{z}$ also solves them. Thus uniqueness is equivalent to $\ker(A) = \{\mathbf{0}\}$, equivalently $\operatorname{rank}(A) = n$.
Structural Commentary: Least squares is orthogonal projection onto the column space of A.
Key Insight: The normal equations say exactly that the residual is orthogonal to every column of A. □

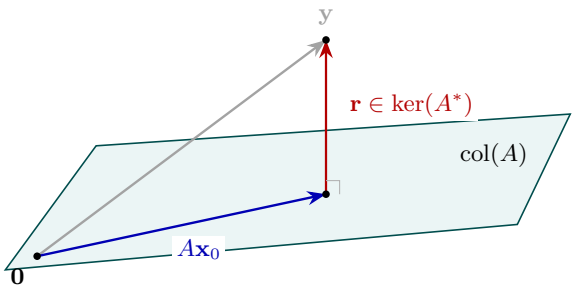


Figure 32: Fitted vector and orthogonal residual: $A\mathbf{x}_0 = P_{\operatorname{col}(A)}\mathbf{y}$ and $\mathbf{r} = \mathbf{y} - A\mathbf{x}_0 \in \operatorname{col}(A)^\perp = \ker(A^*)$.

Proposition 2.3.5. *Least-Squares Solution Set and Minimum Norm*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$, $\mathbf{y} \in \mathbb{F}^m$, *and* $\operatorname{LS}(A, \mathbf{y}) = \arg\min_{\mathbf{x} \in \mathbb{F}^n} \|A\mathbf{x} - \mathbf{y}\|_2$. *If* $\mathbf{x}_0 \in \operatorname{LS}(A, \mathbf{y})$, *then* $\operatorname{LS}(A, \mathbf{y}) = \mathbf{x}_0 + \ker(A)$.
Also $\operatorname{LS}(A, \mathbf{y}) \cap (\ker A)^\perp = \{\mathbf{s}\}$, *where* $\mathbf{s} = P_{(\ker A)^\perp}\mathbf{x}_0 \in \operatorname{col}(A^*)$, *and* $\mathbf{s}$ *is unique minimizer of* $\|\cdot\|_2$ *on* $\operatorname{LS}(A, \mathbf{y})$.

Proof. Set $\mathbf{b} = P_{\operatorname{col}(A)}\mathbf{y}$. By Theorem 2.3.4, $\operatorname{LS}(A, \mathbf{y}) = \{\mathbf{x} \in \mathbb{F}^n : A\mathbf{x} = \mathbf{b}\}$.
Thus, for $\mathbf{x}_0 \in \operatorname{LS}(A, \mathbf{y})$, $\operatorname{LS}(A, \mathbf{y}) = \{\mathbf{x} : A(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}\} = \mathbf{x}_0 + \ker(A)$.
Write $\mathbf{x}_0 = \mathbf{s} + \mathbf{k}$ with $\mathbf{s} \in (\ker A)^\perp$ and $\mathbf{k} \in \ker(A)$. Since $A\mathbf{s} = A\mathbf{x}_0$, $\mathbf{s} \in \operatorname{LS}(A, \mathbf{y}) \cap (\ker A)^\perp$.
If $\mathbf{u}$ is another element of this intersection, then $\mathbf{u} - \mathbf{s} \in \ker(A) \cap (\ker A)^\perp = \{\mathbf{0}\}$.
Finally, any $\mathbf{x} \in \operatorname{LS}(A, \mathbf{y})$ has $\mathbf{x} = \mathbf{s} + \mathbf{z}$ with $\mathbf{z} \in \ker(A)$, so $\|\mathbf{x}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{z}\|_2^2 \geq \|\mathbf{s}\|_2^2$, with equality iff $\mathbf{z} = \mathbf{0}$.
Structural Commentary: Least-squares solutions form an affine translate of the nullspace.
Key Insight: The minimum-norm representative is the one with no nullspace component. □

Corollary 2.3.6. *Projection Formula for Full Column Rank*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ *have rank* n.
Then A^*A *is invertible and the orthogonal projection onto* $\operatorname{col}(A)$ *is* $P_{\operatorname{col}(A)} = A(A^*A)^{-1}A^*$.

Proof. If $A^*A\mathbf{x} = \mathbf{0}$, then $\|A\mathbf{x}\|_2^2 = \langle \mathbf{x}, A^*A\mathbf{x} \rangle = 0$, so $A\mathbf{x} = \mathbf{0}$.
Since $\operatorname{rank}(A) = n$, we get $\mathbf{x} = \mathbf{0}$. Hence A^*A is invertible. By Theorem 2.3.4, the least-squares solution is $\mathbf{x}_0 = (A^*A)^{-1}A^*\mathbf{y}$. The fitted vector is $P_{\operatorname{col}(A)}\mathbf{y} = A\mathbf{x}_0 = A(A^*A)^{-1}A^*\mathbf{y}$.
Since this holds for every $\mathbf{y}$, the displayed operator formula follows.
Structural Commentary: Full column rank makes the normal-equation coordinates unique.
Key Insight: Solve for coefficients, then map back by A. □

Definition 2.3.7. *Minimum Norm Solution*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ and $\mathbf{y} \in \operatorname{col}(A)$.
A solution $\mathbf{s}$ of $A\mathbf{x} = \mathbf{y}$ is a *minimum norm solution* if $\|\mathbf{s}\|_2 \leq \|\mathbf{x}\|_2$ for every solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{y}$.

Theorem 2.3.8. *Minimum Norm Solution of a Consistent Linear System*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$, $\mathbf{y} \in \mathbb{F}^m$, *and suppose* $A\mathbf{x} = \mathbf{y}$ *is consistent.*

i. There is a unique minimum norm solution $\mathbf{s} \in \mathbb{F}^n$, *and* $\mathbf{s} \in \operatorname{col}(A^*)$.
ii. $\mathbf{s}$ *is the only solution of* $A\mathbf{x} = \mathbf{y}$ *that lies in* $\operatorname{col}(A^*)$.
iii. The system $AA^*\mathbf{u} = \mathbf{y}$ *is consistent. If* $AA^*\mathbf{u}_0 = \mathbf{y}$, *then* $\mathbf{s} = A^*\mathbf{u}_0$.

Proof. First note that $\operatorname{col}(A^*) = (\ker A)^\perp$. Indeed, if $\mathbf{z} = A^*\mathbf{u}$ and $\mathbf{x} \in \ker(A)$, then $\langle \mathbf{x}, \mathbf{z} \rangle = \langle A\mathbf{x}, \mathbf{u} \rangle = 0$, so $\mathbf{z}$ is orthogonal to every vector in $\ker(A)$. Thus $\operatorname{col}(A^*) \subseteq (\ker A)^\perp$. Also,

$$\dim(\operatorname{col}(A^*)) = \operatorname{rank}(A^*) = \operatorname{rank}(A) = n - \dim(\ker(A)) = \dim((\ker A)^\perp),$$

so the two subspaces are equal. Hence $\mathbb{F}^n = \operatorname{col}(A^*) \oplus \ker(A)$ orthogonally.

(i) Choose a solution $\mathbf{x}_0$ of $A\mathbf{x} = \mathbf{y}$ and write $\mathbf{x}_0 = \mathbf{s} + \mathbf{v}$, with $\mathbf{s} \in \operatorname{col}(A^*)$ and $\mathbf{v} \in \ker(A)$. Then $A\mathbf{s} = A\mathbf{x}_0 = \mathbf{y}$. If $\mathbf{x}$ is any solution, then $\mathbf{x} - \mathbf{s} \in \ker(A)$, so every solution has the form $\mathbf{s} + \mathbf{w}$ with $\mathbf{w} \in \ker(A)$. Since $\mathbf{s} \perp \mathbf{w}$, $\|\mathbf{s} + \mathbf{w}\|_2^2 = \|\mathbf{s}\|_2^2 + \|\mathbf{w}\|_2^2 \geq \|\mathbf{s}\|_2^2$, with equality only when $\mathbf{w} = \mathbf{0}$. Thus $\mathbf{s}$ is the unique solution of minimum norm.

(ii) If $\mathbf{s}' \in \operatorname{col}(A^*)$ is another solution, then $\mathbf{s}' - \mathbf{s} \in \ker(A) \cap \operatorname{col}(A^*) = \{\mathbf{0}\}$, so $\mathbf{s}' = \mathbf{s}$.

(iii) Since $\mathbf{s} \in \operatorname{col}(A^*)$, write $\mathbf{s} = A^*\mathbf{z}$. Then $AA^*\mathbf{z} = A\mathbf{s} = \mathbf{y}$, so $AA^*\mathbf{u} = \mathbf{y}$ is consistent. Conversely, if $AA^*\mathbf{u}_0 = \mathbf{y}$, then $A^*\mathbf{u}_0$ is a solution in $\operatorname{col}(A^*)$, so $A^*\mathbf{u}_0 = \mathbf{s}$ by (ii).

Structural Commentary: Solutions are $\mathbf{s} + \ker(A)$; $\operatorname{col}(A^*)$ meets this set orthogonally.
Key Insight: The minimum norm solution has no nullspace component. □

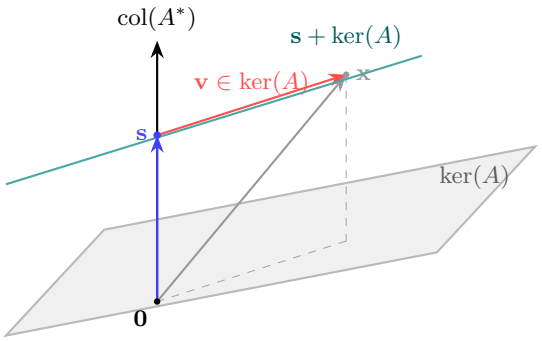


Figure 33: Among all solutions $\mathbf{s} + \ker(A)$, the unique one in $\operatorname{col}(A^*) = (\ker A)^\perp$ has minimum norm.

Perspective 2.3.9. *Roadmap to the Pseudoinverse*
Theorem 2.3.4 projects $\mathbf{y}$ onto $\operatorname{col}(A)$; Proposition 2.3.5 selects the minimum-norm preimage in $\operatorname{col}(A^*)$. Definition 5.1.16 and Theorem 5.1.21 package both choices as $A^\dagger\mathbf{y}$.

2.4 Adjoints and Orthogonal Operators

Adjoints express dual action in inner-product language. They become conjugate transposes in orthonormal coordinates and lead to self-adjoint maps, projections, isometries, and unitary or orthogonal operators.

Definition 2.4.1. *Adjoint of a Linear Map*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
A function $S : \mathcal{W} \to \mathcal{V}$ is an *adjoint* of T if $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, S\mathbf{w}\rangle_{\mathcal{V}}$, $\mathbf{v} \in \mathcal{V}$, $\mathbf{w} \in \mathcal{W}$.

Perspective 2.4.2. *Dual Map and Inner-Product Adjoint*
For $T : \mathcal{V} \to \mathcal{W}$, the pullback $\varphi \mapsto \varphi \circ T$ is the *dual map* $T' : \mathcal{W}' \to \mathcal{V}'$. Also called *operator adjoint* and written $T^\times$. It is distinct from the inner-product adjoint T^* of Definition 2.4.1, which requires inner products.

Lemma 2.4.3. *Uniqueness of Adjoints*
A linear map has at most one adjoint.

Proof. Suppose $S_1, S_2 : \mathcal{W} \to \mathcal{V}$ are adjoints of T. For each $\mathbf{w} \in \mathcal{W}$ and every $\mathbf{v} \in \mathcal{V}$, $\langle \mathbf{v}, S_1\mathbf{w} - S_2\mathbf{w}\rangle_{\mathcal{V}} = 0$.
Taking $\mathbf{v} = S_1\mathbf{w} - S_2\mathbf{w}$ gives $S_1\mathbf{w} = S_2\mathbf{w}$. Hence $S_1 = S_2$.
Structural Commentary: Same tests force candidate adjoints to agree.
Key Insight: Inner products separate vectors. □

Theorem 2.4.4. *Existence of the Adjoint*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $\mathcal{W}$ be an inner-product space, and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. Then T has a unique adjoint $T^* : \mathcal{W} \to \mathcal{V}$, and T^* is linear.

Proof. For each $\mathbf{w} \in \mathcal{W}$, the map $\varphi_{\mathbf{w}} : \mathcal{V} \to \mathbb{F}$ defined by $\varphi_{\mathbf{w}}(\mathbf{v}) = \langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}}$ is a linear functional. By Theorem 1.3.27, there is a unique vector $T^*\mathbf{w} \in \mathcal{V}$ such that $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}}$ for every $\mathbf{v} \in \mathcal{V}$.
Thus T^* is an adjoint, and it is unique by Lemma 2.4.3. For $\mathbf{w}_1, \mathbf{w}_2 \in \mathcal{W}$ and $a \in \mathbb{F}$,

$$\langle \mathbf{v}, T^*(a\mathbf{w}_1 + \mathbf{w}_2)\rangle_{\mathcal{V}} = \langle T\mathbf{v}, a\mathbf{w}_1 + \mathbf{w}_2\rangle_{\mathcal{W}} = \overline{a}\,\langle T\mathbf{v}, \mathbf{w}_1\rangle_{\mathcal{W}} + \langle T\mathbf{v}, \mathbf{w}_2\rangle_{\mathcal{W}} = \langle \mathbf{v}, aT^*\mathbf{w}_1 + T^*\mathbf{w}_2\rangle_{\mathcal{V}}\,.$$

for every $\mathbf{v} \in \mathcal{V}$. Taking the difference of the two second-slot vectors and then setting $\mathbf{v}$ equal to that difference gives $T^*(a\mathbf{w}_1 + \mathbf{w}_2) = aT^*\mathbf{w}_1 + T^*\mathbf{w}_2$.
Structural Commentary: Each $\mathbf{w}$ gives a Riesz-represented functional on $\mathcal{V}$.
Key Insight: The adjoint is pointwise Riesz representation for T. □

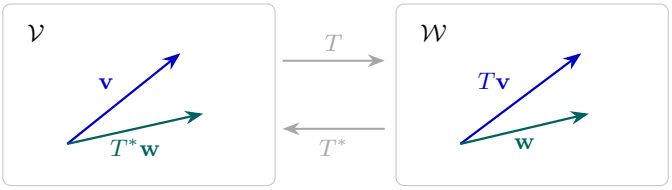


Figure 34: Adjoint relation: $\langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}}$.

Perspective 2.4.5. *Hilbert Adjoints as Riesz-Transported Dual Maps*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces, and let $R_{\mathcal{V}} : \mathcal{V}' \to \mathcal{V}$ and $R_{\mathcal{W}} : \mathcal{W}' \to \mathcal{W}$ send each functional to its Riesz vector from Definition 1.3.28. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then the dual map $T' : \mathcal{W}' \to \mathcal{V}'$ from Definition 5.1.11 and the Hilbert-space adjoint T^* satisfy $T' = \big(R_{\mathcal{V}}\big)^{-1} \circ T^* \circ R_{\mathcal{W}}$.
Indeed, for $\varphi \in \mathcal{W}'$ and $\mathbf{v} \in \mathcal{V}$, $(T'\varphi)(\mathbf{v}) = \varphi(T\mathbf{v}) = \langle T\mathbf{v}, R_{\mathcal{W}}\varphi\rangle_{\mathcal{W}} = \langle \mathbf{v}, T^*R_{\mathcal{W}}\varphi\rangle_{\mathcal{V}}$. Thus the algebraic pullback of functionals is the same operation as moving Riesz vectors by T^* and then returning to functionals.
Structural Commentary: The dual map belongs to the algebraic theory of Definition 5.1.11; the adjoint appears only after the Riesz identification from Theorem 1.3.27.
Key Insight: T' and T^* encode the same reversed action after functionals have been identified with vectors.

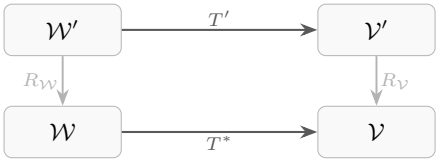


Figure 35: Riesz transports the dual map to the adjoint: $R_{\mathcal{V}}T' = T^*R_{\mathcal{W}}$.

Definition 2.4.6. *Conjugate Transpose*
For $A = [a_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$, the *conjugate transpose* of A is $A^* = [\overline{a_{ji}}] \in \mathrm{Mat}(n, m, \mathbb{F})$.
Over $\mathbb{R}$, this is the ordinary transpose.

Theorem 2.4.7. *Matrix of the Adjoint*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces. Let $\beta = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ be an orthonormal basis of $\mathcal{V}$ and let $\gamma = (\mathbf{f}_1, \dots, \mathbf{f}_m)$ be an orthonormal basis of $\mathcal{W}$. If $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, then $[T^*]_{\beta\leftarrow\gamma} = \big([T]_{\gamma\leftarrow\beta}\big)^*$.

Proof. Let $A = [T]_{\gamma\leftarrow\beta}$ and $B = [T^*]_{\beta\leftarrow\gamma}$.
By Theorem 1.3.25, $B_{i,j} = \langle T^*\mathbf{f}_j, \mathbf{e}_i\rangle_{\mathcal{V}} = \overline{\langle \mathbf{e}_i, T^*\mathbf{f}_j\rangle_{\mathcal{V}}} = \overline{\langle T\mathbf{e}_i, \mathbf{f}_j\rangle_{\mathcal{W}}} = \overline{A_{j,i}}$. Thus $B = A^*$.
Structural Commentary: Entries swap by moving T across the inner product.
Key Insight: In orthonormal coordinates, taking adjoints is exactly taking conjugate transposes. □

Proposition 2.4.8. *Basic Adjoint Identities*
For finite-dimensional inner-product spaces and composable linear maps:

i. $(aS + T)^* = \overline{a}S^* + T^*$ *for* $S, T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ *and* $a \in \mathbb{F}$;
ii. $(SR)^* = R^*S^*$ *for* $R \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ *and* $S \in \mathcal{L}(\mathcal{V}, \mathcal{W})$;
iii. $(T^*)^* = T$;
iv. $I_{\mathcal{V}}^* = I_{\mathcal{V}}$.

Proof. Verify the defining identity; uniqueness gives equality.

(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{w} \in \mathcal{W}$, $\langle (aS + T)\mathbf{v}, \mathbf{w}\rangle = a\langle S\mathbf{v}, \mathbf{w}\rangle + \langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, \overline{a}S^*\mathbf{w} + T^*\mathbf{w}\rangle$, so $(aS+T)^* = \overline{a}S^* + T^*$.
(ii) For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, $\langle SR\mathbf{u}, \mathbf{w}\rangle = \langle R\mathbf{u}, S^*\mathbf{w}\rangle = \langle \mathbf{u}, R^*S^*\mathbf{w}\rangle$, so $(SR)^* = R^*S^*$.
(iii) Conjugating $\langle T\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, T^*\mathbf{w}\rangle$ gives $\langle T^*\mathbf{w}, \mathbf{v}\rangle = \langle \mathbf{w}, T\mathbf{v}\rangle$, so T is the adjoint of T^*.
(iv) Since $\langle I\mathbf{v}, \mathbf{w}\rangle = \langle \mathbf{v}, I\mathbf{w}\rangle$, the identity map is its own adjoint.

Structural Commentary: Put both sides in one inner-product slot, then use uniqueness.
Key Insight: Like matrix transposes, adjoints reverse products and conjugate scalars. □

Proposition 2.4.9. *Adjoints of Inverses and Polynomials*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space. For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$, set $p^{\#}(z) = \sum_{j=0}^{m} \overline{a_j} z^j$.

(i) If T is invertible, then T^* is invertible and $(T^{-1})^* = (T^*)^{-1}$.
(ii) $p(T)^* = p^{\#}(T^*)$. In particular, $p \in \mathbb{R}[z] \Rightarrow p(T)^* = p(T^*)$.

Proof.

(i) Proposition 2.4.8 gives $(T^{-1})^*T^* = (TT^{-1})^* = I_{\mathcal{V}}$ and $T^*(T^{-1})^* = (T^{-1}T)^* = I_{\mathcal{V}}$.

(ii) By conjugate-linearity in scalars and product reversal, $p(T)^* = \left(\sum_{j=0}^{m} a_j T^j\right)^* = \sum_{j=0}^{m} \overline{a_j}(T^*)^j = p^{\#}(T^*)$.

Structural Commentary: Adjoint reverses multiplication and conjugates scalars.
Key Insight: Adjoints send $p(T)$ to $p^{\#}(T^*)$. □

Theorem 2.4.10. *Adjoints and Invariant Orthogonal Complements*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space and let $\mathcal{U} \leq \mathcal{V}$.

(i) $\mathcal{U} \in \mathrm{Inv}(T) \iff \mathcal{U}^\perp \in \mathrm{Inv}(T^*)$.
(ii) $(\mathcal{U}, \mathcal{U}^\perp)$ *reduces* $T \iff \mathcal{U} \in \mathrm{Inv}(T) \cap \mathrm{Inv}(T^*)$.
(iii) If $(\mathcal{U}, \mathcal{U}^\perp)$ *reduces* T, *then* $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$.

Proof.

(i) Since $\mathcal{U} = (\mathcal{U}^\perp)^\perp$, $\mathcal{U} \in \mathrm{Inv}(T) \iff \langle T\mathbf{u}, \mathbf{z}\rangle = 0 \quad (\mathbf{u} \in \mathcal{U},\ \mathbf{z} \in \mathcal{U}^\perp)$
$\iff \langle \mathbf{u}, T^*\mathbf{z}\rangle = 0$, where $\mathbf{u} \in \mathcal{U},\ \mathbf{z} \in \mathcal{U}^\perp \iff \mathcal{U}^\perp \in \mathrm{Inv}(T^*)$.
(ii) By definition and then (i) applied to T^*, $(\mathcal{U}, \mathcal{U}^\perp)$ reduces $T \iff \mathcal{U}, \mathcal{U}^\perp \in \mathrm{Inv}(T) \iff \mathcal{U} \in \mathrm{Inv}(T) \cap \mathrm{Inv}(T^*)$.
(iii) By (ii), $(T^*)|_{\mathcal{U}}$ is well-defined. For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$,
$\langle T|_{\mathcal{U}}\mathbf{u}, \mathbf{v}\rangle_{\mathcal{U}} = \langle T\mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}} = \langle \mathbf{u}, T^*\mathbf{v}\rangle_{\mathcal{V}} = \langle \mathbf{u}, (T^*)|_{\mathcal{U}}\mathbf{v}\rangle_{\mathcal{U}}$, so uniqueness of adjoints on $\mathcal{U}$ gives the claim.

Structural Commentary: Adjoints trade invariance with orthogonal-complement invariance.
Key Insight: A subspace reduces T precisely when it is stable under both T and T^*. □

Proposition 2.4.11. *Adjoint Kernel-Range Orthogonality*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then $\ker(T^*) = (\mathrm{range}(T))^\perp$, $\mathrm{range}(T^*) = (\ker(T))^\perp$, *and* $\ker(T) = (\mathrm{range}(T^*))^\perp$, $\mathrm{range}(T) = (\ker(T^*))^\perp$.
Equivalently, the nullspace of an adjoint is the orthogonal complement of the range of the original map.

Proof. For $\mathbf{w} \in \mathcal{W}$,

$$\begin{aligned} \mathbf{w} \in \ker(T^*) &\iff \langle \mathbf{v}, T^*\mathbf{w}\rangle_{\mathcal{V}} = 0 \quad \text{for every } \mathbf{v} \in \mathcal{V} \\ &\iff \langle T\mathbf{v}, \mathbf{w}\rangle_{\mathcal{W}} = 0 \quad \text{for every } \mathbf{v} \in \mathcal{V} \\ &\iff \mathbf{w} \in (\mathrm{range}(T))^\perp. \end{aligned}$$

Applying this identity to T^* gives $\ker(T) = (\mathrm{range}(T^*))^\perp$.
Since all spaces are finite-dimensional, the ranges and kernels are finite-dimensional subspaces.
Taking orthogonal complements and using Corollary 2.1.7 gives

$$(\ker(T^*))^\perp = ((\mathrm{range}(T))^\perp)^\perp = \mathrm{range}(T), \qquad (\ker(T))^\perp = ((\mathrm{range}(T^*))^\perp)^\perp = \mathrm{range}(T^*)$$

Structural Commentary: Adjoints trade ranges for kernels, and orthogonal complements reverse that trade.
Key Insight: A vector is killed by T^* exactly when it is orthogonal to everything T can produce. □

Proposition 2.4.12. *Adjoint Products and Algebraic Projections*
Let $\mathcal{V}$ and $\mathcal{W}$ be finite-dimensional inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) $\ker(T^*T) = \ker(T)$ *and* $\ker(TT^*) = \ker(T^*)$.
(ii) $\mathrm{range}(T^*T) = \mathrm{range}(T^*)$ *and* $\mathrm{range}(TT^*) = \mathrm{range}(T)$.
(iii) If $\mathcal{V} = \mathcal{U} \oplus \mathcal{Z}$, *then* $\left(\rho_{\mathcal{U},\mathcal{Z}}\right)^* = \rho_{\mathcal{Z}^\perp,\mathcal{U}^\perp}$.

Proof.

(i) For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} \in \ker(T^*T) \iff 0 = \langle T^*T\mathbf{v}, \mathbf{v}\rangle_{\mathcal{V}} = \|T\mathbf{v}\|^2 \iff \mathbf{v} \in \ker(T)$.
Apply this to T^* for $\ker(TT^*) = \ker(T^*)$.
(ii) The inclusions $\mathrm{range}(T^*T) \subseteq \mathrm{range}(T^*)$ and $\mathrm{range}(TT^*) \subseteq \mathrm{range}(T)$ are immediate.
By (i), rank-nullity, and Proposition 2.4.11, $\dim \mathrm{range}(T^*T) = \dim \mathcal{V} - \dim \ker(T) = \dim \mathrm{range}(T^*)$, so equality holds; the TT^* identity follows similarly.
(iii) Put $P = \rho_{\mathcal{U},\mathcal{Z}}$ and $Q = \rho_{\mathcal{Z}^\perp,\mathcal{U}^\perp}$. Since $\mathcal{Z}^\perp \cap \mathcal{U}^\perp = (\mathcal{U} + \mathcal{Z})^\perp = \{\mathbf{0}\}$ and the dimensions add, $\mathcal{V} = \mathcal{Z}^\perp \oplus \mathcal{U}^\perp$.
Write $\mathbf{x} = \mathbf{u} + \mathbf{z}$ and $\mathbf{y} = \mathbf{a} + \mathbf{b}$ with $\mathbf{u} \in \mathcal{U}$, $\mathbf{z} \in \mathcal{Z}$, $\mathbf{a} \in \mathcal{Z}^\perp$, $\mathbf{b} \in \mathcal{U}^\perp$.
Then $\langle P\mathbf{x}, \mathbf{y}\rangle = \langle \mathbf{u}, \mathbf{a} + \mathbf{b}\rangle = \langle \mathbf{u}, \mathbf{a}\rangle = \langle \mathbf{u} + \mathbf{z}, \mathbf{a}\rangle = \langle \mathbf{x}, Q\mathbf{y}\rangle$, so $P^* = Q$.

Structural Commentary: Oblique projection adjoints swap orthogonal-complement summands.
Key Insight: Orthogonal projections are exactly the self-adjoint special case $\mathcal{Z} = \mathcal{U}^\perp$. □

Corollary 2.4.13. *Matrix Fundamental Subspaces as Orthogonal Complements*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ act as $A:\mathbb{F}^n \to \mathbb{F}^m$ with standard inner products.
Then $\ker(A) = \mathrm{col}(A^*)^\perp$ and $\mathrm{col}(A^*) = \ker(A)^\perp$; $\ker(A^*) = \mathrm{col}(A)^\perp$ and $\mathrm{col}(A) = \ker(A^*)^\perp$.
Over $\mathbb{R}$, this says $\ker(A) \perp \mathrm{row}(A)$ in $\mathbb{R}^n$ and $\ker(A^t) \perp \mathrm{col}(A)$ in $\mathbb{R}^m$, with both pairs complementary.

Proof. By Theorem 2.4.7, the adjoint of A is represented by A^*.
Proposition 2.4.11 gives $\ker(A^*) = \mathrm{col}(A)^\perp$ and $\ker(A) = \mathrm{col}(A^*)^\perp$. Taking orthogonal complements and using Corollary 2.1.7 gives the two column-space identities. Over $\mathbb{R}$, $A^* = A^t$ and $\mathrm{col}(A^t) = \mathrm{row}(A)$.
Structural Commentary: The four fundamental subspaces form two orthogonal-complement pairs.
Key Insight: Adjoints turn missing directions into kernels. □

Definition 2.4.14. *Self-Adjoint and Hermitian*
An operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space is *self-adjoint* if $T^* = T$. A square matrix A is *Hermitian* if $A^* = A$. In orthonormal coordinates, these are the same condition by Theorem 2.4.7.

Lemma 2.4.15. *Zero Complex Quadratic Form*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
If $\langle T\mathbf{v}, \mathbf{v}\rangle = 0$ for every $\mathbf{v} \in \mathcal{V}$, then $T = 0$.

Proof. Set $Q(\mathbf{x}) = \langle T\mathbf{x}, \mathbf{x}\rangle$.
For $\mathbf{u}, \mathbf{w} \in \mathcal{V}$, expansion gives the polarization identity for the sesquilinear form $(\mathbf{u},\mathbf{w}) \mapsto \langle T\mathbf{u}, \mathbf{w}\rangle$:

$$\langle T\mathbf{u}, \mathbf{w}\rangle = \frac{Q(\mathbf{u}+\mathbf{w}) - Q(\mathbf{u}-\mathbf{w})}{4} + i\,\frac{Q(\mathbf{u}+i\mathbf{w}) - Q(\mathbf{u}-i\mathbf{w})}{4}.$$

The hypothesis makes every term on the right zero, so $\langle T\mathbf{u}, \mathbf{w}\rangle = 0$ for all $\mathbf{u}, \mathbf{w}$.
Taking $\mathbf{w} = T\mathbf{u}$ gives $\|T\mathbf{u}\|^2 = 0$, hence $T\mathbf{u} = \mathbf{0}$ for every $\mathbf{u}$.
Structural Commentary: Over $\mathbb{C}$, diagonal values of a sesquilinear form determine all mixed values.
Key Insight: Polarization turns the one-vector condition into a two-vector condition. □

Proposition 2.4.16. *Self-Adjointness from Real Quadratic Values*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$.
Then T is self-adjoint if and only if $\langle T\mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. For each $\mathbf{v}$, $\langle (T - T^*)\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle - \langle T^*\mathbf{v}, \mathbf{v}\rangle = \langle T\mathbf{v}, \mathbf{v}\rangle - \overline{\langle T\mathbf{v}, \mathbf{v}\rangle}$.
If all values $\langle T\mathbf{v}, \mathbf{v}\rangle$ are real, this expression is zero for every $\mathbf{v}$, and Lemma 2.4.15 gives $T - T^* = 0$.
Conversely, if $T = T^*$, then $\langle T\mathbf{v}, \mathbf{v}\rangle = \langle \mathbf{v}, T\mathbf{v}\rangle = \overline{\langle T\mathbf{v}, \mathbf{v}\rangle}$, so $\langle T\mathbf{v}, \mathbf{v}\rangle$ is real.
Structural Commentary: The imaginary part of $\langle T\mathbf{v}, \mathbf{v}\rangle$ is exactly the diagonal trace of $T - T^*$.
Key Insight: Over $\mathbb{C}$, real quadratic values force the operator to equal its adjoint. □

Proposition 2.4.17. *Zero Quadratic Form for Self-Adjoint Operators*
Let T be a self-adjoint operator on a finite-dimensional inner-product space $\mathcal{V}$.
If $\langle T\mathbf{v}, \mathbf{v}\rangle = 0$ for every $\mathbf{v} \in \mathcal{V}$, then $T = 0$.

Proof. If $\mathbb{F} = \mathbb{C}$, this is Lemma 2.4.15. Assume $\mathbb{F} = \mathbb{R}$.
For $\mathbf{u}, \mathbf{w} \in \mathcal{V}$, self-adjointness gives $\langle T\mathbf{w}, \mathbf{u}\rangle = \langle \mathbf{w}, T\mathbf{u}\rangle = \langle T\mathbf{u}, \mathbf{w}\rangle$, so the real polarization identity gives

$$\langle T\mathbf{u}, \mathbf{w}\rangle = \frac{\langle T(\mathbf{u}+\mathbf{w}), \mathbf{u}+\mathbf{w}\rangle - \langle T(\mathbf{u}-\mathbf{w}), \mathbf{u}-\mathbf{w}\rangle}{4} = 0.$$

Taking $\mathbf{w} = T\mathbf{u}$ gives $T\mathbf{u} = \mathbf{0}$ for every $\mathbf{u}$.
Structural Commentary: Self-adjointness supplies the missing symmetry needed for real polarization.
Key Insight: A self-adjoint operator is determined by its quadratic form. □

Definition 2.4.18. *Orthogonal Projection Matrix*
A square matrix $P \in \mathrm{Mat}(n,n,\mathbb{F})$ is an *orthogonal projection matrix* if $P^2 = P$ and $P^* = P$.

Theorem 2.4.19. *Orthogonal Projections as Self-Adjoint Idempotents*
Let $\mathcal{V}$ be finite-dimensional and let $P \in \mathcal{L}(\mathcal{V})$.
Then P is the orthogonal projection onto $\mathrm{range}(P)$ if and only if $P^2 = P$ and $P^* = P$.

Proof.

(⇒) If $P = P_{\mathcal{U}}$ for $\mathcal{U} = \mathrm{range}(P)$, then Proposition 2.2.3 gives $P^2 = P$. For $\mathbf{v} = \mathbf{u} + \mathbf{w}$ and $\mathbf{x} = \mathbf{a} + \mathbf{b}$ with $\mathbf{u}, \mathbf{a} \in \mathcal{U}$ and $\mathbf{w}, \mathbf{b} \in \mathcal{U}^\perp$, $\langle P\mathbf{v}, \mathbf{x}\rangle = \langle \mathbf{u}, \mathbf{a}+\mathbf{b}\rangle = \langle \mathbf{u}, \mathbf{a}\rangle = \langle \mathbf{u}+\mathbf{w}, \mathbf{a}\rangle = \langle \mathbf{v}, P\mathbf{x}\rangle$, so $P^* = P$.

($\Leftarrow$) Suppose $P^2 = P$ and $P^* = P$. For each $\mathbf{v}$, $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ with $P\mathbf{v} \in \text{range}(P)$ and $P(\mathbf{v} - P\mathbf{v}) = \mathbf{0}$. Thus the residual lies in $\ker(P)$. Since $P = P^*$, Proposition 2.4.11 gives $\ker(P) = (\text{range}(P))^\perp$. Hence every $\mathbf{v}$ has the orthogonal decomposition $\mathbf{v} = P\mathbf{v} + (\mathbf{v} - P\mathbf{v})$ with first part in $\text{range}(P)$, so P is exactly the orthogonal projection onto $\text{range}(P)$.

Structural Commentary: Idempotence splits by range and kernel; self-adjointness makes the split orthogonal.
Key Insight: An idempotent is orthogonal exactly when $\ker(P) = (\text{range}(P))^\perp$. □

Theorem 2.4.20. *Unitary Coordinate Form of an Orthogonal Projection*
Let $P \in \text{Mat}(n, n, \mathbb{F})$ be an orthogonal projection matrix and let $r = \text{rank}(P)$. Then P is unitarily similar to $I_r \oplus 0_{n-r}$. Equivalently, there is a unitary matrix U such that

$$U^*PU = \begin{bmatrix} I_r & 0 \\ 0 & 0_{n-r} \end{bmatrix}.$$

Proof. Let $\mathcal{U} = \text{col}(P)$. Identifying P with its induced operator on $\mathbb{F}^n$, Theorem 2.4.19 says P is the orthogonal projection onto $\mathcal{U}$, so P is the identity on $\mathcal{U}$ and zero on $\mathcal{U}^\perp$. Choose an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_r$ of $\mathcal{U}$ and extend it to an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ of $\mathbb{F}^n$. Let $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$.
Then U is unitary. For $j \le r$, $P\mathbf{u}_j = \mathbf{u}_j$, and for $j > r$, $\mathbf{u}_j \in \mathcal{U}^\perp$ so $P\mathbf{u}_j = \mathbf{0}$. Therefore the jth column of U^*PU is the coordinate column of $P\mathbf{u}_j$ in this orthonormal basis, giving the displayed block diagonal form.
Structural Commentary: Orthogonal projection splits the space into range and orthogonal kernel.
Key Insight: In a basis adapted to $\text{col}(P) \oplus \text{col}(P)^\perp$, projection keeps the first coordinates and kills the rest. □

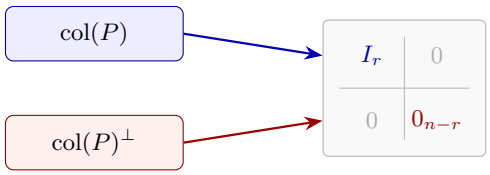


Figure 36: Adapted basis: $U^*PU = I_r \oplus 0_{n-r}$, identity on $\text{col}(P)$ and zero on $\text{col}(P)^\perp$.

Definition 2.4.21. *Isometry and Isometric Isomorphism*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.

(i) T is an *isometry* if $\|T\mathbf{v}\| = \|\mathbf{v}\|$ for all $\mathbf{v} \in \mathcal{V}$.
(ii) T is an *isometric isomorphism* if T is bijective and isometric.
The spaces $\mathcal{V}$ and $\mathcal{W}$ are *isometrically isomorphic* if such a T exists.

Proposition 2.4.22. *Basic Properties of Isometries*
Let $\mathcal{V}$ be an inner-product space and let $S, T \in \mathcal{L}(\mathcal{V})$ be isometries.

i. ST is an isometry;
ii. T is injective;
iii. if $\mathcal{V}$ is finite-dimensional, then T is invertible and T^{-1} is an isometry.

Proof.

(i) For every $\mathbf{v}$, $\|ST\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{v}\|$.
(ii) If $T\mathbf{v} = \mathbf{0}$, then $\|\mathbf{v}\| = \|T\mathbf{v}\| = 0$, so $\mathbf{v} = \mathbf{0}$. Thus $\ker(T) = \{\mathbf{0}\}$.
(iii) In finite dimensions, injectivity implies invertibility by Theorem 3.2.38.
If $\mathbf{u} = T\mathbf{v}$, then $\left\|T^{-1}\mathbf{u}\right\| = \|\mathbf{v}\| = \|T\mathbf{v}\| = \|\mathbf{u}\|$.

Structural Commentary: Isometries preserve length, so they cannot collapse nonzero vectors.
Key Insight: In finite dimensions, length preservation already forces an invertible geometry-preserving map. □

Example 2.4.23. *An Isometry Need Not Be Onto*
Define $T : \ell^2(\mathbb{F}) \to \ell^2(\mathbb{F})$ by $T(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$.
For $\mathbf{x} = (x_j)_{j \ge 1}$, $\|T\mathbf{x}\|_2^2 = \sum_{j \ge 1} |x_j|^2 = \|\mathbf{x}\|_2^2$, so T is an isometry.
Since every vector in $\text{range}(T)$ has first coordinate 0, $(1, 0, 0, \ldots) \notin \text{range}(T)$; hence T is not surjective.
Structural Commentary: Infinite-dimensional isometries can preserve all lengths while missing whole directions.
Key Insight: Injective does not imply surjective outside finite dimensions.

Theorem 2.4.24. *Isometries Preserve Inner Products*
Let $\mathcal{V}$ and $\mathcal{W}$ be inner-product spaces and let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$.
Then T is an isometry if and only if $\langle T\mathbf{u}, T\mathbf{v} \rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{v} \rangle_{\mathcal{V}}$ for every $\mathbf{u}, \mathbf{v} \in \mathcal{V}$.

Proof.

($\Rightarrow$) Suppose T is an isometry. Then every norm appearing in the polarization identities of Theorem 1.1.23 is unchanged after applying T. Therefore $\langle T\mathbf{u}, T\mathbf{v}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{v}\rangle_{\mathcal{V}}$.

($\Leftarrow$) Suppose T preserves inner products. Then $\|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle_{\mathcal{W}} = \langle \mathbf{u}, \mathbf{u}\rangle_{\mathcal{V}} = \|\mathbf{u}\|^2$, so T is an isometry.

Structural Commentary: Polarization converts preservation of norms into preservation of inner products.
Key Insight: Length preservation for all vectors also preserves angle information. □

Remark 2.4.25. *Isometry Conventions*

(i) For linear maps between inner-product spaces, norm preservation and inner-product preservation are equivalent by Theorem 2.4.24; hence either convention defines the same isometries.

(ii) For general bilinear forms, an isometry is often required to be bijective and form-preserving; if the form is singular, preservation alone need not imply injectivity.

Theorem 2.4.26. *Isometries and the Adjoint Product*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is an isometry if and only if $T^*T = I_{\mathcal{V}}$.

Proof.

($\Rightarrow$) Suppose T is an isometry. By Theorem 2.4.24, $\langle T^*T\mathbf{u}, \mathbf{v}\rangle = \langle T\mathbf{u}, T\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle$ for all $\mathbf{u}, \mathbf{v}$. Thus $\langle (T^*T - I_{\mathcal{V}})\mathbf{u}, \mathbf{v}\rangle = 0$ for every $\mathbf{v}$, so $T^*T\mathbf{u} = \mathbf{u}$ for every $\mathbf{u}$.

($\Leftarrow$) Suppose $T^*T = I_{\mathcal{V}}$. Then $\|T\mathbf{u}\|^2 = \langle T\mathbf{u}, T\mathbf{u}\rangle = \langle T^*T\mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2$ for every $\mathbf{u}$, so T is an isometry.

Structural Commentary: The operator T^*T records inner products after applying T.
Key Insight: T is an isometry exactly when this Gram operator is the identity. □

Definition 2.4.27. *Unitary and Orthogonal*
An operator $U \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space is *unitary* if $U^*U = UU^* = I_{\mathcal{V}}$.
A matrix $U \in \mathrm{Mat}(n, n, \mathbb{F})$ is *unitary* if $U^*U = I_n$. If $\mathbb{F} = \mathbb{R}$, a unitary matrix is called *orthogonal*.

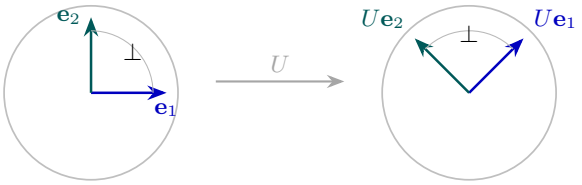


Figure 37: Unitary invariance: $\|U\mathbf{x}\| = \|\mathbf{x}\|$ and $\langle U\mathbf{x}, U\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$.

Theorem 2.4.28. *Equivalent Conditions for a Unitary Matrix*
Let $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n] \in \mathrm{Mat}(n, n, \mathbb{F})$. The following are equivalent:

i. U is unitary, i.e. $U^*U = I_n$;
ii. the columns $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are orthonormal in $\mathbb{F}^n$;
iii. $UU^* = I_n$;
iv. U^* is unitary;
v. the rows of U are orthonormal;
vi. U is invertible and $U^{-1} = U^*$;
vii. $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for every $\mathbf{x} \in \mathbb{F}^n$;
viii. $\langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} = \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n}$ for every $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$.

Proof.

(i) $\Rightarrow$ *(ii)*. Since $(U^*U)_{ij} = \mathbf{u}_i^*\mathbf{u}_j = \langle \mathbf{u}_j, \mathbf{u}_i\rangle_{\mathbb{F}^n}$, equation $U^*U = I_n$ states that columns of U are orthonormal.

(ii) $\Rightarrow$ *(iii)*. Orthonormal columns give $U^*U = I_n$. A left inverse of a square matrix is also a right inverse by Theorem 3.3.30, so $UU^* = I_n$.

(iii) $\Rightarrow$ *(iv)*. The equation $UU^* = I_n$ is $(U^*)^*U^* = I_n$, so U^* is unitary.

(iv) $\Rightarrow$ *(v)*. Apply *(i)* $\Rightarrow$ *(ii)* to U^*; the columns of U^* are the conjugates of the rows of U.

(v) $\Rightarrow$ *(vi)*. Orthonormal rows give $UU^* = I_n$. Thus U^* is the inverse of U, so $U^{-1} = U^*$.

(vi) $\Rightarrow$ *(vii)*. If $U^{-1} = U^*$, then $U^*U = I_n$. Therefore $\|U\mathbf{x}\|_2^2 = \langle U\mathbf{x}, U\mathbf{x}\rangle_{\mathbb{F}^n} = \langle U^*U\mathbf{x}, \mathbf{x}\rangle_{\mathbb{F}^n} = \|\mathbf{x}\|_2^2$ for every $\mathbf{x} \in \mathbb{F}^n$. Since both norms are nonnegative, $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$.

(vii) $\Rightarrow$ *(viii)*. Apply Theorem 2.4.24 to $\mathbf{x} \mapsto U\mathbf{x}$.

(viii) $\Rightarrow$ *(i)*. For every $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, $\langle (U^*U - I_n)\mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = \langle U\mathbf{x}, U\mathbf{y}\rangle_{\mathbb{F}^n} - \langle \mathbf{x}, \mathbf{y}\rangle_{\mathbb{F}^n} = 0$. Taking $\mathbf{y} = (U^*U - I_n)\mathbf{x}$ shows $(U^*U - I_n)\mathbf{x} = \mathbf{0}$ for every $\mathbf{x}$, so $U^*U = I_n$.

Structural Commentary: A unitary matrix is an isometry written in orthonormal coordinates.
Key Insight: Columns, rows, inverses, norms, and inner products are different faces of the same condition. □

Proposition 2.4.29. *Algebra of Unitary Matrices*
Let $U, V \in \mathrm{Mat}(n, n, \mathbb{F})$ and $W \in \mathrm{Mat}(m, m, \mathbb{F})$, where $V \oplus W$ denotes the block diagonal matrix with blocks V and W.

i. If U and V are unitary, then UV is unitary.
ii. $V \oplus W$ is unitary if and only if V and W are unitary.

Proof.

(i) Compute $(UV)^*(UV) = V^*U^*UV = V^*V = I_n$.
(ii) Since $(V \oplus W)^*(V \oplus W) = V^*V \oplus W^*W$, this product equals I_{n+m} exactly when $V^*V = I_n$ and $W^*W = I_m$.

Structural Commentary: Unitary structure is stable under composition and blockwise orthogonal sums.
Key Insight: The equation $U^*U = I$ makes every verification a short adjoint calculation. □

Theorem 2.4.30. *Extending Isometric Columns to a Unitary Matrix*
If $X \in \mathrm{Mat}(m, n, \mathbb{F})$ and $X^*X = I_n$, then there exists $X' \in \mathrm{Mat}(m, m-n, \mathbb{F})$ such that $[X\ X']$ is unitary.

Proof. Write $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_n]$. Since $(X^*X)_{ij} = \langle \mathbf{x}_j, \mathbf{x}_i \rangle_{\mathbb{F}^m}$, the condition $X^*X = I_n$ says $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are orthonormal in $\mathbb{F}^m$. By Corollary 1.3.13, extend this list to an orthonormal basis $\mathbf{x}_1, \ldots, \mathbf{x}_m$ of $\mathbb{F}^m$.
Let $X' = [\mathbf{x}_{n+1}\ \cdots\ \mathbf{x}_m]$. Then the columns of $[X\ X']$ are orthonormal, so $[X\ X']$ is unitary.
Structural Commentary: Isometric columns are a partial orthonormal basis.
Key Insight: Complete the columns by extending the orthonormal list. □

Definition 2.4.31. *Unitary and Permutation Similarity*
Square matrices $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ are *unitarily similar* if $A = UBU^*$ for some unitary U. They are *orthogonally similar* if $\mathbb{F} = \mathbb{R}$ and U can be chosen real orthogonal.
A *permutation matrix* is a square matrix with exactly one entry equal to 1 in each row and each column, and all other entries equal to 0. Matrices are *permutation similar* if $A = PBP^*$ for some permutation matrix P.

Proposition 2.4.32. *Permutation Matrices Are Orthogonal*
If $P = [\mathbf{e}_{\sigma(1)}\ \cdots\ \mathbf{e}_{\sigma(n)}]$ is a permutation matrix, then $P^*P = PP^* = I_n$. Hence P is real orthogonal.

Proof. The columns of P are a rearrangement of the standard orthonormal basis of $\mathbb{R}^n$, hence they are orthonormal. Theorem 2.4.28 gives $P^*P = PP^* = I_n$.
Structural Commentary: A permutation matrix only reorders the standard basis.
Key Insight: Reordering orthonormal vectors preserves orthonormality. □

2.5 Householder Reflections

Definition 2.5.1. *Householder Matrix*
Let $\mathbf{w} \in \mathbb{F}^n$ be nonzero and set $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|_2$.
The *Householder matrix* associated to $\mathbf{w}$ is $U_{\mathbf{w}} := I_n - 2P_{\mathbf{u}} = I_n - 2\frac{\mathbf{w}\mathbf{w}^*}{\|\mathbf{w}\|_2^2}$.
It acts by $\mathbf{x} \mapsto U_{\mathbf{w}}\mathbf{x} = \mathbf{x} - 2\langle \mathbf{x}, \mathbf{u} \rangle_{\mathbb{F}^n} \mathbf{u}$.

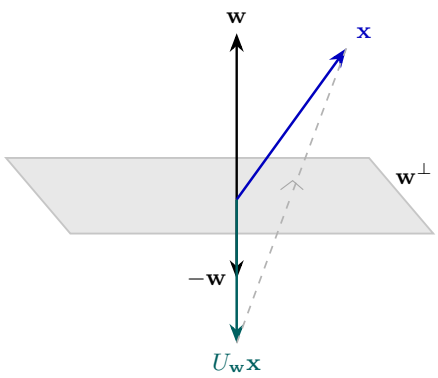


Figure 38: A Householder matrix fixes $\mathbf{w}^\perp$ and flips the span($\mathbf{w}$) component.

Theorem 2.5.2. *Householder Matrices Are Unitary Reflections*
Every Householder matrix $U_{\mathbf{w}}$ is unitary, Hermitian, and involutive: $U_{\mathbf{w}}^* U_{\mathbf{w}} = I_n$, $U_{\mathbf{w}}^* = U_{\mathbf{w}}$, $U_{\mathbf{w}}^2 = I_n$.
If $\mathbf{w} \in \mathbb{R}^n$, then $U_{\mathbf{w}}$ is real orthogonal and symmetric.

Proof. Let $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|_2$. By Proposition 2.2.5, $P_{\mathbf{u}}^* = P_{\mathbf{u}}$ and $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$.
Therefore $U_{\mathbf{w}}^* = I_n - 2P_{\mathbf{u}} = U_{\mathbf{w}}$, $U_{\mathbf{w}}^2 = (I_n - 2P_{\mathbf{u}})^2 = I_n - 4P_{\mathbf{u}} + 4P_{\mathbf{u}}^2 = I_n$.
Thus $U_{\mathbf{w}}^* U_{\mathbf{w}} = U_{\mathbf{w}}^2 = I_n$, so $U_{\mathbf{w}}$ is unitary. The real case is the same statement with conjugation removed.
Structural Commentary: A Householder matrix reflects across an orthogonal hyperplane.
Key Insight: The identity $P_{\mathbf{u}}^2 = P_{\mathbf{u}}$ makes $(I - 2P_{\mathbf{u}})^2 = I$. □

Theorem 2.5.3. *Householder Map Between Equal-Norm Vectors*
Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ satisfy $\|\mathbf{x}\|_2 = \|\mathbf{y}\|_2 \neq 0$. Define $\sigma = \begin{cases} 1, & \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n} = 0, \\ -\dfrac{\overline{\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n}}}{|\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n}|}, & \langle \mathbf{x}, \mathbf{y} \rangle_{\mathbb{F}^n} \neq 0, \end{cases}$, where $\mathbf{w} = \mathbf{y} - \sigma\mathbf{x}$.
Then $\mathbf{w} \neq \mathbf{0}$, $|\sigma| = 1$, and $\sigma U_{\mathbf{w}} \mathbf{x} = \mathbf{y}$.

Proof. Clearly $|\sigma| = 1$. Let $z = \langle \mathbf{x}, \mathbf{y} \rangle$ and $r = \|\mathbf{x}\|_2 = \|\mathbf{y}\|_2$. By construction, $\sigma z = -|z|$, with the convention that this equals 0 when $z = 0$. Hence $\|\mathbf{w}\|_2^2 = \|\mathbf{y} - \sigma\mathbf{x}\|_2^2 = 2r^2 - \sigma z - \overline{\sigma z} = 2\left(r^2 + |z|\right) > 0$, so $\mathbf{w} \neq \mathbf{0}$.
Since $\mathbf{y} = \mathbf{w} + \sigma\mathbf{x}$, $\sigma U_{\mathbf{w}} \mathbf{x} = \sigma\mathbf{x} - 2\sigma \frac{\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|_2^2} \mathbf{w} = \mathbf{y} - \left(1 + 2\sigma \frac{\langle \mathbf{x}, \mathbf{w} \rangle}{\|\mathbf{w}\|_2^2}\right) \mathbf{w}$.
But $\langle \mathbf{x}, \mathbf{w} \rangle = z - \overline{\sigma} r^2$, so $2\sigma \langle \mathbf{x}, \mathbf{w} \rangle = 2\sigma z - 2r^2 = -2(r^2 + |z|) = -\|\mathbf{w}\|_2^2$.
Coefficient of $\mathbf{w}$ is hence zero, proving $\sigma U_{\mathbf{w}} \mathbf{x} = \mathbf{y}$.
Structural Commentary: The scalar σ chooses the phase so that $\mathbf{y} - \sigma\mathbf{x}$ gives a stable reflection direction.
Key Insight: Householder reflections connect equal-norm vectors up to a unit scalar. □

Corollary 2.5.4. *Householder Unitary Annihilator*
Let $\mathbf{x} = [x_1 \ \cdots \ x_n]^t \in \mathbb{F}^n$ be nonzero. Define $\sigma = \begin{cases} 1, & x_1 = 0, \\ -\dfrac{\overline{x_1}}{|x_1|}, & x_1 \neq 0, \end{cases}$, where $\mathbf{w} = \|\mathbf{x}\|_2 \mathbf{e}_1 - \sigma\mathbf{x}$.
Then $\sigma U_{\mathbf{w}} \mathbf{x} = \|\mathbf{x}\|_2 \mathbf{e}_1$. If $\mathbf{x}$ is a unit vector, then $\overline{\sigma} U_{\mathbf{w}}$ is unitary and has $\mathbf{x}$ as its first column.

Proof. Apply Theorem 2.5.3 with $\mathbf{y} = \|\mathbf{x}\|_2 \mathbf{e}_1$, noting that $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 \|\mathbf{x}\|_2$. If $\|\mathbf{x}\|_2 = 1$, then $(\sigma U_{\mathbf{w}})\mathbf{x} = \mathbf{e}_1$, so $(\sigma U_{\mathbf{w}})^* \mathbf{e}_1 = \mathbf{x}$. Since $(\sigma U_{\mathbf{w}})^* = \overline{\sigma} U_{\mathbf{w}}$, this says the first column of $\overline{\sigma} U_{\mathbf{w}}$ is $\mathbf{x}$.
Structural Commentary: The reflector concentrates all mass of $\mathbf{x}$ into the first coordinate.
Key Insight: Householder matrices produce explicit unitary matrices with a prescribed first column. □

3 Bilinear and Quadratic Forms

3.1 Bilinear Pairings and Radicals

Metric vector spaces keep the algebraic part of inner-product geometry and drop positivity. Throughout this subsection $\mathbb{F}$ is an arbitrary field and vector spaces are finite-dimensional unless explicitly stated otherwise. The bilinear form is written $\langle \mathbf{x}, \mathbf{y} \rangle$ when it is fixed.

Definition 3.1.1. *Bilinear Pairings and Their Radicals*
Let $\mathcal{U}, \mathcal{W}$ be vector spaces over $\mathbb{F}$.

(i) A *bilinear pairing* is a map $B : \mathcal{U} \times \mathcal{W} \to \mathbb{F}$ linear in each variable.
(ii) The *left radical* is $\operatorname{rad}_L(B) := \{\mathbf{u} \in \mathcal{U} : B(\mathbf{u}, \mathbf{w}) = 0 \ \forall \mathbf{w} \in \mathcal{W}\}$.
The *right radical* is $\operatorname{rad}_R(B) := \{\mathbf{w} \in \mathcal{W} : B(\mathbf{u}, \mathbf{w}) = 0 \ \forall \mathbf{u} \in \mathcal{U}\}$.
(iii) The *right orthogonal* of $S \subseteq \mathcal{U}$ is $S^{\perp_R} := \{\mathbf{w} \in \mathcal{W} : B(\mathbf{s}, \mathbf{w}) = 0 \ \forall \mathbf{s} \in S\}$.
The *left orthogonal* of $T \subseteq \mathcal{W}$ is ${}^{\perp_L}T := \{\mathbf{u} \in \mathcal{U} : B(\mathbf{u}, \mathbf{t}) = 0 \ \forall \mathbf{t} \in T\}$.
(iv) B is *left nonsingular* if $\operatorname{rad}_L(B) = \{\mathbf{0}\}$, *right nonsingular* if $\operatorname{rad}_R(B) = \{\mathbf{0}\}$, and *perfect* if both hold.

Proposition 3.1.2. *Matrices of Bilinear Pairings*
Let $\alpha = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$, $\beta = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ be bases of $\mathcal{U}$, $\mathcal{W}$. Let $M = M_{\alpha,\beta} := [B(\mathbf{u}_i, \mathbf{w}_j)]_{i,j} \in \operatorname{Mat}(m, n, \mathbb{F})$.

(i) $B(\mathbf{u}, \mathbf{w}) = [\mathbf{u}]_\alpha^t M [\mathbf{w}]_\beta$.
(ii) $\{[\mathbf{u}]_\alpha : \mathbf{u} \in \operatorname{rad}_L(B)\} = \ker(M^t)$ *and* $\{[\mathbf{w}]_\beta : \mathbf{w} \in \operatorname{rad}_R(B)\} = \ker(M)$.
(iii) $\dim \operatorname{rad}_L(B) = m - \operatorname{rank}(M)$ *and* $\dim \operatorname{rad}_R(B) = n - \operatorname{rank}(M)$.
(iv) If B is perfect, then $m = n$; when $m = n$, B is perfect if and only if M is invertible.

Proof.

(i) If $\mathbf{u} = \sum_i x_i \mathbf{u}_i$ and $\mathbf{w} = \sum_j y_j \mathbf{w}_j$, then $B(\mathbf{u}, \mathbf{w}) = \sum_{i,j} x_i M_{ij} y_j = [\mathbf{u}]_\alpha^t M [\mathbf{w}]_\beta$.

(ii) For $\mathbf{x} = [\mathbf{u}]_\alpha$ and $\mathbf{y} = [\mathbf{w}]_\beta$, $\mathbf{u} \in \operatorname{rad}_L(B) \iff \mathbf{x}^t M\mathbf{y} = 0\ \forall \mathbf{y} \iff M^t\mathbf{x} = \mathbf{0}$.
Similarly, $\mathbf{w} \in \operatorname{rad}_R(B) \iff M\mathbf{y} = \mathbf{0}$.
(iii) By (ii) and rank-nullity, $\dim \operatorname{rad}_L(B) = \dim \ker(M^t) = m - \operatorname{rank}(M^t) = m - \operatorname{rank}(M)$,
$\dim \operatorname{rad}_R(B) = \dim \ker(M) = n - \operatorname{rank}(M)$.
(iv) By (iii), perfection is equivalent to $m - \operatorname{rank}(M) = 0$ and $n - \operatorname{rank}(M) = 0$. Hence $m = n$; when $m = n$, this is equivalent to M being invertible.

Structural Commentary: A pairing matrix has one radical for each input slot.
Key Insight: Nonsingularity of a pairing is ordinary full-rank behavior of its representing matrix. □

Proposition 3.1.3. *Two-Sided Basis Change for Pairings*
With notation as above, let α' and β' be bases, and set $P = C_{\alpha\leftarrow\alpha'}$, $Q = C_{\beta\leftarrow\beta'}$. Then $M_{\alpha',\beta'} = P^t M_{\alpha,\beta} Q$. For square forms this is congruence; for rectangular pairings it is two-sided equivalence.

Proof. By Proposition 3.1.2, $[\mathbf{u}]_{\alpha'}^t M_{\alpha',\beta'} [\mathbf{w}]_{\beta'} = B(\mathbf{u},\mathbf{w}) = (P[\mathbf{u}]_{\alpha'})^t M_{\alpha,\beta}(Q[\mathbf{w}]_{\beta'}) = [\mathbf{u}]_{\alpha'}^t P^t M_{\alpha,\beta} Q [\mathbf{w}]_{\beta'}$.
Thus $M_{\alpha',\beta'} = P^t M_{\alpha,\beta} Q$.
Structural Commentary: Pairing matrices change independently in each input slot.
Key Insight: The number of input slots determines how basis changes act on the matrix. □

Perspective 3.1.4. *Square Pairings*
Henceforth $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ is a square pairing, usually symmetric, skew-symmetric, or alternating.
In these cases $\operatorname{rad}_L(B) = \operatorname{rad}_R(B)$, so one radical and one orthogonality notation suffice.

Definition 3.1.5. *Symmetric, Skew-Symmetric, and Alternating Forms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $B : \mathcal{V} \times \mathcal{V} \to \mathbb{F}$ be bilinear.

(i) B is a *bilinear form* on $\mathcal{V}$; when fixed, write $B(\mathbf{x},\mathbf{y}) = \langle \mathbf{x},\mathbf{y}\rangle$.
(ii) $\operatorname{Sym}(B)$ means $B(\mathbf{x},\mathbf{y}) = B(\mathbf{y},\mathbf{x})$ for all $\mathbf{x},\mathbf{y} \in \mathcal{V}$; then B is *symmetric*.
(iii) $\operatorname{Skew}(B)$ means $B(\mathbf{x},\mathbf{y}) = -B(\mathbf{y},\mathbf{x})$ for all $\mathbf{x},\mathbf{y} \in \mathcal{V}$; then B is *skew-symmetric*.
(iv) $\operatorname{Alt}(B)$ means $B(\mathbf{x},\mathbf{x}) = 0$ for all $\mathbf{x} \in \mathcal{V}$; then B is *alternating*.
(v) A *metric vector space* is a pair $(\mathcal{V}, B)$ with $\operatorname{Sym}(B)$, $\operatorname{Skew}(B)$, or $\operatorname{Alt}(B)$.
The symmetric case is an *orthogonal geometry*; the alternating case is a *symplectic geometry*.

Proposition 3.1.6. *Characteristic Controls the Form Types*
Let $B(\mathbf{x},\mathbf{y}) = \langle \mathbf{x},\mathbf{y}\rangle$ be a bilinear form on $\mathcal{V}$.

(i) $\operatorname{Alt}(B) \Rightarrow \operatorname{Skew}(B)$.
(ii) If $\operatorname{char}(\mathbb{F}) \neq 2$, *then* $\operatorname{Alt}(B) \iff \operatorname{Skew}(B)$, *and* $\operatorname{Sym}(B) \wedge \operatorname{Alt}(B) \Rightarrow B = 0$.
(iii) If $\operatorname{char}(\mathbb{F}) = 2$, *then* $\operatorname{Sym}(B) \iff \operatorname{Skew}(B)$ *and* $\operatorname{Alt}(B) \Rightarrow \operatorname{Sym}(B)$.

Proof.

(i) If $\operatorname{Alt}(B)$, then $0 = \langle \mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y}\rangle = \langle \mathbf{x},\mathbf{y}\rangle + \langle \mathbf{y},\mathbf{x}\rangle$, so $\langle \mathbf{x},\mathbf{y}\rangle = -\langle \mathbf{y},\mathbf{x}\rangle$.
(ii) Assume $\operatorname{char}(\mathbb{F}) \neq 2$. By (i), $\operatorname{Alt}(B) \Rightarrow \operatorname{Skew}(B)$. Conversely, $\operatorname{Skew}(B)$ gives $\langle \mathbf{x},\mathbf{x}\rangle = -\langle \mathbf{x},\mathbf{x}\rangle$, hence $2\langle \mathbf{x},\mathbf{x}\rangle = 0$ and $\operatorname{Alt}(B)$. If also $\operatorname{Sym}(B)$ and $\operatorname{Alt}(B)$, then $\langle \mathbf{x},\mathbf{y}\rangle = -\langle \mathbf{x},\mathbf{y}\rangle$, so $\langle \mathbf{x},\mathbf{y}\rangle = 0$ for all $\mathbf{x},\mathbf{y}$.
(iii) If $\operatorname{char}(\mathbb{F}) = 2$, then $-1 = 1$, so $\operatorname{Sym}(B) \iff \operatorname{Skew}(B)$.
Moreover $\operatorname{Alt}(B) \Rightarrow \operatorname{Skew}(B)$ by (i), hence $\operatorname{Alt}(B) \Rightarrow \operatorname{Sym}(B)$.

Structural Commentary: Skew-symmetry collapses by characteristic.
Key Insight: The equation $\langle \mathbf{x}+\mathbf{y}, \mathbf{x}+\mathbf{y}\rangle = 0$ polarizes alternation. □

Definition 3.1.7. *Minkowski Space*
Minkowski space is the orthogonal geometry $\mathbb{R}^{3,1} := \mathbb{R}^4$ with standard basis $(\mathbf{e}_1,\dots,\mathbf{e}_4)$
and bilinear form $\langle \mathbf{x},\mathbf{y}\rangle = x_1y_1 + x_2y_2 + x_3y_3 - x_4y_4$.
Equivalently, its Gram matrix is $\operatorname{diag}(1,1,1,-1)$. It is nonsingular but not positive definite.

Example 3.1.8. *Null Vectors in Minkowski Space*
In $\mathbb{R}^{3,1}$, set $q(\mathbf{x}) := \langle \mathbf{x},\mathbf{x}\rangle = x_1^2 + x_2^2 + x_3^2 - x_4^2$.
Null cone is $\mathcal{N} := \{\mathbf{x} \neq \mathbf{0} : q(\mathbf{x}) = 0\} = \{\mathbf{x} \neq \mathbf{0} : x_1^2 + x_2^2 + x_3^2 = x_4^2\}$. Every $\mathbf{x} \in \mathcal{N}$ is isotropic; i.e., $\mathbf{e}_1 + \mathbf{e}_4 \in \mathcal{N}$.
As $\operatorname{diag}(1,1,1,-1)$, I_4 have different real inertias, $(3,1,0)$ and $(4,0,0)$, $\mathbb{R}^{3,1}$ is not isometric to Euclidean $\mathbb{R}^4$.

Definition 3.1.9. *Matrix, Rank, and Discriminant of a Form*
Let $\beta = (\mathbf{v}_1,\dots,\mathbf{v}_n)$ be an ordered basis of a metric vector space $\mathcal{V}$.

(i) The *matrix of the form* in β is $M_\beta := [\langle \mathbf{v}_i, \mathbf{v}_j\rangle]_{i,j}$.
(ii) The *rank* of the form is $\operatorname{rank}(M_\beta)$, independent of β.
(iii) The *discriminant set* is $\Delta(\mathcal{V}) := \{r^2 \det(M_\beta) : r \in \mathbb{F}^\times\}$.
(iv) If the form is nonsingular, then $\Delta(\mathcal{V})$ is the square class of $\det(M_\beta)$ in $\mathbb{F}^\times/(\mathbb{F}^\times)^2$.

Proposition 3.1.10. *Matrices Represent Forms up to Congruence*
Let β, γ be ordered bases of $\mathcal{V}$ and put $C = C_{\beta\leftarrow\gamma}$.

(i) $\langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\beta^t M_\beta [\mathbf{y}]_\beta$.
(ii) $M_\gamma = C^t M_\beta C$.
(iii) Matrices A, N represent the same form on $\mathcal{V}$ if and only if $N = P^t A P$ for some invertible P.

Proof.

(i) If $\mathbf{x} = \sum_i a_i \mathbf{v}_i$ and $\mathbf{y} = \sum_j b_j \mathbf{v}_j$, then $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i,j} a_i b_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle = [\mathbf{x}]_\beta^t M_\beta [\mathbf{y}]_\beta$.
(ii) Since $[\mathbf{x}]_\beta = C[\mathbf{x}]_\gamma$ and $[\mathbf{y}]_\beta = C[\mathbf{y}]_\gamma$, $[\mathbf{x}]_\gamma^t M_\gamma [\mathbf{y}]_\gamma = \langle \mathbf{x}, \mathbf{y} \rangle = [\mathbf{x}]_\gamma^t C^t M_\beta C [\mathbf{y}]_\gamma$, so $M_\gamma = C^t M_\beta C$.
(iii) If $A = M_\beta$ and $N = M_\gamma$ represent one form, then (ii) gives $N = C_{\beta\leftarrow\gamma}^t A C_{\beta\leftarrow\gamma}$.
Conversely, suppose $N = P^t A P$ with $P = [p_{ij}]$ invertible and $A = M_\beta$.
Define $\gamma = (\mathbf{w}_1, \ldots, \mathbf{w}_n)$ by $\mathbf{w}_j = \sum_i p_{ij} \mathbf{v}_i$. Then $C_{\beta\leftarrow\gamma} = P$, so (ii) gives $M_\gamma = P^t A P = N$.

Structural Commentary: Congruence is basis change for forms.
Key Insight: Basis change acts in both slots: $M \mapsto P^t M P$. □

3.2 Quadratic Forms and Polar Forms

Definition 3.2.1. *Quadratic Forms and Polar Forms*
A *quadratic form* on $\mathcal{V}$ is a function $Q : \mathcal{V} \to \mathbb{F}$ such that:

(i) $Q(a\mathbf{v}) = a^2 Q(\mathbf{v})$ for all $a \in \mathbb{F}$ and $\mathbf{v} \in \mathcal{V}$;
(ii) the *polar form* $B_Q(\mathbf{u}, \mathbf{v}) := Q(\mathbf{u} + \mathbf{v}) - Q(\mathbf{u}) - Q(\mathbf{v})$ is bilinear.

Proposition 3.2.2. *Quadratic Forms and Symmetric Bilinear Forms*
For every quadratic form Q, its polar form B_Q is symmetric.

If $\mathrm{char}(\mathbb{F}) \neq 2$, *maps* $\{B : \mathrm{Sym}(B)\} \underset{Q \mapsto B_Q}{\overset{B \mapsto Q_B}{\rightleftarrows}} \{Q : Q \text{ quadratic}\}$ *and* $Q_B(\mathbf{x}) := \frac{1}{2} B(\mathbf{x}, \mathbf{x})$ *are inverse correspondences between symmetric bilinear forms and quadratic forms.*

Proof. First, $B_Q(\mathbf{u}, \mathbf{v}) = Q(\mathbf{u} + \mathbf{v}) - Q(\mathbf{u}) - Q(\mathbf{v}) = B_Q(\mathbf{v}, \mathbf{u})$, so B_Q is symmetric.
($\Rightarrow$) If $\mathrm{Sym}(B)$, then $Q_B(a\mathbf{x}) = a^2 Q_B(\mathbf{x})$ and $B_{Q_B}(\mathbf{u}, \mathbf{v}) = \frac{1}{2}\big(B(\mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v}) - B(\mathbf{u}, \mathbf{u}) - B(\mathbf{v}, \mathbf{v})\big) = B(\mathbf{u}, \mathbf{v})$.
($\Leftarrow$) If Q is quadratic, then $Q_{B_Q}(\mathbf{x}) = \frac{1}{2} B_Q(\mathbf{x}, \mathbf{x}) = \frac{1}{2}\big(Q(2\mathbf{x}) - 2Q(\mathbf{x})\big) = Q(\mathbf{x})$.

Structural Commentary: Away from characteristic 2, quadratic and symmetric bilinear data coincide.
Key Insight: Dividing by 2 recovers diagonal data from polar data. □

Proposition 3.2.3. *Polar Forms in Characteristic Two*
If $\mathrm{char}(\mathbb{F}) = 2$, *then* $\mathrm{Alt}(B_Q)$ *for every quadratic form Q on $\mathcal{V}$.*

Proof. For $\mathbf{x} \in \mathcal{V}$, $B_Q(\mathbf{x}, \mathbf{x}) = Q(2\mathbf{x}) - Q(\mathbf{x}) - Q(\mathbf{x}) = Q(\mathbf{0}) = 0$. Thus $\mathrm{Alt}(B_Q)$.
Structural Commentary: In characteristic 2, every polar form is alternating.
Key Insight: The diagonal of the polar form vanishes automatically because both $2\mathbf{x}$ and $2Q(\mathbf{x})$ vanish. □

Example 3.2.4. *Characteristic Two Loses Diagonal Data*
Assume $\mathrm{char}(\mathbb{F}) = 2$. On $\mathbb{F}^2$, let $Q_0(x, y) = xy$ and $Q_1(x, y) = xy + x^2$.
Then $B_{Q_0}((x, y), (u, v)) = xv + uy = B_{Q_1}((x, y), (u, v))$, and $Q_0(1, 0) = 0 \neq 1 = Q_1(1, 0)$.
Indeed, the added square has zero polar contribution, where $(x + u)^2 - x^2 - u^2 = 2xu = 0$.
More generally, for any linear $L : \mathcal{V} \to \mathbb{F}$, Q and $Q + L^2$ have the same polar form.
Structural Commentary: In characteristic 2, squares are invisible to polarization.
Key Insight: Characteristic 2 quadratic forms need data beyond their polar forms.

Example 3.2.5. *Alternating versus Nonalternating Symmetric Forms*
Assume $\mathrm{char}(\mathbb{F}) = 2$.
The symmetric matrix $H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ defines a nonsingular alternating form on $\mathbb{F}^2$, since $\begin{bmatrix} x & y \end{bmatrix} H \begin{bmatrix} x \\ y \end{bmatrix} = 2xy = 0$.
Thus H is not congruent to any nonzero diagonal form: alternating means all self-pairings vanish, while a nonzero diagonal form has some basis vector with nonzero self-pairing. By contrast, [1] is symmetric and nonalternating.
Structural Commentary: In characteristic 2, symmetric forms split into alternating and nonalternating parts.
Key Insight: Nonsingular alternating blocks are invisible on the diagonal.

Proposition 3.2.6. *Isometries from Quadratic Values*
Let $\mathcal{V}$ be an orthogonal geometry with $\mathrm{char}(\mathbb{F}) \neq 2$, *let $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$, and let $T : \mathcal{V} \to \mathcal{V}$ be bijective. Then T is an isometry if and only if $q(T\mathbf{x}) = q(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{V}$.*

Proof.

($\Rightarrow$) If T is an isometry, then $q(T\mathbf{x}) = \langle T\mathbf{x}, T\mathbf{x}\rangle = \langle \mathbf{x}, \mathbf{x}\rangle = q(\mathbf{x})$.
($\Leftarrow$) If $q \circ T = q$, then $\langle T\mathbf{x}, T\mathbf{y}\rangle = \frac{q(T\mathbf{x}+T\mathbf{y})-q(T\mathbf{x})-q(T\mathbf{y})}{2} = \frac{q(\mathbf{x}+\mathbf{y})-q(\mathbf{x})-q(\mathbf{y})}{2} = \langle \mathbf{x}, \mathbf{y}\rangle$.
Thus T preserves the form, so it is an isometry.

Structural Commentary: In odd characteristic, q determines the orthogonal form.
Key Insight: Polarization turns preservation of lengths into preservation of inner products. □

Definition 3.2.7. *Orthogonality, Isotropy, and Radical*
Let $\mathcal{V}$ be a metric vector space.

(i) $\mathbf{x} \perp \mathbf{y}$ means $\langle \mathbf{x}, \mathbf{y}\rangle = 0$; then $\mathbf{x}, \mathbf{y}$ are *orthogonal*. For $S \subseteq \mathcal{V}$, $S^\perp := \{\mathbf{v} \in \mathcal{V} : \langle \mathbf{v}, \mathbf{s}\rangle = 0 \ \forall \mathbf{s} \in S\}$
(ii) $\mathcal{S} \leq \mathcal{V}$ is *totally isotropic* or *totally degenerate* if $\mathcal{S} \subseteq \mathcal{S}^\perp$, and *nonsingular as a subspace* if $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$.
(iii) $\mathbf{x} \neq \mathbf{0}$ is *isotropic* or *null* if $\langle \mathbf{x}, \mathbf{x}\rangle = 0$, and *nonisotropic* otherwise.
The space $\mathcal{V}$ is *isotropic* if such an $\mathbf{x}$ exists, and *anisotropic* otherwise.
(iv) $\mathbf{x}$ is *degenerate* if $\mathbf{x} \perp \mathcal{V}$. The *radical* is $\operatorname{rad}(\mathcal{V}) := \mathcal{V}^\perp$
(v) $\mathcal{V}$ is *nonsingular* or *nondegenerate* if $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, *singular* otherwise.
(vi) $\mathcal{V}$ is *totally singular* or *totally degenerate* if $\operatorname{rad}(\mathcal{V}) = \mathcal{V}$.

Proposition 3.2.8. *Metric Radical as a Bilinear Radical*
Let $B(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x}, \mathbf{y}\rangle$ be the form of a metric vector space $\mathcal{V}$. Then $\operatorname{rad}_L(B) = \operatorname{rad}_R(B) = \operatorname{rad}(\mathcal{V})$.
Thus $\mathcal{V}$ is nonsingular if and only if B is perfect.

Proof.

(i) By definition, $\operatorname{rad}(\mathcal{V}) = \{\mathbf{x} \in \mathcal{V} : \langle \mathbf{x}, \mathbf{y}\rangle = 0 \ \forall \mathbf{y} \in \mathcal{V}\} = \operatorname{rad}_L(B)$
(ii) If $\mathbf{x} \in \operatorname{rad}_L(B)$, then $\langle \mathbf{x}, \mathbf{y}\rangle = 0$ for all $\mathbf{y}$. Since B is symmetric or skew-symmetric, $\langle \mathbf{y}, \mathbf{x}\rangle = \pm \langle \mathbf{x}, \mathbf{y}\rangle = 0$, so $\mathbf{x} \in \operatorname{rad}_R(B)$. The reverse inclusion is identical.
(iii) B is perfect if and only if $\operatorname{rad}_L(B) = \operatorname{rad}_R(B) = \{\mathbf{0}\}$, if and only if $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.

Structural Commentary: Metric forms are the square bilinear forms whose two null directions agree.
Key Insight: Metric geometry uses the common left-right radical. □

Proposition 3.2.9. *Radical and Representing Matrices*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be an ordered basis of $\mathcal{V}$.
Then $\{[\mathbf{x}]_\beta : \mathbf{x} \in \operatorname{rad}(\mathcal{V})\} = \ker(M_\beta^t)$, and $\dim \operatorname{rad}(\mathcal{V}) = \dim(\mathcal{V}) - \operatorname{rank}(M_\beta)$.

Proof. For $\mathbf{c} = [\mathbf{x}]_\beta$, $\mathbf{x} \in \operatorname{rad}(\mathcal{V}) \iff \langle \mathbf{x}, \mathbf{v}_j\rangle = 0 \ \forall j \iff \mathbf{c}^t M_\beta = 0 \iff \mathbf{c} \in \ker(M_\beta^t)$.
Rank-nullity gives $\dim \operatorname{rad}(\mathcal{V}) = n - \operatorname{rank}(M_\beta^t) = n - \operatorname{rank}(M_\beta)$.
Structural Commentary: Degenerate vectors are precisely the coordinate null vectors of the representing matrix.
Key Insight: The radical is the coordinate nullspace of M_β^t. □

Proposition 3.2.10. *Matrix Criterion for Nonsingularity*
Let β be an ordered basis of $\mathcal{V}$. Then $\mathcal{V}$ is nonsingular if and only if M_β is invertible.

Proof.

($\Rightarrow$) If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, then $\ker(M_\beta^t) = \{\mathbf{0}\}$. Since M_β is square, M_β is invertible.
($\Leftarrow$) If M_β is invertible, then $\ker(M_\beta^t) = \{\mathbf{0}\}$, so $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.

Structural Commentary: Zero radical is matrix invertibility in coordinates.
Key Insight: Nonsingularity of a form is ordinary matrix nonsingularity in any basis. □

Remark 3.2.11. *Subspace Radicals*
For $\mathcal{S} \leq \mathcal{V}$, compute $\operatorname{rad}(\mathcal{S})$ using $\langle \cdot, \cdot \rangle|_{\mathcal{S}\times\mathcal{S}}$, where $\operatorname{rad}(\mathcal{S}) = \{\mathbf{s} \in \mathcal{S} : \langle \mathbf{s}, \mathbf{t}\rangle = 0 \ \forall \mathbf{t} \in \mathcal{S}\} = \mathcal{S} \cap \mathcal{S}^\perp$.
In general, $\operatorname{rad}(\mathcal{S}) \neq \mathcal{S}^\perp$; the latter is ambient.

Proposition 3.2.12. *Riesz Map Isomorphism Criterion*
Let $\mathcal{V}$ be a finite-dimensional metric vector space and define $\tau : \mathcal{V} \to \mathcal{V}'$ where $\tau(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x}\rangle$.
Then $\ker(\tau) = \operatorname{rad}(\mathcal{V})$, and τ is an isomorphism $\iff \operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.

Proof. Bilinearity gives $\tau \in \mathcal{L}(\mathcal{V}, \mathcal{V}')$, and
$\mathbf{x} \in \ker(\tau) \iff \langle \mathbf{v}, \mathbf{x}\rangle = 0 \ \forall \mathbf{v} \in \mathcal{V} \iff \langle \mathbf{x}, \mathbf{v}\rangle = 0 \ \forall \mathbf{v} \in \mathcal{V} \iff \mathbf{x} \in \mathcal{V}^\perp = \operatorname{rad}(\mathcal{V})$

($\Rightarrow$) If τ is an isomorphism, then $\ker(\tau) = \{\mathbf{0}\}$, hence $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$.
($\Leftarrow$) If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, then $\ker(\tau) = \{\mathbf{0}\}$. Since $\dim \mathcal{V} = \dim \mathcal{V}'$, τ is an isomorphism.

Structural Commentary: The Riesz map is invertible exactly off the radical.
Key Insight: Nonsingularity is injectivity of the Riesz map. □

Theorem 3.2.13. *Riesz Map for Nonsingular Metric Spaces*
Let $\mathcal{V}$ be a finite-dimensional nonsingular metric vector space. Then $\forall f \in \mathcal{V}'\ \exists!\, \mathbf{x} \in \mathcal{V},\ f(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle\ \ \forall\, \mathbf{v} \in \mathcal{V}$.

Proof. Let $\tau : \mathcal{V} \to \mathcal{V}'$ be given by $\tau(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \mathbf{x} \rangle$. By Proposition 3.2.12, τ is an isomorphism.
Hence each $f \in \mathcal{V}'$ has a unique $\mathbf{x} \in \mathcal{V}$ with $\tau(\mathbf{x}) = f$.
Structural Commentary: Nonsingularity identifies vectors with covectors; positivity is irrelevant.
Key Insight: Riesz representation is fundamentally nonsingularity, not order or length. □

Definition 3.2.14. *Adjoint for a Nonsingular Bilinear Form*
Let $\mathcal{V}$ be nonsingular.
For $T \in \mathcal{L}(\mathcal{V})$, the *metric adjoint* is the unique $T^* \in \mathcal{L}(\mathcal{V})$ such that $\langle T\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, T^*\mathbf{y} \rangle\ \forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V}$.

Proposition 3.2.15. *Metric Adjoints and Isometries*
Let $\mathcal{V}$ be nonsingular and let $T \in \mathcal{L}(\mathcal{V})$.

(i) $\exists!\, T^* \in \mathcal{L}(\mathcal{V})$.
(ii) If β is ordered, $M = M_\beta$, and $A = [T]_\beta$, then $[T^*]_\beta = M^{-1}A^tM$.
(iii) T is an isometry $\iff$ T is bijective and $T^*T = I$. For bijective T, this is equivalent to $T^{-1} = T^*$.

Proof.

(i) For $\mathbf{y} \in \mathcal{V}$, set $f_{\mathbf{y}}(\mathbf{x}) = \langle T\mathbf{x}, \mathbf{y} \rangle$. Theorem 3.2.13 gives a unique $T^*\mathbf{y}$ such that $f_{\mathbf{y}}(\mathbf{x}) = \langle \mathbf{x}, T^*\mathbf{y} \rangle$ for all $\mathbf{x}$. Uniqueness gives linearity of $\mathbf{y} \mapsto T^*\mathbf{y}$.
(ii) With $A = [T]_\beta$ and $B = [T^*]_\beta$, the adjoint identity gives $A^tM = MB$, so $B = M^{-1}A^tM$.
(iii) $\forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$, $\langle T\mathbf{x}, T\mathbf{y} \rangle = \langle \mathbf{x}, T^*T\mathbf{y} \rangle$. so form preservation $\iff \langle \mathbf{x}, (T^*T - I)\mathbf{y} \rangle = 0\ \forall\, \mathbf{x}, \mathbf{y} \iff T^*T = I$. This gives the stated bijective criterion; if T is bijective, then $T^*T = I \iff T^* = T^{-1}$.

Structural Commentary: Adjoint theory needs nonsingularity, not positivity.
Key Insight: Isometry means the adjoint is the inverse. □

3.3 Orthogonal Decomposition and Witt Theory

Theorem 3.3.1. *Dimensions of Orthogonal Complements*
Let $\mathcal{S} \leq \mathcal{V}$. If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$ or $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, then $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = \dim(\mathcal{V})$.
If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, then $\mathcal{S}^{\perp\perp} = \mathcal{S}$, $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\} \iff \operatorname{rad}(\mathcal{S}^\perp) = \{\mathbf{0}\}$.

Proof.

i. Define $\rho : \mathcal{V} \to \mathcal{S}'$ by $\rho(\mathbf{x})(\mathbf{s}) = \langle \mathbf{s}, \mathbf{x} \rangle$. Then $\ker(\rho) = \mathcal{S}^\perp$.
ii. If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, extend $f \in \mathcal{S}'$ to $\widetilde{f} \in \mathcal{V}'$ and use Theorem 3.2.13. If $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, represent f inside $\mathcal{S}$. Thus ρ is onto, so rank-nullity gives $\dim \mathcal{V} = \dim \mathcal{S}^\perp + \dim \mathcal{S}$.
iii. If $\operatorname{rad}(\mathcal{V}) = \{\mathbf{0}\}$, then $\mathcal{S} \subseteq \mathcal{S}^{\perp\perp}$ and $\dim \mathcal{S}^{\perp\perp} = \dim \mathcal{V} - \dim \mathcal{S}^\perp = \dim \mathcal{S}$, hence $\mathcal{S}^{\perp\perp} = \mathcal{S}$.
($\Rightarrow$) If $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, then $\operatorname{rad}(\mathcal{S}^\perp) = \mathcal{S}^\perp \cap \mathcal{S}^{\perp\perp} = \mathcal{S}^\perp \cap \mathcal{S}$, so $\operatorname{rad}(\mathcal{S}^\perp) = \{\mathbf{0}\}$.
($\Leftarrow$) If $\operatorname{rad}(\mathcal{S}^\perp) = \{\mathbf{0}\}$, then $\mathcal{S}^\perp \cap \mathcal{S} = \{\mathbf{0}\}$, so $\operatorname{rad}(\mathcal{S}) = \mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$.

Structural Commentary: Complement dimensions come from surjectivity of the restriction-pairing map.
Key Insight: Orthogonal complement dimensions are rank-nullity for the restriction-of-pairing map. □

Definition 3.3.2. *Orthogonal Direct Sum*
For subspaces $\mathcal{S}, \mathcal{T} \leq \mathcal{V}$, write $\mathcal{V} = \mathcal{S} \odot \mathcal{T} \iff \mathcal{V} = \mathcal{S} \oplus \mathcal{T}$ and $\mathcal{S} \perp \mathcal{T}$.
Then $\mathcal{V}$ is the *orthogonal direct sum* of $\mathcal{S}$ and $\mathcal{T}$.

Theorem 3.3.3. *Splitting off the Radical*
Let $R = \operatorname{rad}(\mathcal{V})$. Then $\exists \mathcal{S} \leq \mathcal{V}$ such that $\mathcal{V} = R \odot \mathcal{S}$, $\operatorname{rad}(\mathcal{S}) = \{\mathbf{0}\}$, and R is totally singular.

Proof. Choose $\mathcal{S}$ with $\mathcal{V} = R \oplus \mathcal{S}$. Since $R \perp \mathcal{V}$, $\mathcal{V} = R \odot \mathcal{S}$ and $R \subseteq R^\perp$, so R is totally singular. If $\mathbf{s} \in \operatorname{rad}(\mathcal{S})$, then $\mathbf{s} \perp \mathcal{S}$ and $\mathbf{s} \perp R$, hence $\mathbf{s} \in R \cap \mathcal{S} = \{\mathbf{0}\}$.
Structural Commentary: Singularity splits as an invisible summand.
Key Insight: Once the radical is removed, the remaining form is nonsingular. □

Proposition 3.3.4. *Metric Quotients*
Let $\mathcal{U} \leq \mathcal{V}$. The rule $\langle \mathbf{x} + \mathcal{U}, \mathbf{y} + \mathcal{U} \rangle_{\mathcal{V}/\mathcal{U}} := \langle \mathbf{x}, \mathbf{y} \rangle$ is well-defined if and only if $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$.
Then $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\mathcal{U}$. In particular, $\mathcal{V}/\operatorname{rad}(\mathcal{V})$ is nonsingular.

Proof.

($\Rightarrow$) If the rule is well-defined, then $0 = \langle \mathbf{u} + \mathcal{U}, \mathbf{y} + \mathcal{U} \rangle_{\mathcal{V}/\mathcal{U}} = \langle \mathbf{u}, \mathbf{y} \rangle\ \forall\, \mathbf{u} \in \mathcal{U},\ \mathbf{y} \in \mathcal{V}$, so $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$.
($\Leftarrow$) If $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$, replacing $\mathbf{x}, \mathbf{y}$ by $\mathbf{x}+\mathbf{u}, \mathbf{y}+\mathbf{v}$ changes $\langle \mathbf{x}, \mathbf{y} \rangle$ by $\langle \mathbf{u}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle = 0$, rule well-defined.

If $\mathcal{U} \leq \operatorname{rad}(\mathcal{V})$, then $\mathbf{x}+\mathcal{U} \in \operatorname{rad}(\mathcal{V}/\mathcal{U}) \iff \langle \mathbf{x}, \mathbf{y}\rangle = 0 \ \forall \mathbf{y} \in \mathcal{V} \iff \mathbf{x} \in \operatorname{rad}(\mathcal{V})$, hence $\operatorname{rad}(\mathcal{V}/\mathcal{U}) = \operatorname{rad}(\mathcal{V})/\mathcal{U}$.
Structural Commentary: Quotients preserve the form exactly along invisible directions.
Key Insight: The radical is the largest subspace that can be collapsed without making the form ambiguous. □

Proposition 3.3.5. *Radicals and Orthogonal Direct Sums*
If $\mathcal{V} = \mathcal{S} \odot \mathcal{T}$, then $\operatorname{rad}(\mathcal{V}) = \operatorname{rad}(\mathcal{S}) \odot \operatorname{rad}(\mathcal{T})$; $\mathcal{V}/\operatorname{rad}(\mathcal{V}) \cong \mathcal{S}/\operatorname{rad}(\mathcal{S}) \odot \mathcal{T}/\operatorname{rad}(\mathcal{T})$ as metric vector spaces. Hence $\mathcal{V}$ is nonsingular if and only if $\mathcal{S}$ and $\mathcal{T}$ are nonsingular.

Proof. For $\mathbf{v} = \mathbf{s}+\mathbf{t}$, $\mathbf{v} \in \operatorname{rad}(\mathcal{V}) \iff \mathbf{v} \perp \mathcal{V} \iff \mathbf{s} \perp \mathcal{S}$ and $\mathbf{t} \perp \mathcal{T} \iff \mathbf{s} \in \operatorname{rad}(\mathcal{S})$, $\mathbf{t} \in \operatorname{rad}(\mathcal{T})$.
Hence $\operatorname{rad}(\mathcal{V}) = \operatorname{rad}(\mathcal{S}) \odot \operatorname{rad}(\mathcal{T})$. The isomorphism is induced by $\Phi((\mathbf{s}+\mathbf{t})+\operatorname{rad}(\mathcal{V})) = (\mathbf{s}+\operatorname{rad}(\mathcal{S}), \mathbf{t}+\operatorname{rad}(\mathcal{T}))$.
It is well-defined and bijective by the radical formula, and it preserves the form since cross terms vanish.

($\Rightarrow$) If $\mathcal{V}$ is nonsingular, then $\operatorname{rad}(\mathcal{S}) \odot \operatorname{rad}(\mathcal{T}) = \{\mathbf{0}\}$, so $\mathcal{S}$ and $\mathcal{T}$ are nonsingular.
($\Leftarrow$) If $\mathcal{S}$ and $\mathcal{T}$ are nonsingular, then $\operatorname{rad}(\mathcal{V}) = \operatorname{rad}(\mathcal{S}) \odot \operatorname{rad}(\mathcal{T}) = \{\mathbf{0}\}$.

Structural Commentary: Orthogonal direct sums split both visible and invisible parts.
Key Insight: The regular quotient of an orthogonal sum is the orthogonal sum of the regular quotients. □

Corollary 3.3.6. *Recovering a Singular Space from Its Regular Quotient*
If $\dim \mathcal{V} = \dim \mathcal{W}$ and $\mathcal{V}/\operatorname{rad}(\mathcal{V}) \cong \mathcal{W}/\operatorname{rad}(\mathcal{W})$, then $\mathcal{V} \cong \mathcal{W}$ as metric vector spaces.

Proof. By Theorem 3.3.3, choose $\mathcal{V} = \operatorname{rad}(\mathcal{V}) \odot \mathcal{S}$, $\mathcal{W} = \operatorname{rad}(\mathcal{W}) \odot \mathcal{T}$, with $\mathcal{S}, \mathcal{T}$ nonsingular.
The quotient isometry gives $\mathcal{S} \cong \mathcal{T}$. Since $\dim \mathcal{V} = \dim \mathcal{W}$, also $\dim \operatorname{rad}(\mathcal{V}) = \dim \operatorname{rad}(\mathcal{W})$; choose any vector-space isomorphism between the radicals, which is an isometry because both radicals are totally singular.
The orthogonal direct sum of these two isometries gives $\mathcal{V} \cong \mathcal{W}$.
Structural Commentary: A singular metric space is its regular quotient plus invisible dimension.
Key Insight: Once the regular quotient and total dimension are fixed, the radical has no further metric data. □

Definition 3.3.7. *Hyperbolic Pairs and Hyperbolic Spaces*

(i) A *hyperbolic pair* is $(\mathbf{u}, \mathbf{v})$ with $\langle \mathbf{u}, \mathbf{u}\rangle = \langle \mathbf{v}, \mathbf{v}\rangle = 0$ and $\langle \mathbf{u}, \mathbf{v}\rangle = 1$.
(ii) The plane $\operatorname{span}(\mathbf{u}, \mathbf{v})$ of a hyperbolic pair is a *hyperbolic plane*.
(iii) A *hyperbolic space* is an orthogonal direct sum of hyperbolic planes.
(iv) A basis $(\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_k, \mathbf{v}_k)$ is a *hyperbolic basis* if each $(\mathbf{u}_i, \mathbf{v}_i)$ is hyperbolic and the planes $\operatorname{span}(\mathbf{u}_i, \mathbf{v}_i)$ are mutually orthogonal. In the symplectic case it is a *symplectic basis*.

Theorem 3.3.8. *Capturing Isotropic Vectors*
Let $\mathcal{V}$ be nonsingular. If $\mathcal{V}$ is orthogonal, assume $\operatorname{char}(\mathbb{F}) \neq 2$.
For every isotropic $\mathbf{u} \in \mathcal{V}$, $\exists \mathbf{v} \in \mathcal{V}$ such that $(\mathbf{u}, \mathbf{v})$ is hyperbolic.

Proof. Choose $\mathbf{x}$ with $c := \langle \mathbf{u}, \mathbf{x}\rangle \neq 0$. If the form is alternating, take $\mathbf{v} = c^{-1}\mathbf{x}$. Then $\langle \mathbf{u}, \mathbf{v}\rangle = 1$ and $\langle \mathbf{v}, \mathbf{v}\rangle = 0$.
If the form is symmetric, set $\mathbf{v} = c^{-1}\mathbf{x} - \frac{\langle \mathbf{x}, \mathbf{x}\rangle}{2c^2}\mathbf{u}$. Using $\langle \mathbf{u}, \mathbf{u}\rangle = 0$ gives $\langle \mathbf{u}, \mathbf{v}\rangle = 1$ and $\langle \mathbf{v}, \mathbf{v}\rangle = 0$.
Structural Commentary: Nonsingularity supplies a partner; one adjustment makes it isotropic.
Key Insight: Isotropic directions in nonsingular spaces come in hyperbolic pairs. □

Proposition 3.3.9. *Recognizing Orthogonal Hyperbolic Planes*
Let $\mathcal{H}$ be a 2-dimensional nonsingular orthogonal geometry with $\operatorname{char}(\mathbb{F}) \neq 2$. The following are equivalent:

(i) $\mathcal{H}$ is a hyperbolic plane;
(ii) $\exists\, 0 \neq \mathbf{u} \in \mathcal{H}$ with $\langle \mathbf{u}, \mathbf{u}\rangle = 0$;
(iii) $\Delta(\mathcal{H}) = (\mathbb{F}^\times)^2(-1)$ as a square class.

Proof.

(*i*) $\Rightarrow$ (*ii*) A hyperbolic pair consists of nonzero isotropic vectors.
(*ii*) $\Rightarrow$ (*i*) By Theorem 3.3.8, choose $\mathbf{v}$ with $(\mathbf{u}, \mathbf{v})$ hyperbolic. Since $\dim \mathcal{H} = 2$, $\mathcal{H} = \operatorname{span}(\mathbf{u}, \mathbf{v})$.
(*i*) $\Rightarrow$ (*iii*) In a hyperbolic basis, $M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, so $\det M = -1$ and $\Delta(\mathcal{H}) = (\mathbb{F}^\times)^2(-1)$.
(*iii*) $\Rightarrow$ (*ii*) Diagonalize $M = \operatorname{diag}(a, b)$ with $a, b \neq 0$. If $\Delta(\mathcal{H}) = (\mathbb{F}^\times)^2(-1)$, then $ab = -r^2$ for some $r \neq 0$. Hence $a(r/a)^2 + b = 0$. Thus $(r/a, 1)$ is isotropic in this basis.

Structural Commentary: In dimension 2, isotropy and discriminant detect hyperbolicity.
Key Insight: Hyperbolic means the form represents 0 nontrivially. □

Theorem 3.3.10. *Nonsingular Completions*
Let $\mathcal{U}$ be a subspace of a nonsingular metric vector space $\mathcal{V}$, and assume $\operatorname{char}(\mathbb{F}) \neq 2$ in the orthogonal case. Write $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) \odot \mathcal{W}$, where $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ is a basis of $\operatorname{rad}(\mathcal{U})$ and $\mathcal{W}$ is nonsingular. Then $\exists\, \mathbf{v}_1, \dots, \mathbf{v}_r$ such that $\widehat{\mathcal{U}} = \mathcal{H}_r \odot \mathcal{W}$; $\mathcal{H}_r = \operatorname{span}(\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{u}_r, \mathbf{v}_r)$ is nonsingular, contains $\mathcal{U}$, and $\dim \widehat{\mathcal{U}} = \dim \mathcal{U} + \dim \operatorname{rad}(\mathcal{U})$.
The minimal nonsingular extensions of $\mathcal{U}$ are exactly such hyperbolic extensions, up to isometry.

Proof.

(i) Since $\mathcal{W}$ is nonsingular, Theorem 3.3.1 gives $\mathcal{V} = \mathcal{W} \odot \mathcal{W}^\perp$. Thus $\mathbf{u}_i \in \mathcal{W}^\perp$ for all i.

(ii) Construct $\mathbf{v}_j$ inductively. Suppose $\mathcal{H}_{j-1}$ is built and set $\mathcal{S}_j = \operatorname{span}(\mathbf{u}_{j+1}, \dots, \mathbf{u}_r) \odot \mathcal{W} \odot \mathcal{H}_{j-1}$. Then $\mathbf{u}_j \perp \mathcal{S}_j$ and $\mathbf{u}_j \notin \mathcal{S}_j = \mathcal{S}_j^{\perp\perp}$, so $\exists \mathbf{x} \in \mathcal{S}_j^\perp$ with $c := \langle \mathbf{u}_j, \mathbf{x} \rangle \neq 0$. Put

$$\mathbf{v}_j = \begin{cases} c^{-1}\mathbf{x}, & \text{alternating case,} \\ c^{-1}\mathbf{x} - \dfrac{\langle \mathbf{x}, \mathbf{x} \rangle}{2c^2}\mathbf{u}_j, & \text{orthogonal case.} \end{cases}$$

Then $\mathbf{v}_j \in \mathcal{S}_j^\perp$ and $(\mathbf{u}_j, \mathbf{v}_j)$ is hyperbolic; set $\mathcal{H}_j = \mathcal{H}_{j-1} \odot \operatorname{span}(\mathbf{u}_j, \mathbf{v}_j)$.

(iii) After r steps, $\widehat{\mathcal{U}} = \mathcal{H}_r \odot \mathcal{W}$ is nonsingular and $\dim \widehat{\mathcal{U}} = \dim \mathcal{U} + r$.

(iv) Let $\mathcal{X} \supseteq \mathcal{U}$ be nonsingular and choose $\mathcal{X} = \mathcal{U} \oplus \mathcal{Y}$. If $0 \neq \mathbf{u} \in \operatorname{rad}(\mathcal{U})$ paired trivially with $\mathcal{Y}$, then $\mathbf{u} \in \operatorname{rad}(\mathcal{X})$, impossible. Hence the pairing $\operatorname{rad}(\mathcal{U}) \times \mathcal{Y} \to \mathbb{F}$ has zero left radical, so $\dim \mathcal{Y} \geq r$ and $\dim \mathcal{X} \geq \dim \mathcal{U} + r$. If equality holds, the construction inside $\mathcal{X}$ gives $\widehat{\mathcal{U}} \subseteq \mathcal{X}$ with equal dimension, hence $\widehat{\mathcal{U}} = \mathcal{X}$. Any two minimal completions are isometric since each is $\mathcal{W} \odot$ a hyperbolic r-space.

Structural Commentary: Add one hyperbolic partner per radical direction.
Key Insight: Minimal nonsingular completion is radical repair by hyperbolic planes. □

Principle 3.3.11. *Witt Extension and Cancellation*
Let $\mathcal{V} \cong \mathcal{V}'$ be nonsingular metric vector spaces of same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$. Then the following equivalent principles hold:

(i) *Witt extension*: every isometry $\tau : \mathcal{S} \to \mathcal{T}$ between subspaces $\mathcal{S} \leq \mathcal{V}$ and $\mathcal{T} \leq \mathcal{V}'$ extends to an isometry $\overline{\tau} : \mathcal{V} \to \mathcal{V}'$ with $\overline{\tau}|_{\mathcal{S}} = \tau$;

(ii) *Witt cancellation*: for nonsingular subspaces $\mathcal{S} \leq \mathcal{V}$ and $\mathcal{T} \leq \mathcal{V}'$, $\mathcal{S} \cong \mathcal{T} \Rightarrow \mathcal{S}^\perp \cong \mathcal{T}^\perp$.

Proposition 3.3.12. *Witt Extension–Cancellation Bridge*
Let $\mathcal{V} \cong \mathcal{V}'$ be nonsingular metric vector spaces of same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$. Using Theorem 3.3.10, Witt extension $\iff$ Witt cancellation.

Proof.

($\Rightarrow$) Let $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^\perp$, $\mathcal{V}' = \mathcal{T} \odot \mathcal{T}^\perp$, and let $\sigma : \mathcal{S} \to \mathcal{T}$ be an isometry. Extend σ to $\widehat{\sigma} : \mathcal{V} \to \mathcal{V}'$. Then $\widehat{\sigma}(\mathcal{S}^\perp) = \mathcal{T}^\perp$, hence $\mathcal{S}^\perp \cong \mathcal{T}^\perp$.

($\Leftarrow$) Let $\sigma : \mathcal{S} \to \mathcal{T}$ be an isometry. By Theorem 3.3.10, choose minimal nonsingular completions $\widehat{\mathcal{S}} \supseteq \mathcal{S}$ and $\widehat{\mathcal{T}} \supseteq \mathcal{T}$, and extend σ to $\widehat{\sigma} : \widehat{\mathcal{S}} \to \widehat{\mathcal{T}}$ by matching hyperbolic partners of $\operatorname{rad}(\mathcal{S})$ and $\operatorname{rad}(\mathcal{T})$. Then $\mathcal{V} = \widehat{\mathcal{S}} \odot \widehat{\mathcal{S}}^\perp$, $\mathcal{V}' = \widehat{\mathcal{T}} \odot \widehat{\mathcal{T}}^\perp$. Cancellation gives $\eta : \widehat{\mathcal{S}}^\perp \to \widehat{\mathcal{T}}^\perp$, so $\widehat{\sigma} \odot \eta : \mathcal{V} \to \mathcal{V}'$ extends σ.

Structural Commentary: Complete first, then cancel complements.
Key Insight: Complete singular subspaces first; after that, the complement is the only missing piece. □

Theorem 3.3.13. *Witt Extension Theorem*
Let $\mathcal{V} \cong \mathcal{W}$ be nonsingular metric vector spaces of same kind, either symplectic or orthogonal with $\operatorname{char}(\mathbb{F}) \neq 2$. For every subspace isometry $\sigma : \mathcal{S} \to \mathcal{T}$, $\mathcal{S} \leq \mathcal{V}$, $\mathcal{T} \leq \mathcal{W}$, there exists an isometry $\widehat{\sigma} : \mathcal{V} \to \mathcal{W}$ with $\widehat{\sigma}|_{\mathcal{S}} = \sigma$. Thus, if $\mathcal{V} = \mathcal{W}$, every subspace isometry is the restriction of an ambient isometry.

Proof. Induct simultaneously on $(\dim \mathcal{V}, \dim \mathcal{S})$. If $\mathcal{S} = 0$, take any fixed isometry $\mathcal{V} \cong \mathcal{W}$.
Let $\mathcal{R} = \operatorname{rad}(\mathcal{S})$, $\mathcal{R}' = \sigma\mathcal{R} = \operatorname{rad}(\mathcal{T})$. By Theorem 3.3.10, choose minimal nonsingular completions $\widehat{\mathcal{S}} \supseteq \mathcal{S}$ and $\widehat{\mathcal{T}} \supseteq \mathcal{T}$; matching hyperbolic partners over $\mathcal{R}$ and $\mathcal{R}'$ extends σ to $\widehat{\sigma} : \widehat{\mathcal{S}} \to \widehat{\mathcal{T}}$. Hence assume $\mathcal{S}$, $\mathcal{T}$ nonsingular. Choose a nonsingular summand $\mathcal{A} \leq \mathcal{S}$, where

$$\mathcal{A} = \begin{cases} \operatorname{span}(\mathbf{s}, \mathbf{u}) \text{ hyperbolic}, & \text{symplectic case,} \\ \operatorname{span}(\mathbf{s}),\ \langle \mathbf{s}, \mathbf{s} \rangle \neq 0, & \text{orthogonal case.} \end{cases}$$

Set $\mathcal{B} = \sigma\mathcal{A}$, $\mathcal{S}_0 = \mathcal{S} \cap \mathcal{A}^\perp$, and $\mathcal{T}_0 = \mathcal{T} \cap \mathcal{B}^\perp$. One-summand case extends $\sigma|_{\mathcal{A}}$ to $\alpha : \mathcal{V} \to \mathcal{W}$; hence $\alpha(\mathcal{A}^\perp) = \mathcal{B}^\perp$ and $\mathcal{V} = \mathcal{A} \odot \mathcal{A}^\perp$, $\mathcal{W} = \mathcal{B} \odot \mathcal{B}^\perp$, $\mathcal{S} = \mathcal{A} \odot \mathcal{S}_0$, and $\mathcal{T} = \mathcal{B} \odot \mathcal{T}_0$. Since $\dim \mathcal{A}^\perp < \dim \mathcal{V}$, induction gives an isometry $\eta : \mathcal{A}^\perp \to \mathcal{B}^\perp$ with $\eta|_{\mathcal{S}_0} = \sigma|_{\mathcal{S}_0}$. Then $(\sigma|_{\mathcal{A}}) \odot \eta : \mathcal{V} \to \mathcal{W}$ extends σ.

Structural Commentary: Subspace isometries extend ambiently.
Key Insight: Complete radicals, then peel matched summands. □

Corollary 3.3.14. *Witt Cancellation*
Let $\mathcal{V} = \mathcal{S} \odot \mathcal{A}$, $\mathcal{W} = \mathcal{T} \odot \mathcal{B}$ be nonsingular metric vector spaces of the same kind, either symplectic or orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$. *If $\mathcal{V} \cong \mathcal{W}$ and $\mathcal{S} \cong \mathcal{T}$, then $\mathcal{A} \cong \mathcal{B}$.*

Proof. Let $\sigma : \mathcal{S} \to \mathcal{T}$ be an isometry. By Theorem 3.3.13, $\exists\, \widehat{\sigma} : \mathcal{V} \to \mathcal{W}$ with $\widehat{\sigma}|_{\mathcal{S}} = \sigma$.
Since $\mathcal{A} = \mathcal{S}^{\perp}$ and $\mathcal{B} = \mathcal{T}^{\perp}$, $\widehat{\sigma}(\mathcal{A}) = \widehat{\sigma}(\mathcal{S}^{\perp}) = (\widehat{\sigma}\mathcal{S})^{\perp} = \mathcal{T}^{\perp} = \mathcal{B}$. Thus $\widehat{\sigma}|_{\mathcal{A}} : \mathcal{A} \to \mathcal{B}$ is an isometry.
Structural Commentary: Matched summands cancel.
Key Insight: Extend, then restrict to complements. □

Perspective 3.3.15. *The Classification Problem*
Metric vector spaces are isometric exactly when their representing matrices are congruent.
Thus classification up to isometry is equivalent to classification of the associated matrices up to congruence, i.e. to specifying canonical representatives or complete congruence invariants.

3.4 Symplectic Geometry

Theorem 3.4.1. *Classification of Symplectic Geometries*
Let $\mathcal{V}$ be a symplectic geometry, $R = \mathrm{rad}(\mathcal{V})$, and $n = \dim(\mathcal{V})$.

(i) If $R = 0$, then $\mathcal{V}$ is hyperbolic; hence n is even.
(ii) $\mathcal{V} = R \odot \mathcal{H}$, where R is totally singular and $\mathcal{H}$ is hyperbolic.
(iii) If $r = \mathrm{rank}(\mathcal{V}) = 2k$, then some basis has representing matrix $M_{\mathcal{V}} = X_{2k,n-2k} := \mathrm{diag}(J_{2k}, 0_{n-2k})$, $J_{2k} := \bigoplus_{i=1}^{k} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. For symplectic geometries, $\mathcal{V}, \mathcal{W}$, $\mathcal{V} \cong \mathcal{W} \iff (\dim \mathcal{V}, \mathrm{rank}\, \mathcal{V}) = (\dim \mathcal{W}, \mathrm{rank}\, \mathcal{W})$.

Proof.

(i) The case $\mathcal{V} = 0$ is clear. If $R = 0$ and $\mathcal{V} \neq 0$, then every $0 \neq x \in \mathcal{V}$ is isotropic.
By Theorem 3.3.8, x lies in a hyperbolic plane $\mathcal{H}_1 \subset \mathcal{V}$. Since $\mathcal{H}_1$ is nonsingular, Theorem 3.3.1 gives $\mathcal{V} = \mathcal{H}_1 \odot \mathcal{H}_1^{\perp}$. Induction on n gives the claim.
(ii) By Theorem 3.3.3, $\mathcal{V} = R \odot \mathcal{H}$ with $\mathrm{rad}(\mathcal{H}) = 0$.
Hence $\mathcal{H}$ is hyperbolic by (i), and R is totally singular since $R \perp \mathcal{V}$.
(iii) In (ii), $\dim \mathcal{H} = r = 2k$ and $\dim R = n - 2k$.
A symplectic basis of $\mathcal{H}$ followed by any basis of R gives $M_{\mathcal{V}} = X_{2k,n-2k}$. Dimension and rank are isometry invariants; conversely, equal (n, r) gives the same normal matrix, hence an isometry.

Structural Commentary: Alternating forms have only one nonsingular building block: the hyperbolic plane.
Key Insight: Symplectic classification is rank plus dimension. □

Theorem 3.4.2. *Witt Theorems for Symplectic Geometries*
Let $\mathcal{V} \cong \mathcal{V}'$ be nonsingular symplectic geometries. Then Witt extension and cancellation hold for all subspaces.

Proof. By Proposition 3.3.12, it suffices to prove cancellation for nonsingular $\mathcal{S} \subset \mathcal{V}$ and $\mathcal{T} \subset \mathcal{V}'$ with $\mathcal{S} \cong \mathcal{T}$.
Let $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^{\perp}$, $\mathcal{V}' = \mathcal{T} \odot \mathcal{T}^{\perp}$. Then $\dim \mathcal{S}^{\perp} = \dim \mathcal{V} - \dim \mathcal{S} = \dim \mathcal{V}' - \dim \mathcal{T} = \dim \mathcal{T}^{\perp}$.
By Theorem 3.3.1, $\mathcal{S}^{\perp}$ and $\mathcal{T}^{\perp}$ are nonsingular symplectic geometries; by Theorem 3.4.1, equal dimension gives $\mathcal{S}^{\perp} \cong \mathcal{T}^{\perp}$. Thus cancellation holds, and Proposition 3.3.12 gives extension.
Structural Commentary: Cancellation reduces to matching hyperbolic summand counts.
Key Insight: For symplectic forms, dimension alone identifies nonsingular complements. □

Definition 3.4.3. *Symplectic Transvection*
Let $\mathcal{V}$ be a nonsingular symplectic geometry.
For $\mathbf{v} \in \mathcal{V} \setminus \{\mathbf{0}\}$, $a \in \mathbb{F}$, *symplectic transvection* with direction $\mathbf{v}$, parameter a is $\tau_{\mathbf{v},a} : \mathcal{V} \to \mathcal{V}$, $\mathbf{x} \mapsto \mathbf{x} + a \langle \mathbf{x}, \mathbf{v} \rangle \mathbf{v}$.

Proposition 3.4.4. *Basic Properties of Symplectic Transvections*
Let $\mathcal{V}$ be a nonsingular symplectic geometry, $0 \neq \mathbf{v} \in \mathcal{V}$, $a, b \in \mathbb{F}$, $c \in \mathbb{F}^{\times}$, and $\sigma \in \mathrm{Sp}(\mathcal{V})$. Then:

(i) $\tau_{\mathbf{v},a} \in \mathrm{Sp}(\mathcal{V})$;
(ii) $\tau_{\mathbf{v},a} = I$ if and only if $a = 0$;
(iii) $\tau_{\mathbf{v},a}\tau_{\mathbf{v},b} = \tau_{\mathbf{v},a+b}$ and $\tau_{\mathbf{v},a}^{-1} = \tau_{\mathbf{v},-a}$;
(iv) $\sigma\tau_{\mathbf{v},a}\sigma^{-1} = \tau_{\sigma\mathbf{v},a}$;
(v) $\tau_{c\mathbf{v},a} = \tau_{\mathbf{v},ac^2}$.

Proof.

(i) For $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, alternation and skew-symmetry give
$\langle \tau_{\mathbf{v},a}\mathbf{x}, \tau_{\mathbf{v},a}\mathbf{y}\rangle = \langle \mathbf{x},\mathbf{y}\rangle + a\langle \mathbf{x},\mathbf{v}\rangle\langle \mathbf{v},\mathbf{y}\rangle + a\langle \mathbf{y},\mathbf{v}\rangle\langle \mathbf{x},\mathbf{v}\rangle + a^2\langle \mathbf{x},\mathbf{v}\rangle\langle \mathbf{y},\mathbf{v}\rangle\langle \mathbf{v},\mathbf{v}\rangle = \langle \mathbf{x},\mathbf{y}\rangle$.
Also $\tau_{\mathbf{v},-a}\tau_{\mathbf{v},a}(\mathbf{x}) = \mathbf{x} = \tau_{\mathbf{v},a}\tau_{\mathbf{v},-a}(\mathbf{x})$, so $\tau_{\mathbf{v},a}$ is invertible and hence symplectic.

(ii) If $a = 0$, then $\tau_{\mathbf{v},a} = I$. If $a \neq 0$, nonsingularity gives $\mathbf{x} \in \mathcal{V}$ with $\langle \mathbf{x},\mathbf{v}\rangle \neq 0$, so $\tau_{\mathbf{v},a}\mathbf{x} - \mathbf{x} = a\langle \mathbf{x},\mathbf{v}\rangle\,\mathbf{v} \neq \mathbf{0}$.

(iii) For $\mathbf{x} \in \mathcal{V}$, since $\langle \mathbf{v},\mathbf{v}\rangle = 0$, $\tau_{\mathbf{v},a}\tau_{\mathbf{v},b}(\mathbf{x}) = \mathbf{x} + (a+b)\langle \mathbf{x},\mathbf{v}\rangle\,\mathbf{v} = \tau_{\mathbf{v},a+b}(\mathbf{x})$. Taking $b = -a$ gives $\tau_{\mathbf{v},a}^{-1} = \tau_{\mathbf{v},-a}$.

(iv) For $\mathbf{x} \in \mathcal{V}$, $\sigma\tau_{\mathbf{v},a}\sigma^{-1}(\mathbf{x}) = \mathbf{x} + a\langle \sigma^{-1}\mathbf{x},\mathbf{v}\rangle\,\sigma\mathbf{v} = \mathbf{x} + a\langle \mathbf{x},\sigma\mathbf{v}\rangle\,\sigma\mathbf{v} = \tau_{\sigma\mathbf{v},a}(\mathbf{x})$.

(v) For $\mathbf{x} \in \mathcal{V}$, $\tau_{c\mathbf{v},a}(\mathbf{x}) = \mathbf{x} + a\langle \mathbf{x},c\mathbf{v}\rangle\,c\mathbf{v} = \mathbf{x} + ac^2\langle \mathbf{x},\mathbf{v}\rangle\,\mathbf{v} = \tau_{\mathbf{v},ac^2}(\mathbf{x})$

Structural Commentary: Fix $\mathbf{v}^\perp$; shear along $\mathbb{F}\mathbf{v}$.
Key Insight: Alternation cancels the possible quadratic error term. □

Theorem 3.4.5. *Symplectic Transvections Generate*
For a nonsingular symplectic geometry $\mathcal{V}$, $\mathrm{Sp}(\mathcal{V}) = \langle \tau_{\mathbf{v},a} : 0 \neq \mathbf{v} \in \mathcal{V},\ a \in \mathbb{F}\rangle$.

Proof. Let G denote the right-hand side. By Proposition 3.4.4, $G \leq \mathrm{Sp}(\mathcal{V})$.
If $\mathbf{x},\mathbf{y} \neq \mathbf{0}$, $\langle \mathbf{x},\mathbf{y}\rangle \neq 0$, then $\tau_{\mathbf{y}-\mathbf{x},\,\langle \mathbf{x},\mathbf{y}\rangle^{-1}}(\mathbf{x}) = \mathbf{y}$. If $\langle \mathbf{x},\mathbf{y}\rangle = 0$, choose $\mathbf{z}$ with $\langle \mathbf{x},\mathbf{z}\rangle\langle \mathbf{z},\mathbf{y}\rangle \neq 0$, $\mathbf{x} \mapsto \mathbf{z} \mapsto \mathbf{y}$.
Hence G is transitive on nonzero vectors; the same argument in $\mathbf{u}^\perp/\mathbb{F}\mathbf{u}$ gives transitivity on hyperbolic pairs.
Let $\sigma \in \mathrm{Sp}(\mathcal{V})$ and induct on $n = \dim \mathcal{V}$. For $n = 2$, hyperbolic-pair transitivity gives $\sigma \in G$.
For $n > 2$, choose a hyperbolic plane $\mathcal{H} = \mathrm{span}(\mathbf{u},\mathbf{v})$. Pair transitivity gives $\gamma \in G$ with $\gamma\sigma|_{\mathcal{H}} = I_{\mathcal{H}}$.
Then $\gamma\sigma$ restricts to an element of $\mathrm{Sp}(\mathcal{H}^\perp)$, which is a product of transvections by induction; extended by the identity on $\mathcal{H}$, these are transvections of $\mathcal{V}$. Thus $\sigma \in G$.
Structural Commentary: Symplectic transformations are built from elementary shears along null directions.
Key Insight: Hyperbolic pairs are the movable frames of symplectic geometry. □

Proposition 3.4.6. *Symplectic Transformations Have Determinant One*
If $\mathcal{V}$ is nonsingular symplectic, then $\mathrm{Sp}(\mathcal{V}) \subseteq \mathrm{SL}(\mathcal{V})$.

Proof. By Theorem 3.4.5, it suffices to show $\det(\tau_{\mathbf{v},a}) = 1$. Let $\tau_{\mathbf{v},a} = I + a\,\mathbf{v} \otimes \lambda_{\mathbf{v}}$, $\lambda_{\mathbf{v}}(\mathbf{x}) = \langle \mathbf{x},\mathbf{v}\rangle$.
Since $\lambda_{\mathbf{v}}(\mathbf{v}) = \langle \mathbf{v},\mathbf{v}\rangle = 0$, $\tau_{\mathbf{v},a}$ fixes $\mathbb{F}\mathbf{v}$ and induces I on $\mathcal{V}/\mathbb{F}\mathbf{v}$. Hence its matrix in any basis starting with $\mathbf{v}$ is unipotent triangular, so $\det(\tau_{\mathbf{v},a}) = 1$.
Structural Commentary: Generators rule out $\det = -1$.
Key Insight: Symplectic transvections have determinant 1, and they generate the symplectic group. □

3.5 Classical Groups and Isometries

Theorem 3.5.1. *Orthogonal Bases and Diagonal Congruence*
Let $(\mathcal{V}, B)$ be an orthogonal geometry, $q(\mathbf{x}) := B(\mathbf{x},\mathbf{x})$, $n = \dim \mathcal{V}$, and $r = \operatorname{rank} B$.
Then $\exists$ basis β such that $M_\beta = \mathrm{diag}(a_1,\dots,a_r,0_{n-r})$, $\quad a_i \in \mathbb{F}^\times \iff \neg\big(\mathrm{char}(\mathbb{F}) = 2 \ \wedge\ q \equiv 0 \ \wedge\ B \neq 0\big)$.
Equivalently, a symmetric matrix is congruent to a diagonal matrix iff it is not a nonzero alternating matrix in characteristic 2.

Proof.

($\Rightarrow$) If M_β is diagonal and $\mathrm{char}(\mathbb{F}) = 2$ with $q \equiv 0$, then all diagonal entries of M_β vanish. Hence $M_\beta = 0$, so $B = 0$.

($\Leftarrow$) If $B = 0$, any basis works. Otherwise, the hypothesis and Proposition 3.1.6 give $q \not\equiv 0$.
Peel off nonisotropic lines, where $\mathcal{V} = \mathbb{F}\mathbf{u}_1 \odot \cdots \odot \mathbb{F}\mathbf{u}_s \odot \mathcal{W}$, $q(\mathbf{u}_i) = a_i \in \mathbb{F}^\times$, $q|_{\mathcal{W}} \equiv 0$.
If $\mathrm{char}(\mathbb{F}) \neq 2$, Proposition 3.1.6 gives $B|_{\mathcal{W}} = 0$, so append any basis of $\mathcal{W}$.
If $\mathrm{char}(\mathbb{F}) = 2$, then $\mathcal{W} = R \odot \mathcal{H}_1 \odot \cdots \odot \mathcal{H}_t$ by Theorem 3.4.1.
For each hyperbolic plane $\mathcal{H}_j = \mathrm{span}(\mathbf{e},\mathbf{f})$, rescale so $B(\mathbf{e},\mathbf{f}) = a = q(\mathbf{u})$ for some current nonisotropic $\mathbf{u}$.
Then $\mathbb{F}\mathbf{u} \odot \mathcal{H}_j = \mathbb{F}(\mathbf{u}+\mathbf{e}) \odot \mathbb{F}(\mathbf{u}+\mathbf{f}) \odot \mathbb{F}(\mathbf{u}+\mathbf{e}+\mathbf{f})$ is orthogonal with all three norms a.
Absorb all $\mathcal{H}_j$ and append any basis of R.

Structural Commentary: Only nonzero alternating characteristic-2 blocks obstruct diagonals.
Key Insight: Peel off nonisotropic lines until only the radical or characteristic-2 alternating residue remains. □

Example 3.5.2. *Characteristic Two Orthogonal-Symplectic Plane*
Assume $\mathrm{char}(\mathbb{F}) = 2$. Let $\mathcal{H} = \mathbb{F}\mathbf{e} \oplus \mathbb{F}\mathbf{f}$ with Gram matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
Then B is symmetric, alternating, and nonsingular. For every $a,b \in \mathbb{F}$, $B(a\mathbf{e}+b\mathbf{f}, a\mathbf{e}+b\mathbf{f}) = 2ab = 0$, so $\mathcal{H}$ is both orthogonal and symplectic. If $0 \neq \mathbf{x} \in \mathcal{H}$, then $\{\mathbf{x}\}^\perp = \mathbb{F}\mathbf{x}$; hence no two independent nonzero vectors are orthogonal. Thus $\mathcal{H}$ has no orthogonal basis, the obstruction excluded in Theorem 3.5.1.

Method 3.5.3. *Lagrange Orthogonalization*
Let $\mathcal{V}$ be an orthogonal geometry with $\mathrm{char}(\mathbb{F}) \neq 2$, and set $q(\mathbf{x}) = \langle \mathbf{x}, \mathbf{x} \rangle$.
From a generating list $L = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, construct an orthogonal basis as follows.

(i) In the current summand with generators $(\mathbf{x}_i)$, choose a pivot $\mathbf{u}$ with $q(\mathbf{u}) \neq 0$; if none exists, stop with a totally degenerate remainder. If all $\mathbf{x}_i$ are isotropic but the summand is not totally degenerate, take $\mathbf{u} = \mathbf{x}_i + \mathbf{x}_j$ with $\langle \mathbf{x}_i, \mathbf{x}_j \rangle \neq 0$.
(ii) Given mutually orthogonal pivots $\mathbf{u}_1, \dots, \mathbf{u}_k$, let $S_k = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$.
Replace each generator $\mathbf{x}_i$ by $\mathbf{w}_i = \mathbf{x}_i - \sum_{j=1}^{k} \frac{\langle \mathbf{x}_i, \mathbf{u}_j \rangle}{q(\mathbf{u}_j)} \mathbf{u}_j$. Then the nonzero $\mathbf{w}_i$ generate $S_k^{\perp}$.
(iii) Iterate inside $S_k^{\perp}$ until the remaining summand R is totally degenerate.
Output $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{z}_1, \dots, \mathbf{z}_m)$, where $(\mathbf{z}_1, \dots, \mathbf{z}_m)$ is any basis of R.

Only nonisotropic pivots are allowed, since the projection step divides by $q(\mathbf{u}_j) \neq 0$.

Theorem 3.5.4. *Canonical Orthogonal Forms over Common Fields*
Let $M = M^t \in M_n(\mathbb{F})$, let $r = \operatorname{rank} M$, and write $M \sim N$ for congruence. If $\mathrm{char}(\mathbb{F}) = 2$, assume M is not nonzero alternating.

(i) If $\mathbb{F}$ is algebraically closed, then $M \sim I_r \oplus 0_{n-r}$; r is complete.
(ii) If $\mathbb{F} = \mathbb{R}$, then $M \sim I_p \oplus (-I_m) \oplus 0_k$ with $p + m + k = n$; (p, m) is complete.
(iii) If $\mathbb{F} = \mathbb{F}_q$ and $\mathrm{char}(\mathbb{F}) = 2$, then $M \sim I_r \oplus 0_{n-r}$.
(iv) If $\mathbb{F} = \mathbb{F}_q$ has odd characteristic and $d \in \mathbb{F}_q^{\times}$ is a fixed nonsquare, then for $r > 0$, $M \sim I_r \oplus 0_{n-r}$ or $M \sim I_{r-1} \oplus [d] \oplus 0_{n-r}$ with exactly one alternative; for $r = 0$, $M \sim 0_n$.

Proof.

(i) By Theorem 3.5.1, write $M \sim \mathrm{diag}(a_1, \dots, a_r, 0_{n-r})$ with $a_i \neq 0$.
If $\mathbb{F}$ is algebraically closed, $a_i = s_i^2$, so scaling by s_i^{-1} gives $M \sim I_r \oplus 0_{n-r}$.
(ii) Write $M \sim \mathrm{diag}(a_1, \dots, a_r, 0_k)$ over $\mathbb{R}$. Rescaling gives $a_i = \pm 1$, hence $M \sim I_p \oplus (-I_m) \oplus 0_k$.
Uniqueness of (p, m) is Sylvester's inertia theorem, Theorem 2.2.14.
(iii) Write $M \sim \mathrm{diag}(a_1, \dots, a_r, 0_{n-r})$ over $\mathbb{F}_q$ with $\mathrm{char}(\mathbb{F}_q) = 2$.
Since $x \mapsto x^2$ is an automorphism, each a_i is a square; hence $M \sim I_r \oplus 0_{n-r}$.
(iv) Write $M \sim \mathrm{diag}(a_1, \dots, a_r, 0_{n-r})$ over odd $\mathbb{F}_q$. Since $\mathbb{F}_q^{\times}/(\mathbb{F}_q^{\times})^2 = \{1, d\}$, the nonsingular diagonal part is determined by its discriminant square class $\delta(M) := a_1 \cdots a_r \pmod{(\mathbb{F}_q^{\times})^2}$, giving I_r when $\delta(M) = 1$ and $I_{r-1} \oplus [d]$ when $\delta(M) = d$. Appending the zero block gives the stated forms, and δ separates them.

Structural Commentary: Once diagonalized, congruence classification becomes square-class bookkeeping.
Key Insight: The base field determines how many diagonal coefficients can be normalized to 1. □

Definition 3.5.5. *Universal Orthogonal Geometry*
An orthogonal geometry $\mathcal{V}$ over $\mathbb{F}$ is *universal* if its norm map $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$ represents every scalar $q(\mathcal{V}) = \mathbb{F}$.

Proposition 3.5.6. *Finite-Field Square Classes and Universality*
Let $\mathbb{F}_q$ be a finite field and let $S = (\mathbb{F}_q^{\times})^2$.

(i) If $\mathrm{char}(\mathbb{F}_q) = 2$, then every element of $\mathbb{F}_q$ is a square.
(ii) If $\mathrm{char}(\mathbb{F}_q) \neq 2$, then $|S| = (q-1)/2$ and $\mathbb{F}_q^{\times} \setminus S = dS$ for any nonsquare d.
(iii) If $\mathcal{V}$ is orthogonal over odd $\mathbb{F}_q$ and contains a nonsingular subspace of dimension ≥ 2, then $\mathcal{V}$ is universal.

Proof.

(i) The map $\varphi : \mathbb{F}_q \to \mathbb{F}_q$, $\varphi(x) = x^2$, is the Frobenius automorphism in characteristic 2.
(ii) For $\mathrm{char}(\mathbb{F}_q) \neq 2$, the homomorphism $\varphi : \mathbb{F}_q^{\times} \to \mathbb{F}_q^{\times}$, $\varphi(x) = x^2$, has kernel $\{\pm 1\}$.
Hence $|S| = (q-1)/2$, and the nonsquares are the other coset dS.
(iii) Let $q(\mathbf{x}) = \langle \mathbf{x}, \mathbf{x} \rangle$. By Theorem 3.5.1, the nonsingular subspace contains orthogonal nonisotropic $\mathbf{u}, \mathbf{v}$; set $a = q(\mathbf{u})$, $b = q(\mathbf{v})$. For fixed $c \in \mathbb{F}_q$, the sets $A = a\{\alpha^2 : \alpha \in \mathbb{F}_q\}$ and $B = c - b\{\beta^2 : \beta \in \mathbb{F}_q\}$ both have $(q+1)/2$ elements, so $A \cap B \neq \varnothing$. Thus $c = a\alpha^2 + b\beta^2 = q(\alpha \mathbf{u} + \beta \mathbf{v})$ for some $\alpha, \beta \in \mathbb{F}_q$.

Structural Commentary: Over finite fields, cardinality replaces order arguments.
Key Insight: Two half-sized subsets of a finite field must overlap. □

Example 3.5.7. *Two Rank-Two Orthogonal Planes over $\mathbb{F}_5$*
In $\mathbb{F}_5$, $(\mathbb{F}_5^{\times})^2 = \{1, 4\}$ and the nonsquare class is $\{2, 3\}$. Let $q_1(x, y) = x^2 + y^2$, $q_2(x, y) = x^2 + 2y^2$.
Then q_1 has Gram matrix I_2 and is hyperbolic, since $q_1(1, 2) = 1 + 4 = 0$.

The form q_2 has Gram matrix $\mathrm{diag}(1, 2)$ and is anisotropic: if $q_2(x, y) = 0$ with $y \neq 0$, then $\left(\frac{x}{y}\right)^2 = -2 = 3$, impossible since 3 is nonsquare; if $y = 0$, then $x = 0$.

Thus I_2 and $\mathrm{diag}(1,2)$ realize the two nonsingular rank-2 classes in Theorem 3.5.4.
Structural Commentary: Odd finite-field classes are square-class classes.
Key Insight: In rank 2, the square class of the determinant decides whether the plane is hyperbolic or anisotropic.

Lemma 3.5.8. *Odd Finite-Field Isotropy Bound*
Let $\mathbb{F}_q$ have odd characteristic, $\mathcal{V}$ be nonsingular orthogonal with $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$. Let $\mathcal{H}$ be hyperbolic plane.

(i) If $\dim \mathcal{V} \geq 3$, then $\mathcal{V}$ is isotropic.
(ii) $\mathcal{V} \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$, where $\mathcal{R}$ is nonsingular anisotropic and $\dim \mathcal{R} \leq 2$.

Proof.

(i) By Theorem 3.5.1, choose a nonsingular plane $\mathcal{P} \subset \mathcal{V}$ and $0 \neq \mathbf{z} \in \mathcal{P}^{\perp}$. If $q(\mathbf{z}) = 0$, then $\mathbf{z}$ is isotropic. Otherwise Proposition 3.5.6 gives $\mathbf{p} \in \mathcal{P}$ with $q(\mathbf{p}) = -q(\mathbf{z})$, so $\mathbf{p} + \mathbf{z}$ is isotropic.
(ii) Induct on $\dim \mathcal{V}$. If $\mathcal{V}$ is anisotropic, take $s = 0$ and $\mathcal{R} = \mathcal{V}$; then (i) gives $\dim \mathcal{R} \leq 2$. Otherwise Theorem 3.3.8 gives $\mathcal{V} \cong \mathcal{H} \odot \mathcal{W}$ with $\mathcal{W}$ nonsingular and $\dim \mathcal{W} = \dim \mathcal{V} - 2$, so induction gives the decomposition.

Structural Commentary: Odd finite fields force isotropy in dimension ≥ 3.
Key Insight: Hyperbolic splitting leaves only a residue of dimension at most 2. □

Theorem 3.5.9. *Odd Finite-Field Orthogonal Decomposition*
Let $\mathbb{F}_q$ have odd characteristic, fix a nonsquare $d \in \mathbb{F}_q^{\times}$, and let $\mathcal{V}$ be nonsingular orthogonal. Write $\mathcal{H}$ for a hyperbolic plane, $\langle a \rangle$ for a line of value a, and $\mathcal{A}_d := \langle 1 \rangle \odot \langle -d \rangle$.

(i) If $\dim \mathcal{V} = 2m+1$, then $\exists\, a \in \mathbb{F}_q^{\times}$ such that $\mathcal{V} \cong \mathcal{H}^{\odot m} \odot \langle a \rangle$, and $a(\mathbb{F}_q^{\times})^2 = (-1)^m \Delta(\mathcal{V})$.
(ii) If $\dim \mathcal{V} = 2m$, then exactly one holds: $\mathcal{V} \cong \mathcal{H}^{\odot m} \iff \Delta(\mathcal{V}) = (-1)^m (\mathbb{F}_q^{\times})^2$ or $\mathcal{V} \cong \mathcal{H}^{\odot (m-1)} \odot \mathcal{A}_d \iff \Delta(\mathcal{V}) = (-1)^m d (\mathbb{F}_q^{\times})^2$.

Proof.

(i) By Lemma 3.5.8, $\mathcal{V} \cong \mathcal{H}^{\odot m} \odot \mathcal{R}$ with $\dim \mathcal{R} = 1$, so $\mathcal{R} \cong \langle a \rangle$.
Since $\Delta(\mathcal{H}) = -(\mathbb{F}_q^{\times})^2$, $\Delta(\mathcal{V}) = (-1)^m a (\mathbb{F}_q^{\times})^2$, hence $a(\mathbb{F}_q^{\times})^2 = (-1)^m \Delta(\mathcal{V})$.
(ii) By Lemma 3.5.8, $\mathcal{V} \cong \mathcal{H}^{\odot s} \odot \mathcal{R}$ with $\dim \mathcal{R} = 0$ or 2. If $\dim \mathcal{R} = 0$, then $s = m$ and $\Delta(\mathcal{V}) = (-1)^m (\mathbb{F}_q^{\times})^2$. If $\dim \mathcal{R} = 2$, then $s = m - 1$ and Proposition 3.3.9 shows that the anisotropic residue is the nonsquare class $\mathcal{A}_d$, with $\Delta(\mathcal{A}_d) = -d(\mathbb{F}_q^{\times})^2$; hence $\Delta(\mathcal{V}) = (-1)^m d (\mathbb{F}_q^{\times})^2$.

Structural Commentary: The anisotropic residue has dimension ≤ 2.
Key Insight: Parity and discriminant determine the residue. □

Example 3.5.10. *Quadratic Refinements over $\mathbb{F}_2$*
On $V = \mathbb{F}_2^2$, set $B((x,y),(u,v)) = xv + yu$, $Q_0(x,y) = xy$, $Q_1(x,y) = x^2 + xy + y^2$. Then $B_{Q_0} = B_{Q_1} = B$, the unique nonsingular alternating plane over $\mathbb{F}_2$. However, $Q_0^{-1}(0) \setminus \{\mathbf{0}\} = \{(1,0),(0,1)\}$, $Q_1^{-1}(0) = \{\mathbf{0}\}$.
Hence $Q_0 \ncong Q_1$ although they have the same polar form.
Structural Commentary: One polar form, two refinements.
Key Insight: Track Q, not only B_Q.

Definition 3.5.11. *Arf Invariant over $\mathbb{F}_2$*
Let $Q : \mathcal{V} \to \mathbb{F}_2$ be quadratic.

(i) Its *polar form* is $B_Q(\mathbf{x},\mathbf{y}) = Q(\mathbf{x}+\mathbf{y}) + Q(\mathbf{x}) + Q(\mathbf{y})$ and is assumed nonsingular.
(ii) For a *symplectic basis* $\beta = (\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_m, \mathbf{v}_m)$ of B_Q,
the *Arf invariant* is $\mathrm{Arf}(Q) := \sum_{i=1}^{m} Q(\mathbf{u}_i) Q(\mathbf{v}_i) \in \mathbb{F}_2$, independent of β.

Theorem 3.5.12. *Quadratic Refinements over $\mathbb{F}_2$*
Let $Q : \mathcal{V} \to \mathbb{F}_2$ be quadratic with nonsingular polar form B_Q. Then $\dim \mathcal{V} = 2m$, and $Q \cong Q_{\mathrm{Arf}(Q)}$.
Here $Q_0(x_1, y_1, \ldots, x_m, y_m) := \sum_{i=1}^{m} x_i y_i$ and, for $m \geq 1$, $Q_1(x_1, y_1, \ldots, x_m, y_m) := x_1^2 + x_1 y_1 + y_1^2 + \sum_{i=2}^{m} x_i y_i$.
Q_0 has Arf invariant 0, and Q_1 has Arf invariant 1.

Proof. As B_Q nonsingular alternating, Theorem 3.4.1 gives symplectic basis $(\mathbf{u}_1, \mathbf{v}_1, \ldots, \mathbf{u}_m, \mathbf{v}_m)$, $\dim \mathcal{V} = 2m$.
On each hyperbolic pair $(\mathbf{u}, \mathbf{v})$, if $Q(\mathbf{u})Q(\mathbf{v}) = 0$, a change of pair gives the refined plane xy.
If $Q(\mathbf{u}) = Q(\mathbf{v}) = 1$, the plane is $x^2 + xy + y^2$.
Thus Q is an orthogonal sum of refined planes of types $H_Q := xy$ and $A_Q := x^2 + xy + y^2$.
Since $A_Q \odot A_Q \cong H_Q \odot H_Q$, all pairs of anisotropic refinements cancel.
Remaining parity is $\mathrm{Arf}(Q) = \sum_{i=1}^{m} Q(\mathbf{u}_i) Q(\mathbf{v}_i) \in \mathbb{F}_2$, so $Q \cong Q_0$ when $\mathrm{Arf}(Q) = 0$ and $Q \cong Q_1$ when $\mathrm{Arf}(Q) = 1$.

Structural Commentary: In characteristic 2, the same symplectic polar form supports two quadratic geometries.
Key Insight: The Arf invariant is the missing one-bit datum that the polar bilinear form cannot see. □

Definition 3.5.13. *Isometries and Classical Groups*
Let $\mathcal{V}, \mathcal{W}$ be metric vector spaces.

(i) A bijective linear map $\sigma : \mathcal{V} \to \mathcal{W}$ is an *isometry* if $\langle \sigma\mathbf{x}, \sigma\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle \ \ \forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$.
(ii) The group of isometries $\mathcal{V} \to \mathcal{V}$ is denoted $\mathrm{Iso}(\mathcal{V})$. If $\mathcal{V}$ is nonsingular orthogonal, this is the *orthogonal group* $\mathcal{O}(\mathcal{V})$; if $\mathcal{V}$ is nonsingular symplectic, this is the *symplectic group* $\mathrm{Sp}(\mathcal{V})$.

Proposition 3.5.14. *Matrix Criterion for Isometries*
Let β be a basis of a metric vector space $\mathcal{V}$, let $M = M_\beta$, and let $A = [\sigma]_{\beta \leftarrow \beta}$.
Then $\sigma \in \mathrm{Iso}(\mathcal{V}) \iff \sigma$ is bijective and $A^t M A = M$. If M is nonsingular, then $\det A = \pm 1$.

Proof.

($\Rightarrow$) If σ is an isometry, then it is bijective and, for all coordinate vectors X, Y, $X^t A^t M A Y = (AX)^t M (AY) = X^t M Y$. Hence $A^t M A = M$.
($\Leftarrow$) If σ is bijective and $A^t M A = M$, then for all $\mathbf{x}, \mathbf{y} \in \mathcal{V}$ with coordinate vectors X, Y, $\langle \sigma\mathbf{x}, \sigma\mathbf{y}\rangle = (AX)^t M (AY) = X^t M Y = \langle \mathbf{x}, \mathbf{y}\rangle$, so σ is an isometry.

If M is invertible, taking determinants in $A^t M A = M$ gives $(\det A)^2 = 1$, hence $\det A = \pm 1$.
Structural Commentary: Isometries are exactly the coordinate changes that stabilize the representing matrix.
Key Insight: Orthogonal and symplectic groups are stabilizer groups under congruence. □

Proposition 3.5.15. *Fixed Spaces of Isometries*
Let $\mathcal{V}$ be nonsingular and let $\tau \in \mathrm{Iso}(\mathcal{V})$. Then $\ker(I - \tau) = \mathrm{im}(I - \tau)^\perp$, and $\dim \ker(I - \tau) = \dim \mathrm{im}(I - \tau)^\perp$.

Proof. If $\mathbf{x}$ is fixed by τ, then for every $\mathbf{y}$, $\langle \mathbf{x}, (I - \tau)\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle - \langle \tau\mathbf{x}, \tau\mathbf{y}\rangle = 0$, so $\ker(I - \tau) \subseteq \mathrm{im}(I - \tau)^\perp$.
The dimensions agree by rank-nullity and Theorem 3.3.1, hence equality holds.
Structural Commentary: The fixed space of an isometry is the orthogonal complement of its displacement space.
Key Insight: Fixed vectors are exactly vectors orthogonal to all displacements. □

Proposition 3.5.16. *Symmetries and Fixed Hyperplanes*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, and let $\tau \in \mathcal{O}(\mathcal{V})$ with $\tau \neq I$. Then $\tau = \sigma_{\mathbf{u}}$ for some nonisotropic $\mathbf{u} \iff \ker(I - \tau)$ is a nonsingular hyperplane. In this case $\ker(I - \sigma_{\mathbf{u}}) = \mathrm{span}(\mathbf{u})^\perp$.

Proof.

($\Rightarrow$) The formula for $\sigma_{\mathbf{u}}$ gives $\sigma_{\mathbf{u}}\mathbf{u} = -\mathbf{u}$ and fixes $\mathrm{span}(\mathbf{u})^\perp$ pointwise. Hence $\ker(I - \sigma_{\mathbf{u}}) = \mathrm{span}(\mathbf{u})^\perp$, a nonsingular hyperplane.
($\Leftarrow$) Let $\mathcal{H} = \ker(I - \tau)$ be a nonsingular hyperplane and write $\mathcal{V} = \mathcal{H} \odot \mathcal{L}$ with $\dim \mathcal{L} = 1$. Since τ fixes $\mathcal{H}$ and is an isometry, $\tau\mathcal{L} = \mathcal{L}$. Thus $\tau|_{\mathcal{L}} = \lambda I$ with $\lambda^2 = 1$; as $\tau \neq I$ and $\mathrm{char}(\mathbb{F}) \neq 2$, $\lambda = -1$. For $0 \neq \mathbf{u} \in \mathcal{L}$, $\tau = \sigma_{\mathbf{u}}$.

Structural Commentary: Symmetries are exactly fixed-hyperplane isometries.
Key Insight: A symmetry fixes a nonsingular hyperplane and flips its orthogonal line. □

Proposition 3.5.17. *No Unipotent Isometries in Anisotropic Spaces*
Let $\mathcal{V}$ be nonsingular anisotropic. If $\tau \in \mathrm{Iso}(\mathcal{V})$ is unipotent, then $\tau = I$.

Proof. Put $N = \tau - I$. Then N is nilpotent and, by Proposition 3.2.15, $N^* = \tau^* - I = \tau^{-1} - I \in N\mathbb{F}[N]$.
If $N \neq 0$, choose maximal $k \geq 1$ with $N^k \neq 0$ and choose $\mathbf{x}$ with $\mathbf{u} = N^k\mathbf{x} \neq 0$.
Since $N^{k+1} = 0$ and $(N^*)^k \in N^k\mathbb{F}[N]$, $\langle \mathbf{u}, \mathbf{u}\rangle = \langle N^k\mathbf{x}, N^k\mathbf{x}\rangle = \langle \mathbf{x}, (N^*)^k N^k \mathbf{x}\rangle = 0$, contradicting anisotropy.
Hence $N = 0$ and $\tau = I$.
Structural Commentary: Top nilpotent displacements are isotropic.
Key Insight: An anisotropic form leaves no room for unipotent shear. □

Definition 3.5.18. *Rotations, Reflections, and Symmetries*
Let $\mathcal{V}$ be a nonsingular orthogonal geometry.

(i) For $\sigma \in \mathcal{O}(\mathcal{V})$, $\det \sigma = 1$ means σ is a *rotation*, and $\det \sigma = -1$ means σ is a *reflection*.
(ii) If $\mathrm{char}(\mathbb{F}) \neq 2$ and $q(\mathbf{u}) = \langle \mathbf{u}, \mathbf{u}\rangle \neq 0$, the *symmetry* in the line $\mathbb{F}\mathbf{u}$ is $\sigma_{\mathbf{u}}(\mathbf{x}) = \mathbf{x} - 2\frac{\langle \mathbf{x}, \mathbf{u}\rangle}{\langle \mathbf{u}, \mathbf{u}\rangle}\mathbf{u}$.
It satisfies $\sigma_{\mathbf{u}}\mathbf{u} = -\mathbf{u}$ and $\sigma_{\mathbf{u}}|_{\mathrm{span}(\mathbf{u})^\perp} = I$.

Proposition 3.5.19. *Matching Equal-Length Vectors by a Symmetry*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, and set $q(\mathbf{x}) = \langle \mathbf{x}, \mathbf{x}\rangle$.
If $q(\mathbf{u}) = q(\mathbf{v}) \neq 0$, then $\exists$ a symmetry σ with $\sigma\mathbf{u} = \mathbf{v}$ or $\sigma\mathbf{u} = -\mathbf{v}$.

Proof. Let $\mathbf{a} = \mathbf{u} - \mathbf{v}$ and $\mathbf{b} = \mathbf{u} + \mathbf{v}$. Then $\langle \mathbf{a}, \mathbf{b} \rangle = q(\mathbf{u}) - q(\mathbf{v}) = 0$.
They cannot both be isotropic: if $q(\mathbf{a}) = q(\mathbf{b}) = 0$, then $q(2\mathbf{u}) = 0$, contradicting $q(\mathbf{u}) \neq 0$.
If $q(\mathbf{a}) \neq 0$, then $\sigma_{\mathbf{a}}\mathbf{u} = \mathbf{v}$; if $q(\mathbf{b}) \neq 0$, then $\sigma_{\mathbf{b}}\mathbf{u} = -\mathbf{v}$.
Structural Commentary: One symmetry matches equal-length lines.
Key Insight: Use $\mathbf{u} \pm \mathbf{v}$ as the reflecting direction. □

Theorem 3.5.20. *Orthogonal Transformations Are Products of Symmetries*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$, and set $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$. Then $\mathcal{O}(\mathcal{V}) = \langle \sigma_{\mathbf{u}} : q(\mathbf{u}) \neq 0 \rangle$.

Proof. Induct on $n = \dim \mathcal{V}$.
If $n = 0$, this is trivial; if $n = 1$, any $\tau \in \mathcal{O}(\mathcal{V})$ acts on a nonisotropic line by ± 1, hence is I or the symmetry of that line. For $n > 1$, choose $\mathbf{u}$ with $q(\mathbf{u}) \neq 0$. Since $q(\tau\mathbf{u}) = q(\mathbf{u})$, Proposition 3.5.19 gives a symmetry σ with $\sigma\tau\mathbf{u} = \mathbf{u}$ or $-\mathbf{u}$. Composing further with $\sigma_{\mathbf{u}}$ if necessary, obtain a product ρ of symmetries such that $\rho\tau$ fixes $\mathbf{u}$. Hence $\rho\tau$ preserves $\mathrm{span}(\mathbf{u})^{\perp}$, and its restriction lies in $\mathcal{O}(\mathrm{span}(\mathbf{u})^{\perp})$.
By induction, this restriction is a product of symmetries in $\mathrm{span}(\mathbf{u})^{\perp}$, extended by I on $\mathbb{F}\mathbf{u}$; these are symmetries of $\mathcal{V}$. Thus τ is a product of symmetries.
Structural Commentary: Symmetries are the orthogonal analogue of elementary generators.
Key Insight: Match one nonisotropic line, then reduce to its orthogonal complement. □

Theorem 3.5.21. *Witt Theorems for Orthogonal Geometries*
Let $\mathcal{V} \cong \mathcal{V}'$ be nonsingular orthogonal geometries over $\mathbb{F}$ with $\mathrm{char}(\mathbb{F}) \neq 2$, and write $q(\mathbf{x}) := \langle \mathbf{x}, \mathbf{x} \rangle$.
If $\mathcal{S} \subseteq \mathcal{V}$ and $\mathcal{T} \subseteq \mathcal{V}'$ are nonsingular and $\tau : \mathcal{S} \to \mathcal{T}$ is an isometry, then for $\mathcal{S}^{\perp} \cong \mathcal{T}^{\perp}$ $\exists$ Φ such that $\mathcal{V} \to \mathcal{V}'$ isometric with $\Phi|_{\mathcal{S}} = \tau$.

Proof. By Proposition 3.3.12, it suffices to prove cancellation. Transport along a fixed isometry $\mathcal{V} \cong \mathcal{V}'$ and assume $\mathcal{V}' = \mathcal{V}$. Thus $\mathcal{V} = \mathcal{S} \odot \mathcal{S}^{\perp} = \mathcal{T} \odot \mathcal{T}^{\perp}$, with $\tau : \mathcal{S} \to \mathcal{T}$ an isometry.
Induct on $r = \dim \mathcal{S}$. If $r = 0$, the claim is trivial. If $r = 1$, write $\mathcal{S} = \mathbb{F}\mathbf{s}$ with $q(\mathbf{s}) \neq 0$. Since $q(\tau\mathbf{s}) = q(\mathbf{s})$, Proposition 3.5.19 gives $\sigma \in \mathcal{O}(\mathcal{V})$ with $\sigma\mathcal{S} = \mathcal{T}$; hence $\sigma(\mathcal{S}^{\perp}) = \mathcal{T}^{\perp}$.
For $r > 1$, choose $\mathbf{s} \in \mathcal{S}$ with $q(\mathbf{s}) \neq 0$ and write $\mathcal{S} = \mathbb{F}\mathbf{s} \odot \mathcal{U}$. By case $r = 1$, choose $\sigma \in \mathcal{O}(\mathcal{V})$ with $\sigma(\mathbb{F}\mathbf{s}) = \mathbb{F}\tau\mathbf{s}$. Inside $\mathcal{W} := (\mathbb{F}\tau\mathbf{s})^{\perp}$, $\mathcal{W} = \sigma\mathcal{U} \odot \sigma(\mathcal{S}^{\perp}) = \tau\mathcal{U} \odot \mathcal{T}^{\perp}$, $\sigma\mathcal{U} \cong \tau\mathcal{U}$. Since $\dim \mathcal{U} < r$, induction in $\mathcal{W}$ gives $\sigma(\mathcal{S}^{\perp}) \cong \mathcal{T}^{\perp}$. Thus $\mathcal{S}^{\perp} \cong \mathcal{T}^{\perp}$. Cancellation follows, and extension follows from Proposition 3.3.12.
Structural Commentary: Orthogonal Witt cancellation reduces one nonisotropic line at a time.
Key Insight: Symmetries align the first line; induction handles the remaining orthogonal complement. □

Definition 3.5.22. *Witt Index*
Let $\mathcal{V}$ be nonsingular orthogonal with $\mathrm{char}(\mathbb{F}) \neq 2$. The *Witt index* is $w(\mathcal{V}) := \max\{\dim \mathcal{L} : \mathcal{L} \leq \mathcal{V},\ \mathcal{L} \subseteq \mathcal{L}^{\perp}\}$.

Theorem 3.5.23. *Witt Index and Anisotropic Decomposition*
Let $\mathcal{V}$ be orthogonal over $\mathbb{F}$ with $\mathrm{char}(\mathbb{F}) \neq 2$. Then:

(i) if $\mathcal{V}$ is nonsingular, every maximal totally isotropic subspace has dimension $w(\mathcal{V})$;
(ii) if $\mathcal{V}$ is nonsingular, every maximal hyperbolic subspace has dimension $2w(\mathcal{V})$;
(iii) for $R = \mathrm{rad}(\mathcal{V})$ and a nonsingular complement $\mathcal{U}$ of R, there exist $\mathcal{H} \leq \mathcal{U}$ hyperbolic and $\mathcal{S} \leq \mathcal{U}$ anisotropic such that $\mathcal{V} = \mathcal{S} \odot \mathcal{H} \odot R$, $\dim \mathcal{H} = 2w(\mathcal{U})$.

Proof.

(i) Let $\mathcal{L}$ be maximal totally isotropic and choose $\mathcal{M}$ totally isotropic with $\dim \mathcal{M} = w(\mathcal{V})$.
Since $\dim \mathcal{L} \leq w(\mathcal{V})$, choose $\mathcal{M}_0 \leq \mathcal{M}$ with $\dim \mathcal{M}_0 = \dim \mathcal{L}$.
Any isomorphism $\mathcal{L} \to \mathcal{M}_0$ is an isometry, so Theorem 3.3.13 extends it to $\Phi \in \mathcal{O}(\mathcal{V})$.
Then $\mathcal{L} \subseteq \Phi^{-1}\mathcal{M}$; maximality gives $\mathcal{L} = \Phi^{-1}\mathcal{M}$, hence $\dim \mathcal{L} = w(\mathcal{V})$.

(ii) If $\mathcal{H}$ has k hyperbolic planes, then $\mathcal{H}$ contains a k-dimensional totally isotropic subspace, so $k \leq w(\mathcal{V})$.
Conversely, Theorem 3.3.10 completes a $w(\mathcal{V})$-dimensional totally isotropic subspace to a hyperbolic subspace $\mathcal{H}_0$ of dimension $2w(\mathcal{V})$. If a maximal hyperbolic $\mathcal{H}$ had dimension $2k < 2w(\mathcal{V})$, choose hyperbolic $\mathcal{H}_1 \leq \mathcal{H}_0$ with $\dim \mathcal{H}_1 = 2k$ and extend an isometry $\mathcal{H} \to \mathcal{H}_1$ to $\Phi \in \mathcal{O}(\mathcal{V})$; then $\Phi^{-1}\mathcal{H}_0$ is a larger hyperbolic subspace containing $\mathcal{H}$, contradiction.

(iii) By Theorem 3.3.3, $\mathcal{V} = R \odot \mathcal{U}$ with $\mathcal{U}$ nonsingular. Choose maximal hyperbolic $\mathcal{H} \leq \mathcal{U}$ and set $\mathcal{S} = \mathcal{H}^{\perp} \cap \mathcal{U}$.
Then $\dim \mathcal{H} = 2w(\mathcal{U})$ by (ii), and $\mathcal{U} = \mathcal{H} \odot \mathcal{S}$. If $\mathcal{S}$ had a nonzero isotropic vector, Theorem 3.3.8 would give a hyperbolic plane in $\mathcal{S}$, enlarging $\mathcal{H}$; hence $\mathcal{S}$ is anisotropic.

Structural Commentary: Orthogonal spaces split into radical, hyperbolic part, and anisotropic residue.
Key Insight: The Witt index counts how many hyperbolic planes can be split off. □

4 Affine, Projective, and Convex Geometry

4.1 Affine Geometry

Affine geometry studies cosets of subspaces as geometric objects in their own right. Throughout this subsection, vector spaces are finite-dimensional over $\mathbb{F}$ unless explicitly stated otherwise.

Definition 4.1.1. *Flats* Let $\mathcal{S} \leq \mathcal{V}$ and $\mathbf{x} \in \mathcal{V}$, and set $\mathbf{x}+\mathcal{S} = \{\mathbf{x}+\mathbf{s} : \mathbf{s} \in \mathcal{S}\}$.

(i) The affine subset $\mathbf{x}+\mathcal{S}$ from Definition 2.2.1 is a *flat* in $\mathcal{V}$; $\mathcal{S}$ is its *base*, and $\mathbf{x}$ is a *flat representative*.
(ii) Let $\mathcal{A}(\mathcal{V})$ denote the set of all flats in $\mathcal{V}$, with $\dim(\mathcal{A}(\mathcal{V})) := \dim(\mathcal{V})$.
(iii) The *dimension* of $\mathbf{x}+\mathcal{S}$ is $\dim(\mathcal{S})$.
A 0-flat is a *point*, a 1-flat is a *line*, a 2-flat is a *plane*, and a codimension-one flat is a *hyperplane*.
(iv) Two flats $X = \mathbf{x}+\mathcal{S}$ and $Y = \mathbf{y}+\mathcal{T}$ are *parallel*, written $X \parallel Y$, if $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$.

Structural Commentary: A flat has many representatives but one base: $\mathbf{x}+\mathcal{S} = \mathbf{y}+\mathcal{T} \Rightarrow \mathcal{S} = \mathcal{T}$.

Proposition 4.1.2. *Basic Incidence of Flats*
Let $X = \mathbf{x}+\mathcal{S}$ and $Y = \mathbf{y}+\mathcal{T}$ be flats in $\mathcal{V}$, and write $\mathbf{w}+X := \{\mathbf{w}+\mathbf{u} : \mathbf{u} \in X\}$.

(i) $(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+X \subseteq Y) \iff \mathcal{S} \subseteq \mathcal{T}$.
(ii) $(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+X = Y) \iff \mathcal{S} = \mathcal{T}$.
(iii) If $X \cap Y \neq \emptyset$, then $X \subseteq Y \iff \mathcal{S} \subseteq \mathcal{T}$.
(iv) If $X \cap Y \neq \emptyset$, then $X = Y \iff \mathcal{S} = \mathcal{T}$.
(v) If $X \parallel Y$, then $X \subseteq Y$, or $Y \subseteq X$, or $X \cap Y = \emptyset$.
(vi) $X \parallel Y \iff [(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+X \subseteq Y)] \vee [(\exists\, \mathbf{w} \in \mathcal{V})(\mathbf{w}+Y \subseteq X)]$.

Proof.

(i) Since $\mathbf{w}+X = (\mathbf{w}+\mathbf{x})+\mathcal{S}$, every translate of X has base $\mathcal{S}$. If $\mathbf{w}+X \subseteq Y$, choose $\mathbf{z} \in \mathbf{w}+X$; then $\mathbf{w}+X = \mathbf{z}+\mathcal{S}$ and $Y = \mathbf{z}+\mathcal{T}$, hence $\mathcal{S} \subseteq \mathcal{T}$. Conversely, if $\mathcal{S} \subseteq \mathcal{T}$, then $(\mathbf{y}-\mathbf{x})+X = \mathbf{y}+\mathcal{S} \subseteq Y$.
(ii) If $\mathbf{w}+X = Y$, then the bases are equal. Conversely, if $\mathcal{S} = \mathcal{T}$, then $(\mathbf{y}-\mathbf{x})+X = Y$.
(iii) Choose $\mathbf{z} \in X \cap Y$. Then $X = \mathbf{z}+\mathcal{S}$ and $Y = \mathbf{z}+\mathcal{T}$, so translating by $-\mathbf{z}$ gives $X \subseteq Y \iff \mathcal{S} \subseteq \mathcal{T}$.
(iv) With the same $\mathbf{z}$, translate by $-\mathbf{z}$ to get $X = Y \iff \mathcal{S} = \mathcal{T}$.
(v) If $X \parallel Y$, then $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$. If $X \cap Y \neq \emptyset$, part (iii) gives $X \subseteq Y$ or $Y \subseteq X$; otherwise $X \cap Y = \emptyset$.
(vi) If $X \parallel Y$, then either $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$, so part (i) gives a translate $X' \subseteq Y$ or a translate $Y' \subseteq X$. Conversely, either containment gives $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$ by part (i), hence $X \parallel Y$.

Structural Commentary: Intersecting flats reduce affine incidence to subspace incidence.
Key Insight: The representative may move, but the base controls containment, equality, and parallelism. □

Definition 4.1.3. *Affine Combinations*

(i) For $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathcal{V}$ and $r_1, \ldots, r_n \in \mathbb{F}$, the sum $\sum_{j=1}^n r_j\mathbf{x}_j$ is an *affine combination* if $\sum_{j=1}^n r_j = 1$.
(ii) A nonempty $E \subseteq \mathcal{V}$ is *affine closed* if $\sum_{j=1}^n r_j\mathbf{x}_j \in E$ whenever $\mathbf{x}_j \in E$, $r_j \in \mathbb{F}$, and $\sum_{j=1}^n r_j = 1$.
(iii) A nonempty $E \subseteq \mathcal{V}$ is *two-affine closed* if $r\mathbf{x}+(1-r)\mathbf{y} \in E$ for all $\mathbf{x}, \mathbf{y} \in E$ and $r \in \mathbb{F}$.
(iv) For $\mathbf{x} \neq \mathbf{y}$, the affine line through them is $\overline{\mathbf{xy}} := \{r\mathbf{x}+(1-r)\mathbf{y} : r \in \mathbb{F}\} = \mathbf{y}+\operatorname{span}(\mathbf{x}-\mathbf{y})$.
Equivalently, a nonempty $E \subseteq \mathcal{V}$ is two-affine closed iff $\overline{\mathbf{xy}} \subseteq E$ for all distinct $\mathbf{x}, \mathbf{y} \in E$.

Theorem 4.1.4. *Two-Point Affine Closure*
Assume $\operatorname{char}(\mathbb{F}) \neq 2$. For every $\emptyset \neq E \subseteq \mathcal{V}$, E is affine closed $\iff$ E is two-affine closed.

Proof.

(⇒) A two-point affine combination is an affine combination, so affine closure implies two-affine closure.
(⇐) Assume E is two-affine closed.

Prove by induction on n that every affine combination $\sum_{j=1}^n r_j\mathbf{x}_j$, with $\mathbf{x}_j \in E$ and $\sum_{j=1}^n r_j = 1$, lies in E.

The cases $n = 1, 2$ are immediate. Let $n \geq 3$ and set $\mathbf{z} := \sum_{j=1}^n r_j\mathbf{x}_j$, $\sum_{j=1}^n r_j = 1$.

If some $r_j \neq 1$, relabel so $r_1 \neq 1$ and put $\mathbf{u} := \sum_{j=2}^n \frac{r_j}{1-r_1}\mathbf{x}_j$. Then $\sum_{j=2}^n \frac{r_j}{1-r_1} = 1$, so $\mathbf{u} \in E$ by induction, and $\mathbf{z} = r_1\mathbf{x}_1 + (1-r_1)\mathbf{u} \in E$ by two-affine closure.
Otherwise all $r_j = 1$; since 2 is invertible, $\mathbf{z} = 2\left(\frac{1}{2}\mathbf{x}_1 + \frac{1}{2}\mathbf{x}_2\right) + r_3\mathbf{x}_3 + \cdots + r_n\mathbf{x}_n$, where $\frac{1}{2}\mathbf{x}_1 + \frac{1}{2}\mathbf{x}_2 \in E$ by two-affine closure; this is an $(n-1)$-term affine combination, hence lies in E by induction.

Structural Commentary: For $\operatorname{char}(\mathbb{F}) \neq 2$, two-point closure generates finite affine closure.
Key Insight: The only exceptional induction case is handled by averaging two coefficient-1 terms. □

Remark 4.1.5. *Characteristic Two Warning*
Condition $\operatorname{char}(\mathbb{F}) \neq 2$ is necessary: in $\mathbb{F}_2^2$, $E := \{(0,0),(1,0),(0,1)\}$ is two-affine closed, but $\operatorname{aff}(E) = \mathbb{F}_2^2 \neq E$.

Theorem 4.1.6. *Flats Are the Affine-Closed Sets*
For every $\emptyset \neq E \subseteq \mathcal{V}$, $E \in \mathcal{A}(\mathcal{V}) \iff E$ *is affine closed.*

Proof.

(i) Suppose $E = \mathbf{x} + \mathcal{S}$ with $\mathcal{S} \leq \mathcal{V}$. If $\mathbf{x}_j = \mathbf{x} + \mathbf{s}_j \in E$ and $\sum_j r_j = 1$,
then $\sum_j r_j \mathbf{x}_j = \sum_j r_j(\mathbf{x} + \mathbf{s}_j) = \mathbf{x} + \sum_j r_j \mathbf{s}_j \in \mathbf{x} + \mathcal{S}$. Thus E is affine closed.

(ii) Suppose E is affine closed. Choose $\mathbf{x} \in E$ and set $\mathcal{S} := E - \mathbf{x} = \{\mathbf{e} - \mathbf{x} : \mathbf{e} \in E\}$.
For $\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}$, write $\mathbf{s}_j = \mathbf{x}_j - \mathbf{x}$ with $\mathbf{x}_j \in E$.
For $r_1, r_2 \in \mathbb{F}$, $r_1\mathbf{s}_1 + r_2\mathbf{s}_2 + \mathbf{x} = r_1\mathbf{x}_1 + r_2\mathbf{x}_2 + (1 - r_1 - r_2)\mathbf{x} \in E$, so $r_1\mathbf{s}_1 + r_2\mathbf{s}_2 \in \mathcal{S}$.
Hence $\mathcal{S} \leq \mathcal{V}$ by Theorem 1.2.2, and $E = \mathbf{x} + \mathcal{S}$ is a flat.

Structural Commentary: An affine-closed set becomes a subspace after choosing one point as an origin.
Key Insight: Flats are precisely translated subspaces. □

Theorem 4.1.7. *Flats Are the Two-Affine-Closed Sets*
Assume $\operatorname{char}(\mathbb{F}) \neq 2$. *For every* $\emptyset \neq E \subseteq \mathcal{V}$, $E \in \mathcal{A}(\mathcal{V}) \iff E$ *is two-affine closed.*

Proof.

(i) If $E \in \mathcal{A}(\mathcal{V})$, then E is affine closed by Theorem 4.1.6, hence two-affine closed by Theorem 4.1.4.

(ii) If E is two-affine closed, then E is affine closed by Theorem 4.1.4, hence $E \in \mathcal{A}(\mathcal{V})$ by Theorem 4.1.6.

Structural Commentary: In characteristic not 2, two-point closure detects flats.
Key Insight: Two-affine closure equals finite affine closure, hence flatness. □

Definition 4.1.8. *Affine Hull and Affine Span*
Let $\emptyset \neq E \subseteq \mathcal{V}$.

(i) The *affine hull* $\operatorname{affhull}(E)$ is the least flat $X \in \mathcal{A}(\mathcal{V})$ with $E \subseteq X$.

(ii) The *affine span* $\operatorname{affspan}(E)$ is the set of all finite affine combinations of points of E.

Theorem 4.1.9. *Affine Hull Formula*
Let $\emptyset \neq E \subseteq \mathcal{V}$, $\mathbf{x} \in E$, *and* $\mathcal{S}_\mathbf{x} := \operatorname{span}(E - \mathbf{x})$.
Then $\operatorname{affhull}(E) = \operatorname{affspan}(E) = \mathbf{x} + \mathcal{S}_\mathbf{x}$, *and hence* $\dim(\operatorname{affhull}(E)) = \dim(\mathcal{S}_\mathbf{x})$.

Proof. If $\mathbf{z} = \sum_{j=1}^{n} r_j \mathbf{e}_j$ with $\mathbf{e}_j \in E$ and $\sum_j r_j = 1$, then $\mathbf{z} = \mathbf{x} + \sum_{j=1}^{n} r_j(\mathbf{e}_j - \mathbf{x}) \in \mathbf{x} + \mathcal{S}_\mathbf{x}$, so $\operatorname{affspan}(E) \subseteq \mathbf{x} + \mathcal{S}_\mathbf{x}$.
Conversely, for $\sum_{j=1}^{n} a_j(\mathbf{e}_j - \mathbf{x}) \in \mathcal{S}_\mathbf{x}$, $\mathbf{x} + \sum_{j=1}^{n} a_j(\mathbf{e}_j - \mathbf{x}) = \left(1 - \sum_{j=1}^{n} a_j\right)\mathbf{x} + \sum_{j=1}^{n} a_j \mathbf{e}_j \in \operatorname{affspan}(E)$, hence $\operatorname{affspan}(E) = \mathbf{x} + \mathcal{S}_\mathbf{x}$. This set is a flat containing E; any flat containing E is affine closed by Theorem 4.1.6, hence contains $\operatorname{affspan}(E)$. Therefore it is $\operatorname{affhull}(E)$. The dimension formula follows from Definition 4.1.1.
Structural Commentary: Affine span is ordinary span after temporarily choosing one point as the origin.
Key Insight: Differences from a base point record all directions in the affine hull. □

Lemma 4.1.10. *Intersections of Flats*
Let $X_i = \mathbf{x}_i + \mathcal{S}_i$ *be flats indexed by* $i \in I$. *If* $\bigcap_{i \in I} X_i \neq \emptyset$ *and* $\mathbf{z} \in \bigcap_{i \in I} X_i$, *then* $\bigcap_{i \in I} X_i = \mathbf{z} + \bigcap_{i \in I} \mathcal{S}_i$.

Proof. Since $\mathbf{z} \in X_i$, each flat has the form $X_i = \mathbf{z} + \mathcal{S}_i$. Hence $\mathbf{u} \in \bigcap_{i \in I} X_i \iff \mathbf{u} - \mathbf{z} \in \bigcap_{i \in I} \mathcal{S}_i$, QED.
Structural Commentary: A common point turns affine intersection into subspace intersection.
Key Insight: Translate to the common point; only the bases remain. □

Theorem 4.1.11. *Lattice of Flats*
Let $\mathcal{A}_0(\mathcal{V}) := \mathcal{A}(\mathcal{V}) \cup \{\emptyset\}$, *ordered by inclusion. Then* $\mathcal{A}_0(\mathcal{V})$ *is a complete lattice.*
For every family $\mathcal{C} \subseteq \mathcal{A}_0(\mathcal{V})$, $\bigwedge \mathcal{C} = \bigcap_{X \in \mathcal{C}} X$, $\bigvee \mathcal{C} = \begin{cases} \operatorname{affhull}\left(\bigcup_{X \in \mathcal{C}} X\right), & \bigcup_{X \in \mathcal{C}} X \neq \emptyset, \\ \emptyset, & \bigcup_{X \in \mathcal{C}} X = \emptyset. \end{cases}$

Proof.

(i) Arbitrary intersections of elements of $\mathcal{A}_0(\mathcal{V})$ lie in $\mathcal{A}_0(\mathcal{V})$: they are either empty, equal to $\mathcal{V}$ for the empty family, or flats by Lemma 4.1.10.

(ii) If $\bigcup_{X\in\mathcal{C}} X = \emptyset$, then $\emptyset$ is the least upper bound of $\mathcal{C}$.
Otherwise, affhull$(\bigcup_{X\in\mathcal{C}} X)$ is the least flat containing every $X \in \mathcal{C}$, so it is the join.

(iii) Since arbitrary meets and joins exist, $\mathcal{A}_0(\mathcal{V})$ is complete.

Structural Commentary: Adding the empty set repairs the failure of affine flats to be closed under intersection.
Key Insight: Meet is common incidence; join is the affine hull forced by union. □

Remark 4.1.12. *Empty Affine Hull Convention*
Within $\mathcal{A}_0(\mathcal{V})$, one may set affhull$(\emptyset) := \emptyset$ to write the join formula uniformly.

Corollary 4.1.13. *Join of Two Flats*
Let $X = \mathbf{x} + \mathcal{S}$ and $Y = \mathbf{y} + \mathcal{T}$ be flats. Then $X \vee Y = \mathbf{x} + \operatorname{span}\big(\{\mathbf{x}-\mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T}\big)$.
If $X \cap Y \neq \emptyset$, then $X \vee Y = \mathbf{x} + (\mathcal{S} + \mathcal{T})$.

Proof. By Theorem 4.1.11, $X \vee Y = \text{affhull}(X \cup Y)$. Apply Theorem 4.1.9 to $X \cup Y$ with base point $\mathbf{x}$, then $(X \cup Y) - \mathbf{x} = \mathcal{S} \cup \big((\mathbf{y} - \mathbf{x}) + \mathcal{T}\big)$, whose span is $\operatorname{span}(\{\mathbf{x}-\mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T})$.
If $X \cap Y \neq \emptyset$, then $\mathbf{x} - \mathbf{y} \in \mathcal{S} + \mathcal{T}$, so this span is $\mathcal{S} + \mathcal{T}$.
Structural Commentary: The join adds both bases and the displacement between representatives.
Key Insight: The displacement direction disappears exactly when the flats meet. □

Corollary 4.1.14. *Dimension of the Join of Two Flats*
Let $X = \mathbf{x} + \mathcal{S}$ and $Y = \mathbf{y} + \mathcal{T}$ be flats.
(i) If $X \cap Y \neq \emptyset$, then $\dim(X \vee Y) = \dim(X) + \dim(Y) - \dim(X \cap Y)$.
(ii) If $X \cap Y = \emptyset$, then $\dim(X \vee Y) = \dim(\mathcal{S} + \mathcal{T}) + 1$.

Proof.

(i) If $X \cap Y \neq \emptyset$, choose $\mathbf{z} \in X \cap Y$.
By Corollary 4.1.13 and Lemma 4.1.10, $X \vee Y = \mathbf{x} + (\mathcal{S} + \mathcal{T})$; $X \cap Y = \mathbf{z} + (\mathcal{S} \cap \mathcal{T})$. Thus the dimension formula for subspace sums gives $\dim(X \vee Y) = \dim(\mathcal{S} + \mathcal{T}) = \dim(\mathcal{S}) + \dim(\mathcal{T}) - \dim(\mathcal{S} \cap \mathcal{T})$.
Since $\dim X = \dim \mathcal{S}$, $\dim Y = \dim \mathcal{T}$, and $\dim(X \cap Y) = \dim(\mathcal{S} \cap \mathcal{T})$, this is (i).

(ii) If $X \cap Y = \emptyset$, Corollary 4.1.13 gives base $\mathcal{U} := \operatorname{span}(\{\mathbf{x}-\mathbf{y}\} \cup \mathcal{S} \cup \mathcal{T})$.
Since $\mathbf{x} - \mathbf{y} \notin \mathcal{S} + \mathcal{T}$, we have $\dim \mathcal{U} = \dim(\mathcal{S} + \mathcal{T}) + 1$, hence $\dim(X \vee Y) = \dim(\mathcal{S} + \mathcal{T}) + 1$.

Structural Commentary: Intersections subtract overlap; disjoint joins add one connecting direction.
Key Insight: The vector between representatives is new exactly when the flats are disjoint. □

Definition 4.1.15. *Affinely Independent Sets*
A nonempty set $E \subseteq \mathcal{V}$ is *affinely independent* if, for every $\mathbf{x} \in E$, the set $(E - \mathbf{x}) \setminus \{\mathbf{0}\}$ is linearly independent.

Theorem 4.1.16. *Affine Independence Criteria*
For $\emptyset \neq E \subseteq \mathcal{V}$, conditions (i)–(iv) are equivalent. If E is finite, they are also equivalent to (v).
(i) E is affinely independent.
(ii) For every $\mathbf{x} \in E$, $\mathbf{x} \notin \text{affhull}(E \setminus \{\mathbf{x}\})$, with affhull$(\emptyset) = \emptyset$ in the singleton case.
(iii) $\forall$ distinct $\mathbf{x}_1, \ldots, \mathbf{x}_n \in E$ and $r_1, \ldots, r_n \in \mathbb{F}$, $\sum_{j=1}^{n} r_j \mathbf{x}_j = \mathbf{0}$ and $\sum_{j=1}^{n} r_j = 0$ imply $r_1 = \cdots = r_n = 0$.
(iv) $\forall$ distinct $\mathbf{x}_1, \ldots, \mathbf{x}_n \in E$ and affine coefficient vectors $\mathbf{r}, \mathbf{s} \in \mathbb{F}^n$,
$\sum_{j=1}^{n} r_j \mathbf{x}_j = \sum_{j=1}^{n} s_j \mathbf{x}_j$ and $\sum_{j=1}^{n} r_j = \sum_{j=1}^{n} s_j = 1$ imply $\mathbf{r} = \mathbf{s}$.
(v) If $E = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ is finite, then $\dim(\text{affhull}(E)) = n - 1$.

Proof.

(i) $\Rightarrow$ (iii) Let $\sum_{j=1}^{n} r_j \mathbf{x}_j = \mathbf{0}$ with distinct $\mathbf{x}_j \in E$ and $\sum_j r_j = 0$. Then $\sum_{j=1}^{n-1} r_j(\mathbf{x}_j - \mathbf{x}_n) = \mathbf{0}$.
By (i), $r_1 = \cdots = r_{n-1} = 0$, and then $r_n = 0$.

(iii) $\Rightarrow$ (i) Fix $\mathbf{x} \in E$. Any relation $\sum_{j=1}^{m} a_j(\mathbf{y}_j - \mathbf{x}) = \mathbf{0}$, $\mathbf{y}_j \in E \setminus \{\mathbf{x}\}$ is the zero-sum relation
$\sum_{j=1}^{m} a_j \mathbf{y}_j - (\sum_j a_j)\mathbf{x} = \mathbf{0}$, so (iii) gives $a_1 = \cdots = a_m = 0$.

(iii) $\Rightarrow$ (ii) If $\mathbf{x}_k \in \text{affhull}(E \setminus \{\mathbf{x}_k\})$, then $\mathbf{x}_k = \sum_j r_j \mathbf{x}_j$ for $\mathbf{x}_j \neq \mathbf{x}_k$ and $\sum_j r_j = 1$,
giving a nontrivial zero-sum relation, contradiction.

(*ii*) $\Rightarrow$ (*iii*) If a zero-sum relation is nontrivial, choose $r_k \neq 0$.
Then $\mathbf{x}_k = \sum_{j\neq k} \left(-\frac{r_j}{r_k}\right)\mathbf{x}_j$, $\sum_{j\neq k} -\frac{r_j}{r_k} = 1$, so $\mathbf{x}_k \in \operatorname{affhull}(E \setminus \{\mathbf{x}_k\})$, contradicting (ii).
(*iii*) $\Rightarrow$ (*iv*) Subtract the two affine representations; the coefficient vector $\mathbf{r}-\mathbf{s}$ has sum 0, so (iii) gives $\mathbf{r}=\mathbf{s}$.
(*iv*) $\Rightarrow$ (*iii*) Let $\sum_j r_j\mathbf{x}_j = \mathbf{0}$ and $\sum_j r_j = 0$. Choose $\mathbf{a} \in \mathbb{F}^n$ with $\sum_j a_j = 1$.
Then $\mathbf{a}$ and $\mathbf{a}+\mathbf{r}$ are affine coefficient vectors for the same point. By (iv), $\mathbf{r}=\mathbf{0}$.
(*i*) $\Leftrightarrow$ (*v*) If $E = \{\mathbf{x}_1,\dots,\mathbf{x}_n\}$ and $\mathbf{x} \in E$, then Theorem 4.1.9 gives $\dim(\operatorname{affhull}(E)) = \dim(\operatorname{span}(E-\mathbf{x}))$.
Since $(E-\mathbf{x})\setminus\{\mathbf{0}\}$ has $n-1$ vectors, this dimension is $n-1$ iff those vectors are linearly independent.

Structural Commentary: Affine independence is linear independence after forgetting one chosen origin.
Key Insight: Uniqueness is absence of nontrivial zero-sum affine relations. □

Definition 4.1.17. *Affine Bases and Barycentric Coordinates* Let $A = \mathbf{x}+\mathcal{S}$ be an n-flat.

(i) A subset $\mathcal{B} \subseteq A$ is an *affine basis* of A if $\mathcal{B}$ is affinely independent and $\operatorname{affspan}(\mathcal{B}) = A$; an *ordered affine basis* is an ordering of such a basis.
(ii) If $(\mathbf{s}_1,\dots,\mathbf{s}_n)$ is an ordered basis of $\mathcal{S}$, then $(\mathbf{x}+\mathbf{s}_1,\dots,\mathbf{x}+\mathbf{s}_n,\mathbf{x})$ is an ordered affine basis of A.
(iii) For an ordered affine basis $(\mathbf{p}_1,\dots,\mathbf{p}_{n+1})$, every $\mathbf{v} \in A$ has a unique expression $\mathbf{v} = \sum_{j=1}^{n+1} r_j\mathbf{p}_j$, $\sum_{j=1}^{n+1} r_j = 1$.
The scalars $(r_1,\dots,r_{n+1})$ are the *barycentric coordinates* of $\mathbf{v}$ in this basis.

Example 4.1.18. *Barycentric Coordinates on a Plane*
Let $A = \mathbf{x}+\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2) \subseteq \mathbb{R}^3$, with $(\mathbf{v}_1,\mathbf{v}_2)$ linearly independent.
Set $\mathbf{p}_1 = \mathbf{x}+\mathbf{v}_1$, $\mathbf{p}_2 = \mathbf{x}+\mathbf{v}_2$, and $\mathbf{p}_3 = \mathbf{x}$. Then $(\mathbf{p}_1,\mathbf{p}_2,\mathbf{p}_3)$ is an ordered affine basis of A, and every $\mathbf{a} \in A$ has a unique barycentric expansion $\mathbf{a} = r_1\mathbf{p}_1 + r_2\mathbf{p}_2 + r_3\mathbf{p}_3$, $r_1+r_2+r_3 = 1$, with $r_1,r_2,r_3 \in \mathbb{R}$.

Definition 4.1.19. *Affine Transformation*
A map $f : \mathcal{V} \to \mathcal{V}$ is an *affine transformation* if it preserves finite affine combinations:
for all $\mathbf{x}_j \in \mathcal{V}$ and $r_j \in \mathbb{F}$ with $\sum_j r_j = 1$, $f\left(\sum_j r_j\mathbf{x}_j\right) = \sum_j r_j f(\mathbf{x}_j)$.

Theorem 4.1.20. *Two-Point Test for Affine Maps*
Assume $\operatorname{char}(\mathbb{F}) \neq 2$.
A map $f : \mathcal{V} \to \mathcal{V}$ is affine if and only if $f(r\mathbf{x}+(1-r)\mathbf{y}) = rf(\mathbf{x})+(1-r)f(\mathbf{y})$ for all $\mathbf{x},\mathbf{y} \in \mathcal{V}$ and $r \in \mathbb{F}$.

Proof.
($\Rightarrow$) Since $r+(1-r)=1$, affinity gives $f(r\mathbf{x}+(1-r)\mathbf{y}) = rf(\mathbf{x})+(1-r)f(\mathbf{y})$.
($\Leftarrow$) Assume displayed identity. Prove by induction on n that $f\left(\sum_{j=1}^n r_j\mathbf{x}_j\right) = \sum_{j=1}^n r_j f(\mathbf{x}_j)$; $\sum_{j=1}^n r_j = 1$.
The cases $n = 1, 2$ are immediate. For $n \geq 3$, set $\mathbf{z} := \sum_{j=1}^n r_j\mathbf{x}_j$. If some $r_j \neq 1$, relabel so $r_1 \neq 1$ and put $\mathbf{u} := \sum_{j=2}^n \frac{r_j}{1-r_1}\mathbf{x}_j$. Then $f(\mathbf{z}) = f(r_1\mathbf{x}_1+(1-r_1)\mathbf{u}) = r_1f(\mathbf{x}_1)+(1-r_1)f(\mathbf{u}) = \sum_{j=1}^n r_j f(\mathbf{x}_j)$, by the two-point identity and induction. Otherwise all $r_j = 1$; since 2 is invertible, $\mathbf{z} = 2\left(\frac{1}{2}\mathbf{x}_1+\frac{1}{2}\mathbf{x}_2\right)+r_3\mathbf{x}_3+\cdots+r_n\mathbf{x}_n$, an $(n-1)$-term affine combination, so induction plus two-point identity gives same formula for $f(\mathbf{z})$.

Structural Commentary: In characteristic not 2, line preservation gives affine preservation.
Key Insight: The same two-point assembly that builds affine-closed sets also tests affine maps. □

Definition 4.1.21. *Translations*
For $\mathbf{a} \in \mathcal{V}$, the *translation by* $\mathbf{a}$ is the map $T_\mathbf{a} : \mathcal{V} \to \mathcal{V}$ defined by $T_\mathbf{a}(\mathbf{x}) = \mathbf{x}+\mathbf{a}$.

Theorem 4.1.22. *Affine Maps Are Linear Maps Followed by Translations*
A map $f : \mathcal{V} \to \mathcal{V}$ is affine iff $f = T_\mathbf{a} \circ \tau$ for unique $\mathbf{a} \in \mathcal{V}$ and $\tau \in \mathcal{L}(\mathcal{V})$, namely $\mathbf{a} = f(\mathbf{0})$ and $\tau = T_{-\mathbf{a}} \circ f$.

Proof.
($\Rightarrow$) Set $\mathbf{a} := f(\mathbf{0})$ and $\tau(\mathbf{x}) := f(\mathbf{x})-\mathbf{a}$. For $r,s \in \mathbb{F}$ and $\mathbf{x},\mathbf{y} \in \mathcal{V}$,
$\tau(r\mathbf{x}+s\mathbf{y}) = f(r\mathbf{x}+s\mathbf{y}+(1-r-s)\mathbf{0})-\mathbf{a} = r\tau(\mathbf{x})+s\tau(\mathbf{y})$, so $\tau \in \mathcal{L}(\mathcal{V})$ and $f = T_\mathbf{a} \circ \tau$.
Uniqueness follows from $\mathbf{a} = f(\mathbf{0})$ and $\tau(\mathbf{x}) = f(\mathbf{x})-\mathbf{a}$.
($\Leftarrow$) If $f = T_\mathbf{a} \circ \tau$ with τ linear,
then for $\sum_j r_j = 1$, $f\left(\sum_j r_j\mathbf{x}_j\right) = \mathbf{a}+\tau\left(\sum_j r_j\mathbf{x}_j\right) = \sum_j r_j(\mathbf{a}+\tau\mathbf{x}_j) = \sum_j r_j f(\mathbf{x}_j)$, hence f is affine.

Structural Commentary: An affine map is linear after moving its image of the origin back to the origin.
Key Insight: The failure of an affine map to be linear is exactly a translation. □

Definition 4.1.23. *Affine Group Data*

(i) The *general linear group* is $\mathrm{GL}(\mathcal{V}) := \{\tau \in \mathcal{L}(\mathcal{V}) : \tau \text{ is invertible}\}$.
(ii) The *affine group* is $\mathrm{Aff}(\mathcal{V}) := \{T_{\mathbf{a}} \circ \tau : \mathbf{a} \in \mathcal{V}, \tau \in \mathrm{GL}(\mathcal{V})\}$.
(iii) The *translation group* is $\mathrm{Trans}(\mathcal{V}) := \{T_{\mathbf{a}} : \mathbf{a} \in \mathcal{V}\}$.
(iv) The *linear-part map* is $\Lambda : \mathrm{Aff}(\mathcal{V}) \to \mathrm{GL}(\mathcal{V})$; $\Lambda(T_{\mathbf{a}} \circ \tau) = \tau$, by Theorem 4.1.22.

Corollary 4.1.24. *Bijective Affine Maps*
For $\mathbf{a} \in \mathcal{V}$ and $\tau \in \mathcal{L}(\mathcal{V})$, $T_{\mathbf{a}} \circ \tau$ is bijective $\iff$ $\tau \in \mathrm{GL}(\mathcal{V})$.

Proof.
($\Rightarrow$) If $T_{\mathbf{a}} \circ \tau$ is bijective, then $\tau = T_{-\mathbf{a}} \circ (T_{\mathbf{a}} \circ \tau)$ is bijective, hence $\tau \in \mathrm{GL}(\mathcal{V})$.
($\Leftarrow$) If $\tau \in \mathrm{GL}(\mathcal{V})$, then $T_{\mathbf{a}} \circ \tau$ is bijective with inverse $\tau^{-1} \circ T_{-\mathbf{a}}$.
Structural Commentary: Translation never affects invertibility; only the linear part does.
Key Insight: Bijective affine maps are exactly translations of linear isomorphisms. □

Corollary 4.1.25. *Affine Group and Translations*

(i) For $\mathbf{a}, \mathbf{b} \in \mathcal{V}$ and $\tau, \sigma \in \mathcal{L}(\mathcal{V})$, $(T_{\mathbf{a}} \circ \tau) \circ (T_{\mathbf{b}} \circ \sigma) = T_{\mathbf{a}+\tau\mathbf{b}} \circ (\tau\sigma)$.
(ii) $\mathrm{Aff}(\mathcal{V})$ is a group, $\mathrm{Trans}(\mathcal{V}) \trianglelefteq \mathrm{Aff}(\mathcal{V})$, and Λ is a surjective group homomorphism with $\ker \Lambda = \mathrm{Trans}(\mathcal{V})$.
(iii) Consequently, $\mathrm{Aff}(\mathcal{V})/\mathrm{Trans}(\mathcal{V}) \cong \mathrm{GL}(\mathcal{V})$.

Proof.
i. For $\mathbf{x} \in \mathcal{V}$, $((T_{\mathbf{a}} \circ \tau) \circ (T_{\mathbf{b}} \circ \sigma))(\mathbf{x}) = \mathbf{a} + \tau\mathbf{b} + (\tau\sigma)\mathbf{x}$.
ii. For $\tau, \sigma \in \mathrm{GL}(\mathcal{V})$, (i) gives closure in $\mathrm{Aff}(\mathcal{V})$; the identity is $T_{\mathbf{0}} \circ I_{\mathcal{V}}$, and $(T_{\mathbf{a}} \circ \tau)^{-1} = T_{-\tau^{-1}\mathbf{a}} \circ \tau^{-1}$. Thus $\mathrm{Aff}(\mathcal{V})$ is a group. Also (i) gives $\Lambda(gh) = \Lambda(g)\Lambda(h)$, $\ker \Lambda = \{T_{\mathbf{a}} \circ I_{\mathcal{V}} : \mathbf{a} \in \mathcal{V}\} = \mathrm{Trans}(\mathcal{V})$, and Λ is surjective since $\tau = T_{\mathbf{0}} \circ \tau$ for all $\tau \in \mathrm{GL}(\mathcal{V})$. Hence $\mathrm{Trans}(\mathcal{V}) \trianglelefteq \mathrm{Aff}(\mathcal{V})$.
iii. Since Λ is surjective and $\ker \Lambda = \mathrm{Trans}(\mathcal{V})$, the first isomorphism theorem gives the induced isomorphism $\overline{\Lambda} : \mathrm{Aff}(\mathcal{V})/\mathrm{Trans}(\mathcal{V}) \to \mathrm{GL}(\mathcal{V})$; $\overline{\Lambda}(g\ \mathrm{Trans}(\mathcal{V})) = \Lambda(g)$.

Structural Commentary: The affine group is the translation part together with the invertible linear part.
Key Insight: Forgetting the translation records a homomorphism onto the linear automorphism group. □

4.2 Projective Geometry

Definition 4.2.1. *Projective Geometry of a Vector Space*

(i) Let $\mathrm{Sub}(\mathcal{V})$ be the set of subspaces of $\mathcal{V}$.
The *projective geometry* of $\mathcal{V}$ is $\mathrm{Sub}(\mathcal{V})$ with $\mathrm{pdim}(\mathcal{U}) := \dim(\mathcal{U}) - 1$.
(ii) A *projective point* is a 1-dimensional subspace; a *projective line* is a 2-dimensional subspace.
(iii) If $H = \mathbf{x} + \mathcal{K}$ is an affine hyperplane with $\mathbf{0} \notin H$ and $X \subseteq H$ is an affine flat, then X is represented projectively by $\mathrm{span}(X)$, with $\mathrm{pdim}(\mathrm{span}(X)) = \dim(X)$.

Perspective 4.2.2. *Projective Completion of an Affine Hyperplane*
Let $H = \mathbf{x} + \mathcal{K}$ be a hyperplane with $\mathbf{0} \notin H$. The map $X \subseteq H \mapsto \mathrm{span}(X) \leq \mathcal{V}$ adjoins the missing directions in $\mathcal{K}$. Thus, when $\dim H = 2$, parallel affine lines in H acquire a common projective point at infinity.

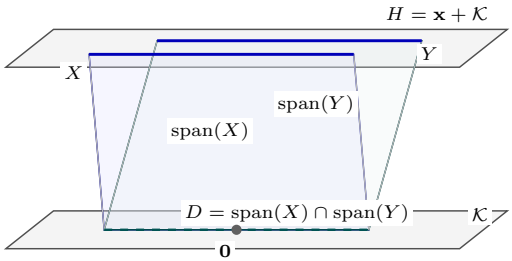


Figure 39: In the projective completion of $H = \mathbf{x} + \mathcal{K}$, an affine flat $X \subseteq H$ is represented by $\mathrm{span}(X)$; parallel flats meet in their common direction $D \leq \mathcal{K}$.

Theorem 4.2.3. *Linear Span Map from Affine to Projective Geometry*
Let $H = \mathbf{x} + \mathcal{K}$ be an affine hyperplane with $\mathbf{0} \notin H$, and set $\mathcal{A}(H) := \{X \in \mathcal{A}(\mathcal{V}) : X \subseteq H\}$.
The map $P : \mathcal{A}(H) \to \mathrm{Sub}(\mathcal{V})$, $P(X) = \mathrm{span}(X)$ satisfies:

(i) *P is injective and $P^{-1}(U) = U \cap H$ for $U \in \mathrm{im}\, P$.*
(ii) *$\mathrm{im}\, P = \{\mathcal{U} \leq \mathcal{V} : \mathcal{U} \not\subseteq \mathcal{K}\}$.*
(iii) *$X \subseteq Y$ if and only if $P(X) \subseteq P(Y)$.*
(iv) *If $\bigcap_{i\in I} X_i \neq \emptyset$, then $P\left(\bigcap_{i\in I} X_i\right) = \bigcap_{i\in I} P(X_i)$.*
(v) *For any family $\{X_i : i \in I\} \subseteq \mathcal{A}(H)$, $P\left(\bigvee_{i\in I} X_i\right) = \bigvee_{i\in I} P(X_i)$, where $\bigvee_{i\in I} P(X_i)$ is computed in $\mathrm{Sub}(\mathcal{V})$, i.e. $\bigvee_{i\in I} P(X_i) = \sum_{i\in I} P(X_i)$.*
(vi) *$\mathrm{pdim}(P(X)) = \dim(X)$.*
(vii) *$X \parallel Y$ if and only if one of $P(X) \cap \mathcal{K}$ and $P(Y) \cap \mathcal{K}$ is contained in the other.*

Proof.

(i) Let $X = \mathbf{u} + \mathcal{S} \subseteq H$. Then $\mathcal{S} \leq \mathcal{K}$, $\mathbf{u} \notin \mathcal{K}$, and $P(X) = \mathbb{F}\mathbf{u} \oplus \mathcal{S}$.
If $\mathbf{z} \in P(X) \cap H$, then $\mathbf{z} = r\mathbf{u} + \mathbf{s} = \mathbf{u} + \mathbf{k}$ with $\mathbf{s} \in \mathcal{S}$ and $\mathbf{k} \in \mathcal{K}$. Hence $(1-r)\mathbf{u} \in \mathcal{K}$, so $r = 1$, and $P(X) \cap H = X$. Thus P is injective and $P^{-1}(U) = U \cap H$ on $\mathrm{im}\, P$.
(ii) If $\mathcal{U} \not\subseteq \mathcal{K}$, choose $\mathbf{u} \in \mathcal{U} \setminus \mathcal{K}$. Since $\mathcal{V}/\mathcal{K}$ is 1-dimensional and $H = \mathbf{x} + \mathcal{K}$, there is $r \neq 0$ with $\mathbf{h} := r\mathbf{u} \in H$. Then $\mathcal{U} \cap H = \mathbf{h} + (\mathcal{U} \cap \mathcal{K})$; $P(\mathcal{U} \cap H) = \mathbb{F}\mathbf{h} + (\mathcal{U} \cap \mathcal{K}) = \mathcal{U}$. Thus $\mathrm{im}\, P = \{\mathcal{U} \leq \mathcal{V} : \mathcal{U} \not\subseteq \mathcal{K}\}$.
(iii) If $X \subseteq Y$, then $P(X) \subseteq P(Y)$. Conversely, if $P(X) \subseteq P(Y)$, then $X = P(X) \cap H \subseteq P(Y) \cap H = Y$.
(iv) If $\mathbf{z} \in \bigcap_i X_i$ and $X_i = \mathbf{z} + \mathcal{S}_i$, then Lemma 4.1.10 gives $\bigcap_i X_i = \mathbf{z} + \bigcap_i \mathcal{S}_i$.
Since $P(X_i) = \mathbb{F}\mathbf{z} \oplus \mathcal{S}_i$, $\bigcap_i P(X_i) = \mathbb{F}\mathbf{z} \oplus \bigcap_i \mathcal{S}_i = P\left(\bigcap_i X_i\right)$.
(v) By Theorem 4.1.11, $P\left(\bigvee_i X_i\right) = \mathrm{span}\left(\mathrm{affhull}\left(\bigcup_i X_i\right)\right) = \mathrm{span}\left(\bigcup_i X_i\right) = \bigvee_i P(X_i)$.
(vi) For $X = \mathbf{u} + \mathcal{S} \subseteq H$, $\mathrm{pdim}(P(X)) = \dim(\mathbb{F}\mathbf{u} \oplus \mathcal{S}) - 1 = \dim \mathcal{S} = \dim X$.
(vii) If $X = \mathbf{u} + \mathcal{S}$ and $Y = \mathbf{v} + \mathcal{T}$, then $P(X) \cap \mathcal{K} = \mathcal{S}$ and $P(Y) \cap \mathcal{K} = \mathcal{T}$.
Hence $X \parallel Y \iff \mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S} \iff P(X) \cap \mathcal{K} \subseteq P(Y) \cap \mathcal{K}$ or $P(Y) \cap \mathcal{K} \subseteq P(X) \cap \mathcal{K}$.

Structural Commentary: H selects affine representatives for projective subspaces outside $\mathcal{K}$.
Key Insight: Intersecting with H undoes the linear span exactly because H does not pass through the origin. □

4.3 Positive Systems, Convexity, and Separation

This subsection uses finite-dimensional real inner-product geometry to decide when linear systems have positive solutions. The guiding theme is that a missing positive solution is often detected by a separating hyperplane, hence by a dual vector certificate.

Definition 4.3.1. *Positive Orthants*
For $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, use the following order and orthant notation:

(i) $\mathbf{x} \geq \mathbf{0}$ iff $\mathbf{x}$ is *coordinatewise nonnegative*: $x_i \geq 0$ for every i.
(ii) $\mathbf{x} > \mathbf{0}$ iff $\mathbf{x}$ is *nonzero nonnegative*: $\mathbf{x} \geq \mathbf{0}$ and $\mathbf{x} \neq \mathbf{0}$.
(iii) $\mathbf{x} \gg \mathbf{0}$ iff $\mathbf{x}$ is *coordinatewise positive*: $x_i > 0$ for every i.
(iv) $\mathbb{R}^n_+ := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} > \mathbf{0}\}$ is the *nonzero nonnegative orthant*.
(v) $\mathbb{R}^n_{++} := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} \gg \mathbf{0}\}$ is the *coordinatewise positive interior*.

Motivation 4.3.2. *Positive Null Vectors as Probabilities*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{R})$ and let $\mathbf{x} = (x_1, \ldots, x_n) > \mathbf{0}$ satisfy $A\mathbf{x} = \mathbf{0}$.

(i) Set $s := x_1 + \cdots + x_n > 0$ and $\boldsymbol{\pi} := s^{-1}\mathbf{x}$.
(ii) Then $\pi_i \geq 0$ and $\sum_{i=1}^n \pi_i = 1$, so $\boldsymbol{\pi}$ is a probability vector.
(iii) Since $A\boldsymbol{\pi} = s^{-1}A\mathbf{x} = \mathbf{0}$, equivalently $\sum_{i=1}^n \pi_i \mathbf{a}_i = \mathbf{0}$.
(iv) If $\mathbf{x} \gg \mathbf{0}$, then $\pi_i > 0$ for every i.

Thus a positive null vector normalizes to probabilities under which the columns of A have zero expected value.

Principle 4.3.3. *Positive Solutions Are Orthant Intersections*
For $A \in \mathrm{Mat}(m, n, \mathbb{R})$, $\ker(A) = \mathrm{row}(A)^\perp$ by Corollary 2.4.13.
Hence $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0},\ \mathbf{x} > \mathbf{0}\} = \mathrm{row}(A)^\perp \cap \mathbb{R}^n_+$; and $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0},\ \mathbf{x} \gg \mathbf{0}\} = \mathrm{row}(A)^\perp \cap \mathbb{R}^n_{++}$.
Thus *positive null vectors* are precisely orthant intersections with $\mathrm{row}(A)^\perp$.

Definition 4.3.4. *Convex Combinations, Cones, and Hulls*
Let $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$.

i. A *convex combination* of these vectors is a vector $t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k$, $t_i \geq 0$, $\sum_{i=1}^{k} t_i = 1$.
ii. A set $C \subseteq \mathbb{R}^n$ is *convex* if the line segment between any two of its points lies in C.
iii. A set $K \subseteq \mathbb{R}^n$ is a *cone* if $\mathbf{x} \in K$ and $a \geq 0$ imply $a\mathbf{x} \in K$.
iv. *Convex hull* of finite set $S = \{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is smallest convex subset of $\mathbb{R}^n$ containing S, denoted $\operatorname{conv}(S)$.

Closedness and compactness use the Euclidean metric conventions from Definitions 3.1.14 and 4.1.1.

Proposition 4.3.5. *Finite Convex Hulls*
Let $S = \{\mathbf{x}_1, \dots, \mathbf{x}_k\} \subseteq \mathbb{R}^n$ *and* $\Delta_{k-1} := \left\{(t_1, \dots, t_k) \in \mathbb{R}^k : t_i \geq 0,\ \sum_{i=1}^{k} t_i = 1\right\}$.
Then $\operatorname{conv}(S) = \left\{\sum_{i=1}^{k} t_i\mathbf{x}_i : (t_1, \dots, t_k) \in \Delta_{k-1}\right\}$. *Moreover,* $\operatorname{conv}(S)$ *is compact.*

Proof. Let $T : \Delta_{k-1} \to \mathbb{R}^n$ be $T(t_1, \dots, t_k) = \sum_{i=1}^{k} t_i\mathbf{x}_i$, and set $C := T(\Delta_{k-1})$. Taking $t_i = 1$ and $t_j = 0$ for $j \neq i$ gives $S \subseteq C$. If $K \subseteq \mathbb{R}^n$ is convex and $S \subseteq K$, then $C \subseteq K$, hence $C \subseteq \operatorname{conv}(S)$.
Conversely, Δ_{k-1} is convex and T is affine, so C is convex; since $S \subseteq C$, $\operatorname{conv}(S) \subseteq C$. Thus $\operatorname{conv}(S) = C$.
The simplex Δ_{k-1} is closed and bounded, hence compact by Heine–Borel, Theorem 4.2.5.
Since T is continuous, $\operatorname{conv}(S) = T(\Delta_{k-1})$ is compact by Theorem 5.1.33.
Structural Commentary: A finite convex hull is the continuous image of a compact coefficient simplex.
Key Insight: Convexity is carried by the coefficients t_i, not by the ambient coordinates. □

Proposition 4.3.6. *Sums of Convex and Closed Sets*
For $X, Y \subseteq \mathbb{R}^n$, *set* $X + Y := \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in X,\ \mathbf{y} \in Y\}$. *Then:*

(i) X, Y *convex* $\implies$ $X + Y$ *convex.*
(ii) X *compact and* Y *closed* $\implies$ $X + Y$ *closed.*

Proof.

(i) Assume X, Y are convex. Let $\mathbf{z}_r = \mathbf{x}_r + \mathbf{y}_r \in X + Y$ for $r = 0, 1$, and let $0 \leq t \leq 1$. Then $t\mathbf{z}_0 + (1-t)\mathbf{z}_1 = \big(t\mathbf{x}_0 + (1-t)\mathbf{x}_1\big) + \big(t\mathbf{y}_0 + (1-t)\mathbf{y}_1\big) \in X + Y$. Thus $X + Y$ is convex.
(ii) Assume X is compact and Y is closed. Let $\mathbf{z}_j \in X + Y$ and $\mathbf{z}_j \to \mathbf{z}$.
Let $\mathbf{z}_j = \mathbf{x}_j + \mathbf{y}_j$ with $\mathbf{x}_j \in X$ and $\mathbf{y}_j \in Y$. Since X is compact, a subsequence satisfies $\mathbf{x}_{j_\ell} \to \mathbf{x} \in X$.
Then $\mathbf{y}_{j_\ell} = \mathbf{z}_{j_\ell} - \mathbf{x}_{j_\ell} \to \mathbf{z} - \mathbf{x} \in Y$, as Y is closed. Hence $\mathbf{z} = \mathbf{x} + (\mathbf{z} - \mathbf{x}) \in X + Y$, so $X + Y$ is closed.

Structural Commentary: Compactness lets one hold the X-part still enough for closedness of Y to apply.
Key Insight: A limit in a sum can be split by passing to a convergent compact subsequence. □

Definition 4.3.7. *Hyperplanes, Half-Spaces, and Separation*
Let $\mathbf{N} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $b \in \mathbb{R}$.

(i) The *affine hyperplane* with normal $\mathbf{N}$ and level b is $\mathcal{H}(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x} \rangle = b\}$.
(ii) The case $b = 0$ gives the *linear hyperplane* $\mathcal{H}(\mathbf{N}, 0) = \operatorname{span}(\mathbf{N})^{\perp}$.
(iii) The *closed half-spaces* determined by $\mathcal{H}(\mathbf{N}, b)$ are
$\mathcal{H}_+(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x} \rangle \geq b\}$ and $\mathcal{H}_-(\mathbf{N}, b) := \{\mathbf{x} \in \mathbb{R}^n : \langle \mathbf{N}, \mathbf{x} \rangle \leq b\}$.
The *open half-spaces* are obtained by replacing $\geq, \leq$ with $>, <$.
(iv) For $X \subseteq \mathbb{R}^n$ and $\star \in \{<, \leq, >, \geq\}$, use the *set-inequality convention* $\langle \mathbf{N}, X \rangle \star b \iff \langle \mathbf{N}, \mathbf{x} \rangle \star b\ \forall \mathbf{x} \in X$.
(v) An *orientation* of $\mathcal{H}(\mathbf{N}, b)$ is $\sigma(\mathbf{N}, b) := (\sigma\mathbf{N}, \sigma b)$ for $\sigma \in \{\pm 1\}$.
(vi) Sets $X, Y \subseteq \mathbb{R}^n$ are *strictly separated* by $\mathcal{H}(\mathbf{N}, b)$ if $\exists \sigma \in \{\pm 1\}$ such that $\langle \sigma\mathbf{N}, X \rangle < \sigma b < \langle \sigma\mathbf{N}, Y \rangle$
(vii) Sets $X, Y \subseteq \mathbb{R}^n$ are *strongly separated* by $\mathcal{H}(\mathbf{N}, b)$ if $\exists \sigma \in \{\pm 1\}$ and $\varepsilon > 0$ such that $\langle \sigma\mathbf{N}, X \rangle < \sigma b - \varepsilon < \sigma b + \varepsilon < \langle \sigma\mathbf{N}, Y \rangle$.

Remark 4.3.8. *Hyperplane Geometry*
The half-spaces satisfy $\mathcal{H}_+(\mathbf{N}, b) \cap \mathcal{H}_-(\mathbf{N}, b) = \mathcal{H}(\mathbf{N}, b)$, and two open half-spaces with $\mathcal{H}(\mathbf{N}, b)$ partition $\mathbb{R}^n$.
Moreover, $\mathbf{N} \in \mathcal{H}(\mathbf{N}, \|\mathbf{N}\|^2)$, by Cauchy–Schwarz (Theorem 1.1.13), $\mathbf{N}$ is closest point of this hyperplane to $\mathbf{0}$.

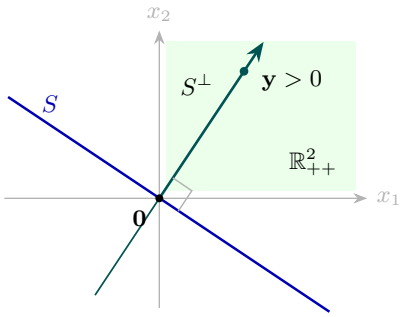


Figure 40: In $\mathbb{R}^2$, a negative-slope line S has $S^\perp$ meeting the strongly positive orthant.

Theorem 4.3.9. *Nearest Points and Separating a Point*
Let $C \subseteq \mathbb{R}^n$ be nonempty, closed, and convex.

(i) There exists a unique $\mathbf{N}_C \in C$ such that $\|\mathbf{N}_C\| = \min_{\mathbf{x}\in C} \|\mathbf{x}\|$.
(ii) If $\mathbf{b} \notin C$, set $D := C - \mathbf{b}$ and let $\mathbf{N} \in D$ satisfy $\|\mathbf{N}\| = \min_{\mathbf{z}\in D} \|\mathbf{z}\|$.
Then $\mathbf{N} \neq \mathbf{0}$ and $\langle \mathbf{N}, C\rangle \geq \|\mathbf{N}\|^2 + \langle \mathbf{N}, \mathbf{b}\rangle > \langle \mathbf{N}, \mathbf{b}\rangle$.
With $\alpha := \langle \mathbf{N}, \mathbf{b}\rangle + \frac{1}{2}\|\mathbf{N}\|^2$, the hyperplane $\mathcal{H}(\mathbf{N}, \alpha)$ strongly separates $\{\mathbf{b}\}$ and C.

Proof.

(i) Choose $\mathbf{x}_0 \in C$ and $s \geq \|\mathbf{x}_0\|$. Then $K := C \cap \overline{B}_s(\mathbf{0})$ is nonempty and compact by Theorem 4.2.5. Since $\|\cdot\|$ is continuous, Theorem 5.1.35 gives $\mathbf{N}_C \in K$ minimizing $\|\cdot\|$ on K. If $\mathbf{x} \in C$ and $\|\mathbf{x}\| < \|\mathbf{N}_C\|$, then $\mathbf{x} \in K$, a contradiction; hence $\mathbf{N}_C$ minimizes on C. If $\mathbf{u} \neq \mathbf{v}$ are minimizers of norm m, then $\mathbf{w} := (\mathbf{u}+\mathbf{v})/2 \in C$ and, by the parallelogram identity, Theorem 1.1.20, $\|\mathbf{w}\|^2 = m^2 - \frac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2 < m^2$, a contradiction.

(ii) The set $D = C - \mathbf{b}$ is nonempty, closed, and convex, so (i) gives its unique minimum-norm vector $\mathbf{N}$. Since $\mathbf{b} \notin C$, $\mathbf{0} \notin D$, hence $\mathbf{N} \neq \mathbf{0}$. For $\mathbf{z} \in D$, define $\phi(t) := \|\mathbf{N} + t(\mathbf{z} - \mathbf{N})\|^2$, $0 \leq t \leq 1$. Minimality gives $\phi'_+(0) \geq 0$, hence $\langle \mathbf{N}, \mathbf{z}\rangle \geq \|\mathbf{N}\|^2$. Taking $\mathbf{z} = \mathbf{x} - \mathbf{b}$ yields $\langle \mathbf{N}, \mathbf{x}\rangle \geq \|\mathbf{N}\|^2 + \langle \mathbf{N}, \mathbf{b}\rangle$ for all $\mathbf{x} \in C$. For $\alpha := \langle \mathbf{N}, \mathbf{b}\rangle + \|\mathbf{N}\|^2/2$ and $\varepsilon := \|\mathbf{N}\|^2/4$, then $\langle \mathbf{N}, \mathbf{b}\rangle < \alpha - \varepsilon < \alpha + \varepsilon < \langle \mathbf{N}, C\rangle$, so $\mathcal{H}(\mathbf{N}, \alpha)$ strongly separates $\{\mathbf{b}\}$ and C.

Structural Commentary: The nearest point in $C - \mathbf{b}$ supplies the normal vector to a separating hyperplane.
Key Insight: Minimal norm turns a metric problem into the linear inequality $\langle \mathbf{N}, \mathbf{z}\rangle \geq \|\mathbf{N}\|^2$. □

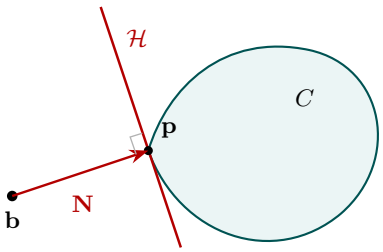


Figure 41: A nearest vector in $C - \mathbf{b}$ gives a normal direction and a strongly separating hyperplane.

Theorem 4.3.10. *Separating a Compact Convex Set from a Subspace*
Let $C \subseteq \mathbb{R}^n$ be nonempty, compact, and convex, and let $S \leq \mathbb{R}^n$ satisfy $C \cap S = \emptyset$.
Then there exists $\mathbf{N} \in S^\perp \setminus \{\mathbf{0}\}$ such that $\langle \mathbf{N}, C\rangle \geq \|\mathbf{N}\|^2 > 0$.
Consequently, $\mathcal{H}(\mathbf{N}, \|\mathbf{N}\|^2/2)$ strongly separates S and C.

Proof.

(i) Set $D := S + C$. Since S is closed in finite dimension, Proposition 4.3.6 gives D closed and convex. Also $\mathbf{0} \notin D$, for $\mathbf{0} = \mathbf{s} + \mathbf{c}$ would imply $\mathbf{c} = -\mathbf{s} \in C \cap S$.

(ii) Apply Theorem 4.3.9 to D and $\mathbf{0}$. Then $\exists \mathbf{N} \neq \mathbf{0}$ such that $\langle \mathbf{N}, \mathbf{s} + \mathbf{c}\rangle \geq \|\mathbf{N}\|^2$, $\mathbf{s} \in S$, $\mathbf{c} \in C$. Fix $\mathbf{c}_0 \in C$. If $a := \langle \mathbf{N}, \mathbf{s}\rangle \neq 0$ for some $\mathbf{s} \in S$, then $\langle \mathbf{N}, \lambda\mathbf{s} + \mathbf{c}_0\rangle = \lambda a + \langle \mathbf{N}, \mathbf{c}_0\rangle \to -\infty$ for λ of opposite sign and $|\lambda| \to \infty$, a contradiction. Hence $\mathbf{N} \in S^\perp$. Taking $\mathbf{s} = \mathbf{0}$ gives $\langle \mathbf{N}, C\rangle \geq \|\mathbf{N}\|^2 > 0$. Since $\langle \mathbf{N}, \mathbf{s}\rangle = 0$ for all $\mathbf{s} \in S$, the choice $\varepsilon := \|\mathbf{N}\|^2/4$ gives strong separation by $\mathcal{H}(\mathbf{N}, \|\mathbf{N}\|^2/2)$.

Structural Commentary: The sum $S + C$ is a closed convex set disjoint from $\mathbf{0}$, so point separation applies.
Key Insight: Since the separating inequality holds for all translates by S, scaling inside S forces $\mathbf{N}$ to annihilate S. □

Theorem 4.3.11. *Positive Orthant Alternatives for Subspaces*
Let $S \leq \mathbb{R}^n$. Then:

(i) $S \cap \mathbb{R}^n_+ = \emptyset \iff S^\perp \cap \mathbb{R}^n_{++} \neq \emptyset$.
(ii) $S \cap \mathbb{R}^n_{++} = \emptyset \iff S^\perp \cap \mathbb{R}^n_+ \neq \emptyset$.

Proof.

(*i*) ($\Rightarrow$) Let $\Delta := \operatorname{conv}\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$. By Proposition 4.3.5, Δ is compact and convex, and $\Delta \subseteq \mathbb{R}^n_+$.
Hence $\Delta \cap S = \emptyset$. Theorem 4.3.10 gives $\mathbf{N} \in S^\perp \setminus \{\mathbf{0}\}$ with $\langle \mathbf{N}, \mathbf{e}_i \rangle \geq \|\mathbf{N}\|^2 > 0$, $1 \leq i \leq n$. QED.
(*i*) ($\Leftarrow$) If $\mathbf{x} \in S \cap \mathbb{R}^n_+$ and $\mathbf{N} \in S^\perp \cap \mathbb{R}^n_{++}$, then $\langle \mathbf{N}, \mathbf{x} \rangle > 0$, contradicting $\mathbf{N} \perp S$.
(*ii*) ($\Rightarrow$) If $S = \{\mathbf{0}\}$, then $\mathbf{e}_1 \in S^\perp \cap \mathbb{R}^n_+$. Assume $S \neq \{\mathbf{0}\}$.
Let $\mathbf{B}_1, \dots, \mathbf{B}_k$ be a basis of S and set $M = [\mathbf{B}_1 \ \cdots \ \mathbf{B}_k] \in \operatorname{Mat}(n, k, \mathbb{R})$, $\operatorname{rows}(M) = \{\mathbf{R}_1, \dots, \mathbf{R}_n\} \subseteq \mathbb{R}^k$.
For $\mathbf{a} = (a_1, \dots, a_n)$, $\mathbf{a} \in S^\perp \iff \sum_{i=1}^n a_i \mathbf{R}_i = \mathbf{0}$.
Set $K := \operatorname{conv}\{\mathbf{R}_1, \dots, \mathbf{R}_n\}$. Hence $S^\perp \cap \mathbb{R}^n_+ \neq \emptyset \iff \mathbf{0} \in K$. If $\mathbf{0} \notin K$, Theorem 4.3.9 gives $\mathbf{u} \in \mathbb{R}^k$ with $\langle \mathbf{u}, \mathbf{R}_i \rangle > 0$ for all i. Then $\mathbf{v} := M\mathbf{u} \in S$ has coordinates $v_i = \langle \mathbf{u}, \mathbf{R}_i \rangle > 0$, so $\mathbf{v} \in S \cap \mathbb{R}^n_{++}$, a contradiction. Thus $\mathbf{0} \in K$, and $S^\perp \cap \mathbb{R}^n_+ \neq \emptyset$.

Structural Commentary: Use the simplex and row hull as compact witnesses.
Key Insight: Missing positivity yields an orthogonal positive certificate. □

Theorem 4.3.12. *Homogeneous Positive Alternatives*
Let $A \in \operatorname{Mat}(m, n, \mathbb{R})$. Exactly one alternative holds in each pair:

(i) $\ker(A) \cap \mathbb{R}^n_{++} \neq \emptyset$ *or* $\operatorname{col}(A^t) \cap \mathbb{R}^n_+ \neq \emptyset$.
(ii) $\ker(A) \cap \mathbb{R}^n_+ \neq \emptyset$ *or* $\operatorname{col}(A^t) \cap \mathbb{R}^n_{++} \neq \emptyset$.

Proof.

(i) Let $S := \operatorname{row}(A) = \operatorname{col}(A^t) \subseteq \mathbb{R}^n$. By Principle 4.3.3, $\ker(A) = S^\perp$.
Moreover, $A^t\mathbf{v} \in \operatorname{col}(A^t) = S$ for every $\mathbf{v} \in \mathbb{R}^m$, and every vector of S is of this form.
Hence the first pair is equivalent to $S^\perp \cap \mathbb{R}^n_{++} \neq \emptyset$ or $S \cap \mathbb{R}^n_+ \neq \emptyset$.
Theorem 4.3.11(i) says exactly that these two alternatives are mutually exclusive and exhaustive.
(ii) With the same S, one has $\ker(A) \cap \mathbb{R}^n_+ = S^\perp \cap \mathbb{R}^n_+$, and $\{A^t\mathbf{v} : \mathbf{v} \in \mathbb{R}^m\} \cap \mathbb{R}^n_{++} = S \cap \mathbb{R}^n_{++}$.
Thus the second pair is equivalent to $S^\perp \cap \mathbb{R}^n_+ \neq \emptyset$ or $S \cap \mathbb{R}^n_{++} \neq \emptyset$.
Theorem 4.3.11(ii) says exactly that these two alternatives are mutually exclusive and exhaustive.

Structural Commentary: The row space gives the certificate side; the nullspace gives the solution side.
Key Insight: Positive solvability is an orthogonal-complement alternative. □

Method 4.3.13. *Certifying Positive Homogeneous Systems*
Let $A \in \operatorname{Mat}(m, n, \mathbb{R})$ and $S := \operatorname{row}(A) = \operatorname{col}(A^t)$.

(i) $\ker(A) \cap \mathbb{R}^n_{++} = \emptyset$ is certified by $\exists\, \mathbf{v} \in \mathbb{R}^m$ with $A^t\mathbf{v} > \mathbf{0}$.
(ii) $\ker(A) \cap \mathbb{R}^n_+ = \emptyset$ is certified by $\exists\, \mathbf{v} \in \mathbb{R}^m$ with $A^t\mathbf{v} \gg \mathbf{0}$.
(iii) Existence follows from row-space avoidance,
$S \cap \mathbb{R}^n_+ = \emptyset \Rightarrow \ker(A) \cap \mathbb{R}^n_{++} \neq \emptyset$; or $S \cap \mathbb{R}^n_{++} = \emptyset \Rightarrow \ker(A) \cap \mathbb{R}^n_+ \neq \emptyset$.

Lemma 4.3.14. *Nonnegative Matrix Cone*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \operatorname{Mat}(m, n, \mathbb{R})$ *and* $C_A := A(\mathbb{R}^n_{\geq 0}) = \left\{ \sum_{i=1}^n t_i \mathbf{a}_i : t_i \geq 0 \right\}$.
Then C_A is a closed convex cone in $\mathbb{R}^m$.

Proof. For $\mathbf{x}_r = A\mathbf{y}_r \in C_A$ with $\mathbf{y}_r \geq \mathbf{0}$.
For $0 \leq t \leq 1$, $a \geq 0$, $t\mathbf{x}_1 + (1-t)\mathbf{x}_2 = A(t\mathbf{y}_1 + (1-t)\mathbf{y}_2) \in C_A$, where $a\mathbf{x}_1 = A(a\mathbf{y}_1) \in C_A$, C_A is convex cone.
For closedness, choose for $\mathbf{x} \in C_A$ a representation $\mathbf{x} = \sum_{i \in I} t_i \mathbf{a}_i, \qquad t_i > 0$ with I minimal.
If $\{\mathbf{a}_i : i \in I\}$ were dependent, choose $(d_i)_{i \in I} \neq 0$ with $\sum_{i \in I} d_i \mathbf{a}_i = \mathbf{0}$.
Replacing d by $-d$ if necessary, some $d_i < 0$; then $\lambda := \min_{d_i < 0} \frac{t_i}{-d_i}$ makes $t_i + \lambda d_i \geq 0$ for all i and zero for at least one i, contradicting minimality. Hence the active columns are independent.
Thus $C_A = \bigcup_{\substack{I \subseteq \{1,\dots,n\} \\ \{\mathbf{a}_i : i \in I\} \text{ independent}}} K_I$, where $K_I := \left\{ \sum_{i \in I} t_i \mathbf{a}_i : t_i \geq 0 \right\}$.

For such I, the map $T_I : \mathbb{R}^I \to \operatorname{span}\{\mathbf{a}_i : i \in I\}$, where $T_I((t_i)) = \sum_{i\in I} t_i\mathbf{a}_i$, is a linear isomorphism, so $K_I = T_I(\mathbb{R}^I_{\geq 0})$ is closed in its finite-dimensional range, and the range is closed in $\mathbb{R}^m$.
Hence each K_I is closed in $\mathbb{R}^m$, and the finite union above is closed.
Structural Commentary: Reduce C_A to finitely many independent-column cones.
Key Insight: Minimal support turns generators independent, making each cone closed. □

Theorem 4.3.15. *Farkas's Lemma*
Let $A \in \operatorname{Mat}(m, n, \mathbb{R})$, $\mathbf{b} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$, and $C_A := \{A\mathbf{y} : \mathbf{y} \in \mathbb{R}^n,\ \mathbf{y} \geq \mathbf{0}\}$.
Exactly one of the following holds:

(i) $\mathbf{b} \in C_A$, equivalently $\exists\, \mathbf{u} > \mathbf{0}$ with $A\mathbf{u} = \mathbf{b}$.
(ii) $\exists\, \mathbf{v} \in \mathbb{R}^m$ such that $A^t\mathbf{v} \leq \mathbf{0}$ and $\langle \mathbf{v}, \mathbf{b}\rangle > 0$.

Proof.

(i) If $\mathbf{b} = A\mathbf{u}$ with $\mathbf{u} \geq \mathbf{0}$ and $A^t\mathbf{v} \leq \mathbf{0}$, then $\langle \mathbf{v}, \mathbf{b}\rangle = \langle \mathbf{v}, A\mathbf{u}\rangle = \langle A^t\mathbf{v}, \mathbf{u}\rangle \leq 0$.
(ii) Suppose $\mathbf{b} \notin C_A$. By Lemma 4.3.14, C_A is closed and convex. Since $\mathbf{0} \in C_A$, Theorem 4.3.9, after changing sign if necessary, gives $\mathbf{v} \in \mathbb{R}^m$ and $\alpha \in \mathbb{R}$ such that $\langle \mathbf{v}, C_A\rangle < \alpha < \langle \mathbf{v}, \mathbf{b}\rangle$.
Thus $\alpha > 0$ and $\langle \mathbf{v}, \mathbf{b}\rangle > 0$. For every $\mathbf{y} \geq \mathbf{0}$, $\langle A^t\mathbf{v}, \mathbf{y}\rangle = \langle \mathbf{v}, A\mathbf{y}\rangle < \alpha$.
If $(A^t\mathbf{v})_i > 0$, taking $\mathbf{y} = \lambda\mathbf{e}_i$ and $\lambda \to \infty$ contradicts the bounded inequality. Hence $A^t\mathbf{v} \leq \mathbf{0}$.

Structural Commentary: Nonrepresentability means $\mathbf{b} \notin C_A$.
Key Insight: Separation turns nonmembership in a cone into the dual inequality $A^t\mathbf{v} \leq 0$. □

Example 4.3.16. *Why the Weak Inequality Matters*
The conclusion $A^t\mathbf{v} \leq \mathbf{0}$ cannot generally be strengthened to $A^t\mathbf{v} \ll \mathbf{0}$. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.
If $\mathbf{u} \geq \mathbf{0}$, then $(A\mathbf{u})_1 = u_1 \geq 0$, so $A\mathbf{u} \neq \mathbf{b}$.
The certificate $\mathbf{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ satisfies $A^t\mathbf{v} = (-1, 0, 0)^t \leq \mathbf{0}$ and $\langle \mathbf{v}, \mathbf{b}\rangle = 1 > 0$.
For arbitrary $\mathbf{w} = (r, s)^t$, $A^t\mathbf{w} = (r, s, -s)^t$, so $A^t\mathbf{w} \ll \mathbf{0}$ would force $s < 0 < -s$, impossible.
Structural Commentary: Opposite columns force a zero in every certificate.
Key Insight: Boundary separation is sometimes the best possible separation.

5 Eigenstructure and Invariant Subspaces

5.1 Eigenvalues and Eigenvectors

Spectral linear algebra understands an operator through subspaces it preserves. Eigenstructure, triangularization, diagonalization, and commuting families all refine this invariant-subspace viewpoint.

Definition 5.1.1. *Eigenvalue and Eigenvector*
Let $T \in \mathcal{L}(\mathcal{V})$. A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of T if there exists $\mathbf{v} \in \mathcal{V}$ with $\mathbf{v} \neq \mathbf{0}$ and $T\mathbf{v} = \lambda\mathbf{v}$. Such a vector $\mathbf{v}$ is an *eigenvector* of T corresponding to λ.

Definition 5.1.2. *Eigenspace*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\lambda \in \mathbb{F}$. The *eigenspace* of T corresponding to λ is $E_\lambda(T) := \ker(T - \lambda I_\mathcal{V})$. Thus λ is an eigenvalue of T if and only if $E_\lambda(T) \neq \{\mathbf{0}\}$. Equivalently, $E_\lambda(T)$ is the nullspace of the shifted operator $T - \lambda I_\mathcal{V}$.

Definition 5.1.3. *Spectrum and Geometric Multiplicity*
Let $T \in \mathcal{L}(\mathcal{V})$. The *spectrum* of T is $\operatorname{spec}(T) := \{\lambda \in \mathbb{F} : \lambda \text{ is an eigenvalue of } T\}$.
The *geometric multiplicity* of $\lambda \in \mathbb{F}$ is $\dim E_\lambda(T)$. For a square matrix $A \in \operatorname{Mat}(n, n, \mathbb{F})$, $E_\lambda(A) := \ker(A - \lambda I_n)$ and $\operatorname{spec}(A)$ denotes the spectrum of the induced operator on $\mathbb{F}^n$.

Proposition 5.1.4. *Eigenvectors as Kernel Vectors*
Let $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$. Vector $\mathbf{v}$ is eigenvector of T corresponding to λ if and only if $\mathbf{v} \in \ker(T - \lambda I_\mathcal{V})$, $\mathbf{v} \neq \mathbf{0}$.

Proof.

($\Rightarrow$) If $\mathbf{v}$ is an eigenvector corresponding to λ, then $\mathbf{v} \neq \mathbf{0}$ and $T\mathbf{v} = \lambda\mathbf{v}$, so $(T - \lambda I_\mathcal{V})\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $\mathbf{v} \in \ker(T - \lambda I_\mathcal{V})$ and $\mathbf{v} \neq \mathbf{0}$, then $(T - \lambda I_\mathcal{V})\mathbf{v} = \mathbf{0}$, so $T\mathbf{v} = \lambda\mathbf{v}$.

Structural Commentary: Eigenvectors are kernels of shifted operators.
Key Insight: Subtract $\lambda I_\mathcal{V}$ to convert $T\mathbf{v} = \lambda\mathbf{v}$ into homogeneous form. □

Theorem 5.1.5. *Equivalent Conditions for an Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{F}$. The following are equivalent:

(i) λ *is an eigenvalue of* T*;*
(ii) $T - \lambda I_{\mathcal{V}}$ *is not injective;*
(iii) $T - \lambda I_{\mathcal{V}}$ *is not surjective;*
(iv) $T - \lambda I_{\mathcal{V}}$ *is not invertible.*

Proof.

$(i) \Rightarrow (ii)$ If λ is an eigenvalue, Proposition 5.1.4 gives nonzero vector in $\ker(T-\lambda I_{\mathcal{V}})$, so $T-\lambda I_{\mathcal{V}}$ not injective.
$(ii) \Rightarrow (i)$ If $T - \lambda I_{\mathcal{V}}$ not injective, then $\ker(T - \lambda I_{\mathcal{V}})$ contains nonzero vector, so λ is eigenvalue (Prop 5.1.4).
$(ii) \Rightarrow (iii)$ If $T - \lambda I_{\mathcal{V}}$ were surjective, Theorem 3.2.38 would make it injective, a contradiction.
$(iii) \Rightarrow (iv)$ If $T - \lambda I_{\mathcal{V}}$ were invertible, then it would be surjective, a contradiction.
$(iv) \Rightarrow (ii)$ If $T - \lambda I_{\mathcal{V}}$ were injective, Theorem 3.2.38 would make it invertible, a contradiction.

Structural Commentary: In finite dimensions, noninvertibility has equivalent forms.
Key Insight: A scalar λ is spectral precisely when $T - \lambda I_{\mathcal{V}}$ has a nontrivial kernel. □

Theorem 5.1.6. *Eigenvectors for Distinct Eigenvalues Are Independent*
Let $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Suppose the list is linearly dependent. Let k be smallest positive integer s.t. $\mathbf{v}_k \in \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$. Then $k \geq 2$, $(\mathbf{v}_1, \ldots, \mathbf{v}_{k-1})$ is linearly independent, and $\exists$ scalars $a_1, \ldots, a_{k-1}$ s.t. $\mathbf{v}_k = a_1\mathbf{v}_1 + \cdots + a_{k-1}\mathbf{v}_{k-1}$. Applying T gives $\lambda_k\mathbf{v}_k = a_1\lambda_1\mathbf{v}_1 + \cdots + a_{k-1}\lambda_{k-1}\mathbf{v}_{k-1}$. Multiplying first displayed equation by λ_k and subtracting gives $0 = a_1(\lambda_k - \lambda_1)\mathbf{v}_1 + \cdots + a_{k-1}(\lambda_k - \lambda_{k-1})\mathbf{v}_{k-1}$. Since $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$ are linearly independent and $\lambda_k \neq \lambda_i$ for $i < k$, all $a_i = 0$. Hence $\mathbf{v}_k = \mathbf{0}$, contradicting that $\mathbf{v}_k$ is an eigenvector.
Structural Commentary: A smallest dependence isolates the first new eigenvector that depends on earlier ones. Applying T and subtracting separates the distinct eigenvalue factors.
Key Insight: Distinct eigenvalues make the same dependence relation impossible after applying T. □

Corollary 5.1.7. *Number of Distinct Eigenvalues*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$, then T has at most $\dim(\mathcal{V})$ distinct eigenvalues.

Proof. If T had more than $\dim(\mathcal{V})$ distinct eigenvalues, choose $m > \dim(\mathcal{V})$ of them, one eigenvector for each. By Theorem 5.1.6, these eigenvectors form linearly independent list of length m, contradicting Theorem 1.3.9.
Structural Commentary: Distinct eigenvalues give independent directions.
Key Insight: Count one eigenvector per eigenspace. □

Corollary 5.1.8. *Invertibility and the Zero Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then T is invertible if and only if $0 \notin \operatorname{spec}(T)$.

Proof. Apply Theorem 5.1.5 with $\lambda = 0$. Then 0 is an eigenvalue exactly when $T - 0I_{\mathcal{V}} = T$ is not invertible.
Structural Commentary: The zero eigenvalue is the spectral form of singularity.
Key Insight: Invertibility is equivalent to having trivial kernel, and $E_0(T) = \ker T$. □

Corollary 5.1.9. *Matrix Invertibility and the Spectrum*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$. Then A is invertible if and only if $0 \notin \operatorname{spec}(A)$.

Proof.

$(\Rightarrow)$ If A is invertible, then $\ker(A) = \{\mathbf{0}\}$, so 0 is not an eigenvalue of A.
$(\Leftarrow)$ If $0 \notin \operatorname{spec}(A)$, then $\ker(A) = \{\mathbf{0}\}$. Corollary 3.3.22 gives full rank, hence A is invertible.

Structural Commentary: Matrix invertibility is the coordinate form of trivial kernel.
Key Insight: A zero eigenvalue is exactly a nonzero solution of $A\mathbf{x} = \mathbf{0}$. □

5.2 Invariant Subspaces

Definition 5.2.1. *Invariant Subspace*
Let $T \in \mathcal{L}(\mathcal{V})$. A subspace $\mathcal{U}$ of $\mathcal{V}$ is *invariant under T* if $T\mathbf{u} \in \mathcal{U}$ for every $\mathbf{u} \in \mathcal{U}$.
The set of all T-invariant subspaces is $\operatorname{Inv}(T) := \{\mathcal{U} \leq \mathcal{V} : T\mathcal{U} \subseteq \mathcal{U}\}$.
Key Insight: $\mathcal{U}$ is invariant under T iff $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$.

Proposition 5.2.2. *One-Dimensional Invariant Subspaces and Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, $\mathbf{v} \neq \mathbf{0}$. Then $\operatorname{span}(\mathbf{v})$ is invariant under T if and only if there exists $\lambda \in \mathbb{F}$ such that $T\mathbf{v} = \lambda\mathbf{v}$.

Proof.

($\Rightarrow$) If $\operatorname{span}(\mathbf{v})$ is invariant under T, then $T\mathbf{v} \in \operatorname{span}(\mathbf{v})$, so $T\mathbf{v} = \lambda\mathbf{v}$ for some $\lambda \in \mathbb{F}$.
($\Leftarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{u} = a\mathbf{v} \in \operatorname{span}(\mathbf{v})$, then $T\mathbf{u} = aT\mathbf{v} = a\lambda\mathbf{v} \in \operatorname{span}(\mathbf{v})$.

Structural Commentary: One-dimensional invariance means rescaling one direction.
Key Insight: In dimension one, staying inside the subspace means being a scalar multiple of the same vector. □

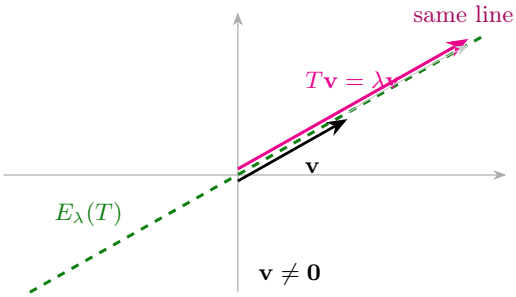


Figure 42: An eigendirection is a one-dimensional invariant subspace.

Proposition 5.2.3. *Invariant Subspaces and Block Form*
Let $T \in \mathcal{L}(\mathcal{V})$ and suppose $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

(i) If $\mathcal{U}$ is invariant under T, then in a basis adapted to $\mathcal{U} \oplus \mathcal{W}$, matrix of T has upper block-triangular form

$$\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}.$$

(ii) If both $\mathcal{U}$ and $\mathcal{W}$ are invariant under T, then the matrix of T has the block-diagonal form

$$\begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix}.$$

Proof. Choose basis $(\mathbf{u}_1, \ldots, \mathbf{u}_p)$ of $\mathcal{U}$ and basis $(\mathbf{w}_1, \ldots, \mathbf{w}_q)$ of $\mathcal{W}$, and use concatenation as basis of $\mathcal{V}$.

(i) If $\mathcal{U}$ is invariant, then each $T\mathbf{u}_j$ lies in $\mathcal{U}$, so its coordinates along $\mathbf{w}_1, \ldots, \mathbf{w}_q$ are zero. These entries form the lower-left block.
(ii) If $\mathcal{W}$ is also invariant, then each $T\mathbf{w}_j$ lies in $\mathcal{W}$, so its coordinates along $\mathbf{u}_1, \ldots, \mathbf{u}_p$ are zero. These entries form the upper-right block.

Structural Commentary: Block form records how an operator moves vectors between chosen summands. Invariant summands force the corresponding off-diagonal movement to vanish.
Key Insight: Zero blocks are coordinate evidence of invariant subspaces. □

Definition 5.2.4. *Reducing Pair*
Let $T \in \mathcal{L}(\mathcal{V})$. A pair of subspaces $(\mathcal{U}, \mathcal{W})$ *reduces* T if $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$; $T(\mathcal{U}) \subseteq \mathcal{U}$; and $T(\mathcal{W}) \subseteq \mathcal{W}$.
Then $T = T|_{\mathcal{U}} \oplus T|_{\mathcal{W}}$ with respect to $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$.

Proposition 5.2.5. *Reducing Pairs and Projections*
Let $T \in \mathcal{L}(\mathcal{V})$, let $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, and let $\rho = \rho_{\mathcal{U},\mathcal{W}}$. Then $(\mathcal{U}, \mathcal{W})$ reduces $T \iff T\rho = \rho T$.

Proof.

($\Rightarrow$) Let $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$.
Since both summands are invariant, $T\rho\mathbf{v} = T\mathbf{u}$ and $\rho T\mathbf{v} = \rho(T\mathbf{u} + T\mathbf{w}) = T\mathbf{u}$. Thus $T\rho = \rho T$.
($\Leftarrow$) Suppose $T\rho = \rho T$. If $\mathbf{u} \in \mathcal{U}$, then $\rho(T\mathbf{u}) = T(\rho\mathbf{u}) = T\mathbf{u}$, so $T\mathbf{u} \in \operatorname{range}(\rho) = \mathcal{U}$.
If $\mathbf{w} \in \mathcal{W}$, then $\rho(T\mathbf{w}) = T(\rho\mathbf{w}) = \mathbf{0}$, so $T\mathbf{w} \in \ker(\rho) = \mathcal{W}$. Hence $(\mathcal{U}, \mathcal{W})$ reduces T.

Structural Commentary: Commutation with ρ means T preserves the two projection components.
Key Insight: Reducing pairs are exactly decompositions whose projection commutes with T. □

Proposition 5.2.6. *Invariant Subspaces and Algebraic Projections*
Let $T \in \mathcal{L}(\mathcal{V})$, let $\mathcal{V} = \mathcal{U} \oplus \mathcal{W}$, and let $\rho = \rho_{\mathcal{U},\mathcal{W}}$. Then $\mathcal{U} \in \operatorname{Inv}(T) \iff \rho T\rho = T\rho$.
Consequently, if $\mathcal{U} \in \operatorname{Inv}(T)$, then $\rho_{\mathcal{U},\mathcal{Z}} T \rho_{\mathcal{U},\mathcal{Z}} = T\rho_{\mathcal{U},\mathcal{Z}}$ for every complement $\mathcal{Z}$ of $\mathcal{U}$.

Proof. For $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in \mathcal{U}$ and $\mathbf{w} \in \mathcal{W}$, $\mathcal{U} \in \operatorname{Inv}(T)$ gives $(\rho T\rho - T\rho)\mathbf{v} = \rho T\mathbf{u} - T\mathbf{u} = \mathbf{0}$.
Thus $\rho T\rho = T\rho$. Conversely, if $\rho T\rho = T\rho$ and $\mathbf{u} \in \mathcal{U}$, then $\rho T\mathbf{u} = \rho T\rho\mathbf{u} = T\rho\mathbf{u} = T\mathbf{u}$.
Hence $T\mathbf{u} \in \operatorname{range}(\rho) = \mathcal{U}$, so $\mathcal{U} \in \operatorname{Inv}(T)$.

Structural Commentary: A projection onto $\mathcal{U}$ tests whether the $\mathcal{W}$-component of $T\mathbf{u}$ vanishes.
Key Insight: Invariance says applying T after projecting into $\mathcal{U}$ needs no further correction by ρ. □

Perspective 5.2.7. *Matrix Viewpoint on Structure*
Basic block matrices already capture important geometry. Upper block-triangular form records the presence of an invariant subspace; block diagonal form records a genuine decomposition into independent invariant pieces.

Definition 5.2.8. *Diagonal and Upper-Triangular Matrices*
The *diagonal* of a square matrix is its upper-left to lower-right entries. A square matrix is *diagonal* if all off-diagonal entries are zero, and *upper triangular* if all entries below the diagonal are zero.

Theorem 5.2.9. *Invariant Subspaces, Blocks, and Projections*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, let $\mathcal{U}$ be an r-dimensional subspace of $\mathbb{F}^n$ with $1 \le r \le n-1$, and let $Q = [Q_1\ Q_2]$ be unitary, where the columns of Q_1 and Q_2 are orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$, respectively. Let $P = Q_1 Q_1^*$ be the orthogonal projection onto $\mathcal{U}$. The following are equivalent:

(i) $\mathcal{U}$ *is invariant under* A;
(ii) $Q^*AQ = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}$ *for some compatible matrices* B, X, C;
(iii) $PAP = AP$.

Proof. Let $M = Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$ and $E = Q^*PQ = I_r \oplus 0_{n-r}$. Since Q is unitary and $P = Q_1Q_1^*$, the matrix of P in the Q-coordinates is E. Also $Y = Q_2^*AQ_1$, and $\ker(Q_2^*) = \mathrm{col}(Q_1) = \mathcal{U}$.

(i) $\Rightarrow$ *(ii)* If $A\mathcal{U} \subseteq \mathcal{U}$, then each column of AQ_1 lies in $\mathcal{U} = \ker(Q_2^*)$, so $Y = Q_2^*AQ_1 = 0$.

(ii) $\Rightarrow$ *(iii)* If $Y = 0$, then $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$.
Thus $Q^*(PAP)Q = EME = ME = Q^*(AP)Q$, so $PAP = AP$.

(iii) $\Rightarrow$ *(i)* If $PAP = AP$, then $EME = ME$. Since $EME = \begin{bmatrix} B & 0 \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $Y = 0$.
Hence the columns of AQ_1 lie in $\ker(Q_2^*) = \mathcal{U}$. Since the columns of Q_1 span $\mathcal{U}$, linearity gives $A\mathcal{U} \subseteq \mathcal{U}$.

Structural Commentary: In adapted coordinates, the lower-left block is the component of $A\mathcal{U}$ in $\mathcal{U}^\perp$.
Key Insight: $PAP = AP$ exactly kills this component. □

Corollary 5.2.10. *Reducing Subspaces and Commuting Projections*
Maintain the notation of Theorem 5.2.9. The following are equivalent:

(i) $\mathcal{U}$ *is invariant under both* A *and* A^*;
(ii) $Q^*AQ = B \oplus C$ *for some* $B \in \mathrm{Mat}(r,r,\mathbb{F})$ *and* $C \in \mathrm{Mat}(n-r, n-r, \mathbb{F})$;
(iii) $PA = AP$.

Proof. Let $M = Q^*AQ = \begin{bmatrix} B & X \\ Y & C \end{bmatrix}$ and $E = Q^*PQ = I_r \oplus 0_{n-r}$.
Then $Q^*A^*Q = M^*$, $Q^*(PA)Q = EM$, and $Q^*(AP)Q = ME$.

(i) $\Rightarrow$ *(ii)* By Theorem 5.2.9, A-invariance gives $Y = 0$. Applying the same theorem to A^* gives that the lower-left block of M^* is zero, namely $X^* = 0$, so $X = 0$. Thus $M = B \oplus C$.

(ii) $\Rightarrow$ *(iii)* If $M = B \oplus C$, then $EM = ME$, hence $PA = AP$.

(iii) $\Rightarrow$ *(i)* If $PA = AP$, then $EM = ME$. Since $EM = \begin{bmatrix} B & X \\ 0 & 0 \end{bmatrix}$ and $ME = \begin{bmatrix} B & 0 \\ Y & 0 \end{bmatrix}$, we get $X = Y = 0$.
The equation $Y = 0$ gives A-invariance by Theorem 5.2.9. Since the lower-left block of M^* is $X^* = 0$, applying the same theorem to A^* gives A^*-invariance.

Structural Commentary: Reducing subspaces are exactly block diagonal decompositions in adapted coordinates.
Key Insight: Commutation with P kills both off-diagonal blocks. □

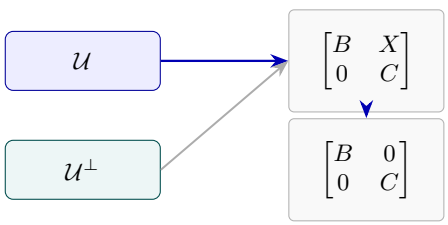


Figure 43: Adapted coordinates: invariance gives upper block form; $PA = AP$ gives block diagonal form.

Definition 5.2.11. *Restriction and Quotient Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\mathcal{U}$ be invariant under T.

(i) The *restriction operator* $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is defined by $T|_{\mathcal{U}}(\mathbf{u}) = T\mathbf{u}$ for $\mathbf{u} \in \mathcal{U}$.
(ii) The *quotient operator* $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$ is defined by $(T/\mathcal{U})(\mathbf{v} + \mathcal{U}) = T\mathbf{v} + \mathcal{U}$, $\mathbf{v} \in \mathcal{V}$.

Proposition 5.2.12. *Restriction and Quotient Operators Are Well-Defined*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\mathcal{U}$ be invariant under T. Then $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ and $T/\mathcal{U} \in \mathcal{L}(\mathcal{V}/\mathcal{U})$.

Proof. Since $\mathcal{U}$ is invariant under T, $T\mathbf{u} \in \mathcal{U}$ whenever $\mathbf{u} \in \mathcal{U}$, so $T|_{\mathcal{U}}$ maps $\mathcal{U}$ into itself. Its linearity is inherited from T. For the quotient operator, suppose $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$. Then $\mathbf{v} - \mathbf{w} \in \mathcal{U}$.
Invariance gives $T(\mathbf{v} - \mathbf{w}) \in \mathcal{U}$, so $T\mathbf{v} - T\mathbf{w} \in \mathcal{U}$ and hence $T\mathbf{v} + \mathcal{U} = T\mathbf{w} + \mathcal{U}$. Thus $T/\mathcal{U}$ is well-defined.
For $a \in \mathbb{F}$, $(T/\mathcal{U})(a(\mathbf{v} + \mathcal{U}) + (\mathbf{w} + \mathcal{U})) = T(a\mathbf{v} + \mathbf{w}) + \mathcal{U} = a(T\mathbf{v} + \mathcal{U}) + (T\mathbf{w} + \mathcal{U})$, so $T/\mathcal{U}$ is linear.
Structural Commentary: Restriction stays inside $\mathcal{U}$; quotient collapses $\mathcal{U}$.
Key Insight: Invariance makes both operators well-defined. □

Perspective 5.2.13. *Two Smaller Views of an Operator*
When $\mathcal{U}$ is invariant under T, the operators $T|_{\mathcal{U}}$ and $T/\mathcal{U}$ split attention between what happens inside $\mathcal{U}$ and what remains after $\mathcal{U}$ is collapsed. These two views often simplify an operator, even though they need not determine the whole operator by themselves.

5.3 Triangularization and Decomposition

Theorem 5.3.1. *Complex Operators Have Eigenvalues*
Every operator on a finite-dimensional, nonzero complex vector space has an eigenvalue.

Proof. Let $T \in \mathcal{L}(\mathcal{V})$. By Proposition 6.1.4, T has a nonconstant annihilating polynomial. Since $\mathcal{V}$ is complex and nonzero, Theorem 6.3.15 implies that one root of this polynomial is an eigenvalue of T.
Structural Commentary: Finite dimension gives an annihilating polynomial; over $\mathbb{C}$, it factors linearly.
Key Insight: One noninjective factor produces an eigenvector. □

Definition 5.3.2. *Commutant*
For $\emptyset \neq \mathcal{F} \subseteq \mathrm{Mat}(n, n, \mathbb{F})$, the *commutant* of $\mathcal{F}$ is $\mathrm{Comm}(\mathcal{F}) := \{X \in \mathrm{Mat}(n, n, \mathbb{F}) : XA = AX \ \forall A \in \mathcal{F}\}$.
For a single matrix A, write $\mathrm{Comm}(A)$.

Lemma 5.3.3. *Commuting with a Block Scalar Diagonal Matrix*
Let $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ and let $\Lambda = \lambda_1 I_{n_1} \oplus \cdots \oplus \lambda_k I_{n_k}$.
Partition $A = [A_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ conformally with Λ, so $A_{ij} \in \mathrm{Mat}(n_i, n_j, \mathbb{F})$.

(i) If A is block diagonal, then $A\Lambda = \Lambda A$.
(ii) If $A\Lambda = \Lambda A$ and $\lambda_1, \dots, \lambda_k$ are distinct, then A is block diagonal.

Proof. Block multiplication gives $(A\Lambda)_{ij} = \lambda_j A_{ij}$ and $(\Lambda A)_{ij} = \lambda_i A_{ij}$. If A is block diagonal, then $A_{ij} = 0$ for $i \neq j$, and these two expressions agree for every block. Conversely, if $A\Lambda = \Lambda A$, then $(\lambda_j - \lambda_i)A_{ij} = 0$ for every i, j. Distinctness of the λ_i forces $A_{ij} = 0$ whenever $i \neq j$, so A is block diagonal.
Structural Commentary: Distinct scalar blocks force commuting matrices to respect the block decomposition.
Key Insight: Distinct block scalars make every off-diagonal block visible and force it to vanish. □

Example 5.3.4. *Distinctness Is Necessary*
The distinctness hypothesis in Lemma 5.3.3 cannot be dropped. If $\lambda_1 = \cdots = \lambda_k = 1$, then $\Lambda = I_n$ commutes with every matrix in $\mathrm{Mat}(n, n, \mathbb{F})$, including matrices that are not block diagonal for the chosen partition.

Theorem 5.3.5. *Common Eigenvector for a Commuting Family*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $\mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be a nonempty commuting family: $ST = TS$ for all $S, T \in \mathcal{F}$. Then there exists $\mathbf{v} \neq \mathbf{0}$ that is an eigenvector of every operator in $\mathcal{F}$.

Proof. First suppose $\mathcal{F} = \{T_1, \dots, T_m\}$ is finite, and induct on m. The case $m = 1$ is Theorem 5.3.1.
For $m > 1$, choose $\lambda \in \mathrm{spec}(T_1)$ and set $\mathcal{U} = E_\lambda(T_1)$. Then $\mathcal{U} \neq \{\mathbf{0}\}$.
If $\mathbf{u} \in \mathcal{U}$ and $j \geq 2$, commutation gives $(T_1 - \lambda I_{\mathcal{V}})T_j\mathbf{u} = T_j(T_1 - \lambda I_{\mathcal{V}})\mathbf{u} = \mathbf{0}$, so $\mathcal{U}$ is invariant under $T_2, \dots, T_m$.
Their restrictions to $\mathcal{U}$ still commute, and by induction have a common eigenvector $\mathbf{v} \in \mathcal{U}$, $\mathbf{v} \neq \mathbf{0}$.
Since $\mathbf{v} \in E_\lambda(T_1)$, it is also an eigenvector of T_1. For general $\mathcal{F}$, the subspace $\mathrm{span}(\mathcal{F})$ of $\mathcal{L}(\mathcal{V})$ is finite-dimensional. Choose a basis $T_1, \dots, T_m$ of this span from the spanning set $\mathcal{F}$.
The finite case gives a nonzero $\mathbf{v}$ and scalars μ_j with $T_j\mathbf{v} = \mu_j\mathbf{v}$. If $S \in \mathcal{F}$, write $S = \sum_j a_j T_j$.
Then $S\mathbf{v} = (\sum_j a_j\mu_j)\mathbf{v}$, so $\mathbf{v}$ is an eigenvector of every member of $\mathcal{F}$.
Structural Commentary: Commutation makes one eigenspace invariant, reducing the family by one.
Key Insight: Find the common eigenvector inside an eigenspace. □

Theorem 5.3.6. *Criteria for an Upper-Triangular Operator Matrix*
Let $T \in \mathcal{L}(\mathcal{V})$ and let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of $\mathcal{V}$. The following are equivalent:

(i) $[T]_{\beta \leftarrow \beta}$ *is upper triangular;*
(ii) $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ *for each* $j = 1, \dots, n$;
(iii) $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ *is invariant under* T *for each* $j = 1, \dots, n$.

Proof.

(*i*) $\Rightarrow$ (*ii*) The jth column of $[T]_{\beta \leftarrow \beta}$ is the coordinate column of $T\mathbf{v}_j$ in the basis β. Since the matrix is upper triangular, this column has zero entries below position j, so $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$.
(*ii*) $\Rightarrow$ (*i*) If $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for every j, then the jth coordinate column of $T\mathbf{v}_j$ has zero entries below position j. Hence $[T]_{\beta \leftarrow \beta}$ is upper triangular.
(*ii*) $\Rightarrow$ (*iii*) If $\mathbf{u} \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$, then linearity gives $T\mathbf{u}$ as a linear combination of $T\mathbf{v}_1, \dots, T\mathbf{v}_j$, all of which lie in $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$. Hence this span is invariant.
(*iii*) $\Rightarrow$ (*ii*) Apply invariance of $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ to $\mathbf{v}_j$.

Structural Commentary: Upper triangularity records an invariant flag.
Key Insight: Columns below the diagonal vanish exactly when $T\mathbf{v}_j$ uses only earlier basis vectors and itself. □

Corollary 5.3.7. *Upper-Triangular Form in an Orthonormal Basis*
Let $\mathcal{V}$ be a finite-dimensional inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. If T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$, then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis for which $[T]_{\beta \leftarrow \beta}$ is upper triangular. By Theorem 5.3.6, each subspace

$$\mathcal{U}_j = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j), \qquad j = 1, \dots, n,$$

is invariant under T. Apply Gram–Schmidt to β and obtain an orthonormal basis $\gamma = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ with

$$\operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j) = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j) = \mathcal{U}_j.$$

Thus each $\operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$ is invariant under T, and Theorem 5.3.6 shows that $[T]_{\gamma \leftarrow \gamma}$ is upper triangular.
Structural Commentary: Upper triangularity is preservation of a nested flag of subspaces. Gram–Schmidt preserves that flag while making the basis orthonormal.
Key Insight: Triangular form depends on the flag, not on the particular basis used inside the flag. □

Theorem 5.3.8. *Complex Upper Triangularisation*
Let $\mathcal{V}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some basis of $\mathcal{V}$.

Proof. Induct on $n = \dim(\mathcal{V})$. If $n \le 1$, the result is immediate.
Suppose $n > 1$ and the result holds in smaller dimensions. Choose an eigenvector $\mathbf{v}_1$ of T by Theorem 5.3.1, and set $\mathcal{U} = \operatorname{span}(\mathbf{v}_1)$. Then $\mathcal{U}$ is T-invariant, so the quotient operator $T/\mathcal{U}$ is defined on $\mathcal{V}/\mathcal{U}$, which has dimension $n-1$. By induction, choose a basis $(\mathbf{v}_2 + \mathcal{U}, \dots, \mathbf{v}_n + \mathcal{U})$ of $\mathcal{V}/\mathcal{U}$ with respect to which $T/\mathcal{U}$ is upper triangular, and choose representatives $\mathbf{v}_2, \dots, \mathbf{v}_n \in \mathcal{V}$.
The lifted list $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$: any relation descends to the quotient basis, forcing all coefficients of $\mathbf{v}_2, \dots, \mathbf{v}_n$ to be zero, and then the coefficient of $\mathbf{v}_1$ is zero. Now $T\mathbf{v}_1 \in \operatorname{span}(\mathbf{v}_1)$.
For $j \ge 2$, upper triangularity of $T/\mathcal{U}$ gives $T\mathbf{v}_j + \mathcal{U} = (T/\mathcal{U})(\mathbf{v}_j + \mathcal{U}) \in \operatorname{span}(\mathbf{v}_2 + \mathcal{U}, \dots, \mathbf{v}_j + \mathcal{U})$, so $T\mathbf{v}_j \in \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$. Hence Theorem 5.3.6 gives an upper-triangular matrix for T in the basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$.
Structural Commentary: Lift an upper-triangular quotient basis above an eigenline.
Key Insight: The quotient builds the remaining invariant flag. □

Definition 5.3.9. *Almost Upper-Triangular Real Matrices*
Let $A \in \operatorname{Mat}(n, n, \mathbb{R})$. The matrix A is *almost upper triangular* if there exist $d_1, \dots, d_m \in \{1, 2\}$ with $\sum_j d_j = n$ such that, in the corresponding block form $A = [A_{ij}]_{i,j=1}^{m}$,

(i) $A_{ij} = 0$ for $i > j$;
(ii) each diagonal block is either $A_{jj} = [\lambda_j]$, $\lambda_j \in \mathbb{R}$, or $A_{jj} = \begin{bmatrix} a_j & -b_j \\ b_j & a_j \end{bmatrix}$, $a_j, b_j \in \mathbb{R}$, $b_j \neq 0$.

The matrix A is in *real Schur block form* if only (i) holds for some such block sizes $d_j \in \{1, 2\}$.

Theorem 5.3.10. *Real Almost Upper Triangularisation*
Let $\dim_{\mathbb{R}} \mathcal{V} < \infty$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\exists$ a basis β of $\mathcal{V}$ such that $[T]_{\beta \leftarrow \beta}$ is almost upper triangular.

Proof. Induct on $n = \dim_{\mathbb{R}} \mathcal{V}$. If $n \le 1$, clear.
For $n \ge 2$, Theorem 5.3.1 gives an eigenpair $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$, where $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$.
Thus $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $T\mathbf{v} = b\mathbf{u} + a\mathbf{v}$.

If $b = 0$, choose $\mathbf{0} \neq \mathbf{x} \in \{\mathbf{u}, \mathbf{v}\}$, set $\mathcal{U} = \operatorname{span}(\mathbf{x})$, $\alpha = (\mathbf{x})$, so $[T|_{\mathcal{U}}]_{\alpha \leftarrow \alpha} = [a]$.
If $b \neq 0$, set $\mathcal{U} = \operatorname{span}(\mathbf{u}, \mathbf{v})$, $\alpha = (\mathbf{u}, -\mathbf{v})$; then $\mathbf{u}, \mathbf{v}$ are independent and $[T|_{\mathcal{U}}]_{\alpha \leftarrow \alpha} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.
In either case, $T(\mathcal{U}) \subseteq \mathcal{U}$, $\dim \mathcal{U} \in \{1, 2\}$, and the first diagonal block is admissible.
By induction, $T/\mathcal{U}$ has an almost upper-triangular matrix in some quotient basis $\overline{\gamma}$.
Lift $\overline{\gamma}$ to $\gamma \subseteq \mathcal{V}$ and set $\beta = (\alpha, \gamma)$. Then $[T]_{\beta \leftarrow \beta}$ is block upper triangular with first block $[T|_{\mathcal{U}}]_{\alpha \leftarrow \alpha}$ and quotient block $[T/\mathcal{U}]_{\overline{\gamma} \leftarrow \overline{\gamma}}$, hence is almost upper triangular.
Structural Commentary: Real triangularisation starts with an invariant line or plane.
Key Insight: Complex eigenvectors split into real and imaginary parts; those parts form the first real block. □

Corollary 5.3.11. *Orthonormal Real Schur Block Form*
Let $\dim_{\mathbb{R}} \mathcal{V} < \infty$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\exists$ an orthonormal basis γ of $\mathcal{V}$ s.t. $[T]_{\gamma \leftarrow \gamma}$ is in real Schur block form.

Proof. By Theorem 5.3.10, choose $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ with block sizes $d_j \in \{1, 2\}$.
Put $s_k = d_1 + \cdots + d_k$ and $\mathcal{U}_k = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_{s_k})$. Then $T(\mathcal{U}_k) \subseteq \mathcal{U}_k$ for all k.
Apply Gram–Schmidt to β and obtain an orthonormal basis $\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ with $\operatorname{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{s_k}) = \mathcal{U}_k$, $1 \leq k \leq m$. Thus $[T]_{\gamma \leftarrow \gamma}$ is block upper triangular for the same $d_j \in \{1, 2\}$, hence is in real Schur block form.
Structural Commentary: Gram–Schmidt preserves the invariant block flag.
Key Insight: Orthonormalization changes blocks, not the flag. □

Proposition 5.3.12. *Invertibility from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then T is invertible if and only if $\lambda_j \neq 0$ for every j.

Proof. Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be such a basis.

($\Leftarrow$) If every $\lambda_j \neq 0$, then upper-triangular matrix $[T]_{\beta \leftarrow \beta}$ is invertible by Theorem 4.2.5, so T is invertible.
($\Rightarrow$) Suppose T is invertible. If $\lambda_j = 0$ for some j, set $\mathcal{U}_j = \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_j)$ and $\mathcal{U}_0 = \{\mathbf{0}\}$. Upper triangularity gives $T(\mathcal{U}_j) \subseteq \mathcal{U}_{j-1}$ because the coefficient of $\mathbf{v}_j$ in $T\mathbf{v}_j$ is $\lambda_j = 0$ and $T\mathbf{v}_i \in \mathcal{U}_{j-1}$ for $i < j$.
Since $\dim(\mathcal{U}_j) > \dim(\mathcal{U}_{j-1})$, the map $\mathcal{U}_j \to \mathcal{U}_{j-1}$ induced by T is not injective. Thus there is a nonzero $\mathbf{u} \in \mathcal{U}_j$ with $T\mathbf{u} = \mathbf{0}$, contradicting invertibility. Hence every $\lambda_j \neq 0$.

Structural Commentary: Diagonal entries control each flag layer.
Key Insight: A zero diagonal entry forces a kernel vector. □

Proposition 5.3.13. *Eigenvalues from an Upper-Triangular Matrix*
Suppose $T \in \mathcal{L}(\mathcal{V})$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to some basis of $\mathcal{V}$. Then the eigenvalues of T are precisely $\lambda_1, \ldots, \lambda_n$.

Proof. For $\lambda \in \mathbb{F}$, the matrix of $T - \lambda I_{\mathcal{V}}$ in the same basis is upper triangular with diagonal entries $\lambda_1 - \lambda, \ldots, \lambda_n - \lambda$. By Proposition 5.3.12, $T - \lambda I_{\mathcal{V}}$ is not invertible if and only if $\lambda = \lambda_j$ for some j. By Theorem 5.1.5, this is equivalent to λ being an eigenvalue of T.
Structural Commentary: Scalar shifts preserve triangular form.
Key Insight: Eigenvalues occur exactly when one shifted diagonal entry becomes zero. □

Perspective 5.3.14. *Why Triangular Form Matters*
Upper-triangular form is the first useful simplification beyond finding one eigenvector. It preserves enough of the operator to read eigenvalues from the diagonal while still allowing nontrivial coupling above the diagonal.

Definition 5.3.15. *Diagonalizable Operator*
An operator $T \in \mathcal{L}(\mathcal{V})$ is *diagonalizable* if T has a diagonal matrix with respect to some basis of $\mathcal{V}$.

Proposition 5.3.16. *Eigenspaces for Distinct Eigenvalues Form a Direct Sum*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T. Then $E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)$ is a direct sum. Consequently, $\dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T) \leq \dim(\mathcal{V})$.

Proof. Suppose $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in E_{\lambda_j}(T)$. If some $\mathbf{u}_j$ were nonzero, then the nonzero vectors among $\mathbf{u}_1, \ldots, \mathbf{u}_m$ would be eigenvectors corresponding to distinct eigenvalues, and Theorem 5.1.6 would make them linearly independent. This contradicts their sum being $\mathbf{0}$. Hence every $\mathbf{u}_j = \mathbf{0}$, so the sum is direct.
By Theorem 2.3.3, $\dim(E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T)) = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T)$.
The left side is the dimension of a subspace of $\mathcal{V}$, so Proposition 1.4.2 gives the stated inequality.
Structural Commentary: Distinct eigenspaces form independent pieces.
Key Insight: A zero sum across them would contradict eigenvector independence. □

Theorem 5.3.17. *Conditions Equivalent to Diagonalizability*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \dots, \lambda_m$ denote the distinct eigenvalues of T. Set $n = \dim(\mathcal{V})$. The following are equivalent:

(i) *T is diagonalizable;*
(ii) *$\mathcal{V}$ has a basis consisting of eigenvectors of T;*
(iii) *$\exists$ one-dimensional subspaces $\mathcal{U}_1, \dots, \mathcal{U}_n$ of $\mathcal{V}$, each invariant under T, such that $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$;*
(iv) *$\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$;*
(v) *$\dim(\mathcal{V}) = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T)$.*

Proof.

(i) $\Rightarrow$ *(ii)* If T is diagonalizable, then T has a diagonal matrix with respect to some basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$. Hence $T\mathbf{v}_j = a_j\mathbf{v}_j$ for each j, so this basis consists of eigenvectors of T.

(ii) $\Rightarrow$ *(i)* If $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of eigenvectors of T, then $T\mathbf{v}_j = a_j\mathbf{v}_j$ for each j. Thus the matrix of T with respect to this basis is diagonal.

(ii) $\Rightarrow$ *(iii)* Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ be an eigenbasis, and put $\mathcal{U}_j = \operatorname{span}(\mathbf{v}_j)$. Each $\mathcal{U}_j$ is one-dimensional and invariant under T by Proposition 5.2.2; the basis decomposition gives $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_n$.

(iii) $\Rightarrow$ *(ii)* Choose nonzero $\mathbf{v}_j \in \mathcal{U}_j$. Proposition 5.2.2 makes each $\mathbf{v}_j$ an eigenvector of T, and the direct-sum decomposition makes $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ a basis of $\mathcal{V}$.

(ii) $\Rightarrow$ *(iv)* Every vector in an eigenbasis lies in one of $E_{\lambda_1}(T), \dots, E_{\lambda_m}(T)$, so these eigenspaces span $\mathcal{V}$. Proposition 5.3.16 shows their sum is direct.

(iv) $\Rightarrow$ *(v)* If $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$, then dimensions add by Theorem 2.3.3.

(v) $\Rightarrow$ *(ii)* For each j, choose a basis β_j of $E_{\lambda_j}(T)$, and concatenate these bases into a list β. The length of β is $\dim(\mathcal{V})$. If a linear combination of vectors in β is zero, grouping terms by eigenspace gives $\mathbf{u}_1 + \cdots + \mathbf{u}_m = \mathbf{0}$ with $\mathbf{u}_j \in E_{\lambda_j}(T)$. Proposition 5.3.16 gives $\mathbf{u}_j = \mathbf{0}$ for every j; since each β_j is independent, all coefficients are zero. Thus β is independent, Proposition 1.4.3 makes β a basis of $\mathcal{V}$ consisting of eigenvectors of T.

Structural Commentary: Diagonalizability means the eigenspaces exhaust $\mathcal{V}$. Directness is automatic for distinct eigenspaces, so the real test is spanning, or equivalently the dimension count.
Key Insight: A diagonal basis is exactly an eigenbasis. □

5.4 Diagonalization and Simultaneous Forms

Perspective 5.4.1. *Diagonalization as Exhaustion by Eigenspaces*
Distinct eigenspaces are already separated by Proposition 5.3.16. Thus diagonalizability is the stronger statement that these separated pieces account for every direction in $\mathcal{V}$.

Theorem 5.4.2. *Algebraic Spectral Resolution*
Let $0 \neq \mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\Lambda = \{\lambda_1, \dots, \lambda_m\} \subseteq \mathbb{F}$ with $\lambda_i \neq \lambda_j$ for $i \neq j$. The following are equivalent:

(i). *T is diagonalizable and $\operatorname{spec}(T) = \Lambda$.*
(ii). *$\exists$ nonzero projections $\rho_1, \dots, \rho_m \in \mathcal{L}(\mathcal{V})$ s.t. $\rho_i\rho_j = 0$ $(i \neq j)$, $\sum_{i=1}^{m} \rho_i = I_{\mathcal{V}}$, $T = \sum_{i=1}^{m} \lambda_i\rho_i$.*

In this case, for each i, $\operatorname{range}(\rho_i) = E_{\lambda_i}(T)$; $\ker(\rho_i) = \bigoplus_{j\neq i} E_{\lambda_j}(T)$; and $\rho_i = \prod_{j\neq i} \frac{T-\lambda_j I_{\mathcal{V}}}{\lambda_i-\lambda_j}$

Proof.

(i) $\Rightarrow$ *(ii)* By Theorem 5.3.17, $\mathcal{V} = \bigoplus_{i=1}^{m} E_{\lambda_i}(T)$. Let $\rho_i = \rho_{E_{\lambda_i}(T), \bigoplus_{j\neq i} E_{\lambda_j}(T)}$.

Then $\rho_i\rho_j = 0$ for $i \neq j$, $\sum_{i=1}^{m} \rho_i = I_{\mathcal{V}}$, and $T = \sum_{i=1}^{m} \lambda_i\rho_i$. Also $\operatorname{range}(\rho_i)$ and $\ker(\rho_i)$ are as displayed.

For $\ell_i(z) = \prod_{j\neq i} \frac{z-\lambda_j}{\lambda_i-\lambda_j}$, we have $\ell_i(\lambda_k) = \delta_{ik}$, hence $\ell_i(T) = \rho_i$.

(ii) $\Rightarrow$ *(i)* Put $\mathcal{V}_i = \operatorname{range}(\rho_i)$. By Theorem 3.2.8, $\mathcal{V} = \bigoplus_{i=1}^{m} \mathcal{V}_i$.

If $\mathbf{x} \in \mathcal{V}_i$, then $\rho_i\mathbf{x} = \mathbf{x}$ and $\rho_j\mathbf{x} = \mathbf{0}$ for $j \neq i$, so $T\mathbf{x} = \lambda_i\mathbf{x}$. Thus $\mathcal{V}_i \subseteq E_{\lambda_i}(T)$.

Conversely, if $T\mathbf{x} = \lambda_i\mathbf{x}$ and $\mathbf{x} = \sum_{j=1}^{m} \mathbf{x}_j$ with $\mathbf{x}_j \in \mathcal{V}_j$, then $\sum_{j=1}^{m} (\lambda_j - \lambda_i)\mathbf{x}_j = \mathbf{0}$, so $\mathbf{x}_j = \mathbf{0}$ for $j \neq i$.

Hence $\mathcal{V}_i = E_{\lambda_i}(T)$. Since each $\mathcal{V}_i \neq 0$, each $\lambda_i \in \operatorname{spec}(T)$; concatenating bases of the $\mathcal{V}_i$ gives an eigenbasis, so T is diagonalizable and $\operatorname{spec}(T) = \Lambda$.

The kernel and polynomial formulas follow from $\mathcal{V} = \bigoplus_{i=1}^{m} \mathcal{V}_i$ and $\ell_i(\lambda_k) = \delta_{ik}$.

Structural Commentary: A diagonalizable operator is a weighted sum of the projections onto its eigenspaces.
Key Insight: Spectral projections are Lagrange polynomials in the operator. □

Lemma 5.4.3. *Diagonalizable Restrictions to Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ be diagonalizable and let $\mathcal{U} \in \mathrm{Inv}(T)$. For $\lambda \in \mathrm{spec}(T)$, set $\mathcal{U}_\lambda = \mathcal{U} \cap E_\lambda(T)$. Then $\mathcal{U} = \bigoplus_{\lambda \in \mathrm{spec}(T)} \mathcal{U}_\lambda$, where $T|_\mathcal{U}$ is diagonalizable.

Proof. Let $\rho_\lambda = p_\lambda(T)$ be the spectral projection from Theorem 5.4.2. Since $\mathcal{U} \in \mathrm{Inv}(T)$, $\rho_\lambda(\mathcal{U}) \subseteq \mathcal{U}$. Hence, for $\mathbf{u} \in \mathcal{U}$, $\mathbf{u} = \sum_{\lambda \in \mathrm{spec}(T)} \rho_\lambda \mathbf{u}$, where $\rho_\lambda \mathbf{u} \in \mathcal{U}_\lambda$. Thus $\mathcal{U} = \sum_{\lambda \in \mathrm{spec}(T)} \mathcal{U}_\lambda$, and the sum is direct since $\mathcal{U}_\lambda \subseteq E_\lambda(T)$. Bases of the nonzero $\mathcal{U}_\lambda$ concatenate to an eigenbasis for $T|_\mathcal{U}$.
Structural Commentary: Invariant subspaces inherit spectral projections.
Key Insight: Polynomial spectral projections cannot move vectors out of a T-invariant subspace. □

Theorem 5.4.4. *Simultaneous Diagonalisation of Commuting Diagonalizable Operators*
Let $\dim \mathcal{V} < \infty$, and let $\emptyset \neq \mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be finite with every $T \in \mathcal{F}$ diagonalizable. Then $\forall S, T \in \mathcal{F}$, $ST = TS \iff \exists$ a basis β of $\mathcal{V}$ such that $[T]_{\beta \leftarrow \beta}$ is diagonal $\forall T \in \mathcal{F}$.

Proof.

($\Rightarrow$) Induct on $n = \dim \mathcal{V}$. If every $T \in \mathcal{F}$ is scalar, any basis works. Otherwise choose nonscalar $S \in \mathcal{F}$.
By Theorem 5.4.2, $\mathcal{V} = \bigoplus_{\lambda \in \mathrm{spec}(S)} E_\lambda(S)$, with at least two nonzero summands of dimension $< n$.
If $T \in \mathcal{F}$ and $\mathbf{x} \in E_\lambda(S)$, then $S(T\mathbf{x}) = T(S\mathbf{x}) = \lambda T\mathbf{x}$, so $E_\lambda(S) \in \mathrm{Inv}(T)$.
By Lemma 5.4.3, each $T|_{E_\lambda(S)}$ is diagonalizable; the restricted family still commutes.
Induction gives a common eigenbasis of each $E_\lambda(S)$; concatenate over λ.

($\Leftarrow$) For $S, T \in \mathcal{F}$, set $D_S = [S]_{\beta \leftarrow \beta}$ and $D_T = [T]_{\beta \leftarrow \beta}$.
Since D_S, D_T are diagonal, $[ST]_{\beta \leftarrow \beta} = D_S D_T = D_T D_S = [TS]_{\beta \leftarrow \beta}$. Thus $ST = TS$.

Structural Commentary: Commutation preserves eigenspaces.
Key Insight: Diagonalize one operator, then restrict. □

Corollary 5.4.5. *Enough Eigenvalues Imply Diagonalizability*
If $\mathcal{V}$ is finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$ has $\dim(\mathcal{V})$ distinct eigenvalues, then T is diagonalizable.

Proof. Let $n = \dim(\mathcal{V})$, and let $\lambda_1, \ldots, \lambda_n$ be distinct eigenvalues of T. For each j, choose an eigenvector $\mathbf{v}_j$ corresponding to λ_j. By Theorem 5.1.6, $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is linearly independent. Since this list has length n, Proposition 1.4.3 makes it a basis of $\mathcal{V}$. Thus $\mathcal{V}$ has an eigenbasis, so T is diagonalizable by Theorem 5.3.17.
Structural Commentary: One eigenvector from each distinct eigenvalue already gives a full-size independent list.
Key Insight: Once the independent eigenvectors reach $\dim(\mathcal{V})$, they must form a basis. □

Theorem 5.4.6. *Commutant of a Diagonalizable Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonalizable. Suppose its distinct eigenvalues are $\mu_1, \ldots, \mu_d$, and set $n_i = \dim E_{\mu_i}(A)$. Write $A = S(\mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d})S^{-1}$. Then $\mathrm{Comm}(A) = \{S(X_1 \oplus \cdots \oplus X_d)S^{-1} : X_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})\}$. In particular, $\dim \mathrm{Comm}(A) = n_1^2 + \cdots + n_d^2$.

Proof. Let $\Lambda = \mu_1 I_{n_1} \oplus \cdots \oplus \mu_d I_{n_d}$. Matrix B commutes with A if and only if $S^{-1}BS$ commutes with Λ. By Lemma 5.3.3, this happens exactly when $S^{-1}BS = X_1 \oplus \cdots \oplus X_d$ with $X_i \in \mathrm{Mat}(n_i, n_i, \mathbb{C})$. Dimension count follows because conjugation by S is a vector-space isomorphism and $\dim \mathrm{Mat}(n_i, n_i, \mathbb{C}) = n_i^2$.
Structural Commentary: Diagonalizing A reduces commutation to respecting equal-eigenvalue blocks.
Key Insight: Distinct eigenvalue blocks cannot mix under a commuting matrix. □

Theorem 5.4.7. *Simultaneous Schur Triangularisation*
Let $\emptyset \neq \mathcal{F} \subseteq \mathrm{Mat}(n, n, \mathbb{C})$ be a commuting family. Then there is a unitary $U \in \mathrm{Mat}(n, n, \mathbb{C})$ such that U^*AU is upper triangular for every $A \in \mathcal{F}$. If every matrix in $\mathcal{F}$ is real and has only real eigenvalues, then U can be chosen real orthogonal.

Proof. Induct on n. The case $n = 1$ is immediate.
For $n > 1$, Theorem 5.3.5 gives a common eigenvector; normalize it to a unit vector $\mathbf{x}$ and extend to an orthonormal basis. Let $V = [\mathbf{x}\ V_2]$ be the corresponding unitary matrix.
For every $A \in \mathcal{F}$, since $A\mathbf{x} = \lambda_A \mathbf{x}$, we have $V^*AV = \begin{bmatrix} \lambda_A & * \\ 0 & A' \end{bmatrix}$.
If $A, B \in \mathcal{F}$, then $AB = BA$. The lower-right block of $V^*ABV = (V^*AV)(V^*BV)$ is $A'B'$, while the lower-right block of V^*BAV is $B'A'$. Hence $A'B' = B'A'$, so the compressed family is still commuting.
By induction, choose a unitary W that simultaneously upper triangularises all compressed blocks.

Then $U = V(1 \oplus W)$ is unitary and U^*AU is upper triangular for every $A \in \mathcal{F}$.
For the real case, run the same induction over $\mathbb{R}$. A common real eigenvector exists as follows. If all matrices are scalar, any real unit vector works. Otherwise choose a nonscalar $A_0 \in \mathcal{F}$.
Since A_0 has only real eigenvalues, some real eigenspace $\mathcal{U} = E_\lambda(A_0)$ is nonzero; since A_0 is not scalar, $\mathcal{U} \neq \mathbb{R}^n$.
Commutation gives $(A_0 - \lambda I)B\mathbf{u} = B(A_0 - \lambda I)\mathbf{u} = \mathbf{0}$ for $B \in \mathcal{F}$ and $\mathbf{u} \in \mathcal{U}$, so $\mathcal{U}$ is invariant under every B.
The restricted family on $\mathcal{U}$ is commuting and still has only real eigenvalues: in a basis adapted to $\mathcal{U}$ each B has block form $\begin{bmatrix} B|_{\mathcal{U}} & * \\ 0 & * \end{bmatrix}$, so the characteristic polynomial of $B|_{\mathcal{U}}$ divides that of B. Induction on $\dim(\mathcal{U})$ gives a common real eigenvector. Extending by real orthonormal bases gives a real orthogonal U.
Structural Commentary: A common eigenline creates the first block; commutation descends to the compression.
Key Insight: Simultaneous triangularisation is Schur induction with a common eigenvector at every step. □

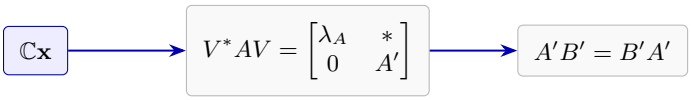


Figure 44: A shared eigenline gives a simultaneous first block; lower-right compressions still commute.

Corollary 5.4.8. *Eigenvalues of Sums and Products of Commuting Matrices*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ and $AB = BA$. Their eigenvalues can be ordered as $\lambda_1, \ldots, \lambda_n$ and $\mu_1, \ldots, \mu_n$ so that $A + B$ has eigenvalues $\lambda_i + \mu_i$ and AB has eigenvalues $\lambda_i\mu_i$, counted with multiplicity.

Proof. Choose simultaneous Schur forms $T = U^*AU$ and $R = U^*BU$. Set $\lambda_i = t_{ii}$ and $\mu_i = r_{ii}$. Then $T + R$ and TR are upper triangular with diagonals $\lambda_i + \mu_i$ and $\lambda_i\mu_i$, and are similar to $A + B$ and AB.
Structural Commentary: Simultaneous Schur form aligns both spectra on one diagonal.
Key Insight: In common triangular coordinates, diagonals add and multiply. □

6 Polynomial Structure and Canonical Forms

6.1 Operator Polynomials

Polynomial structure records the equations an operator satisfies and turns them into invariant pieces through characteristic and minimal polynomials, Cayley–Hamilton, generalized eigenspaces, and nilpotent parts.

Definition 6.1.1. *Operator Powers and Polynomial Evaluation*
Let $T \in \mathcal{L}(\mathcal{V})$.

(i) The *powers* of T are $T^m := \underbrace{T \cdots T}_{m \text{ times}}$ for $m \in \mathbb{N}$, with $T^0 := I_{\mathcal{V}}$. If T is invertible, set $T^{-m} := (T^{-1})^m$.

(ii) For $p(z) = \sum_{j=0}^{m} a_j z^j \in \mathbb{F}[z]$, the *polynomial p applied to T* is $p(T) := \sum_{j=0}^{m} a_j T^j$.

(iii) $p \in \mathbb{F}[z]$ is an *annihilating polynomial* for T if $p(T) = 0_{\mathcal{L}(\mathcal{V})}$; equivalently, p *annihilates* T.

Definition 6.1.2. *Product of Polynomials*
If $p, q \in \mathbb{F}[z]$, their product $pq \in \mathbb{F}[z]$ is defined by $(pq)(z) = p(z)q(z)$ for $z \in \mathbb{F}$.

Proposition 6.1.3. *Multiplicative Properties of Operator Polynomials*
Let $p, q \in \mathbb{F}[z]$ and $T \in \mathcal{L}(\mathcal{V})$. Then

(i) $(pq)(T) = p(T)q(T)$;
(ii) $p(T)q(T) = q(T)p(T)$.

Proof.

(i) Write $p(z) = \sum_{j=0}^{m} a_j z^j$ and $q(z) = \sum_{k=0}^{n} b_k z^k$. Then $(pq)(z) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k z^{j+k}$.

Substituting T gives $(pq)(T) = \sum_{j=0}^{m} \sum_{k=0}^{n} a_j b_k T^{j+k} = \left(\sum_{j=0}^{m} a_j T^j \right) \left(\sum_{k=0}^{n} b_k T^k \right) = p(T)q(T)$.

(ii) Since polynomial multiplication is commutative, $pq = qp$.
By part (i), $p(T)q(T) = (pq)(T) = (qp)(T) = q(T)p(T)$.

Structural Commentary: Polynomial evaluation preserves products.
Key Insight: Expanding a polynomial product works the same with T as with z. □

Proposition 6.1.4. *Existence of Annihilating Polynomials*
Let $\mathcal{V}$ be finite-dimensional with $n = \dim(\mathcal{V}) > 0$, and let $T \in \mathcal{L}(\mathcal{V})$.
Then T has a nonconstant annihilating polynomial of degree at most n^2.

Proof. By Theorem 3.3.32, the vector space $\mathcal{L}(\mathcal{V})$ has dimension n^2, so the n^2+1 operators $I_{\mathcal{V}}, T, T^2, \dots, T^{n^2}$ are linearly dependent. Hence there are scalars $a_0, \dots, a_{n^2}$, not all zero, such that $\sum_{j=0}^{n^2} a_j T^j = 0$.
If $a_1 = \cdots = a_{n^2} = 0$, then $a_0 I_{\mathcal{V}} = 0$, forcing $a_0 = 0$ because $\mathcal{V} \neq \{\mathbf{0}\}$, a contradiction. Thus $p(z) = \sum_{j=0}^{n^2} a_j z^j$ is a nonconstant annihilating polynomial.
Structural Commentary: Finite-dimensionality makes the powers of one operator linearly dependent inside $\mathcal{L}(\mathcal{V})$.
Key Insight: A dependence among $I_{\mathcal{V}}, T, \dots, T^{n^2}$ is exactly a polynomial equation satisfied by T. □

Proposition 6.1.5. *Polynomial Evaluation on Eigenvectors*
Let $T \in \mathcal{L}(\mathcal{V})$, let $p \in \mathbb{F}[z]$, and suppose $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $p(T)\mathbf{v} = p(\lambda)\mathbf{v}$.

Proof. Induct on j to get $T^j\mathbf{v} = \lambda^j\mathbf{v}$ for every $j \geq 0$.
If $p(z) = \sum_{j=0}^{m} a_j z^j$, then $p(T)\mathbf{v} = \sum_{j=0}^{m} a_j T^j\mathbf{v} = \sum_{j=0}^{m} a_j \lambda^j\mathbf{v} = p(\lambda)\mathbf{v}$.
Structural Commentary: On a one-dimensional eigenspace, T acts like the scalar λ.
Key Insight: Polynomial expressions in T reduce to the same polynomial evaluated at λ. □

6.2 Characteristic and Minimal Polynomials

Motivation 6.2.1. *Two Spectral Polynomials*
The characteristic polynomial records the whole primary decomposition with dimensions.
The minimal polynomial records the shortest polynomial relation satisfied by the operator.

Definition 6.2.2. *Characteristic Polynomial*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, the *characteristic polynomial* of A is $p_A(z) := \det(zI_n - A)$.

Theorem 6.2.3. *Basic Form of the Characteristic Polynomial*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$. Then p_A is monic of degree n and $p_A(z) = z^n - (\operatorname{tr} A)z^{n-1} + \cdots + (-1)^n \det A$.
Moreover, for $\lambda \in \mathbb{F}$, $\lambda \in \operatorname{spec}(A)$ if and only if $p_A(\lambda) = 0$.

Proof. The only determinant term of degree greater than $n-2$ comes from the identity permutation:

$$(z - a_{11})(z - a_{22}) \cdots (z - a_{nn}) = z^n - (a_{11} + \cdots + a_{nn})z^{n-1} + \cdots .$$

Thus p_A is monic and the coefficient of z^{n-1} is $-\operatorname{tr} A$; also $p_A(0) = \det(-A) = (-1)^n \det A$. Finally, $p_A(\lambda) = 0$ if and only if $\lambda I_n - A$ is singular, equivalently $\ker(A - \lambda I_n) \neq \{\mathbf{0}\}$.
Structural Commentary: The diagonal product determines the top two coefficients.
Key Insight: Eigenvalues are precisely the roots of $\det(zI_n - A)$. □

Definition 6.2.4. *Algebraic Multiplicity*
For $\lambda \in \mathbb{F}$, let $m_A(\lambda)$ be the multiplicity of λ as a root of p_A, with $m_A(\lambda) = 0$ if $\lambda \notin \operatorname{spec}(A)$.
For $\lambda \in \operatorname{spec}(A)$, this is the *algebraic multiplicity* of λ. Unqualified multiplicity means algebraic multiplicity.

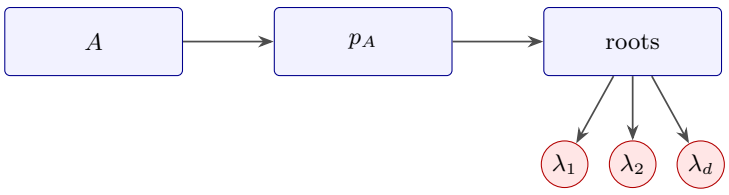


Figure 45: Form $p_A(z) = \det(zI_n - A)$ and factor it; each root λ is counted by $m_A(\lambda)$.

Method 6.2.5. *Characteristic Equation and Eigenspace Computation*
To compute eigenvalues and eigenspaces of $A \in \mathrm{Mat}(n, n, \mathbb{F})$:

(i) Form the characteristic polynomial $p_A(z) = \det(zI_n - A)$ from Definition 6.2.2.
(ii) Solve the characteristic equation $p_A(\lambda) = 0$. By Theorem 6.2.3, the solutions are exactly the eigenvalues in $\mathbb{F}$.
(iii) For each eigenvalue λ, row-reduce $A - \lambda I_n$ and compute

$$E_\lambda(A) = \ker(A - \lambda I_n).$$

(iv) Choose a basis of each eigenspace. The dimension of this basis is the geometric multiplicity of λ.

Algebraic multiplicity comes from the factorisation of p_A, while geometric multiplicity comes from the nullspace computation.

Proposition 6.2.6. *Upper-Triangular Eigenvalues*
If $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ *is upper triangular, then* $p_A(z) = \prod_{j=1}^{n} (z - a_{jj})$.
Thus $\mathrm{spec}(A) = \{a_{jj} : 1 \le j \le n\}$, *and for* $\lambda \in \mathrm{spec}(A)$, $m_A(\lambda) = \#\{j : a_{jj} = \lambda\}$.

Proof. Since $zI_n - A$ is upper triangular with diagonal entries $z - a_{jj}$, $p_A(z) = \det(zI_n - A) = \prod_{j=1}^{n} (z - a_{jj})$.
Thus the roots of p_A are exactly the diagonal entries, and the exponent of $(z - \lambda)$ is $\#\{j : a_{jj} = \lambda\}$.
Structural Commentary: Triangular form removes all off-diagonal determinant contributions.
Key Insight: Once a matrix is triangular, the spectrum is visible on the diagonal. □

Theorem 6.2.7. *Similarity Preserves Characteristic Data*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ *and suppose* $B = SAS^{-1}$ *for some invertible* S. *Then* $p_B = p_A$. *In particular, for each* $\lambda \in \mathrm{spec}(A)$, $m_B(\lambda) = m_A(\lambda)$ *and* $\dim E_\lambda(B) = \dim E_\lambda(A)$.

Proof. For every z, $p_B(z) = \det(zI_n - SAS^{-1}) = \det(S(zI_n - A)S^{-1}) = \det(S)\det(zI_n - A)\det(S)^{-1} = p_A(z)$.
Thus roots and root multiplicities agree, so $m_B(\lambda) = m_A(\lambda)$. Also $B - \lambda I_n = S(A - \lambda I_n)S^{-1}$.
Hence $\mathbf{x} \in \ker(A - \lambda I_n)$ if and only if $S\mathbf{x} \in \ker(B - \lambda I_n)$, S restricts to isomorphism between two eigenspaces.
Structural Commentary: Similarity changes coordinates, not the shifted determinant or shifted kernel dimension.
Key Insight: Conjugation inserts S and S^{-1}, which cancel in determinants and identify kernels. □

Proposition 6.2.8. *Block Upper-Triangular Characteristic Polynomial*
Let $A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$, B *and* D *are square. Then* $p_A(z) = p_B(z)p_D(z)$, *so* $m_A(\lambda) = m_B(\lambda) + m_D(\lambda)$ $\forall \lambda \in \mathbb{F}$.

Proof. The shifted matrix is block upper triangular, $zI - A = \begin{bmatrix} zI - B & -C \\ 0 & zI - D \end{bmatrix}$.
Theorem 1.1.11 gives $p_A(z) = \det(zI - B)\det(zI - D) = p_B(z)p_D(z)$.
Roots of a product are the roots of the factors, with multiplicity added.
Structural Commentary: The off-diagonal block couples vectors but not determinant factors.
Key Insight: In block upper-triangular form, spectral data lives on the diagonal blocks. □

Theorem 6.2.9. *Geometric Multiplicity Is at Most Algebraic Multiplicity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ *and let* $\lambda \in \mathrm{spec}(A)$. *Then* $\dim E_\lambda(A) \le m_A(\lambda)$.

Proof. Let $k = \dim E_\lambda(A)$ and choose a basis $\mathbf{x}_1, \dots, \mathbf{x}_k$ of $E_\lambda(A)$. Extend it to a basis of $\mathbb{F}^n$, and let S be the matrix whose columns are this basis. In the new basis,

$$S^{-1}AS = \begin{bmatrix} \lambda I_k & * \\ 0 & C \end{bmatrix}.$$

By Theorem 6.2.7, Proposition 6.2.8, $p_A(z) = p_{S^{-1}AS}(z) = (z - \lambda)^k p_C(z)$. Thus $m_A(\lambda) \ge k = \dim E_\lambda(A)$.
Structural Commentary: An eigenspace-first basis exposes a λI_k block.
Key Insight: Each independent eigenvector forces one visible factor $z - \lambda$ in the characteristic polynomial. □

Theorem 6.2.10. *Diagonalizability and Multiplicity*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* A *is diagonalizable if and only if* $\dim E_\lambda(A) = m_A(\lambda)$ *for every* $\lambda \in \mathrm{spec}(A)$.

Proof.
(⇒) If A is diagonalizable, then A is similar to a diagonal matrix D. Theorem 6.2.7 gives $m_A(\lambda) = m_D(\lambda)$ and $\dim E_\lambda(A) = \dim E_\lambda(D)$. For diagonal D, both numbers count diagonal entries equal to λ.
(⇐) Assume $\dim E_\lambda(A) = m_A(\lambda)$ for every eigenvalue.

If $\mu_1, \dots, \mu_d$ are distinct eigenvalues, then $\sum_{i=1}^{d} m_A(\mu_i) = n$. Hence $\dim E_{\mu_1}(A) + \cdots + \dim E_{\mu_d}(A) = n$.

Theorem 5.3.17 gives an eigenbasis, so A is diagonalizable.

Structural Commentary: Diagonalizability means root counts match eigenspace dimensions.
Key Insight: No missing eigendirections means an eigenbasis. □

Definition 6.2.11. *Characteristic Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $T \in \mathcal{L}(\mathcal{V})$. If $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T and $d_j = \dim \mathcal{G}_{\lambda_j}(T)$, then the *characteristic polynomial* of T is $q_T(z) := \prod_{j=1}^{m} (z - \lambda_j)^{d_j}$.

Proposition 6.2.12. *Degree and Roots of q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be the characteristic polynomial of T.

(i) $\deg q_T = \dim(\mathcal{V})$.
(ii) The roots of q_T are precisely the eigenvalues of T.

Proof. Write $q_T(z) = \prod_{j=1}^{m} (z - \lambda_j)^{d_j}$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues and $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.

(i) Proposition 6.5.4 gives $\deg q_T = \sum_j d_j = \dim(\mathcal{V})$.
(ii) The displayed factorization has exactly the roots $\lambda_1, \ldots, \lambda_m$.

Structural Commentary: Each primary summand contributes one root factor.
Key Insight: Multiplicity is dimension. □

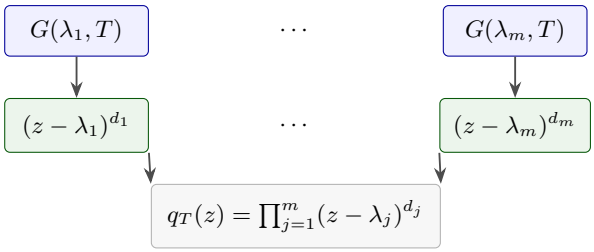


Figure 46: Each primary summand contributes $(z - \lambda_j)^{d_j}$, where $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.

Definition 6.2.13. *Monic Polynomial*
A nonzero polynomial is *monic* if its highest-degree coefficient is 1.

Theorem 6.2.14. *Existence of the Minimal Polynomial*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then T has a unique least-degree monic annihilator μ_T.

Proof. Proposition 6.1.4 gives a nonzero annihilator; scaling makes it monic, so a least-degree monic annihilator exists. If p and q are two of them, then $(p - q)(T) = 0$ and $\deg(p - q) < \deg p$.
If $p \neq q$, scaling $p - q$ gives a smaller monic annihilator, impossible. Hence $p = q$.
Structural Commentary: Existence comes from finite-dimensional dependence; uniqueness from minimality.
Key Insight: Two least monic annihilators cannot differ. □

Definition 6.2.15. *Minimal Polynomial of an Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. The *minimal polynomial* of T, denoted μ_T, is the unique monic polynomial of smallest degree such that $\mu_T(T) = 0$.

Definition 6.2.16. *Minimal Polynomial of a Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$. The *minimal polynomial* of A, denoted μ_A, is the unique monic polynomial of smallest positive degree such that $\mu_A(A) = 0$.

6.3 Cayley-Hamilton and Annihilators

Theorem 6.3.1. *Cayley–Hamilton Theorem for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, q_T be characteristic polynomial of $T \in \mathcal{L}(\mathcal{V})$. Then $q_T(T) = 0$.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues and set $d_j = \dim \mathcal{G}_{\lambda_j}(T)$.
It suffices, by primary decomposition, to show $q_T(T)$ kills each $\mathcal{G}_{\lambda_j}(T)$.
On $\mathcal{G}_{\lambda_j}(T)$, the operator $T - \lambda_j I_{\mathcal{V}}$ is nilpotent, so Corollary 6.4.11 gives $(T - \lambda_j I_{\mathcal{V}})^{d_j} = 0$ on $\mathcal{G}_{\lambda_j}(T)$.
Since the factors of $q_T(T)$ commute, this factor kills the whole product on $\mathcal{G}_{\lambda_j}(T)$. Hence $q_T(T) = 0$.
Structural Commentary: Check the polynomial on each primary summand.
Key Insight: One factor of $q_T(T)$ kills each summand. □

Theorem 6.3.2. *Cayley–Hamilton Theorem*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have characteristic polynomial $p_A(z) = \det(zI_n - A)$. Then $p_A(A) = 0$.

Proof. Choose a Schur form $A = UTU^*$ with diagonal $\lambda_1, \dots, \lambda_n$.
Since $p_A(A) = Up_A(T)U^*$ and $p_A(z) = \prod_{i=1}^{n}(z-\lambda_i)$, it suffices to prove $(T-\lambda_1 I_n)(T-\lambda_2 I_n)\cdots(T-\lambda_n I_n) = 0$.
Let $Q_j := \prod_{i=1}^{j}(T-\lambda_i I_n)$. We show that the first j columns of Q_j are zero. For $j = 1$, the first column of $T-\lambda_1 I_n$ is zero as T is upper triangular with first diagonal entry λ_1. Assume the first j columns of Q_j are zero.
Since $T-\lambda_{j+1} I_n$ is upper triangular and its $(j+1, j+1)$ entry is zero, its first $j+1$ columns lie in $\operatorname{span}(\mathbf{e}_1, \dots, \mathbf{e}_j)$.
By the induction hypothesis, Q_j kills this span, so $Q_{j+1} = Q_j(T - \lambda_{j+1} I_n)$ has first $j+1$ zero columns.
At $j = n$, all columns of Q_n are zero, so $Q_n = 0$.
Structural Commentary: Schur form reduces the theorem to a triangular column-killing induction.
Key Insight: Each factor $T - \lambda_i I_n$ kills the next leading column. □

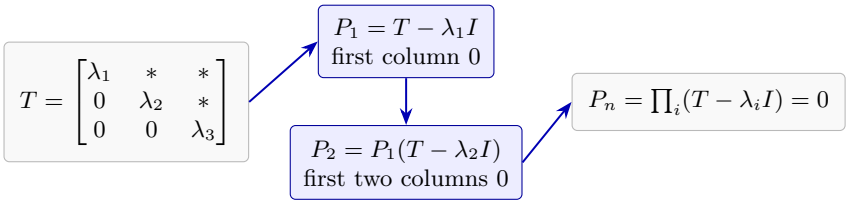


Figure 47: In Schur form, the factors $T - \lambda_i I$ kill the leading columns one layer at a time.

Theorem 6.3.3. *Basic Properties of the Minimal Polynomial*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$, and let $\lambda_1, \dots, \lambda_d$ be the distinct eigenvalues of A.

(i) μ_A exists, is unique, and $\deg \mu_A \le n$.
(ii) If a nonzero polynomial f annihilates A, then μ_A divides f. In particular, μ_A divides p_A.
(iii) There are integers q_i satisfying $1 \le q_i \le m_A(\lambda_i)$ such that $\mu_A(z) = \prod_{i=1}^{d}(z-\lambda_i)^{q_i}$.

Proof.

(i) Cayley–Hamilton supplies the monic annihilator p_A, so a least-degree monic annihilator μ_A exists and has degree at most n. If g is another least-degree monic annihilator, write $g = h\mu_A + r$, where $r = 0$ or $\deg r < \deg \mu_A$. Evaluating at A gives $r(A) = 0$. If $r \neq 0$, scaling r to be monic would give a smaller-degree monic annihilator, a contradiction. Hence $r = 0$, so $g = h\mu_A$.
Since g and μ_A are monic of the same degree, $h = 1$ and $g = \mu_A$.

(ii) If $f(A) = 0$, divide $f = q\mu_A + r$ with $\deg r < \deg \mu_A$. Evaluating at A gives $r(A) = 0$, hence $r = 0$ and $\mu_A \mid f$. Taking $f = p_A$ gives $\mu_A \mid p_A$.

(iii) Since $\mu_A \mid p_A = \prod_{i=1}^{d}(z-\lambda_i)^{m_A(\lambda_i)}$, its roots are eigenvalues with exponents at most the algebraic multiplicities. If $A\mathbf{v} = \lambda_i \mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $\mu_A(\lambda_i)\mathbf{v} = \mu_A(A)\mathbf{v} = 0$, so every λ_i occurs.

Structural Commentary: Polynomial division makes μ_A the least monic generator of annihilators.
Key Insight: Cayley–Hamilton gives existence; eigenvectors force the roots. □

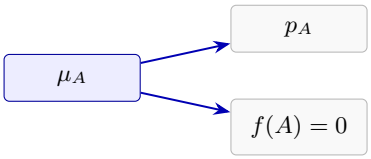


Figure 48: μ_A is the least monic annihilator; arrows indicate that μ_A divides p_A and every annihilator f.

Theorem 6.3.4. *Minimal Polynomial of a Spectrally Disjoint Direct Sum*
Let $A = A_1 \oplus \cdots \oplus A_d$, where the spectra of $A_1, \dots, A_d$ are pairwise disjoint. Then $\mu_A = \mu_{A_1}\mu_{A_2}\cdots\mu_{A_d}$.

Proof. Let $f = \prod_i \mu_{A_i}$. Since $f(A_i) = 0$ for all i, blockwise evaluation gives $f(A) = 0$, and therefore $\mu_A \mid f$.
Conversely, $\mu_A(A_i) = 0$ for each i, so $\mu_{A_i} \mid \mu_A$ for each i. The roots of μ_{A_i} are exactly $\operatorname{spec}(A_i)$ by Theorem 6.3.12, so the disjointness of the spectra makes the polynomials μ_{A_i} pairwise relatively prime.
Hence their product f divides μ_A. Since $\mu_A \mid f$ and $f \mid \mu_A$, and both are monic, $\mu_A = f$.
Structural Commentary: Direct sums are annihilated blockwise; disjoint spectra make the block factors coprime.
Key Insight: Each spectral component contributes its own minimal-polynomial factor. □

Theorem 6.3.5. *Similarity Preserves the Minimal Polynomial*
If $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ are similar, then $p_A = p_B$ and $\mu_A = \mu_B$.

Proof. The identity $p_A = p_B$ is Theorem 6.2.7. Let $A = SBS^{-1}$, then $f(A) = Sf(B)S^{-1}$ for every polynomial f, so A and B have the same annihilating polynomials. Hence $\mu_A = \mu_B$.
Structural Commentary: Similarity preserves polynomial equations.
Key Insight: Conjugation commutes with polynomial evaluation. □

Theorem 6.3.6. *Diagonalizability and the Minimal Polynomial*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have distinct eigenvalues $\lambda_1, \ldots, \lambda_d$.
Then A is diagonalizable if and only if $\mu_A(z) = (z - \lambda_1) \cdots (z - \lambda_d)$.
Equivalently, A is diagonalizable if and only if μ_A is square-free.

Proof. Set $p(z) := \prod_{i=1}^{d} (z - \lambda_i)$.
($\Rightarrow$) If $A = S\Lambda S^{-1}$ is diagonalizable, then $p(\Lambda) = 0$, so $p(A) = 0$.
Since μ_A contains every factor $z - \lambda_i$ and divides p, we have $\mu_A = p$.
($\Leftarrow$) Suppose $\mu_A = p$. On the primary summand $\mathcal{G}_{\lambda_i}(A)$, the restriction of $A - \lambda_j I_n$ for $j \neq i$ is $(A - \lambda_i I_n) + (\lambda_i - \lambda_j)I$, a nonzero scalar plus a nilpotent operator. By Lemma 6.4.12, these restricted factors are invertible. Since $0 = p(A)|_{\mathcal{G}_{\lambda_i}(A)} = \left(\prod_{j \neq i} (A - \lambda_j I_n) \right) |_{\mathcal{G}_{\lambda_i}(A)} (A - \lambda_i I_n)|_{\mathcal{G}_{\lambda_i}(A)}$, the invertible product of the $j \neq i$ factors can be cancelled, giving $(A - \lambda_i I_n) = 0$ on $\mathcal{G}_{\lambda_i}(A)$.
Thus every primary summand is an eigenspace, so A is diagonalizable.
Structural Commentary: Repeated factors of μ_A record nilpotent parts in primary blocks.
Key Insight: Square-free μ_A kills each primary block in one step. □

Corollary 6.3.7. *Square-Free Annihilator Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. If a polynomial f with no repeated roots satisfies $f(A) = 0$, then A is diagonalizable.

Proof. Since $\mu_A \mid f$, the minimal polynomial is square-free; Theorem 6.3.6 applies.
Structural Commentary: A square-free annihilator forces square-free μ_A.
Key Insight: Square-free μ_A means diagonalizable. □

Method 6.3.8. *Computing μ_T from a Matrix*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be the matrix of T in some basis.
(i) For $m = 1, 2, \ldots$, test the monic candidate $p_m(z) = a_0 + a_1 z + \cdots + a_{m-1} z^{m-1} + z^m$.
(ii) Solve linear system by comparing matrix entries in $a_0 I + a_1 A + \cdots + a_{m-1} A^{m-1} = -A^m$ for $a_0, \ldots, a_{m-1}$.
(iii) If no solution exists, increase m. The first solvable candidate is μ_T.

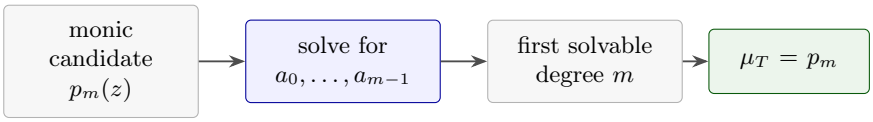


Figure 49: Solve $a_0 I + \cdots + a_{m-1} A^{m-1} = -A^m$; the first solvable monic candidate is μ_T.

Method 6.3.9. *Testing a Minimal Polynomial from p_A*
Suppose $p_A(z) = \prod_i (z - \lambda_i)^{a_i}$ is known.
i. Write each candidate as $m_{\mathbf{q}}(z) = \prod_i (z - \lambda_i)^{q_i}$, where $1 \leq q_i \leq a_i$.
ii. Test exponent vectors $\mathbf{q} = (q_i)$ in increasing total degree $\sum_i q_i$.
iii. For each $\mathbf{q}$, compute $m_{\mathbf{q}}(A) = \prod_i (A - \lambda_i I)^{q_i}$.
iv. The first candidate with $m_{\mathbf{q}}(A) = 0$ is μ_A, by Theorem 6.3.3.

Theorem 6.3.10. *Annihilators Are Multiples of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim, $T \in \mathcal{L}(\mathcal{V})$, and $f \in \mathbb{F}[z]$ be nonzero. Then $f(T) = 0$ if and only if μ_T divides f.

Proof.
($\Rightarrow$) Suppose $f(T) = 0$. Divide $f = s\mu_T + r$, where $r = 0$ or $\deg r < \deg \mu_T$.
Then $r(T) = 0$, so minimality forces $r = 0$. Hence μ_T divides f.
($\Leftarrow$) If $f = \mu_T s$, then $f(T) = \mu_T(T)s(T) = 0$.
Structural Commentary: Polynomial division reduces every annihilator to its remainder modulo μ_T.
Key Insight: Minimality forces that remainder to be zero. □

Corollary 6.3.11. *q_T Is a Multiple of μ_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then the characteristic polynomial q_T is a polynomial multiple of the minimal polynomial μ_T.

Proof. Theorem 6.3.1 gives $q_T(T) = 0$. Theorem 6.3.10 then implies that μ_T divides q_T.
Structural Commentary: Cayley–Hamilton turns the characteristic polynomial into an annihilator.
Key Insight: The minimal polynomial divides every annihilator, hence divides q_T. □

Theorem 6.3.12. *Roots of μ_T Are Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$. Then the roots of μ_T are precisely the eigenvalues of T.

Proof.

($\Rightarrow$) If $\mu_T(\lambda) = 0$, write $\mu_T(z) = (z - \lambda)g(z)$. Since $\deg g < \deg \mu_T$, minimality gives $g(T) \neq 0$; choose $\mathbf{v}$ with $\mathbf{u} = g(T)\mathbf{v} \neq \mathbf{0}$. Then $(T - \lambda I_{\mathcal{V}})\mathbf{u} = \mu_T(T)\mathbf{v} = \mathbf{0}$.
($\Leftarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $0 = \mu_T(T)\mathbf{v} = \mu_T(\lambda)\mathbf{v}$, so $\mu_T(\lambda) = 0$.

Structural Commentary: A root splits off one shifted operator.
Key Insight: The quotient cannot already annihilate T. □

Corollary 6.3.13. *Minimal and Characteristic Polynomials Have the Same Roots*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$.
Then $\{\lambda \in \mathbb{C} : \mu_T(\lambda) = 0\} = \{\lambda \in \mathbb{C} : q_T(\lambda) = 0\} = \operatorname{spec}(T)$.

Proof. By Theorem 6.3.12, $\{\lambda \in \mathbb{C} : \mu_T(\lambda) = 0\} = \operatorname{spec}(T)$.
By Proposition 6.2.12, $\{\lambda \in \mathbb{C} : q_T(\lambda) = 0\} = \operatorname{spec}(T)$.
Structural Commentary: Different multiplicities, same spectral scalars.
Key Insight: Minimal polynomial roots and characteristic polynomial roots are both just eigenvalues. □

Perspective 6.3.14. *Characteristic vs. Minimal Polynomial*
The characteristic polynomial records every primary block with its dimension. The minimal polynomial records the least polynomial needed to annihilate all blocks at once.

Theorem 6.3.15. *Annihilating Polynomials Control the Spectrum*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let p be a nonconstant annihilating polynomial for T.

(i) Every eigenvalue of T is a root of p.
(ii) If $\mathbb{F} = \mathbb{C}$, then at least one root of p is an eigenvalue of T.

Proof.

(i) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then Proposition 6.1.5 gives $0 = p(T)\mathbf{v} = p(\lambda)\mathbf{v}$. Since $\mathbf{v} \neq \mathbf{0}$, $p(\lambda) = 0$.
(ii) Factor $p(z) = c(z - \lambda_1)\cdots(z - \lambda_m)$ over $\mathbb{C}$, with $c \neq 0$. Then $0 = p(T) = c(T - \lambda_1 I_{\mathcal{V}})\cdots(T - \lambda_m I_{\mathcal{V}})$. At least one factor $T - \lambda_j I_{\mathcal{V}}$ is not injective; otherwise product would be injective. Hence λ_j is an eigenvalue.

Structural Commentary: Annihilating polynomial restricts spectrum to its roots; over $\mathbb{C}$, one root must appear.
Key Insight: Eigenvectors turn $p(T)$ into $p(\lambda)$. □

Proposition 6.3.16. *Kernel and Range of a Polynomial in an Operator Are Invariant*
Let $T \in \mathcal{L}(\mathcal{V})$ and $p \in \mathbb{F}[z]$. Then $\ker(p(T))$ and $\operatorname{range}(p(T))$ are invariant under T.

Proof. If $\mathbf{v} \in \ker(p(T))$, then $p(T)\mathbf{v} = \mathbf{0}$.
Since $p(T)$ commutes with T by Proposition 6.1.3, $p(T)(T\mathbf{v}) = T(p(T)\mathbf{v}) = T\mathbf{0} = \mathbf{0}$.
If $\mathbf{v} \in \operatorname{range}(p(T))$, write $\mathbf{v} = p(T)\mathbf{u}$. Then $T\mathbf{v} = Tp(T)\mathbf{u} = p(T)(T\mathbf{u})$, so $T\mathbf{v} \in \operatorname{range}(p(T))$.
Structural Commentary: Polynomial kernels and ranges are stable under the original operator.
Key Insight: Since T commutes with $p(T)$, it preserves both $\ker p(T)$ and $\operatorname{range} p(T)$. □

Theorem 6.3.17. *Sylvester Linear Matrix Equation*
Let $A \in \operatorname{Mat}(m, m, \mathbb{C})$ and $B \in \operatorname{Mat}(n, n, \mathbb{C})$. If $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$, then for every $C \in \operatorname{Mat}(m, n, \mathbb{C})$ the equation $AX - XB = C$ has a unique solution $X \in \operatorname{Mat}(m, n, \mathbb{C})$.

Proof. Let $L(X) = AX - XB$. Since L has equal finite-dimensional domain and codomain, it suffices to show $\ker L = \{0\}$. If $AX = XB$, then $p(A)X = Xp(B)$ for every polynomial p.
With $p = p_B$, Cayley–Hamilton gives $p_B(A)X = Xp_B(B) = 0$. For every $\lambda \in \operatorname{spec}(A)$, disjointness gives $p_B(\lambda) \neq 0$; hence $p_B(A)$ is invertible by Corollary 7.3.22. Thus $X = 0$, so L is bijective.
Structural Commentary: Polynomial calculus pass through homogeneous solutions; disjoint spectra invert $p_B(A)$.
Key Insight: Spectral separation turns an intertwining relation into the zero relation. □

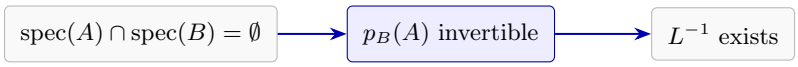


Figure 50: Disjoint spectra make $p_B(A)$ invertible, so the Sylvester map $L(X) = AX - XB$ is invertible.

Corollary 6.3.18. *Removing a Spectrally Separated Off-Diagonal Block*
Let $M = \begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$ *and* $\operatorname{spec}(A) \cap \operatorname{spec}(B) = \emptyset$. *Then* M *is similar to* $A \oplus B$.

Proof. By Theorem 6.3.17, choose X such that $AX - XB = -C$. Then

$$\begin{bmatrix} I & -X \\ 0 & I \end{bmatrix} \begin{bmatrix} A & C \\ 0 & B \end{bmatrix} \begin{bmatrix} I & X \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & AX + C - XB \\ 0 & B \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}.$$

Structural Commentary: A block shear cancels the off-diagonal block via Sylvester's equation.
Key Insight: Spectrally separated diagonal blocks cannot carry essential upper-right coupling. □

Theorem 6.3.19. *Schur Form Grouped by Eigenvalues*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ *have distinct eigenvalues* $\lambda_1, \ldots, \lambda_d$ *with algebraic multiplicities* $n_1, \ldots, n_d$.
Then A *is unitarily similar to a block upper-triangular matrix*

$$T = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1d} \\ 0 & T_{22} & \cdots & T_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & T_{dd} \end{bmatrix}, \qquad T_{ii} \in \operatorname{Mat}(n_i, n_i, \mathbb{C}),$$

where each T_{ii} *is upper triangular with all diagonal entries equal to* λ_i. *Moreover,* A *is similar to* $T_{11} \oplus \cdots \oplus T_{dd}$.

Proof. Order the Schur diagonal as $\lambda_1, \ldots, \lambda_1, \lambda_2, \ldots, \lambda_2, \ldots, \lambda_d, \ldots, \lambda_d$.
Theorem 7.1.10 gives the displayed block Schur form. Since $\operatorname{spec}(T_{11})$ is disjoint from the trailing block spectrum, Corollary 6.3.18 splits off T_{11}. Iterating splits off every diagonal block.
Structural Commentary: Grouped Schur form isolates spectral blocks; Sylvester shears split disjoint groups.
Key Insight: Cross-group upper blocks are coordinate artefacts. □

Corollary 6.3.20. *Commuting with Disjoint Blocks*
Let $A = A_{11} \oplus \cdots \oplus A_{dd}$, *where the spectra of* $A_{11}, \ldots, A_{dd}$ *are pairwise disjoint.*
Partition $B = [B_{ij}]$ *conformally with* A. *If* $AB = BA$, *then* $B_{ij} = 0$ *for all* $i \neq j$.

Proof. Equating (i, j) blocks in $AB = BA$ gives $A_{ii}B_{ij} - B_{ij}A_{jj} = 0$.
For $i \neq j$, the spectra of A_{ii} and A_{jj} are disjoint, so Theorem 6.3.17 with $C = 0$ gives $B_{ij} = 0$.
Structural Commentary: A commuting matrix cannot mix direct-sum blocks with disjoint spectra.
Key Insight: The off-diagonal block solves a homogeneous Sylvester equation, whose only solution is zero. □

Generalized eigenvectors repair the main defect of ordinary eigenvectors: an operator may not have enough eigendirections to form a basis. The replacement is to study vectors killed by powers of shifted operators.

6.4 Generalized Eigenspaces

Definition 6.4.1. *Generalized Eigenvectors and Generalized Eigenspaces*
Let $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$. Nonzero vector $\mathbf{v} \in \mathcal{V}$ is a *generalized eigenvector* of T to λ if $(T - \lambda I_{\mathcal{V}})^j \mathbf{v} = \mathbf{0}$ for some positive integer j. The *generalized eigenspace* corresponding to λ is

$$\mathcal{G}_\lambda(T) := \{\mathbf{0}\} \cup \{\mathbf{v} \neq \mathbf{0} : \mathbf{v} \text{ is a generalized eigenvector of } T \text{ corresponding to } \lambda\}.$$

Every eigenvector is a generalized eigenvector by taking $j = 1$. The point is that higher powers of $T - \lambda I_{\mathcal{V}}$ may reveal directions that are not eigenvectors but are eventually pushed into the eigenspace.

Proposition 6.4.2. *Kernels of Powers Increase*
Let $T \in \mathcal{L}(\mathcal{V})$. *Then* $\{\mathbf{0}\} = \ker(T^0) \subseteq \ker(T) \subseteq \ker(T^2) \subseteq \cdots \subseteq \ker(T^k) \subseteq \ker(T^{k+1}) \subseteq \cdots$.

Proof. If $\mathbf{v} \in \ker(T^k)$, then $T^k \mathbf{v} = \mathbf{0}$, hence $T^{k+1}\mathbf{v} = T(T^k \mathbf{v}) = \mathbf{0}$. Thus $\ker(T^k) \subseteq \ker(T^{k+1})$ for every $k \geq 0$.
Structural Commentary: Powers of T form an increasing chain of kernels.
Key Insight: Once a vector is killed by T^k, applying one more T cannot revive it. □

Proposition 6.4.3. *Equality Forces Stabilization*
Let $T \in \mathcal{L}(\mathcal{V})$ *and* $m \geq 0$. *If* $\ker(T^m) = \ker(T^{m+1})$, *then* $\ker(T^m) = \ker(T^{m+1}) = \ker(T^{m+2}) = \cdots$.

Proof. By Proposition 6.4.2, it suffices to prove $\ker(T^{m+k+1}) \subseteq \ker(T^{m+k})$ for every $k \geq 0$.
Take $\mathbf{v} \in \ker(T^{m+k+1})$. Then $T^{m+1}(T^k\mathbf{v}) = T^{m+k+1}\mathbf{v} = \mathbf{0}$, so $T^k\mathbf{v} \in \ker(T^{m+1}) = \ker(T^m)$.
Hence $T^{m+k}\mathbf{v} = T^m(T^k\mathbf{v}) = \mathbf{0}$, proving the reverse inclusion.
Structural Commentary: Stabilization at one step propagates through the whole tail.
Key Insight: A later kernel vector can be shifted down into the already-stable kernel. □

Theorem 6.4.4. *Kernels Stop Growing by $\dim(\mathcal{V})$*
If $\mathcal{V}$ is finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, and $n = \dim(\mathcal{V})$, then $\ker(T^n) = \ker(T^{n+1}) = \ker(T^{n+2}) = \cdots$.

Proof. If $\ker(T^n) \neq \ker(T^{n+1})$, Prop 6.4.3 implies all inclusions $\{\mathbf{0}\} = \ker(T^0) \subsetneq \ker(T) \subsetneq \cdots \subsetneq \ker(T^{n+1})$ are strict. Each strict inclusion raises dimension by at least 1, so $\dim\ker(T^{n+1}) \geq n+1$, impossible because $\ker(T^{n+1}) \subseteq \mathcal{V}$. Thus $\ker(T^n) = \ker(T^{n+1})$, and Proposition 6.4.3 gives the tail equality.
Structural Commentary: Strict kernel growth is dimension-limited.
Key Insight: Dimension bounds force the kernel chain to stabilize no later than step $\dim(\mathcal{V})$. □

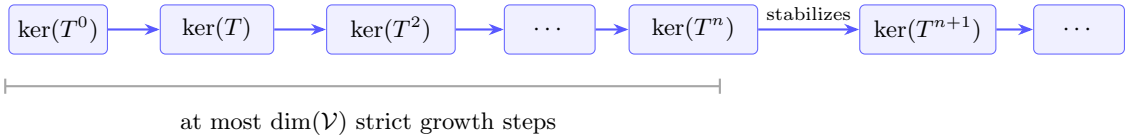


Figure 51: The kernel filtration grows until it stabilizes; after equality appears once, the tail is constant.

Theorem 6.4.5. *Ranks of Matrix Powers Stabilize*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$. Then:

(i) $\mathrm{rank}(A^k) \geq \mathrm{rank}(A^{k+1})$ *for* $k = 1, 2, \ldots$;
(ii) if $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+1})$, *then* $\mathrm{col}(A^k) = \mathrm{col}(A^{k+p})$ *and* $\mathrm{rank}(A^k) = \mathrm{rank}(A^{k+p})$ *for every* $p = 1, 2, \ldots$;
(iii) if A is not invertible, then there is a least positive integer $q \leq n$ such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Proof.

(i) Since $A^{k+1} = A^kA$, each column of A^{k+1} is a linear combination of the columns of A^k. Hence $\mathrm{col}(A^{k+1}) \subseteq \mathrm{col}(A^k)$, so $\mathrm{rank}(A^{k+1}) \leq \mathrm{rank}(A^k)$.
(ii) The inclusion $\mathrm{col}(A^{k+1}) \subseteq \mathrm{col}(A^k)$ and equality of dimensions give $\mathrm{col}(A^{k+1}) = \mathrm{col}(A^k)$. Let $U = \mathrm{col}(A^k)$. Then multiplication by A maps U onto U, so induction gives $\mathrm{col}(A^{k+p}) = U$ for every $p \geq 1$.
(iii) If A is not invertible, then Corollary 3.5.5 gives $\mathrm{rank}(A) \leq n-1$. If all inequalities $\mathrm{rank}(A^k) > \mathrm{rank}(A^{k+1})$ held for $k = 1, \ldots, n$, then $\mathrm{rank}(A^{n+1}) \leq \mathrm{rank}(A) - n < 0$, impossible. Hence equality occurs for some $q \leq n$; choose the least such q.

Structural Commentary: The column spaces of powers form a decreasing chain. In finite dimensions, a decreasing chain cannot keep losing dimension forever.
Key Insight: Once two consecutive ranks agree, the image has stabilized under multiplication by A. □

Definition 6.4.6. *Index of a Matrix*
If $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is invertible, define its *index* to be 0. If A is not invertible, its index is the least positive integer q such that $\mathrm{rank}(A^q) = \mathrm{rank}(A^{q+1})$.

Theorem 6.4.7. *Stable Kernel-Range Decomposition*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and set $n = \dim(\mathcal{V})$. Then $\mathcal{V} = \ker(T^n) \oplus \mathrm{range}(T^n)$.

Proof. First show the intersection is trivial. If $\mathbf{v} \in \ker(T^n) \cap \mathrm{range}(T^n)$, then $\mathbf{v} = T^n\mathbf{u}$ and $T^n\mathbf{v} = \mathbf{0}$. Hence $T^{2n}\mathbf{u} = \mathbf{0}$, so $\mathbf{u} \in \ker(T^{2n}) = \ker(T^n)$ by Theorem 6.4.4. Thus $\mathbf{v} = T^n\mathbf{u} = \mathbf{0}$.
By Rank-Nullity Theorem 3.2.16, $\dim\ker(T^n) + \dim\mathrm{range}(T^n) = \dim(\mathcal{V})$.
The zero intersection and the dimension count imply the sum is direct and equals $\mathcal{V}$.
Structural Commentary: At the stable power, killed directions and surviving image directions separate cleanly.
Key Insight: Stabilization turns the usual rank-nullity count into an actual direct-sum decomposition. □

Theorem 6.4.8. *Description of Generalized Eigenspaces*
Let $\mathcal{V}$ be finite-dimensional, $T \in \mathcal{L}(\mathcal{V})$, $\lambda \in \mathbb{F}$, and $n = \dim(\mathcal{V})$. Then $\mathcal{G}_\lambda(T) = \ker\big((T - \lambda I_\mathcal{V})^n\big)$.
In particular, $\mathcal{G}_\lambda(T)$ is a subspace of $\mathcal{V}$.

Proof. Set $S = T - \lambda I_\mathcal{V}$.

($\Rightarrow$) If $\mathbf{v} \in \mathcal{G}_\lambda(T)$, then $S^j\mathbf{v} = \mathbf{0}$ for some $j \geq 1$. If $j \leq n$, Proposition 6.4.2 gives $\mathbf{v} \in \ker(S^n)$; if $j > n$, Theorem 6.4.4 gives $\ker(S^j) = \ker(S^n)$.

($\Leftarrow$) If $\mathbf{v} \in \ker(S^n)$, then either $\mathbf{v} = \mathbf{0}$ or $\mathbf{v}$ is a generalized eigenvector of T corresponding to λ.

Thus $\mathcal{G}_\lambda(T) = \ker(S^n)$, which is a subspace.
Structural Commentary: Generalized eigenspaces are stable shifted kernels.
Key Insight: Finite dimension makes exponent $\dim(\mathcal{V})$ enough. □

Proposition 6.4.9. *Generalized Eigenspaces Are Invariant*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{G}_\lambda(T)$ is invariant under T for every $\lambda \in \mathbb{F}$.

Proof. By Theorem 6.4.8, $\mathcal{G}_\lambda(T) = \ker((T - \lambda I_\mathcal{V})^n)$, where $n = \dim(\mathcal{V})$.
This is the kernel of a polynomial in T, so it is invariant under T by Proposition 6.3.16.
Structural Commentary: A generalized eigenspace is cut out by a polynomial equation in T.
Key Insight: Polynomial kernels commute with the operator that defines them. □

Definition 6.4.10. *Nilpotent Operator*
An operator $N \in \mathcal{L}(\mathcal{V})$ is *nilpotent* if $N^m = 0$ for some positive integer m.

Corollary 6.4.11. *Nilpotent Power Bound*
If $\mathcal{V}$ is finite-dimensional, $N \in \mathcal{L}(\mathcal{V})$ is nilpotent, and $n = \dim(\mathcal{V})$, then $N^n = 0$.

Proof. Since N is nilpotent, $\ker(N^m) = \mathcal{V}$ for some m.
The kernel chain has stabilized by the nth step by Theorem 6.4.4, so $\ker(N^n) = \mathcal{V}$. Hence $N^n = 0$.
Structural Commentary: Nilpotence says some kernel in the chain is already the whole space.
Key Insight: The stabilization bound pulls that full-kernel conclusion back to exponent $\dim(\mathcal{V})$. □

Lemma 6.4.12. *Nonzero Scalar Plus Nilpotent Is Invertible*
Let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent and let $a \in \mathbb{F}$ with $a \neq 0$. Then $aI_\mathcal{V} + N$ is invertible.

Proof. Choose m such that $N^m = 0$. Then $(aI_\mathcal{V} + N)\left(a^{-1}\sum_{r=0}^{m-1}(-a^{-1}N)^r\right) = I_\mathcal{V} - (-a^{-1}N)^m = I_\mathcal{V}$, and the same computation in the reverse order gives a two-sided inverse.
Structural Commentary: A nilpotent perturbation has a finite geometric-series inverse.
Key Insight: Nilpotence truncates the Neumann series exactly. □

Theorem 6.4.13. *Generalized Eigenvectors for Distinct Eigenvalues Are Independent*
Let $\mathcal{V}$ be finite-dimensional and $T \in \mathcal{L}(\mathcal{V})$. Suppose $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of T, and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be corresponding generalized eigenvectors. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_m)$ is linearly independent.

Proof. Let $n = \dim(\mathcal{V})$ and suppose $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.
Fix $r \in \{1, \ldots, m\}$ and apply $Q_r(T) := \prod_{i \neq r}(T - \lambda_i I_\mathcal{V})^n$ to the relation.
For $i \neq r$, Theorem 6.4.8 gives $Q_r(T)\mathbf{v}_i = \mathbf{0}$, so $a_r Q_r(T)\mathbf{v}_r = \mathbf{0}$.
On $\mathcal{G}_{\lambda_r}(T)$, the operator $T - \lambda_i I_\mathcal{V}$ equals $(T - \lambda_r I_\mathcal{V}) + (\lambda_r - \lambda_i)I_\mathcal{V}$. The first term is nilpotent on $\mathcal{G}_{\lambda_r}(T)$ and $\lambda_r - \lambda_i \neq 0$, so Lemma 6.4.12 makes each factor invertible on $\mathcal{G}_{\lambda_r}(T)$. Thus $Q_r(T)\mathbf{v}_r \neq \mathbf{0}$, forcing $a_r = 0$. Since r was arbitrary, all coefficients vanish.
Structural Commentary: Polynomial filters isolate one generalized eigenspace at a time.
Key Insight: Distinct eigenvalues let polynomial filters kill the other components and preserve the target one. □

Theorem 6.4.14. *Nilpotent Operators and Strictly Upper-Triangular Matrices*
Let $\mathcal{V}$ be finite-dimensional and $N \in \mathcal{L}(\mathcal{V})$. Then N is nilpotent if and only if N has a strictly upper-triangular matrix in some basis.

Proof. Let $n = \dim(\mathcal{V})$. By Corollary 6.4.11, $\{\mathbf{0}\} \subseteq \ker(N) \subseteq \ker(N^2) \subseteq \cdots \subseteq \ker(N^n) = \mathcal{V}$.

($\Rightarrow$) Choose a basis of $\ker(N)$, extend it to a basis of $\ker(N^2)$, and continue until obtaining a basis of $\mathcal{V}$. If $\mathbf{v}$ first appears in $\ker(N^k)$, then $N\mathbf{v} \in \ker(N^{k-1})$, which is spanned by earlier basis vectors. Hence the matrix of N is strictly upper triangular.

($\Leftarrow$) If the matrix of N is $n \times n$ strictly upper triangular, then N sends each basis vector into the span of earlier basis vectors. After n applications every basis vector is sent to $\mathbf{0}$, so $N^n = 0$.

Structural Commentary: Kernel filtration gives adapted basis; triangular form records motion to earlier layers.
Key Insight: Nilpotence is strict upper-triangularity in a suitable basis. □

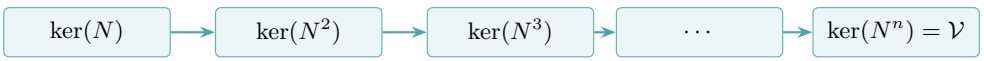


Figure 52: Extending bases through the kernel filtration makes N move vectors to earlier layers.

Perspective 6.4.15. *Eigenvalue Plus Nilpotent Part*
On $\mathcal{G}_\lambda(T)$, the operator T behaves as $\lambda I_\mathcal{V}$ plus the nilpotent operator $T - \lambda I_\mathcal{V}$. Generalized eigenvectors are therefore ordinary eigenvectors enriched by nilpotent chain data.

6.5 Primary Decomposition

Theorem 6.5.1. *Primary Decomposition over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. Then

(i) $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$;
(ii) each $(T - \lambda_j I_\mathcal{V})|_{\mathcal{G}_{\lambda_j}(T)}$ *is nilpotent.*

Proof. Let $n = \dim(\mathcal{V})$.

(i) Proceed by induction on n. If $n = 1$, then T has single eigenvalue λ_1 and $\mathcal{V} = \mathcal{G}_{\lambda_1}(T)$.
Assume $n > 1$. Let $S = T - \lambda_1 I_\mathcal{V}$, $\mathcal{U} := \operatorname{range}(S^n)$.
By Theorems 6.4.7 and 6.4.8, $\mathcal{V} = \ker(S^n) \oplus \operatorname{range}(S^n) = \mathcal{G}_{\lambda_1}(T) \oplus \mathcal{U}$. The subspace $\mathcal{U}$ is T-invariant by Proposition 6.3.16, and $\dim(\mathcal{U}) < n$ because $\mathcal{G}_{\lambda_1}(T) \neq \{\mathbf{0}\}$. If $\mathcal{U} = \{\mathbf{0}\}$, then (i) follows.
Otherwise distinct eigenvalues of $T|_\mathcal{U}$ are $\lambda_2, \ldots, \lambda_m$: every eigenvalue of $T|_\mathcal{U}$ is an eigenvalue of T but not λ_1, while an eigenvector $\mathbf{v}$ of T for λ_j, $j \geq 2$, satisfies $S^n\mathbf{v} = (\lambda_j - \lambda_1)^n\mathbf{v}$ and hence lies in $\mathcal{U}$.
Induction gives $\mathcal{U} = \mathcal{G}_{\lambda_2}(T|_\mathcal{U}) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T|_\mathcal{U})$. For $j \geq 2$, $\mathcal{G}_{\lambda_j}(T|_\mathcal{U}) = \mathcal{G}_{\lambda_j}(T)$.
The inclusion $\subseteq$ follows from the definition. Conversely, if $\mathbf{v} \in \mathcal{G}_{\lambda_j}(T)$, write $\mathbf{v} = \mathbf{g} + \mathbf{u}$ with $\mathbf{g} \in \mathcal{G}_{\lambda_1}(T)$ and $\mathbf{u} \in \mathcal{U}$. For some r, $(T - \lambda_j I_\mathcal{V})^r\mathbf{v} = \mathbf{0}$. Since both summands are T-invariant, the two terms $(T - \lambda_j I_\mathcal{V})^r\mathbf{g}$ and $(T - \lambda_j I_\mathcal{V})^r\mathbf{u}$ remain in the two direct summands and have sum zero, so both are zero. Thus $\mathbf{g} \in \mathcal{G}_{\lambda_1}(T) \cap \mathcal{G}_{\lambda_j}(T) = \{\mathbf{0}\}$ by Theorem 6.4.13, and $\mathbf{v} = \mathbf{u} \in \mathcal{G}_{\lambda_j}(T|_\mathcal{U})$. Hence (i) follows.

(ii) By Proposition 6.4.9, the restriction is an operator on $\mathcal{G}_{\lambda_j}(T)$. By Theorem 6.4.8, $(T - \lambda_j I_\mathcal{V})^n\mathbf{v} = \mathbf{0}\ \forall \mathbf{v} \in \mathcal{G}_{\lambda_j}(T)$. Hence nth power of $(T - \lambda_j I_\mathcal{V})|_{\mathcal{G}_{\lambda_j}(T)}$ is zero.

Structural Commentary: Peel off one generalized eigenspace, then induct on the invariant range.
Key Insight: Stable kernel-range splitting isolates one eigenvalue at a time. □

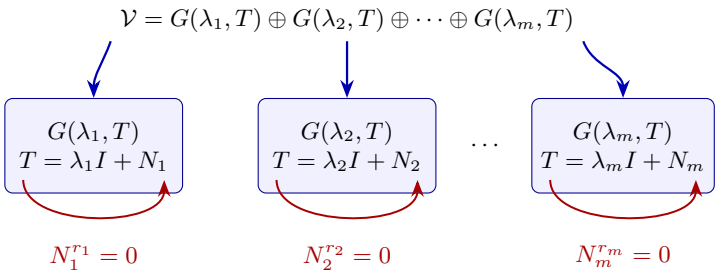


Figure 53: Primary decomposition: on $\mathcal{G}_{\lambda_j}(T)$, $T = \lambda_j I$ plus nilpotent.

Corollary 6.5.2. *Basis of Generalized Eigenvectors*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a generalized-eigenvector basis for T.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 6.5.1, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$. Choose a basis of each generalized eigenspace and concatenate the bases. Directness makes the concatenated list a basis of $\mathcal{V}$, and every nonzero vector in each list is a generalized eigenvector.
Structural Commentary: A direct sum basis is obtained by choosing bases in each summand.
Key Insight: Bases of the direct-sum pieces concatenate. □

Definition 6.5.3. *Multiplicity of an Eigenvalue*
Let $\mathcal{V}$ be finite-dimensional and let $T \in \mathcal{L}(\mathcal{V})$.
The *multiplicity* of an eigenvalue λ of T is $\dim \mathcal{G}_\lambda(T) = \dim \ker\big((T - \lambda I_\mathcal{V})^{\dim(\mathcal{V})}\big)$.
This is the algebraic multiplicity in generalized-eigenspace language.

Proposition 6.5.4. *Sum of Multiplicities*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional complex vector space and let $T \in \mathcal{L}(\mathcal{V})$. Then the sum of the multiplicities of all eigenvalues of T equals $\dim(\mathcal{V})$.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 6.5.1, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$.
Taking dimensions, by Corollary 2.3.2, $\dim(\mathcal{V}) = \sum_{j=1}^{m} \dim \mathcal{G}_{\lambda_j}(T)$, exactly the sum of the multiplicities.

Structural Commentary: Multiplicity is dimension of the generalized-eigenspace summand.
Key Insight: Multiplicities are dimensions of the direct-sum pieces. □

Theorem 6.5.5. *Block Diagonal Form with Upper-Triangular Primary Blocks*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, and let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T \in \mathcal{L}(\mathcal{V})$, with multiplicities $d_1, \ldots, d_m$. Then T has a matrix of the form

$$\begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_m \end{bmatrix},$$

where each $A_j \in \mathrm{Mat}(d_j, d_j, \mathbb{C})$ is upper triangular with diagonal $\lambda_j, \ldots, \lambda_j$.

Proof. Set $\mathcal{U}_j = \mathcal{G}_{\lambda_j}(T)$ and $N_j = (T - \lambda_j I_{\mathcal{V}})|_{\mathcal{U}_j}$.
By Theorem 6.5.1, the $\mathcal{U}_j$ form a T-invariant direct-sum decomposition and each N_j is nilpotent.
Choose a basis of each $\mathcal{U}_j$ in which N_j is strictly upper triangular. Then $T|_{\mathcal{U}_j} = \lambda_j I_{\mathcal{U}_j} + N_j$ has an upper triangular matrix with diagonal $\lambda_j, \ldots, \lambda_j$. Concatenating these bases gives the desired block diagonal matrix.
Structural Commentary: Primary decomposition gives the blocks; nilpotent triangularisation shapes them.
Key Insight: On each generalized eigenspace, T is eigenvalue plus nilpotent. □

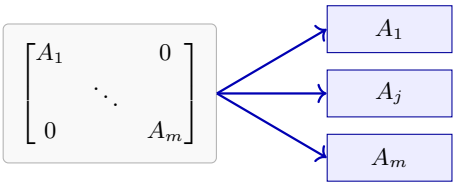


Figure 54: Primary blocks: $A_j = \lambda_j I + N_j$, with N_j strictly upper triangular.

6.6 Jordan and Canonical Forms

Jordan and canonical forms turn primary decomposition into explicit blocks. Jordan chains, complexification, and real block forms organize the nilpotent and real spectral structure. Jordan blocks are elementary divisors $(z - \lambda)^k$ after the relevant polynomial factors split into linear factors.

Definition 6.6.1. *Jordan Chains, Blocks, and Matrices*
For $k \geq 1$, the *nilpotent Jordan block* $J_k(0)$ is the $k \times k$ matrix with 1 on the first superdiagonal and 0 elsewhere:

$$J_k(0) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & 0 \end{bmatrix}.$$

For $\lambda \in \mathbb{C}$, set $J_k(\lambda) := \lambda I_k + J_k(0)$. A *Jordan matrix* is a direct sum of Jordan blocks. A basis is a *Jordan basis* for an operator T if the matrix of T in that basis is a Jordan matrix.

Perspective 6.6.2. *Jordan Strings as Columns*
A length-k Jordan string for A at λ is $(\mathbf{x}_1, \ldots, \mathbf{x}_k)$ with $(A - \lambda I)\mathbf{x}_1 = \mathbf{0}$; $(A - \lambda I)\mathbf{x}_j = \mathbf{x}_{j-1}$, $2 \leq j \leq k$.
For $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$, $AX = XJ_k(\lambda)$.
If all strings form the columns of a basis matrix M, then $AM = MJ$ and $M^{-1}AM = J$.

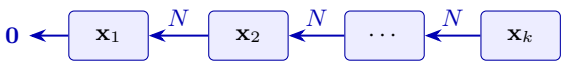


Figure 55: Jordan chain for $N = A - \lambda I$; columns give $AX = XJ_k(\lambda)$.

Method 6.6.3. *Filippov Column-Space Jordan Strings*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, let $\lambda \in \mathrm{spec}(A)$, and set $N = A - \lambda I_n$ on the generalized eigenspace $\mathcal{G}_\lambda(A) = \ker(N^n)$.

(i) Work first inside $\mathrm{range}(N) \cap \mathcal{G}_\lambda(A)$, where N is again nilpotent.
(ii) Build Jordan strings for this smaller nilpotent operator.
(iii) For each string ending in $\mathbf{u}$, solve $N\mathbf{v} = \mathbf{u}$ in $\mathcal{G}_\lambda(A)$ and lift the string by appending $\mathbf{v}$.

(iv) Add enough vectors from $\ker(N)$ to complete a basis of $\mathcal{G}_\lambda(A)$; these added vectors are length-one strings.
(v) Repeat the process recursively until all of $\mathcal{G}_\lambda(A)$ is covered, then concatenate the strings for all eigenvalues.

Theorem 6.6.5 justifies the lifts; Perspective 6.6.2 turns the resulting columns into $AM = MJ$.

Proposition 6.6.4. *Powers of a Jordan Block*
For $p \geq 0$, $J_k(\lambda)^p = \sum_{r=0}^{\min(p,k-1)} \binom{p}{r} \lambda^{p-r} J_k(0)^r$. Thus $J_k(\lambda)^p \to 0$ if and only if $|\lambda| < 1$.

Proof.

($\Rightarrow$) If $J_k(\lambda)^p \to 0$, then its diagonal entries λ^p tend to 0, so $|\lambda| < 1$.
($\Leftarrow$) If $|\lambda| < 1$, then for each fixed $0 \leq r < k$ the scalar $\binom{p}{r}\lambda^{p-r}$ tends to 0, because a polynomial in p is dominated by the exponential decay of $|\lambda|^p$.

Structural Commentary: Jordan powers are exponential-polynomial terms.
Key Insight: Polynomial growth cannot defeat exponential decay when $|\lambda| < 1$. □

Theorem 6.6.5. *Nilpotent Jordan Chain Basis*
Let $\mathcal{V}$ be finite-dim, $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $\exists$ vectors $\mathbf{v}_1, \ldots, \mathbf{v}_r \in \mathcal{V}$ and nonnegative integers $m_1, \ldots, m_r$ such that $\mathcal{B}_N = \left(N^{m_1}\mathbf{v}_1, \ldots, N\mathbf{v}_1, \mathbf{v}_1, \ldots, N^{m_r}\mathbf{v}_r, \ldots, N\mathbf{v}_r, \mathbf{v}_r\right)$ is a basis of $\mathcal{V}$ and $N^{m_j+1}\mathbf{v}_j = \mathbf{0}$ for every j.

Proof. Induct on $n = \dim(\mathcal{V})$. If $N = 0$, any basis works with all $m_j = 0$.
Now assume $N \neq 0$. Then N is not injective, so $\text{range}(N)$ is a proper subspace, and $N|_{\text{range}(N)}$ is nilpotent.
By induction, $\text{range}(N)$ has a chain basis $\left(N^{m_1}\mathbf{u}_1, \ldots, N\mathbf{u}_1, \mathbf{u}_1, \ldots, N^{m_r}\mathbf{u}_r, \ldots, N\mathbf{u}_r, \mathbf{u}_r\right)$ with $N^{m_j+1}\mathbf{u}_j = \mathbf{0}$.
Choose $\mathbf{v}_j \in \mathcal{V}$ such that $N\mathbf{v}_j = \mathbf{u}_j$, and lift the chains to $\mathcal{L} = \left(N^{m_1+1}\mathbf{v}_1, \ldots, N\mathbf{v}_1, \mathbf{v}_1, \ldots, N^{m_r+1}\mathbf{v}_r, \ldots, N\mathbf{v}_r, \mathbf{v}_r\right)$.
The list $\mathcal{L}$ is linearly independent. Indeed, applying N to a dependence among vectors in $\mathcal{L}$ gives a dependence among the range-chain basis, so all coefficients except possibly those on the leading vectors $N^{m_j+1}\mathbf{v}_j$ vanish. The remaining relation is a relation among $N^{m_j+1}\mathbf{v}_j = N^{m_j}\mathbf{u}_j$, which are members of the range-chain basis, so those coefficients vanish too. Also $N(\text{span}(\mathcal{L})) = \text{range}(N)$: each range-chain basis vector $N^q\mathbf{u}_j$ is $N(N^q\mathbf{v}_j)$ for the corresponding lifted vector. Extend $\mathcal{L}$ to a basis by adding $\mathbf{w}_1, \ldots, \mathbf{w}_s$. For each i, choose $\mathbf{x}_i \in \text{span}(\mathcal{L})$ such that $N\mathbf{x}_i = N\mathbf{w}_i$, and set $\mathbf{z}_i = \mathbf{w}_i - \mathbf{x}_i$. Then $N\mathbf{z}_i = \mathbf{0}$. Since each $\mathbf{x}_i$ lies in $\text{span}(\mathcal{L})$, replacing $\mathbf{w}_i$ by $\mathbf{z}_i$ preserves the span and linear independence of the full basis. Thus $\mathcal{L}, \mathbf{z}_1, \ldots, \mathbf{z}_s$ is the desired chain basis.
Structural Commentary: Build chains in $\text{range}(N)$, lift them once, then add kernel chains.
Key Insight: Applying N turns lifted-chain dependences into known range-chain dependences. □

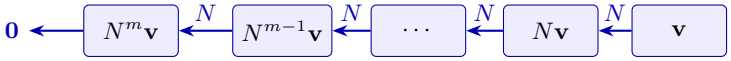


Figure 56: A nilpotent chain shifts left under N; in this basis order it becomes one Jordan block.

Theorem 6.6.6. *Jordan Form over $\mathbb{C}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}$ has a Jordan basis for T.

Proof. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. By Theorem 6.5.1, $\mathcal{V} = \mathcal{G}_{\lambda_1}(T) \oplus \cdots \oplus \mathcal{G}_{\lambda_m}(T)$, and each $N_j := (T - \lambda_j I_\mathcal{V})|_{\mathcal{G}_{\lambda_j}(T)}$ is nilpotent. Apply Theorem 6.6.5 to N_j on each generalized eigenspace.
The resulting chain basis of $\mathcal{G}_{\lambda_j}(T)$ is a basis of that summand, and the direct-sum decomposition makes the concatenation of these bases a basis of $\mathcal{V}$. In a chain of length r for N_j, the matrix of N_j is $J_r(0)$, so the matrix of $T|_{\mathcal{G}_{\lambda_j}(T)} = \lambda_j I + N_j$ on the same chain is $J_r(\lambda_j)$. Therefore the concatenated basis is a Jordan basis for T.
Structural Commentary: Primary pieces decompose into nilpotent chains.
Key Insight: Jordan form is "eigenvalue plus nilpotent chains" on every generalized eigenspace. □

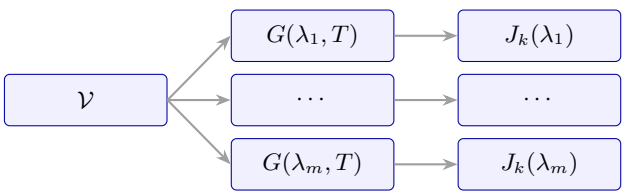


Figure 57: Primary pieces $\mathcal{G}_{\lambda_j}(T)$ give nilpotent chains, hence Jordan blocks.

Corollary 6.6.7. *Diagonalizability in Jordan Form*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have Jordan form J. For each eigenvalue λ, $\dim E_\lambda(A)$ equals the number of λ-blocks in J. Hence A is diagonalizable if and only if every Jordan block has size 1.

Proof. Write $A = SJS^{-1}$. Then $S : E_\lambda(J) \to E_\lambda(A)$ is an isomorphism. A block $J_k(\mu)$ contributes to $E_\lambda(J)$ only when $\mu = \lambda$, and then $\dim \ker(J_k(\lambda) - \lambda I_k) = \dim \ker J_k(0) = 1$.
Thus $\dim E_\lambda(A)$ is the number of λ-blocks. By Theorem 6.2.10, diagonalizability means geometric and algebraic multiplicities agree for every λ; here this holds exactly when every block has size 1.
Structural Commentary: Each Jordan block contributes one eigenvector direction.
Key Insight: Missing eigenvectors are measured by block sizes greater than 1. □

Lemma 6.6.8. *Powers of a Nilpotent Jordan Block*
For $k \geq 1$ and $p \geq 0$, $\mathrm{rank}\big(J_k(0)^p\big) = \max(k - p, 0)$ and $\dim \ker\big(J_k(0)^p\big) = \min(p, k)$.

Proof. Let $N = J_k(0)$. In the standard basis, $N\mathbf{e}_1 = \mathbf{0}$ and $N\mathbf{e}_j = \mathbf{e}_{j-1}$ for $2 \leq j \leq k$. Iterating gives

$$N^p \mathbf{e}_j = \begin{cases} \mathbf{0}, & j \leq p, \\ \mathbf{e}_{j-p}, & j > p. \end{cases}$$

If $p < k$, the image is $\mathrm{span}(\mathbf{e}_1, \ldots, \mathbf{e}_{k-p})$, so the rank is $k - p$; if $p \geq k$, the image is $\{\mathbf{0}\}$, so the rank is 0.
Hence $\mathrm{rank}(N^p) = \max(k - p, 0)$. Rank-nullity gives $\dim \ker(N^p) = k - \max(k - p, 0) = \min(p, k)$.
Structural Commentary: A nilpotent Jordan block is a left-shift on the standard basis.
Key Insight: Each power deletes one more leading layer of the chain. □

Definition 6.6.9. *Weyr Characteristic*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ and $\lambda \in \mathrm{spec}(A)$.
For $p \geq 1$, define $w_p(\lambda, A) := \mathrm{rank}\big((A - \lambda I_n)^{p-1}\big) - \mathrm{rank}\big((A - \lambda I_n)^p\big)$.
The positive part of the sequence $w_1, w_2, \ldots$ is the *Weyr characteristic* of A associated with λ.

Theorem 6.6.10. *Jordan Block Sizes from Rank Drops*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$, $\lambda \in \mathrm{spec}(A)$, and $p \geq 1$. Then $w_p(\lambda, A)$ is the number of λ-Jordan blocks of size at least p; hence $w_p(\lambda, A) - w_{p+1}(\lambda, A)$ is the number of λ-blocks of size exactly p.

Proof. Rank is similarity-invariant, so assume A is in Jordan form.
Blocks $J_k(\mu)$ with $\mu \neq \lambda$ contribute 0 to the rank drop because $J_k(\mu) - \lambda I_k$ is invertible.
A block $J_k(\lambda)$ contributes $\mathrm{rank}\big(J_k(0)^{p-1}\big) - \mathrm{rank}\big(J_k(0)^p\big)$, which is 1 exactly when $p \leq k$ by Lemma 6.6.8.
Summing over blocks proves the first claim; subtracting the $p + 1$ count gives the exact-size count.
Structural Commentary: Rank drops count surviving chain lengths.
Key Insight: A length-k block contributes to drops $1, \ldots, k$. □

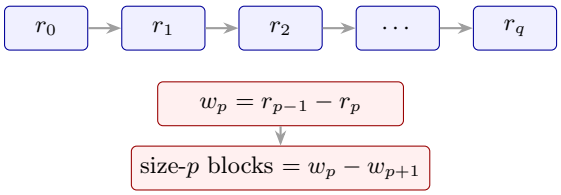


Figure 58: For $r_p = \mathrm{rank}((A - \lambda I)^p)$, drops count surviving chains; $w_p - w_{p+1}$ counts size-p blocks.

Method 6.6.11. *Computing Jordan Blocks by Rank Drops*
For each eigenvalue λ of $A \in \mathrm{Mat}(n, n, \mathbb{C})$:

(i) Compute $r_p = \mathrm{rank}\big((A - \lambda I_n)^p\big)$ for $p = 0, 1, 2, \ldots$ until the differences stabilize at 0.
(ii) Set $w_p = r_{p-1} - r_p$. Then w_p is the number of λ-blocks of size at least p.
(iii) The number of blocks $J_p(\lambda)$ is $w_p - w_{p+1}$.
(iv) Assemble the Jordan form by taking the direct sum over all eigenvalues.

Theorem 6.6.12. *Nonzero Eigenvalues of AB and BA*
Let $A \in \mathrm{Mat}(m, n, \mathbb{C})$, $B \in \mathrm{Mat}(n, m, \mathbb{C})$, and suppose $n \geq m$. Then $p_{BA}(z) = z^{n-m} p_{AB}(z)$.
Consequently, for $\lambda \neq 0$, $m_{AB}(\lambda) = m_{BA}(\lambda)$. If $m_{AB}(0) = k$, then $m_{BA}(0) = k + n - m$.

Proof. For $z \neq 0$, Sylvester's determinant identity gives

$$\det(zI_n - BA) = z^n \det(I_n - z^{-1}BA) = z^n \det(I_m - z^{-1}AB) = z^{n-m}\det(zI_m - AB).$$

Both sides are polynomials in z, so the identity holds for all z. The claimed multiplicities follow by factoring p_{AB} and comparing with $p_{BA} = z^{n-m}p_{AB}$.
Structural Commentary: Sylvester's determinant identity moves the rectangular product across the determinant.
Key Insight: Passing from AB to BA only adds the necessary extra zero roots when the sizes differ. □

Theorem 6.6.13. *Nonzero Jordan Blocks of AB and BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. For every $\lambda \neq 0$, the Jordan blocks of AB and BA at λ coincide.

Proof. Set $X = \begin{bmatrix} AB & A \\ 0 & 0 \end{bmatrix}$ and $Y = \begin{bmatrix} 0 & A \\ 0 & BA \end{bmatrix}$. Then $\begin{bmatrix} I_n & 0 \\ B & I_n \end{bmatrix} X \begin{bmatrix} I_n & 0 \\ -B & I_n \end{bmatrix} = Y$, so X and Y are similar.
For $\lambda \neq 0$ and $p \geq 1$, the shifted powers have block triangular forms

$$(X - \lambda I_{2n})^p = \begin{bmatrix} (AB - \lambda I_n)^p & * \\ 0 & (-\lambda)^p I_n \end{bmatrix}, \qquad (Y - \lambda I_{2n})^p = \begin{bmatrix} (-\lambda)^p I_n & * \\ 0 & (BA - \lambda I_n)^p \end{bmatrix}$$

The diagonal block $(-\lambda)^p I_n$ is invertible.
Block Gaussian elimination by invertible row and column operations removes the starred block, so

$$\operatorname{rank}\big((X - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((AB - \lambda I_n)^p\big),$$
$$\operatorname{rank}\big((Y - \lambda I_{2n})^p\big) = n + \operatorname{rank}\big((BA - \lambda I_n)^p\big).$$

Since X and Y are similar, these left sides agree for all p.
Hence the shifted rank drops of AB and BA at λ agree, so Theorem 6.6.10 gives the same Jordan blocks.
Structural Commentary: Similar block embeddings make the nonzero shifted ranks match.
Key Insight: At $\lambda \neq 0$, the extra block is invertible and splits off. □

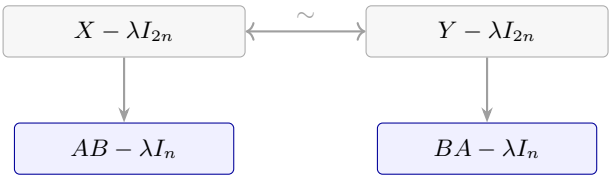


Figure 59: For $\lambda \neq 0$, the auxiliary similarity reduces the comparison to shifted ranks of AB and BA.

Corollary 6.6.14. *Uniqueness of Jordan Form*
If J and J' are Jordan matrices similar to $A \in \mathrm{Mat}(n, n, \mathbb{C})$, they differ only by a permutation of Jordan blocks.

Proof. Similarity preserves spectra and the ranks of $(X - \lambda I_n)^p$. Thus J and J' have the same Weyr characteristic for each eigenvalue. Theorem 6.6.10 then gives the same block multiplicities, so only block order may differ.
Structural Commentary: Weyr data is similarity-invariant and fixes block multiplicities.
Key Insight: Jordan form is unique modulo block order. □

Corollary 6.6.15. *Similarity Criterion for AB and BA*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$. AB is similar to BA if and only if $\operatorname{rank}\big((AB)^p\big) = \operatorname{rank}\big((BA)^p\big)$, $p = 1, \ldots, n$.

Proof.

(⇒) If $AB \sim BA$, then $(AB)^p \sim (BA)^p$ for all p, so their ranks are equal.
(⇐) Assume the rank equalities hold for $p = 1, \ldots, n$. Theorem 6.6.13 gives the same nonzero Jordan blocks. Together with the equality at $p = 0$, the rank equalities give the same Weyr characteristic at $\lambda = 0$ by Theorem 6.6.10. Hence the nilpotent blocks also agree, so the Jordan forms coincide up to block order.

Structural Commentary: Nonzero blocks agree automatically; only the nilpotent blocks must be tested.
Key Insight: Powers of AB and BA measure exactly the zero-Jordan-block data. □

Method 6.6.16. *Checking Whether AB and BA Are Similar*
For $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$:

(i) The nonzero Jordan blocks of AB and BA already match.
(ii) Compute $\operatorname{rank}\big((AB)^p\big)$ and $\operatorname{rank}\big((BA)^p\big)$ for $p = 1, \ldots, n$.
(iii) If all ranks agree, then $AB \sim BA$; if some rank differs, then only the nilpotent Jordan structure differs.

Definition 6.6.17. *Nonderogatory Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *nonderogatory* if its Jordan form has exactly one Jordan block for each distinct eigenvalue. Equivalently, $\dim E_\lambda(A) = 1$ for every $\lambda \in \mathrm{spec}(A)$.

Lemma 6.6.18. *Commutant of One Jordan Block*
If $J = J_m(\lambda)$ and $XJ = JX$, then $X = p(J)$ for some polynomial p with $\deg p < m$.

Proof. It suffices to treat $N = J_m(0)$, since $J_m(\lambda) = \lambda I_m + N$.
The vector $\mathbf{e}_m$ is cyclic for N: $(\mathbf{e}_m, N\mathbf{e}_m, \ldots, N^{m-1}\mathbf{e}_m)$ is the standard basis in reverse order.
Choose the unique polynomial p with $\deg p < m$ such that $p(N)\mathbf{e}_m = X\mathbf{e}_m$.
For every $r = 0, \ldots, m-1$, $p(N)N^r\mathbf{e}_m = N^r p(N)\mathbf{e}_m = N^r X\mathbf{e}_m = XN^r\mathbf{e}_m$.
Thus $p(N)$ and X agree on a basis, so $X = p(N) = q(J)$ for $q(z) = p(z - \lambda)$.
Structural Commentary: A single Jordan block is generated by one cyclic vector.
Key Insight: The cyclic vector determines the commutant. □

Theorem 6.6.19. *Nonderogatory Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is nonderogatory, then $\mathrm{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$.

Proof. The inclusion $\{p(A) : p \in \mathbb{C}[z]\} \subseteq \mathrm{Comm}(A)$ follows from Proposition 6.1.3.
For reverse inclusion, first replace A by its Jordan form $J = J_{m_1}(\lambda_1) \oplus \cdots \oplus J_{m_d}(\lambda_d)$, where λ_i are distinct.
If $XJ = JX$ and $X = [X_{ij}]$ is partitioned conformally with J, the block equation is $X_{ij}J_{m_j}(\lambda_j) = J_{m_i}(\lambda_i)X_{ij}$.
Corollary 6.3.20 gives $X_{ij} = 0$ for $i \neq j$. Each diagonal block X_{ii} commutes with $J_{m_i}(\lambda_i)$, so Lemma 6.6.18 gives $X_{ii} = p_i(J_{m_i}(\lambda_i))$ for some polynomial p_i with $\deg p_i < m_i$. The ideals generated by $(z - \lambda_i)^{m_i}$ are pairwise coprime, so Hermite interpolation gives a polynomial p with $p \equiv p_i \pmod{(z - \lambda_i)^{m_i}}$ for every i.
On the ith block, $p - p_i = (z - \lambda_i)^{m_i} q_i$ for some q_i, and $(J_{m_i}(\lambda_i) - \lambda_i I)^{m_i} = 0$. Hence $p(J_{m_i}(\lambda_i)) = p_i(J_{m_i}(\lambda_i)) = X_{ii}$ for every i, so $X = p(J)$. Conjugating back to A proves the claim.
Structural Commentary: Disjoint eigenvalue blocks force block diagonality; one Jordan block per eigenvalue makes each diagonal commutant polynomial.
Key Insight: Hermite interpolation glues the separate block polynomials into one polynomial in A. □

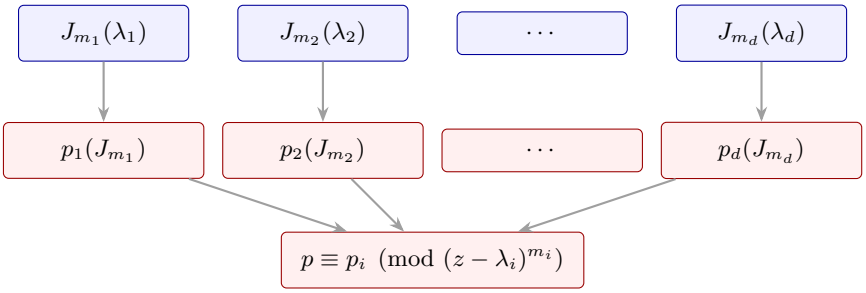


Figure 60: Blockwise commutant polynomials glue by interpolation.

6.7 Transpose and Conjugate Similarity

Theorem 6.7.1. *Conjugate Similarity Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. Then A is similar to $\overline{A}$ if and only if each $J_k(\lambda)$ occurs in the Jordan form of A with the same multiplicity as $J_k(\overline{\lambda})$.

Proof. Let J be a Jordan form of A. Then $\overline{J}$ is a Jordan form of $\overline{A}$, obtained by replacing each $J_k(\lambda)$ by $J_k(\overline{\lambda})$.
($\Rightarrow$) If A is similar to $\overline{A}$, then J and $\overline{J}$ are similar.
By Corollary 6.6.14, their Jordan blocks have same multiplicities.
($\Leftarrow$) If the multiplicities agree, then J and $\overline{J}$ differ only by a permutation of blocks. Hence A is similar to $\overline{A}$.
Structural Commentary: Conjugation changes eigenvalue labels but not block sizes.
Key Insight: Non-real Jordan blocks must occur in conjugate pairs. □

Corollary 6.7.2. *Hermitian Products and Conjugate Products*
Let $A, H, K \in \mathrm{Mat}(n, n, \mathbb{C})$, and suppose H and K are Hermitian.
Then HK is similar to KH, and $A\overline{A}$ is similar to $\overline{A}A$.

Proof. For each $p = 1, \dots, n$, adjoints preserves rank, $((HK)^p)^* = (KH)^p$. Hence $\operatorname{rank}\big((HK)^p\big) = \operatorname{rank}\big((KH)^p\big)$. Similarly, entrywise conjugation preserves rank and gives $\overline{(A\overline{A})^p} = (\overline{A}A)^p$, so $\operatorname{rank}\big((A\overline{A})^p\big) = \operatorname{rank}\big((\overline{A}A)^p\big)$. Apply Corollary 6.6.15 in both cases.
Structural Commentary: Adjoint and conjugation preserve rank and reverse product order.
Key Insight: The rank-power test turns order reversal into similarity. □

Definition 6.7.3. *Reversal Matrix*
For $k \geq 1$, *reversal matrix* $R_k \in \operatorname{Mat}(k, k, \mathbb{C})$ defined $(R_k)_{ij} = \begin{cases} 1, & i+j = k+1, \\ 0, & i+j \neq k+1 \end{cases}$, thus $R_k^t = R_k$, $R_k^2 = I_k$.

Lemma 6.7.4. *Reversal Conjugates a Jordan Block to Its Transpose*
For every $k \geq 1$ and $\lambda \in \mathbb{C}$, $J_k(\lambda)^t = R_k J_k(\lambda) R_k$. Moreover, $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.

Proof. Let $N = J_k(0)$. Since R_k reverses standard basis order, $R_k N R_k = N^t$.
Adding scalar gives $R_k J_k(\lambda) R_k = J_k(\lambda)^t$. As $R_k = R_k^t = R_k^{-1}$, this identity gives $J_k(\lambda)^t R_k = R_k J_k(\lambda)$ and $R_k J_k(\lambda)^t = J_k(\lambda) R_k$, so $R_k J_k(\lambda)$ and $J_k(\lambda) R_k$ are symmetric.
Structural Commentary: Reversal changes the superdiagonal shift into the subdiagonal shift.
Key Insight: Transposition reads the same Jordan chain backwards. □

Theorem 6.7.5. *Transpose Similarity and Symmetric Factorisations*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. Then:

(i) A is similar to A^t via a symmetric similarity matrix.
(ii) $A = BC = DE$, where B, C, D, E are symmetric and B, E are invertible.

Proof. Choose a Jordan form $A = SJS^{-1}$, and let R be the direct sum of the reversal matrices matching the Jordan blocks of J. By Lemma 6.7.4, $J^t = RJR$, $R^t = R$, $R^2 = I_n$, and both RJ and JR are symmetric.

(i) Set $P = (S^{-1})^t R S^{-1}$. Then $P^t = P$, $P^{-1} = SRS^t$, and $PAP^{-1} = (S^{-1})^t RJRS^t = (S^{-1})^t J^t S^t = A^t$.
(ii) Using $R^2 = I_n$, $A = (SRS^t)\big((S^{-1})^t RJS^{-1}\big)$ and $A = (SJRS^t)\big((S^{-1})^t RS^{-1}\big)$.
The products equal SJS^{-1} because $S^t(S^{-1})^t = I_n$ and $R^2 = I_n$. All four factors are symmetric because R, RJ, JR are symmetric; SRS^t and $(S^{-1})^t RS^{-1}$ are invertible.

Structural Commentary: Reduce to Jordan blocks, reverse chains, then conjugate back.
Key Insight: Symmetric involutions produce both transpose similarity and symmetric factors. □

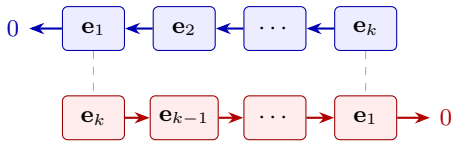


Figure 61: Reversal gives $R_k J_k(0) R_k = J_k(0)^t$: the same chain read backwards.

Theorem 6.7.6. *Conjugate Similarity and Hermitian Products*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) A is similar to $\overline{A}$.
(ii) A is similar to A^*.
(iii) A is similar to A^* via a Hermitian similarity matrix.
(iv) $A = HK = LM$, where H, K, L, M are Hermitian, and H, M are invertible.
(v) $A = HK$ for some Hermitian matrices H, K.

Proof.

(i) $\Rightarrow$ *(ii)* Theorem 6.7.5 applied to $\overline{A}$ gives $\overline{A} \sim (\overline{A})^t = A^*$. Thus $A \sim A^*$.
(ii) $\Rightarrow$ *(iii)* Choose invertible S such that $A = SA^*S^{-1}$, so $AS = SA^*$. For $\theta \in \mathbb{R}$, put $S_\theta = e^{i\theta} S$.
Then $AS_\theta = S_\theta A^*$ and, by taking adjoints, $AS_\theta^* = S_\theta^* A^*$. Hence $A(S_\theta + S_\theta^*) = (S_\theta + S_\theta^*)A^*$.
The matrix $H_\theta = S_\theta + S_\theta^*$ is Hermitian. Also $S_\theta^{-1} H_\theta = I_n + e^{-2i\theta} S^{-1} S^*$. Thus H_θ is singular exactly when $-e^{2i\theta}$ is an eigenvalue of $S^{-1}S^*$. There are only finitely many such values, so choose θ for which H_θ is invertible. Then $A = H_\theta A^* H_\theta^{-1}$.
(iii) $\Rightarrow$ *(iv)* Let R be invertible Hermitian and satisfy $A = RA^*R^{-1}$. Then $AR = RA^*$ and $R^{-1}A = A^*R^{-1}$.
Thus A^*R^{-1} and RA^* are Hermitian, and $A = R(A^*R^{-1}) = (RA^*)R^{-1}$.
This gives the two asserted Hermitian factorizations, with the first factor in the first product and the second factor in the second product invertible.

(*iv*) $\Rightarrow$ (*v*) The first factorization in (iv) already has the form $A = HK$ with H and K Hermitian.
(*v*) $\Rightarrow$ (*i*) If $A = HK$ with H, K Hermitian, then Corollary 6.7.2 gives $HK \sim KH$.
By Theorem 6.7.5, $KH \sim (KH)^t = H^t K^t = \overline{H}\,\overline{K} = \overline{A}$. Hence $A \sim \overline{A}$.

Structural Commentary: The equivalences move between conjugation, adjoints, and Hermitian products by combining transpose similarity with a Hermitianized similarity matrix.
Key Insight: A phase-rotated average $e^{i\theta}S + e^{-i\theta}S^*$ turns an arbitrary similarity to A^* into a Hermitian one. □

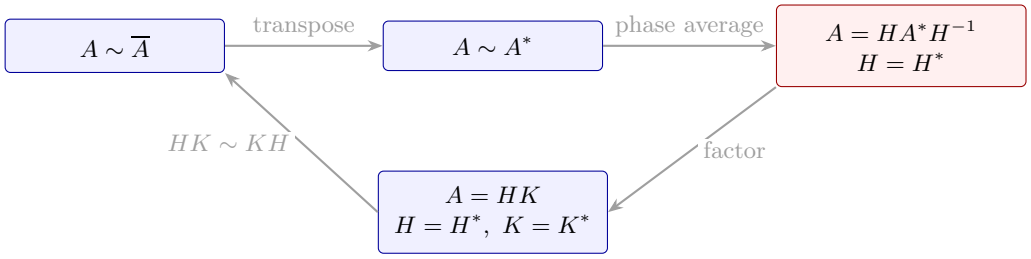


Figure 62: Conjugate similarity, adjoint similarity, and Hermitian product factorisations are equivalent.

7 Spectral Theorems and Operator Functions

7.1 Spectral Theory of Structured Operators

Structured spectral theory studies operators whose geometry and spectrum align. Normal, self-adjoint, positive, and unitary maps admit orthogonal decompositions and spectral resolutions.

Definition 7.1.1. *Normal Operator*
An operator $T \in \mathcal{L}(\mathcal{V})$ on a finite-dim inner-product space is *normal* if it commutes with its adjoint: $TT^* = T^*T$.
A square matrix A is *normal* if $A^*A = AA^*$.
Every self-adjoint operator, every Hermitian matrix, and every unitary matrix is normal.

Theorem 7.1.2. *Normal Operators and Equal Adjoint Norms*
An operator $T \in \mathcal{L}(\mathcal{V})$ is normal if and only if $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. The operator $A = T^*T - TT^*$ is self-adjoint.
For every $\mathbf{v} \in \mathcal{V}$, $\langle A\mathbf{v}, \mathbf{v}\rangle = \langle T^*T\mathbf{v}, \mathbf{v}\rangle - \langle TT^*\mathbf{v}, \mathbf{v}\rangle = \|T\mathbf{v}\|^2 - \|T^*\mathbf{v}\|^2$.
Thus T is normal exactly when $A = 0$, which by Proposition 2.4.17 is equivalent to the displayed quadratic form being zero for every $\mathbf{v}$. This is equivalent to $\|T\mathbf{v}\| = \|T^*\mathbf{v}\|$ for every $\mathbf{v}$.
Structural Commentary: Normality is measured by the self-adjoint commutator $T^*T - TT^*$.
Key Insight: A normal operator and its adjoint move every vector the same distance. □

Theorem 7.1.3. *Normal Operators Share Eigenvectors with Their Adjoints*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. If $T\mathbf{v} = \lambda\mathbf{v}$ for some $\mathbf{v} \neq \mathbf{0}$, then $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Proof. The operator $T - \lambda I_{\mathcal{V}}$ is normal by Proposition 2.4.8 and scalar multiples of $I_{\mathcal{V}}$ commute with T and T^*. Applying Theorem 7.1.2 to $T - \lambda I_{\mathcal{V}}$ gives $0 = \|(T - \lambda I_{\mathcal{V}})\mathbf{v}\| = \|(T - \lambda I_{\mathcal{V}})^*\mathbf{v}\| = \|(T^* - \overline{\lambda} I_{\mathcal{V}})\mathbf{v}\|$.
Structural Commentary: For normal operators, kernels of $T - \lambda I$ and $T^* - \overline{\lambda} I$ coincide.
Key Insight: Normality makes eigenvector information visible from both T and T^*. □

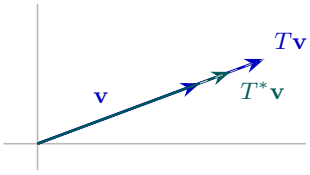


Figure 63: For normal T, $T\mathbf{v} = \lambda\mathbf{v}$ implies $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.

Theorem 7.1.4. *Orthogonal Eigenvectors for Normal Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal. Eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose $T\mathbf{u} = \alpha\mathbf{u}$ and $T\mathbf{v} = \beta\mathbf{v}$ with $\alpha \neq \beta$. By Theorem 7.1.3, $T^*\mathbf{v} = \overline{\beta}\mathbf{v}$.
Hence $\alpha \langle \mathbf{u}, \mathbf{v} \rangle = \langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T^*\mathbf{v} \rangle = \langle \mathbf{u}, \overline{\beta}\mathbf{v} \rangle = \beta \langle \mathbf{u}, \mathbf{v} \rangle$. Since $\alpha - \beta \neq 0$, $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.
Structural Commentary: Normality lets the operator move from one eigenvector to the other through the adjoint.
Key Insight: Distinct eigenvalue scalings are incompatible unless the inner product is zero. □

Theorem 7.1.5. *Eigenvalues of Self-Adjoint Operators Are Real*
Every eigenvalue of a self-adjoint operator on a finite-dimensional inner-product space is real.

Proof. Let $T = T^*$ and $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then $\lambda \|\mathbf{v}\|^2 = \langle \lambda\mathbf{v}, \mathbf{v} \rangle = \langle T\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{v}, T\mathbf{v} \rangle = \langle \mathbf{v}, \lambda\mathbf{v} \rangle = \overline{\lambda} \|\mathbf{v}\|^2$.
Since $\|\mathbf{v}\|^2 > 0$, $\lambda = \overline{\lambda}$, so $\lambda \in \mathbb{R}$.
Structural Commentary: Self-adjointness moves T between inner-product slots without changing the value.
Key Insight: An eigenvalue must equal its conjugate because the eigenvector has positive squared norm. □

Proposition 7.1.6. *Invertible Quadratic Expressions*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint on a finite-dimensional inner-product space.
If $b, c \in \mathbb{R}$ and $b^2 < 4c$, then $T^2 + bT + cI_{\mathcal{V}}$ is invertible.

Proof. Let $S = T^2 + bT + cI_{\mathcal{V}}$. For $\mathbf{v} \neq \mathbf{0}$,

$$\begin{aligned} \langle S\mathbf{v}, \mathbf{v} \rangle &= \|T\mathbf{v}\|^2 + b \langle T\mathbf{v}, \mathbf{v} \rangle + c \|\mathbf{v}\|^2 \geq \|T\mathbf{v}\|^2 - |b| \, \|T\mathbf{v}\| \, \|\mathbf{v}\| + c \|\mathbf{v}\|^2 \\ &= \left(\|T\mathbf{v}\| - \frac{|b|}{2} \|\mathbf{v}\| \right)^2 + \left(c - \frac{b^2}{4} \right) \|\mathbf{v}\|^2 > 0 \end{aligned}$$

Thus $S\mathbf{v} \neq \mathbf{0}$ for every nonzero $\mathbf{v}$, so $\ker(S) = \{\mathbf{0}\}$. In finite dimensions, injectivity implies invertibility.
Structural Commentary: Negative discriminant keeps this self-adjoint quadratic positive.
Key Insight: Self-adjointness converts $\langle T^2\mathbf{v}, \mathbf{v} \rangle$ into $\|T\mathbf{v}\|^2$, so completing the square controls the kernel. □

Lemma 7.1.7. *Self-Adjoint Operators Have Eigenvalues*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real inner-product space. Every self-adjoint $T \in \mathcal{L}(\mathcal{V})$ has an eigenvalue.

Proof. Choose $\mathbf{v} \neq \mathbf{0}$ and set $n = \dim(\mathcal{V})$.
The list $\mathbf{v}, T\mathbf{v}, \ldots, T^n\mathbf{v}$ is linearly dependent, so some nonzero real polynomial p satisfies $p(T)\mathbf{v} = \mathbf{0}$.
Factor p over $\mathbb{R}$ as $p(x) = a \prod_{j=1}^{M} (x^2 + b_j x + c_j) \prod_{k=1}^{m} (x - \lambda_k)$, where $a \neq 0$, $b_j^2 < 4c_j$, $\lambda_k \in \mathbb{R}$.
By Proposition 7.1.6, each quadratic factor $Q_j(T) := T^2 + b_j T + c_j I_{\mathcal{V}}$ is invertible. Let $R(T) = \prod_j Q_j(T)$ and $L(T) = \prod_k (T - \lambda_k I_{\mathcal{V}})$. The factors commute because they are polynomials in T, and $R(T)$ is invertible.
If $m = 0$, then $p(T) = aR(T)$ would be invertible, contradicting $p(T)\mathbf{v} = \mathbf{0}$ with $\mathbf{v} \neq \mathbf{0}$. Hence $m \geq 1$.
Since $\mathbf{0} = p(T)\mathbf{v} = aR(T)L(T)\mathbf{v}$ and $aR(T)$ is invertible, $L(T)\mathbf{v} = \mathbf{0}$. Thus $L(T)$ has nontrivial kernel.
If every $T - \lambda_k I_{\mathcal{V}}$ were injective, their product $L(T)$ would be injective, a contradiction. Therefore some $T - \lambda_k I_{\mathcal{V}}$ is not injective, and λ_k is an eigenvalue of T.
Structural Commentary: An annihilating polynomial remains; quadratic factors are invertible.
Key Insight: The nontrivial kernel must come from a real linear factor, hence from an eigenvalue. □

Proposition 7.1.8. *Self-Adjoint Operators and Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ be self-adjoint, and let $\mathcal{U}$ be a subspace of $\mathcal{V}$ invariant under T. Then:

i. $\mathcal{U}^\perp$ is invariant under T;
ii. $T|_{\mathcal{U}} \in \mathcal{L}(\mathcal{U})$ is self-adjoint;
iii. $T|_{\mathcal{U}^\perp} \in \mathcal{L}(\mathcal{U}^\perp)$ is self-adjoint.

Proof.

(i) If $\mathbf{v} \in \mathcal{U}^\perp$ and $\mathbf{u} \in \mathcal{U}$, then $\langle T\mathbf{v}, \mathbf{u} \rangle = \langle \mathbf{v}, T\mathbf{u} \rangle = 0$ because $T\mathbf{u} \in \mathcal{U}$. Hence $T\mathbf{v} \in \mathcal{U}^\perp$.
(ii) For $\mathbf{u}, \mathbf{v} \in \mathcal{U}$, $\langle (T|_{\mathcal{U}})\mathbf{u}, \mathbf{v} \rangle = \langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T\mathbf{v} \rangle = \langle \mathbf{u}, (T|_{\mathcal{U}})\mathbf{v} \rangle$. Thus $T|_{\mathcal{U}}$ is self-adjoint.
(iii) By (i), $\mathcal{U}^\perp$ is invariant, so (ii)'s argument applies to $T|_{\mathcal{U}^\perp}$.

Structural Commentary: Self-adjointness preserves orthogonal complements of invariant subspaces.
Key Insight: Move T across the inner product; invariance of $\mathcal{U}$ then forces orthogonality to persist. □

Theorem 7.1.9. *Schur Triangularisation*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. Then T has an upper-triangular matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof. Proceed by induction on $n = \dim(\mathcal{V})$. The cases $n = 0$ and $n = 1$ are immediate.
Suppose $n \geq 2$ and the result holds for smaller dimensions. By Theorem 5.3.1, choose a unit eigenvector $\mathbf{e}_1$ of T, say $T\mathbf{e}_1 = \lambda\mathbf{e}_1$. Let $\mathcal{U} = \operatorname{span}(\mathbf{e}_1)$ and define $S \in \mathcal{L}(\mathcal{U}^\perp)$ by $S\mathbf{w} = P_{\mathcal{U}^\perp}(T\mathbf{w})$.
By induction, choose an orthonormal basis $(\mathbf{e}_2, \ldots, \mathbf{e}_n)$ of $\mathcal{U}^\perp$ with respect to which S has an upper-triangular matrix. Then $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ is an orthonormal basis of $\mathcal{V}$.
For $j \geq 2$, the coordinates of $T\mathbf{e}_j$ along $\mathbf{e}_2, \ldots, \mathbf{e}_n$ are exactly the coordinates of $S\mathbf{e}_j$, hence have no entries below row j. Also $T\mathbf{e}_1 = \lambda\mathbf{e}_1$. Therefore $[T]_{\gamma \leftarrow \gamma}$ is upper triangular for $\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$.
Structural Commentary: Schur triangularisation builds an eigenline first, then triangularises the projected action on its orthogonal complement.
Key Insight: The lower-right block is controlled by the compression $P_{\mathcal{U}^\perp} T|_{\mathcal{U}^\perp}$. □

Theorem 7.1.10. *Matrix Schur Triangularisation with Ordered Diagonal*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$ in a prescribed order. If $A\mathbf{x} = \lambda_1\mathbf{x}$ and $\|\mathbf{x}\|_2 = 1$, then there is a unitary $U = [\mathbf{x}\ U_2]$ such that U^*AU is upper triangular with diagonal $\lambda_1, \ldots, \lambda_n$.
If A is real, all $\lambda_i \in \mathbb{R}$, and $\mathbf{x} \in \mathbb{R}^n$, then U may be chosen real orthogonal.

Proof. Induct on n. The case $n = 1$ is immediate. Extend $\mathbf{x}$ to an orthonormal basis and set $V = [\mathbf{x}\ V_2]$.
Since $A\mathbf{x} = \lambda_1\mathbf{x}$, the first column of V^*AV is $\lambda_1\mathbf{e}_1$, so $V^*AV = \begin{bmatrix} \lambda_1 & * \\ 0 & A' \end{bmatrix}$.
Block triangularity gives $p_{V^*AV}(z) = (z - \lambda_1)p_{A'}(z)$, so A' has eigenvalues $\lambda_2, \ldots, \lambda_n$, counted with algebraic multiplicity. By induction, choose unitary W with $W^*A'W = T'$ upper triangular and diagonal $\lambda_2, \ldots, \lambda_n$.
Then $U := V(I_1 \oplus W)$ has first column $\mathbf{x}$ and $U^*AU = \begin{bmatrix} \lambda_1 & * \\ 0 & T' \end{bmatrix}$.
The real case uses real orthonormal extensions; the block A' is real, and the same characteristic-polynomial factorization leaves the real eigenvalue list $\lambda_2, \ldots, \lambda_n$ for the induction step.
Structural Commentary: Isolate the chosen eigenline, then repeat on the lower-right block.
Key Insight: The diagonal order is controlled by which eigenline is peeled off first. □

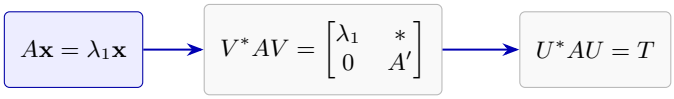


Figure 64: Schur induction: isolate an eigenline, then triangularise the compressed block.

Corollary 7.1.11. *Eigenvalues of the Inverse*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ be invertible with eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity. Then A^{-1} has eigenvalues $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$, counted with algebraic multiplicity.

Proof. Take a Schur form $A = UTU^*$ with diagonal $\lambda_1, \ldots, \lambda_n$.
Then $A^{-1} = UT^{-1}U^*$, and T^{-1} is upper triangular with diagonal $\lambda_1^{-1}, \ldots, \lambda_n^{-1}$.
Structural Commentary: Invert the Schur form and read the diagonal.
Key Insight: The inverse spectrum is the reciprocal spectrum. □

Theorem 7.1.12. *Complex Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

i. T is normal;
ii. $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
iii. T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) By Schur triangularisation (Theorem 7.1.9), choose an orthonormal basis $\gamma = (\mathbf{e}_1, \ldots, \mathbf{e}_n)$ in which $A = [T]_{\gamma \leftarrow \gamma} = [a_{ij}]$ is upper triangular. Show by induction on k that the kth row has no entries to the right of the diagonal. Assume this has been proved for rows $1, \ldots, k-1$.
Upper triangularity gives $T\mathbf{e}_k = \sum_{i \leq k} a_{ik}\mathbf{e}_i$ and $T^*\mathbf{e}_k = \sum_{j \geq k} \overline{a_{kj}}\mathbf{e}_j$.
The induction hypothesis gives $a_{ik} = 0$ for $i < k$, so $\|T\mathbf{e}_k\|^2 = |a_{kk}|^2$.
Since T is normal, Theorem 7.1.2 gives $|a_{kk}|^2 = \|T\mathbf{e}_k\|^2 = \|T^*\mathbf{e}_k\|^2 = |a_{kk}|^2 + \sum_{j>k} |a_{kj}|^2$.
Thus each a_{kj} with $j > k$ is zero. Induction makes A diagonal, so γ is an orthonormal basis of eigenvectors.

(*ii*) $\Rightarrow$ (*iii*) In an orthonormal eigenbasis, the matrix of T is diagonal.

(*iii*) $\Rightarrow$ (*i*) If $[T]$ is diagonal in an orthonormal basis, then $[T^*] = [T]^*$ is diagonal as well.
Diagonal matrices commute, so $TT^* = T^*T$; hence T is normal.

Structural Commentary: Schur gives triangular form; normality kills the upper entries.
Key Insight: Equal norms of $T\mathbf{e}_j$ and $T^*\mathbf{e}_j$ force each row tail to vanish. □

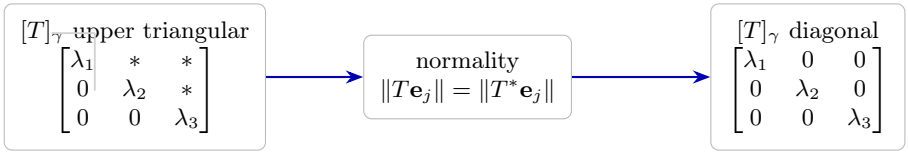


Figure 65: Normality turns Schur triangular form diagonal by forcing row tails to vanish.

Theorem 7.1.13. *Real Spectral Theorem*
Let $\mathcal{V}$ be a finite-dimensional real inner-product space and let $T \in \mathcal{L}(\mathcal{V})$. The following are equivalent:

i. T is self-adjoint;
ii. $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of T;
iii. T has a diagonal matrix with respect to some orthonormal basis of $\mathcal{V}$.

Proof.

(*i*) $\Rightarrow$ (*ii*) Assume $T = T^*$ and induct on $n = \dim(\mathcal{V})$. For $n = 0$ there is nothing to prove, and for $n = 1$ every nonzero vector is an eigenvector after normalization.
Let $n \geq 2$. Lemma 7.1.7 gives a unit eigenvector $\mathbf{u}$, and set $\mathcal{U} = \operatorname{span}(\mathbf{u})$. The subspace $\mathcal{U}$ is T-invariant. By Proposition 7.1.8, $\mathcal{U}^\perp$ is T-invariant and the restriction $T|_{\mathcal{U}^\perp}$ is self-adjoint.
Since $\dim(\mathcal{U}^\perp) = n - 1$, the induction hypothesis gives an orthonormal eigenbasis of $\mathcal{U}^\perp$.
Adjoining $\mathbf{u}$ gives an orthonormal eigenbasis of $\mathcal{V}$ by Theorem 2.1.4.
(*ii*) $\Rightarrow$ (*iii*) In an orthonormal eigenbasis, the matrix of T is diagonal.
(*iii*) $\Rightarrow$ (*i*) In an orthonormal basis where $[T]$ is real diagonal, $[T^*] = [T]^t = [T]$. Hence $T = T^*$.

Structural Commentary: Split off one eigendirection, then restrict to its orthogonal complement.
Key Insight: One eigenvector reduces the dimension. □

Perspective 7.1.14. *Normal Means Diagonal in Orthogonal Coordinates*
Complex spectral theorem states that normal operators are precisely the operators whose full structure is visible in an orthonormal eigenbasis. This is why normality is the natural coordinate-free version of being diagonal.

Perspective 7.1.15. *Real Normal Operators Use Planes*
Over $\mathbb{R}$, a normal operator need not be orthogonally diagonalizable. Its real spectral form is an orthogonal sum of invariant lines and planes, with each plane represented by $\begin{bmatrix}a & -b \\ b & a\end{bmatrix}$. See Theorem 4.5.11.

Proposition 7.1.16. *Basic Spectral Properties of Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, and let $p \in \mathbb{C}[z]$.

(i) $p(T)$ is normal and commutes with T.
(ii) T^* is normal and $E_\lambda(T) = E_{\overline{\lambda}}(T^*)$ for every $\lambda \in \mathbb{C}$.
(iii) $\ker T = \ker T^*$ and $\operatorname{range}(T) = \operatorname{range}(T^*)$.

Proof. Choose an orthonormal eigenbasis $\mathbf{e}_1, \ldots, \mathbf{e}_n$ with $T\mathbf{e}_j = \mu_j \mathbf{e}_j$.

(i) $p(T)\mathbf{e}_j = p(\mu_j)\mathbf{e}_j$, $p(T)$ is diagonal in this basis, hence normal. It commutes with T by Proposition 6.1.3.
(ii) Since $T^*\mathbf{e}_j = \overline{\mu_j}\mathbf{e}_j$, T^* is diagonal in the same basis, hence normal. Also $T\mathbf{v} = \lambda\mathbf{v}$ iff $T^*\mathbf{v} = \overline{\lambda}\mathbf{v}$.
(iii) Part (ii) with $\lambda = 0$ gives $\ker T = \ker T^*$. Both ranges are $\operatorname{span}\{\mathbf{e}_j : \mu_j \neq 0\}$.

Structural Commentary: Normality gives one orthonormal basis that diagonalises both T and T^*.
Key Insight: Kernel and range are read from the nonzero diagonal entries in that basis. □

Corollary 7.1.17. *Normal Operators Have No Nilpotent Primary Part*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$ be normal.

(i) If $S = p(T)$ for some $p \in \mathbb{C}[z]$, then $\ker(S^k) = \ker(S)$ for every $k \geq 1$.
(ii) The minimal polynomial of T is square-free, where $\mu_T(z) = \prod_{\lambda \in \operatorname{spec}(T)} (z - \lambda)$.

Proof. Let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be an orthonormal eigenbasis with $T\mathbf{e}_j = \lambda_j \mathbf{e}_j$.

(i) For $S = p(T)$, $k \geq 1$, $S^k\mathbf{e}_j = p(\lambda_j)^k \mathbf{e}_j$; $\ker(S^k) = \operatorname{span}\{\mathbf{e}_j : p(\lambda_j)^k = 0\} = \operatorname{span}\{\mathbf{e}_j : p(\lambda_j) = 0\} = \ker(S)$.

(ii) Let $m(z) = \prod_{\lambda \in \operatorname{spec}(T)} (z - \lambda)$. Since $m(T)\mathbf{e}_j = 0$ for all j, $m(T) = 0$, so $\mu_T \mid m$. Conversely, every $\lambda \in \operatorname{spec}(T)$ is a root of μ_T by Theorem 6.3.12, hence $m \mid \mu_T$. Since both polynomials are monic, $\mu_T = m$.

Structural Commentary: Normality makes every primary component an eigenspace.
Key Insight: A diagonal operator has no hidden nilpotent part. □

Theorem 7.1.18. *Spectral Resolution of a Normal Operator*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $\mu_1, \dots, \mu_d$ be distinct scalars. For $T \in \mathcal{L}(\mathcal{V})$, consider:

(i) T is normal and $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$.
(ii) $\exists$ pairwise orthogonal projections $P_1, \dots, P_d \in \mathcal{L}(\mathcal{V})$ such that $P_i P_j = 0$ for $i \neq j$, $P_1 + \cdots + P_d = I_{\mathcal{V}}$, and $T = \mu_1 P_1 + \cdots + \mu_d P_d$
(iii) Any projections satisfying (ii) obey $\operatorname{range}(P_j) = E_{\mu_j}(T)$ whenever $\mu_j \in \operatorname{spec}(T)$.

Then (i) and (ii) are equivalent, and any projections satisfying (ii) satisfy (iii).

Proof.

(i) $\Rightarrow$ (ii). By Theorem 7.1.12, $\mathcal{V} = \bigoplus_{\lambda \in \operatorname{spec}(T)} E_\lambda(T)$ as an orthogonal direct sum.

Let P_j be the orthogonal projection onto $E_{\mu_j}(T)$, and set $P_j = 0$ when $\mu_j \notin \operatorname{spec}(T)$. Since $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$, these projections are pairwise orthogonal, sum to $I_{\mathcal{V}}$, and satisfy $T = \sum_j \mu_j P_j$.

(ii) $\Rightarrow$ (i). Since $P_j^* = P_j$ and $P_i P_j = 0$ for $i \neq j$, $T^* = \sum_{j=1}^{d} \overline{\mu_j} P_j$ and $T^* T = \sum_{j=1}^{d} |\mu_j|^2 P_j = T T^*$.

Thus T is normal. If $\lambda \in \operatorname{spec}(T)$ and $\mathbf{0} \neq \mathbf{x} \in E_\lambda(T)$, then $0 = (T - \lambda I_{\mathcal{V}})\mathbf{x} = \sum_{j=1}^{d} (\mu_j - \lambda) P_j \mathbf{x}$.

The summands lie in the pairwise orthogonal ranges $\operatorname{range}(P_j)$, hence $0 = \sum_{j=1}^{d} |\mu_j - \lambda|^2 \, \|P_j \mathbf{x}\|^2$.

Because $\sum_j P_j = I_{\mathcal{V}}$, at least one $P_k \mathbf{x}$ is nonzero, so $\mu_k = \lambda$. Thus $\operatorname{spec}(T) \subseteq \{\mu_1, \dots, \mu_d\}$.

(ii) $\Rightarrow$ (iii). For $\mathbf{x} \in \mathcal{V}$, $\mathbf{x} = \sum_{i=1}^{d} P_i \mathbf{x}$ and $(T - \mu_j I_{\mathcal{V}})\mathbf{x} = \sum_{i \neq j} (\mu_i - \mu_j) P_i \mathbf{x}$.

The summands are orthogonal, so $\|(T - \mu_j I_{\mathcal{V}})\mathbf{x}\|^2 = \sum_{i \neq j} |\mu_i - \mu_j|^2 \, \|P_i \mathbf{x}\|^2$, which vanishes exactly when $P_i \mathbf{x} = 0$ for all $i \neq j$, equivalently $\mathbf{x} = P_j \mathbf{x}$. Thus $E_{\mu_j}(T) = \operatorname{range}(P_j)$.

Structural Commentary: A normal operator decomposes into scalar actions on orthogonal eigenspaces.
Key Insight: Spectral projections isolate eigenspaces; the operator is the weighted sum of those projections. □



Figure 66: Spectral resolution: $\mathcal{V} = \bigoplus_j \operatorname{range}(P_j)$, $P_i P_j = 0$ for $i \neq j$, $\sum_j P_j = I_{\mathcal{V}}$, and $T = \sum_j \mu_j P_j$.

Perspective 7.1.19. *Spectral Resolutions Are Orthogonal Resolutions*
For the projections P_j in Theorem 7.1.18, $P_j = P_j^* = P_j^2$; $P_i P_j = 0$ $(i \neq j)$; $\sum_j P_j = I_{\mathcal{V}}$.
Hence $\mathcal{V} = \bigoplus_j^{\perp} \operatorname{range}(P_j)$ is the orthogonal decomposition into T-invariant spectral subspaces.
Structural Commentary: Each P_j is forced by its eigenspace: it is the orthogonal projection onto it.
Key Insight: Normality turns diagonalization into a projection-valued decomposition.

Corollary 7.1.20. *Functional Calculus from Spectral Resolution*
Let $T = \sum_{j=1}^{d} \mu_j P_j$ be the spectral resolution of a normal operator and let $f \in \mathbb{C}[z]$. Then $f(T) = \sum_{j=1}^{d} f(\mu_j) P_j$. In particular, for each j there is a polynomial p_j such that $p_j(T) = P_j$.

Proof. Since $P_i P_j = 0$ for $i \neq j$ and $P_j^2 = P_j$, induction gives $T^m = \sum_j \mu_j^m P_j$ for every $m \geq 0$.
Linearity then gives the displayed formula for every polynomial f.
For the final assertion, choose the Lagrange polynomial p_j satisfying $p_j(\mu_i) = \delta_{ij}$. Then $p_j(T) = P_j$.
Structural Commentary: Orthogonal spectral projections multiply like coordinate idempotents.
Key Insight: Polynomial functions of a normal operator act by scalar functions on each eigenspace. □

7.2 Normal Operator Consequences

Lemma 7.2.1. *Adjoints of Normal Matrices as One Polynomial*
Let $A_1, \dots, A_k$ be normal complex matrices. Then $\exists$ a polynomial $p \in \mathbb{C}[z]$ s.t. $p(A_j) = A_j^* \ \forall j = 1, \dots, k$.

Proof. Let $\Omega = \bigcup_{j=1}^{k} \operatorname{spec}(A_j)$. Since Ω is finite, choose a Lagrange interpolation polynomial p satisfying $p(\lambda) = \overline{\lambda}$ for every $\lambda \in \Omega$. For each A_j, Corollary 7.1.20 gives $p(A_j) = A_j^*$.
Structural Commentary: One polynomial interpolates conjugation on all spectra.
Key Insight: For normal matrices, A^* is $p(A)$ with $p(\lambda) = \overline{\lambda}$. □

Theorem 7.2.2. *Fuglede–Putnam Theorem*
Let $A \in \operatorname{Mat}(m, m, \mathbb{C})$ and $B \in \operatorname{Mat}(n, n, \mathbb{C})$ be normal, and let $X \in \operatorname{Mat}(m, n, \mathbb{C})$.
Then $AX = XB \iff A^*X = XB^*$.

Proof.
(⇒) Assume $AX = XB$. Then $A^rX = XB^r$ for every $r \geq 0$, so $p(A)X = Xp(B)$ for every polynomial p. By Lemma 7.2.1, choose p with $p(A) = A^*$ and $p(B) = B^*$. Thus $A^*X = XB^*$.
(⇐) Assume $A^*X = XB^*$. Apply the same argument to the normal matrices A^* and B^*, choosing q with $q(A^*) = A$ and $q(B^*) = B$. Hence $AX = XB$.

Structural Commentary: An intertwining relation passes through every polynomial in the two normal matrices.
Key Insight: Normality makes adjoints polynomial, so intertwining transfers to adjoints. □

Method 7.2.3. *Finding a Spectral Decomposition of a Normal Matrix*
For a normal matrix $A \in \operatorname{Mat}(n, n, \mathbb{C})$:

i. Find the distinct eigenvalues $\mu_1, \dots, \mu_d$.
ii. For each j, solve $(A - \mu_j I_n)\mathbf{x} = 0$ to get $E_{\mu_j}(A)$.
iii. Orthonormalise a basis of each eigenspace.
iv. Concatenate these orthonormal bases as the columns of U; then U is unitary and $A = U\Lambda U^*$, where Λ repeats each μ_j according to $\dim E_{\mu_j}(A)$.

Theorem 7.2.4. *Spectral Classification of Normal Operators*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $T \in \mathcal{L}(\mathcal{V})$ be normal.

(i) T is unitary if and only if $|\lambda| = 1$ for every $\lambda \in \operatorname{spec}(T)$.
(ii) T is self-adjoint if and only if $\operatorname{spec}(T) \subseteq \mathbb{R}$.
(iii) T is an orthogonal projection if and only if $\operatorname{spec}(T) \subseteq \{0, 1\}$.
(iv) T is skew-adjoint, meaning $T^* = -T$, if and only if every $\lambda \in \operatorname{spec}(T)$ is purely imaginary.

Proof. Write the spectral resolution $T = \sum_j \mu_j P_j$, where the μ_j are the distinct eigenvalues and $P_j \neq 0$.
Then $I_{\mathcal{V}} = \sum_j P_j$, $T^* = \sum_j \overline{\mu_j} P_j$, $T^*T = \sum_j |\mu_j|^2 P_j$, and $T^2 = \sum_j \mu_j^2 P_j$.
Since ranges of P_j are nonzero and independent, an identity $\sum_j \alpha_j P_j = 0$ holds only when every $\alpha_j = 0$.
(i) T is unitary iff $T^*T = I_{\mathcal{V}} = \sum_j P_j$, iff $|\mu_j| = 1$ for every j.
(ii) T is self-adjoint iff $T = T^*$, iff $\mu_j = \overline{\mu_j}$ for every j.
(iii) T is an orthogonal projection iff $T = T^*$ and $T^2 = T$, iff $\mu_j = \overline{\mu_j}$ and $\mu_j^2 = \mu_j \ \forall j$, iff $\mu_j \in \{0, 1\} \ \forall j$.
(iv) $T^* = -T$ iff $\overline{\mu_j} = -\mu_j$ for every j, iff every μ_j is purely imaginary.

Structural Commentary: For normal operators, structural classes are scalar conditions on spectral blocks.
Key Insight: For normal T, these properties reduce to scalar tests. □

Proposition 7.2.5. *Cartesian Decomposition and Normality*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space.
For every $T \in \mathcal{L}(\mathcal{V})$, there are unique self-adjoint operators $H, K \in \mathcal{L}(\mathcal{V})$ such that $T = H + iK$.
They are $H = \frac{1}{2}(T + T^*)$ and $K = \frac{1}{2i}(T - T^*)$. Moreover, T is normal if and only if $HK = KH$.

Proof. The displayed formulas give self-adjoint H, K and $T = H + iK$. If $T = X + iY$ with X, Y self-adjoint, then $T^* = X - iY$, so $X = \frac{1}{2}(T + T^*)$ and $Y = \frac{1}{2i}(T - T^*)$; uniqueness follows.
For $T = H + iK$, $T^*T = (H - iK)(H + iK) = H^2 + K^2 + i(HK - KH)$,
whereas $TT^* = (H + iK)(H - iK) = H^2 + K^2 - i(HK - KH)$. Thus $T^*T = TT^*$ if and only if $HK = KH$.
Structural Commentary: Every complex operator splits into Hermitian and skew-Hermitian parts.
Key Insight: Normality says the two Hermitian coordinates of T commute. □

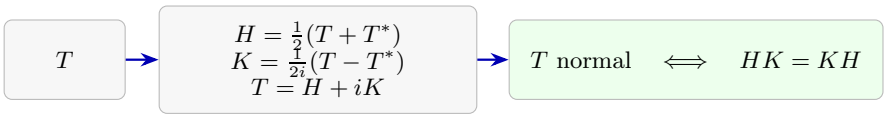


Figure 67: Cartesian decomposition: T is normal exactly when $HK = KH$.

Lemma 7.2.6. *Upper-Triangular Normal Matrices Are Diagonal*
If $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is upper triangular and normal, then A is diagonal.

Proof. Induct on n. The case $n = 1$ is immediate.
Since A is upper triangular, $A\mathbf{e}_1 = a_{11}\mathbf{e}_1$, while $A^*\mathbf{e}_1 = \overline{a_{11}}\mathbf{e}_1 + \overline{a_{12}}\mathbf{e}_2 + \cdots + \overline{a_{1n}}\mathbf{e}_n$.
Normality gives $\|A\mathbf{e}_1\|_2 = \|A^*\mathbf{e}_1\|_2$, hence $a_{12} = \cdots = a_{1n} = 0$. Thus $A = a_{11} \oplus B$ with B upper triangular.
The block diagonal normality equation gives $B^*B = BB^*$, so B is normal. By induction, B is diagonal.
Structural Commentary: Normality forbids upper-triangular row tails.
Key Insight: Compare the lengths of $A\mathbf{e}_1$ and $A^*\mathbf{e}_1$ to kill the first row, then induct. □

Proposition 7.2.7. *Normal Block Upper-Triangular Matrices Collapse*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $A = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix} \in \mathrm{Mat}(n, n, \mathbb{F})$, where B and C are square.
Then A is normal if and only if $X = 0$ and both B and C are normal.

Proof.
($\Rightarrow$) Suppose A is normal. Comparing the $(1,1)$ blocks of A^*A and AA^* gives $B^*B = BB^* + XX^*$.
Taking traces and using cyclicity, $\mathrm{tr}(XX^*) = \mathrm{tr}(B^*B) - \mathrm{tr}(BB^*) = 0$. Since $\mathrm{tr}(XX^*) = \|X\|_F^2$, we get $X = 0$. Thus $A = B \oplus C$, and $A^*A = AA^*$ reduces blockwise to $B^*B = BB^*$ and $C^*C = CC^*$.
($\Leftarrow$) If $X = 0$ and B, C are normal, then $A = B \oplus C$, so blockwise multiplication gives $A^*A = AA^*$.
Structural Commentary: A normal operator cannot have one-way coupling between orthogonal blocks.
Key Insight: The off-diagonal block contributes the nonnegative trace $\|X\|_F^2$, so normality forces it to vanish. □

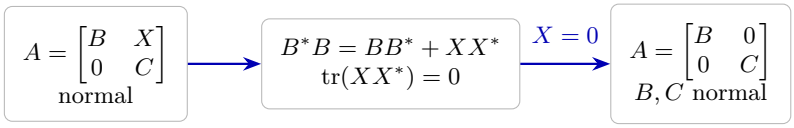


Figure 68: Normality removes the upper-right block in a block upper-triangular matrix.

Theorem 7.2.8. *Invariant Subspaces of Normal Matrices Reduce*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ be normal, and let $\mathcal{U} \subseteq \mathbb{F}^n$ be invariant under A. Then $\mathcal{U}$ is invariant under A^*. If P is the orthogonal projection onto $\mathcal{U}$, then $AP = PA$.

Proof. The cases $\mathcal{U} = \{0\}$ and $\mathcal{U} = \mathbb{F}^n$ are immediate. Otherwise choose a unitary $Q = [Q_1\ Q_2]$ whose columns form orthonormal bases of $\mathcal{U}$ and $\mathcal{U}^\perp$. Invariance of $\mathcal{U}$ says AQ_1 has columns in $\mathcal{U} = \mathrm{col}(Q_1)$, hence $Q_2^*AQ_1 = 0$.
Therefore $Q^*AQ = \begin{bmatrix} B & X \\ 0 & C \end{bmatrix}$. This matrix is normal because it is unitarily similar to A.
Proposition 7.2.7 gives $X = 0$, so $Q^*AQ = B \oplus C$. Hence $Q^*A^*Q = B^* \oplus C^*$, which shows $\mathcal{U}$ is A^*-invariant.
Also $Q^*PQ = I_{\dim(\mathcal{U})} \oplus 0$, so Q^*AQ commutes with Q^*PQ; conjugating back gives $AP = PA$.
Structural Commentary: In adapted coordinates, normal invariance becomes block diagonality.
Key Insight: Normal invariant subspaces are automatically reducing subspaces. □

Corollary 7.2.9. *Invariant Subspaces of Normal Operators Reduce*
Let $\mathcal{V}$ be a finite-dimensional inner-product space, let $T \in \mathcal{L}(\mathcal{V})$ be normal, let $\mathcal{U}$ be invariant under T. Then:

(i) $\mathcal{U}^\perp$ is invariant under T;
(ii) $\mathcal{U}$ is invariant under T^*;
(iii) $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$;
(iv) $T|_{\mathcal{U}}$ and $T|_{\mathcal{U}^\perp}$ are normal.

Proof. Choose an orthonormal basis adapted to $\mathcal{U} \oplus \mathcal{U}^\perp$.
In these coordinates, Theorem 7.2.8 makes the matrix of T block diagonal, say $A = B \oplus C$.
The block diagonal form shows $\mathcal{U}^\perp$ is invariant under T and $\mathcal{U}$ is invariant under T^*. It also gives $(T|_{\mathcal{U}})^* = (T^*)|_{\mathcal{U}}$ because the adjoint block on $\mathcal{U}$ is B^*. Finally, normality of A reduces blockwise to normality of B and C.

Structural Commentary: Normal operator turns every invariant subspace into an orthogonal reducing summand.
Key Insight: Normality kills the one-way coupling between $\mathcal{U}$ and $\mathcal{U}^{\perp}$. □

Proposition 7.2.10. *Coprime Normal Primary Pieces Are Orthogonal*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space, and let $\mathcal{S}, \mathcal{R} \leq \mathcal{V}$ be T-invariant. If $p, q \in \mathbb{C}[z]$, $(p, q) = 1$, $p(T)|_{\mathcal{S}} = 0$, and $q(T)|_{\mathcal{R}} = 0$, then $\mathcal{S} \perp \mathcal{R}$.

Proof. Choose $a, b \in \mathbb{C}[z]$ with $ap + bq = 1$. For $\mathbf{s} \in \mathcal{S}$, $\mathbf{s} = a(T)p(T)\mathbf{s} + b(T)q(T)\mathbf{s} = b(T)q(T)\mathbf{s}$.
By Corollary 7.2.9, $\mathcal{R}$ reduces T. Hence $b(T)^*\mathcal{R} \subseteq \mathcal{R}$ and $q(T)^*|_{\mathcal{R}} = (q(T)|_{\mathcal{R}})^* = 0$.
Thus, for $\mathbf{r} \in \mathcal{R}$, $\langle \mathbf{s}, \mathbf{r} \rangle = \langle b(T)q(T)\mathbf{s}, \mathbf{r} \rangle = \langle \mathbf{s}, q(T)^* b(T)^* \mathbf{r} \rangle = 0$.
Structural Commentary: Coprime annihilators orthogonally separate normal invariant pieces.
Key Insight: Normality makes invariant subspaces reduce, so annihilation also holds after taking adjoints. □

Theorem 7.2.11. *Simultaneous Unitary Diagonalisation of Commuting Normal Operators*
Let $\emptyset \neq \mathcal{F} \subseteq \mathcal{L}(\mathcal{V})$ be a normal family on a finite-dimensional complex inner-product space. Then $\mathcal{F}$ is pairwise commuting if and only if $\mathcal{V}$ has an orthonormal basis of common eigenvectors for $\mathcal{F}$.

Proof.

(⇐) In a common orthonormal eigenbasis, every $A \in \mathcal{F}$ is diagonal. Hence all operators in $\mathcal{F}$ commute.

(⇒) Induct on $\dim(\mathcal{V})$. The zero-dimensional case is vacuous. If every $A \in \mathcal{F}$ is scalar, any orthonormal basis works. Otherwise choose a nonscalar $A_0 \in \mathcal{F}$ and write its spectral resolution as $A_0 = \sum_j \mu_j P_j$.
Since A_0 is nonscalar, at least two nonzero ranges $\operatorname{range}(P_j)$ occur, and each has dimension strictly smaller than $\dim(\mathcal{V})$. By Corollary 7.1.20, each P_j is a polynomial in A_0.
If $B \in \mathcal{F}$, then $BA_0 = A_0B$, so B commutes with every polynomial in A_0, in particular with P_j. Hence $B(\operatorname{range}(P_j)) \subseteq \operatorname{range}(P_j)$ because $P_jB\mathbf{x} = BP_j\mathbf{x} = B\mathbf{x}$ whenever $P_j\mathbf{x} = \mathbf{x}$.
The same commutation preserves $\ker(P_j) = \operatorname{range}(P_j)^{\perp}$.
In the decomposition $\mathcal{V} = \operatorname{range}(P_j) \oplus \ker(P_j)$, the matrix of B is therefore block diagonal; since B is normal, its diagonal restrictions are normal block by block.
Thus, on each nonzero $\operatorname{range}(P_j)$, the restricted family is again pairwise commuting and normal.
By induction it has a common orthonormal eigenbasis. Concatenating these bases over all j gives a common orthonormal eigenbasis of $\mathcal{V}$.

Structural Commentary: Spectral projections reduce the whole commuting family.
Key Insight: Diagonalise one nonscalar member, then induct on its eigenspaces. □

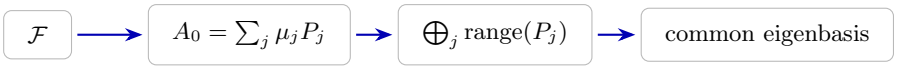


Figure 69: For $A_0 = \sum_j \mu_j P_j$, commutativity gives $BP_j = P_jB$; the ranges reduce $\mathcal{F}$, and induction gives a common orthonormal eigenbasis.

Corollary 7.2.12. *Sums and Products of Commuting Normal Operators*
Let $S, T \in \mathcal{L}(\mathcal{V})$ be commuting normal operators on a finite-dimensional complex inner-product space. Then $S + T$ and ST are normal.

Proof. By Theorem 7.2.11, S and T are diagonal in a common orthonormal basis.
Then $S + T$ and ST are also diagonal in that basis, hence normal.
Structural Commentary: Commuting normal operators share orthonormal coordinates.
Key Insight: In common diagonal form, sums and products remain diagonal. □

Theorem 7.2.13. *Normality Transfers from AB to BA*
Let $A, B \in \operatorname{Mat}(n, n, \mathbb{C})$. If A, B, and AB are normal, then BA is normal.

Proof. By Theorem 6.6.12, AB and BA have the same eigenvalues counted with algebraic multiplicity. Also,

$$\begin{aligned}\operatorname{tr}\big((BA)^*(BA)\big) &= \operatorname{tr}(A^*B^*BA) = \operatorname{tr}(AA^*B^*B) \\ &= \operatorname{tr}(A^*AB^*B) = \operatorname{tr}(A^*ABB^*) = \operatorname{tr}(B^*A^*AB) = \operatorname{tr}\big((AB)^*(AB)\big),\end{aligned}$$

using cyclicity of trace and normality of A and B. Thus $\|BA\|_F^2 = \|AB\|_F^2$, so $\Delta(BA) = \Delta(AB)$.
Since AB is normal, $\Delta(AB) = 0$; hence $\Delta(BA) = 0$, and Theorem 5.1.9 gives that BA is normal.
Structural Commentary: Same spectrum and Frobenius norm give the same Schur defect.
Key Insight: Schur defect turns normality of AB into normality of BA. □

Proposition 7.2.14. *Permutation Similarity Reorders Diagonal Entries*
Let $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ and let $P = [\mathbf{e}_{\sigma(1)} \ \cdots \ \mathbf{e}_{\sigma(n)}]$ be a permutation matrix. Then $P^*\Lambda P = \operatorname{diag}(\lambda_{\sigma(1)}, \dots, \lambda_{\sigma(n)})$.

Proof. The (i,j) entry of $P^*\Lambda P$ is $\mathbf{e}_{\sigma(i)}^* \Lambda \mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)} \mathbf{e}_{\sigma(i)}^* \mathbf{e}_{\sigma(j)} = \lambda_{\sigma(j)} \delta_{ij}$.
Thus all off-diagonal entries are zero, and the jth diagonal entry is $\lambda_{\sigma(j)}$.
Structural Commentary: Permutation similarity applies the same basis reordering to rows and columns.
Key Insight: Diagonal entries move exactly as the standard basis vectors are rearranged. □

Corollary 7.2.15. *Normal Matrices with the Same Spectrum Are Unitarily Similar*
If $A, B \in \operatorname{Mat}(n, n, \mathbb{C})$ are normal and have the same eigenvalues counted with algebraic multiplicity, then A and B are unitarily similar.

Proof. By Theorem 7.1.12, write $A = U\Lambda U^*$ and $B = V\Gamma V^*$ with U, V unitary and Λ, Γ diagonal.
Since eigenvalues agree with multiplicity, Proposition 7.2.14 gives a permutation matrix P such that $\Gamma = P^*\Lambda P$.
Hence $B = (VP^*U^*)A(UPV^*)$, and UPV^* is unitary.
Structural Commentary: Normal matrices are classified up to unitary similarity by their eigenvalue multiset.
Key Insight: After diagonalising both matrices, only a permutation of diagonal entries remains. □

Theorem 7.2.16. *Description of Complex Isometries*
Let $\mathcal{V}$ be a finite-dimensional complex inner-product space and let $S \in \mathcal{L}(\mathcal{V})$. Then S is an isometry if and only if $\mathcal{V}$ has an orthonormal basis consisting of eigenvectors of S whose eigenvalues all have absolute value 1.

Proof. If $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$ in an orthonormal basis with $|\lambda_j| = 1$, then for $\mathbf{v} = \sum_j c_j \mathbf{e}_j$,
$\|S\mathbf{v}\|^2 = \sum_j |\lambda_j c_j|^2 = \sum_j |c_j|^2 = \|\mathbf{v}\|^2$, so S is an isometry. Conversely, if S is an isometry, then $S^*S = I_{\mathcal{V}}$ by Theorem 2.4.26. Finite-dimensional invertibility gives $SS^* = I_{\mathcal{V}}$, so S is normal. By Complex Spectral Theorem, choose orthonormal eigenbasis $S\mathbf{e}_j = \lambda_j \mathbf{e}_j$. Since $\mathbf{e}_j$ is unit and S preserves norms, $|\lambda_j| = \|S\mathbf{e}_j\| = 1$.
Structural Commentary: Complex isometries are normal with unit-circle spectrum.
Key Insight: In an eigenbasis, isometry means $|\lambda_j| = 1$. □

7.3 Operator Functions

Operator functions are defined by applying scalar functions to spectral data.
Corollary 7.1.20 gives $f(T) = \sum_j f(\mu_j) P_j$ for normal operators; this subsection extends that rule to diagonalizable and Jordan forms, with nilpotent terms where needed.

Theorem 7.3.1. *Polynomial Functional Calculus for Diagonalizable Matrices*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be diagonalizable and write $A = S\Lambda S^{-1}$, where $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$.
If $p \in \mathbb{F}[z]$, then $p(A) = S \operatorname{diag}(p(\lambda_1), \dots, p(\lambda_n)) S^{-1}$.

Proof. For every $j \geq 0$, $A^j = S\Lambda^j S^{-1}$.

If $p(z) = \sum_j c_j z^j$, then $p(A) = \sum_j c_j A^j = S \left(\sum_j c_j \Lambda^j \right) S^{-1} = Sp(\Lambda) S^{-1}$.

Since Λ is diagonal, $p(\Lambda) = \operatorname{diag}(p(\lambda_1), \dots, p(\lambda_n))$.
Structural Commentary: Diagonalization reduces $p(A)$ to scalar evaluation on eigenvalues.
Key Insight: Compute functions of A by applying the function to the eigenvalues. □

Definition 7.3.2. *Spectral Resolution Notation*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space.
Define $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$, where $\Omega_T = \operatorname{spec}(T)$.

(i) *Spectral Set*: $\Omega_T = \operatorname{spec}(T)$.
(ii) *Spectral Projection*: P_λ is the orthogonal projection onto $E_\lambda(T)$.
(iii) *Spectral Resolution*: $\{P_\lambda\}_{\lambda \in \Omega_T}$, equivalently $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$.

Theorem 7.3.3. *Finite Spectral Functional Calculus for Normal Operators*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal, with spectral resolution $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$.
The finite spectral functional calculus is $\mathbb{C}^{\Omega_T} \to \mathcal{L}(\mathcal{V})$, where $f \mapsto f(T) := \sum_{\lambda \in \Omega_T} f(\lambda) P_\lambda$.
If $p \in \mathbb{C}[z]$ and $p|_{\Omega_T} = f$, then $p(T) = f(T)$. Hence every $f \in \mathbb{C}^{\Omega_T}$ has a polynomial representative in T.

Proof. For $p|_{\Omega_T} = f$, Corollary 7.1.20 gives $p(T) = \sum_{\lambda\in\Omega_T} p(\lambda)P_\lambda = \sum_{\lambda\in\Omega_T} f(\lambda)P_\lambda = f(T)$.
Since Ω_T is finite, Lagrange interpolation gives $p \in \mathbb{C}[z]$ with $p|_{\Omega_T} = f$.
Structural Commentary: Spectral sums reduce finite functions to polynomial evaluation.
Key Insight: Finite spectral data is interpolated by one polynomial. □

Theorem 7.3.4. *Rules for Normal Spectral Functions*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal, with spectral resolution $T = \sum_{\lambda\in\Omega_T} \lambda P_\lambda$. For $f, g : \Omega_T \to \mathbb{C}$:

(i) $(f+g)(T) = f(T) + g(T)$ *and* $(fg)(T) = f(T)g(T)$;
(ii) $f(T)^* = \overline{f}(T)$, *where* $\overline{f}(\lambda) = \overline{f(\lambda)}$;
(iii) $f(T)$ *is normal and* $\operatorname{spec}(f(T)) = f(\Omega_T)$.

Proof.

(i) Since $P_\lambda P_\mu = \delta_{\lambda\mu}P_\lambda$, $f(T)g(T) = \sum_{\lambda\in\Omega_T} f(\lambda)g(\lambda)P_\lambda = (fg)(T)$; and $(f+g)(T) = f(T) + g(T)$.

(ii) Since $P_\lambda^* = P_\lambda$, $f(T)^* = \sum_{\lambda\in\Omega_T} \overline{f(\lambda)}P_\lambda = \overline{f}(T)$.

(iii) For each $\lambda \in \Omega_T$, $f(T)|_{\operatorname{range}(P_\lambda)} = f(\lambda)I$.
Thus $f(T)$ is diagonal in an orthonormal eigenbasis, hence normal, and $\operatorname{spec}(f(T)) = f(\Omega_T)$.

Structural Commentary: Spectral projections make operator rules pointwise.
Key Insight: Spectral functions act as scalars on spectral subspaces. □

Definition 7.3.5. *Positive Operator*
$T \in \mathcal{L}(\mathcal{V})$ is *positive* if $T = T^*$ and $\langle T\mathbf{v}, \mathbf{v}\rangle \geq 0$ for all $\mathbf{v} \in \mathcal{V}$.

Definition 7.3.6. *Square Root of an Operator*
An operator $R \in \mathcal{L}(\mathcal{V})$ is a *square root* of $T \in \mathcal{L}(\mathcal{V})$ if $R^2 = T$.

Theorem 7.3.7. *Characterization of Positive Operators*
For $T \in \mathcal{L}(\mathcal{V})$, the following are equivalent:

i. T *is positive;*
ii. T *is self-adjoint and every eigenvalue of T is nonnegative;*
iii. T *has a positive square root;*
iv. T *has a self-adjoint square root;*
v. there exists $R \in \mathcal{L}(\mathcal{V})$ such that $T = R^*R$.

Proof.

(i) $\Rightarrow$ *(ii)* If $T\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$, then $\lambda \in \mathbb{R}$ by Theorem 7.1.5, and $0 \leq \langle T\mathbf{v}, \mathbf{v}\rangle = \lambda\|\mathbf{v}\|^2$, so $\lambda \geq 0$.
(ii) $\Rightarrow$ *(iii)* By the real or complex spectral theorem, choose an orthonormal eigenbasis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ with $T\mathbf{e}_j = \lambda_j\mathbf{e}_j$ and $\lambda_j \geq 0$. Define $S\mathbf{e}_j = \sqrt{\lambda_j}\mathbf{e}_j$ and extend linearly.
Then S is diagonal with nonnegative real diagonal entries in an orthonormal basis, hence self-adjoint and positive, and $S^2\mathbf{e}_j = \lambda_j\mathbf{e}_j = T\mathbf{e}_j$ for every j. Thus $S^2 = T$.
(iii) $\Rightarrow$ *(iv)* A positive square root is self-adjoint.
(iv) $\Rightarrow$ *(v)* If $T = R^2$ with $R = R^*$, then $T = R^*R$.
(v) $\Rightarrow$ *(i)* If $T = R^*R$, then $T^* = T$ and $\langle T\mathbf{v}, \mathbf{v}\rangle = \langle R\mathbf{v}, R\mathbf{v}\rangle = \|R\mathbf{v}\|^2 \geq 0$ for every $\mathbf{v}$, so T is positive.

Structural Commentary: Positivity equals nonnegative spectrum and square-root factorization.
Key Insight: In an eigenbasis, the proof is scalar: $\lambda \geq 0$ iff $\sqrt{\lambda}$ exists. □

Proposition 7.3.8. *Unique Positive Square Root*
Every positive operator on $\mathcal{V}$ has a unique positive square root.

Proof. Existence follows from Theorem 7.3.7. For uniqueness, decompose $\mathcal{V} = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T)$.
Let R be a positive square root of T. Since $R^2 = T$, we have $RT = R^3 = TR$, so each $E_{\lambda_j}(T)$ is invariant under R. On $E_{\lambda_j}(T)$, the restriction $R_j := R|_{E_{\lambda_j}(T)}$ is positive and satisfies $R_j^2 = \lambda_j I$.
By the spectral theorem for the positive operator R_j, every eigenvalue ρ of R_j satisfies $\rho \geq 0$ and $\rho^2 = \lambda_j$, hence $\rho = \sqrt{\lambda_j}$. Therefore $R_j = \sqrt{\lambda_j}I$ on $E_{\lambda_j}(T)$.
Thus every positive square root agrees with the same scalar operator on each eigenspace $E_{\lambda_j}(T)$, so it is unique.
Structural Commentary: Each eigenspace determines its own scalar root.
Key Insight: Positivity selects the nonnegative root. □

Definition 7.3.9. *Positive Square Root Notation*
If T is positive, then $\sqrt{T}$ denotes the unique positive square root of T.

Example 7.3.10. *Basic Functions from the Spectral Calculus*
Let $T \in \mathcal{L}(\mathcal{V})$ be normal, with spectral resolution $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$. Then $T^* = \sum_{\lambda \in \Omega_T} \overline{\lambda} P_\lambda$.
If $0 \notin \Omega_T$, then $T^{-1} = \sum_{\lambda \in \Omega_T} \lambda^{-1} P_\lambda$. If $T \geq 0$, then $T^{1/2} = \sum_{\lambda \in \Omega_T} \lambda^{1/2} P_\lambda$, where $\lambda^{1/2} \geq 0$.

Theorem 7.3.11. *Commutativity via Spectral Projections*
Let $T, S \in \mathcal{L}(\mathcal{V})$ be normal, with spectral resolutions $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$ and $S = \sum_{\mu \in \Omega_S} \mu Q_\mu$.

(i) For $R \in \mathcal{L}(\mathcal{V})$, $RT = TR$ if and only if $RP_\lambda = P_\lambda R$ for all $\lambda \in \Omega_T$.
(ii) $TS = ST$ if and only if $P_\lambda Q_\mu = Q_\mu P_\lambda$ for all $\lambda \in \Omega_T$ and $\mu \in \Omega_S$.
(iii) If $f : \Omega_T \to \mathbb{C}$ and $g : \Omega_S \to \mathbb{C}$ are injective, then $f(T)g(S) = g(S)f(T)$ if and only if $TS = ST$.

Proof.

(i). If $RT = TR$, then $Rp(T) = p(T)R$ for all $p \in \mathbb{C}[z]$. Since $P_\lambda \in \mathbb{C}[T]$ by Corollary 7.1.20, $RP_\lambda = P_\lambda R$. Conversely, if $RP_\lambda = P_\lambda R$ for all $\lambda \in \Omega_T$, then $RT = \sum_{\lambda \in \Omega_T} \lambda R P_\lambda = \sum_{\lambda \in \Omega_T} \lambda P_\lambda R = TR$.

(ii). If $TS = ST$, then (i) with $R = S$ gives $SP_\lambda = P_\lambda S$. Since $Q_\mu \in \mathbb{C}[S]$, $P_\lambda Q_\mu = Q_\mu P_\lambda$ for all λ, μ. Conversely, if $P_\lambda Q_\mu = Q_\mu P_\lambda$ for all λ, μ, then $TS = \sum_{\lambda,\mu} \lambda\mu P_\lambda Q_\mu = \sum_{\lambda,\mu} \lambda\mu Q_\mu P_\lambda = ST$

(iii). Suppose f and g are injective. If $f(T)g(S) = g(S)f(T)$, choose $h : f(\Omega_T) \to \Omega_T$ and $k : g(\Omega_S) \to \Omega_S$ with $h \circ f = \mathrm{id}_{\Omega_T}$ and $k \circ g = \mathrm{id}_{\Omega_S}$. By Theorem 7.3.3, $T = h(f(T))$ and $S = k(g(S))$, so $TS = ST$. Conversely, if $TS = ST$, then (ii) gives $P_\lambda Q_\mu = Q_\mu P_\lambda$,
and $f(T)g(S) = \sum_{\lambda,\mu} f(\lambda)g(\mu)P_\lambda Q_\mu = \sum_{\lambda,\mu} f(\lambda)g(\mu)Q_\mu P_\lambda = g(S)f(T)$.

Structural Commentary: Spectral projections are the coordinate idempotents of a normal operator.
Key Insight: To commute with a normal operator is to preserve each of its eigenspaces. □

Corollary 7.3.12. *A Commuting Normal Pair Comes from One Normal Operator*
Let $T, S \in \mathcal{L}(\mathcal{V})$ be normal on a finite-dimensional complex inner-product space.
Then $TS = ST \iff \exists\, H \in \mathcal{L}(\mathcal{V})$ normal, $p, q \in \mathbb{C}[z]$ such that $T = p(H)$, $S = q(H)$.
If $TS = ST$, then H may be chosen in $\mathbb{C}[T, S]$.

Proof.

($\Rightarrow$) Suppose $TS = ST$. Write $T = \sum_{\lambda \in \Omega_T} \lambda P_\lambda$ and $S = \sum_{\mu \in \Omega_S} \mu Q_\mu$.
By Theorem 7.3.11, $P_\lambda Q_\mu = Q_\mu P_\lambda$. Let $\mathcal{I} = \{(\lambda, \mu) \in \Omega_T \times \Omega_S : P_\lambda Q_\mu \neq 0\}$.
The projections $P_\lambda Q_\mu$, $(\lambda, \mu) \in \mathcal{I}$, are pairwise orthogonal and sum to $I_\mathcal{V}$.
Choose distinct scalars $\alpha_{\lambda\mu}$ for $(\lambda, \mu) \in \mathcal{I}$ and set $H = \sum_{(\lambda,\mu) \in \mathcal{I}} \alpha_{\lambda\mu} P_\lambda Q_\mu$.
Then H is normal, and $H \in \mathbb{C}[T, S]$ since $P_\lambda \in \mathbb{C}[T]$ and $Q_\mu \in \mathbb{C}[S]$.
Choose $p, q \in \mathbb{C}[z]$ with $p(\alpha_{\lambda\mu}) = \lambda$ and $q(\alpha_{\lambda\mu}) = \mu$ on $\mathcal{I}$.
Hence $p(H) = \sum_{(\lambda,\mu) \in \mathcal{I}} \lambda P_\lambda Q_\mu = T$, and $q(H) = \sum_{(\lambda,\mu) \in \mathcal{I}} \mu P_\lambda Q_\mu = S$.

($\Leftarrow$) If $T = p(H)$ and $S = q(H)$, then $TS = p(H)q(H) = q(H)p(H) = ST$.

Structural Commentary: One scalar labels each common spectral block.
Key Insight: Joint spectral projections reduce a pair of commuting normal operators to finite scalar data. □

Definition 7.3.13. *Matrix Exponential*
For $A \in \mathrm{Mat}(n, n, \mathbb{C})$ and $t \in \mathbb{C}$, define $e^{tA} := \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}$.

Proposition 7.3.14. *Pure Modes for Powers and Exponentials*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be diagonalizable with $A = S\Lambda S^{-1}$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and the columns of S are $\mathbf{v}_1, \ldots, \mathbf{v}_n$. If $\mathbf{x}_0 = \sum_{j=1}^{n} c_j \mathbf{v}_j$, then $A^k \mathbf{x}_0 = \sum_{j=1}^{n} c_j \lambda_j^k \mathbf{v}_j$ for $k \geq 0$; and $e^{tA}\mathbf{x}_0 = \sum_{j=1}^{n} c_j e^{\lambda_j t}\mathbf{v}_j$ for $t \in \mathbb{C}$.

Proof. Let $c = (c_1, \ldots, c_n)^t$, so $\mathbf{x}_0 = Sc$.

Then $A^k = S\Lambda^k S^{-1}$, and Definition 7.3.13 gives $e^{tA} = S\left(\sum_{r=0}^{\infty} \frac{t^r \Lambda^r}{r!}\right) S^{-1} = S\,\mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})S^{-1}$.
Applying these two formulas to Sc gives the stated identities.
Structural Commentary: Eigenbasis coordinates decouple powers and exponentials.
Key Insight: Diagonalization turns coupled evolution into independent scalar modes. □

Corollary 7.3.15. *Simple Spectrum Commutants Are Polynomials*
If $A \in \mathrm{Mat}(n,n,\mathbb{C})$ has n distinct eigenvalues, then $\mathrm{Comm}(A) = \{p(A) : p \in \mathbb{C}[z]\}$, $\dim \mathrm{Comm}(A) = n$.

Proof. Write $A = S\,\mathrm{diag}(\lambda_1,\dots,\lambda_n)S^{-1}$ with distinct λ_i. Theorem 5.4.6 says every $B \in \mathrm{Comm}(A)$ has the form $S\,\mathrm{diag}(\xi_1,\dots,\xi_n)S^{-1}$. Choose the Lagrange interpolation polynomial $p(z) = \sum_{i=1}^{n} \xi_i \prod_{j\neq i} \frac{z-\lambda_j}{\lambda_i-\lambda_j}$.
Then $p(\lambda_i) = \xi_i$ for every i, so Theorem 7.3.1 gives $B = p(A)$.
The dimension statement is the case $n_1 = \cdots = n_n = 1$ of Theorem 5.4.6.
Structural Commentary: Simple spectrum leaves one scalar choice per eigendirection.
Key Insight: Interpolation converts arbitrary diagonal data into a polynomial in A. □

Corollary 7.3.16. *Reduction Modulo the Characteristic Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and let $f \in \mathbb{C}[z]$. If $f = p_A q + r$ with $\deg r < n$, then $f(A) = r(A)$.

Proof. Evaluate division identity at A: $f(A) = p_A(A)q(A) + r(A) = 0q(A) + r(A) = r(A)$.
Theorem 6.3.2 gives $p_A(A) = 0$.
Structural Commentary: Cayley–Hamilton makes polynomial expressions in A reducible modulo p_A.
Key Insight: Every high power of A can be rewritten using powers below n. □

Corollary 7.3.17. *Powers Span at Most n Dimensions*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. Then $\mathrm{span}(I_n, A, A^2, \dots) = \mathrm{span}(I_n, A, \dots, A^{n-1})$, so this has dimension at most n.

Proof. For each $k \geq 0$, Corollary 7.3.16 applied to $f(z) = z^{n+k}$ gives a polynomial r_k of degree at most $n-1$ such that $A^{n+k} = r_k(A)$. Thus every power of A lies in $\mathrm{span}(I_n, A, \dots, A^{n-1})$.
Structural Commentary: Polynomial reduction collapses the infinite power list to its first n terms.
Key Insight: Cayley–Hamilton gives the recurrence relation for powers of A. □

Corollary 7.3.18. *Inverse from the Characteristic Polynomial*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ have $p_A(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$.
If A is invertible, then $A^{-1} = -c_0^{-1}(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_2 A + c_1 I_n)$.

Proof. Since A is invertible, $0 \notin \mathrm{spec}(A)$, so $c_0 = p_A(0) = (-1)^n \det A \neq 0$.
Cayley–Hamilton gives $A^n + c_{n-1}A^{n-1} + \cdots + c_1 A + c_0 I_n = 0$.
Move $c_0 I_n$ to the other side and multiply by $-c_0^{-1}A^{-1}$.
Structural Commentary: The characteristic equation contains an identity term exactly when A is invertible.
Key Insight: Solve Cayley–Hamilton for A^{-1} after factoring out one A. □

Proposition 7.3.19. *Eigenvalues of a Polynomial in a Matrix*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $p \in \mathbb{C}[z]$, and let $\lambda_1,\dots,\lambda_n$ be the eigenvalues of A, counted with algebraic multiplicity. Then the eigenvalues of $p(A)$ are $p(\lambda_1),\dots,p(\lambda_n)$, counted with algebraic multiplicity.

Proof. Choose a Schur form $U^*AU = T$ with diagonal $\lambda_1,\dots,\lambda_n$. Then $p(A) = Up(T)U^*$, and $p(T)$ is upper triangular with diagonal $p(\lambda_1),\dots,p(\lambda_n)$. Hence $p(A)$ has precisely these eigenvalues.
Structural Commentary: Schur form reduces the claim to the diagonal.
Key Insight: A polynomial preserves triangular form and applies p to each diagonal entry. □

Corollary 7.3.20. *Polynomial Spectral Mapping*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and $p \in \mathbb{C}[z]$. Then $\mathrm{spec}(p(A)) = \{p(\lambda) : \lambda \in \mathrm{spec}(A)\}$.

Proof. Let $\lambda_1,\dots,\lambda_n$ be the eigenvalues of A, counted with algebraic multiplicity.
By Proposition 7.3.19, the eigenvalues of $p(A)$, counted with algebraic multiplicity, are $p(\lambda_1),\dots,p(\lambda_n)$. Taking underlying sets gives the claim.
Structural Commentary: Polynomial evaluation pushes the spectrum forward through the scalar polynomial.
Key Insight: Once Schur form has exposed the diagonal, p acts entry by entry there. □

Corollary 7.3.21. *Polynomial Spectral Mapping for Operators*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{C}$, let $T \in \mathcal{L}(\mathcal{V})$, and let $p \in \mathbb{C}[z]$.
Then $\mathrm{spec}(p(T)) = \{p(\lambda) : \lambda \in \mathrm{spec}(T)\}$.

Proof. Choose a basis β and set $A = [T]_{\beta\leftarrow\beta}$.
Then $[p(T)]_{\beta\leftarrow\beta} = p(A)$, and coordinate isomorphism gives $\mathrm{spec}(T) = \mathrm{spec}(A)$ and $\mathrm{spec}(p(T)) = \mathrm{spec}(p(A))$.
Corollary 7.3.20 gives $\mathrm{spec}(p(T)) = \mathrm{spec}(p(A)) = \{p(\lambda) : \lambda \in \mathrm{spec}(A)\} = \{p(\lambda) : \lambda \in \mathrm{spec}(T)\}$.
Structural Commentary: Coordinates reduce the operator statement to the matrix statement.
Key Insight: Polynomial spectral mapping is an operator fact seen through any convenient basis. □

Corollary 7.3.22. *Polynomial Invertibility Test*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ and $p \in \mathbb{C}[z]$. If $p(\lambda) \neq 0$ for every $\lambda \in \mathrm{spec}(A)$, then $p(A)$ is invertible.

Proof. By Proposition 7.3.19, $0 \notin \mathrm{spec}(p(A))$. Corollary 5.1.9 gives invertibility.
Structural Commentary: Invertibility of $p(A)$ is checked on spectral values of A.
Key Insight: Avoiding zero on the spectrum prevents $p(A)$ from having a kernel. □

Proposition 7.3.23. *Exponential of a Jordan Block*
For $k \geq 1$, $\lambda \in \mathbb{C}$, and $t \in \mathbb{C}$, $e^{tJ_k(\lambda)} = e^{\lambda t} \sum_{r=0}^{k-1} \frac{t^r}{r!} J_k(0)^r$.

Proof. Since $J_k(\lambda) = \lambda I_k + J_k(0)$ and the two summands commute, $e^{tJ_k(\lambda)} = e^{\lambda t I_k} e^{tJ_k(0)}$.
The first factor is $e^{\lambda t} I_k$, and the second exponential truncates because $J_k(0)^k = 0$.
Structural Commentary: The scalar part supplies $e^{\lambda t}$; the nilpotent part supplies a finite polynomial in t.
Key Insight: Nilpotence turns the exponential series into a finite sum. □

Lemma 7.3.24. *Identity Plus Nilpotent Has a Square Root*
Let $\mathcal{V}$ be a complex vector space and let $N \in \mathcal{L}(\mathcal{V})$ be nilpotent. Then $I_\mathcal{V} + N$ has a square root.

Proof. Choose $m \geq 1$ such that $N^m = 0$. If $m = 1$, then $N = 0$ and $I_\mathcal{V}$ is a square root.
Otherwise seek $R = I_\mathcal{V} + a_1 N + \cdots + a_{m-1} N^{m-1}$. With $a_0 = 1$, choose $a_1, \ldots, a_{m-1}$ recursively so the coefficients of R^2 match those of $I_\mathcal{V} + N$: $2a_1 = 1$, and $2a_k + \sum_{r=1}^{k-1} a_r a_{k-r} = 0 \quad (2 \leq k \leq m-1)$
Terms of degree $\geq m$ vanish because $N^m = 0$, so $R^2 = I_\mathcal{V} + N$.
Structural Commentary: Nilpotence reduces the square-root series to a finite polynomial.
Key Insight: Match coefficients recursively. □



Figure 70: Nilpotence truncates the square-root series; coefficients are chosen so $R^2 = I + N$.

Theorem 7.3.25. *Invertible Complex Operators Have Square Roots*
Let $\mathcal{V}$ be finite-dimensional over $\mathbb{C}$ and let $T \in \mathcal{L}(\mathcal{V})$ be invertible. Then T has a square root.

Proof. Let $\mathcal{U}_j = \mathcal{G}_{\lambda_j}(T)$ for the distinct eigenvalues of T. Primary decomposition gives $\mathcal{V} = \mathcal{U}_1 \oplus \cdots \oplus \mathcal{U}_m$, and invertibility gives $\lambda_j \neq 0$. Put $N_j = (T - \lambda_j I_\mathcal{V})|_{\mathcal{U}_j}$.
Then $T|_{\mathcal{U}_j} = \lambda_j \left(I_{\mathcal{U}_j} + \lambda_j^{-1} N_j\right)$ with $\lambda_j^{-1} N_j$ nilpotent. By Lemma 7.3.24, choose $B_j^2 = I_{\mathcal{U}_j} + \lambda_j^{-1} N_j$; choose $\alpha_j^2 = \lambda_j$ and set $R_j = \alpha_j B_j$. Then $R_j^2 = T|_{\mathcal{U}_j}$. Define $R = R_1 \oplus \cdots \oplus R_m$ on the direct sum. Then $R^2 = T$.
Structural Commentary: Take square roots on primary blocks, then recombine.
Key Insight: Invertibility makes every scalar block nonzero. □

7.4 The Umbral Calculus

The umbral calculus is the formal linear algebra of polynomial sequences. It identifies a formal power series with three related objects: an algebra element, a linear functional on $\mathbb{C}[x]$, and an operator on $\mathbb{C}[x]$. Refines duality of Definition 5.1.11 and polynomial operator calculus of Definition 6.1.1, without invoking convergence.

Definition 7.4.1. *Formal Power Series and Order*
Let $\mathscr{F} := \mathbb{C}[[t]]$.

(i) $f \in \mathscr{F}$ are *formal power series* $f(t) = \sum_{k \geq 0} a_k t^k$, w. coefficientwise addition, Cauchy product multiplication.
(ii) The *order* of f is $o(f) := \min\{k : a_k \neq 0\}$, where $o(0) := +\infty$.
(iii) A series is *invertible* if it has a multiplicative inverse in $\mathscr{F}$, and is a *delta series* if $o(f) = 1$.
(iv) If $o(f) = 1$, its *compositional inverse* is denoted $\overline{f}$ and satisfies $f(\overline{f}(t)) = \overline{f}(f(t)) = t$.
(v) The *lower factorial polynomials* are $x)_0 := 1$ and $(x)_n := x(x-1)\cdots(x-n+1)$, for $n \geq 1$

Proposition 7.4.2. *Formal Order Calculus*
For $f, g \in \mathscr{F}$:

(i) $f \in \mathscr{F}^\times \iff o(f) = 0$;
(ii) $o(fg) = o(f) + o(g)$ *and* $o(f+g) \geq \min\{o(f), o(g)\}$;
(iii) if $g(t) = \sum_{k \geq 0} b_k t^k$ *and* $o(f) \geq 1$, *then* $g \circ f := g(f(t)) := \sum_{k \geq 0} b_k f(t)^k \in \mathscr{F}$;

(iv) *if* $o(f) \geq 1$ *and* $g \neq 0$, *then* $o(g \circ f) = o(g)o(f)$;
(v) *if* $o(f) = 1$, *then for every* $g \in \mathscr{F}$ *there are unique* $c_k \in \mathbb{C}$ *such that* $g(t) = \sum_{k\geq 0} c_k f(t)^k$.

Proof.

(i) Let $[t^n] : \mathscr{F} \to \mathbb{C}$ denote coefficient extraction. Then $f \in \mathscr{F}^\times \iff [t^0]f \in \mathbb{C}^\times \iff o(f) = 0$.
(ii) Let $r = o(f)$ and $s = o(g)$. Then $[t^{r+s}]fg = [t^r]f\,[t^s]g \neq 0$, and all lower coefficients vanish; hence $o(fg) = r + s$. Also $[t^m](f + g) = 0$ for $m < \min\{r, s\}$, giving the stated lower bound.
(iii) Since $o(f) \geq 1$, $o(f^k) \to \infty$; hence $\forall n$, coefficient $[t^n] \sum_{k\geq 0} b_k f^k$ has only finitely many nonzero terms.
(iv) Let $s = o(g)$. Then $g \circ f = f^s \left(b_s + \sum_{k>s} b_k f^{k-s} \right)$, where second factor has order 0; hence $o(g \circ f) = s\, o(f)$.
(v) If $o(f) = 1$ and $f(t) = a_1 t + \cdots$ with $a_1 \neq 0$, then $f^k = a_1^k t^k +$ higher terms. Thus $\{t^k\}_{k\geq 0} \mapsto \{f^k\}_{k\geq 0}$ is triangular with nonzero diagonal, so recursive coefficient matching gives existence and uniqueness.

Structural Commentary: Delta series behave like a formal change of coordinate at $t = 0$.
Key Insight: Order replaces convergence as the bookkeeping device. □

Definition 7.4.3. *Derivation*
Let $\mathcal{A}$ be an algebra over a field. Linear operator $\partial : \mathcal{A} \to \mathcal{A}$ is a *derivation* if $\partial(ab) = (\partial a)b + a(\partial b)$ $\forall a, b \in \mathcal{A}$.

Proposition 7.4.4. *Formal Differentiation*
Define $\partial_t : \mathscr{F} \to \mathscr{F}$ *by* $\partial_t \left(\sum_{k\geq 0} a_k t^k \right) := \sum_{k\geq 1} k a_k t^{k-1}$.
Then $\partial_t \in \mathrm{Der}_{\mathbb{C}}(\mathscr{F})$; *equivalently, for all* $f, g \in \mathscr{F}$, $\partial_t(fg) = (\partial_t f)g + f(\partial_t g)$.

Proof. Linearity is immediate. Let $f(t) = \sum_{k\geq 0} a_k t^k$ and $g(t) = \sum_{k\geq 0} b_k t^k$.

For $n \geq 0$, $[t^n]\partial_t(fg) = (n+1) \sum_{j=0}^{n+1} a_j b_{n+1-j}$, while $[t^n]\big((\partial_t f)g + f(\partial_t g)\big) = \sum_{j=1}^{n+1} j a_j b_{n+1-j} + \sum_{j=0}^{n} (n+1-j) a_j b_{n+1-j}$.
The last two sums combine to the preceding coefficient, so the Leibniz identity holds.
Structural Commentary: Formal differentiation is the canonical derivation of $\mathbb{C}[[t]]$.
Key Insight: The Leibniz rule is a Cauchy-product coefficient identity. □

Definition 7.4.5. *The Umbral Algebra*

(i) *Polynomial algebra*: set $\mathscr{P} := \mathbb{C}[x]$ and let $\mathscr{P}' := \mathrm{Hom}_{\mathbb{C}}(\mathscr{P}, \mathbb{C})$.
(ii) *Umbral pairing*: for $L \in \mathscr{P}'$ and $p(x) \in \mathscr{P}$, write $\langle L \mid p(x) \rangle := L(p(x))$.
(iii) *Exponential normalization*: write each $f \in \mathscr{F}$ as $f(t) = \sum_{k\geq 0} \frac{a_k}{k!} t^k$.
(iv) *Associated functional*: set $\langle f(t) \mid x^n \rangle := a_n$ for $n \geq 0$, extending $\mathbb{C}$-linearly.

Theorem 7.4.6. *Formal Series Represent All Algebraic Functionals*
The $\mathbb{C}$*-linear map* $\Phi : \mathscr{P}' \xrightarrow{\sim} \mathscr{F}$; $L \longmapsto \sum_{k\geq 0} \frac{\langle L | x^k \rangle}{k!} t^k$ *is an isomorphism with inverse*

$$\Phi^{-1}\left(\sum_{k\geq 0} \frac{a_k}{k!} t^k \right)\left(\sum_{k=0}^{d} c_k x^k \right) := \sum_{k=0}^{d} c_k a_k.$$

Thus, under Φ^{-1}, $\langle t^k \mid x^n \rangle = n!\delta_{n,k}$ *and* $\langle t^k \mid p \rangle = p^{(k)}(0)$, $(p \in \mathscr{P})$.
Thus $t^0 = \mathrm{ev}_0$ *and* $t^k = D^k|_0$.

Proof. Since $\{x^k\}_{k\geq 0}$ is a Hamel basis of $\mathscr{P}$, $\mathscr{P}' \cong \mathbb{C}^{\mathbb{N}}$ via $L \mapsto (\langle L \mid x^k \rangle)_{k\geq 0}$.
The inverse sends the coefficient sequence (a_k) to the unique functional with $L(x^k) = a_k$, so Φ is bijective.
Moreover, $t^k = k!\, t^k/k!$ gives $\langle t^k \mid x^n \rangle = n!\delta_{n,k}$, and linearity gives $\langle t^k \mid p \rangle = p^{(k)}(0)$.
Structural Commentary: $\mathscr{P}'$ is the sequence space encoded by exponentially normalised formal series.
Key Insight: A formal series is a compact record of the functional's values on all monomials. □

Definition 7.4.7. *Delta and Invertible Functionals*
Under the identification $\mathscr{F} \simeq \mathscr{P}'$:

(i) *Delta functional*: a functional represented by $f \in \mathscr{F}$ with $o(f) = 1$.
(ii) *Invertible functional*: a functional represented by $f \in \mathscr{F}^\times$, equivalently $o(f) = 0$.

Proposition 7.4.8. *Basic Umbral Identities*
For $f, g \in \mathscr{F}$, $p \in \mathscr{P}$, *and* $n \geq 0$:

(i) $f(t) = \sum_{k\geq 0} \frac{\langle f(t) \mid x^k \rangle}{k!} t^k$;

(ii) $p(x) = \sum_{k=0}^{\deg p} \frac{\langle t^k \mid p\rangle}{k!} x^k$;

(iii) $\langle fg \mid x^n\rangle = \sum_{k=0}^{n} \binom{n}{k} \langle f \mid x^k\rangle \langle g \mid x^{n-k}\rangle$;

(iv) *if* $o(f) > \deg p$, *then* $\langle f \mid p\rangle = 0$;

(v) *if* $o(f_k) = k$ *for all* $k \geq 0$, *then* $\left\langle \sum_{k\geq 0} a_k f_k \middle| p \right\rangle = \sum_{k=0}^{\deg p} a_k \langle f_k \mid p\rangle$, *with* $\sum_{k\geq 0} a_k f_k$ *well-defined in* $\mathscr{F}$.

Proof.

(i) If $f(t) = \sum_{k\geq 0} a_k t^k/k!$, then $a_k = \langle f \mid x^k\rangle$ by definition.

(ii) Since $\langle t^k \mid p\rangle = p^{(k)}(0)$, this is Taylor's formula for the polynomial p.

(iii) Write $f = \sum a_k t^k/k!$ and $g = \sum b_k t^k/k!$.

The Cauchy product is $fg = \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \right) \frac{t^n}{n!}$, and $a_k = \langle f \mid x^k\rangle$, $b_k = \langle g \mid x^k\rangle$.

(iv) If $p = \sum_{j=0}^{d} c_j x^j$ and $o(f) > d$, then $\langle f \mid x^j\rangle = 0$ for $0 \leq j \leq d$.

(v) Since $o(f_k) = k$, each coefficient of $\sum_{k\geq 0} a_k f_k$ has only finitely many contributions.

With $d = \deg p$, (iv) gives $\langle f_k \mid p\rangle = 0$ for $k > d$, proving the finite formula.

Structural Commentary: The umbral pairing dualizes coefficient calculus.
Key Insight: Order makes infinite formal sums act finitely on each polynomial. □

Theorem 7.4.9. *Multiplication by x Moves to Formal Differentiation*
For all $f \in \mathscr{F}$ and $p \in \mathscr{P}$, $\langle f \mid xp\rangle = \langle \partial_t f \mid p\rangle$.
Equivalently, multiplication by x on $\mathscr{P}$ has adjoint ∂_t on $\mathscr{F}$.

Proof. By linearity it suffices to take $p = x^n$.
For $f = \sum_{k\geq 0} a_k t^k/k!$, $\partial_t f = \sum_{k\geq 0} a_{k+1} \frac{t^k}{k!}$, where $\langle \partial_t f \mid x^n\rangle = a_{n+1} = \langle f \mid x^{n+1}\rangle$.
Structural Commentary: Under the umbral pairing, $x^* = \partial_t$.
Key Insight: The factorial normalisation shifts coefficients by one without extra constants. □

Definition 7.4.10. *Basic Umbral Functionals*
For $a \in \mathbb{C}$ and $p \in \mathscr{P}$:

(i) *Evaluation*: $\epsilon_a = e^{at}$ and $\langle e^{at} \mid p\rangle = p(a)$; hence $\epsilon_a \epsilon_b = \epsilon_{a+b}$.
(ii) *Forward difference*: $\Delta_a = e^{at} - 1$ and $\langle \Delta_a \mid p\rangle = p(a) - p(0)$.
(iii) *Abel functional*: te^{at} and $\langle te^{at} \mid p\rangle = p'(a)$.
(iv) *Gamma functional*: $(1-t)^{-1}$ and $\left\langle (1-t)^{-1} \middle| p \right\rangle = \int_0^\infty p(u) e^{-u}\, du$.
(v) *Integral functional*: $\left\langle \frac{e^{at}-1}{t} \middle| p \right\rangle = \int_0^a p(u)\, du$
(vi) *Bernoulli numbers*: for $a = 1$, $B_n = \left\langle \frac{t}{e^t - 1} \middle| x^n \right\rangle$.

Definition 7.4.11. *Formal Series as Polynomial Operators*
For $f(t) = \sum_{k\geq 0} a_k t^k/k! \in \mathscr{F}$:

(i) *Operator action*: define $f(t) \in \mathrm{End}_{\mathbb{C}}(\mathscr{P})$ by $f(t)p(x) := \sum_{k\geq 0} \frac{a_k}{k!} p^{(k)}(x)$, a finite sum for each $p \in \mathscr{P}$.

(ii) *Monomial action*: for $n \geq 0$, $f(t)x^n = \sum_{k=0}^{n} \binom{n}{k} a_k x^{n-k}$.

Theorem 7.4.12. *Functional-Operator Compatibility*
For $f, g \in \mathscr{F}$ and $p \in \mathscr{P}$, $\langle fg \mid p\rangle = \langle f \mid gp\rangle$. In particular, $\langle f \mid p\rangle = \langle t^0 \mid fp\rangle$.

Proof. By linearity take $p = x^n$. Write $f = \sum_{i\geq 0} a_i t^i/i!$ and $g = \sum_{j\geq 0} b_j t^j/j!$.
Then $\langle fg \mid x^n\rangle = \sum_{j=0}^{n} \binom{n}{j} a_{n-j} b_j$. Also $gx^n = \sum_{j=0}^{n} \binom{n}{j} b_j x^{n-j}$ and $\langle f \mid gx^n\rangle = \sum_{j=0}^{n} \binom{n}{j} b_j a_{n-j}$.
Thus $\langle fg \mid x^n\rangle = \langle f \mid gx^n\rangle$. Taking $(f, g) = (t^0, f)$ in the first identity gives $\langle f \mid p\rangle = \langle t^0 \mid fp\rangle$.
Structural Commentary: Product in $\mathscr{F}$ is adjoint to operator action.
Key Insight: The same series acts on either side of the umbral pairing. □

Definition 7.4.13. *Basic Umbral Operators*
For $a \in \mathbb{C}$ and $p \in \mathscr{P}$:

(i) *Translation*: $e^{at}p(x) = p(x+a)$.
(ii) *Forward difference*: $(e^{at}-1)p(x) = p(x+a) - p(x)$.
(iii) *Abel operator*: $te^{at}p(x) = p'(x+a)$.
(iv) *Gamma operator*: $(1-t)^{-1}p(x) = \int_0^\infty p(x+u)e^{-u}\,du$
(v) *Integral operator*: $\frac{e^{at}-1}{t}p(x) = \int_x^{x+a} p(u)\,du$

Theorem 7.4.14. *Shift-Invariant Operators*
For $T \in \mathrm{End}_{\mathbb{C}}(\mathscr{P})$, the following are equivalent:

(i) $T = f(t)$ *for some* $f \in \mathscr{F}$;
(ii) $[T,t] = 0$;
(iii) $[T, h(t)] = 0$ *for every delta series* h;
(iv) $[T, e^{at}] = 0$ *for all* $a \in \mathbb{C}$.

Proof.

(i) $\Rightarrow$ *(ii), (iii), (iv)* If $T = f(t)$, then $\forall u(t) \in \mathscr{F}$, $[T, u(t)] = [f(t), u(t)] = 0$, since $\mathscr{F}$ is commutative.
(iii) $\Rightarrow$ *(ii)* Since $o(t) = 1$, $(\forall h \in \mathscr{F},\ o(h) = 1 \Rightarrow [T, h(t)] = 0) \Rightarrow [T,t] = 0$.
(iv) $\Rightarrow$ *(ii)* For each $p \in \mathscr{P}$, $0 = [a^1]\,[T, e^{at}]p = [T,t]p$, so $[T,t] = 0$.
(ii) $\Rightarrow$ *(i)* Set $g(t) := \sum_{k\geq 0} \frac{\langle t^0 | Tx^k\rangle}{k!} t^k$. Let $a_k := \langle t^0 \mid Tx^k\rangle$.

Since $[T,t] = 0$, for $m, k \geq 0$, $\langle t^m \mid Tx^k\rangle = \langle t^0 \mid t^m Tx^k\rangle = \langle t^0 \mid Tt^m x^k\rangle = (k)_m a_{k-m}$, with the convention $a_{k-m} = 0$ if $m > k$. The same computation from Definition 7.4.11 gives $\langle t^m \mid g(t)x^k\rangle = (k)_m a_{k-m}$.
Thus $\langle t^m \mid Tx^k - g(t)x^k\rangle = 0$ for all m, k. Since $\{t^m : m \geq 0\}$ separates $\mathscr{P}$, $T = g(t)$.

Structural Commentary: The commutant of t on $\mathscr{P}$ is $\mathscr{F}$.
Key Insight: The derivative is enough to recover every coefficient of the operator. □

Definition 7.4.15. *Sheffer, Associated, and Appell Sequences*
Let f be a delta series, g an invertible series, and $\deg s_n = n$.

(i) *Sheffer sequence*: $(s_n)_{n\geq 0}$ is Sheffer for (g,f) if $\langle g(t)f(t)^k \mid s_n(x)\rangle = n!\delta_{n,k}$ for $n, k \geq 0$.
(ii) *Associated sequence*: a Sheffer sequence for $(1, f)$.
(iii) *Appell sequence*: a Sheffer sequence for (g, t).

Theorem 7.4.16. *Existence and Expansion Theorems*
Let f be a delta series, g an invertible series, and $q_k := g(t)f(t)^k$.

(i) There is a unique sequence $(s_n)_{n\geq 0}$ with $\deg s_n = n$ and $\langle q_k \mid s_n\rangle = n!\delta_{n,k}$.
(ii) If $p_n := g(t)s_n$, then (s_n) is Sheffer for (g,f) iff (p_n) is associated with f.
(iii) For all $h \in \mathscr{F}$, $h = \sum_{k\geq 0} \frac{\langle h | s_k\rangle}{k!} q_k$.
(iv) For all $p \in \mathscr{P}$, $p = \sum_{k=0}^{\deg p} \frac{\langle q_k | p\rangle}{k!} s_k$.

Proof.

(i) Since $o(q_k) = k$, the matrix $A_n = (\langle q_i \mid x^j\rangle)_{0\leq i,j\leq n}$ is triangular with nonzero diagonal.
Hence the equations $\langle q_k \mid s_n\rangle = n!\delta_{n,k}$ have a unique solution $s_n \in \mathscr{P}_{\leq n}$, necessarily with $\deg s_n = n$.
(ii) By Theorem 7.4.12, $\langle f(t)^k \mid p_n\rangle = \langle f(t)^k \mid g(t)s_n\rangle = \langle q_k \mid s_n\rangle$.
(iii) Since $o(q_k) = k$, every $h \in \mathscr{F}$ has a unique expansion $h = \sum c_k q_k$.
Pairing with s_n gives $\langle h \mid s_n\rangle = c_n n!$, so $c_n = \langle h \mid s_n\rangle / n!$.
(iv) Let $d = \deg p$ and $r := p - \sum_{k=0}^{d} \langle q_k \mid p\rangle s_k/k!$.
Then $\langle q_j \mid r\rangle = 0$ for all $j \leq d$, and also for $j > d$ by $o(q_j) > d$. Since the q_j separate $\mathscr{P}$, $r = 0$.

Structural Commentary: (s_n) is dual to $q_k = gf^k$.
Key Insight: The order filtration makes the orthogonality equations triangular. □

Theorem 7.4.17. *Generating and Operator Characterizations*
Let f be a delta series with compositional inverse $\overline{f}$, let $g \in \mathscr{F}^\times$, and let $(p_n), (s_n)$ be polynomial sequences with $\deg p_n = \deg s_n = n$. Set $P(y,u) := \sum_{n\geq 0} \frac{p_n(y)}{n!}u^n$ and $S(y,u) := \sum_{n\geq 0} \frac{s_n(y)}{n!}u^n$.

(i) (p_n) is associated with f iff $P(y,u) = e^{y\overline{f}(u)}$

(ii) (s_n) *is Sheffer for* (g,f) *iff* $S(y,u) = g(\overline{f}(u))^{-1}e^{y\overline{f}(u)}$
(iii) (p_n) *is associated with* f *iff* $p_n(0) = \delta_{n,0}$, $f(t)p_n = np_{n-1}$, *where* $p_{-1} := 0$.
(iv) (s_n) *is Sheffer for* (γ, f) *for some* $\gamma \in \mathscr{F}^\times$ *iff* $f(t)s_n = ns_{n-1}$, *where* $s_{-1} := 0$.
(v) If (p_n) *is associated with* f, *then* $p_n(x) = \sum_{k=0}^{n} \frac{\langle \overline{f}(t)^k | x^n \rangle}{k!} x^k$
(vi) If (s_n) *is Sheffer for* (g,f), *then* $s_n(x) = \sum_{k=0}^{n} \frac{1}{k!} \langle g(\overline{f}(t))^{-1}\overline{f}(t)^k | x^n \rangle x^k$.

Proof.

(*i*) ($\Rightarrow$) By Theorem 7.4.16, $e^{yt} = \sum_{n\geq 0} \frac{p_n(y)}{n!} f(t)^n$. Substitute $t = \overline{f}(u)$ to obtain $P(y,u) = e^{y\overline{f}(u)}$.

(*i*) ($\Leftarrow$) If $P(x,u) = e^{x\overline{f}(u)}$, then for all $k \geq 0$, $\sum_{n\geq 0} \frac{\langle f(t)^k | p_n \rangle}{n!} u^n = \langle f(t)^k \mid P(x,u) \rangle = f(\overline{f}(u))^k = u^k$.

Thus $\langle f(t)^k \mid p_n \rangle = n!\delta_{n,k}$.

(*ii*) ($\Rightarrow$) By Theorem 7.4.16, $e^{yt} = \sum_{n\geq 0} \frac{s_n(y)}{n!} g(t) f(t)^n$. Substitute $t = \overline{f}(u)$ and divide by $g(\overline{f}(u))$.

(*ii*) ($\Leftarrow$) If $S(x,u) = g(\overline{f}(u))^{-1}e^{x\overline{f}(u)}$, then $\sum_{n\geq 0} \frac{\langle g(t)f(t)^k | s_n \rangle}{n!} u^n = \langle g(t)f(t)^k \mid S(x,u) \rangle = u^k$,

so $\langle g(t)f(t)^k \mid s_n \rangle = n!\delta_{n,k}$.

(*iii*) ($\Rightarrow$) If (p_n) is associated, then $p_n(0) = \langle 1 \mid p_n \rangle = \delta_{n,0}$ and, for all k, $\langle f(t)^k \mid f(t)p_n - np_{n-1} \rangle = 0$.
Since $\{f(t)^k : k \geq 0\}$ separates $\mathscr{P}$, $f(t)p_n = np_{n-1}$.

(*iii*) ($\Leftarrow$) The lowering relation gives $f(t)^k p_n = (n)_k p_{n-k}$.
Hence $\langle f(t)^k \mid p_n \rangle = \langle 1 \mid f(t)^k p_n \rangle = (n)_k p_{n-k}(0) = n!\delta_{n,k}$.

(*iv*) ($\Rightarrow$) If (s_n) is Sheffer for (γ, f), then for all k, $\langle \gamma(t)f(t)^k \mid f(t)s_n - ns_{n-1} \rangle = 0$.
Since $\{\gamma(t)f(t)^k : k \geq 0\}$ separates $\mathscr{P}$, $f(t)s_n = ns_{n-1}$.

(*iv*) ($\Leftarrow$) Define $L \in \mathscr{P}'$ by $L(s_0) = 1$ and $L(s_n) = 0$ for $n > 0$, and put $\gamma := \Phi(L)$. Then $\langle \gamma \mid 1 \rangle = L(1) \neq 0$, so $\gamma \in \mathscr{F}^\times$, and $\langle \gamma(t)f(t)^k \mid s_n \rangle = \langle \gamma(t) \mid f(t)^k s_n \rangle = (n)_k L(s_{n-k}) = n!\delta_{n,k}$.

(*v*) From (i), $e^{x\overline{f}(u)} = \sum_{k\geq 0} \frac{\overline{f}(u)^k}{k!} x^k$. Applying $n![u^n]$ gives the stated formula for p_n.

(*vi*) From (ii), $g(\overline{f}(u))^{-1}e^{x\overline{f}(u)} = \sum_{k\geq 0} \frac{g(\overline{f}(u))^{-1}\overline{f}(u)^k}{k!} x^k$. Applying $n![u^n]$ gives the stated formula for s_n.

Structural Commentary: $\overline{f}$ converts f-lowering to exponential generating functions.
Key Insight: Generating functions package the whole triangular duality into one formal identity. □

Method 7.4.18. *Building Sheffer Sequences*
To construct the Sheffer sequence for (g,f):

(i) Verify $o(f) = 1$ and $o(g) = 0$.
(ii) If $\overline{f}$ is explicit, compute $S(y,t) := g(\overline{f}(t))^{-1}e^{y\overline{f}(t)}$, where $s_n(y) = n![t^n]S(y,t)$.
(iii) If $\overline{f}$ is unavailable, solve recursively from $f(t)s_n = ns_{n-1}$, where $\langle g(t)f(t)^k \mid s_n \rangle = n!\delta_{n,k}$.
(iv) For associated sequences, set $g = 1$ and use Theorem 7.5.1.

Definition 7.4.19. *Binomial Type and Sheffer Identity*
Let (p_n) and (s_n) be polynomial sequences.

(i) *Binomial type*: (p_n) is of binomial type if $p_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} p_k(y)p_{n-k}(x)$.

(ii) *Sheffer identity*: (s_n) satisfies the Sheffer identity relative to (p_n) if $s_n(x+y) = \sum_{k=0}^{n} \binom{n}{k} p_k(y)s_{n-k}(x)$.

Theorem 7.4.20. *Action and Binomial Identities*
Let (s_n) *be Sheffer for* (g,f), *let* (p_n) *be associated with* f, *and write* $E_y := e^{yt}$. *Then:*

(i) for $h \in \mathscr{F}$ *and* $n \geq 0$, $h(t)s_n = \sum_{k=0}^{n} \binom{n}{k} \langle h \mid s_k \rangle p_{n-k}$;
(ii) (p_n) *is associated with a delta series iff* (p_n) *is of binomial type;*
(iii) (s_n) *is Sheffer for some* $g \in \mathscr{F}^\times$ *and same* f *iff* (s_n) *satisfies the Sheffer identity relative to* (p_n).

Proof.

(*i*) By Theorem 7.4.16, $h = \sum_{k\geq 0} \frac{\langle h | s_k \rangle}{k!} g(t)f(t)^k$.

Since $g(t)s_n = p_n$ and $f(t)^k p_n = (n)_k p_{n-k}$, $h(t)s_n = \sum_{k=0}^{n} \frac{\langle h | s_k \rangle}{k!} (n)_k p_{n-k}$.

(ii) If (p_n) is associated, apply (i) with $s_n = p_n$, $g = 1$, and $h = E_y$ to get the binomial identity.
Conversely, if the binomial identity holds, then $p_n(0) = \delta_{n,0}$ and the operator $Qp_n := np_{n-1}$ commutes with every E_y; hence $Q = f(t)$ by Theorem 7.4.14.
Since Q lowers degree by one, $o(f) = 1$, and Theorem 7.4.17 gives that (p_n) is associated with f.
(iii) If (s_n) is Sheffer, apply (i) with $h = E_y$ to obtain the Sheffer identity.
Conversely, applying $f(t)$ in y-variable to Sheffer identity and then setting $y = 0$ gives $f(t)s_n = ns_{n-1}$; Theorem 7.4.17 yields a $g \in \mathscr{F}^\times$ such that (s_n) is Sheffer for (g, f).

Structural Commentary: Translation turns the operator formula into addition laws for polynomial sequences.
Key Insight: Binomial identities are the visible polynomial shadow of a delta operator. □

Example 7.4.21. *Standard Sheffer Examples*

(i) *Monomials*: $p_n(x) = x^n$ is associated with $f(t) = t$; the binomial identity is the usual binomial theorem.
(ii) *Lower factorials*: $(x)_n$ is associated with $f(t) = e^t - 1$, with $(1+t)^y = e^{y\log(1+t)} = \sum_{k\geq 0} \frac{(y)_k}{k!} t^k$,
and Vandermonde identity $(x+y)_n = \sum_{k=0}^{n} \binom{n}{k} (x)_k (y)_{n-k}$.
(iii) *Abel polynomials*: associated with $f(t) = te^{at}$, $A_n(x; a) = x(x - an)^{n-1}$.
(iv) *Hermite polynomials*: for $g(t) = e^{t^2/2}$, the polynomial is $H_n(x) = e^{-t^2/2} x^n$, where $e^{yt - t^2/2} = \sum_{k\geq 0} \frac{H_k(y)}{k!} t^k$.
(v) *Laguerre polynomials*: order α, Sheffer for $g(t) = (1-t)^{-\alpha-1}$, $f(t) = \frac{t}{t-1}$, $\frac{1}{(1-t)^{\alpha+1}} e^{yt/(t-1)} = \sum_{k\geq 0} \frac{L_k^{(\alpha)}(y)}{k!} t^k$.

Definition 7.4.22. *Umbral Operators, Continuity, and Adjoints*
Let (s_n) be Sheffer for (g, f), and let (p_n) be associated with f.
(i) *Sheffer operator*: $\lambda_{g,f} \in \mathcal{L}(\mathscr{P})$ is defined by $\lambda_{g,f}(x^n) = s_n(x)$.
(ii) *Umbral operator*: $\lambda_f \in \mathcal{L}(\mathscr{P})$ is defined by $\lambda_f(x^n) = p_n(x)$.
(iii) *Sheffer/umbral shift*: $\theta_{g,f} s_n = s_{n+1}$; in the associated case, $\theta_f p_n = p_{n+1}$.
(iv) *Continuous operator*: $T \in \mathcal{L}(\mathscr{F})$ is continuous iff $o(f_k) \to \infty \Longrightarrow o(Tf_k) \to \infty$.
Equivalently, for every defined formal sum with $o(f_k) \to \infty$, $T\left(\sum_{k\geq 0} a_k f_k\right) = \sum_{k\geq 0} a_k T(f_k)$
(v) *Algebraic adjoint*: for $\tau \in \mathcal{L}(\mathscr{P})$, $\tau^\times \in \mathcal{L}(\mathscr{F})$ is defined by $\langle \tau^\times h \mid p\rangle = \langle h \mid \tau p\rangle$ for $h \in \mathscr{F}$, $p \in \mathscr{P}$.
This is the dual-map adjoint of Definition 5.1.11, not an inner-product adjoint.

Theorem 7.4.23. *Continuous Operators Are Exactly Operator Adjoints*
Let $\mathcal{L}_{\text{cont}}(\mathscr{F})$ denote the continuous linear operators on $\mathscr{F}$.
Then $(-)^\times : \mathcal{L}(\mathscr{P}) \xrightarrow{\sim} \mathcal{L}_{\text{cont}}(\mathscr{F})$, $\tau \longmapsto \tau^\times$ is a vector-space isomorphism.

Proof.
(⇒) Let $\tau \in \mathcal{L}(\mathscr{P})$ and $o(h_j) \to \infty$. For fixed n, $\tau x^n \in \mathscr{P}$, hence $\langle \tau^\times h_j \mid x^n\rangle = \langle h_j \mid \tau x^n\rangle = 0$ for all sufficiently large j. Thus $o(\tau^\times h_j) \to \infty$, so $\tau^\times \in \mathcal{L}_{\text{cont}}(\mathscr{F})$.
If $\tau^\times = 0$, then $\langle h \mid \tau p\rangle = 0$ for all h, p, hence $\tau = 0$; the map is injective.
(⇐) Let $T \in \mathcal{L}_{\text{cont}}(\mathscr{F})$. Define $\tau_T x^n := \sum_{k\geq 0} \frac{\langle Tt^k \mid x^n\rangle}{k!} x^k$. Since $o(Tt^k) \to \infty$, the sum is finite for each n.
For all m, n, $\langle \tau_T^\times t^m \mid x^n\rangle = \langle t^m \mid \tau_T x^n\rangle = \langle Tt^m \mid x^n\rangle$. Thus $\tau_T^\times t^m = Tt^m$ for every m.
By continuity, $\tau_T^\times = T$, so the map is surjective.

Structural Commentary: Continuity is the finiteness condition for dualising polynomial operators.
Key Insight: Finiteness on each polynomial degree is the hidden topology. □

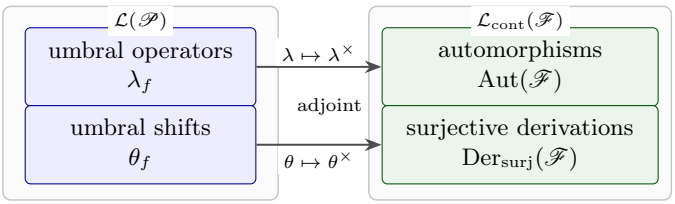


Figure 71: Adjoints identify $\mathcal{L}(\mathscr{P})$ with $\mathcal{L}_{\text{cont}}(\mathscr{F})$ and map the umbral classes shown.

Theorem 7.4.24. *Umbral Operators and Adjoint Automorphisms*
Let f be a delta series with associated sequence (p_n) and umbral operator $\lambda_f(x^n) = p_n(x)$.
Let $C_{\overline{f}} : \mathscr{F} \to \mathscr{F}$ be substitution by $\overline{f}$. Then $\lambda_f^\times = C_{\overline{f}}$, $C_{\overline{f}}(h) = h(\overline{f})$, so $\lambda_f^\times \in \mathrm{Aut}_{\mathbb{C}\text{-}alg}(\mathscr{F})$ and $\lambda_f^\times f = t$.
Moreover, $\lambda \in \mathcal{L}(\mathscr{P})$ is umbral iff $\lambda^\times \in \mathrm{Aut}_{\mathbb{C}\text{-}alg}(\mathscr{F})$.

Proof.
($\Rightarrow$) If $\lambda = \lambda_f$, then for $k, n \geq 0$, $\langle \lambda_f^\times f^k \mid x^n \rangle = \langle f^k \mid p_n \rangle = n!\delta_{n,k} = \langle t^k \mid x^n \rangle$.
Thus $\lambda_f^\times f^k = t^k$. For $h = q(f)$, equivalently $q = h(\overline{f})$, continuity gives $\lambda_f^\times h = q(t) = h(\overline{f}(t))$.
Hence $\lambda_f^\times = C_{\overline{f}} \in \mathrm{Aut}_{\mathbb{C}\text{-alg}}(\mathscr{F})$.
($\Leftarrow$) Let $A := \lambda^\times \in \mathrm{Aut}_{\mathbb{C}\text{-alg}}(\mathscr{F})$ and set $f := A^{-1}t$. Since A^{-1} preserves (t) and $(t)^2$, $o(f) = 1$.
With $p_n := \lambda x^n$, $\langle f^k \mid p_n \rangle = \langle Af^k \mid x^n \rangle = \langle t^k \mid x^n \rangle = n!\delta_{n,k}$.
Thus (p_n) is associated with the delta series f, so $\lambda = \lambda_f$.
Structural Commentary: Umbral operators are formal coordinate changes on $\mathscr{F}$.
Key Insight: The adjoint remembers the delta series as the coordinate sent to t. □

Definition 7.4.25. *Umbral Composition*
For polynomial sequences (p_n) and (q_n) with $p_n(x) = \sum_{k=0}^n p_{n,k}x^k$, define $(p \circ_U q)_n(x) := \sum_{k=0}^n p_{n,k}q_k(x)$.

Theorem 7.4.26. *Group Law for Umbral Operators*
If f, g are delta series, then $\lambda_g \circ \lambda_f = \lambda_{f\circ g}$ and $\lambda_f^{-1} = \lambda_{\overline{f}}$.
Equivalently, associated sequences form a group under umbral composition.

Proof. Using $(AB)^\times = B^\times A^\times$ and Theorem 7.4.24, $(\lambda_g\lambda_f)^\times h = \lambda_f^\times \lambda_g^\times h = h(\overline{g}(\overline{f})) = h(\overline{f \circ g})$,
so $\lambda_g\lambda_f = \lambda_{f\circ g}$. Taking $g = \overline{f}$ gives $\lambda_f^{-1} = \lambda_{\overline{f}}$. If (p_n) and (q_n) are associated with f and g,
then $(\lambda_g\lambda_f)x^n = \lambda_g p_n = \sum_{k=0}^n p_{n,k}q_k = (p \circ_U q)_n$, so umbral composition corresponds to $f \circ g$.
Structural Commentary: The umbral group is the substitution group of delta series.
Key Insight: Composition of associated sequences is composition of delta series in reverse adjoint form. □

Proposition 7.4.27. *Sheffer Operators*
Let f, k be delta series and let $g, h \in \mathscr{F}^\times$.
If (s_n) is Sheffer for (g, f) and (p_n) is associated with f, then $s_n = g(t)^{-1}p_n$, and $\lambda_{g,f} = g(t)^{-1}\lambda_f$.
Moreover, $\lambda_{g,f} \circ \lambda_{h,k} = \lambda_{g\cdot(h\circ f),\, k\circ f}$ and $\lambda_{g,f}^{-1} = \lambda_{g^{-1}(\overline{f}),\, \overline{f}}$.

Proof. Theorem 7.4.16 gives $g(t)s_n = p_n$, hence $s_n = g(t)^{-1}p_n$ and $\lambda_{g,f} = g(t)^{-1}\lambda_f$.
For $a \in \mathscr{F}$, the adjoint substitution identity gives $\lambda_f a(t) = a(f(t))\lambda_f$.
Therefore $\lambda_{g,f}\lambda_{h,k} = g(t)^{-1}\lambda_f h(t)^{-1}\lambda_k = g(t)^{-1}h(f(t))^{-1}\lambda_{k\circ f}$.
This is $\lambda_{g\cdot(h\circ f),\, k\circ f}$. Taking $(h, k) = (g^{-1}(\overline{f}), \overline{f})$ gives the inverse formula.
Structural Commentary: Sheffer operators are invertible weights followed by umbral substitutions.
Key Insight: Their group law is the semidirect product law $(g, f)(h, k) = (g(h \circ f), k \circ f)$. □

Theorem 7.4.28. *Umbral Shift Derivations*
Let f be a delta series and let θ_f be its umbral shift.
Define the continuous derivation $\partial_f : \mathscr{F} \to \mathscr{F}$ by $\partial_f(f^k) = kf^{k-1}$, $k \geq 0$. Then $\theta_f^\times = \partial_f$, $\partial_f f = 1$.
If f, g are delta series, then $\partial_g = (\partial_g f)\partial_f$.

Proof. Let (p_n) be associated with f.
For $k, n \geq 0$, $\langle \theta_f^\times f^k \mid p_n \rangle = \langle f^k \mid p_{n+1} \rangle = (n+1)!\delta_{n+1,k} = \langle kf^{k-1} \mid p_n \rangle$.
Hence $\theta_f^\times f^k = kf^{k-1}$ for all k, so $\theta_f^\times = \partial_f$ by continuity.
For the chain rule, test on the coordinate powers f^k, which is $(\partial_g f)\partial_f(f^k) = k(\partial_g f)f^{k-1} = \partial_g(f^k)$.
Continuity gives the identity on $\mathscr{F}$.
Structural Commentary: An umbral shift is differentiation in its delta coordinate.
Key Insight: The associated basis turns shifting into formal differentiation. □

Theorem 7.4.29. *Umbral Shifts and Surjective Derivations*
For $\theta \in \mathcal{L}(\mathscr{P})$, θ is an umbral shift $\iff$ $\theta^\times$ is a surjective derivation of $\mathscr{F}$.

Proof.
($\Rightarrow$) If $\theta = \theta_f$, then $\theta^\times = \partial_f$ by Theorem 7.4.28. Since $\partial_f\left(\frac{f^{k+1}}{k+1}\right) = f^k$, $k \geq 0$, ∂_f is a surjective derivation.

($\Leftarrow$) Let $D := \theta^\times$ be a surjective derivation. Choose $u \in \mathscr{F}$ with $Du = 1$ and set $f := u - [t^0]u$.
Then $Df = 1$ and, since D lowers order by one, $o(f) = 1$. Let (p_n) be associated with f.
For $k, n \geq 0$, $\langle f^k \mid \theta p_n\rangle = \langle Df^k \mid p_n\rangle = \langle kf^{k-1} \mid p_n\rangle = \langle f^k \mid p_{n+1}\rangle$.
The functionals f^k separate $\mathscr{P}$, so $\theta p_n = p_{n+1}$ and $\theta = \theta_f$.

Structural Commentary: Surjective derivations are derivatives in delta coordinates.
Key Insight: Surjectivity chooses the coordinate whose derivative is 1. □

7.5 Umbral Recurrences and Variants

Theorem 7.5.1. *Recurrence and Transfer Formulas*
Let (p_n) be associated with the delta series f, and set $u(t) := f(t)/t \in \mathscr{F}^\times$.
Then $p_{n+1} = x[f'(t)]^{-1}p_n$, and, for $n \geq 1$, $p_n = f'(t)u(t)^{-n-1}x^n = xu(t)^{-n}x^{n-1}$.

Proof. By the chain rule for umbral shifts with coordinate t, $\theta_t = (\partial_t f)\theta_f = f'(t)\theta_f$.
Since $\theta_t = x$ and $\theta_f p_n = p_{n+1}$, $p_{n+1} = x[f'(t)]^{-1}p_n$. The transfer identity follows by applying this recurrence inductively after writing $f = tu$, hence $p_n = f'(t)u(t)^{-n-1}x^n = xu(t)^{-n}x^{n-1}$.
Structural Commentary: The recurrence generates (p_n) from $f'(t)^{-1}$.
Key Insight: Transfer formulas convert the difficult inverse coordinate problem into formal operator algebra. □

Corollary 7.5.2. *Sheffer Shift Formula*
If (s_n) is Sheffer for (g, f), then $\theta_{g,f} = g(t)^{-1}\theta_f g(t) = \left[x - \frac{g'(t)}{g(t)}\right]\frac{1}{f'(t)}$, and hence $s_{n+1} = \left(x - \frac{g'}{g}\right)(f')^{-1}s_n$.

Proof. Since $g(t)s_n$ is associated with f, $\theta_{g,f} = g^{-1}\theta_f g$. By Theorem 7.5.1, $\theta_f = x(f')^{-1}$.
Using the Pincherle commutator $g'(t) = g(t)x - xg(t)$, we get $g^{-1}x(f')^{-1}g = g^{-1}xg(f')^{-1} = \left(x - \frac{g'}{g}\right)(f')^{-1}$.
Applying this operator to s_n gives the recurrence.
Structural Commentary: The Sheffer shift is a conjugated associated shift.
Key Insight: The logarithmic derivative g'/g is the correction term. □

Definition 7.5.3. *Stirling and Exponential Polynomials*

(i) *Stirling numbers*: $S(n,k)$ are defined by $x^n = \sum_{k=0}^{n} S(n,k)(x)_k$.
(ii) *Exponential polynomials*: define $\phi_n(x) := \sum_{k=0}^{n} S(n,k)x^k$.
(iii) *Boundary convention*: $S(n,-1) = 0$ and $S(n,k) = 0$ for $k > n$.

Example 7.5.4. *Lower Factorials and Stirling Numbers*
For $f(t) = e^t - 1$, $f'(t) = e^t$, hence $p_{n+1}(x) = xe^{-t}p_n(x) = xp_n(x-1)$. With $p_0 = 1$, this gives $p_n = (x)_n$.
For $f(t) = \log(1+t)$, $\overline{f}(t) = e^t - 1$, and the associated sequence is (ϕ_n).
The inverse umbral relation is $x^n = \sum_{k=0}^{n} S(n,k)(x)_k$, and $\phi_n(x) = \sum_{k=0}^{n} S(n,k)x^k$.
The recurrence formula gives $\phi_{n+1}(x) = x(1+t)\phi_n(x) = x\big(\phi_n(x) + \phi_n'(x)\big)$,
so coefficient comparison yields $S(n+1,k) = S(n,k-1) + kS(n,k)$.

Definition 7.5.5. *q- and Logarithmic Umbral Variants*

(i) *Gaussian coefficient*: for $0 \leq k \leq n$, $\binom{n}{k}_q := \prod_{j=1}^{n}(1-q^j)/\left(\prod_{j=1}^{k}(1-q^j)\prod_{j=1}^{n-k}(1-q^j)\right)$.
(ii) *q-umbral calculus*: the deformation $\binom{n}{k} \rightsquigarrow \binom{n}{k}_q$.
(iii) *Logarithmic umbral calculus*: the logarithmic-type variant based on harmonic logarithms.

Remark 7.5.6. *Further Directions*
Classical umbral methods extend to many polynomial families and operations:

(i) *Families*: lower factorial, Stirling, exponential, Gould, central factorial, Abel, Mittag–Leffler, Bessel, Bell, Hermite, Bernoulli, Euler, Laguerre, Poisson–Charlier, actuarial, Meixner, Pidduck, Narumi, Boole, Peters, Mahler, and Mott polynomials.
(ii) *Operations*: connection constants, duplication formulas, Lagrange inversion, cross sequences, Steffensen sequences, operational formulas, inverse relations, recurrence solving, and binomial convolution.
(iii) *Variants*: q-umbral and logarithmic umbral calculi replace or modify the binomial structure through the objects in Definition 7.5.5.

Part IV
Matrix Analysis and Numerical Linear Algebra

1 Trace, Determinant, Rank, and Matrix Invariants

1.1 Trace and Determinant

Matrix classes package operator structure in coordinates. Trace, determinant, rank, positivity, idempotents, nilpotents, spectral radius, and structured products give computable invariants.

Definition 1.1.1. *Trace and Determinant of a Matrix*
For $A = [A_{j,k}] \in \mathrm{Mat}(n, n, \mathbb{F})$, the *trace* of A is $\mathrm{tr}(A) := A_{1,1} + \cdots + A_{n,n}$.
The *determinant* of A is $\det A := \sum_{\sigma \in \mathfrak{S}_n} \mathrm{sign}(\sigma) A_{\sigma(1),1} \cdots A_{\sigma(n),n}$.

Definition 1.1.2. *Gram Determinant*
Vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ in inner-product space has *Gram determinant* $\mathrm{GramDet}(\mathbf{u}_1, \ldots, \mathbf{u}_n) := \det \mathrm{Gram}(\mathbf{u}_1, \ldots, \mathbf{u}_n)$.

Example 1.1.3. *Trace Functional and the Frobenius Inner Product*
On $\mathrm{Mat}(n, n, \mathbb{F})$ with the Frobenius inner product from Example 1.1.6, the Riesz vector of the trace functional $A \mapsto \mathrm{tr}(A)$ is I_n, since $\langle A, I_n \rangle_F = \sum_{j=1}^{n} a_{jj} = \mathrm{tr}(A)$.

Proposition 1.1.4. *Trace Is Cyclic for Two Factors*
If $A \in \mathrm{Mat}(m, n, \mathbb{F})$ and $B \in \mathrm{Mat}(n, m, \mathbb{F})$, then $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.

Proof. Expand both traces by entries: $\mathrm{tr}(AB) = \sum_{j=1}^{m} \sum_{k=1}^{n} A_{j,k} B_{k,j} = \sum_{k=1}^{n} \sum_{j=1}^{m} B_{k,j} A_{j,k} = \mathrm{tr}(BA)$.
Structural Commentary: Cyclicity is just reindexing the same double sum.
Key Insight: The two double sums contain exactly the same scalar terms. □

Proposition 1.1.5. *Trace Arithmetic*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$ and $r \in \mathbb{F}$. Then
(i) $\mathrm{tr}(rA) = r\,\mathrm{tr}(A)$;
(ii) $\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B)$.

Proof. We prove the two identities separately.
(i) Summing the diagonal entries of rA gives $\mathrm{tr}(rA) = \sum_{j=1}^{n} r a_{jj} = r \sum_{j=1}^{n} a_{jj} = r\,\mathrm{tr}(A)$.
(ii) Summing the diagonal entries of $A+B$ gives $\mathrm{tr}(A+B) = \sum_{j=1}^{n} (a_{jj}+b_{jj}) = \sum_{j=1}^{n} a_{jj} + \sum_{j=1}^{n} b_{jj} = \mathrm{tr}(A)+\mathrm{tr}(B)$.
Structural Commentary: Trace is linear because it is the sum of diagonal entries.
Key Insight: Scalar multiplication and addition pass through the diagonal sum. □

Proposition 1.1.6. *Cyclic Products*
Let $m \geq 2$, $A_j \in \mathrm{Mat}(n_j, n_{j+1}, \mathbb{F})$ for $j = 1, \ldots, m$, $n_{m+1} = n_1$, $P_{p,q} := A_p A_{p+1} \cdots A_q$.
Then $\mathrm{tr}(P_{1,m}) = \mathrm{tr}(A_m P_{1,m-1})$. Consequently, for $A, B, C \in \mathrm{Mat}(n, n, \mathbb{F})$, $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$

Proof. Since $P_{1,m-1} \in \mathrm{Mat}(n_1, n_m, \mathbb{F})$ and $A_m \in \mathrm{Mat}(n_m, n_1, \mathbb{F})$,
Proposition 1.1.4 gives $\mathrm{tr}(P_{1,m}) = \mathrm{tr}(P_{1,m-1} A_m) = \mathrm{tr}(A_m P_{1,m-1})$.
Applying this cyclic move twice to ABC gives the three-factor identity.
Structural Commentary: Trace preserves cyclic order, not arbitrary order.
Key Insight: One cyclic move is $\mathrm{tr}(XY) = \mathrm{tr}(YX)$ with $X = P_{1,m-1}$ and $Y = A_m$. □

Example 1.1.7. *Cyclicity Is Not Arbitrary Commutativity*
Let $A = E_{1,2}$, $B = E_{2,1}$, and $C = E_{1,1}$ in $\mathrm{Mat}(2, 2, \mathbb{F})$. Then $\mathrm{tr}(ABC) = 1$, while $\mathrm{tr}(ACB) = 0$.
Thus trace permits cyclic rotations of factors, but not arbitrary rearrangements.

Theorem 1.1.8. *Basic Determinant Laws*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$.
(i) $\det I_n = 1$.
(ii) $\det(A)$ *is linear as a function of each column of A when the other columns are fixed.*
(iii) Swapping two columns negates the determinant; equal columns force $\det A = 0$.

(iv) *If A is triangular, then* $\det A = A_{1,1}\cdots A_{n,n}$.
(v) $\det(AB) = \det(A)\det(B)$.
(vi) *A is invertible if and only if* $\det A \neq 0$. *In that case* $\det(A^{-1}) = (\det A)^{-1}$.
(vii) $\det(A^t) = \det(A)$.

Proof.

(i) $\det I_n = \sum_{\sigma\in\mathfrak{S}_n} \operatorname{sign}(\sigma)(I_n)_{\sigma(1),1}\cdots(I_n)_{\sigma(n),n} = 1$ as product is nonzero only when $\sigma(k) = k\ \forall k$.

(ii) For fixed k, $\det(A_{\cdot,1},\dots,\mathbf{x},\dots,A_{\cdot,n}) = \sum_{\sigma\in\mathfrak{S}_n} \operatorname{sign}(\sigma)x_{\sigma(k)} \prod_{\ell\neq k} A_{\sigma(\ell),\ell}$, which is linear in $\mathbf{x}$.

(iii) If $B_{\cdot,k} = A_{\cdot,\tau(k)}$ for transposition τ, then $\det B = \sum_{\rho\in\mathfrak{S}_n} \operatorname{sign}(\rho\tau) \prod_{k=1}^{n} A_{\rho(k),k} = -\det A$.
If the swapped columns are equal, then $B = A$, so $\det A = -\det A$ and $\det A = 0$.

(iv) If A is upper triangular and $\prod_{k=1}^{n} A_{\sigma(k),k} \neq 0$, then $\sigma(k) \leq k$ for all k, hence $\sigma(k) = k$ for all k.
Thus $\det A = A_{1,1}\cdots A_{n,n}$. If A is lower triangular, nonzero products require $\sigma(k) \geq k$ for all k, again forcing $\sigma(k) = k$ for all k, so the same diagonal product formula holds.

(v) If F is alternating and multilinear in columns, then expanding each column in the standard basis gives $F(A_{\cdot,1},\dots,A_{\cdot,n}) = \sum_{i_1,\dots,i_n} A_{i_1,1}\cdots A_{i_n,n}F(\mathbf{e}_{i_1},\dots,\mathbf{e}_{i_n}) = \sum_{\sigma\in\mathfrak{S}_n} \operatorname{sign}(\sigma)F(I_n)A_{\sigma(1),1}\cdots A_{\sigma(n),n}$.
The terms with repeated i_j vanish by alternation, and the surviving ordered bases differ from $(\mathbf{e}_1,\dots,\mathbf{e}_n)$ by sign of permutation. For $H(B) = \det(AB)$, map H is alternating multilinear and $H(I_n) = \det A$.
Hence $\det(AB) = H(B) = H(I_n)\det B = \det(A)\det(B)$.

(vi) If A is invertible, then $1 = \det I_n = \det(AA^{-1}) = \det(A)\det(A^{-1})$.
If $A_{\cdot,k} = \sum_{\ell\neq k} c_\ell A_{\cdot,\ell}$, then $\det A = \sum_{\ell\neq k} c_\ell \det(A_{\cdot,1},\dots,A_{\cdot,\ell},\dots,A_{\cdot,\ell},\dots,A_{\cdot,n}) = 0$.
Thus $\det A \neq 0$ iff the columns of A are independent, iff A is invertible.

(vii) Using the permutation definition and replacing σ by σ^{-1},
$\det(A^t) = \sum_{\sigma\in\mathfrak{S}_n} \operatorname{sign}(\sigma)A_{1,\sigma(1)}\cdots A_{n,\sigma(n)} = \sum_{\tau\in\mathfrak{S}_n} \operatorname{sign}(\tau)A_{\tau(1),1}\cdots A_{\tau(n),n} = \det(A)$; $\operatorname{sign}(\tau^{-1}) = \operatorname{sign}(\tau)$.

Structural Commentary: The determinant is the normalized alternating multilinear measure of columns.
Key Insight: Multilinearity expands columns; alternation kills repeated directions. □

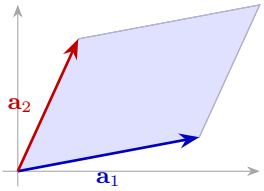


Figure 72: Determinant is signed column-volume; swaps reverse orientation and dependence gives zero volume.

Method 1.1.9. *Determinant by Structure*
To compute or simplify a determinant:

(i) first look for triangular or block triangular form;
(ii) use column linearity to split complicated columns only when it reduces structure;
(iii) use column swaps to move useful pivots into position, tracking a factor of -1 for each swap;
(iv) detect zero determinants from dependent columns or repeated columns;
(v) for products or changes of basis, use multiplicativity instead of expanding permutations.

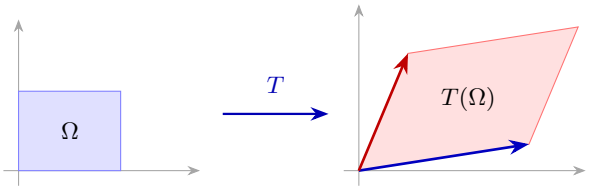


Figure 73: A linear map scales oriented volume by its determinant; singular maps collapse volume to zero.

Lemma 1.1.10. *Identity Blocks Do Not Change Determinant*
Let $A \in \operatorname{Mat}(m,m,\mathbb{F})$ *and let* $n \geq 0$. *Then* $\det(I_n \oplus A) = \det(A) = \det(A \oplus I_n)$.

Proof. In the permutation formula for $I_n \oplus A$, a nonzero term must satisfy $\sigma(k) = k$ for $1 \le k \le n$. Remaining choices are exactly permutations of $\{n+1, \dots, n+m\}$, after relabeling, gives $\det(A)$. Hence $\det(I_n \oplus A) = \det(A)$. The same argument applied to the last n coordinates gives $\det(A \oplus I_n) = \det(A)$.
The case $n = 0$ is the empty identity block convention.
Structural Commentary: Identity blocks add directions without changing volume.
Key Insight: In the determinant expansion, the identity block forces the matching coordinates to be fixed. □

Theorem 1.1.11. *Determinant of an Upper Block-Triangular Matrix*
Let $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $B \in \mathrm{Mat}(r, s, \mathbb{F})$, *and* $D \in \mathrm{Mat}(s, s, \mathbb{F})$. *Then* $\det \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det(A) \det(D)$.

Proof. Factor $\begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} I_r & B \\ 0 & I_s \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & I_s \end{bmatrix}$.
The first and last factors have determinants $\det(D)$ and $\det(A)$ by Lemma 1.1.10.
The middle factor is upper triangular with every diagonal entry equal to 1, so its determinant is 1.
Multiplicativity of determinant gives the formula.
Structural Commentary: Block upper triangular matrices separate determinant contribution into diagonal blocks. The off-diagonal block does not affect the determinant.
Key Insight: Factor the matrix into two block diagonal pieces and one unit upper triangular piece. □

Proposition 1.1.12. *Schur Complement Determinant Formula*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$.
If A *is invertible, then the Schur complement of* A *in* M *is* $M/A := D - CA^{-1}B$, *and* $\det(M) = \det(A) \det(M/A)$.

Proof. Block Gaussian elimination gives $\begin{bmatrix} I_r & 0 \\ -CA^{-1} & I_s \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}$.
The elimination matrix is lower block-triangular with identity diagonal blocks, so its determinant is 1. Take determinants, apply Theorem 1.1.11 to right-hand block triangular matrix, $\det(M) = \det(A) \det(D - CA^{-1}B)$.
Structural Commentary: Schur complementation is block Gaussian elimination.
Key Insight: Eliminating C splits the determinant into $\det(A)$ times the remaining block determinant. □

Proposition 1.1.13. *Schur Complement of the Lower-Right Block*
Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, $A \in \mathrm{Mat}(r, r, \mathbb{F})$, $D \in \mathrm{Mat}(s, s, \mathbb{F})$.
If D *is invertible, then the Schur complement of* D *in* M *is* $M/D := A - BD^{-1}C$, *and* $\det(M) = \det(D) \det(M/D)$.

Proof. Block Gaussian elimination gives $\begin{bmatrix} I_r & -BD^{-1} \\ 0 & I_s \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A - BD^{-1}C & 0 \\ C & D \end{bmatrix}$.
The elimination matrix has determinant 1. The transpose of the matrix on the right is upper block-triangular, and $\det(X^t) = \det(X)$ by Theorem 1.1.8. Therefore Theorem 1.1.11 gives $\det(M) = \det(A - BD^{-1}C) \det(D)$.
Structural Commentary: This is the lower-right version of Schur complementation. Eliminating the upper-right block isolates the determinant contribution of D.
Key Insight: Eliminate by blocks, then read the determinant from the diagonal blocks. □

Theorem 1.1.14. *Sylvester Determinant Identity*
Let $A \in \mathrm{Mat}(n, m, \mathbb{F})$ *and* $B \in \mathrm{Mat}(m, n, \mathbb{F})$. *Then* $\det(I_n + AB) = \det(I_m + BA)$.

Proof. Let $M = \begin{bmatrix} I_n & -A \\ B & I_m \end{bmatrix}$.
Using Proposition 1.1.13 with lower-right block I_m gives $\det(M) = \det(I_n + AB)$. Using Proposition 1.1.12 with upper-left block I_n gives $\det(M) = \det(I_m + BA)$. Hence the two determinants are equal.
Structural Commentary: Two Schur complements of one block matrix give Sylvester's identity.
Key Insight: Compare the same determinant after eliminating different identity blocks. □

Theorem 1.1.15. *Cauchy–Binet Formula*
Let $A \in \mathrm{Mat}(k, n, \mathbb{F})$, $B \in \mathrm{Mat}(n, k, \mathbb{F})$, *and* $k \le n$. *For* $J = \{j_1 < \cdots < j_k\} \subseteq \{1, \dots, n\}$, *let* A^J *use the columns of* A *in* J *and* B_J *use the rows of* B *in* J.
Then $\det(AB) = \sum_J \det(A^J) \det(B_J)$, *where* J *ranges over all* k*-subsets.*

Proof. Put $S(A,B) := \sum_J \det(A^J)\det(B_J)$. For an elementary $n \times n$ matrix E, determinant multilinearity and row/column operation rules give $S(A,B) = S(AE, E^{-1}B)$, and $AB = (AE)(E^{-1}B)$.
If $\operatorname{rank} A < k$, both sides vanish. Otherwise reduce A by elementary column operations to $A' = [C\ 0]$ with C invertible, and transform B oppositely to B'.
Then $AB = A'B'$ and $\det(A'B') = \det(C)\det(B'_{J_0}) = S(A',B')$, $J_0 = \{1,\dots,k\}$, as every other k-minor of A' is zero. Hence $\det(AB) = S(A,B)$.
Structural Commentary: Cauchy–Binet is determinant multiplicativity for rectangular middle dimensions.
Key Insight: The middle space contributes one k-coordinate shadow at a time. □

Proposition 1.1.16. *Bordered Determinant Formulas*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, let $\mathbf{x},\mathbf{y} \in \mathbb{F}^n$, and let $c \in \mathbb{F}$.

(i) If $c \neq 0$, then $\det \begin{bmatrix} c & \mathbf{x}^t \\ \mathbf{y} & A \end{bmatrix} = c \det\left(A - \frac{1}{c}\mathbf{y}\mathbf{x}^t\right)$.

(ii) If A is invertible, then $\det \begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = \left(c - \mathbf{y}^t A^{-1}\mathbf{x}\right)\det(A)$

Proof.

(i) This is Proposition 1.1.12 with the scalar block c in the upper-left corner.
(ii) This is Proposition 1.1.12 with A as the upper-left block and $D = c$.

Structural Commentary: Bordered determinants are Schur complements with scalar borders.
Key Insight: Treat the border as a 1×1 block and apply the Schur complement formula. □

1.2 Minors, Cofactors, and Rank

Definition 1.2.1. *Minors, Cofactors, and Adjugate*
For $A \in \mathrm{Mat}(n,n,\mathbb{F})$, let M_{ij} delete row i and column j from A; set $\det(\emptyset_{0\times 0}) = 1$. Define $C_{ij} := (-1)^{i+j}\det(M_{ij})$. Then $C = [C_{ij}]$ is the *cofactor matrix*, and $\operatorname{adj}(A) := C^t$ is the *adjugate*.

Lemma 1.2.2. *Cofactor Expansion*
For $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{F})$, $\det A = \sum_{k=1}^{n} a_{ik}C_{ik}$ *and* $\det A = \sum_{k=1}^{n} a_{kj}C_{kj}$.

Proof. Using $\det A = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{r=1}^{n} a_{r,\sigma(r)}$, group terms by $\sigma(i) = k$.
For fixed k, the remaining permutation sum equals $(-1)^{i+k}\det(M_{ik}) = C_{ik}$, giving the row formula. Applying this to A^t and using Theorem 1.1.8 gives the column formula.
Structural Commentary: Cofactor expansion is the big determinant formula collected around one row or column.
Key Insight: Fix one entry; the rest is its minor. □

Remark 1.2.3. *Adjugate Form of the Bordered Formula*
Proposition 1.1.16 gives the adjugate form $\det \begin{bmatrix} A & \mathbf{x} \\ \mathbf{y}^t & c \end{bmatrix} = c\det(A) - \mathbf{y}^t \operatorname{adj}(A)\mathbf{x}$. No invertibility of A is required.

Theorem 1.2.4. *Adjugate Identity and Cramer's Rule*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and let $\mathbf{b} \in \mathbb{F}^n$. Then $A\operatorname{adj}(A) = \operatorname{adj}(A)A = (\det A)I_n$. If $\det A \neq 0$, then, for $A\mathbf{x} = \mathbf{b}$, $A^{-1} = \frac{1}{\det A}\operatorname{adj}(A)$ and $x_j = \frac{\det A_j(\mathbf{b})}{\det A}$ for $1 \le j \le n$, where $A_j(\mathbf{b})$ replaces column j of A by $\mathbf{b}$.

Proof. Since $(\operatorname{adj} A)_{kj} = C_{jk}$, $(A\operatorname{adj} A)_{ij} = \sum_k a_{ik}C_{jk}$.
For $i = j$ this is $\det A$ by Lemma 1.2.2; for $i \neq j$ it is the cofactor expansion of a matrix with rows i and j equal, hence 0. Thus $A\operatorname{adj}(A) = (\det A)I_n$. The same argument applied to A^t gives $\operatorname{adj}(A)A = (\det A)I_n$. If $\det A \neq 0$, division gives the inverse formula. Finally, $(\operatorname{adj} A\,\mathbf{b})_j = \sum_i C_{ij}b_i = \det A_j(\mathbf{b})$, by expansion down column j, so $x = A^{-1}\mathbf{b}$ gives Cramer's rule.
Structural Commentary: Cofactors assemble all one-column replacements of A into a single inverse formula.
Key Insight: Off-diagonal products vanish because they are determinants with two equal rows or columns. □

Theorem 1.2.5. *Trace and Determinant Are Similarity Invariants*
If $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ are similar, then $\operatorname{tr}(A) = \operatorname{tr}(B)$, $\det(A) = \det(B)$.

Proof. Write $B = PAP^{-1}$ with P invertible.
Proposition 1.1.4 gives $\operatorname{tr}(B) = \operatorname{tr}\big(P(AP^{-1})\big) = \operatorname{tr}\big((AP^{-1})P\big) = \operatorname{tr}(A)$.
Theorem 1.1.8 gives $\det(B) = \det(P)\det(A)\det(P^{-1}) = \det(A)$.

Structural Commentary: Trace and determinant ignore the coordinate system used to write an operator.
Key Insight: Conjugating by P inserts P and P^{-1}, which cancel inside trace and determinant. □

Principle 1.2.6. *Trace and Determinant Are Coordinate-Free*
Trace and determinant are first computed from a matrix, but their real structural meaning is operator-level: changing basis conjugates the matrix, and conjugation preserves both quantities.

Theorem 1.2.7. *Trace and Determinant of an Operator Are Basis-Independent*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let β, γ be bases of $\mathcal{V}$.
Then $\operatorname{tr}\big([T]_{\beta\leftarrow\beta}\big) = \operatorname{tr}\big([T]_{\gamma\leftarrow\gamma}\big)$, $\det\big([T]_{\beta\leftarrow\beta}\big) = \det\big([T]_{\gamma\leftarrow\gamma}\big)$.

Proof. By Theorem 3.6.7, the two matrices are similar: $[T]_{\gamma\leftarrow\gamma} = C_{\gamma\leftarrow\beta}[T]_{\beta\leftarrow\beta}C_{\beta\leftarrow\gamma}$.
Theorem 1.2.5 now gives equality of traces and determinants.
Structural Commentary: Basis changes conjugate matrices; trace and determinant remain operator invariants.
Key Insight: Similar matrices have the same trace and determinant. □

Definition 1.2.8. *Trace and Determinant of an Operator*
Let $\mathcal{V}$ be finite-dim, let $T \in \mathcal{L}(\mathcal{V})$. Choose any basis β of $\mathcal{V}$. Define $\operatorname{tr} T := \operatorname{tr}\big([T]_{\beta\leftarrow\beta}\big)$, $\det T := \det\big([T]_{\beta\leftarrow\beta}\big)$.
Theorem 1.2.7 shows these definitions do not depend on chosen basis.

Proposition 1.2.9. *Unitary Matrices and Isometries Have Determinant of Modulus One*
Let $U \in \operatorname{Mat}(n, n, \mathbb{F})$ be unitary. Then $|\det U| = 1$. More generally, if $S \in \mathcal{L}(\mathcal{V})$ is an isometry on a finite-dimensional inner-product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $|\det S| = 1$.

Proof. The permutation definition gives $\det(U^*) = \overline{\det U}$. By Theorem 1.1.8, taking determinants in $U^*U = I_n$ gives $1 = \det(U^*)\det(U) = \overline{\det U}\det U = |\det U|^2$. Hence $|\det U| = 1$.
For an isometry S, choose an orthonormal basis and let U be the matrix of S in that basis. Then U is unitary by Theorem 2.4.26, and $\det S = \det U$.
Structural Commentary: In orthonormal coordinates, isometries are unitary or orthogonal, so the determinant calculation is the matrix one.
Key Insight: Determinants turn $U^*U = I$ into $|\det U|^2 = 1$. □

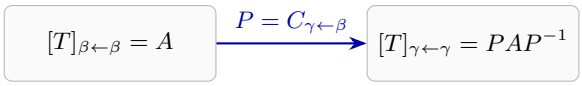


Figure 74: Basis changes conjugate matrices; trace and determinant are unchanged.

Proposition 1.2.10. *Trace and Determinant from q_T*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dim over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $T \in \mathcal{L}(\mathcal{V})$, $q_T(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$, $n = \dim \mathcal{V}$.
Then $\operatorname{tr} T = -c_{n-1}$ and $\det T = (-1)^n c_0$. If $\mathbb{F} = \mathbb{C}$ and $\lambda_1, \ldots, \lambda_n$ list the eigenvalues of T with algebraic multiplicity, then $\operatorname{tr} T = \lambda_1 + \cdots + \lambda_n$, $\det T = \lambda_1 \cdots \lambda_n$.

Proof. Choose a basis β and set $A = [T]_{\beta\leftarrow\beta}$.
Then $q_T = p_A$, and by Definition 1.2.8, $\operatorname{tr} T = \operatorname{tr} A$ and $\det T = \det A$.
Theorem 6.2.3 gives $\operatorname{tr} T = -c_{n-1}$ and $\det T = (-1)^n c_0$.
If $\mathbb{F} = \mathbb{C}$, then comparing $q_T(z) = (z - \lambda_1)\cdots(z - \lambda_n)$ with the displayed form of q_T gives the formulas.
Structural Commentary: Characteristic polynomial stores trace and determinant in first and last coefficients.
Key Insight: Compare coefficients of $\det(zI - T)$. □

Proposition 1.2.11. *Trace and Determinant from Eigenvalues*
Let $\lambda_1, \ldots, \lambda_n$ list the eigenvalues of $A \in \operatorname{Mat}(n, n, \mathbb{C})$ according to m_A. Then $\operatorname{tr} A = \sum_{j=1}^{n} \lambda_j$ and $\det A = \prod_{j=1}^{n} \lambda_j$.

Proof. By Fundamental Theorem of Algebra, $p_A(z) = \prod_{j=1}^{n}(z - \lambda_j) = z^n - \left(\sum_{j=1}^{n} \lambda_j\right) z^{n-1} + \cdots + (-1)^n \prod_{j=1}^{n} \lambda_j$.
Theorem 6.2.3 gives $p_A(z) = z^n - (\operatorname{tr} A)z^{n-1} + \cdots + (-1)^n \det A$. Uniqueness of coefficients gives the identities.
Structural Commentary: Coefficients of p_A encode symmetric functions of the eigenvalues.
Key Insight: Trace is the first root sum and determinant is the full root product. □

Proposition 1.2.12. *Transpose, Conjugate, and Adjoint Spectra*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. Then $p_{A^t}(z) = p_A(z)$, $p_{\overline{A}}(z) = \overline{p_A(\overline{z})}$, and $p_{A^*}(z) = \overline{p_A(\overline{z})}$.
Consequently, if $\lambda \in \operatorname{spec}(A)$, then $m_{A^t}(\lambda) = m_A(\lambda)$ and $m_{\overline{A}}(\overline{\lambda}) = m_{A^*}(\overline{\lambda}) = m_A(\lambda)$.

Proof. Use determinant identities:

$$p_{A^t}(z) = \det(zI_n - A^t) = \det((zI_n - A)^t) = p_A(z),$$
$$p_{\overline{A}}(z) = \det(zI_n - \overline{A}) = \det(\overline{\overline{z}I_n - A}) = \overline{\det(\overline{z}I_n - A)} = \overline{p_A(\overline{z})}.$$

Since $A^* = \overline{A}^t$, the first identity applied to $\overline{A}$ gives the formula for A^*.
Multiplicity is preserved because conjugating a polynomial sends a factor $(z-\lambda)^k$ to $(z-\overline{\lambda})^k$.
Structural Commentary: Transpose preserves p_A; conjugation conjugates its roots.
Key Insight: The adjoint combines conjugation with transposition, so its spectrum is conjugated. □

Proposition 1.2.13. *Transpose Preserves Spectrum and Geometric Multiplicity*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then $\lambda \in \mathrm{spec}(A) \iff \lambda \in \mathrm{spec}(A^t)$, and $\dim E_\lambda(A) = \dim E_\lambda(A^t)$.

Proof. Since $(A-\lambda I_n)^t = A^t - \lambda I_n$, Theorem 3.3.23 gives $\mathrm{rank}(A-\lambda I_n) = \mathrm{rank}(A^t - \lambda I_n)$.
Matrix rank-nullity then gives $\dim E_\lambda(A) = n - \mathrm{rank}(A-\lambda I_n) = n - \mathrm{rank}(A^t - \lambda I_n) = \dim E_\lambda(A^t)$.
($\Rightarrow$) If $\lambda \in \mathrm{spec}(A)$, then $E_\lambda(A) \neq \{\mathbf{0}\}$, so $E_\lambda(A^t) \neq \{\mathbf{0}\}$ and $\lambda \in \mathrm{spec}(A^t)$.
($\Leftarrow$) The same argument with A and A^t interchanged gives the converse.
Structural Commentary: Transposition preserves rank, so shifted kernels have the same dimension.
Key Insight: Spectrum and geometric multiplicity are kernel facts for $A - \lambda I_n$. □

Corollary 1.2.14. *Square Matrix Algebraic Invertibility Dictionary*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(n,n,\mathbb{F})$. The following are equivalent:

(i) *A is invertible;*
(ii) *$\mathbf{a}_1, \ldots, \mathbf{a}_n$ form a basis of $\mathbb{F}^n$;*
(iii) *$\mathbf{a}_1, \ldots, \mathbf{a}_n$ are linearly independent;*
(iv) *$\mathbf{a}_1, \ldots, \mathbf{a}_n$ span $\mathbb{F}^n$;*
(v) *for every $\mathbf{b} \in \mathbb{F}^n$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution;*
(vi) *the homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the zero solution.*

Proof.

(i) $\Rightarrow$ *(ii)* By Theorem 3.3.29, the columns of an invertible matrix form a basis.
(ii) $\Rightarrow$ *(iii)* A basis is linearly independent.
(iii) $\Rightarrow$ *(iv)* In $\mathbb{F}^n$, n linearly independent vectors form a basis, hence span.
(iv) $\Rightarrow$ *(v)* In $\mathbb{F}^n$, n spanning vectors form a basis, so every $\mathbf{b}$ has unique coordinates $\mathbf{x}$ in $A\mathbf{x} = \mathbf{b}$.
(v) $\Rightarrow$ *(vi)* Apply (v) to $\mathbf{b} = \mathbf{0}$; since $\mathbf{x} = \mathbf{0}$ is a solution, it is the only one.
(vi) $\Rightarrow$ *(i)* Condition (vi) says the columns are independent.
Hence they form a basis of $\mathbb{F}^n$, and Corollary 3.5.3 gives invertibility.

Structural Commentary: Square invertibility is column basishood.
Key Insight: Invertible columns give unique coordinates. □

Corollary 1.2.15. *Nonsingularity by Rank, Determinant, and Spectrum*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$. The following are equivalent:

(i) *A is invertible;*
(ii) *$\mathrm{rank}(A) = n$;*
(iii) *$\det A \neq 0$;*
(iv) *$0 \notin \mathrm{spec}(A)$.*

Over $\mathbb{C}$, this is also equivalent to all eigenvalues of A being nonzero.

Proof.

(i) $\Rightarrow$ *(ii)* This is Corollary 3.5.5.
(ii) $\Rightarrow$ *(iii)* Full rank gives invertibility by Corollary 3.5.5, hence $\det A \neq 0$ by Theorem 1.1.8.
(iii) $\Rightarrow$ *(iv)* If $\det A \neq 0$, then A is invertible by Theorem 1.1.8; hence $0 \notin \mathrm{spec}(A)$ by Corollary 5.1.9.
(iv) $\Rightarrow$ *(i)* This is Corollary 5.1.9.

Over $\mathbb{C}$, Theorem 6.2.3 identifies $\lambda_1, \ldots, \lambda_n$ as the roots of p_A, so (iv) is equivalent to $\lambda_j \neq 0$ for all j.
Structural Commentary: Rank, determinant, and spectrum detect singularity.
Key Insight: Singularity can be seen as rank loss, zero determinant, or the zero eigenvalue. □

Method 1.2.16. *Trace–Determinant Invariant Method*
To get eigenvalue data without solving for eigenvalues:

(i) choose a basis in which $[T]_{\beta\leftarrow\beta}$ is easiest to compute;
(ii) compute $\mathrm{tr}\, T$ and $\det T$ from this matrix;

(iii) if the eigenvalues are $\lambda_1, \ldots, \lambda_n$, conclude $\lambda_1 + \cdots + \lambda_n = \operatorname{tr} T$, and $\lambda_1 \cdots \lambda_n = \det T$;
(iv) for equations $ST - TS = A$, first test $\operatorname{tr} A = 0$; otherwise no solution exists.

Theorem 1.2.17. *Rank Bounds for Products and Concatenations*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$, $B \in \mathrm{Mat}(k,n,\mathbb{F})$, *and* $C \in \mathrm{Mat}(m,p,\mathbb{F})$. *Then* $\operatorname{rank}(AB) \le \min(\operatorname{rank}(A), \operatorname{rank}(B))$ *and* $\operatorname{rank}(AB) = \operatorname{rank}(A)$ *if and only if* $\operatorname{col}(AB) = \operatorname{col}(A)$.
Also, $\max(\operatorname{rank}(A), \operatorname{rank}(C)) \le \operatorname{rank}\begin{bmatrix} A & C \end{bmatrix}$ *and* $\operatorname{rank}\begin{bmatrix} A & C \end{bmatrix} = \operatorname{rank}(A)$ *if and only if* $\operatorname{col}(A) = \operatorname{col}(A) + \operatorname{col}(C)$.

Proof. Each column of AB is a linear combination of the columns of A, so $\operatorname{col}(AB) \subseteq \operatorname{col}(A)$. Therefore $\operatorname{rank}(AB) \le \operatorname{rank}(A)$; since both spaces are finite-dimensional, equality holds exactly when $\operatorname{col}(AB) = \operatorname{col}(A)$.
Applying this same argument to $(AB)^t = B^t A^t$ and using Theorem 3.3.23 gives $\operatorname{rank}(AB) \le \operatorname{rank}(B)$.
For the block matrix, its columns are precisely the columns of A and C, hence $\operatorname{col}\begin{bmatrix} A & C \end{bmatrix} = \operatorname{col}(A) + \operatorname{col}(C)$.
This space contains both $\operatorname{col}(A)$ and $\operatorname{col}(C)$, giving the lower bound. Equality with $\operatorname{rank}(A)$ occurs exactly when the containing subspace has the same dimension as $\operatorname{col}(A)$, equivalently when $\operatorname{col}(A) + \operatorname{col}(C) = \operatorname{col}(A)$.
Structural Commentary: Products can only reduce accessible column directions, while horizontal concatenation can only add column directions.
Key Insight: Rank tracks whether a matrix operation preserves, loses, or adds column-space dimension. □

Theorem 1.2.18. *Sylvester Rank Inequality*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$ *and* $B \in \mathrm{Mat}(k,n,\mathbb{F})$. *Then* $\operatorname{rank}(A) + \operatorname{rank}(B) - k \le \operatorname{rank}(AB)$.

Proof. Let $R = \operatorname{col}(B) \subseteq \mathbb{F}^k$. The columns of AB are A applied to the columns of B, so $\operatorname{col}(AB) = A(R)$.
Apply rank-nullity to the restriction $T_A|_R : R \to \mathbb{F}^m$. Its range is $A(R) = \operatorname{col}(AB)$ and its kernel is $R \cap \ker(A)$.
Hence $\operatorname{rank}(AB) = \dim(A(R)) = \dim(R) - \dim(R \cap \ker(A)) \ge \operatorname{rank}(B) - \dim(\ker(A))$.
Since A has k columns, Corollary 3.3.22 gives $\dim(\ker(A)) = k - \operatorname{rank}(A)$. Substitution gives desired inequality.
Structural Commentary: Multiplication by A can lose only the part of $\operatorname{col}(B)$ lying in $\ker(A)$. Sylvester's inequality is this loss estimate in dimensions.
Key Insight: The shared middle dimension k bounds how much rank can disappear in the product. □

Theorem 1.2.19. *Frobenius Rank Inequality*
Let $A \in \mathrm{Mat}(m,k,\mathbb{F})$, $B \in \mathrm{Mat}(k,p,\mathbb{F})$, *and* $C \in \mathrm{Mat}(p,n,\mathbb{F})$.
Then $\operatorname{rank}(AB) + \operatorname{rank}(BC) \le \operatorname{rank}(B) + \operatorname{rank}(ABC)$.

Proof. Let $R = \operatorname{col}(B)$ and $S = \operatorname{col}(BC)$. Since every column of BC is a linear combination of columns of B, we have $S \subseteq R$. The induced map T_A maps R onto $\operatorname{col}(AB)$ and maps S onto $\operatorname{col}(ABC)$.
Rank-nullity on the restrictions $T_A|_R$ and $T_A|_S$ gives

$$\operatorname{rank}(AB) = \dim R - \dim(R \cap \ker(A)), \quad \operatorname{rank}(ABC) = \dim S - \dim(S \cap \ker(A)).$$

Since $S \cap \ker(A) \subseteq R \cap \ker(A)$, we have $\dim(S \cap \ker(A)) \le \dim(R \cap \ker(A))$.
Thus $\dim R - \dim(R \cap \ker(A)) + \dim S \le \dim R + \dim S - \dim(S \cap \ker(A))$.
Substituting $\dim R = \operatorname{rank}(B)$ and $\dim S = \operatorname{rank}(BC)$ gives the stated inequality.
Structural Commentary: Frobenius tracks A on nested column spaces.
Key Insight: The smaller column space cannot lose more dimensions to $\ker(A)$ than the larger one. □

Corollary 1.2.20. *Rank Inequalities from Frobenius*
The following are special cases of Theorem 1.2.19:
(i) $\operatorname{rank}(AB) \le \min(\operatorname{rank}(A), \operatorname{rank}(B))$;
(ii) $\operatorname{rank}(A) + \operatorname{rank}(B) - k \le \operatorname{rank}(AB)$.

Proof.
(i) Apply Frobenius to conformable factors X, Y, Z. Taking $Z = 0$ gives $\operatorname{rank}(XY) \le \operatorname{rank}(Y)$, because $\operatorname{rank}(Y0) = \operatorname{rank}(XY0) = 0$. Taking $X = 0$ gives $\operatorname{rank}(YZ) \le \operatorname{rank}(Y)$.
Renaming the factors gives $\operatorname{rank}(AB) \le \operatorname{rank}(B)$ and $\operatorname{rank}(AB) \le \operatorname{rank}(A)$.
(ii) Taking the middle matrix to be I_k gives $\operatorname{rank}(A) + \operatorname{rank}(B) \le k + \operatorname{rank}(AB)$, which is Sylvester's inequality.

Structural Commentary: Frobenius is the three-factor parent inequality. The usual upper and lower product-rank bounds appear by making one factor zero or identity.
Key Insight: Zero factors isolate upper bounds; an identity middle factor yields the Sylvester lower bound. □

Proposition 1.2.21. *Sylvester's Law of Nullity*
For a matrix X, *let* $\operatorname{nullity}(X) = \dim(\ker(X))$; *this is the dimension of its nullspace.*
(i) If $A \in \mathrm{Mat}(m,k,\mathbb{F})$ *and* $B \in \mathrm{Mat}(k,n,\mathbb{F})$, *then* $\operatorname{nullity}(AB) \le \operatorname{nullity}(A) + \operatorname{nullity}(B)$.
(ii) If $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$, *then* $\max(\operatorname{nullity}(A), \operatorname{nullity}(B)) \le \operatorname{nullity}(AB)$.

Proof.

(i) By rank-nullity and Theorem 1.2.18,

$$\begin{aligned}\text{nullity}(AB) &= n - \text{rank}(AB) \le n - \text{rank}(A) - \text{rank}(B) + k \\ &= (k - \text{rank}(A)) + (n - \text{rank}(B)) = \text{nullity}(A) + \text{nullity}(B).\end{aligned}$$

(ii) For square A, B, $\text{rank}(AB) \le \text{rank}(A)$, $\text{rank}(AB) \le \text{rank}(B)$. Subtract from n gives desired lower bounds.

Structural Commentary: Nullity is rank loss from the domain.
Key Insight: Use rank-nullity to translate rank bounds into kernel bounds. □

Theorem 1.2.22. *Rank Triangle Inequality*
If $A, B \in \text{Mat}(m, n, \mathbb{F})$, then $|\text{rank}(A) - \text{rank}(B)| \le \text{rank}(A + B) \le \text{rank}(A) + \text{rank}(B)$.

Proof. Since $\text{col}(A + B) \subseteq \text{col}(A) + \text{col}(B)$, Theorem 2.3.1 gives $\text{rank}(A + B) \le \dim(\text{col}(A) + \text{col}(B)) \le \text{rank}(A) + \text{rank}(B)$. Applying this upper bound to $A = (A + B) - B$ gives $\text{rank}(A) \le \text{rank}(A + B) + \text{rank}(B)$, so $\text{rank}(A) - \text{rank}(B) \le \text{rank}(A + B)$. Applying it to $B = (A + B) - A$ gives $\text{rank}(B) - \text{rank}(A) \le \text{rank}(A + B)$. Together these are exactly $|\text{rank}(A) - \text{rank}(B)| \le \text{rank}(A + B)$.
Structural Commentary: The rank of a sum is controlled by the sum of the two column spaces. The reverse bounds come by solving for either summand.
Key Insight: Rank behaves like a dimension-valued size under addition. □

1.3 Commutators and Special Matrix Classes

Definition 1.3.1. *Commutator*
For $S, T \in \mathcal{L}(\mathcal{V})$, the *commutator* of S and T is $[S, T] := ST - TS$.
For $A, B \in \text{Mat}(n, n, \mathbb{F})$, the *matrix commutator* is $[A, B] := AB - BA$.

Proposition 1.3.2. *Trace Obstruction for Commutators*
If $A, B \in \text{Mat}(n, n, \mathbb{F})$, then $\text{tr}(AB - BA) = 0$. Hence any matrix similar to a commutator has trace zero.

Proof. By linearity of trace and Proposition 1.1.4, $\text{tr}(AB - BA) = \text{tr}(AB) - \text{tr}(BA) = 0$.
If C is similar to $AB - BA$, then Theorem 1.2.5 gives $\text{tr}(C) = \text{tr}(AB - BA) = 0$.
Structural Commentary: Trace kills commutators because it is cyclic.
Key Insight: AB and BA have the same trace, so their difference has trace zero. □

Lemma 1.3.3. *Scalar Matrices and One-Step Orbits*
Let $A \in \text{Mat}(n, n, \mathbb{F})$ with $n \ge 2$.
Then $(\mathbf{x}, A\mathbf{x})$ is linearly dependent for every $\mathbf{x} \in \mathbb{F}^n$ if and only if A is a scalar matrix.

Proof.

($\Rightarrow$) Suppose $A = cI_n$. Then $A\mathbf{x} = c\mathbf{x}$ for every $\mathbf{x}$, so $(\mathbf{x}, A\mathbf{x})$ is linearly dependent.

($\Leftarrow$) Suppose $(\mathbf{x}, A\mathbf{x})$ is linearly dependent for every $\mathbf{x}$.
For each standard basis vector $\mathbf{e}_i$, $\exists a_i \in \mathbb{F}$ such that $A\mathbf{e}_i = a_i\mathbf{e}_i$. For $i \ge 2$, dependence of $\mathbf{e}_1 + \mathbf{e}_i$ and $A(\mathbf{e}_1 + \mathbf{e}_i) = a_1\mathbf{e}_1 + a_i\mathbf{e}_i$ means $a_1\mathbf{e}_1 + a_i\mathbf{e}_i = b(\mathbf{e}_1 + \mathbf{e}_i)$ for some $b \in \mathbb{F}$. Comparing the independent vectors $\mathbf{e}_1, \mathbf{e}_i$ gives $a_1 = b = a_i$. Hence every a_i equals a_1, so $A = a_1 I_n$.

Structural Commentary: A scalar matrix preserves every one-dimensional coordinate direction and every diagonal line between two coordinates. These diagonal lines force all coordinate scalars to agree.
Key Insight: If every vector is an eigenvector, then there can be only one eigenvalue. □

Lemma 1.3.4. *Nonscalar Matrices Have a Zero Diagonal Entry After Similarity*
Let $A \in \text{Mat}(n, n, \mathbb{F})$ with $n \ge 2$.
If A is not a scalar matrix, then A is similar over $\mathbb{F}$ to a matrix whose $(1, 1)$ entry is 0.

Proof. By Lemma 1.3.3, there is $\mathbf{x} \in \mathbb{F}^n$ such that $(\mathbf{x}, A\mathbf{x})$ is linearly independent. Extend $(\mathbf{x}, A\mathbf{x})$ to a basis β of $\mathbb{F}^n$. In the basis β, the first column of the matrix of A is the coordinate vector of $A\mathbf{x}$, namely $(0, 1, 0, \ldots, 0)^t$. Thus the $(1, 1)$ entry is 0, and this matrix is similar to A.
Structural Commentary: Nonscalarity produces a vector whose first iterate points in a genuinely new direction. Using these two vectors as the start of a basis forces a zero in the first diagonal position.
Key Insight: Put $\mathbf{x}$ first and $A\mathbf{x}$ second in the basis. □

Theorem 1.3.5. *Similarity to a Constant-Diagonal Matrix*
Let $n \ge 1$, let $A \in \text{Mat}(n, n, \mathbb{F})$, and assume $k \cdot 1_{\mathbb{F}} \ne 0$ for every $1 \le k \le n$. Then A is similar over $\mathbb{F}$ to a matrix whose diagonal entries are all equal to $\frac{1}{n}\text{tr}(A)$.

Proof. Set $\tau = \frac{1}{n}\operatorname{tr}(A)$ and $B = A - \tau I_n$. Then $\operatorname{tr}(B) = 0$. It is enough to show that every trace-zero matrix is similar to a matrix with zero diagonal, because Theorem 3.6.15 then adds τI_n back to the diagonal.
To prove the trace-zero claim by induction on n. If $n = 1$, a trace-zero matrix is $[0]$. Let $n \geq 2$ and assume the claim holds for smaller sizes. Let $\operatorname{tr}(B) = 0$. If B is scalar, then $B = cI_n$ and $0 = \operatorname{tr}(B) = nc$.
Since $n \cdot 1_{\mathbb{F}} \neq 0$, we get $c = 0$, so B already has zero diagonal.
If B is not scalar, Lemma 1.3.4 makes B similar to $\begin{bmatrix} 0 & * \\ * & B_1 \end{bmatrix}$, and $\operatorname{tr}(B_1) = 0$.
By induction, choose an invertible S_1 such that $S_1^{-1}B_1S_1$ has zero diagonal. Conjugating the displayed block matrix by $1 \oplus S_1$ leaves the first diagonal entry equal to 0 and replaces the lower-right block by $S_1^{-1}B_1S_1$.
Hence the resulting similar matrix has every diagonal entry equal to 0.
Structural Commentary: The proof removes one zero diagonal entry at a time.
Key Insight: Similarity creates the first zero; trace zero handles the smaller block. □

Theorem 1.3.6. *Shoda's Theorem in the Distinct-Diagonal Case*
Let $n \geq 1$ and assume $k \cdot 1_{\mathbb{F}} \neq 0$ for every $1 \leq k \leq n$. For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, the matrix A is a commutator in $\operatorname{Mat}(n, n, \mathbb{F})$ if and only if $\operatorname{tr}(A) = 0$. In particular, no nonzero scalar matrix is a commutator.

Proof.

(⇒) Proposition 1.3.2 gives $\operatorname{tr}(A) = 0$.
Thus cI_n can be a commutator only when $nc = \operatorname{tr}(cI_n) = 0$, hence only when $c = 0$.

(⇐) Suppose $\operatorname{tr}(A) = 0$. By Theorem 1.3.5, there is an invertible S such that $B = S^{-1}AS = [b_{ij}]$ has zero diagonal. Let $X = \operatorname{diag}(1, 2, \ldots, n)$, viewing the integers as scalars in $\mathbb{F}$, and define $Y = [y_{ij}]$ by

$$y_{ij} = \begin{cases} (i-j)^{-1}b_{ij}, & i \neq j, \\ 0, & i = j. \end{cases}$$

The hypothesis ensures that $i - j \neq 0$ in $\mathbb{F}$ whenever $i \neq j$. Thus $XY - YX = [(i-j)y_{ij}] = B$.
Hence $A = (SXS^{-1})(SYS^{-1}) - (SYS^{-1})(SXS^{-1})$, so A is a commutator.

Structural Commentary: Trace is the obstruction; similarity exposes zero diagonal form.
Key Insight: Cyclicity forces trace zero; distinct diagonals realize it. □

Remark 1.3.7. *Field Hypothesis in the Diagonal Proof*
The proof above uses that the integers $1, \ldots, n$ are distinct and that each smaller block size remains nonzero in $\mathbb{F}$. In small characteristic, this diagonal proof must be replaced by a more delicate argument.

Corollary 1.3.8. *The Identity Is Not a Commutator*
Assume $\dim \mathcal{V} \cdot 1_{\mathbb{F}} \neq 0$. There do not exist $S, T \in \mathcal{L}(\mathcal{V})$ such that $ST - TS = I_{\mathcal{V}}$.

Proof. If $ST - TS = I_{\mathcal{V}}$, choose a basis and let $A = [S]$ and $B = [T]$ in that basis. Then $AB - BA = I_n$.
Taking traces and using Proposition 1.1.4, $0 = \operatorname{tr}(AB) - \operatorname{tr}(BA) = \operatorname{tr}(I_n) = n \cdot 1_{\mathbb{F}}$, contradicting the hypothesis.
Structural Commentary: Commutators have zero trace, while the identity has dimension as its trace.
Key Insight: Trace detects a scalar obstruction invisible from multiplication alone. □

Definition 1.3.9. *Idempotent and Nilpotent Matrices*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$.

(i) A is *idempotent* if $A^2 = A$.
(ii) A is *nilpotent* if $A^k = 0$ for some positive integer k.

An idempotent matrix is an algebraic projection: it fixes its range and kills its kernel.

Proposition 1.3.10. *Basic Invariants of Idempotents*
If $P \in \operatorname{Mat}(n, n, \mathbb{F})$ satisfies $P^2 = P$, then $\mathbb{F}^n = \operatorname{range}(P) \oplus \ker(P)$, P is diagonalizable, and its only possible eigenvalues are 0 and 1. Moreover $\operatorname{tr}(P) = \operatorname{rank}(P)$.
If $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then P is the orthogonal projection onto $\operatorname{range}(P)$ if and only if $P = P^*$.

Proof. For every $\mathbf{x}$, write $\mathbf{x} = P\mathbf{x} + (\mathbf{x} - P\mathbf{x})$. The first term lies in $\operatorname{range}(P)$, while $P(\mathbf{x} - P\mathbf{x}) = P\mathbf{x} - P^2\mathbf{x} = 0$, so the second term lies in $\ker(P)$. The intersection is zero because $\mathbf{x} = P\mathbf{y}$ and $P\mathbf{x} = 0$ imply $\mathbf{x} = P^2\mathbf{y} = P\mathbf{x} = 0$.
In a basis adapted to $\operatorname{range}(P) \oplus \ker(P)$, the matrix of P is $I_r \oplus 0$, where $r = \operatorname{rank}(P)$.
Hence P is diagonalizable, its eigenvalues lie in $\{0, 1\}$, and $\operatorname{tr}(P) = r$.
For the final claim, self-adjoint idempotents are exactly orthogonal projections by Theorem 2.4.19.
Structural Commentary: Idempotents split the space into fixed directions and killed directions.
Key Insight: In adapted coordinates, every idempotent is just $I_r \oplus 0$. □

Proposition 1.3.11. *Basic Invariants of Nilpotents*
For $N \in \mathrm{Mat}(n, n, \mathbb{F})$, the following are equivalent:

(i) N is nilpotent;
(ii) the characteristic polynomial of N is t^n;
(iii) $N^n = 0$.

In this case, $\mathrm{tr}(N) = 0$, $\det(N) = 0$ for $n \geq 1$, and the spectral radius of N over $\mathbb{C}$ is 0.

Proof. If $N^k = 0$, then every eigenvalue λ of N over an algebraic closure satisfies $\lambda^k = 0$, hence $\lambda = 0$. Thus the characteristic polynomial is t^n. If the characteristic polynomial is t^n, Cayley–Hamilton gives $N^n = 0$. The implication $N^n = 0 \Rightarrow$ nilpotent is immediate. The trace, determinant, and spectral-radius statements follow from the eigenvalue data and from Definitions 1.1.1 and 3.3.1.
Structural Commentary: Nilpotent matrices have no spectral mass away from zero.
Key Insight: Cayley–Hamilton turns the characteristic polynomial t^n into the equation $N^n = 0$. □

2 Positive Definite and Structured Matrices

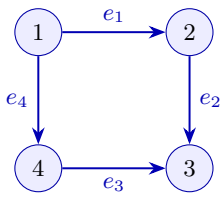
2.1 Positive Semidefinite and Positive Definite Matrices

Definition 2.1.1. *Positive Semidefinite and Positive Definite Matrices*
Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $A \in \mathrm{Mat}(n, n, \mathbb{F})$.
The matrix A is *positive semidefinite*, written $A \succeq 0$, if $A = A^*$ and $\mathbf{x}^* A\mathbf{x} \geq 0$ for every $\mathbf{x} \in \mathbb{F}^n$.
It is *positive definite*, written $A \succ 0$, if $A = A^*$ and $\mathbf{x}^* A\mathbf{x} > 0$ for every nonzero $\mathbf{x} \in \mathbb{F}^n$.
Negative semidefinite and negative definite mean $-A \succeq 0$ and $-A \succ 0$, respectively.

Proposition 2.1.2. *Rectangular Gram Products Are Positive Semidefinite*
Let $R \in \mathrm{Mat}(m, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Then $R^*R \succeq 0$, $RR^* \succeq 0$, and $(R^*R)_{j,j} = \|R_{\cdot,j}\|_2^2$; $(RR^*)_{i,i} = \|R_{i,\cdot}\|_2^2$. Full column rank gives $R^*R \succ 0$; full row rank gives $RR^* \succ 0$.

Proof. The matrices are self-adjoint by Proposition 2.4.8.
For $\mathbf{x} \in \mathbb{F}^n$ and $\mathbf{y} \in \mathbb{F}^m$, $\mathbf{x}^* R^* R\mathbf{x} = \|R\mathbf{x}\|_2^2 \geq 0$ and $\mathbf{y}^* RR^*\mathbf{y} = \|R^*\mathbf{y}\|_2^2 \geq 0$. Hence both products are positive semidefinite. Taking $\mathbf{x} = \mathbf{e}_j$ and $\mathbf{y} = \mathbf{e}_i$ gives the diagonal formulas. If R has full column rank, then $R\mathbf{x} \neq 0$ for $\mathbf{x} \neq 0$, so $R^*R \succ 0$. If R has full row rank, apply the same argument to R^*.
Structural Commentary: Multiplying a matrix by its adjoint produces a Gram matrix of columns or rows.
Key Insight: The quadratic form is a squared norm, so no cancellation can make it negative. □

Definition 2.1.3. *Directed Incidence Matrix*
For edges $e = u \to v$ on vertices $1, \ldots, n$, $B \in \mathrm{Mat}(m, n, \mathbb{R})$ has row e equal to $-\mathbf{e}_u^t + \mathbf{e}_v^t$.



$$B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

Figure 75: A directed incidence matrix records oriented edge differences from vertices to edges.

Example 2.1.4. *A Directed Incidence Matrix*
For Figure 75, $B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{bmatrix}$. For node potentials $\mathbf{x} = (x_1, x_2, x_3, x_4)^t$, $B\mathbf{x} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_4 \\ x_4 - x_1 \end{bmatrix}$.
Hence $\ker(B) = \mathrm{span}\{(1, 1, 1, 1)^t\}$, and $\mathrm{rank}(B) = 3$. The loop vector $\mathbf{y} = (1, 1, -1, -1)^t$ spans $\ker(B^t)$.
Thus $\mathbf{b} \in \mathrm{col}(B)$ if and only if $\mathbf{y}^t\mathbf{b} = b_1 + b_2 - b_3 - b_4 = 0$. This is Kirchhoff loop compatibility.
Structural Commentary: Incidence subspaces encode graph constraints.
Key Insight: A sparse matrix can encode both the algebra of a linear map and the combinatorics of a graph.

Perspective 2.1.5. *Incidence Data and Least Squares*
For an incidence matrix B as in Example 2.1.4, $B\mathbf{x} = \mathbf{b}$ seeks node potentials matching edge differences. If loop consistency fails, least squares projects $\mathbf{b}$ onto $\mathrm{col}(B)$, where $B^*(B\mathbf{x}_0 - \mathbf{b}) = \mathbf{0}$ (Theorem 2.3.4 on B). The constant-vector kernel means potentials are unique only after fixing one node or choosing minimum norm.

Proposition 2.1.6. *Incidence Laplacians Are Positive Semidefinite*
Let B be a directed incidence matrix and set $L = B^tB$. Then $L \succeq 0$ and $\mathbf{x}^tL\mathbf{x} = \sum_{e=(u\to v)} (x_v - x_u)^2$, with $\ker(L)$ equal to the vectors constant on each connected component.

Proof. Since $L = B^tB$, Proposition 2.1.2 gives $L \succeq 0$.
For $e = (u \to v)$, $(B\mathbf{x})_e = x_v - x_u$, so $\mathbf{x}^tL\mathbf{x} = \|B\mathbf{x}\|_2^2 = \sum_{e=(u\to v)} (x_v - x_u)^2$. Thus $\mathbf{x} \in \ker(L)$ if and only if $B\mathbf{x} = \mathbf{0}$, equivalently $x_u = x_v$ on every edge, which means $\mathbf{x}$ is constant on each connected component.
Structural Commentary: Laplacians sum squared edge differences.
Key Insight: A graph Laplacian is a Gram product, so positivity is automatic. □

Theorem 2.1.7. *Equivalent Forms of Positive Semidefiniteness*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. The following are equivalent:

(i) $A \succeq 0$.
(ii) *A is Hermitian and all eigenvalues of A are nonnegative.*
(iii) $A = B^*B$ *for some* $B \in \mathrm{Mat}(m, n, \mathbb{F})$.
(iv) *A has a positive semidefinite square root.*

In this case the positive semidefinite square root is unique.
If $A = U\Lambda U^*$ with $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ and $\lambda_i \geq 0$, then $A^{1/2} = U\,\mathrm{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})U^*$.

Proof.

(i) $\Rightarrow$ *(ii)* This is the matrix form of Theorem 7.3.7.
(ii) $\Rightarrow$ *(iv)* By the spectral theorem, write $A = U\Lambda U^*$, where U is unitary and $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_i \geq 0$. Then $A^{1/2} = U\,\mathrm{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})U^*$ is positive semidefinite and satisfies $(A^{1/2})^2 = A$.
(iv) $\Rightarrow$ *(iii)* If $C \succeq 0$ and $C^2 = A$, then $C = C^*$ and $A = C^*C$, so (iii) holds with $m = n$.
(iii) $\Rightarrow$ *(i)* If $A = B^*B$, then $A^* = A$ and $\mathbf{x}^*A\mathbf{x} = \mathbf{x}^*B^*B\mathbf{x} = \|B\mathbf{x}\|_2^2 \geq 0$, so $A \succeq 0$.

The positive semidefinite square root is unique by Proposition 7.3.8.
Structural Commentary: Positive semidefinite matrices are Gram-type matrices with nonnegative spectrum.
Key Insight: Positivity is best checked either on eigenvalues or by finding a factor B with $A = B^*B$. □

Method 2.1.8. *PSD Spectral Factorisation Method*
To analyse a Hermitian matrix A:

(i) diagonalise $A = U\Lambda U^*$;
(ii) check the signs of the diagonal entries of Λ;
(iii) if $\Lambda \geq 0$, set $A^{1/2} = U\Lambda^{1/2}U^*$ and use $A = (A^{1/2})^*A^{1/2}$;
(iv) translate kernel, rank, and zero-quadratic-form questions into the factor $A^{1/2}$.

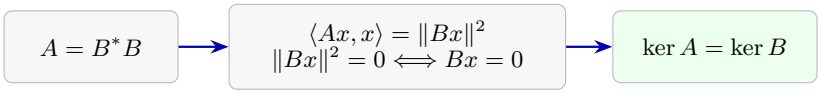


Figure 76: For $A = B^*B$, the squared-norm identity gives $\ker A = \ker B$.

Theorem 2.1.9. *Equivalent Forms of Positive Definiteness*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. The following are equivalent:

(i) $A \succ 0$.
(ii) *A is Hermitian and all eigenvalues of A are positive.*
(iii) $A \succeq 0$ *and A is invertible.*
(iv) $A = B^*B$ *for some invertible* $B \in \mathrm{Mat}(n, n, \mathbb{F})$.
(v) *A is invertible and $A^{-1} \succ 0$.*

Proof.

(i) $\Rightarrow$ *(ii)* By definition A is Hermitian. If $(\lambda, \mathbf{v})$ is an eigenpair with $\mathbf{v} \neq \mathbf{0}$, then $\lambda\|\mathbf{v}\|_2^2 = \mathbf{v}^*A\mathbf{v} > 0$, so every eigenvalue is positive.
(ii) $\Rightarrow$ *(iii)* Positive eigenvalues imply $A \succeq 0$ by Theorem 2.1.7; they also exclude 0, so A is invertible.
(iii) $\Rightarrow$ *(iv)* By Theorem 2.1.7, A has a positive semidefinite square root $A^{1/2}$.
Since A is invertible, $A^{1/2}$ is invertible, and $A = (A^{1/2})^*A^{1/2}$.
(iv) $\Rightarrow$ *(i)* If $A = B^*B$ and B is invertible, then for every nonzero $\mathbf{x}$, $\mathbf{x}^*A\mathbf{x} = \|B\mathbf{x}\|_2^2 > 0$.

(*ii*) $\Leftrightarrow$ (*v*) Hermitian-ness is preserved by inversion, and eigenvalues of A^{-1} are reciprocal eigenvalues of A. Thus positive eigenvalues of A are equivalent to positive eigenvalues of A^{-1}.

Structural Commentary: Positive definite matrices are the invertible part of the positive semidefinite cone.
Key Insight: Positive definite means every eigenvalue is strictly on the positive side of zero. □

Proposition 2.1.10. *Trace and Determinant of Positive Semidefinite Matrices*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ be positive semidefinite, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$. Then $\operatorname{tr} A \geq 0$, with equality if and only if $A = 0$, and $\det A \geq 0$, with strict inequality if and only if A is positive definite.

Proof. List the eigenvalues of A as $\lambda_1, \ldots, \lambda_n$. By Theorem 2.1.7, each $\lambda_j \geq 0$.
Hence $\operatorname{tr} A = \sum_{j=1}^{n} \lambda_j \geq 0$ and $\det A = \prod_{j=1}^{n} \lambda_j \geq 0$.
The trace is zero exactly when every $\lambda_j = 0$, which for a Hermitian matrix means $A = 0$. The determinant is positive exactly when every $\lambda_j > 0$, which is positive definiteness by Theorem 2.1.9.
Structural Commentary: Trace and determinant aggregate the nonnegative eigenvalues by sum and product.
Key Insight: Equality is controlled by which eigenvalues vanish. □

Theorem 2.1.11. *Kernel, Rank, and Columns of B^*B*
Let $B \in \mathrm{Mat}(m,n,\mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, and set $A = B^*B$. Then:

(i) $A \succeq 0$;
(ii) $\ker(A) = \ker(B)$;
(iii) $\operatorname{rank}(A) = \operatorname{rank}(B)$;
(iv) $\operatorname{col}(A) = \operatorname{col}(B^*)$;
(v) $A = 0$ *if and only if* $B = 0$.

Proof.

(i) Positive semidefiniteness follows from Theorem 2.1.7.
(ii) If $B\mathbf{x} = 0$, then $A\mathbf{x} = 0$. Conversely, if $A\mathbf{x} = 0$, then $0 = \mathbf{x}^*A\mathbf{x} = \|B\mathbf{x}\|_2^2$ so $B\mathbf{x} = 0$.
(iii) Since A and B both have n columns, rank-nullity and (ii) give $\operatorname{rank}(A) = \operatorname{rank}(B)$.
(iv) Since $A = A^*$, Proposition 2.4.11 and (ii) give $\operatorname{col}(A) = \ker(A)^\perp = \ker(B)^\perp = \operatorname{col}(B^*)$.
(v) By (iii), $\operatorname{rank}(A) = 0$ if and only if $\operatorname{rank}(B) = 0$.
A matrix has rank zero exactly when it is the zero matrix, so $A = 0$ if and only if $B = 0$.

Structural Commentary: The Gram product B^*B preserves the kernel of B.
Key Insight: The identity $\mathbf{x}^*B^*B\mathbf{x} = \|B\mathbf{x}\|_2^2$ is the whole mechanism. □

Corollary 2.1.12. *Principal Blocks and Column Inclusion*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ be PSD, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, and partition $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^* & A_{22} \end{bmatrix}$, $A_{11} \in \mathrm{Mat}(k,k,\mathbb{F})$.
Then $A_{11} \succeq 0$, $\operatorname{col}(A_{12}) \subseteq \operatorname{col}(A_{11})$, and $\operatorname{rank}(A_{12}) \leq \operatorname{rank}(A_{11})$. If $A \succ 0$, then $A_{11} \succ 0$.

Proof. Write $A = B^*B$ and partition $B = [B_1 \; B_2]$ conformally. Then $A_{11} = B_1^*B_1$ and $A_{12} = B_1^*B_2$.
Hence $A_{11} \succeq 0$ and $\operatorname{col}(A_{12}) = \operatorname{col}(B_1^*B_2) \subseteq \operatorname{col}(B_1^*) = \operatorname{col}(B_1^*B_1) = \operatorname{col}(A_{11})$, equality $\operatorname{col}(B_1^*) = \operatorname{col}(B_1^*B_1)$ follows from Theorem 2.1.11. The rank inequality follows by taking dimensions
If $A \succ 0$ and $\mathbf{x} \neq 0$ in $\mathbb{F}^k$, then $\mathbf{x}^*A_{11}\mathbf{x} = \begin{bmatrix}\mathbf{x}\\0\end{bmatrix}^* A \begin{bmatrix}\mathbf{x}\\0\end{bmatrix} > 0$, so $A_{11} \succ 0$.

Structural Commentary: In a positive block matrix, cross-block columns lie in principal block's column space.
Key Insight: Factor first; then column inclusion is immediate from $A_{12} = B_1^*B_2$. □

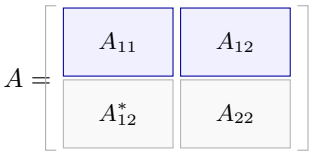


Figure 77: For $A \succeq 0$, factorisation forces $\operatorname{col}(A_{12}) \subseteq \operatorname{col}(A_{11})$.

2.2 Principal Minors and Inertia

Definition 2.2.1. *Principal and Leading Principal Minors*
For $A \in \mathrm{Mat}(n,n,\mathbb{C})$, a *principal minor* is the determinant of a principal submatrix of A.
A *leading principal minor* is the determinant of the leading $k \times k$ principal submatrix for some $k = 1, \ldots, n$.

Theorem 2.2.2. *Sylvester's Criterion*
A Hermitian matrix $A \in \mathrm{Mat}(n,n,\mathbb{C})$ is positive definite if and only if all leading principal minors are positive.

Proof.
($\Rightarrow$) If $A \succ 0$, then each leading principal submatrix is positive definite by Corollary 2.1.12. Its eigenvalues are positive, so its determinant is positive.
($\Leftarrow$) Induct on n. If $n = 1$, the single leading principal minor is the only diagonal entry, so positivity of that minor is exactly positive definiteness. Assume $n \geq 2$ and write $A = \begin{bmatrix} B & \mathbf{y} \\ \mathbf{y}^* & a \end{bmatrix}$, $B \in \mathrm{Mat}(n-1, n-1, \mathbb{C})$
The leading principal minors of B are positive, so $B \succ 0$.
Since $\det(B) > 0$ and $\det(A) > 0$, Proposition 1.1.12 gives $a - \mathbf{y}^* B^{-1} \mathbf{y} = \frac{\det(A)}{\det(B)} > 0$.
For $\mathbf{u} \in \mathbb{C}^{n-1}$ and $t \in \mathbb{C}$, $\begin{bmatrix} \mathbf{u} \\ t \end{bmatrix}^* A \begin{bmatrix} \mathbf{u} \\ t \end{bmatrix} = (\mathbf{u} + B^{-1}\mathbf{y}t)^* B (\mathbf{u} + B^{-1}\mathbf{y}t) + \left(a - \mathbf{y}^* B^{-1} \mathbf{y}\right) |t|^2$.
First term is nonnegative as $B \succ 0$, and second coefficient is positive. If displayed vector is nonzero, then either $t \neq 0$ and second term is positive, or $t = 0$ and then $\mathbf{u} \neq 0$, making first term positive. Thus $A \succ 0$.

Structural Commentary: Leading minors make each leading block positive and final Schur complement positive.
Key Insight: Complete the square with the leading block. □

Method 2.2.3. *2×2 Quadratic Form Test*
Let $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ be real symmetric, $\Delta = ac - b^2$, and $q(x,y) = \begin{bmatrix} x & y \end{bmatrix} A \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2$.
For $a \neq 0$, $q(x,y) = a\left(x + \frac{b}{a}y\right)^2 + \frac{\Delta}{a}y^2$.
If $a = 0$, swap variables when $c \neq 0$; if $a = c = 0$, use $q = 2bxy$. Thus:
(i) $q \succ 0$ if and only if $a > 0$ and $\Delta > 0$;
(ii) $q \prec 0$ if and only if $a < 0$ and $\Delta > 0$;
(iii) q is indefinite if and only if $\Delta < 0$;
(iv) if $\Delta = 0$, then $q \succeq 0$ iff $a, c \geq 0$, and $q \preceq 0$ iff $a, c \leq 0$.
This is the 2×2 form of Sylvester's criterion and inertia.
Structural Commentary: Completing the square is the 2×2 shadow of the LDL^* and Schur-complement tests.
Key Insight: Δ and the leading sign classify the form.

Corollary 2.2.4. *Zero Quadratic Form Detects the Kernel*
If $A \in \mathrm{Mat}(n,n,\mathbb{F})$ is positive semidefinite, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, then $\mathbf{x}^* A \mathbf{x} = 0$ if and only if $A\mathbf{x} = 0$.

Proof. Write $A = B^* B$.
($\Rightarrow$) If $\mathbf{x}^* A \mathbf{x} = 0$, then $\|B\mathbf{x}\|_2^2 = 0$, so $B\mathbf{x} = 0$ and hence $A\mathbf{x} = B^* B \mathbf{x} = 0$.
($\Leftarrow$) If $A\mathbf{x} = 0$, then $\mathbf{x}^* A \mathbf{x} = 0$.

Structural Commentary: A zero PSD quadratic value has no hidden cancellation.
Key Insight: Zero length-square means the factored vector itself is zero. □

Theorem 2.2.5. *Stability Rules for Positive Semidefinite Matrices*
Let $A, B \succeq 0$ over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Whenever the expressions are defined:
(i) $aA + bB \succeq 0$ for $a, b \geq 0$; if $a, b > 0$ and one of A, B is positive definite, then $aA + bB \succ 0$.
(ii) $S^* A S \succeq 0$; moreover, $S^* A S \succ 0$ if and only if $S\mathbf{x} \notin \ker(A)$ for all $\mathbf{x} \neq \mathbf{0}$.
(iii) $A \oplus B \succeq 0$, with $A \oplus B \succ 0$ if and only if $A \succ 0$ and $B \succ 0$.

Proof.
(i) $\mathbf{x}^*(aA + bB)\mathbf{x} = a\mathbf{x}^* A \mathbf{x} + b\mathbf{x}^* B \mathbf{x} \geq 0$. If, for example, $A \succ 0$ and $a > 0$, then for every nonzero $\mathbf{x}$ the first summand is positive and the second is nonnegative, so $aA + bB \succ 0$; the case $B \succ 0$ is identical.
(ii) $\mathbf{x}^* S^* A S \mathbf{x} = (S\mathbf{x})^* A (S\mathbf{x}) \geq 0$. By Corollary 2.2.4, equality occurs exactly when $S\mathbf{x} \in \ker(A)$.
(iii) $(\mathbf{x} \oplus \mathbf{y})^*(A \oplus B)(\mathbf{x} \oplus \mathbf{y}) = \mathbf{x}^* A \mathbf{x} + \mathbf{y}^* B \mathbf{y} \geq 0$. This expression is positive for every nonzero $\mathbf{x} \oplus \mathbf{y}$ exactly when each summand is positive on its own nonzero vectors, namely when $A \succ 0$ and $B \succ 0$.

Structural Commentary: The PSD cone is stable under sums, pullbacks, and direct sums.
Key Insight: Test the quadratic form after each construction. □

Proposition 2.2.6. *Explicit Square Root for a 2×2 Positive Semidefinite Matrix*
Let $A \in \mathrm{Mat}(2,2,\mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, be positive semidefinite and nonzero. Set $\tau = \big(\operatorname{tr} A + 2\sqrt{\det A}\big)^{1/2}$. Then $\tau > 0$ and $A^{1/2} = \frac{1}{\tau}\big(A + \sqrt{\det A}\, I_2\big)$.

Proof. Let $C = \frac{1}{\tau}\big(A + \sqrt{\det A}\, I_2\big)$. By Proposition 2.1.10, $\det A \geq 0$ and $\operatorname{tr} A > 0$, hence $\tau > 0$; by Theorem 2.2.5, $C \succeq 0$. Cayley–Hamilton gives $A^2 - (\operatorname{tr} A)A + (\det A)I_2 = 0$, so

$$C^2 = \frac{1}{\tau^2}\big(A^2 + 2\sqrt{\det A}\, A + (\det A)I_2\big) = \frac{1}{\tau^2}\big((\operatorname{tr} A)A + 2\sqrt{\det A}\, A\big) = A.$$

Thus $C = A^{1/2}$ by uniqueness of the positive semidefinite square root.
Structural Commentary: In dimension 2, Cayley–Hamilton makes $A^{1/2}$ linear in A.
Key Insight: Square the linear candidate and use Cayley–Hamilton. □

Theorem 2.2.7. *Simultaneous Congruence with One Positive Definite Form*
Let $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$ be Hermitian, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, and let $A \succ 0$.
Then, for some invertible S and real diagonal Λ, $A = SS^*$ and $B = S\Lambda S^*$.
The diagonal entries of Λ are the eigenvalues of $A^{-1}B$.

Proof. Let $C = A^{-1/2}BA^{-1/2} = U\Lambda U^*$ be spectral decomposition, $S = A^{1/2}U$. Then $SS^* = A$ and $S\Lambda S^* = B$. Also $A^{-1}B = A^{-1/2}CA^{1/2}$, so $A^{-1}B$ is similar to C and has eigenvalues given by the diagonal entries of Λ.
Structural Commentary: The positive definite form A normalizes B to a Hermitian matrix.
Key Insight: Normalize, diagonalise, then transform back. □

Corollary 2.2.8. *Products with One Positive Definite Factor*
Let $A, B \in \mathrm{Mat}(n,n,\mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, and let $A \succ 0$.

(i) If B is Hermitian, then AB is diagonalizable with real spectrum.
(ii) If $B \succeq 0$, then AB is diagonalizable with nonnegative real spectrum; if $B \neq 0$, one eigenvalue is positive.
(iii) If $B \succ 0$, then AB is diagonalizable with positive real spectrum.

Proof. Let $C = A^{1/2}BA^{1/2}$. Since $A^{-1/2}(AB)A^{1/2} = C$, the matrices AB and C are similar.

(i) If B is Hermitian, then C is Hermitian, hence diagonalizable with real spectrum.
(ii) If $B \succeq 0$, then $C \succeq 0$ by Theorem 2.2.5, so its spectrum is nonnegative.
If $B \neq 0$, then $C \neq 0$, hence some eigenvalue is positive.
(iii) If $B \succ 0$, then $C \succ 0$ by Theorem 2.2.5, so all eigenvalues are positive.

Structural Commentary: A positive definite factor makes AB similar to a Hermitian congruence.
Key Insight: Use $AB \sim A^{1/2}BA^{1/2}$. □

Theorem 2.2.9. *Principal Axes for Real Quadratic Forms*
Let $A \in \mathrm{Mat}(n,n,\mathbb{R})$ be symmetric. Then:

(i) there are a real orthogonal Q and real $\lambda_1,\dots,\lambda_n$ such that, with $\mathbf{y} = Q^t\mathbf{x}$, $\mathbf{x}^tA\mathbf{x} = \sum_{j=1}^{n} \lambda_j y_j^2$.
(ii) $A \succeq 0$ if and only if every $\lambda_j \geq 0$;
(iii) $A \succ 0$ if and only if every $\lambda_j > 0$;
(iv) A is indefinite if and only if some $\lambda_j > 0$ and some $\lambda_k < 0$.

Proof.

(i) By the real spectral theorem, write $A = Q\operatorname{diag}(\lambda_1,\dots,\lambda_n)Q^t$.
Then $\mathbf{x}^tA\mathbf{x} = \mathbf{x}^tQ\operatorname{diag}(\lambda_j)Q^t\mathbf{x} = \mathbf{y}^t\operatorname{diag}(\lambda_j)\mathbf{y} = \sum_{j=1}^{n} \lambda_j y_j^2$.
(ii) The displayed sum is nonnegative for all $\mathbf{y}$ exactly when each coefficient λ_j is nonnegative.
(iii) The displayed sum is positive for all nonzero $\mathbf{y}$ exactly when each coefficient λ_j is positive.
(iv) The diagonal form takes both signs exactly when its coefficients include both signs.

Structural Commentary: Principal axes turn the form into signed squares.
Key Insight: Eigenvalue signs classify the quadratic form. □

Method 2.2.10. *Quadratic Form Principal-Axes Method*
To analyse a real quadratic form:

(i) write it as $\mathbf{x}^tA\mathbf{x}$ with $A = A^t$; split each cross coefficient equally between a_{ij} and a_{ji};
(ii) diagonalise $A = Q\Lambda Q^t$;
(iii) change variables by $\mathbf{y} = Q^t\mathbf{x}$;
(iv) classify the form from the signs of the eigenvalues in Λ.

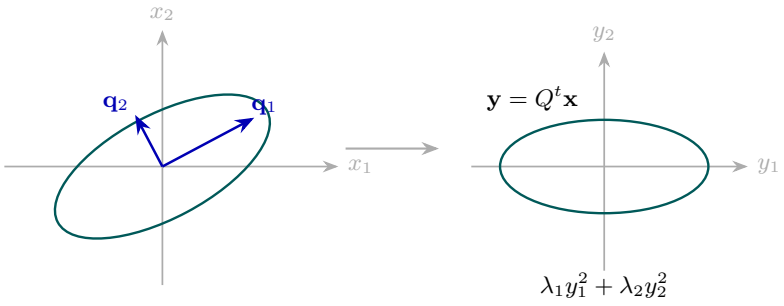


Figure 78: Orthogonal coordinates reveal the principal axes of a quadratic form.

Definition 2.2.11. *$*$-Congruence and Hermitian Inertia*
Two matrices $A, B \in \mathrm{Mat}(n, n, \mathbb{C})$ are *$*$-congruent* if $A = SBS^*$ for some invertible $S \in \mathrm{Mat}(n, n, \mathbb{C})$.
For Hermitian A, its *inertia* is $\mathrm{in}(A) = (n_+(A), n_-(A), n_0(A))$, where these are the numbers of positive, negative, and zero eigenvalues of A, counted with multiplicity.

Perspective 2.2.12. *Congruence versus Similarity*
Similarity $B = P^{-1}AP$ is change of basis for an operator. Congruence $N = P^tMP$ is change of basis for a bilinear form, with the same basis change in both slots. Over $\mathbb{R}$ this is the real case of $*$-congruence. Hence inertia classifies forms, not similarity classes of operators.

Lemma 2.2.13. *Positive Subspaces and Inertia*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ be Hermitian. The maximal dimensions of subspaces on which $\mathbf{x}^*A\mathbf{x}$ is positive or negative for every nonzero $\mathbf{x}$ are $n_+(A)$ and $n_-(A)$, respectively.

Proof. Let P_+, $P_{\leq 0}$ be positive, nonpositive spectral subspaces. Then $\dim P_+ = n_+(A)$, P_+ is positive for A. If $\dim \mathcal{U} > n_+(A)$, then $\dim \mathcal{U} + \dim P_{\leq 0} > n_+(A) + n_-(A) + n_0(A) = n$, so $\mathcal{U} \cap P_{\leq 0}$ contains a nonzero vector. On this vector quadratic form is nonpositive, so $\mathcal{U}$ cannot be positive. Apply this to $-A$ for negative case.
Structural Commentary: The positive spectral subspace is maximal; the nonpositive part blocks enlargement.
Key Insight: Inertia counts maximal sign-definite dimensions. □

Theorem 2.2.14. *Sylvester's Inertia Theorem*
Hermitian matrices A, B of the same size are $*$-congruent if and only if $n_+(A) = n_+(B)$ and $n_-(A) = n_-(B)$.

Proof.
($\Rightarrow$) Suppose $A = SBS^*$ with S invertible. Then $\mathbf{x}^*A\mathbf{x} = (S^*\mathbf{x})^*B(S^*\mathbf{x})$.
Thus S^* transports positive and negative subspaces, so Lemma 2.2.13 gives $n_\pm(A) = n_\pm(B)$.
($\Leftarrow$) Let the common inertia be (p, q, r) and set $D = I_p \oplus (-I_q) \oplus 0_r$.
By the spectral theorem, each of A and B is $*$-congruent to D, hence they are $*$-congruent to each other.
Structural Commentary: $*$-congruence is a coordinate change for Hermitian quadratic forms.
Key Insight: Spectral decomposition reduces the form to signed squares and zero directions. □

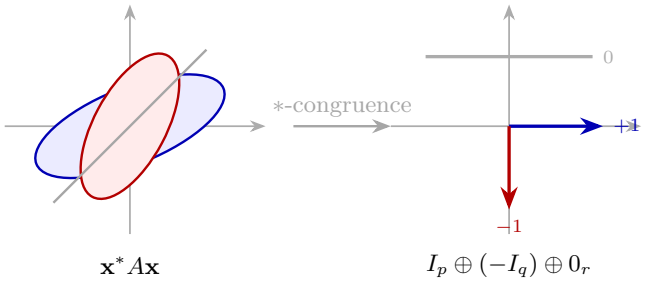


Figure 79: $*$-congruence reduces a Hermitian form to signed coordinate axes.

2.3 Hermitian and Normal Matrix Structure

Definition 2.3.1. *Phase Inertia Matrix*
A *phase inertia matrix* is a diagonal complex matrix whose nonzero diagonal entries have modulus 1.

Proposition 2.3.2. *Phase Inertia Form for Normal Matrices*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be normal.
Then A is $*$-congruent to a phase inertia matrix D, whose entries record the spectral rays of A.

Proof. Write $A = U\,\mathrm{diag}(\lambda_j)U^*$. Write $\lambda_j = r_j d_j$, where $r_j > 0$ and $|d_j| = 1$ if $\lambda_j \neq 0$, while $r_j = 1$ and $d_j = 0$ if $\lambda_j = 0$. Set $D = \mathrm{diag}(d_j)$, $S = U\,\mathrm{diag}(\sqrt{r_j})$. Then S is invertible and $A = SDS^*$.
The entries $d_j = e^{i\theta}$ count exactly the eigenvalues on the ray $\{\rho e^{i\theta} : \rho > 0\}$.
Structural Commentary: Normal diagonalization separates modulus from phase; $*$-congruence absorbs moduli.
Key Insight: Only spectral phases survive. □

Perspective 2.3.3. *Hermitian Inertia as Real Phase Inertia*
For Hermitian matrices, the only nonzero spectral rays are the positive and negative real axes.
Sylvester's inertia theorem is therefore the real-ray version of phase inertia.

3 Matrix Products, Vectorization, and Spectral Radius

3.1 Kronecker Products and Vectorization

Definition 3.1.1. *Kronecker Product*
Let $A = [a_{ij}] \in \mathrm{Mat}(m,n,\mathbb{F})$ and $B \in \mathrm{Mat}(p,q,\mathbb{F})$. The *Kronecker product* of A and B is the block matrix

$$A \otimes B := \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix} \in \mathrm{Mat}(mp, nq, \mathbb{F}).$$

The Kronecker product is also called the tensor product of matrices.

Proposition 3.1.2. *Basic Kronecker Identities*
Whenever the expressions are defined,

(i) $c(A \otimes B) = (cA) \otimes B = A \otimes (cB)$;
(ii) $(A+B) \otimes C = A \otimes C + B \otimes C$ *and* $A \otimes (B+C) = A \otimes B + A \otimes C$;
(iii) $(A \otimes B) \otimes C = A \otimes (B \otimes C)$;
(iv) $(A \otimes B)^t = A^t \otimes B^t$;
(v) $I_m \otimes I_n = I_{mn}$.

If $\mathbb{F} = \mathbb{C}$, then also $\overline{A \otimes B} = \overline{A} \otimes \overline{B}$ and $(A \otimes B)^* = A^* \otimes B^*$.

Proof.

(i) The (i,j) block of $c(A \otimes B)$ is $c(a_{ij}B) = (ca_{ij})B = a_{ij}(cB)$, which is (i,j) block of $(cA) \otimes B$ and $A \otimes (cB)$.
(ii) The (i,j) block of $(A+B) \otimes C$ is $(a_{ij} + b_{ij})C = a_{ij}C + b_{ij}C$, so $(A+B) \otimes C = A \otimes C + B \otimes C$.
The proof of $A \otimes (B+C) = A \otimes B + A \otimes C$ is identical inside each block.
(iii) If $B = [b_{rs}]$, then both $(A \otimes B) \otimes C$ and $A \otimes (B \otimes C)$ have blocks $a_{ij}b_{rs}C$ in the same relative positions.
(iv) The (j,i) block of $(A \otimes B)^t$ is $(a_{ij}B)^t = a_{ij}B^t$, which is the (j,i) block of $A^t \otimes B^t$.
(v) The block matrix $I_m \otimes I_n$ has I_n on its diagonal blocks and zero elsewhere, hence equals I_{mn}.

If $\mathbb{F} = \mathbb{C}$, conjugation is checked blockwise, adjoint identity follows from conjugation together with part (iv).
Structural Commentary: Kronecker products replace scalar entries by whole scaled blocks. Standard algebraic rules survive because each block is still controlled by scalar distributivity and transposition.
Key Insight: Check identities block-by-block; each block reduces to an ordinary scalar-matrix identity. □

Theorem 3.1.3. *Mixed Product Property*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $B \in \mathrm{Mat}(p,q,\mathbb{F})$, $C \in \mathrm{Mat}(n,r,\mathbb{F})$, $D \in \mathrm{Mat}(q,s,\mathbb{F})$. $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.

Proof. Write $A = [a_{ik}]$ and $C = [c_{kj}]$. The (i,j) block of $(A \otimes B)(C \otimes D)$ is
$\sum_{k=1}^{n} (a_{ik}B)(c_{kj}D) = \left(\sum_{k=1}^{n} a_{ik}c_{kj}\right) BD = (AC)_{ij}BD$, which is exactly the (i,j) block of $(AC) \otimes (BD)$.
Structural Commentary: The outer indices multiply as AC, while the inner block action multiplies as BD.
Key Insight: Kronecker multiplication separates into ordinary multiplication in each factor. □

Corollary 3.1.4. *Inverse of a Kronecker Product*
If $A \in \mathrm{Mat}(m,m,\mathbb{F})$ and $B \in \mathrm{Mat}(n,n,\mathbb{F})$ are invertible, then $A \otimes B$ is invertible and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

Proof. By Theorem 3.1.3, $(A \otimes B)(A^{-1} \otimes B^{-1}) = I_m \otimes I_n = I_{mn}$ and the reverse product is identical.
Structural Commentary: Inverting a Kronecker product inverts each independent factor.
Key Insight: The mixed product property turns the inverse check into two ordinary inverse checks. □

Proposition 3.1.5. *Kronecker Product and Sum Spectra in the Diagonalizable Case*
Let $A = P\Lambda P^{-1}$, $B = QMQ^{-1}$, *where* $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ *and* $M = \operatorname{diag}(\mu_1, \ldots, \mu_q)$.
Then $A \otimes B \sim \Lambda \otimes M$; *and* $A \otimes I_q + I_n \otimes B \sim \Lambda \otimes I_q + I_n \otimes M$.
Their diagonal entries are respectively $\lambda_i\mu_j$ *and* $\lambda_i + \mu_j$ *for* $1 \le i \le n$, $1 \le j \le q$.

Proof. By Theorem 3.1.3 and $(P \otimes Q)^{-1} = P^{-1} \otimes Q^{-1}$, $A \otimes B = (P \otimes Q)(\Lambda \otimes M)(P^{-1} \otimes Q^{-1})$.
Similarly, $A \otimes I_q + I_n \otimes B = (P \otimes Q)(\Lambda \otimes I_q + I_n \otimes M)(P^{-1} \otimes Q^{-1})$.
The middle matrices are diagonal with the stated (i,j)-indexed entries.
Structural Commentary: Eigenvectors tensor together, and Kronecker sums add the two scalar eigenvalues.
Key Insight: Diagonalize each factor first; the Kronecker operation becomes entrywise on eigenvalue pairs. □

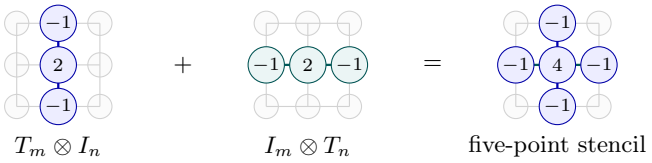


Figure 80: Vertical plus horizontal second differences give the five-point stencil.

Definition 3.1.6. *Vectorisation*
Let $X = [\mathbf{x}_1\ \mathbf{x}_2\ \cdots\ \mathbf{x}_n] \in \operatorname{Mat}(m, n, \mathbb{F})$. The *vectorisation* of X is

$$\operatorname{vec}(X) := \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \in \mathbb{F}^{mn}.$$

Thus vec stacks the columns of X vertically.

Theorem 3.1.7. *Vectorisation of a Triple Product*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$, $X \in \operatorname{Mat}(n, p, \mathbb{F})$, *and* $B \in \operatorname{Mat}(p, q, \mathbb{F})$. *Then* $\operatorname{vec}(AXB) = (B^t \otimes A)\operatorname{vec}(X)$.

Proof. Write $B = [\mathbf{b}_1\ \cdots\ \mathbf{b}_q]$ and $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_p]$.
The kth column of AXB is $AX\mathbf{b}_k = A\sum_{i=1}^{p} b_{ik}\mathbf{x}_i = \sum_{i=1}^{p} b_{ik}A\mathbf{x}_i = (\mathbf{b}_k^t \otimes A)\operatorname{vec}(X)$.
Stacking these q columns gives $(B^t \otimes A)\operatorname{vec}(X)$.
Structural Commentary: Vectorisation rewrites two-sided multiplication as one matrix action.
Key Insight: Left multiplication contributes A; right multiplication contributes B^t. □

Perspective 3.1.8. *Vectorising a Minimal-Polynomial Test*
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$ and a monic candidate $p_m(z) = a_0 + a_1 z + \cdots + a_{m-1}z^{m-1} + z^m$,
the matrix equation $a_0 I + a_1 A + \cdots + a_{m-1}A^{m-1} = -A^m$ can be written as the ordinary linear system

$$\begin{bmatrix} \operatorname{vec}(I) & \operatorname{vec}(A) & \cdots & \operatorname{vec}(A^{m-1}) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{m-1} \end{bmatrix} = -\operatorname{vec}(A^m)$$

Structural Commentary: Vectorisation turns matrix identities into linear systems.
Key Insight: Polynomial membership becomes a coefficient system in vectorized powers.

3.2 Hadamard Products

Definition 3.2.1. *Hadamard Product*
Let $A = [a_{ij}], B = [b_{ij}] \in \operatorname{Mat}(m, n, \mathbb{F})$.
The *Hadamard product*, or *entrywise product*, of A and B is $A \circ B := [a_{ij}b_{ij}] \in \operatorname{Mat}(m, n, \mathbb{F})$.

Proposition 3.2.2. *Basic Hadamard Identities*
For matrices of compatible sizes:

(i) $A \circ B = B \circ A$;

(ii) $A \circ (\gamma B + C) = \gamma(A \circ B) + A \circ C$;
(iii) $(A \circ B)^* = A^* \circ B^*$;
(iv) *if* E *is the all-ones matrix, then* $A \circ E = A$;
(v) *if* $D = \operatorname{diag}(d_1, \ldots, d_n)$, *then* $A \circ D = \operatorname{diag}(a_{11}d_1, \ldots, a_{nn}d_n)$.

Proof.

(i) $(A \circ B)_{ij} = a_{ij}b_{ij} = b_{ij}a_{ij} = (B \circ A)_{ij}$.
(ii) $\big(A \circ (\gamma B + C)\big)_{ij} = a_{ij}(\gamma b_{ij} + c_{ij}) = \gamma a_{ij}b_{ij} + a_{ij}c_{ij}$.
(iii) $\big((A \circ B)^*\big)_{ij} = \overline{a_{ji}b_{ji}} = \overline{a_{ji}}\,\overline{b_{ji}} = (A^* \circ B^*)_{ij}$.
(iv) $(A \circ E)_{ij} = a_{ij}$.
(v) Since D has zero off-diagonal entries, $A \circ D$ is diagonal with entries $a_{ii}d_i$.

Structural Commentary: Hadamard multiplication is entrywise, with no row-column summation.
Key Insight: Prove Hadamard identities entry by entry. □

Theorem 3.2.3. *Schur Product Theorem*
If $A, B \in \operatorname{Mat}(n, n, \mathbb{F})$ *are positive semidefinite, where* $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, *then* $A \circ B$ *is positive semidefinite.*

Proof. By Theorem 2.1.7, regard A and B as Gram matrices: $a_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle$, $b_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$ for suitable vectors $\mathbf{u}_1, \ldots, \mathbf{u}_n$ and $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Then $(A \circ B)_{ij} = \langle \mathbf{u}_j, \mathbf{u}_i \rangle \langle \mathbf{v}_j, \mathbf{v}_i \rangle = \langle \mathbf{u}_j \otimes \mathbf{v}_j, \mathbf{u}_i \otimes \mathbf{v}_i \rangle$.
Thus $A \circ B$ is the Gram matrix of the vectors $\mathbf{u}_j \otimes \mathbf{v}_j$, hence is positive semidefinite by Proposition 1.4.2.
Structural Commentary: The Schur product is the Gram matrix of paired tensor vectors.
Key Insight: Pair the jth vector from each Gram representation and take tensor products. □

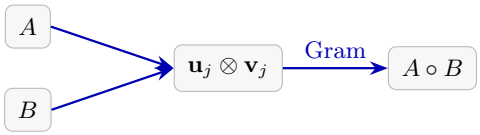


Figure 81: For Gram matrices, entrywise products become tensor inner products, so $A \circ B$ is Gram.

Corollary 3.2.4. *Positive Definite Hadamard Products*
Let $A, B \in \operatorname{Mat}(n, n, \mathbb{F})$ *be positive semidefinite, where* $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

(i) *If* A *is positive definite and every diagonal entry of* B *is positive, then* $A \circ B$ *is positive definite.*
(ii) *If* A *and* B *are positive definite, then* $A \circ B$ *is positive definite.*

Proof.

(i) Let $\lambda_{\min} > 0$ be the smallest eigenvalue of A. Then $A - \lambda_{\min} I_n \succeq 0$, so $(A - \lambda_{\min} I_n) \circ B \succeq 0$ by Theorem 3.2.3. Thus, for $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^*(A \circ B)\mathbf{x} = \mathbf{x}^*((A - \lambda_{\min} I_n) \circ B)\mathbf{x} \lambda_{\min} \sum_{j=1}^{n} b_{jj} |x_j|^2 > 0$.
(ii) If $B \succ 0$, then $b_{jj} = \mathbf{e}_j^* B \mathbf{e}_j > 0$ for every j, so (i) applies.

Structural Commentary: Subtract the spectral floor of A; the diagonal of B gives strictness.
Key Insight: Use $I_n \circ B = \operatorname{diag}(b_{11}, \ldots, b_{nn})$. □

Method 3.2.5. *Schur Product Positivity Method*
To prove an entrywise product is positive semidefinite:

(i) factor or interpret each matrix as a Gram matrix;
(ii) pair the corresponding vectors;
(iii) take tensor products of the paired vectors;
(iv) identify the Hadamard product as the Gram matrix of those tensor products.

3.3 Spectral Radius and Matrix Powers

Definition 3.3.1. *Spectral Radius and Convergent Matrix*
For $A \in \operatorname{Mat}(n, n, \mathbb{C})$, the *spectral radius* is $\rho(A) := \max\{|\lambda| : \lambda \in \operatorname{spec}(A)\}$.
We say $A^p \to 0$ if every entry of A^p converges to 0 as $p \to \infty$.

Theorem 3.3.2. *Matrix Powers Converge to Zero*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$. *Then* $A^p \to 0$ *if and only if* $\rho(A) < 1$.

Proof. Choose a Jordan form $A = SJS^{-1}$.
Since S and S^{-1} are fixed, $A^p \to 0$ if and only if every Jordan block power in J^p converges to 0.

($\Rightarrow$) Each block power tends to 0, so Proposition 6.6.4 gives $|\lambda| < 1$ for every eigenvalue. Thus $\rho(A) < 1$.
($\Leftarrow$) If $\rho(A) < 1$, then every eigenvalue satisfies $|\lambda| < 1$. Proposition 6.6.4 makes each block power tend to 0.

Structural Commentary: Similarity reduces matrix-power convergence to Jordan block-power convergence.
Key Insight: The largest eigenvalue modulus controls the asymptotic size of all blocks. □

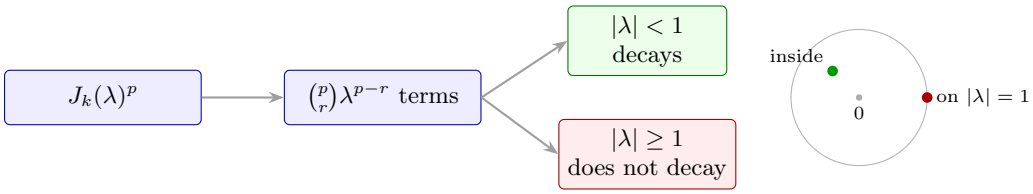


Figure 82: The gray circle is $|\lambda| = 1$: inside it, powers decay; on or outside it, they do not.

Definition 3.3.3. *Power Bounded Matrix*
A matrix $A \in \mathrm{Mat}(n, n, \mathbb{C})$ is *power bounded* if for each entry position (i, j), the sequence of entries of A^p in that position is bounded for $p = 1, 2, \ldots$.

Theorem 3.3.4. *Power Bounded Jordan Criterion*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$. The following are equivalent:

(i) A is power bounded.
(ii) $\rho(A) \leq 1$, and every eigenvalue λ with $|\lambda| = 1$ has only 1×1 Jordan blocks.
(iii) $\rho(A) \leq 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Work in Jordan form.

(*i*) $\Rightarrow$ (*ii*): If $|\lambda| > 1$, the diagonal entries λ^p are unbounded.
If $|\lambda| = 1$ and $J_k(\lambda)$ has $k \geq 2$, then the first superdiagonal contains $p\lambda^{p-1}$, also unbounded.
(*ii*) $\Rightarrow$ (*i*): Blocks with $|\lambda| < 1$ converge to 0 by Proposition 6.6.4; 1×1 blocks with $|\lambda| = 1$ are bounded. Finite direct sums and fixed similarities preserve boundedness.
(*ii*) $\Leftrightarrow$ (*iii*): For fixed λ, $\dim E_\lambda(A)$ counts Jordan blocks, while $m_A(\lambda)$ counts their total size. Equality holds exactly when every block has size 1.

Structural Commentary: Unit-circle eigenvalues are harmless only when their nilpotent parts vanish.
Key Insight: Nontrivial Jordan blocks on the unit circle create polynomial growth. □

4 Matrix Decompositions

4.1 LU, QR, and Cholesky-Type Decompositions

Decompositions expose usable structure in a matrix. LU, QR, Cholesky, SVD, compact SVD, and polar decomposition separate elimination, geometry, positivity, rank, and stretching.

Perspective 4.1.1. *Matrix Factorisation Roadmap*
Factorisations expose structure by task: elimination gives LU and LDL^t (Theorem 4.1.5, Corollary 4.1.6); positivity gives Cholesky (Theorem 4.1.12); geometry gives QR (Theorem 4.1.8); singular directions give SVD, compact SVD, and the pseudoinverse (Theorems 4.2.10, 4.2.14, Definition 5.1.16); polar decomposition separates motion from stretch (Theorem 4.2.19). Spectral factorisations appeared earlier: diagonalisation, spectral theorems, Jordan form, and Schur form (Theorems 5.3.17, 7.1.12, 7.1.13, 6.6.6, 7.1.10).
Key Insight: Choose the factorisation that exposes the needed structure.

Definition 4.1.2. *Full-Rank Factorization*
A factorization $A = XY$ with X full column rank and Y full row rank is called a *full-rank factorization* of A.

Theorem 4.1.3. *Full-Rank Factorization*
Let $A \in \mathrm{Mat}(m, n, \mathbb{F})$ have rank $r \geq 1$, and let the columns of $X \in \mathrm{Mat}(m, r, \mathbb{F})$ be a basis of $\mathrm{col}(A)$. Then there is a unique $Y \in \mathrm{Mat}(r, n, \mathbb{F})$ such that $A = XY$, and $\mathrm{rank}(X) = \mathrm{rank}(Y) = r$.

Proof. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Since the columns of X form a basis of $\mathrm{col}(A)$, for each j there is a unique $\mathbf{y}_j \in \mathbb{F}^r$ such that $\mathbf{a}_j = X\mathbf{y}_j$. Set $Y = [\mathbf{y}_1 \ \cdots \ \mathbf{y}_n]$. Then $A = [X\mathbf{y}_1 \ \cdots \ X\mathbf{y}_n] = XY$ and uniqueness of the

coordinates $\mathbf{y}_j$ gives uniqueness of Y. The columns of X are independent, so $\operatorname{rank}(X) = r$.
By Theorem 1.2.17, $r = \operatorname{rank}(A) = \operatorname{rank}(XY) \leq \operatorname{rank}(Y) \leq r$ because Y has only r rows. Hence $\operatorname{rank}(Y) = r$.
Structural Commentary: A matrix factors through a basis of its column space.
Key Insight: Full-rank factorization separates column directions from their coordinates. □

Definition 4.1.4. *LU Factorisation*
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, an *LU factorisation* is a product $A = LU$, where L is lower triangular and U is upper triangular. It is *unit lower triangular* if every diagonal entry of L is 1.

Theorem 4.1.5. *LU Factorisation from Nonzero Leading Principal Minors*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$. If every leading principal minor of A is nonzero, then $A = LU$ for a unique unit lower triangular L and a unique upper triangular U.

Proof. Induct on n. The case $n = 1$ is immediate. Write $A = \begin{bmatrix} a & \mathbf{r} \\ \mathbf{c} & B \end{bmatrix}$, $a \neq 0$.
Set $\boldsymbol{\ell} = a^{-1}\mathbf{c}$ and $S = B - a^{-1}\mathbf{c}\mathbf{r}$. For each leading $k \times k$ principal block of A, the block determinant formula gives determinant a times the corresponding leading $(k-1) \times (k-1)$ principal minor of S. Hence all leading principal minors of S are nonzero. By induction, $S = L_1 U_1$ with L_1 unit lower triangular and U_1 upper triangular.
Then $A = \begin{bmatrix} 1 & 0 \\ \boldsymbol{\ell} & L_1 \end{bmatrix} \begin{bmatrix} a & \mathbf{r} \\ 0 & U_1 \end{bmatrix}$. For uniqueness, comparing the first row and first column of LU determines a, $\mathbf{r}$, and $\boldsymbol{\ell}$, then reduces uniqueness to the lower-right Schur complement.
Structural Commentary: Gaussian elimination peels off one pivot and passes to a Schur complement.
Key Insight: Nonzero leading principal minors are exactly the no-row-swap pivot condition. □

Corollary 4.1.6. *Symmetric LU Gives an LDL^t Factorisation*
Let $A = A^t \in \operatorname{Mat}(n, n, \mathbb{R})$ have nonzero leading principal minors.
Then $A = LDL^t$, with L unit lower triangular and D diagonal invertible.

Proof. By Theorem 4.1.5, write $A = LU$ uniquely with L unit lower triangular and U upper triangular.
Since the leading principal minors are nonzero, each $u_{ii} \neq 0$. Put $D = \operatorname{diag}(u_{11}, \ldots, u_{nn})$ and $U_0 = D^{-1}U$; then U_0 is unit upper triangular and $A = LDU_0$. Since $A = A^t$, also $A = U_0^t D L^t$. This is another LU factorisation with unit lower factor U_0^t, so uniqueness gives $L = U_0^t$. Hence $A = LDL^t$.
Structural Commentary: Symmetry makes the upper triangular unit factor the transpose of the lower one.
Key Insight: Once the pivots are separated into D, the two triangular halves mirror each other. □

Definition 4.1.7. *Wide and Narrow QR Factorisation*
For $A \in \operatorname{Mat}(m, n, \mathbb{F})$ with $m \geq n$, factorisation $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$, $V \in \operatorname{Mat}(m, m, \mathbb{F})$ unitary, $R \in \operatorname{Mat}(n, n, \mathbb{F})$ upper triangular is called a *wide QR factorisation*. Writing $V = [Q\ Q']$ gives the *narrow QR factorisation* $A = QR$.

Theorem 4.1.8. *QR Factorisation*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ and suppose $m \geq n$.

(i) $\exists$ a unitary V and an upper triangular R with real nonnegative diagonal entries such that $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$. Writing $V = [Q\ Q']$ with Q the first n columns gives $A = QR$, with Q orthonormal columns.
(ii) If $\operatorname{rank}(A) = n$, then narrow $A = QR$ is unique when R has positive diagonal entries.

Proof.

(i) At step k, after the first $k-1$ columns have been triangularized, apply Corollary 2.5.4 to the tail of the kth column. This gives a unitary $I_{k-1} \oplus U_k$ that sends that tail to $r_{kk}\mathbf{e}_1$ with $r_{kk} \geq 0$.
Since the first $k-1$ entries of every earlier column already contain all possible nonzero entries, acting only on rows $k, \ldots, m$ preserves the earlier triangular columns.
After n steps, $U_n \cdots U_1 A = \begin{bmatrix} R \\ 0 \end{bmatrix}$, where R is upper triangular with nonnegative diagonal entries.
Let $V = (U_n \cdots U_1)^*$. Then $A = V \begin{bmatrix} R \\ 0 \end{bmatrix}$, and $V = [Q\ Q']$ gives $A = QR$ with Q orthonormal columns.
If $\operatorname{rank}(A) = n$, then $n = \operatorname{rank}(A) = \operatorname{rank}(R)$, so the triangular matrix R is invertible and its nonnegative diagonal entries are positive.

(ii) Suppose $A = Q_1 R_1 = Q_2 R_2$, where $Q_i^* Q_i = I_n$ and the R_i are upper triangular with positive diagonal entries. Then $A^*A = R_1^* R_1 = R_2^* R_2$. Let $D = R_1 R_2^{-1}$. Substituting $R_1 = D R_2$ gives $D^* D = I_n$.
Thus the columns of D are orthonormal. Since D is upper triangular, the first column has only one possible nonzero entry; orthogonality with it forces the rest of the first row to vanish.
Inducting on the lower-right principal submatrix shows D is diagonal, and $D^*D = I_n$ gives diagonal

entries of modulus 1. These diagonal entries are ratios of positive diagonal entries of R_1 and R_2, so they are positive. Hence $D = I_n$. Therefore $R_1 = R_2$, and then $Q_1 = Q_2$.

Structural Commentary: Householder steps triangularise columns; full rank makes the diagonal positive.
Key Insight: Positive diagonals remove the sign/phase ambiguity. □

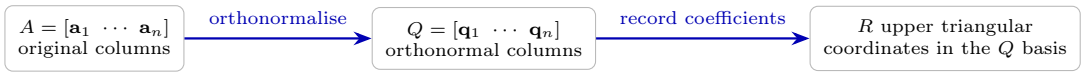


Figure 83: QR writes $A = QR$ while preserving prefix column spaces: $\mathrm{col}(A_k) = \mathrm{col}(Q_k)$.

Perspective 4.1.9. *Gram–Schmidt and Householder QR*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ have full column rank.
Gram–Schmidt gives orthonormal columns $\mathbf{q}_1, \ldots, \mathbf{q}_n$ s.t. $\mathrm{span}(\mathbf{a}_1, \ldots, \mathbf{a}_k) = \mathrm{span}(\mathbf{q}_1, \ldots, \mathbf{q}_k)$, $k = 1, \ldots, n$. Hence $A = QR$ with $Q^*Q = I$ and R upper triangular. The Householder construction gives the same factorization via unitary transformations, which is the numerically stable form.

Perspective 4.1.10. *Rank-Deficient QR from Gram–Schmidt*
Let $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n] \in \mathrm{Mat}(m, n, \mathbb{F})$, $m \geq n$, and let $\mathbf{u}_j$ be the arbitrary-list Gram–Schmidt remainders from Theorem 1.3.8. Set $\mathcal{J} = \{j : \mathbf{u}_j \neq \mathbf{0}\}$. For $j \in \mathcal{J}$ put $\mathbf{q}_j = \mathbf{u}_j / \|\mathbf{u}_j\|$; for $j \notin \mathcal{J}$ choose $\mathbf{q}_j \in \mathrm{col}(A)^\perp$ so that $Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_n]$ has orthonormal columns. With $R = Q^*A$, one has $A = QR$, R upper triangular, and $r_{jj} = 0$ exactly when $\mathbf{u}_j = \mathbf{0}$. Thus uniqueness fails unless $\mathcal{J} = \{1, \ldots, n\}$ and $r_{jj} > 0$ for all j.
Structural Commentary: Rank defect becomes zero QR pivots.
Key Insight: The full-rank hypothesis in uniqueness is a statement about nonzero orthogonal remainders.

Corollary 4.1.11. *Prefix Column Spaces in QR Factorisation*
Let $A = QR$ be the narrow QR factorisation of $A \in \mathrm{Mat}(m, n, \mathbb{F})$ with $\mathrm{rank}(A) = n$ and positive diagonal entries in R. For $k = 1, \ldots, n$, let A_k and Q_k be the first k columns of A and Q, respectively. Then $\mathrm{col}(A_k) = \mathrm{col}(Q_k)$. Thus the columns of Q_k form an orthonormal basis of $\mathrm{col}(A_k)$.

Proof. Let R_k be the leading $k \times k$ principal submatrix of R. Since R is upper triangular with positive diagonal entries, R_k is invertible. The first k columns of $A = QR$ satisfy $A_k = Q_k R_k$, because upper triangularity prevents later columns of Q from contributing. Hence $\mathrm{col}(A_k) \subseteq \mathrm{col}(Q_k)$ and $Q_k = A_k R_k^{-1}$ gives the reverse inclusion. The columns of Q_k are orthonormal because Q has orthonormal columns.
Structural Commentary: QR preserves prefix column spaces while orthonormalising them.
Key Insight: Invertible R_k only changes coordinates within that space. □

Theorem 4.1.12. *Cholesky Factorisation*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$, where $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. If $A \succeq 0$, then $A = LL^*$ for some lower triangular L with nonnegative diagonal entries. If $A \succ 0$, then L is unique and has positive diagonal entries.

Proof. Let $B = A^{1/2}$ and take a QR factorisation $B = QR$ with R upper triangular and diagonal entries nonnegative. Then $A = B^*B = R^*R$, so $L = R^*$ works.
If $A \succ 0$, then L is invertible, so its diagonal entries are positive. For uniqueness, suppose $A = L_1 L_1^* = L_2 L_2^*$ with L_i lower triangular and positive diagonal. Set $R_i = L_i^*$. Then R_i is upper triangular with positive diagonal and $B^2 = A = R_i^* R_i$. Define $Q_i = B R_i^{-1}$. Then $Q_i^* Q_i = (R_i^{-1})^* B^2 R_i^{-1} = (R_i^{-1})^* R_i^* R_i R_i^{-1} = I_n$, so $B = Q_i R_i$ are QR factorisations with positive diagonal. Uniqueness of QR gives $R_1 = R_2$, hence $L_1 = L_2$.
Structural Commentary: Cholesky is QR applied to $A^{1/2}$.
Key Insight: Triangularise the square root, then set $L = R^*$. □

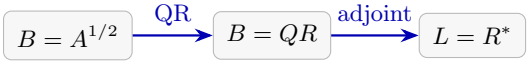


Figure 84: Apply QR to $B = A^{1/2}$; then $A = R^*R = LL^*$, with uniqueness in the positive definite case.

4.2 Spectral and Singular Value Decompositions

Theorem 4.2.1. *Polar Decomposition*
For every $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, $\exists$ an isometry $S \in \mathcal{L}(\mathcal{V})$ s.t. $T = S\sqrt{T^*T}$.

Proof. Let $P = \sqrt{T^*T}$. Since T^*T is positive, P is well defined by Proposition 7.3.8.
For every $\mathbf{v} \in \mathcal{V}$, $\|T\mathbf{v}\|^2 = \langle T^*T\mathbf{v}, \mathbf{v}\rangle = \langle P^2\mathbf{v}, \mathbf{v}\rangle = \|P\mathbf{v}\|^2$.
Define $S_1 : \operatorname{range}(P) \to \operatorname{range}(T)$ by $S_1(P\mathbf{v}) = T\mathbf{v}$. The norm identity shows this is well defined: if $P\mathbf{v}_1 = P\mathbf{v}_2$, then $\|T(\mathbf{v}_1 - \mathbf{v}_2)\| = \|P(\mathbf{v}_1 - \mathbf{v}_2)\| = 0$. It also shows S_1 is an isometry onto $\operatorname{range}(T)$.
The same norm identity gives $\ker(P) = \ker(T)$, so $\operatorname{rank}(P) = \operatorname{rank}(T)$ by rank-nullity. Hence $\operatorname{range}(P)^\perp$ and $\operatorname{range}(T)^\perp$ have the same dimension. Choose any isometry between these complements and combine it orthogonally with S_1 to obtain an isometry S on all of $\mathcal{V}$.
Then $SP\mathbf{v} = S_1(P\mathbf{v}) = T\mathbf{v}$ for every $\mathbf{v}$, so $T = S\sqrt{T^*T}$.
Structural Commentary: The positive factor $\sqrt{T^*T}$ reproduces exactly the lengths produced by T.
Key Insight: Once the metric stretching has been extracted, the remaining map is forced to be an isometry. □

Corollary 4.2.2. *Uniqueness in Operator Polar Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, and put $|T| := \sqrt{T^*T}$ and $|T^*| := \sqrt{TT^*}$.

(i) If $T = SP$, $S^*S = I_{\mathcal{V}}$, and $P \succeq 0$, then $P = |T|$. Hence the positive right polar factor is unique.
(ii) $\exists$ left polar decomposition $T = |T^*|\, R$ with R an isometry, and $|T^*|$ is unique positive left polar factor.
(iii) If T is invertible, then both isometric factors are unique: $S = T\,|T|^{-1}$ and $R = |T^*|^{-1}\, T$.

Proof.

(i) If $T = SP$, then $T^*T = PS^*SP = P^2$. Since $P \succeq 0$, Proposition 7.3.8 gives $P = \sqrt{T^*T} = |T|$.
(ii) Apply Theorem 4.2.1 to T^*. There is an isometry S_0 such that $T^* = S_0\,|T^*|$.
Taking adjoints gives $T = |T^*|\, S_0^*$. Set $R = S_0^*$. If also $T = QR$ with $Q \succeq 0$ and R an isometry, then $RR^* = I_{\mathcal{V}}$ in finite dimensions, so $TT^* = QRR^*Q = Q^2$. Hence $Q = |T^*|$.
(iii) If T is invertible, then $|T|$ and $|T^*|$ are positive definite, hence invertible.
Therefore any right and left polar decompositions force $S = T\,|T|^{-1}$, $R = |T^*|^{-1}\, T$.

Structural Commentary: The positive polar factor is canonical.
Key Insight: Invertibility removes the kernel complement, so the whole factorisation becomes unique. □

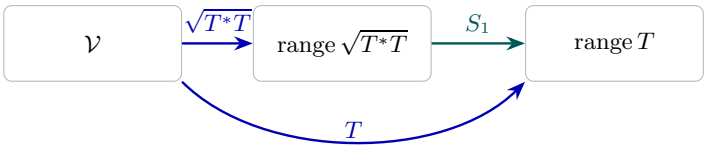


Figure 85: Polar decomposition builds S_1 using $\|T\mathbf{v}\| = \left\|\sqrt{T^*T}\mathbf{v}\right\|$.

Perspective 4.2.3. *Polar Decomposition Separates Stretch and Direction*
In $T = S\sqrt{T^*T}$, the positive factor $\sqrt{T^*T}$ determines the metric stretch and the isometry S determines the direction. Thus polar decomposition is the operator analogue of modulus times phase.
Structural Commentary: For normal T, the polar factors commute and share spectral data.
Key Insight: Polar decomposition factors an operator into positive geometry and isometric motion.

Definition 4.2.4. *Singular Values*
For finite-dimensional inner-product spaces $\mathcal{U}, \mathcal{V}$ and $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, the *singular values* of T are the eigenvalues of $\sqrt{T^*T} \in \mathcal{L}(\mathcal{U})$, where each eigenvalue λ is repeated $\dim E_\lambda(\sqrt{T^*T})$.

Theorem 4.2.5. *Singular Value Decomposition*
Let $T \in \mathcal{L}(\mathcal{V})$ have singular values $s_1, \ldots, s_n$. Then there exist orthonormal bases $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ and $(\mathbf{f}_1, \ldots, \mathbf{f}_n)$ of $\mathcal{V}$ such that $T\mathbf{v} = s_1 \langle \mathbf{v}, \mathbf{e}_1\rangle \mathbf{f}_1 + \cdots + s_n \langle \mathbf{v}, \mathbf{e}_n\rangle \mathbf{f}_n$ for every $\mathbf{v} \in \mathcal{V}$.

Proof. Let $|T| = \sqrt{T^*T}$. Since $|T|$ is positive, the real or complex spectral theorem gives an orthonormal basis $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ with $|T|\,\mathbf{e}_j = s_j\mathbf{e}_j$. Thus $|T|\,\mathbf{v} = \sum_{j=1}^{n} s_j \langle \mathbf{v}, \mathbf{e}_j\rangle \mathbf{e}_j$.
By Theorem 4.2.1, $T = S\,|T|$ for some isometry S. Put $\mathbf{f}_j = S\mathbf{e}_j$. The vectors $\mathbf{f}_1, \ldots, \mathbf{f}_n$ form an orthonormal basis, and applying S to the displayed expansion gives the desired formula.
Structural Commentary: SVD diagonalises $|T|$, then transports its eigenvectors by S.
Key Insight: The s_j are stretches; the $\mathbf{f}_j$ are their landing directions. □

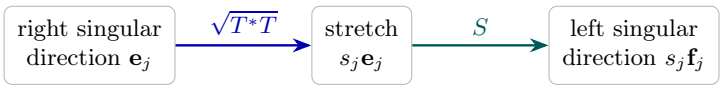


Figure 86: Since $T = S\sqrt{T^*T}$, each $\mathbf{e}_j$ maps to $s_j\mathbf{f}_j$.

Proposition 4.2.6. *Singular Values from T^*T*
The singular values of $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ are the nonnegative square roots of the eigenvalues of T^*T, with the same eigenspace multiplicities.

Proof. By the spectral theorem, choose an orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ and nonnegative scalars $\lambda_1, \dots, \lambda_n$ such that $T^*T\mathbf{e}_j = \lambda_j \mathbf{e}_j$. The unique positive square root of T^*T sends $\mathbf{e}_j$ to $\sqrt{\lambda_j}\mathbf{e}_j$.
Thus the singular values are $\sqrt{\lambda_j}$, with the same eigenspace multiplicities.
Structural Commentary: The positive square root keeps eigenspaces and sends λ to $\sqrt{\lambda}$.
Key Insight: Singular values can be read from T^*T directly. □

Theorem 4.2.7. *Singular Vectors for a Linear Map*
Let $\mathcal{U}, \mathcal{V}$ be finite-dimensional inner-product spaces over $\mathbb{R}$ or $\mathbb{C}$, and let $T \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ have rank r.
Put $n = \dim \mathcal{U}$, $m = \dim \mathcal{V}$, and $\mathcal{U}_+ = \operatorname{range}(T^*)$, $\mathcal{U}_0 = \ker(T)$; $\mathcal{V}_+ = \operatorname{range}(T)$, $\mathcal{V}_0 = \ker(T^*)$.
Then $\exists$ ordered orthonormal bases $\beta = (\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$; $\gamma = (\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}, \dots, \mathbf{v}_m)$
with $\operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{U}_+$; $\operatorname{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n) = \mathcal{U}_0$; $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) = \mathcal{V}_+$; $\operatorname{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_m) = \mathcal{V}_0$,
and numbers $s_1, \dots, s_r > 0$ such that $T\mathbf{u}_k = s_k\mathbf{v}_k$; $T^*\mathbf{v}_k = s_k\mathbf{u}_k$, for $1 \le k \le r$.
The s_k are the positive singular values; $\mathbf{u}_k$ and $\mathbf{v}_k$ are right and left singular vectors.

Proof. Let $A = T^*T$. Then $A \succeq 0$, $A = A^*$, and $\ker(A) = \ker(T)$. By spectral theorem, choose rthonormal eigenbasis $\mathbf{u}_1, \dots, \mathbf{u}_n$ s.t. $A\mathbf{u}_k = s_k^2\mathbf{u}_k$ for $1 \le k \le r$; $A\mathbf{u}_k = 0$ for $k > r$; $s_k > 0$.
Thus $\operatorname{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n) = \mathcal{U}_0$, and Proposition 2.4.11 gives $\operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{U}_+$. For $1 \le k \le r$,
set $\mathbf{v}_k = s_k^{-1}T\mathbf{u}_k$. Then $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \frac{1}{s_i s_j}\langle T\mathbf{u}_i, T\mathbf{u}_j \rangle = \frac{1}{s_i s_j}\langle T^*T\mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij}$; $T^*\mathbf{v}_k = s_k^{-1}A\mathbf{u}_k = s_k\mathbf{u}_k$.
Hence $\mathbf{v}_1, \dots, \mathbf{v}_r$ are orthonormal and span $\mathcal{V}_+ = T(\mathcal{U}_+)$. Complete them by an orthonormal basis of $\mathcal{V}_0 = \mathcal{V}_+^{\perp}$.
Structural Commentary: T^*T gives right directions; T gives left directions.
Key Insight: Away from the two kernels, T and T^* act symmetrically by the same positive scalars. □

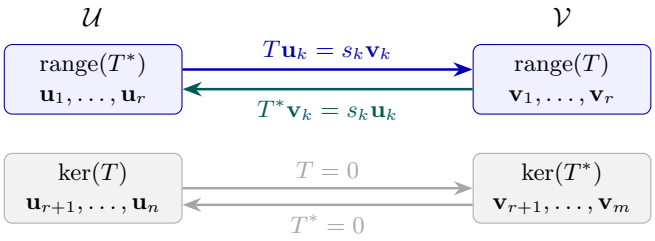


Figure 87: The singular-vector bases split the domain and codomain into active singular subspaces and kernels.

Perspective 4.2.8. *Operator-to-Matrix SVD Bridge*
In the bases from Theorem 4.2.7, the matrix of T is rectangular diagonal with diagonal entries $s_1, \dots, s_r$ and zeros elsewhere. Returning to standard orthonormal bases gives the matrix form $A = V\Sigma W^*$ in Theorem 4.2.10.

Definition 4.2.9. *Matrix Singular Values*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$, let $q = \min(m, n)$, and let $r = \operatorname{rank}(A)$. The *positive singular values* of A are the positive eigenvalues of $|A| = (A^*A)^{1/2}$, written $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0$.
The full list of singular values is $\sigma_1, \dots, \sigma_r, \sigma_{r+1}, \dots, \sigma_q$, where $\sigma_{r+1} = \cdots = \sigma_q = 0$.

Theorem 4.2.10. *Matrix Singular Value Decomposition*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \ge \cdots \ge \sigma_q \ge 0$, where $q = \min(m, n)$. Then $\exists$ unitary matrices $V \in \operatorname{Mat}(m, m, \mathbb{F})$, $W \in \operatorname{Mat}(n, n, \mathbb{F})$ such that $A = V\Sigma W^*$, $\Sigma = \begin{bmatrix} \operatorname{diag}(\sigma_1, \dots, \sigma_q) & 0 \\ 0 & 0 \end{bmatrix} \in \operatorname{Mat}(m, n, \mathbb{R})$ with the evident deletion of empty blocks.

Proof. If $A = 0$, take $\Sigma = 0$ and choose any unitary V, W. Assume $r = \operatorname{rank}(A) \ge 1$.
Diagonalise A^*A as $W^*A^*AW = \operatorname{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0)$, where $\mathbf{w}_1, \dots, \mathbf{w}_n$ are the columns of W.
For $1 \le j \le r$, set $\mathbf{v}_j = \sigma_j^{-1}A\mathbf{w}_j$. Then $\mathbf{v}_i^*\mathbf{v}_j = \frac{1}{\sigma_i\sigma_j}\mathbf{w}_i^*A^*A\mathbf{w}_j = \delta_{ij}$ so $\mathbf{v}_1, \dots, \mathbf{v}_r$ are orthonormal.
Complete them to an orthonormal basis of $\mathbb{F}^m$ and let V be the matrix with these columns.

For $j > r$, $\|A\mathbf{w}_j\|_2^2 = \mathbf{w}_j^* A^* A \mathbf{w}_j = 0$, hence $A\mathbf{w}_j = 0$. Thus $AW = V\Sigma$, equivalently $A = V\Sigma W^*$.
Structural Commentary: A^*A determines the right singular directions and singular values.
Key Insight: Apply A and normalize to get the left singular directions. □

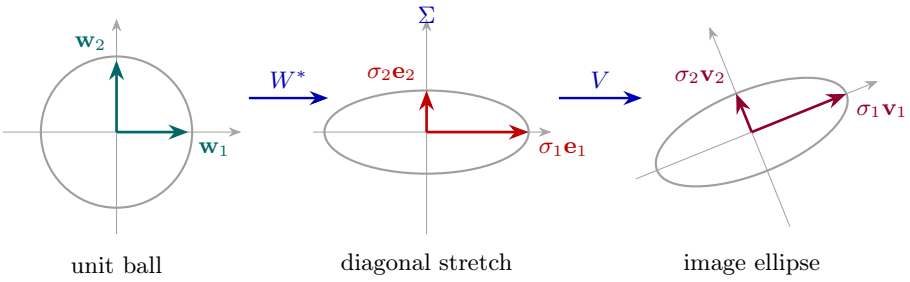


Figure 88: SVD: right-unitary motion, diagonal stretch, left-unitary motion.

Proposition 4.2.11. *Basic Singular Value Identities*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, where $q = \min(m,n)$ and $r = \mathrm{rank}(A)$. Let $c \in \mathbb{F}$.

(i) The positive eigenvalues of A^*A and AA^* are $\sigma_1^2, \ldots, \sigma_r^2$.
(ii) $\sum_{j=1}^{q} \sigma_j^2 = \mathrm{tr}(A^*A) = \mathrm{tr}(AA^*) = \|A\|_F^2$, where $\|A\|_F^2 := \mathrm{tr}(A^*A)$.
(iii) A and A^* have the same singular values. Over $\mathbb{C}$, so do A^t and $\overline{A}$.
(iv) The singular values of cA are $|c|\,\sigma_1, \ldots, |c|\,\sigma_q$.

Proof. Let $A = V\Sigma W^*$ be an SVD. Then $A^*A = W(\Sigma^*\Sigma)W^*$ and $AA^* = V(\Sigma\Sigma^*)V^*$.

(i) The nonzero diagonal entries of $\Sigma^*\Sigma$ and $\Sigma\Sigma^*$ are $\sigma_1^2, \ldots, \sigma_r^2$.
(ii) Since A^*A and $\Sigma^*\Sigma$ are unitarily similar, and AA^* and $\Sigma\Sigma^*$ are unitarily similar,
$\mathrm{tr}(A^*A) = \mathrm{tr}(\Sigma^*\Sigma) = \sum_{j=1}^{q} \sigma_j^2 = \mathrm{tr}(\Sigma\Sigma^*) = \mathrm{tr}(AA^*)$. The final equality is the definition of $\|A\|_F$.
(iii) For A^*, the two Gram matrices are interchanged. Over $\mathbb{C}$, $(A^t)^*A^t = (AA^*)^t$ and $(\overline{A})^*\overline{A} = (A^*A)^t$. Since a matrix and its transpose have the same characteristic polynomial, the spectra agree.
(iv) Since $(cA)^*(cA) = |c|^2 A^*A$, singular values scale by $|c|$.

Structural Commentary: Unitary factors do not change Gram spectra; Σ contains all singular data.
Key Insight: Reduce each claim to $\Sigma^*\Sigma$ and $\Sigma\Sigma^*$. □

Corollary 4.2.12. *Square Invertibility and Singular Values*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$, $\mathbb{F} \in \{\mathbb{R},\mathbb{C}\}$, have singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. The following are equivalent:

(i) A is invertible;
(ii) $\mathrm{rank}(A) = n$;
(iii) $\sigma_n > 0$;
(iv) $A^*A \succ 0$.

If $r = \mathrm{rank}(A) < n$, then $\sigma_1, \ldots, \sigma_r > 0$ and $\sigma_{r+1} = \cdots = \sigma_n = 0$.

Proof.

(i) ⇒ *(ii)* By Corollary 3.5.5, invertibility gives full rank.
(ii) ⇒ *(iii)* By Definition 4.2.9, the number of positive singular values is $\mathrm{rank}(A)$.
(iii) ⇒ *(iv)* If $\sigma_n > 0$, then A has full column rank, so Proposition 2.1.2 gives $A^*A \succ 0$.
(iv) ⇒ *(i)* If $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}^*A^*A\mathbf{x} = 0$; PSD gives $\mathbf{x} = \mathbf{0}$. Thus $\ker(A) = \{\mathbf{0}\}$, so A is invertible.

The rank-defect statement follows again from Definition 4.2.9.
Structural Commentary: SVD sees nonsingularity as the absence of zero stretch directions.
Key Insight: A zero singular value is exactly a nonzero vector sent to zero. □

Definition 4.2.13. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$.
A factorisation $A = X\Sigma_r Y^*$ is a *compact singular value decomposition* if $X \in \mathrm{Mat}(m,r,\mathbb{F})$ and $Y \in \mathrm{Mat}(n,r,\mathbb{F})$ have orthonormal columns and $\Sigma_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$.

Theorem 4.2.14. *Compact Singular Value Decomposition*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have rank $r \geq 1$.

(i) If $A = V\Sigma W^*$ is an SVD and X, Y are the first r columns of V, W, then $A = X\Sigma_r Y^*$.

(ii) Conversely, if X, Y have orthonormal columns and $\Sigma_r = \operatorname{diag}(\sigma_1, \dots, \sigma_r)$ with $\sigma_1 \geq \cdots \geq \sigma_r > 0$, then $X\Sigma_r Y^*$ has positive singular values $\sigma_1, \dots, \sigma_r$.

Proof.

(i) Partition $V = [X\ X_0]$, $W = [Y\ Y_0]$, and $\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}$. Multiplication gives $A = V\Sigma W^* = X\Sigma_r Y^*$.

(ii) Let $B = X\Sigma_r Y^*$. Since $X^*X = Y^*Y = I_r$, $B^*B = Y\Sigma_r^2 Y^*$.
Completing Y to a unitary matrix $\widetilde{Y} = [Y\ Y_0]$ gives $\widetilde{Y}^* B^* B \widetilde{Y} = \Sigma_r^2 \oplus 0$.
Thus the positive eigenvalues of B^*B are $\sigma_1^2, \dots, \sigma_r^2$, so the positive singular values of B are $\sigma_1, \dots, \sigma_r$.

Structural Commentary: Compact SVD keeps only the nonzero singular directions.
Key Insight: The rank-r data X, Σ_r, Y reconstructs A. □

Theorem 4.2.15. *Column and Kernel Spaces from Compact SVD*
Let $A = X\Sigma_r Y^*$ be a compact SVD. Then $\operatorname{col}(A) = \operatorname{col}(X) = (\ker A^*)^\perp$, and $\ker(A) = \operatorname{col}(Y)^\perp = (\operatorname{col}(A^*))^\perp$.

Proof. Since $A = X\Sigma_r Y^*$ and $AY\Sigma_r^{-1} = X$, $\operatorname{col}(A) = \operatorname{col}(X)$.
For $\mathbf{z} \in \mathbb{F}^n$, $A\mathbf{z} = 0 \iff X\Sigma_r Y^*\mathbf{z} = 0 \iff Y^*\mathbf{z} = 0 \iff \mathbf{z} \perp \operatorname{col}(Y)$, because X has independent columns and Σ_r is invertible. The orthogonal-complement identities follow from Proposition 2.4.11.
Structural Commentary: Compact SVD exposes image directions X and null directions $\operatorname{col}(Y)^\perp$.
Key Insight: A acts only on the right singular subspace $\operatorname{col}(Y)$. □

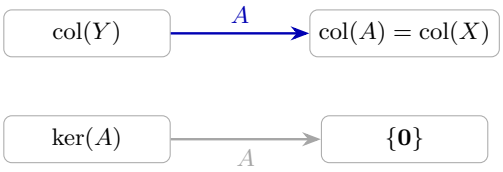


Figure 89: Compact SVD: A maps $\operatorname{col}(Y)$ onto $\operatorname{col}(X)$ and kills $\ker(A) = \operatorname{col}(Y)^\perp$.

Theorem 4.2.16. *Uniqueness of Compact SVD up to Singular Subspaces*
Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$ be compact SVDs with the same Σ_r. Then $X_1 = X_2U$, $Y_1 = Y_2U$ for some unitary U satisfying $U\Sigma_r = \Sigma_r U$. Hence $U = U_1 \oplus \cdots \oplus U_d$ along the equal singular-value blocks of Σ_r.

Proof. Theorem 4.2.15 gives $\operatorname{col}(X_1) = \operatorname{col}(X_2)$ and $\operatorname{col}(Y_1) = \operatorname{col}(Y_2)$. Since each X_i and Y_i has orthonormal columns, there are unitary $U, V \in \operatorname{Mat}(r, r, \mathbb{F})$ such that $X_1 = X_2U$ and $Y_1 = Y_2V$. Substitution and multiplication on the left by X_2^* and on the right by Y_2 give $\Sigma_r = U\Sigma_r V^*$, hence $U\Sigma_r = \Sigma_r V$.
Therefore $U\Sigma_r^2 U^* = \Sigma_r^2$, so U commutes with Σ_r^2. Lemma 5.3.3 makes U block diagonal on equal singular-value blocks. Hence $U\Sigma_r = \Sigma_r U$, and $U\Sigma_r = \Sigma_r V$ forces $V = U$ because Σ_r is invertible.
Structural Commentary: Compact SVD is unique up to unitary changes inside equal singular-value blocks.
Key Insight: Distinct singular values fix subspaces; repeated ones only fix the subspace, not its basis. □

Definition 4.2.17. *Matrix Modulus*
For $A \in \operatorname{Mat}(m, n, \mathbb{F})$, the *modulus* of A is $|A| := (A^*A)^{1/2} \in \operatorname{Mat}(n, n, \mathbb{F})$.
Thus $|A^*| = (AA^*)^{1/2} \in \operatorname{Mat}(m, m, \mathbb{F})$.

Proposition 4.2.18. *Modulus from SVD*
Let $A = V\Sigma W^*$ be an SVD. Then $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$ and $|A^*| = V(\Sigma\Sigma^*)^{1/2}V^*$.

Proof. Since $A^*A = W(\Sigma^*\Sigma)W^*$ and $AA^* = V(\Sigma\Sigma^*)V^*$, the displayed matrices are positive semidefinite square roots of A^*A and AA^*, respectively. Uniqueness of the positive semidefinite square root gives the claim.
Structural Commentary: The modulus keeps only the singular-value stretch.
Key Insight: In an SVD, $|A|$ is the right singular-vector version of the diagonal stretch. □

Theorem 4.2.19. *Rectangular Polar Decomposition*
Let $A = V\Sigma W^* \in \operatorname{Mat}(m, n, \mathbb{F})$ be an SVD.

(i) If $m \geq n$, let V_n be the first n columns of V and set $U = V_n W^*$. Then $U^*U = I_n$ and $A = U\,|A|$.
(ii) If $m \leq n$, let W_m be the first m columns of W and set $U = VW_m^*$. Then $UU^* = I_m$ and $A = |A^*|\,U$.
(iii) If $m = n$, then $U = VW^*$ is unitary and $A = U\,|A| = |A^*|\,U$.

Proof.

(i) Proposition 4.2.18 gives $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$.
Since $V_n(\Sigma^*\Sigma)^{1/2} = V\Sigma$, $U|A| = V_nW^*W(\Sigma^*\Sigma)^{1/2}W^* = V\Sigma W^* = A$. Also $U^*U = WV_n^*V_nW^* = I_n$.
(ii) Apply (i) to A^* with the SVD $A^* = W\Sigma^*V^*$, obtaining $A^* = U_0|A^*|$ with $U_0 = W_mV^*$ and $U_0^*U_0 = I_m$.
Taking adjoints gives $A = |A^*|U_0^* = |A^*|VW_m^*$, and $U_0^*(U_0^*)^* = U_0^*U_0 = I_m$. Thus $U = VW_m^*$ works.
(iii) This is the common case of (i) and (ii), with $U = VW^*$ unitary.

Structural Commentary: Polar decomposition compresses the two SVD unitary factors into one partial isometry.
Key Insight: SVD records every singular direction; polar form records stretch plus motion. □

Method 4.2.20. *SVD and Polar Computation Method*
To compute an SVD and the associated polar factors of $A \in \mathrm{Mat}(m,n,\mathbb{F})$:

(i) Diagonalise A^*A; the square roots of its positive eigenvalues are the positive singular values.
(ii) Choose orthonormal right singular vectors $\mathbf{w}_j$ for the positive eigenvalues and set $\mathbf{v}_j = \sigma_j^{-1}A\mathbf{w}_j$.
(iii) Complete the $\mathbf{v}_j$ and $\mathbf{w}_j$ to unitary matrices V, W to obtain $A = V\Sigma W^*$.
Omit zero-singular directions for compact SVD.
(iv) Use $|A| = W(\Sigma^*\Sigma)^{1/2}W^*$ and merge V, W as in Theorem 4.2.19 to obtain the polar factor.

Theorem 4.2.21. *Uniqueness of the Positive Polar Factor*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$.

(i) If $m \geq n$ and $A = UB$ with $U^*U = I_n$ and $B \succeq 0$, then $B = |A|$.
If $\mathrm{rank}(A) = n$, then $U = A|A|^{-1}$ is unique.
(ii) If $m \leq n$ and $A = CU$ with $UU^* = I_m$ and $C \succeq 0$, then $C = |A^*|$.
If $\mathrm{rank}(A) = m$, then $U = |A^*|^{-1}A$ is unique.

Proof.

(i) $A^*A = BU^*UB = B^2$, $B = |A|$ by uniqueness of positive square root. If $\mathrm{rank}(A) = n$, then $\ker(A) = \{\mathbf{0}\}$.
Since $\mathbf{x}^*A^*A\mathbf{x} = \|A\mathbf{x}\|_2^2$, this gives $\ker(A^*A) = \{\mathbf{0}\}$ and $|A|$ is invertible. Thus $U = A|A|^{-1}$.
(ii) $AA^* = CUU^*C = C^2$, so $C = |A^*|$. If $\mathrm{rank}(A) = m$, then A^* has full column rank, so $\ker(A^*) = \{\mathbf{0}\}$.
Since $\mathbf{y}^*AA^*\mathbf{y} = \|A^*\mathbf{y}\|_2^2$, this gives $\ker(AA^*) = \{\mathbf{0}\}$ and $|A^*|$ is invertible. Thus $U = |A^*|^{-1}A$.

Structural Commentary: The positive factor is fixed by the Gram matrix.
Key Insight: Full rank makes the partial isometry unique. □

Theorem 4.2.22. *Normality and Polar Commutation*
Let $A \in \mathrm{Mat}(n,n,\mathbb{F})$ and suppose $A = U|A|$ with U unitary. Then A is normal if and only if $U|A| = |A|U$.

Proof.

($\Rightarrow$) If A is normal, then $|A|^2 = A^*A = AA^* = U|A|^2U^*$. Thus $U|A|^2U^* = |A|^2$. The matrix $U|A|U^*$ is PSD and squares to $|A|^2$, so uniqueness of positive square root gives $U|A|U^* = |A|$. Hence $U|A| = |A|U$.
($\Leftarrow$) If $U|A| = |A|U$, then $AA^* = U|A|^2U^* = |A|^2 = A^*A$.

Structural Commentary: Normality says that the polar motion preserves the stretching eigenspaces.
Key Insight: Normality is commutation of polar motion and stretch. □

Proposition 4.2.23. *Absolute Determinant Comes from the Positive Polar Factor*
For $T \in \mathcal{L}(\mathcal{V})$ on a finite-dimensional inner-product space, $|\det T| = \det\sqrt{T^*T}$.

Proof. By Theorem 4.2.1, $T = S\sqrt{T^*T}$ for some isometry S. Hence $|\det T| = |\det S|\left|\det\sqrt{T^*T}\right| = \det\sqrt{T^*T}$, where Proposition 1.2.9 gives $|\det S| = 1$, and positivity of $\sqrt{T^*T}$ gives nonnegative determinant.
Structural Commentary: Polar decomposition isolates stretch; only the positive factor changes volume.
Key Insight: Determinant sign or phase belongs to the isometric part; volume belongs to the positive part. □

5 Norms, Conditioning, Perturbation, and Stability

5.1 Matrix Norms

Matrix norms measure stretch, energy, approximation quality, and sensitivity. Singular values organize norms, low-rank approximation, pseudoinverses, conditioning, and error bounds.

Definition 5.1.1. *Spectral Norm and Extreme Singular Values*
For $A \in \mathrm{Mat}(m,n,\mathbb{F})$, define the *spectral norm* $\|A\|_2 := \max_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2$.

If the singular values of A are $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$, $q = \min(m,n)$, set $\sigma_{\max}(A) := \sigma_1$.
When A is square, $\sigma_{\min}(A) := \sigma_n$.

Theorem 5.1.2. *Spectral Norm is the Largest Singular Value*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$. Then $\|A\|_2 = \sigma_{\max}(A)$.
Moreover, if $A = V\Sigma W^*$ is an SVD and $\mathbf{w}_1$ is the first column of W, then $\|A\mathbf{w}_1\|_2 = \sigma_1$.

Proof. Let $\mathbf{w}_1, \ldots, \mathbf{w}_n$ be the columns of W. For $\mathbf{x} = \sum_{j=1}^n c_j \mathbf{w}_j$ with $\|\mathbf{x}\|_2 = 1$, unitary invariance of the Euclidean norm gives $\|A\mathbf{x}\|_2^2 = \|\Sigma W^*\mathbf{x}\|_2^2 = \sum_{j=1}^{q} \sigma_j^2 |c_j|^2 \leq \sigma_1^2 \sum_{j=1}^{n} |c_j|^2 = \sigma_1^2$.
Thus $\|A\|_2 \leq \sigma_1$. Taking $\mathbf{x} = \mathbf{w}_1$ gives equality.
Structural Commentary: The right singular basis diagonalises the stretching computation.
Key Insight: The largest stretch occurs in a top right singular direction. □

Proposition 5.1.3. *Basic Properties of the Spectral Norm*
For compatible matrices A, B and scalar c:

(i) $\|A\|_2 \geq 0$, *and* $\|A\|_2 = 0$ *if and only if* $A = 0$;
(ii) $\|cA\|_2 = |c| \, \|A\|_2$;
(iii) $\|A + B\|_2 \leq \|A\|_2 + \|B\|_2$;
(iv) $\|AB\|_2 \leq \|A\|_2 \, \|B\|_2$.

Proof.

(i) Since $\|A\mathbf{x}\|_2 \geq 0$, $\|A\|_2 \geq 0$. Also $\|A\|_2 = 0$ iff $A\mathbf{x} = 0$ for every unit $\mathbf{x}$, iff $A = 0$.
(ii) $\|cA\|_2 = \max_{\|\mathbf{x}\|_2=1} \|cA\mathbf{x}\|_2 = |c| \, \|A\|_2$.
(iii) For $\|\mathbf{x}\|_2 = 1$, $\|(A+B)\mathbf{x}\|_2 \leq \|A\mathbf{x}\|_2 + \|B\mathbf{x}\|_2 \leq \|A\|_2 + \|B\|_2$. Maximize over unit $\mathbf{x}$.
(iv) For $\|\mathbf{x}\|_2 = 1$, $\|AB\mathbf{x}\|_2 \leq \|A\|_2 \, \|B\mathbf{x}\|_2 \leq \|A\|_2 \, \|B\|_2$. Maximize over unit $\mathbf{x}$.

Structural Commentary: The spectral norm is the operator norm induced by Euclidean geometry.
Key Insight: Bound each output vector first, then take the worst case. □

Definition 5.1.4. *Operator Norm on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$*
For $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, define $\|A\|_{\mathrm{op}} := \sup_{\|\mathbf{h}\|_2 \leq 1} \|A\mathbf{h}\|_2$.
Equivalently, after standard bases identify A with its matrix, $\|A\|_{\mathrm{op}}$ is the spectral norm from Definition 5.1.1.
Thus $\|A\mathbf{h}\|_2 \leq \|A\|_{\mathrm{op}} \, \|\mathbf{h}\|_2$, and sums and products obey the norm estimates in Proposition 5.1.3.
The induced metric on $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ is $d(A,B) := \|A - B\|_{\mathrm{op}}$; this is the metric used for continuity of derivative maps in Definition 3.2.2.

Proposition 5.1.5. *Frobenius and Spectral Norm Comparison*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$ have singular values $\sigma_1 \geq \cdots \geq \sigma_q \geq 0$. Then $\|A\|_F^2 = \sum_{j=1}^q \sigma_j^2$ and $\|A\|_2 \leq \|A\|_F$.
If $A \neq 0$, equality $\|A\|_2 = \|A\|_F$ holds if and only if $\mathrm{rank}(A) = 1$.

Proof. The Frobenius identity is Proposition 4.2.11. Hence $\|A\|_2^2 = \sigma_1^2 \leq \sum_{j=1}^q \sigma_j^2 = \|A\|_F^2$. Assume $A \neq 0$.

($\Rightarrow$) If $\|A\|_2 = \|A\|_F$, then $\sum_{j=2}^q \sigma_j^2 = 0$, so only σ_1 is nonzero and $\mathrm{rank}(A) = 1$.
($\Leftarrow$) If $\mathrm{rank}(A) = 1$, then $\sigma_2 = \cdots = \sigma_q = 0$, so $\|A\|_2 = \|A\|_F$.

Structural Commentary: Frobenius counts all singular energy; spectral keeps the largest part.
Key Insight: Equality means exactly one nonzero singular value. □

Theorem 5.1.6. *Submultiplicativity of Frobenius and Spectral Norms*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $B \in \mathrm{Mat}(n,k,\mathbb{F})$. Then $\|AB\|_F \leq \|A\|_2 \, \|B\|_F \leq \|A\|_F \, \|B\|_F$ and $\|AB\|_2 \leq \|A\|_2 \, \|B\|_2$.

Proof. Write $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_k]$. Then $\|AB\|_F^2 = \sum_{j=1}^{k} \|A\mathbf{b}_j\|_2^2 \leq \|A\|_2^2 \sum_{j=1}^{k} \|\mathbf{b}_j\|_2^2 = \|A\|_2^2 \, \|B\|_F^2$.
Proposition 5.1.5 gives $\|A\|_2 \leq \|A\|_F$. The spectral-norm inequality is Proposition 5.1.3.
Structural Commentary: Frobenius norm is column energy; each column of AB is A applied to a column of B.
Key Insight: Control every column by the spectral norm of A. □

Proposition 5.1.7. *Unitary Invariance of Singular Values*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, and let $U \in \mathrm{Mat}(m,m,\mathbb{F})$ and $V \in \mathrm{Mat}(n,n,\mathbb{F})$ be unitary. Then A and UAV have the same singular values. In particular, $\|UAV\|_2 = \|A\|_2$ and $\|UAV\|_F = \|A\|_F$.

Proof. Since $(UAV)^*(UAV) = V^*A^*AV$, the Gram matrices of A and UAV are unitarily similar, hence have the same eigenvalues. Singular values are the nonnegative square roots of these eigenvalues. The two norm identities follow from Theorem 5.1.2 and Proposition 5.1.5.
Structural Commentary: Left and right unitary factors change coordinates, not singular-value data.
Key Insight: Singular values are spectral data of A^*A, and unitary conjugation preserves that spectrum. □

Example 5.1.8. *Similarity Need Not Preserve Singular Values*
Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Then $B = DAD^{-1}$ with $D = \operatorname{diag}(2, 1)$, so A and B are similar.
However $A^*A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $B^*B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ have different eigenvalues, so A and B have different singular values.
Singular values are preserved by unitary left-right equivalence, not by arbitrary similarity.

Theorem 5.1.9. *Schur's Inequality and Normality*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted algebraically.
Then $\sum_{j=1}^{n} |\lambda_j|^2 \leq \|A\|_F^2$ with equality if and only if A is normal.

Proof. Let $A = UTU^*$ be a Schur form, where $T = [t_{ij}]$ is upper triangular and $t_{jj} = \lambda_j$.
Then $\|A\|_F^2 = \operatorname{tr}(T^*T) = \sum_{j=1}^{n} |\lambda_j|^2 + \sum_{i<j} |t_{ij}|^2$. Inequality holds. Equality holds exactly when T is diagonal.
($\Rightarrow$) If equality holds, then T is diagonal, so A is unitarily diagonalizable and hence normal.
($\Leftarrow$) If A is normal, then T is normal and upper triangular, hence diagonal by Lemma 7.2.6.
Structural Commentary: Schur form splits Frobenius norm into eigenvalue mass plus off-diagonal mass.
Key Insight: Normality means the Schur off-diagonal part vanishes. □

Definition 5.1.10. *Defect from Normality*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$, counted with algebraic multiplicity.
The *defect from normality* of A is $\Delta(A) := \|A\|_F^2 - \sum_{j=1}^{n} |\lambda_j|^2$.
By Theorem 5.1.9, $\Delta(A) \geq 0$, with equality if and only if A is normal.

Theorem 5.1.11. *Eigenvalues Versus Singular Values*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ have eigenvalues $\lambda_1, \ldots, \lambda_n$ ordered so that $|\lambda_1| \geq \cdots \geq |\lambda_n|$, and singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Then:
(i) $|\det A| = \prod_{j=1}^{n} |\lambda_j| = \prod_{j=1}^{n} \sigma_j$;
(ii) $\rho(A) \leq \|A\|_2$;
(iii) $\sigma_n \leq |\lambda_n|$;
(iv) *A is normal if and only if $|\lambda_j| = \sigma_j$ for every j;*
(v) *A is positive semidefinite if and only if $\lambda_j = \sigma_j$ for every j.*

Proof.
(i) The eigenvalue product gives $|\det A| = \prod_j |\lambda_j|$.
If $A = V\Sigma W^*$ is an SVD, then $|\det V| = |\det W| = 1$, so $|\det A| = |\det \Sigma| = \prod_j \sigma_j$.
(ii) If $A\mathbf{x} = \lambda\mathbf{x}$ with $\|\mathbf{x}\|_2 = 1$, then $|\lambda| = \|A\mathbf{x}\|_2 \leq \|A\|_2$. Maximize over eigenvalues.
(iii) If A is singular, both sides are 0 or the inequality is immediate.
If A is invertible, apply (ii) to A^{-1}: $\frac{1}{|\lambda_n|} = \rho(A^{-1}) \leq \|A^{-1}\|_2 = \frac{1}{\sigma_n}$.
(iv) If A is normal, the spectral theorem gives $\sigma_j = |\lambda_j|$. Conversely, equality gives $\sum_j |\lambda_j|^2 = \sum_j \sigma_j^2 = \|A\|_F^2$, so Theorem 5.1.9 gives normality.
(v) If $A \succeq 0$, then A is normal and $\lambda_j \geq 0$, so (iv) gives $\lambda_j = \sigma_j$. Conversely, equality gives normality with nonnegative real eigenvalues, hence $A \succeq 0$ by Theorem 2.1.7.
Structural Commentary: Normality aligns eigenvalue moduli with singular values; positivity removes the moduli.
Key Insight: Normality is the bridge that turns spectral data into singular-value data. □

Theorem 5.1.12. *Trace Bound by Singular Values*
If $A \in \operatorname{Mat}(n, n, \mathbb{C})$, then $|\operatorname{tr} A| \leq \sum_{j=1}^{n} \sigma_j(A)$. Consequently, $\sum_{j=1}^{n} |\lambda_j(A)| \leq \sum_{j=1}^{n} \sigma_j(A)$ and $\sum_{j=1}^{n} |a_{jj}| \leq \sum_{j=1}^{n} \sigma_j(A)$.

Proof. Let $A = V\Sigma W^*$ be an SVD.
Writing $\mathbf{v}_k, \mathbf{w}_k$ for the columns of V, W, $\operatorname{tr} A = \sum_{k=1}^{n} \sigma_k \mathbf{w}_k^* \mathbf{v}_k$, $|\mathbf{w}_k^* \mathbf{v}_k| \leq 1$, so $|\operatorname{tr} A| \leq \sum_k \sigma_k$.
For eigenvalues, take a Schur form T of A and a diagonal unitary D with $\operatorname{tr}(DT) = \sum_j |\lambda_j|$.

For diagonal entries, take diagonal unitary D with $\operatorname{tr}(DA) = \sum_j |a_{jj}|$.

In both cases, unitary invariance of singular values and the trace bound give the result.
Structural Commentary: Phase rotations turn trace estimates into eigenvalue and diagonal estimates.
Key Insight: Prove $|\operatorname{tr} A| \le \sum_j \sigma_j$ once, then rotate phases. □

Proposition 5.1.13. *Rank-One SVD Terms and Truncation Error*
Let $A = X\Sigma_r Y^* = \sum_{j=1}^r \sigma_j \mathbf{x}_j \mathbf{y}_j^*$ be a compact SVD, with $\sigma_1 \ge \cdots \ge \sigma_r > 0$.
For $0 \le k \le r$, set $A_k := \sum_{j=1}^k \sigma_j \mathbf{x}_j \mathbf{y}_j^*$, $A_0 := 0$.
Then the rank-one summands are mutually orthogonal in the Frobenius inner product, and

$$\|A\|_F^2 = \sum_{j=1}^r \sigma_j^2, \qquad \|A - A_k\|_F^2 = \sum_{j=k+1}^r \sigma_j^2, \qquad \|A - A_k\|_2 = \begin{cases} \sigma_{k+1}, & k < r, \\ 0, & k = r. \end{cases}$$

Proof. For $i, j \in \{1, \dots, r\}$, $\langle \sigma_i \mathbf{x}_i \mathbf{y}_i^*, \sigma_j \mathbf{x}_j \mathbf{y}_j^* \rangle_F = \sigma_i \sigma_j \operatorname{tr}(\mathbf{y}_i \mathbf{x}_i^* \mathbf{x}_j \mathbf{y}_j^*) = \sigma_i \sigma_j \delta_{ij}$.
Pythagoras in the Frobenius inner product gives the two Frobenius formulas. The residual $A - A_k$ has compact SVD $\sum_{j=k+1}^r \sigma_j \mathbf{x}_j \mathbf{y}_j^*$, so Theorem 5.1.2 gives the spectral-norm formula.
Structural Commentary: Compact SVD decomposes a matrix into orthogonal rank-one energy packets.
Key Insight: Truncation error is exactly the energy left in the discarded singular directions. □

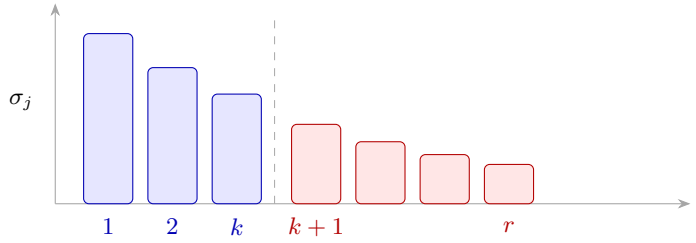


Figure 90: Blue bars form A_k; red bars are discarded, with $\|A - A_k\|_F^2 = \sum_{j>k} \sigma_j^2$.

Theorem 5.1.14. *Eckart–Young Low-Rank Approximation*
Let $A \in \operatorname{Mat}(m, n, \mathbb{F})$ have singular values $\sigma_1 \ge \cdots \ge \sigma_q \ge 0$, where $q = \min(m, n)$ and $\sigma_j = 0$ for $j > q$. If A_k is the rank-k truncated SVD, $0 \le k < q$, and $B \in \operatorname{Mat}(m, n, \mathbb{F})$ has $\operatorname{rank}(B) \le k$, then

$$\|A - A_k\|_2 \le \|A - B\|_2, \qquad \|A - A_k\|_F \le \|A - B\|_F.$$

Proof. Let $\mathbf{w}_1, \dots, \mathbf{w}_n$ be right singular vectors for A.

i. Since $\operatorname{rank}(B) \le k$, rank-nullity gives $\dim \ker(B) \ge n - k$. The subspaces $\ker(B)$ and $\operatorname{span}(\mathbf{w}_1, \dots, \mathbf{w}_{k+1})$ in $\mathbb{F}^n$ have dimensions summing at least $n + 1$, so their intersection contains a nonzero vector.
Choose unit $\mathbf{x}$ in this intersection. Let $\mathbf{x} = \sum_{i=1}^{k+1} c_i \mathbf{w}_i$ gives $\sum_i |c_i|^2 = 1$ and $\|A\mathbf{x}\|_2^2 = \sum_{i=1}^{k+1} \sigma_i^2 |c_i|^2 \ge \sigma_{k+1}^2$.
Thus $\|A - B\|_2 \ge \|(A - B)\mathbf{x}\|_2 = \|A\mathbf{x}\|_2 \ge \sigma_{k+1} = \|A - A_k\|_2$.

ii. Choose an $(n-k)$-dimensional subspace $\mathcal{U} \subseteq \ker(B)$, with projection P and orthonormal basis $\mathbf{z}_1, \dots, \mathbf{z}_{n-k}$.
Since $B\mathbf{z}_j = 0$, $\|A - B\|_F^2 \ge \sum_{j=1}^{n-k} \|(A - B)\mathbf{z}_j\|_2^2 = \sum_{j=1}^{n-k} \|A\mathbf{z}_j\|_2^2 = \sum_{i=1}^{n} \sigma_i^2 \|P\mathbf{w}_i\|_2^2$.
Put $\alpha_i = \|P\mathbf{w}_i\|_2^2$. Then $0 \le \alpha_i \le 1$ and $\sum_i \alpha_i = \operatorname{tr}(P) = n - k$.
Among all such weights, the sum $\sum_i \sigma_i^2 \alpha_i$ is minimized by placing weight 1 on the $n - k$ smallest values of the decreasing sequence σ_i^2, hence $\sum_i \sigma_i^2 \alpha_i \ge \sum_{i=k+1}^{n} \sigma_i^2 = \|A - A_k\|_F^2$.

Structural Commentary: Rank-k competitors vanish on at least $n - k$ directions.
Key Insight: The optimal loss is precisely the singular-value tail. □

Method 5.1.15. *Truncated SVD Approximation Method*
To compress or approximate A by rank k:

(i) compute or estimate the leading singular triples $(\sigma_j, \mathbf{x}_j, \mathbf{y}_j)$;
(ii) keep $A_k = \sum_{j=1}^k \sigma_j \mathbf{x}_j \mathbf{y}_j^*$;
(iii) use $\|A - A_k\|_F^2 = \sum_{j>k} \sigma_j^2$ and $\|A - A_k\|_2 = \sigma_{k+1}$ to quantify the lost information.

Definition 5.1.16. *Moore–Penrose Pseudoinverse*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. If $A = 0$, define $A^\dagger = 0_{n,m}$.
If $\mathrm{rank}(A) = r \geq 1$ and $A = X\Sigma_r Y^*$ is a compact SVD, define $A^\dagger := Y\Sigma_r^{-1}X^* \in \mathrm{Mat}(n,m,\mathbb{F})$.
This is the *pseudoinverse* or *Moore–Penrose pseudoinverse* of A.

Proposition 5.1.17. *Pseudoinverse is Well Defined*
The definition of $A^\dagger$ does not depend on the chosen compact SVD of A.

Proof. Let $A = X_1\Sigma_r Y_1^* = X_2\Sigma_r Y_2^*$. By Theorem 4.2.16, $X_1 = X_2U$, $Y_1 = Y_2U$, and $U\Sigma_r^{-1}U^* = \Sigma_r^{-1}$.
Hence $Y_1\Sigma_r^{-1}X_1^* = Y_2U\Sigma_r^{-1}U^*X_2^* = Y_2\Sigma_r^{-1}X_2^*$.
Structural Commentary: Compact SVD ambiguity cancels in $Y\Sigma_r^{-1}X^*$.
Key Insight: Inverting the singular values preserves the same block symmetries. □

Corollary 5.1.18. *Pseudoinverse from a Full SVD*
Let $A = V\Sigma W^*$ be a full SVD of $A \in \mathrm{Mat}(m,n,\mathbb{F})$, with rank r and positive singular values $\sigma_1,\dots,\sigma_r$.
Write $V = [X\ X_0]$, $W = [Y\ Y_0]$, and $\Sigma_r = \mathrm{diag}(\sigma_1,\dots,\sigma_r)$.
In conformal blocks, $\Sigma = \begin{bmatrix}\Sigma_r & 0\\ 0 & 0\end{bmatrix}$, $\Sigma^\dagger := \begin{bmatrix}\Sigma_r^{-1} & 0\\ 0 & 0\end{bmatrix} \in \mathrm{Mat}(n,m,\mathbb{F})$. Then $A^\dagger = W\Sigma^\dagger V^*$.

Proof. If $r = 0$, then $A = 0$ and $W\Sigma^\dagger V^* = 0$.
If $r > 0$, the block form gives $A = X\Sigma_r Y^*$; $A^\dagger = Y\Sigma_r^{-1}X^*$; and $W\Sigma^\dagger V^* = Y\Sigma_r^{-1}X^*$. Thus $A^\dagger = W\Sigma^\dagger V^*$.
Structural Commentary: Full SVD pads compact SVD by zero blocks.
Key Insight: The pseudoinverse flips V and W and inverts only the positive singular values. □

Theorem 5.1.19. *Basic Properties of the Pseudoinverse*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$. Then:

(i) $\mathrm{col}(A^\dagger) = \mathrm{col}(A^*)$ *and* $\ker(A^\dagger) = \ker(A^*)$;
(ii) $AA^\dagger$ *is the orthogonal projection of* $\mathbb{F}^m$ *onto* $\mathrm{col}(A)$;
(iii) $A^\dagger A$ *is the orthogonal projection of* $\mathbb{F}^n$ *onto* $\mathrm{col}(A^*)$;
(iv) $AA^\dagger A = A$ *and* $A^\dagger AA^\dagger = A^\dagger$;
(v) $(A^\dagger)^\dagger = A$;
(vi) if $m = n$ *and* A *is invertible, then* $A^\dagger = A^{-1}$.

Proof. The zero case is immediate. Let $A = X\Sigma_r Y^*$ be a compact SVD; then $A^\dagger = Y\Sigma_r^{-1}X^*$.

(i) $\mathrm{col}(A^\dagger) = \mathrm{col}(Y) = \mathrm{col}(A^*)$ and $\ker(A^\dagger) = \ker(X^*) = \ker(A^*)$.
(ii) $AA^\dagger = XX^*$, the orthogonal projection onto $\mathrm{col}(X) = \mathrm{col}(A)$.
(iii) $A^\dagger A = YY^*$, the orthogonal projection onto $\mathrm{col}(Y) = \mathrm{col}(A^*)$.
(iv) These follow from $XX^*X = X$ and $YY^*Y = Y$.
(v) The compact SVD of $A^\dagger$ is $Y\Sigma_r^{-1}X^*$, so $(A^\dagger)^\dagger = A$.
(vi) If A is square and invertible, then $AA^\dagger = A^\dagger A = I_n$, hence $A^\dagger = A^{-1}$.

Structural Commentary: The pseudoinverse inverts nonzero singular directions and projects away null directions.
Key Insight: Replace each $\sigma_j > 0$ by σ_j^{-1} on the same singular subspaces. □

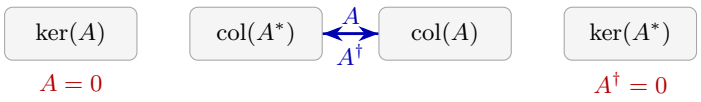


Figure 91: A and $A^\dagger$ invert the nonzero singular subspaces; they vanish on $\ker(A)$ and $\ker(A^*)$, respectively.

Theorem 5.1.20. *Penrose Characterization of the Pseudoinverse*
Let $A \in \mathrm{Mat}(m,n,\mathbb{F})$, $\mathcal{C} = \mathrm{col}(A)$, and $\mathcal{D} = \mathrm{col}(A^*) = (\ker A)^\perp$. Then $A^\dagger$ is unique $B \in \mathrm{Mat}(n,m,\mathbb{F})$ satisfying

(i) $ABA = A$;
(ii) $BAB = B$;
(iii) AB *is Hermitian;*
(iv) BA *is Hermitian.*

Proof.

(i) Theorem 5.1.19 gives $AA^\dagger A = A$; $A^\dagger AA^\dagger = A^\dagger$, and $AA^\dagger$, $A^\dagger A$ are orthogonal projections.
Hence $A^\dagger$ satisfies the four equations.

(ii) Let B satisfy the four equations. Then $(AB)^2 = ABAB = AB$; $(AB)^* = AB$.
Also $\text{range}(AB) = \mathcal{C}$, since $\text{range}(AB) \subseteq \mathcal{C}$ and $AB(A\mathbf{x}) = ABA\mathbf{x} = A\mathbf{x}$. By Theorem 2.4.19, $AB = P_{\mathcal{C}}$.
(iii) $(BA)^2 = BABA = BA$; $(BA)^* = BA$. Also $\ker(BA) = \ker(A)$, as $BA\mathbf{x} = 0 \Rightarrow A\mathbf{x} = ABA\mathbf{x} = 0$ and the reverse inclusion is immediate. By Theorem 2.4.19, $BA = P_{\mathcal{D}}$.
(iv) For $\mathbf{y} \in \mathbb{F}^m$, write $\mathbf{y} = \mathbf{p} + \mathbf{z}$ with $\mathbf{p} \in \mathcal{C}$ and $\mathbf{z} \in \mathcal{C}^\perp = \ker(A^*)$. Since $AB\mathbf{z} = 0$, $B\mathbf{z} = BAB\mathbf{z} = 0$.
If $\mathbf{p} = A\mathbf{x}$, then $B\mathbf{p} = BA\mathbf{x} = P_{\mathcal{D}}\mathbf{x}$, independent of the choice of $\mathbf{x}$ because $\ker(P_{\mathcal{D}}) = \mathcal{D}^\perp = \ker(A)$.
Thus the four equations determine B uniquely. Since $A^\dagger$ satisfies them, $B = A^\dagger$.

Structural Commentary: Penrose equations force the two projection factors.
Key Insight: Once AB and BA are projections, B is forced. □

Theorem 5.1.21. *Pseudoinverse and Minimum-Norm Least Squares*
Let $A \in \text{Mat}(m, n, \mathbb{F})$, let $\mathbf{y} \in \mathbb{F}^m$, and set $\mathbf{x}_0 = A^\dagger\mathbf{y}$.

(i) If $A\mathbf{x} = \mathbf{y}$ is consistent, then $A\mathbf{x}_0 = \mathbf{y}$ and $\mathbf{x}_0$ is the unique solution of smallest Euclidean norm.
(ii) For arbitrary $\mathbf{y}$, $\mathbf{x}_0$ is a least-squares solution of $A\mathbf{x} \approx \mathbf{y}$, and it is the unique least-squares solution of smallest Euclidean norm.

Proof. Let $P = AA^\dagger = P_{\text{col}(A)}$ and note that $\mathbf{x}_0 \in \text{col}(A^\dagger) = \text{col}(A^*) = (\ker A)^\perp$.

(i) If $\mathbf{y} \in \text{col}(A)$, then $A\mathbf{x}_0 = P\mathbf{y} = \mathbf{y}$. For any solution $\mathbf{x}$, $\mathbf{x} - \mathbf{x}_0 \in \ker(A)$,
so $\|\mathbf{x}\|_2^2 = \|\mathbf{x}_0 + (\mathbf{x} - \mathbf{x}_0)\|_2^2 = \|\mathbf{x}_0\|_2^2 + \|\mathbf{x} - \mathbf{x}_0\|_2^2$ with equality only when $\mathbf{x} = \mathbf{x}_0$.
(ii) For arbitrary $\mathbf{y}$, $A\mathbf{x}_0 = P\mathbf{y}$, so $\mathbf{y} - A\mathbf{x}_0 \perp \text{col}(A)$ and $\mathbf{x}_0$ is least-squares optimal by Theorem 2.3.4. Any least-squares solution $\mathbf{x}$ satisfies $A\mathbf{x} = P\mathbf{y} = A\mathbf{x}_0$, hence the same orthogonal decomposition gives $\|\mathbf{x}\|_2 \geq \|\mathbf{x}_0\|_2$, with equality only for $\mathbf{x} = \mathbf{x}_0$.

Structural Commentary: Least squares projects onto $\text{col}(A)$, then removes the $\ker(A)$ component.
Key Insight: $A^\dagger\mathbf{y}$ is the fitted solution orthogonal to $\ker(A)$. □

Method 5.1.22. *Pseudoinverse Solution Method*
For a system $A\mathbf{x} \approx \mathbf{y}$:

(i) compute a compact SVD $A = X\Sigma_r Y^*$;
(ii) form $A^\dagger = Y\Sigma_r^{-1}X^*$;
(iii) set $\mathbf{x}_0 = A^\dagger\mathbf{y}$;
(iv) interpret $A\mathbf{x}_0 = P_{\text{col}(A)}\mathbf{y}$ and $\mathbf{x}_0 \in (\ker A)^\perp$.

5.2 Conditioning

Definition 5.2.1. *Spectral Condition Number*
If $A \in \text{Mat}(n, n, \mathbb{F})$ is invertible, its *spectral condition number* is $\kappa_2(A) := \|A\|_2 \left\|A^{-1}\right\|_2$.
Equivalently, $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$. If A is singular, $\kappa_2(A)$ is undefined.

Proposition 5.2.2. *Properties of the Spectral Condition Number*
Let $A, B \in \text{Mat}(n, n, \mathbb{F})$ be invertible.

(i) $\kappa_2(AB) \leq \kappa_2(A)\kappa_2(B)$;
(ii) $\kappa_2(A) \geq 1$, with equality if and only if A is a nonzero scalar multiple of a unitary matrix;
(iii) $\kappa_2(cA) = \kappa_2(A)$ for every nonzero scalar c;
(iv) $\kappa_2(A) = \kappa_2(A^*) = \kappa_2(\overline{A}) = \kappa_2(A^t) = \kappa_2(A^{-1})$ over $\mathbb{C}$, with the evident real special cases;
(v) $\kappa_2(A^*A) = \kappa_2(A)^2$;
(vi) if U, V are unitary, then $\kappa_2(UAV) = \kappa_2(A)$.

Proof.

(i) Use submultiplicativity: $\kappa_2(AB) = \|AB\|_2 \left\|B^{-1}A^{-1}\right\|_2 \leq \|A\|_2 \|B\|_2 \left\|B^{-1}\right\|_2 \left\|A^{-1}\right\|_2$.
(ii) Since $I_n = AA^{-1}$, $1 = \|I_n\|_2 \leq \kappa_2(A)$. Equality means $\sigma_{\max}(A) = \sigma_{\min}(A)$, equivalently every singular value is the same positive number s. If $A = V\Sigma W^*$, then $\Sigma = sI_n$, so $A = sVW^*$ is a nonzero scalar multiple of a unitary matrix. The converse is immediate.
(iii) This follows from $\|cA\|_2 = |c| \|A\|_2$ and $(cA)^{-1} = c^{-1}A^{-1}$.
(iv) These matrices have the same singular values as A, except A^{-1} has singular values $\sigma_n^{-1} \geq \cdots \geq \sigma_1^{-1}$.
(v) The singular values of A^*A are $\sigma_1(A)^2, \ldots, \sigma_n(A)^2$.
(vi) This follows from Proposition 5.1.7.

Structural Commentary: Condition number compares largest and smallest singular stretches.
Key Insight: A well-conditioned matrix stretches every direction by nearly the same factor. □

Corollary 5.2.3. *Positive Definite Condition Number by Eigenvalues*
Let $A \succ 0$ have eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_n$. Then $\|A\|_2 = \lambda_n$; $\left\|A^{-1}\right\|_2 = \frac{1}{\lambda_1}$; and $\kappa_2(A) = \frac{\lambda_n}{\lambda_1}$.

Proof. By Theorems 2.1.9 and 5.1.11, the singular values of A are $\lambda_1, \dots, \lambda_n$.
Hence Theorem 5.1.2 gives $\|A\|_2 = \lambda_n$. Since $\operatorname{spec}(A^{-1}) = \{\lambda_1^{-1}, \dots, \lambda_n^{-1}\}$, same argument gives $\left\|A^{-1}\right\|_2 = \lambda_1^{-1}$, and multiplying yields $\kappa_2(A) = \lambda_n/\lambda_1$.
Structural Commentary: For $A \succ 0$, singular and eigenvalue stretches coincide.
Key Insight: Conditioning compares the largest and smallest positive eigendirections. □

Perspective 5.2.4. *Determinant is Not Conditioning*
Determinants measure volume, not sensitivity.
$\kappa_2(cA) = \kappa_2(A)$ for $c \neq 0$ (Proposition 5.2.2), but $\det(cA) = c^n \det(A)$.

Theorem 5.2.5. *Condition Number and Linear-System Error*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible, let $\mathbf{y} \neq \mathbf{0}$, and let $A\mathbf{x} = \mathbf{y}$.
If $A(\Delta\mathbf{x}) = \Delta\mathbf{y}$, then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta\mathbf{y}\|_2}{\|\mathbf{y}\|_2}$

Proof. From $\mathbf{y} = A\mathbf{x}$, we have $\|\mathbf{y}\|_2 \le \|A\|_2 \|\mathbf{x}\|_2$, hence $1/\|\mathbf{x}\|_2 \le \|A\|_2 / \|\mathbf{y}\|_2$. Also $\Delta\mathbf{x} = A^{-1}\Delta\mathbf{y}$, so $\|\Delta\mathbf{x}\|_2 \le \left\|A^{-1}\right\|_2 \|\Delta\mathbf{y}\|_2$. Multiplying the two estimates gives the displayed bound.
Structural Commentary: Solution error is controlled by the ratio of extreme singular stretches.
Key Insight: $\kappa_2(A)$ measures worst-case error magnification. □

Theorem 5.2.6. *Condition Number and Matrix Perturbation*
Let $A \in \operatorname{Mat}(n, n, \mathbb{F})$ be invertible. Suppose $A\mathbf{x} = \mathbf{y}$; $(A + \Delta A)(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{y}$, and $\mathbf{x} + \Delta\mathbf{x} \neq \mathbf{0}$.
Then $\frac{\|\Delta\mathbf{x}\|_2}{\|\mathbf{x}+\Delta\mathbf{x}\|_2} \le \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2}$.

Proof. Subtracting gives $A\Delta\mathbf{x} + \Delta A(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{0}$, so $\Delta\mathbf{x} = -A^{-1}\Delta A(\mathbf{x} + \Delta\mathbf{x})$.
Thus $\|\Delta\mathbf{x}\|_2 \le \left\|A^{-1}\right\|_2 \|\Delta A\|_2 \|\mathbf{x} + \Delta\mathbf{x}\|_2 = \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2} \|\mathbf{x} + \Delta\mathbf{x}\|_2$. Divide by $\|\mathbf{x} + \Delta\mathbf{x}\|_2 \neq 0$.
Structural Commentary: Perturbing the matrix changes the operator applied to the perturbed solution.
Key Insight: The same condition number controls matrix errors and right-hand-side errors. □

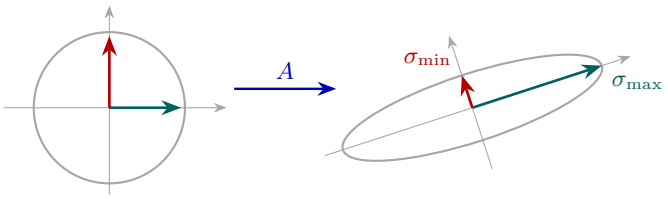


Figure 92: A maps a unit error ball to an ellipse; $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$ is the long/short semiaxis ratio.

5.3 Perturbation Bounds

Perturbation theory studies how matrix information changes under small errors. Bauer–Fike, Gersgorin, interlacing, Weyl inequalities, and pseudospectra measure robustness.

Lemma 5.3.1. *Small Spectral-Norm Perturbations of the Identity*
If $E \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\|E\|_2 < 1$, then $I_n + E$ is invertible.

Proof. Theorem 5.1.11 gives $\rho(E) \le \|E\|_2 < 1$, so $-1 \notin \operatorname{spec}(E)$.
Hence $0 \notin \operatorname{spec}(I_n + E)$, and $I_n + E$ is invertible.
Structural Commentary: Spectral-norm smallness traps every eigenvalue inside the unit disk.
Key Insight: If -1 is not an eigenvalue of E, then $I + E$ has no kernel. □

Theorem 5.3.2. *Bauer–Fike Eigenvalue Perturbation Bound*
Let $A \in \operatorname{Mat}(n, n, \mathbb{C})$ be diagonalizable, with $A = S\Lambda S^{-1}$ and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$.
If $\Delta A \in \operatorname{Mat}(n, n, \mathbb{C})$ and $\mu \in \operatorname{spec}(A + \Delta A)$, then $\min_{1 \le j \le n} |\mu - \lambda_j| \le \kappa_2(S) \|\Delta A\|_2$.

Proof. If $\mu = \lambda_j$ for some j, the claim is immediate. Otherwise $\Lambda - \mu I_n$ is invertible.
Since μ is an eigenvalue of $A + \Delta A$, the matrix $A + \Delta A - \mu I_n$ is singular.
Conjugating gives $S^{-1}(A + \Delta A - \mu I_n)S = (\Lambda - \mu I_n)\left(I_n + (\Lambda - \mu I_n)^{-1} S^{-1} \Delta A S\right)$ so the second factor is singular.
Lemma 5.3.1 implies $1 \le \left\|(\Lambda - \mu I_n)^{-1} S^{-1} \Delta A S\right\|_2 \le \frac{\left\|S^{-1}\right\|_2 \|\Delta A\|_2 \|S\|_2}{\min_j |\mu - \lambda_j|}$. Rearranging gives the bound.
Structural Commentary: Diagonalize, then test how perturbation breaks shifted invertibility.
Key Insight: Eigenvalue sensitivity is controlled by how ill-conditioned the eigenvector matrix S is. □

Method 5.3.3. *Bauer–Fike Perturbation Method*
To bound eigenvalue movement for a diagonalizable matrix:

(i) diagonalize $A = S\Lambda S^{-1}$;
(ii) estimate $\|\Delta A\|_2$;
(iii) compute or bound $\kappa_2(S)$;
(iv) place every eigenvalue of $A + \Delta A$ inside the union of disks $|z - \lambda_j| \le \kappa_2(S)\,\|\Delta A\|_2$.

Perspective 5.3.4. *Pseudospectral Intuition*
For a matrix A, the ε-pseudospectrum records complex numbers z for which $A - zI$ is singular or nearly singular. Equivalently, it asks where a perturbation of size at most ε can move the spectrum. Bauer–Fike gives a clean enclosure for diagonalizable matrices, while Gersgorin and norm bounds give cruder but often cheaper enclosures. The main lesson is that eigenvalues of nonnormal matrices can be much more sensitive than their locations alone suggest.

Definition 5.3.5. *Gersgorin Disks*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. The *deleted absolute row sum* in row k is $R_k^{\mathrm{off}}(A) := \sum_{j \neq k} |a_{kj}|$.

The *kth Gersgorin disk* of A is $\mathrm{Ger}_k(A) := \{z \in \mathbb{C} : |z - a_{kk}| \le R_k^{\mathrm{off}}(A)\}$, where $\mathrm{Ger}(A) := \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.

Theorem 5.3.6. *Gersgorin Disk Theorem*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) = \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.

Proof. Let $\lambda \in \mathrm{spec}(A)$ and choose an eigenvector $\mathbf{x} = [x_1 \ \cdots \ x_n]^t$.
Pick k such that $|x_k| = \max_j |x_j|$; then $x_k \neq 0$. The kth row of $A\mathbf{x} = \lambda\mathbf{x}$ gives $(\lambda - a_{kk})x_k = \sum_{j \neq k} a_{kj} x_j$.

Divide by x_k and use $|x_j/x_k| \le 1$: $|\lambda - a_{kk}| = \left|\sum_{j \neq k} a_{kj} \frac{x_j}{x_k}\right| \le \sum_{j \neq k} |a_{kj}| \left|\frac{x_j}{x_k}\right| \le R_k^{\mathrm{off}}(A)$.

Thus $\lambda \in \mathrm{Ger}_k(A) \subseteq \mathrm{Ger}(A)$.
Structural Commentary: Normalize by a largest component; all row ratios have modulus at most one.
Key Insight: Row k traps λ in the disk centered at a_{kk} with radius $R_k^{\mathrm{off}}(A)$. □

Method 5.3.7. *Gersgorin Localisation Method*
For $A = [a_{ij}]$, Gersgorin localisation gives a disk enclosure for the spectrum.

(i) For each row k, place a disk at the diagonal entry a_{kk} with radius $R_k^{\mathrm{off}}(A) = \sum_{j \neq k} |a_{kj}|$.
The row enclosure is $\mathrm{Ger}(A) = \bigcup_{k=1}^{n} \mathrm{Ger}_k(A)$.
(ii) Theorem 5.3.6 guarantees that every eigenvalue of A lies in $\mathrm{Ger}(A)$.
(iii) For a sharper enclosure, repeat the same construction for A^t. The centers are still a_{kk}, while the radii becomes deleted column sums of A. Then use $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$. This is the intersection of two unions of disks; an eigenvalue need not lie in the overlap of each individual row-column disk pair.

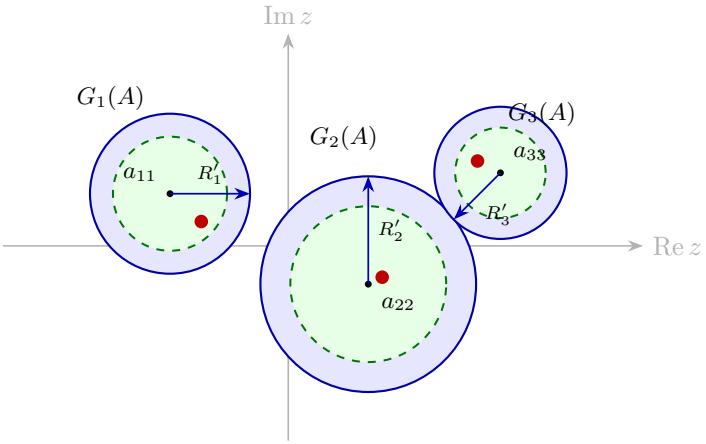


Figure 93: Blue row disks contain the spectrum; green dashed column disks give the sharper row-column enclosure $\mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$. Red points illustrate possible eigenvalue locations.

Corollary 5.3.8. *Row-Column Gersgorin Refinement*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$. *Then* $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A) \cap \mathrm{Ger}(A^t)$.
If $R_k(A) = \sum_{j=1}^{n} |a_{kj}|$ *and* $R_{\max}(A) = \max_{1\le k\le n} R_k(A)$, *then* $\mathrm{spec}(A) \subseteq \{z \in \mathbb{C} : |z| \le \min(R_{\max}(A), R_{\max}(A^t))\}$.

Proof. Theorem 5.3.6 gives $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A)$.
Proposition 1.2.13 gives $\mathrm{spec}(A) = \mathrm{spec}(A^t)$, so applying Gersgorin to A^t gives $\mathrm{spec}(A) \subseteq \mathrm{Ger}(A^t)$.
Also $\mathrm{Ger}_k(A) \subseteq \{z : |z| \le |a_{kk}| + R_k^{\mathrm{off}}(A) = R_k(A)\}$, and similarly for A^t.
Structural Commentary: Row disks and column disks are both valid because A and A^t have the same spectrum.
Key Insight: Intersecting independent enclosures usually sharpens the eigenvalue search region. □

5.4 Stability

Definition 5.4.1. *Diagonal Dominance*
Let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{C})$. Matrix A is *diagonally dominant* if $|a_{kk}| \ge R_k^{\mathrm{off}}(A)$ for every k, and *strictly diagonally dominant* if $|a_{kk}| > R_k^{\mathrm{off}}(A)$ for every k.

Corollary 5.4.2. *Strict Diagonal Dominance Implies Invertibility*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$.

(i) If A is strictly diagonally dominant, then A is invertible.
(ii) If A has real spectrum and nonnegative diagonal entries, then diagonal dominance gives $\mathrm{spec}(A) \subseteq [0,\infty)$*; strict diagonal dominance gives* $\mathrm{spec}(A) \subseteq (0,\infty)$.

Proof.

(i) Strict dominance gives $0 \notin \mathrm{Ger}_k(A)$ for every k, since $|a_{kk}| > R_k^{\mathrm{off}}(A)$.
Thus $0 \notin \mathrm{spec}(A)$ by Gersgorin, so A is invertible by Corollary 5.1.9.
(ii) If $a_{kk} \ge 0$ and $a_{kk} \ge R_k^{\mathrm{off}}(A)$, then each $\mathrm{Ger}_k(A)$ lies in the closed right half-plane. In the strict case, each disk lies in the open right half-plane. Gersgorin and the real-spectrum hypothesis give the claim.

Structural Commentary: Gersgorin converts diagonal dominance into disk placement.
Key Insight: The disks control whether eigenvalues can reach 0 or the left half-plane. □

Corollary 5.4.3. *Diagonal Dominance and Positive Semidefiniteness*
Let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{C})$ *be Hermitian with nonnegative diagonal entries.*

(i) If A is diagonally dominant, then A is positive semidefinite.
(ii) If A is diagonally dominant and invertible, then A is positive definite.
(iii) If A is strictly diagonally dominant, then A is positive definite.

Proof. Hermitian matrices have real spectrum.

(i) Corollary 5.4.2 gives $\mathrm{spec}(A) \subseteq [0,\infty)$; hence $A \succeq 0$ by Theorem 2.1.7.
(ii) If A is invertible, then $0 \notin \mathrm{spec}(A)$, so every eigenvalue is positive; hence $A \succ 0$.
(iii) Strict diagonal dominance implies invertibility by Corollary 5.4.2; apply (ii).

Structural Commentary: Gersgorin traps the real spectrum in $[0,\infty)$; invertibility removes 0.
Key Insight: For Hermitian matrices, positivity is spectral location. □

Theorem 5.4.4. *Weak Diagonal Dominance Criterion*
Let $A = [a_{ij}] \in \mathrm{Mat}(n,n,\mathbb{C})$ *be diagonally dominant with no zero entries.*
If $|a_{kk}| > R_k^{\mathrm{off}}(A)$ *for some k, then A is invertible.*

Proof. Assume $A\mathbf{x} = 0$ with $\mathbf{x} \ne \mathbf{0}$, and choose k with $|x_k| = \|\mathbf{x}\|_\infty$.
Row k gives $|a_{kk}|\,\|\mathbf{x}\|_\infty = \left|-\sum_{j\ne k} a_{kj}x_j\right| \le \sum_{j\ne k} |a_{kj}|\,|x_j| \le R_k^{\mathrm{off}}(A)\,\|\mathbf{x}\|_\infty \le |a_{kk}|\,\|\mathbf{x}\|_\infty$.
Hence all inequalities are equalities. Since no a_{kj} vanishes, $|x_j| = \|\mathbf{x}\|_\infty$ for every j. Applying the same chain to any row i gives $|a_{ii}| = R_i'(A)$ for every i, contradicting the strict row. Thus $\ker(A) = \{\mathbf{0}\}$, so A is invertible.
Structural Commentary: A zero vector in the kernel would force equality in every Gersgorin row.
Key Insight: One strict row breaks that equality pattern. □

Definition 5.4.5. *Increasing Hermitian Eigenvalue Order*
For a Hermitian matrix $A \in \mathrm{Mat}(n,n,\mathbb{C})$, write $\lambda_1(A) \le \lambda_2(A) \le \cdots \le \lambda_n(A)$ for its eigenvalues counted with algebraic multiplicity. If only one Hermitian matrix is under discussion, we write λ_i instead of $\lambda_i(A)$.

Theorem 5.4.6. *Rayleigh Quotient on a Spectral Interval*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian with increasing eigenvalues $\lambda_1 \le \cdots \le \lambda_n$ and corresponding orthonormal eigenvectors $\mathbf{u}_1,\dots,\mathbf{u}_n$. For a unit vector, write $\mathrm{Ray}_A(\mathbf{x}) := \mathbf{x}^*A\mathbf{x}$. If $1 \le p \le q \le n$ and $\mathbf{x} \in \mathrm{span}(\mathbf{u}_p,\dots,\mathbf{u}_q)$ is a unit vector, then $\lambda_p \le \mathrm{Ray}_A(\mathbf{x}) \le \lambda_q$.

Proof. Write $\mathbf{x} = \sum_{i=p}^{q} c_i\mathbf{u}_i$. Orthonormality gives $\sum_{i=p}^{q} |c_i|^2 = 1$, and $\mathbf{x}^*A\mathbf{x} = \sum_{i=p}^{q} \lambda_i |c_i|^2$.
This is a convex combination of $\lambda_p,\dots,\lambda_q$, so it lies between the smallest and largest of them.
Structural Commentary: In an eigenbasis, the Rayleigh quotient is an eigenvalue average.
Key Insight: Spectral coordinates turn a quadratic form into weighted scalar data. □

Corollary 5.4.7. *Rayleigh Extrema and Diagonal Entry Bounds*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian.

(i) If $\mathcal{U}_{p,q} = \mathrm{span}(\mathbf{u}_p,\dots,\mathbf{u}_q)$, then $\min_{\substack{\mathbf{x}\in\mathcal{U}_{p,q}\\ \|\mathbf{x}\|_2=1}} \mathbf{x}^*A\mathbf{x} = \lambda_p(A)$, and $\max_{\substack{\mathbf{x}\in\mathcal{U}_{p,q}\\ \|\mathbf{x}\|_2=1}} \mathbf{x}^*A\mathbf{x} = \lambda_q(A)$.
(ii) For every unit vector $\mathbf{x}$, $\lambda_1(A) \le \mathbf{x}^*A\mathbf{x} \le \lambda_n(A)$.
(iii) If $A = [a_{ij}]$, then $\lambda_1(A) \le a_{ii} \le \lambda_n(A)$ for $i = 1,\dots,n$.

Proof.
(i) Theorem 5.4.6 gives the inequalities on $\mathcal{U}_{p,q}$, with equality at $\mathbf{u}_p$ and $\mathbf{u}_q$.
(ii) This is (i) with $p = 1$ and $q = n$.
(iii) Apply (ii) to $\mathbf{x} = \mathbf{e}_i$, since $\mathbf{e}_i^*A\mathbf{e}_i = a_{ii}$.
Structural Commentary: Rayleigh extrema on extreme eigendirections; diagonal entries are coordinate quotients.
Key Insight: The whole numerical range of a Hermitian matrix sits inside its spectral interval. □

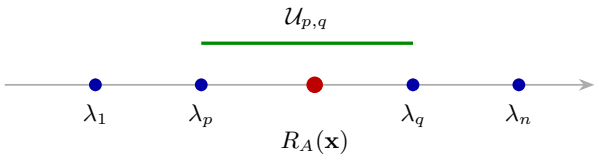


Figure 94: For $\mathbf{x} \in \mathcal{U}_{p,q}$, $\mathrm{Ray}_A(\mathbf{x})$ is a convex combination of $\lambda_p,\dots,\lambda_q$.

Theorem 5.4.8. *Rayleigh Minimax Principle*
Let $A = A^* \in \mathrm{Mat}(n,n,\mathbb{C})$ have eigenvalues $\lambda_1 \le \cdots \le \lambda_n$, and set $R_A(\mathbf{x}) = \mathbf{x}^*A\mathbf{x}/(\mathbf{x}^*\mathbf{x})$ for $\mathbf{x} \neq \mathbf{0}$.
For $1 \le k \le n$, $\lambda_k = \min_{\substack{\mathcal{S}\subseteq\mathbb{C}^n\\ \dim\mathcal{S}=k}} \max_{\mathbf{0}\neq\mathbf{x}\in\mathcal{S}} R_A(\mathbf{x}) = \max_{\substack{\mathcal{S}\subseteq\mathbb{C}^n\\ \dim\mathcal{S}=n-k+1}} \min_{\mathbf{0}\neq\mathbf{x}\in\mathcal{S}} R_A(\mathbf{x})$.

Proof. Let $\mathbf{u}_1,\dots,\mathbf{u}_n$ be an orthonormal eigenbasis in increasing order.
Set $E_k = \mathrm{span}(\mathbf{u}_1,\dots,\mathbf{u}_k)$; $F_k = \mathrm{span}(\mathbf{u}_k,\dots,\mathbf{u}_n)$.
By Theorem 5.4.6, $\max_{\mathbf{0}\neq\mathbf{x}\in E_k} R_A(\mathbf{x}) = \lambda_k$ and $\min_{\mathbf{0}\neq\mathbf{x}\in F_k} R_A(\mathbf{x}) = \lambda_k$.
If $\dim\mathcal{S} = k$, then $\mathcal{S} \cap F_k \neq \{\mathbf{0}\}$, so $\max_{\mathbf{0}\neq\mathbf{x}\in\mathcal{S}} R_A(\mathbf{x}) \ge \lambda_k$; equality is attained by E_k.
If $\dim\mathcal{T} = n-k+1$, then $\mathcal{T} \cap E_k \neq \{\mathbf{0}\}$, so $\min_{\mathbf{0}\neq\mathbf{x}\in\mathcal{T}} R_A(\mathbf{x}) \le \lambda_k$; equality is attained by F_k.
Structural Commentary: Dimension forces intersections with spectral flags.
Key Insight: Minimax is Rayleigh's quotient plus dimension counting. □

Method 5.4.9. *Rayleigh Eigenbasis Method*
To bound a Hermitian quadratic form:

(i) diagonalise A with an orthonormal eigenbasis;
(ii) expand the unit vector in that basis;
(iii) rewrite $\mathbf{x}^*A\mathbf{x}$ as $\sum_i \lambda_i |c_i|^2$;
(iv) use the eigenvalue range allowed by the support of $\mathbf{x}$.

Theorem 5.4.10. *Rank-One Positive Perturbation Interlacing*
Let $A, E \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian, with $E \succeq 0$ and $\mathrm{rank}(E) \le 1$.
Then $\lambda_i(A) \le \lambda_i(A+E) \le \lambda_{i+1}(A)$ $(1 \le i < n)$, $\lambda_n(A) \le \lambda_n(A+E)$
Equivalently, $\lambda_i(A-E) \le \lambda_i(A) \le \lambda_{i+1}(A-E)$ $(1 \le i < n)$, $\lambda_n(A-E) \le \lambda_n(A)$.

Proof. Let $\mathbf{u}_j$ and $\mathbf{v}_j$ be orthonormal eigenvectors of A and $A+E$, respectively, in increasing eigenvalue order. For the lower bound, set $\mathcal{U} = \operatorname{span}(\mathbf{u}_i, \dots, \mathbf{u}_n)$, $\mathcal{V} = \operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_i)$. Since $\dim \mathcal{U} + \dim \mathcal{V} = n+1$, choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V}$. Then $\lambda_i(A) \le \mathbf{x}^* A \mathbf{x} \le \mathbf{x}^*(A+E)\mathbf{x} \le \lambda_i(A+E)$ by Theorem 5.4.6 and $E \succeq 0$.
For the upper bound, set $\mathcal{U} = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$, $\mathcal{V} = \operatorname{span}(\mathbf{v}_i, \dots, \mathbf{v}_n)$, $\mathcal{W} = \ker(E)$. Here $\dim(\mathcal{U} \cap \mathcal{V}) \ge 2$ and $\dim \mathcal{W} \ge n-1$, so choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$. Thus $\lambda_i(A+E) \le \mathbf{x}^*(A+E)\mathbf{x} = \mathbf{x}^* A \mathbf{x} \le \lambda_{i+1}(A)$.
The statements for $A - E$ follow by applying the proved result to $A - E$ and E.
Structural Commentary: Interlacing comes from test vectors in overlapping spectral subspaces.
Key Insight: Rank one leaves a large kernel, so the upper-bound vector can avoid the perturbation. □

Corollary 5.4.11. *Rank-Two Indefinite Interlacing*
Let $A, H \in \operatorname{Mat}(n, n, \mathbb{C})$ be Hermitian, $n \ge 2$. Suppose $\operatorname{rank}(H) \le 2$ and H has at most one positive and one negative eigenvalue. Then

$$\lambda_1(A+H) \le \lambda_2(A), \qquad \lambda_{i-1}(A) \le \lambda_i(A+H) \le \lambda_{i+1}(A) \quad (2 \le i < n), \qquad \lambda_{n-1}(A) \le \lambda_n(A+H).$$

Proof. Write $H = E_+ - E_-$ with $E_\pm \succeq 0$ and $\operatorname{rank}(E_\pm) \le 1$. Set $B = A - E_-$; then $A = B + E_-$ and $A + H = B + E_+$. For $2 \le i < n$, rank-one interlacing gives $\lambda_{i-1}(A) \le \lambda_i(B) \le \lambda_i(A+H) \le \lambda_{i+1}(B) \le \lambda_{i+1}(A)$.
The endpoints follow similarly: $\lambda_1(A+H) \le \lambda_2(B) \le \lambda_2(A)$, and $\lambda_{n-1}(A) \le \lambda_n(B) \le \lambda_n(A+H)$.
Structural Commentary: Such an H is a difference of two rank-one PSD perturbations.
Key Insight: Interlace once with E_-, then once with E_+. □

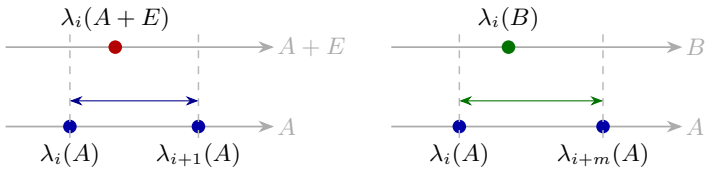


Figure 95: Rank-one and principal-submatrix interlacing place new eigenvalues between old ones.

Theorem 5.4.12. *Cauchy Interlacing: One Principal Deletion*
Let $A = \begin{bmatrix} B & \mathbf{y} \\ \mathbf{y}^* & c \end{bmatrix} \in \operatorname{Mat}(n+1, n+1, \mathbb{C})$ be Hermitian, with $B \in \operatorname{Mat}(n, n, \mathbb{C})$.
Then, for $1 \le i \le n$, $\lambda_i(A) \le \lambda_i(B) \le \lambda_{i+1}(A)$.

Proof. Let $\mathbf{u}_j$ and $\mathbf{v}_j$ be orthonormal eigenvectors of A and B, respectively, in increasing eigenvalue order. Identify $\mathbb{C}^n$ with $\{(\mathbf{z}, 0)\} \subseteq \mathbb{C}^{n+1}$, so $(\mathbf{z}, 0)^* A (\mathbf{z}, 0) = \mathbf{z}^* B \mathbf{z}$.
For the lower bound, set $\mathcal{S} = \operatorname{span}((\mathbf{v}_1, 0), \dots, (\mathbf{v}_i, 0))$ and $\mathcal{T} = \operatorname{span}(\mathbf{u}_i, \dots, \mathbf{u}_{n+1})$. Their dimensions are i and $n+2-i$, so $\dim \mathcal{S} + \dim \mathcal{T} > n+1$ and $\mathcal{S} \cap \mathcal{T}$ contains a nonzero vector.
Choose a unit $\mathbf{x} = (\mathbf{z}, 0)$ in the intersection. The spectral expansion in the $\mathbf{u}_j$ gives $\mathbf{x}^* A \mathbf{x} \ge \lambda_i(A)$, while the expansion in the $\mathbf{v}_j$ gives $\mathbf{z}^* B \mathbf{z} \le \lambda_i(B)$. Hence $\lambda_i(A) \le \lambda_i(B)$.
For the upper bound, take $\mathcal{S}' = \operatorname{span}((\mathbf{v}_i, 0), \dots, (\mathbf{v}_n, 0))$ and $\mathcal{T}' = \operatorname{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+1})$. Again the dimensions sum to $n+2$, so choose a unit $\mathbf{x} = (\mathbf{z}, 0) \in \mathcal{S}' \cap \mathcal{T}'$.
Now $\mathbf{z}^* B \mathbf{z} \ge \lambda_i(B)$ and $\mathbf{x}^* A \mathbf{x} \le \lambda_{i+1}(A)$, giving $\lambda_i(B) \le \lambda_{i+1}(A)$.
Structural Commentary: A principal submatrix is a coordinate slice of the larger quadratic form.
Key Insight: Deleting one coordinate costs one dimension, so eigenvalues shift by at most one index. □

Corollary 5.4.13. *Singular Values Interlace After Deleting a Row or Column*
Let B be obtained from $A \in \operatorname{Mat}(n, n, \mathbb{C})$ by deleting one row or column.
With singular values ordered decreasingly, $\sigma_i(A) \ge \sigma_i(B) \ge \sigma_{i+1}(A)$ $(1 \le i < n)$.

Proof. For a row deletion, permute rows so that $PA = \begin{bmatrix} B \\ \mathbf{y}^* \end{bmatrix}$.
Then PA has the same singular values as A, and BB^* is a principal submatrix of $(PA)(PA)^*$.
Cauchy interlacing applied to these Gram matrices gives the result after taking square roots and reversing the increasing eigenvalue order. A column deletion follows by applying the row case to A^*.
Structural Commentary: Row deletion makes BB^* a principal submatrix of a permuted AA^*.
Key Insight: Singular-value interlacing is Cauchy interlacing applied to Gram matrices. □

Corollary 5.4.14. *Cauchy Interlacing for Principal Submatrices*
Let $B \in \operatorname{Mat}(n, n, \mathbb{C})$ be a principal submatrix of Hermitian $A \in \operatorname{Mat}(n+m, n+m, \mathbb{C})$.
Then, for $1 \le i \le n$, $\lambda_i(A) \le \lambda_i(B) \le \lambda_{i+m}(A)$.

Proof. After a permutation similarity, B is leading.
Delete the other m coordinates one at a time and apply Theorem 5.4.12 at each step.
Structural Commentary: Principal submatrices come from repeated coordinate deletion.
Key Insight: Each deletion shifts eigenvalue indices by at most one. □

Theorem 5.4.15. *Schur Majorization of the Diagonal*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian, with increasing eigenvalues $\lambda_i(A)$ and increasing diagonal entries d_i. Then $\sum_{i=1}^{k} \lambda_i(A) \le \sum_{i=1}^{k} d_i$, $(1 \le k \le n)$ with equality for $k = n$.

Proof. Permute coordinates so that $a_{ii} = d_i$, and let A_k be the leading $k \times k$ principal submatrix.
Cauchy interlacing gives $\lambda_i(A) \le \lambda_i(A_k)$ for $1 \le i \le k$, hence $\sum_{i=1}^{k} \lambda_i(A) \le \sum_{i=1}^{k} \lambda_i(A_k) = \mathrm{tr}(A_k) = \sum_{i=1}^{k} d_i$.
For $k = n$, this is $\mathrm{tr}(A) = \mathrm{tr}(A)$.
Structural Commentary: Ordered diagonal sums are traces of leading principal blocks.
Key Insight: Diagonal majorization is Cauchy interlacing plus trace. □

Method 5.4.16. *Eigenvalue Interlacing Method*
To prove or use an interlacing estimate:

(i) order Hermitian eigenvalues increasingly;
(ii) identify dimension lost: rank-one perturbation, one deleted coordinate, m-coordinate principal submatrix;
(iii) choose test vectors by intersecting spectral subspaces with the unaffected subspace;
(iv) convert the Rayleigh quotient bounds into eigenvalue inequalities.

5.5 Weyl Bounds and Rank-One Spectral Shifts

Theorem 5.5.1. *Weyl Eigenvalue Inequalities*
Let $A, B \in \mathrm{Mat}(n,n,\mathbb{C})$ be Hermitian. For $1 \le i \le n$,

$$\lambda_i(A+B) \le \lambda_{i+j}(A) + \lambda_{n-j}(B), \qquad 0 \le j \le n-i,$$
$$\lambda_{i-j+1}(A) + \lambda_j(B) \le \lambda_i(A+B), \qquad 1 \le j \le i.$$

Proof. Let $\mathbf{u}_k, \mathbf{v}_k, \mathbf{w}_k$ be increasing eigenbases for $A, A+B, B$.
Fix i and $0 \le j \le n-i$, and set $\mathcal{U} = \mathrm{span}(\mathbf{u}_1, \dots, \mathbf{u}_{i+j})$, $\mathcal{V} = \mathrm{span}(\mathbf{v}_i, \dots, \mathbf{v}_n)$, $\mathcal{W} = \mathrm{span}(\mathbf{w}_1, \dots, \mathbf{w}_{n-j})$.
Their dimensions sum to $2n+1$, so choose a unit $\mathbf{x} \in \mathcal{U} \cap \mathcal{V} \cap \mathcal{W}$.
Then $\lambda_i(A+B) \le \mathbf{x}^*(A+B)\mathbf{x} = \mathbf{x}^*A\mathbf{x} + \mathbf{x}^*B\mathbf{x} \le \lambda_{i+j}(A) + \lambda_{n-j}(B)$.
The lower bound follows by applying this upper bound to $-A$ and $-B$ and using $\lambda_k(-C) = -\lambda_{n-k+1}(C)$.
Structural Commentary: Weyl bounds come from intersecting the spectral flags of A, B, and $A+B$.
Key Insight: One shared test vector controls all three Rayleigh quotients. □

Method 5.5.2. *Weyl Bound Method*
To bound eigenvalues of $A+B$:

(i) choose the target index i;
(ii) use the upper Weyl family with $j = 0, \dots, n-i$ to get upper bounds;
(iii) use the lower Weyl family with $j = 1, \dots, i$ to get lower bounds;
(iv) for positive semidefinite or low-rank B, insert the known zero or nonnegative eigenvalues to recover monotonicity or interlacing.

Theorem 5.5.3. *Brauer Rank-One Eigenvalue Shift*
Let $A \in \mathrm{Mat}(n,n,\mathbb{C})$, let $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, and let $\mathbf{y} \in \mathbb{C}^n$. If $\lambda, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A with multiplicity, then the eigenvalues of $A + \mathbf{x}\mathbf{y}^*$ are $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \dots, \lambda_n$.
Moreover, $\mathbf{x}$ is an eigenvector of $A + \mathbf{x}\mathbf{y}^*$ corresponding to $\lambda + \mathbf{y}^*\mathbf{x}$.

Proof. Put $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|_2$. By Schur triangularisation, choose a unitary $U = [\mathbf{u}\ U_2]$ such that $U^*AU = \begin{bmatrix} \lambda & * \\ 0 & T' \end{bmatrix}$, where the diagonal entries of T' are $\lambda_2, \dots, \lambda_n$. Also $U^*(\mathbf{x}\mathbf{y}^*)U = \begin{bmatrix} \mathbf{y}^*\mathbf{x} & * \\ 0 & 0 \end{bmatrix}$. Therefore $U^*(A + \mathbf{x}\mathbf{y}^*)U$ is upper triangular with diagonal entries $\lambda + \mathbf{y}^*\mathbf{x}, \lambda_2, \dots, \lambda_n$. Finally, $(A + \mathbf{x}\mathbf{y}^*)\mathbf{x} = A\mathbf{x} + \mathbf{x}(\mathbf{y}^*\mathbf{x}) = (\lambda + \mathbf{y}^*\mathbf{x})\mathbf{x}$.
Structural Commentary: A rank-one eigenvector update changes only the first Schur diagonal entry.
Key Insight: Align the update with the first coordinate; only one diagonal term shifts. □

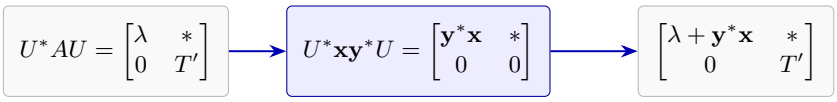


Figure 96: Brauer's update shifts one Schur diagonal entry.

Corollary 5.5.4. *Damped Rank-One Eigenvalue Reset*
Let $A \in \mathrm{Mat}(n, n, \mathbb{C})$ have eigenpair $(\lambda, \mathbf{x})$, let $\mathbf{y}^*\mathbf{x} = 1$, and let $\tau \in \mathbb{C}$. If $\lambda, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A, then $A_\tau := \tau A + (1-\tau)\lambda\mathbf{x}\mathbf{y}^*$ has eigenvalues $\lambda, \tau\lambda_2, \ldots, \tau\lambda_n$.

Proof. The eigenvalues of τA are $\tau\lambda, \tau\lambda_2, \ldots, \tau\lambda_n$. Since $A_\tau = \tau A + \mathbf{x}((1-\overline{\tau})\overline{\lambda}\mathbf{y})^*$, Theorem 5.5.3 shifts only $\tau\lambda$ by $(1-\tau)\lambda\mathbf{y}^*\mathbf{x} = (1-\tau)\lambda$. Thus $\tau\lambda$ becomes λ, while the other eigenvalues stay $\tau\lambda_i$.
Structural Commentary: Damping scales the spectrum; Brauer's update restores one eigenvalue.
Key Insight: Brauer's theorem changes one spectral coordinate without moving the rest. □

6 Numerical Linear Algebra

6.1 Direct Methods

Numerical linear algebra turns matrix structure into algorithms for direct solves, least squares, eigenvalues, SVD, structured transforms, Krylov methods, iteration, and preconditioning.

Perspective 6.1.1. *Solve with Factors Before Forming Inverses*
For $A\mathbf{x} = \mathbf{b}$, use available factors such as LU, QR, or Cholesky to solve triangular or orthogonal subproblems. This avoids forming A^{-1} and preserves sparse or triangular structure.

Perspective 6.1.2. *Iterative Refinement by Residual Correction*
Given an approximate solution $\mathbf{x}_k$ of $A\mathbf{x} = \mathbf{b}$, set $\mathbf{r}_k = \mathbf{b} - A\mathbf{x}_k$.
Solve $A\mathbf{d}_k = \mathbf{r}_k$ with the same factors or preconditioner, then update $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{d}_k$.
Exact arithmetic corrects in one step; floating point helps when $\mathbf{r}_k$ is accurate and conditioning is moderate.

Method 6.1.3. *Augmented Inverse Computation Preview*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$:

(i) Form $[\,A\ I_n\,]$.
(ii) Apply Gauss–Jordan reduction to the left block.
(iii) If $[\,A\ I_n\,] \sim [\,I_n\ B\,]$, then $B = A^{-1}$.
(iv) Otherwise A is not invertible, by Corollary 3.5.5.

Proposition 6.1.4. *QR Route for Least Squares*
Let $A = QR \in \mathrm{Mat}(m, n, \mathbb{F})$ be a full-column-rank narrow QR factorisation. For $\mathbf{y} \in \mathbb{F}^m$, the least-squares solution solves $R\mathbf{x} = Q^*\mathbf{y}$. Moreover $P_{\mathrm{col}(A)} = QQ^*$.

Proof. The normal equations reduce by $A = QR$ and $Q^*Q = I_n$ to $R^*R\mathbf{x} = R^*Q^*\mathbf{y}$. Since R is invertible, this is equivalent to $R\mathbf{x} = Q^*\mathbf{y}$. Also $\mathrm{col}(A) = \mathrm{col}(Q)$, and the columns of Q are orthonormal, so $P_{\mathrm{col}(A)} = QQ^*$.
Structural Commentary: QR replaces $\mathrm{col}(A)$ by an orthonormal basis Q.
Key Insight: Project onto Q, then solve triangularly. □

Perspective 6.1.5. *Why Normal Equations Can Be Numerically Fragile*
For full-column-rank least squares, solving $A^*A\mathbf{x} = A^*\mathbf{y}$ squares the spectral condition number because $\kappa_2(A^*A) = \kappa_2(A)^2$. QR-based methods avoid this squaring and therefore preserve conditioning better.

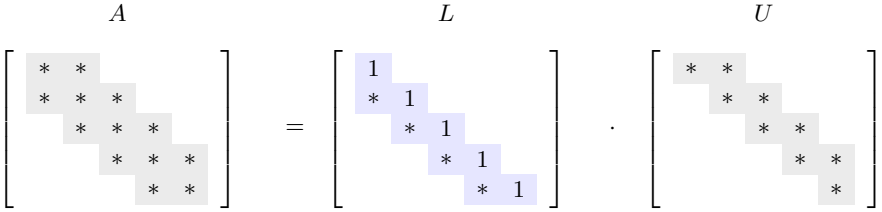


Figure 97: A tridiagonal matrix can factor as bidiagonal L times bidiagonal U, retaining local band structure.

Definition 6.1.6. *Band and Tridiagonal Matrices*
For $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$ and $1 \le w \le n$, say that A has *half-bandwidth* w if $a_{ij} = 0$ whenever $|i - j| \ge w$. Thus $w = 1$ gives a diagonal matrix, while $w = 2$ gives a *tridiagonal* matrix.

Method 6.1.7. *Banded Elimination Preview*
Let $A \in \mathrm{Mat}(n, n, \mathbb{F})$ have half-bandwidth w.

(i) Eliminate without row exchanges that leave the band.
(ii) At pivot k, update only entries with $|i - k| < w$ and $|j - k| < w$.
(iii) Obtain $A = LU$ with L and U of comparable bandwidth.

Cost: $O(nw^2)$ operations; for tridiagonal matrices ($w = 2$), $O(n)$.

Perspective 6.1.8. *Five-Point Stencils Are Banded but Less Local*
The 1D second-difference pattern $-1, 2, -1$ is tridiagonal; the row-ordered $N \times N$ five-point stencil is still sparse, but vertical neighbors create diagonals offset by about N. Thus bandwidth grows with grid width, and incomplete LU keeps a prescribed sparse pattern for a cheap approximate solve or preconditioner.

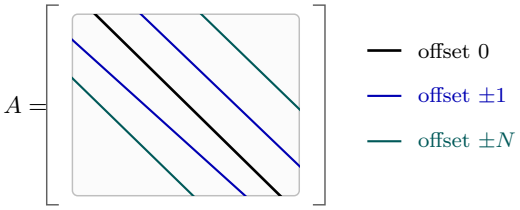


Figure 98: Row ordering gives matrix bands at offsets $0, \pm 1, \pm N$.

6.2 Iterative Methods

Definition 6.2.1. *Krylov Subspace*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$ and a vector $\mathbf{b} \in \mathbb{F}^n$, the mth *Krylov subspace* generated by $(A, \mathbf{b})$ is
$\mathcal{K}_m(A, \mathbf{b}) := \mathrm{span}\{\mathbf{b}, A\mathbf{b}, \ldots, A^{m-1}\mathbf{b}\}$.

Perspective 6.2.2. *Krylov Methods Preview*
Krylov methods work in $\mathcal{K}_m(A, \mathbf{b})$: Arnoldi and Lanczos build bases, while CG and GMRES impose projection or residual-minimization conditions using only matrix-vector products.

Method 6.2.3. *Arnoldi, Lanczos, and CG Preview*
Let $\mathcal{K}_m = \mathcal{K}_m(A, \mathbf{b})$ and let $Q_m = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_m]$ be an orthonormal basis.

(i) *Arnoldi*: $AQ_m = Q_m H_m + h_{m+1,m}\mathbf{q}_{m+1}\mathbf{e}_m^t$, with H_m upper Hessenberg.
(ii) *Lanczos*: if $A = A^*$, then H_m is tridiagonal and $A\mathbf{q}_j = \beta_{j-1}\mathbf{q}_{j-1} + \alpha_j\mathbf{q}_j + \beta_j\mathbf{q}_{j+1}$.
(iii) *CG*: if $A \succ 0$, choose $\mathbf{x}_k \in \mathbf{x}_0 + \mathcal{K}_k(A, \mathbf{r}_0)$ with $\mathbf{r}_k \perp \mathcal{K}_k(A, \mathbf{r}_0)$; search directions satisfy $\mathbf{p}_i^* A\mathbf{p}_j = 0$, $i \ne j$.

Each step uses one matrix-vector product.

Definition 6.2.4. *Matrix Splitting Iteration*
Let $A = S - T$ with S invertible, and set $B := S^{-1}T$.
The *splitting iteration* for $A\mathbf{x} = \mathbf{b}$ is $S\mathbf{x}_{k+1} = T\mathbf{x}_k + \mathbf{b}$, and $\mathbf{x}_{k+1} = B\mathbf{x}_k + S^{-1}\mathbf{b}$.
For a solution $\mathbf{x}_*$, the error $\mathbf{e}_k := \mathbf{x}_k - \mathbf{x}_*$ satisfies $\mathbf{e}_{k+1} = B\mathbf{e}_k$.

Theorem 6.2.5. *Splitting Convergence Criterion*
Let $A = S - T$, with S invertible, and set $B = S^{-1}T$. For $\mathbf{x}_{k+1} = B\mathbf{x}_k + S^{-1}\mathbf{b}$:

(i) if A is invertible, then $\mathbf{x}_k \to A^{-1}\mathbf{b}$ for all $\mathbf{x}_0, \mathbf{b}$ if and only if $\rho(B) < 1$;
(ii) $\rho(B) < 1$ implies A is invertible.

Proof.

(i) Let $\mathbf{x}_* = A^{-1}\mathbf{b}$ and $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_*$. Then $\mathbf{e}_k = B^k\mathbf{e}_0$, so convergence for all $\mathbf{x}_0, \mathbf{b}$ is equivalent to $B^k \to 0$, equivalently $\rho(B) < 1$ by Theorem 3.3.2.
(ii) If $\rho(B) < 1$, then $1 \notin \mathrm{spec}(B)$, so $I_n - B$ is invertible. Since $A = S - T = S(I_n - B)$, A is invertible.

Structural Commentary: A stationary method is a matrix-power problem in disguise.
Key Insight: The iteration matrix $B = S^{-1}T$ controls the errors. □

Corollary 6.2.6. *Singular Matrices Obstruct Splitting Convergence*
Let $A = S - T$, with S invertible. If A is singular, then $1 \in \mathrm{spec}(S^{-1}T)$.
Hence $\rho(S^{-1}T) \ge 1$, so $\rho(S^{-1}T) < 1$ cannot hold.

Proof. Choose $\mathbf{0} \neq \mathbf{x} \in \ker A$. Then $(S-T)\mathbf{x} = \mathbf{0}$, so $S\mathbf{x} = T\mathbf{x}$ and $S^{-1}T\mathbf{x} = \mathbf{x}$. Thus 1 is eigenvalue of $S^{-1}T$.
Structural Commentary: Singular A forces an undamped error mode.
Key Insight: A null vector of A becomes a fixed error mode. □

Method 6.2.7. *Stationary Iterative Method*
For $A\mathbf{x} = \mathbf{b}$, a stationary iteration is:

(i) Choose fixed M and $\mathbf{c}$.
(ii) Update $\mathbf{x}_{k+1} = M\mathbf{x}_k + \mathbf{c}$.
(iii) If $\mathbf{x}_*$ exists, use $\mathbf{x}_{k+1} - \mathbf{x}_* = M(\mathbf{x}_k - \mathbf{x}_*)$ and Theorem 3.3.2 for convergence.

Method 6.2.8. *Jacobi, Gauss–Seidel, and SOR Splittings*
Let $A = D+L+U$, with D diagonal, L strictly lower triangular, and U strictly upper triangular. For a splitting $A = S - T$, use $S\mathbf{x}_{k+1} = T\mathbf{x}_k + \mathbf{b}$ and $B = S^{-1}T$.

(i) *Jacobi*: $S = D$, $T = -(L+U)$, $B_J = -D^{-1}(L+U)$.
(ii) *Gauss–Seidel*: $S = D+L$, $T = -U$, $B_{\mathrm{GS}} = -(D+L)^{-1}U$.
(iii) *SOR*: for $\omega \neq 0$, $S_\omega = \frac{1}{\omega}D + L$; $T_\omega = \left(\frac{1}{\omega} - 1\right)D - U$; and $B_\omega = S_\omega^{-1}T_\omega$.

Convergence is controlled by $\rho(B)$; $\omega = 1$ gives Gauss–Seidel.

Corollary 6.2.9. *Strict Diagonal Dominance Gives Jacobi Convergence*
If $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{C})$ is strictly row diagonally dominant, then Jacobi converges for every right-hand side.

Proof. Here $a_{ii} \neq 0$ and $B_J = -D^{-1}(L+U) = I_n - D^{-1}A$. Thus $(B_J)_{ii} = 0$ and $\sum_{j\neq i} |(B_J)_{ij}| = \sum_{j\neq i} |a_{ij}| \,/\, |a_{ii}| < 1$.
Gersgorin gives $\mathrm{spec}(B_J) \subset \{z : |z| < 1\}$, so $\rho(B_J) < 1$; Theorem 6.2.5 applies.
Structural Commentary: Diagonal dominance puts the Jacobi spectrum in the unit disk.
Key Insight: Small off-diagonal row sums force the error matrix to have spectral radius below 1. □

Example 6.2.10. *Two-by-Two Jacobi and Gauss–Seidel*
For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{b} \in \mathbb{R}^2$, write $A = D + L + U$.
The Jacobi iteration matrix is $B_J = -D^{-1}(L+U) = \begin{bmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{bmatrix}$, so $\rho(B_J) = \frac{1}{2}$.
The Gauss–Seidel iteration matrix is $B_{\mathrm{GS}} = -(D+L)^{-1}U = \begin{bmatrix} 0 & -\frac{1}{2} \\ 0 & \frac{1}{4} \end{bmatrix}$, so $\rho(B_{\mathrm{GS}}) = \frac{1}{4}$.
Both converge by Theorem 6.2.5, and smaller spectral radius predicts faster asymptotic decay for Gauss–Seidel.

Example 6.2.11. *Poisson Tridiagonal Stationary Rates*
For $A_n = \mathrm{tridiag}(-1, 2, -1)$, Jacobi gives $B_J = I_n - \frac{1}{2}A_n$; $\lambda_k(B_J) = \cos\frac{k\pi}{n+1}$; and $\rho(B_J) = \cos\frac{\pi}{n+1}$.
For the same ordering, Gauss–Seidel satisfies $\rho(B_{\mathrm{GS}}) = \rho(B_J)^2 = \cos^2\frac{\pi}{n+1}$. As $n \to \infty$, both radii tend to 1; fine grids need acceleration or preconditioning.

Perspective 6.2.12. *Preconditioning Preview*
A preconditioner replaces $A\mathbf{x} = \mathbf{b}$ by an equivalent system such as $M^{-1}A\mathbf{x} = M^{-1}\mathbf{b}$ or $AM^{-1}\mathbf{y} = \mathbf{b}$, with $\mathbf{x} = M^{-1}\mathbf{y}$. It should improve conditioning or convergence while being much cheaper to apply than A^{-1}.

6.3 Numerical Spectral Methods

Definition 6.3.1. *Upper Hessenberg Matrix*
Let $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{F})$. Then A is *upper Hessenberg* if $a_{ij} = 0$ whenever $i > j + 1$.
Equivalently, all entries below the first subdiagonal are zero.

Numerical linear algebra also replaces a matrix by a unitarily similar one with many forced zero entries. Upper Hessenberg form is the standard preliminary reduction before QR-type eigenvalue algorithms.

$$\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \end{bmatrix}$$

Figure 99: Upper Hessenberg pattern: red is the first subdiagonal; blue entries below it are forced zero.

Theorem 6.3.2. *Unitary Upper Hessenberg Reduction*
Every $A \in \mathrm{Mat}(n, n, \mathbb{F})$ is unitarily similar to upper Hessenberg matrix with real nonnegative subdiagonal entries.

Proof. Set $A_0 = A$. For $k = 1, \ldots, n-2$, suppose columns $1, \ldots, k-1$ of A_{k-1} already have zeros below the first subdiagonal. Let $\mathbf{w}$ be the tail of column k in rows $k+1, \ldots, n$.
By Corollary 2.5.4, choose a unitary V_k with $V_k^* \mathbf{w} = \alpha \mathbf{e}_1$, $\alpha \geq 0$.
Put $U_k = I_k \oplus V_k$ and $A_k = U_k^* A_{k-1} U_k$. Right multiplication by U_k changes only columns $k+1, \ldots, n$, so columns $1, \ldots, k-1$ keep their existing Hessenberg zeros. Left multiplication by U_k^* acts only on rows $k+1, \ldots, n$; those rows are already zero in columns $1, \ldots, k-1$, so the zeros remain. In column k, the tail becomes $V_k^* \mathbf{w} = \alpha \mathbf{e}_1$, so entries below row $k+1$ are zero and the new subdiagonal entry is real nonnegative.
After $n-2$ steps, $A_{n-2} = U^* A U$ is upper Hessenberg, where $U = U_1 \cdots U_{n-2}$ is unitary.
Structural Commentary: Each trailing unitary creates one Hessenberg column and preserves earlier ones.
Key Insight: Kill one column tail at a time. □

Perspective 6.3.3. *Why Hessenberg Form Matters*
Hessenberg form preserves spectral data by unitary similarity, making QR-type eigenvalue algorithms cheaper.

Method 6.3.4. *QR Eigenvalue Algorithm Preview*
For a square matrix A_0, the basic QR eigenvalue iteration is:

(i) In practice, first reduce A_0 to Hessenberg form by Theorem 6.3.2.
(ii) Factor $A_k = Q_k R_k$, with Q_k unitary and R_k upper triangular.
(iii) Set $A_{k+1} = R_k Q_k = Q_k^* A_k Q_k$, using shifts to speed convergence.

Each step is a unitary similarity, so the spectrum is preserved.

Method 6.3.5. *Hessenberg QR Step by Plane Rotations*
Let $H \in \mathrm{Mat}(n, n, \mathbb{C})$ be upper Hessenberg and let $\mu \in \mathbb{C}$.

(i) Set $B_0 = H - \mu I_n$.
(ii) For $j = 1, \ldots, n-1$, choose a unitary plane rotation G_j acting on coordinates $j, j+1$, so that $(G_j^* B_{j-1})_{j+1,j} = 0$ and $B_j = G_j^* B_{j-1}$.
(iii) Put $Q = G_1 \cdots G_{n-1}$ and $R = B_{n-1}$; then $H - \mu I_n = QR$.
(iv) Define the shifted QR update $H_+ = RQ + \mu I_n = Q^* H Q$

Adjacent rotations implement the Hessenberg step by local bulge chasing.

Theorem 6.3.6. *Power Method Dominant-Eigenvector Preview*
Let $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$, where $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is a basis of $\mathbb{C}^n$, and assume $|\lambda_1| > \max_{j \geq 2} |\lambda_j|$.
If $\mathbf{u}_0 = \sum_{j=1}^{n} c_j \mathbf{v}_j$ with $c_1 \neq 0$, then $\lambda_1^{-k} A^k \mathbf{u}_0 \longrightarrow c_1 \mathbf{v}_1$. Thus the directions of $A^k \mathbf{u}_0$ converge to $\mathrm{span}\{\mathbf{v}_1\}$.

Proof. Set $\rho = \max_{j \geq 2} |\lambda_j / \lambda_1| < 1$. Since $A^k \mathbf{v}_j = \lambda_j^k \mathbf{v}_j$, $\lambda_1^{-k} A^k \mathbf{u}_0 = c_1 \mathbf{v}_1 + \sum_{j=2}^{n} c_j \left(\frac{\lambda_j}{\lambda_1}\right)^k \mathbf{v}_j$.

The remainder has norm at most $\rho^k \sum_{j=2}^{n} |c_j| \, \|\mathbf{v}_j\|$, hence tends to 0.
Structural Commentary: The power method is spectral filtering by repeated multiplication.
Key Insight: A spectral gap makes every non-dominant component decay relative to the dominant one. □

Method 6.3.7. *Inverse and Shifted Power Iteration Preview*
Let $\mathcal{N}(\mathbf{z}) := \mathbf{z} / \|\mathbf{z}\|_2$ for $\mathbf{z} \neq \mathbf{0}$.

(i) *Inverse*: if $0 \notin \mathrm{spec}(A)$, solve $A\mathbf{y}_{k+1} = \mathbf{x}_k$ and set $\mathbf{x}_{k+1} = \mathcal{N}(\mathbf{y}_{k+1})$.
This is power iteration on A^{-1} and targets minimal $|\lambda|$.
(ii) *Shift-invert*: for $\sigma \notin \mathrm{spec}(A)$, $(A - \sigma I)\mathbf{y}_{k+1} = \mathbf{x}_k$, $\mathbf{x}_{k+1} = \mathcal{N}(\mathbf{y}_{k+1})$.
Eigenvalues become $(\lambda - \sigma)^{-1}$, targeting λ nearest σ.
(iii) *Shifted power*: $\mathbf{x}_{k+1} = \mathcal{N}\big((A - \sigma I)\mathbf{x}_k\big)$; eigenvalues become $\lambda - \sigma$, emphasizing large $|\lambda - \sigma|$.

Use factors of A or $A - \sigma I$ for solves; do not form explicit inverses.

Method 6.3.8. *SVD Numerics Preview*
Stable numerical SVD computation proceeds as:

(i) Avoid forming $A^* A$ when possible, since it can square the condition number.
(ii) Use orthogonal or unitary transformations to reduce A to bidiagonal form.
(iii) Compute singular values from the bidiagonal problem, then use them for low-rank approximation, pseudoinverse truncation, and least-squares diagnostics through Theorem 5.1.14 and Definition 5.2.1.

Definition 6.3.9. *Discrete Fourier Matrix*
For $n \geq 1$, set $\omega_n = e^{2\pi i/n}$ and $F_n := [\omega_n^{jk}]_{j,k=0}^{n-1}$.
Then $F_n\mathbf{c}$ is the unnormalised DFT of $\mathbf{c} \in \mathbb{C}^n$; $Q_n := n^{-1/2}F_n$ is the normalized Fourier matrix.

Definition 6.3.10. *Circulant Matrix*
On the 0-indexed basis of $\mathbb{C}^n$, define the *cyclic shift* $P_n\mathbf{e}_j = \mathbf{e}_{(j-1) \bmod n}$.
A matrix $C \in \mathrm{Mat}(n, n, \mathbb{C})$ is *circulant* if, for some $a_0, \ldots, a_{n-1} \in \mathbb{C}$,

$$C = a_0 I_n + a_1 P_n + \cdots + a_{n-1} P_n^{n-1} = p(P_n),$$
$$p(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}$$

Theorem 6.3.11. *Fourier Basis for Cyclic Shifts and Circulants*
Let $n \geq 1$, let $\omega_n := e^{2\pi i/n}$, and let P_n be the cyclic shift.
For $0 \leq k < n$, set $\mathbf{f}_k := (\omega_n^{jk})_{j=0}^{n-1}$; $F_n := [\mathbf{f}_0 \ \cdots \ \mathbf{f}_{n-1}]$; and $\Omega_n := \mathrm{diag}(1, \omega_n, \ldots, \omega_n^{n-1})$.
Then $F_n^* F_n = nI_n$, so $Q_n := n^{-1/2}F_n$ is unitary, and $P_n F_n = F_n \Omega_n$.
If $C = p(P_n)$ is circulant, then $Q_n^* C Q_n = p(\Omega_n) = \mathrm{diag}\big(p(1), p(\omega_n), \ldots, p(\omega_n^{n-1})\big)$.

Proof. For $0 \leq k, \ell \leq n-1$, $\langle \mathbf{f}_k, \mathbf{f}_\ell \rangle = \sum_{r=0}^{n-1} (\omega_n^{k-\ell})^r = n\delta_{k\ell}$. For $k = \ell$ the sum is n; for $k \neq \ell$, $W = \omega_n^{k-\ell}$ has $W \neq 1$ and $W^n = 1$, so $(1-W)\sum_{r=0}^{n-1} W^r = 0$. Hence $F_n^* F_n = nI_n$, and Theorem 2.4.28 makes Q_n unitary.
Also $P_n \mathbf{f}_k = \omega_n^k \mathbf{f}_k$, so $Q_n^* P_n Q_n = \Omega_n$; applying p gives the formula.
Structural Commentary: Fourier coordinates diagonalize every circulant.
Key Insight: Orthogonality of roots of unity turns a cyclic matrix problem into scalar evaluation. □

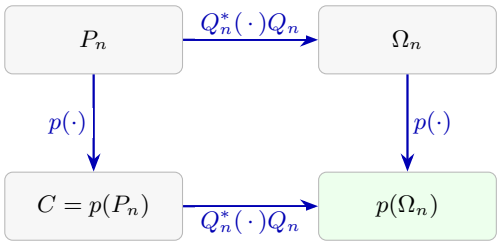


Figure 100: Fourier diagonalizes P_n; polynomial evaluation diagonalizes $C = p(P_n)$.

Proposition 6.3.12. *Inverse of the Fourier Matrix*
The Fourier matrix satisfies $F_n^* F_n = nI_n$, and therefore $F_n^{-1} = \frac{1}{n}F_n^*$.

Proof. Theorem 6.3.11 gives $Q_n^* Q_n = I_n$ for $Q_n = n^{-1/2}F_n$. Multiplying by n gives $F_n^* F_n = nI_n$, so $n^{-1}F_n^*$ is a left inverse of F_n. Since F_n is square, Theorem 3.3.30 makes it the inverse.
Structural Commentary: Fourier inversion is conjugate transpose plus one scalar normalization.
Key Insight: The inverse is simple because the Fourier columns are orthogonal. □

Method 6.3.13. *Radix-Two FFT Split*
Let $n = 2m$ and $\mathbf{y} = F_n\mathbf{c}$.

(i) Split $\mathbf{c}' := (c_{2r})_{r=0}^{m-1}$ and $\mathbf{c}'' := (c_{2r+1})_{r=0}^{m-1}$.
(ii) Compute $\mathbf{y}' := F_m\mathbf{c}'$ and $\mathbf{y}'' := F_m\mathbf{c}''$.
(iii) With $d_j := \omega_n^j$, and using $\omega_n^2 = \omega_m$ and $\omega_n^m = -1$, set $y_j = y'_j + d_j y''_j$; and $y_{j+m} = y'_j - d_j y''_j$ for $0 \leq j < m$.
(iv) For $n = 2^\ell$, repeat (i)–(iii) for ℓ stages.
n^2 dense products become about $\frac{1}{2}n\log_2 n$ twiddle multiplications plus additions.

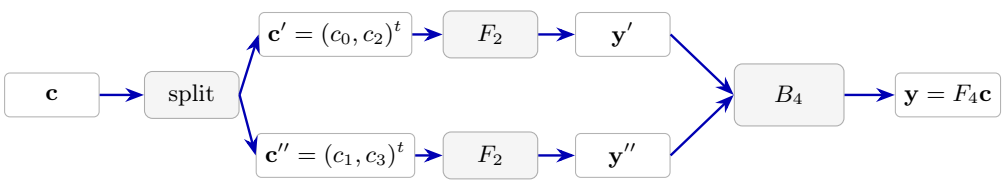


Figure 101: For $n = 4$: split, apply two F_2 transforms, then $B_4\binom{\mathbf{y}'}{\mathbf{y}''} = \binom{\mathbf{y}'+D\mathbf{y}''}{\mathbf{y}'-D\mathbf{y}''}$ with $D = \mathrm{diag}(1, i)$.

Part V
Groups, Rings, and Fields

1 Algebraic Language and Categories

1.1 Sets, Maps, Relations, and Universal Properties

The elementary set, order, choice, function, and bijection vocabulary used here is collected in the foundational prerequisites. This subsection continues with set algebra, universal properties, and equivalence-based structure.

Theorem 1.1.1. *Universal Property of Quotients*
Let $\pi : \mathcal{V} \to \mathcal{V}/\mathcal{U}$ be the quotient map.
If $L \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $\mathcal{U} \subseteq \ker(L)$, then $\exists!\, \overline{L} \in \mathcal{L}(\mathcal{V}/\mathcal{U}, \mathcal{W})$ such that $\overline{L}\pi = L$, $\overline{L}(\mathbf{v}+\mathcal{U}) = L\mathbf{v}$.
Moreover, $\ker(\overline{L}) = \ker(L)/\mathcal{U}$ and $\operatorname{range}(\overline{L}) = \operatorname{range}(L)$.

Proof. If $\mathbf{v}+\mathcal{U} = \mathbf{w}+\mathcal{U}$, then $\mathbf{v}-\mathbf{w} \in \mathcal{U} \subseteq \ker(L)$, so $L\mathbf{v} = L\mathbf{w}$. Hence the formula defines a linear map $\overline{L}$ and gives $\overline{L}\pi = L$. If $M\pi = L$, then $M(\mathbf{v}+\mathcal{U}) = M\pi(\mathbf{v}) = L\mathbf{v} = \overline{L}(\mathbf{v}+\mathcal{U})$, so $M = \overline{L}$.
Finally, $\mathbf{v}+\mathcal{U} \in \ker(\overline{L}) \iff L\mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \ker(L)$, giving $\ker(\overline{L}) = \ker(L)/\mathcal{U}$.
The range equality follows from $\overline{L}(\mathbf{v}+\mathcal{U}) = L\mathbf{v}$.
Structural Commentary: Maps killing $\mathcal{U}$ descend uniquely to $\mathcal{V}/\mathcal{U}$.
Key Insight: Quotients are the natural domain for maps whose ambiguity contains $\mathcal{U}$. □

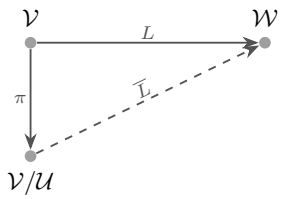


Figure 102: If $\mathcal{U} \subseteq \ker(L)$, then $L = \overline{L}\pi$ uniquely.

Definition 1.1.2. *Equivalence Relations, Classes, Invariants, and Canonical Forms*
Let $\sim$ be a relation on a set X.

(i) $\sim$ is an *equivalence relation* if, for all $x, y, z \in X$, $x \sim x$, $x \sim y \Rightarrow y \sim x$, and $(x \sim y$ and $y \sim z) \Rightarrow x \sim z$.
(ii) If $\sim$ is an equivalence relation, the *equivalence class* of $x \in X$ is $[x] := \{y \in X : y \sim x\}$.
(iii) A function $f : X \to Y$ is an *invariant* of $\sim$ if $x \sim y \Rightarrow f(x) = f(y)$, and a *complete invariant* if $x \sim y \iff f(x) = f(y)$.
(iv) A subset $\mathcal{C} \subseteq X$ is an *set of canonical forms* for $\sim$ if each equivalence class meets $\mathcal{C}$ in exactly one element.

Perspective 1.1.3. *Similarity Invariants and Canonical Forms*
Similarity is equality up to change of basis. Trace, determinant, characteristic polynomials, and minimal polynomials are similarity invariants by Theorems 1.2.5, 6.2.7, and 6.3.5, but they are not generally complete. Canonical forms, such as the Jordan form of Theorem 6.6.6, choose one structured representative of each similarity class when possible.

1.2 Categories and Functors

1.3 Algebraic Structures as Objects

Definition 1.3.1. *Fields and Ordered Fields*
A *field* is a set F with addition and multiplication satisfying closure, commutativity, associativity, identities $0, 1$ with $1 \neq 0$, additive inverses, multiplicative inverses for nonzero elements, and distributivity.
An *ordered field* is a field such that $y < z$ implies $x + y < x + z$, and $x > 0$ and $y > 0$ imply $xy > 0$.

Remark 1.3.2. *Fields as Objects and Scalars*
Unlike their scalar role in Definition 1.1.1, fields are now Analysis objects: $\mathbb{Q}$ is ordered but incomplete, while $\mathbb{R}$ has the least-upper-bound property.

Proposition 1.3.3. *Basic Consequences of the Field Axioms*
Every field has cancellation, unique identities and inverses, $0x = 0$, no zero divisors, and the sign rules

$$(-x)y = -(xy) = x(-y), \qquad (-x)(-y) = xy.$$

Proof. Cancellation comes from adding or multiplying by the relevant inverse; uniqueness follows from cancellation. Also $0x + 0x = (0+0)x = 0x$, so $0x = 0$. If $x, y \neq 0$ and $xy = 0$, multiplying by $x^{-1}y^{-1}$ gives $1 = 0$, impossible. The sign rules follow because the displayed negatives are additive inverses of xy.
Structural Commentary: The field axioms license ordinary algebra.
Key Insight: Inverses force cancellation and signs. □

Proposition 1.3.4. *Basic Inequality Rules in Ordered Fields*
In every ordered field,

(i) $x > 0$ *iff* $-x < 0$;
(ii) multiplying an inequality by a positive element preserves the inequality;
(iii) multiplying an inequality by a negative element reverses the inequality;
(iv) if $x \neq 0$, *then* $x^2 > 0$, *so in particular* $1 > 0$;
(v) if $0 < x < y$, *then* $0 < 1/y < 1/x$.

Proof.

(i) If $x > 0$, then $0 = -x + x > -x + 0$, so $-x < 0$; the converse is the same with $-x$.
(ii) If $y < z$ and $x > 0$, then $z - y > 0$, so $x(z-y) > 0$ and hence $xy < xz$.
(iii) Apply (ii) to $-x > 0$.
(iv) Either $x > 0$ or $-x > 0$, so $x^2 > 0$; take $x = 1$ for $1 > 0$.
(v) Since $x, y > 0$, multiply $x < y$ by $(1/x)(1/y) > 0$ to get $1/y < 1/x$.

Structural Commentary: Ordered fields make algebraic signs compatible with order.
Key Insight: Positivity decides whether multiplication preserves order. □

Perspective 1.3.5. *Finite-Support Coordinate Model*
Proposition 3.3.16 identifies an n-dimensional vector space with $\mathbb{F}^n$, and Theorem 3.3.32 identifies linear maps with matrices after bases are chosen. The space $\mathbb{F}^{(\mathbb{N})}$ is the countable standard-coordinate model with finite support. This is the field analogue of the free-module notation $(R^B)_0$.

1.4 Exactness and Diagrams

Theorem 1.4.1. *First Isomorphism Theorem*
Let $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$. *Then* $\widehat{T} : \mathcal{V}/\ker(T) \to \operatorname{range}(T)$; $\widehat{T}(\mathbf{v} + \ker(T)) = T\mathbf{v}$ *is a well-defined linear isomorphism.*

Proof. If $\mathbf{v} + \ker(T) = \mathbf{w} + \ker(T)$, then $\mathbf{v} - \mathbf{w} \in \ker(T)$, so $T\mathbf{v} = T\mathbf{w}$; hence $\widehat{T}$ is well-defined.
Linearity follows from the quotient operations, $\widehat{T}(a(\mathbf{v} + \ker(T)) + (\mathbf{w} + \ker(T))) = aT\mathbf{v} + T\mathbf{w}$.
Also, $\widehat{T}(\mathbf{v} + \ker(T)) = \mathbf{0} \iff T\mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \ker(T) \iff \mathbf{v} + \ker(T) = \ker(T)$, so $\widehat{T}$ is injective.
If $\mathbf{y} \in \operatorname{range}(T)$, then $\mathbf{y} = T\mathbf{v} = \widehat{T}(\mathbf{v} + \ker(T))$ for some $\mathbf{v} \in \mathcal{V}$, so $\widehat{T}$ is surjective. Thus $\widehat{T}$ is linear isomorphism.
Structural Commentary: The quotient by the kernel is the part of the domain actually seen by T.
Key Insight: Image is domain modulo kernel. □

Theorem 1.4.2. *Second Isomorphism Theorem*
Let $\mathcal{S}$ *and* $\mathcal{T}$ *be subspaces of* $\mathcal{V}$. *Then* $(\mathcal{S} + \mathcal{T})/\mathcal{T} \cong \mathcal{S}/(\mathcal{S} \cap \mathcal{T})$.

Proof. Define $L : \mathcal{S} \to (\mathcal{S} + \mathcal{T})/\mathcal{T}$ by $L\mathbf{s} = \mathbf{s} + \mathcal{T}$. Then L is linear by the quotient operations.
Moreover, $\ker(L) = \{\mathbf{s} \in \mathcal{S} : \mathbf{s} + \mathcal{T} = \mathcal{T}\} = \mathcal{S} \cap \mathcal{T}$. If $(\mathbf{s} + \mathbf{t}) + \mathcal{T} \in (\mathcal{S} + \mathcal{T})/\mathcal{T}$, with $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$, then $(\mathbf{s} + \mathbf{t}) + \mathcal{T} = \mathbf{s} + \mathcal{T} = L\mathbf{s}$, so L is surjective. Theorem 1.4.1 gives the stated isomorphism.
Structural Commentary: Modding out by $\mathcal{T}$ leaves only the part of $\mathcal{S}$ not already inside $\mathcal{T}$.
Key Insight: In a quotient by $\mathcal{T}$, adding $\mathcal{T}$ to $\mathcal{S}$ costs nothing. □

Theorem 1.4.3. *Third Isomorphism Theorem*
If $\mathcal{U} \subseteq \mathcal{T} \subseteq \mathcal{V}$ *are subspaces, then* $(\mathcal{V}/\mathcal{U})/(\mathcal{T}/\mathcal{U}) \cong \mathcal{V}/\mathcal{T}$.

Proof. Define $L : \mathcal{V}/\mathcal{U} \to \mathcal{V}/\mathcal{T}$ by $L(\mathbf{v} + \mathcal{U}) = \mathbf{v} + \mathcal{T}$.
If $\mathbf{v} + \mathcal{U} = \mathbf{w} + \mathcal{U}$, then $\mathbf{v} - \mathbf{w} \in \mathcal{U} \subseteq \mathcal{T}$, so L is well-defined.
It is linear by the quotient operations, and it is surjective since $L(\mathbf{v} + \mathcal{U}) = \mathbf{v} + \mathcal{T}$ for every coset $\mathbf{v} + \mathcal{T}$.
Moreover, $\ker(L) = \{\mathbf{v} + \mathcal{U} : \mathbf{v} + \mathcal{T} = \mathcal{T}\} = \{\mathbf{v} + \mathcal{U} : \mathbf{v} \in \mathcal{T}\} = \mathcal{T}/\mathcal{U}$. Theorem 1.4.1 gives $(\mathcal{V}/\mathcal{U})/(\mathcal{T}/\mathcal{U}) \cong \mathcal{V}/\mathcal{T}$.
Structural Commentary: Collapsing $\mathcal{U}$ and then collapsing $\mathcal{T}/\mathcal{U}$ is the same as collapsing $\mathcal{T}$ at once.
Key Insight: Quotients of quotients cancel only when the nesting is explicit. □

2 Group Theory

2.1 Groups, Subgroups, and Homomorphisms

Groups are the algebraic language of reversible operations. They appear here as the additive structure underlying rings and modules, and later as symmetry groups acting on mathematical objects.

Definition 2.1.1. *Groups*
A *group* is a nonempty set G with a binary operation $(a, b) \mapsto ab$ such that:

(i) $(ab)c = a(bc)$ for all $a, b, c \in G$;
(ii) there is an element $e \in G$ with $eg = ge = g$ for all $g \in G$;
(iii) for each $g \in G$, there is $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.

The group is *abelian* if $ab = ba$ for all $a, b \in G$.
In an abelian group the operation is often written additively, with identity 0 and inverse $-a$.

Example 2.1.2. *Basic Groups*

(i) The bijections $S \to S$ form a group under composition; it is usually nonabelian.
(ii) $\mathrm{Mat}(m, n, \mathbb{F})$ is an abelian group under matrix addition.
(iii) $\mathrm{GL}_n(\mathbb{F})$, the invertible $n \times n$ matrices, is a group under multiplication.
(iv) $\mathbb{Z}/n\mathbb{Z}$ is a finite abelian group under addition modulo n.

Definition 2.1.3. *Subgroup*
A *subgroup* of a group G is a nonempty subset $H \subseteq G$ that is itself a group under operation inherited from G.

Proposition 2.1.4. *Subgroup Test*
Let G be a group and let $H \subseteq G$ be nonempty. Then $H \leq G \iff ab^{-1} \in H \ \forall a, b \in H$.

Proof.

($\Rightarrow$) If $H \leq G$, then $b^{-1} \in H$ and $ab^{-1} \in H$ for all $a, b \in H$.
($\Leftarrow$) Assume $ab^{-1} \in H$ for all $a, b \in H$. Choose $h \in H$. Then $e_G = hh^{-1} \in H$. For $b \in H$, $b^{-1} = e_G b^{-1} \in H$. For $a, c \in H$, since $c^{-1} \in H$, we have $ac = a(c^{-1})^{-1} \in H$. Hence $H \leq G$.

Structural Commentary: Closure under one expression packages identity, inverses, and products.
Key Insight: In a group, subtraction-style closure is enough. □

Definition 2.1.5. *Generated Groups and Homomorphisms*
Let G, H be groups.

(i) For $a \in G$, the *subgroup generated by a* is $\langle a \rangle := \{a^k : k \in \mathbb{Z}\} \leq G$.
(ii) G is *cyclic* if $G = \langle a \rangle$ for some $a \in G$.
If it exists, $\mathrm{ord}(a) := \min\{n \in \mathbb{N}_{>0} : a^n = e_G\}$; then $\langle a \rangle = \{e_G, a, \ldots, a^{\mathrm{ord}(a)-1}\}$.
(iii) A *homomorphism* $\varphi : G \to H$ satisfies $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in G$.
(iv) For a homomorphism $\varphi : G \to H$, $\ker \varphi := \{g \in G : \varphi(g) = e_H\}$; and $\mathrm{im}\, \varphi := \varphi(G)$.
An *isomorphism* is a bijective homomorphism.

Proposition 2.1.6. *Basic Homomorphism Properties*
Let $\varphi : G \to H$ be a group homomorphism. Then:

(i) $\varphi(e_G) = e_H$;
(ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ *for all* $g \in G$;
(iii) $\ker \varphi \leq G$;
(iv) $\mathrm{im}\, \varphi \leq H$.

Proof.

(i) Since $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)^2$, cancellation gives $\varphi(e_G) = e_H$.
(ii) For $g \in G$, $\varphi(g)\varphi(g^{-1}) = \varphi(gg^{-1}) = e_H$, hence $\varphi(g^{-1}) = \varphi(g)^{-1}$.
(iii) If $a, b \in \ker \varphi$, then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H$, so $\ker \varphi \leq G$ by Proposition 2.1.4.
(iv) If $x = \varphi(a)$ and $y = \varphi(b)$ lie in $\mathrm{im}\, \varphi$, then $xy^{-1} = \varphi(ab^{-1}) \in \mathrm{im}\, \varphi$, so $\mathrm{im}\, \varphi \leq H$ by Proposition 2.1.4.

Structural Commentary: Homomorphisms preserve all group structure once they preserve multiplication.
Key Insight: $\varphi(ab) = \varphi(a)\varphi(b)$ preserves e, inverses, kernels, and images. □

Proposition 2.1.7. *Kernel Criterion for Injectivity*
Let $\varphi : G \to H$ be a group homomorphism. Then φ injective $\iff \ker \varphi = \{e_G\}$.

Proof.
($\Rightarrow$) If φ injective, $g \in \ker\varphi$, then $\varphi(g) = e_H = \varphi(e_G)$ by Proposition 2.1.6, so $g = e_G$. Hence $\ker\varphi = \{e_G\}$.
($\Leftarrow$) If $\ker\varphi = \{e_G\}$ and $\varphi(a) = \varphi(b)$, then $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H$, so $ab^{-1} = e_G$ and $a = b$.
Structural Commentary: Injectivity is detected by the identity fiber $\ker\varphi$.
Key Insight: The kernel measures exactly the failure of injectivity. □

2.2 Group Actions

A group action packages the idea that elements of a group act as reversible motions of a set. The conjugation action is the finite-group ingredient used later in Wedderburn's theorem.

Definition 2.2.1. *Actions, Orbits, and Stabilizers*
Let G be a group and let $\mathcal{X}$ be a set.

(i) *Left action* of G on $\mathcal{X}$ is map $G \times \mathcal{X} \to \mathcal{X}$, $(g, x) \mapsto gx$, s.t. $e_G x = x$ and $(gh)x = g(hx)$ $\forall g, h \in G$, $x \in \mathcal{X}$.
(ii) The *orbit* of $x \in \mathcal{X}$ is $Gx := \{gx : g \in G\}$.
(iii) The *stabilizer* of $x \in \mathcal{X}$ is $G_x := \{g \in G : gx = x\}$.

Theorem 2.2.2. *Orbit-Stabilizer*
Let a finite group G act on $\mathcal{X}$. For each $x \in \mathcal{X}$, $|Gx| = [G : G_x] = \frac{|G|}{|G_x|}$.

Proof. Define $\Phi : G/G_x \to Gx$, $\Phi(gG_x) = gx$.
For $g, h \in G$, $gG_x = hG_x \iff h^{-1}g \in G_x \iff h^{-1}gx = x \iff gx = hx$. Thus Φ is well defined and injective. It is surjective by definition of Gx, hence bijective, so $|Gx| = |G/G_x| = [G : G_x]$.
Structural Commentary: Orbits are parametrized by cosets of the stabilizer.
Key Insight: Same orbit point $\iff$ same stabilizer coset. □

2.3 Normality and Structure Theorems

2.4 Sylow Theory and Finite Groups

Definition 2.4.1. *Conjugacy Classes and Centralizers*
Let G be a group.

(i) The *conjugation action* of G on itself is $g \cdot x := gxg^{-1}$.
(ii) The *conjugacy class* of $x \in G$ is $\{x\}^G := \{gxg^{-1} : g \in G\}$.
(iii) The *centralizer* of $x \in G$ is $C_G(x) := \{g \in G : gx = xg\}$.
(iv) The *center* of G is $Z(G) := \{g \in G : gx = xg \text{ for all } x \in G\}$.

Theorem 2.4.2. *Class Equation*
Let G be finite, $S \subseteq G$ contain one representative of each noncentral conjugacy class.
Then $|G| = |Z(G)| + \sum_{x \in S} [G : C_G(x)]$.

Proof. The conjugacy classes form a partition $G = Z(G) \sqcup \bigsqcup_{x \in S} \{x\}^G$.
Indeed, $\left|\{x\}^G\right| = 1 \iff gxg^{-1} = x$ for all $g \in G \iff x \in Z(G)$. For $x \in S$, Theorem 2.2.2 applied to conjugation gives $\left|\{x\}^G\right| = [G : C_G(x)]$. Taking cardinalities yields the formula.
Structural Commentary: G decomposes into central singleton classes and noncentral conjugacy classes.
Key Insight: Count conjugacy classes by orbit-stabilizer. □

2.5 Representation-Theoretic Preview

Definition 2.5.1. *Group Algebra*
For a group G and field $\mathbb{F}$, the *group algebra* is $\mathbb{F}[G] := \{\sum_{g \in G} r_g g : r_g \in \mathbb{F},\ |\{g : r_g \neq 0\}| < \infty\}$.
It has coefficientwise vector-space operations and product $(\sum_g r_g g)(\sum_h s_h h) := \sum_{g,h} r_g s_h (gh)$.

Example 2.5.2. *Group Algebras*
In $\mathbb{F}[G]$, G is an $\mathbb{F}$-basis, e_G is the algebra identity, and associativity follows from $(gh)k = g(hk)$ by bilinearity. If G is finite, then $\dim_{\mathbb{F}} \mathbb{F}[G] = |G|$. Moreover, $\mathbb{F}[G]$ is commutative iff G is abelian.

Example 2.5.3. *Cyclic Group Algebras and Circulant Shifts*
Let $G = \langle\alpha\rangle$, $|G| = n$, and let $\mathcal{B} = (1, \alpha, \ldots, \alpha^{n-1})$ be the ordered $\mathbb{F}$-basis of $\mathbb{F}[G]$.

For $0 \le j,k < n$, $L_{\alpha^k}(\alpha^j) = \alpha^{j+k}$ with exponents read modulo n, so L_{α^k} is the k-fold cyclic shift of $\mathcal{B}$.
Let $P := [L_\alpha]_{\mathcal{B}}$, one has $[L_{\alpha^k}]_{\mathcal{B}} = P^k$, where

$$P = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}.$$

Hence $[L_{\sum_{k=0}^{n-1} c_k \alpha^k}]_{\mathcal{B}} = \sum_{k=0}^{n-1} c_k P^k$, precisely the circulant matrices.

3 Rings, Ideals, and Polynomial Rings

3.1 Rings and Ideals

Rings add multiplication to an abelian group. Ideals are the subsets by which quotient rings can be formed, just as subspaces are the subsets by which quotient vector spaces can be formed.

Definition 3.1.1. *Rings, Identities, and Fields*
Let R be a set with operations $+$ and $\cdot$.

(i) R is a *ring* if $(R,+)$ is an abelian group, multiplication is associative, and $a(b+c) = ab+ac$, $(a+b)c = ac+bc$ for all $a,b,c \in R$.
(ii) A ring R is *commutative* if $ab = ba$ for all $a,b \in R$.
(iii) A ring R has *identity* if there is $1_R \in R$ such that $1_R a = a 1_R = a$ for all $a \in R$.
(iv) A *field* is a commutative ring with identity $1_R \neq 0$ such that every $a \in R \setminus \{0\}$ has $a^{-1} \in R$.

Example 3.1.2. *Basic Rings*

(i) $(\mathbb{Z},+,\cdot)$ and $(\mathbb{F}[x],+,\cdot)$ are commutative rings with identity 1.
(ii) $\mathrm{Mat}(n,n,\mathbb{F})$ is a unital ring under matrix addition and multiplication, with identity I_n; it is noncommutative for $n \ge 2$.
(iii) $2\mathbb{Z}$ is a commutative ring under inherited addition and multiplication, but has no identity.
(iv) $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring under addition and multiplication modulo n, with identity $1+n\mathbb{Z}$.

Definition 3.1.3. *Ring Homomorphisms and Subrings*
Let R, S be rings with identity.

(i) A *ring homomorphism* $\varphi : R \to S$ preserves $+$, $\cdot$, and 1, where $\varphi(a+b) = \varphi(a)+\varphi(b)$, $\varphi(ab) = \varphi(a)\varphi(b)$, $\varphi(1_R) = 1_S$ for all $a,b \in R$.
(ii) A *subring* of R is a subset $T \subseteq R$ that is a ring under the inherited operations and has identity 1_R.

Proposition 3.1.4. *Subring Test*
Let R be a ring with identity and let $S \subseteq R$. Then $S \le R \iff 1_R \in S$, $a-b \in S$, $ab \in S$ $\forall a,b \in S$.

Proof.
($\Rightarrow$) If $S \le R$, then $1_R \in S$ and S is closed under subtraction and multiplication.
($\Leftarrow$) Assume $1_R \in S$, $a-b \in S$, and $ab \in S$ for all $a,b \in S$. Then S is nonempty, $(S,+) \le (R,+)$ by Proposition 2.1.4, and multiplication restricts to S. All ring laws and the identity 1_R are inherited from R, so $S \le R$.

Structural Commentary: A subring is a subset stable under the operations and containing the ambient unit.
Key Insight: Subtraction handles the additive group part in one stroke. □

Definition 3.1.5. *Ideals and Generated Ideals*
Let R be a ring.

(i) A nonempty subset $\mathcal{I} \subseteq R$ is a *two-sided ideal* if $a-b \in \mathcal{I}$ $\forall a,b \in \mathcal{I}$, and $ra, ar \in \mathcal{I}$ $\forall a \in \mathcal{I}$, $r \in R$.
(ii) If R is commutative with identity, $S \subseteq R$, *ideal generated by S* is $\langle S \rangle := \left\{ \sum_{j=1}^{n} r_j s_j : n \ge 1,\ r_j \in R,\ s_j \in S \right\}$.
(iii) *Finite generated ideal notation*: if $S = \{s_1, \dots, s_n\}$, write $\langle s_1, \dots, s_n \rangle := \langle S \rangle$.
(iv) In $\mathbb{F}[x]$, $\langle p(x) \rangle = \{q(x)p(x) : q(x) \in \mathbb{F}[x]\}$ is the *principal* ideal generated by $p(x)$.

Proposition 3.1.6. *Generated Ideals Are Smallest*
Let R be a commutative ring with identity and let $S \subseteq R$.
Then $\langle S \rangle$ is an ideal, $S \subseteq \langle S \rangle$, and for every ideal $\mathcal{J} \subseteq R$, $S \subseteq \mathcal{J} \implies \langle S \rangle \subseteq \mathcal{J}$.

Proof. Let $x = \sum_{j=1}^{m} r_j s_j$ and $y = \sum_{k=1}^{n} t_k u_k$ lie in $\langle S \rangle$.
Then $x - y = \sum_{j=1}^{m} r_j s_j + \sum_{k=1}^{n} (-t_k) u_k \in \langle S \rangle$; $ax = \sum_{j=1}^{m} (a r_j) s_j \in \langle S \rangle$ for all $a \in R$, so $\langle S \rangle$ is an ideal.
Since $s = 1_R s \in \langle S \rangle$, $S \subseteq \langle S \rangle$. If $\mathcal{J}$ is an ideal with $S \subseteq \mathcal{J}$, then $r_j s_j \in \mathcal{J}$ and finite sums remain in $\mathcal{J}$, QED.
Structural Commentary: Generated ideals close a set under exactly the operations ideals must allow.
Key Insight: To force S into an ideal, one must include all ring-linear combinations of elements of S. □

Proposition 3.1.7. *Intersections and Chain Unions of Ideals*
Let R be a ring.

(i) If $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ is a nonempty family of ideals of R, then $\bigcap_{\alpha \in A} \mathcal{I}_\alpha$ is an ideal of R.
(ii) If $\{\mathcal{I}_\alpha\}_{\alpha \in A}$ is a chain of ideals of R under inclusion, then $\bigcup_{\alpha \in A} \mathcal{I}_\alpha$ is an ideal of R.

Proof.

(i) Let $\mathcal{I} = \bigcap_{\alpha \in A} \mathcal{I}_\alpha$. If $a, b \in \mathcal{I}$ and $r \in R$, then $a - b, ra, ar \in \mathcal{I}_\alpha$ for every $\alpha \in A$, hence $a - b, ra, ar \in \mathcal{I}$.
(ii) Let $\mathcal{U} = \bigcup_{\alpha \in A} \mathcal{I}_\alpha$. If $a, b \in \mathcal{U}$, then $a \in \mathcal{I}_\alpha$ and $b \in \mathcal{I}_\beta$ for some α, β. Since the family is a chain, one of $\mathcal{I}_\alpha, \mathcal{I}_\beta$ contains both a, b; hence $a - b \in \mathcal{U}$. If $a \in \mathcal{U}$ and $r \in R$, then $a \in \mathcal{I}_\alpha$ for some α, so $ra, ar \in \mathcal{I}_\alpha \subseteq \mathcal{U}$.

Structural Commentary: Chains avoid the usual failure of unions to preserve closure.
Key Insight: Comparability places any finite test inside one ideal. □

Definition 3.1.8. *Quotient Rings*
Let R be a ring and let $\mathcal{I} \trianglelefteq R$ be a two-sided ideal.

(i) The underlying set is the additive quotient group $R/\mathcal{I} = \{a + \mathcal{I} : a \in R\}$.
(ii) Addition and multiplication are $(a + \mathcal{I}) + (b + \mathcal{I}) = (a + b) + \mathcal{I}$ and $(a + \mathcal{I})(b + \mathcal{I}) = ab + \mathcal{I}$.
(iii) With these operations, $R/\mathcal{I}$ is the *quotient ring* of R by $\mathcal{I}$.

Proposition 3.1.9. *Ideals Make Quotient Multiplication Well Defined*
Let $\mathcal{I} \leq (R, +)$. The rule $(a + \mathcal{I})(b + \mathcal{I}) := ab + \mathcal{I}$ is well defined on $R/\mathcal{I}$ iff $\mathcal{I} \trianglelefteq R$.

Proof.

(⇒) If the rule is well defined and $i \in \mathcal{I}$, then $i + \mathcal{I} = 0 + \mathcal{I}$.
For $r \in R$, $ri + \mathcal{I} = r0 + \mathcal{I}$, and $ir + \mathcal{I} = 0r + \mathcal{I}$, so $ri, ir \in \mathcal{I}$. Thus $\mathcal{I} \trianglelefteq R$.
(⇐) If $\mathcal{I} \trianglelefteq R$ and $a' = a + i$, $b' = b + j$ with $i, j \in \mathcal{I}$, then $a'b' - ab = aj + ib + ij \in \mathcal{I}$.
Hence $a'b' + \mathcal{I} = ab + \mathcal{I}$, so the product is independent of representatives.

Structural Commentary: Ideals are exactly the additive subgroups compatible with multiplication of cosets.
Key Insight: Coset multiplication is well defined exactly for ideals. □

Definition 3.1.10. *Maximal Ideals*
Let R be a ring and let $\mathcal{M} \trianglelefteq R$.

(i) $\mathcal{M}$ is *proper* if $\mathcal{M} \neq R$.
(ii) $\mathcal{M}$ is *maximal* if $\mathcal{M} \neq R$ and, for every ideal $\mathcal{J} \trianglelefteq R$, $\mathcal{M} \subseteq \mathcal{J} \subseteq R \implies \mathcal{J} = \mathcal{M}$ or $\mathcal{J} = R$.

Theorem 3.1.11. *Maximal Ideals and Fields*
Let R be a commutative ring with identity and let $\mathcal{I} \trianglelefteq R$. Then $R/\mathcal{I}$ is a field $\iff$ $\mathcal{I}$ is maximal

Proof.

(⇒) If $R/\mathcal{I}$ is a field and $\mathcal{I} \subsetneq \mathcal{J} \trianglelefteq R$, then $\mathcal{J}/\mathcal{I}$ is a nonzero ideal of $R/\mathcal{I}$.
Hence $\mathcal{J}/\mathcal{I} = R/\mathcal{I}$, so $\mathcal{J} = R$. Thus $\mathcal{I}$ is maximal.
(⇐) If $\mathcal{I}$ is maximal and $0 \neq a + \mathcal{I} \in R/\mathcal{I}$, then $a \notin \mathcal{I}$.
The ideal $\mathcal{I} + \langle a \rangle$ strictly contains $\mathcal{I}$, so maximality gives $\mathcal{I} + \langle a \rangle = R$. Thus $1 = i + ra$ for some $i \in \mathcal{I}$, $r \in R$, and $(r + \mathcal{I})(a + \mathcal{I}) = 1 + \mathcal{I}$. Hence every nonzero coset is invertible.

Structural Commentary: Maximal ideals are exactly kernels whose quotients have no residual ideal structure.
Key Insight: Maximality makes every nonzero coset invertible. □

Theorem 3.1.12. *Existence of Maximal Ideals*
If R is a commutative ring with identity and $R \neq 0$, then R has a maximal ideal.

Proof. Let $\mathcal{P} := \{\mathcal{I} \trianglelefteq R : \mathcal{I} \neq R\}$, ordered by inclusion. Since $R \neq 0$, $\{0\} \in \mathcal{P}$.
If $\mathcal{C} \subseteq \mathcal{P}$ is a chain, then $\mathcal{U} := \bigcup_{\mathcal{I} \in \mathcal{C}} \mathcal{I}$ is an ideal by Proposition 3.1.7.
Also $\mathcal{U} \neq R$: if $1_R \in \mathcal{U}$, then $1_R \in \mathcal{I}$ for some $\mathcal{I} \in \mathcal{C}$, forcing $\mathcal{I} = R$, contradiction. Thus $\mathcal{U} \in \mathcal{P}$ is an upper bound for $\mathcal{C}$. By Zorn's lemma, Principle 6.5.2, $\mathcal{P}$ has a maximal element, i.e. R has a maximal ideal.
Structural Commentary: Maximal ideals are produced by maximality, not by an explicit formula in general.
Key Insight: Chain unions keep properness because containing 1 is a finite obstruction. □

Definition 3.1.13. *Zero Divisors and Integral Domains*
Let R be a ring.

(i) A *zero divisor* is a nonzero $a \in R$ such that $\exists\, 0 \neq b \in R$ with $ab = 0$ or $ba = 0$.
(ii) An *integral domain* is a commutative ring with identity $1_R \neq 0$ and no zero divisors.

Proposition 3.1.14. *Cancellation in Integral Domains*
Let R be a commutative ring with identity $1_R \neq 0$.
Then R is an integral domain $\iff$ $(\forall r, x, y \in R)(r \neq 0 \wedge rx = ry \Rightarrow x = y)$.

Proof.
($\Rightarrow$) If $rx = ry$ and $r \neq 0$, then $r(x - y) = 0$; no zero divisors gives $x - y = 0$.
($\Leftarrow$) If $ab = 0$ and $a \neq 0$, then $ab = a0$; cancellation gives $b = 0$, so no zero divisors exist.
Structural Commentary: Domains are exactly cancellative commutative rings.
Key Insight: Zero divisors obstruct cancellation. □

Definition 3.1.15. *Field of Quotients*
Let R be an integral domain. On $R \times (R \setminus \{0\})$, set $(a, b) \sim (c, d) \iff ad = bc$.
The *field of quotients* of R is $(R \times (R \setminus \{0\}))/\sim$; write $[(a, b)] = a/b$. For $b, d \neq 0$, $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$; $\frac{a}{b}\frac{c}{d} = \frac{ac}{bd}$.

Proposition 3.1.16. *Domains Embed in Fields*
Let R be an integral domain with field of quotients K.
Then $\iota : R \hookrightarrow K$; $\iota(a) = a/1$ is an injective ring homomorphism.

Proof. By Proposition 3.1.14, the relation and operations in Definition 3.1.15 are well defined.
For $a, b \in R$, $\iota(a + b) = \frac{a+b}{1} = \frac{a}{1} + \frac{b}{1}$; $\iota(ab) = \frac{ab}{1} = \frac{a}{1}\frac{b}{1}$; and $\iota(1_R) = \frac{1_R}{1}$..
Thus ι is a ring homomorphism. If $\iota(a) = \iota(b)$, then $a/1 = b/1$, so $a = b$; hence ι is injective.
Structural Commentary: The field of quotients formally adds denominators to a domain.
Key Insight: Integral domains are exactly the setting where fraction arithmetic is consistent. □

Definition 3.1.17. *Characteristic*
Let R be a ring with identity. Its *characteristic* is $\operatorname{char}(R) := \begin{cases} \min\{n \in \mathbb{N}_{>0} : n1_R = 0_R\}, & \text{if this set is nonempty,} \\ 0, & \text{otherwise.} \end{cases}$.

Proposition 3.1.18. *Finite Domains Have Prime Characteristic*
Let R be a finite ring with identity. Then $\operatorname{char}(R) \neq 0$. If R is an integral domain, then $\operatorname{char}(R)$ is prime.

Proof. Since R is finite, the map $n \mapsto n1_R$ from $\mathbb{N}$ to R is not injective.
Thus $i1_R = j1_R$ for some $i < j$, so $(j - i)1_R = 0_R$ and $\operatorname{char}(R) > 0$.
Assume R is a domain and set $c = \operatorname{char}(R)$. Since $1_R \neq 0_R$, $c > 1$.
If $c = ab$ with $1 < a, b < c$, then $(a1_R)(b1_R) = ab1_R = c1_R = 0_R$.
By minimality of c, $a1_R \neq 0_R$ and $b1_R \neq 0_R$, contradicting that R has no zero divisors. Hence c is prime.
Structural Commentary: Characteristic records the additive order of the unit.
Key Insight: In a domain, a composite characteristic would factor zero nontrivially. □

Definition 3.1.19. *Algebras over a Field*
Let $\mathbb{F}$ be a field.
An *$\mathbb{F}$-algebra* is an $\mathbb{F}$-vector space $\mathcal{A}$ with a ring product satisfying $\lambda(ab) = (\lambda a)b = a(\lambda b)$ $\forall \lambda \in \mathbb{F}$ and $a, b \in \mathcal{A}$.

Example 3.1.20. *Matrix and Polynomial Algebras*
$\operatorname{Mat}(n, n, \mathbb{F})$ and $\mathbb{F}[x]$ are algebras over $\mathbb{F}$. The first is generally noncommutative; the second is commutative.

3.2 Polynomial Rings and Factorization

Polynomial rings over fields behave like the integers in the ways linear algebra needs: division with remainder, gcds, Bezout identities, irreducible factorization, and principal ideals.

Definition 3.2.1. *Polynomials over a Field*
Let $\mathbb{F}$ be a field.

(i) $\mathbb{F}[x]$ denotes the ring of polynomials in one variable over $\mathbb{F}$.
(ii) If $0 \neq p \in \mathbb{F}[x]$ and $p(x) = a_0 + a_1x + \cdots + a_nx^n$, $a_n \neq 0$, then $\operatorname{lc}(p) := a_n$, $\deg p := n$.
Also, p is *monic* iff $\operatorname{lc}(p) = 1$. Set $\deg 0 = -\infty$.
(iii) A nonzero $p \in \mathbb{F}[x]$ *splits over* $\mathbb{F}$ iff $p = \alpha \prod_{i=1}^{m} (x - \lambda_i)$ for some $m \geq 0$, $\alpha \in \mathbb{F}^\times$, and $\lambda_i \in \mathbb{F}$.

Theorem 3.2.2. *Polynomial Division Algorithm*
Let $f, g \in \mathbb{F}[x]$ with $g \neq 0$. Then $\exists!\,(q, r) \in \mathbb{F}[x]^2$ such that $f = qg + r$, $r = 0$ or $\deg r < \deg g$.

Proof. Let $d = \deg g$ and $\ell = \mathrm{lc}(g)$. Existence follows by induction on $\deg f$. If $f = 0$ or $\deg f < d$, take $(q, r) = (0, f)$. Otherwise, let $n = \deg f$, $a = \mathrm{lc}(f)$, and $f_1 = f - a\ell^{-1}x^{n-d}g$. Then $\deg f_1 < n$.
By induction, $f_1 = q_1 g + r$ with $r = 0$ or $\deg r < d$, hence $f = (q_1 + a\ell^{-1}x^{n-d})g + r$.
For uniqueness, if $qg + r = q'g + r'$ with admissible remainders, then $(q - q')g = r' - r$.
If $q \neq q'$, then $\deg((q - q')g) \geq d$, while $\deg(r' - r) < d$, contradiction. Hence $q = q'$ and then $r = r'$.
Structural Commentary: Division works because the leading coefficient of g is invertible in the field.
Key Insight: Cancel leading terms until the remainder is too small to divide again. □

Definition 3.2.3. *Divisibility, GCDs, and Coprimality*
For $p, q \in \mathbb{F}[x]$, write $p \mid q$ if $q = hp$ for some $h \in \mathbb{F}[x]$.
A *greatest common divisor* of $f, g \in \mathbb{F}[x]$, not both zero, is the unique monic polynomial $\gcd(f, g)$ such that:

(i) $\gcd(f, g) \mid f$ and $\gcd(f, g) \mid g$;
(ii) every common divisor of f and g divides $\gcd(f, g)$.

The polynomials f, g are *relatively prime* if $\gcd(f, g) = 1$.

Theorem 3.2.4. *Polynomial GCD and Bezout Identity*
For $f, g \in \mathbb{F}[x]$, not both zero, $\exists$ a unique monic $d \in \mathbb{F}[x]$ s.t. $d \mid f$, $d \mid g$, $h \mid f$, $h \mid g \Longrightarrow h \mid d$
Moreover, $d = \gcd(f, g)$ satisfies $d = af + bg$ for some $a, b \in \mathbb{F}[x]$.
Hence $\gcd(f, g) = 1$ iff $\exists\, a, b \in \mathbb{F}[x]$ with $af + bg = 1$.

Proof. If one of f, g is 0, the claim is immediate from the monic associate of the other.
Assume $f, g \neq 0$ and run the Euclidean algorithm: $r_{i-1} = q_i r_i + r_{i+1}$, where $r_{i+1} < \deg r_i$, starting with $(r_0, r_1) = (f, g)$ and ending with $r_N \neq 0$, $r_{N+1} = 0$. Then $r_N \mid r_{N-1}$, and upward substitution gives $r_N \mid f, g$.
Conversely, any common divisor of f, g divides every r_i, hence divides r_N. Thus $d := \mathrm{lc}(r_N)^{-1} r_N$ is a monic gcd; uniqueness follows since two monic gcds divide each other.
Back-substitution gives $d = af + bg$ for some $a, b \in \mathbb{F}[x]$. Finally, $d = 1$ gives $af + bg = 1$; conversely, if $af + bg = 1$, every common divisor divides 1, so $d = 1$.
Structural Commentary: The Euclidean algorithm produces both the gcd and its Bezout certificate.
Key Insight: GCD data is ideal data in $\mathbb{F}[x]$. □

Definition 3.2.5. *Irreducible Polynomials*
A nonconstant polynomial $p \in \mathbb{F}[x]$ is *irreducible over* $\mathbb{F}$ if every factorization $p = ab$ in $\mathbb{F}[x]$ has a or b constant.

Proposition 3.2.6. *Irreducible Polynomials Are Prime in $\mathbb{F}[x]$*
Let $p \in \mathbb{F}[x]$ be nonconstant. Then p is irreducible over $\mathbb{F}$ iff $\forall f, g \in \mathbb{F}[x]$, $p \mid fg \Longrightarrow p \mid f$ or $p \mid g$.

Proof.

(⇒) Assume p irreducible and $p \mid fg$. If $p \mid f$, done. Otherwise $\gcd(p, f) = 1$, so Theorem 3.2.4 gives $a, b \in \mathbb{F}[x]$ with $ap + bf = 1$. Multiplying by g gives $apg + bfg = g$; since $p \mid apg$ and $p \mid bfg$, we get $p \mid g$.
(⇐) Assume the divisibility property and write $p = ab$. Since $p \mid ab$, either $p \mid a$ or $p \mid b$. If $p \mid a$, then $a = ph$, so $p = phb$ and cancellation gives $1 = hb$, hence $b \in \mathbb{F}^\times$. Similarly, $p \mid b$ forces $a \in \mathbb{F}^\times$. Thus p is irreducible.

Structural Commentary: Over $\mathbb{F}[x]$, irreducible and prime are the same notion.
Key Insight: Bezout turns coprimality into a divisibility test. □

Theorem 3.2.7. *Unique Factorization in $\mathbb{F}[x]$*

For every nonconstant $f \in \mathbb{F}[x]$, $\exists\, c \in \mathbb{F}^\times$ and monic irreducibles $p_1, \ldots, p_m \in \mathbb{F}[x]$ such that $f = c \prod_{i=1}^{m} p_i$.

If $f = d \prod_{j=1}^{n} q_j$ is another such factorization, then $m = n$, $c = d$, and $p_i = q_{\sigma(i)}$ for some $\sigma \in S_m$.

Proof.

(i) *Existence*: Induct on $\deg f$. If f is irreducible, write $f = \mathrm{lc}(f)\big(\mathrm{lc}(f)^{-1} f\big)$.
If $f = uv$ with $0 < \deg u, \deg v < \deg f$, apply induction to u and v and concatenate the factors.
(ii) *Uniqueness*: Suppose $c \prod_{i=1}^{m} p_i = d \prod_{j=1}^{n} q_j$ with all p_i, q_j monic irreducible. Since $p_1 \mid \prod_{j=1}^{n} q_j$, iterating Proposition 3.2.6 gives $p_1 \mid q_j$ for some j. As p_1, q_j are monic irreducible, $p_1 = q_j$.
Cancel and continue by induction on m; the final scalar equality follows from leading coefficients.

Structural Commentary: Polynomial factorization over a field mirrors prime factorization in $\mathbb{Z}$.
Key Insight: The prime property of irreducibles drives uniqueness. □

Definition 3.2.8. *Units, Associates, Prime and Irreducible Elements*
Let R be an integral domain.

(i) A *unit* is an element $u \in R$ with $uv = 1$ for some $v \in R$.
(ii) Nonzero elements $a, b \in R$ are *associates*, written $a \sim b$, if $a = ub$ for some unit u.
(iii) A nonzero nonunit $p \in R$ is *prime* if $p \mid ab$ implies $p \mid a$ or $p \mid b$.
(iv) A nonzero nonunit $r \in R$ is *irreducible* if $r = ab$ implies a or b is a unit.

Proposition 3.2.9. *Divisibility and Principal Ideals*
Let R be an integral domain and let $r, s \in R$ be nonzero.

(i) $r \in R^\times \iff \langle r \rangle = R$.
(ii) $r \sim s \iff \langle r \rangle = \langle s \rangle$.
(iii) $r \mid s \iff \langle s \rangle \subseteq \langle r \rangle$.

Proof.

(i) ($\Rightarrow$) If $rv = 1$, then $1 \in \langle r \rangle$, so $\langle r \rangle = R$.
(i) ($\Leftarrow$) If $\langle r \rangle = R$, then $1 = ar$ for some $a \in R$, hence $r \in R^\times$.
(ii) ($\Rightarrow$) If $r = us$ with $u \in R^\times$, then $r \in \langle s \rangle$ and $s = u^{-1}r \in \langle r \rangle$, so $\langle r \rangle = \langle s \rangle$.
(ii) ($\Leftarrow$) If $\langle r \rangle = \langle s \rangle$, then $r = as$ and $s = br$ for some $a, b \in R$.
Thus $r = abr$, so $ab = 1$ by cancellation; hence $a \in R^\times$ and $r \sim s$.
(iii) ($\Rightarrow$) If $s = ar$, then every element of $\langle s \rangle$ is of the form $xs = xar \in \langle r \rangle$.
(iii) ($\Leftarrow$) If $\langle s \rangle \subseteq \langle r \rangle$, then $s \in \langle r \rangle$, so $s = ar$ for some $a \in R$; hence $r \mid s$.

Structural Commentary: Divisibility reverses inclusion of principal ideals.
Key Insight: Principal ideals translate multiplicative divisibility into order by containment. □

Definition 3.2.10. *PIDs and UFDs*
Let R be an integral domain.

(i) R is a *principal ideal domain* if every ideal $\mathcal{I} \trianglelefteq R$ has $\mathcal{I} = \langle a \rangle$ for some $a \in R$.
(ii) R is a *unique factorization domain* if every $0 \neq a \in R \setminus R^\times$ factors into irreducibles, uniquely up to reordering and associates.

Theorem 3.2.11. *$\mathbb{Z}$ and $\mathbb{F}[x]$ Are Principal Ideal Domains*

(i) If $0 \neq \mathcal{I} \trianglelefteq \mathbb{Z}$, then $\mathcal{I} = \langle d \rangle$ for $d = \min(\mathcal{I} \cap \mathbb{N}_{>0})$.
(ii) If $0 \neq \mathcal{I} \trianglelefteq \mathbb{F}[x]$, then $\mathcal{I} = \langle m \rangle$ for the unique monic $m \in \mathcal{I}$ of minimal degree.

Proof.

(i) If $\mathcal{I} = 0$, then $\mathcal{I} = \langle 0 \rangle$. Otherwise choose $d = \min(\mathcal{I} \cap \mathbb{N}_{>0})$. For $a \in \mathcal{I}$, write $a = qd + r$ with $0 \leq r < d$. Since $r = a - qd \in \mathcal{I}$, minimality gives $r = 0$, so $\mathcal{I} = \langle d \rangle$.
(ii) If $\mathcal{I} = 0$, then $\mathcal{I} = \langle 0 \rangle$. Otherwise choose monic $m \in \mathcal{I}$ of minimal degree. For $p \in \mathcal{I}$, divide $p = qm + r$ with $r = 0$ or $\deg r < \deg m$. Since $r = p - qm \in \mathcal{I}$, minimality gives $r = 0$, so $\mathcal{I} = \langle m \rangle$.
If m' is another monic minimal generator, then $m \mid m'$ and $m' \mid m$, hence $m = m'$.

Structural Commentary: Euclidean division turns arbitrary ideals into one-generator ideals.
Key Insight: Choose the smallest positive integer or smallest-degree monic polynomial and divide by it. □

Corollary 3.2.12. *GCDs Generate Polynomial Ideals*
For $0 \neq p_1, \ldots, p_n \in \mathbb{F}[x]$ and monic $m = \gcd(p_1, \ldots, p_n)$, $\langle p_1, \ldots, p_n \rangle = \langle m \rangle$.

Proof. By Theorem 3.2.11, $\langle p_1, \ldots, p_n \rangle = \langle g \rangle$ for a unique monic g. Since $p_i \in \langle g \rangle$, $g \mid p_i$ for all i.
If $q \mid p_i$ for all i, then $\langle p_1, \ldots, p_n \rangle \subseteq \langle q \rangle$, so $g \in \langle q \rangle$ and $q \mid g$. Hence $g = m$, and the claim follows.
Structural Commentary: In $\mathbb{F}[x]$, gcds and finitely generated ideals are the same data.
Key Insight: The gcd is the monic generator of the ideal generated by the polynomials. □

Example 3.2.13. *A Polynomial Ring That Is Not a PID*
In $\mathbb{F}[x, y]$, the ideal $\langle x, y \rangle$ is not principal. If $\langle x, y \rangle = \langle p \rangle$, then $p \mid x$ and $p \mid y$, so $p \in \mathbb{F}^\times$. Hence $\langle x, y \rangle = \mathbb{F}[x, y]$, contradicting $1 \notin \langle x, y \rangle$ since every element of $\langle x, y \rangle$ has zero constant term.

Proposition 3.2.14. *PIDs Satisfy the Ascending Chain Condition*
If R is a PID and $\mathcal{I}_1 \subseteq \mathcal{I}_2 \subseteq \cdots$ is a chain of ideals, then $\exists N$ such that

$$\mathcal{I}_j = \mathcal{I}_N \qquad \forall j \geq N.$$

Proof. Let $\mathcal{I} = \bigcup_{j\geq 1} \mathcal{I}_j$. By Proposition 3.1.7, $\mathcal{I}$ is an ideal. Since R is a PID, $\mathcal{I} = \langle a \rangle$. Choose N with $a \in \mathcal{I}_N$. Then $\mathcal{I} = \langle a \rangle \subseteq \mathcal{I}_N \subseteq \mathcal{I}$, so $\mathcal{I}_j = \mathcal{I}_N$ for all $j \geq N$.
Structural Commentary: The generator appears at a finite stage.
Key Insight: Principal generation converts infinite ideal growth into a finite containment. □

Theorem 3.2.15. *Prime, Irreducible, and Bezout in a PID*
Let R be a principal ideal domain.

(i) For $0 \neq r \in R \setminus R^\times$, r is irreducible iff $\langle r \rangle$ is maximal among proper ideals.
(ii) Every irreducible $r \in R$ is prime.
(iii) For $a, b \in R$, $\gcd(a, b) = 1$ iff $\exists x, y \in R$ such that $xa + yb = 1$.

Proof.

(i) If r is irreducible and $\langle r \rangle \subseteq \mathcal{I} \subseteq R$, write $\mathcal{I} = \langle a \rangle$. Then $r = xa$, so irreducibility gives $\mathcal{I} = \langle r \rangle$ or $\mathcal{I} = R$. Conversely, if $r = ab$ with $a, b \notin R^\times$, then $\langle r \rangle \subsetneq \langle a \rangle \subsetneq R$.
(ii) Let r be irreducible and $r \mid ab$. By (i), $\langle r, a \rangle = \langle r \rangle$ or R. In the first case $r \mid a$. In the second, $1 = xr + ya$, so $b = xrb + yab$ and $r \mid b$.
(iii) If $xa + yb = 1$, every common divisor of a, b is a unit. Conversely, let $\langle a, b \rangle = \langle d \rangle$. If $\gcd(a, b) = 1$, then $d \in R^\times$, so $\langle a, b \rangle = R$ and $1 = xa + yb$ for some $x, y \in R$.

Structural Commentary: In a PID, factorization statements become ideal-containment statements.
Key Insight: Maximal principal ideals are the bridge from irreducible to prime. □

Theorem 3.2.16. *Every PID Is a UFD*
If R is a PID, then R is a UFD.

Proof. Let $0 \neq r \in R \setminus R^\times$. If factorization into nonunits never terminates, obtain $s_1, s_2, \ldots$ with $s_{j+1} \mid s_j$ properly. By Proposition 3.2.9,

$$\langle s_1 \rangle \subsetneq \langle s_2 \rangle \subsetneq \cdots,$$

contradicting Proposition 3.2.14; hence irreducible factorizations exist.
For uniqueness, let $p_1 \cdots p_m = q_1 \cdots q_n$ with all p_i, q_j irreducible. By Theorem 3.2.15, p_1 is prime, so $p_1 \mid q_j$ for some j; irreducibility gives $p_1 \sim q_j$. Cancel associates and iterate. Thus $m = n$ and the factors agree up to order and associates.
Structural Commentary: ACC gives existence; irreducible-is-prime gives uniqueness.
Key Insight: PIDs have enough ideal control to force unique factorization. □

Definition 3.2.17. *Roots of Unity and Cyclotomic Polynomials*
For $n \geq 1$:

(i) The complex nth roots of unity are $U_n := \{z \in \mathbb{C} : z^n = 1\}$.
(ii) A root $\omega \in U_n$ is *primitive* iff $\operatorname{ord}(\omega) = n$; set $\Omega_n := \{\omega \in U_n : \operatorname{ord}(\omega) = n\}$.
(iii) The nth *cyclotomic polynomial* is $\Phi_n(x) := \prod_{\omega \in \Omega_n} (x - \omega)$.
(iv) The number of primitive nth roots is $\varphi(n) := |\Omega_n|$.

Proposition 3.2.18. *Roots of Unity Factorization*
Let $\zeta_n = e^{2\pi i/n}$. Then $U_n = \langle \zeta_n \rangle$; $|U_n| = n$; $\Omega_n = \{\zeta_n^k : 1 \leq k \leq n,\ \gcd(k, n) = 1\}$; $|\Omega_n| = \varphi(n)$. Moreover, $U_n = \coprod_{d \mid n} \Omega_d$ and $x^n - 1 = \prod_{d \mid n} \Phi_d(x)$.

Proof. The roots of $x^n - 1$ are precisely ζ_n^k for $0 \leq k < n$, hence $U_n = \langle \zeta_n \rangle$ and $|U_n| = n$.
In a cyclic group, $\operatorname{ord}(\zeta_n^k) = n/\gcd(k, n)$, so ζ_n^k is primitive iff $\gcd(k, n) = 1$; thus $|\Omega_n| = \varphi(n)$.
Each $z \in U_n$ has unique order $d \mid n$, giving $U_n = \coprod_{d \mid n} \Omega_d$.
Therefore $x^n - 1 = \prod_{z \in U_n} (x - z) = \prod_{d \mid n} \prod_{\omega \in \Omega_d} (x - \omega) = \prod_{d \mid n} \Phi_d(x)$ *Structural Commentary*: Orders partition roots of unity by divisors.
Key Insight: Cyclicity turns root counting into order counting. □

Proposition 3.2.19. *Cyclotomic Polynomials Have Integer Coefficients*
For every $n \geq 1$, $\Phi_n(x) \in \mathbb{Z}[x]$ is monic.

Proof. Induct on n. For $n = 1$, $\Phi_1(x) = x - 1$.
Assume $\Phi_d \in \mathbb{Z}[x]$ for every proper divisor $d \mid n$, and set $P_n(x) := \prod_{\substack{d \mid n \\ d < n}} \Phi_d(x) \in \mathbb{Z}[x]$.
Then P_n is monic, and Proposition 3.2.18 gives $x^n - 1 = \Phi_n(x) P_n(x)$.

Divide $x^n - 1$ by P_n in $\mathbb{Z}[x]$: $x^n - 1 = hP_n + r$, where $h, r \in \mathbb{Z}[x]$ and $r = 0$ or $\deg r < \deg P_n$.
Uniqueness of division over $\mathbb{C}[x]$ gives $h = \Phi_n$ and $r = 0$. Hence $\Phi_n \in \mathbb{Z}[x]$, and Φ_n is monic by definition.
Structural Commentary: Proper divisors isolate the next cyclotomic factor.
Key Insight: Monic division keeps the quotient in $\mathbb{Z}[x]$. □

Lemma 3.2.20. *A Cyclotomic Size Estimate*
For $n > 1$ and $z \in \mathbb{Z}_{\geq 2}$, $|\Phi_n(z)| > z - 1$.

Proof. For $\omega \in \Omega_n$, $\omega \neq 1$ and $|\omega| = 1$, hence $|z - \omega| > z - 1$. Thus $|\Phi_n(z)| = \prod_{\omega \in \Omega_n} |z - \omega| > (z-1)^{\varphi(n)} \geq z - 1$.
Structural Commentary: Primitive factors all lie outside the closest point 1.
Key Insight: Each primitive root contributes a factor larger than $z - 1$. □

4 Fields and Galois Theory

4.1 Field Extensions

4.2 Splitting Fields and Normality

4.3 Galois Groups

4.4 Classical Applications

Perspective 4.4.1. *Finite Fields and Characteristic*
If $\mathrm{char}(F) = 2$, then $(\forall a \in F)\ 2a = 0$, so $a = -a$. A finite field has $|F| = p^n$ with p prime and $n \geq 1$; conversely, each prime power $q = p^n$ determines a unique field $\mathbb{F}_q$ up to isomorphism.
Thus the classification is field-theoretic, while the characteristic constraint is ring-theoretic.

4.5 Scalar Extension in Linear Algebra

Definition 4.5.1. *Complexification of a Real Vector Space*
Let $\mathcal{V}$ be a real vector space.

(i) The *complexification* of $\mathcal{V}$ is $\mathcal{V}_{\mathbb{C}} := \mathcal{V} \times \mathcal{V}$, with $(\mathbf{u}, \mathbf{v})$ denoted $\mathbf{u} + i\mathbf{v}$.
(ii) Addition and $\mathbb{C}$-scalar multiplication are

$$(\mathbf{u}_1 + i\mathbf{v}_1) + (\mathbf{u}_2 + i\mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{u}_2) + i(\mathbf{v}_1 + \mathbf{v}_2),$$
$$(a + bi)(\mathbf{u} + i\mathbf{v}) = (a\mathbf{u} - b\mathbf{v}) + i(b\mathbf{u} + a\mathbf{v}), \qquad a, b \in \mathbb{R}.$$

(iii) *Complexification map* is real-linear embedding $\mathrm{cpx} : \mathcal{V} \hookrightarrow \mathcal{V}_{\mathbb{C}}$; $\mathrm{cpx}(\mathbf{u}) = \mathbf{u} + i\mathbf{0}$, where $\{\mathbf{u} + i\mathbf{0} : \mathbf{u} \in \mathcal{V}\}$.
(iv) For $\mathbf{z} = \mathbf{u} + i\mathbf{v} \in \mathcal{V}_{\mathbb{C}}$, define $\mathrm{Re}\,\mathbf{z} := \mathbf{u}$ and $\mathrm{Im}\,\mathbf{z} := \mathbf{v}$. Hence $\mathbf{z} = \mathbf{0}$ iff $\mathrm{Re}\,\mathbf{z} = \mathrm{Im}\,\mathbf{z} = \mathbf{0}$.

Proposition 4.5.2. *Complexification Preserves Bases*
Let $\mathcal{V}$ be real. Then $\mathcal{V}_{\mathbb{C}}$ is complex, and every real basis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of $\mathcal{V}$ is a complex basis of $\mathcal{V}_{\mathbb{C}}$.
Thus $\dim_{\mathbb{C}}(\mathcal{V}_{\mathbb{C}}) = \dim_{\mathbb{R}}(\mathcal{V})$.

Proof. The vector-space axioms follow componentwise from those of $\mathcal{V}$.
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, write $\mathbf{u} = \sum_{j=1}^{n} a_j \mathbf{v}_j$ and $\mathbf{v} = \sum_{j=1}^{n} b_j \mathbf{v}_j$, $a_j, b_j \in \mathbb{R}$. Then $\mathbf{u} + i\mathbf{v} = \sum_{j=1}^{n} (a_j + ib_j)\mathbf{v}_j$, so the list spans.
If $\sum_{j=1}^{n} \lambda_j \mathbf{v}_j = \mathbf{0}$ and $\lambda_j = a_j + ib_j$, then $\sum_{j=1}^{n} a_j \mathbf{v}_j = \mathbf{0}$ and $\sum_{j=1}^{n} b_j \mathbf{v}_j = \mathbf{0}$.
Real linear independence gives $a_j = b_j = 0$ for all j, hence $\lambda_j = 0$ for all j.
Structural Commentary: Complexification keeps the basis length unchanged.
Key Insight: Separate real and imaginary parts. □

Corollary 4.5.3. *Coordinates Survive Complexification*
Let $\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a real basis of $\mathcal{V}$ and set $\beta_{\mathbb{C}} := \mathrm{cpx}(\beta)$. Then, for all $\mathbf{v} \in \mathcal{V}$, $[\mathrm{cpx}(\mathbf{v})]_{\beta_{\mathbb{C}}} = [\mathbf{v}]_{\beta} \in \mathbb{R}^n \subseteq \mathbb{C}^n$.

Proof. If $[\mathbf{v}]_{\beta} = (a_1, \ldots, a_n)^t$, then $a_j \in \mathbb{R}$ and $\mathrm{cpx}(\mathbf{v}) = \mathbf{v} + i\mathbf{0} = \sum_{j=1}^{n} a_j(\mathbf{v}_j + i\mathbf{0}) = \sum_{j=1}^{n} a_j \,\mathrm{cpx}(\mathbf{v}_j)$.
Thus $[\mathrm{cpx}(\mathbf{v})]_{\beta_{\mathbb{C}}} = (a_1, \ldots, a_n)^t = [\mathbf{v}]_{\beta}$.
Structural Commentary: The embedded real copy keeps exactly the same coordinate columns.
Key Insight: Complexification enlarges the scalar field, not the coordinates of real vectors. □

Proposition 4.5.4. *Complexification as a Real Space*
If $\mathcal{V}$ has real basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, then $\mathcal{V}_{\mathbb{C}}$, as a real vector space, has basis $(\mathbf{v}_1, \dots, \mathbf{v}_n, i\mathbf{v}_1, \dots, i\mathbf{v}_n)$.
Hence $\dim_{\mathbb{R}}(\mathcal{V}_{\mathbb{C}}) = 2\dim_{\mathbb{R}}(\mathcal{V})$, and $\operatorname{cpx} : \mathcal{V} \to \mathcal{V}_{\mathbb{C}}$, $\operatorname{cpx}(\mathbf{v}) = \mathbf{v} + i\mathbf{0}$, is real-linear and injective.

Proof. For $\mathbf{u} = \sum a_j\mathbf{v}_j$ and $\mathbf{w} = \sum b_j\mathbf{v}_j$, $\mathbf{u} + i\mathbf{w} = \sum a_j\mathbf{v}_j + \sum b_j(i\mathbf{v}_j)$, so the displayed list spans $\mathcal{V}_{\mathbb{C}}$ over $\mathbb{R}$.
If $\sum a_j\mathbf{v}_j + \sum b_j(i\mathbf{v}_j) = \mathbf{0}$, then Re and Im give $\sum a_j\mathbf{v}_j = \sum b_j\mathbf{v}_j = \mathbf{0}$; all a_j, b_j vanish. Thus this is real basis.
For $r \in \mathbb{R}$, $\operatorname{cpx}(r\mathbf{v} + \mathbf{w}) = r\operatorname{cpx}(\mathbf{v}) + \operatorname{cpx}(\mathbf{w})$, and $\ker(\operatorname{cpx}) = \{\mathbf{0}\}$, so cpx is real-linear and injective.
Structural Commentary: Complexification is two real copies of $\mathcal{V}$.
Key Insight: The real and imaginary parts are independent real copies of the same basis. □

Definition 4.5.5. *Complexification of a Real Operator*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
The *complexification* of T is the operator $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$ defined by $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) := T\mathbf{u} + iT\mathbf{v}$.

Proposition 4.5.6. *Complexification Preserves Operator Data*
Let $\mathcal{V}$ be a real vector space and let $T \in \mathcal{L}(\mathcal{V})$.

(i) $T_{\mathbb{C}} \in \mathcal{L}(\mathcal{V}_{\mathbb{C}})$.
(ii) $(T_{\mathbb{C}})^k(\mathbf{u} + i\mathbf{v}) = T^k\mathbf{u} + iT^k\mathbf{v}$ for every integer $k \geq 0$.
(iii) If β is a real basis of $\mathcal{V}$, then $[T_{\mathbb{C}}]_{\beta \leftarrow \beta} = [T]_{\beta \leftarrow \beta}$, now regarded as a complex matrix.

Proof.

(i) Additivity is immediate. For $\alpha = a + bi \in \mathbb{C}$,
$$T_{\mathbb{C}}\big(\alpha(\mathbf{u} + i\mathbf{v})\big) = T(a\mathbf{u} - b\mathbf{v}) + iT(b\mathbf{u} + a\mathbf{v}) = (aT\mathbf{u} - bT\mathbf{v}) + i(bT\mathbf{u} + aT\mathbf{v}) = \alpha T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}),$$
so $T_{\mathbb{C}}$ is complex-linear.

(ii) For $k = 0$, the claim is the identity. If it holds for k, then
$$(T_{\mathbb{C}})^{k+1}(\mathbf{u} + i\mathbf{v}) = T_{\mathbb{C}}(T^k\mathbf{u} + iT^k\mathbf{v}) = T^{k+1}\mathbf{u} + iT^{k+1}\mathbf{v}.$$
Thus the formula holds for all $k \geq 0$.

(iii) If $T\mathbf{v}_j = \sum_i a_{ij}\mathbf{v}_i$ with $a_{ij} \in \mathbb{R}$, then $T_{\mathbb{C}}\mathbf{v}_j = \sum_i a_{ij}\mathbf{v}_i$. Hence the coordinate matrix is unchanged.

Structural Commentary: Complexification keeps the coordinate matrix unchanged.
Key Insight: Same real entries; larger scalar field. □



Figure 103: Complexification packages two real vectors: $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = T\mathbf{u} + iT\mathbf{v}$.

Method 4.5.7. *Complexifying a Real Matrix*
To use complex eigenvalues for a real matrix $A \in \operatorname{Mat}(n, n, \mathbb{R})$:

(i) Regard the same real matrix A as a complex matrix on $\mathbb{C}^n$.
(ii) Find complex eigenpairs or invariant data for this complexified operator.
(iii) If $A(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$, separate parts: $A\mathbf{u} = a\mathbf{u} - b\mathbf{v}$, $A\mathbf{v} = b\mathbf{u} + a\mathbf{v}$
(iv) Conclude that $\operatorname{span}(\mathbf{u}, \mathbf{v}) \subseteq \mathbb{R}^n$ is an invariant line or plane.

Theorem 4.5.8. *Invariant Line or Plane over $\mathbb{R}$*
Every operator on a nonzero finite-dimensional real vector space has an invariant subspace of dimension 1 or 2.

Proof. Let $\mathcal{V}$ be a nonzero finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
By Theorem 5.3.1, $T_{\mathbb{C}}$ has an eigenvalue $\lambda = a + bi$ with eigenvector $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$.
The equation $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = (a + bi)(\mathbf{u} + i\mathbf{v})$ gives $T\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $T\mathbf{v} = b\mathbf{u} + a\mathbf{v}$.
Thus $\mathcal{U} = \operatorname{span}(\mathbf{u}, \mathbf{v})$ is nonzero, has dimension 1 or 2, and is invariant under T.
Structural Commentary: A complex eigenvector decomposes into two real directions that are closed under T.
Key Insight: Nonreal eigenvalues become real invariant planes; real eigenvalues become invariant lines. □

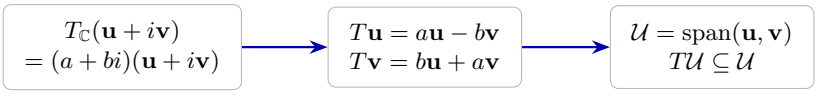


Figure 104: Separate a complex eigenpair into a real invariant plane.

Perspective 4.5.9. *Real Normal Operators Use Planes*
Over $\mathbb{R}$, normal operators need not have enough eigenvectors. The replacement for a missing conjugate eigenpair is a two-dimensional invariant plane carrying a rotation-dilation block.

Proposition 4.5.10. *Two-Dimensional Real Normal Blocks*
Let $\dim(\mathcal{V}) = 2$ *over* $\mathbb{R}$ *and let* $T \in \mathcal{L}(\mathcal{V})$.

Then T *is normal but not self-adjoint if and only if, in every orthonormal basis,* $[T] = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, $b \neq 0$.

Moreover, some orthonormal basis makes $b > 0$.

Proof.

($\Rightarrow$) In an orthonormal basis, write $A = [T] = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$.
Normality gives $A^tA = AA^t$. Comparing the $(1,1)$ and $(2,2)$ entries gives $b^2 = c^2$. If $c = b$, then $A = A^t$, impossible because T is not self-adjoint. Thus $c = -b$ and $b \neq 0$. Comparing off-diagonal entries gives $-ab + bd = ab - bd$, so $d = a$. Since the original orthonormal basis was arbitrary, every orthonormal basis has the displayed form.

($\Leftarrow$) A matrix $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ with $b \neq 0$ satisfies $A^tA = AA^t$ and is not symmetric. Hence T is normal but not self-adjoint. If $b < 0$, replace the second basis vector by its negative to make $b > 0$.

Structural Commentary: In real dimension 2, non-self-adjoint normality is exactly a scalar plus a rotation block.
Key Insight: Normality forces $c = -b$ and $d = a$. □

Theorem 4.5.11. *Real Normal Operator Block Form*
Let $\mathcal{V}$ *be finite-dimensional over* $\mathbb{R}$ *and let* $T \in \mathcal{L}(\mathcal{V})$. *Then* T *is normal if and only if some orthonormal basis makes* $[T]$ *block diagonal, with each diagonal block either* 1*-by-*1 *or of the form* $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, $b > 0$.

Proof.

($\Rightarrow$) Induct on $n = \dim(\mathcal{V})$. The cases $n \leq 1$ are immediate.
For $n \geq 2$, Theorem 4.5.8 gives an invariant subspace $\mathcal{U}$ with $\dim(\mathcal{U}) \in \{1, 2\}$.
By Corollary 7.2.9, $\mathcal{U}^\perp$ is invariant under T and both restrictions are normal.
If $\dim(\mathcal{U}) = 1$, take a unit vector in $\mathcal{U}$. If $\dim(\mathcal{U}) = 2$, use the Real Spectral Theorem 7.1.13 when $T|_{\mathcal{U}}$ is self-adjoint, giving scalar 1-by-1 blocks on $\mathcal{U}$, and use Proposition 4.5.10 otherwise.
By induction, choose the required orthonormal block basis on $\mathcal{U}^\perp$. Concatenating it with the basis chosen on $\mathcal{U}$ gives an orthonormal basis of $\mathcal{V}$ because $\mathcal{V} = \mathcal{U} \oplus \mathcal{U}^\perp$.

($\Leftarrow$) Each allowed block is normal. If $[T]$ is the block diagonal sum of these blocks, then $[T]^*[T]$ and $[T][T]^*$ are the corresponding block diagonal sums of B^*B and BB^*, hence are equal block by block.

Structural Commentary: Normality lets invariant lines or planes split orthogonally.
Key Insight: Real normal operators split into scalar lines and rotation-dilation planes. □

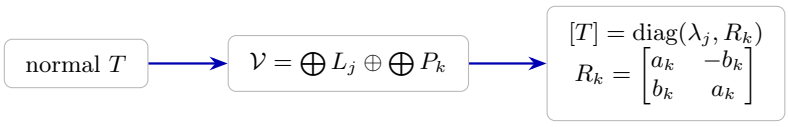


Figure 105: Orthogonal line/plane split into scalar and rotation blocks.

Example 4.5.12. *Plane Rotations*
For $\theta \in \mathbb{R}$, the real matrix $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is an isometry of $\mathbb{R}^2$. If θ is not an integer multiple of π, then R_θ has no real eigenvalue, which is why real isometries require two-dimensional rotation blocks.

Theorem 4.5.13. *Real Isometry Block Form*
Let $\mathcal{V}$ *be finite-dimensional over* $\mathbb{R}$ *and let* $S \in \mathcal{L}(\mathcal{V})$. *Then* S *is an isometry if and only if some orthonormal basis makes* $[S]$ *block diagonal, with each diagonal block one of* $[1]$, $[-1]$, $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ *with* $0 < \theta < \pi$.

Proof.

($\Rightarrow$) Since S is an isometry, $S^*S = I_{\mathcal{V}}$ by Theorem 2.4.26. Finite-dimensional invertibility gives $SS^* = I_{\mathcal{V}}$, so S is normal. Apply Theorem 4.5.11. A 1-by-1 block $[\lambda]$ has $\lambda^2 = 1$, so $\lambda = \pm 1$.
A 2-by-2 block $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ has $a^2 + b^2 = 1$ and $b > 0$, so $a = \cos\theta$ and $b = \sin\theta$ for a unique $\theta \in (0, \pi)$.

($\Leftarrow$) Each displayed block B satisfies $B^tB = I$ on its coordinate subspace. Therefore the block diagonal matrix satisfies $[S]^t[S] = I$, so S preserves norms and is an isometry.

Structural Commentary: Isometries are the unit-length special case of real normal blocks.
Key Insight: Scalar blocks reduce to ± 1, and planar normal blocks reduce to rotations. □

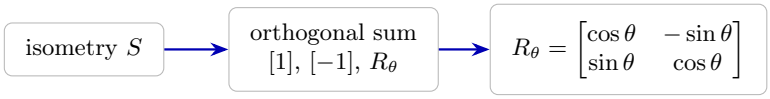


Figure 106: Normal form: fixed/flipped lines and rotation planes.

Proposition 4.5.14. *Real Eigenvalues of a Complexification*
Let $\mathcal{V}$ be real, let $T \in \mathcal{L}(\mathcal{V})$, and let $\lambda \in \mathbb{R}$. Then λ is an eigenvalue of T if and only if it is an eigenvalue of $T_{\mathbb{C}}$.

Proof.
($\Rightarrow$) If $T\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then $T_{\mathbb{C}}(\mathbf{v} + i\mathbf{0}) = \lambda(\mathbf{v} + i\mathbf{0})$.
($\Leftarrow$) If $T_{\mathbb{C}}(\mathbf{u} + i\mathbf{v}) = \lambda(\mathbf{u} + i\mathbf{v})$ and $\mathbf{u} + i\mathbf{v} \neq \mathbf{0}$, then $T\mathbf{u} = \lambda\mathbf{u}$ and $T\mathbf{v} = \lambda\mathbf{v}$. Since $\mathbf{u}$ or $\mathbf{v}$ is nonzero, T has eigenvalue λ.

Structural Commentary: Real eigenvalues survive complexification exactly.
Key Insight: For real λ, real and imaginary parts decouple. □

Lemma 4.5.15. *Conjugation Transfers Generalized Eigenspaces*
Let $\mathcal{V}$ be real, let $T \in \mathcal{L}(\mathcal{V})$, let $\lambda \in \mathbb{C}$, and let $k \geq 0$. Then $(T_{\mathbb{C}} - \lambda I)^k(\mathbf{u} + i\mathbf{v}) = \mathbf{0} \iff (T_{\mathbb{C}} - \overline{\lambda} I)^k(\mathbf{u} - i\mathbf{v}) = \mathbf{0}$.

Proof. Let $\Gamma(\mathbf{u} + i\mathbf{v}) = \mathbf{u} - i\mathbf{v}$.
Then Γ is a conjugate-linear bijection, $\Gamma T_{\mathbb{C}} = T_{\mathbb{C}}\Gamma$, and $\Gamma\big((T_{\mathbb{C}} - \lambda I)^k\mathbf{x}\big) = (T_{\mathbb{C}} - \overline{\lambda} I)^k\Gamma\mathbf{x}$.
($\Rightarrow$) Apply the displayed identity to $\mathbf{x} = \mathbf{u} + i\mathbf{v}$.
($\Leftarrow$) Apply the first direction with $\overline{\lambda}$ in place of λ and $\mathbf{u} - i\mathbf{v}$ in place of $\mathbf{u} + i\mathbf{v}$.

Structural Commentary: Conjugation commutes with $T_{\mathbb{C}}$ and conjugates scalars.
Key Insight: Conjugation pairs generalized eigenspaces. □

Corollary 4.5.16. *Conjugate Eigenvalues Have Equal Multiplicity*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$.
If λ is an eigenvalue of $T_{\mathbb{C}}$, then $\overline{\lambda}$ is an eigenvalue of $T_{\mathbb{C}}$, and $\dim_{\mathbb{C}} \mathcal{G}_\lambda(T_{\mathbb{C}}) = \dim_{\mathbb{C}} \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$.

Proof. Lemma 4.5.15 with $k = 1$ sends a nonzero eigenvector for λ to a nonzero eigenvector for $\overline{\lambda}$. Thus $\overline{\lambda}$ is an eigenvalue. With $k = \dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}}$, Theorem 6.4.8 identifies generalized eigenspaces with the corresponding stabilized kernels. By Lemma 4.5.15, $\Gamma(\mathbf{u}+i\mathbf{v}) = \mathbf{u}-i\mathbf{v}$ restricts to conjugate-linear bijection $\mathcal{G}_\lambda(T_{\mathbb{C}}) \longrightarrow \mathcal{G}_{\overline{\lambda}}(T_{\mathbb{C}})$. Conjugate-linear bijections preserve complex dimension, so the multiplicities agree.
Structural Commentary: Conjugation mirrors generalized eigenspaces.
Key Insight: Nonreal spectral data of a real operator arrives in equal-sized conjugate pairs. □

Corollary 4.5.17. *Odd-Dimensional Real Operators Have Eigenvalues*
Every operator on a nonzero odd-dimensional real vector space has a real eigenvalue.

Proof. Let $T \in \mathcal{L}(\mathcal{V})$, where $\dim_{\mathbb{R}} \mathcal{V}$ is odd. By Proposition 4.5.2, $\dim_{\mathbb{C}} \mathcal{V}_{\mathbb{C}}$ is odd.
The sum of the multiplicities of all eigenvalues of $T_{\mathbb{C}}$ equals this odd dimension by Proposition 6.5.4. Nonreal eigenvalues contribute in conjugate pairs of equal multiplicity by Corollary 4.5.16, hence contribute an even total. Therefore $T_{\mathbb{C}}$ has at least one real eigenvalue, which is an eigenvalue of T by Proposition 4.5.14.
Structural Commentary: Odd dimension forces a real eigenvalue.
Key Insight: Parity forces at least one real spectral direction. □

Definition 4.5.18. *Characteristic Polynomial of a Real Operator*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
The *characteristic polynomial* of T is $q_T := q_{T_{\mathbb{C}}}$, the characteristic polynomial of its complexification.

Theorem 4.5.19. *Characteristic Polynomial over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space, let $T \in \mathcal{L}(\mathcal{V})$, and let q_T be its characteristic polynomial.
(i) $q_T \in \mathbb{R}[z]$.
(ii) $\deg q_T = \dim_{\mathbb{R}} \mathcal{V}$.
(iii) The real zeros of q_T are precisely the eigenvalues of T.

(iv) $q_T(T) = 0$.

Proof.

(i) Factor $q_{T_\mathbb{C}}$ over $\mathbb{C}$. Real eigenvalues give real linear factors. If $\lambda \notin \mathbb{R}$ has multiplicity m, then $\overline{\lambda}$ has multiplicity m by Corollary 4.5.16, and $(z-\lambda)^m(z-\overline{\lambda})^m = \left(z^2 - 2\operatorname{Re}(\lambda)z + |\lambda|^2\right)^m$, which is real.
(ii) Proposition 6.2.12 and Proposition 4.5.2 give $\deg q_T = \dim_\mathbb{R} \mathcal{V}$.
(iii) Real zeros of $q_T = q_{T_\mathbb{C}}$ are real eigenvalues of $T_\mathbb{C}$ (Prop 6.2.12), hence eigenvalues of T (Prop 4.5.14).
(iv) Cayley–Hamilton over $\mathbb{C}$ gives $q_T(T_\mathbb{C}) = 0$. By (i) and Proposition 4.5.6, $q_T(T_\mathbb{C}) = (q_T(T))_\mathbb{C}$. If $q_T(T)$ were nonzero on some $\mathbf{u} \in \mathcal{V}$, then $(q_T(T))_\mathbb{C}(\mathbf{u} + i\mathbf{0}) = q_T(T)\mathbf{u} + i\mathbf{0}$ would be nonzero, a contradiction. Hence $q_T(T) = 0$.

Structural Commentary: Nonreal factors pair into real quadratics.
Key Insight: Complexify, then pair conjugates. □

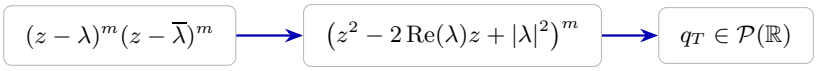


Figure 107: Conjugate factors pair into real quadratics.

Corollary 4.5.20. *q_T Is a Multiple of μ_T over $\mathbb{R}$*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be a finite-dimensional real vector space and let $T \in \mathcal{L}(\mathcal{V})$.
Then $\deg \mu_T \leq \dim_\mathbb{R} \mathcal{V}$, and q_T is a polynomial multiple of μ_T in $\mathbb{R}[z]$.

Proof. Theorem 4.5.19 gives $q_T(T) = 0$.
By Theorem 6.3.10, μ_T divides q_T in $\mathbb{R}[z]$. The degree bound follows from $\deg q_T = \dim_\mathbb{R} \mathcal{V}$.
Structural Commentary: Cayley–Hamilton makes q_T a real annihilator of T.
Key Insight: The minimal polynomial divides every real annihilator. □

Proposition 4.5.21. *Minimal Polynomial Survives Complexification*
Let $\mathcal{V} \neq \{\mathbf{0}\}$ be finite-dimensional over $\mathbb{R}$ and let $T \in \mathcal{L}(\mathcal{V})$. Then $\mu_{T_\mathbb{C}} = \mu_T$ in $\mathbb{C}[z]$.

Proof.

($\Rightarrow$) Since $\mu_T(T) = 0$, Proposition 4.5.6 gives $\mu_T(T_\mathbb{C}) = (\mu_T(T))_\mathbb{C} = 0$. Hence $\mu_{T_\mathbb{C}}$ divides μ_T in $\mathbb{C}[z]$.
($\Leftarrow$) Let $q = \mu_{T_\mathbb{C}} = r + is$, with $r, s \in \mathbb{R}[z]$. Since q is monic, r is monic and $\deg r = \deg q$. For every $\mathbf{u} \in \mathcal{V}$, $0 = q(T_\mathbb{C})(\mathbf{u} + i\mathbf{0}) = r(T)\mathbf{u} + is(T)\mathbf{u}$. Thus $r(T) = 0$, so $\deg \mu_T \leq \deg r = \deg q$. The first part gives opposite degree inequality. As q divides μ_T and both are monic with equal degree, $q = \mu_T$.

Structural Commentary: Annihilation passes to complexification and back by real-part extraction.
Key Insight: Evaluate $q(T_\mathbb{C})$ on $\mathbf{u} + i\mathbf{0}$. □

Part VI
Modules, Multilinear Algebra, and Homological Methods

1 Module Theory

1.1 Modules and Submodules

Throughout this subsection, R denotes a nonzero commutative ring with identity 1.
Modules are vector spaces with the scalar field replaced by a ring; the price is that nonzero scalars may fail to be invertible and may even kill nonzero vectors.

Perspective 1.1.1. *Polynomial Scalars from One Operator*
Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and let $T \in \mathcal{L}(\mathcal{V})$. Every polynomial $p(z) \in \mathbb{F}[z]$ acts on $\mathcal{V}$ by $p \cdot \mathbf{v} := p(T)\mathbf{v}$. Thus one operator turns $\mathcal{V}$ into an object with scalar ring $\mathbb{F}[z]$.

Definition 1.1.2. *R-Module*
An *R-module* is an abelian group $(M, +)$ equipped with a scalar action $R \times M \to M$, $(r, u) \mapsto ru$, such that for all $r, s \in R$ and $u, v \in M$,

(i) $r(u + v) = ru + rv$.
(ii) $(r + s)u = ru + su$.
(iii) $(rs)u = r(su)$.
(iv) $1u = u$.

The ring R is the *base ring* of M.

Example 1.1.3. *Basic Module Models*

(i) R^n is an R-module under componentwise addition and scalar multiplication.
(ii) $\mathrm{Mat}(m, n, R)$ is an R-module under entrywise operations.
(iii) R is an R-module over itself by ring multiplication.
(iv) Every vector space over a field $\mathbb{F}$ is an $\mathbb{F}$-module.

Structural Commentary: Modules are abelian groups with compatible scalar actions.
Key Insight: The new behavior comes from the scalar ring, not from changing addition.

Perspective 1.1.4. *The Base Ring Controls the Geometry*
The structure of an R-module is governed by the arithmetic of R. When R is a field, every nonzero scalar is invertible, giving the usual theory of bases, dimension, and complements. For a general ring, nonunits and zero divisors obstruct division and cancellation, so these notions need not retain their vector-space properties.

Definition 1.1.5. *Submodule*
A *submodule* of an R-module M is a nonempty subset $N \subseteq M$ that is an R-module under the operations inherited from M, written $N \leq M$.

Theorem 1.1.6. *Submodule Test*
Let M be an R-module and let $\emptyset \neq N \subseteq M$. Then $N \leq M \iff \forall r, s \in R,\ \forall u, v \in N,\ ru + sv \in N$.

Proof.

(⇒) If $N \leq M$, then $ru, sv \in N$, hence $ru + sv \in N$.
(⇐) Assume the displayed condition. Choose $n \in N$. Then $0 = 0n + 0n \in N$.
For $x, y \in N$ and $r \in R$, $x + y = 1x + 1y \in N$, $-x = (-1)x + 0x \in N$, $rx = rx + 0x \in N$.
Thus N is an additive subgroup of M closed under R-scalar multiplication, so the remaining module axioms are inherited from M and $N \leq M$.

Structural Commentary: One closure test packages addition, additive inverses, and scalar multiplication.
Key Insight: Submodules are exactly the subsets closed under all two-term R-linear combinations. □

Proposition 1.1.7. *Intersections and Sums of Submodules*
Let M be an R-module.
If $A, B \leq M$, then $A \cap B \leq M$; $A + B := \{a + b : a \in A,\ b \in B\} \leq M$, and $A + B = \bigcap_{\substack{N \leq M \\ A \cup B \subseteq N}} N$.

Proof. Since $0 \in A \cap B$ and $rx+sy \in A \cap B$ for all $x, y \in A \cap B$ and $r, s \in R$, the submodule test gives $A \cap B \leq M$.
If $x = a + b$ and $y = a' + b'$ with $a, a' \in A$ and $b, b' \in B$, then $rx + sy = (ra + sa') + (rb + sb') \in A + B$.
Thus $A + B \leq M$. Moreover, $A \cup B \subseteq A + B$, and if $N \leq M$ contains $A \cup B$, then $a + b \in N$ for all $a \in A$, $b \in B$; hence $A + B \subseteq N$. This proves the displayed intersection formula.
Structural Commentary: $A \cap B$ is the meet; $A + B$ is the join.
Key Insight: Sums are forced by closure under addition. □

Definition 1.1.8. *Ideal*
An *ideal* of R is an additive subgroup $\mathcal{I} \leq (R, +)$ such that $R\mathcal{I} := \{ri : r \in R,\ i \in \mathcal{I}\} \subseteq \mathcal{I}$, written $\mathcal{I} \trianglelefteq R$.

Proposition 1.1.9. *Ideals Are Submodules of the Regular Module*
View R as the regular R-module ${}_RR$. For every $N \subseteq R$,

$$N \leq {}_RR \quad \Longleftrightarrow \quad N \trianglelefteq R.$$

Proof. For $N \subseteq R$,

$$N \leq {}_RR \quad \Longleftrightarrow \quad N \leq (R, +) \text{ and } RN \subseteq N \quad \Longleftrightarrow \quad N \trianglelefteq R,$$

by Definition 1.1.8.
Structural Commentary: Ideals are the module-theoretic subobjects of the ring acting on itself.
Key Insight: Ring multiplication becomes scalar multiplication. □

1.2 Generated, Cyclic, and Finitely Generated Modules

Definition 1.2.1. *Generated Submodule*

(i) For $S \subseteq M$, *submodule generated by S* is $\langle\langle S \rangle\rangle_R := \left\{ \sum_{j=1}^{n} r_j s_j : n \geq 1,\ r_j \in R,\ s_j \in S \right\}$; $\langle\langle \emptyset \rangle\rangle_R = \{0\}$.

(ii) A subset $S \subseteq M$ *spans* or *generates* M if $\langle\langle S \rangle\rangle_R = M$.

Proposition 1.2.2. *Generated Submodule Is Smallest*
For every $S \subseteq M$, $\langle\langle S \rangle\rangle_R = \bigcap_{\substack{N \leq M \\ S \subseteq N}} N$.

Proof. Let $G := \langle\langle S \rangle\rangle_R$. By construction, $S \subseteq G$ and the submodule test gives $G \leq M$.
If $N \leq M$ and $S \subseteq N$, then $\sum_{j=1}^{n} r_j s_j \in N$, where $r_j \in R,\ s_j \in S$, so $G \subseteq N$.
Structural Commentary: Generation means closing a set under the operations the module axioms require.
Key Insight: The generated submodule adds no elements except those forced by finite R-linear combination. □

Definition 1.2.3. *Cyclic and Finitely Generated Modules*

(i) For $v \in M$, the *cyclic submodule generated by v* is $\langle\langle v \rangle\rangle_R = Rv = \{rv : r \in R\}$.
(ii) M is *finitely generated* if $M = \langle\langle v_1, \ldots, v_n \rangle\rangle_R$ for some $v_1, \ldots, v_n \in M$.
(iii) M is *cyclic* if $M = Rv$ for some $v \in M$.

Example 1.2.4. *A Finitely Generated Module with a Non-Finitely Generated Submodule*
Let $R = \mathbb{F}[x_1, x_2, \ldots]$. Then $R = \langle\langle 1 \rangle\rangle_R$ as an R-module.
Set $I := (x_1, x_2, \ldots) = \{f \in R : f(0, 0, \ldots) = 0\}$. Then $I \trianglelefteq R$, hence $I \leq R$, but I is not finitely generated.
If $I = \langle\langle p_1, \ldots, p_n \rangle\rangle_R$, choose k such that x_k occurs in no p_i. Let $\varphi_k : R \to \mathbb{F}[x_k]$ send $x_k \mapsto x_k$ and $x_j \mapsto 0$ for $j \neq k$. Since $\varphi_k(p_i) = 0$ for all i, every element of $\langle\langle p_1, \ldots, p_n \rangle\rangle_R$ maps to 0, but $\varphi_k(x_k) = x_k \neq 0$, contradiction.

Definition 1.2.5. *Linear Independence in a Module*

(i) Subset $S \subseteq M$ is *linearly independent over R* if,
$\forall$ distinct $v_1, \ldots, v_n \in S$, $r_1 v_1 + \cdots + r_n v_n = 0 \Longrightarrow r_1 = \cdots = r_n = 0$.
(ii) A subset $S \subseteq M$ is *linearly dependent* if it is not linearly independent.

Example 1.2.6. *Dependence Without Solving for One Vector*
In the $\mathbb{Z}$-module $\mathbb{Z}$, $3 \cdot 2 - 2 \cdot 3 = 0$. Thus $\{2, 3\}$ is linearly dependent.
However, $2 \notin 3\mathbb{Z}$; $3 \notin 2\mathbb{Z}$, so dependence need not express one element as a scalar multiple of another.

Proposition 1.2.7. *Scaling Independent Sets over a Domain*
Assume R is an integral domain. If $\{v_1, \ldots, v_n\}$ is R-linearly independent and $0 \neq r \in R$, then $\{rv_1, \ldots, rv_n\}$ is R-linearly independent.

Proof. Let $a_1, \dots, a_n \in R$ and suppose $a_1(rv_1) + \cdots + a_n(rv_n) = 0$. Then $(ra_1)v_1 + \cdots + (ra_n)v_n = 0$.
Independence gives $ra_i = 0$ for all i. Since R is a domain and $r \neq 0$, $a_i = 0$ for all i.
Structural Commentary: Domains preserve independence under nonzero scaling.
Key Insight: The failure of this statement is exactly zero-divisor behavior. □

1.3 Torsion and Quotients

Definition 1.3.1. *Torsion*

(i) A nonzero element $v \in M$ is a *torsion element* if $\exists\, 0 \neq r \in R$ such that $rv = 0$.
(ii) M is *torsion-free* if it has no torsion elements.
(iii) M is a *torsion module* if every nonzero element of M is torsion.
(iv) The *torsion set* of M is $M_{\text{tor}} := \{0\} \cup \{v \in M : v \text{ is torsion}\}$

Example 1.3.2. *$\mathbb{Z}/n\mathbb{Z}$ as a Torsion Module*
For $n > 1$, $\mathbb{Z}/n\mathbb{Z}$ is a $\mathbb{Z}$-module via $k \cdot \overline{a} = \overline{ka}$. For every $\overline{a} \in \mathbb{Z}/n\mathbb{Z}$, $n\overline{a} = \overline{0}$.
Hence $\mathbb{Z}/n\mathbb{Z}$ is a torsion module, and $\{\overline{a}\}$ is linearly dependent whenever $\overline{a} \neq \overline{0}$.

Proposition 1.3.3. *Torsion Submodule over an Integral Domain*
If R is an integral domain, then $M_{\text{tor}} \leq M$, $(M/M_{\text{tor}})_{\text{tor}} = \{0\}$.

Proof. Let $x, y \in M_{\text{tor}}$ and $r, s \in R$.
Choose $a, b \in R \setminus \{0\}$ with $ax = 0$ and $by = 0$, taking $a = 1$ if $x = 0$ and $b = 1$ if $y = 0$.
Since R is a domain, $ab \neq 0$, and $ab(rx + sy) = rb(ax) + sa(by) = 0$. Thus $rx + sy \in M_{\text{tor}}$, so $M_{\text{tor}} \leq M$.
Now let $0 \neq r \in R$ and suppose $r(m + M_{\text{tor}}) = M_{\text{tor}}$. Then $rm \in M_{\text{tor}}$, so $srm = 0$ for some $0 \neq s \in R$; if $rm = 0$, take $s = 1$. Since $sr \neq 0$, $m \in M_{\text{tor}}$, hence $m + M_{\text{tor}} = M_{\text{tor}}$.
Structural Commentary: Domains let annihilators be multiplied without becoming zero.
Key Insight: Domains make torsion closed and quotients torsion-free. □

Definition 1.3.4. *Annihilators*

(i) For $v \in M$, the *annihilator* of v is $\text{Ann}_R(v) := \{r \in R : rv = 0\}$.
(ii) For $N \leq M$, the *annihilator* of N is $\text{Ann}_R(N) := \{r \in R : rN = \{0\}\}$.

Proposition 1.3.5. *Annihilator Properties*
For $v \in M$ and $N \leq M$:

(i) $\text{Ann}_R(v) \trianglelefteq R$ *and* $\text{Ann}_R(N) \trianglelefteq R$.
(ii) v *is torsion* $\iff v \neq 0$ *and* $\text{Ann}_R(v) \neq \{0\}$.
(iii) $A \leq B \leq M \implies \text{Ann}_R(B) \subseteq \text{Ann}_R(A)$.

Proof.

(i) Since $0 \in \text{Ann}_R(v)$ and, for $a, b \in \text{Ann}_R(v)$ and $r, s \in R$, $(ra + sb)v = r(av) + s(bv) = 0$, $\text{Ann}_R(v) \trianglelefteq R$. The same computation for every $n \in N$ gives $\text{Ann}_R(N) \trianglelefteq R$.
(ii) ($\Rightarrow$) If v is torsion, then $v \neq 0$ and $\exists\, 0 \neq r \in R$ with $rv = 0$, so $\text{Ann}_R(v) \neq \{0\}$.
(ii) ($\Leftarrow$) If $v \neq 0$ and $\text{Ann}_R(v) \neq \{0\}$, choose $0 \neq r \in \text{Ann}_R(v)$. Then $rv = 0$, so v is torsion.
(iii) If $A \leq B \leq M$ and $r \in \text{Ann}_R(B)$, then $rA = \{0\}$; hence $r \in \text{Ann}_R(A)$.

Structural Commentary: Annihilators reverse inclusion.
Key Insight: Killing more vectors gives fewer scalars. □

Proposition 1.3.6. *Torsion Generators Give a Nonzero Annihilator over a Domain*
Assume R is an integral domain. If $M = \langle\!\langle u_1, \dots, u_n \rangle\!\rangle_R$, $u_i \in M_{\text{tor}}$ $\forall i$, then $\text{Ann}_R(M) \neq \{0\}$.

Proof. Choose $0 \neq a_i \in R$ with $a_i u_i = 0$, and set $a := \prod_{i=1}^{n} a_i$. Since R is a domain, $a \neq 0$.
For each i, $au_i = 0$, hence $aM = \{0\}$ because the u_i generate M. Thus $0 \neq a \in \text{Ann}_R(M)$.
Structural Commentary: Finite generation lets many annihilators be compressed into one product.
Key Insight: A finite torsion generating set gives a global order for the module. □

Definition 1.3.7. *Module Homomorphism*
Let M, N be R-modules and let $\tau : M \to N$.

(i) τ is *R-linear* if, $\forall r, s \in R$ and $u, v \in M$, $\tau(ru + sv) = r\tau(u) + s\tau(v)$, and is then called an *R-homomorphism*.
(ii) The set of all R-homomorphisms $M \to N$ is $\text{Hom}_R(M, N)$.
(iii) An R-homomorphism $M \to M$ is an *endomorphism*.
(iv) An injective, surjective, or bijective R-homomorphism is a *monomorphism*, *epimorphism*, or *isomorphism*.

Proposition 1.3.8. *Hom, Kernel, and Image*
Let M, N be R-modules and $\tau \in \mathrm{Hom}_R(M,N)$.

(i) With $(a\sigma + b\rho)(m) := a\sigma(m) + b\rho(m)$, $\mathrm{Hom}_R(M,N)$ is an R-module.
(ii) $\ker\tau := \tau^{-1}(\{0\}) \leq M$.
(iii) $\mathrm{im}\,\tau := \tau(M) \leq N$.
(iv) τ is a monomorphism $\iff \ker\tau = \{0\}$.

Proof. By Theorem 1.1.6, for $A \subseteq P$ in an R-module P, $A \leq P \iff A \neq \emptyset$ and $\forall a,b \in R, \forall x,y \in A, ax+by \in A$.

(i) For $\sigma,\rho \in \mathrm{Hom}_R(M,N)$ and $a,b,r,s \in R$, $(a\sigma + b\rho)(ru+sv) = r(a\sigma+b\rho)(u) + s(a\sigma+b\rho)(v)$, where $u,v \in M$, using commutativity of R. Since $0 \in \mathrm{Hom}_R(M,N)$, the displayed criterion gives the claim inside the pointwise module of maps $M \to N$.
(ii) Since $0 \in \ker\tau$, and $x,y \in \ker\tau$, $a,b \in R$ imply $\tau(ax+by) = a\tau(x) + b\tau(y) = 0$, so the displayed criterion gives $\ker\tau \leq M$.
(iii) Since $0 = \tau(0) \in \mathrm{im}\,\tau$, and $y_i = \tau(x_i) \in \mathrm{im}\,\tau$ $(i = 1,2)$, $a,b \in R$ imply $ay_1 + by_2 = \tau(ax_1 + bx_2) \in \mathrm{im}\,\tau$, so the displayed criterion gives $\mathrm{im}\,\tau \leq N$.
(iv) If τ is injective, then $\tau(m) = 0 = \tau(0)$ implies $m = 0$, so $\ker\tau = \{0\}$. Conversely, if $\ker\tau = \{0\}$ and $\tau(m) = \tau(m')$, then $\tau(m-m') = 0$, hence $m = m'$.

Structural Commentary: Kernels and images are the canonical submodules attached to a module map.
Key Insight: Homomorphisms preserve precisely the finite R-linear combinations that define submodules. □

Definition 1.3.9. *Quotient Module*
Let $S \leq M$. Define $u \equiv v \pmod S \iff u - v \in S$.
The *quotient module* is $M/S := \{u + S : u \in M\}$; $u + S := \{u + s : s \in S\}$,
with operations $(u+S) + (v+S) := (u+v) + S$; $r(u+S) := ru + S$.

Proposition 1.3.10. *Quotient Operations Are Well-Defined*
Let $S \leq M$. The operations of Definition 1.3.9 are well-defined and make M/S an R-module. Moreover, $0_{M/S} = S$, $\pi : M \twoheadrightarrow M/S$, $\pi(u) := u + S$, $\ker\pi = S$..

Proof. If $u \equiv u' \pmod S$ and $v \equiv v' \pmod S$, then $(u+v) - (u'+v') = (u-u') + (v-v') \in S$ and $ru - ru' = r(u-u') \in S$. Thus $+$ and R-scaling on cosets are representative-independent. The module axioms descend from M. Finally, $0_{M/S} = 0 + S = S$, π is surjective, and $\ker\pi = \{u \in M : u + S = S\} = \{u \in M : u \in S\} = S$. *Structural Commentary*: Quotients collapse exactly the submodule S to zero.
Key Insight: Two representatives define the same coset precisely when their difference is invisible modulo S. □

Example 1.3.11. *A Quotient of a Free Module Need Not Be Free*
As a $\mathbb{Z}$-module, $\mathbb{Z}$ is free on $\{1\}$. For $n > 1$, the submodule $n\mathbb{Z}$ is cyclic, but the quotient $\mathbb{Z}/n\mathbb{Z}$ is torsion by Example 1.3.2. Hence $\mathbb{Z}/n\mathbb{Z}$ is not free as a $\mathbb{Z}$-module.

Theorem 1.3.12. *Correspondence Theorem for Modules*
Let $S \leq M$ and let $\pi : M \twoheadrightarrow M/S$ be the quotient map. Then $\Phi : \{T \leq M : S \leq T\} \longrightarrow \{U \leq M/S\}$, where $\Phi(T) := T/S$ is an order isomorphism, with inverse $\Psi(U) := \pi^{-1}(U)$.

Proof. If $S \leq T \leq M$, then $\Phi(T) = \pi(T) \leq M/S$.
If $U \leq M/S$, then $\Psi(U) = \pi^{-1}(U) \leq M$ and $S = \ker\pi \leq \Psi(U)$.
Moreover, $\Psi\Phi(T) = \pi^{-1}(T/S) = T$ and $\Phi\Psi(U) = \pi(\pi^{-1}(U)) = U$, where the second equality uses surjectivity of π. Finally, $T_1 \leq T_2 \Longrightarrow T_1/S \leq T_2/S$, and $U_1 \leq U_2 \Longrightarrow \pi^{-1}(U_1) \leq \pi^{-1}(U_2)$. Thus Φ is an order isomorphism.
Structural Commentary: Submodules above S are exactly the submodules visible after S is collapsed.
Key Insight: Pull back along the quotient map to recover the original submodule. □

Theorem 1.3.13. *Module Isomorphism Theorems*
The following canonical maps are R-module isomorphisms.

(i) If $\tau \in \mathrm{Hom}_R(M,N)$, then $\overline{\tau} : M/\ker\tau \xrightarrow{\sim} \mathrm{im}\,\tau$ and $\overline{\tau}(m + \ker\tau) := \tau(m)$.
(ii) If $S,T \leq M$, then $\theta : S/(S\cap T) \xrightarrow{\sim} (S+T)/T$ and $\theta\big(s + (S\cap T)\big) := s + T$.
(iii) If $S \leq T \leq M$, then $\eta : (M/S)/(T/S) \xrightarrow{\sim} M/T$ and $\eta\big((m+S) + T/S\big) := m + T$.

Proof.

(i) If $m - m' \in \ker\tau$, then $\tau(m) = \tau(m')$, so $\overline{\tau}$ is well-defined and R-linear. Also $\mathrm{im}\,\overline{\tau} = \mathrm{im}\,\tau$ and $\ker\overline{\tau} = \{m + \ker\tau : \tau(m) = 0\} = \{0_{M/\ker\tau}\}$.
(ii) Apply (i) to $\varphi \in \mathrm{Hom}_R(S, (S+T)/T)$, $\varphi(s) := s + T$. Then $\mathrm{im}\,\varphi = (S+T)/T$ and $\ker\varphi = \{s \in S : s \in T\} = S \cap T$.

(iii) Apply (i) to $\psi \in \mathrm{Hom}_R(M/S, M/T)$, $\psi(m+S) := m+T$.
Then $\operatorname{im}\psi = M/T$ and $\ker\psi = \{m+S : m \in T\} = T/S$.

Structural Commentary: Each theorem identifies a quotient by the exact ambiguity in a natural map.
Key Insight: Image is domain modulo kernel, repeatedly. □

Definition 1.3.14. *Direct Sums and Direct Summands*

(i) For R-modules $M_1, \dots, M_n$, their *external direct sum* is $\bigoplus_{i=1}^{n} M_i := \{(m_i)_{i=1}^n : m_i \in M_i\}$ with componentwise R-module operations.
(ii) For submodules $\{S_i : i \in I\}$ of M, their *internal direct sum* is denoted $M = \bigoplus_{i\in I} S_i$, which means $M = \sum_{i\in I} S_i$ and $\forall i \in I$, $S_i \cap \sum_{j\neq i} S_j = \{0\}$.
(iii) A submodule $S \le M$ is a *direct summand* if $\exists\, T \le M$ such that $M = S \oplus T$. Such a T is a *complement* of S in M.

Proposition 1.3.15. *Direct Sum Characterizations*
For a family $\{S_i : i \in I\}$ of distinct submodules of M, the following conditions are equivalent:

(i) $\sum_{i\in I} S_i$ *is direct, i.e.* $\forall k \in I,\ S_k \cap \sum_{i\neq k} S_i = \{0\}$.
(ii) For every $F \subseteq_{\text{fin}} I$ *and* $x_i \in S_i$ $(i \in F)$, $\sum_{i\in F} x_i = 0 \Longrightarrow x_i = 0\ \forall i \in F$.
(iii) Every $0 \neq x \in \sum_{i\in I} S_i$ *has a unique finite nonzero decomposition,* $\exists!\,(F, (x_i)_{i\in F})$ *where* $F \subseteq_{\text{fin}} I$; $0 \neq x_i \in S_i$; $x = \sum_{i\in F} x_i$.

Proof.

(i) $\Rightarrow$ *(ii)*. If (i) holds and $\sum_{i\in F} x_i = 0$, then any $x_k \neq 0$ gives $x_k = -\sum_{i\in F\setminus\{k\}} x_i \in S_k \cap \sum_{i\neq k} S_i$, a contradiction.
(ii) $\Rightarrow$ *(i)*. If $x \in S_k \cap \sum_{i\neq k} S_i$, write $x = \sum_{i\in F} x_i$ with $F \subseteq_{\text{fin}} I \setminus \{k\}$ and $x_i \in S_i$.
Then $x - \sum_{i\in F} x_i = 0$, so (ii) gives $x = 0$.
(ii) $\Rightarrow$ *(iii)*. Existence follows from the definition of $\sum_i S_i$ after deleting zero terms.
If $\sum_{i\in F} x_i = \sum_{j\in G} y_j$, $0 \neq x_i \in S_i$, $0 \neq y_j \in S_j$, then, over $H = F \cup G$, missing terms set to 0, $\sum_{h\in H}(x_h - y_h) = 0$.
By (ii), $x_h = y_h$ for all $h \in H$, hence $F = G$ and the terms agree.
(iii) $\Rightarrow$ *(ii)*. If $\sum_{i\in F} x_i = 0$, set $F_0 := \{i \in F : x_i \neq 0\}$.
If $F_0 \neq \emptyset$, choose $k \in F_0$. If $F_0 = \{k\}$, then $x_k = 0$, contradiction; otherwise $0 \neq x_k = -\sum_{i\in F_0\setminus\{k\}} x_i$, giving two finite nonzero decompositions of x_k, contradiction. Hence $F_0 = \emptyset$.

Structural Commentary: Directness is the absence of hidden cancellation among different summands.
Key Insight: Unique decomposition is controlled by the zero vector. □

Example 1.3.16. *Most Submodules of $\mathbb{Z}$ Are Not Complemented*
As a $\mathbb{Z}$-module, every submodule of $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \ge 0$. For $m, n > 0$, $m\mathbb{Z} \cap n\mathbb{Z} = \operatorname{lcm}(m,n)\mathbb{Z} \neq \{0\}$. Thus $m\mathbb{Z} \oplus n\mathbb{Z} \neq \mathbb{Z}$ for all $m, n > 0$, so the only complemented submodules of $\mathbb{Z}$ are $\{0\}$ and $\mathbb{Z}$.

Theorem 1.3.17. *Complements and Quotients*
If $M = S \oplus T$, then $\pi|_T : T \xrightarrow{\sim} M/S$, $t \mapsto t + S$, where $\pi : M \twoheadrightarrow M/S$ is the quotient map.
Hence any two complements of S are isomorphic.

Proof. $\ker(\pi|_T) = T \cap S = \{0\}$. Also, for $m = s + t$ with $s \in S$, $t \in T$, one has $m + S = t + S$; hence $\operatorname{im}(\pi|_T) = M/S$. Thus $\pi|_T$ is an isomorphism, and all complements are isomorphic to M/S.
Structural Commentary: A complement gives a concrete copy of the quotient inside the original module.
Key Insight: Quotienting by S remembers exactly the complementary coordinate. □

Theorem 1.3.18. *Direct Summands Extend Epimorphisms*
Let $N \le M$ and let M_1 be an R-module.

(i) If $M = N \oplus H$ and $\sigma : N \twoheadrightarrow M_1$, then $\exists$ a unique $\overline{\sigma} \in \mathrm{Hom}_R(M, M_1)$ such that $\overline{\sigma}(n+h) := \sigma(n)$, $n \in N$, $h \in H$, $\overline{\sigma}|_N = \sigma$, $\overline{\sigma} : M \twoheadrightarrow M_1$, *and* $\ker\overline{\sigma} = \ker\sigma \oplus H$.
(ii) If $\sigma : N \xrightarrow{\sim} M_1$, then $\mathcal{E}_\sigma := \{\overline{\sigma} : M \twoheadrightarrow M_1 : \overline{\sigma}|_N = \sigma\} \longleftrightarrow \mathcal{C}_N(M) := \{H \le M : M = N \oplus H\}$ via $\overline{\sigma} \mapsto \ker\overline{\sigma}$ and $H \mapsto \overline{\sigma}_H$, where $\overline{\sigma}_H(n+h) := \sigma(n)$.

Proof.

(i) Direct-sum uniqueness makes $\overline{\sigma}$ well-defined, and componentwise computation gives R-linearity. Moreover, $\operatorname{im}\overline{\sigma} = \operatorname{im}\sigma = M_1$; $\ker\overline{\sigma} = \{n + h : \sigma(n) = 0\} = \ker\sigma \oplus H$. This shows uniqueness.

(ii) If $H \in \mathcal{C}_N(M)$, then (i) gives $\overline{\sigma}_H \in \mathcal{E}_\sigma$ and, since $\ker\sigma = \{0\}$, $\ker\overline{\sigma}_H = H$. Conversely, if $\overline{\sigma} \in \mathcal{E}_\sigma$, then $N \cap \ker\overline{\sigma} = \{0\}$, and for each $m \in M$, $n := \sigma^{-1}(\overline{\sigma}(m)) \in N$, $m - n \in \ker\overline{\sigma}$. Thus $M = N \oplus \ker\overline{\sigma}$, so the two displayed assignments are inverse.

Structural Commentary: A complement supplies the directions on which an extension may vanish.
Key Insight: Extending an isomorphism is equivalent to choosing its kernel complement. □

Definition 1.3.19. *Retract*
Let $N \leq M$.

(i) A *retraction* onto N is a map $\rho \in \operatorname{Hom}_R(M, N)$ such that $\rho|_N = \mathrm{id}_N$.
(ii) The submodule N is a *retract* of M iff $\exists\, \rho \in \operatorname{Hom}_R(M, N)$ with $\rho|_N = \mathrm{id}_N$.

Corollary 1.3.20. *Retracts Are Direct Summands*
For $N \leq M$, N is a retract of $M \iff \exists\, H \leq M$ such that $M = N \oplus H$.

Proof.

($\Rightarrow$) Let $\rho : M \to N$ satisfy $\rho|_N = \mathrm{id}_N$. For every $m \in M$, $m = \rho(m) + (m - \rho(m))$, $\rho(m) \in N$, $m - \rho(m) \in \ker\rho$. Moreover $N \cap \ker\rho = \{0\}$, so $M = N \oplus \ker\rho$.

($\Leftarrow$) If $M = N \oplus H$, define $\rho : M \to N$ by $\rho(n + h) := n$, $n \in N$, $h \in H$. Then $\rho \in \operatorname{Hom}_R(M, N)$ and $\rho|_N = \mathrm{id}_N$, so N is a retract.

Structural Commentary: Retractions are projections onto submodules.
Key Insight: A submodule is split exactly when it can be recovered by an idempotent map. □

Definition 1.3.21. *One-Sided Inverses*
Let $\tau \in \operatorname{Hom}_R(A, B)$.

(i) A *left inverse* of τ is $\lambda \in \operatorname{Hom}_R(B, A)$ such that $\lambda\tau = \mathrm{id}_A$.
(ii) A *right inverse* of τ is $\rho \in \operatorname{Hom}_R(B, A)$ such that $\tau\rho = \mathrm{id}_B$.
(iii) A *two-sided inverse* of τ is $\upsilon \in \operatorname{Hom}_R(B, A)$ such that $\upsilon\tau = \mathrm{id}_A$ and $\tau\upsilon = \mathrm{id}_B$.

Theorem 1.3.22. *One-Sided Inverses and Complements*
Let $\sigma : M \to M_1$ be an R-homomorphism.
For $S \leq N$, set $\mathcal{C}_S(N) := \{H \leq N : N = S \oplus H\}$, $\mathcal{L}_\sigma := \{\lambda \in \operatorname{Hom}_R(M_1, M) : \lambda\sigma = \mathrm{id}_M\}$, and $\mathcal{R}_\sigma := \{\rho \in \operatorname{Hom}_R(M_1, M) : \sigma\rho = \mathrm{id}_{M_1}\}$.

(i) If σ is injective, then $\mathcal{L}_\sigma \xrightarrow{\sim} \mathcal{C}_{\operatorname{im}\sigma}(M_1)$ by $\lambda \mapsto \ker\lambda$.
(ii) If σ is surjective, then $\mathcal{R}_\sigma \xrightarrow{\sim} \mathcal{C}_{\ker\sigma}(M)$ by $\rho \mapsto \operatorname{im}\rho$.

Proof.

(i) For $H \in \mathcal{C}_{\operatorname{im}\sigma}(M_1)$, define $\lambda_H(\sigma m + h) := m$. Directness and injectivity give well-definedness; $\lambda_H\sigma = \mathrm{id}_M$ and $\ker\lambda_H = H$. Conversely, if $\lambda \in \mathcal{L}_\sigma$, then $\operatorname{im}\sigma \cap \ker\lambda = \{0\}$, since $\sigma m \in \ker\lambda \Rightarrow m = 0$, and $y = \sigma\lambda y + (y - \sigma\lambda y)$ with $y - \sigma\lambda y \in \ker\lambda$. Thus $M_1 = \operatorname{im}\sigma \oplus \ker\lambda$, and the constructions are inverse.

(ii) For $H \in \mathcal{C}_{\ker\sigma}(M)$, $\sigma|_H : H \to M_1$ is an isomorphism; let $\rho_H = (\sigma|_H)^{-1}$ followed by $H \hookrightarrow M$. Then $\rho_H \in \mathcal{R}_\sigma$ and $\operatorname{im}\rho_H = H$. Conversely, if $\rho \in \mathcal{R}_\sigma$, then ρ is injective, $\operatorname{im}\rho \cap \ker\sigma = \{0\}$, and $m = (m - \rho\sigma m) + \rho\sigma m$ with $m - \rho\sigma m \in \ker\sigma$, $\rho\sigma m \in \operatorname{im}\rho$. Thus $M = \ker\sigma \oplus \operatorname{im}\rho$, and the constructions are inverse.

Structural Commentary: One-sided inverses exist exactly when the relevant image or kernel splits.
Key Insight: Injective maps need a complemented image; surjective maps need a complemented kernel. □

Proposition 1.3.23. *Split Epimorphisms Split the Domain*
Let $\sigma : M \twoheadrightarrow N$ be an R-homomorphism and set $\mathcal{R}_\sigma := \{\rho \in \operatorname{Hom}_R(N, M) : \sigma\rho = \mathrm{id}_N\}$. These are equivalent:

(i) $\mathcal{R}_\sigma \neq \emptyset$.
(ii) $\exists\, H \leq M$ such that $M = \ker\sigma \oplus H$.

Moreover, $\rho \in \mathcal{R}_\sigma \Longrightarrow M = \ker\sigma \oplus \operatorname{im}\rho$.

Proof.

($\Rightarrow$) Let $\rho \in \mathcal{R}_\sigma$. If $\rho n = 0$, then $n = \sigma\rho n = 0$, so ρ is injective.
If $\rho n \in \ker\sigma$, then $n = \sigma\rho n = 0$, hence $\operatorname{im}\rho \cap \ker\sigma = \{0\}$.
For $m \in M$, $m = (m - \rho\sigma m) + \rho\sigma m$, with $m - \rho\sigma m \in \ker\sigma$ and $\rho\sigma m \in \operatorname{im}\rho$. Thus $M = \ker\sigma \oplus \operatorname{im}\rho$.

($\Leftarrow$) Let $M = \ker\sigma \oplus H$. Then $\ker(\sigma|_H) = H \cap \ker\sigma = \{0\}$, and $\operatorname{im}(\sigma|_H) = N$ since $n = \sigma m$, $m = k + h$, gives $n = \sigma h$. Thus $\sigma|_H : H \xrightarrow{\sim} N$, and $\rho := \iota_H(\sigma|_H)^{-1} \in \mathcal{R}_\sigma$, where $\iota_H : H \hookrightarrow M$.

Structural Commentary: A split epimorphism is a quotient map with complemented kernel.
Key Insight: The right inverse chooses canonical representatives for the cosets of $\ker\sigma$. □

Principle 1.3.24. *Modules Are Not Just Vector Spaces with New Letters*
For a general ring R, the following vector-space implications fail:

(i) $N \leq M \not\Rightarrow \exists H \leq M$ with $M = N \oplus H$.
(ii) M finitely generated and $N \leq M \not\Rightarrow N$ finitely generated.
(iii) $M \neq 0 \not\Rightarrow \exists \emptyset \neq S \subseteq M$ linearly independent.
(iv) S minimal spanning or maximal independent $\not\Rightarrow S$ is a basis.
(v) M free and $N \leq M \not\Rightarrow N$ free.
(vi) F free and $S \leq F \not\Rightarrow F/S$ free.

Over a field, these obstructions vanish because $R \setminus \{0\} \subseteq R^\times$.

2 Noetherian Modules and PID Structure Theorems

This subsection begins with the structure facts that make free modules and finite generation usable: rank, splitting maps onto free modules, Noetherian modules, and Hilbert's basis theorem. It then develops the classification of finitely generated modules over a principal ideal domain. The guiding model is the abelian-group theorem over $\mathbb{Z}$ and, later, the polynomial-module classification behind rational canonical form.

2.1 Free Modules and Rank

Definition 2.1.1. *Free Module and Basis*

(i) A subset $\mathcal{B} \subseteq M$ is a *basis* of M if $\mathcal{B}$ is R-linearly independent and $\langle\langle \mathcal{B} \rangle\rangle_R = M$.
(ii) An R-module M is *free* if $M = \{0\}$ or M has a basis.
(iii) If $\mathcal{B}$ is a basis of M, then M is *free on* $\mathcal{B}$.

Theorem 2.1.2. *Basis Means Unique Finite Coordinates*
A subset $\mathcal{B} \subseteq M$ is a basis if and only if $\forall v \in M,\ \exists!\,(F, (r_b)_{b\in F})$, with $F \subseteq_{\text{fin}} \mathcal{B}$, $r_b \in R \setminus \{0\}$, $v = \sum_{b\in F} r_b b$.

The case $F = \emptyset$ gives the empty sum 0.

Proof.

($\Rightarrow$) Spanning gives a finite expression; collecting equal basis elements and deleting zero coefficients gives the displayed form. If $\sum_{b\in F} r_b b = \sum_{b\in G} s_b b$, then $\sum_{b\in F\cup G} (r_b - s_b)b = 0$, where missing coefficients are 0. Independence gives $r_b = s_b$ for all b, hence $F = G$ and the coefficients agree.
($\Leftarrow$) Existence gives $\langle\langle \mathcal{B} \rangle\rangle_R = M$. If $\sum_{b\in F} r_b b = 0$ and $F_0 := \{b \in F : r_b \neq 0\}$ is nonempty, then $0 = \sum_{b\in F_0} r_b b$ contradicts uniqueness of the empty expression. Thus all $r_b = 0$, so $\mathcal{B}$ is independent.

Structural Commentary: Bases give unique finite coordinates.
Key Insight: Existence is spanning; uniqueness is independence. □

Proposition 2.1.3. *Bases Are Minimal Spanning and Maximal Independent*
If $\mathcal{B}$ is a basis of M, then $\forall b \in \mathcal{B}$, $\langle\langle \mathcal{B} \setminus \{b\} \rangle\rangle_R \neq M$, and $\forall v \in M \setminus \mathcal{B}$, $\mathcal{B} \cup \{v\}$ is R-linearly dependent.

Proof. If $b \in \mathcal{B}$ and $b \in \langle\langle \mathcal{B} \setminus \{b\} \rangle\rangle_R$, then b has two finite coordinate expressions, contradicting Theorem 2.1.2. Hence deleting b destroys spanning. If $v \notin \mathcal{B}$, write $v = \sum_{b\in F} r_b b$ with $F \subseteq_{\text{fin}} \mathcal{B}$.
Then $v - \sum_{b\in F} r_b b = 0$ is a nontrivial relation in $\mathcal{B} \cup \{v\}$, so $\mathcal{B} \cup \{v\}$ is dependent.

Structural Commentary: Basis still implies the familiar minimal/maximal properties.
Key Insight: The converses are vector-space privileges, not general module facts. □

Example 2.1.4. *A Free Module with a Nonfree Submodule*
Let $R = \mathbb{Z} \times \mathbb{Z}$ with componentwise multiplication. As an R-module, $R = R(1,1)$, so R is free on $\{(1,1)\}$.
Set $N := \mathbb{Z} \times \{0\} \leq R$. For every $0 \neq (a,0) \in N$, $(0,1)(a,0) = 0$.
Thus every nonzero element of N is torsion, so N has no nonempty independent subset and is not free.

Perspective 2.1.5. *Dimension as the Field Case of Rank*
An $\mathbb{F}$-vector space is a free $\mathbb{F}$-module. By Theorem 1.3.14, finite bases have invariant cardinality; over general rings, rank invariance requires additional hypotheses.

Theorem 2.1.6. *Maps Out of a Free Module*
Let M be free on $\mathcal{B}$ and let N be an R-module.
For every function $f : \mathcal{B} \to N$, there exists a unique $\widetilde{f} \in \mathrm{Hom}_R(M,N)$ such that $\widetilde{f}|_{\mathcal{B}} = f$.

Proof. For $v = \sum_{b\in F} r_b b$ with $F \subseteq_{\text{fin}} \mathcal{B}$, define $\widetilde{f}(v) := \sum_{b\in F} r_b f(b)$.
Theorem 2.1.2 makes the expression of v unique, so $\widetilde{f}$ is well-defined.
For $x = \sum a_b b$ and $y = \sum c_b b$, combining supports gives $\widetilde{f}(rx+sy) = r\widetilde{f}(x) + s\widetilde{f}(y)$, so $\widetilde{f}$ is R-linear and clearly extends f. If $\phi \in \mathrm{Hom}_R(M,N)$ also extends f, then $\phi\left(\sum_{b\in F} r_b b\right) = \sum_{b\in F} r_b f(b) = \widetilde{f}\left(\sum_{b\in F} r_b b\right)$, hence $\phi = \widetilde{f}$.
Structural Commentary: A free module is controlled entirely by its basis values.
Key Insight: Freeness is the universal permission to choose images of basis elements arbitrarily. □

Theorem 2.1.7. *Rank Is Well Defined fo r Free Modules*
Let A benonzero commutative ring with identity, M be free A-module, $\mathrm{Bas}_A(M)$ denote set of A-bases of M.
(i) $B, B' \in \mathrm{Bas}_A(M) \Longrightarrow |B| = |B'|$.
(ii) $B \in \mathrm{Bas}_A(M)$, $S \subseteq M$, *and* $\langle\langle S\rangle\rangle_A = M \Longrightarrow |B| \le |S|$.

Proof.
(i) Choose a maximal ideal $\mathcal{I} \lhd A$ by Theorem 3.1.12.
Put $K := A/\mathcal{I}$, a field by Theorem 3.1.11, $\mathcal{I}M := \langle\langle im : i \in \mathcal{I},\ m \in M\rangle\rangle_A$, and $V := M/\mathcal{I}M$.
Since $\mathcal{I}V = 0$, V is a K-vector space via $(a+\mathcal{I})(m+\mathcal{I}M) := am + \mathcal{I}M$.
(ii) Let $B \in \mathrm{Bas}_A(M)$ and set $\overline{B} := \{\overline{b} := b + \mathcal{I}M : b \in B\}$.
In B-coordinates, $\mathcal{I}M = \{\sum_{b\in F} i_b b : F \subseteq_{\text{fin}} B,\ i_b \in \mathcal{I}\}$; hence $\overline{B}$ spans V.
If $\sum_{j=1}^{n} (a_j + \mathcal{I})\overline{b_j} = 0$ with distinct $b_j \in B$, then $\sum_{j=1}^{n} a_j b_j \in \mathcal{I}M$, so $a_j \in \mathcal{I}\ \forall j$; hence $\overline{B}$ is K-independent.
If $\overline{b} = \overline{b'}$ with $b \neq b'$, then $b - b' \in \mathcal{I}M$, forcing the B-coefficient $1 \in \mathcal{I}$, impossible. Thus $B \xrightarrow{\sim} \overline{B}$.
(iii) Therefore $|B| = \left|\overline{B}\right| = \dim_K V$ for every $B \in \mathrm{Bas}_A(M)$, so (i) follows from Theorem 1.4.7.
If $\langle\langle S\rangle\rangle_A = M$, then $\overline{S} := \{s + \mathcal{I}M : s \in S\}$ spans V, so $|B| = \dim_K V \le \left|\overline{S}\right| \le |S|$.
Structural Commentary: Modulo a maximal ideal, free module becomes a vector space with same basis size.
Key Insight: The field quotient $A/\mathcal{I}$ lets vector-space dimension certify module rank. □

Definition 2.1.8. *Rank of a Free Module*
Let A be a nonzero commutative ring with identity. The *rank* of a free A-module M, denoted $\mathrm{rk}_A(M)$, is the cardinality of any basis of M. The zero module has rank 0.

Example 2.1.9. *Why Commutativity Matters for Rank*
Let $V = \bigoplus_{k\ge 0} \mathbb{F} e_k$ and $E = \mathcal{L}(V)$. As a left E-module, $E = E\,\mathrm{id}_V$, so $\{\mathrm{id}_V\}$ is a basis.
For each $n \ge 1$, define $\beta_s \in E$, $0 \le s < n$, by $\beta_s(e_{kn+t}) = \begin{cases} e_k, & t = s, \\ 0, & t \neq s, \end{cases}$ where $k \ge 0$, $0 \le t < n$.
If $\sum_{s=0}^{n-1} \alpha_s \beta_s = 0$, then evaluating at e_{kn+t} gives $\alpha_t(e_k) = 0$ for all k, t, hence $\alpha_t = 0$ for all t.
Conversely, for $h \in E$, define $\alpha_s(e_k) := h(e_{kn+s})$; then $h = \sum_{s=0}^{n-1} \alpha_s \beta_s$.
Thus $\{\beta_0, \dots, \beta_{n-1}\}$ is an E-basis of E, so E has bases of all positive finite cardinalities.
Structural Commentary: Noncommutative scalar rings can make a free module have incompatible basis sizes.
Key Insight: The rank theorem is a commutative-ring theorem, not a formal consequence of freeness alone.

Theorem 2.1.10. *Finite-Support Coordinate Modules*
Let A be a commutative ring with identity and let B be a set. Put $A^{(B)} := \{f \in A^B : |\mathrm{supp}(f)| < \infty\}$.
For $b \in B$, let $\delta_b \in A^{(B)}$ be given by $\delta_b(b) = 1$ and $\delta_b(x) = 0$ for $x \neq b$.
Then $A^{(B)} = \bigoplus_{b\in B} A\delta_b$, so $\{\delta_b : b \in B\}$ is an A-basis. If $A \neq 0$, then $\mathrm{rk}_A(A^{(B)}) = |B|$.

Proof. For $f, g \in A^{(B)}$ and $a \in A$, $\mathrm{supp}(f+g) \subseteq \mathrm{supp}(f) \cup \mathrm{supp}(g)$ and $\mathrm{supp}(af) \subseteq \mathrm{supp}(f)$, hence $A^{(B)} \le A^B$.
For $f \in A^{(B)}$, $f = \sum_{b\in \mathrm{supp}(f)} f(b)\delta_b$, so the δ_b span. If $\sum_{j=1}^{m} a_j \delta_{b_j} = 0$ with $b_1, \dots, b_m$ distinct, then evaluation at b_i gives $a_i = 0$ for every i. Thus the δ_b are independent, and the rank statement follows from the displayed basis.
Structural Commentary: Free modules are finite-support coordinate modules on their bases.
Key Insight: Finite support is the module-theoretic version of finite linear combination. □

Proposition 2.1.11. *Free Modules Are Coordinate Modules*
Let A be a commutative ring with identity and let M be free on a basis B.
Then the coordinate map $\Phi_B : M \to A^{(B)}$ is an isomorphism.
If $A \neq 0$ and M, N are free A-modules, then $M \cong N \iff \mathrm{rk}_A(M) = \mathrm{rk}_A(N)$.

Proof. For $m = \sum_{b \in F} a_b b$, set $\Phi_B(m) = f_m$, with $f_m(b) = a_b$ for $b \in F$ and $f_m(b) = 0$ for $b \notin F$.
Unique coordinates give $\Phi_B : M \xrightarrow{\sim} A^{(B)}$, with inverse $f \mapsto \sum_{b \in \mathrm{supp}(f)} f(b)b$.

($\Rightarrow$) If $\psi : M \xrightarrow{\sim} N$ and C is a basis of M, then $\psi(C)$ is a basis of N, so $\mathrm{rk}_A(M) = |C| = |\psi(C)| = \mathrm{rk}_A(N)$.
($\Leftarrow$) If $\mathrm{rk}_A(M) = \mathrm{rk}_A(N)$, choose bases C, D with $|C| = |D|$.
Any bijection $C \to D$ extends, by Theorem 2.1.6, to $M \xrightarrow{\sim} N$.

Structural Commentary: A basis identifies a free module with a standard coordinate model.
Key Insight: Over a nonzero commutative ring, rank classifies free modules up to isomorphism. □

Theorem 2.1.12. *Independent Sets Are Bounded by Rank over a Domain*
Let A be integral domain with field of quotients Q, M free over A, $S \subseteq M$ A-independent. Then $|S| \leq \mathrm{rk}_A(M)$.

Proof. Choose an A-basis B and identify $M = A^{(B)} \subseteq Q^{(B)}$.
If $\sum_{i=1}^{n} q_i s_i = 0$ with $q_i \in Q$ and distinct $s_i \in S$, choose $d \in A \setminus \{0\}$ with $dq_i \in A$ for all i.
Then $\sum_{i=1}^{n} (dq_i)s_i = 0$ in M, so $dq_i = 0$ by A-independence, hence $q_i = 0$ for all i.
Thus S is Q-independent in $Q^{(B)}$, and $|S| \leq \dim_Q Q^{(B)} = |B| = \mathrm{rk}_A(M)$.
Structural Commentary: The field of quotients lets ordinary dimension bound module independence.
Key Insight: Domains allow denominator clearing without creating false zero relations. □

Theorem 2.1.13. *Free Quotients Split*
Let A be a commutative ring with identity.

(i) If $\sigma : M \twoheadrightarrow F$ and F is free, then $\exists N \leq M$ such that $M = \ker(\sigma) \oplus N$ and $N \cong F$.
(ii) If $S \leq M$ and M/S is free, then $\exists T \leq M$ such that $M = S \oplus T$ and $T \cong M/S$; hence $M \cong S \oplus M/S$.
(iii) If $A \neq 0$ and $M, S, M/S$ are free, then $\mathrm{rk}_A(M) = \mathrm{rk}_A(S) + \mathrm{rk}_A(M/S)$, with cardinal addition; in finite rank, $\mathrm{rk}_A(M/S) = \mathrm{rk}_A(M) - \mathrm{rk}_A(S)$.

Proof.

(i) Let B be a basis of F. For $b \in B$, choose $m_b \in M$ with $\sigma(m_b) = b$. By Theorem 2.1.6, $b \mapsto m_b$ extends to $\tau : F \to M$. Since $\sigma\tau = \mathrm{id}_F$, Theorem 1.3.22 gives $M = \ker(\sigma) \oplus \mathrm{im}(\tau)$ and $\mathrm{im}(\tau) \cong F$.
(ii) Apply (i) to $\pi : M \twoheadrightarrow M/S$. Then $M = S \oplus T$ for some $T \cong M/S$, so $M \cong S \oplus M/S$.
(iii) From (ii), $M = S \oplus T$ with $T \cong M/S$. If B_S, B_T are bases of S, T, then $B_S \sqcup B_T$ is a basis of M, hence $\mathrm{rk}_A(M) = |B_S| + |B_T| = \mathrm{rk}_A(S) + \mathrm{rk}_A(M/S)$. The finite formula follows.

Structural Commentary: Free modules are projective: maps onto them have linear sections.
Key Insight: Choosing representatives for basis vectors of the quotient is enough to split the quotient map. □

2.2 Noetherian Modules

Definition 2.2.1. *Noetherian Modules and Rings*
Let A be a commutative ring with identity.

(i) An A-module M satisfies *ACC* if every chain $M_1 \leq M_2 \leq \cdots$ of submodules satisfies $\exists n_0, \forall n \geq n_0, M_n = M_{n_0}$. Such M is *Noetherian*.
(ii) The ring A is *Noetherian* if the regular A-module A is Noetherian, equivalently if every ideal-chain $I_1 \subseteq I_2 \subseteq \cdots$ stabilizes.

Theorem 2.2.2. *Noetherian Means Finitely Generated Submodules*
Let M be an A-module.

(i) M is Noetherian $\iff$ every $N \leq M$ is f.g.
(ii) If A is commutative with identity, then A is Noetherian $\iff$ every $\mathcal{I} \lhd A$ is f.g.

Proof.

(i) ($\Rightarrow$) Let $N \leq M$. If N is not f.g., choose $u_{n+1} \in N \setminus \langle\!\langle u_1, \ldots, u_n \rangle\!\rangle_A$ inductively.
Then $\langle\!\langle u_1 \rangle\!\rangle_A < \langle\!\langle u_1, u_2 \rangle\!\rangle_A < \cdots$, contradicting ACC.

(*i*) ($\Leftarrow$) Let $M_1 \le M_2 \le \cdots$ and put $U = \bigcup_{n\ge 1} M_n$. Then $U \le M$, so $U = \langle\!\langle u_1, \dots, u_r \rangle\!\rangle_A$. Choose k with $u_1, \dots, u_r \in M_k$. Hence $U = M_k$ and $M_n = M_k$ for all $n \ge k$.
(*ii*) Apply (i) to the regular A-module A: its submodules are exactly the ideals of A.

Structural Commentary: ACC and finite generation are two faces of the same finiteness condition.
Key Insight: Failure of finite generation builds an infinite increasing chain one new generator at a time. □

Theorem 2.2.3. *Finitely Generated Modules over Noetherian Rings*
Let A be a commutative ring with identity. Then A is Noetherian $\iff$ every f.g. A-module is Noetherian.

Proof.
($\Rightarrow$) Assume A Noetherian. First prove by induction on n that every $S \le A^n$ is f.g. The case $n = 1$ is Theorem 2.2.2. For $n > 1$, let $\pi : S \to A$ be last-coordinate projection. Then $\pi(S) \lhd A$ is f.g. and $\ker(\pi) \le A^{n-1}$ is f.g. by induction. Lifts of generators of $\pi(S)$, together with generators of $\ker(\pi)$, generate S. Thus A^n is Noetherian. If M is f.g., choose $A^n \twoheadrightarrow M$; for $N \le M$, its preimage in A^n is f.g., hence N is f.g. Therefore M is Noetherian.
($\Leftarrow$) Since $A = \langle\!\langle 1 \rangle\!\rangle_A$, the regular A-module A is f.g.; by hypothesis it is Noetherian. Hence A is Noetherian.

Structural Commentary: Noetherianity passes from rings to finite free modules to quotients.
Key Insight: Finitely generated modules are quotients of finite free modules. □

Theorem 2.2.4. *Hilbert Basis Theorem*
Let A be commutative with identity. If A is Noetherian, then $A[x]$ is Noetherian.

Proof. It suffices to show every ideal $\mathcal{J} \lhd A[x]$ is f.g. If $\mathcal{J} = 0$, done.
For $g \ne 0$, write $\mathrm{lc}(g)$ and $\mathrm{lt}(g)$. Set $L = \langle \mathrm{lc}(g) : g \in \mathcal{J}, g \ne 0 \rangle \lhd A$. Since A is Noetherian, choose $a_1, \dots, a_m$ generating L and $f_i \in \mathcal{J}$ with $\mathrm{lc}(f_i) = a_i$. Replacing f_i by powers of x times f_i, assume $\deg f_i = d$ for all i.
For $0 \le k < d$, set $L_k = \langle \mathrm{lc}(g) : g \in \mathcal{J}, \deg g = k \rangle$ and choose finite $P_k \subseteq \mathcal{J}$ with $\deg p = k$ for $p \in P_k$ and $\{\mathrm{lc}(p) : p \in P_k\}$ generating L_k. Put $G = \{f_1, \dots, f_m\} \cup \bigcup_{k=0}^{d-1} P_k$ and $\mathcal{A} = \langle G \rangle \lhd A[x]$. Then $\mathcal{A} \subseteq \mathcal{J}$.
For $g \in \mathcal{J} \setminus \mathcal{A}$ of minimal degree, set $r = \deg g$. If $r < d$, write $\mathrm{lc}(g) = \sum_{p \in P_r} c_p \,\mathrm{lc}(p)$ and set $h = \sum_{p \in P_r} c_p p$.
If $r \ge d$, write $\mathrm{lc}(g) = \sum_{i=1}^m c_i a_i$ and set $h = \sum_{i=1}^m c_i x^{r-d} f_i$. In either case $h \in \mathcal{A}$ and $\mathrm{lt}(h) = \mathrm{lt}(g)$.
Then $g - h \in \mathcal{J} \setminus \mathcal{A}$ and $\deg(g - h) < \deg g$, contradiction. Thus $\mathcal{J} = \mathcal{A}$ is f.g.
Structural Commentary: Ideals in $A[x]$ are controlled by finitely many leading coefficients in A.
Key Insight: Match the leading term, subtract, and induct on degree. □

2.3 Smith Normal Form

Definition 2.3.1. *Standing PID Notation*
In the PID-structure results, R denotes a principal ideal domain unless otherwise stated.
For $r \in R$, write $\langle r \rangle = Rr$; for an R-module M and $v \in M$, write $\langle\!\langle v \rangle\!\rangle_R = Rv$.

Perspective 2.3.2. *PID Arithmetic Controls Module Structure*
Fields have $F \setminus \{0\} \subseteq F^\times$. PIDs replace invertibility by factorization, gcds, and lcms.
This divisibility data classifies finitely generated modules.

Definition 2.3.3. *Order of a Submodule or Element*
Let M be an R-module and $N \le M$. Since $\mathrm{Ann}_R(N) = \{r \in R : rN = 0\} = \langle \alpha \rangle$, any such α is an *order* of N; write $o(N) = \alpha$, up to associates. For $v \in M$, set $o(v) := o(\langle\!\langle v \rangle\!\rangle_R)$.

Proposition 2.3.4. *Orders and Direct Sums*
Let $M = A \oplus B$. If $o(A) = \alpha$ and $o(B) = \beta$, then $o(M) \sim \mathrm{lcm}(\alpha, \beta)$.
More generally, $o(\bigoplus_{i=1}^n M_i) \sim \mathrm{lcm}(o(M_1), \dots, o(M_n))$.

Proof. $r(A \oplus B) = 0 \iff rA = 0$ and $rB = 0$.
Hence $\mathrm{Ann}_R(A \oplus B) = \mathrm{Ann}_R(A) \cap \mathrm{Ann}_R(B) = \langle \alpha \rangle \cap \langle \beta \rangle = \langle \mathrm{lcm}(\alpha, \beta) \rangle$.
Thus $o(A \oplus B) \sim \mathrm{lcm}(\alpha, \beta)$, and induction gives the finite case.
Structural Commentary: Direct sums intersect annihilator conditions.
Key Insight: Lcms appear because one scalar must kill every summand at once. □

Theorem 2.3.5. *Cyclic Modules over a PID*
Let M be an R-module and let $v \in M$.
(i) If $\mathrm{Ann}_R(v) = \langle \alpha \rangle$, then $R/\langle \alpha \rangle \xrightarrow{\sim} \langle\!\langle v \rangle\!\rangle_R$, $r + \langle \alpha \rangle \mapsto rv$.
(ii) If $C = \langle\!\langle v \rangle\!\rangle_R$ and $S \le C$, then $S = \langle\!\langle sv \rangle\!\rangle_R$ for some $s \in R$.
(iii) If $o(v) = \alpha$ and $\beta \in R$, then $o(\beta v) \sim \alpha / \gcd(\alpha, \beta)$, and $\langle\!\langle \beta v \rangle\!\rangle_R = \langle\!\langle v \rangle\!\rangle_R \iff \gcd(\alpha, \beta) \sim 1$.

Proof.

(i) The map $\phi : R \to \langle\!\langle v \rangle\!\rangle_R$, $\phi(r) = rv$, is onto and $\ker \phi = \mathrm{Ann}_R(v) = \langle \alpha \rangle$. Hence $R/\langle \alpha \rangle \xrightarrow{\sim} \langle\!\langle v \rangle\!\rangle_R$.

(ii) For $S \leq C$, set $I = \{r \in R : rv \in S\} \lhd R$.
Since R is a PID, $I = \langle s \rangle$ for some $s \in R$, and $S = \{rv : r \in I\} = \langle\!\langle sv \rangle\!\rangle_R$.

(iii) Put $d = \gcd(\alpha, \beta)$, $\alpha = d\alpha'$, and $\beta = d\beta'$ with $\gcd(\alpha', \beta') \sim 1$.
For $r \in R$, $r\beta v = 0 \iff \alpha \mid r\beta \iff \alpha' \mid r\beta' \iff \alpha' \mid r$, the last step by Bezout.
Thus $\mathrm{Ann}_R(\beta v) = \langle \alpha' \rangle = \langle \alpha / \gcd(\alpha, \beta) \rangle$. Finally, $\langle\!\langle \beta v \rangle\!\rangle_R = \langle\!\langle v \rangle\!\rangle_R \iff (\exists c \in R)(v = c\beta v)$
$\iff (\exists c \in R)(\alpha \mid 1 - c\beta) \iff \gcd(\alpha, \beta) \sim 1$.

Structural Commentary: A cyclic module is the base ring modulo the first relation satisfied by its generator.
Key Insight: Multiplying a cyclic generator divides its order by the gcd of the multiplier and the old order. □

Theorem 2.3.6. *Composing and Decomposing Cyclic Modules*
Let M be an R-module.

(i) If $o(u_i) = \alpha_i \neq 0$ and $\gcd(\alpha_i, \alpha_j) \sim 1$ for $i \neq j$, then $o(\sum_{i=1}^n u_i) \sim \prod_{i=1}^n \alpha_i$ and $\bigoplus_{i=1}^n \langle\!\langle u_i \rangle\!\rangle_R = \langle\!\langle \sum_{i=1}^n u_i \rangle\!\rangle_R$.

(ii) If $o(v) = \alpha = \prod_{i=1}^n \alpha_i \neq 0$ and $\gcd(\alpha_i, \alpha_j) \sim 1$ for $i \neq j$,
then $\exists v_1, \ldots, v_n \in M$ such that $v = \sum_{i=1}^n v_i$, $o(v_i) \sim \alpha_i$, and $\langle\!\langle v \rangle\!\rangle_R = \bigoplus_{i=1}^n \langle\!\langle v_i \rangle\!\rangle_R$.

Proof.

(i) It suffices to prove $n = 2$ and iterate. Let $o(u) = \alpha$, $o(w) = \beta$, and $\gcd(\alpha, \beta) \sim 1$.
If $x \in \langle\!\langle u \rangle\!\rangle_R \cap \langle\!\langle w \rangle\!\rangle_R$, then $\alpha x = \beta x = 0$, so Bezout gives $x = 0$. If $r(u + w) = 0$, then $ru = -rw$ lies in this intersection; hence $ru = rw = 0$, so $\alpha \mid r$ and $\beta \mid r$, hence $\alpha\beta \mid r$. Since $\alpha\beta(u+w) = 0$, $o(u+w) \sim \alpha\beta$.
Choose $a, b \in R$ with $a\alpha + b\beta = 1$. Then $u = b\beta(u+w)$ and $w = a\alpha(u+w)$, so $\langle\!\langle u \rangle\!\rangle_R \oplus \langle\!\langle w \rangle\!\rangle_R = \langle\!\langle u+w \rangle\!\rangle_R$.

(ii) Put $\beta_i = \alpha/\alpha_i$. Since $\gcd(\beta_1, \ldots, \beta_n) \sim 1$, choose $c_i \in R$ with $\sum_{i=1}^n c_i\beta_i = 1$, and set $v_i = c_i\beta_i v$.
Then $v = \sum_{i=1}^n v_i$. Also $\gcd(c_i, \alpha_i) \sim 1$ and $\gcd(\alpha, \beta_i) \sim \beta_i$, so Theorem 2.3.5 gives $o(v_i) \sim \alpha_i$.
Part (i) gives $\langle\!\langle v \rangle\!\rangle_R = \bigoplus_{i=1}^n \langle\!\langle v_i \rangle\!\rangle_R$.

Structural Commentary: Relatively prime cyclic pieces merge into one cyclic piece.
Key Insight: Bezout identities recover each component from the single sum. □

Perspective 2.3.7. *Noetherian Modules over a PID*
Over a PID, $N \leq R^n$ implies N free with $\mathrm{rk}_R(N) \leq n$; Noetherianity is the f.g. shadow.

Theorem 2.3.8. *Submodules of Finite Free PID Modules*
Let F be free over R, $\mathrm{rk}_R(F) = n < \infty$, and $N \leq F$. Then $\exists m \leq n$ such that $N \cong R^m$.
Consequently, for $\mu_R(M) := \min\{|X| : \langle\!\langle X \rangle\!\rangle_R = M\}$, $\mu_R(M) \leq n$ implies $S \leq M \Longrightarrow \mu_R(S) \leq n$.

Proof. Identify $F \simeq R^n$. For $1 \leq k \leq n$, set $I_k := \{x_k : x = (x_1, \ldots, x_k, 0, \ldots, 0) \in N\} \lhd R$ and write $I_k = \langle r_k \rangle$..
If $r_k \neq 0$, choose $u_k \in N \cap (R^k \times \{0\}^{n-k})$ with $(u_k)_k = r_k$. Let $K = \{k : r_k \neq 0\}$.
If $\sum_{k \in K} c_k u_k = 0$ and $j = \max\{k : c_k \neq 0\}$, then the jth coordinate gives $c_j r_j = 0$, hence $c_j = 0$, contradiction.
Thus $\{u_k : k \in K\}$ is R-independent. Set $\ell(0) = 0$ and $\ell(x) = \max \mathrm{supp}(x)$ for $0 \neq x \in N$.
If $j = \ell(x) > 0$, then $x_j \in I_j = \langle r_j \rangle$, say $x_j = cr_j$, and $\ell(x - cu_j) < j$.
Induction on $\ell(x)$ gives $N = \bigoplus_{k \in K} Ru_k \cong R^{|K|}$, so $\mathrm{rk}_R(N) = |K| \leq n$.
If $\mu_R(M) \leq n$, choose $\pi : R^n \twoheadrightarrow M$. For $S \leq M$, $P := \pi^{-1}(S) \leq R^n$ has $P = \langle\!\langle v_1, \ldots, v_m \rangle\!\rangle_R$ with $m \leq n$.
Hence $S = \langle\!\langle \pi(v_1), \ldots, \pi(v_m) \rangle\!\rangle_R$ and $\mu_R(S) \leq m \leq n$.
Structural Commentary: PIDs let one choose pivot generators in a submodule of R^n.
Key Insight: Pivot ideals I_k yield triangular free generators. □

Corollary 2.3.9. *Finitely Generated PID Modules Are Noetherian*
If $\mu_R(M) < \infty$, then M is Noetherian.

Proof. Let $n = \mu_R(M)$. By Theorem 2.3.8, $N \leq M \Longrightarrow \mu_R(N) \leq n < \infty$.
Thus every submodule of M is f.g.; Theorem 2.2.2 gives M Noetherian.
Structural Commentary: Finite generation of every submodule prevents infinite strict growth.
Key Insight: Generators of the union appear at a finite stage. □

Definition 2.3.10. *Rank of a Finite Free Module*
If F is a finite free R-module and $F \cong R^n$, the integer n is the *rank* of F, denoted $\mathrm{rk}(F)$.

Proposition 2.3.11. *Finite Free Rank Is Well-Defined over a Domain*
Let R be an integral domain and $m, n \geq 0$. Then $R^m \cong_R R^n \Longrightarrow m = n$.

Proof. Let $K = \mathrm{Frac}(R)$ and $\phi : R^m \xrightarrow{\sim} R^n$. Then $K \otimes_R \phi : K \otimes_R R^m \xrightarrow{\sim} K \otimes_R R^n$, i.e. $K^m \xrightarrow{\sim} K^n$.
Hence $m = \dim_K K^m = \dim_K K^n = n$.
Structural Commentary: Passing to the fraction field recovers ordinary dimension.
Key Insight: Rank over a domain is detected after denominators are allowed. □

Proposition 2.3.12. *A Full-Size Spanning Set Is a Basis*
Let $\mathrm{rk}_R(F) = n < \infty$. If $F = \langle\!\langle s_1, \ldots, s_n \rangle\!\rangle_R$, then $(s_1, \ldots, s_n)$ is an R-basis of F.

Proof. Choose a basis $(b_i)_{i=1}^n$ and define $\tau \in \mathrm{End}_R(F)$ by $\tau(b_i) = s_i$. Then τ is onto.
Since F is free, $\exists \rho : F \to F$ with $\tau\rho = \mathrm{id}_F$; hence $F = \ker\tau \oplus \mathrm{im}\,\rho$ and $\mathrm{im}\,\rho \cong F$.
By Theorem 2.3.8, $\ker\tau \cong R^r$ for some $r \leq n$. Thus $R^n \cong F \cong \ker\tau \oplus F \cong R^{r+n}$, so $r = 0$ by Proposition 2.3.11. Hence $\ker\tau = 0$, τ is an isomorphism, and $(s_i) = \tau(b_i)$ is a basis.
Structural Commentary: Surjective finite-free endomorphisms over a PID are injective.
Key Insight: Rank leaves no room for a split kernel. □

Perspective 2.3.13. *Smith Form as Diagonalized Relations*
For a presentation $R^n \xrightarrow{A} R^m \to M \to 0$, Smith form replaces A by a diagonal relation matrix, decoupling finite relations into cyclic constraints.

Theorem 2.3.14. *Smith Normal Form over a PID*
Let $A \in \mathrm{Mat}(m, n, R)$, viewed as $R^n \to R^m$.
$\exists$ invertible $P \in \mathrm{Mat}(m, m, R)$ and $Q \in \mathrm{Mat}(n, n, R)$ such that $PAQ = \mathrm{diag}(d_1, \ldots, d_r, 0, \ldots, 0)$,
where $d_i \in R \setminus \{0\}$, $d_1 \mid \cdots \mid d_r$, and $r = \mathrm{rk}_R(\mathrm{im}\,A)$. The ideals $\langle d_i \rangle$ are unique.

Proof.

(i) *Existence*: If $A = 0$, done. Set $I(A) = \langle a_{ij} \rangle = \langle d \rangle$. Elementary operations move a Bezout combination equal to d to position $(1,1)$. If a pivot $p \nmid a$, Bezout on (p, a) replaces p by $\gcd(p, a)$, so $\langle p \rangle$ strictly increases; ACC in a PID forces termination. Then p divides every entry, so clear its row and column. Induct on the complementary submatrix. Finally replace each bad adjacent pair (a, b) by $(\gcd(a, b), \mathrm{lcm}(a, b))$ until $d_1 \mid \cdots \mid d_r$.

(ii) *Uniqueness*: Let $\Delta_k(A)$ be the ideal of $k \times k$ minors and $\Delta_0(A) = R$. Matrix equivalence preserves each Δ_k. In Smith form, $\Delta_k(A) = \langle d_1 \cdots d_k \rangle$ for $1 \leq k \leq r$, and $\Delta_k(A) = 0$ for $k > r$. Since R is a domain, $(\Delta_k(A) : \Delta_{k-1}(A)) = \langle d_k \rangle$, so d_k is determined up to associates.

Structural Commentary: Smith form preserves rank plus divisibility data.
Key Insight: Determinantal ideals encode invariant factors. □

Theorem 2.3.15. *Adapted Bases for Submodules of Free Modules*
Let $N \leq F$, $\mathrm{rk}_R(F) = n$, and $\mathrm{rk}_R(N) = k$.
Then $\exists (v_i)_{i=1}^n \in \mathrm{Bas}_R(F)$ and $r_1, \ldots, r_k \in R \setminus \{0\}$ such that $r_1 \mid \cdots \mid r_k$ and $(r_i v_i)_{i=1}^k \in \mathrm{Bas}_R(N)$.

Proof. Choose bases of N and F, and let A represent $\iota : N \hookrightarrow F$.
Smith form gives $PAQ = \mathrm{diag}(r_1, \ldots, r_k, 0, \ldots, 0)$ with $r_i \neq 0$ and $r_1 \mid \cdots \mid r_k$, since ι is injective.
The row operations give a basis $(v_i)_{i=1}^n$ of F and the column operations give a basis $(w_i)_{i=1}^k$ of N with $\iota(w_i) = r_i v_i$. Hence $(r_i v_i)_{i=1}^k = (w_i)_{i=1}^k$ is a basis of N.
Structural Commentary: Inclusions diagonalize after choosing bases.
Key Insight: Diagonal entries are the required rescalings. □

Corollary 2.3.16. *Smith Form Presents Finitely Generated PID Modules*
Let $R^n \xrightarrow{A} R^m \to M \to 0$ be a finite presentation.
If $A \sim \mathrm{diag}(d_1, \ldots, d_r, 0, \ldots, 0)$ in Smith form. $I = \{i : 1 \leq i \leq r,\ d_i \notin R^\times\}$, then $M \cong R^{m-r} \oplus \bigoplus_{i \in I} R/\langle d_i \rangle$.

Proof. Equivalence $A \sim PAQ$ changes bases in R^m, R^n, so $\mathrm{coker}\,A \cong \mathrm{coker}(PAQ)$.
For the diagonal map, $d_i \in R^\times$ gives 0, $d_i \notin R^\times$ gives $R/\langle d_i \rangle$, and each zero row gives R.
Structural Commentary: Smith form turns presentations into direct sums.
Key Insight: Diagonal relations create cyclic torsion summands, and zero rows create free summands. □

Method 2.3.17. *Reading a Module from Smith Form*
For finite presentation $R^n \xrightarrow{A} R^m \to M \to 0$ over PID, put $A \sim D = \operatorname{diag}(d_1, \dots, d_r, 0, \dots, 0)$, $d_1 \mid \cdots \mid d_r$:

(i) $d_i \in R^\times \implies$ no summand;
(ii) $d_i \notin R^\times \implies R/\langle d_i \rangle$;
(iii) each zero target coordinate contributes one R;
(iv) factor nonunit d_i into prime powers for elementary divisors.

Key Insight: Smith form changes arbitrary relations into independent one-coordinate relations.

Example 2.3.18. *Smith Form for a Finite Abelian Group*
Let $M = \mathbb{Z}^2/\langle\!\langle (6,0), (0,10), (2,4) \rangle\!\rangle_{\mathbb{Z}}$, with relation matrix $A = \begin{bmatrix} 6 & 0 & 2 \\ 0 & 10 & 4 \end{bmatrix}$. Then $\Delta_1(A) = \langle 6, 10, 2, 4 \rangle = \langle 2 \rangle$ and $\Delta_2(A) = \langle 60, 24, -20 \rangle = \langle 4 \rangle$. Hence $d_1 \sim 2$, $d_1 d_2 \sim 4$, so $d_2 \sim 2$ and $M \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.
Key Insight: Determinantal ideals read off cyclic summands.

Example 2.3.19. *Recombining Elementary Divisors*
Let $G = \mathbb{Z}/12\mathbb{Z} \oplus \mathbb{Z}/18\mathbb{Z}$. Then $\operatorname{ElemDiv}(G) = \{2, 2^2, 3, 3^2\}$. Recombining equal-height primary blocks gives $2 \cdot 3 = 6$ and $2^2 \cdot 3^2 = 36$; hence $G \cong \mathbb{Z}/6\mathbb{Z} \oplus \mathbb{Z}/36\mathbb{Z}$, with $6 \mid 36$.
Structural Commentary: Elementary divisors split by primes; invariant factors recombine them.
Key Insight: Prime-wise data equals one divisibility chain.

2.4 Finitely Generated PID Modules

Theorem 2.4.1. *Finitely Generated Torsion-Free Modules Are Free*
For a f.g. R-module M, M is free $\iff$ M is torsion-free.

Proof.
($\Rightarrow$) If $M \cong R^n$, then $am = 0$ with $a \neq 0$ forces every coordinate of m to be 0, since R is a domain.
($\Leftarrow$) Let M be torsion-free and $G = \{u_1, \dots, u_n\}$ generate M.
Choose maximal R-independent $S = \{v_1, \dots, v_k\} \subseteq G$. For $w \in G \setminus S$, dependence of $S \cup \{w\}$ gives $a_w \neq 0$ and $b_{w,i} \in R$ with $a_w w + \sum_i b_{w,i} v_i = 0$. Put $a = \prod_{w \in G \setminus S} a_w$. Then $aM \subseteq \langle\!\langle S \rangle\!\rangle_R$, so aM is free by Theorem 2.3.8. Since M is torsion-free, $M \xrightarrow{\sim} aM$, $m \mapsto am$; hence M is free.
Structural Commentary: Clear denominators into a free submodule.
Key Insight: One scalar embeds M into a free sp an. □

Theorem 2.4.2. *Free Plus Torsion Decomposition*
For f.g. M, $M = M_{\text{tor}} \oplus F$ with F f.g. free. Moreover M_{tor} is intrinsic and $F \cong M/M_{\text{tor}}$.

Proof. By Proposition 1.3.3, $T := M_{\text{tor}} \leq M$ and $Q := M/T$ is f.g. torsion-free.
Thus Q is free by Theorem 2.4.1. Since Q is free, the quotient map $\pi : M \twoheadrightarrow Q$ splits; hence $M = T \oplus \operatorname{im}\sigma$ for any section $\sigma : Q \to M$, and $\operatorname{im}\sigma \cong Q$. If $M = T' \oplus G$ with T' torsion and G free, then $m = t' + g \in M_{\text{tor}}$ forces $g = 0$, so $T' = M_{\text{tor}}$. Every complement is isomorphic to M/M_{tor}.
Structural Commentary: F.g. PID modules split into torsion plus free.
Key Insight: The quotient by torsion is free, and freeness makes the quotient map split. □

Definition 2.4.3. *Primary Module*
Let $p \in R$ be prime. A torsion R-module M is *p-primary* if $o(M) \sim p^e$ for some $e \geq 1$.

Theorem 2.4.4. *Primary Decomposition*
Let M be f.g. torsion with $o(M) = \mu \sim \prod_{i=1}^{n} p_i^{e_i}$, where the p_i are distinct nonassociate primes.

Put $q_i := p_i^{e_i}$, $\mu_i := \mu/q_i$, and $M_i := \mu_i M$. Then $M = \bigoplus_{i=1}^{n} M_i$, $M_i = \{v \in M : q_i v = 0\}$, and $o(M_i) \sim q_i$.

The decomposition is unique up to ordering.

Proof. Since $\gcd(\mu_1, \dots, \mu_n) \sim 1$, choose $a_i \in R$ with $\sum_i a_i \mu_i = 1$. Hence $x = \sum_i a_i \mu_i x$, so $M = \sum_i M_i$.
Also $q_i M_i = \mu M = 0$. Conversely, if $q_i v = 0$, choose $a, b \in R$ with $a\mu_i + bq_i = 1$; then $v = a\mu_i v \in M_i$.
Thus $M_i = \{v : q_i v = 0\}$. If $\sum_i x_i = 0$ with $x_i \in M_i$, applying μ_i gives $\mu_i x_i = 0$ as $q_j \mid \mu_i$ for $j \neq i$.
Since $q_i x_i = 0$ and $\gcd(q_i, \mu_i) \sim 1$, Bezout gives $x_i = 0$; hence the sum is direct.
Now $o(M_i) \mid q_i$ and $\operatorname{lcm}_i o(M_i) \sim o(M) \sim \prod_i q_i$, so $o(M_i) \sim q_i$.
Finally each summand is the intrinsic set $\{v : q_i v = 0\}$, where q_i is the p_i-part of $o(M)$.
Structural Commentary: Primary decomposition separates a torsion module prime by prime.
Key Insight: Bezout idempotents built from the coprime factors of the order project onto the primary pieces. □

Lemma 2.4.5. *The First p-Torsion Layer*
Let $p \in R$ be prime and $M[p] := \ker(p : M \to M) = \{v \in M : pv = 0\}$.
Then $M[p]$ is an $R/\langle p\rangle$-vector space. If $M = S \oplus T$, then $M[p] = S[p] \oplus T[p]$.

Proof. $\langle p\rangle$ is maximal by Theorem 3.2.15 and Theorem 3.1.11; hence $\kappa := R/\langle p\rangle$ is a field. Define $(r+\langle p\rangle)v := rv$. If $r - r' \in \langle p\rangle$ and $v \in M[p]$, then $(r - r')v = 0$, so the action is well-defined; the axioms are inherited from M. If $v = s + t \in M[p]$ under $M = S \oplus T$, then $0 = pv = ps + pt$ gives $ps = pt = 0$ by directness. Thus $v \in S[p] \oplus T[p]$, and the reverse inclusion is immediate.
Structural Commentary: Killing by p turns scalar multiplication modulo p into field scalar multiplication.
Key Insight: The layer $M[p]$ counts the number of cyclic primary summands. □

Theorem 2.4.6. *Cyclic Decomposition of a Primary Module*
Let M be f.g. p-primary torsion with $o(M) \sim p^e$.
Then $M = \bigoplus_{i=1}^{m} \langle\!\langle v_i \rangle\!\rangle_R$ with $\mathrm{Ann}_R(v_i) = \langle p^{e_i}\rangle$ and $e = e_1 \geq e_2 \geq \cdots \geq e_m \geq 1$.
The annihilator chain $\langle p^{e_1}\rangle \subseteq \cdots \subseteq \langle p^{e_m}\rangle$ is intrinsic.

Proof.

(i) *Existence*: Choose a finite presentation of M.
Corollary 2.3.16 gives $M \cong R^s \oplus \bigoplus_{\ell=1}^{t} R/\langle d_\ell\rangle$, with d_ℓ nonunit. Since M is torsion, $s = 0$.
Since M is p-primary, $d_\ell \sim p^{e_\ell}$ with $e_\ell \geq 1$. Reorder so $e_1 \geq \cdots \geq e_m$.
Proposition 2.3.4 gives $o(M) \sim p^{e_1}$, hence $e_1 = e$.

(ii) *Uniqueness*: Put $\kappa = R/\langle p\rangle$. By Lemma 2.4.5, each summand $R/\langle p^{e_i}\rangle$ contributes one κ-dimension to $M[p]$, so $\dim_\kappa M[p] = m$. Passing from M to pM replaces each $e_i > 1$ by $e_i - 1$ and removes each $e_i = 1$. Induction on e recovers the exponent multiset, hence the annihilator chain.

Structural Commentary: A primary module splits into cyclic strings whose lengths are the exponents e_i.
Key Insight: The dimensions of the layers $M[p], pM[p], p^2M[p], \ldots$ determine the string lengths. □

Proposition 2.4.7. *Layer Counts Recover Primary Exponents*
Let $\kappa = R/\langle p\rangle$ and $\lambda_k(M) := \dim_\kappa(p^{k-1}M/p^kM)$ for $k \geq 1$.
If $M \cong \bigoplus_{i=1}^{m} R/\langle p^{e_i}\rangle$ with $e_1 \geq \cdots \geq e_m \geq 1$, then $\lambda_k(M) = \#\{i : e_i \geq k\}$.
Hence the number of summands of exact exponent k is $\lambda_k(M) - \lambda_{k+1}(M)$.

Proof. For $C = R/\langle p^e\rangle$, $p^{k-1}C/p^kC$ is 0 if $k > e$ and is κ if $k \leq e$, generated by the class of p^{k-1}. Since direct sums commute with $p^{k-1}M$ and p^kM, dimensions add, giving $\lambda_k(M) = \#\{i : e_i \geq k\}$. Thus $\lambda_k - \lambda_{k+1}$ counts exactly the summands ending at exponent k.
Structural Commentary: Successive p-layers determine the exponent list.
Key Insight: A cyclic summand of order p^e contributes one dimension to exactly the first e layers. □

2.5 Elementary Divisors and Invariant Factors

Definition 2.5.1. *Elementary Divisors*
For finitely generated torsion M, if its primary cyclic summands have annihilators $\langle p_i^{e_{i,j}}\rangle$, set $\mathrm{ElemDiv}(M) := \{p_i^{e_{i,j}}\}_{i,j}$ as a multiset, up to associates.

Theorem 2.5.2. *Primary Cyclic Decomposition*
Every f.g. torsion R-module M admits $M = \bigoplus_i \bigoplus_{j=1}^{k_i} \langle\!\langle v_{i,j} \rangle\!\rangle_R$, where $\mathrm{Ann}_R(v_{i,j}) = \langle p_i^{e_{i,j}}\rangle$, p_i are distinct nonassociate primes, and $e_{i,1} \geq \cdots \geq e_{i,k_i} \geq 1$. For f.g. torsion M, N, $M \cong N \iff \mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$.

Proof.

($\Rightarrow$) Theorem 2.4.4 gives $M = \bigoplus_i M_{p_i}$ with each M_{p_i} f.g. p_i-primary. Theorem 2.4.6 gives $M_{p_i} = \bigoplus_{j=1}^{k_i} \langle\!\langle v_{i,j} \rangle\!\rangle_R$ with $\mathrm{Ann}_R(v_{i,j}) = \langle p_i^{e_{i,j}}\rangle$. If $M \cong N$, the same uniqueness statements identify the primary pieces and their exponent chains; hence $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$.

($\Leftarrow$) If $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$, then both modules are isomorphic to the same direct sum $\bigoplus_{q \in \mathrm{ElemDiv}(M)} R/\langle q\rangle$, since a cyclic summand with annihilator $\langle q\rangle$ is $R/\langle q\rangle$ by Theorem 2.3.5. Thus $M \cong N$.

Structural Commentary: A torsion module is a direct sum of prime-power cyclic blocks.
Key Insight: Elementary divisors are the atoms of the PID module classification. □

Proposition 2.5.3. *Elementary Divisors of a Direct Sum*
For f.g. torsion A, B, $\mathrm{ElemDiv}(A \oplus B) = \mathrm{ElemDiv}(A) \uplus \mathrm{ElemDiv}(B)$*, where $\uplus$ is multiset union.*

Proof. If $A = \bigoplus_i A_i$ and $B = \bigoplus_j B_j$ are primary cyclic decompositions, then $A \oplus B = (\bigoplus_i A_i) \oplus (\bigoplus_j B_j)$ is one for $A \oplus B$. Thus the prime-power annihilator multiset is the multiset union.
Structural Commentary: Direct sums concatenate cyclic primary blocks.
Key Insight: Elementary divisors count blocks with multiplicity. □

Definition 2.5.4. *Invariant Factors*
For f.g. torsion M, write $\mathrm{ElemDiv}(M) = \{p_i^{e_{i,j}} : 1 \le j \le k_i\}_i$ with $e_{i,1} \ge \cdots \ge e_{i,k_i} \ge 1$.
Let $m = \max_i k_i$, set $e_{i,j} = 0$ for $j > k_i$, and define $d_j := \prod_i p_i^{e_{i,j}}$.
The nonunit d_j, up to associates, are the *invariant factors*; they satisfy $d_m \mid \cdots \mid d_1$.

Method 2.5.5. *From Elementary Divisors to Invariant Factors*

(i) Group $\mathrm{ElemDiv}(M)$ by prime: $\{p_i^{e_{i,j}}\}_j$.
(ii) Reorder each list so $e_{i,1} \ge e_{i,2} \ge \cdots$ and pad by $e_{i,j} = 0$ to common length.
(iii) Form $d_j = \prod_i p_i^{e_{i,j}}$.
(iv) Discard unit d_j; the rest are the invariant factors from Definition 2.5.4.

Example 2.5.6. *Packing Elementary Divisors into Invariant Factors*
Over $R = \mathbb{F}[z]$, let $\mathrm{ElemDiv}(M) = \{(z-1)^3, z-1, (z^2+1)^2, z^2+1\}$. The exponent rows are $(3,1)$ for $z-1$ and $(2,1)$ for z^2+1, so $d_1 = (z-1)^3(z^2+1)^2$ and $d_2 = (z-1)(z^2+1)$, with $d_2 \mid d_1$. Hence $M \cong R/\langle d_1\rangle \oplus R/\langle d_2\rangle$.
Structural Commentary: Elementary divisors separate primes; invariant factors pack rows.
Key Insight: Line up equal-height primary strings and multiply across the row.

Theorem 2.5.7. *Invariant Factor Decomposition*
For f.g. torsion M, $M \cong \bigoplus_{j=1}^m R/\langle d_j\rangle$ with $d_j \notin R^\times$ and $d_m \mid \cdots \mid d_1$.
Writing $\mathrm{Inv}(M) := (d_m, \ldots, d_1)$ up to associates, for f.g. torsion M, N, $M \cong N \iff \mathrm{Inv}(M) = \mathrm{Inv}(N)$.

Proof. By Theorem 2.5.2, $M \cong \bigoplus_{q \in \mathrm{ElemDiv}(M)} R/\langle q\rangle$.
Let $\mathrm{ElemDiv}(M) = \{p_i^{e_{i,j}} : 1 \le j \le k_i\}_i$ with $e_{i,1} \ge \cdots \ge e_{i,k_i}$, pad by $e_{i,j} = 0$, and set $d_j = \prod_i p_i^{e_{i,j}}$.
For fixed j, the nonunit $p_i^{e_{i,j}}$ are pairwise coprime, so Theorem 2.3.6 gives $\bigoplus_{e_{i,j}>0} R/\langle p_i^{e_{i,j}}\rangle \cong R/\langle d_j\rangle$.
Hence $M \cong \bigoplus_j R/\langle d_j\rangle$ with $d_m \mid \cdots \mid d_1$.
($\Rightarrow$) If $M \cong N$, then $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$ by Theorem 2.5.2.
Grouping by height gives $\mathrm{Inv}(M) = \mathrm{Inv}(N)$.
($\Leftarrow$) If $\mathrm{Inv}(M) = \mathrm{Inv}(N)$, by factorization of each d_j, $\mathrm{ElemDiv}(M) = \mathrm{ElemDiv}(N)$; QED (Theorem 2.5.2).
Structural Commentary: Invariant factors pack prime-power blocks.
Key Insight: Same data, different packaging. □

Corollary 2.5.8. *Classification of Finitely Generated PID Modules*
For f.g. M, $M \cong R^r \oplus \bigoplus_{j=1}^m R/\langle d_j\rangle$, where $r \ge 0$, $d_j \notin R^\times$, and $d_m \mid \cdots \mid d_1$.
The pair $(r, \mathrm{Inv}(M_{\mathrm{tor}}))$, equivalently $(r, \mathrm{ElemDiv}(M_{\mathrm{tor}}))$, is a complete invariant.

Proof. Let $T := M_{\mathrm{tor}}$. Theorem 2.4.2 gives $M = T \oplus F$ with $F \cong R^r$.
Theorem 2.5.7 gives $T \cong \bigoplus_{j=1}^m R/\langle d_j\rangle$ with $d_m \mid \cdots \mid d_1$. Since T is intrinsic and $F \cong M/T$, uniqueness and completeness follow from rank uniqueness and the invariant-factor theorem.
Structural Commentary: F.g. PID modules are free plus cyclic torsion.
Key Insight: The free rank records the zero-annihilator part; invariant factors record the torsion part. □

Perspective 2.5.9. *Largest Invariant Factor and Product*
If $M \cong \bigoplus_{j=1}^m R/\langle d_j\rangle$ with $d_m \mid \cdots \mid d_1$, then $\mathrm{Ann}_R(M) = \langle d_1\rangle$ and $\prod_{j=1}^m d_j \sim \prod_{q \in \mathrm{ElemDiv}(M)} q$.
For $R = \mathbb{F}[z]$ and $M = (\mathbb{F}^n)_A$, after monic normalisation, $d_1 = m_A$ and $\prod_j d_j = \chi_A$.

Theorem 2.5.10. *Characterizing Cyclic Torsion Modules*
Let M be f.g. torsion with order $\mu \sim \prod_{i=1}^n p_i^{e_i}$, p_i distinct nonassociate primes. These are equivalent:

(i) M is cyclic.
(ii) $\forall i$, the primary component M_{p_i} is cyclic.
(iii) $\mathrm{ElemDiv}(M) = \{p_i^{e_i} : 1 \le i \le n\}$.
(iv) $M \cong R/\langle \mu\rangle$.

Proof.

(*i*) $\Rightarrow$ (*ii*) If $M = \langle\!\langle x \rangle\!\rangle_R$, Theorem 2.3.6 gives $M = \bigoplus_i \langle\!\langle x_i \rangle\!\rangle_R$ with $o(x_i) \sim p_i^{e_i}$; hence $M_{p_i} = \langle\!\langle x_i \rangle\!\rangle_R$.
(*ii*) $\Rightarrow$ (*iii*) By Theorem 2.5.2, cyclic M_{p_i} of order $p_i^{e_i}$ has sole elementary divisor $p_i^{e_i}$; use Proposition 2.5.3.
(*iii*) $\Rightarrow$ (*iv*) Method 2.5.5 packs the displayed elementary divisors into one invariant factor $d_1 \sim \prod_i p_i^{e_i} \sim \mu$.
(*iv*) $\Rightarrow$ (*i*) Since $R/\langle\mu\rangle = \langle\!\langle 1 + \langle\mu\rangle \rangle\!\rangle_R$, any isomorphism $R/\langle\mu\rangle \xrightarrow{\sim} M$ sends $1 + \langle\mu\rangle$ to a generator of M.

Structural Commentary: Cyclic torsion means one string per prime.
Key Insight: Multiple elementary divisors for the same prime are the obstruction to one generator. □

Definition 2.5.11. *Indecomposable Module*
A nonzero R-module M is *indecomposable* if $M = A \oplus B \Longrightarrow A = 0$ or $B = 0$.

Theorem 2.5.12. *Indecomposable Finitely Generated Torsion Modules*
Let $0 \neq M$ be f.g. torsion. The following are equivalent:

(*i*) *M is indecomposable.*
(*ii*) *M is primary cyclic.*
(*iii*) $\mathrm{ElemDiv}(M) = \{p^e\}$ *for some prime p and $e \geq 1$.*

Proof.

(*i*) $\Rightarrow$ (*ii*) By Theorem 2.5.2, write $M \cong \bigoplus_{\alpha \in A} C_\alpha$, where $0 \neq C_\alpha \cong R/\langle p_\alpha^{e_\alpha} \rangle$.
If $|A| \geq 2$, then $M \cong C_{\alpha_0} \oplus \bigoplus_{\alpha \neq \alpha_0} C_\alpha$ is a nontrivial decomposition, contradiction.
Since $M \neq 0$, $|A| = 1$; hence M is primary cyclic.
(*ii*) $\Rightarrow$ (*iii*) If $M \cong R/\langle p^e \rangle$, then $\mathrm{ElemDiv}(M) = \{p^e\}$.
(*iii*) $\Rightarrow$ (*i*) If $M = A \oplus B$, Proposition 2.5.3 gives $\{p^e\} = \mathrm{ElemDiv}(A) \sqcup \mathrm{ElemDiv}(B)$.
Hence one multiset is empty, so one summand is 0.

Structural Commentary: Indecomposable torsion modules are precisely the single primary cyclic blocks.
Key Insight: Elementary divisors are the indecomposable pieces. □

Proposition 2.5.13. *Prime Divisors Give Prime-Order Cyclic Submodules*
Let M be f.g. torsion with order μ. If $p \mid \mu$ is prime, then $\exists w \in M$ with $o(w) \sim p$; equivalently $Rw \cong R/\langle p \rangle$.

Proof. Write $\mu = pq$. Since $q \notin \mathrm{Ann}_R(M) = \langle\mu\rangle$, choose $v \in M$ with $w := qv \neq 0$.
Then $pw = \mu v = 0$, so $o(w) \mid p$. Since $w \neq 0$, $o(w)$ is nonunit; as p is prime, $o(w) \sim p$.
Structural Commentary: Every prime in the global order appears in some one-dimensional primary layer.
Key Insight: Divide the order by one prime and apply the remaining scalar to an element it does not kill. □

3 Linear Operators as Modules

3.1 Polynomial Modules from Operators

Throughout this subsection, $\mathcal{V}$ is finite-dimensional over $\mathbb{F}$ unless explicitly stated otherwise.

Perspective 3.1.1. *An Operator as a Polynomial Action*
Fix $T \in \mathcal{L}(\mathcal{V})$. Evaluation at T defines an $\mathbb{F}[z]$-action on $\mathcal{V}$ by $p \cdot \mathbf{v} := p(T)\mathbf{v}$.
Thus the orbit $\mathbf{v}, T\mathbf{v}, T^2\mathbf{v}, \ldots$ is encoded by the cyclic subspace $\{p(T)\mathbf{v} : p \in \mathbb{F}[z]\} = \mathrm{span}\{T^j\mathbf{v} : j \geq 0\}$.

Definition 3.1.2. *The Module Attached to an Operator*
Let $T \in \mathcal{L}(\mathcal{V})$. The *operator module* attached to T is the $\mathbb{F}[z]$-module $\mathcal{V}_T$ with underlying additive group $\mathcal{V}$ and action $p(z) \cdot \mathbf{v} := p(T)\mathbf{v}$. For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, define $(\mathbb{F}^n)_A$ analogously by $p(z) \cdot \mathbf{x} := p(A)\mathbf{x}$.

Proposition 3.1.3. *Operator Modules Are Finitely Generated Torsion Modules*
Let $0 < \dim \mathcal{V} < \infty$ and $T \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}_T$ is a finitely generated torsion $\mathbb{F}[z]$-module.

Proof. Let $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be an $\mathbb{F}$-basis of $\mathcal{V}$. Since $\mathbb{F} \subseteq \mathbb{F}[z]$, this basis generates $\mathcal{V}_T$.
By Proposition 6.1.4, there exists $0 \neq p \in \mathbb{F}[z]$ with $p(T) = 0$. Thus $p \cdot \mathbf{v} = p(T)\mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathcal{V}$, so $\mathcal{V}_T$ is torsion. Finally, $\mathbb{F}[z]$ is a PID by Theorem 3.2.11.
Structural Commentary: Finite-dimensionality gives a global polynomial relation for the operator.
Key Insight: Rational canonical form is the coordinate form of the structure theorem for $\mathcal{V}_T$. □

Theorem 3.1.4. *Submodules Are Invariant Subspaces*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathcal{S} \subseteq \mathcal{V}$. Then $\mathcal{S} \leq_{\mathbb{F}[z]} \mathcal{V}_T$ if and only if $\mathcal{S} \leq_{\mathbb{F}} \mathcal{V}$ and $T\mathcal{S} \subseteq \mathcal{S}$.

Proof.

($\Rightarrow$) If $\mathcal{S} \leq_{\mathbb{F}[z]} \mathcal{V}_T$, then $\mathbb{F} \subseteq \mathbb{F}[z]$ gives $\mathcal{S} \leq_{\mathbb{F}} \mathcal{V}$. Since $z \cdot \mathbf{s} = T\mathbf{s}$ for $\mathbf{s} \in \mathcal{S}$, we have $T\mathcal{S} \subseteq \mathcal{S}$.

($\Leftarrow$) If $\mathcal{S} \leq_{\mathbb{F}} \mathcal{V}$ and $T\mathcal{S} \subseteq \mathcal{S}$, then $T^j\mathcal{S} \subseteq \mathcal{S}$ for all $j \geq 0$.
Hence for $p(z) = \sum_j a_j z^j$ and $\mathbf{s} \in \mathcal{S}$, $p \cdot \mathbf{s} = p(T)\mathbf{s} = \sum_j a_j T^j \mathbf{s} \in \mathcal{S}$, so $\mathcal{S} \leq_{\mathbb{F}[z]} \mathcal{V}_T$.

Structural Commentary: The single scalar z remembers the operator.
Key Insight: Invariance under T is exactly closure under all polynomial scalars. □

Theorem 3.1.5. *Similarity Is Module Isomorphism*
Let $T, S \in \mathcal{L}(\mathcal{V})$. Then $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S \iff T \sim S$. More precisely, for $\Phi \in \mathrm{GL}(\mathcal{V})$, $\Phi : \mathcal{V}_T \to \mathcal{V}_S$ is $\mathbb{F}[z]$-linear if and only if $\Phi T = S\Phi$, equivalently $S = \Phi T \Phi^{-1}$.

Proof.
($\Rightarrow$) If $\Phi : \mathcal{V}_T \to \mathcal{V}_S$ is an $\mathbb{F}[z]$-isomorphism, then $\Phi \in \mathrm{GL}(\mathcal{V})$, $\Phi(T\mathbf{v}) = \Phi(z \cdot \mathbf{v}) = z \cdot \Phi(\mathbf{v}) = S\Phi(\mathbf{v})$ $\forall \mathbf{v} \in \mathcal{V}$.
Hence $\Phi T = S\Phi$, so $S = \Phi T \Phi^{-1}$.
($\Leftarrow$) If $\Phi \in \mathrm{GL}(\mathcal{V})$ and $S = \Phi T \Phi^{-1}$, then $\Phi T = S\Phi$, hence $\Phi T^j = S^j \Phi$ for all $j \geq 0$.
Thus for $p(z) = \sum_j a_j z^j$, $\Phi(p(T)\mathbf{v}) = p(S)\Phi(\mathbf{v})$, so $\Phi : \mathcal{V}_T \to \mathcal{V}_S$ is an $\mathbb{F}[z]$-isomorphism.

Structural Commentary: A module isomorphism is precisely an intertwiner for the action of z.
Key Insight: Similarity is isomorphism of the associated $\mathbb{F}[z]$-modules. □

Corollary 3.1.6. *Matrix Similarity Is Matrix-Module Isomorphism*
Let $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$. Then $A \sim B \iff (\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$. More precisely, for $X \in \mathrm{GL}_n(\mathbb{F})$, $X : (\mathbb{F}^n)_A \to (\mathbb{F}^n)_B$ is $\mathbb{F}[z]$-linear if and only if $XA = BX$, equivalently $B = XAX^{-1}$.

Proof.
($\Rightarrow$) If $X : (\mathbb{F}^n)_A \to (\mathbb{F}^n)_B$ is an $\mathbb{F}[z]$-isomorphism, then $X \in \mathrm{GL}_n(\mathbb{F})$ and $X(A\mathbf{x}) = X(z \cdot \mathbf{x}) = z \cdot X\mathbf{x} = BX\mathbf{x}$ for all $\mathbf{x} \in \mathbb{F}^n$. Hence $XA = BX$, so $B = XAX^{-1}$.
($\Leftarrow$) If $X \in \mathrm{GL}_n(\mathbb{F})$ and $B = XAX^{-1}$, then $XA = BX$, hence $XA^j = B^j X$ for all $j \geq 0$. Thus $Xp(A) = p(B)X$ for all $p \in \mathbb{F}[z]$, so $X : (\mathbb{F}^n)_A \to (\mathbb{F}^n)_B$ is an $\mathbb{F}[z]$-isomorphism.

Structural Commentary: Similarity is invertible module intertwining.
Key Insight: The matrix is remembered by the scalar action of z. □

Definition 3.1.7. *Module Annihilator and Minimal Polynomial*
(i) The *annihilator* of $\mathcal{V}_T$ is $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) := \{p \in \mathbb{F}[z] : p(T)\mathcal{V} = \{\mathbf{0}\}\}$.
(ii) *Minimal polynomial* μ_T is unique monic generator of this nonzero principal ideal, so $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$.
(iii) For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, define $\mu_A := \mu_{T_A}$, where $T_A(\mathbf{x}) = A\mathbf{x}$.

Proposition 3.1.8. *Minimal Polynomial as Module Order*
Let $T \in \mathcal{L}(\mathcal{V})$. Then μ_T is the unique monic polynomial of minimal degree in $\{p \in \mathbb{F}[z] : p(T) = 0\}$. If $S \sim T$, then $\mu_S = \mu_T$; hence similar matrices have the same minimal polynomial.

Proof. By Definition 3.1.7, $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T) = \langle \mu_T \rangle$ and $p \in \mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T)$ iff $p(T) = 0$. Thus μ_T is the unique monic annihilator of minimal degree. If $S = \Phi T \Phi^{-1}$, then $p(S) = \Phi p(T) \Phi^{-1}$ for all $p \in \mathbb{F}[z]$, so $\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_S) = \mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T)$ and $\mu_S = \mu_T$. The matrix case follows by applying this to the associated operators.
Structural Commentary: The minimal polynomial is the order of the whole operator module.
Key Insight: Similarity preserves the module and therefore preserves its annihilator ideal. □

Definition 3.1.9. *Cyclic Subspace and Vector Annihilator*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V}$.
(i) The *T-cyclic subspace generated by $\mathbf{v}$* is $\mathcal{C}_T(\mathbf{v}) := \mathrm{span}\{T^j\mathbf{v} : j \geq 0\}$
(ii) The vector $\mathbf{v}$ is a *cyclic vector* for T if $\mathcal{C}_T(\mathbf{v}) = \mathcal{V}$.
(iii) The *annihilator of $\mathbf{v}$ for T* is $\mathrm{ann}_T(\mathbf{v}) := \{p \in \mathbb{F}[z] : p(T)\mathbf{v} = \mathbf{0}\}$

Proposition 3.1.10. *Cyclic Subspaces Are Smallest Invariant Hosts*
Let $T \in \mathcal{L}(\mathcal{V})$ and $\mathbf{v} \in \mathcal{V}$. Then $\mathcal{C}_T(\mathbf{v}) = \{p(T)\mathbf{v} : p \in \mathbb{F}[z]\}$.
Moreover, $\mathcal{C}_T(\mathbf{v})$ is the smallest T-invariant subspace containing $\mathbf{v}$, and $\mathrm{ann}_T(\mathbf{v})$ is an ideal of $\mathbb{F}[z]$.

Proof. Equality follows as $p(T)\mathbf{v}$ is a finite linear combination of the orbit vectors $T^j\mathbf{v}$, and every such finite linear combination equals $p(T)\mathbf{v}$ for some $p \in \mathbb{F}[z]$. Also $T(p(T)\mathbf{v}) = (zp)(T)\mathbf{v}$, so $\mathcal{C}_T(\mathbf{v})$ is T-invariant.
If $\mathcal{U}$ is T-invariant and contains $\mathbf{v}$, then $T^j\mathbf{v} \in \mathcal{U}$ for every $j \geq 0$, hence $\mathcal{C}_T(\mathbf{v}) \subseteq \mathcal{U}$. Finally, if $p, q \in \mathrm{ann}_T(\mathbf{v})$ and $r \in \mathbb{F}[z]$, then $(p+q)(T)\mathbf{v} = \mathbf{0}$ and $(rp)(T)\mathbf{v} = r(T)p(T)\mathbf{v} = \mathbf{0}$ by Proposition 6.1.3. $\mathrm{ann}_T(\mathbf{v})$ is an ideal.
Structural Commentary: A cyclic subspace is the span of the orbit generated by repeatedly applying T.
Key Insight: Invariance is exactly what lets the whole orbit stay inside the same host subspace. □

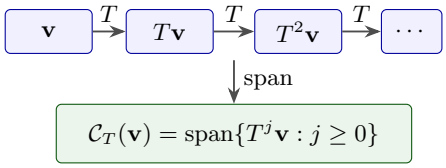


Figure 108: The T-orbit of $\mathbf{v}$ spans $\mathcal{C}_T(\mathbf{v})$.

Definition 3.1.11. *Order of a Vector under an Operator*
Let $\mathcal{V}$ be finite-dimensional, let $T \in \mathcal{L}(\mathcal{V})$, and let $\mathbf{v} \neq \mathbf{0}$. The *order of* $\mathbf{v}$ *under* T is the unique monic polynomial $m_{\mathbf{v}} \in \operatorname{ann}_T(\mathbf{v})$ of least degree; equivalently, $\operatorname{ann}_T(\mathbf{v}) = (m_{\mathbf{v}})$

Definition 3.1.12. *Companion Matrix*
Let $f(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0$ be monic. For $n \geq 2$, the *companion matrix* of f is

$$C_f := \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{bmatrix}.$$

For $n = 1$, set $C_f = [-c_0]$.

Theorem 3.1.13. *Companion Matrix Realises a Polynomial*
Let $f(z) = z^n + \sum_{i=0}^{n-1} c_i z^i$ be monic and let C_f be its companion matrix. Then $p_{C_f} = \mu_{C_f} = f$.

Proof. Let $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ be the standard basis.

For $0 \leq j < n$, $C_f^j \mathbf{e}_1 = \mathbf{e}_{j+1}$, and last column gives $C_f^n \mathbf{e}_1 = -\sum_{i=0}^{n-1} c_i C_f^i \mathbf{e}_1$. Hence $f(C_f)\mathbf{e}_1 = 0$.

Since $\mathbf{e}_j = C_f^{j-1}\mathbf{e}_1$ and $f(C_f)C_f = C_f f(C_f)$, $f(C_f)\mathbf{e}_j = 0\ \forall j$, so $f(C_f) = 0$.

If $0 \neq g(z) = \sum_{i=0}^{m} b_i z^i$ with $m < n$, then $g(C_f)\mathbf{e}_1 = \sum_{i=0}^{m} b_i \mathbf{e}_{i+1} \neq \mathbf{0}$.

Thus no nonzero polynomial of degree $< n$ annihilates C_f, so $\deg \mu_{C_f} = n$.

Since $\mu_{C_f} \mid f$ and both are monic of degree n, $\mu_{C_f} = f$.

By Cayley–Hamilton, $\mu_{C_f} \mid p_{C_f}$; as p_{C_f} and f are monic of degree n, $p_{C_f} = f$.

Structural Commentary: The cyclic vector $\mathbf{e}_1$ generates the basis, and last column imposes relation $f(C_f) = 0$.

Key Insight: The companion matrix makes f the first possible annihilating relation. □

Theorem 3.1.14. *Cyclic Subspaces as Polynomial Quotients*
Let $\dim \mathcal{V} < \infty$, $T \in \mathcal{L}(\mathcal{V})$, $0 \neq \mathbf{v} \in \mathcal{V}$, and let $m_{\mathbf{v}}(z) = z^d + \sum_{i=0}^{d-1} c_i z^i$ be the order of $\mathbf{v}$.

Define $\Phi_{\mathbf{v}} : \mathbb{F}[z] \to \mathcal{C}_T(\mathbf{v})$ by $\Phi_{\mathbf{v}}(p) = p(T)\mathbf{v}$ and $\mathcal{B}_{\mathbf{v}} := (\mathbf{v}, T\mathbf{v}, \ldots, T^{d-1}\mathbf{v})$. Then

(i) $\operatorname{im} \Phi_{\mathbf{v}} = \mathcal{C}_T(\mathbf{v})$, $\ker \Phi_{\mathbf{v}} = \langle m_{\mathbf{v}} \rangle$, and $\mathcal{C}_T(\mathbf{v}) \cong \mathbb{F}[z]/\langle m_{\mathbf{v}} \rangle$;
(ii) $\mathcal{B}_{\mathbf{v}}$ is a basis of $\mathcal{C}_T(\mathbf{v})$;
(iii) $[T|_{\mathcal{C}_T(\mathbf{v})}]_{\mathcal{B}_{\mathbf{v}}} = C_{m_{\mathbf{v}}}$.

Proof.

(i) By Proposition 3.1.10, $\operatorname{im} \Phi_{\mathbf{v}} = \mathcal{C}_T(\mathbf{v})$. $\ker \Phi_{\mathbf{v}} = \operatorname{ann}_T(\mathbf{v}) = \langle m_{\mathbf{v}} \rangle$, by Theorem 3.2.30 $\mathcal{C}_T(\mathbf{v}) \cong \mathbb{F}[z]/\langle m_{\mathbf{v}} \rangle$.

(ii) If $\sum_{j=0}^{d-1} a_j T^j \mathbf{v} = \mathbf{0}$, then $q(z) = \sum_{j=0}^{d-1} a_j z^j \in \operatorname{ann}_T(\mathbf{v})$ and $\deg q < d$, hence $q = 0$; thus $\mathcal{B}_{\mathbf{v}}$ is independent.
For $p = hm_{\mathbf{v}} + r$ with $\deg r < d$ or $r = 0$, $p(T)\mathbf{v} = r(T)\mathbf{v} \in \operatorname{span} \mathcal{B}_{\mathbf{v}}$, so $\mathcal{B}_{\mathbf{v}}$ spans.

(iii) For $0 \leq j < d-1$, $T(T^j\mathbf{v}) = T^{j+1}\mathbf{v}$. Since $m_{\mathbf{v}}(T)\mathbf{v} = \mathbf{0}$, $T^d\mathbf{v} = -\sum_{i=0}^{d-1} c_i T^i \mathbf{v}$, last-column relation of $C_{m_{\mathbf{v}}}$.

Structural Commentary: The first relation cuts out the quotient and companion block.

Key Insight: One orbit carries the whole cyclic subspace. □

Perspective 3.1.15. *The Translation Dictionary*
The module vocabulary for $\mathcal{V}_T$ and the vector-space vocabulary for T match as follows:

$\mathbb{F}[z]$-module $\mathcal{V}_T$	$\mathbb{F}$-vector space $\mathcal{V}$
$p(z)\cdot\mathbf{v}$	$p(T)\mathbf{v}$
submodule	T-invariant subspace
$\mathrm{Ann}_{\mathbb{F}[z]}(\mathcal{V}_T)=\langle\mu_T\rangle$	minimal polynomial of T
$\langle\!\langle\mathbf{v}\rangle\!\rangle_{\mathbb{F}[z]}$	$\mathcal{C}_T(\mathbf{v})$
$\mathbb{F}[z]/\langle m_{\mathbf{v}}\rangle$	cyclic block with companion matrix $C_{m_{\mathbf{v}}}$

Key Insight: Once z is interpreted as T, module structure becomes operator structure.

Method 3.1.16. *Moving Between Module and Spectral Data*
For $T\in\mathcal{L}(\mathcal{V})$, translate the $\mathbb{F}[z]$-module data of $\mathcal{V}_T$ into spectral data as follows.

(i) Factor $\mu_T=\prod_{i=1}^{s}p_i^{e_i}$ over $\mathbb{F}$; the primary components are $\ker p_i(T)^{e_i}$.
(ii) Decompose each $\ker p_i(T)^{e_i}$ into cyclic summands; their orders $p_i^{e_{i,j}}$ are the elementary divisors.
(iii) Over $\mathbb{F}$, each elementary divisor $p_i^{e_{i,j}}$ gives a companion block $C_{p_i^{e_{i,j}}}$ in rational canonical form.
(iv) Over splitting field for p_i, same primary cyclic summands refine into Jordan blocks (Perspective 3.3.6).
(v) T is diagonalizable exactly when every elementary divisor is linear, i.e., Corollary 3.4.12 applies.

Key Insight: Rational and Jordan forms encode the same elementary-divisor data over different fields.

3.2 Invariant Factors

Definition 3.2.1. *Invariant Factors and Elementary Divisors*
Let $T\in\mathcal{L}(\mathcal{V})$.

(i) A *cyclic decomposition* is $\mathcal{V}=\bigoplus_{j=1}^{r}\mathcal{C}_j$, where each $\mathcal{C}_j$ is nonzero, T-invariant, and cyclic.
(ii) $\mathrm{InvFact}(T):=(f_1,\dots,f_r)$, where f_j are monic nonconstant invariant factors of $\mathcal{V}_T$; $f_r\mid f_{r-1}\mid\cdots\mid f_1$.
(iii) $\mathrm{ElemDiv}(T)$ is the multiset of monic prime powers in the f_j, counted with multiplicity.
(iv) *Elementary divisor basis* is $\mathcal{B}_{\mathrm{ED}}:=\bigsqcup_\alpha(\mathbf{v}_\alpha,T\mathbf{v}_\alpha,\dots,T^{d_\alpha-1}\mathbf{v}_\alpha)$, where $d_\alpha=\deg m_{\mathbf{v}_\alpha}$; $\{m_{\mathbf{v}_\alpha}\}_\alpha=\mathrm{ElemDiv}(T)$.
(v) For $A\in\mathrm{Mat}(n,n,\mathbb{F})$, set $\mathrm{InvFact}(A):=\mathrm{InvFact}(T_A)$ and $\mathrm{ElemDiv}(A):=\mathrm{ElemDiv}(T_A)$.

Theorem 3.2.2. *Primary Cyclic Decomposition for an Operator*
Let $\mu_T=\prod_{i=1}^{s}p_i^{e_i}$ with $p_i\in\mathbb{F}[z]$ distinct monic irreducible, and set $\mathcal{V}_{p_i}:=\ker p_i(T)^{e_i}$. Then

(i) $\mathcal{V}_T=\bigoplus_{i=1}^{s}\mathcal{V}_{p_i}$;
(ii) $\mathcal{V}_{p_i}=\bigoplus_{j=1}^{k_i}\mathcal{C}_T(\mathbf{v}_{i,j})$, with $m_{\mathbf{v}_{i,j}}=p_i^{e_{i,j}}$ and $e_i=e_{i,1}\ge e_{i,2}\ge\cdots\ge e_{i,k_i}$;
(iii) $\mathrm{ElemDiv}(T)=\{p_i^{e_{i,j}}:1\le i\le s,\ 1\le j\le k_i\}$, corresponding cyclic bases form elementary divisor basis.

Proof.

(i) By Theorem 2.5.2, $\mathcal{V}_T$, $\mathcal{V}_T=\bigoplus_i\mathcal{V}_T[p_i^\infty]$; $\mathcal{V}_T[p_i^\infty]:=\{\mathbf{x}\in\mathcal{V}:\exists N\ge1,\ p_i(T)^N\mathbf{x}=0\}=\ker p_i(T)^{e_i}=\mathcal{V}_{p_i}$.
(ii) By Theorem 2.5.2, $\mathcal{V}_{p_i}=\bigoplus_{j=1}^{k_i}\mathcal{C}_T(\mathbf{v}_{i,j})$, $m_{\mathbf{v}_{i,j}}=p_i^{e_{i,j}}$ and $e_i=e_{i,1}\ge\cdots\ge e_{i,k_i}$.
(iii) Definition 3.2.1 gives $\mathrm{ElemDiv}(T)=\{p_i^{e_{i,j}}:1\le i\le s,\ 1\le j\le k_i\}$, and concatenating the bases of $\mathcal{C}_T(\mathbf{v}_{i,j})$ gives an elementary divisor basis.

Structural Commentary: Primary module pieces are polynomial-kernel pieces for the operator.
Key Insight: Elementary divisors are exactly the orders of the cyclic primary summands. □

Theorem 3.2.3. *Invariant-Factor Skeleton for Operators*
Let $0<\dim\mathcal{V}<\infty$, $T\in\mathcal{L}(\mathcal{V})$, and $\mathrm{InvFact}(T)=(f_1,\dots,f_r)$.
Then $f_r\mid\cdots\mid f_1$ and there are T-cyclic subspaces $\mathcal{C}_j$ such that $\mathcal{V}=\bigoplus_{j=1}^{r}\mathcal{C}_j$, $\mathcal{C}_j\cong\mathbb{F}[z]/\langle f_j\rangle$, $[T]=\bigoplus_{j=1}^{r}C_{f_j}$ in the concatenated cyclic bases, $\mu_T=f_1$, and $p_T=\prod_{j=1}^{r}f_j$.

Proof. By Proposition 3.1.3, $\mathcal{V}_T$ is finitely generated torsion over $\mathbb{F}[z]$.
Theorem 2.5.7 gives $\mathcal{V}_T\cong\bigoplus_{j=1}^{r}\mathbb{F}[z]/\langle f_j\rangle$ with $f_r\mid\cdots\mid f_1$, uniquely.
Transport cyclic generators to T-cyclic subspaces $\mathcal{C}_j$; Theorem 3.1.14 gives $\mathcal{C}_j\cong\mathbb{F}[z]/\langle f_j\rangle$ and block C_{f_j}.
Since $\mathrm{Ann}(\mathcal{V}_T)=\langle\mathrm{lcm}(f_1,\dots,f_r)\rangle=\langle f_1\rangle=\langle\mu_T\rangle$, $f_1=\mu_T$.
Finally, Theorem 3.1.13 gives $p_{C_{f_j}}=f_j$, so $p_T=\prod_{j=1}^{r}f_j$.
Structural Commentary: Torsion module decomposition produces companion blocks.
Key Insight: Invariant factors are the polynomial-module version of choosing canonical blocks for similarity. □

3.3 Elementary Divisors

Theorem 3.3.1. *Reducing Decompositions Combine the Data*
Let $\mathcal{V}=\mathcal{A}\oplus\mathcal{B}$ with $\mathcal{A},\mathcal{B}$ T-invariant, and set $T_{\mathcal{A}}:=T|_{\mathcal{A}}$, $T_{\mathcal{B}}:=T|_{\mathcal{B}}$.
Then $\mu_T=\mathrm{lcm}(\mu_{T_{\mathcal{A}}},\mu_{T_{\mathcal{B}}})$ and $\mathrm{ElemDiv}(T)=\mathrm{ElemDiv}(T_{\mathcal{A}})\sqcup\mathrm{ElemDiv}(T_{\mathcal{B}})$, where $\sqcup$ is multiset union.
The invariant factors are obtained by recombining the elementary divisors into a divisibility chain.

Proof. By Theorem 3.1.4, $\mathcal{V}_T = \mathcal{A}_{T_\mathcal{A}} \oplus \mathcal{B}_{T_\mathcal{B}}$ as $\mathbb{F}[z]$-modules.
Hence $\operatorname{Ann}(\mathcal{V}_T) = \operatorname{Ann}(\mathcal{A}_{T_\mathcal{A}}) \cap \operatorname{Ann}(\mathcal{B}_{T_\mathcal{B}}) = \langle \operatorname{lcm}(\mu_{T_\mathcal{A}}, \mu_{T_\mathcal{B}}) \rangle$.
Proposition 2.5.3 gives elementary-divisor identity.
Structural Commentary: Reducing decompositions split the operator module.
Key Insight: Direct sums take lcm on global orders and multiset union on elementary divisors. □

Definition 3.3.2. *Characteristic Polynomial from Elementary Divisors*
For $T \in \mathcal{L}(\mathcal{V})$, define $\chi_T(z) := \prod_{q \in \operatorname{ElemDiv}(T)} q(z)$, with product over the multiset $\operatorname{ElemDiv}(T)$.
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, set $\chi_A := \chi_{T_A}$.

Theorem 3.3.3. *Minimal and Characteristic Polynomials from Elementary Divisors*
Let $\operatorname{ElemDiv}(T) = \{p_i^{e_{i,j}} : 1 \le i \le s,\ 1 \le j \le k_i\}$.
Then $\mu_T = \prod_i p_i^{\max_j e_{i,j}}$, $\chi_T = \prod_{i,j} p_i^{e_{i,j}}$, *and* $\deg \chi_T = \dim \mathcal{V}$.
Hence $\mu_T \mid \chi_T$, *and* μ_T, χ_T *have the same irreducible factors.*

Proof. By Theorem 3.2.2, $\mathcal{V}_T = \bigoplus_{i,j} \mathcal{C}_T(\mathbf{v}_{i,j})$ with $m_{\mathbf{v}_{i,j}} = p_i^{e_{i,j}}$.
Hence $\operatorname{Ann}(\mathcal{V}_T) = \langle \operatorname{lcm}_{i,j} p_i^{e_{i,j}} \rangle = \langle \prod_i p_i^{\max_j e_{i,j}} \rangle$, so $\mu_T = \prod_i p_i^{\max_j e_{i,j}}$. Definition 3.3.2 gives $\chi_T = \prod_{i,j} p_i^{e_{i,j}}$.
By Theorem 3.1.14, $\dim \mathcal{C}_T(\mathbf{v}_{i,j}) = \deg p_i^{e_{i,j}}$; summing gives $\deg \chi_T = \dim \mathcal{V}$.
Divisibility and common irreducible factors are immediate.
Structural Commentary: μ_T records maximal exponents; χ_T records all exponents.
Key Insight: The pair (μ_T, χ_T) forgets the block distribution. □

Definition 3.3.4. *Nonderogatory Operators*
$T \in \mathcal{L}(\mathcal{V})$ is *nonderogatory* if $\mu_T = \chi_T$, equivalently $\deg \mu_T = \dim \mathcal{V}$.
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, A is nonderogatory if T_A is.

Theorem 3.3.5. *Cyclic and Nonderogatory Are the Same*
For $T \in \mathcal{L}(\mathcal{V})$, *the following are equivalent:*

(i) $\mathcal{V}_T$ *is cyclic;*
(ii) T *has a cyclic vector;*
(iii) $\operatorname{ElemDiv}(T) = \{p_1^{e_1}, \dots, p_s^{e_s}\}$, *one elementary divisor for each irreducible factor of* μ_T*;*
(iv) T *is nonderogatory.*

Proof.

(i) $\Rightarrow$ (ii) If $\mathcal{V}_T = \mathbb{F}[z]\mathbf{v}$, then $\mathcal{V} = \{p(T)\mathbf{v} : p \in \mathbb{F}[z]\} = \mathcal{C}_T(\mathbf{v})$, so $\mathbf{v}$ is cyclic.
(ii) $\Rightarrow$ (i) If $\mathcal{C}_T(\mathbf{v}) = \mathcal{V}$, then $\mathcal{V}_T = \mathbb{F}[z]\mathbf{v}$.
(i) $\Leftrightarrow$ (iii) Theorem 2.5.10 on $\mathcal{V}_T$. Cyclic torsion module has one cyclic summand for each primary factor.
(iii) $\Rightarrow$ (iv) If $\operatorname{ElemDiv}(T) = \{p_i^{e_i}\}_{i=1}^s$, then $\chi_T = \prod_i p_i^{e_i} = \mu_T$, so T is nonderogatory.
(iv) $\Rightarrow$ (iii) Let $\operatorname{ElemDiv}(T) = \{p_i^{e_{i,j}}\}$. If $\mu_T = \chi_T$, then $\prod_i p_i^{\max_j e_{i,j}} = \prod_{i,j} p_i^{e_{i,j}}$, so $k_i = 1$ for all i.

Structural Commentary: Nonderogatory means all primary information can be generated by one vector.
Key Insight: Equality $\mu_T = \chi_T$ says that product and lcm of the cyclic orders coincide. □

Perspective 3.3.6. *Elementary Divisors and Jordan Blocks*
Over $\mathbb{C}$, elementary divisors are exactly $(z - \lambda)^k$, and $\mathbb{C}[z]/\langle (z-\lambda)^k \rangle$ corresponds to the Jordan block $J_k(\lambda)$.
Thus $\operatorname{ElemDiv}(T)$ is the multiset of Jordan blocks.

Perspective 3.3.7. *Crosswalk with the Spectral Presentation*
The module and spectral presentations encode the same data at different splitting levels:

(i) diagonalizable $\iff$ $\operatorname{ElemDiv}(T)$ consists of linear factors, by Theorem 6.3.6;
(ii) $(z - \lambda)^k \iff J_k(\lambda)$ over an algebraically closed field, by Perspective 3.3.6 and Theorem 6.6.6;
(iii) rank drops recover the exponents k via Proposition 2.4.7, with $p = z - \lambda$ and $M = \mathcal{V}_T$.

Key Insight: Rational form is Jordan form before irreducibles split.

3.4 Operator Classification

Perspective 3.4.1. *Operator Modules and Similarity*
For $T \in \mathcal{L}(\mathcal{V})$, $\mathcal{V}_T$ has action $p \cdot \mathbf{v} = p(T)\mathbf{v}$. Then $S \sim T \iff \mathcal{V}_S \cong_{\mathbb{F}[z]} \mathcal{V}_T$; $\mathbb{F}[z]/\langle f \rangle \iff C_f$ (Theorem 3.1.14).

Theorem 3.4.2. *Complete Similarity Invariants*
Let $T, S \in \mathcal{L}(\mathcal{V})$ *and* $A, B \in \operatorname{Mat}(n, n, \mathbb{F})$.

(i) $T \sim S \iff \mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S \iff \operatorname{ElemDiv}(T) = \operatorname{ElemDiv}(S) \iff \operatorname{InvFact}(T) = \operatorname{InvFact}(S)$.

(ii) $A \sim B \iff (\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B \iff \mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B) \iff \mathrm{InvFact}(A) = \mathrm{InvFact}(B)$.
(iii) *If* $A = [T]_{\mathcal{B}}$ *for some basis* $\mathcal{B}$ *of* $\mathcal{V}$, *then* $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(T)$ *and* $\mathrm{InvFact}(A) = \mathrm{InvFact}(T)$.

Proof.

(*i*) ($\Rightarrow$) If $T \sim S$, then $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S$ by Theorem 3.1.5.
Module isomorphism preserves the PID decomposition, hence preserves ElemDiv and InvFact.
(*i*) ($\Leftarrow$) Equality of either ElemDiv or InvFact gives $\mathcal{V}_T \cong_{\mathbb{F}[z]} \mathcal{V}_S$ by uniqueness of the PID decomposition.
Thus $T \sim S$ by Theorem 3.1.5.
(*ii*) ($\Rightarrow$) If $A \sim B$, then $(\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$ by Corollary 3.1.6; hence the divisor data agree.
(*ii*) ($\Leftarrow$) Equality of either matrix divisor datum gives $(\mathbb{F}^n)_A \cong_{\mathbb{F}[z]} (\mathbb{F}^n)_B$, so $A \sim B$ by Corollary 3.1.6.
(*iii*) If $A = [T]_{\mathcal{B}}$, then $\Phi(\mathbf{v}) = [\mathbf{v}]_{\mathcal{B}}$ satisfies $\Phi(p(T)\mathbf{v}) = p(A)\Phi(\mathbf{v})$.
Hence $\Phi : \mathcal{V}_T \to (\mathbb{F}^n)_A$ is an $\mathbb{F}[z]$-isomorphism, so the divisor data agree.

Structural Commentary: Similarity is module isomorphism plus PID data.
Key Insight: Elementary divisors classify operator modules. □

Lemma 3.4.3. *Determinant of a Companion Block*
Let $f(z) = z^n + \sum_{i=0}^{n-1} c_i z^i$ *be monic. Then* $\det(zI_n - C_f) = f(z)$.

Proof. Set $D_n(c_0, \dots, c_{n-1}) := \det(zI_n - C_f)$. For $n = 1$, $D_1(c_0) = z + c_0$.
For $n \geq 2$, first-row expansion gives $D_n(c_0, \dots, c_{n-1}) = zD_{n-1}(c_1, \dots, c_{n-1}) + c_0$, since the $(1,1)$-minor is again companion-shaped and the last first-row cofactor contributes c_0. Induction gives $D_n = z^n + \sum_{i=0}^{n-1} c_i z^i = f(z)$.
Structural Commentary: The last column of C_f stores the coefficients of the polynomial relation.
Key Insight: The shifted determinant reads that stored relation back as f. □

Definition 3.4.4. *Rational Canonical Block Forms*

(i) *Invariant-factor RCF*: $A = \mathrm{diag}(C_{s_1}, \dots, C_{s_r})$, where $s_j \in \mathbb{F}[z]$ are monic nonconstant and $s_r \mid \cdots \mid s_1$.
(ii) *Elementary-divisor RCF*: $A = \mathrm{diag}(C_{q_1}, \dots, C_{q_m})$, where each q_j is power of monic irreducible polynomial.

Theorem 3.4.5. *Rational Canonical Form Skeleton*
For $A \in \mathrm{Mat}(n, n, \mathbb{F})$, *there is a unique tuple* $\mathrm{InvFact}(A) = (s_1, \dots, s_r)$ *with* $s_j \in \mathbb{F}[z]$ *monic nonconstant,* $s_r \mid \cdots \mid s_1$, *and* $A \sim_{\mathbb{F}} R_A := \mathrm{diag}(C_{s_1}, \dots, C_{s_r})$. *Moreover,* $\mu_A = s_1$ *and* $p_A = \prod_{j=1}^{r} s_j$.
Equivalently, for $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, $A \sim_{\mathbb{F}} B \iff \mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$ *as multisets.*

Proof. Let $T_A : \mathbb{F}^n \to \mathbb{F}^n$, $\mathbf{x} \mapsto A\mathbf{x}$.
Theorem 3.2.3 gives $\mathbb{F}^n = \bigoplus_{j=1}^{r} \mathcal{C}_j$, $\mathcal{C}_j \cong_{\mathbb{F}[z]} \mathbb{F}[z]/\langle s_j \rangle$, $s_r \mid \cdots \mid s_1$, $[T_A] = R_A$, $\mu_{T_A} = s_1$, and $p_{T_A} = \prod_{j=1}^{r} s_j$.
Hence $A \sim_{\mathbb{F}} R_A$, $\mu_A = s_1$, and $p_A = \prod_{j=1}^{r} s_j$; uniqueness of $\mathrm{InvFact}(A)$ is uniqueness in the same theorem.
($\Rightarrow$) If $A \sim_{\mathbb{F}} B$, then Theorem 3.4.2 gives $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$.
($\Leftarrow$) If $\mathrm{ElemDiv}(A) = \mathrm{ElemDiv}(B)$, then Theorem 3.4.2 gives $A \sim_{\mathbb{F}} B$.

Structural Commentary: Companion blocks classify similarity without requiring p_A to split.
Key Insight: Similarity classes can be represented by polynomial data rather than by choosing roots. □

Theorem 3.4.6. *Rational Canonical Form: Two Versions*
Let $T \in \mathcal{L}(\mathcal{V})$.

(i) *If* $\mathrm{ElemDiv}(T) = \{q_1, \dots, q_m\}$, *then* $\exists$ *basis* $\mathcal{R}$ *with* $[T]_{\mathcal{R}} = \mathrm{diag}(C_{q_1}, \dots, C_{q_m})$.
The form is unique up to block permutation.
(ii) *If* $\mathrm{InvFact}(T) = (s_1, \dots, s_r)$ *and* $s_r \mid \cdots \mid s_1$, *then* $\exists$ *basis* $\mathcal{B}$ *with* $[T]_{\mathcal{B}} = \mathrm{diag}(C_{s_1}, \dots, C_{s_r})$.
The form is unique in this order.
(iii) *If* $A \in \mathrm{Mat}(n, n, \mathbb{F})$, *then the similarity class* $[A]_{\sim_{\mathbb{F}}}$ *contains both forms.*
Elementary-divisor form is unique up to block order, invariant-factor form by its divisibility chain.

Proof.

(i) Theorem 3.2.2 gives $\mathcal{V} = \bigoplus_{\alpha=1}^{m} \mathcal{C}_\alpha$ with orders q_α; Theorem 3.1.14 gives cyclic bases and blocks C_{q_α}.
Permuting summands conjugates by a permutation matrix.
(ii) Theorem 3.2.3 gives $\mathcal{V} = \bigoplus_{j=1}^{r} \mathcal{C}_j$, $\mathcal{C}_j \cong_{\mathbb{F}[z]} \mathbb{F}[z]/\langle s_j \rangle$, and cyclic bases with block matrix $\mathrm{diag}(C_{s_1}, \dots, C_{s_r})$.
(iii) For A, apply (i),(ii) to $T_A : \mathbf{x} \mapsto A\mathbf{x}$; any chosen basis gives a similarity matrix.
Uniqueness follows from Theorem 3.4.2.

Structural Commentary: Elementary divisors split primary blocks; invariant factors recombine them.
Key Insight: Both forms classify the same similarity classes, but they expose different polynomial data. □

Method 3.4.7. *Constructing an Operator from Elementary Divisors*
Let $Q = \{q_1, \dots, q_m\}$ be a finite multiset of monic prime powers in $\mathbb{F}[z]$, and set $n := \sum_{j=1}^{m} \deg q_j$.

(i) Set $R_Q := \operatorname{diag}(C_{q_1}, \dots, C_{q_m}) \in \operatorname{Mat}(n, n, \mathbb{F})$.
(ii) Then $\operatorname{ElemDiv}(T_{R_Q}) = Q$; hence every such Q is realised on $\mathbb{F}^n$.
(iii) Apply Method 2.5.5 to recombine Q into $\operatorname{InvFact}(T_{R_Q}) = (s_1, \dots, s_r)$, then $\operatorname{diag}(C_{s_1}, \dots, C_{s_r}) \sim_{\mathbb{F}} R_Q$.

Structural Commentary: RCF constructs operators from divisor data.
Key Insight: Companion blocks realise cyclic summands; block sums realise direct sums.

Example 3.4.8. *Rational Canonical Possibilities from Degree Counts*
Let $T \in \mathcal{L}(\mathbb{R}^7)$ and $\mu_T = (z-1)(z^2+1)^2$. For $Q = \operatorname{ElemDiv}(T)$, $\operatorname{lcm} Q = \mu_T$ and $\sum_{q \in Q} \deg q = 7$.
Hence $Q_1 = \{z-1, (z^2+1)^2, z^2+1\}$ or $Q_2 = \{z-1, z-1, z-1, (z^2+1)^2\}$.
Thus the elementary-divisor RCF is $R_i = \bigoplus_{q \in Q_i} C_q$ for $i = 1, 2$, up to block order.

Theorem 3.4.9. *Determinant Form of the Characteristic Polynomial*
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, $\chi_A(z) = \det(zI_n - A)$. If $A = [T]_{\mathcal{B}}$, then $\chi_T = \chi_A$.

Proof. Let $Q := \operatorname{ElemDiv}(A)$ and choose elementary-divisor RCF $R = \bigoplus_{q \in Q} C_q \sim_{\mathbb{F}} A$.
By Lemma 3.4.3, $\det(zI - R) = \prod_{q \in Q} \det(zI - C_q) = \prod_{q \in Q} q = \chi_R$.
Since similarity preserves both Q and $\det(zI - \cdot)$, $\chi_A = \chi_R = \det(zI - R) = \det(zI - A)$.
If $A = [T]_{\mathcal{B}}$, then $\operatorname{ElemDiv}(A) = \operatorname{ElemDiv}(T)$, hence $\chi_A = \chi_T$.
Structural Commentary: Rational canonical form reduces the determinant computation to companion blocks.
Key Insight: The abstract product of elementary divisors is the concrete shifted determinant. □

Theorem 3.4.10. *Changing the Base Field*
Let $\mathbb{F} \subseteq \mathbb{K}$, $A \in \operatorname{Mat}(n, n, \mathbb{F})$, and $\operatorname{ElemDiv}_{\mathbb{F}}(A) = \{p_i^{e_{i,j}}\}_{i,j}$ with $p_i \in \mathbb{F}[z]$ monic irreducible.
If $p_i = \prod_{\ell=1}^{m_i} a_{i,\ell}^{d_{i,\ell}}$ in $\mathbb{K}[z]$, with $a_{i,\ell}$ pairwise distinct monic irreducible over $\mathbb{K}$, then $\operatorname{ElemDiv}_{\mathbb{K}}(A) = \{a_{i,\ell}^{d_{i,\ell} e_{i,j}}\}_{i,j,\ell}$.
Moreover, $\operatorname{InvFact}_{\mathbb{K}}(A) = \operatorname{InvFact}_{\mathbb{F}}(A)$.

Proof. Take elementary-divisor RCF over $\mathbb{F}$: $R = \bigoplus_{i,j} C_{p_i^{e_{i,j}}} \sim_{\mathbb{F}} A$.
Since $R \in \operatorname{Mat}(n, n, \mathbb{K})$ and $p_i^{e_{i,j}} = \prod_{\ell=1}^{m_i} a_{i,\ell}^{d_{i,\ell} e_{i,j}}$ with coprime factors in $\mathbb{K}[z]$,
Theorem 2.3.6 gives $C_{p_i^{e_{i,j}}} \sim_{\mathbb{K}} \bigoplus_{\ell=1}^{m_i} C_{a_{i,\ell}^{d_{i,\ell} e_{i,j}}}$. Summing over (i, j) yields $\operatorname{ElemDiv}_{\mathbb{K}}(A)$.
Recombination into the divisibility chain gives the original invariant factors.
Structural Commentary: Field extension refines elementary divisors.
Key Insight: Elementary divisors split; invariant factors persist. □

Corollary 3.4.11. *Nonderogatory Means One Companion Block*
For $A \in \operatorname{Mat}(n, n, \mathbb{F})$, the following are equivalent:

(i) A is nonderogatory, i.e. $\mu_A = \chi_A$;
(ii) $(\mathbb{F}^n)_A$ is cyclic;
(iii) $A \sim_{\mathbb{F}} C_{\chi_A}$.

Proof.

(i) $\Rightarrow$ (ii) This is Theorem 3.3.5.
(ii) $\Rightarrow$ (iii) Cyclicity gives $\operatorname{InvFact}(A) = (s)$. By Theorem 3.2.3, $s = \mu_A = \chi_A$; hence $A \sim_{\mathbb{F}} C_{\chi_A}$.
(iii) $\Rightarrow$ (i) Since $\mu_{C_f} = f$ by Theorem 3.1.13, similarity gives $\mu_A = \mu_{C_{\chi_A}} = \chi_A$.

Structural Commentary: Nonderogatory means one cyclic block.
Key Insight: One cyclic generator gives one companion block. □

Corollary 3.4.12. *When the Elementary-Divisor Form Is Diagonal*
Let $T \in \mathcal{L}(\mathcal{V})$, $Q = \operatorname{ElemDiv}(T)$, and $R_{\text{ed}} = \bigoplus_{q \in Q} C_q$.
Then R_{ed} is diagonal iff $\deg q = 1$ for all $q \in Q$, iff $\mu_T = \prod_{\lambda \in \Lambda} (z - \lambda)$ for some finite $\Lambda \subseteq \mathbb{F}$.

Proof.

($\Rightarrow$) If R_{ed} is diagonal, each block C_q has size 1, hence $\deg q = 1$ for all $q \in Q$.
Thus $\mu_T = \operatorname{lcm} Q$ is a product of distinct linear factors.
($\Leftarrow$) If $\mu_T = \prod_{\lambda \in \Lambda} (z - \lambda)$, then each prime power $q \in Q$ divides μ_T, so $q = z - \lambda$ for some $\lambda \in \Lambda$.
Hence each C_q is 1×1, and R_{ed} is diagonal.

Structural Commentary: Diagonal RCF means all companion blocks are 1×1.
Key Insight: Nonlinear or repeated factors obstruct diagonal RCF. □

Proposition 3.4.13. *AB and BA Have the Same Characteristic Polynomial*
For $A, B \in \mathrm{Mat}(n, n, \mathbb{F})$, $\chi_{AB} = \chi_{BA}$.
If $A \in \mathrm{GL}(n, \mathbb{F})$ or $B \in \mathrm{GL}(n, \mathbb{F})$, then $AB \sim_{\mathbb{F}} BA$; otherwise not necessarily.

Proof. In $\mathbb{F}(z)$, by Sylvester's identitym $\det(zI - AB) = z^n \det(I - z^{-1}AB) = z^n \det(I - z^{-1}BA) = \det(zI - BA)$, hence $\chi_{AB} = \chi_{BA}$. If A is invertible, $BA = A^{-1}(AB)A$; if B is invertible, $AB = B^{-1}(BA)B$.
Without invertibility, take $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$.
Then $AB = B$, $BA = 0$, $\chi_{AB} = \chi_{BA} = z^2$, but $\mathrm{rank}(AB) = 1 \neq 0 = \mathrm{rank}(BA)$, so $AB \not\sim_{\mathbb{F}} BA$.
Structural Commentary: Characteristic polynomials see less than similarity.
Key Insight: AB and BA share determinant data; invertibility yields similarity. □

Example 3.4.14. *The Same Polynomial over Different Fields*
Let $\dim V = 4$ and $\mu_T = (z^2 + 1)(z^2 - 2)$. Since $\deg \mu_T = 4$, T is nonderogatory and $\mathrm{InvFact}_K(T) = (\mu_T)$ for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}$; hence $R_{\mathrm{inv},K} = C_{\mu_T}$. Meanwhile

K	$\mathrm{ElemDiv}_K(T)$
$\mathbb{Q}$	$\{z^2 + 1,\ z^2 - 2\}$
$\mathbb{R}$	$\{z^2 + 1,\ z - \sqrt{2},\ z + \sqrt{2}\}$
$\mathbb{C}$	$\{z - i,\ z + i,\ z - \sqrt{2},\ z + \sqrt{2}\}$

Thus $R_{\mathrm{ed},K} = \bigoplus_{q \in \mathrm{ElemDiv}_K(T)} C_q$: scalar extension refines elementary divisors while preserving the invariant factor. Over $\mathbb{C}$, $R_{\mathrm{ed},\mathbb{C}}$ is diagonal.

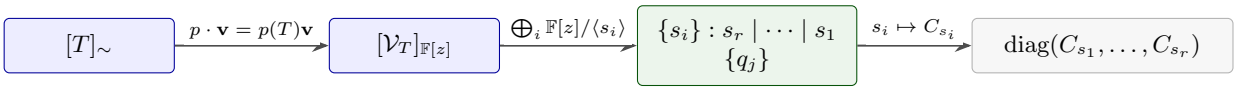


Figure 109: Rational-canonical dictionary: $[T]_\sim$, $[\mathcal{V}_T]_{\mathbb{F}[z]}$, $\{s_i\}$ or $\{q_j\}$, and $\mathrm{diag}(C_{s_i})$ are equivalent data.

4 Associative Algebras

4.1 Associative Algebra Structures

The module viewpoint in Definition 3.1.2 uses one operator at a time. An algebra keeps track of a whole vector space of operators or products at once. The guiding example is $\mathcal{L}(\mathcal{V})$, where addition and scalar multiplication are pointwise and multiplication is composition.

Definition 4.1.1. *Unital Associative Algebra*
Let $\mathbb{F}$ be a field.

(i) An *associative $\mathbb{F}$-algebra* is an $\mathbb{F}$-vector space $\mathcal{A}$ and unital ring $(\mathcal{A}, +, \cdot, 1_{\mathcal{A}})$ such that $r(ab) = (ra)b = a(rb)$ for all $r \in \mathbb{F}$ and $a, b \in \mathcal{A}$.
(ii) $\mathcal{A}$ is *finite-dimensional* iff $\dim_{\mathbb{F}} \mathcal{A} < \infty$, and *commutative* iff $ab = ba$ for all $a, b \in \mathcal{A}$.
(iii) $a \in \mathcal{A}$ is *invertible* iff $\exists b \in \mathcal{A}$ such that $ab = ba = 1_{\mathcal{A}}$.
(iv) In this subsection, *algebra* means unital associative $\mathbb{F}$-algebra unless stated otherwise.

Remark 4.1.2. *Convention*
Relative to Definition 3.1.19, assume multiplication is associative and unital.
Thus, *algebra* means unital associative $\mathbb{F}$-algebra.

Definition 4.1.3. *Center and Central Algebras*

(i) The *center* of an $\mathbb{F}$-algebra $\mathcal{A}$ is $Z(\mathcal{A}) := \{a \in \mathcal{A} : ax = xa\ \forall x \in \mathcal{A}\}$.
(ii) $\mathcal{A}$ is *central over $\mathbb{F}$* iff $Z(\mathcal{A}) = \mathbb{F}1_{\mathcal{A}} := \{r1_{\mathcal{A}} : r \in \mathbb{F}\}$.

Proposition 4.1.4. *The Scalar Field Sits in the Center*
If $\mathcal{A}$ is an $\mathbb{F}$-algebra with $1_{\mathcal{A}} \neq 0$, then $\iota_{\mathcal{A}} : \mathbb{F} \hookrightarrow Z(\mathcal{A}), \quad r \mapsto r1_{\mathcal{A}}$, is an injective unital ring homomorphism.
Conversely, if R is a unital ring and $\mathbb{F} \leq Z(R)$ is a subfield, then R is an $\mathbb{F}$-algebra via $r \cdot x := rx$.

Proof.

(⇒) Write $\iota = \iota_{\mathcal{A}}$. For $r, s \in \mathbb{F}$, $\iota(r + s) = (r + s)1_{\mathcal{A}} = \iota(r) + \iota(s)$, $\iota(rs) = (rs)1_{\mathcal{A}} = (r1_{\mathcal{A}})(s1_{\mathcal{A}})$, $\iota(1) = 1_{\mathcal{A}}$.
For $r \in \mathbb{F}$ and $a \in \mathcal{A}$, $\iota(r)a = (r1_{\mathcal{A}})a = r(1_{\mathcal{A}}a) = ra = r(a1_{\mathcal{A}}) = a(r1_{\mathcal{A}}) = a\iota(r)$, so $\mathrm{im}\,\iota \subseteq Z(\mathcal{A})$.
If $\iota(r) = 0$ and $r \neq 0$, then $1_{\mathcal{A}} = \iota(r^{-1})\iota(r) = 0$, contradiction; hence $\ker \iota = \{0\}$.

($\Leftarrow$) Define $r \cdot x = rx$. The field laws in $\mathbb{F} \subseteq R$ give the vector-space axioms, and $1_{\mathbb{F}} \cdot x = x$.
For $r \in \mathbb{F}$ and $x, y \in R$, centrality gives $r \cdot (xy) = rxy = (rx)y = x(ry) = x(r \cdot y)$, so the scalar action is compatible with the ring product.

Structural Commentary: An algebra structure is a choice of central scalar copy $\mathbb{F}1_{\mathcal{A}} \subseteq Z(\mathcal{A})$.
Key Insight: Scalars are precisely central multipliers of the unit. □

Method 4.1.5. *Building Multiplication from Structure Constants*
Let $V = \bigoplus_{i \in I} \mathbb{F} b_i$. Choose $c_{ij}^k \in \mathbb{F}$ with $\{k : c_{ij}^k \neq 0\}$ finite for each (i, j).

(i) Define $b_i b_j := \sum_{k \in I} c_{ij}^k b_k$ and extend bilinearly: $(\sum_i r_i b_i)(\sum_j s_j b_j) := \sum_{i,j,k} r_i s_j c_{ij}^k b_k$.
(ii) The product is associative iff $\sum_{\ell \in I} c_{ij}^\ell c_{\ell k}^m = \sum_{\ell \in I} c_{jk}^\ell c_{i\ell}^m$ for all $i, j, k, m \in I$.
(iii) The product is commutative iff $c_{ij}^k = c_{ji}^k$ for all $i, j, k \in I$.
(iv) A basis vector b_e is an identity iff $c_{ej}^m = \delta_j^m = c_{je}^m$ for all $j, m \in I$.

Example 4.1.6. *Basic Algebras*

(i) If $\mathbb{F} \subseteq \mathbb{E}$ are fields, then $\mathbb{E}$ is an $\mathbb{F}$-algebra by $r \cdot x = rx$ and multiplication in $\mathbb{E}$.
(ii) $\mathbb{F}[x]$ is a commutative $\mathbb{F}$-algebra with unit 1 and scalar embedding $r \mapsto r$.
(iii) $\mathrm{Mat}(n, n, \mathbb{F})$ is an $\mathbb{F}$-algebra with unit I_n and product AB.
(iv) For an $\mathbb{F}$-vector space $\mathcal{V}$, $\mathcal{L}(\mathcal{V}) = \mathrm{End}_{\mathbb{F}}(\mathcal{V})$ is an $\mathbb{F}$-algebra with unit $I_{\mathcal{V}}$ and product $S \circ T$.

Proposition 4.1.7. *Endomorphisms and Matrices Are the Same Algebra after Choosing a Basis*
Let $\dim \mathcal{V} = n$ *and fix an ordered basis* $\mathcal{B}$ *of* $\mathcal{V}$.
Then $\mu_{\mathcal{B}} : \mathcal{L}(\mathcal{V}) \to \mathrm{Mat}(n, n, \mathbb{F})$, $T \mapsto [T]_{\mathcal{B}}$, *is an* $\mathbb{F}$*-algebra isomorphism.*

Proof. By Theorem 3.3.32, $\mu_{\mathcal{B}}$ is an $\mathbb{F}$-linear bijection.
For $S, T \in \mathcal{L}(\mathcal{V})$ and $x \in \mathcal{V}$, $[STx]_{\mathcal{B}} = [S]_{\mathcal{B}}[Tx]_{\mathcal{B}} = [S]_{\mathcal{B}}[T]_{\mathcal{B}}[x]_{\mathcal{B}}$, hence $\mu_{\mathcal{B}}(ST) = \mu_{\mathcal{B}}(S)\mu_{\mathcal{B}}(T)$.
Also $\mu_{\mathcal{B}}(I_{\mathcal{V}}) = I_n$; therefore $\mu_{\mathcal{B}}$ is an $\mathbb{F}$-algebra isomorphism.
Structural Commentary: A basis is a coordinate bridge from abstract endomorphisms to concrete matrices.
Key Insight: Composition becomes matrix multiplication after taking coordinate columns. □

4.2 Modules over Algebras

Definition 4.2.1. *Subalgebras, Ideals, Quotients, Homomorphisms, and Regular Representation*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra.

(i) A *subalgebra* is a subset $\mathcal{B} \subseteq \mathcal{A}$ such that $1_{\mathcal{A}} \in \mathcal{B}$, $\mathcal{B} \leq_{\mathbb{F}} \mathcal{A}$, and $b, b' \in \mathcal{B} \Rightarrow bb' \in \mathcal{B}$.
(ii) For $X \subseteq \mathcal{A}$, the *subalgebra generated by* X is $\langle X \rangle_{\mathrm{subalg}} := \bigcap \{\mathcal{B} : \mathcal{B} \text{ is a subalgebra of } \mathcal{A},\ X \subseteq \mathcal{B}\}$.
Equivalently, $\langle X \rangle_{\mathrm{subalg}} = \mathrm{span}_{\mathbb{F}}\{1_{\mathcal{A}},\ x_1 \cdots x_m : m \geq 1,\ x_i \in X\}$.
(iii) A *two-sided ideal* is a subset $\mathcal{I} \subseteq \mathcal{A}$ such that $\mathcal{I} \neq \emptyset$, $u, v \in \mathcal{I} \Rightarrow u - v \in \mathcal{I}$, and $a \in \mathcal{A},\ u \in \mathcal{I} \Rightarrow au, ua \in \mathcal{I}$.
(iv) For nonempty $X \subseteq \mathcal{A}$, the *ideal generated by* X is $\langle X \rangle_{\mathrm{ideal}} := \{\sum_{j=1}^n a_j x_j b_j : n \geq 1,\ x_j \in X,\ a_j, b_j \in \mathcal{A}\}$.
(v) $\mathcal{A}$ is *simple* iff $\mathcal{A}^2 \neq 0$ and its only two-sided ideals are 0 and $\mathcal{A}$.
(vi) If $\mathcal{I} \triangleleft \mathcal{A}$, then $\mathcal{A}/\mathcal{I}$ is an $\mathbb{F}$-algebra by $(a + \mathcal{I})(b + \mathcal{I}) := ab + \mathcal{I}$ and $r(a + \mathcal{I}) := ra + \mathcal{I}$.
(vii) For $\mathbb{F}$-algebras $\mathcal{A}, \mathcal{B}$, an *algebra homomorphism* is a map $\sigma : \mathcal{A} \to \mathcal{B}$ such that, for all $a, a' \in \mathcal{A}$ and $r \in \mathbb{F}$, $\sigma(a + a') = \sigma(a) + \sigma(a')$, $\sigma(aa') = \sigma(a)\sigma(a')$, $\sigma(1_{\mathcal{A}}) = 1_{\mathcal{B}}$, and $\sigma(ra) = r\sigma(a)$.
(viii) For $a \in \mathcal{A}$, the *left multiplication operator* is $L_a : \mathcal{A} \to \mathcal{A}$, $x \mapsto ax$.
The map $L_{\mathcal{A}} : \mathcal{A} \to \mathcal{L}(\mathcal{A})$, $a \mapsto L_a$, is the *left regular representation* of $\mathcal{A}$.

Proposition 4.2.2. *Kernel and First Isomorphism Theorem for Algebras*
If $\sigma : \mathcal{A} \to \mathcal{B}$ *is an* $\mathbb{F}$*-algebra homomorphism,*
then $\ker \sigma \triangleleft \mathcal{A}$ *and* $\overline{\sigma} : \mathcal{A}/\ker \sigma \to \mathrm{im}\, \sigma$, $a + \ker \sigma \mapsto \sigma(a)$, *is an* $\mathbb{F}$*-algebra isomorphism.*

Proof. Set $K := \ker \sigma$. Linearity gives $K \leq_{\mathbb{F}} \mathcal{A}$.
If $u \in K$ and $a \in \mathcal{A}$, then $\sigma(au) = \sigma(a)\sigma(u) = 0 = \sigma(u)\sigma(a) = \sigma(ua)$, hence $au, ua \in K$ and $K \triangleleft \mathcal{A}$.
Define $\overline{\sigma}(a + K) := \sigma(a)$. Since $a + K = a' + K \iff a - a' \in K \iff \sigma(a) = \sigma(a')$, $\overline{\sigma}$ is well defined and injective; it is surjective onto $\mathrm{im}\, \sigma$.
For $a, a' \in \mathcal{A}$ and $r \in \mathbb{F}$, $\overline{\sigma}((a + K)(a' + K)) = \sigma(aa') = \sigma(a)\sigma(a')$, $\overline{\sigma}(r(a + K)) = r\sigma(a)$, and $\overline{\sigma}(1_{\mathcal{A}} + K) = 1_{\mathcal{B}}$.
Thus $\overline{\sigma}$ is an $\mathbb{F}$-algebra isomorphism.
Structural Commentary: Quotienting by the kernel removes exactly the relations invisible to σ.
Key Insight: The image is the faithful algebra obtained after collapsing $\ker \sigma$. □

Theorem 4.2.3. *Left Regular Representation*
For every $\mathbb{F}$*-algebra* $\mathcal{A}$, $L_{\mathcal{A}} : \mathcal{A} \hookrightarrow \mathcal{L}(\mathcal{A})$, $a \mapsto L_a$, *is an* $\mathbb{F}$*-algebra monomorphism.*

Proof. For $a, b \in \mathcal{A}$, $r \in \mathbb{F}$, and $x \in \mathcal{A}$, $L_{a+b}(x) = (a+b)x = L_a(x) + L_b(x)$ and $L_{ra}(x) = (ra)x = rL_a(x)$, so $L_{\mathcal{A}}$ is $\mathbb{F}$-linear. Also $L_{ab}(x) = (ab)x = a(bx) = (L_aL_b)(x)$ and $L_{1_{\mathcal{A}}} = I_{\mathcal{A}}$, so products and units are preserved. If $L_a = 0$, then $a = L_a(1_{\mathcal{A}}) = 0$, so $L_{\mathcal{A}}$ is injective.
Structural Commentary: Left multiplication realises $\mathcal{A}$ inside its own operator algebra.
Key Insight: The unit detects elements, since $L_a(1_{\mathcal{A}}) = a$. □

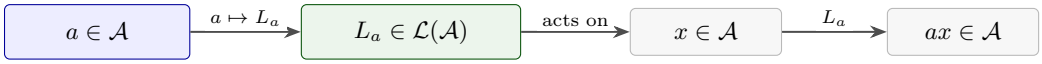


Figure 110: The left regular representation sends $a \in \mathcal{A}$ to $L_a : x \mapsto ax$; $L_{ab} = L_aL_b$ and $L_a(1_{\mathcal{A}}) = a$ give multiplicativity and injectivity.

Theorem 4.2.4. *Center and Ideals of an Endomorphism Algebra*
Let $0 \neq \mathcal{V}$ be an $\mathbb{F}$-vector space, and set $\mathcal{F}(\mathcal{V}) := \{T \in \mathcal{L}(\mathcal{V}) : \dim \operatorname{im} T < \infty\}$.

(i) $Z(\mathcal{L}(\mathcal{V})) = \mathbb{F}I_{\mathcal{V}}$, so $\mathcal{L}(\mathcal{V})$ is central.
(ii) $\mathcal{F}(\mathcal{V}) \triangleleft \mathcal{L}(\mathcal{V})$, and $0 \neq \mathcal{J} \triangleleft \mathcal{L}(\mathcal{V}) \Rightarrow \mathcal{F}(\mathcal{V}) \subseteq \mathcal{J}$.
(iii) $\mathcal{L}(\mathcal{V})$ is simple iff $\dim \mathcal{V} < \infty$.

Proof.

(i) Let $T \in Z(\mathcal{L}(\mathcal{V}))$. For $0 \neq v \in \mathcal{V}$ and $\varphi \in \mathcal{V}^*$ with $\varphi(v) = 1$, commutation with $v \otimes \varphi : x \mapsto \varphi(x)v$ gives $Tv = T(v \otimes \varphi)(v) = (v \otimes \varphi)T(v) = \varphi(Tv)v$, so every line is T-stable. Thus $Tv = \lambda_v v$ for $v \neq 0$, and applying this to independent u, v and to $u + v$ gives $\lambda_u = \lambda_v$; hence $T = rI_{\mathcal{V}}$.
Scalars commute with all endomorphisms.

(ii) If $R, S \in \mathcal{L}(\mathcal{V})$ and $T \in \mathcal{F}(\mathcal{V})$, then $\operatorname{im}(RTS) \subseteq R(\operatorname{im} T)$, so $RTS \in \mathcal{F}(\mathcal{V})$ and $\mathcal{F}(\mathcal{V}) \triangleleft \mathcal{L}(\mathcal{V})$. If $0 \neq \mathcal{J} \triangleleft \mathcal{L}(\mathcal{V})$, choose $0 \neq S \in \mathcal{J}$, e with $Se \neq 0$, and $\eta \in \mathcal{V}^*$ with $\eta(Se) = 1$. For any $v \in \mathcal{V}$ and $\varphi \in \mathcal{V}^*$, $v \otimes \varphi = (v \otimes \eta)S(e \otimes \varphi) \in \mathcal{J}$, so $\mathcal{J}$ contains all rank-one maps, hence all finite-rank maps.

(iii) If $\dim \mathcal{V} < \infty$, then $\mathcal{F}(\mathcal{V}) = \mathcal{L}(\mathcal{V})$, so (ii) makes every nonzero two-sided ideal equal to $\mathcal{L}(\mathcal{V})$.
If $\dim \mathcal{V} = \infty$, then $0 \neq \mathcal{F}(\mathcal{V}) \neq \mathcal{L}(\mathcal{V})$, so $\mathcal{L}(\mathcal{V})$ is not simple.

Structural Commentary: Rank-one operators are the atoms controlling both centrality and ideals in $\mathcal{L}(\mathcal{V})$.
Key Insight: Sandwich one nonzero map to obtain every rank-one map. □

Definition 4.2.5. *Algebraic Elements and Minimal Polynomials*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and $a \in \mathcal{A}$.

(i) The *evaluation map at a* is $\operatorname{ev}_a : \mathbb{F}[x] \to \mathcal{A}$, $p \mapsto p(a)$.
(ii) a is *algebraic over $\mathbb{F}$* iff $\ker \operatorname{ev}_a \neq 0$.
(iii) If a is algebraic, its *minimal polynomial* m_a is the unique monic generator of $\ker \operatorname{ev}_a$, equivalently the monic polynomial of least degree with $m_a(a) = 0$.

Theorem 4.2.6. *Finite-Dimensional Algebras Are Algebraic*
If $\dim_{\mathbb{F}} \mathcal{A} < \infty$, then every $a \in \mathcal{A}$ is algebraic over $\mathbb{F}$. Moreover, the algebraic closure $\overline{\mathbb{Q}}_{\mathbb{C}}$ of $\mathbb{Q}$ inside $\mathbb{C}$ is an infinite-dimensional $\mathbb{Q}$-algebra whose elements are all algebraic.

Proof. Let $\dim_{\mathbb{F}} \mathcal{A} = n$ and $a \in \mathcal{A}$. The $n+1$ vectors $1, a, \dots, a^n$ are linearly dependent, so $\sum_{j=0}^{n} c_j a^j = 0$ for some $(c_0, \dots, c_n) \neq 0$; hence $p(x) = \sum_{j=0}^{n} c_j x^j \neq 0$ satisfies $p(a) = 0$.
If $\alpha, \beta \in \overline{\mathbb{Q}}_{\mathbb{C}}$, then $\mathbb{Q}(\alpha, \beta)/\mathbb{Q}$ is finite, so $\alpha + \beta$, $\alpha\beta$, and $r\alpha$ for $r \in \mathbb{Q}$ are algebraic; hence $\overline{\mathbb{Q}}_{\mathbb{C}}$ is a $\mathbb{Q}$-algebra.
For each $n \geq 1$, $x^n - 2$ is Eisenstein at 2, so it has a root of degree n in $\overline{\mathbb{Q}}_{\mathbb{C}}$; therefore $\dim_{\mathbb{Q}} \overline{\mathbb{Q}}_{\mathbb{C}} = \infty$.
Structural Commentary: Finite-dimensionality forces powers of each element into a linear relation.
Key Insight: Algebraic need not mean finite-dimensional as an algebra. □

Proposition 4.2.7. *The Algebra Generated by One Algebraic Element*
Let $a \in \mathcal{A}$ be algebraic with minimal polynomial m_a. Then $\mathbb{F}[a] := \operatorname{im} \operatorname{ev}_a = \{p(a) : \deg p < \deg m_a\}$ and $\mathbb{F}[a] \cong \mathbb{F}[x]/\langle m_a \rangle$.

Proof. $\operatorname{ev}_a : \mathbb{F}[x] \to \mathcal{A}$ is an $\mathbb{F}$-algebra homomorphism with image $\mathbb{F}[a]$.
Since $\ker \operatorname{ev}_a = \{p : p(a) = 0\}$ and m_a is the monic generator of this kernel, $\ker \operatorname{ev}_a = \langle m_a \rangle$.
Proposition 4.2.2 gives $\mathbb{F}[a] \cong \mathbb{F}[x]/\langle m_a \rangle$. $\forall p \in \mathbb{F}[x]$, by division $p = qm_a + r$, $\deg r < \deg m_a$, hence $p(a) = r(a)$.
Structural Commentary: One algebraic element gives one polynomial quotient.
Key Insight: Evaluation turns polynomial division modulo m_a into normal forms in $\mathbb{F}[a]$. □

Theorem 4.2.8. *Minimal Polynomial, Annihilator, and Invertibility*
Let $a \in \mathcal{A}$ be algebraic and write $m_a(x) = x^n + \sum_{j=0}^{n-1} \alpha_j x^j$.
Then $\operatorname{ann}_{\mathbb{F}[x]}(a) := \{f \in \mathbb{F}[x] : f(a) = 0\} = \langle m_a \rangle$, and $a \in \mathcal{A}^\times \iff \alpha_0 \neq 0$.

Proof. By Definition 4.2.5, $\operatorname{ann}_{\mathbb{F}[x]}(a) = \ker \operatorname{ev}_a = \langle m_a \rangle$.

($\Rightarrow$) If $a \in \mathcal{A}^\times$ and $\alpha_0 = 0$, then $m_a = xq$ with $\deg q < \deg m_a$, and $0 = m_a(a) = aq(a)$ gives $q(a) = 0$, contradicting minimality of m_a; hence $\alpha_0 \neq 0$.

($\Leftarrow$) If $\alpha_0 \neq 0$, then $0 = m_a(a) = a(a^{n-1} + \sum_{j=1}^{n-1} \alpha_j a^{j-1}) + \alpha_0$, so $b := -\alpha_0^{-1}(a^{n-1} + \sum_{j=1}^{n-1} \alpha_j a^{j-1})$ satisfies $ab = 1_{\mathcal{A}}$. Since $b \in \mathbb{F}[a]$, $ab = ba$, so $ba = 1_{\mathcal{A}}$ and $a \in \mathcal{A}^\times$.

Structural Commentary: Invertibility is read from the constant term of the minimal polynomial.
Key Insight: A nonzero constant term solves for a^{-1} as a polynomial in a. □

Definition 4.2.9. *Zero Divisors and Spectrum*
Let $\mathcal{A}$ be an $\mathbb{F}$-algebra and $0 \neq a \in \mathcal{A}$.

(i) a is a *left zero divisor* iff $\exists b \in \mathcal{A} \setminus \{0\}$ with $ab = 0$.
(ii) a is a *right zero divisor* iff $\exists c \in \mathcal{A} \setminus \{0\}$ with $ca = 0$.
(iii) If a is algebraic, *spectrum* is $\operatorname{spec}_{\mathcal{A}}(a) := \{r \in \mathbb{F} : m_a(r) = 0\}$; elements are algebraic *eigenvalues* of a.

Theorem 4.2.10. *Invertible if and only if Not a Zero Divisor*
Let $a \in \mathcal{A}$ be algebraic. Then $a \in \mathcal{A}^\times$ iff a is neither a left nor a right zero divisor.

Proof.

($\Rightarrow$) If $a \in \mathcal{A}^\times$ and $ab = 0$, then $b = a^{-1}ab = 0$; if $ca = 0$, then $c = caa^{-1} = 0$.
Hence a is neither a left nor a right zero divisor.

($\Leftarrow$) Prove the contrapositive. If $a \notin \mathcal{A}^\times$, Theorem 4.2.8 gives $m_a(x) = xp(x)$ with $p \neq 0$ and $\deg p < \deg m_a$.
Then $0 = m_a(a) = ap(a) = p(a)a$, while minimality gives $p(a) \neq 0$. Thus a is both left, right zero divisor.

Structural Commentary: For algebraic elements, noninvertibility forces a polynomial kernel witness.
Key Insight: Factoring m_a by x manufactures the zero-divisor witness $p(a)$. □

Corollary 4.2.11. *One-Sided Inverses Become Two-Sided*
Let $a \in \mathcal{A}$ be algebraic. If $\exists b \in \mathcal{A}$ with $ba = 1_{\mathcal{A}}$ or $ab = 1_{\mathcal{A}}$, then $a \in \mathcal{A}^\times$.

Proof. Assume $ba = 1_{\mathcal{A}}$. If $ac = 0$, then $c = bac = 0$; if $ax = 0$, then $x = bax = 0$. Hence a is neither a right nor a left zero divisor, so Theorem 4.2.10 gives $a \in \mathcal{A}^\times$. The case $ab = 1_{\mathcal{A}}$ is symmetric.
Structural Commentary: Algebraicity upgrades one-sided cancellation to two-sided invertibility.
Key Insight: A one-sided inverse rules out both zero-divisor obstructions. □

Theorem 4.2.12. *Roots Detected by Invertibility*
Let $a \in \mathcal{A}$ be algebraic and $r \in \mathbb{F}$. Then $m_a(r) = 0 \iff a - r1_{\mathcal{A}} \notin \mathcal{A}^\times$. In particular, $a \in \mathcal{A}^\times \iff 0 \notin \operatorname{spec}_{\mathcal{A}}(a)$.

Proof.

($\Rightarrow$) If $m_a(r) = 0$, write $m_a = (x - r)q$ with $\deg q < \deg m_a$.
Then $0 = m_a(a) = (a - r1_{\mathcal{A}})q(a) = q(a)(a - r1_{\mathcal{A}})$ and $q(a) \neq 0$ by minimality, so $a - r1_{\mathcal{A}} \notin \mathcal{A}^\times$.

($\Leftarrow$) If $m_a(r) \neq 0$, then $\gcd(x - r, m_a) = 1$, so $\exists u, v \in \mathbb{F}[x]$ with $u(x)(x - r) + v(x)m_a(x) = 1$.
Evaluating at a gives $u(a)(a - r1_{\mathcal{A}}) = 1_{\mathcal{A}} = (a - r1_{\mathcal{A}})u(a)$, hence $a - r1_{\mathcal{A}} \in \mathcal{A}^\times$, contradiction.

Taking $r = 0$ gives $a \in \mathcal{A}^\times \iff m_a(0) \neq 0 \iff 0 \notin \operatorname{spec}_{\mathcal{A}}(a)$.
Structural Commentary: Roots of m_a are precisely the nonunit scalar shifts of a.
Key Insight: Bézout turns $\gcd(x - r, m_a) = 1$ into an explicit inverse. □

Theorem 4.2.13. *Polynomial Spectral Mapping for Algebraic Elements*
Let $\mathbb{F}$ be algebraically closed, $a \in \mathcal{A}$ algebraic, and $p \in \mathbb{F}[x]$.
Then $p(a)$ is algebraic and $\operatorname{spec}_{\mathcal{A}}(p(a)) = p(\operatorname{spec}_{\mathcal{A}}(a)) := \{p(r) : r \in \operatorname{spec}_{\mathcal{A}}(a)\}$.

Proof. Since $\mathbb{F}[p(a)] \subseteq \mathbb{F}[a]$ and $\dim_{\mathbb{F}} \mathbb{F}[a] < \infty$, $p(a)$ is algebraic. If p is constant, the claim is immediate.
If $\lambda \in \operatorname{spec}_{\mathcal{A}}(p(a))$, then $p(a) - \lambda 1_{\mathcal{A}} \notin \mathcal{A}^\times$. Factor $p(x) - \lambda = c\prod_{j=1}^{m}(x - r_j)^{e_j}$ over $\mathbb{F}$.
Since a product of units is a unit, $a - r_j 1_{\mathcal{A}} \notin \mathcal{A}^\times$ for some j. Theorem 4.2.12 gives $r_j \in \operatorname{spec}_{\mathcal{A}}(a)$ and $\lambda = p(r_j)$.
Conversely, if $r \in \operatorname{spec}_{\mathcal{A}}(a)$, then $p(x) - p(r) = (x - r)q(x)$, so $p(a) - p(r)1_{\mathcal{A}} = (a - r1_{\mathcal{A}})q(a)$ is noninvertible; otherwise $a - r1_{\mathcal{A}}$ has a one-sided inverse and Corollary 4.2.11 makes it invertible. Thus $p(r) \in \operatorname{spec}_{\mathcal{A}}(p(a))$.
Structural Commentary: Polynomial evaluation carries spectra by applying the same polynomial to roots.
Key Insight: Factor $p(x) - \lambda$ and detect a nonunit scalar shift of a. □

Proposition 4.2.14. *Spectra of ab and ba*
If $\dim_{\mathbb{F}} \mathcal{A} < \infty$, $\mathbb{F}$ *is algebraically closed, and* $a, b \in \mathcal{A}$, *then* $\operatorname{spec}_{\mathcal{A}}(ab) = \operatorname{spec}_{\mathcal{A}}(ba)$.

Proof. All elements are algebraic by Theorem 4.2.6. Let $r \in \mathbb{F}^\times$.
If $ba - r1_{\mathcal{A}} \in \mathcal{A}^\times$, set $u := a(ba - r1_{\mathcal{A}})^{-1}b - 1_{\mathcal{A}}$; then $(ab - r1_{\mathcal{A}})u = r1_{\mathcal{A}}$, so $ab - r1_{\mathcal{A}}$ has a right inverse, hence is invertible by Corollary 4.2.11. Thus $r \notin \operatorname{spec}_{\mathcal{A}}(ba) \Rightarrow r \notin \operatorname{spec}_{\mathcal{A}}(ab)$, and symmetry gives equality off 0.
If $ba \in \mathcal{A}^\times$ with inverse y, then $b(ay) = 1_{\mathcal{A}}$ and $(yb)a = 1_{\mathcal{A}}$, so $b, a \in \mathcal{A}^\times$ by Corollary 4.2.11; hence $ab \in \mathcal{A}^\times$.
The converse is symmetric, so 0 belongs to both spectra or neither.
Structural Commentary: $\operatorname{spec}(ab) \setminus \{0\} = \operatorname{spec}(ba) \setminus \{0\}$, and algebraicity resolves the 0-case.
Key Insight: The resolvent identity transfers invertibility between $ba - r1_{\mathcal{A}}$ and $ab - r1_{\mathcal{A}}$ for $r \neq 0$. □

Definition 4.2.15. *Division Algebras*
An associative $\mathbb{F}$-algebra $\mathcal{D}$ is a *division algebra* iff $\mathcal{D} \setminus \{0\} \subseteq \mathcal{D}^\times$.

Example 4.2.16. *Hamilton's Quaternions*
Hamilton's quaternion algebra is $\mathbb{H} := \mathbb{R}1 \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$ with unit 1,
relations $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, and $ki = j = -ik$, extended $\mathbb{R}$-bilinearly.
It is noncommutative since $ij = -ji$. For $q = r_0 + r_1 i + r_2 j + r_3 k$, define $\overline{q} := r_0 - r_1 i - r_2 j - r_3 k$ and $N(q) := q\overline{q} = r_0^2 + r_1^2 + r_2^2 + r_3^2$. If $q \neq 0$, then $N(q) > 0$ and $q^{-1} = N(q)^{-1}\overline{q}$, so $\mathbb{H}$ is a real division algebra.

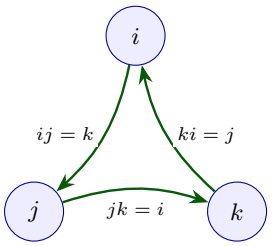


Figure 111: Quaternion basis products: the cycle $i \to j \to k \to i$ gives $ij = k$, $jk = i$, and $ki = j$; reversed order negates the product. Also $i^2 = j^2 = k^2 = -1$, and 1 is the unit.

Theorem 4.2.17. *Division Algebras over Algebraically Closed Fields*
If $\dim_{\mathbb{F}} \mathcal{D} < \infty$, $\mathcal{D}$ *is a division* $\mathbb{F}$*-algebra, and* $\mathbb{F}$ *is algebraically closed, then* $\mathcal{D} = \mathbb{F}1_{\mathcal{D}} \cong \mathbb{F}$.

Proof. Let $a \in \mathcal{D}$. By Theorem 4.2.6, a is algebraic with minimal polynomial m_a.
If $m_a = pq$ with $0 < \deg p, \deg q < \deg m_a$, then $0 = m_a(a) = p(a)q(a)$, while minimality gives $p(a), q(a) \neq 0$; this contradicts $\mathcal{D} \setminus \{0\} \subseteq \mathcal{D}^\times$. Thus m_a is irreducible, hence linear over algebraically closed $\mathbb{F}$: $m_a = x - r$.
Therefore $a = r1_{\mathcal{D}}$, so $\mathcal{D} \subseteq \mathbb{F}1_{\mathcal{D}}$, and the reverse inclusion is the scalar copy.
Structural Commentary: Algebraic closure leaves only scalars.
Key Insight: Irreducible minimal polynomials collapse to linear polynomials. □

Theorem 4.2.18. *Wedderburn's Theorem*
Every finite division algebra $\mathcal{D}$ *is commutative; equivalently* $\mathcal{D} = Z(\mathcal{D})$, *hence* $\mathcal{D}$ *is a finite field.*

Proof. Let $F := Z(\mathcal{D})$, $|F| = z$, $[\mathcal{D} : F] = n$, and $G := \mathcal{D}^\times$.
Suppose $\mathcal{D}$ is noncommutative, and let B be representatives of the noncentral conjugacy classes of G.
For $\beta \in B$, $C_{\mathcal{D}}(\beta)$ is a proper division subalgebra containing F, so $|C_{\mathcal{D}}(\beta)| = z^{k_\beta}$ with $1 < k_\beta < n$ and $k_\beta \mid n$.
By the class equation, Theorem 2.4.2, $z^n - 1 = z - 1 + \sum_{\beta \in B} (z^n - 1)/(z^{k_\beta} - 1)$.
Since $k_\beta \mid n$ and $k_\beta < n$, the primitive nth roots occur in $(x^n - 1)/(x^{k_\beta} - 1)$, so Propositions 3.2.18 and 3.2.19 give $Q_n(z) \mid (z^n - 1)/(z^{k_\beta} - 1)$ for each β, and also $Q_n(z) \mid z^n - 1$.
The class equation therefore gives $Q_n(z) \mid z - 1$, contradicting Lemma 3.2.20, which gives $|Q_n(z)| > z - 1$.
Thus G has no noncentral conjugacy classes, so G is abelian and $\mathcal{D}$ is commutative.
Structural Commentary: The class equation converts noncommutativity into cyclotomic divisibility.
Key Insight: The cyclotomic lower bound is too large to divide $z - 1$. □

Theorem 4.2.19. *Frobenius Classification over* $\mathbb{R}$
If $\dim_{\mathbb{R}} \mathcal{D} < \infty$ *and* $\mathcal{D}$ *is a real division algebra, then* $\mathcal{D} \cong \mathbb{R}$, $\mathcal{D} \cong \mathbb{C}$, *or* $\mathcal{D} \cong \mathbb{H}$.

Proof. For each $d \in \mathcal{D}$, m_d is linear or irreducible quadratic over $\mathbb{R}$; completing the square gives $d = r + u$ with $r \in \mathbb{R}$ and $u^2 \in \mathbb{R}_{\leq 0}$. Set $\mathcal{D}_0 := \{u \in \mathcal{D} : u^2 \in \mathbb{R}_{\leq 0}\}$; the preceding decomposition and uniqueness give $\mathcal{D} = \mathbb{R} \oplus \mathcal{D}_0$. If $\mathcal{D}_0 = 0$, then $\mathcal{D} \cong \mathbb{R}$. If $\dim \mathcal{D}_0 = 1$, choose $i \in \mathcal{D}_0$ with $i^2 = -1$, so $\mathcal{D} = \mathbb{R} \oplus \mathbb{R}i \cong \mathbb{C}$.
If $\dim \mathcal{D}_0 > 1$, use $\langle u, v \rangle := -\frac{1}{2}(uv + vu)$ on $\mathcal{D}_0$ to choose orthonormal $i, j \in \mathcal{D}_0$ with $i^2 = j^2 = -1$; $ij + ji = 0$.

Then $k := ij$ satisfies $k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, and reversed products change sign.
If $\ell \in \mathcal{D}_0$ is orthogonal to i, j, k, the same identities force $\ell = 0$, so $\mathcal{D} = \mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k \cong \mathbb{H}$.
Structural Commentary: A real division algebra splits into scalars plus a Euclidean pure part.
Key Insight: Orthogonal imaginary units generate exactly $\mathbb{C}$ or $\mathbb{H}$. □

Perspective 4.2.20. *Operator Connection*
$T \in \mathcal{L}(\mathcal{V})$ gives an $\mathbb{F}[x]$-module by $x \cdot v = T(v)$. For any algebra $\mathcal{A}$, the left regular map $L_{\mathcal{A}} : \mathcal{A} \hookrightarrow \mathcal{L}(\mathcal{A})$ realizes $\mathcal{A}$ as a faithful endomorphism algebra.

5 Tensor Products

5.1 Tensor Products and Universal Properties

Tensor products are the device that turns bilinear data into linear data. The guiding rule is the same one already used for quotient spaces in Theorem 1.1.1: a universal object is recognised by the unique way every compatible map factors through it.

Definition 5.1.1. *Finite-Coordinate Tensor Product Model*
Let $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ be ordered bases of $\mathcal{V}$ and $\mathcal{W}$.
Define the *coordinate tensor model* $\mathcal{V} \otimes_{\beta,\gamma} \mathcal{W} := \mathrm{Mat}(m, n, \mathbb{F})$ by $\mathbf{v}_i \otimes_{\beta,\gamma} \mathbf{w}_j := E_{i,j}$.
If $\mathbf{v} = \sum_{i=1}^{m} x_i \mathbf{v}_i$ and $\mathbf{w} = \sum_{j=1}^{n} y_j \mathbf{w}_j$, then $\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w} := \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j E_{i,j} \in \mathrm{Mat}(m, n, \mathbb{F})$.

Proposition 5.1.2. *Tensor Model Basis and Bilinear Evaluation*
In the coordinate tensor model associated to bases $\beta = (\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $\gamma = (\mathbf{w}_1, \dots, \mathbf{w}_n)$:

(i) the matrices $E_{i,j} = \mathbf{v}_i \otimes_{\beta,\gamma} \mathbf{w}_j$ form a basis;
(ii) the map $(\mathbf{v}, \mathbf{w}) \mapsto \mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}$ is bilinear;
(iii) every bilinear form $B : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$ determines a unique linear functional Θ_B on $\mathrm{Mat}(m, n, \mathbb{F})$ such that $\Theta_B(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) = B(\mathbf{v}, \mathbf{w})$ for all $\mathbf{v} \in \mathcal{V}$ and $\mathbf{w} \in \mathcal{W}$.

Proof.

(i) This is the standard entry-matrix basis of $\mathrm{Mat}(m, n, \mathbb{F})$ from Proposition 3.3.4.
(ii) The coordinates of $a\mathbf{v}_1 + \mathbf{v}_2$ are the same linear combination of the coordinates of $\mathbf{v}_1$ and $\mathbf{v}_2$, and the displayed formula is linear in those coordinates. The same argument applies in the second variable.
(iii) Define $\Theta_B(E_{i,j}) = B(\mathbf{v}_i, \mathbf{w}_j)$ on the basis from (i), and extend linearly by Proposition 3.1.11. If $\mathbf{v} = \sum_i x_i \mathbf{v}_i$ and $\mathbf{w} = \sum_j y_j \mathbf{w}_j$, then $\Theta_B(\mathbf{v} \otimes_{\beta,\gamma} \mathbf{w}) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_i y_j B(\mathbf{v}_i, \mathbf{w}_j) = B(\mathbf{v}, \mathbf{w})$, by bilinearity of B.
Uniqueness follows since the values on basis matrices $E_{i,j}$ are forced by $\mathbf{v}_i, \mathbf{w}_j$.

Structural Commentary: The coordinate tensor model stores each pair of basis directions in its own matrix slot, so a bilinear form can be evaluated by a single linear functional on those slots.
Key Insight: Tensor coordinates turn products of coordinates into linear coordinates. □

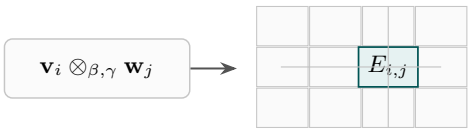


Figure 112: A basis pair occupies the (i, j) matrix slot.

Definition 5.1.3. *Bilinear Maps*
Let $\mathcal{U}, \mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces and $f : \mathcal{U} \times \mathcal{V} \to \mathcal{W}$.

(i) f is *bilinear* if $\forall r, s \in \mathbb{F}$, $\mathbf{u}, \mathbf{u}' \in \mathcal{U}$, $\mathbf{v}, \mathbf{v}' \in \mathcal{V}$, $f(r\mathbf{u} + s\mathbf{u}', \mathbf{v}) = rf(\mathbf{u}, \mathbf{v}) + sf(\mathbf{u}', \mathbf{v})$ and $f(\mathbf{u}, r\mathbf{v} + s\mathbf{v}') = rf(\mathbf{u}, \mathbf{v}) + sf(\mathbf{u}, \mathbf{v}')$.
(ii) $\mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W}) := \{f : \mathcal{U} \times \mathcal{V} \to \mathcal{W} : f \text{ bilinear}\}$.
(iii) If $\mathcal{W} = \mathbb{F}$, then $\mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathbb{F})$ is the space of *bilinear forms* from Definition 5.3.1.

Example 5.1.4. *Basic Bilinear Maps*

(i) If A is an $\mathbb{F}$-algebra, then $\mu : A \times A \to A$, $\mu(a, b) := ab$, is bilinear.
(ii) Evaluation $\mathrm{ev} : \mathcal{L}(\mathcal{V}, \mathcal{W}) \times \mathcal{V} \to \mathcal{W}$, $\mathrm{ev}(T, \mathbf{v}) := T\mathbf{v}$, is bilinear.
(iii) If $\varphi \in \mathcal{V}'$ and $\psi \in \mathcal{W}'$, then $B_{\varphi,\psi} : \mathcal{V} \times \mathcal{W} \to \mathbb{F}$, $B_{\varphi,\psi}(\mathbf{v}, \mathbf{w}) := \varphi(\mathbf{v})\psi(\mathbf{w})$, is bilinear.

Definition 5.1.5. *Tensor Product by Universal Property*
Let $\mathcal{U}, \mathcal{V}$ be $\mathbb{F}$-vector spaces.

(i) A pair $(\mathcal{T}, t)$ is a *tensor product* of $\mathcal{U}$ and $\mathcal{V}$ if $\mathcal{T}$ is an $\mathbb{F}$-vector space, $t \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{T})$, and for every $\mathbb{F}$-vector space $\mathcal{W}$ and $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$, $\exists! \widetilde{f} \in \mathcal{L}(\mathcal{T}, \mathcal{W})$ such that $f = \widetilde{f} \circ t$.
(ii) The map $\widetilde{f}$ is the *mediating map* for f.
(iii) Write $\mathcal{T} = \mathcal{U} \otimes \mathcal{V}$ and $\mathbf{u} \otimes \mathbf{v} := t(\mathbf{u}, \mathbf{v})$; such tensors are *decomposable*.

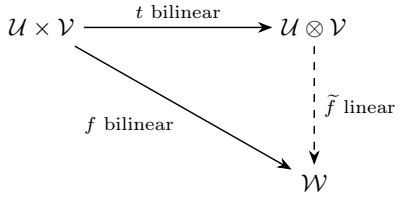


Figure 113: A tensor product is universal for bilinear maps out of $\mathcal{U} \times \mathcal{V}$.

Theorem 5.1.6. *Coordinate-Free Quotient Construction*
Let $\mathcal{U}, \mathcal{V}$ be $\mathbb{F}$-vector spaces and let $\mathbb{F}^{(\mathcal{U}\times\mathcal{V})}$ have basis $\{[\mathbf{u}, \mathbf{v}] : (\mathbf{u}, \mathbf{v}) \in \mathcal{U} \times \mathcal{V}\}$.
Let $\mathcal{R}$ be the span of all $r[\mathbf{u}_1, \mathbf{v}] + s[\mathbf{u}_2, \mathbf{v}] - [r\mathbf{u}_1 + s\mathbf{u}_2, \mathbf{v}]$; and $r[\mathbf{u}, \mathbf{v}_1] + s[\mathbf{u}, \mathbf{v}_2] - [\mathbf{u}, r\mathbf{v}_1 + s\mathbf{v}_2]$, where $r, s \in \mathbb{F}$, $\mathbf{u}, \mathbf{u}_1, \mathbf{u}_2 \in \mathcal{U}$, and $\mathbf{v}, \mathbf{v}_1, \mathbf{v}_2 \in \mathcal{V}$. Then $\mathcal{U} \otimes \mathcal{V} := \mathbb{F}^{(\mathcal{U}\times\mathcal{V})}/\mathcal{R}$, $t(\mathbf{u}, \mathbf{v}) = [\mathbf{u}, \mathbf{v}] + \mathcal{R}$ is a tensor product of $\mathcal{U}$ and $\mathcal{V}$.

Proof. Let $\pi : \mathbb{F}^{(\mathcal{U}\times\mathcal{V})} \to \mathcal{U} \otimes \mathcal{V}$ be the quotient map.
Since the generators of $\mathcal{R}$ vanish in the quotient, $t(\mathbf{u}, \mathbf{v}) = \pi[\mathbf{u}, \mathbf{v}]$ is bilinear.
Let $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ and define $\alpha : \mathbb{F}^{(\mathcal{U}\times\mathcal{V})} \to \mathcal{W}$ by $\alpha[\mathbf{u}, \mathbf{v}] := f(\mathbf{u}, \mathbf{v})$.
Bilinearity gives $\mathcal{R} \subseteq \ker \alpha$, so Theorem 1.1.1 gives a unique $\widetilde{f} \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$ with $\widetilde{f} \circ \pi = \alpha$. Hence $\widetilde{f} \circ t = f$.
If $L \circ t = f$, then L and $\widetilde{f}$ agree on the spanning set $\{[\mathbf{u}, \mathbf{v}] + \mathcal{R}\}$, so $L = \widetilde{f}$.
Structural Commentary: The quotient kills precisely the failure of the raw ordered-pair basis to be bilinear.
Key Insight: A tensor product is a free bilinear map, followed by no extra identifications. □

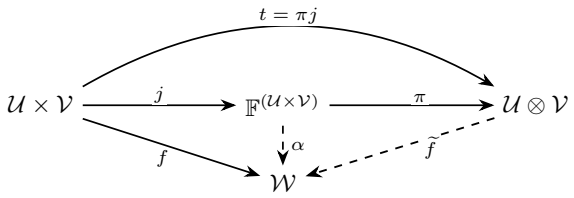


Figure 114: The free linear map α kills $\mathcal{R}$, so it descends uniquely to $\widetilde{f} : \mathcal{U} \otimes \mathcal{V} \to \mathcal{W}$.

Corollary 5.1.7. *Tensor Products Are Unique up to Isomorphism*
If $(\mathcal{T}, t)$ and $(\mathcal{S}, s)$ are tensor products of $\mathcal{U}$ and $\mathcal{V}$,
then there is a unique isomorphism $\Phi : \mathcal{T} \to \mathcal{S}$ satisfying $\Phi(t(\mathbf{u}, \mathbf{v})) = s(\mathbf{u}, \mathbf{v})$, where $\mathbf{u} \in \mathcal{U}$, $\mathbf{v} \in \mathcal{V}$.

Proof. Apply universal property of $(\mathcal{T}, t)$ to bilinear map s and universal property of $(\mathcal{S}, s)$ to bilinear map t.
This gives linear maps $\Phi : \mathcal{T} \to \mathcal{S}$ and $\Psi : \mathcal{S} \to \mathcal{T}$. The composites $\Psi\Phi$ and $I_{\mathcal{T}}$ are both mediating maps for t, while $\Phi\Psi$ and $I_{\mathcal{S}}$ are both mediating maps for s. Uniqueness forces $\Psi\Phi = I_{\mathcal{T}}$ and $\Phi\Psi = I_{\mathcal{S}}$.
Structural Commentary: Universal objects are unique up to canonical isomorphism.
Key Insight: Two universal factorisations force the comparison maps to be inverse. □

5.2 Tensor Product Bases and Maps

Theorem 5.2.1. *Tensor Bases, Dimension, and Coordinate Rank*
Let $\mathcal{U}$ and $\mathcal{V}$ be finite-dimensional with bases $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ and $\gamma = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

(i) The tensors $\mathbf{u}_i \otimes \mathbf{v}_j$ form a basis of $\mathcal{U} \otimes \mathcal{V}$.
(ii) $\dim(\mathcal{U} \otimes \mathcal{V}) = \dim(\mathcal{U}) \dim(\mathcal{V})$.
(iii) Every $\mathbf{z} \in \mathcal{U} \otimes \mathcal{V}$ has a unique matrix $R_{\beta,\gamma}(\mathbf{z}) = [r_{ij}] \in \mathrm{Mat}(m, n, \mathbb{F})$ such that $\mathbf{z} = \sum_{i=1}^{m} \sum_{j=1}^{n} r_{ij}(\mathbf{u}_i \otimes \mathbf{v}_j)$.
If β', γ' are other bases, then $\operatorname{rank} R_{\beta',\gamma'}(\mathbf{z}) = \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$.

Proof.

(i) The coordinate tensor model of Definition 5.1.1 has the universal property in Definition 5.1.5. Hence Corollary 5.1.7 identifies $\mathcal{U} \otimes \mathcal{V} \cong \mathrm{Mat}(m, n, \mathbb{F})$ with $\mathbf{u}_i \otimes \mathbf{v}_j \mapsto E_{i,j}$, and $\{E_{i,j}\}$ is a basis.
(ii) Counting the mn basis tensors gives $\dim(\mathcal{U} \otimes \mathcal{V}) = mn = \dim(\mathcal{U}) \dim(\mathcal{V})$.
(iii) The basis in (i) gives unique coefficients r_{ij}, hence $R_{\beta,\gamma}(\mathbf{z})$. If β', γ' are other bases, then $R_{\beta',\gamma'}(\mathbf{z}) = P R_{\beta,\gamma}(\mathbf{z}) Q$ for invertible change-of-basis matrices P, Q, so $\operatorname{rank} R_{\beta',\gamma'}(\mathbf{z}) = \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$.

Structural Commentary: Tensor rank is coordinate-free matrix rank.
Key Insight: Changing bases performs row and column changes, which preserve matrix rank. □

Definition 5.2.2. *Tensor Coordinate Matrix and Rank*
Let $\mathcal{U}, \mathcal{V}$ be finite-dimensional, with bases $\beta = (\mathbf{u}_1, \ldots, \mathbf{u}_m)$ and $\gamma = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

(i) For $\mathbf{z} = \sum_{i=1}^{m} \sum_{j=1}^{n} r_{ij} (\mathbf{u}_i \otimes \mathbf{v}_j)$, define the *coordinate matrix* $R_{\beta,\gamma}(\mathbf{z}) := [r_{ij}]$.
(ii) The *rank of the tensor* $\mathbf{z}$ is $\operatorname{rank}(\mathbf{z}) := \operatorname{rank} R_{\beta,\gamma}(\mathbf{z})$, independent of β, γ by Theorem 5.2.1.

Theorem 5.2.3. *Zero Tests and Decomposable Rank*
Let $\mathcal{U}$ and $\mathcal{V}$ be vector spaces.

(i) $\forall \mathbf{u} \in \mathcal{U}, \mathbf{v} \in \mathcal{V}$, $\mathbf{0} \otimes \mathbf{v} = \mathbf{u} \otimes \mathbf{0} = \mathbf{0}$.
(ii) If $\mathbf{u}_1, \ldots, \mathbf{u}_r$ are linearly independent and $\sum_{i=1}^{r} \mathbf{u}_i \otimes \mathbf{v}_i = \mathbf{0}$, then $\mathbf{v}_1 = \cdots = \mathbf{v}_r = \mathbf{0}$.
(iii) If $\dim \mathcal{U}, \dim \mathcal{V} < \infty$, then $\mathbf{z} \neq \mathbf{0}$ is decomposable $\iff \operatorname{rank}(\mathbf{z}) = 1$.
More generally, $\operatorname{rank}(\mathbf{z}) = \min \left\{ r \geq 0 : \mathbf{z} = \sum_{i=1}^{r} \mathbf{x}_i \otimes \mathbf{y}_i,\ \mathbf{x}_i \in \mathcal{U}, \mathbf{y}_i \in \mathcal{V} \right\}$.

Proof.

(i) The tensor map $t : \mathcal{U} \times \mathcal{V} \to \mathcal{U} \otimes \mathcal{V}$ is bilinear, hence linear in each coordinate; therefore $t(\mathbf{0}, \mathbf{v}) = t(\mathbf{u}, \mathbf{0}) = \mathbf{0}$.
(ii) For each k, extend the dual basis functional on $\operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ to $\varphi_k \in \mathcal{U}'$ with $\varphi_k(\mathbf{u}_i) = \delta_{ki}$. The bilinear map $(\mathbf{u}, \mathbf{v}) \mapsto \varphi_k(\mathbf{u})\mathbf{v}$ induces $T_k \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{V})$.
Applying T_k gives $\mathbf{0} = T_k \left(\sum_{i=1}^{r} \mathbf{u}_i \otimes \mathbf{v}_i \right) = \sum_{i=1}^{r} \delta_{ki} \mathbf{v}_i = \mathbf{v}_k$.
(iii) Choose bases β, γ. If $\mathbf{z} = \mathbf{x} \otimes \mathbf{y} \neq \mathbf{0}$, then $R_{\beta,\gamma}(\mathbf{z}) = [\mathbf{x}]_\beta [\mathbf{y}]_\gamma^{\mathsf{T}}$, so $\operatorname{rank}(\mathbf{z}) = 1$.
Conversely, if $\operatorname{rank} R_{\beta,\gamma}(\mathbf{z}) = r$, write $R_{\beta,\gamma}(\mathbf{z}) = \sum_{i=1}^{r} a_i b_i^{\mathsf{T}}$ with columns a_i, b_i.
Translating a_i, b_i back to vectors $\mathbf{x}_i, \mathbf{y}_i$ gives $\mathbf{z} = \sum_{i=1}^{r} \mathbf{x}_i \otimes \mathbf{y}_i$ as a sum of r decomposable tensors.
Conversely, any sum of s decomposable tensors gives a sum of s rank-one coordinate matrices, so $\operatorname{rank}(\mathbf{z}) \leq s$; hence the displayed minimum equals $\operatorname{rank}(\mathbf{z})$.

Structural Commentary: Linear functionals can test one tensor coordinate at a time.
Key Insight: Tensor rank is matrix rank after choosing bases, but the number itself survives changing bases. □

Theorem 5.2.4. *Bilinearity Equals Linearity on a Tensor Product*
For $\mathbb{F}$-vector spaces $\mathcal{U}, \mathcal{V}, \mathcal{W}$, the map $\Phi : \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W}) \to \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$, $\Phi(f) = \widetilde{f}$, where $\widetilde{f}(\mathbf{u} \otimes \mathbf{v}) = f(\mathbf{u}, \mathbf{v})$, is an isomorphism.

Proof. The universal property gives existence and uniqueness of $\widetilde{f}$.
Define $\Psi : \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W}) \to \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$ by $\Psi(L) := L \circ t$.
Then $\Psi(\Phi(f)) = \widetilde{f} \circ t = f$, and $\Phi(\Psi(L)) = L$ by uniqueness of the mediating map for $L \circ t$. Hence Φ is bijective.
For $a \in \mathbb{F}$ and f, g bilinear, $\Phi(af + g)(\mathbf{u} \otimes \mathbf{v}) = af(\mathbf{u}, \mathbf{v}) + g(\mathbf{u}, \mathbf{v}) = (a\Phi(f) + \Phi(g))(\mathbf{u} \otimes \mathbf{v})$, so Φ is linear.
Structural Commentary: Tensor products package bilinear maps as ordinary linear maps.
Key Insight: To define a linear map out of $\mathcal{U} \otimes \mathcal{V}$, define a bilinear rule on $\mathcal{U} \times \mathcal{V}$. □

Method 5.2.5. *Defining Maps on Tensor Products*
To define $L \in \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{W})$:

(i) define a candidate rule $f : \mathcal{U} \times \mathcal{V} \to \mathcal{W}$ by $f(\mathbf{u}, \mathbf{v})$;
(ii) prove $f \in \mathrm{Bil}_{\mathbb{F}}(\mathcal{U}, \mathcal{V}; \mathcal{W})$;
(iii) set $L := \Phi(f)$ from Theorem 5.2.4, so $L(\mathbf{u} \otimes \mathbf{v}) = f(\mathbf{u}, \mathbf{v})$.

Do not assign arbitrary values on decomposable tensors: they span $\mathcal{U} \otimes \mathcal{V}$ but are usually linearly dependent.

Theorem 5.2.6. *Tensor Product of Linear Maps*
Let $A \in \mathcal{L}(\mathcal{U},\mathcal{U}')$ and $B \in \mathcal{L}(\mathcal{V},\mathcal{V}')$.
There exists a unique $A \otimes B : \mathcal{U} \otimes \mathcal{V} \to \mathcal{U}' \otimes \mathcal{V}'$ such that $(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$.
Moreover, $(A_2A_1) \otimes (B_2B_1) = (A_2 \otimes B_2)(A_1 \otimes B_1)$, $I_{\mathcal{U}} \otimes I_{\mathcal{V}} = I_{\mathcal{U}\otimes\mathcal{V}}$.
In finite dimensions, $[A \otimes B]^{\beta'\otimes\gamma'}_{\beta\otimes\gamma} = [A]^{\beta'}_{\beta} \otimes [B]^{\gamma'}_{\gamma}$ for lexicographically ordered tensor-product bases, where the right-hand $\otimes$ is the matrix Kronecker product.

Proof. The map $(\mathbf{u},\mathbf{v}) \mapsto A\mathbf{u} \otimes B\mathbf{v}$ is bilinear, so Theorem 5.2.4 gives the unique $A \otimes B$.
The two functorial identities hold on decomposable tensors $((A_2 \otimes B_2)(A_1 \otimes B_1))(\mathbf{u} \otimes \mathbf{v}) = A_2A_1\mathbf{u} \otimes B_2B_1\mathbf{v}$ and $(I_{\mathcal{U}} \otimes I_{\mathcal{V}})(\mathbf{u} \otimes \mathbf{v}) = \mathbf{u} \otimes \mathbf{v}$, and decomposable tensors span. If $A\mathbf{u}_j = \sum_i a_{ij}\mathbf{u}'_i$ and $B\mathbf{v}_\ell = \sum_k b_{k\ell}\mathbf{v}'_k$, then $(A \otimes B)(\mathbf{u}_j \otimes \mathbf{v}_\ell) = \sum_{i,k} a_{ij}b_{k\ell}(\mathbf{u}'_i \otimes \mathbf{v}'_k)$, which is the Kronecker block-entry rule.
Structural Commentary: Tensoring maps acts independently in the two tensor coordinates.
Key Insight: The mixed product property for Kronecker products is the coordinate shadow of functoriality. □

Theorem 5.2.7. *Tensor Product of Spaces of Linear Maps*
Let $\mathcal{U},\mathcal{V},\mathcal{X},\mathcal{Y}$ be $\mathbb{F}$-vector spaces. The assignment $(A,B) \mapsto A \otimes B$ induces a natural injective map $\Theta : \mathcal{L}(\mathcal{U},\mathcal{X}) \otimes \mathcal{L}(\mathcal{V},\mathcal{Y}) \to \mathcal{L}(\mathcal{U} \otimes \mathcal{V}, \mathcal{X} \otimes \mathcal{Y})$ with $\Theta(A \otimes B)(\mathbf{u} \otimes \mathbf{v}) = A\mathbf{u} \otimes B\mathbf{v}$.
If $\mathcal{U},\mathcal{V},\mathcal{X},\mathcal{Y}$ are finite-dimensional, then Θ is an isomorphism.
In particular, $\mathcal{U}' \otimes \mathcal{V}' \hookrightarrow (\mathcal{U} \otimes \mathcal{V})'$; $(\varphi \otimes \psi)(\mathbf{u} \otimes \mathbf{v}) = \varphi(\mathbf{u})\psi(\mathbf{v})$ is an isomorphism when $\dim\mathcal{U}, \dim\mathcal{V} < \infty$.

Proof.

(i) By Theorem 5.2.6, $(A,B) \mapsto A \otimes B$ is bilinear, hence induces Θ.

(ii) Let $0 \neq Z = \sum_{i=1}^{r} A_i \otimes B_i$ with r minimal. Then $A_i \neq 0$ and (B_i) is linearly independent.
Choose $\mathbf{u}$ with some $A_i\mathbf{u} \neq 0$, after reindexing let $A_1\mathbf{u},\dots,A_m\mathbf{u}$ be basis of $\operatorname{span}\{A_i\mathbf{u} : 1 \le i \le r\}$.
Absorb remaining coefficients into B_i gives $C_1,\dots,C_m$, not all zero, s.t. $\Theta(Z)(\mathbf{u} \otimes \mathbf{v}) = \sum_{i=1}^{m} (A_i\mathbf{u}) \otimes C_i\mathbf{v}$.
If $\Theta(Z) = 0$, Theorem 5.2.3 gives $C_i\mathbf{v} = 0$ for all $i,\mathbf{v}$, contradiction. Thus $\ker\Theta = 0$.

(iii) If $\mathcal{U},\mathcal{V},\mathcal{X},\mathcal{Y}$ are finite-dimensional, then the domain and codomain of Θ have equal dimension by Theorem 5.2.1, so injectivity implies bijectivity. Taking $\mathcal{X} = \mathcal{Y} = \mathbb{F}$, $\mathcal{L}(\mathcal{U},\mathbb{F}) = \mathcal{U}'$, $\mathcal{L}(\mathcal{U} \otimes \mathcal{V},\mathbb{F}) = (\mathcal{U} \otimes \mathcal{V})'$ gives the dual embedding and finite-dimensional isomorphism.

Structural Commentary: Tensoring maps yields maps on tensor products.
Key Insight: The rule on decomposable tensors is enough, and finite dimensions make the embedding onto. □

Proposition 5.2.8. *Scalar Extension by Tensor Product*
Let $K/\mathbb{F}$ be a field extension and let $\mathcal{V}_{\mathbb{F}}$ be an $\mathbb{F}$-vector space.
Then $K \otimes_{\mathbb{F}} \mathcal{V}_{\mathbb{F}}$ is a K-vector space via $\alpha \cdot (\beta \otimes \mathbf{v}) := (\alpha\beta) \otimes \mathbf{v}$. With $\mu : \mathcal{V}_{\mathbb{F}} \to K \otimes_{\mathbb{F}} \mathcal{V}_{\mathbb{F}}$, $\mu(\mathbf{v}) := 1 \otimes \mathbf{v}$, every $\mathbb{F}$-linear $f : \mathcal{V}_{\mathbb{F}} \to \mathcal{Y}$ into a K-vector space $\mathcal{Y}$ admits a unique K-linear $\widetilde{f} : K \otimes_{\mathbb{F}} \mathcal{V}_{\mathbb{F}} \to \mathcal{Y}$ such that $\widetilde{f} \circ \mu = f$.

Proof. For $\alpha \in K$, $(\beta,\mathbf{v}) \mapsto (\alpha\beta) \otimes \mathbf{v}$ is $\mathbb{F}$-bilinear, hence induces $M_\alpha \in \mathcal{L}_{\mathbb{F}}(K \otimes_{\mathbb{F}} \mathcal{V}_{\mathbb{F}})$.
On decomposable tensors, $M_\alpha M_\lambda = M_{\alpha\lambda}$, $M_1 = I$, and distributivity holds; since decomposable tensors span, these define a K-vector-space structure.
For f, set $g(\beta,\mathbf{v}) := \beta f(\mathbf{v})$. Then g is $\mathbb{F}$-bilinear, so it induces $\widetilde{f}$ with $\widetilde{f}(\beta \otimes \mathbf{v}) = \beta f(\mathbf{v})$. This is K-linear and gives $\widetilde{f} \circ \mu = f$. If h is K-linear and $h \circ \mu = f$, then $h(\beta \otimes \mathbf{v}) = \beta h(1 \otimes \mathbf{v}) = \beta f(\mathbf{v})$, so $h = \widetilde{f}$.
Structural Commentary: Scalar extension tensors in the larger field.
Key Insight: Tensoring with K upgrades which scalars are allowed to act. □

6 Symmetric and Exterior Algebras

6.1 Symmetric Algebras

The previous subsection shows that tensor products are the natural place where bilinear maps become linear. The same idea extends to maps with more variables, and then to tensor spaces, symmetric tensors, exterior tensors, and the determinant.

Definition 6.1.1. *Multilinear Maps and Iterated Tensor Products*
Let $\mathcal{V}_1,\dots,\mathcal{V}_n,\mathcal{W}$ be $\mathbb{F}$-vector spaces.

(i) A map $f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to \mathcal{W}$ is *multilinear*, or *n-linear*, if for each k, the map $\mathbf{v}_k \mapsto f(\mathbf{v}_1,\dots,\mathbf{v}_k,\dots,\mathbf{v}_n)$ is linear with the other entries fixed.

(ii) $\operatorname{Mult}_{\mathbb{F}}(\mathcal{V}_1,\dots,\mathcal{V}_n;\mathcal{W}) := \{f : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to \mathcal{W} : f \text{ multilinear}\}$.

(iii) If $\mathcal{W} = \mathbb{F}$, elements of $\operatorname{Mult}_{\mathbb{F}}(\mathcal{V}_1,\dots,\mathcal{V}_n;\mathbb{F})$ are *multilinear forms*.

(iv) A pair $(\mathcal{T}, t)$ is an *iterated tensor product* of $\mathcal{V}_1, \dots, \mathcal{V}_n$ if $t \in \mathrm{Mult}_{\mathbb{F}}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{T})$ and, for every $\mathbb{F}$-vector space $\mathcal{Y}$ and $f \in \mathrm{Mult}_{\mathbb{F}}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{Y})$, $\exists! \widetilde{f} \in \mathcal{L}(\mathcal{T}, \mathcal{Y})$ such that $f = \widetilde{f} \circ t$.
(v) Write $\mathcal{T} = \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$ and $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n := t(\mathbf{v}_1, \dots, \mathbf{v}_n)$.

Example 6.1.2. *Basic Multilinear Maps*

(i) If A is an associative $\mathbb{F}$-algebra, then $m_n : A^n \to A$, $m_n(a_1, \dots, a_n) := a_1 \cdots a_n$, is n-linear.
(ii) If $\dim \mathcal{V} = n$, then $\det : \mathcal{V}^n \to \mathbb{F}$, applied to coordinate columns in any basis, is alternating n-linear form.

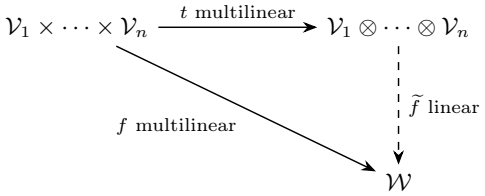


Figure 115: The iterated tensor product is universal for multilinear maps.

Theorem 6.1.3. *Basic Laws for Iterated Tensor Products*
Let $\mathcal{V}_1, \dots, \mathcal{V}_n, \mathcal{V}, \mathcal{W}$ be $\mathbb{F}$-vector spaces.

(i) $\mathrm{Mult}_{\mathbb{F}}(\mathcal{V}_1, \dots, \mathcal{V}_n; \mathcal{W}) \cong \mathcal{L}(\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n, \mathcal{W})$ *naturally.*
(ii) For $1 \le r < n$, the canonical α_r is $\alpha_r : (\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_r) \otimes (\mathcal{V}_{r+1} \otimes \cdots \otimes \mathcal{V}_n) \cong \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$, where $\alpha_r((\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_r) \otimes (\mathbf{v}_{r+1} \otimes \cdots \otimes \mathbf{v}_n)) = \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n$.
(iii) For $\pi \in S_n$, the canonical τ_π is $\tau_\pi : \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n \cong \mathcal{V}_{\pi(1)} \otimes \cdots \otimes \mathcal{V}_{\pi(n)}$, where $\tau_\pi(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n) = \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}$.
(iv) $\lambda : \mathbb{F} \otimes \mathcal{V} \cong \mathcal{V}$, $\lambda(r \otimes \mathbf{v}) = r\mathbf{v}$, and $\rho : \mathcal{V} \otimes \mathbb{F} \cong \mathcal{V}$, $\rho(\mathbf{v} \otimes r) = r\mathbf{v}$.
(v) If $\dim \mathcal{V}_i < \infty$ for all i, then $\dim(\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n) = \prod_{i=1}^n \dim(\mathcal{V}_i)$.

Proof.

(i) Let $t : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$ be canonical. Define $\Phi(f) = \widetilde{f}$, where $\widetilde{f}$ is the unique linear map satisfying $\widetilde{f}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n) = f(\mathbf{v}_1, \dots, \mathbf{v}_n)$; conversely define $\Psi(T) = T \circ t$. Then $\Psi \circ \Phi$ and $\Phi \circ \Psi$ are identity maps, so Φ is a linear bijection.
(ii) Set $A_r = (\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_r) \otimes (\mathcal{V}_{r+1} \otimes \cdots \otimes \mathcal{V}_n)$ and $B = \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$. The rules defining $\alpha_r : A_r \to B$ and $\beta_r : B \to A_r$, $\beta_r(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n) = (\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_r) \otimes (\mathbf{v}_{r+1} \otimes \cdots \otimes \mathbf{v}_n)$, are induced by multilinearity; $\alpha_r \circ \beta_r$ and $\beta_r \circ \alpha_r$ fix the spanning decomposable tensors, hence are identities.
(iii) Set $C_\pi = \mathcal{V}_{\pi(1)} \otimes \cdots \otimes \mathcal{V}_{\pi(n)}$. The map $p_\pi : \mathcal{V}_1 \times \cdots \times \mathcal{V}_n \to C_\pi$, $p_\pi(\mathbf{v}_1, \dots, \mathbf{v}_n) = \mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}$, is multilinear, so the universal property induces the unique τ_π with $\tau_\pi \circ t = p_\pi$.
The map $\sigma_\pi : C_\pi \to \mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n$, $\sigma_\pi(\mathbf{v}_{\pi(1)} \otimes \cdots \otimes \mathbf{v}_{\pi(n)}) = \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_n$, is induced similarly; $\sigma_\pi \tau_\pi$ and $\tau_\pi \sigma_\pi$ fix decomposable spanning tensors, hence are identities.
(iv) The maps $m_L : \mathbb{F} \times \mathcal{V} \to \mathcal{V}$, $m_L(r, \mathbf{v}) = r\mathbf{v}$, and $m_R : \mathcal{V} \times \mathbb{F} \to \mathcal{V}$, $m_R(\mathbf{v}, r) = r\mathbf{v}$, are bilinear, hence induce λ and ρ. Since $\lambda(1 \otimes \mathbf{v}) = \mathbf{v}$ and $1 \otimes r\mathbf{v} = r \otimes \mathbf{v}$, $\lambda^{-1}(\mathbf{v}) = 1 \otimes \mathbf{v}$.
Since $\rho(\mathbf{v} \otimes 1) = \mathbf{v}$ and $r\mathbf{v} \otimes 1 = \mathbf{v} \otimes r$, $\rho^{-1}(\mathbf{v}) = \mathbf{v} \otimes 1$.
(v) Choose finite bases B_i of $\mathcal{V}_i$ and set $\mathcal{B} = \{\mathbf{b}_1 \otimes \cdots \otimes \mathbf{b}_n : \mathbf{b}_i \in B_i\}$. By repeated application of Theorem 5.2.1, $\mathcal{B}$ is a basis; hence $\dim(\mathcal{V}_1 \otimes \cdots \otimes \mathcal{V}_n) = \#\mathcal{B} = \prod_{i=1}^n \#B_i = \prod_{i=1}^n \dim \mathcal{V}_i$.

Structural Commentary: Tensor parenthesising and ordering are canonical bookkeeping.
Key Insight: Universal properties extend pure-tensor rules uniquely. □

Definition 6.1.4. *Tensor Spaces of Type (p, q)*
Let $\dim \mathcal{V} < \infty$.

(i) For $(p, q) \in \mathbb{Z}^2_{\ge 0} \setminus \{(0, 0)\}$, with empty tensor powers equal to $\mathbb{F}$, $\mathcal{T}^p_q(\mathcal{V}) := \mathcal{V}^{\otimes p} \otimes (\mathcal{V}')^{\otimes q}$.
(ii) $\mathcal{T}^p_q(\mathcal{V})$ is the *tensor space of type (p, q)*; p is the *contravariant type*, q the *covariant type*.
(iii) $\mathcal{T}^0_0(\mathcal{V}) := \mathbb{F}$.

Proposition 6.1.5. *Tensor Spaces as Multilinear Functionals*
Let $\dim \mathcal{V} = n < \infty$ and $p, q \in \mathbb{Z}_{\ge 0}$.

(i) $\dim \mathcal{T}^p_q(\mathcal{V}) = n^{p+q}$.
(ii) Let $\iota : \mathcal{V} \to \mathcal{V}''$, $\iota(\mathbf{v})(\eta) = \eta(\mathbf{v})$. Via ι, natural isomorphism is $\Theta_{p,q} : \mathcal{T}^p_q(\mathcal{V}) \xrightarrow{\sim} \mathrm{Mult}_{\mathbb{F}}((\mathcal{V}')^{\times p} \times \mathcal{V}^{\times q}; \mathbb{F})$; where $\Theta_{p,q}(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p \otimes \varphi_1 \otimes \cdots \otimes \varphi_q)(\eta_1, \dots, \eta_p, \mathbf{x}_1, \dots, \mathbf{x}_q) = \prod_{i=1}^p \eta_i(\mathbf{v}_i) \prod_{j=1}^q \varphi_j(\mathbf{x}_j)$, with empty products equal to 1.

Proof.

(i) By Theorem 6.1.3, $\dim \mathcal{V}' = n$, $\dim \mathcal{T}_q^p(\mathcal{V}) = \dim(\mathcal{V}^{\otimes p}) \dim((\mathcal{V}')^{\otimes q}) = n^p n^q = n^{p+q}$, with $n^0 = 1$.

(ii) Set $X = (\mathcal{V}')^{\otimes p} \otimes \mathcal{V}^{\otimes q}$. Since $\iota : \mathcal{V} \cong \mathcal{V}''$ and all factors are finite-dimensional, $\mathcal{T}_q^p(\mathcal{V}) = \mathcal{V}^{\otimes p} \otimes (\mathcal{V}')^{\otimes q} \cong ((\mathcal{V}')')^{\otimes p} \otimes \mathcal{V}'^{\otimes q} \cong X' \cong \mathrm{Mult}_{\mathbb{F}}((\mathcal{V}')^{\times p} \times \mathcal{V}^{\times q}; \mathbb{F})$. On decomposables, the composite is $\Theta_{p,q}$; spanning gives uniqueness.

Structural Commentary: Tensor elements are multilinear functionals after finite-dimensional duality.
Key Insight: Choosing which factors are active turns tensor factors into input slots. □

Definition 6.1.6. *Contraction*
Let $p, q \geq 1$. The *contraction* is the unique linear map $C : \mathcal{T}_q^p(\mathcal{V}) \to \mathcal{T}_{q-1}^{p-1}(\mathcal{V})$, where $C(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p \otimes \varphi_1 \otimes \cdots \otimes \varphi_q) = \varphi_1(\mathbf{v}_1)\mathbf{v}_2 \otimes \cdots \otimes \mathbf{v}_p \otimes \varphi_2 \otimes \cdots \otimes \varphi_q$.
For (i, j), C_{ij} is C conjugated by the factor permutation placing $\mathbf{v}_i, \varphi_j$ first.

Example 6.1.7. *Rank-One Operators and Trace*
Under $\mathcal{V} \otimes \mathcal{V}' \cong \mathcal{L}(\mathcal{V})$, $\mathbf{v} \otimes \varphi \mapsto T_{\mathbf{v},\varphi}$, where $T_{\mathbf{v},\varphi}(\mathbf{x}) = \varphi(\mathbf{x})\mathbf{v}$. Hence $C(\mathbf{v} \otimes \varphi) = \varphi(\mathbf{v}) = \mathrm{tr}(T_{\mathbf{v},\varphi})$.

Definition 6.1.8. *Symmetric, Skew-Symmetric, and Alternating Maps*
Let $f : \mathcal{V}^{\times n} \to \mathcal{W}$ be multilinear.

(i) f is *symmetric* if interchanging any two coordinate positions does not change its value.
(ii) f is *skew-symmetric* if interchanging any two coordinate positions multiplies its value by -1.
(iii) f is *alternating* if $f(\mathbf{v}_1, \ldots, \mathbf{v}_n) = 0$ whenever two coordinate entries are equal.

If $\mathrm{char}(\mathbb{F}) \neq 2$, alternating and skew-symmetric are equivalent.
If $\mathrm{char}(\mathbb{F}) = 2$, skew-symmetric is the same as symmetric, and alternating is the stronger condition.

Definition 6.1.9. *Permutation Action and Symmetric/Exterior Tensor Spaces*
Let $\dim \mathcal{V} < \infty$, $p \in \mathbb{Z}_{\geq 0}$, $\mathcal{V}^{\otimes 0} = \mathbb{F}$, $T_p := \mathcal{V}^{\otimes p}$, and $G_p := \mathfrak{S}_p$.

(i) For $\sigma \in G_p$, the *permutation action* is $\lambda_\sigma : T_p \to T_p$, determined by $\lambda_\sigma(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p) = \mathbf{v}_{\sigma(1)} \otimes \cdots \otimes \mathbf{v}_{\sigma(p)}$.
(ii) The *symmetric tensors* are $\mathrm{ST}^p(\mathcal{V}) := \{\mathbf{z} \in T_p : \forall \sigma \in G_p, \lambda_\sigma \mathbf{z} = \mathbf{z}\}$.
(iii) The *antisymmetric tensors* are $\mathrm{AT}^p(\mathcal{V}) := \{\mathbf{z} \in T_p : \forall \sigma \in G_p, \lambda_\sigma \mathbf{z} = \mathrm{sign}(\sigma)\mathbf{z}\}$.
(iv) Set $R_\vee := \mathrm{span}\{\lambda_\sigma \mathbf{z} - \mathbf{z} : \sigma \in G_p, \mathbf{z} \in T_p\}$.
(v) The *symmetric power* is $\mathrm{Sym}^p(\mathcal{V}) := T_p / R_\vee$.
(vi) Set $A_p := \mathrm{span}\{\lambda_\sigma \mathbf{z} - \mathrm{sign}(\sigma)\mathbf{z} : \sigma \in G_p, \mathbf{z} \in T_p\}$ and $D_p := \mathrm{span}\{\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p : \exists i < j, \mathbf{v}_i = \mathbf{v}_j\}$. The *exterior power* is $\bigwedge^p \mathcal{V} := T_p / (A_p + D_p)$.
(vii) For quotient map $q_\vee$, the *symmetric product* is $q_\vee(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p) = \mathbf{v}_1 \vee \cdots \vee \mathbf{v}_p$
(viii) For quotient map $q_\wedge$, the *exterior product* is $q_\wedge(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p) = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_p$.

Theorem 6.1.10. *Universal Properties of Symmetric and Exterior Powers*
Let $\mathcal{U}$ be a vector space, $M = \mathrm{Mult}_{\mathbb{F}}(\mathcal{V}, \ldots, \mathcal{V}; \mathcal{U})$, and $f^\sigma(\mathbf{v}_1, \ldots, \mathbf{v}_p) := f(\mathbf{v}_{\sigma(1)}, \ldots, \mathbf{v}_{\sigma(p)})$.

(i) $\mathcal{L}(\mathrm{Sym}^p(\mathcal{V}), \mathcal{U}) \cong M_\vee$, where $M_\vee := \{f \in M : \forall \sigma \in \mathfrak{S}_p,\ f^\sigma = f\}$.
(ii) $\mathcal{L}(\bigwedge^p \mathcal{V}, \mathcal{U}) \cong M_\wedge$, where $M_\wedge := \{f \in M : \forall \sigma \in \mathfrak{S}_p,\ f^\sigma = \mathrm{sign}(\sigma) f,\ \forall (\mathbf{v}_1, \ldots, \mathbf{v}_p), (\exists i < j, \mathbf{v}_i = \mathbf{v}_j) \Rightarrow f(\mathbf{v}_1, \ldots, \mathbf{v}_p) = 0\}$.

Proof.

(i) Let $t : \mathcal{V}^{\times p} \to \mathcal{V}^{\otimes p}$ and $q_\vee : \mathcal{V}^{\otimes p} \to \mathrm{Sym}^p(\mathcal{V})$ be canonical. The map $\Phi_\vee(T) = T q_\vee t$ sends $\mathcal{L}(\mathrm{Sym}^p(\mathcal{V}), \mathcal{U})$ into $M_\vee$. Conversely, for $f \in M_\vee$, there is a unique $\widetilde{f} : \mathcal{V}^{\otimes p} \to \mathcal{U}$ with $f = \widetilde{f} t$; since $\widetilde{f}(R_\vee) = 0$, it factors uniquely through $q_\vee$ as $\widetilde{f} = \overline{f} q_\vee$, where $\overline{f} : \mathrm{Sym}^p(\mathcal{V}) \to \mathcal{U}$. Thus $f \mapsto \overline{f}$ is $\Phi_\vee^{-1}$.

(ii) Let $t : \mathcal{V}^{\times p} \to \mathcal{V}^{\otimes p}$ and $q_\wedge : \mathcal{V}^{\otimes p} \to \bigwedge^p \mathcal{V}$ be canonical. The map $\Phi_\wedge(T) = T q_\wedge t$ sends $\mathcal{L}(\bigwedge^p \mathcal{V}, \mathcal{U})$ into $M_\wedge$. Conversely, for $f \in M_\wedge$, the induced $\widetilde{f} : \mathcal{V}^{\otimes p} \to \mathcal{U}$ satisfies $\widetilde{f}(R_\wedge) = 0$, hence factors uniquely as $\widetilde{f} = \overline{f} q_\wedge$; thus $f \mapsto \overline{f}$ is $\Phi_\wedge^{-1}$.

Structural Commentary: Symmetric and exterior powers are tensor quotients by permutation relations.
Key Insight: Symmetric and alternating multilinear maps are ordinary linear maps out of the right quotient. □

Theorem 6.1.11. *Dimensions of Symmetric and Exterior Powers*
Let $\dim \mathcal{V} = n < \infty$ and $p \in \mathbb{Z}_{\geq 0}$. Set $I_\vee = \{1 \leq i_1 \leq \cdots \leq i_p \leq n\}$ and $I_\wedge = \{1 \leq i_1 < \cdots < i_p \leq n\}$.

(i) $\dim \mathrm{Sym}^p(\mathcal{V}) = \#I_\vee = \binom{n+p-1}{p}$.
(ii) $\dim \bigwedge^p \mathcal{V} = \#I_\wedge = \binom{n}{p}$, with $\binom{n}{p} = 0$ for $p > n$.

Proof. Let $E = (\mathbf{e}_1, \dots, \mathbf{e}_n)$ be a basis. For $I = (i_1, \dots, i_p)$, set $\mathbf{e}_I^\vee = \mathbf{e}_{i_1} \vee \cdots \vee \mathbf{e}_{i_p}$ and $\mathbf{e}_I^\wedge = \mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_p}$.

(i) $R_\vee$ identifies each word with its nondecreasing representative. Thus $B_\vee = \{\mathbf{e}_I^\vee : I \in I_\vee\}$ is the quotient basis, and $\dim \operatorname{Sym}^p(\mathcal{V}) = \#B_\vee = \#I_\vee = \binom{n+p-1}{p}$.

(ii) $R_\wedge$ kills repeated words and identifies each remaining word with its increasing representative up to sign. Thus $B_\wedge = \{\mathbf{e}_I^\wedge : I \in I_\wedge\}$ is the quotient basis, and $\dim \bigwedge^p \mathcal{V} = \#B_\wedge = \#I_\wedge = \binom{n}{p}$.

Structural Commentary: Symmetric powers count repetitions; exterior powers forbid repetitions.
Key Insight: Multisets give symmetric dimensions, subsets give exterior dimensions. □

Proposition 6.1.12. *Symmetrization in Characteristic Zero*
Assume $\operatorname{char}(\mathbb{F}) = 0$. *Define* $S, A \in \mathcal{L}(\mathcal{V}^{\otimes p})$ *by* $S := \frac{1}{p!} \sum_{\sigma \in \mathfrak{S}_p} \lambda_\sigma$ *and* $A := \frac{1}{p!} \sum_{\sigma \in \mathfrak{S}_p} \operatorname{sign}(\sigma) \lambda_\sigma$.

(i) $S^2 = S$ *and* $\operatorname{im} S = \operatorname{ST}^p(\mathcal{V})$.
(ii) $A^2 = A$ *and* $\operatorname{im} A = \operatorname{AT}^p(\mathcal{V})$.

Proof.

(i) For $\tau \in \mathfrak{S}_p$, $\lambda_\tau S = p!^{-1} \sum_\sigma \lambda_{\tau\sigma} = S$, hence $\operatorname{im} S \subseteq \operatorname{ST}^p(\mathcal{V})$.
If $\mathbf{z} \in \operatorname{ST}^p(\mathcal{V})$, then $S\mathbf{z} = \mathbf{z}$, so $\operatorname{im} S = \operatorname{ST}^p(\mathcal{V})$ and $S^2 = S$.

(ii) For $\tau \in \mathfrak{S}_p$, $\lambda_\tau A = p!^{-1} \sum_\sigma \operatorname{sign}(\sigma) \lambda_{\tau\sigma} = \operatorname{sign}(\tau) A$, hence $\operatorname{im} A \subseteq \operatorname{AT}^p(\mathcal{V})$.
If $\mathbf{z} \in \operatorname{AT}^p(\mathcal{V})$, then $A\mathbf{z} = p!^{-1} \sum_\sigma \operatorname{sign}(\sigma)^2 \mathbf{z} = \mathbf{z}$, so $\operatorname{im} A = \operatorname{AT}^p(\mathcal{V})$ and $A^2 = A$.

Structural Commentary: Averaging gives symmetry projections.
Key Insight: The factor $1/p!$ makes the averaging map fix the tensors it projects onto. □

6.2 Exterior Algebras

Definition 6.2.1. *Tensor, Symmetric, and Exterior Algebras*
Let $\dim \mathcal{V} < \infty$.

(i) The *tensor algebra* is $\mathcal{T}(\mathcal{V}) := \bigoplus_{p \geq 0} \mathcal{V}^{\otimes p}$, with $\mathcal{V}^{\otimes 0} := \mathbb{F}$ and product induced by tensor concatenation.
(ii) The *symmetric algebra* is $\operatorname{Sym}(\mathcal{V}) := \bigoplus_{p \geq 0} \operatorname{Sym}^p(\mathcal{V})$, with product induced by $\vee$.
(iii) The *exterior algebra* is $\bigwedge \mathcal{V} := \bigoplus_{0 \leq p \leq \dim \mathcal{V}} \bigwedge^p \mathcal{V}$, with product induced by $\wedge$.
(iv) Each is *graded*: degree p times degree q lies in degree $p + q$.

Corollary 6.2.2. *Determinant from the Exterior Universal Property*
Let $\dim \mathcal{V} = n$, $E = (\mathbf{e}_1, \dots, \mathbf{e}_n)$, $\Omega_E := \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n$, *and* $A_E(\mathbf{v}_1, \dots, \mathbf{v}_n) = ([\mathbf{v}_j]_i)_{i,j}$.

(i) There is a unique $\ell_E \in (\bigwedge^n \mathcal{V})'$ *with* $\ell_E(\Omega_E) = 1$*; hence* $d_E(\mathbf{v}_1, \dots, \mathbf{v}_n) := \ell_E(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n)$ *is the unique alternating form satisfying* $d_E(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$.

(ii) For all $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathcal{V}$, $d_E(\mathbf{v}_1, \dots, \mathbf{v}_n) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sign}(\sigma) \prod_{j=1}^n [\mathbf{v}_j]_{\sigma(j)} = \det A_E(\mathbf{v}_1, \dots, \mathbf{v}_n)$.

Proof.

(i) By Theorem 6.1.11, $\bigwedge^n \mathcal{V} = \mathbb{F}\Omega_E$, so $\ell_E(\Omega_E) = 1$ fixes a unique $\ell_E \in (\bigwedge^n \mathcal{V})'$; via Theorem 6.1.10, ℓ_E corresponds to $d_E(\mathbf{v}_1, \dots, \mathbf{v}_n) = \ell_E(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n)$, hence d_E is unique.

(ii) Let $\mathbf{v}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$, $a_{ij} = [\mathbf{v}_j]_i$. Repeated index tuples have zero wedge; distinct tuples are $(\sigma(1), \dots, \sigma(n))$, with wedge $\operatorname{sign}(\sigma)\Omega_E$. Hence $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sign}(\sigma) \prod_{j=1}^n a_{\sigma(j),j} \Omega_E$, and

$$d_E(\mathbf{v}_1, \dots, \mathbf{v}_n) = \ell_E(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_n) = \sum_{\sigma \in \mathfrak{S}_n} \operatorname{sign}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}$$
$$= \det(a_{ij}) = \det A_E(\mathbf{v}_1, \dots, \mathbf{v}_n).$$

Structural Commentary: Determinant is the basis-normalised alternating form.
Key Insight: Top exterior dimension one turns normalisation into determinant. □

Corollary 6.2.3. *Determinant Properties from Exterior Powers*
Let $A, B \in \operatorname{Mat}(n, n, \mathbb{F})$, *and set* $d(A) := d_E(A_{\bullet 1}, \dots, A_{\bullet n})$ *for* E *the standard basis of* $\mathbb{F}^n$.

(i) $d(A^t) = d(A)$.
(ii) $d(AB) = d(A)d(B)$.
(iii) $A \in \operatorname{GL}(n, \mathbb{F}) \iff d(A) \neq 0$.

Proof. Let $\omega = \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n$ and let T_A denote the linear map with matrix A.

(i) By Corollary 6.2.2, $d(A^t) = \sum_\sigma \operatorname{sign}(\sigma) \prod_{j=1}^n A_{j,\sigma(j)}$. With $\tau = \sigma^{-1}$, $\sum_\tau \operatorname{sign}(\tau) \prod_{i=1}^n A_{\tau(i),i} = d(A)$.

(ii) Since $(\bigwedge^n T_A)\omega = d(A)\omega$ and $T_{AB} = T_A \circ T_B$, applying $\bigwedge^n(T_A \circ T_B) = (\bigwedge^n T_A)(\bigwedge^n T_B)$ to ω gives $d(AB)\omega = d(A)d(B)\omega$, hence $d(AB) = d(A)d(B)$.

(iii) ($\Rightarrow$) If $A \in \mathrm{GL}(n, \mathbb{F})$, then $I = AA^{-1}$, so $1 = d(I) = d(A)d(A^{-1})$ by (ii); hence $d(A) \neq 0$.

(iii) ($\Leftarrow$) Let $C_j = A_{\bullet j}$. Suppose $\sum_{j=1}^n c_j C_j = 0$ and $c_k \neq 0$. Then $C_k = -c_k^{-1} \sum_{j \neq k} c_j C_j$, so multilinearity in the kth slot gives $d(A) = d(C_1, \ldots, C_k, \ldots, C_n) = -c_k^{-1} \sum_{j \neq k} c_j d(C_1, \ldots, C_{k-1}, C_j, C_{k+1}, \ldots, C_n) = 0$, since each summand has equal jth and kth entries.
Thus $d(A) \neq 0$ implies $C_1, \ldots, C_n$ are independent, hence $A \in \mathrm{GL}(n, \mathbb{F})$.

Structural Commentary: Determinant laws are top-exterior scalar laws.
Key Insight: Top exterior dimension one makes determinants multiplicative. □

6.3 Multilinear Algebra Revisited

Definition 6.3.1. *Double Dual Space*
The *double dual space* of $\mathcal{V}$ is $\mathcal{V}'' := (\mathcal{V}')'$. Define $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ by $(\Lambda_{\mathcal{V}}\mathbf{v})(\varphi) = \varphi(\mathbf{v}), \mathbf{v} \in \mathcal{V},\ \varphi \in \mathcal{V}'$.

Theorem 6.3.2. *Natural Map to the Double Dual*
The map $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ satisfies:

(i) *$\Lambda_{\mathcal{V}}$ is linear;*
(ii) *if $T \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ and $T'' = (T')'$, then $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}}T$;*
(iii) *if $\mathcal{V}$ is finite-dimensional, then $\Lambda_{\mathcal{V}}$ is an isomorphism.*

Proof.

(i) For fixed $\mathbf{v} \in \mathcal{V}$, the map $\varphi \mapsto \varphi(\mathbf{v})$ is linear on $\mathcal{V}'$, so $\Lambda_{\mathcal{V}}\mathbf{v} \in \mathcal{V}''$.
For $\mathbf{u}, \mathbf{v} \in \mathcal{V}$, $a \in \mathbb{F}$, and $\varphi \in \mathcal{V}'$, $(\Lambda_{\mathcal{V}}(a\mathbf{u} + \mathbf{v}))(\varphi) = \varphi(a\mathbf{u} + \mathbf{v}) = a(\Lambda_{\mathcal{V}}\mathbf{u})(\varphi) + (\Lambda_{\mathcal{V}}\mathbf{v})(\varphi)$, so $\Lambda_{\mathcal{V}}$ is linear.

(ii) If $\psi \in \mathcal{W}'$, then $(T''(\Lambda_{\mathcal{V}}\mathbf{v}))(\psi) = (\Lambda_{\mathcal{V}}\mathbf{v})(T'\psi) = (T'\psi)(\mathbf{v}) = \psi(T\mathbf{v}) = (\Lambda_{\mathcal{W}}(T\mathbf{v}))(\psi)$; $T''\Lambda_{\mathcal{V}} = \Lambda_{\mathcal{W}}T$.

(iii) Suppose $\mathcal{V}$ is finite-dimensional. If $\mathbf{v} \neq \mathbf{0}$, extend $(\mathbf{v})$ to a basis of $\mathcal{V}$ and let φ be the first functional of the corresponding dual basis. Then $\varphi(\mathbf{v}) = 1$, so $(\Lambda_{\mathcal{V}}\mathbf{v})(\varphi) = 1$ and $\Lambda_{\mathcal{V}}\mathbf{v} \neq 0$. Hence $\ker(\Lambda_{\mathcal{V}}) = \{\mathbf{0}\}$ and $\Lambda_{\mathcal{V}}$ is injective. Since $\dim(\mathcal{V}'') = \dim(\mathcal{V}') = \dim(\mathcal{V})$, Corollary 3.2.19 shows that $\Lambda_{\mathcal{V}}$ is an isomorphism.

Structural Commentary: The double-dual map sends a vector to its evaluation action on functionals. It is natural because it commutes with every linear map.
Key Insight: Finite-dimensional vectors are completely recovered by how all linear functionals evaluate them. □

Perspective 6.3.3. *Why the Double Dual Is Natural*
A finite-dimensional space $\mathcal{V}$ is isomorphic to $\mathcal{V}'$, but such an isomorphism usually requires choosing a basis. The map $\Lambda_{\mathcal{V}} : \mathcal{V} \to \mathcal{V}''$ requires no choice of basis.

7 Chain Complexes and Homology

7.1 Chain Complexes

7.2 Homology

7.3 Derived Functors Preview

Part VII
Commutative Algebra

1 Prime Ideals and Spectra

1.1 Prime Ideals

1.2 Spectra

2 Localization and Local Rings

2.1 Localization

2.2 Local Rings

3 Noetherian Rings and Modules

3.1 Noetherian Rings

3.2 Noetherian Modules

4 Integral Dependence and Dimension

4.1 Integral Dependence

4.2 Dimension Theory

5 Regularity and Singularities

5.1 Regularity

5.2 Singularities Preview

Part VIII
Algebraic Geometry

1 Affine Algebraic Sets

1.1 Affine Algebraic Sets

1.2 Ideals and Coordinate Rings

2 Morphisms and Coordinate Rings

2.1 Regular Maps

2.2 Coordinate Ring Functoriality

3 Projective Geometry

3.1 Projective Space

3.2 Projective Varieties

4 Schemes

4.1 Affine Schemes

4.2 Gluing and Scheme Morphisms

5 Sheaves and Cohomology

5.1 Sheaves

5.2 Cohomology Preview

6 Smoothness, Tangent Spaces, and Dimension

6.1 Smoothness

6.2 Tangent Spaces

6.3 Dimension

7 Divisors, Line Bundles, and Curves

7.1 Divisors

7.2 Line Bundles

7.3 Curves

Part IX
Real and Topological Analysis

1 Number Systems and Completeness

1.1 Ordered Fields, Suprema, and Real Structure

Number systems supply the ordered and algebraic background for Analysis: $\mathbb{Q}$ has gaps, $\mathbb{R}$ repairs them by completeness, and $\mathbb{C}$ keeps field algebra without order.

Proposition 1.1.1. *The Rational Gap at $\sqrt{2}$*
There is no rational p such that $p^2 = 2$.

Proof. Let $p = m/n$. Then $m^2 = 2n^2$, so m is even; $m = 2k$. Hence $n^2 = 2k^2$, so n is even, a contradiction.
Structural Commentary: This exhibits a gap in $\mathbb{Q}$.
Key Insight: $\sqrt{2}$ is not rational. □

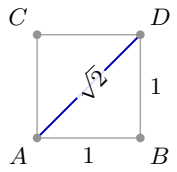


Figure 116: The diagonal of a unit square has length $\sqrt{2}$, so the rational gap is geometric as well as algebraic.

Example 1.1.2. *No Largest Lower Side and No Smallest Upper Side*
Let $A := \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}$, and $B := \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}$.
For every $p > 0$ define $q := p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2}$. Then $q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2}$. Hence if $p \in A$, then $q \in A$ and $q > p$; if $p \in B$, then $q \in B$ and $0 < q < p$. Thus A has no largest element in $\mathbb{Q}$, and B has no smallest element in $\mathbb{Q}$.

Remark 1.1.3. *Density Does Not Mean Completeness*
Between two rationals $r < s$ there is another rational, namely $(r + s)/2$. Nevertheless, $\mathbb{Q}$ is not complete.

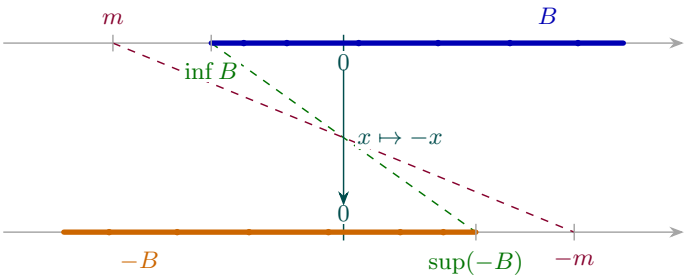


Figure 117: Reflecting B through 0 turns lower bounds of B into upper bounds of $-B$, so $\inf B = -\sup(-B)$.

Corollary 1.1.4. *Infimum by Reflection*
If $B \subseteq \mathbb{R}$ is nonempty and bounded below, then $\inf B = -\sup(-B)$ and $-B := \{-b : b \in B\}$.

Proof. Since B is nonempty and any lower bound m of B gives an upper bound $-m$ of $-B$, let $s = \sup(-B)$. For $b \in B$, $-b \le s$, so $-s \le b$; hence $-s$ is a lower bound of B. If t is any lower bound of B, then $-t$ is an upper bound of $-B$, so $s \le -t$ and $t \le -s$. Thus $-s = \inf B$.
Structural Commentary: Reflection swaps lower bounds of B with upper bounds of $-B$.
Key Insight: The infimum of B is the negative supremum of its reflection. □

Definition 1.1.5. *Real Absolute Value and Distance*
For $x \in \mathbb{R}$, define $|x| := \begin{cases} x, & x \ge 0, \\ -x, & x < 0. \end{cases}$
For $a, b \in \mathbb{R}$, the distance between a and b on the real line is $|a - b|$.

Proposition 1.1.6. *Real Absolute-Value Laws*
For $x, y, a, b, c \in \mathbb{R}$,

(i) $|x| \geq 0$, $|x| = 0$ *iff* $x = 0$, *and* $-|x| \leq x \leq |x|$;
(ii) $|xy| = |x|\,|y|$;
(iii) $|x + y| \leq |x| + |y|$;
(iv) $|a - b| \leq |a - c| + |c - b|$.

Proof.

(i) Immediate from Definition 1.1.5.
(ii) Check the four sign cases for x and y.
(iii) Add $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$, then $-(|x| + |y|) \leq x + y \leq |x| + |y|$. Since $|x| + |y| \geq 0$, this gives $|x + y| \leq |x| + |y|$.
(iv) Apply (iii) to $a - b = (a - c) + (c - b)$.

Structural Commentary: Absolute value turns order bounds into distance estimates.
Key Insight: The triangle inequality on $\mathbb{R}$ is the inequality $|(a - c) + (c - b)| \leq |a - c| + |c - b|$. □

Perspective 1.1.7. *Completeness Becomes a Theorem*
Dedekind cuts build completeness from $\mathbb{Q}$: real numbers are rational cuts, rational cuts embed $\mathbb{Q}$, and least upper bounds arise by taking unions of cuts.

Definition 1.1.8. *Dedekind Cuts and Their Order*
A *cut* is a set $\alpha \subseteq \mathbb{Q}$ such that

(i) α is nonempty and $\alpha \neq \mathbb{Q}$;
(ii) if $p \in \alpha$, $q \in \mathbb{Q}$, and $q < p$, then $q \in \alpha$;
(iii) if $p \in \alpha$, then some $r \in \alpha$ satisfies $p < r$.

Let $\mathscr{D}$ be the set of all cuts. For $\alpha, \beta \in \mathscr{D}$, define $\alpha < \beta$ to mean $\alpha \subsetneq \beta$.
Rational letters $p, q, r, \ldots$ denote elements of $\mathbb{Q}$; Greek letters $\alpha, \beta, \gamma, \ldots$ denote cuts.
A cut is the rational left side of a real number: a proper initial segment with no last point.

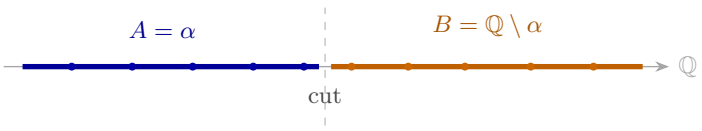


Figure 118: A Dedekind cut splits $\mathbb{Q}$ into α and $\mathbb{Q} \setminus \alpha$, with no last rational in α.

Remark 1.1.9. *Left Cuts Versus Two-Sided Cuts*
Two-sided notation writes a cut as $A|B$, with $A = \alpha$ and $B = \mathbb{Q} \setminus \alpha$; Definition 1.1.8 records only A.
Thus "no largest element" is the same left-side condition.

Definition 1.1.10. *Arithmetic of Cuts*
For cuts α, β, define $\alpha + \beta := \{r + s : r \in \alpha,\ s \in \beta\}$ and $0^* := \{q \in \mathbb{Q} : q < 0\}$.
Let candidate additive inverse of α be $-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q},\ r > 0,\ -p - r \notin \alpha\}$. Let $\mathscr{D}^+ := \{\alpha \in \mathscr{D} : \alpha > 0^*\}$.
For $\alpha, \beta \in \mathscr{D}^+$, define $\alpha\beta := \{p \in \mathbb{Q} : \exists r \in \alpha,\ \exists s \in \beta,\ r > 0,\ s > 0,\ p \leq rs\}$ and $1^* := \{q \in \mathbb{Q} : q < 1\}$.
Extend multiplication to all cuts by setting $\alpha 0^* = 0^*\alpha = 0^*$ and, for nonzero signs,

$$\alpha\beta := \begin{cases} (-\alpha)(-\beta), & \alpha < 0^*,\ \beta < 0^*, \\ -\big((-\alpha)\beta\big), & \alpha < 0^*,\ \beta > 0^*, \\ -\big(\alpha(-\beta)\big), & \alpha > 0^*,\ \beta < 0^*. \end{cases}$$

Motivation: arithmetic uses rational approximations inside cuts; signs come after positive products.

Definition 1.1.11. *Rational Cuts*
For $r \in \mathbb{Q}$, define the *rational cut*, $r^* := \{p \in \mathbb{Q} : p < r\}$.
The map $r \mapsto r^*$ is intended to identify $\mathbb{Q}$ with a subfield of the cut field.

Theorem 1.1.12. *Existence of the Real Field*
There exists an ordered field $\mathbb{R}$ with the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Proof. Construct $\mathbb{R}$ from $\mathbb{Q}$ using Definitions 1.1.8–1.1.11.

(i) *Order.* The relation $\alpha < \beta$ is transitive, and at most one of $\alpha < \beta$, $\alpha = \beta$, and $\beta < \alpha$ can hold. For trichotomy, suppose $\alpha \not\subseteq \beta$ and $\alpha \neq \beta$. Choose $p \in \alpha$ with $p \notin \beta$. If $q \in \beta$, then $q < p$, since $p \notin \beta$ and β

is downward closed. Hence $q \in \alpha$, so $\beta \subseteq \alpha$. Since $\alpha \neq \beta$, $\beta < \alpha$. Thus $\mathscr{D}$ is ordered.
Commentary: One missing rational decides containment.

(ii) *Least Upper Bounds.* Let $A \subseteq \mathscr{D}$ be nonempty and bounded above by a cut β. Define $\gamma := \bigcup_{\alpha \in A} \alpha$.
Since some $\alpha_0 \in A$ is nonempty, γ is nonempty. Since every $\alpha \in A$ lies in β, $\gamma \subseteq \beta$, so $\gamma \neq \mathbb{Q}$. If $p \in \gamma$ and $q < p$, then $p \in \alpha_1$ for some $\alpha_1 \in A$, hence $q \in \alpha_1 \subseteq \gamma$. If $p \in \gamma$, choose $\alpha_1 \in A$ with $p \in \alpha_1$ and then choose $r \in \alpha_1$ with $p < r$; then $r \in \gamma$. Thus γ is a cut.
Every $\alpha \in A$ satisfies $\alpha \leq \gamma$. If $\delta < \gamma$, choose $s \in \gamma \setminus \delta$. Then $s \in \alpha$ for some $\alpha \in A$, so $\delta < \alpha$ and δ is not an upper bound of A. Hence $\gamma = \sup A$.
Motivation: Supremum is union.
Commentary: Bounded unions stay proper and give the smallest upper bound.

(iii) *Addition.* The set $\alpha + \beta$ is a cut. It is nonempty by choosing one element from each summand.
If $r' \notin \alpha$ and $s' \notin \beta$, then every $r \in \alpha$ and $s \in \beta$ satisfies $r < r'$ and $s < s'$, so $r + s < r' + s'$; hence $r' + s' \notin \alpha + \beta$, and $\alpha + \beta$ is proper. If $p = r + s \in \alpha + \beta$ and $q < p$, then $q = r + (s + q - p)$ with $s+q-p < s$, so $s+q-p \in \beta$ and $q \in \alpha+\beta$. If $p = r+s$, choose $r_1 \in \alpha$ with $r < r_1$; then $p < r_1+s \in \alpha+\beta$. Thus $\alpha + \beta$ is a cut. Commutativity and associativity follow by rewriting representatives in $\mathbb{Q}$.
The cut 0^* is the additive identity: $\alpha + 0^* \subseteq \alpha$ by downward closure, while if $p \in \alpha$, choose $r \in \alpha$ with $p < r$ and write $p = r + (p - r)$ with $p - r < 0$.
The inverse $-\alpha$ is a cut, and $\alpha + (-\alpha) = 0^*$. The inclusion $\alpha + (-\alpha) \subseteq 0^*$ follows because $s \in -\alpha$ forces $-s \notin \alpha$, so any $r \in \alpha$ satisfies $r < -s$. Conversely, if $v < 0$, put $w = -v/2 > 0$ and choose an integer n with $nw \in \alpha$ but $(n+1)w \notin \alpha$; then $p = -(n+2)w$ lies in $-\alpha$ and $v = nw + p \in \alpha + (-\alpha)$.
Commentary: Addition inherits its laws from $\mathbb{Q}$ once closure and inverses are checked.

(iv) *Order Compatibility with Addition.* The additive laws make $\mathscr{D}$ an additive group.
If $\beta < \gamma$, then $\alpha+\beta \subseteq \alpha+\gamma$. Equality would imply $\beta = \gamma$ after adding $-\alpha$ to both sides, so $\alpha+\beta < \alpha+\gamma$. Consequently, $\alpha > 0^*$ if and only if $-\alpha < 0^*$.
Commentary: Cancellation turns inclusion into strict inclusion.

(v) *Positive Multiplication.* If $\alpha, \beta \in \mathscr{D}^+$, then $\alpha\beta$ is a cut. Since each positive cut contains some positive rational, $\alpha\beta$ is nonempty. Choose positive $r' \notin \alpha$ and $s' \notin \beta$; then every positive $r \in \alpha$ and $s \in \beta$ satisfies $rs < r's'$, so $r's' \notin \alpha\beta$, and the product is proper. Downward closure is built into the condition $p \leq rs$. If $p \leq rs$ with $r, s > 0$, then either choose q with $p < q < rs$, or, when needed, choose $r_1 \in \alpha$ with $r < r_1$ and take q with $p < q \leq r_1 s$; hence the product has no largest element.
Commutativity follows from $rs = sr$. Associativity and distributivity are proved by unfolding membership and using the rational identities $(rs)t = r(st)$ and $r(s+t) = rs+rt$, with a positive rational slack inserted whenever a representative must be made strictly larger inside a cut. The identity is 1^*: if $p \leq rs$ with $r \in \alpha$, $s < 1$, and $r, s > 0$, then $p < r$, so $p \in \alpha$; conversely, if $p \in \alpha$, choose $r \in \alpha$ with $p < r$ and then choose $s \in \mathbb{Q}$ with $0 < s < 1$ and $p \leq rs$. In particular, if $\alpha > 0^*$ and $\beta > 0^*$, then $\alpha\beta > 0^*$.
Commentary: Positivity avoids sign reversals.

(vi) *Signed Multiplication.* The definitions extend multiplication to all cuts by reducing to positive factors and additive inverses. Since $\delta = -(-\delta)$ and the sign rules for addition have already been proved, commutativity, associativity, and inverses reduce to the positive case.
For distributivity, split into signs of $\alpha, \beta, \gamma, \beta + \gamma$. For example, if $\alpha > 0^*$, $\beta < 0^*$, and $\beta + \gamma > 0^*$, then $\gamma = (\beta + \gamma) + (-\beta)$, so positive distributivity gives $\alpha\gamma = \alpha(\beta + \gamma) + \alpha(-\beta)$. Because $\alpha(-\beta) = -(\alpha\beta)$, adding $\alpha\beta$ to both sides gives $\alpha\beta + \alpha\gamma = \alpha(\beta + \gamma)$. The other sign cases are identical reductions after replacing negative terms by their additive inverses.
Commentary: Distributivity reduces to positive cases plus signs.

(vii) *Embedding $\mathbb{Q}$.* Each r^* is a cut. For $r, s \in \mathbb{Q}$, $r^* + s^* = (r + s)^*$, $r^*s^* = (rs)^*$, and $r^* < s^* \iff r < s$.
For addition, if $p = u+v$ with $u < r$ and $v < s$, then $p < r+s$. Conversely, if $p < r+s$, choose $t > 0$ with $2t = r + s - p$ and put $r' = r - t$, $s' = s - t$; then $r' < r$, $s' < s$, and $p = r' + s'$. For positive products, if $p \leq uv$ with $0 < u < r$ and $0 < v < s$, then $p < rs$. Conversely, if $0 < p < rs$, choose positive $u < r$ and $v < s$ close enough to r and s that $p \leq uv$; if $p \leq 0$, use any positive $u < r$, $v < s$. The signed product cases follow from the definitions by additive inverses. The order equivalence follows because $r < s$ puts r in s^* but not in r^*, while $r^* < s^*$ gives some $p < s$ with $p \notin r^*$, hence $r \leq p < s$.
Commentary: The map $r \mapsto r^*$ preserves arithmetic and order.

Identify $\mathbb{Q}$ with the subfield $\mathbb{Q}^* := \{r^* : r \in \mathbb{Q}\} \subseteq \mathscr{D}$, and write $\mathbb{R}$ for $\mathscr{D}$. Items (i), (iii)–(vi) give an ordered field; item (ii) gives the least-upper-bound property; item (vii) embeds $\mathbb{Q}$ as a subfield.
Any two ordered fields with the least-upper-bound property are isomorphic, so this construction characterizes the real field up to ordered-field structure. □

Perspective 1.1.13. *Number Systems from Representatives*
Build by representatives, then pass to classes:

$$\mathbb{Z}: \ (a,b) \mapsto [(a,b)], \qquad \mathbb{Q}: \ (a,b),\ b \neq 0 \mapsto [(a,b)], \qquad \mathbb{R}: \ \text{cuts or Cauchy data} \mapsto [\cdot].$$

The pattern is: choose $\sim$, prove $+, \cdot, <, \ldots$ are well defined on $X/\sim$, then use the classes as the number system; compare Definition 4.1.1 and Proposition 2.2.6.
Key Insight: Representatives are scaffolding; the quotient laws are the interface.

Perspective 1.1.14. *Cauchy Construction as a Parallel Model*
Cuts build $\mathbb{R}$ here; Cauchy representatives give parallel model:

$$(a_n) \sim (b_n) \iff |a_n - b_n| \to 0, \qquad \mathbb{R} \simeq \{\text{rational Cauchy sequences}\}/\sim$$

This is the representative pattern from Perspective 1.1.13; compare the completion skeleton in Perspective 4.1.22. Definition 4.1.1 is the first well-definedness checkpoint.

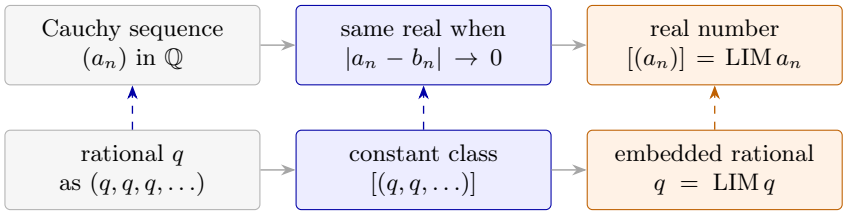


Figure 119: Real numbers as equivalence classes of rational Cauchy representatives.

Theorem 1.1.15. *Archimedean Property and Density of $\mathbb{Q}$*

(i) If $x, y \in \mathbb{R}$ and $x > 0$, then there is a positive integer n such that $nx > y$.
(ii) If $x, y \in \mathbb{R}$ and $x < y$, then there exists $p \in \mathbb{Q}$ such that $x < p < y$.

Proof.

(i) Let $A = \{nx : n \in \mathbb{N}\}$. If no nx exceeded y, then A would be bounded above. By Theorem 1.1.12, let $\alpha = \sup A$. Since $x > 0$, $\alpha - x < \alpha$, so $\alpha - x$ is not an upper bound.
Hence $\alpha - x < mx$ for some $m \in \mathbb{N}$, and $\alpha < (m+1)x \in A$, a contradiction.

(ii) Since $y - x > 0$, part (i) gives $n \in \mathbb{N}$ with $n(y-x) > 1$. Apply part (i) again to get $m_1, m_2 \in \mathbb{N}$ with $m_1 > nx$ and $m_2 > -nx$. Then some $m \in \mathbb{Z}$ satisfies $m - 1 \le nx < m$.
Thus $nx < m \le nx + 1 < ny$, so $x < m/n < y$ and $m/n \in \mathbb{Q}$.

Structural Commentary: Completeness makes $\{nx : n \in \mathbb{N}\}$ unbounded.
Key Insight: Archimedean unboundedness gives rational density. □

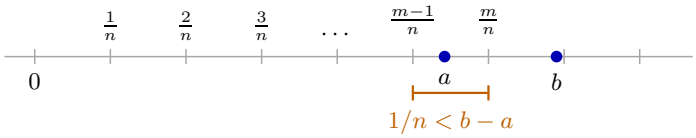


Figure 120: Choose n so the spacing $1/n$ is smaller than $b - a$; then one grid point m/n falls inside (a, b).

Theorem 1.1.16. *Existence and Uniqueness of Positive nth Roots*
For every $x \in \mathbb{R}$ with $x > 0$ and every positive integer n, $\exists$ a unique positive real number y such that $y^n = x$.

Proof.

(i) *Uniqueness.* If $0 < y_1 < y_2$, then $y_1^n < y_2^n$.

(ii) *Existence.* Let $E := \{t \in \mathbb{R} : t > 0,\ t^n < x\}$. $E \neq \varnothing$: for $t = x/(1+x)$, $0 < t < 1$ and $t^n \le t < x$.
Also E is bounded above by $1 + x$, since $t > 1 + x$ implies $t^n \ge t > x$. By Theorem 1.1.12, let $y = \sup E$.
For $0 < a < b$, $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1}) < (b-a)nb^{n-1}$.
If $y^n < x$, choose h with $0 < h < 1$ and $h < \frac{x - y^n}{n(y+1)^{n-1}}$.
Then $(y+h)^n - y^n < hn(y+h)^{n-1} < hn(y+1)^{n-1} < x - y^n$.
Thus $y + h \in E$, contradicting that y is an upper bound. If $y^n > x$, set $k := \frac{y^n - x}{ny^{n-1}}$. Then $0 < k < y$.
For $t \ge y - k$, $y^n - t^n \le y^n - (y-k)^n < kny^{n-1} = y^n - x$, so $t^n > x$ and $t \notin E$.
Hence $y - k$ is an upper bound for E, contradicting $y = \sup E$. Therefore $y^n = x$.

Structural Commentary: The root is $\sup\{t > 0 : t^n < x\}$.
Key Insight: Completeness supplies the missing root. □

Corollary 1.1.17. *Density of Rational and Irrational Points*
If $x < y$ in $\mathbb{R}$, then there exist $r \in \mathbb{Q}$ and $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $x < r < y$ and $x < s < y$.

Proof. The rational point r is supplied by Theorem 1.1.15. Let $\eta = \sqrt{2}$.
Choose $n \in \mathbb{N}$ with $0 < \eta/n < y - x$, and choose $q \in \mathbb{Q}$ with $x - \eta/n < q < y - \eta/n$.
Then $s := q + \eta/n$ lies in (x, y) and is irrational, since $s \in \mathbb{Q}$ would imply $\eta = n(s - q) \in \mathbb{Q}$.
Structural Commentary: Irrational density is rational density shifted by a small irrational amount.
Key Insight: Once one irrational is known, every interval contains irrational translates of rational points. □

Corollary 1.1.18. *Product Rule for Positive Roots*
If $a, b > 0$ and n is a positive integer, then $(ab)^{1/n} = a^{1/n} b^{1/n}$.

Proof. Let $\alpha = a^{1/n}$ and $\beta = b^{1/n}$. Then $ab = \alpha^n \beta^n = (\alpha\beta)^n$ by commutativity of multiplication.
The uniqueness part of Theorem 1.1.16 gives the identity.
Structural Commentary: Use uniqueness after checking nth powers.
Key Insight: Root identities are uniqueness arguments. □

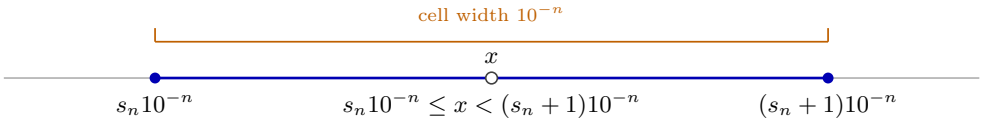


Figure 121: At precision 10^{-n}, the truncation $s_n 10^{-n}$ traps x in a cell of width 10^{-n}.

Proposition 1.1.19. *Decimal Truncations Squeeze a Real*
Let $x \ge 0$. For $n \ge 0$, set $s_n := \max\{k \in \mathbb{Z}_{\ge 0} : k10^{-n} \le x\}$.
Then $s_n 10^{-n} \le x < (s_n + 1)10^{-n}$; and $0 \le x - s_n 10^{-n} < 10^{-n}$; so $s_n 10^{-n} \to x$.
Moreover $s_{n+1} = 10s_n + a_{n+1}$ for some $a_{n+1} \in \{0, \dots, 9\}$.

Proof. The maximum exists by the Archimedean property, Theorem 1.1.15, and well-ordering, Principle 2.6.2.
Maximality gives the displayed inequalities. The squeeze and Theorem 2.2.9 give $s_n 10^{-n} \to x$.
At levels n and $n + 1$, $10s_n \le s_{n+1} < 10s_n + 10$, so $a_{n+1} := s_{n+1} - 10s_n \in \{0, \dots, 9\}$.
Structural Commentary: Decimal existence is an Archimedean grid argument followed by a squeeze.
Key Insight: Refining the mesh by a factor of 10 adds exactly one digit. □

Definition 1.1.20. *Extended Real Number System*
The *extended real number system* is $\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$.
The order on $\mathbb{R}$ is preserved, and $-\infty < x < +\infty$, $(x \in \mathbb{R})$.
Every nonempty subset of $\overline{\mathbb{R}}$ has least upper bound; if $E \subseteq \mathbb{R}$ is nonempty and unbounded above in $\mathbb{R}$, then $\sup E = +\infty$. Same convention applies to greatest lower bounds.

Remark 1.1.21. *Extended Arithmetic Conventions*
$\overline{\mathbb{R}}$ is not a field. For real x, $x + \infty = +\infty$, $x - \infty = -\infty$, and $\frac{x}{+\infty} = \frac{x}{-\infty} = 0$.
If $x > 0$, then $x(+\infty) = +\infty$ and $x(-\infty) = -\infty$; if $x < 0$, the signs reverse.
Real numbers are called *finite* when one wants to distinguish them from $\pm\infty$.

1.2 Complex Numbers and Absolute Value

Definition 1.2.1. *Complex Numbers*
A *complex number* is an ordered pair (a, b) of real numbers. For $x = (a, b)$ and $y = (c, d)$, define equality componentwise and set $x + y := (a + c, b + d)$ and $xy := (ac - bd, ad + bc)$

Theorem 1.2.2. *The Complex Numbers Form a Field*
With the operations in Definition 1.2.1, the set of complex numbers is a field.
Its additive and multiplicative identities are $(0, 0)$ and $(1, 0)$.

Proof. The field axioms are checked directly from the two coordinate formulas and the field structure of $\mathbb{R}$.
If $x = (a, b) \neq (0, 0)$, then $a^2 + b^2 > 0$ by Proposition 1.3.4, and $x^{-1} = \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$ satisfies $xx^{-1} = (1, 0)$.
Structural Commentary: Inverses use conjugate-over-square.
Key Insight: Positivity of $a^2 + b^2$ supplies the denominator for every nonzero complex number. □

Proposition 1.2.3. *Real Numbers Inside Complex Numbers*
For all real a, b, $(a,0)+(b,0)=(a+b,0)$, and $(a,0)(b,0)=(ab,0)$.
Thus complex numbers of form $(a,0)$ have same arithmetic as real numbers, and identify $(a,0)$ with a.
The complex unit is then $i := (0,1)$.

Proposition 1.2.4. *Coordinate Form and i^2*
We have $i^2=-1$. Moreover, if $a,b\in\mathbb{R}$, then $(a,b)=a+bi$.

Proof. First, $i^2=(0,1)(0,1)=(-1,0)=-1$. Also, $a+bi=(a,0)+(b,0)(0,1)=(a,0)+(0,b)=(a,b)$.
Structural Commentary: The ordered-pair construction recovers the usual notation after $i=(0,1)$ is named.
Key Insight: The symbol i is not assumed; it is produced by the pair model. □

Definition 1.2.5. *Conjugate, Real Part, and Imaginary Part*
If $z=a+bi$ with $a,b\in\mathbb{R}$, then the *complex conjugate* of z is $\overline{z} := a-bi$.
The numbers a and b are the *real part* and *imaginary part*, written $a=\operatorname{Re}(z)$ and $b=\operatorname{Im}(z)$.

Proposition 1.2.6. *Basic Conjugation Identities*
If $z,w\in\mathbb{C}$, then

(i) $\overline{z+w}=\overline{z}+\overline{w}$;
(ii) $\overline{zw}=\overline{z}\,\overline{w}$;
(iii) $z+\overline{z}=2\operatorname{Re}(z)$ *and* $z-\overline{z}=2i\operatorname{Im}(z)$;
(iv) $z\overline{z}$ *is real and positive unless* $z=0$.

Proof. Write $z=a+bi$ and $w=c+di$.

(i) $\overline{z+w}=(a+c)-(b+d)i=\overline{z}+\overline{w}$.
(ii) $\overline{zw}=(ac-bd)-(ad+bc)i=(a-bi)(c-di)=\overline{z}\,\overline{w}$.
(iii) $z+\overline{z}=2a=2\operatorname{Re}(z)$ and $z-\overline{z}=2bi=2i\operatorname{Im}(z)$.
(iv) $z\overline{z}=a^2+b^2$, positive unless $a=b=0$, i.e. unless $z=0$.

Structural Commentary: Conjugation fixes Re and negates Im.
Key Insight: $z\overline{z}=a^2+b^2\geq 0$. □

Definition 1.2.7. *Absolute Value of a Complex Number*
For $z\in\mathbb{C}$, define $|z| := (z\overline{z})^{1/2}$, the nonnegative square root supplied by Theorem 1.1.16 and Proposition 1.2.6.

Theorem 1.2.8. *Absolute Value Identities*
For $z,w\in\mathbb{C}$,

(i) $|z|>0$ *unless* $z=0$, *and* $|0|=0$;
(ii) $|\overline{z}|=|z|$;
(iii) $|zw|=|z|\,|w|$;
(iv) $|\operatorname{Re}(z)|\leq|z|$;
(v) $|z+w|\leq|z|+|w|$.

Proof.

(i) If $z=a+bi$, then $|z|=(a^2+b^2)^{1/2}$; this is positive unless $a=b=0$, and $|0|=0$.
(ii) $|\overline{z}|^2=\overline{z}z=z\overline{z}=|z|^2$, so nonnegative square roots give $|\overline{z}|=|z|$.
(iii) $|zw|^2=zw\overline{zw}=z\overline{z}w\overline{w}=|z|^2\,|w|^2$; take nonnegative square roots.
(iv) If $z=a+bi$, then $|\operatorname{Re}(z)|=|a|\leq(a^2+b^2)^{1/2}=|z|$.
(v) $z\overline{w}$ and $\overline{z}w$ are conjugates, $|z+w|^2=|z|^2+2\operatorname{Re}(z\overline{w})+|w|^2\leq|z|^2+2\,|z\overline{w}|+|w|^2=(|z|+|w|)^2$, using (iii) and (iv). Take nonnegative square roots.

Structural Commentary: The complex absolute value behaves like length because $z\overline{z}$ is a squared length.
Key Insight: Triangle inequality is a real-part estimate plus multiplicativity. □

Theorem 1.2.9. *Finite Schwarz Inequality for Complex Scalars*
If $a_1,\dots,a_n$ and $b_1,\dots,b_n$ are complex numbers, then $\left|\sum_{j=1}^{n} a_j\overline{b_j}\right|^2\leq\left(\sum_{j=1}^{n}|a_j|^2\right)\left(\sum_{j=1}^{n}|b_j|^2\right)$.

Proof. This is Theorem 1.1.13 in $\mathbb{C}^n$ with the standard inner product.
Equivalently, set $A=\sum|a_j|^2$, $B=\sum|b_j|^2$, and $C=\sum a_j\overline{b_j}$; when $B>0$, $\sum|Ba_j-Cb_j|^2=B(AB-|C|^2)\geq 0$.
Structural Commentary: This scalar inequality is the coordinate model for geometric Cauchy–Schwarz inequality.
Key Insight: A nonnegative square sum controls the mixed term. □

2 Real Sequences and Completeness Methods

2.1 Sequences, Cauchy Criteria, and Completeness

Sequences are the first testing ground for limits, order, and completeness. The main tools are subsequences, Cauchy criteria, monotone convergence, limsup/liminf, and Bolzano–Weierstrass.

Definition 2.1.1. *Convergent Real and Complex Sequences*
A real or complex sequence (s_n) *converges* to s if, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $n \geq N$ implies $|s_n - s| < \varepsilon$.
Notation is $s_n \to s$ or $\lim_{n\to\infty} s_n = s$. If no such s exists, the sequence *diverges*.

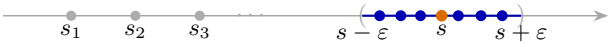


Figure 122: For all $n \geq N$, the terms s_n stay inside $(s-\varepsilon, s+\varepsilon)$.

Definition 2.1.2. *Range and Boundedness of a Real Sequence*
The *range* of (s_n) is set $\{s_n : n \in \mathbb{N}\}$. Real sequence is *bounded* if its range is bounded above and below in $\mathbb{R}$.

Theorem 2.1.3. *Limit Algebra for Complex Sequences*
Suppose (s_n) and (t_n) are complex sequences with $s_n \to s$ and $t_n \to t$. Then, for every $c \in \mathbb{C}$,

(i) $s_n + t_n \to s + t$;
(ii) $cs_n \to cs$ *and* $c + s_n \to c + s$;
(iii) $s_n t_n \to st$;
(iv) if $s \neq 0$ *and* $s_n \neq 0$ *for all* n, *then* $1/s_n \to 1/s$.

Proof.

(i) Given $\varepsilon > 0$, choose N such that $|s_n - s| < \varepsilon/2$, $|t_n - t| < \varepsilon/2$ for $n \geq N$.
Then $|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \varepsilon$.
(ii) Scalar multiplication uses $|cs_n - cs| = |c|\,|s_n - s|$; translation is (i) with constant sequence c.
(iii) Since $s_n \to s$, choose $N \geq 2$ with $|s_n - s| < 1$ for $n \geq N$ and set $M := \max\{|s_1|, \ldots, |s_{N-1}|, |s| + 1\}$.
Then $|s_n t_n - st| \leq |s_n|\,|t_n - t| + |t|\,|s_n - s| \leq M\,|t_n - t| + |t|\,|s_n - s| \to 0$.
(iv) If $s \neq 0$, then for all sufficiently large n, $|s_n| > |s|/2$. Hence $\left|\frac{1}{s_n} - \frac{1}{s}\right| = \frac{|s_n - s|}{|s_n||s|} \leq \frac{2}{|s|^2}\,|s_n - s| \to 0$.

Structural Commentary: Boundedness controls products; nonzero limits control reciprocals.
Key Insight: Product and reciprocal rules are estimates after the easy sum rule. □

Theorem 2.1.4. *Coordinatewise Convergence in $\mathbb{R}^k$*
Let $\mathbf{x}_n = (\alpha_{j,n})_{j=1}^k$ and $\mathbf{x} = (\alpha_j)_{j=1}^k$ in $\mathbb{R}^k$. Then $\mathbf{x}_n \to \mathbf{x} \iff \alpha_{j,n} \to \alpha_j$ $(1 \leq j \leq k)$.
Moreover, if $\mathbf{x}_n \to \mathbf{x}$, $\mathbf{y}_n \to \mathbf{y}$, and $\beta_n \to \beta$ in $\mathbb{R}$, then $\mathbf{x}_n + \mathbf{y}_n \to \mathbf{x} + \mathbf{y}$; $\mathbf{x}_n \cdot \mathbf{y}_n \to \mathbf{x} \cdot \mathbf{y}$; $\beta_n \mathbf{x}_n \to \beta \mathbf{x}$.

Proof.

($\Rightarrow$) Since $|\alpha_{j,n} - \alpha_j| \leq \|\mathbf{x}_n - \mathbf{x}\|_2$ for each j, each coordinate converges.
($\Leftarrow$) Given $\varepsilon > 0$, choose N so that $|\alpha_{j,n} - \alpha_j| < \varepsilon/\sqrt{k}$ for all j and $n \geq N$. Then $\|\mathbf{x}_n - \mathbf{x}\|_2 < \varepsilon$.

The algebraic conclusions follow coordinatewise from Theorem 2.1.3.
Structural Commentary: Euclidean convergence is exactly simultaneous coordinate convergence.
Key Insight: The norm packages finitely many coordinate estimates into one estimate. □

Remark 2.1.5. *Coordinate Metrics Give the Same Limits*
For $k \in \mathbb{N}$ and $x, y \in \mathbb{R}^k$, set $d_1(x,y) = \sum_{j=1}^k |x_j - y_j|$, $d_2(x,y) = \left(\sum_{j=1}^k |x_j - y_j|^2\right)^{1/2}$, $d_\infty(x,y) = \max_{1\leq j\leq k} |x_j - y_j|$.
Since $d_\infty \leq d_2 \leq d_1 \leq k d_\infty$, these have same convergent sequences, equivalently same coordinatewise criterion.

Definition 2.1.6. *Subsequences and Subsequential Limits*
Let (p_n) be a sequence. If $n_1 < n_2 < n_3 < \cdots$ is a strictly increasing sequence of positive integers, then (p_{n_k}) is a *subsequence* of (p_n). If $p_{n_k} \to p$, then p is a *subsequential limit* of (p_n).

Remark 2.1.7. *Subsequences as Selection Maps*
Equivalently, a subsequence is the composite of the original sequence, viewed as a function on $\mathbb{N}$ as in Definition 6.1.4, with a strictly increasing map $\sigma : \mathbb{N} \to \mathbb{N}$. In this notation the subsequence is $k \mapsto p_{\sigma(k)}$.
Key Insight: A subsequence keeps the original order but is allowed to skip terms.

Remark 2.1.8. *Subsequences of Convergent Sequences*
If $p_n \to p$, then every subsequence of (p_n) also converges to p. Conversely, if every subsequence of (p_n) converges to p, then (p_n) itself converges to p, since (p_n) is one of its subsequences.

Definition 2.1.9. *Monotone Real Sequence*
A real sequence (s_n) is *monotonically increasing* if $s_n \leq s_{n+1}$ for all n, and *monotonically decreasing* if $s_n \geq s_{n+1}$ for all n. A *monotonic* sequence is either increasing or decreasing.

Definition 2.1.10. *Dominant Index*
For a real sequence (s_n), an index n is *dominant* if $s_n \geq s_m$ for every $m > n$.

Theorem 2.1.11. *Monotone Subsequence Theorem*
Every real sequence has a monotone subsequence.

Proof. If the dominant indices $n_1 < n_2 < \cdots$ are infinite, then $s_{n_1} \geq s_{n_2} \geq \cdots$, giving a decreasing subsequence. Otherwise choose N beyond all dominant indices and set $n_1 > N$.
Given $n_k > N$, nondominance gives $n_{k+1} > n_k$ with $s_{n_{k+1}} > s_{n_k}$; hence (s_{n_k}) is increasing.
Structural Commentary: Dominant indices give decreasing case; nondominance recursively gives increasing case.
Key Insight: The proof organizes arbitrary real behavior into one ordered thread. □

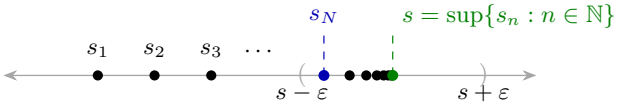


Figure 123: For an increasing bounded sequence, $s = \sup\{s_n\}$ is the limit.

Theorem 2.1.12. *Monotone Convergence Theorem for Sequences*
A monotonic real sequence converges if and only if it is bounded.

Proof.

(⇒) If $s_n \to s$, choose N with $|s_n - s| < 1$ for $n \geq N$. The tail lies in $(s-1, s+1)$ and the initial set $\{s_1, \ldots, s_{N-1}\}$ is finite, so (s_n) is bounded.

(⇐) Let $E := \{s_n : n \in \mathbb{N}\}$. If (s_n) is increasing, set $s := \sup E$. Given $\varepsilon > 0$, choose N with $s - \varepsilon < s_N \leq s$; then $s - \varepsilon < s_n \leq s$ for $n \geq N$, so $s_n \to s$. If (s_n) is decreasing, set $s := \inf E$. Given $\varepsilon > 0$, choose N with $s \leq s_N < s + \varepsilon$; then $s \leq s_n \leq s_N < s + \varepsilon$ for $n \geq N$, so $s_n \to s$.

Structural Commentary: Monotonicity traps bounded tails at $\sup E$ or $\inf E$.
Key Insight: The least upper bound is forced to be the limit. □

Theorem 2.1.13. *Bolzano–Weierstrass for Real Sequences*
Every bounded real sequence has a convergent subsequence.

Proof. By Theorem 2.1.11, choose a monotone subsequence. It is bounded because the original sequence is bounded, so Theorem 2.1.12 gives convergence.
Structural Commentary: Boundedness prevents escape; monotone subsequences force one ordered thread.
Key Insight: Monotone subsequence plus monotone convergence is the real-line compactness mechanism. □

Definition 2.1.14. *Cauchy Real Sequence*
A real sequence (s_n) is *Cauchy* if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m, n \geq N$ implies $|s_n - s_m| < \varepsilon$.

Lemma 2.1.15. *Cauchy Plus One Convergent Subsequence*
If a real Cauchy sequence has a subsequence converging to $s \in \mathbb{R}$, then the whole sequence converges to s.

Proof. Given $\varepsilon > 0$, choose N_0 so that $m, n \geq N_0$ implies $|s_m - s_n| < \varepsilon/2$. Choose j with $n_j \geq N_0$ and $|s_{n_j} - s| < \varepsilon/2$. Then for every $n \geq N_0$, $|s_n - s| \leq |s_n - s_{n_j}| + |s_{n_j} - s| < \varepsilon$.
Structural Commentary: Cauchy tails follow any convergent thread.
Key Insight: Mutual tail closeness turns one subsequential limit into the full limit. □

Theorem 2.1.16. *Real Cauchy Criterion*
A real sequence converges in $\mathbb{R}$ if and only if it is Cauchy.

Proof. If $s_n \to s$, choose N such that $|s_n - s| < \varepsilon/2$ for $n \geq N$; then $|s_n - s_m| < \varepsilon$ for $m, n \geq N$. Conversely, suppose (s_n) is Cauchy. Choose N so that $m, n \geq N$ implies $|s_m - s_n| < 1$. Then the tail lies in $(s_N - 1, s_N + 1)$, and the finitely many initial terms are bounded, so (s_n) is bounded. By Theorem 2.1.13, some subsequence converges; Lemma 2.1.15 makes the whole sequence converge.
Structural Commentary: Completeness of $\mathbb{R}$ means Cauchy behavior has a real endpoint.
Key Insight: Cauchy gives boundedness, Bolzano–Weierstrass gives a subsequential endpoint, and the Cauchy property pulls the whole sequence there. □

2.2 Divergence, Upper and Lower Limits, and Sequence Estimates

Definition 2.2.1. *Divergence to $\pm\infty$*
For a real sequence (s_n),

$$s_n \to +\infty \iff \forall M \in \mathbb{R},\ \exists N \in \mathbb{N} :\ n \geq N \Rightarrow s_n \geq M,$$
$$s_n \to -\infty \iff \forall M \in \mathbb{R},\ \exists N \in \mathbb{N} :\ n \geq N \Rightarrow s_n \leq M.$$

This extends $\to$ notation for divergent real sequences; Definition 2.1.1 is unchanged.

Proposition 2.2.2. *Unbounded Real Sequences Have Infinite Subsequences*
A real sequence is unbounded iff some subsequence tends to $+\infty$ or $-\infty$.

Proof. Set $n_0 = 0$. If (s_n) is unbounded above, every tail is unbounded above, so choose $n_k > n_{k-1}$ with $s_{n_k} \geq k$; then $s_{n_k} \to +\infty$. The unbounded-below case similarly gives $s_{n_k} \leq -k$, hence $s_{n_k} \to -\infty$.
Conversely, a subsequence tending to $\pm\infty$ is unbounded, so (s_n) is unbounded.
Structural Commentary: Unbounded tails supply the thresholds $\pm k$.
Key Insight: Large late terms form a divergent subsequence. □

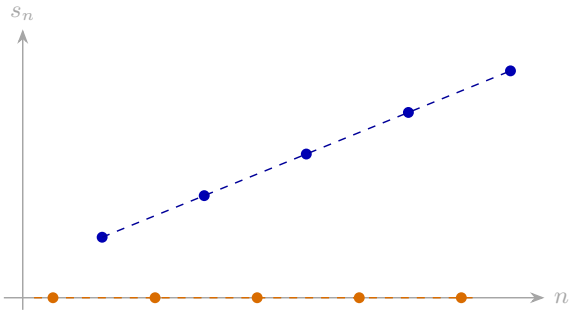


Figure 124: For $s_n = [(-1)^n + 1]n$, odd terms (orange) stay at 0, while even terms (blue) diverge to $+\infty$.

Definition 2.2.3. *Upper and Lower Limits*
For real sequence (s_n), let $E := \{x \in \overline{\mathbb{R}} : \exists\, n_1 < n_2 < \cdots,\ s_{n_k} \to x\}$, allowing $x = \pm\infty$.
Set $s^* := \sup E$ and $s_* := \inf E$; equivalently, $\limsup_{n\to\infty} s_n = s^*$ and $\liminf_{n\to\infty} s_n = s_*$.

Theorem 2.2.4. *Characterization of the Upper Limit*
Let (s_n) be a real sequence, and let E, s^*, s_* be as in Definition 2.2.3. Then:

(i) $s^* \in E$.
(ii) If $x > s^*$, then $s_n < x$ eventually.
(iii) These two properties characterize s^* uniquely.

Dually, s_* is uniquely characterized by $s_* \in E$ and $x < s_*$ implies $s_n > x$ eventually.

Proof.

(i) Since $s^* = \sup E$: if $s^* = +\infty$, use Proposition 2.2.2; if $s^* \in \mathbb{R}$, use Theorem 4.1.4. If $s^* = -\infty$ and $s_n \not\to -\infty$, infinitely many terms are bounded below; they are either unbounded above or have a bounded subsequence, giving an element of E above $-\infty$, a contradiction.
(ii) If $s_n \geq x > s^*$ infinitely often, that subsequence is unbounded above or has a finite subsequential limit $y \geq x$; both contradict $s^* = \sup E < x$. Hence $s_n < x$ eventually.
(iii) If $p < q$ both satisfy (i)–(ii), choose $p < x < q$. Then $s_n < x$ eventually, contradicting $q \in E$.

Apply the same argument to $(-s_n)$ for s_*.
Structural Commentary: Limsup is the maximal subsequential limit and eventual ceiling.
Key Insight: Limsup combines subsequences with eventual upper bounds. □

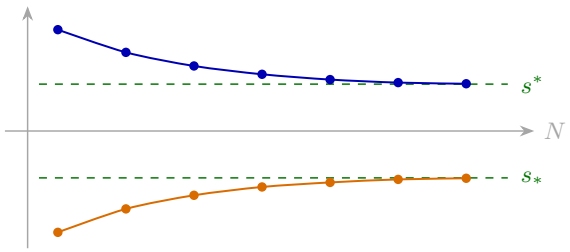


Figure 125: Blue tail suprema u_N decrease to $s^* = \limsup s_n$; orange tail infima ℓ_N increase to $s_* = \liminf s_n$.

Proposition 2.2.5. *Tail Suprema Formula*
For a real sequence (s_n), put $T_N := \{s_n : n \geq N\}$ and, in $\overline{\mathbb{R}}$, $u_N := \sup T_N$, $\ell_N := \inf T_N$.
Then $u_N \downarrow$, $\ell_N \uparrow$, and $\limsup_{n\to\infty} s_n = \inf_N u_N$ and $\liminf_{n\to\infty} s_n = \sup_N \ell_N$.

Proof. Since $T_{N+1} \subseteq T_N$, $u_N \downarrow$ and $\ell_N \uparrow$. Let $s^* = \limsup s_n$ and $u = \inf_N u_N$.
If $x > s^*$, then $s_n < x$ eventually, so $u \leq u_N \leq x$ for some N. If $y < s^*$, then $s^* \in E$ gives a subsequence tending to s^*; hence every tail meets (y, ∞), so $u_N \geq y$ for all N and $u \geq y$. Thus $u = s^*$.
Apply the same formula to $(-s_n)$ and use Corollary 1.1.4.
Structural Commentary: Tail envelopes converge to lim sup and lim inf.
Key Insight: Shrinking tails compute the extremal subsequential limits. □

Theorem 2.2.6. *Convergence and Equal Upper/Lower Limits*
A real sequence (s_n) converges to a finite number $s \in \mathbb{R}$ if and only if $\limsup_{n\to\infty} s_n = \liminf_{n\to\infty} s_n = s$.

Proof.
($\Rightarrow$) If $s_n \to s$, every subsequence converges to s, and the tail is bounded, so no subsequence tends to $\pm\infty$. Thus $E = \{s\}$, hence $\limsup s_n = \liminf s_n = s$.
($\Leftarrow$) Suppose $\limsup s_n = \liminf s_n = s$. Given $\varepsilon > 0$, Theorem 2.2.4 gives $s_n < s+\varepsilon$ and $s_n > s-\varepsilon$ eventually. Hence $|s_n - s| < \varepsilon$ eventually, so $s_n \to s$.

Structural Commentary: Equality of the two extremal subsequential limits traps the whole tail.
Key Insight: Limsup supplies an eventual ceiling; liminf supplies an eventual floor. □

Theorem 2.2.7. *Order Comparison for Upper and Lower Limits*
If $s_n \leq t_n$ for all $n \geq N$, where N is fixed, then:
(i) $\limsup_{n\to\infty} s_n \leq \limsup_{n\to\infty} t_n$.
(ii) $\liminf_{n\to\infty} s_n \leq \liminf_{n\to\infty} t_n$.

Proof. Discard finitely many terms, so $s_n \leq t_n$ for all n.
(i) Put $a = \limsup s_n$ and $b = \limsup t_n$. If $x < a$, then $s_n > x$ infinitely often, hence $t_n > x$ infinitely often; Theorem 2.2.4 gives $x \leq b$. Thus $a \leq b$.
(ii) Apply (i) to $-t_n \leq -s_n$ to get $\limsup(-t_n) \leq \limsup(-s_n)$, equivalently $\liminf s_n \leq \liminf t_n$.

Structural Commentary: Eventual order is inherited by the extremal subsequential limits.
Key Insight: Limsup and liminf ignore finitely many initial terms. □

Theorem 2.2.8. *Pinching Theorem for Real Sequences*
Let (a_n), (s_n), and (b_n) be real sequences with $a_n \leq s_n \leq b_n$ for all sufficiently large n.
If $a_n \to s$ and $b_n \to s$, then $s_n \to s$.

Proof. Let $\varepsilon > 0$. Choose N_0 s.t. $a_n \leq s_n \leq b_n$ for $n \geq N_0$; N_1 s.t. $s-\varepsilon < a_n$ for $n \geq N_1$; and N_2 s.t. $b_n < s+\varepsilon$ for $n \geq N_2$. If $N = \max\{N_0, N_1, N_2\}$, then $s - \varepsilon < s_n < s + \varepsilon$ for all $n \geq N$. Hence $s_n \to s$.
Structural Commentary: Two convergent bounds squeeze the middle sequence to the same limit.
Key Insight: Once both walls are inside the ε-neighborhood, the middle term has no room to escape. □

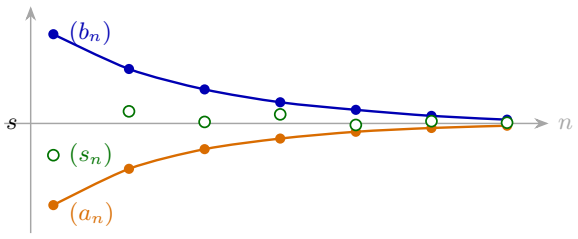


Figure 126: With $a_n \le s_n \le b_n$ and $a_n, b_n \to s$, the middle sequence also tends to s.

Theorem 2.2.9. *Special Sequence Limits*

(i) If $p > 0$, then $\lim_{n\to\infty} n^{-p} = 0$.
(ii) If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.
(iii) $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
(iv) If $p > 0$ and $\alpha \in \mathbb{R}$, then $\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$.
(v) If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.

Proof.

(i) Given $\varepsilon > 0$, choose $N > (1/\varepsilon)^{1/p}$ by Theorem 1.1.15. Then $n \ge N$ gives $n^{-p} < \varepsilon$.
(ii) If $p > 1$, set $x_n := \sqrt[n]{p} - 1$. Then $x_n > 0$ and $1 + nx_n \le (1 + x_n)^n = p$, so $0 < x_n \le (p-1)/n \to 0$. By Theorem 2.2.8, $x_n \to 0$. If $p = 1$, then $\sqrt[n]{p} = 1$ for every n.
If $0 < p < 1$, then $1/p > 1$, so $\sqrt[n]{1/p} \to 1$ by the previous case.
Since $\sqrt[n]{p} = 1/\sqrt[n]{1/p}$ and limits preserve reciprocals away from 0, $\sqrt[n]{p} \to 1$.
(iii) Put $x_n := \sqrt[n]{n} - 1$. Then $x_n \ge 0$, and for $n \ge 2$ the binomial theorem gives $n = (1 + x_n)^n \ge \frac{n(n-1)}{2} x_n^2$.
Thus $0 \le x_n \le \sqrt{2/(n-1)}$, so Theorem 2.2.8 gives $x_n \to 0$.
(iv) Choose an integer $k > \alpha$ with $k > 0$. For $n > 2k$, $(1+p)^n \ge \binom{n}{k} p^k = \frac{n(n-1)\cdots(n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}$.
Hence $0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k}$, and $n^{\alpha-k} \to 0$ by (i). Apply Theorem 2.2.8.
(v) If $x \ne 0$, write $|x| = (1+p)^{-1}$ with $p > 0$ and apply (iv) with $\alpha = 0$ to get $|x|^n \to 0$. Since $-|x|^n \le x^n \le |x|^n$, Theorem 2.2.8 gives $x^n \to 0$; $x = 0$ is immediate.

Structural Commentary: Each limit is reduced to an elementary comparison sequence.
Key Insight: The binomial theorem supplies the right upper bounds for roots and exponential domination. □

Method 2.2.10. *Sequence Proof Patterns*

(i) *Convergence*: fix $\varepsilon > 0$, then find N so that all later terms lie in the relevant ε-ball.
(ii) *Uniqueness*: compare two proposed limits with $\varepsilon/2$ and use the triangle inequality.
(iii) *Subsequences*: choose $n_k \uparrow$ recursively with distances $< 1/k$.
(iv) *Cauchy arguments*: estimate the diameter of tails rather than guessing the limit first.
(v) *Monotone sequences*: use sup or inf and then prove eventual trapping.

Key Insight: Most sequence proofs are bookkeeping for quantifiers, indices, and one controlling inequality.

3 Metric Spaces and Euclidean Topology

3.1 Metrics, Open Sets, and Euclidean Topology

Metric topology studies sets through balls: openness, closure, and limit points come from distance. Euclidean estimates rely on norms, Cauchy–Schwarz, and the triangle inequality.

Definition 3.1.1. *Metric Space*
A *metric space* is a set X equipped with a function $d : X \times X \to \mathbb{R}$ such that, for all $p, q, r \in X$,

i. $d(p, q) > 0$ if $p \ne q$, and $d(p, p) = 0$;
ii. $d(p, q) = d(q, p)$;
iii. $d(p, q) \le d(p, r) + d(r, q)$.

The function d is called a *distance function* or *metric*.

Perspective 3.1.2. *ε-Closeness Becomes Metric Language*
On $\mathbb{R}$, ε-close means $|x - y| < \varepsilon$, i.e. $d(x, y) < \varepsilon$ with $d(x, y) = |x - y|$; may be abstracted to metric balls.
Key Insight: Absolute value is distance on $\mathbb{R}$.

Example 3.1.3. *Euclidean Metric and Subspaces*
The Euclidean space $\mathbb{R}^k$ is a metric space with $d(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_2$, $\mathbf{x},\mathbf{y}\in\mathbb{R}^k$.
Every subset Y of a metric space X is a metric space by restricting the same distance function to points of Y.

Example 3.1.4. *Discrete Metric*
On any set X, define $d(x,y)=\begin{cases}0, & x=y,\\ 1, & x\neq y.\end{cases}$.
This is the *discrete metric*: distinct points are all distance 1 apart.

Example 3.1.5. *Hamming Distance*
For $x,y\in\{0,1\}^n$, define $h(x,y) := |\{i : x_i \neq y_i\}|$. Then h is a metric.

Proof. $h\geq 0$, $h(x,y)=h(y,x)$, and $h(x,y)=0 \iff x_i=y_i\ \forall i \iff x=y$.
For $x,y,z\in\{0,1\}^n$, $\{i : x_i\neq y_i\}\subseteq\{i : x_i\neq z_i\}\cup\{i : z_i\neq y_i\}$, so $h(x,y)\leq h(x,z)+h(z,y)$.
Structural Commentary: Hamming distance counts coordinate disagreements.
Key Insight: Each disagreement between x and y must be paid for through z in at least one leg. □

Example 3.1.6. *Coordinate Norm Metrics on $\mathbb{R}^k$*
By Proposition 1.1.21 and Theorem 1.1.13, $d_1(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_1$; $d_2(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_2$; $d_\infty(\mathbf{x},\mathbf{y}) := \|\mathbf{x}-\mathbf{y}\|_\infty$ are metrics on $\mathbb{R}^k$, and $\forall \mathbf{x},\mathbf{y}\in\mathbb{R}^k$, $d_\infty(\mathbf{x},\mathbf{y})\leq d_2(\mathbf{x},\mathbf{y})\leq d_1(\mathbf{x},\mathbf{y})\leq \sqrt{k}\, d_2(\mathbf{x},\mathbf{y})$; $d_2(\mathbf{x},\mathbf{y})\leq \sqrt{k}\, d_\infty(\mathbf{x},\mathbf{y})$.
Hence d_1, d_2, d_∞ have the same convergent sequences.

Theorem 3.1.7. *Hölder and Minkowski for Sequence Spaces*
Let $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$.

(i) If $p,q>1$ and $p^{-1}+q^{-1}=1$, then $\mathbf{x}\in\ell^p(\mathbb{F})$, $\mathbf{y}\in\ell^q(\mathbb{F})$ imply $\sum_{n\geq 1}|x_n y_n|\leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q$.
(ii) If $1\leq p<\infty$ and $\mathbf{x},\mathbf{y}\in\ell^p(\mathbb{F})$, then $\mathbf{x}+\mathbf{y}\in\ell^p(\mathbb{F})$ and $\|\mathbf{x}+\mathbf{y}\|_p\leq \|\mathbf{x}\|_p+\|\mathbf{y}\|_p$.

Proof.

(i) If $\|\mathbf{x}\|_p\|\mathbf{y}\|_q=0$, the claim is immediate.
Otherwise set $X:=\|\mathbf{x}\|_p$, $Y:=\|\mathbf{y}\|_q$, $u_n:=|x_n|/X$, and $v_n:=|y_n|/Y$. Young's inequality gives $u_n v_n\leq u_n^p/p+v_n^q/q$; summing over $1\leq n\leq N$ and letting $N\to\infty$ gives $\sum_{n\geq 1}|x_n y_n|\leq XY$.

(ii) For $p=1$, apply $|x_n+y_n|\leq|x_n|+|y_n|$ and sum. For $p>1$, put $s_N:=(\sum_{n=1}^N|x_n+y_n|^p)^{1/p}$. If $s_N=0$, the bound is trivial; if $s_N>0$, then $\sum_{n=1}^N|x_n+y_n|^p\leq\sum_{n=1}^N(|x_n|+|y_n|)\,|x_n+y_n|^{p-1}\leq(\|\mathbf{x}\|_p+\|\mathbf{y}\|_p)s_N^{p-1}$ by (i), so $s_N\leq\|\mathbf{x}\|_p+\|\mathbf{y}\|_p$. Let $N\to\infty$.

Structural Commentary: Hölder controls products; Minkowski controls sums.
Key Insight: The metric property of ℓ^p is exactly Minkowski's inequality. □

Example 3.1.8. *Function and Sequence Metric Spaces*
Let $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$.

(i) On $C_{\mathbb{F}}[a,b]$, $d_\infty(f,g):=\sup_{t\in[a,b]}|f(t)-g(t)|$ and $d_1(f,g):=\int_a^b|f(t)-g(t)|\,dt$ are metrics. Definiteness of d_1 uses continuity.
(ii) On $\ell^\infty(\mathbb{F})$, $d_\infty(\mathbf{x},\mathbf{y}):=\sup_{n\geq1}|x_n-y_n|$ is a metric.
(iii) For $1\leq p<\infty$, $d_p(\mathbf{x},\mathbf{y}):=\|\mathbf{x}-\mathbf{y}\|_p=(\sum_{n\geq1}|x_n-y_n|^p)^{1/p}$ is a metric on $\ell^p(\mathbb{F})$ by Theorem 3.1.7.

Structural Commentary: Norm estimates produce metrics on function and sequence spaces.
Key Insight: Infinite-dimensional examples require estimates, not just coordinate formulas.

Definition 3.1.9. *Euclidean Intervals, Cells, Balls, and Convexity*

(i) In $\mathbb{R}$, $(a,b):=\{x : a<x<b\}$, $[a,b]:=\{x : a\leq x\leq b\}$, with $[a,b)$ and $(a,b]$ defined by mixed inequalities.
(ii) For $a_i<b_i$, a *k-cell* is $\prod_{i=1}^k[a_i,b_i]=\{\mathbf{x}\in\mathbb{R}^k : a_i\leq x_i\leq b_i,\ 1\leq i\leq k\}$.
(iii) For $\mathbf{x}\in\mathbb{R}^k$ and $r>0$, $B_r(\mathbf{x}):=\{\mathbf{y} : \|\mathbf{y}-\mathbf{x}\|_2<r\}$ and $\overline{B}_r(\mathbf{x}):=\{\mathbf{y} : \|\mathbf{y}-\mathbf{x}\|_2\leq r\}$.
(iv) $E\subseteq\mathbb{R}^k$ is *convex* iff $\lambda\mathbf{x}+(1-\lambda)\mathbf{y}\in E$ for all $\mathbf{x},\mathbf{y}\in E$ and $0\leq\lambda\leq1$.
(v) The *integer lattice* is $\mathbb{Z}^k$, and the *first orthant* is $\{\mathbf{x}\in\mathbb{R}^k : x_i\geq 0\ \forall i\}$.
(vi) The *unit ball* is $B^k:=\{\mathbf{x}\in\mathbb{R}^k : \|\mathbf{x}\|_2\leq 1\}$, and the *unit sphere* is $S^{k-1}:=\{\mathbf{x}\in\mathbb{R}^k : \|\mathbf{x}\|_2=1\}$.
(vii) A *convex combination* of $\mathbf{x},\mathbf{y}$ is $\lambda\mathbf{x}+(1-\lambda)\mathbf{y}$ with $0\leq\lambda\leq1$; these points form the line segment $[\mathbf{x},\mathbf{y}]$.

Proposition 3.1.10. *Balls and Cells Are Convex*
Open balls, closed balls, and k-cells in $\mathbb{R}^k$ are convex.

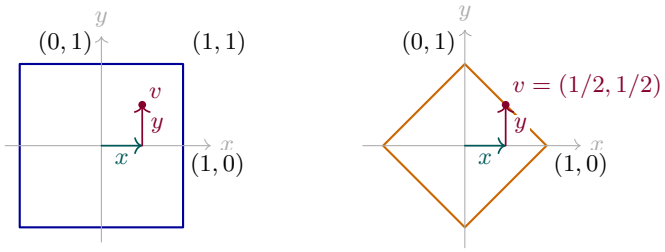


Figure 127: Unit balls in $\mathbb{R}^2$: $\max\{|x|, |y|\} < 1$ and $|x| + |y| < 1$.

Proof. Let $\mathbf{y}, \mathbf{z} \in B_r(\mathbf{x})$ and $0 < \lambda < 1$.
Then $\|\lambda\mathbf{y} + (1-\lambda)\mathbf{z} - \mathbf{x}\|_2 = \|\lambda(\mathbf{y} - \mathbf{x}) + (1-\lambda)(\mathbf{z} - \mathbf{x})\|_2 \leq \lambda \|\mathbf{y} - \mathbf{x}\|_2 + (1-\lambda) \|\mathbf{z} - \mathbf{x}\|_2 < r$.
The same proof with $\leq r$ gives closed balls. For a k-cell, each coordinate of $\lambda\mathbf{y} + (1-\lambda)\mathbf{z}$ lies between the corresponding coordinates of $\mathbf{y}$ and $\mathbf{z}$, hence remains in the same interval $[a_i, b_i]$.
Structural Commentary: Convexity is checked by following the line segment between two points.
Key Insight: Balls use the triangle inequality; cells use coordinate inequalities. □

Definition 3.1.11. *Product Metrics*
For metric spaces (X, d_X) and (Y, d_Y), and $p = (x, y)$, $p' = (x', y')$ in $X \times Y$, define

$$d_E(p, p') := \sqrt{d_X(x, x')^2 + d_Y(y, y')^2},$$
$$d_{\max}(p, p') := \max\{d_X(x, x'), d_Y(y, y')\},$$
$$d_{\text{sum}}(p, p') := d_X(x, x') + d_Y(y, y')$$

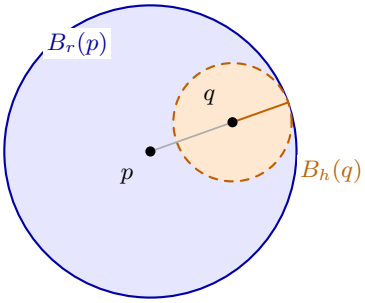
These are the Euclidean, maximum, and sum product metrics; the sum metric is often simplest for estimates.

Proposition 3.1.12. *Comparing Product Metrics*
On $X \times Y$, $d_{\max} \leq d_E \leq d_{\text{sum}} \leq 2d_{\max}$. Hence the three product metrics have same convergent sequences.

Proof. Let $a = d_X(x, x')$ and $b = d_Y(y, y')$.
Then $\max\{a, b\} \leq \sqrt{a^2 + b^2} \leq a + b \leq 2\max\{a, b\}$; convergence follows by squeezing.
Structural Commentary: The three product metrics differ by controlled constants.
Key Insight: Equivalent metric estimates preserve convergence. □

Proposition 3.1.13. *Bounded Compression of a Metric*
If (X, d) is a metric space, then $\rho(x, y) := d(x, y)/(1 + d(x, y))$ is a bounded metric on X, and $\tau_\rho = \tau_d$.

Proof. $0 \leq \rho < 1$, and definiteness and symmetry follow from d.
For $a, b \geq 0$, $(a + b)/(1 + a + b) \leq a/(1 + a) + b/(1 + b)$.
Since $t/(1 + t)$ is increasing, $d(x, z) \leq d(x, y) + d(y, z)$ gives $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.
For balls, $B_d(x, \eta) \subseteq B_\rho(x, \eta)$ for $0 < \eta < 1$, and $B_\rho(x, \eta) \subseteq B_d(x, \eta/(1 - \eta))$; hence $\tau_\rho = \tau_d$.
Structural Commentary: Compressing distances into $[0, 1)$ changes boundedness but not local neighborhoods.
Key Insight: Topology depends on small-scale control, not on the large-scale size of distances. □



$h = r - d(p, q) > 0$

Figure 128: Every point inside a metric ball has a smaller ball still contained in the original ball.

Definition 3.1.14. *Metric Topology Terminology*
Let X be a metric space, and let all points and sets below be understood as points of X and subsets of X.

(i) A *neighborhood* of p with radius $r > 0$ is $B_r(p) := \{q \in X : d(p,q) < r\}$.
The corresponding *closed ball* is $\overline{B}_r(p) := \{q \in X : d(p,q) \leq r\}$, and the *sphere* of radius r about p is $\{q \in X : d(p,q) = r\}$.
(ii) A point p is a *limit point* of E if every neighborhood of p contains a point $q \neq p$ with $q \in E$.
(iii) If $p \in E$ and p is not a limit point of E, then p is an *isolated point* of E.
(iv) A set E is *closed* if every limit point of E is a point of E.
(v) A point p is an *interior point* of E if some neighborhood of p is contained in E.
(vi) A set E is *open* if every point of E is an interior point of E.
(vii) The *complement* of E is $E^c := X \setminus E$.
(viii) A set E is *dense* in X if every point of X is either a limit point of E, a point of E, or both.

For $X = \mathbb{R}$, this gives $B_\varepsilon(a) = (a - \varepsilon, a + \varepsilon)$.
Structural Commentary: Openness and closedness are ambient notions, not opposites.

Example 3.1.15. *Open, Closed, Both, or Neither*
In $\mathbb{R}$, $\emptyset, \mathbb{R}$ are open and closed, while $[a,b)$, $a < b$, is neither. If d is discrete on X, then $B_{1/2}(x) = \{x\}$ for all $x \in X$, so every $E \subseteq X$ and E^c are open; hence every $E \subseteq X$ is closed.

Theorem 3.1.16. *Every Neighborhood Is Open*
Every neighborhood in a metric space is an open set.

Proof. Let $E = B_r(p)$, and choose $q \in E$. Put $h := r - d(p,q) > 0$.
If $s \in B_h(q)$, then $d(p,s) \leq d(p,q) + d(q,s) < d(p,q) + h = r$.
Thus $B_h(q) \subseteq E$, so q is an interior point of E. Since q was arbitrary, E is open.
Structural Commentary: Around any point inside a ball, the leftover distance to boundary gives a smaller ball.
Key Insight: The triangle inequality turns radius slack into openness. □

Corollary 3.1.17. *Closed Balls Are Closed*
For every $p \in X$ and $r > 0$, the closed ball $\overline{B}_r(p)$ is closed.

Proof. Let $q \notin \overline{B}_r(p)$, so $d(p,q) > r$. Put $h := (d(p,q) - r)/2$.
If $s \in B_h(q)$, then $d(p,s) \geq d(p,q) - d(q,s) > d(p,q) - h > r$, so $s \notin \overline{B}_r(p)$.
Hence the complement of $\overline{B}_r(p)$ is open, and $\overline{B}_r(p)$ is closed.
Structural Commentary: Outside a closed ball, positive distance remains to the boundary.
Key Insight: Reverse triangle estimates certify closedness. □

Theorem 3.1.18. *Limit Points Force Infinitely Many Nearby Points*
If p is a limit point of E, then every neighborhood of p contains infinitely many points of E.

Proof. Suppose some neighborhood N of p contains only finitely many points of E distinct from p, say $q_1, \ldots, q_n$.
If there are none, then N itself contradicts that p is a limit point. Otherwise put $\rho := \min_{1 \leq m \leq n} d(p, q_m) > 0$.
The neighborhood $B_\rho(p)$ contains no point $q \in E$ with $q \neq p$, contradicting the definition of limit point.
Structural Commentary: A finite list of nearby exceptions has a smallest positive distance from p.
Key Insight: Limit points cannot be witnessed by only finitely many neighboring points. □

Corollary 3.1.19. *Finite Point Sets Have No Limit Points*
A finite point set in a metric space has no limit points.

Proof. Let $E \subseteq X$ be finite. If p were a limit point of E, then by Theorem 3.1.18, every neighborhood of p would contain infinitely many distinct points of E, impossible as E is finite. Hence E has no limit points.
Structural Commentary: This corollary is the finite obstruction to accumulation.
Key Insight: A limit point requires infinitely many nearby witnesses. □

Theorem 3.1.20. *Open Sets and Closed Complements*
A set E is open if and only if its complement E^c is closed.

Proof.

(⇒) Suppose E is open. If x is a limit point of E^c and $x \in E$, then some neighborhood N of x satisfies $N \subseteq E$, contradicting $N \cap (E^c \setminus \{x\}) \neq \emptyset$. Hence $x \in E^c$, so E^c is closed.
(⇐) Suppose E^c is closed. If $x \in E$, then $x \notin E^c$, hence x is not a limit point of E^c. Thus some neighborhood N of x has $N \cap E^c = \emptyset$, i.e. $N \subseteq E$. Hence E is open.

Structural Commentary: Openness of a set is equivalent to closure of the outside.
Key Insight: Interior points of E cannot be limit points of E^c. □

Corollary 3.1.21. *Closed Sets and Open Complements*
A set F is closed if and only if F^c is open.

Proof. Apply Theorem 3.1.20 with $E = F^c$. Then F^c is open if and only if $(F^c)^c$ is closed. Since $(F^c)^c = F$, this is exactly the assertion that F^c is open if and only if F is closed.
Structural Commentary: This is the same open–closed duality after taking complements.
Key Insight: Closedness of F is openness of the outside F^c. □

Theorem 3.1.22. *Basic Topological Calculus*
In a metric space:

(i) any union of open sets is open;
(ii) any intersection of closed sets is closed;
(iii) any finite intersection of open sets is open;
(iv) any finite union of closed sets is closed.

Proof.

(i) Let $G := \bigcup_\alpha G_\alpha$. If $x \in G$, then $x \in G_\alpha$ for some α.
Since G_α is open, some neighborhood of x lies in $G_\alpha \subseteq G$. Hence G is open.

(ii) Let $F := \bigcap_\alpha F_\alpha$. By Theorem 1.4.3, $F^c = \bigcup_\alpha F_\alpha^c$.
Each F_α^c is open by Corollary 3.1.21, so F^c is open by (i). Hence F is closed.

(iii) Let $H := \bigcap_{i=1}^n G_i$. If $x \in H$, let $r_i > 0$, $B_{r_i}(x) \subseteq G_i$, $r := \min_{1\le i\le n} r_i$. Then $B_r(x) \subseteq H$, so H is open.

(iv) Let $F := \bigcup_{i=1}^n F_i$. By Theorem 1.4.3, $F^c = \bigcap_{i=1}^n F_i^c$.
Each F_i^c is open by Corollary 3.1.21, so F^c is open by (iii). Hence F is closed.

Structural Commentary: Arbitrary unions are easy for open sets; finite intersections require a minimum radius.
Key Insight: The finite minimum is the reason infinite intersections of open sets need not be open. □

Definition 3.1.23. *Closure*
If $E \subseteq X$ and E^{acc} denotes the set of all limit points of E in X, then the *closure* of E is $\overline{E} := E \cup E^{\text{acc}}$.

Theorem 3.1.24. *Closure Is the Smallest Closed Superset*
If X is a metric space and $E \subseteq X$, then:

(i) $\overline{E}$ is closed;
(ii) $\overline{E} = E$ if and only if E is closed;
(iii) $\overline{E} \subseteq F$ for every closed set $F \subseteq X$ such that $E \subseteq F$.

Thus $\overline{E}$ is the smallest closed subset of X that contains E.

Proof.

(i) Let $p \notin \overline{E}$. Then $p \notin E$ and p is not a limit point of E, so $B_r(p) \cap E = \emptyset$ for some $r > 0$. If $q \in B_r(p)$, then $q \notin E$ and $B_{r-d(p,q)}(q) \subseteq B_r(p)$, so $q \notin E^{\text{acc}}$.
Hence $B_r(p) \subseteq (\overline{E})^c$, and $\overline{E}$ is closed by Corollary 3.1.21.

(ii) If $\overline{E} = E$, then E is closed by (i). Conversely, if E is closed, then $E^{\text{acc}} \subseteq E$, so $\overline{E} = E \cup E^{\text{acc}} = E$.

(iii) If F is closed and $E \subseteq F$, then $E^{\text{acc}} \subseteq F^{\text{acc}} \subseteq F$. Thus $\overline{E} = E \cup E^{\text{acc}} \subseteq F$.

Structural Commentary: Closure adds exactly the limit points forced by E.
Key Insight: Closed supersets must contain all limit points, so they must contain $\overline{E}$. □

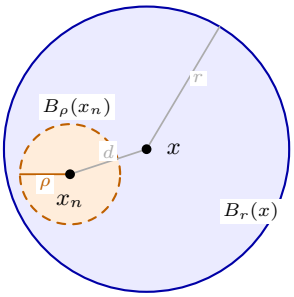


Figure 129: If $d = d(x, x_n) < r$, then any $0 < \rho < r - d$ gives $B_\rho(x_n) \subseteq B_r(x)$.

Definition 3.1.25. *Interior, Boundary, Adherence, and Distance*
Let (X, d) be a metric space and $E, F \subseteq X$.

(i) $E^\circ := \{p \in E : \exists r > 0,\ B_r(p) \subseteq E\}$ is the *interior*
(ii) $\text{ext}(E) := (E^c)^\circ = \{p \in X : \exists r > 0,\ B_r(p) \cap E = \emptyset\}$ is the *exterior*.
(iii) $\text{bd}(E) := \overline{E} \setminus E^\circ = \{p \in X : \forall r > 0,\ B_r(p) \cap E \neq \emptyset,\ B_r(p) \cap E^c \neq \emptyset\}$ is the *boundary*.
(iv) $p \in X$ is *adherent* to E iff $\forall r > 0,\ B_r(p) \cap E \neq \emptyset$, equivalently $p \in \overline{E}$.
(v) For $p \in X$, nonempty E, F, $\text{dist}(p, E) := \inf\{d(p, q) : q \in E\}$' $\text{dist}(E, F) := \inf\{d(p, q) : p \in E,\ q \in F\}$.

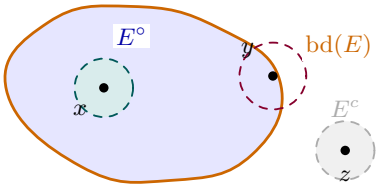


Figure 130: Interior, exterior, and boundary points are distinguished by how small balls meet E and E^c.

Proposition 3.1.26. *Distance-to-Set Closure Criterion*
Let $\emptyset \neq E \subseteq X$. For $p \in X$, $p \in \overline{E} \iff \text{dist}(p, E) = 0$.

Proof.
($\Rightarrow$) If $p \in \overline{E}$, then $\forall \varepsilon > 0$, $B_\varepsilon(p) \cap E \neq \emptyset$; choose $q_\varepsilon \in E$ with $d(p, q_\varepsilon) < \varepsilon$.
Thus $0 \leq \text{dist}(p, E) \leq \varepsilon$ for all $\varepsilon > 0$, so $\text{dist}(p, E) = 0$.
($\Leftarrow$) If $\text{dist}(p, E) = 0$, then $\forall r > 0$, $\exists q \in E$ with $d(p, q) < r$.
Hence $B_r(p) \cap E \neq \emptyset$ for all $r > 0$, so p is adherent to E, equivalently $p \in \overline{E}$.
Structural Commentary: Closure can be detected by vanishing distance to the set.
Key Insight: The infimum packages all small-ball tests into one number. □

Remark 3.1.27. *Set Distance Is Not a Metric on Subsets*
The function $\text{dist}(E, F)$ is useful, but it is not a metric on nonempty subsets.
In $\mathbb{R}$, for example, $\text{dist}((0, 1), (1, 2)) = 0$ although the sets are distinct and disjoint.



Figure 131: Adherent endpoints need not lie in E; isolated points are adherent but not limit points.

3.2 Relative Topology

Definition 3.2.1. *Relative Openness*
Suppose $E \subseteq Y \subseteq X$, where X is a metric space. The set E is *open relative to Y* if for each $p \in E$ there is an $r > 0$ such that $q \in E$ whenever $q \in Y$ and $d(p, q) < r$.

Theorem 3.2.2. *Relative Opens Are Intersections*
Suppose $Y \subseteq X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X.

Proof.

($\Rightarrow$) For each $p \in E$, choose $r_p > 0$ with $B_{r_p}(p) \cap Y \subseteq E$, and set $G := \bigcup_{p \in E} B_{r_p}(p)$. Then G is open in X, and $E \subseteq G \cap Y$. If $q \in G \cap Y$, then $q \in B_{r_p}(p) \cap Y$ for some $p \in E$, so $q \in E$. Thus $E = G \cap Y$.

($\Leftarrow$) If $E = G \cap Y$ with G open in X, then for each $p \in E$ choose $r > 0$ with $B_r(p) \subseteq G$. Hence $B_r(p) \cap Y \subseteq G \cap Y = E$, so E is open relative to Y.

Structural Commentary: Relative openness is ordinary openness viewed through the ambient subset Y.
Key Insight: Intersect open sets in X with Y to get exactly the open sets of Y. □

Example 3.2.3. *Relative Topology Depends on the Ambient Space*
Let $X = \{(x, 0) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the metric inherited from $\mathbb{R}^2$. The set $E = \{(x, 0) : -1 < x < 1\}$ is not open as a subset of $\mathbb{R}^2$, but it is open relative to X, since $E = X \cap \big((-1, 1) \times (-1, 1)\big)$.
Likewise, $[0, 1)$ is not closed in $\mathbb{R}$, but it is closed relative to $(-1, 1)$ because $[0, 1) = (-1, 1) \cap [0, \infty)$.

3.3 Density, Separability, and Closure Tests

Proposition 3.3.1. *Density and Closure*
Let X be a metric space and let $E \subseteq X$. Then E is dense in X if and only if $\overline{E} = X$.

Proof.

($\Rightarrow$) Suppose E is dense in X. Let $p \in X$. By Definition 3.1.14, p is either a point of E or a limit point of E. Hence $p \in E \cup E^{\text{acc}} = \overline{E}$ by Definition 3.1.23. Thus $X \subseteq \overline{E}$. Since $\overline{E} \subseteq X$, $\overline{E} = X$.

($\Leftarrow$) Suppose $\overline{E} = X$. Let $p \in X$. Then $p \in \overline{E} = E \cup E^{\text{acc}}$, so p is either a point of E or a limit point of E. Therefore E is dense in X by Definition 3.1.14.

Structural Commentary: Density says every point is already in E or is forced by nearby points of E.
Key Insight: The closure packages exactly those two possibilities. □

Proposition 3.3.2. *Real-Line Density Interval Test*
Let $E \subseteq \mathbb{R}$. Then E is dense in $\mathbb{R}$ if and only if every nonempty open interval (a, b) satisfies $(a, b) \cap E \neq \emptyset$.

Proof.

($\Rightarrow$) Let $(a, b) \neq \emptyset$ and choose $p \in (a, b)$, $r := \frac{1}{2}\min\{p - a, b - p\}$. If $p \in E$, done. If $p \notin E$, density gives $p \in E^{\text{acc}}$, so choose $q \in (E \setminus \{p\}) \cap B_r(p)$. Since $B_r(p) \subseteq (a, b)$, $q \in (a, b) \cap E$.

($\Leftarrow$) Suppose $(a, b) \cap E \neq \emptyset$ for every nonempty open interval (a, b). For $p \in \mathbb{R}$, either $p \in E$, or $p \notin E$ and $\forall \varepsilon > 0$, choose $q \in (p - \varepsilon, p + \varepsilon) \cap E$. Then $q \neq p$ and $q \in B_\varepsilon(p)$, so $p \in E^{\text{acc}}$. Hence $\mathbb{R} = E \cup E^{\text{acc}} = \overline{E}$, so E is dense in $\mathbb{R}$.

Structural Commentary: On $\mathbb{R}$, balls and open intervals test the same local condition.
Key Insight: Density means no interval can avoid the set. □

Definition 3.3.3. *Separable Metric Space*
A metric space X is *separable* if it contains a countable dense subset.

Example 3.3.4. *Separable and Nonseparable Metric Spaces*

(i) $\mathbb{R}$ is separable: $\mathbb{Q}$ is countable and dense. Hence $\mathbb{R}^k$ is separable via $\mathbb{Q}^k$, and $\mathbb{C}^k$ via $\mathbb{C}^k \simeq \mathbb{R}^{2k}$.

(ii) If X is discrete, then X is separable $\iff$ X is countable, since $B_{1/2}(x) = \{x\}$ forces every dense $D \subseteq X$ to contain x.

(iii) $\ell^\infty(\mathbf{F})$ is not separable. The set $\{0, 1\}^{\mathbb{N}}$ is uncountable, and $s \neq t$ gives $d_\infty(s, t) = 1$; the balls $B_{1/3}(s)$ are pairwise disjoint, so no countable dense set can meet all of them.

(iv) For $1 \leq p < \infty$, $\ell^p(\mathbf{F})$ is separable. Let D be the countable set of finite-support sequences with rational coordinates, using rational real and imaginary parts if $\mathbf{F} = \mathbb{C}$.
For $x \in \ell^p$ and $\varepsilon > 0$, choose N with $\sum_{n>N} |x_n|^p < (\varepsilon/2)^p$, then choose $y \in D$ with support in $\{1, \ldots, N\}$ and $\sum_{n=1}^{N} |x_n - y_n|^p < (\varepsilon/2)^p$. Thus $\|x - y\|_p < \varepsilon$.

Structural Commentary: Separability is the existence of a countable set meeting every nonempty open ball.
Key Insight: Finite rational data are dense in ℓ^p, while ℓ^∞ contains an uncountable 1-separated set.

Theorem 3.3.5. *Suprema Belong to Closures*
Let $\emptyset \neq E \subseteq \mathbb{R}$ be bounded above and $y := \sup E$. Then $y \in \overline{E}$; if E is closed, then $y \in E$.

Proof. If $y \in E$, then $y \in \overline{E}$. If $y \notin E$, then $\forall h > 0$, $y - h$ is not an upper bound; choose $x_h \in E$ with $y - h < x_h \leq y$. Thus $x_h \neq y$, $|x_h - y| < h$, and $B_h(y) \cap (E \setminus \{y\}) \neq \emptyset$, so $y \in E^{\text{acc}} \subseteq \overline{E}$.
If E is closed, $\overline{E} = E$ by Theorem 3.1.24(ii).

Structural Commentary: LUB gives points of E arbitrarily close below $\sup E$.
Key Insight: A supremum is either attained or approached. □

Method 3.3.6. *Metric Proof Patterns*

(i) To prove G is open, fix $x \in G$ and find $r > 0$ with $B_r(x) \subseteq G$.
(ii) To prove F is closed, show $F^{\text{acc}} \subseteq F$, or equivalently show F^c is open.
(iii) To prove $C = \overline{E}$, show $E \subseteq C$, C is closed, and $C \subseteq F$ for every closed $F \supseteq E$.
(iv) To prove $p \in \overline{E}$, show $\operatorname{dist}(p, E) = 0$, or build a sequence in E converging to p.

3.4 Bounded Sets and Metric Sequences

Definition 3.4.1. *Bounded Sets and Metric Sequences*
Let (X, d) be a metric space, $E \subseteq X$, and (p_n) a sequence in X.

(i) E is *bounded* iff $\exists q \in X, \exists M > 0$ such that $E \subseteq B_M(q)$
(ii) E is *perfect* iff E is closed and every $p \in E$ is a limit point of E.
(iii) (p_n) *converges* to $p \in X$ iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $n \geq N \Longrightarrow d(p_n, p) < \varepsilon$; write $p_n \to p$ or $\lim_{n\to\infty} p_n = p$. If no such p exists, (p_n) *diverges*.
(iv) The *range* of (p_n) is $\operatorname{ran}(p_n) := \{p_n : n \in \mathbb{N}\}$; (p_n) is *bounded* iff $\operatorname{ran}(p_n)$ is bounded.
(v) (p_n) is *eventually in* E iff $\exists N \in \mathbb{N}$ such that $n \geq N \Longrightarrow p_n \in E$
(vi) (p_n) is *frequently in* E iff $\forall N \in \mathbb{N}, \exists n \geq N$ such that $p_n \in E$.

Structural Commentary: Boundedness, compactness, and completeness are intrinsic to restricted metrics; openness and closedness are ambient.
Key Insight: Convergence is tail control; frequent membership is arbitrarily late recurrence.

Proposition 3.4.2. *Limit Points Are Subsequence Limits*
For $(p_n) \subseteq X$ and $p \in X$, the following are equivalent:

(i) $\forall r > 0, \forall N \in \mathbb{N}, \exists n \geq N$ *such that* $p_n \in B_r(p)$.
(ii) $\exists n_1 < n_2 < \cdots$ *such that* $p_{n_k} \to p$.

Proof.

(i) $\Rightarrow$ (ii) Set $n_0 = 0$. By (i), recurse $n_k > n_{k-1}$ with $p_{n_k} \in B_{1/k}(p)$; then $d(p_{n_k}, p) < 1/k$, so $p_{n_k} \to p$.
(ii) $\Rightarrow$ (i) If $p_{n_k} \to p$, then for any $r > 0$ and $N \in \mathbb{N}$, choose k with $n_k \geq N$ and $d(p_{n_k}, p) < r$. Thus $p_{n_k} \in B_r(p)$, so (i) holds.

Structural Commentary: Frequent visits are exactly enough to extract a convergent subsequence.
Key Insight: Limits require all sufficiently late terms; limit points require arbitrarily late witnesses. □

Theorem 3.4.3. *Basic Properties of Metric Convergence*
Let X be a metric space and $(p_n) \subseteq X$.

(i) $p_n \to p \Longleftrightarrow \forall r > 0, \exists N \in \mathbb{N}$ *such that* $n \geq N \Longrightarrow p_n \in B_r(p)$.
(ii) If $p_n \to p$ *and* $p_n \to p'$, *then* $p = p'$.
(iii) If $E \subseteq X$ *and* $p \in E^{\text{acc}}$, *then* $\exists (q_n) \subseteq E \setminus \{p\}$ *such that* $q_n \to p$.

Proof.

(i) ($\Rightarrow$) If $p_n \to p$ and $r > 0$, take $\varepsilon = r$ in the definition.
(i) ($\Leftarrow$) If the ball condition holds, then for every $\varepsilon > 0$ take $r = \varepsilon$, giving $d(p_n, p) < \varepsilon$ eventually.
(ii) Given $\varepsilon > 0$, choose N, N' with $n \geq N \Longrightarrow d(p_n, p) < \varepsilon/2$ and $n \geq N' \Longrightarrow d(p_n, p') < \varepsilon/2$. If $n \geq \max\{N, N'\}$, then $d(p, p') \leq d(p, p_n) + d(p_n, p') < \varepsilon$, so $d(p, p') = 0$.
(iii) Since $p \in E^{\text{acc}}$, for each $n \in \mathbb{N}$ choose $q_n \in E \setminus \{p\}$ with $d(q_n, p) < 1/n$. Then $q_n \to p$.

Structural Commentary: Eventual containment gives convergence; triangle inequality gives uniqueness.
Key Insight: A limit point can always be approached by a sequence of points from the set. □

Corollary 3.4.4. *Subsequences Preserve Metric Limits*
If $p_n \to p$ in a metric space, then every subsequence (p_{n_k}) also converges to p.

Proof. Given $\varepsilon > 0$, choose N such that $n \geq N$ implies $d(p_n, p) < \varepsilon$.
Since $n_k \to \infty$, all sufficiently large k satisfy $n_k \geq N$, hence $d(p_{n_k}, p) < \varepsilon$.
Structural Commentary: A subsequence is only a delayed tail of the same sequence.
Key Insight: Eventual estimates survive deleting terms. □

Proposition 3.4.5. *Discrete-Metric Convergence*
Let X carry the discrete metric $d(x,y) = 0$ if $x = y$ and $d(x,y) = 1$ if $x \neq y$.
For $(p_n) \subset X$ and $p \in X$, $p_n \to p$ if and only if $\exists N \in \mathbb{N}$ such that $n \geq N \Longrightarrow p_n = p$.

Proof.

($\Rightarrow$) Suppose $p_n \to p$. For $\varepsilon = 1/2$, choose $N \in \mathbb{N}$ such that $n \geq N \Longrightarrow d(p_n, p) < 1/2$.
Since d takes only the values $0, 1$, $d(p_n, p) = 0$, hence $p_n = p$ for all $n \geq N$.
($\Leftarrow$) Suppose $\exists N \in \mathbb{N}$ s.t. $n \geq N \Longrightarrow p_n = p$. Then $\forall \varepsilon > 0$, $n \geq N$ gives $d(p_n, p) = 0 < \varepsilon$, so $p_n \to p$.

Structural Commentary: The discrete metric has no room for distinct points to approach one another.
Key Insight: In the discrete metric, $d(x,y) < 1 \iff x = y$. □

Proposition 3.4.6. *Sequential Closure Criterion*
Let $E \subseteq X$, where X is a metric space, and let $p \in X$. Then $p \in \overline{E} \iff \exists (p_n)_{n \geq 1}$ with $p_n \in E$ $\forall n$, $p_n \to p$.

Proof.

($\Rightarrow$) If $p \in \overline{E}$, choose $p_n \in E \cap B_{1/n}(p)$ for $n \geq 1$ by Principle 6.5.1. Then $d(p_n, p) < 1/n$, so $p_n \to p$.
($\Leftarrow$) If $p_n \in E$ and $p_n \to p$, then every ball $B_r(p)$ contains some p_n.
Thus p is adherent to E by Definition 3.1.25, so $p \in \overline{E}$.

Structural Commentary: Closure can be read either by neighborhoods or by sequences.
Key Insight: Adherence supplies one witness at every scale. □

Theorem 3.4.7. *Closed Sets and Limit Points via Sequences*
Let X be a metric space and $E \subseteq X$.

(i) $p \in E^{\text{acc}} \iff \exists (p_n) \subseteq E \setminus \{p\}$ *such that* $p_n \to p$.
(ii) E *is closed* $\iff$ *every* $(p_n) \subseteq E$ *with* $p_n \to p \in X$ *satisfies* $p \in E$.

Proof.

(i) ($\Rightarrow$) If $p \in E^{\text{acc}}$, choose $p_n \in (E \setminus \{p\}) \cap B_{1/n}(p)$ for each n. Then $d(p_n, p) < 1/n$, so $p_n \to p$.
(i) ($\Leftarrow$) If $(p_n) \subseteq E \setminus \{p\}$ and $p_n \to p$, then every $B_r(p)$ contains some $p_n \neq p$. Hence $p \in E^{\text{acc}}$.
(ii) ($\Rightarrow$) If E is closed, $(p_n) \subseteq E$, and $p_n \to p$, then Proposition 3.4.6 gives $p \in \overline{E} = E$.
(ii) ($\Leftarrow$) If every convergent sequence in E has limit in E, then Proposition 3.4.6 gives $\overline{E} \subseteq E$. QED.

Structural Commentary: In metric spaces, closedness can be tested by sequences.
Key Insight: Balls at radii $1/n$ convert local information into a sequence. □

4 Compactness and Completeness

4.1 Compactness, Completeness, and Total Boundedness

Compactness extracts finite or convergent structure from infinite behavior; completeness asks whether Cauchy approximations converge internally.

Definition 4.1.1. *Open Covers and Compact Sets*
Let $E \subseteq X$, where X is a metric space.
An *open cover* of E is a collection $\{G_\alpha\}_{\alpha \in A}$ of open subsets of X such that $E \subseteq \bigcup_{\alpha \in A} G_\alpha$. A set $K \subseteq X$ is *compact* if every open cover of K has a finite subcover: $K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}$ for some finite subcollection.



Figure 132: Compactness extracts a finite subcover.

Definition 4.1.2. *Sequential Compactness*
A metric space is *sequentially compact* if every sequence has a convergent subsequence.

Theorem 4.1.3. *Compactness Produces Convergent Subsequences*

(i) If (p_n) is a sequence in a compact metric space X, then $p_{n_i} \to p$ for some subsequence and some $p \in X$.

(ii) If $(\mathbf{x}_n)$ is bounded in $\mathbb{R}^k$, then it has a convergent subsequence.

Proof.

(i) Let $E = \{p_n : n \in \mathbb{N}\}$. If E is finite, choose a value p attained infinitely often; the corresponding constant subsequence converges. If E is infinite, Theorem 4.1.34 gives a limit point $p \in X$ of E. By Theorem 3.1.18, choose $n_i > n_{i-1}$ with $d(p_{n_i}, p) < 1/i$. Then $p_{n_i} \to p$.

(ii) The range of $(\mathbf{x}_n)$ has compact closure by Theorem 4.2.5; apply (i) in that closure.

Structural Commentary: Compactness turns arbitrary infinite behavior into an accumulating subsequence.
Key Insight: In $\mathbb{R}^k$, boundedness is enough because bounded sets sit inside compact closed cells. □

Theorem 4.1.4. *Subsequential Limits Form a Closed Set*
For a sequence (p_n) in a metric space X, the set of all subsequential limits is closed.

Proof. Let E^* be the set of subsequential limits and let q be a limit point of E^*. Set $n_0 = 0$.
For each i, choose $x_i \in E^*$ with $d(x_i, q) < 2^{-i}$. Since x_i is a subsequential limit, choose $n_i > n_{i-1}$ such that $d(p_{n_i}, x_i) < 2^{-i}$. Then $d(p_{n_i}, q) < 2^{1-i}$, so $p_{n_i} \to q$, and $q \in E^*$.
Structural Commentary: A limit of subsequential limits is reached by diagonalizing through those subsequences.
Key Insight: Choose later and closer terms at the same time. □

Definition 4.1.5. *Cauchy Sequence*
A sequence (p_n) in a metric space X is *Cauchy* if $\forall \varepsilon > 0$ $\exists$ integer N s.t. $n \geq N,\ m \geq N \implies d(p_n, p_m) < \varepsilon$.

Definition 4.1.6. *Diameter*
Let E be a nonempty subset of a metric space X. The *diameter* of E is $\operatorname{diam} E := \sup\{d(p, q) : p, q \in E\}$, with the supremum interpreted in the extended real sense when the set of distances is unbounded.

Proposition 4.1.7. *Basic Diameter Properties*
Let $\emptyset \neq E, F \subseteq X$.

(i) E is bounded $\iff \operatorname{diam} E < \infty \iff \exists q \in X, \exists M > 0$ such that $E \subseteq B_M(q)$.
(ii) $\operatorname{diam} E = 0 \iff |E| = 1$.
(iii) $E \subseteq F \implies \operatorname{diam} E \leq \operatorname{diam} F$.
(iv) E, F bounded $\implies E \cup F$ bounded.

Proof.

(i) If $E \subseteq B_M(q)$, then $d(x, y) \leq d(x, q) + d(q, y) < 2M$ for $x, y \in E$, so $\operatorname{diam} E < \infty$.
If $\operatorname{diam} E < \infty$ and $q \in E$, then $E \subseteq B_{\operatorname{diam} E + 1}(q)$.

(ii) $\operatorname{diam} E = 0 \iff d(x, y) = 0$ for all $x, y \in E \iff x = y$ for all $x, y \in E$; since $E \neq \emptyset$, this is $|E| = 1$.

(iii) $\{d(x, y) : x, y \in E\} \subseteq \{d(x, y) : x, y \in F\}$, so monotonicity of sup gives $\operatorname{diam} E \leq \operatorname{diam} F$.

(iv) Choose $e \in E$, $f \in F$, and set $D := \operatorname{diam} E + d(e, f) + \operatorname{diam} F < \infty$. For $x, y \in E \cup F$, either x, y lie in the same set, or after swapping, $x \in E$, $y \in F$, and $d(x, y) \leq d(x, e) + d(e, f) + d(f, y) \leq D$.
Thus $\operatorname{diam}(E \cup F) < \infty$, so $E \cup F$ is bounded by (i).

Structural Commentary: Diameter is a single-number form of boundedness for nonempty sets.
Key Insight: Cross-distances between two bounded sets are controlled by one bridge distance. □

Proposition 4.1.8. *Cauchy Sequences and Tail Diameters*
For (p_n) in a metric space X, set $E_N := \{p_j : j \geq N\}$. Then (p_n) is Cauchy if and only if $\operatorname{diam} E_N \to 0$.

Proof.

($\Rightarrow$) Given $\varepsilon > 0$, choose N_0 so that $m, n \geq N_0$ implies $d(p_m, p_n) < \varepsilon/2$.
If $N \geq N_0$, then every pair in E_N has distance $< \varepsilon/2$, so $\operatorname{diam} E_N < \varepsilon$.

($\Leftarrow$) Given $\varepsilon > 0$, choose N_0 with $\operatorname{diam} E_{N_0} < \varepsilon$.
If $m, n \geq N_0$, then $p_m, p_n \in E_{N_0}$, hence $d(p_m, p_n) \leq \operatorname{diam} E_{N_0} < \varepsilon$.

Structural Commentary: Cauchy behavior is shrinking diameter of sequence tails.
Key Insight: The limit is not named; only mutual closeness is measured. □

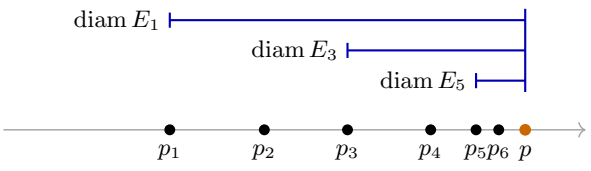


Figure 133: Tail diameters $\operatorname{diam} E_N$ shrink toward 0.

Theorem 4.1.9. *Nested Compact Sets with Shrinking Diameter*
Let X be a metric space.

(i) If $\emptyset \neq E \subseteq X$, then $\operatorname{diam} \overline{E} = \operatorname{diam} E$.
(ii) If $K_1 \supseteq K_2 \supseteq \cdots$ are nonempty compact subsets of X and $\operatorname{diam} K_n \to 0$, then $|\bigcap_{n=1}^{\infty} K_n| = 1$.

Proof.

(i) Since $E \subseteq \overline{E}$, $\operatorname{diam} E \leq \operatorname{diam} \overline{E}$. For $p, q \in \overline{E}$ and $\varepsilon > 0$, choose $p', q' \in E$ with $d(p, p'), d(q, q') < \varepsilon$; then $d(p, q) \leq \operatorname{diam} E + 2\varepsilon$. Hence $\operatorname{diam} \overline{E} \leq \operatorname{diam} E$.
(ii) Set $I := \bigcap_{n \geq 1} K_n$. Corollary 4.1.33 gives $I \neq \emptyset$. If $x, y \in I$, then $d(x, y) \leq \operatorname{diam} K_n$ for all n, so $d(x, y) = 0$ as $\operatorname{diam} K_n \to 0$. Thus $x = y$, hence $|I| = 1$.

Structural Commentary: Closure preserves diam; $K_n \downarrow$ with $\operatorname{diam} K_n \to 0$ forces a singleton.
Key Insight: Compactness gives existence; shrinking diameter gives uniqueness. □

Theorem 4.1.10. *Cauchy Criterion and Completeness Results*

(i) In any metric space, every convergent sequence is Cauchy.
(ii) If X is a compact metric space, then every Cauchy sequence in X converges to a point of X.
(iii) In $\mathbb{R}^k$, every Cauchy sequence converges.

Proof.

(i) If $p_n \to p$, choose N with $d(p_n, p) < \varepsilon/2$ for $n \geq N$. For $m, n \geq N$, $d(p_n, p_m) < \varepsilon$.
(ii) Let $E_N := \{p_j : j \geq N\}$ and $F_N := \overline{E_N}$. Then $\operatorname{diam} F_N \to 0$, the F_N are compact and nested, and Theorem 4.1.9 gives $\bigcap_N F_N = \{p\}$. If $\operatorname{diam} F_N < \varepsilon$, then $p, p_n \in F_N$ for $n \geq N$, so $d(p_n, p) < \varepsilon$.
(iii) For Cauchy $(\mathbf{x}_n) \subseteq \mathbb{R}^k$, choose N with $\operatorname{diam}\{\mathbf{x}_j : j \geq N\} < 1$. Its range is bounded, hence has compact closure by Theorem 4.2.5; apply (ii).

Structural Commentary: Cauchy controls $\operatorname{diam} E_N$; compactness turns $\bigcap_N F_N$ into the limit.
Key Insight: In $\mathbb{R}^k$, the Cauchy criterion is equivalent to convergence. □

Lemma 4.1.11. *A Cauchy Sequence with a Convergent Subsequence Converges*
Let (p_n) be a Cauchy sequence in a metric space X. If some subsequence (p_{n_j}) converges to $p \in X$, then $p_n \to p$.

Proof. Given $\varepsilon > 0$, choose N_0 so that $m, n \geq N_0$ implies $d(p_m, p_n) < \varepsilon/2$.
Choose j with $n_j \geq N_0$ and $d(p_{n_j}, p) < \varepsilon/2$. Then for every $n \geq N_0$, $d(p_n, p) \leq d(p_n, p_{n_j}) + d(p_{n_j}, p) < \varepsilon$.
Structural Commentary: The Cauchy tail follows any convergent subsequence to the same limit.
Key Insight: Mutual closeness plus one convergent thread forces the whole tail to converge. □

Proposition 4.1.12. *Boundedness and Distances of Cauchy Sequences*
Let $(p_n), (q_n)$ be Cauchy sequences in a metric space X.

(i) Cauchy boundedness: the range of $n \mapsto p_n$, $\operatorname{ran}(p_n) := \{p_n : n \in \mathbb{N}\}$ is bounded.
(ii) distance convergence: the sequence $n \mapsto d(p_n, q_n)$ converges in $\mathbb{R}$.
(iii) nonconverse: there exists a bounded sequence in $\mathbb{R}$ that is not Cauchy.

Proof.

(i) Choose N with $m, n \geq N \Longrightarrow d(p_m, p_n) < 1$.
Then $\{p_n : n \geq N\} \subseteq B_1(p_N)$, and $\{p_1, \ldots, p_{N-1}\}$ is finite, so $\operatorname{ran}(p_n)$ is bounded.
(ii) For $m, n \in \mathbb{N}$, $|d(p_n, q_n) - d(p_m, q_m)| \leq d(p_n, p_m) + d(q_n, q_m)$. Hence $(d(p_n, q_n))$ Cauchy in $\mathbb{R}$, converges.
(iii) Let $a_n := (-1)^n$. Then $|a_n| \leq 1$ for all n, but with $\varepsilon = 1$, for every N there exist $m, n \geq N$ with $|a_m - a_n| = 2$; hence (a_n) is bounded but not Cauchy.

Structural Commentary: Cauchy tails are bounded; Cauchy-pair distances converge.
Key Insight: The four-point triangle estimate controls distance sequences. □

Definition 4.1.13. *Complete Metric Space*
A metric space X is *complete* if every Cauchy sequence in X converges to a point of X.
Compact metric spaces and $\mathbb{R}^k$ are complete (by Theorem 4.1.10).

Proposition 4.1.14. *Completeness and Closed Subspaces*
Let (X, d) be a metric space, $E \subseteq X$, and $d_E := d|_{E \times E}$.

(i) If (E, d_E) is complete, then E is closed in X, equivalently $E = \overline{E}^X$.
(ii) If (X, d) is complete, then (E, d_E) is complete if and only if E is closed in X, equivalently $E = \overline{E}^X$.

Proof.

(i) Assume (E, d_E) complete. Let $p \in \overline{E}^X$. Choose $(p_n) \subseteq E$ with $d(p_n, p) < 1/n$.
Then $p_n \to p$ in X, hence (p_n) is Cauchy in (E, d_E). Thus $p_n \to q \in E$ in (E, d_E), hence also in X.
Uniqueness in X gives $p = q \in E$. Therefore $\overline{E}^X \subseteq E$, so $E = \overline{E}^X$.

(ii) Assume (X, d) complete. By (i), (E, d_E) complete $\Rightarrow E = \overline{E}^X$.
Conversely, if $E = \overline{E}^X$ and $(p_n) \subseteq E$ is Cauchy in (E, d_E), then (p_n) is Cauchy in X, so $p_n \to p \in X$.
Since $p \in \overline{E}^X = E$, every Cauchy sequence in (E, d_E) converges in (E, d_E).

Structural Commentary: Completeness gives the limit in X; closedness keeps it in E.
Key Insight: In a complete ambient space, E is complete exactly when $\overline{E} = E$. □

Example 4.1.15. *Finite-Dimensional Complete Metric Spaces*
For $k \in \mathbb{N}$, $(\mathbb{R}^k, d_2)$ and $(\mathbb{C}^k, d_2)$ are complete; in particular, so are $\mathbb{R}$ and $\mathbb{C}$.

Proof. Completeness of $\mathbb{R}$ is Theorem 2.1.16.
If $z_n = x_n + iy_n$ is Cauchy in $\mathbb{C}$, then (x_n) and (y_n) are Cauchy in $\mathbb{R}$, so $z_n \to x + iy \in \mathbb{C}$.
Theorem 4.1.10 gives completeness of $\mathbb{R}^k$, and the isometry $\mathbb{C}^k \cong \mathbb{R}^{2k}$ gives completeness of $\mathbb{C}^k$.
Structural Commentary: Finite-dimensional completeness is coordinatewise completeness.
Key Insight: A Cauchy sequence in finitely many coordinates converges coordinate by coordinate. □

Example 4.1.16. *Two Metrics on $C[0,1]$*
For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $C_{\mathbb{F}} := C([0,1], \mathbb{F})$ is complete under $d_\infty(f,g) := \sup_{x\in[0,1]} |f(x) - g(x)|$ by Theorem 8.1.1, but not under $d_1(f,g) := \int_0^1 |f(x) - g(x)|\, dx$.

Proof. For $n \geq 2$, define $f_n \in C_{\mathbb{F}}$ by

$$f_n(x) := \begin{cases} 0, & 0 \leq x \leq \frac{1}{2}, \\ n(x - \frac{1}{2}), & \frac{1}{2} \leq x \leq \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \leq x \leq 1. \end{cases}$$

Let $s := \mathbf{1}_{(1/2,1]}$. Then $d_1(f_n, s) = 1/(2n)$, hence (f_n) is d_1-Cauchy. If $f_n \to g \in C_{\mathbb{F}}$ in d_1, then $d_1(g, s) = 0$, so $g = s$ a.e. Continuity forces $g = 0$ on $[0, 1/2]$ and $g = 1$ on $(1/2, 1]$, impossible at $1/2$.
Structural Commentary: The integral metric permits a continuous ramp to collapse to a jump.
Key Insight: Supremum control preserves continuity; integral control can miss jumps. □

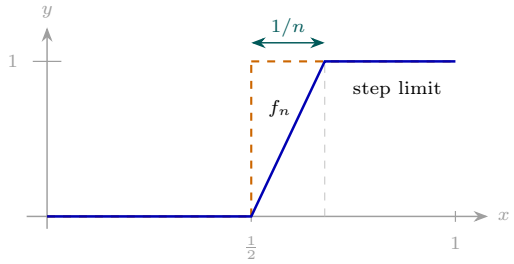


Figure 134: The continuous ramps f_n converge in the integral metric to a discontinuous step function.

Example 4.1.17. *Completeness of ℓ^∞ and ℓ^p*
For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $(\ell^\infty(\mathbb{F}), \|\cdot\|_\infty)$ and $(\ell^p(\mathbb{F}), \|\cdot\|_p)$, $1 \leq p < \infty$, are complete.

Proof. For ℓ^∞, let $\mathbf{x}^{(n)} = (x_j^{(n)})_j$ be Cauchy. Then $x_j^{(n)} \to y_j \in \mathbb{F}$ for each j.
If $\left\|\mathbf{x}^{(n)} - \mathbf{x}^{(m)}\right\|_\infty < \varepsilon$ for $m, n \geq N$, then $\left|x_j^{(n)} - y_j\right| \leq \varepsilon$ for all j, so $\left\|\mathbf{x}^{(n)} - \mathbf{y}\right\|_\infty \leq \varepsilon$.
Taking $n = N$ gives $\mathbf{y} \in \ell^\infty$, hence $\mathbf{x}^{(n)} \to \mathbf{y}$.
For ℓ^p, let $\mathbf{x}^{(n)}$ be Cauchy and define $\mathbf{y} = (y_j)$ by coordinatewise limits.
If $\left\|\mathbf{x}^{(n)} - \mathbf{x}^{(m)}\right\|_p < \varepsilon$ for $m, n \geq N$, then for each r, $\sum_{j=1}^{r} \left|x_j^{(n)} - y_j\right|^p = \lim_{m\to\infty} \sum_{j=1}^{r} \left|x_j^{(n)} - x_j^{(m)}\right|^p \leq \varepsilon^p$.
Letting $r \to \infty$ gives $\left\|\mathbf{x}^{(n)} - \mathbf{y}\right\|_p \leq \varepsilon$ for $n \geq N$. Taking $n = N$ gives $\mathbf{y} \in \ell^p$, hence $\mathbf{x}^{(n)} \to \mathbf{y}$.
Structural Commentary: Completeness of scalar coordinates combines with uniform tail control.
Key Insight: Norm-Cauchy control keeps coordinatewise limits inside the space. □

Definition 4.1.18. *Contraction*
Let X be a metric space with metric d.
A map $\varphi : X \to X$ is a *contraction* if $\exists$ a constant $0 \le c < 1$ such that $d(\varphi(x), \varphi(y)) \le c\, d(x, y)$ for $x, y \in X$.
Structural Commentary: Bound $c < 1$ excludes nonexpansive maps and is the hypothesis in Theorem 4.1.19.

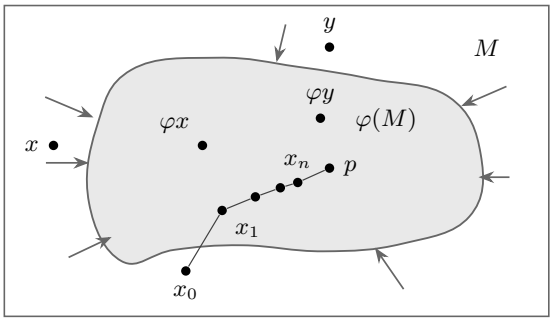


Figure 135: A contraction pulls iterates toward the unique fixed point.

Theorem 4.1.19. *Contraction Principle*
If X is a complete metric space and $\varphi : X \to X$ is a contraction, then $\exists$ a unique $x \in X$ such that $\varphi(x) = x$.

Proof. Choose $x_0 \in X$ and define $x_{n+1} = \varphi(x_n)$.
If $c < 1$ is a contraction constant, induction gives $d(x_{n+1}, x_n) \le c^n d(x_1, x_0)$.
For $n < m$, the triangle inequality and the finite geometric sum give $d(x_n, x_m) \le \sum_{j=n}^{m-1} d(x_j, x_{j+1}) \le \frac{c^n}{1-c} d(x_1, x_0)$.
Thus (x_n) is Cauchy, so completeness gives $x_n \to x \in X$.
Since a contraction is continuous, $\varphi(x) = \lim \varphi(x_n) = \lim x_{n+1} = x$.
If $\varphi(x) = x$ and $\varphi(y) = y$, then $d(x, y) \le c\, d(x, y)$, hence $d(x, y) = 0$ and $x = y$.
Structural Commentary: Iteration turns the contraction estimate into a geometric Cauchy estimate.
Key Insight: Completeness supplies the limiting point; the strict constant $c < 1$ forces uniqueness. □

Perspective 4.1.20. *Fixed Points as Existence Proofs*
A contraction proof solves $x = \varphi(x)$ by iteration in a complete metric space. This is the template behind the inverse function theorem, Theorem 4.1.4.

Example 4.1.21. *A Noncomplete Metric Space*
The rational numbers with $d(x, y) = |x - y|$ are not complete.
Let (r_n) be the decimal truncation sequence $1.4,\ 1.41,\ 1.414,\ 1.4142, \ldots$, for $\sqrt{2}$.
This is a concrete instance of the truncation squeeze in Proposition 1.1.19. If $k, n \ge N$, then $|r_k - r_n| \le 10^{-N}$, so (r_n) is Cauchy in $\mathbb{Q}$. As a sequence in $\mathbb{R}$, it converges to $\sqrt{2}$.
If it converged to some $r \in \mathbb{Q}$ in the inherited metric, uniqueness of limits in $\mathbb{R}$ would force $r = \sqrt{2}$, contradicting Proposition 1.1.1. Thus $\mathbb{Q}$ is not complete in the inherited metric. *Structural Commentary*: The decimal approximants are Cauchy before their missing limit is available inside $\mathbb{Q}$.
Key Insight: Completeness asks whether every internally Cauchy approximation has an internal endpoint.

Perspective 4.1.22. *Completion by Cauchy Sequences*
Complete by identifying Cauchy sequences with $d(p_n, q_n) \to 0$ and setting $D([(p_n)], [(q_n)]) = \lim_{n\to\infty} d(p_n, q_n)$.
Constant sequences embed the original space; triangle inequalities and diagonals give a complete metric.

Theorem 4.1.23. *Metric Completion*

For every metric space (X, d), $\exists$ complete metric space $(\widehat{X}, \widehat{d})$, isometry $\iota : X \to \widehat{X}$ with $\overline{\iota(X)}^{\widehat{X}} = \widehat{X}$.
If (Y, d_Y, τ), (Z, d_Z, σ) are two such completions, then $\exists!$ bijective isometry $\Phi : Y \to Z$ s.t. $\Phi \circ \tau = \sigma$.

Proof.

i. *Existence*. Let $\mathrm{CS}(X)$ be the Cauchy sequences in X.
For $a = (a_n), b = (b_n) \in \mathrm{CS}(X)$, $|d(a_n, b_n) - d(a_m, b_m)| \le d(a_n, a_m) + d(b_n, b_m)$, so $D(a, b) := \lim_{n\to\infty} d(a_n, b_n)$ exists. Put $a \sim b \Leftrightarrow D(a, b) = 0$, $\widehat{X} := \mathrm{CS}(X)/\!\sim$, and $\widehat{d}([a], [b]) := D(a, b)$.
The estimate gives well-definedness; metric axioms pass to limits. Map $\iota(x) := [(x, x, \ldots)]$ is an isometry.
If $a = (a_n)$ and $d(a_n, a_N) < \varepsilon$ for $n \ge N$, then $\widehat{d}(\iota(a_N), [a]) \le \varepsilon$, hence $\overline{\iota(X)}^{\widehat{X}} = \widehat{X}$.
Let $([a^{(k)}])$ be Cauchy in $\widehat{X}$. Choose $x_k \in X$ with $\widehat{d}(\iota(x_k), [a^{(k)}]) < 1/k$.

Since $d(x_k, x_l) \le 1/k + \widehat{d}([a^{(k)}], [a^{(l)}]) + 1/l$, (x_k) is Cauchy in X,
and $\widehat{d}([a^{(k)}], [(x_j)]) \le 1/k + \limsup_{j\to\infty} d(x_k, x_j) \to 0$. Thus $([a^{(k)}]) \to [(x_j)]$, so $(\widehat{X}, \widehat{d})$ is complete.

ii. *Uniqueness*. Let (Y, d_Y, τ) and (Z, d_Z, σ) be completions.
Define $\rho_0 : \tau(X) \to \sigma(X)$ by $\rho_0(\tau x) := \sigma x$; it is an isometry.
For $y \in Y$, choose $y_n \in \tau(X)$ with $y_n \to y$. Since $\rho_0(y_n)$ is Cauchy in Z, define $\Phi(y) := \lim_{n\to\infty} \rho_0(y_n)$.
If $z_n \in \tau(X)$ and $z_n \to y$, then $d_Z(\rho_0(y_n), \rho_0(z_n)) = d_Y(y_n, z_n) \to 0$, so Φ is well defined.
If $y'_n \to y'$, then $d_Z(\Phi(y), \Phi(y')) = \lim_{n\to\infty} d_Z(\rho_0(y_n), \rho_0(y'_n)) = \lim_{n\to\infty} d_Y(y_n, y'_n) = d_Y(y, y')$.
Hence Φ is an isometry extending ρ_0. Its image is complete, hence closed in Z, and contains the dense set $\sigma(X)$; therefore $\Phi(Y) = Z$. Uniqueness follows from continuity and density of $\tau(X)$.

Structural Commentary: Missing limits are installed as equivalence classes of Cauchy approximations.
Key Insight: Completion adds Cauchy limits and is unique over the dense copy of X. □

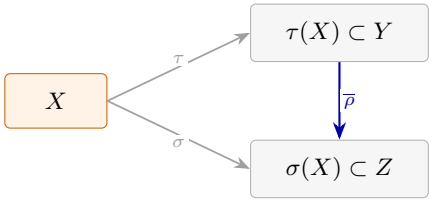


Figure 136: For completions Y, Z, the dense-copy isometry $\sigma\tau^{-1} : \tau(X) \to \sigma(X)$ extends uniquely to $\overline{\rho} : Y \cong Z$.

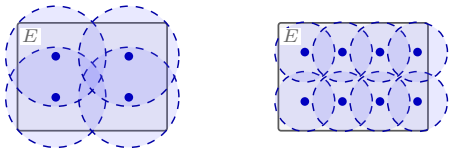


Figure 137: Finite ε- and δ-nets cover E, with $0 < \delta < \varepsilon$.

Definition 4.1.24. *Total Boundedness*
Let (X, d) be a metric space. It is *totally bounded* if, $\forall \varepsilon > 0$, $\exists$ a finite set $F \subseteq X$ such that $X \subseteq \bigcup_{x\in F} B_\varepsilon(x)$.
F is a finite ε-net. $E \subseteq X$ totally bounded if same condition holds with finite $F \subseteq E$ and X.

Theorem 4.1.25. *Total Boundedness Implies Boundedness*
Let X be a metric space. If X is totally bounded, then X is bounded.

Proof. If $X = \emptyset$, this is trivial. Otherwise cover X by finitely many 1-balls $B_1(x_i)$ and fix x_1.
Then $d(x, x_1) < 1 + \max_i d(x_i, x_1)$ for every $x \in X$.
Structural Commentary: One finite cover at scale 1 gives one global bound.
Key Insight: Total boundedness is boundedness with finite control at every scale. □

Theorem 4.1.26. *Compactness as Complete Finite Resolution*
Let X be a metric space. Then, X is compact $\iff$ X is complete and totally bounded.

Proof.

($\Rightarrow$) Let (p_k) be Cauchy in X. By Theorem 4.1.10, compactness forces (p_k) to converge, so X is complete.
Given $\varepsilon > 0$, cover $\{B_\varepsilon(x) : x \in X\}$ has finite subcover $B_\varepsilon(y_1), \dots, B_\varepsilon(y_m)$; hence X is totally bounded.

($\Leftarrow$) Let $(x_n) \subseteq X$. Choose nested subsequences $(x_k^{(j)})_{k\ge1}$ and points $a_j \in X$ with $x_k^{(j)} \in B_{1/j}(a_j)$ $\forall k$.
Diagonal $y_j := x_j^{(j)}$ satisfies $y_k, y_\ell \in B_{1/j}(a_j)$ for $k, \ell \ge j$, so it is Cauchy and converges by completeness; hence X is sequentially compact. Now let $\{U_\alpha\}$ cover X. If no $\delta > 0$ has every $B_\delta(x)$ contained in some U_α, choose x_n with $B_{1/n}(x_n) \nsubseteq U_\alpha$ $\forall \alpha$. Passing to $x_{n_k} \to x \in U_{\alpha_0}$, and openness gives $B_{1/n_k}(x_{n_k}) \subseteq U_{\alpha_0}$ for large k, a contradiction. Thus such δ exists.
Choose a finite $\delta/2$-net $\{z_i\}_{i=1}^m$ and α_i with $B_\delta(z_i) \subseteq U_{\alpha_i}$; then $\{U_{\alpha_i}\}_{i=1}^m$ covers X.

Structural Commentary: Finite nets plus completeness force compactness.
Key Insight: Compact metric spaces are exactly complete spaces visible at every finite scale. □

Example 4.1.27. *An Open Cover Without a Finite Subcover*
For $x \in (0, 1)$, set $O_x := (x/2, 1)$. These sets cover $(0, 1)$, since $y \in O_x$ whenever $0 < x < y < 1$.
No finite subcollection covers $(0, 1)$: if $x' = \min\{x_1, \dots, x_n\}$, then $(0, x'/2]$ is missed by $O_{x_1} \cup \cdots \cup O_{x_n}$.

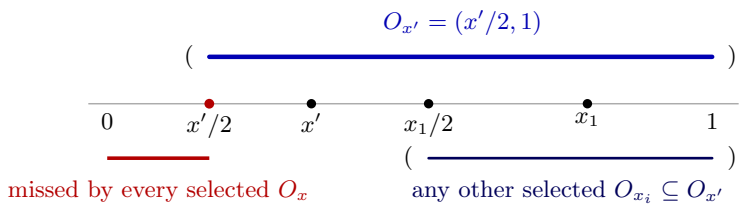


Figure 138: For finite choice, $x' = \min_i x_i$ makes $O_{x'}$ largest selected interval, but $(0, x'/2]$ remains uncovered.

Theorem 4.1.28. *Compactness Is Independent of the Ambient Space*
Suppose $K \subseteq Y \subseteq X$, where X is a metric space.
Then K is compact relative to X if and only if K is compact relative to Y.

Proof.
($\Rightarrow$) Let $\{V_\alpha\}_{\alpha \in A}$ be a relative-open cover of K in Y.
By Theorem 3.2.2, write $V_\alpha = Y \cap G_\alpha$ with G_α open in X. Since $\{G_\alpha\}_{\alpha \in A}$ covers K, compactness in X gives a finite $F \subseteq A$ such that $K \subseteq \bigcup_{\alpha \in F} G_\alpha$. As $K \subseteq Y$, $K \subseteq \bigcup_{\alpha \in F} V_\alpha$.
($\Leftarrow$) Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of K in X. Then $\{Y \cap G_\alpha\}_{\alpha \in A}$ is a relative-open cover of K in Y.
Compactness in Y gives a finite $F \subseteq A$ such that $K \subseteq \bigcup_{\alpha \in F}(Y \cap G_\alpha) \subseteq \bigcup_{\alpha \in F} G_\alpha$.
Structural Commentary: Relative covers are ambient covers intersected with Y.
Key Insight: Openness depends on the ambient space, but compactness does not. □

Theorem 4.1.29. *Compact Sets Are Closed*
Every compact subset of a metric space is closed.

Proof. Let K be compact in X. To prove K^c is open.
Fix $p \in K^c$. For each $q \in K$, put $V_q := B_{\frac{1}{2}d(p,q)}(p)$, $W_q := B_{\frac{1}{2}d(p,q)}(q)$.
The sets W_q form an open cover of K, so compactness gives $q_1, \ldots, q_n \in K$ with $K \subseteq W_{q_1} \cup \cdots \cup W_{q_n}$.
Let $V := V_{q_1} \cap \cdots \cap V_{q_n}$, which is open by Theorem 3.1.22.
No point of V lies in any W_{q_i}, by the triangle inequality. Hence $V \cap K = \emptyset$, so p is an interior point of K^c.
Structural Commentary: Compactness makes the separation finite.
Key Insight: A finite subcover lets local point-separations become one open neighborhood of the complement. □

Theorem 4.1.30. *Closed Subsets of Compact Sets Are Compact*
Suppose $F \subseteq K \subseteq X$, where F is closed in X and K is compact. Then F is compact.

Proof. Let $\{V_\alpha\}_{\alpha \in A}$ cover F. Since F^c is open, $\{V_\alpha\}_{\alpha \in A} \cup \{F^c\}$ covers K.
Thus $K \subseteq F^c \cup \bigcup_{\alpha \in F_0} V_\alpha$ for some finite $F_0 \subseteq A$, so $F \subseteq \bigcup_{\alpha \in F_0} V_\alpha$.
Structural Commentary: The open complement of a closed set covers exactly the part of K not under discussion.
Key Insight: Cover K, then discard the complement piece. □

Corollary 4.1.31. *Closed Intersections with Compact Sets*
If F is closed in X and K is compact in X, then $F \cap K$ is compact.

Proof. By Theorem 4.1.29, K is closed in X. Since F is closed, Theorem 3.1.22 gives that $F \cap K$ is closed in X. Also $F \cap K \subseteq K$, so Theorem 4.1.30 implies $F \cap K$ is compact.
Structural Commentary: The intersection is closed inside a compact set.
Key Insight: Closed subsets of compact sets remain compact. □

Theorem 4.1.32. *Finite Intersection Property for Compact Sets*
Let $\{K_\alpha\}_{\alpha \in A}$ be compact subsets of a metric space X, with $A \neq \emptyset$.
If $\bigcap_{\alpha \in F} K_\alpha \neq \emptyset$ for every finite nonempty $F \subseteq A$, then $\bigcap_{\alpha \in A} K_\alpha \neq \emptyset$.

Proof. Choose $\alpha_0 \in A$ and suppose $\bigcap_{\alpha \in A} K_\alpha = \emptyset$. Then $K_{\alpha_0} \subseteq \bigcup_{\alpha \in A} K_\alpha^c$. Since each K_α^c is open by Theorem 4.1.29, compactness of K_{α_0} gives a finite $F \subseteq A$ with $K_{\alpha_0} \subseteq \bigcup_{\alpha \in F} K_\alpha^c$. Hence $\bigcap_{\alpha \in F \cup \{\alpha_0\}} K_\alpha = \emptyset$, a contradiction.
Structural Commentary: Complements turn failure into a finite cover.
Key Insight: For compact sets, failure of intersection is already witnessed finitely. □

Corollary 4.1.33. *Nested Nonempty Compact Sets Intersect*
If $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ is a sequence of nonempty compact sets, then $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

Proof. Let $\emptyset \neq F \subseteq \mathbb{N}$ be finite and put $m := \max F$. Since the sequence is nested, $\bigcap_{n\in F} K_n = K_m \neq \emptyset$.

Thus $\{K_n\}_{n\in\mathbb{N}}$ has the finite intersection property, so Theorem 4.1.32 gives $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

Structural Commentary: Nestedness reduces finite intersections to one term.
Key Insight: A nested compact sequence has the finite intersection property. □

Theorem 4.1.34. *Infinite Subsets of Compact Sets Have Limit Points*
Let $K \subseteq X$ be compact in a metric space X. If $E \subseteq K$ is infinite, then $E^{\text{acc}} \cap K \neq \emptyset$.

Proof. Suppose $E^{\text{acc}} \cap K = \emptyset$. For each $q \in K$, choose $r_q > 0$ such that $B_{r_q}(q) \cap (E \setminus \{q\}) = \emptyset$, so $\#(B_{r_q}(q) \cap E) \leq 1$.
Since $\{B_{r_q}(q) : q \in K\}$ covers K, compactness gives $q_1, \ldots, q_n \in K$ with $K \subseteq \bigcup_{i=1}^{n} B_{r_{q_i}}(q_i)$.

As $E \subseteq K$, $E \subseteq \bigcup_{i=1}^{n} B_{r_{q_i}}(q_i)$, finite union of sets meeting E in at most one point. Thus E is finite, contradiction.
Structural Commentary: Compactness makes the local "at most one point" obstruction finite.
Key Insight: Infinite subsets of compact sets must accumulate somewhere inside the compact set. □

4.2 Heine–Borel and Euclidean Compactness

Theorem 4.2.1. *Nested Closed Intervals Intersect*
If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ is a sequence of closed intervals in $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. Let $I_n = [a_n, b_n]$. The set $\{a_n : n \in \mathbb{N}\}$ is nonempty and bounded above by each b_m.
Let $x = \sup\{a_n : n \in \mathbb{N}\}$. Then $a_m \leq x \leq b_m$ for every m, so $x \in I_m$ for all m.
Structural Commentary: Completeness traps a common point.
Key Insight: The least upper bound of the left endpoints lies in every interval. □

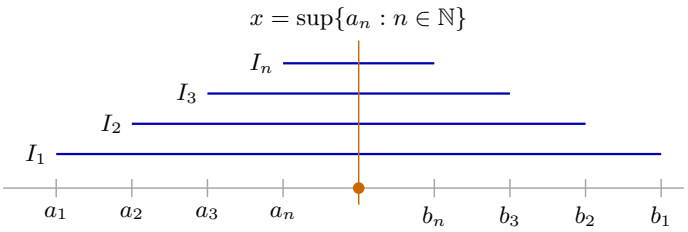


Figure 139: For nested closed intervals $[a_n, b_n]$, the supremum of the left endpoints lies between every a_n and b_n.

Theorem 4.2.2. *Nested k-Cells Intersect*
If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$ is a sequence of k-cells in $\mathbb{R}^k$, then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Proof. Let $I_n = \{\mathbf{x} \in \mathbb{R}^k : a_{n,j} \leq x_j \leq b_{n,j},\ 1 \leq j \leq k\}$. For each coordinate j, the intervals $[a_{n,j}, b_{n,j}]$ are nested, so Theorem 4.2.1 gives a point x_j^* lying in all of them. Then $\mathbf{x}^* = (x_1^*, \ldots, x_k^*)$ lies in every I_n.
Structural Commentary: A nested cell problem reduces coordinatewise to nested intervals.
Key Insight: Choose one persistent coordinate at a time. □

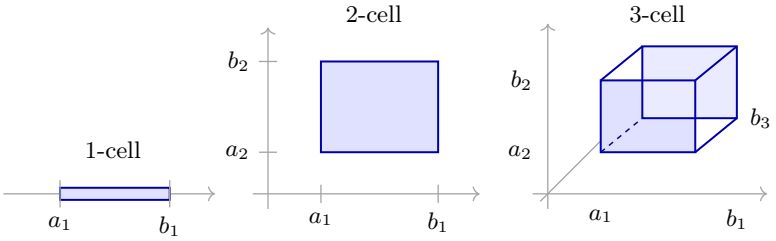


Figure 140: A 1-cell is a closed interval, a 2-cell is a coordinate rectangle, and a 3-cell is a coordinate box.

Theorem 4.2.3. *k-Cells Are Compact*
Every k-cell in $\mathbb{R}^k$ is compact.

Proof. Let $I = \{\mathbf{x} : a_j \le x_j \le b_j,\ 1 \le j \le k\}$ and set $\delta := \left(\sum_{j=1}^{k} (b_j - a_j)^2\right)^{1/2}$. Suppose an open cover $\{G_\alpha\}$ of I has no finite subcover. Bisect each coordinate interval. Among the resulting 2^k subcells, at least one cannot be finitely covered by the G_α; call it I_1. Continue inductively, obtaining nested cells $I \supseteq I_1 \supseteq I_2 \supseteq \cdots$ such that no I_n is finitely covered and every two points of I_n are within $2^{-n}\delta$ of each other.
By Theorem 4.2.2, choose $\mathbf{x}^* \in \bigcap_n I_n$. Since $\{G_\alpha\}$ covers I, some G_α contains $\mathbf{x}^*$. Because G_α is open, $B_r(\mathbf{x}^*) \subseteq G_\alpha$ for some $r > 0$. By the Archimedean property in Theorem 1.1.15, choose n with $n > \delta/r$; since $2^n \ge n$, we have $2^{-n}\delta < r$. Then $I_n \subseteq G_\alpha$, contradicting the construction of I_n.
Structural Commentary: Repeated bisection traps a point in cells whose diameters shrink to zero.
Key Insight: An open set around the trapped point eventually contains the whole small cell. □

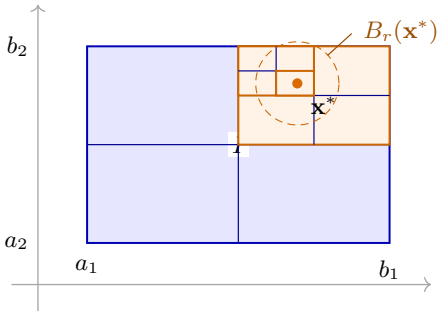


Figure 141: Repeated bisection traps a point.

Lemma 4.2.4. *Bounded Euclidean Sets Fit in k-Cells*
Every bounded subset $E \subseteq \mathbb{R}^k$ is contained in some k-cell.

Proof. If $E = \emptyset$, any k-cell contains E. Suppose $E \neq \emptyset$. By Definition 3.4.1, choose $\mathbf{q} = (q_1, \ldots, q_k) \in \mathbb{R}^k$ and $M > 0$ such that $\|\mathbf{p} - \mathbf{q}\|_2 < M$ for all $\mathbf{p} \in E$. Let $I := \{\mathbf{x} = (x_1, \ldots, x_k) : q_j - M \le x_j \le q_j + M,\ 1 \le j \le k\}$. For $\mathbf{p} = (p_1, \ldots, p_k) \in E$, $|p_j - q_j| \le \|\mathbf{p} - \mathbf{q}\|_2 < M$ for each j, so $\mathbf{p} \in I$.
Thus $E \subseteq I$, and I is a k-cell by Definition 3.1.9.
Structural Commentary: A Euclidean ball centered at $\mathbf{q}$ sits inside the coordinate box with side half-length M.
Key Insight: Boundedness supplies one radius; coordinates turn it into interval bounds. □

Theorem 4.2.5. *Heine–Borel in $\mathbb{R}^k$*
For $E \subseteq \mathbb{R}^k$, any one of the following three properties implies the other two:

(i) E is closed and bounded;
(ii) E is compact;
(iii) every infinite subset of E has a limit point in E.

Proof.

(*i*) $\Rightarrow$ (*ii*): Since E is bounded, Lemma 4.2.4 gives $E \subseteq I$ for some k-cell I. By Theorem 4.2.3, I is compact; since E is closed, Theorem 4.1.30 gives that E is compact.
(*ii*) $\Rightarrow$ (*iii*): This is Theorem 4.1.34.
(*iii*) $\Rightarrow$ (*i*): If E is unbounded, choose $\mathbf{x}_n \in E$ with $\|\mathbf{x}_n\|_2 > n$ and set $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$. If $\mathbf{x}_n \in B_R(\mathbf{y})$, then $n < \|\mathbf{y}\|_2 + R$; hence S has no limit point, contradicting (iii). Thus E is bounded. If E is not closed, choose $\mathbf{x}_0 \in E^{\text{acc}} \setminus E$ and $\mathbf{x}_n \in E$ with $0 < \|\mathbf{x}_n - \mathbf{x}_0\|_2 < 1/n$. Then $S := \{\mathbf{x}_n : n \in \mathbb{N}\}$ is infinite and $S^{\text{acc}} \subseteq \{\mathbf{x}_0\} \subseteq E^c$, contradicting (iii). Thus E is closed.

Structural Commentary: In $\mathbb{R}^k$, the three tests coincide.
Key Insight: Heine–Borel is the bridge between open-cover compactness and Euclidean bounded geometry. □

Corollary 4.2.6. *Heine–Borel on the Real Line*
For $E \subseteq \mathbb{R}$, the following are equivalent:

(i) E is closed and bounded;
(ii) E is compact;
(iii) every sequence in E has a subsequence converging to a point of E.

Proof.

(*i*) $\Rightarrow$ (*ii*) Since $E \subseteq \mathbb{R} = \mathbb{R}^1$ is closed and bounded, Theorem 4.2.5 with $k = 1$ implies E is compact.

(*ii*) $\Rightarrow$ (*iii*) Compactness and Theorem 4.1.26 give convergent subsequence; compactness keeps limit in E.
(*iii*) $\Rightarrow$ (*i*) Let $F \subseteq E$ be infinite. Choose distinct $x_n \in F$ using Principle 6.5.1. By (iii), some $x_{n_j} \to x \in E$. Since the x_n are distinct, x is a limit point of F in the sense of Definition 3.1.14. Hence every infinite subset of E has a limit point in E, so Theorem 4.2.5 gives that E is closed and bounded.

Structural Commentary: Real-line compactness equals closed-bounded sequential compactness.
Key Insight: Closed bounded sets keep sequences from escaping. □

Theorem 4.2.7. *Weierstrass Limit-Point Theorem*
Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.

Proof. By Lemma 4.2.4, a bounded set E lies in some k-cell I. By Theorem 4.2.3, I is compact.
Since E is an infinite subset of I, Theorem 4.1.34 gives a limit point of E in $I \subseteq \mathbb{R}^k$.
Structural Commentary: Boundedness places the set inside a compact cell.
Key Insight: Compact cells force bounded infinite sets to accumulate. □

4.3 Baire Category

Definition 4.3.1. *F_σ, G_δ, and Nowhere-Dense Sets*
For subsets of a metric space X:

(i) A is an *F_σ set* if $A = \bigcup_{n=1}^{\infty} F_n$ for closed sets F_n.
(ii) B is a *G_δ set* if $B = \bigcap_{n=1}^{\infty} G_n$ for open sets G_n.
(iii) E is *nowhere dense* if $(\overline{E})^\circ = \emptyset$; equivalently, no nonempty open set is contained in $\overline{E}$.

For $X = \mathbb{R}$, this says exactly that $\overline{E}$ contains no nonempty open interval.

Definition 4.3.2. *First Category*

A set $E \subseteq X$ is *of first category*, *meager*, or *thin*, if covered by countably many nowhere-dense sets: $E \subseteq \bigcup_{n=1}^{\infty} E_n$.

The Baire Category Theorem states that a nonempty complete metric space is not first category in itself.

Definition 4.3.3. *Residual, Thick, and Generic Sets*

A set $G \subseteq X$ is *residual*, or *thick*, if it contains a countable intersection $\bigcap_{n=1}^{\infty} O_n$ of dense open subsets of X.

A property is *generic* in X if the set of points satisfying it is residual.
By Theorem 4.3.5, residual subsets of a complete metric space are dense.
In this sense, meager sets are topologically small and residual sets are topologically large.

Theorem 4.3.4. *Dense Open Intersections in $\mathbb{R}$*
If $G_1, G_2, G_3, \ldots$ are dense open subsets of $\mathbb{R}$, then $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$.

Proof. We construct nested closed intervals. Since G_1 is dense and open, $G_1 \cap (0,1)$ contains a nonempty open interval; choose a closed interval $I_1 \subseteq G_1 \cap (0,1)$ with nonempty interior.
Given $I_n = [a_n, b_n]$, density of G_{n+1} gives a point of G_{n+1} in (a_n, b_n). Since G_{n+1} is open, choose a closed interval $I_{n+1} \subseteq G_{n+1} \cap (a_n, b_n)$ with nonempty interior and length $< 1/(n+1)$.
Then $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$, so Theorem 4.2.1 gives $x \in \bigcap_n I_n$. Since $I_n \subseteq G_n$ for every n, this x lies in $\bigcap_n G_n$.
Structural Commentary: Dense openness builds nested intervals; completeness keeps a limit point.
Key Insight: A countable dense-open intersection cannot be destroyed one interval at a time. □

Theorem 4.3.5. *Dense Open Intersections in Complete Metric Spaces*
Let (X,d) be complete, let $O_1, O_2, \ldots$ be dense open subsets of X, and let $U \subseteq X$ be nonempty and open. Then $U \cap \bigcap_{n=1}^{\infty} O_n \neq \emptyset$. Consequently, $\bigcap_{n=1}^{\infty} O_n$ is dense in X.

Proof. Choose $x_1 \in U \cap O_1$ and $0 < r_1 < 1$ with $C_1 := \overline{B}_{r_1}(x_1) \subseteq U \cap O_1$.
Inductively, if $C_n = \overline{B}_{r_n}(x_n) \subseteq U \cap O_1 \cap \cdots \cap O_n$ and $0 < r_n < 2^{-n}$, density and openness of O_{n+1} give $x_{n+1} \in B_{r_n}(x_n) \cap O_{n+1}$ and $0 < r_{n+1} < 2^{-(n+1)}$ such that $C_{n+1} := \overline{B}_{r_{n+1}}(x_{n+1}) \subseteq B_{r_n}(x_n) \cap O_{n+1}$.
Then $C_{n+1} \subseteq C_n$. If $m > n$, then $x_m \in B_{r_n}(x_n)$, so $d(x_m, x_n) < 2^{-n}$; hence (x_n) is Cauchy. Let $x_n \to x$.
Since each C_n is closed and contains all sufficiently late x_m, we have $x \in C_n$ for every n. Thus $x \in U \cap \bigcap_n O_n$.
Since every nonempty open U meets $\bigcap_n O_n$, the intersection is dense.
Structural Commentary: The interval proof becomes a nested closed-ball proof.
Key Insight: Dense open sets choose the next ball; completeness supplies the limiting point. □

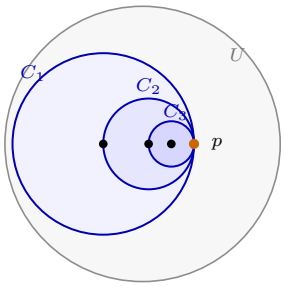


Figure 142: Nested closed balls shrink to p.

Theorem 4.3.6. *Baire Category Theorem for Complete Metric Spaces*
A nonempty complete metric space is not the union of a countable collection of nowhere-dense sets.

Proof. Suppose $X = \bigcup_n E_n$ with each E_n nowhere dense, and set $O_n := (\overline{E_n})^c$.
Then each O_n is dense open: it is open, and every nonempty open set meets it because no such set lies in $\overline{E_n}$.
By Theorem 4.3.5, choose $x \in \bigcap_n O_n$. Then $x \notin E_n$ for all n, contradicting $X = \bigcup_n E_n$.
Structural Commentary: Countably many nowhere-dense sets have dense open complements.
Key Insight: Completeness forbids countable assembly from thin pieces. □

Perspective 4.3.7. *Category in Function Spaces*
Baire category is a theorem about complete metric spaces, not just $\mathbb{R}$.
It will be reused for complete function spaces under the supremum metric.

Theorem 4.3.8. *Baire's Theorem for $\mathbb{R}$*
The real line $\mathbb{R}$ cannot be written as a countable union of nowhere-dense sets.

Proof. Assume $\mathbb{R} = \bigcup_n E_n$ with each E_n nowhere dense, and set $U_n := (\overline{E_n})^c$.
Each U_n is dense open: it is open, and every nonempty interval meets it because no such interval lies in $\overline{E_n}$.
By Theorem 4.3.4, choose $x \in \bigcap_n U_n$. Then $x \notin E_n$ for all n, contradicting $\mathbb{R} = \bigcup_n E_n$.
Structural Commentary: Nowhere-dense sets have dense open complements.
Key Insight: Countably many thin sets cannot exhaust the real line. □

Corollary 4.3.9. *Rationals and Irrationals by Category*
The set $\mathbb{Q}$ is F_σ but not G_δ. The set $\mathbb{R} \setminus \mathbb{Q}$ is G_δ but not F_σ.

Proof. Write $\mathbb{Q} = \{q_1, q_2, \ldots\}$. Since singletons are closed, $\mathbb{Q} = \bigcup_n \{q_n\}$ is F_σ, and $\mathbb{R} \setminus \mathbb{Q} = \bigcap_n (\{q_n\})^c$ is G_δ.
If $\mathbb{Q} = \bigcap_n G_n$ with each G_n open, then each G_n is dense because it contains the dense set $\mathbb{Q}$. Set $H_n := G_n \setminus \{q_n\}$.
Each H_n is dense open, so Theorem 4.3.4 gives a point of $\bigcap_n H_n$. But $\bigcap_n H_n = \emptyset$, since $\bigcap_n G_n = \mathbb{Q}$ and q_n is removed at stage n. Hence $\mathbb{Q}$ is not G_δ. Therefore $\mathbb{R} \setminus \mathbb{Q}$ is not F_σ, or else its complement $\mathbb{Q}$ would be G_δ.
Structural Commentary: Category separates the familiar countable dense set from its dense complement.
Key Insight: Dense is not the same as large in the Baire-category sense. □

Method 4.3.10. *Baire-Category Proof Patterns*

(i) *Dense-open intersection*: build nested closed intervals $I_n \subseteq G_n$ and apply Theorem 4.2.1.
(ii) *Complete metric version*: build nested closed balls $C_{n+1} \subseteq C_n$ with $r_n \to 0$ and use completeness.
(iii) *Nowhere dense*: replace E by the dense open set $(\overline{E})^c$.
(iv) *First category*: show the exceptional set is contained in $\bigcup_n E_n$ with each E_n nowhere dense.
(v) *Residual/generic*: show the desired property contains $\bigcap_n O_n$ with each O_n dense open.
(vi) *F_σ/G_δ complements*: pass between countable unions and intersections by De Morgan's laws.

4.4 Perfect Sets and the Cantor Set

Theorem 4.4.1. *Perfect Sets Are Uncountable*
Every nonempty perfect subset of $\mathbb{R}^k$ is uncountable.

Proof. Assume $P = \{\mathbf{x}_1, \mathbf{x}_2, \ldots\}$. Choose a neighborhood V_1 with $V_1 \cap P \neq \emptyset$.
Inductively, since $P \cap V_n$ contains infinitely many points by Theorem 3.1.18, choose $\mathbf{y}_n \in P \cap V_n$ with $\mathbf{y}_n \neq \mathbf{x}_n$ and then a neighborhood V_{n+1} of $\mathbf{y}_n$ such that $\overline{V}_{n+1} \subseteq V_n$ and $\mathbf{x}_n \notin \overline{V}_{n+1}$.
Put $K_n := \overline{V}_n \cap P$. Then K_n is nonempty, compact, and $K_{n+1} \subseteq K_n$. By Corollary 4.1.33, choose $\mathbf{z} \in \bigcap_{n=1}^{\infty} K_n$.
For each n, $\mathbf{z} \in K_{n+1}$ but $\mathbf{x}_n \notin K_{n+1}$, so $\mathbf{z} \neq \mathbf{x}_n$ for all n, contradicting the enumeration.

Structural Commentary: Nested compact sets dodge the list.
Key Insight: Perfectness gives new points; compactness keeps one. □

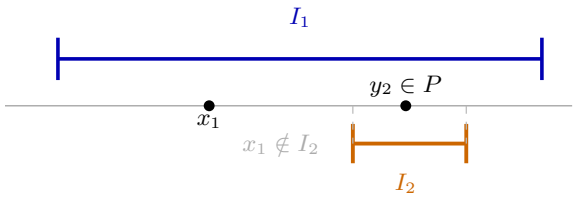


Figure 143: Choose $y_2 \in P \cap I_1$ with $y_2 \neq x_1$, then take a smaller interval $I_2 \subseteq I_1$ around y_2 that avoids x_1.

Corollary 4.4.2. *Intervals and the Real Line Are Uncountable*
Every interval $[a, b] \subseteq \mathbb{R}$ with $a < b$ is uncountable. In particular, $\mathbb{R}$ is uncountable.

Proof. The interval $I := [a, b]$ is a nonempty 1-cell, hence compact by Theorem 4.2.3 and closed by Theorem 4.1.29. For every $x \in I$ and $\varepsilon > 0$, $I \cap \big(B_\varepsilon(x) \setminus \{x\}\big) \neq \emptyset$, so I is perfect. Theorem 4.4.1 gives that I is uncountable. If $\mathbb{R}$ were countable, then $I \subseteq \mathbb{R}$ would be countable, a contradiction.
Structural Commentary: Closed intervals are perfect compact sets.
Key Insight: One closed interval already forces $\mathbb{R}$ to be uncountable. □

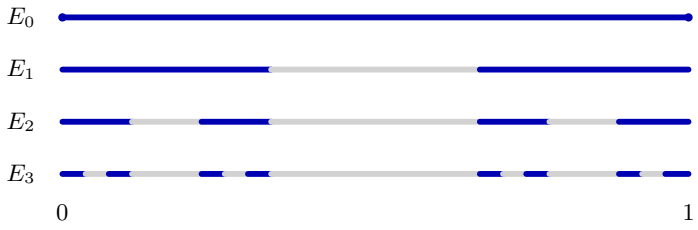


Figure 144: The first stages E_0, E_1, E_2, E_3 in the Cantor-set construction.

Example 4.4.3. *The Cantor Set*
Let $E_0 = [0, 1]$. Remove the open middle third $(1/3, 2/3)$ and call the remaining set E_1. Continue by removing the open middle third from each interval component, so

$$E_1 = [0, 1/3] \cup [2/3, 1],$$
$$E_2 = [0, 1/9] \cup [2/9, 3/9] \cup [6/9, 7/9] \cup [8/9, 1].$$

Then E_n is a union of 2^n compact intervals of length 3^{-n} and $E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots$.
The *Cantor set* is $P := \bigcap_{n=1}^{\infty} E_n$. By Corollary 4.1.33, $P \neq \emptyset$. Since each E_n is closed and $P \subseteq E_1$, P is a closed subset of the compact set E_1, hence compact.
If $J \subseteq [0, 1]$ is nonempty open interval, for some $m \in \mathbb{N}$ and $k \geq 0$, $R_{m,k} := \left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right) \subseteq J$ and $R_{m,k} \cap P = \emptyset$. Thus P contains no nontrivial interval. Finally, let $x \in P$ and let U be an interval neighborhood of x. If $I_n(x)$ is the component of E_n containing x, choose n with $I_n(x) \subseteq U$. An endpoint $y \neq x$ of $I_n(x)$ lies in P, so $U \cap (P \setminus \{x\}) \neq \emptyset$. Hence P is perfect.
Structural Commentary: The Cantor set is built by nested compact deletion, not by listing points.
Key Insight: It is compact and perfect, but contains no interval.

5 Connectedness and Continuity

5.1 Connectedness and Continuity

Connectedness prevents a space from splitting into separated pieces; continuity transports topological structure through maps. This subsection also records IVT/EVT and real-line refinements.

Definition 5.1.1. *Separated and Connected Sets*
Let X be a metric space.

(i) Subsets $A, B \subseteq X$ are *separated* if $A \cap \overline{B} = \overline{A} \cap B = \emptyset$.
(ii) A subset $E \subseteq X$ is *connected* if no nonempty separated subsets $A, B \subseteq E$ satisfy $E = A \cup B$.

(iii) A subset $U \subseteq X$ is *clopen* if it is both open and closed.

Structural Commentary: Separated sets avoid each other's closures.
Key Insight: Disconnected means a nontrivial clopen split; continuous preimages preserve such splits.

Theorem 5.1.2. *Connected Subsets of the Real Line*
A set $E \subseteq \mathbb{R}$ is connected if and only if $x, y \in E,\ x < y \Longrightarrow [x, y] \subseteq E$.

Proof.

($\Rightarrow$) If the interval property fails, choose $x, y \in E$ and $x < z < y$ with $z \notin E$. Then $A := E \cap (-\infty, z)$ and $B := E \cap (z, \infty)$ are nonempty separated sets with $E = A \cup B$, contradicting connectedness.
($\Leftarrow$) Suppose $E = A \cup B$ with A, B nonempty separated. Choose $x \in A$, $y \in B$, $x < y$, and set $S := A \cap [x, y]$, $z := \sup S$. By Theorem 3.3.5, $z \in \overline{A}$; hence $z \notin B$ and $z < y$. If $z \notin A$, then $z \in [x, y] \setminus E$.
If $z \in A$, then $z \notin \overline{B}$, so choose $z_1 \in (z, y)$ with $z_1 \notin B$; by maximality of z, also $z_1 \notin A$.
Thus $z_1 \in [x, y] \setminus E$, contradicting $[x, y] \subseteq E$.

Structural Commentary: On the real line, connectedness is exactly having no gaps between any two of its points.
Key Insight: A separation creates a missing point by taking a supremum at the boundary of one side. □

Method 5.1.3. *Compactness and Connectedness Proof Patterns*

(i) *Open-cover compactness*: for every cover $\{G_\alpha\}_{\alpha \in A}$, find finite $F \subseteq A$ with $K \subseteq \bigcup_{\alpha \in F} G_\alpha$.
(ii) *Euclidean closed-bounded compactness*: if $E \subseteq \mathbb{R}^k$ is closed and bounded, choose a compact k-cell $I \supseteq E$ and apply Theorem 4.1.30.
(iii) *Connectedness on $\mathbb{R}$*: replace connectedness of $E \subseteq \mathbb{R}$ by $x, y \in E, x < y \Longrightarrow [x, y] \subseteq E$.

Definition 5.1.4. *Limit of a Function Between Metric Spaces*
Let X, Y be metric spaces, $E \subseteq X$, $f : E \to Y$, and let p be a limit point of E. For $q \in Y$, $\lim_{x \to p} f(x) = q$ means that, for every $\varepsilon > 0$, there exists $\delta > 0$ such that $x \in E$, $0 < d_X(x, p) < \delta \Longrightarrow d_Y(f(x), q) < \varepsilon$.
The point p need not belong to E, and $f(p)$ is irrelevant when it does.
Key Insight: Domain E controls approach: $J \setminus \{a\}$ gives two-sided limit, (a, b) or (c, a) gives one-sided limits.

Remark 5.1.5. *Adherent-Point Limit Convention*
The adherent-point convention takes $p \in \overline{E}$ and allows $x = p$ in the δ-test; Definition 5.1.4 instead assumes p is a limit point and requires $0 < d_X(x, p)$. If $p \in E$, the unpunctured convention also tests $f(p)$, forcing $q = f(p)$; in the punctured convention this is exactly continuity at p by Theorem 5.1.12.
Key Insight: Punctured limits study nearby values; continuity adds compatibility with the value at the point.

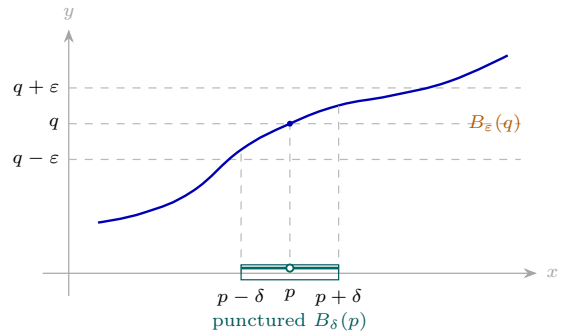


Figure 145: The epsilon–delta geometry for $\lim_{x \to p} f(x) = q$.

Theorem 5.1.6. *Sequential Characterization of Function Limits*
Let X, Y be metric spaces, $E \subseteq X$, $f : E \to Y$, and let p be a limit point of E. For $q \in Y$, $\lim_{x \to p} f(x) = q$ if and only if $f(p_n) \to q$ for every sequence $(p_n) \subseteq E \setminus \{p\}$ with $p_n \to p$.

Proof.

($\Rightarrow$) If $p_n \to p$ with $p_n \in E \setminus \{p\}$, then eventually $0 < d_X(p_n, p) < \delta$; hence $d_Y(f(p_n), q) < \varepsilon$, so $f(p_n) \to q$.
($\Leftarrow$) If the limit fails, then for some $\varepsilon > 0$ and each n there is $x_n \in E$ with $0 < d_X(x_n, p) < 1/n$ and $d_Y(f(x_n), q) \geq \varepsilon$. Then $x_n \to p$, $x_n \neq p$, but $f(x_n) \not\to q$.

Structural Commentary: Function limits can be tested by all sequences approaching the input point.
Key Insight: The negation of an epsilon–delta limit manufactures a bad sequence. □

Principle 5.1.7. *Challenge–Response Order*
Limits and continuity are controlled by quantifier order. In Definition 5.1.4, the ε-challenge at the range point must be answered by a δ-restriction near the domain point. Uniform continuity, Definition 5.2.1, strengthens the same pattern by requiring one δ to work for all relevant domain points at once.

Corollary 5.1.8. *Uniqueness of Function Limits*
If a function has a limit at p, then that limit is unique.

Proof. Let q and r be limits of f at p. Choose $p_n \in E \setminus \{p\}$ with $d_X(p_n, p) < 1/n$. Then $p_n \to p$, so Theorem 5.1.6 gives $f(p_n) \to q$ and $f(p_n) \to r$. By Theorem 3.4.3, $q = r$.
Structural Commentary: Limit uniqueness is inherited from metric sequence uniqueness.
Key Insight: One approaching sequence cannot have two different limits. □

Theorem 5.1.9. *Algebra of Function Limits*
Let X be a metric space, $E \subseteq X$, p be limit point of E.
For $f, g : E \to \mathbb{C}$, assume $\lim_{x\to p} f(x) = A$ and $\lim_{x\to p} g(x) = B$. Then:

(i) $\lim\limits_{x\to p}(f+g)(x) = A + B$;
(ii) $\lim\limits_{x\to p}(fg)(x) = AB$;
(iii) if $B \neq 0$ and f/g is defined near p, then $\lim\limits_{x\to p}(f/g)(x) = A/B$.

For $f, g : E \to \mathbb{R}^k$ with limits a, b, the corresponding limits are $f + g \to a + b$, $\lambda f \to \lambda a$, and $f \cdot g \to a \cdot b$.

Proof.

(i) For any $p_n \in E \setminus \{p\}$ with $p_n \to p$, Theorem 5.1.6 gives $f(p_n) \to A$ and $g(p_n) \to B$. Hence $(f+g)(p_n) \to A + B$ by Theorem 2.1.3, so Theorem 5.1.6 gives (i).
(ii) The same sequence gives $(fg)(p_n) \to AB$ by Theorem 2.1.3; hence (ii).
(iii) If $B \neq 0$ and f/g is defined near p, then every such sequence is eventually in the domain of f/g, and $(f/g)(p_n) \to A/B$ by Theorem 2.1.3; hence (iii).

The vector laws follow by the same sequential characterization and Theorem 2.1.4.
Structural Commentary: Limit algebra for functions is pulled back to sequence algebra.
Key Insight: Sequential characterization lets previous sequence theorems do the work. □

Proposition 5.1.10. *Function Limits Are Local*
Let X, Y be metric spaces, let $E \subseteq X$, let $f : E \to Y$, let p be a limit point of E, and let $q \in Y$.
For every $r > 0$, $\lim\limits_{x\to p;\, x\in E} f(x) = q \iff \lim\limits_{x\to p;\, x\in E\cap B_r(p)} f(x) = q$.

Proof.

($\Rightarrow$) Since $E \cap B_r(p) \subseteq E$, any δ witnessing the limit on E also witnesses the limit on $E \cap B_r(p)$.
($\Leftarrow$) Assume the restricted limit. Given $\varepsilon > 0$, choose $\delta_0 > 0$ such that $x \in E \cap B_r(p)$ and $0 < d_X(x, p) < \delta_0$ imply $d_Y(f(x), q) < \varepsilon$. Let $\delta := \min\{\delta_0, r\}$.
If $x \in E$ and $0 < d_X(x, p) < \delta$, then $x \in E \cap B_r(p)$, so $d_Y(f(x), q) < \varepsilon$.

Structural Commentary: Values away from p cannot affect the limit at p.
Key Insight: Shrink δ until every tested point lies in the chosen local neighborhood. □

Definition 5.1.11. *Continuity at a Point and on a Set*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, $E \subseteq X$, and $f : E \to Y$.

(i) For $p \in E$, f is *continuous at p* if $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in E$, $d_X(x, p) < \delta \implies d_Y(f(x), f(p)) < \varepsilon$.
(ii) f is *continuous on E* if $\forall p \in E$, f is continuous at p.
(iii) If p is isolated in E, then f is continuous at p.

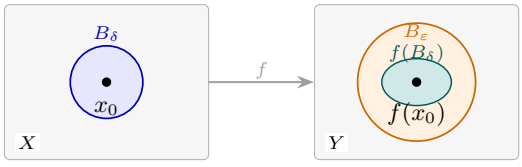


Figure 146: Continuity at x_0: for every $\varepsilon > 0$, choose $\delta > 0$ with $f(B_\delta(x_0)) \subset B_\varepsilon(f(x_0))$.

Theorem 5.1.12. *Continuity as Equality with the Function Limit*
Let X, Y be metric spaces, $E \subseteq X$, $p \in E$, and $f : E \to Y$.
If p is a limit point of E, then f is continuous at p if and only if $\lim_{x \to p} f(x) = f(p)$.

Proof.

($\Rightarrow$) If $0 < d_X(x,p) < \delta$, continuity at p gives $d_Y(f(x), f(p)) < \varepsilon$, hence $\lim_{x \to p} f(x) = f(p)$.
($\Leftarrow$) If $\lim_{x \to p} f(x) = f(p)$, the same δ proves continuity for $x \neq p$; for $x = p$, $d_Y(f(p), f(p)) = 0$.

Structural Commentary: Continuity is the compatibility of the value at p with nearby values.
Key Insight: Limits ignore the point; continuity ties the point back in. □

Remark 5.1.13. *The Sequence-First View*
For functions on subsets of $\mathbb{R}$, continuity at limit points is equivalently sequential: $x_n \to p$ in the domain implies $f(x_n) \to f(p)$; isolated points are automatic by Theorems 5.1.6 and 5.1.12.

Theorem 5.1.14. *Sequential Characterization of Continuity*
Let $(X, d_X), (Y, d_Y)$ be metric spaces, $f : X \to Y$, and $p \in X$.
Then f is continuous at p if and only if $\forall (x_n) \subseteq X$, $x_n \to p \implies f(x_n) \to f(p)$.

Proof.

($\Rightarrow$) Assume f is continuous at p and let $x_n \to p$.
Given $\varepsilon > 0$, choose $\delta > 0$ such that $d_X(x,p) < \delta \implies d_Y(f(x), f(p)) < \varepsilon$.
Since $x_n \to p$, eventually $d_X(x_n, p) < \delta$, hence eventually $d_Y(f(x_n), f(p)) < \varepsilon$. Thus $f(x_n) \to f(p)$.
($\Leftarrow$) Suppose the sequential condition holds. If f is not continuous at p, then $\exists \varepsilon_0 > 0$ such that $\forall \delta > 0$, $\exists x \in X$ with $d_X(x,p) < \delta$ and $d_Y(f(x), f(p)) \geq \varepsilon_0$. Taking $\delta = 1/n$ gives $x_n \in X$ with $d_X(x_n, p) < 1/n$ and $d_Y(f(x_n), f(p)) \geq \varepsilon_0$. Then $x_n \to p$, but $f(x_n) \not\to f(p)$, a contradiction.

Structural Commentary: Failure of epsilon–delta continuity manufactures a bad sequence.
Key Insight: In metric spaces, sequences detect continuity at points. □

Perspective 5.1.15. *Three Views of Continuity*
For metric-space maps, continuity is equivalently ε–δ control, $x_n \to x \Rightarrow f(x_n) \to f(x)$, and $V \subseteq Y$ open $\Rightarrow f^{-1}(V)$ open; see Definition 5.1.11 and Theorems 5.1.14, 5.1.18.
Key Insight: Continuity sends near points forward and pulls open sets back.

Theorem 5.1.16. *Composition of Continuous Functions*
Let $E \subseteq X$ and let $f : E \to Y$, $g : f(E) \to Z$ be maps between metric spaces.
If f is continuous at $p \in E$ and g is continuous at $f(p)$, then $g \circ f$ is continuous at p.

Proof. Given $\varepsilon > 0$, choose $\eta > 0$ for g at $f(p)$, then choose $\delta > 0$ for f at p with tolerance η.
If $x \in E$ and $d_X(x,p) < \delta$, then $d_Y(f(x), f(p)) < \eta$, hence $d_Z(g(f(x)), g(f(p))) < \varepsilon$.
Structural Commentary: The output tolerance for g becomes the input tolerance for f.
Key Insight: Continuity composes by passing epsilon through an intermediate eta. □

Theorem 5.1.17. *The Distance Function Is Continuous*
Let (X, d) be a metric space and $D : X \times X \to \mathbb{R}$, $D(x,y) := d(x,y)$. For $d_\times \in \{d_1, d_2, d_\infty\}$ on $X \times X$, $D : (X \times X, d_\times) \to \mathbb{R}$ is continuous. Equivalently, $x_n \to x$ and $y_n \to y$ in X imply $d(x_n, y_n) \to d(x,y)$.

Proof. For $x, y, u, v \in X$, $|D(x,y) - D(u,v)| = |d(x,y) - d(u,v)| \leq d(x,u) + d(y,v)$. Hence $x_n \to x$ and $y_n \to y$ imply $D(x_n, y_n) \to D(x,y)$. Proposition 3.1.12 and Theorem 5.1.14 give continuity for each product metric.
Structural Commentary: Distance changes continuously when both endpoints move.
Key Insight: The triangle inequality controls distance error by endpoint errors. □

Theorem 5.1.18. *Continuity via Open Preimages*
For metric spaces X, Y, map $f : X \to Y$ is continuous if and only if $f^{-1}(V)$ is open in $X \forall$ open $V \subseteq Y$.

Proof.

($\Rightarrow$) If $p \in f^{-1}(V)$ and V is open, choose $\varepsilon > 0$ with $B_\varepsilon(f(p)) \subseteq V$.
Continuity gives $\delta > 0$ with $B_\delta(p) \subseteq f^{-1}(V)$, so $f^{-1}(V)$ is open.
($\Leftarrow$) For $p \in X$ and $\varepsilon > 0$, the ball $V = B_\varepsilon(f(p))$ is open by Theorem 3.1.16.
Since $p \in f^{-1}(V)$ and $f^{-1}(V)$ is open, some $B_\delta(p) \subseteq f^{-1}(V)$; this is continuity at p.

Structural Commentary: Continuity is exactly preservation of openness under inverse image.
Key Insight: Images are not Boolean-exact, but preimages are; continuity belongs to preimages. □

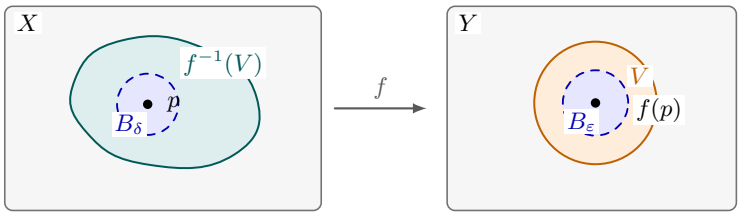


Figure 147: Continuity pulls open sets back to open sets.

Corollary 5.1.19. *Continuity via Closed Preimages*
For metric spaces X, Y, map $f : X \to Y$ is continuous if and only if $f^{-1}(C)$ is closed in $X \forall$ closed $C \subseteq Y$.

Proof.
($\Rightarrow$) If C is closed, then $Y \setminus C$ is open. By Theorem 5.1.18, $f^{-1}(Y \setminus C)$ is open; by Proposition 5.1.2, this is $X \setminus f^{-1}(C)$, so $f^{-1}(C)$ is closed.
($\Leftarrow$) If V is open, then $Y \setminus V$ is closed. Hence $f^{-1}(Y \setminus V)$ is closed, so $f^{-1}(V) = X \setminus f^{-1}(Y \setminus V)$ is open. Theorem 5.1.18 gives continuity.

Structural Commentary: Closed-set continuity is the complement form of open-set continuity.
Key Insight: $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ transfers open and closed formulations. □

Remark 5.1.20. *Preimages, Not Forward Images*
Theorem 5.1.18 is about inverse images. A continuous map need not send open sets to open sets or closed sets to closed sets; compactness is one hypothesis that recovers closedness of images through Theorem 5.1.33.

Definition 5.1.21. *Homeomorphism*
A *homeomorphism* from X to Y is a bijection $f : X \to Y$ such that both f and f^{-1} are continuous.
If such a map exists, X and Y are *homeomorphic*, written $X \simeq Y$.

Proposition 5.1.22. *Homeomorphisms Preserve Topology*
If $f : X \to Y$ is a homeomorphism, then $U \mapsto f(U)$ bijects open subsets of X with open subsets of Y.

Proof. If $U \subseteq X$ is open, then $f(U) = (f^{-1})^{-1}(U)$ is open in Y since f^{-1} is continuous. If $V \subseteq Y$ is open, then $f^{-1}(V)$ is open in X since f is continuous. These two assignments are inverse because f is bijective.
Structural Commentary: A homeomorphism preserves open sets in both directions.
Key Insight: Topological sameness is continuity plus continuous reversibility. □

Definition 5.1.23. *Metric Isometry*
Let $(X, d_X), (Y, d_Y)$ be metric spaces.
(i) A map $f : X \to Y$ is an *isometry* if $\forall x, y \in X$, $d_Y(f(x), f(y)) = d_X(x, y)$.
(ii) X and Y are *isometric* if $\exists$ a bijective isometry $f : X \to Y$.

Theorem 5.1.24. *Basic Properties of Metric Isometries*
Let $f : (X, d_X) \to (Y, d_Y)$ be an isometry.
(i) f is injective.
(ii) f is continuous.
(iii) $f^{-1} : (f(X), d_Y|_{f(X)\times f(X)}) \to (X, d_X)$ is an isometry.

Proof.
(i) If $f(x) = f(y)$, then $d_X(x, y) = d_Y(f(x), f(y)) = 0$, hence $x = y$.
(ii) If $x_n \to x$ in X, then $d_Y(f(x_n), f(x)) = d_X(x_n, x) \to 0$; by Theorem 5.1.14, f is continuous.
(iii) For $u = f(x), v = f(y) \in f(X)$, $d_X(f^{-1}(u), f^{-1}(v)) = d_X(x, y) = d_Y(f(x), f(y)) = d_Y(u, v)$; QED.

Structural Commentary: An isometry preserves all distances, so injectivity and continuity are automatic.
Key Insight: Distance preservation is stronger than topological equivalence. □

Remark 5.1.25. *Metric Versus Linear Isometries*
Metric isometry is purely distance-theoretic. If $T : V \to W$ is linear between inner-product spaces, then $\|Tv\| = \|v\| \ \forall v \in V \iff d_W(Tu, Tv) = d_V(u, v) \ \forall u, v \in V$.

Example 5.1.26. *Continuous Bijection Need Not Be a Homeomorphism*
Let $f : [0, 2\pi) \to S^1$, and $f(t) = (\cos t, \sin t)$. This continuous bijection is not a homeomorphism: points of S^1 approaching $(1, 0)$ from below have preimages tending to 2π, not 0.

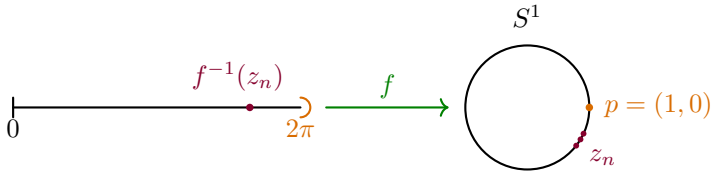


Figure 148: Parametrization $[0, 2\pi) \to S^1$ wraps continuously and bijectively, but inverse jumps at $(1, 0)$.

Remark 5.1.27. *Continuity Only Uses the Domain Metric*
For $f : E \to Y$ with $E \subseteq X$, continuity on E depends only on the restricted metric on E, not on $X \setminus E$.

Definition 5.1.28. *Product Map*
If $f : X \to Y$ and $g : X \to Z$ are maps between metric spaces, their *product map*, or direct sum, is $f \oplus g : X \to Y \times Z$, $(f \oplus g)(x) := (f(x), g(x))$, where $Y \times Z$ carries any of product metrics in Definition 3.1.11.

Proposition 5.1.29. *Continuity of Product Maps*
The product map $f \oplus g : X \to Y \times Z$ is continuous at x_0 iff f and g are continuous at x_0; hence it is continuous on X iff both components are.

Proof. By Proposition 3.1.12, it suffices to use d_{sum}.

($\Rightarrow$) Given $\varepsilon > 0$, choose $\delta > 0$ from continuity of $f \oplus g$ at x_0.
If $d_X(x, x_0) < \delta$, then $d_Y(f(x), f(x_0)), d_Z(g(x), g(x_0)) \le d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) < \varepsilon$.
($\Leftarrow$) Given $\varepsilon > 0$, choose δ_f, δ_g for f, g at tolerance $\varepsilon/2$ and set $\delta = \min\{\delta_f, \delta_g\}$.
Then $d_X(x, x_0) < \delta$ gives $d_{\text{sum}}((f \oplus g)(x), (f \oplus g)(x_0)) = d_Y(f(x), f(x_0)) + d_Z(g(x), g(x_0)) < \varepsilon$.

Structural Commentary: Product continuity is exactly componentwise continuity.
Key Insight: A product metric packages two output estimates into one. □

Theorem 5.1.30. *Continuous Algebra and Components*
Let X be a metric space.

(i) If $f, g : X \to \mathbb{C}$ are continuous, so are $f + g$, fg, and f/g where $g \neq 0$.
(ii) For $f = (f_1, \ldots, f_k) : X \to \mathbb{R}^k$, f is continuous if and only if each f_j is continuous.
(iii) If $f, g : X \to \mathbb{R}^k$ are continuous, so are $f + g$ and $f \cdot g$.

Proof.

(i) At isolated points continuity is automatic; at limit points, use Theorems 5.1.9 and 5.1.12.
(ii) By Theorem 2.1.4, convergence in $\mathbb{R}^k$ is exactly coordinatewise convergence; apply this to $f(x_n) \to f(x)$.
(iii) Apply (ii) to the coordinate functions of $f + g$ and $f \cdot g$, using the scalar algebra from (i).

Structural Commentary: Continuity is stable under the algebraic operations already stable for limits.
Key Insight: Componentwise control is equivalent to Euclidean control in finite dimensions. □

Example 5.1.31. *Coordinate, Polynomial, Rational, and Norm Maps*
Coordinate functions on $\mathbb{R}^k$ are continuous; hence polynomials are continuous, and rational functions are continuous where defined. Also $\mathbf{x} \mapsto \|\mathbf{x}\|_2$ is continuous since $\big|\|\mathbf{x}\|_2 - \|\mathbf{y}\|_2\big| \le \|\mathbf{x} - \mathbf{y}\|_2$.
Thus $p \mapsto \|f(p)\|_2$ is continuous whenever $f : X \to \mathbb{R}^k$ is, by Theorem 5.1.16.

Definition 5.1.32. *Bounded Mapping*
A mapping $f : E \to \mathbb{R}^k$ is *bounded* if there exists $M \in \mathbb{R}$ such that $\|f(x)\|_2 \le M$ for every $x \in E$.

Theorem 5.1.33. *Continuous Images of Compact Sets Are Compact*
If $f : X \to Y$ is continuous and X is compact, then $f(X)$ is compact.

Proof. Let $\{V_\alpha\}$ cover $f(X)$. Then $\{f^{-1}(V_\alpha)\}$ is an open cover of X by Theorem 5.1.18.
Choose $\alpha_1, \ldots, \alpha_n$ with $X \subseteq \bigcup_{i=1}^n f^{-1}(V_{\alpha_i})$; then $f(X) \subseteq \bigcup_{i=1}^n V_{\alpha_i}$.
Structural Commentary: Pull back, take a finite subcover, push forward.
Key Insight: Compactness is preserved by continuous images because inverse images preserve openness. □

Corollary 5.1.34. *Continuous Images into $\mathbb{R}^k$ Are Closed and Bounded*
If $f : X \to \mathbb{R}^k$ is continuous and X is compact, then $f(X)$ is closed and bounded. In particular, f is bounded.

Proof. By Theorem 5.1.33, $f(X)$ is compact.
By Theorem 4.2.5, compact subsets of $\mathbb{R}^k$ are closed and bounded; hence f is bounded.
Structural Commentary: Compactness in the domain becomes closed-bounded geometry in Euclidean range.
Key Insight: Continuous functions on compact spaces cannot escape to infinity. □

Theorem 5.1.35. *Extreme Value Theorem*
Let $f : X \to \mathbb{R}$ be continuous, let X be compact.
If $M := \sup_{p\in X} f(p)$, $m := \inf_{p\in X} f(p)$, then $\exists\, p, q \in X$ such that $f(p) = M$ and $f(q) = m$.

Proof. By Corollary 5.1.34, $f(X)$ is closed and bounded in $\mathbb{R}$. Thus $\sup f(X)$ and $\inf f(X)$ exist and belong to $f(X)$ by Theorems 3.3.5 and 4.3.3. Hence $M = f(p)$ and $m = f(q)$ for some $p, q \in X$.
Structural Commentary: Compactness turns suprema and infima into attained values.
Key Insight: Closed bounded images contain their endpoints. □

Theorem 5.1.36. *Compact Bijective Maps Have Continuous Inverses*
If $f : X \to Y$ is a continuous bijection from compact X to a metric space Y, then $f^{-1} : Y \to X$ is continuous.

Proof. It suffices, by Theorem 5.1.18, to show that $f(V)$ is open in Y for every open $V \subseteq X$. If V is open, then V^c is compact by Theorem 4.1.30, so $f(V^c)$ is compact, hence closed by Theorems 5.1.33 and 4.1.29.
Since f is bijective, $f(V) = Y \setminus f(V^c)$ is open.
Structural Commentary: Compactness turns closed complements into closed images.
Key Insight: A continuous bijection from compact domain has no room for inverse discontinuity. □

Theorem 5.1.37. *Continuous Images of Connected Sets Are Connected*
If $f : X \to Y$ is continuous and $E \subseteq X$ is connected, then $f(E)$ is connected.

Proof. If $f(E) = A \cup B$ with A, B nonempty separated, set $G := E \cap f^{-1}(A)$ and $H := E \cap f^{-1}(B)$. Then $E = G \cup H$ and G, H are nonempty. Since $G \subseteq f^{-1}(\overline{A})$ and $f^{-1}(\overline{A})$ is closed, $\overline{G} \subseteq f^{-1}(\overline{A})$; hence $\overline{G} \cap H = \emptyset$.
Similarly $G \cap \overline{H} = \emptyset$, so G, H separate E, a contradiction.
Structural Commentary: A separation of the image pulls back to a separation of the domain.
Key Insight: Continuous maps cannot create a gap in the image of a connected set. □

Theorem 5.1.38. *Intermediate Value Theorem*
If $f : [a, b] \to \mathbb{R}$ is continuous and c lies strictly between $f(a)$ and $f(b)$, then $f(x) = c$ for some $x \in (a, b)$.

Proof. The interval $[a, b]$ is connected by Theorem 5.1.2, so $f([a, b])$ is connected by Theorem 5.1.37.
Connected subsets of $\mathbb{R}$ contain every value between any two of their points; hence $c = f(x)$ for some $x \in [a, b]$.
Since $c \neq f(a), f(b)$, $x \in (a, b)$.
Structural Commentary: The intermediate value theorem is connectedness of intervals transported by continuity.
Key Insight: Continuous real functions on intervals cannot jump over a value. □

Definition 5.1.39. *Path-Connected Set*
A set E in a metric space is *path-connected* if for every $p, q \in E$ there is a continuous map $\gamma : [a, b] \to E$ with $\gamma(a) = p$ and $\gamma(b) = q$. Such a map γ is a *path* from p to q.

Theorem 5.1.40. *Path-Connected Sets Are Connected*
Every path-connected set is connected.

Proof. Suppose $E = A \cup B$ with A, B nonempty separated.
Choose $p \in A$, $q \in B$, and a path $\gamma : [a, b] \to E$ from p to q. Since $[a, b]$ is connected, so is $\gamma([a, b])$.
But $\gamma([a, b])$ meets both A and B, which separate it, a contradiction.
Structural Commentary: A path transports connectedness from an interval to the set.
Key Insight: To connect two points by a path, the path image cannot be split into two separated pieces. □

Corollary 5.1.41. *Convex Sets Are Path-Connected*
Every convex subset of $\mathbb{R}^k$ is path-connected.

Proof. For $\mathbf{x}, \mathbf{y} \in E$, set $\gamma(t) = (1-t)\mathbf{x}+t\mathbf{y}$ on $[0, 1]$. Convexity gives $\gamma([0, 1]) \subseteq E$, with $\gamma(0) = \mathbf{x}$ and $\gamma(1) = \mathbf{y}$.
Since the coordinates of γ are affine in t, γ is continuous. Hence E is path-connected.
Structural Commentary: Convexity supplies the whole straight path between any two points.
Key Insight: In Euclidean space, line segments are continuous paths. □

Method 5.1.42. *Continuity Proof Patterns*

(i) *ε–δ at p*: prove $\forall \varepsilon > 0\, \exists \delta > 0$ such that $x \in E$ and $d_X(x, p) < \delta$ imply $d_Y(f(x), f(p)) < \varepsilon$.
(ii) *Limit at p*: prove same estimate with $0 < d_X(x, p) < \delta$, target q; equivalently test all $x_n \to p$, $x_n \neq p$.
(iii) *Topological continuity*: prove $V \subseteq Y$ open $\Rightarrow f^{-1}(V)$ open, or $C \subseteq Y$ closed $\Rightarrow f^{-1}(C)$ closed.
(iv) *Compactness*: from open cover $\{V_\alpha\}$ of $f(X)$, pulled-back cover $\{f^{-1}(V_\alpha)\}$ of X extract finite subcover.
(v) *Uniform continuity*: choose local radii r_p, cover compact X by $B_{r_p/2}(p)$, and set $\delta = \frac{1}{2} \min_i r_{p_i}$.

Key Insight: Continuity is local control; compactness turns local controls into finite control.

5.2 Uniform Continuity and Compactness

Definition 5.2.1. *Uniform Continuity*
A mapping $f : X \to Y$ between metric spaces is *uniformly continuous* on X if $\forall\ \varepsilon > 0\ \exists\ \delta > 0$ such that $d_X(p,q) < \delta \implies d_Y(f(p), f(q)) < \varepsilon\ \forall p, q \in X$. The same δ must work for every point of X.
Structural Commentary: Uniform continuity is setwise: for each ε, one δ must work for all pairs in the set.

Proposition 5.2.2. *Sequential Test for Uniform Continuity*
Let X, Y be metric spaces and let $f : X \to Y$. The following are equivalent:

(i) f is uniformly continuous on X;
(ii) whenever (p_n) and (q_n) are sequences in X with $d_X(p_n, q_n) \to 0$, one has $d_Y(f(p_n), f(q_n)) \to 0$.

Proof.

(i) $\Rightarrow$ (ii) Given $\varepsilon > 0$, choose $\delta > 0$ such that $d_X(x,y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$ for all $x, y \in X$. Since $d_X(p_n, q_n) \to 0$, eventually $d_X(p_n, q_n) < \delta$, hence eventually $d_Y(f(p_n), f(q_n)) < \varepsilon$.
(ii) $\Rightarrow$ (i) Prove the contrapositive. If f is not uniformly continuous, then $\exists \varepsilon_0 > 0$ such that $\forall n \geq 1$, $\exists p_n, q_n \in X$ with $d_X(p_n, q_n) < 1/n$ and $d_Y(f(p_n), f(q_n)) \geq \varepsilon_0$.
Thus $d_X(p_n, q_n) \to 0$ but $d_Y(f(p_n), f(q_n)) \not\to 0$, contradicting (ii).

Structural Commentary: Uniform continuity preserves vanishing pairwise distances.
Key Insight: Failure of one global δ manufactures two close input sequences with separated outputs. □

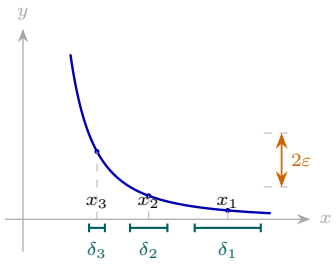


Figure 149: For $f(x) = 1/x^2$ near 0, a fixed vertical tolerance forces smaller pointwise horizontal tolerances.

Theorem 5.2.3. *Continuity on Compact Sets Is Uniform*
If $f : X \to Y$ is continuous and X is compact, then f is uniformly continuous.

Proof. Let $\varepsilon > 0$. For each $a \in X$, choose $r_a > 0$ such that $d_X(x,a) < r_a$ implies $d_Y(f(x), f(a)) < \varepsilon/2$. Compactness gives a finite cover $X \subseteq \bigcup_{i=1}^{n} B_{r_{a_i}/2}(a_i)$. Set $\delta := \frac{1}{2}\min_i r_{a_i} > 0$. If $d_X(p,q) < \delta$ and $p \in B_{r_{a_i}/2}(a_i)$, then $d_X(q, a_i) < r_{a_i}$ also; hence $f(p)$ and $f(q)$ are each within $\varepsilon/2$ of $f(a_i)$. Thus $d_Y(f(p), f(q)) < \varepsilon$.
Structural Commentary: Compactness converts pointwise radii into finitely many radii with positive minimum.
Key Insight: Uniform continuity is continuity with the point-dependence removed by a finite subcover. □

Theorem 5.2.4. *Compactness Hypotheses Are Essential*
Let E be a noncompact subset of $\mathbb{R}$. Then:

(i) there exists a continuous function on E which is not bounded;
(ii) there exists a continuous bounded function on E which has no maximum;
(iii) if E is bounded, then there exists a continuous function on E which is not uniformly continuous.

Proof.

(i) If E is unbounded, $f(x) = x$ works. If E is bounded, Theorem 4.2.5 gives $x_0 \in \overline{E} \setminus E$, and $f(x) = 1/(x - x_0)$ is continuous on E and unbounded.
(ii) If E is unbounded, $h(x) = x^2/(1+x^2)$ has $\sup_E h = 1$ but $h < 1$. If E is bounded, with x_0 as above, $g(x) = 1/(1 + (x - x_0)^2)$ is bounded by 1 and has supremum 1 but no maximum.
(iii) If E is bounded, let $f(x) = 1/(x - x_0)$. Given $\delta > 0$, let $x \in E$ with $|x - x_0| < \delta/2$, then $y \in E$ with $|y - x_0| < \min\{\delta/2, 1/(|f(x)| + 1)\}$. Then $|x - y| < \delta$, $|f(x) - f(y)| > 1$, so f is not uniformly continuous.

Structural Commentary: Compactness is essential in the preceding theorems.
Key Insight: Noncompact sets either escape to infinity or miss an accumulation point. □

5.3 One-Sided Limits, Monotone Functions, and Extended Limits

Definition 5.3.1. *One-Sided Limits and Types of Discontinuity*
Let $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$, and $x_0 \in \mathbb{R}$.

(i) If x_0 is a limit point of $X \cap (x_0, \infty)$, then $f(x_0+) = q$ means $\lim_{x \to x_0;\, x \in X \cap (x_0, \infty)} f(x) = q$.

(ii) If x_0 is a limit point of $X \cap (-\infty, x_0)$, then $f(x_0-) = q$ means $\lim_{x \to x_0;\, x \in X \cap (-\infty, x_0)} f(x) = q$.

(iii) When X approaches x_0 from both sides, the two-sided punctured limit through X exists if and only if the two one-sided limits exist and are equal.

For functions on an interval, a discontinuity with both one-sided limits is of the *first kind*, or *simple*; otherwise it is of the *second kind*. Simple discontinuities are jumps, $f(x_0+) \neq f(x_0-)$, or removable value mismatches.

Example 5.3.2. *Basic Discontinuity Types*
Dirichlet's function $\mathbf{1}_{\mathbb{Q}}$ has a discontinuity of the second kind at every point.
The function $f(x) = x$ on $\mathbb{Q}$ and $f(x) = 0$ off $\mathbb{Q}$ is continuous only at 0; elsewhere its discontinuities are of the second kind. A jump in a piecewise linear function is a simple discontinuity.
Use Corollary 1.1.17: every interval contains rational and irrational points.

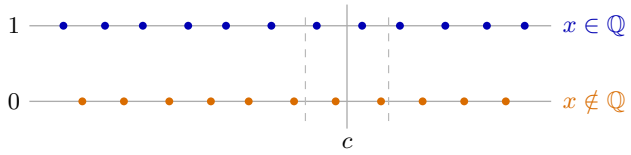


Figure 150: For $g = \mathbf{1}_{\mathbb{Q}}$, interval around c has rational inputs value 1 and irrational inputs value 0.

Definition 5.3.3. *Monotone Functions*
Let f be real-valued on (a, b). The function is *monotonically increasing* if $a < x < y < b$ implies $f(x) \leq f(y)$. Reversing the inequality gives *monotonically decreasing*. A *monotone* function is one of these two types.

Theorem 5.3.4. *One-Sided Limits of Monotone Functions*
If f is increasing on (a, b), then for every $x \in (a, b)$, $\sup_{a<t<x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x<t<b} f(t)$.
Moreover, $a < x < y < b$ implies $f(x+) \leq f(y-)$. For decreasing f, reverse the inequalities.

Proof. Let $A := \sup_{a<t<x} f(t)$.
For every $\varepsilon > 0$, choose $s < x$ with $A - \varepsilon < f(s) \leq A$; then $s < t < x$ implies $A - \varepsilon < f(t) \leq A$, so $f(x-) = A$.
Let $B := \inf_{x<t<b} f(t)$. For every $\varepsilon > 0$, choose $s > x$ with $B \leq f(s) < B + \varepsilon$; then $x < t < s$ implies $B \leq f(t) < B + \varepsilon$, so $f(x+) = B$. If $x < y$, then $\inf_{x<t<y} f(t) \leq \sup_{x<t<y} f(t)$ gives $f(x+) \leq f(y-)$.
For decreasing f, apply the increasing case to $-f$ and multiply the resulting inequalities by -1.
Structural Commentary: Monotonicity converts one-sided limits into suprema and infima.
Key Insight: A monotone function has jumps, but not oscillatory one-sided failure. □

Corollary 5.3.5. *Monotone Functions Have No Second-Kind Discontinuities*
A monotone real function on an interval has no discontinuities of the second kind.

Proof. At every interior point, Theorem 5.3.4 gives both one-sided limits.
Hence any discontinuity is of the first kind by Definition 5.3.1.
Structural Commentary: Monotonicity rules out oscillatory discontinuities.
Key Insight: All monotone discontinuities are jumps or removable value mismatches. □

Theorem 5.3.6. *Monotone Discontinuities Are Countable*
If f is monotone on (a, b), then the set of points of (a, b) at which f is discontinuous is at most countable.

Proof. Assume f is increasing, and let E be its discontinuity set.
For each $x \in E$, Theorem 5.3.4 gives $f(x-) < f(x+)$; choose $r(x) \in \mathbb{Q}$ with $f(x-) < r(x) < f(x+)$.
If $x_1 < x_2$, then $f(x_1+) \leq f(x_2-)$, so $r(x_1) \neq r(x_2)$. Thus E injects into $\mathbb{Q}$, hence is countable by Corollary 6.1.12. If f is decreasing, apply the increasing case to $-f$, which has the same discontinuity set.
Structural Commentary: Each jump contains a rational marker, and monotonicity keeps the markers distinct.
Key Insight: Countability enters by injecting discontinuities into $\mathbb{Q}$. □

Remark 5.3.7. *Monotone Discontinuities Need Not Be Isolated*
For countable $E = \{x_n : n \in \mathbb{N}\} \subset (a, b)$ and positive summable (c_n), $f(x) = \sum_{x_n < x} c_n$ is increasing, has jumps of size c_n at x_n, and is continuous off E. Thus monotone discontinuities may be dense.

Definition 5.3.8. *Extended Limits and Limits at Infinity*
Neighborhoods of $+\infty$ are intervals $(c, +\infty)$; neighborhoods of $-\infty$ are $(-\infty, c)$.
For $f : E \to \mathbb{R}$ and extended real A, x, $f(t) \to A$ as $t \to x$ means: every neighborhood U of A contains $f(t)$ for all punctured $t \in E$ sufficiently near x, i.e. $\exists V \ni x$ such that $t \in V \cap E,\ t \neq x \implies f(t) \in U$.
For real A, x, this agrees with Definition 5.1.4.
Structural Commentary: Extended limits use same neighborhood test, $\pm\infty$ allowed as input or output limits.

Theorem 5.3.9. *Algebra of Extended Limits*
Let $f, g : E \to \mathbb{R}$ and $f(t) \to A$, $g(t) \to B$ as $t \to x$, with A, B, x extended real. Then, whenever RHS is defined:

(i) *if $f(t) \to A'$ as $t \to x$, then $A' = A$;*
(ii) *$(f + g)(t) \to A + B$;*
(iii) *$(fg)(t) \to AB$;*
(iv) *$(f/g)(t) \to A/B$;*

The forms $\infty - \infty$, $0 \cdot \infty$, ∞/∞, and $A/0$ are undefined.

Proof.

(i) If $A' \neq A$, choose disjoint neighborhoods of A and A'.
Definition 5.3.8 would eventually place $f(t)$ in both, a contradiction.
(ii) For any neighborhood W of $A+B$, definedness gives neighborhoods U of A and V of B such that $u+v \in W$ for $u \in U$, $v \in V$. Apply the two limits in a common punctured neighborhood of x.
(iii) The same neighborhood argument applies to multiplication, excluding $0 \cdot \infty$.
(iv) Apply it to division; since $B \neq 0$, choose the neighborhood of B away from 0.

Structural Commentary: Extended limit algebra is ordinary limit algebra in neighborhood form.
Key Insight: The undefined forms are exactly where neighborhood control breaks. □

6 General Topology for Analysis

6.1 Topological Spaces, Bases, and Separation

General topology abstracts the open-set behavior of metric spaces. This selective pass records spaces, bases, products, quotients, and countability axioms for later use.

Definition 6.1.1. *Topological Space*
A *topology* on a set X is a collection $\mathcal{T} \subseteq 2^X$ such that

(i) $\emptyset, X \in \mathcal{T}$;
(ii) arbitrary unions of members of $\mathcal{T}$ belong to $\mathcal{T}$;
(iii) finite intersections of members of $\mathcal{T}$ belong to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is a *topological space*, and the members of $\mathcal{T}$ are the *open sets*. Closed sets are complements of open sets.

Proposition 6.1.2. *Metric Open Sets Form a Topology*
If (X, d) is a metric space, then the open sets from Definition 3.1.14 form a topology on X.

Proof. The empty set and X are open. Arbitrary unions and finite intersections are open by Theorem 3.1.22.
Structural Commentary: Metric balls generate an open-set system with the exact axioms of a topology.
Key Insight: General topology keeps the open-set calculus and forgets the distance function. □

Definition 6.1.3. *Basis*
Let X be a set. A collection $\mathcal{B} \subseteq 2^X$ is a *basis* for a topology if:

(i) for every $x \in X$, some $B \in \mathcal{B}$ contains x;
(ii) if $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then some $B_3 \in \mathcal{B}$ satisfies $x \in B_3 \subseteq B_1 \cap B_2$.

The topology generated by $\mathcal{B}$ consists of all unions of basis elements. A *subbasis* is a collection whose finite intersections form a basis.

Example 6.1.4. *Metric Balls as a Basis*
In a metric space, the collection of all open balls is a basis for the metric topology. On $\mathbb{R}^k$, balls with rational centers and rational radii form a countable basis.

6.2 Products, Quotients, and Countability Axioms

Definition 6.2.1. *Product Topology*
For topological spaces X and Y, the *product topology* on $X \times Y$ is generated by basis sets $U \times V$, where $U \subseteq X$ and $V \subseteq Y$ are open. For a family $\{X_i\}_{i \in I}$, the product topology on $\prod_i X_i$ is generated by finite-coordinate restrictions: sets of the form $\prod_i U_i$ where each U_i is open and $U_i = X_i$ for all but finitely many i.

Proposition 6.2.2. *Product Topology and Coordinate Maps*
A map $f : Z \to X \times Y$ is continuous for the product topology if and only if both coordinate maps $\pi_X \circ f$ and $\pi_Y \circ f$ are continuous.

Proof. If f is continuous, then coordinate maps are continuous because inverse images of open sets are inverse images of product-open cylinders. Conversely, if both coordinates are continuous, then the inverse image of each basis set $U \times V$ is $(\pi_X \circ f)^{-1}(U) \cap (\pi_Y \circ f)^{-1}(V)$, which is open.
Structural Commentary: Product topology is the weakest topology making the projections continuous.
Key Insight: Continuity into a product is checked coordinatewise. □

Definition 6.2.3. *Quotient Topology*
Let $q : X \to Y$ be a surjection from a topological space X onto a set Y.

(i) *Quotient topology* on Y is topology w. open sets are precisely subsets $V \subseteq Y$ s,t, $q^{-1}(V)$ open in X.
(ii) Equivalently, it is the strongest topology on Y for which q is continuous.

Proposition 6.2.4. *Quotient Universal Property*
Let $q : (X, \tau_X) \twoheadrightarrow Y$ and give Y the quotient topology $\tau_Y = \{V \subseteq Y : q^{-1}(V) \in \tau_X\}$.
For any map $g : Y \to (Z, \tau_Z)$, $g : (Y, \tau_Y) \to (Z, \tau_Z)$ is continuous $\iff$ $g \circ q : (X, \tau_X) \to (Z, \tau_Z)$ is continuous.

Proof.

($\Rightarrow$) If g is continuous, then $g \circ q$ is continuous by composition.
($\Leftarrow$) Suppose $g \circ q$ is continuous. For $W \in \tau_Z$, $q^{-1}(g^{-1}(W)) = (g \circ q)^{-1}(W) \in \tau_X$.
Thus $g^{-1}(W) \in \tau_Y$, so g is continuous.

Structural Commentary: Continuity from a quotient is tested by precomposition with the quotient map.
Key Insight: Pull back through the quotient map and use the defining open-set test. □

Definition 6.2.5. *Countability Axioms*
A topological space (X, τ) is:

(i) *first countable* if for each $x \in X$ there is a countable neighborhood basis $\mathcal{B}_x$ at x: for every neighborhood U of x, some $B \in \mathcal{B}_x$ satisfies $B \subseteq U$;
(ii) *second countable* if τ has a countable basis;
(iii) *separable* if there is a countable $D \subseteq X$ with $\overline{D} = X$, agreeing with Definition 3.3.3 for metric spaces.

Proposition 6.2.6. *Second Countable Implies Separable*
If (X, τ) is second countable, then there is a countable $D \subseteq X$ with $\overline{D} = X$.

Proof. Let $\mathcal{B} = \{B_n : n \in I\}$ be a countable basis, with $I \subseteq \mathbb{N}$.
For each $n \in I$ with $B_n \neq \emptyset$, choose $x_n \in B_n$ and set $D = \{x_n : n \in I,\ B_n \neq \emptyset\}$. Then D is countable.
If $U \in \tau \setminus \{\emptyset\}$, choose $x \in U$; some $B_n \in \mathcal{B}$ satisfies $x \in B_n \subseteq U$, so $x_n \in D \cap U$. Hence $\overline{D} = X$.
Structural Commentary: $\mathcal{B}$ countable gives countable D meeting every $U \in \tau \setminus \{\emptyset\}$.
Key Insight: D is dense iff $D \cap U \neq \emptyset$ for every $U \in \tau \setminus \{\emptyset\}$. □

Part X
Calculus, Series, and Approximation

1 One-Variable Differentiation and Riemann Integration

1.1 Differentiation and Basic Theorems

One-variable calculus studies functions on intervals through derivatives, mean-value theorems, Riemann integration, and the FTC, then points toward vector-valued estimates and linear approximation.

Definition 1.1.1. *Derivative of a Real Function*
Let f be real-valued on $[a,b]$ and let $x \in [a,b]$. For $a < t < b$ and $t \neq x$, set $\phi_x(t) := \frac{f(t)-f(x)}{t-x}$.
If $\lim_{t\to x} \phi_x(t)$ exists, denote *derivative* by $f'(x)$; then f is *differentiable at* x.
If this holds for every $x \in E \subseteq [a,b]$, then f is *differentiable on* E.



Figure 151: Domain matters: isolated points give no punctured approach; $[x_0, \infty)$ gives a right-hand one.

Perspective 1.1.2. *Derivative Depends on the Domain*
For $f : X \to \mathbb{R}$, $f'(x_0)$ is the limit of $\frac{f(x)-f(x_0)}{x-x_0}$ as $x \to x_0$ in $X \setminus \{x_0\}$; see Definition 5.1.4. Hence x_0 must be a limit point of X: isolated points give no derivative, while half-interval domains give one-sided derivatives.

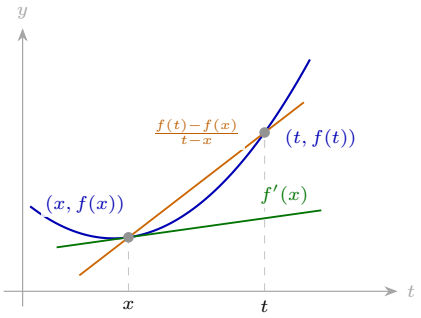


Figure 152: The derivative is the limiting slope of secant lines through $(x, f(x))$.

Proposition 1.1.3. *First-Order Approximation Form*
Let $X \subseteq \mathbb{R}$, let $x_0 \in X$ be a limit point, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. Then $f'(x_0) = L$ relative to X iff $\forall \varepsilon > 0$ $\exists \delta > 0$ such that $|f(x) - f(x_0) - L(x - x_0)| \leq \varepsilon\,|x - x_0|$ whenever $x \in X$ and $|x - x_0| < \delta$.

Proof. For $x \neq x_0$, $\frac{|f(x)-f(x_0)-L(x-x_0)|}{|x-x_0|} = \left|\frac{f(x)-f(x_0)}{x-x_0} - L\right|$, while $x = x_0$ makes approx. estimate automatic.

($\Rightarrow$) If $f'(x_0) = L$, then the displayed right-hand side is at most ε for all $x \in X \setminus \{x_0\}$ sufficiently near x_0. Multiplying by $|x - x_0|$ gives the approximation estimate.

($\Leftarrow$) If the approximation estimate holds, then division by $|x - x_0|$ gives $\left|\frac{f(x)-f(x_0)}{x-x_0} - L\right| \leq \varepsilon$ for all $x \in X\setminus\{x_0\}$ sufficiently near x_0. Hence $f'(x_0) = L$.

Structural Commentary: Differentiability is first-order linear approximation.
Key Insight: Multiply or divide by $|x - x_0|$. □

Theorem 1.1.4. *Differentiability Implies Continuity*
If f is differentiable at x, then f is continuous at x.

Proof. For $t \neq x$, $f(t) - f(x) = \frac{f(t)-f(x)}{t-x}(t - x)$. As $t \to x$, the difference quotient tends to $f'(x)$ and $t - x \to 0$. By Theorem 5.1.9, $f(t) - f(x) \to 0$, so $f(t) \to f(x)$. Theorem 5.1.12 gives continuity at x.
Structural Commentary: A differentiable function has first-order control near the point.
Key Insight: The quotient stays near a finite number while $t - x$ shrinks to zero. □

Theorem 1.1.5. *Algebra Rules for Derivatives*
If f and g are differentiable at x, then so are $f+g$ and fg; if $g(x) \neq 0$, then f/g is differentiable at x. Moreover,

$$(f+g)'(x) = f'(x) + g'(x),$$
$$(fg)'(x) = f'(x)g(x) + f(x)g'(x),$$
$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g(x)^2}.$$

Proof. The sum rule is limit algebra. For $h = fg$, $h(t) - h(x) = f(t)\big(g(t) - g(x)\big) + g(x)\big(f(t) - f(x)\big)$. Dividing by $t - x$ and letting $t \to x$ gives the product rule, using Theorem 1.1.4 for $f(t) \to f(x)$.
For $h = f/g$, continuity of g makes $g(t) \neq 0$ near x, and $\frac{h(t)-h(x)}{t-x} = \frac{1}{g(t)g(x)}\left[g(x)\frac{f(t)-f(x)}{t-x} - f(x)\frac{g(t)-g(x)}{t-x}\right]$. Letting $t \to x$ gives the quotient rule.
Structural Commentary: Product and quotient rules are difference-quotient identities plus continuity.
Key Insight: Difference-quotient identities plus continuity give the algebra rules. □

Theorem 1.1.6. *Chain Rule*
Let f be continuous on $[a,b]$ with $f'(x)$ existing at $x \in [a,b]$, and let g be defined on an interval containing $f([a,b])$ and differentiable at $f(x)$. If $h(t) = g(f(t))$, then $h'(x) = g'(f(x))f'(x)$.

Proof. Let $y = f(x)$. For some $u(t) \to 0$ and $v(s) \to 0$,

$$f(t) - f(x) = (t-x)\big(f'(x) + u(t)\big),$$
$$g(s) - g(y) = (s-y)\big(g'(y) + v(s)\big).$$

With $s = f(t)$, continuity gives $s \to y$, and for $t \neq x$, $\frac{h(t)-h(x)}{t-x} = \big(g'(y) + v(f(t))\big)\big(f'(x) + u(t)\big) \to g'(y)f'(x)$.
Structural Commentary: Each differentiability statement is a linear approximation plus a vanishing error.
Key Insight: Composition multiplies the two first-order parts; the error terms disappear. □

Example 1.1.7. *Oscillation at a Point*
Assuming the usual derivative of $\sin x$, define

$$f_1(x) = \begin{cases} x\sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases} \qquad f_2(x) = \begin{cases} x^2\sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

For f_1, the quotient at 0 is $\sin(1/t)$, so $f_1'(0)$ does not exist. For f_2, $|(f_2(t) - f_2(0))/t| \leq |t|$, so $f_2'(0) = 0$. However, for $x \neq 0$, $f_2'(x) = 2x\sin(1/x) - \cos(1/x)$, so f_2' is not continuous at 0.

Definition 1.1.8. *Local Extrema*
Let f be a real-valued function on a metric space X. The function has a *local maximum* at $p \in X$ if there is $\delta > 0$ such that $f(q) \leq f(p)$ whenever $d(p,q) < \delta$. Local minima are defined by reversing the inequality.

Remark 1.1.9. *Local and Global Extrema*
Global extrema are local, but local extrema depend on domain: isolated points are automatic extrema. Derivative tests therefore need an interior interval point; compactness supplies global extrema (Theorem 5.1.35).

Theorem 1.1.10. *Interior Extremum Has Zero Derivative*
If f has a local extremum at $x \in (a,b)$ and $f'(x)$ exists, then $f'(x) = 0$.

Proof. For a local maximum, $f(t) \leq f(x)$ near x.
Thus the quotient $(f(t) - f(x))/(t-x)$ is nonnegative for $t < x$ and nonpositive for $t > x$. Since the two-sided limit exists, $f'(x) \geq 0$ and $f'(x) \leq 0$, so $f'(x) = 0$. For a local minimum, apply the local-maximum case to $-f$. Since $(-f)'(x) = -f'(x)$, we get $-f'(x) = 0$, hence $f'(x) = 0$.
Structural Commentary: Interior extrema force opposite one-sided quotient signs.
Key Insight: A two-sided derivative compatible with both signs must be zero. □

Theorem 1.1.11. *Rolle's Theorem*
If $f : [a,b] \to \mathbb{R}$ continuous on $[a,b]$, differentiable on (a,b), and $f(a) = f(b)$, then $\exists x \in (a,b)$ s.t. $f'(x) = 0$.

Proof. If f is constant, any $x \in (a,b)$ works. Otherwise, by Theorem 5.1.35, f attains a nonendpoint extremum at some $x \in (a,b)$. Theorem 1.1.10 gives $f'(x) = 0$.
Structural Commentary: Equal endpoints force any nonconstant height change to turn around inside.
Key Insight: Compactness gives the extremum; differentiability makes the interior extremum stationary. □

Theorem 1.1.12. *Cauchy Mean Value Theorem*
Let f, g be continuous on $[a,b]$, differentiable on (a,b). $\exists\, x \in (a,b)$, $\big(f(b) - f(a)\big)g'(x) = \big(g(b) - g(a)\big)f'(x)$.

Proof. Set $h(t) := \big(f(b) - f(a)\big)g(t) - \big(g(b) - g(a)\big)f(t)$.
Then $h(a) = h(b)$, h is continuous on $[a, b]$, and h is differentiable on (a, b). By Theorem 1.1.11, $h'(x) = 0$ for some $x \in (a, b)$. Since $h'(x) = \big(f(b) - f(a)\big)g'(x) - \big(g(b) - g(a)\big)f'(x)$, the identity follows.
Structural Commentary: Build one auxiliary function whose endpoint values agree.
Key Insight: Equal endpoints force an interior horizontal tangent. □

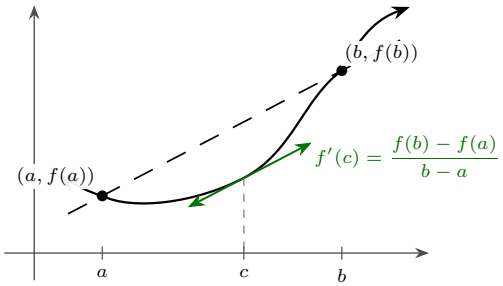


Figure 153: The Mean Value Theorem finds an interior tangent slope equal to the endpoint secant slope.

Corollary 1.1.13. *Mean Value Theorem*
If f is continuous on $[a, b]$, differentiable in (a, b), then $\exists\ x \in (a, b)$ such that $f(b) - f(a) = (b - a)f'(x)$.

Proof. Let $g(t) = t$. Then $g'(x) = 1$, $g(b) - g(a) = b - a$.
Hence Theorem 1.1.12 gives $f(b) - f(a) = (b - a)f'(x)$ for some $x \in (a, b)$.
Structural Commentary: This is Cauchy's theorem with $g(t) = t$.
Key Insight: A secant slope is achieved as a tangent slope somewhere inside. □

Corollary 1.1.14. *Bounded Derivative Gives Lipschitz Control*
If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $|f'| \leq M$ on (a, b), then f is M-Lipschitz on $[a, b]$, hence uniformly continuous.

Proof. For $x < y$, Corollary 1.1.13 gives $f(y) - f(x) = (y - x)f'(c)$ for some $c \in (x, y)$,
so $|f(y) - f(x)| \leq M(y - x)$. Symmetry handles $x > y$, and $x = y$ is trivial.
Structural Commentary: A derivative bound controls every secant slope.
Key Insight: Local slope bounds become global Lipschitz bounds through the Mean Value Theorem. □

Theorem 1.1.15. *Derivative Sign and Monotonicity*
Suppose f is differentiable in (a, b).

(i) If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing.
(ii) If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.
(iii) If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing.

Proof. For $x_1 < x_2$ in (a, b), Corollary 1.1.13 on $[x_1, x_2]$ gives $f(x_2) - f(x_1) = (x_2 - x_1)f'(c)$ for some $c \in (x_1, x_2)$.

(i) If $f' \geq 0$, then $f(x_2) - f(x_1) \geq 0$.
(ii) If $f' = 0$, then $f(x_2) - f(x_1) = 0$.
(iii) If $f' \leq 0$, then $f(x_2) - f(x_1) \leq 0$.

Structural Commentary: Monotonicity is a global reading of the mean-value identity.
Key Insight: The sign of every tangent slope controls every secant slope. □

Theorem 1.1.16. *Derivatives Have the Intermediate Value Property*
If f is differentiable on $[a, b]$ in the one-sided endpoint sense and $f'(a) < \lambda < f'(b)$,
Then $f'(x) = \lambda$ for some $x \in (a, b)$. The same conclusion holds when $f'(b) < \lambda < f'(a)$.

Proof. Set $g(t) = f(t) - \lambda t$. Since $g'(a) < 0 < g'(b)$, there are $t_1, t_2 \in (a, b)$ with $g(t_1) < g(a)$ and $g(t_2) < g(b)$.
Hence the minimum of g on $[a, b]$ occurs at some $x \in (a, b)$. By Theorem 1.1.10, $0 = g'(x) = f'(x) - \lambda$.
Apply the same argument to $-f$ for the reversed case.
Structural Commentary: A derivative need not be continuous, but it cannot jump over values.
Key Insight: Tilt the graph by subtracting λt and force an interior extremum. □

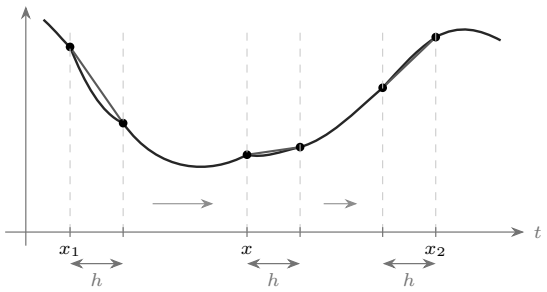


Figure 154: Sliding fixed-width secants in Darboux's theorem.

Perspective 1.1.17. *Sliding Secants*
For fixed small $h > 0$, the secant-slope function $\sigma_h(x) := \frac{f(x+h)-f(x)}{h}$ is continuous. Thus Theorem 5.1.38 finds intermediate secant slopes, and Corollary 1.1.13 converts them to derivative values.
Key Insight: Slide secants, then apply the Mean Value Theorem.

Corollary 1.1.18. *Derivatives Have No Simple Discontinuities*
If f is differentiable on $[a,b]$, then f' cannot have a simple discontinuity on $[a,b]$.

Proof. Suppose a simple discontinuity occurs at an interior point c; endpoint cases are one-sided.
If $f'(c-) \neq f'(c+)$, choose λ strictly between these limits, $\lambda \neq f'(c)$. For $x < c < y$ close to c, Theorem 1.1.16 gives $z \in (x,y)$ with $f'(z) = \lambda$, contradicting the one-sided limits. If $f'(c-) = f'(c+) = L \neq f'(c)$, choose λ between L and $f'(c)$ and apply the same theorem on a sufficiently small one-sided interval with endpoint c.
Structural Commentary: Darboux behavior rules out jump discontinuities for derivatives.
Key Insight: Derivatives may oscillate, but they do not jump. □

Theorem 1.1.19. *Derivative of an Inverse Function*
Let $f : I \to \mathbb{R}$ be one-to-one on an open interval I, let $x_0 \in I$, and put $y_0 = f(x_0)$. Suppose f is differentiable at x_0, $f'(x_0) \neq 0$, and $g = f^{-1} : f(I) \to I$ is continuous at y_0. Then g is differentiable at y_0 and $g'(y_0) = \frac{1}{f'(x_0)}$.

Proof. For $y \in f(I)$ with $y \neq y_0$, write $x = g(y)$. Then $x \neq x_0$ and $\frac{g(y)-g(y_0)}{y-y_0} = \frac{x-x_0}{f(x)-f(x_0)} = \left(\frac{f(x)-f(x_0)}{x-x_0}\right)^{-1}$.
As $y \to y_0$, the assumed continuity of g gives $x = g(y) \to x_0$.
The difference quotient for f tends to $f'(x_0) \neq 0$, so the reciprocal tends to $1/f'(x_0)$ by Theorem 5.1.9.
Structural Commentary: Matched inverse quotients are reciprocal.
Key Insight: Continuity of the inverse converts the limit $y \to y_0$ back into $x \to x_0$. □

Corollary 1.1.20. *One-Dimensional Inverse Function Theorem*
If $f : (a,b) \to (c,d)$ is a differentiable bijection and $f' \neq 0$,
then f is a homeomorphism and f^{-1} is differentiable, with $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$.

Proof. By Theorem 1.1.16, f' has constant sign. Corollary 1.1.13 makes f strictly monotone, hence its continuous interval inverse is continuous. Apply Theorem 1.1.19 pointwise.
Structural Commentary: Darboux gives constant sign; the inverse derivative formula is local.
Key Insight: Nonvanishing derivative is the one-dimensional obstruction to differentiating the inverse. □

Theorem 1.1.21. *L'Hospital's Rule*
Let f, g be differentiable on (a,b), $g' \neq 0$, and $-\infty \leq a < b \leq +\infty$.
If $\frac{f'}{g'} \to A \in \overline{\mathbb{R}}$ as $x \to a$, and either $f, g \to 0$ or $g \to +\infty$, then $\frac{f}{g} \to A$ as $x \to a$.
The same holds as $x \to b$, and for $g \to -\infty$.

Proof. For $a < x < y < b$, Theorem 1.1.12 gives $t \in (x,y)$ with $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(t)}{g'(t)}$.
If $f, g \to 0$, let $x \to a$ with y fixed. If $g \to +\infty$, use $\frac{f(x)}{g(x)} = \frac{f(x)-f(y)}{g(x)-g(y)}\,\frac{g(x)-g(y)}{g(x)} + \frac{f(y)}{g(x)}$ and let $x \to a$.
In both cases, bounds on f'/g' near a transfer to f/g, forcing the limit.
Structural Commentary: CMVT transfers derivative bounds to quotient bounds.
Key Insight: Derivative quotients govern the original indeterminate quotient. □

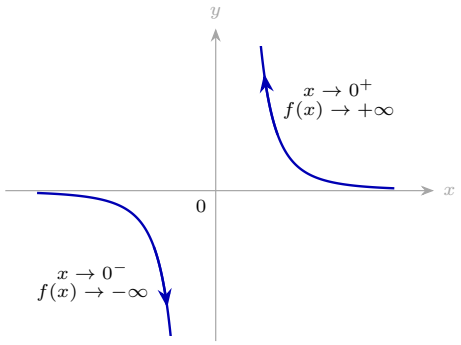


Figure 155: For $f(x) = x^{-3}$, the one-sided limits at 0 are opposite infinities.

Definition 1.1.22. *Higher Derivatives*
If f has derivative f' on an interval and f' is differentiable, then $(f')'$ is denoted f'' and called the *second derivative*. Continuing recursively gives $f^{(n)}$, the derivative of order n.

Definition 1.1.23. *Smoothness Classes*
On an interval, C^0 means continuous, C^r means $f^{(r)}$ exists and continuous, C^∞ means C^r for every finite r. Analytic functions (Definition 9.1.7) form the class C^ω.

Perspective 1.1.24. *Regularity Hierarchy*
The inclusions are $C^\omega \subseteq C^\infty \subseteq \cdots \subseteq C^2 \subseteq C^1 \subseteq C^0$.
The inclusions follow from Theorem 9.1.8 and Definition 1.1.23; they are generally proper.
Key Insight: Higher regularity is extra structure, not just a different notation for continuity.

Theorem 1.1.25. *Higher-Order Leibniz Product Rule*
If $f, g : (a, b) \to \mathbb{R}$ are r times differentiable, $r \geq 1$, then $(fg)^{(r)}(x) = \sum_{k=0}^{r} \binom{r}{k} f^{(k)}(x) g^{(r-k)}(x)$.

Proof. Induct on r. The case $r = 1$ is Theorem 1.1.5.
Differentiating the order-r formula gives $(fg)^{(r+1)} = \sum_{k=0}^{r} \binom{r}{k} f^{(k+1)} g^{(r-k)} + \sum_{k=0}^{r} \binom{r}{k} f^{(k)} g^{(r-k+1)}$.
Reindex and use Pascal's identity.
Structural Commentary: Each derivative chooses whether it lands on f or on g.
Key Insight: The binomial coefficients count those choices. □

Theorem 1.1.26. *Taylor's Theorem with Lagrange Remainder*
Let $f : [a, b] \to \mathbb{R}$, let $n \geq 1$, suppose $f^{(n-1)}$ is continuous on $[a, b]$, and suppose $f^{(n)}$ exists on (a, b).
For distinct $\alpha, \beta \in [a, b]$, set $P(t) := \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t - \alpha)^k$.
Then for some x between α and β, $f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$.

Proof. Choose M so that $f(\beta) = P(\beta) + M(\beta - \alpha)^n$, and set $G(t) := f(t) - P(t) - M(t - \alpha)^n$.
Then $G(\alpha) = G'(\alpha) = \cdots = G^{(n-1)}(\alpha) = G(\beta) = 0$. Repeated applications of Corollary 1.1.13 give $G^{(n)}(x) = 0$ for some x between α and β. Since $G^{(n)}(x) = f^{(n)}(x) - n!M$, the formula follows.
Structural Commentary: The auxiliary error vanishes at β and to order $n - 1$ at α.
Key Insight: Repeated Rolle-style arguments turn endpoint cancellation into a precise remainder. □

Example 1.1.27. *Smooth but Not Analytic*
Define $e(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$. By repeated L'Hospital, Theorem 1.1.21, $e \in C^\infty$ and $e^{(r)}(0) = 0$ for all r.
Its Taylor series at 0 is zero, but $e(x) > 0$ for $x > 0$, so e is not analytic at 0 (Definition 9.1.7).
Key Insight: A function can be infinitely tangent to the x-axis without being locally equal to its Taylor series.

Method 1.1.28. *Newton and Secant Iterations*
To solve $f(x) = 0$:

(i) Newton uses tangent-line zero: $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$, $n \geq 1$ when $f'(x_{n-1}) \neq 0$.

(ii) Secant uses two-point slope: $x_n = x_{n-1} - \frac{f(x_{n-1})(x_{n-2} - x_{n-1})}{f(x_{n-2}) - f(x_{n-1})}$, $n \geq 2$ when the denominator is nonzero.

These define iterates only; local convergence is proved with Taylor's theorem (Theorem 1.1.26).



Figure 156: Newton: tangent-line zero. Secant: two-point line zero.

Definition 1.1.29. *Vector-Valued Derivative*
For $\mathbf{f} : [a, b] \to \mathbb{R}^k$, set $\mathbf{f}'(x) = \mathbf{L}$ when $\lim_{t\to x} \left\| \frac{\mathbf{f}(t)-\mathbf{f}(x)}{t-x} - \mathbf{L} \right\|_2 = 0$.
For $\mathbf{f} = (f_1, \ldots, f_k)$, this is equivalent to componentwise differentiability, with $\mathbf{f}'(x) = (f_1'(x), \ldots, f_k'(x))$.

Theorem 1.1.30. *Vector-Valued Mean Value Inequality*
If $\mathbf{f} : [a, b] \to \mathbb{R}^k$ is continuous on $[a, b]$ and differentiable on (a, b).
Then for some $x \in (a, b)$, $\|\mathbf{f}(b) - \mathbf{f}(a)\|_2 \le (b - a) \|\mathbf{f}'(x)\|_2$.

Proof. Set $\mathbf{z} := \mathbf{f}(b) - \mathbf{f}(a)$ and $\varphi(t) := \mathbf{z} \cdot \mathbf{f}(t)$.
Corollary 1.1.13 gives $\varphi(b) - \varphi(a) = (b-a)\varphi'(x) = (b-a)\mathbf{z} \cdot \mathbf{f}'(x)$ for some $x \in (a, b)$. Since $\varphi(b) - \varphi(a) = \|\mathbf{z}\|_2^2$, Theorem 1.2.9 gives $\|\mathbf{z}\|_2^2 \le (b - a) \|\mathbf{z}\|_2 \|\mathbf{f}'(x)\|_2$, and the result follows by cancellation, with $\mathbf{z} = 0$ trivial.
Structural Commentary: A scalar projection yields the vector norm bound.
Key Insight: Project onto the displacement vector, then use Schwarz to return to norms. □

Method 1.1.31. *Differentiation Proof Patterns*

(i) *Domain*: require x_0 to be a limit point of X; isolated points have no derivative by Perspective 1.1.2.
(ii) *Definition*: factor $f(t) - f(x) = (t - x)Q_x(t)$ and take $Q_x(t) \to f'(x)$.
(iii) *Approximation*: use Proposition 1.1.3 to replace quotient limits by first-order error bounds.
(iv) *Algebra*: prove quotient identities first; use continuity to pass to limits.
(v) *Mean values*: choose h with equal endpoint values and apply Theorem 1.1.10.
(vi) *Taylor*: set $G = f - P - M(t - \alpha)^n$ and iterate the mean value theorem.

Key Insight: Difference quotients and auxiliary functions drive one-variable differentiation.

1.2 Riemann Integration

The integration side of one-variable calculus is organized by Riemann and Riemann–Stieltjes sums; the fundamental theorem connects the two halves.

This subsection develops Riemann and Riemann–Stieltjes integration by partitions, then links integrals to antiderivatives through the Fundamental Theorem of Calculus. Point partitions and interval-envelope partitions are equivalent via common refinement.

Definition 1.2.1. *Partitions and Riemann Upper/Lower Sums*
A *partition* of $[a, b]$ is $\Pi = \{x_0, \ldots, x_n\}$ with $a = x_0 \le x_1 \le \cdots \le x_n = b$, $\Delta x_i := x_i - x_{i-1}$.
For bounded real f on $[a, b]$, set $M_i := \sup_{x_{i-1} \le x \le x_i} f(x)$, $m_i := \inf_{x_{i-1} \le x \le x_i} f(x)$,
and $U(\Pi, f) := \sum_{i=1}^{n} M_i \Delta x_i$, $L(\Pi, f) := \sum_{i=1}^{n} m_i \Delta x_i$.
Define $\overline{\int_a^b} f(x)\,dx := \inf_\Pi U(\Pi, f)$, $\underline{\int_a^b} f(x)\,dx := \sup_\Pi L(\Pi, f)$, with Π ranging over all partitions.
If these agree, f is *Riemann integrable*, written $f \in \mathcal{R}$, and the common value is $\int_a^b f(x)\,dx$ or $\int_a^b f$.
For $m \le f \le M$, $m(b - a) \le L(\Pi, f) \le U(\Pi, f) \le M(b - a)$, upper and lower integrals exist for bounded f.

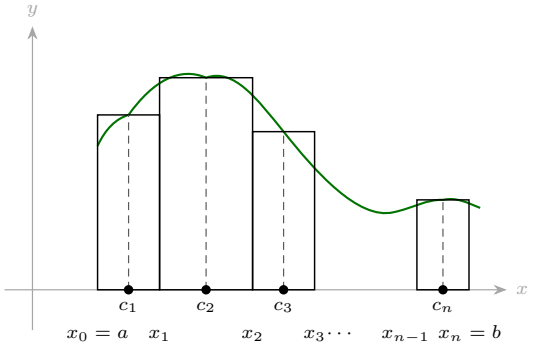


Figure 157: A tagged Riemann sum uses one sample value $f(c_i)$ on each subinterval $[x_{i-1}, x_i]$.

Definition 1.2.2. *Tagged Partitions and Riemann Sums*
A *tagged partition* of $[a,b]$ is a partition $\Pi = \{x_0, \dots, x_n\}$ together with tags $c_i \in [x_{i-1}, x_i]$.
For the ordinary Riemann integral, the corresponding *Riemann sum* is $R(f, \Pi, \{c_i\}) := \sum_{i=1}^{n} f(c_i)\Delta x_i$.
For every tagged partition, $L(\Pi, f) \le R(f, \Pi, \{c_i\}) \le U(\Pi, f)$.

Perspective 1.2.3. *Tagged Sums and Gauges*
Gauge integration keeps tagged sums but replaces one mesh bound δ with a local bound $\delta(c_i)$.
Thus fineness may vary by tag, giving a stronger framework for FTC-type results.

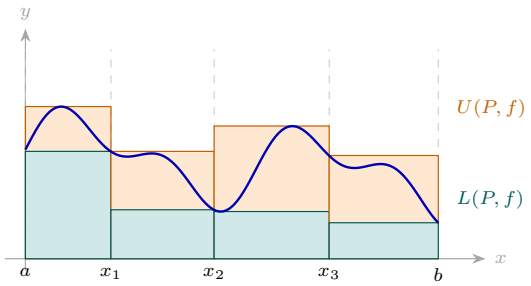


Figure 158: Upper and lower sums record the oscillation of f on each partition interval.

1.3 Fundamental Theorems of Calculus

Theorem 1.3.1. *Integral Functions*
Let $f \in \mathcal{R}$ on $[a,b]$ and define $F(x) := \int_a^x f(t)\,dt$. Then F is continuous.
If f is continuous at x_0, then $F'(x_0) = f(x_0)$, with one-sided derivatives at endpoints.

Proof. Let $|f| \le M$. For $x < y$, Theorem 2.1.16 gives $|F(y) - F(x)| = \left|\int_x^y f(t)\,dt\right| \le M(y-x)$ so F continuous.
If f continuous at x_0, for $x \ne x_0$ near x_0, $\left|\frac{F(x)-F(x_0)}{x-x_0} - f(x_0)\right| \le \sup_{u\in[x,x_0]} |f(u) - f(x_0)| \to 0$. $F'(x_0) = f(x_0)$.
Structural Commentary: Boundedness controls increments; local continuity gives the derivative.
Key Insight: Averaging a continuous function over shrinking intervals returns its value. □

Theorem 1.3.2. *Fundamental Theorem of Calculus*
If $f \in \mathcal{R}$ on $[a,b]$ and $\exists$ a differentiable function F on $[a,b]$ such that $F' = f$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

Proof. Choose $\Pi = \{x_0, \dots, x_n\}$ with $U(\Pi, f) - L(\Pi, f) < \varepsilon$.
By MVT, choose $t_i \in [x_{i-1}, x_i]$ so that $F(b) - F(a) = \sum_i \big(F(x_i) - F(x_{i-1})\big) = \sum_i f(t_i)\Delta x_i$.
Theorem 2.1.9 puts this tagged sum within ε of $\int_a^b f$. Let $\varepsilon \to 0$.
Structural Commentary: Mean-value increments convert antiderivatives into tagged Riemann sums.
Key Insight: The integral is the common limit of all sufficiently fine tagged sums. □

Theorem 1.3.3. *Integration by Parts*
Suppose F, G are differentiable on $[a,b]$, $F' = f \in \mathcal{R}$, and $G' = g \in \mathcal{R}$.
Then $\int_a^b F(x)g(x)\,dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x)\,dx$.

Proof. Set $H := FG$. Since F, G are continuous, $F, G \in \mathcal{R}$; hence $Fg, fG \in \mathcal{R}$ by Theorem 2.1.17.
Thus $H' = Fg + fG \in \mathcal{R}$, and Theorem 1.3.2 gives $F(b)G(b) - F(a)G(a) = H(b) - H(a) = \int_a^b Fg + \int_a^b fG$.
Structural Commentary: Integration by parts is the product rule integrated over the interval.
Key Insight: Move one derivative from G onto F by applying the fundamental theorem to FG. □

2 Riemann-Stieltjes and Improper Integration

2.1 Riemann-Stieltjes Integrability

Definition 2.1.1. *Riemann–Stieltjes Integral*
Let α be increasing on $[a,b]$. For $\Pi = \{x_0, \dots, x_n\}$, put $\Delta\alpha_i := \alpha(x_i) - \alpha(x_{i-1}) \ge 0$.
For bounded real f, with M_i, m_i as in Definition 1.2.1, set $U(\Pi, f, \alpha) := \sum_{i=1}^{n} M_i \Delta\alpha_i$; $L(\Pi, f, \alpha) := \sum_{i=1}^{n} m_i \Delta\alpha_i$.

Define $\overline{\int_a^b f\,d\alpha} := \inf_{\Pi} U(\Pi, f, \alpha)$; $\underline{\int_a^b f\,d\alpha} := \sup_{\Pi} L(\Pi, f, \alpha)$. If value match, f is *Riemann integrable with respect to α*, written $f \in \mathcal{R}(\alpha)$, and the common value is $\int_a^b f\,d\alpha$, also written $\int_a^b f(x)\,d\alpha(x)$.
Taking $\alpha(x) = x$ gives the ordinary Riemann integral; the variable of integration is dummy.

Remark 2.1.2. *Stieltjes Increments as α-Lengths*
Stieltjes sums replace ordinary length Δx_i by $\Delta\alpha_i$, the length measured by α.
Thus $\alpha(x) = x$ gives Riemann sums, while jumps of α place point mass as in Theorem 2.2.2.

Definition 2.1.3. *Refinements*
A partition Π^* is a *refinement* of Π if $\Pi^* \supseteq \Pi$. Given two partitions Π_1, Π_2, their *common refinement* is $\Pi_1 \cup \Pi_2$.

Theorem 2.1.4. *Refinement Monotonicity*
If Π^* is a refinement of Π, then $L(\Pi, f, \alpha) \le L(\Pi^*, f, \alpha)$ and $U(\Pi^*, f, \alpha) \le U(\Pi, f, \alpha)$.

Proof. It suffices to add one point $x^* \in (x_{i-1}, x_i)$. If w_1, w_2 are infima on $[x_{i-1}, x^*]$ and $[x^*, x_i]$, then $w_1, w_2 \ge m_i$ and $L(\Pi^*, f, \alpha) - L(\Pi, f, \alpha) = (w_1 - m_i)\big[\alpha(x^*) - \alpha(x_{i-1})\big] + (w_2 - m_i)\big[\alpha(x_i) - \alpha(x^*)\big] \ge 0$.
Iterate for additional points. The upper-sum inequality is the same argument with suprema.
Structural Commentary: Refinement raises lower sums and lowers upper sums.
Key Insight: Splitting an interval tightens the local oscillation bounds. □

Perspective 2.1.5. *Piecewise-Constant Envelopes*
A partition Π gives lower and upper step envelopes with values m_i and M_i; their integrals are exactly $L(\Pi, f)$ and $U(\Pi, f)$. Common refinements make these envelope sums compatible across partitions.

Theorem 2.1.6. *Lower Integral Below Upper Integral*
For every bounded real f and increasing α on $[a, b]$, $\underline{\int_a^b f\,d\alpha} \le \overline{\int_a^b f\,d\alpha}$.

Proof. Given partitions Π_1, Π_2, let $\Pi^* = \Pi_1 \cup \Pi_2$.
By Theorem 2.1.4, $L(\Pi_1, f, \alpha) \le L(\Pi^*, f, \alpha) \le U(\Pi^*, f, \alpha) \le U(\Pi_2, f, \alpha)$.
Take the supremum over Π_1 and then the infimum over Π_2.
Structural Commentary: Common refinements compare sums from different partitions.
Key Insight: The lower envelope never rises above the upper envelope. □

Theorem 2.1.7. *Upper–Lower Criterion*
A bounded real f is in $\mathcal{R}(\alpha)$ on $[a, b]$ if and only if $\forall \varepsilon > 0\ \exists$ a partition Π with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.

Proof.

($\Rightarrow$) If $f \in \mathcal{R}(\alpha)$, choose Π_1, Π_2 whose lower and upper sums are within $\varepsilon/2$ of $\int_a^b f\,d\alpha$.
Their common refinement Π satisfies the displayed inequality by Theorem 2.1.4.

($\Leftarrow$) If such Π exist, then $L(\Pi, f, \alpha) \le \underline{\int_a^b f\,d\alpha} \le \overline{\int_a^b f\,d\alpha} \le U(\Pi, f, \alpha)$ forces upper and lower integrals to agree.

Structural Commentary: Integrability is exactly the ability to make one partition squeeze the oscillation.
Key Insight: The gap $U - L$ is the quantitative defect of integrability. □

Perspective 2.1.8. *Darboux Gaps and Riemann Sums*
For $\alpha(x) = x$, a small Darboux gap $U(\Pi, f) - L(\Pi, f)$ traps every tagged sum for Π near the integral. This is the local reason the Darboux definition above and the tagged Riemann-sum definition give the same number. The same estimate works with Stieltjes weights $\Delta\alpha_i$ whenever the upper–lower gap is small.

Theorem 2.1.9. *Tagged Sums*
Let $\Pi = \{x_0, \dots, x_n\}$ satisfy $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$, and choose tags $t_i \in [x_{i-1}, x_i]$.

(i) If $s_i, t_i \in [x_{i-1}, x_i]$, then $\sum_{i=1}^n |f(s_i) - f(t_i)|\,\Delta\alpha_i < \varepsilon$.

(ii) If $f \in \mathcal{R}(\alpha)$, then $\left|\sum_{i=1}^n f(t_i)\Delta\alpha_i - \int_a^b f\,d\alpha\right| < \varepsilon$.

Proof.

(i) Since $|f(s_i) - f(t_i)| \le M_i - m_i$, summing gives $\sum_{i=1}^n |f(s_i) - f(t_i)|\,\Delta\alpha_i \le U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.

(ii) The tagged sum and $\int_a^b f\,d\alpha$ both lie between $L(\Pi, f, \alpha)$ and $U(\Pi, f, \alpha)$, so their distance is $< \varepsilon$.

Structural Commentary: A small Darboux gap makes tags and the integral close.
Key Insight: Tags are harmless after oscillation has been controlled. □

Theorem 2.1.10. *Continuous Functions Are Riemann–Stieltjes Integrable*
If f is continuous on $[a,b]$ and α is increasing on $[a,b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. If $\alpha(b) = \alpha(a)$, the Darboux gap is zero.
Otherwise choose $\eta > 0$ with $\eta[\alpha(b) - \alpha(a)] < \varepsilon$. Uniform continuity gives partition Π with $M_i - m_i \le \eta \ \forall i$, so $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \sum_{i=1}^{n} (M_i - m_i)\Delta\alpha_i \le \eta[\alpha(b) - \alpha(a)] < \varepsilon$. Apply Theorem 2.1.7.
Structural Commentary: Uniform continuity controls oscillation uniformly across the partition.
Key Insight: Continuity turns local small intervals into a small total Darboux gap. □

Theorem 2.1.11. *Monotone Functions and Continuous Integrators*
If f is monotone on $[a,b]$ and α is continuous and increasing on $[a,b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. Assume first that f is increasing. If $\alpha(b) = \alpha(a)$, then every $\Delta\alpha_i = 0$, so $U(\Pi, f, \alpha) = L(\Pi, f, \alpha) = 0$ for every partition Π. Choose n and a partition Π such that $\Delta\alpha_i = [\alpha(b) - \alpha(a)]/n$ for all i.
Existence follows from continuity of α and Theorem 5.1.38.
Then $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \frac{\alpha(b)-\alpha(a)}{n} \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) = \frac{\alpha(b)-\alpha(a)}{n}\big(f(b) - f(a)\big) < \varepsilon$ for n large.
If f is decreasing, use the same partition. Then the suprema and infima on $[x_{i-1}, x_i]$ are $f(x_{i-1})$ and $f(x_i)$, so the same computation gives $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) = \frac{\alpha(b)-\alpha(a)}{n}\big(f(a) - f(b)\big)$, which is $< \varepsilon$ for large n.
Structural Commentary: Equal α-increments convert the monotone oscillations into a telescoping sum.
Key Insight: A monotone function's total oscillation is finite and can be spread thinly. □

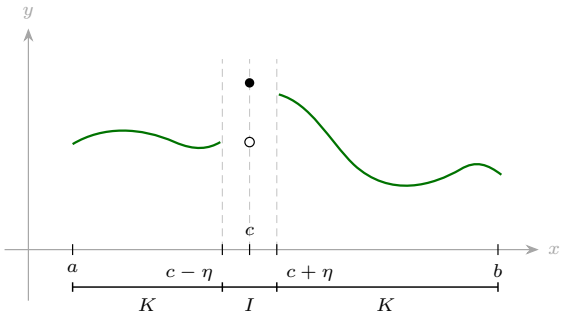


Figure 159: Isolate the discontinuity in a short interval; control the compact pieces outside it.

Theorem 2.1.12. *Finitely Many Discontinuities*
Suppose f is bounded on $[a,b]$, the discontinuity set $E := \{x \in [a,b] : f \text{ is discontinuous at } x\}$ is finite, and α is continuous at each point of E. Then $f \in \mathcal{R}(\alpha)$.

Proof. Let $M = \sup_{[a,b]} |f|$ and let $\eta > 0$. By continuity of α on the finite set E, choose disjoint intervals $I_j = [u_j, v_j]$ covering E, with each point of $E \cap (a,b)$ in some (u_j, v_j), such that $\sum_j \big(\alpha(v_j) - \alpha(u_j)\big) < \eta$.
Set $K = [a,b] \setminus \bigcup_j (u_j, v_j)$. Then $f|_K$ is uniformly continuous by Theorem 5.2.3.
Choose a partition Π containing all u_j, v_j, with no points inside the intervals (u_j, v_j), and fine enough on K that $M_i - m_i \le \eta$ whenever $[x_{i-1}, x_i] \subset K$.
Hence $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) \le \eta\big[\alpha(b) - \alpha(a)\big] + 2M \sum_j \big(\alpha(v_j) - \alpha(u_j)\big) < \eta\big[\alpha(b) - \alpha(a) + 2M\big]$.
Since η is arbitrary, Theorem 2.1.7 applies.
Structural Commentary: Isolate bad points in small α-intervals; use uniform continuity elsewhere.
Key Insight: Finitely many discontinuities do not matter unless α also jumps there. □

Example 2.1.13. *Dirichlet's Function Is Not Riemann Integrable*
Let $g = \mathbf{1}_{\mathbb{Q}}$ on $[0,1]$. Every nondegenerate subinterval contains rationals and irrationals by Corollary 1.1.17, so for every partition Π, $M_i = 1$ and $m_i = 0$ on each nonzero-width subinterval.
Thus $U(\Pi, g) = 1$ and $L(\Pi, g) = 0$, hence $g \notin \mathcal{R}$.

Remark 2.1.14. *Small Discontinuity Sets*
The same idea underlies Theorem 2.5.1: small discontinuity sets, later called measure-zero sets in Definition 1.4.13, do not obstruct integrability. Dirichlet's function fails because every point is discontinuous.

Theorem 2.1.15. *Continuous Composition Preserves Integrability*
Let α be increasing on $[a,b]$. Suppose $f \in \mathcal{R}(\alpha)$, $m \le f(x) \le M$ on $[a,b]$, and $\phi : [m,M] \to \mathbb{R}$ is continuous. Define $h : [a,b] \to \mathbb{R}$ by $h(x) := \phi(f(x))$. Then $h \in \mathcal{R}(\alpha)$.

Proof. Let $C := \sup_{y\in[m,M]} |\phi(y)|$ and let $\varepsilon > 0$. Choose $\eta > 0$ such that $\eta[\alpha(b) - \alpha(a) + 2C] < \varepsilon$. Since ϕ is uniformly continuous on $[m, M]$, choose $\gamma > 0$ such that $|s - t| < \gamma \implies |\phi(s) - \phi(t)| < \eta$ $(s, t \in [m, M])$.
Choose a partition $\Pi = \{x_0, \dots, x_n\}$ with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \gamma\eta$.
Let M_i, m_i denote upper and lower values for f on $[x_{i-1}, x_i]$, M_i^h, m_i^h denote corresponding values for h.
Split $A := \{i : M_i - m_i < \gamma\}$, $B := \{i : M_i - m_i \geq \gamma\}$. For $i \in A$, $M_i^h - m_i^h \leq \eta$; for $i \in B$, $M_i^h - m_i^h \leq 2C$.
Also $\gamma \sum_{i\in B} \Delta\alpha_i \leq \sum_{i\in B} (M_i - m_i)\Delta\alpha_i \leq U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \gamma\eta$, so $\sum_{i\in B} \Delta\alpha_i < \eta$.
$U(\Pi, h, \alpha) - L(\Pi, h, \alpha) \leq \eta \sum_{i\in A} \Delta\alpha_i + 2C \sum_{i\in B} \Delta\alpha_i < \eta[\alpha(b) - \alpha(a) + 2C] < \varepsilon$. Theorem 2.1.7 gives $h \in \mathcal{R}(\alpha)$.
Structural Commentary: Integrability survives continuous remeasurement of the values of f.
Key Insight: Large oscillation intervals have small total α-weight. □

Theorem 2.1.16. *Basic Properties of the Integral*
Let all functions be bounded and real on $[a, b]$, and let all integrators be increasing.

(i) If $f_1, f_2 \in \mathcal{R}(\alpha)$, $c \in \mathbb{R}$, then $f_1 + f_2, cf_1 \in \mathcal{R}(\alpha)$, and $\int_a^b (f_1 + f_2)\, d\alpha = \int_a^b f_1\, d\alpha + \int_a^b f_2\, d\alpha$; $\int_a^b cf_1\, d\alpha = c\int_a^b f_1\, d\alpha$.

(ii) If $f_1 \leq f_2$ and both functions are in $\mathcal{R}(\alpha)$, then $\int_a^b f_1\, d\alpha \leq \int_a^b f_2\, d\alpha$.

(iii) If $f \in \mathcal{R}(\alpha)$ and $a < c < b$, then f is integrable on $[a, c]$ and $[c, b]$, and $\int_a^c f\, d\alpha + \int_c^b f\, d\alpha = \int_a^b f\, d\alpha$.

(iv) If $f \in \mathcal{R}(\alpha)$ and $|f| \leq M$, then $\left|\int_a^b f\, d\alpha\right| \leq M\big[\alpha(b) - \alpha(a)\big]$

(v) If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and $\int_a^b f\, d(\alpha_1 + \alpha_2) = \int_a^b f\, d\alpha_1 + \int_a^b f\, d\alpha_2$.
If $c > 0$ and $f \in \mathcal{R}(\alpha)$, then $f \in \mathcal{R}(c\alpha)$ and $\int_a^b f\, d(c\alpha) = c\int_a^b f\, d\alpha$.

Proof.

(i) $L(\Pi, f_1, \alpha) + L(\Pi, f_2, \alpha) \leq L(\Pi, f_1 + f_2, \alpha) \leq U(\Pi, f_1 + f_2, \alpha) \leq U(\Pi, f_1, \alpha) + U(\Pi, f_2, \alpha)$, where Π is common refinement, and Theorem 2.1.7; scalar multiplication is the same estimate with signs handled by swapping upper and lower sums when $c < 0$.
(ii) If $f_1 \leq f_2$, then $L(\Pi, f_2 - f_1, \alpha) \geq 0$ for every Π; apply (i) to $f_2 - f_1$.
(iii) Refine so that $c \in \Pi$. Then the lower and upper sums split into the sums over $[a, c]$ and $[c, b]$.
(iv) Apply (ii) to $-M \leq f \leq M$ and use the integrals of the constant functions $\pm M$.
(v) Since $\Delta(\alpha_1 + \alpha_2)_i = \Delta\alpha_{1,i} + \Delta\alpha_{2,i}$ and $\Delta(c\alpha)_i = c\,\Delta\alpha_i$, corresponding lower and upper sums split exactly.

Structural Commentary: The algebra of integrals is inherited from the algebra of upper and lower sums.
Key Insight: Common refinements let several integrable functions be approximated by the same partition. □

Theorem 2.1.17. *Products and Absolute Values*
If $f, g \in \mathcal{R}(\alpha)$ on $[a, b]$, then $fg \in \mathcal{R}(\alpha)$, $|f| \in \mathcal{R}(\alpha)$, and $\left|\int_a^b f\, d\alpha\right| \leq \int_a^b |f|\, d\alpha$.

Proof. By Theorem 2.1.16, $f \pm g \in \mathcal{R}(\alpha)$. Applying Theorem 2.1.15 to $\phi(t) = t^2$ gives $(f \pm g)^2 \in \mathcal{R}(\alpha)$, hence $fg \in \mathcal{R}(\alpha)$ from $4fg = (f + g)^2 - (f - g)^2$.
Applying the same theorem to $\phi(t) = |t|$ gives $|f| \in \mathcal{R}(\alpha)$, and $-|f| \leq f \leq |f|$ gives the inequality.
Structural Commentary: Products reduce to squares; absolute values to composition.
Key Insight: Once continuous composition is available, the usual algebra closes. □

2.2 Step Integrators and Change of Variable

Definition 2.2.1. *Unit Step Function*
The *unit step function* I is $I(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases}$.

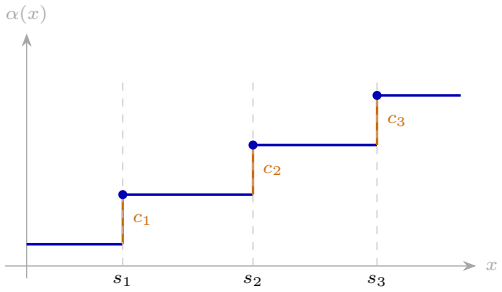


Figure 160: A step integrator concentrates mass at jumps: $\int f\,d\alpha = \sum_i c_i f(s_i)$.

Theorem 2.2.2. *Single Jump Integrator*
If $a < s < b$, f is bounded on $[a,b]$ and continuous at s, and $\alpha(x) = I(x-s)$, then $\int_a^b f\,d\alpha = f(s)$.

Proof. Use partitions $\Pi = \{a, s, x_2, b\}$ with $s < x_2 < b$. The only nonzero $\Delta\alpha_i$ occurs on $[s, x_2]$, so $L(\Pi, f, \alpha)$ and $U(\Pi, f, \alpha)$ are the infimum and supremum of f on $[s, x_2]$. As $x_2 \downarrow s$, continuity at s squeezes both to $f(s)$.
Structural Commentary: A jump of α makes the integral sample f at the jump.
Key Insight: Stieltjes integration includes point masses without changing the Darboux framework. □

Theorem 2.2.3. *Pure Step Integrators Give Series*
Suppose $c_n \geq 0$, $\sum c_n$ converges, and (s_n) is a sequence of distinct points in (a,b).
Define $\alpha(x) := \sum_{n=1}^{\infty} c_n I(x - s_n)$. If f is continuous on $[a,b]$, then $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$.

Proof. By comparison, α is increasing. For $N \in \mathbb{N}$, put $\alpha_N(x) := \sum_{n=1}^{N} c_n I(x - s_n)$ and $\beta_N := \alpha - \alpha_N$..

Theorems 2.2.2 and 2.1.16 give $\int_a^b f\,d\alpha_N = \sum_{n=1}^{N} c_n f(s_n)$. Also $f \in \mathcal{R}(\beta_N)$ by Theorem 2.1.10.

If $|f| \leq M$, then Theorem 2.1.16 gives $\left|\int_a^b f\,d\beta_N\right| \leq M[\beta_N(b) - \beta_N(a)] = M \sum_{n>N} c_n$.

Letting $N \to \infty$ and using additivity in the integrator gives the formula.
Structural Commentary: Finite step integrators sum point samples; summable tails vanish by the integral bound.
Key Insight: A Stieltjes integral can be an ordinary series in disguise. □

Theorem 2.2.4. *Differentiable Integrators Reduce to Riemann Integrals*
Assume α increases monotonically, $\alpha' \in \mathcal{R}$ on $[a,b]$, and f is bounded and real on $[a,b]$.
Then $f \in \mathcal{R}(\alpha) \iff f\alpha' \in \mathcal{R}$, and in that case $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. Let $M = \sup_{[a,b]} |f|$. Choose Π with $U(\Pi, \alpha') - L(\Pi, \alpha') < \varepsilon$.
For any refinement and any tags $s_i \in [x_{i-1}, x_i]$, MVT gives points $t_i \in [x_{i-1}, x_i]$ with $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$.
Hence $|\sum_i f(s_i)\Delta\alpha_i - \sum_i f(s_i)\alpha'(s_i)\Delta x_i| \leq M \sum_i |\alpha'(t_i) - \alpha'(s_i)|\,\Delta x_i \leq M\varepsilon$. Thus every tagged Stieltjes sum for $f\,d\alpha$ is within $M\varepsilon$ of the corresponding tagged Riemann sum for $f\alpha'\,dx$, and conversely. By the tagged-sum criterion, one family has a limit exactly when the other does, and the two limits are equal.
Structural Commentary: The increment $d\alpha$ is replaced by $\alpha'(x)\,dx$ when α' is Riemann integrable.
Key Insight: The mean value theorem turns Stieltjes increments into ordinary Riemann weights. □

Theorem 2.2.5. *Change of Variable*
Let $\varphi : [A,B] \to [a,b]$ be strictly increasing, continuous, and onto.
Let α be increasing on $[a,b]$, $f \in \mathcal{R}(\alpha)$, and set $\beta(y) := \alpha(\varphi(y))$ and $g(y) := f(\varphi(y))$.
Then $g \in \mathcal{R}(\beta)$ and $\int_A^B g\,d\beta = \int_a^b f\,d\alpha$. If also $\alpha(x) = x$ and $\varphi' \in \mathcal{R}$, then $\int_a^b f(x)\,dx = \int_A^B f(\varphi(y))\varphi'(y)\,dy$.

Proof. Partitions $Q = \{y_i\}$ of $[A,B]$ correspond to partitions $\Pi = \{\varphi(y_i)\}$ of $[a,b]$. On corresponding intervals, g has same range as f and $\Delta\beta_i = \Delta\alpha_i$, so $U(Q, g, \beta) = U(\Pi, f, \alpha)$ and $L(Q, g, \beta) = L(\Pi, f, \alpha)$. The upper–lower criterion gives the Stieltjes formula. If $\alpha(x) = x$, then $\beta = \varphi$; apply Theorem 2.2.4 to the left side.
Structural Commentary: A monotone change of variable relabels the same partition intervals.
Key Insight: Stieltjes change of variable is equality of Darboux sums under reparametrization. □

2.3 Improper Integrals

Definition 2.3.1. *Improper Riemann Integrals*
Let $f \in \mathcal{R}$ on every compact subinterval of its domain.
Set $\int_a^b f(x)\,dx := \lim_{d\to b^-} \int_a^d f(x)\,dx$ at a right endpoint, and $\int_a^b f(x)\,dx := \lim_{c\to a^+} \int_c^b f(x)\,dx$ at a left endpoint.
If both endpoints are improper, choose $\xi \in (a,b)$ and set $\int_a^b f(x)\,dx := \int_a^\xi f(x)\,dx + \int_\xi^b f(x)\,dx$, provided both one-sided integrals exist and the sum is not indeterminate.
A finite value means the improper integral *converges*; otherwise it diverges to $+\infty$ or $-\infty$.

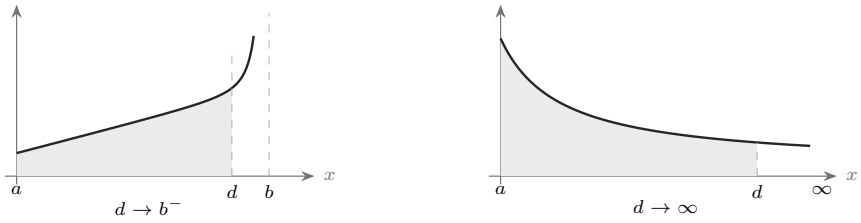


Figure 161: Improper integrals are limits of ordinary integrals over compact truncations.

Theorem 2.3.2. *Nonnegative Improper Integrals*
Let $f \geq 0$ and $f \in \mathcal{R}$ on every $[a,d]$, $a < d < b$. Then $\int_a^b f(x)\,dx = \sup\left\{\int_a^d f(x)\,dx : a < d < b\right\}$.
Thus $\int_a^b f$ exists in $[0,+\infty]$ and converges exactly when this supremum is finite.

Proof. For $a < d_1 < d_2 < b$, $\int_a^{d_2} f - \int_a^{d_1} f = \int_{d_1}^{d_2} f \geq 0$.
Hence $d \mapsto \int_a^d f$ is increasing, so its extended limit exists and equals its supremum.
Structural Commentary: Nonnegative area can only accumulate.
Key Insight: Improper convergence for $f \geq 0$ is boundedness of all finite truncations. □

Perspective 2.3.3. *Principal Value Is Different*
Improper integrals over $(-\infty,\infty)$ require separate one-sided limits.
A symmetric limit such as $\lim_{A\to\infty} \int_{-A}^{A} \sin x\,dx$ may still exist (Cauchy principal value, not Definition 2.3.1).

Definition 2.3.4. *Improper Riemann–Stieltjes Integrals on $\mathbb{R}$*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing and $f \in \mathcal{R}(\alpha)$ on every finite interval.
Define, when the limits exist, $\int_0^\infty f\,d\alpha := \lim_{b\to\infty} \int_0^b f\,d\alpha$; and $\int_{-\infty}^0 f\,d\alpha := \lim_{a\to-\infty} \int_a^0 f\,d\alpha$.
If their sum is not indeterminate, set $\int_{-\infty}^\infty f\,d\alpha := \int_{-\infty}^0 f\,d\alpha + \int_0^\infty f\,d\alpha$.
Finite value means f is *α-integrable on $\mathbb{R}$*. Write $\alpha(\pm\infty) := \lim_{t\to\pm\infty} \alpha(t)$ when these extended limits are used.

Theorem 2.3.5. *Nonnegative and Bounded-Mass Stieltjes Integrals*
Let $\alpha : \mathbb{R} \to \mathbb{R}$ be increasing, and suppose $f \in \mathcal{R}(\alpha)$ on every finite interval.

(i) If $f \geq 0$, then either f is α-integrable on $\mathbb{R}$ or $\int_{-\infty}^\infty f\,d\alpha = +\infty$.
(ii) If $-\infty < \alpha(-\infty) < \alpha(\infty) < +\infty$ and f is bounded, then f is α-integrable on $\mathbb{R}$.

Proof.

(i) By additivity and order preservation, $\int_0^b f\,d\alpha$ and $\int_a^0 f\,d\alpha$ are monotone under outward truncation. Hence both extended one-sided limits exist, so their sum is finite or $+\infty$.

(ii) Let $|f| \leq M$. For finite $a < b$, Theorem 2.1.16 gives $\left|\int_a^b f\,d\alpha\right| \leq M\big[\alpha(b) - \alpha(a)\big]$. Since $\alpha(\infty)$ and $\alpha(-\infty)$ are finite, the right and left tails have vanishing α-increment. Thus the one-sided truncations are Cauchy and have finite limits.

Structural Commentary: Positivity gives monotone limits; finite mass bounds tails.
Key Insight: A bounded increasing integrator behaves like finite total mass. □

2.4 Vector-Valued Integrals and Curves

Definition 2.4.1. *Vector-Valued Riemann–Stieltjes Integral*
For $\mathbf{f} = (f_1,\dots,f_k) : [a,b] \to \mathbb{R}^k$, define $\mathbf{f} \in \mathcal{R}(\alpha)$ iff $f_j \in \mathcal{R}(\alpha)$ $\forall j$, and set $\int_a^b \mathbf{f}\,d\alpha := \left(\int_a^b f_1\,d\alpha, \dots, \int_a^b f_k\,d\alpha\right)$.

Theorem 2.4.2. *Vector-Valued Integral Results*

(i) If $\mathbf{f}, \mathbf{F} : [a,b] \to \mathbb{R}^k$, $\mathbf{f} \in \mathcal{R}$, *and* $\mathbf{F}' = \mathbf{f}$, *then* $\int_a^b \mathbf{f}(t)\,dt = \mathbf{F}(b) - \mathbf{F}(a)$.

(ii) If $\mathbf{f} : [a,b] \to \mathbb{R}^k$ *and* $\mathbf{f} \in \mathcal{R}(\alpha)$, *then* $\|\mathbf{f}\| \in \mathcal{R}(\alpha)$ *and* $\left\|\int_a^b \mathbf{f}\,d\alpha\right\| \le \int_a^b \|\mathbf{f}\|\,d\alpha$

Proof.

(i) Write $\mathbf{f} = (f_1, \dots, f_k)$ and $\mathbf{F} = (F_1, \dots, F_k)$. For each j, Theorem 1.3.2 gives $\int_a^b f_j(t)\,dt = F_j(b) - F_j(a)$. The vector identity follows componentwise.

(ii) Write $\mathbf{f} = (f_1, \dots, f_k)$. Since $\|\mathbf{f}\| = (f_1^2 + \cdots + f_k^2)^{1/2}$, Theorems 2.1.17 and 2.1.15 give $\|\mathbf{f}\| \in \mathcal{R}(\alpha)$. Let $\mathbf{y} := \int_a^b \mathbf{f}\,d\alpha$. If $\mathbf{y} = 0$, the inequality is immediate.

Otherwise, $\|\mathbf{y}\|^2 = \sum_{j=1}^k y_j \int_a^b f_j\,d\alpha = \int_a^b \sum_{j=1}^k y_j f_j\,d\alpha \le \int_a^b \|\mathbf{y}\|\,\|\mathbf{f}\|\,d\alpha = \|\mathbf{y}\| \int_a^b \|\mathbf{f}\|\,d\alpha$, by Theorem 1.2.9 and order preservation. Divide by $\|\mathbf{y}\|$.

Structural Commentary: Coordinatewise integration; Schwarz controls norms.
Key Insight: Integrate the projection of $\mathbf{f}$ onto the resulting vector. □

Definition 2.4.3. *Curves and Rectifiability*
A *curve* in $\mathbb{R}^k$ is a continuous map $\gamma : [a,b] \to \mathbb{R}^k$; it is an *arc* if one-to-one and a *closed curve* if $\gamma(a) = \gamma(b)$. Curves are maps, not just ranges. For a partition $\Pi = \{x_0, \dots, x_n\}$, set $\ell(\Pi, \gamma) := \sum_{i=1}^n \|\gamma(x_i) - \gamma(x_{i-1})\|$. The *length* is $\ell(\gamma) := \sup_\Pi \ell(\Pi, \gamma)$; if $\ell(\gamma) < \infty$, γ is *rectifiable*.

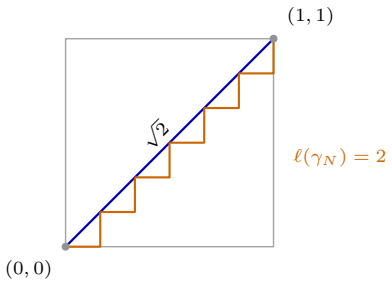


Figure 162: A staircase path can approach the diagonal in position while its polygonal length stays 2, not $\sqrt{2}$.

Perspective 2.4.4. *Length Is Not Just Shape*
Length is a supremum over partitions of one curve. Passing lengths through curve limits needs more than positional convergence; Theorem 2.4.5 uses C^1 speed control.

Theorem 2.4.5. *Length of a Continuously Differentiable Curve*
If $\gamma' : [a,b] \to \mathbb{R}^k$ *is continuous, then* γ *is rectifiable and* $\ell(\gamma) = \int_a^b \|\gamma'(t)\|\,dt$.

Proof. For $\Pi = \{x_0, \dots, x_n\}$, Theorem 2.4.2(i) gives $\|\gamma(x_i) - \gamma(x_{i-1})\| = \left\|\int_{x_{i-1}}^{x_i} \gamma'(t)\,dt\right\| \le \int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\,dt$.
Hence $\ell(\Pi, \gamma) \le \int_a^b \|\gamma'\|$. Conversely, choose Π so fine that $\|\gamma'(s) - \gamma'(t)\| < \varepsilon$ on each subinterval.
Choose $t_i \in [x_{i-1}, x_i]$. Then $\int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\,dt \le \|\gamma'(t_i)\|\,\Delta x_i + \varepsilon \Delta x_i$.
Also $\gamma(x_i) - \gamma(x_{i-1}) = \int_{x_{i-1}}^{x_i} \gamma'(t)\,dt$, so $\|\gamma'(t_i)\|\,\Delta x_i \le \|\gamma(x_i) - \gamma(x_{i-1})\| + \varepsilon \Delta x_i$.
Thus $\int_{x_{i-1}}^{x_i} \|\gamma'(t)\|\,dt \le \|\gamma(x_i) - \gamma(x_{i-1})\| + 2\varepsilon \Delta x_i$.
Summing gives $\int_a^b \|\gamma'\| \le \ell(\gamma) + 2\varepsilon(b-a)$; let $\varepsilon \downarrow 0$.
Structural Commentary: Upper bounds are immediate; uniform continuity gives matching lower bounds.
Key Insight: For C^1 curves, length is accumulated speed. □

Method 2.4.6. *Integration Proof Patterns*

(i) *Definitions*: fix $\Pi = \{x_i\}_{i=0}^n$ before defining $M_i, m_i, \Delta x_i, \Delta \alpha_i$.
(ii) *Refinement*: prove the one-point insertion case; iterate.
(iii) *Integrability*: find Π with $U(\Pi, f, \alpha) - L(\Pi, f, \alpha) < \varepsilon$.
(iv) *Bad sets*: cover E by $I_j = [u_j, v_j]$ with $\sum_j [\alpha(v_j) - \alpha(u_j)] < \varepsilon$; control $K = [a,b] \setminus \bigcup_j (u_j, v_j)$ uniformly.
(v) *Properties*: pass to a common refinement before comparing sums.
(vi) *Calculus*: use the mean value theorem to form tagged sums, then apply Theorem 2.1.9.
(vii) *Improper integrals*: truncate endpoints; use monotonicity or tail bounds.

Key Insight: Riemann–Stieltjes integration is Darboux approximation plus a choice of interval weights $\Delta \alpha_i$.

3 Several-Variable Differentiation

3.1 Partial and Total Derivatives

Several-variable differentiation studies best linear approximations and local geometry, using linear maps, operator norms, determinants, invertibility, and metric estimates.

Definition 3.1.1. *Total Derivative*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} : E \to \mathbb{R}^m$, $\mathbf{x} \in E$, and $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ denote space of linear maps (Definition 3.1.1).
If $\exists A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ s.t. $\lim_{\mathbf{h}\to\mathbf{0}} \frac{\|\mathbf{f}(\mathbf{x}+\mathbf{h})-\mathbf{f}(\mathbf{x})-A\mathbf{h}\|_2}{\|\mathbf{h}\|_2} = 0$, then $\mathbf{f}$ is *differentiable at* $\mathbf{x}$, and A denoted $D\mathbf{f}(\mathbf{x})$ or $\mathbf{f}'(\mathbf{x})$.
Equivalently, $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$, and $\frac{\|\mathbf{r}(\mathbf{h})\|_2}{\|\mathbf{h}\|_2} \to 0$.
If this holds for every $\mathbf{x} \in E$, then $\mathbf{f}$ is differentiable on E.

Perspective 3.1.2. *Derivative First, Matrix Second*
The total derivative is a linear map first; the Jacobian is its matrix in chosen bases. This keeps coordinate formulas subordinate to the intrinsic estimate $\|A\mathbf{h}\|_2 \le \|A\|_{\mathrm{op}} \|\mathbf{h}\|_2$ from Definition 5.1.4.

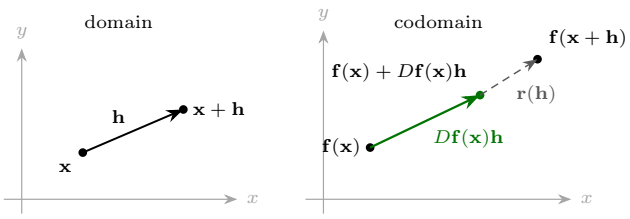


Figure 163: $D\mathbf{f}(\mathbf{x})\mathbf{h}$ is the linear part; $\mathbf{r}(\mathbf{h}) = o(\|\mathbf{h}\|_2)$ is the error.

Theorem 3.1.3. *Basic Consequences of Total Differentiability*
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable at $\mathbf{x} \in E$, then $D\mathbf{f}(\mathbf{x})$ is unique and $\mathbf{f}$ is continuous at $\mathbf{x}$.

Proof. If A_1, A_2 both satisfy Definition 3.1.1, set $B = A_1 - A_2$. Then $\|B\mathbf{h}\|_2 / \|\mathbf{h}\|_2 \to 0$ as $\mathbf{h} \to \mathbf{0}$.
For fixed $\mathbf{h} \neq \mathbf{0}$, applying this to $t\mathbf{h}$ and using linearity makes the quotient independent of t, so $B\mathbf{h} = \mathbf{0}$.
Hence $B = 0$. For continuity, write $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$.
The operator-norm estimate gives $\|D\mathbf{f}(\mathbf{x})\mathbf{h}\|_2 \le \|D\mathbf{f}(\mathbf{x})\|_{\mathrm{op}} \|\mathbf{h}\|_2$, and remainder is $o(\|\mathbf{h}\|_2)$, so $\mathbf{f}(\mathbf{x}+\mathbf{h}) \to \mathbf{f}(\mathbf{x})$.
Structural Commentary: The derivative is a rigid first-order linear part.
Key Insight: A remainder smaller than $\|\mathbf{h}\|$ cannot hide a second linear map. □

Example 3.1.4. *Derivative of a Linear Map*
If $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ and $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$, then $D\mathbf{f}(\mathbf{x}) = A$ for every $\mathbf{x} \in \mathbb{R}^n$.
The remainder in Definition 3.1.1 is identically zero.

Theorem 3.1.5. *Chain Rule*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{x}_0$, $\mathbf{g} : G \subseteq \mathbb{R}^m \to \mathbb{R}^k$ is differentiable at $\mathbf{f}(\mathbf{x}_0)$, and $\mathbf{f}(E) \subseteq G$, then $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}_0$ and $D(\mathbf{g} \circ \mathbf{f})(\mathbf{x}_0) = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))\, D\mathbf{f}(\mathbf{x}_0)$.

Proof. Let $A = D\mathbf{f}(\mathbf{x}_0)$, $B = D\mathbf{g}(\mathbf{f}(\mathbf{x}_0))$, $\mathbf{u}(\mathbf{h}) = \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0) - A\mathbf{h}$, and $\mathbf{v}(\mathbf{k}) = \mathbf{g}(\mathbf{f}(\mathbf{x}_0)+\mathbf{k}) - \mathbf{g}(\mathbf{f}(\mathbf{x}_0)) - B\mathbf{k}$.
For $\mathbf{k} = \mathbf{f}(\mathbf{x}_0+\mathbf{h}) - \mathbf{f}(\mathbf{x}_0)$, differentiability of $\mathbf{f}$ gives $\mathbf{k} \to \mathbf{0}$ and $(\mathbf{g}\circ\mathbf{f})(\mathbf{x}_0+\mathbf{h}) - (\mathbf{g}\circ\mathbf{f})(\mathbf{x}_0) - BA\mathbf{h} = B\mathbf{u}(\mathbf{h}) + \mathbf{v}(\mathbf{k})$, where $\|\mathbf{u}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$ and $\|\mathbf{v}(\mathbf{k})\|_2 = o(\|\mathbf{k}\|_2) = o(\|\mathbf{h}\|_2)$. Divide by $\|\mathbf{h}\|_2$.
Structural Commentary: The first-order parts compose as linear maps.
Key Insight: The output increment of $\mathbf{f}$ is small enough to feed into the remainder estimate for $\mathbf{g}$. □

Theorem 3.1.6. *Componentwise Differentiability*
Let $\mathbf{f} = (f_1, \ldots, f_m) : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and let $\pi_i : \mathbb{R}^m \to \mathbb{R}$ be the ith coordinate projection. Then $\mathbf{f}$ is differentiable at $\mathbf{x} \in E$ iff each f_i is differentiable at $\mathbf{x}$, and $Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$ when these derivatives exist.

Proof.
($\Rightarrow$) If $\mathbf{f}$ is differentiable, write $\mathbf{f}(\mathbf{x}+\mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Applying π_i gives $f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) = \pi_i(D\mathbf{f}(\mathbf{x})\mathbf{h}) + \pi_i(\mathbf{r}(\mathbf{h}))$ and $|\pi_i(\mathbf{r}(\mathbf{h}))| \le \|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
Thus $Df_i(\mathbf{x}) = \pi_i \circ D\mathbf{f}(\mathbf{x})$.
($\Leftarrow$) Suppose $Df_i(\mathbf{x}) = A_i \in \mathcal{L}(\mathbb{R}^n, \mathbb{R})$ for each i. Define $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ by $A\mathbf{h} = (A_1\mathbf{h}, \ldots, A_m\mathbf{h})$.
With $r_i(\mathbf{h}) = f_i(\mathbf{x}+\mathbf{h}) - f_i(\mathbf{x}) - A_i\mathbf{h}$, each $r_i(\mathbf{h}) = o(\|\mathbf{h}\|_2)$.
So $\mathbf{r}(\mathbf{h}) = (r_1(\mathbf{h}), \ldots, r_m(\mathbf{h}))$ satisfies $\|\mathbf{r}(\mathbf{h})\|_2 \le \sum_i |r_i(\mathbf{h})| = o(\|\mathbf{h}\|_2)$. Definition 3.1.1 gives $D\mathbf{f}(\mathbf{x}) = A$.

Structural Commentary: Component derivatives assemble into one linear map.
Key Insight: Components test the derivative; the total derivative packages the tests together. □

Definition 3.1.7. *Partial, Gradient, and Directional Derivatives*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} = (f_1, \dots, f_m) : E \to \mathbb{R}^m$, and let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the standard basis of $\mathbb{R}^n$.
If indicated limits exist, *partial derivative* is $(D_j f_i)(\mathbf{x}) := \lim_{t\to 0} \frac{f_i(\mathbf{x}+t\mathbf{e}_j)-f_i(\mathbf{x})}{t}$ for $1 \le i \le m,\ 1 \le j \le n$.
Directional derivative is $(D_{\mathbf{u}} f)(\mathbf{x}) := \lim_{t\to 0} \frac{f(\mathbf{x}+t\mathbf{u})-f(\mathbf{x})}{t}$ for $\mathbf{u} \in \mathbb{R}^n$; when only the direction is meant, one often assumes $\|\mathbf{u}\|_2 = 1$. If $m = 1$ and all partials exist, the *gradient* is $(\nabla f)(\mathbf{x}) := \sum_{j=1}^{n} (D_j f)(\mathbf{x})\mathbf{e}_j$.

Perspective 3.1.8. *Partials Are Coordinate Tests*
Partials and directional derivatives test selected lines; the total derivative controls all small increments by one linear map. Theorem 3.1.9 extracts partials, and Theorem 3.2.4 gives a continuity-based converse.

Theorem 3.1.9. *Matrix of the Total Derivative*
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable at $\mathbf{x}$, then every $(D_j f_i)(\mathbf{x})$ exists and, in standard bases,

$$[D\mathbf{f}(\mathbf{x})] = \begin{bmatrix} (D_1 f_1)(\mathbf{x}) & \cdots & (D_n f_1)(\mathbf{x}) \\ \vdots & & \vdots \\ (D_1 f_m)(\mathbf{x}) & \cdots & (D_n f_m)(\mathbf{x}) \end{bmatrix}$$

If $m = 1$ and f is differentiable at $\mathbf{x}$, then $(D_{\mathbf{u}} f)(\mathbf{x}) = (\nabla f)(\mathbf{x}) \cdot \mathbf{u}$ for every $\mathbf{u} \in \mathbb{R}^n$.

Proof. Set $\mathbf{h} = t\mathbf{e}_j$ in Definition 3.1.1; limit is $D\mathbf{f}(\mathbf{x})\mathbf{e}_j$, so components form column j (Proposition 3.3.28).
For $m = 1$, set $\mathbf{h} = t\mathbf{u}$, identify linear functional with dot product against $\nabla f(\mathbf{x})$ (Theorem 1.3.27).
Structural Commentary: Partials are coordinate columns of $D\mathbf{f}(\mathbf{x})$.
Key Insight: Differentiability controls all directions at once; partial derivatives record its coordinate columns. □

Remark 3.1.10. *Row and Column Conventions*
With the coordinate-column convention from Proposition 3.3.28, the Jacobian matrix above is the standard-basis matrix of the linear map $D\mathbf{f}(\mathbf{x})$, and its jth column is $D\mathbf{f}(\mathbf{x})\mathbf{e}_j$.

3.2 Differentiability and Linear Approximation

Theorem 3.2.1. *Mean-Value Inequality on Convex Sets*
Let $E \subseteq \mathbb{R}^n$ be open and convex.
If $\mathbf{f} : E \to \mathbb{R}^m$ is differentiable and $\|D\mathbf{f}\|_{op} \le M$ on E, then for all $\mathbf{a}, \mathbf{b} \in E$, $\|\mathbf{f}(\mathbf{b}) - \mathbf{f}(\mathbf{a})\|_2 \le M \|\mathbf{b} - \mathbf{a}\|_2$.
If $D\mathbf{f} = 0$ on E, then $\mathbf{f}$ is constant.

Proof. Fix $\mathbf{a}, \mathbf{b} \in E$, set $\gamma(t) = (1-t)\mathbf{a} + t\mathbf{b}$, and let $\mathbf{g} = \mathbf{f} \circ \gamma$. Then $\gamma([0,1]) \subseteq E$ and $\|\mathbf{g}'(t)\|_2 = \|D\mathbf{f}(\gamma(t))(\mathbf{b} - \mathbf{a})\|_2 \le M \|\mathbf{b} - \mathbf{a}\|_2$. Apply Theorem 1.1.30; the constant case is $M = 0$.
Structural Commentary: Restricting to a line segment reduces the estimate to one variable.
Key Insight: A uniform bound on the derivative bounds every secant displacement. □

Definition 3.2.2. *C^1 Mapping*
Let $E \subseteq \mathbb{R}^n$ be open. A differentiable map $\mathbf{f} : E \to \mathbb{R}^m$ is *continuously differentiable*, or C^1, if $\mathbf{x} \mapsto D\mathbf{f}(\mathbf{x})$ is continuous from E into $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ in operator norm.

Perspective 3.2.3. *C^1 Uses the Operator-Norm Metric*
C^1 continuity measures $\mathbf{x} \mapsto D\mathbf{f}(\mathbf{x})$ in operator norm. In coordinates this is spectral-norm continuity, equivalent here to continuity of the derivative matrix entries.

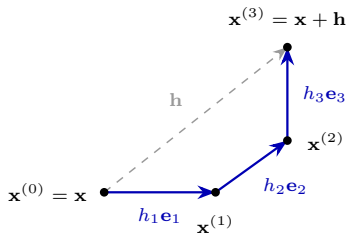


Figure 164: Coordinate increments telescope $\mathbf{x}$ to $\mathbf{x} + \mathbf{h}$.

Theorem 3.2.4. *C^1 Means Continuous Partials*
A map $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is C^1 iff all $D_j f_i$ exist and are continuous on E.

Proof. If $\mathbf{f}$ is C^1, Theorem 3.1.9 makes each $D_j f_i$ a continuous matrix entry of $D\mathbf{f}$.
Conversely reduce componentwise to $m = 1$. Fix $\mathbf{x} \in E$ and take $\mathbf{h}$ small enough that the coordinate polygon from $\mathbf{x}$ to $\mathbf{x} + \mathbf{h}$ lies in E. Put $\mathbf{x}^{(0)} = \mathbf{x}$ and $\mathbf{x}^{(j)} = \mathbf{x} + h_1\mathbf{e}_1 + \cdots + h_j\mathbf{e}_j$. By one-variable MVT on each coordinate segment, $f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) = \sum_{j=1}^{n} h_j (D_j f)(\xi_j)$ for points ξ_j on corresponding segments. Continuity of partials at $\mathbf{x}$ gives $\left| f(\mathbf{x}+\mathbf{h}) - f(\mathbf{x}) - \sum_{j=1}^{n} h_j (D_j f)(\mathbf{x}) \right| \le \sum_{j=1}^{n} |h_j| \, |(D_j f)(\xi_j) - (D_j f)(\mathbf{x})| = o(\|\mathbf{h}\|_2)$.
Hence $Df(\mathbf{x})$ has row matrix $[(D_1 f)(\mathbf{x}) \ \cdots \ (D_n f)(\mathbf{x})]$. Since operator norm is controlled by the Euclidean norm of the matrix entries, continuity of the partials gives continuity of $\mathbf{x} \mapsto Df(\mathbf{x})$.
Structural Commentary: Continuous partial derivatives assemble into a continuous total derivative.
Key Insight: Coordinate-by-coordinate increments work only because the partials vary continuously. □

Higher derivatives repeat the same idea: once $D\mathbf{f}$ is itself a map into the finite-dimensional vector space $\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, we differentiate it again. The linear-algebra viewpoint matters here because the second derivative is naturally bilinear, not merely an array of second partial derivatives.

Definition 3.2.5. *Higher Derivatives and C^r*
Let $E \subseteq \mathbb{R}^n$ be open, $\mathbf{f} : E \to \mathbb{R}^m$, $\mathcal{L}^r(\mathbb{R}^n, \mathbb{R}^m)$ be space of r-linear maps $(\mathbb{R}^n)^r \to \mathbb{R}^m$. Set $D^1\mathbf{f} := D\mathbf{f}$. Recursively, if $D^{r-1}\mathbf{f} : E \to \mathcal{L}^{r-1}(\mathbb{R}^n, \mathbb{R}^m)$ differentiable at $\mathbf{x}$, let $D^r\mathbf{f}(\mathbf{x})(\mathbf{v}_1, \ldots, \mathbf{v}_r) := \big(D(D^{r-1}\mathbf{f})(\mathbf{x})\mathbf{v}_1\big)(\mathbf{v}_2, \ldots, \mathbf{v}_r)$.
For real-valued f, write $D_{ij} f := D_i(D_j f)$ when defined. The map $\mathbf{f}$ is C^r if derivatives through order r exist and $D^r\mathbf{f} : E \to \mathcal{L}^r(\mathbb{R}^n, \mathbb{R}^m)$ is continuous; it is C^∞ if it is C^r for all r.

Theorem 3.2.6. *Second Derivatives and Second Partials*
Let $\mathbf{f} = (f_1, \ldots, f_m) : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ and $p \in E$.
($\Rightarrow$) If $D^2\mathbf{f}(p)$ exists, then each $D_{ij} f_\ell(p)$ exists and $D^2 f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j) = D_{ij} f_\ell(p)$.
($\Leftarrow$) If all $D_{ij} f_\ell$ exist near p and are continuous at p, then $D^2\mathbf{f}(p)$ exists.

Proof.
($\Rightarrow$) Apply Theorem 3.1.6 to $D\mathbf{f} : E \to \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. The entries of $D\mathbf{f}$ are $D_j f_\ell$ by Theorem 3.1.9; differentiating the (ℓ, j) entry in the $\mathbf{e}_i$ direction gives $D_{ij} f_\ell(p)$ and the displayed formula.
($\Leftarrow$) Matrix entries of $D\mathbf{f}$ are functions $D_j f_\ell$. Since first partials $D_i(D_j f_\ell) = D_{ij} f_\ell$ exist near p and are continuous at p, coordinate criterion at p gives differentiability of $D\mathbf{f}$ there. Hence $D^2\mathbf{f}(p)$ exists.
Structural Commentary: Second partials are matrix entries of the derivative of the first-derivative matrix.
Key Insight: Continuous second partials make the first-derivative matrix differentiable. □

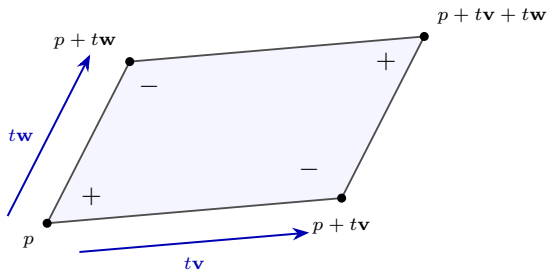


Figure 165: The signed vertex sum on a small parallelogram is symmetric in $\mathbf{v}$ and $\mathbf{w}$.

Theorem 3.2.7. *Symmetry of the Second Derivative*
If $D^2\mathbf{f}(p)$ exists, then $D^2\mathbf{f}(p)(\mathbf{v}, \mathbf{w}) = D^2\mathbf{f}(p)(\mathbf{w}, \mathbf{v})$ for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$.

Proof. Reduce to scalar f by Theorem 3.1.6.
For small t, set $\Delta(t, \mathbf{v}, \mathbf{w}) := f(p + t\mathbf{v} + t\mathbf{w}) - f(p + t\mathbf{v}) - f(p + t\mathbf{w}) + f(p)$.
Then $\Delta(t, \mathbf{v}, \mathbf{w}) = \Delta(t, \mathbf{w}, \mathbf{v})$; see Figure 165. The one-variable mean value theorem and differentiability of Df at p give $D^2 f(p)(\mathbf{v}, \mathbf{w}) = \lim_{t \to 0} \frac{\Delta(t, \mathbf{v}, \mathbf{w})}{t^2}$. The same formula with $\mathbf{v}, \mathbf{w}$ reversed proves symmetry.
Structural Commentary: A pointwise second derivative is already forced to be a symmetric bilinear form.
Key Insight: The parallelogram difference is symmetric in the two directions. □

Corollary 3.2.8. *Mixed Partials from D^2*
If $D^2\mathbf{f}(p)$ exists, then $D_{ij}f_\ell(p) = D_{ji}f_\ell(p)$ for all components and all i, j.

Proof. By Theorem 3.2.6, $D_{ij}f_\ell(p) = D^2f_\ell(p)(\mathbf{e}_i, \mathbf{e}_j)$; and $D_{ji}f_\ell(p) = D^2f_\ell(p)(\mathbf{e}_j, \mathbf{e}_i)$.
Theorem 3.2.7 makes the two right-hand sides equal.
Structural Commentary: Mixed partials are coordinate values of one bilinear map.
Key Insight: Symmetry of $D^2f_\ell(p)$ gives equality of mixed entries. □

Corollary 3.2.9. *Symmetry of Higher Derivatives*
If $D^r\mathbf{f}(p)$ exists, then the r-linear map $D^r\mathbf{f}(p)$ is unchanged by permuting its arguments.

Proof. It suffices to prove invariance under adjacent transpositions. Fix all arguments except positions $k, k+1$. Resulting two-variable second derivative is $(\mathbf{u}, \mathbf{v}) \longmapsto D^r\mathbf{f}(p)(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}, \mathbf{u}, \mathbf{v}, \mathbf{v}_{k+2}, \dots, \mathbf{v}_r)$, so Theorem 3.2.7 allows $\mathbf{u}$ and $\mathbf{v}$ to be swapped. Since adjacent transpositions generate all permutations, $D^r\mathbf{f}(p)$ is symmetric.
Structural Commentary: Higher derivatives are iterated second-derivative symmetries.
Key Insight: Adjacent swaps generate every permutation. □

The next rectangle argument is the classical coordinate route to equality of mixed partials. It is weaker in hypothesis than global C^2, but it proves the same compatibility of coordinate second derivatives.

Theorem 3.2.10. *Rectangle Mean Value Formula*
Let $E \subseteq \mathbb{R}^2$ be open, and suppose D_1f and $D_{21}f$ exist on E. For a closed coordinate rectangle $Q \subseteq E$ with opposite vertices (a, b) and $(a+h, b+k)$, $h, k \neq 0$, set $\Delta(f, Q) := f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b)$. Then some $(x, y) \in \operatorname{int} Q$ satisfies $\Delta(f, Q) = hk(D_{21}f)(x, y)$.

Proof. Set $u(t) = f(t, b+k) - f(t, b)$.
By Corollary 1.1.13, for some $x \in (a, a+h)$, $\Delta(f, Q) = u(a+h) - u(a) = h\big[(D_1f)(x, b+k) - (D_1f)(x, b)\big]$.
Apply same corollary to $s \mapsto (D_1f)(x, s)$; for some $y \in (b, b+k)$, the bracket equals $k(D_{21}f)(x, y)$.
Hence $(x, y) \in \operatorname{int} Q$ and the formula follows.
Structural Commentary: The rectangle difference is a two-step one-variable mean value calculation.
Key Insight: Mixed second derivatives measure the first-order change of a first partial across the other coordinate. □

Theorem 3.2.11. *Equality of Mixed Partials*
Let (a, b) lie in an open set $E \subseteq \mathbb{R}^2$. If D_1f, D_2f, and $D_{21}f$ exist on E, and $D_{21}f$ is continuous at (a, b), then $D_{12}f(a, b)$ exists and $(D_{12}f)(a, b) = (D_{21}f)(a, b)$. Hence $D_{12}f = D_{21}f$ when $f \in C^2(E)$.

Proof. Let $A = (D_{21}f)(a, b)$.
Given $\varepsilon > 0$, continuity of $D_{21}f$ and Theorem 3.2.10 give, for all small $h, k \neq 0$, $\left|\frac{\Delta(f,Q)}{hk} - A\right| < \varepsilon$.
Fix such an h and let $k \to 0$. Since D_2f exists at $(a+h, b)$ and (a, b), $\frac{\Delta(f,Q)}{hk} \to \frac{(D_2f)(a+h,b) - (D_2f)(a,b)}{h}$.
Thus this last quotient differs from A by at most ε for all sufficiently small $h \neq 0$.
Since ε was arbitrary, the quotient tends to A as $h \to 0$, so $D_{12}f(a, b)$ exists and equals A.
Structural Commentary: Continuity of $D_{21}f$ controls local rectangle quotients.
Key Insight: The missing mixed partial is recovered by taking the two one-variable limits in the reverse order. □

Perspective 3.2.12. *Mixed Partials Need Hypotheses*
$D_{12}f$ and $D_{21}f$ are different iterated limits. Theorems 3.2.7 and 3.2.11 give hypotheses for equality.

Theorem 3.2.13. *Differentiation Under a Riemann–Stieltjes Integral*
Let α be increasing on $[a, b]$, let $\varphi : [a, b] \times [c, d] \to \mathbb{R}$, and put $\varphi^t(x) := \varphi(x, t)$.
Assume $\varphi^t \in \mathcal{R}(\alpha)$ for all t and $D_2\varphi(\cdot, t) \rightrightarrows D_2\varphi(\cdot, s)$ on $[a, b]$ as $t \to s \in (c, d)$.
If $F(t) := \int_a^b \varphi(x, t)\, d\alpha(x)$, then $(D_2\varphi)^s \in \mathcal{R}(\alpha)$ and $F'(s) = \int_a^b (D_2\varphi)(x, s)\, d\alpha(x)$.

Proof. For $t \neq s$, set $\psi_t = (\varphi^t - \varphi^s)/(t - s)$. Corollary 1.1.13 and the uniform hypothesis give $\psi_t \rightrightarrows (D_2\varphi)^s$.
Since $\psi_t \in \mathcal{R}(\alpha)$, Theorem 7.2.1 gives $\frac{F(t)-F(s)}{t-s} = \int_a^b \psi_t\, d\alpha \longrightarrow \int_a^b (D_2\varphi)(x, s)\, d\alpha(x)$.
Structural Commentary: Uniform parameter control lets limits pass under the integral.
Key Insight: The theorem is exactly the commutation of a derivative limit with a Riemann–Stieltjes integral. □

Example 3.2.14. *A Gaussian Fourier Integral*
Define $f(t) = \int_{-\infty}^{\infty} e^{-x^2} \cos(xt)\, dx$ and $g(t) = -\int_{-\infty}^{\infty} xe^{-x^2} \sin(xt)\, dx$.
Then $f'(t) = g(t)$, and integration by parts gives $2f'(t) + tf(t) = 0$. Since $f(0) = \sqrt{\pi}$, $f(t) = \sqrt{\pi} \exp\left(-\frac{t^2}{4}\right)$.

Definition 3.2.15. *Jacobian*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is differentiable at $\mathbf{x}$, define $J_{\mathbf{f}}(\mathbf{x}) := \det D\mathbf{f}(\mathbf{x}) = \frac{\partial(y_1,\dots,y_n)}{\partial(x_1,\dots,x_n)}$ when $f = (y_1, \dots, y_n)$.
By Theorem 1.1.8, $D\mathbf{f}(\mathbf{x})$ is invertible if and only if $J_{\mathbf{f}}(\mathbf{x}) \neq 0$. The absolute value $|J_{\mathbf{f}}(\mathbf{x})|$ is the local version of determinant volume scaling from Theorem 5.1.1, used again in the change-of-variables theorem.

4 Inverse, Implicit, and Rank Theorems

4.1 Inverse Function Theorem

Theorem 4.1.1. *Finite-Dimensional Linear Maps Are Lipschitz*
Let $A \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. Then A is $\|A\|_{op}$-Lipschitz: $\|A\mathbf{x}\|_2 \leq \|A\|_{op} \|\mathbf{x}\|_2$ for $\mathbf{x} \in \mathbb{R}^n$.
If $A \in \mathcal{L}(\mathbb{R}^n)$ is invertible, then $\frac{1}{\|A^{-1}\|_{op}} \|\mathbf{x}\|_2 \leq \|A\mathbf{x}\|_2 \leq \|A\|_{op} \|\mathbf{x}\|_2$, so A is a homeomorphism of $\mathbb{R}^n$.

Proof. The upper estimate follows from Definition 5.1.4: apply it to $\mathbf{x}/\|\mathbf{x}\|_2$ when $\mathbf{x} \neq \mathbf{0}$, while $\mathbf{x} = \mathbf{0}$ is trivial.
If A is invertible, apply the same estimate to A^{-1} with $\mathbf{y} = A\mathbf{x}$: $\|\mathbf{x}\|_2 = \|A^{-1}\mathbf{y}\|_2 \leq \|A^{-1}\|_{\text{op}} \|A\mathbf{x}\|_2$.
Thus A and A^{-1} are continuous.
Structural Commentary: Linear maps have finite stretch; invertible ones have positive lower stretch.
Key Insight: The operator norm turns linear algebra into uniform metric estimates. □

Theorem 4.1.2. *Invertible Operators Form an Open Set*
Let $\Omega \subseteq \mathcal{L}(\mathbb{R}^n)$ be the set of invertible linear operators. If $A \in \Omega$, $B \in \mathcal{L}(\mathbb{R}^n)$, $\|B - A\|_{op} \|A^{-1}\|_{op} < 1$, then $B \in \Omega$. Consequently, Ω is open in $\mathcal{L}(\mathbb{R}^n)$ and the map $A \mapsto A^{-1}$ is continuous on Ω.

Proof. Put $\alpha = 1/\|A^{-1}\|_{\text{op}}$ and $\beta = \|B - A\|_{\text{op}}$.
For every $\mathbf{x} \in \mathbb{R}^n$, $\alpha \|\mathbf{x}\|_2 \leq \|A\mathbf{x}\|_2 \leq \|(A-B)\mathbf{x}\|_2 + \|B\mathbf{x}\|_2 \leq \beta \|\mathbf{x}\|_2 + \|B\mathbf{x}\|_2$.
Since $\beta < \alpha$, this shows $B\mathbf{x} \neq 0$ for $\mathbf{x} \neq 0$; hence B is injective and therefore invertible by Theorem 3.2.38.
Replacing $\mathbf{x}$ by $B^{-1}\mathbf{y}$ gives $\|B^{-1}\|_{\text{op}} \leq (\alpha - \beta)^{-1}$.
Finally, $B^{-1} - A^{-1} = B^{-1}(A - B)A^{-1}$, Proposition 5.1.3 gives the continuity of inversion.
Structural Commentary: Small operator-norm perturbations of an invertible map remain invertible.
Key Insight: The inverse changes continuously because the difference of inverses factors through $A - B$. □

Perspective 4.1.3. *Linear Inverse Estimate*
For invertible $A \in \mathcal{L}(\mathbb{R}^n)$, Theorem 4.1.1 gives $c\|\mathbf{h}\|_2 \leq \|A\mathbf{h}\|_2 \leq \|A\|_{\text{op}} \|\mathbf{h}\|_2$; and $c := \frac{1}{\|A^{-1}\|_{\text{op}}}$.
The inverse function theorem preserves this lower stretch under small derivative changes.

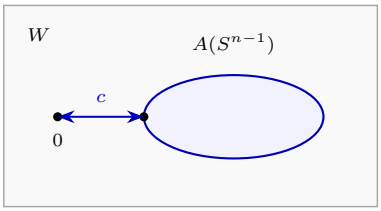


Figure 166: Invertibility keeps $A(S^{n-1})$ a positive distance from 0.

Theorem 4.1.4. *Inverse Function Theorem*
Let $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be C^1, let $\mathbf{a} \in E$, and suppose $D\mathbf{f}(\mathbf{a})$ is invertible. With $\mathbf{b} = \mathbf{f}(\mathbf{a})$, $\mathbf{f}$ restricts to a bijection $U \to V$ between open neighborhoods of $\mathbf{a}$ and $\mathbf{b}$; inverse $\mathbf{g} : V \to U$ is C^1, and $D\mathbf{g}(\mathbf{y}) = \big(D\mathbf{f}(\mathbf{g}(\mathbf{y}))\big)^{-1}$, $\mathbf{y} \in V$.

Proof. Put $A = D\mathbf{f}(\mathbf{a})$ and choose $0 < \theta < 1$.
By continuity of $D\mathbf{f}$, choose $r > 0$ with $\overline{B}_r(\mathbf{a}) \subset E$ and $\|I - A^{-1}D\mathbf{f}(\mathbf{x})\|_{\text{op}} \leq \theta$ on this closed ball.
For fixed $\mathbf{y}$, solutions of $\mathbf{f}(\mathbf{x}) = \mathbf{y}$ are fixed points of $\varphi_{\mathbf{y}}(\mathbf{x}) = \mathbf{x} + A^{-1}(\mathbf{y} - \mathbf{f}(\mathbf{x}))$.
Theorem 3.2.1 gives $\|\varphi_{\mathbf{y}}(\mathbf{x}) - \varphi_{\mathbf{y}}(\mathbf{z})\|_2 \leq \theta \|\mathbf{x} - \mathbf{z}\|_2$. If $\|A^{-1}(\mathbf{y} - \mathbf{b})\|_2 \leq (1 - \theta)r$, then $\varphi_{\mathbf{y}}$ maps $\overline{B}_r(\mathbf{a})$ into itself. Since $\overline{B}_r(\mathbf{a})$ is a closed subset of complete $\mathbb{R}^n$, Proposition 4.1.14 makes it complete.
The contraction principle, Theorem 4.1.19, gives a unique $\mathbf{x} \in \overline{B}_r(\mathbf{a})$ with $\mathbf{f}(\mathbf{x}) = \mathbf{y}$. Thus, after shrinking to open neighborhoods U of $\mathbf{a}$ and V of $\mathbf{b}$, $\mathbf{f} : U \to V$ is bijective with inverse $\mathbf{g}$.
For $\mathbf{y} = \mathbf{f}(\mathbf{x})$ and $\mathbf{y} + \mathbf{k} = \mathbf{f}(\mathbf{x} + \mathbf{h})$ in V, the same derivative bound gives $\|\mathbf{h}\|_2 \leq C\|\mathbf{k}\|_2$ for a local constant C. Moreover $\mathbf{g}(\mathbf{y} + \mathbf{k}) - \mathbf{g}(\mathbf{y}) - \big(D\mathbf{f}(\mathbf{x})\big)^{-1}\mathbf{k} = -\big(D\mathbf{f}(\mathbf{x})\big)^{-1}\big[\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) - D\mathbf{f}(\mathbf{x})\mathbf{h}\big]$.
Since $\mathbf{f}$ is differentiable at $\mathbf{x}$ and $\|\mathbf{h}\|_2 = O(\|\mathbf{k}\|_2)$, division by $\|\mathbf{k}\|_2$ gives $D\mathbf{g}(\mathbf{y}) = (D\mathbf{f}(\mathbf{x}))^{-1}$.
Continuity of $D\mathbf{g}$ follows from continuity of $D\mathbf{f}$, continuity of $\mathbf{g}$, and Theorem 4.1.2.

Structural Commentary: Local inversion is a contraction built from A^{-1}.
Key Insight: Small derivative variation preserves local invertibility. □

Corollary 4.1.5. *Local Diffeomorphisms Are Open*
If $\mathbf{f} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is C^1 and $D\mathbf{f}$ is everywhere invertible, then $\mathbf{f}$ is an open map (locally one-to-one).

Proof. Let $W \subseteq E$ be open and let $\mathbf{y} \in \mathbf{f}(W)$. Choose $\mathbf{x} \in W$ with $\mathbf{f}(\mathbf{x}) = \mathbf{y}$. Applying Theorem 4.1.4 to $\mathbf{f}|_W$ at $\mathbf{x}$ gives open neighborhoods $U \subseteq W$ of $\mathbf{x}$ and V of $\mathbf{y}$ with $\mathbf{f}(U) = V$. Hence $V \subseteq \mathbf{f}(W)$, so $\mathbf{y}$ is interior to $\mathbf{f}(W)$. Since $\mathbf{y}$ was arbitrary, $\mathbf{f}(W)$ is open. Local one-to-one-ness is the same theorem applied pointwise.
Structural Commentary: Openness is checked one image point at a time.
Key Insight: A local inverse turns nearby domain points into a full image neighborhood. □

4.2 Implicit Function Theorem

Definition 4.2.1. *Block Splitting of a Linear Map*
For $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$ and $(\mathbf{h}, \mathbf{k}) \in \mathbb{R}^n \times \mathbb{R}^m$, define $A_x\mathbf{h} := A(\mathbf{h}, \mathbf{0})$ and $A_y\mathbf{k} := A(\mathbf{0}, \mathbf{k})$.
Then $A_x \in \mathcal{L}(\mathbb{R}^n)$, $A_y \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$, and $A(\mathbf{h}, \mathbf{k}) = A_x\mathbf{h} + A_y\mathbf{k}$.

Theorem 4.2.2. *Linear Implicit Function Theorem*
If $A \in \mathcal{L}(\mathbb{R}^{n+m}, \mathbb{R}^n)$, A_x invertible, then $\forall \mathbf{k} \in \mathbb{R}^m$ $\exists$ unique $\mathbf{h} \in \mathbb{R}^n$ s.t. $A(\mathbf{h}, \mathbf{k}) = 0$, that $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k}$.

Proof. Fix $\mathbf{k} \in \mathbb{R}^m$ and set $\mathbf{h}_0 := -(A_x)^{-1}A_y\mathbf{k}$.
By Definition 4.2.1, $A(\mathbf{h}_0, \mathbf{k}) = A_x\mathbf{h}_0 + A_y\mathbf{k} = -A_y\mathbf{k} + A_y\mathbf{k} = 0$, so a solution exists.
If $\mathbf{h} \in \mathbb{R}^n$ also satisfies $A(\mathbf{h}, \mathbf{k}) = 0$, then $A_x\mathbf{h} + A_y\mathbf{k} = 0$, hence $A_x\mathbf{h} = -A_y\mathbf{k}$.
Applying $(A_x)^{-1}$ gives $\mathbf{h} = -(A_x)^{-1}A_y\mathbf{k} = \mathbf{h}_0$. Thus the solution exists and is unique.
Structural Commentary: Splitting A separates the unknown variable from the parameter.
Key Insight: Invertibility of A_x turns constraint $A(\mathbf{h}, \mathbf{k}) = 0$ into a uniquely solvable linear equation for $\mathbf{h}$. □

The implicit theorem below is the nonlinear counterpart of the block equation theorem. The invertible block is the set of variables one can solve for, just as in Theorem 4.2.2.

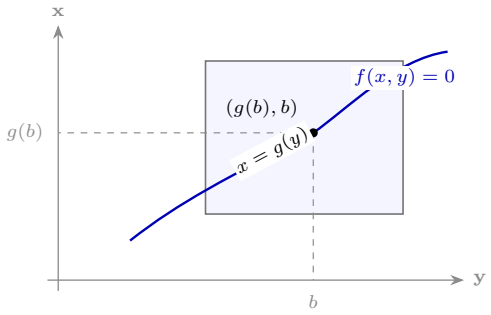


Figure 167: Near a regular point, the zero locus is the graph of an implicit function.

Theorem 4.2.3. *Implicit Function Theorem*
Let $\mathbf{f} : E \subseteq \mathbb{R}^{n+m} \to \mathbb{R}^n$ be C^1, $\mathbf{f}(\mathbf{a}, \mathbf{b}) = 0$, $A = D\mathbf{f}(\mathbf{a}, \mathbf{b})$, and let A_x, A_y be the block maps of Definition 4.2.1. If A_x is invertible, then near $\mathbf{b}$, the equation $\mathbf{f}(\mathbf{x}, \mathbf{y}) = 0$ has unique local solution $\mathbf{x} = \mathbf{g}(\mathbf{y})$, where $\mathbf{g}$ is C^1, $\mathbf{g}(\mathbf{b}) = \mathbf{a}$, $D\mathbf{g}(\mathbf{b}) = -(A_x)^{-1}A_y$.

Proof. Set $\Phi(\mathbf{x}, \mathbf{y}) = (\mathbf{f}(\mathbf{x}, \mathbf{y}), \mathbf{y})$. Then $D\Phi(\mathbf{a}, \mathbf{b})(\mathbf{h}, \mathbf{k}) = (A_x\mathbf{h} + A_y\mathbf{k}, \mathbf{k})$, whose kernel is zero since A_x is invertible; hence $D\Phi(\mathbf{a}, \mathbf{b})$ is invertible by Theorem 3.2.38. Theorem 4.1.4 makes Φ locally invertible.
Points with $\mathbf{f}(\mathbf{x}, \mathbf{y}) = 0$ are exactly those sent to $(0, \mathbf{y})$, so $\Phi^{-1}(0, \mathbf{y}) = (\mathbf{g}(\mathbf{y}), \mathbf{y})$ gives the unique local solution.
Differentiating $\mathbf{f}(\mathbf{g}(\mathbf{y}), \mathbf{y}) = 0$ at $\mathbf{b}$ gives $A_x D\mathbf{g}(\mathbf{b}) + A_y = 0$.
Structural Commentary: Add $\mathbf{y}$ as a coordinate, then invert.
Key Insight: Invertibility of the x-block is precisely the linear ability to solve for x. □

Perspective 4.2.4. *Rank Theorem Linear Model*
Theorem 3.2.30 reduces a linear map to its kernel–range split.
The constant-rank theorem is the local C^1 analogue after coordinate changes.

4.3 Rank Theorem

Theorem 4.3.1. *Rank Theorem*
Let $\mathbf{F} : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be C^1 with constant rank r. At $\mathbf{a} \in E$, put $A = D\mathbf{F}(\mathbf{a})$, $Y_1 = \operatorname{range}(A)$, and let P be the orthogonal projection onto Y_1 (Definition 2.2.2). Locally, after a C^1 coordinate change H with C^1 inverse on a small coordinate neighborhood V_0, there is a C^1 map $\varphi : A(V_0) \subseteq Y_1 \to \ker(P)$ such that $\mathbf{F}(H(\mathbf{x})) = A\mathbf{x}+\varphi(A\mathbf{x})$. In coordinates adapted to $\ker A$ and $\operatorname{range}(A)$, this is the same local geometry as a projection onto r coordinates.

Proof. If $r = 0$, Theorem 3.2.1 makes $\mathbf{F}$ locally constant.
Assume $r > 0$. Choose $S : Y_1 \to \mathbb{R}^n$ with $AS = I_{Y_1}$, possible by choosing preimages of a basis of Y_1, and set $G(\mathbf{x}) = \mathbf{x} + SP(\mathbf{F}(\mathbf{x}) - A\mathbf{x})$. Since $DG(\mathbf{a}) = I$, Theorem 4.1.4 gives a local inverse $H = G^{-1}$.
Let $\Psi = \mathbf{F} \circ H$. From $AG = P\mathbf{F}$ we get $P\Psi(\mathbf{x}) = A\mathbf{x}$. Hence $PD\Psi(\mathbf{x}) = A$. The rank of $D\Psi(\mathbf{x})$ is r because $DH(\mathbf{x})$ is invertible and $\mathbf{F}$ has constant rank r; the rank of $PD\Psi(\mathbf{x})$ is also r.
Therefore P is injective on range $D\Psi(\mathbf{x})$, and $D\Psi(\mathbf{x})\mathbf{v} = 0$ whenever $A\mathbf{v} = 0$.
On a small convex coordinate neighborhood, any two points with the same $A\mathbf{x}$ differ by a vector in $\ker(A)$ and can be joined inside that affine fiber. The preceding derivative vanishing and Theorem 3.2.1 show that Ψ is constant on such fibers. Thus $(I - P)\Psi(\mathbf{x})$ depends only on $A\mathbf{x}$; call it $\varphi(A\mathbf{x})$. Then $\mathbf{F}(H(\mathbf{x})) = \Psi(\mathbf{x}) = A\mathbf{x} + \varphi(A\mathbf{x})$, and φ is C^1 because Ψ is C^1 and the dependence is through the coordinate $A\mathbf{x}$.
Structural Commentary: Constant rank gives a local graph normal form.
Key Insight: Same rank, same local level-set geometry. □

Rank-nullity (Theorem 3.2.16) says that the domain directions split into rank directions and fiber directions. The rank theorem says a constant-rank map has, locally, the same split after C^1 changes of coordinates.

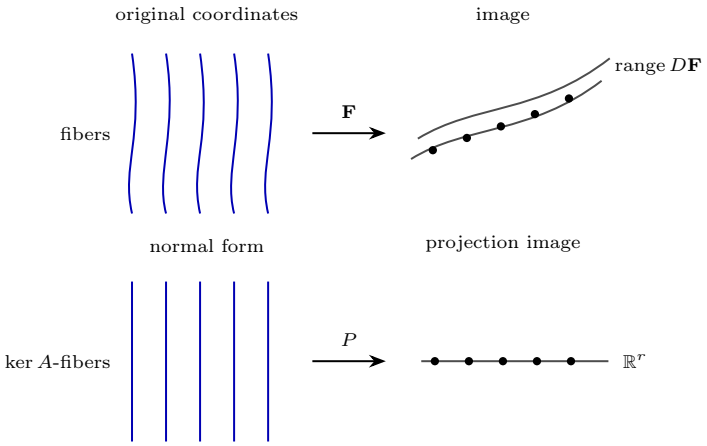


Figure 168: After local coordinate changes, a constant-rank map looks like projection onto the rank directions.

Theorem 4.3.2. *One-Constraint Lagrange Multiplier*
Let $E \subseteq \mathbb{R}^n$ be open, $f, g : E \to \mathbb{R}$ be C^1, and $S = \{\mathbf{x} : g(\mathbf{x}) = c\}$.
If $p \in S$, $\nabla g(p) \neq \mathbf{0}$, and $f|_S$ has a local extremum at p, then $\exists \lambda \in \mathbb{R}$ with $\nabla f(p) = \lambda \nabla g(p)$.

Proof. If $Df(p) = 0$, take $\lambda = 0$. Otherwise suppose $Df(p)$ is not a multiple of $Dg(p)$.
Complete $Dg(p), Df(p)$ to a dual basis $Dg(p)$, $Df(p)$, $\ell_3, \dots, \ell_n$ by Theorem 5.1.4,
and set $\Phi(\mathbf{x}) = \big(g(\mathbf{x}) - c,\ f(\mathbf{x}) - f(p),\ \ell_3(\mathbf{x} - p), \dots, \ell_n(\mathbf{x} - p)\big)$.
Then $D\Phi(p)$ is invertible, so Theorem 4.1.4 makes Φ locally invertible. In these coordinates, S is locally the slice with first coordinate 0, while the second coordinate is $f - f(p)$. Local invertibility gives nearby points of S with this second coordinate of both signs, contradicting extremality.
Hence $Df(p) = \lambda Dg(p)$; Riesz representation, Theorem 1.3.27, gives the gradient form.
Structural Commentary: The constraint kills first-order variation along S.
Key Insight: If Df were independent of Dg, f would be a free constraint coordinate. □

Method 4.3.3. *Several-Variable Differentiation Proof Patterns*

(i) *Linear approximation*: write $\mathbf{f}(\mathbf{x} + \mathbf{h}) - \mathbf{f}(\mathbf{x}) = D\mathbf{f}(\mathbf{x})\mathbf{h} + \mathbf{r}(\mathbf{h})$ with $\|\mathbf{r}(\mathbf{h})\|_2 = o(\|\mathbf{h}\|_2)$.
(ii) *Norm estimates*: use $\|A\mathbf{h}\|_2 \leq \|A\|_{\text{op}} \|\mathbf{h}\|_2$ and the product estimate from Proposition 5.1.3.
(iii) *Line-segment reduction*: on convex sets, restrict to $\gamma(t) = (1 - t)\mathbf{a} + t\mathbf{b}$ and apply one-variable results.
(iv) *Local inversion*: if $A = D\mathbf{f}(\mathbf{a})$ is invertible, write $D\mathbf{f}(\mathbf{x}) = A + E_{\mathbf{x}}$ with $\|E_{\mathbf{x}}\|_{\text{op}}$ small and solve by contraction (Theorem 4.1.19); linear block equations use Theorem 4.2.2.
(v) *Coordinates*: read partial derivatives as the entries of the matrix of $D\mathbf{f}(\mathbf{x})$, using Theorem 3.1.9.

(vi) *Higher derivatives*: regard $D^r\mathbf{f}(p)$ as an r-linear map; recover coordinate partials via Theorem 3.2.6.
(vii) *Rank normal forms*: for $A = D\mathbf{F}(\mathbf{a})$, split domain by $\ker A$ and image by $\operatorname{range} A$; use the quotient-by-kernel picture from Theorem 3.2.30 and project with $P_{\operatorname{range} A}$.

Key Insight: Several-variable calculus is one-variable limiting plus finite-dimensional linear algebra.

5 Differential Forms and Stokes Theory

5.1 Multiple Integration and Change of Variables

This subsection moves from Euclidean-region integration and change of variables to differential forms, pullbacks, and oriented parametrized surfaces.

Theorem 5.1.1. *Linear Maps Scale Volume by $|\det T|$*
Let $T \in \mathcal{L}(\mathbb{R}^n)$ and let $\Omega \subseteq \mathbb{R}^n$ be a sufficiently regular measurable set. Then $\operatorname{vol}(T(\Omega)) = |\det T| \operatorname{vol}(\Omega)$.

Proof. It is enough to prove the formula first for invertible T. Let $T = SP$ by polar decomposition, where S is an isometry and $P = \sqrt{T^*T}$ is positive definite. Isometries preserve Euclidean volume, so $\operatorname{vol}(T(\Omega)) = \operatorname{vol}(P(\Omega))$. By the real spectral theorem, choose orthonormal coordinates with $P = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ and $\lambda_j > 0$. Rectangles are sent to rectangles with side lengths multiplied by λ_j, hence their volumes are multiplied by $\prod_j \lambda_j = \det P$. Finite unions of disjoint rectangles have the same scaling by additivity, and the usual inner-outer rectangle approximation of a sufficiently regular measurable set gives $\operatorname{vol}(P(\Omega)) = (\det P)\operatorname{vol}(\Omega)$.
If T is singular, then $T(\Omega)$ lies in a proper subspace after an orthogonal change of coordinates, so its n-dimensional volume is 0; also $|\det T| = 0$. Finally, Proposition 4.2.23 gives $\det P = |\det T|$ in the invertible case.
Structural Commentary: Polar decomposition reduces volume scaling to diagonal stretching.
Key Insight: The determinant is the product of independent stretch factors. □

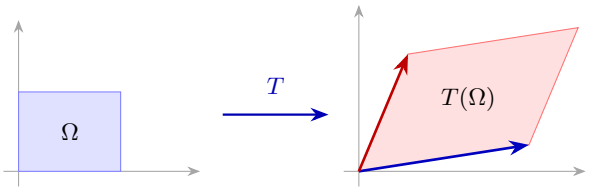


Figure 169: A linear map scales volume by $|\det T|$.

Method 5.1.2. *Change-of-Variables Jacobian Method*
For a differentiable map $\sigma : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$:
(i) compute the derivative matrix $D\sigma(\mathbf{x})$;
(ii) compute the local volume factor $|\det D\sigma(\mathbf{x})|$;
(iii) replace $d\mathbf{y}$ by $|\det D\sigma(\mathbf{x})|\, d\mathbf{x}$ when $\mathbf{y} = \sigma(\mathbf{x})$;
(iv) transform the domain bounds in the new coordinates.

Example 5.1.3. *Polar and Spherical Jacobian Factors*
For polar coordinates $\sigma(r, \theta) = (r\cos\theta, r\sin\theta)$, $D\sigma(r,\theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$, and $|\det D\sigma(r,\theta)| = r$.
Thus area integrals acquire the factor $r\, dr\, d\theta$.
For spherical coordinates $\sigma(\rho,\varphi,\theta) = (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi)$, and $|\det D\sigma(\rho,\varphi,\theta)| = \rho^2\sin\varphi$.
Thus volume integrals acquire the factor $\rho^2\sin\varphi\, d\rho\, d\varphi\, d\theta$.

Definition 5.1.4. *k-Cells and Multiple Integrals*
A *k-cell* is $I^k := \prod_{i=1}^{k} [a_i, b_i] \subseteq \mathbb{R}^k$ or $I^j := \prod_{i=1}^{j} [a_i, b_i]$.
For $f \in C(I^k)$, set $f_k = f$ and, for $j = k, \dots, 1$, define $f_{j-1}(x_1, \dots, x_{j-1}) := \int_{a_j}^{b_j} f_j(x_1, \dots, x_j)\, dx_j$.
The *integral of f over I^k* is $\int_{I^k} f(\mathbf{x})\, d\mathbf{x} := f_0$, also written $\int_{I^k} f$.

For bounded functions on rectangles, the Darboux picture from Definition 1.2.1 generalizes by replacing intervals with grid rectangles.

Definition 5.1.5. *Rectangular Grids and Riemann Sums*
For $R = [a,b] \times [c,d] \subseteq \mathbb{R}^2$ and partitions $P : a = x_0 < \cdots < x_m = b$; and $Q : c = y_0 < \cdots < y_n = d$, put $G = P \times Q$, where $R_{ij} := [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ and $|R_{ij}| := (x_i - x_{i-1})(y_j - y_{j-1})$.

For sample points $(s_{ij}, t_{ij}) \in R_{ij}$, $R(f,G,S) := \sum_{i=1}^{m}\sum_{j=1}^{n} f(s_{ij},t_{ij})\,|R_{ij}|$. For bounded real f, with $m_{ij} := \inf_{R_{ij}} f$ and $M_{ij} := \sup_{R_{ij}} f$, set $L(f,G) := \sum_{i,j} m_{ij}\,|R_{ij}|$ and $U(f,G) := \sum_{i,j} M_{ij}\,|R_{ij}|$.
The lower and upper integrals over R are $\sup_G L(f,G)$ and $\inf_G U(f,G)$. If equal, their common value is $\int_R f$.

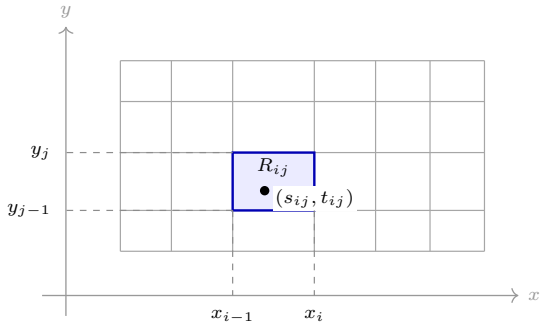


Figure 170: A grid rectangle R_{ij} and sample point (s_{ij}, t_{ij}).

Theorem 5.1.6. *Fubini on a Rectangle, Darboux Form*
Let $R = [a,b]\times[c,d]$ and let $f : R \to \mathbb{R}$ be Riemann integrable.
For $y \in [c,d]$, set $f_y(x) := f(x,y)$ and $\underline{F}(y) := \sup_P L(P,f_y)$; $\overline{F}(y) := \inf_P U(P,f_y)$, where P ranges over partitions of $[a,b]$. Then $\underline{F}, \overline{F}$ are Riemann integrable on $[c,d]$ and $\int_R f = \int_c^d \underline{F}(y)\,dy = \int_c^d \overline{F}(y)\,dy$.
If $\int_a^b f(x,y)\,dx$ exists for every y, this is the usual iterated integral formula.

Proof. For $G = P\times Q$, horizontal slice sums give $L(f,G) \le L(Q,\underline{F}) \le U(Q,\overline{F}) \le U(f,G)$.
As f is integrable on R, the outer terms can be made arbitrarily close. The upper–lower criterion, Theorem 2.1.7, squeezes $\underline{F}$ and $\overline{F}$ to the common value $\int_R f$.
Structural Commentary: Product grids turn area sums into slice sums.
Key Insight: Fubini is a squeeze between lower and upper sums before it is an equality of iterated integrals. □

Theorem 5.1.7. *Order Independence on Cells*
For $f \in C(I^k)$, all iterated orders defining $\int_{I^k} f$ give the same value.

Proof. Let L, L' be two iterated orders. For $h(\mathbf{x}) = h_1(x_1)\cdots h_k(x_k)$, $L(h) = L'(h) = \prod_{i=1}^{k}\int_{a_i}^{b_i} h_i(x_i)\,dx_i$, so $L(g) = L'(g)$ for every finite sum g of such products. These sums are dense in $C(I^k)$ by Theorem 9.2.8.
Choose g with $\|f-g\|_\infty < \varepsilon/V$, where $V = \prod_{i=1}^{k}(b_i - a_i)$.
Since $|L(r)|, |L'(r)| \le V\,\|r\|_\infty$, then $|L(f) - L'(f)| \le |L(f-g)| + |L'(f-g)| < 2\varepsilon$.
Structural Commentary: Product functions plus uniform approximation prove order independence.
Key Insight: Fubini-type order independence is first an algebraic fact, then a uniform approximation fact. □

Perspective 5.1.8. *Order Interchange Requires Hypotheses*
Finite sums are order-free; limiting processes are not.
Theorems 5.1.6 and 5.1.7 give hypotheses for interchanging integral order.

Definition 5.1.9. *Support and Simplex Integrals*
For $f : \mathbb{R}^k \to \mathbb{R}$ or $\mathbb{C}$, define $\operatorname{supp}(f) := \overline{\{\mathbf{x} : f(\mathbf{x}) \neq 0\}}$.
If f is continuous with compact support and $\operatorname{supp}(f) \subseteq I^k$, set $\int_{\mathbb{R}^k} f := \int_{I^k} f$. This is independent of the containing cell I^k. The *standard k-simplex* is $Q^k := \{\mathbf{x} \in \mathbb{R}^k : x_i \ge 0,\ x_1 + \cdots + x_k \le 1\}$.
For $f \in C(Q^k)$, extend f by 0 to unit cube; resulting order-independent iterated integral is denoted $\int_{Q^k} f$.

Definition 5.1.10. *Primitive Mappings and Flips*
Let $E \subseteq \mathbb{R}^n$ be open.
A map $G : E \to \mathbb{R}^n$ is *primitive* if, for some coordinate m and real function g, $G(\mathbf{x}) = \mathbf{x} + \big(g(\mathbf{x}) - x_m\big)\mathbf{e}_m$.
If g is differentiable at $\mathbf{a}$, then $DG(\mathbf{a})$ is the identity matrix with row m replaced by $\big((D_1 g)(\mathbf{a}), \ldots, (D_n g)(\mathbf{a})\big)$, hence $J_G(\mathbf{a}) = \det DG(\mathbf{a}) = (D_m g)(\mathbf{a})$ by Theorem 1.1.8.
A *flip* is the linear map that swaps two standard-basis vectors and fixes the rest.

Theorem 5.1.11. *Local Decomposition into Primitive Mappings*
Let $\mathbf{F} : E \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be C^1, let $\mathbf{0} \in E$, and suppose $\mathbf{F}(\mathbf{0}) = \mathbf{0}$ and $D\mathbf{F}(\mathbf{0})$ is invertible.
Then, near $\mathbf{0}$, $\mathbf{F}(\mathbf{x}) = B_1 \cdots B_{n-1} G_n \circ \cdots \circ G_1(\mathbf{x})$, where each G_i is a primitive C^1 map with $G_i(\mathbf{0}) = \mathbf{0}$ and $DG_i(\mathbf{0})$ invertible, and each B_i is a flip or the identity.

Proof. Let $P_m\mathbf{x} := x_1\mathbf{e}_1 + \cdots + x_m\mathbf{e}_m$. Put $\mathbf{F}_1 = \mathbf{F}$ and inductively assume that $\mathbf{F}_m$ is defined near $\mathbf{0}$, $\mathbf{F}_m(\mathbf{0}) = \mathbf{0}$, $D\mathbf{F}_m(\mathbf{0})$ is invertible, and $P_{m-1}\mathbf{F}_m = P_{m-1}$. Then $\mathbf{F}_m(\mathbf{x}) = P_{m-1}\mathbf{x} + \sum_{i=m}^{n} \alpha_i(\mathbf{x})\mathbf{e}_i$.
Since $D\mathbf{F}_m(\mathbf{0})\mathbf{e}_m \neq \mathbf{0}$, some $k \in \{m, \ldots, n\}$ has $(D_m\alpha_k)(\mathbf{0}) \neq 0$. Let B_m swap coordinates m and k (or be the identity if $k = m$), and set $G_m(\mathbf{x}) := \mathbf{x} + \big(\alpha_k(\mathbf{x}) - x_m\big)\mathbf{e}_m$.
By Definition 5.1.10, G_m is primitive and $DG_m(\mathbf{0})$ is invertible. The inverse function theorem, Theorem 4.1.4, gives a local inverse for G_m, so define $\mathbf{F}_{m+1} := B_m\mathbf{F}_m \circ G_m^{-1}$. Then $P_m\mathbf{F}_{m+1} = P_m$, and the induction continues.
After $m = n-1$, $\mathbf{F}_n$ is primitive with invertible derivative.
Taking $G_n = \mathbf{F}_n$ and using $B_m^2 = I$ gives $\mathbf{F} = \mathbf{F}_1 = B_1 \cdots B_{n-1} G_n \circ G_{n-1} \circ \cdots \circ G_1$ near $\mathbf{0}$.
Structural Commentary: Invertible C^1 maps factor locally into coordinate changes and flips.
Key Insight: Gaussian elimination on $D\mathbf{F}$ becomes nonlinear via the inverse function theorem. □

Theorem 5.1.12. *Partition of Unity on a Compact Set*
Let $K \subseteq \mathbb{R}^n$ be compact and let $\{V_\alpha\}$ cover K by open sets. Then there are $\psi_1, \ldots, \psi_s \in C^1(\mathbb{R}^n)$ such that

(i) $0 \leq \psi_i \leq 1$ *for all i;*
(ii) $\operatorname{supp}(\psi_i) \subseteq V_\alpha$ *for some α;*
(iii) $\sum_{i=1}^{s} \psi_i = 1$ *on K.*

If $f \in C(\mathbb{R}^n)$ and $\operatorname{supp}(f) \subseteq K$, then $f = \sum_{i=1}^{s} \psi_i f$, with each summand supported in one member of the cover.

Proof.

(i) For each $\mathbf{x} \in K$, choose balls with $\overline{B(\mathbf{x})} \subseteq W(\mathbf{x}) \subseteq \overline{W(\mathbf{x})} \subseteq V_{\alpha(\mathbf{x})}$.
By compactness, Definition 4.1.1, finitely many $B(\mathbf{x}_i)$ cover K.
(ii) Choose $\varphi_i \in C^1(\mathbb{R}^n)$ with $\varphi_i = 1$ on $B(\mathbf{x}_i)$ and $\operatorname{supp}(\varphi_i) \subseteq W(\mathbf{x}_i)$. Set $\psi_i = (1-\varphi_1)\cdots(1-\varphi_{i-1})\varphi_i$.
(iii) Then $0 \leq \psi_i \leq 1$, each ψ_i is supported in some V_α, and $\sum_{i=1}^{s} \psi_i = 1 - \prod_{i=1}^{s}(1-\varphi_i)$, which equals 1 on K.
Multiplying by f gives the final assertion.

Structural Commentary: Compactness finite; product cutoffs sum to one.
Key Insight: A partition of unity turns local change-of-variable information into a global formula. □

The nonlinear change-of-variables theorem is the local volume-scaling formula from Theorem 5.1.1 applied cell by cell after a fine grid has been warped by a C^1 diffeomorphism.

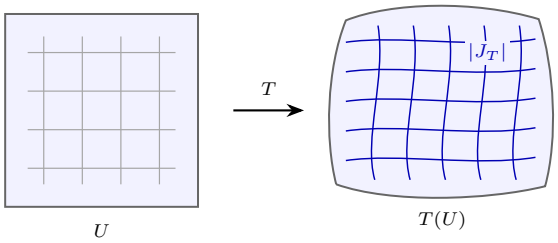


Figure 171: A C^1 change of variables turns a fine grid into curvilinear cells; $|J_T|$ records local volume scaling.

Theorem 5.1.13. *Change of Variables in $\mathbb{R}^k$*
Let $T : E \to \mathbb{R}^k$ be one-to-one and C^1 on open $E \subseteq \mathbb{R}^k$, with $J_T \neq 0$ on E.
If $f \in C(\mathbb{R}^k)$ and $\operatorname{supp}(f)$ is compact in $T(E)$, then $\int_{\mathbb{R}^k} f(\mathbf{y})\,d\mathbf{y} = \int_{\mathbb{R}^k} f(T(\mathbf{x}))\,|J_T(\mathbf{x})|\,d\mathbf{x}$.

Proof. Let $\mathcal{C}$ be the class of maps for which formula holds for all continuous functions with compact support in the image. Primitive maps lie in $\mathcal{C}$ by one-variable substitution, and flips lie in $\mathcal{C}$ by Theorem 5.1.7.
If $P, Q \in \mathcal{C}$, then $P \circ Q \in \mathcal{C}$, since Theorems 3.1.5 and 1.1.8 give $J_{P\circ Q}(\mathbf{x}) = J_P(Q(\mathbf{x}))J_Q(\mathbf{x})$.
After translating coordinates, Theorem 5.1.11 therefore gives the formula on a neighborhood of each point of E. By Theorem 4.1.4, these neighborhoods have open images covering $\operatorname{supp}(f)$.
Choose a partition of unity from Theorem 5.1.12, write $f = \sum_i \psi_i f$ with each $\psi_i f$ supported in one such image, apply the local formula to each summand, and sum.
Structural Commentary: Model cases plus partitions of unity prove the theorem.
Key Insight: The absolute Jacobian is local volume scaling, globalized by compact support. □

Method 5.1.14. *Multiple-Integration Proof Patterns*
For a multiple-integration identity:

(i) *Products*: prove product cases, then pass by uniform approximation;
(ii) *Grids*: compare product-grid upper and lower sums with one-variable slice sums;
(iii) *Support*: enclose compact support in a cell and check cell-independence;
(iv) *Local form*: reduce C^1 coordinate changes to primitive maps and flips;
(v) *Determinants*: use multiplicativity and Theorem 1.1.8;
(vi) *Globalize*: apply the partition of unity, Theorem 5.1.12.

Key Insight: Change of variables is local determinant geometry plus compactness.

5.2 Differential Forms and Manifolds

Differential forms package oriented integration on parametrized cells. They use determinant sign and laws (Definition 1.1.1, Theorems 1.1.8, 1.1.15) and change of variables (Theorem 5.1.13); conceptually, forms test oriented cells.

Perspective 5.2.1. *Forms as Functionals*
The notation $\int f\,dx$ treats dx as directed coordinate variation. Differential forms make this precise: a 1-form such as $f\,dx + g\,dy$ assigns a number to an oriented path; dx records net x-variation, and $f\,dx$ weights it by f.

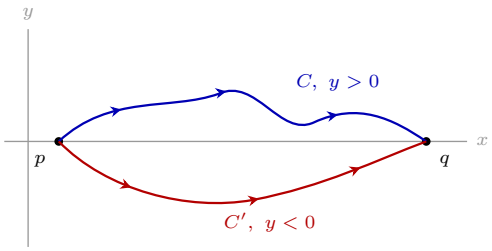


Figure 172: $y\,dx$ records signed, y-weighted x-variation.

Definition 5.2.2. *k-Surfaces and Differential Forms*
Let $E \subseteq \mathbb{R}^n$ be open. A *k-surface* in E is a C^1 map $\Phi : D \to E$ from a compact *parameter domain* $D \subseteq \mathbb{R}^k$, here a k-cell or the simplex Q^k of Definition 5.1.9.
For compact D, C^r means extension to an open neighborhood of D.
A *k-form* in E, $k \geq 1$, is $\omega = \sum a_{i_1\cdots i_k}(\mathbf{x})\,dx_{i_1} \wedge \cdots \wedge dx_{i_k}$, with continuous real coefficients on E, and assigns to each k-surface Φ the number $\int_D \sum a_{i_1\cdots i_k}(\Phi(\mathbf{u}))\frac{\partial(x_{i_1},\ldots,x_{i_k})}{\partial(u_1,\ldots,u_k)}\,d\mathbf{u}$.
A 0-form is a continuous function; a form is C^r if all coefficients are C^r.

Remark 5.2.3. *Parametrized Cells, Not Just Images*
Parametrizations may fail to be injective or regular. Forms therefore act on parametrized cells, not just images; this matters for orientation, multiplicity, and cancellation.

Proposition 5.2.4. *Basic Forms and Standard Presentation*
Swapping neighboring factors in $dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ changes sign, and repeated indices give 0. For an increasing k-index $I = (i_1, \ldots, i_k)$, write $dx_I := dx_{i_1} \wedge \cdots \wedge dx_{i_k}$.
Every k-form has a unique standard presentation $\omega = \sum_I b_I(\mathbf{x})\,dx_I$, where I ranges over increasing k-indices.

Proof. The sign rules follow from the determinant row-swap and repeated-row rules in Theorem 1.1.8, applied to the Jacobian in Definition 5.2.2. The sign needed to sort an index tuple is exactly the permutation sign of Definition 3.2.1; hence every term either vanishes or reduces to an increasing index, giving existence.
For uniqueness, suppose $\omega = \sum_I b_I(\mathbf{x})\,dx_I$ is a zero k-form in standard presentation. If $b_J(\mathbf{v}) \neq 0$ for some increasing $J = (j_1, \ldots, j_k)$, continuity gives a small cell $D = [-h, h]^k$ such that, for $\Phi(\mathbf{u}) = \mathbf{v} + \sum_{r=1}^{k} u_r \mathbf{e}_{j_r}$, the coefficient $b_J(\Phi(\mathbf{u}))$ has constant nonzero sign on D. In the defining integral, the J-Jacobian is 1 and every other standard Jacobian is 0, so $0 = \int_\Phi \omega = \int_D b_J(\Phi(\mathbf{u}))\,d\mathbf{u}$, a contradiction. Thus every b_I is zero.
Structural Commentary: Wedge signs are determinant signs.
Key Insight: Standard presentation removes ordered-index redundancy. □

Theorem 5.2.5. *Naturality Under Reparametrization*
Let $\Phi : D \to E \subseteq \mathbb{R}^n$ be a C^1 k-surface and let $R : D' \to D$ be a C^1 diffeomorphism.
For every k-form ω, $\int_{\Phi\circ R} \omega = \operatorname{sgn}(\det DR) \int_\Phi \omega$, with constant sign on D'.

Proof. It suffices to take $\omega = f\,dx_I$. By Theorem 1.1.15, the relevant minor of $D(\Phi \circ R)$ contributes the corresponding minor of $D\Phi$ times $\det DR$. Change of variables (Theorem 5.1.13) cancels $|\det DR|$ and leaves $\operatorname{sgn}(\det DR)$; linearity handles general forms.
Structural Commentary: Reparametrization changes coordinates, not the oriented cell.
Key Insight: The same determinant sign controls orientation and the sign of the form integral. □

Definition 5.2.6. *Product and Exterior Derivative*
For basic forms, define $dx_I \wedge dx_J$ by concatenating indices and reducing to standard presentation by Proposition 5.2.4; extend bilinearly. For a C^1 0-form f, set $df := \sum_{i=1}^{n} (D_i f)(\mathbf{x})\,dx_i$.
For a C^1 k-form $\omega = \sum_I b_I(\mathbf{x})\,dx_I$, set $d\omega := \sum_I (db_I) \wedge dx_I$.

Theorem 5.2.7. *Exterior Derivative Laws*
If ω, λ are C^1 forms in E and ω is a k-form, then $d(\omega \wedge \lambda) = (d\omega) \wedge \lambda + (-1)^k \omega \wedge d\lambda$. If ω is C^2, then $d^2\omega = 0$.

Proof. By bilinearity, it suffices to take $\omega = f\,dx_I$ and $\lambda = g\,dx_J$.
The ordinary product rule gives $d(fg) = f\,dg + g\,df$; moving dg past k factors in dx_I gives the displayed sign.
For a C^2 0-form f, $d^2 f = \sum_{i,j} (D_{ij} f)(\mathbf{x})\,dx_i \wedge dx_j = 0$, by Theorem 3.2.11 and antisymmetry.
The general case follows from $\omega = \sum_I b_I\,dx_I$ and the product rule just proved.
Structural Commentary: The exterior derivative is built to make mixed partial cancellation visible.
Key Insight: Symmetric second derivatives meet antisymmetric wedge products, so $d^2 = 0$. □

Definition 5.2.8. *Change of Variables for Forms*
Let $T : E \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^m$ be C^1, with $T = (t_1, \ldots, t_m)$. For a k-form $\omega = \sum_I b_I(\mathbf{y})\,dy_I$ in V, define its *pullback* by $\omega_T := T^*\omega := \sum_I b_I(T(\mathbf{x}))\,dt_{i_1} \wedge \cdots \wedge dt_{i_k}$, and $dt_i := \sum_{j=1}^{n} (D_j t_i)(\mathbf{x})\,dx_j$.

Pullback is the differential-form analogue of the dual map in Definition 5.1.11: cells move forward by composition with T, while forms move backward so they can be evaluated before or after that forward motion.

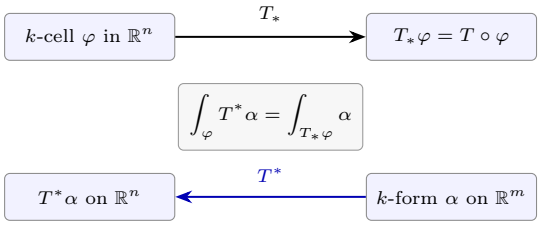


Figure 173: Cells push forward with T; forms pull back with T^*, so $T^*\alpha(\varphi) = \alpha(T_*\varphi)$.

Theorem 5.2.9. *Pullback Commutes with the Basic Operations*
With $T : E \to V$ as in Definition 5.2.8,

(i) $(\omega + \lambda)_T = \omega_T + \lambda_T$ *for forms of the same order;*
(ii) $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$*;*
(iii) $d(\omega_T) = (d\omega)_T$ *when ω is C^1 and T is C^2;*
(iv) if $S : V \to W$ is C^1, then $(\omega_S)_T = \omega_{S \circ T}$.

Moreover, if Φ is a k-surface in E and ω is a k-form in V, then $\int_{T \circ \Phi} \omega = \int_\Phi \omega_T$.

Proof.

(i) Addition is the linearity of Definition 5.2.8.
(ii) Pullback replaces each dy_i by dt_i; bilinearity gives $(\omega \wedge \lambda)_T = \omega_T \wedge \lambda_T$.
(iii) Let $\omega = \sum_I b_I\,dy_I$. Since $d(b_I \circ T) = (db_I)_T$ (Theorem 3.1.5) and $d(dt_i) = d^2 t_i = 0$, $d(\omega_T) = (d\omega)_T$.
(iv) If $S = (s_1, \ldots, s_p)$, both $(\omega_S)_T$ and $\omega_{S \circ T}$ replace dz_q by $d(s_q \circ T)$, by the chain rule.

For the integral identity, expand both sides over D; the Jacobian minors agree by Theorem 1.1.15, since $D(T \circ \Phi) = DT(\Phi) D\Phi$.
Structural Commentary: Pullback translates forms before integration.
Key Insight: Forms are defined so that composition of parameterizations becomes an algebraic substitution. □

5.3 Stokes Theory

Stokes theory relates exterior differentiation to oriented boundary, with Green's theorem, classical Stokes, and divergence as special cases.

Definition 5.3.1. *Chains and Boundaries*

For $\mathbf{p}_0, \dots, \mathbf{p}_k \in \mathbb{R}^n$, oriented affine k-simplex $\sigma = [\mathbf{p}_0, \dots, \mathbf{p}_k]$ is k-surface on Q^k: $\sigma(\mathbf{u}) = \mathbf{p}_0 + \sum_{i=1}^{k} u_i(\mathbf{p}_i - \mathbf{p}_0)$. Permuting vertices multiplies σ by the permutation sign. An *affine k-chain* is a finite formal sum of oriented affine k-simplices; integration is extended linearly.

For $k \geq 1$, define $\partial_{\mathrm{ch}}[\mathbf{p}_0, \dots, \mathbf{p}_k] := \sum_{j=0}^{k} (-1)^j [\mathbf{p}_0, \dots, \mathbf{p}_{j-1}, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k]$, and extend ∂_{ch} linearly. If $\Phi = T \circ \sigma$ is a C^2 oriented simplex, set $\partial_{\mathrm{ch}}\Phi := T(\partial_{\mathrm{ch}}\sigma)$, again extended linearly to chains.

Theorem 5.3.2. *Stokes' Theorem*

If Ψ is a C^2 k-chain in open $V \subseteq \mathbb{R}^m$ and ω is a C^1 $(k-1)$-form in V, then $\int_\Psi d\omega = \int_{\partial_{\mathrm{ch}}\Psi} \omega$.

Proof. By linearity, take $\Psi = \Phi = T \circ \sigma$, where σ is the standard oriented k-simplex.
By Theorem 5.2.9, $\int_\Phi d\omega = \int_\sigma (d\omega)_T = \int_\sigma d(\omega_T)$; and $\int_{\partial_{\mathrm{ch}}\Phi} \omega = \int_{\partial_{\mathrm{ch}}\sigma} \omega_T$, remains to prove formula on σ.
By linearity, reduce to $\lambda = f\, dx_1 \wedge \cdots \wedge \widehat{dx_r} \wedge \cdots \wedge dx_k$.
The boundary formula in Definition 5.3.1 leaves only the two faces in the x_r direction, and the resulting identity is the one-variable fundamental theorem of calculus inside the iterated integral.
Structural Commentary: Pullback reduces Stokes' theorem to a computation on the standard simplex.
Key Insight: The boundary signs are exactly the signs produced by the missing wedge factor. □

Definition 5.3.3. *Closed and Exact Forms*

A k-form ω in E is *exact* if $\omega = d\lambda$ for some $(k-1)$-form λ in E, and *closed* if ω is C^1 with $d\omega = 0$.
By Theorem 5.2.7, exact C^1 forms are closed.

Perspective 5.3.4. *Closed Is Local, Exact Is Global*

Exact $\Rightarrow$ closed is just $d^2 = 0$; the converse depends on the domain.
On $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ the standard warning example is $\omega = \frac{-y}{x^2+y^2}\, dx + \frac{x}{x^2+y^2}\, dy$, which is closed but not exact.
Holes can obstruct global antiderivatives even when local derivative tests vanish.

Theorem 5.3.5. *Poincare Lemma on Convex Sets*

If $E \subseteq \mathbb{R}^n$ is convex and open, then every closed C^1 k-form in E, $k \geq 1$, is exact.
The same holds on every open set C^2-equivalent to a convex open set.

Proof. Choose $\mathbf{a} \in E$ and let $H_t(\mathbf{x}) = \mathbf{a} + t(\mathbf{x} - \mathbf{a})$, $0 \leq t \leq 1$. Convexity gives $H_t(E) \subseteq E$. For a k-form ω, define the $(k-1)$-form as $(K\omega)_{\mathbf{x}}(\mathbf{v}_1, \dots, \mathbf{v}_{k-1}) := \int_0^1 t^{k-1} \omega_{H_t(\mathbf{x})}(\mathbf{x} - \mathbf{a}, \mathbf{v}_1, \dots, \mathbf{v}_{k-1})\, dt$.
A coordinate differentiation under the integral gives the homotopy identity $\omega - H_0^*\omega = dK\omega + Kd\omega$. Since $k \geq 1$, the constant map H_0 pulls every k-form back to 0. If $d\omega = 0$, then $\omega = d(K\omega)$, so ω is exact.
If E is C^2-equivalent to a convex open set C by $\Phi : C \to E$, then $\Phi^*\omega$ is closed by Theorem 5.2.9. The convex case gives $\Phi^*\omega = d\eta$ on C. Applying $(\Phi^{-1})^*$ and using the same pullback law gives $\omega = d((\Phi^{-1})^*\eta)$ on E.
Structural Commentary: Convexity lets coefficient-level antiderivatives be chosen consistently.
Key Insight: Closed is local symmetry; convexity supplies the global primitive. □

Theorem 5.3.6. *Vector Analysis Dictionary in $\mathbb{R}^3$*

For $\mathbf{F} = F_1\mathbf{e}_1 + F_2\mathbf{e}_2 + F_3\mathbf{e}_3$ on open $E \subseteq \mathbb{R}^3$,
set $\lambda_{\mathbf{F}} := F_1\, dx + F_2\, dy + F_3\, dz$ and $\omega_{\mathbf{F}} := F_1\, dy \wedge dz + F_2\, dz \wedge dx + F_3\, dx \wedge dy$. Then
$\mathbf{F} = \nabla u \iff \lambda_{\mathbf{F}} = du$; and $\nabla \times \mathbf{F} = 0 \iff d\lambda_{\mathbf{F}} = 0$; and
$\mathbf{F} = \nabla \times \mathbf{G} \iff \omega_{\mathbf{F}} = d\lambda_{\mathbf{G}}$; and $\nabla \cdot \mathbf{F} = 0 \iff d\omega_{\mathbf{F}} = 0$.
Hence $\nabla \times (\nabla u) = 0$ and $\nabla \cdot (\nabla \times \mathbf{G}) = 0$; on sets C^2-equivalent to convex open sets, the converses hold.

Proof. Expand $d\lambda_{\mathbf{F}}$ and $d\omega_{\mathbf{F}}$ by Definition 5.2.6; their coefficients are exactly $\nabla \times \mathbf{F}$ and $\nabla \cdot \mathbf{F}$. Potentials and vector potentials are exactness of $\lambda_{\mathbf{F}}$ and $\omega_{\mathbf{F}}$; the converses follow from Theorem 5.3.5.
Structural Commentary: Gradient, curl, and divergence are exterior derivative in degrees $0, 1, 2$.
Key Insight: The familiar vector identities are $d^2 = 0$ written in $\mathbb{R}^3$ notation. □

Theorem 5.3.7. *Green, Classical Stokes, and Divergence*

The following are special cases of Theorem 5.3.2.

(i) If $\Omega \subseteq \mathbb{R}^2$ has positively oriented boundary, $\alpha, \beta \in C^1$ near Ω, then $\int_{\mathrm{bd}\,\Omega} (\alpha\, dx + \beta\, dy) = \int_\Omega \left(\frac{\partial \beta}{\partial x} - \frac{\partial \alpha}{\partial y} \right) dA$

(ii) If Φ is a C^2 2-surface in $\mathbb{R}^3$ and $\mathbf{F} \in C^1$ near Φ, then $\int_\Phi (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dA = \int_{\partial_{\mathrm{ch}}\Phi} \mathbf{F} \cdot \mathbf{t}\, ds$.

(iii) If $\Omega \subseteq \mathbb{R}^3$ has positively oriented boundary and $\mathbf{F} \in C^1$ near Ω, then $\int_\Omega (\nabla \cdot \mathbf{F})\, dV = \int_{\mathrm{bd}\,\Omega} \mathbf{F} \cdot \mathbf{n}\, dA$

Proof.

(i) Put $\lambda = \alpha\, dx + \beta\, dy$. Then $d\lambda = (\partial\beta/\partial x - \partial\alpha/\partial y)\, dA$.
(ii) Put $\lambda = \lambda_{\mathbf{F}}$. By Theorem 5.3.6, $d\lambda = \omega_{\nabla \times \mathbf{F}}$.
(iii) Put $\omega = \omega_{\mathbf{F}}$. Then $d\omega = (\nabla \cdot \mathbf{F})\, dx \wedge dy \wedge dz$.

Theorem 5.3.2, together with the line, area, and volume element translations, gives the three displayed formulas.
Structural Commentary: Forms translate Stokes into the classical vector formulas.
Key Insight: Boundary plus d is the content; vector notation is translation. □

Method 5.3.8. *Differential-Forms Proof Patterns*

(i) *Signs*: compute wedge signs by determinant swaps and permutation parity.
(ii) *Standard presentation*: remove repeated indices and sort the rest.
(iii) *Pullback*: substitute dt_i for dy_i and apply the chain rule.
(iv) *Stokes*: pull back to the standard simplex, prove the coordinate case, extend linearly.
(v) *Closed versus exact*: use $d^2 = 0$ for necessity; use convexity or C^2-equivalence for primitives.
(vi) *Vector analysis*: translate fields to forms, apply Stokes, translate back.

Key Insight: Forms package signs, orientation, and parameter changes.

6 Infinite Series and Summability

6.1 Series Tests

Series are sequences of partial sums. Comparison, condensation, root and ratio tests, rearrangements, products, and double sums all refine that viewpoint.

Definition 6.1.1. *Series and Partial Sums*
Let (a_n) be a complex sequence, and for $p \le q$ put $\sum_{n=p}^{q} a_n := a_p + a_{p+1} + \cdots + a_q$.

The index n in the displayed finite sum is a bound variable, in the sense of Principle 2.1.1; it may be renamed when its scope is unchanged.

The *series* $\sum_{n=1}^{\infty} a_n$ (or $a_1 + a_2 + \cdots$) is the partial-sum sequence $s_n := \sum_{k=1}^{n} a_k$. If $s_n \to s$, then $\sum_{n=1}^{\infty} a_n = s$ and series *converges*; otherwise it *diverges*. The same convention applies to series starting at $n = 0$.

Motivation 6.1.2. *Partial Sums Remember Order*
An infinite sum is a limit of partial sums. For alternating harmonic series, odd and even partial sums approach the same value from opposite sides; see Corollary 6.2.3 and Theorem 6.2.13.

Perspective 6.1.3. *Formal Algebra Needs Convergence*
Finite sums are algebraic; infinite sums are partial-sum limits. Algebraic manipulations are justified only by convergence control, as in Theorem 6.1.4, Corollary 6.1.5, and Theorems 6.2.10, 6.2.13.
Key Insight: The symbol $\sum_{n=1}^{\infty}$ is a limit claim before it is an algebraic object.

Theorem 6.1.4. *Cauchy Criterion for Series*
The series $\sum a_n$ converges if and only if $\forall \varepsilon > 0$ $\exists$ an integer N s.t. $m \ge n \ge N$ implies $\left| \sum_{k=n}^{m} a_k \right| \le \varepsilon$.

Proof.

(⇒) If $s_n := \sum_{k=1}^{n} a_k$ converges, then (s_n) is Cauchy. Thus, for $m \ge n$ large enough, $\left| \sum_{k=n}^{m} a_k \right| = |s_m - s_{n-1}| \le \varepsilon$.

(⇐) The tail condition gives, for $m > n$ large enough, $|s_m - s_n| = \left| \sum_{k=n+1}^{m} a_k \right| \le \varepsilon$.

Hence (s_n) is Cauchy and converges by Theorem 4.1.10.

Structural Commentary: A series converges exactly when all sufficiently late finite tails are small.
Key Insight: The Cauchy test measures tails, not the eventual value of the sum. □

Corollary 6.1.5. *Terms of a Convergent Series Tend to Zero*
If $\sum a_n$ converges, then $a_n \to 0$.

Proof. Let $\varepsilon > 0$. By Theorem 6.1.4, there is N such that $m \geq n \geq N$ implies $|\sum_{k=n}^{m} a_k| \leq \varepsilon$.
Taking $m = n$ gives $|a_n| \leq \varepsilon$ for all $n \geq N$, hence $a_n \to 0$.
Structural Commentary: A single term is the shortest possible tail.
Key Insight: The condition $a_n \to 0$ is necessary, but not sufficient for convergence. □

Theorem 6.1.6. *Nonnegative Series and Bounded Partial Sums*
A series of nonnegative real terms converges if and only if its partial sums form a bounded sequence.

Proof.
(⇒) Every convergent sequence is bounded, so the partial sums are bounded.
(⇐) Since $a_n \geq 0$, $s_n = \sum_{k=1}^{n} a_k$ increases; if bounded, it converges by Theorem 2.1.12.
Structural Commentary: Nonnegative series replace cancellation by monotonicity.
Key Insight: For nonnegative terms, convergence is exactly bounded accumulation. □

Theorem 6.1.7. *Comparison Test*
(i) If $|a_n| \leq c_n$ for $n \geq N_0$, and if $\sum c_n$ converges, then $\sum a_n$ converges.
(ii) If $a_n \geq d_n \geq 0$ for $n \geq N_0$, and if $\sum d_n$ diverges, then $\sum a_n$ diverges.

Proof.
(i) Given $\varepsilon > 0$, choose $N \geq N_0$ so that $m \geq n \geq N$ implies $\sum_{k=n}^{m} c_k \leq \varepsilon$.
Then $\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k| \leq \sum_{k=n}^{m} c_k \leq \varepsilon$, so Theorem 6.1.4 applies.
(ii) If $\sum a_n$ converged, then (i) applied to $d_n \leq a_n$ would force $\sum d_n$ to converge, a contradiction.
Structural Commentary: Absolute tail domination transfers Cauchy convergence.
Key Insight: Comparison is a tail test; finitely many early terms do not matter. □

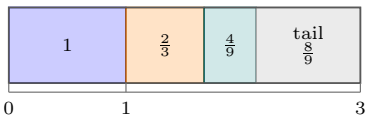


Figure 174: At $x = 2/3$, the strip widths sum to $1 + \frac{2}{3} + \frac{4}{9} + \cdots = 3$.

Theorem 6.1.8. *Geometric Series*
If $0 \leq x < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. If $x \geq 1$, the series diverges.

Proof. For $x \neq 1$, $\sum_{k=0}^{n} x^k = \frac{1-x^{n+1}}{1-x}$. Let $n \to \infty$ and use Theorem 2.2.9.
If $x = 1$, the partial sums are $n + 1$, so they diverge.
Structural Commentary: The finite geometric identity converts convergence into the limit of x^{n+1}.
Key Insight: The geometric series is the model comparison series for ratio-like decay. □

Definition 6.1.9. *Real Decimals as Series*
For digits $b_N, \ldots, b_0, a_1, a_2, \ldots \in \{0, \ldots, 9\}$, with $b_N \neq 0$ unless the integer part is 0,
$b_N \cdots b_0.a_1a_2\cdots := \sum_{i=0}^{N} b_i 10^i + \sum_{j=1}^{\infty} a_j 10^{-j}$. The fractional series converges absolutely since $0 \leq a_j 10^{-j} \leq 9 \cdot 10^{-j}$ and $\sum 9 \cdot 10^{-j}$ is geometric. A minus sign denotes the negative of this value.

Proposition 6.1.10. *Terminal Nines Are Nonunique*
The same real number can have two decimal representations: $1.000\ldots = 0.999\ldots$.

Proof. By Definition 6.1.9 and Theorem 6.1.8, $1.000\ldots = 1$; $0.999\ldots = \sum_{j=1}^{\infty} 9 \cdot 10^{-j} = 9 \cdot \frac{10^{-1}}{1-10^{-1}} = 1$.
Structural Commentary: The two digit strings have the same partial-sum limit.
Key Insight: Uniqueness requires excluding terminal 9 tails. □

Perspective 6.1.11. *Decimal Expansions Are Series*
Proposition 1.1.19 gives decimal truncations that converge to a real number, while Definition 6.1.9 interprets the infinite digit string as a series. Proposition 6.1.10 explains the standard terminal-9 ambiguity.

Theorem 6.1.12. *Cauchy Condensation Test*

Suppose $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

Proof. Let $s_n := a_1 + \cdots + a_n$ and $t_k := a_1 + 2a_2 + \cdots + 2^k a_{2^k}$.
If $n < 2^k$, then grouping by dyadic blocks gives $s_n \leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \leq t_k$.
If $n > 2^k$, grouping from the other side gives $2s_n \geq t_k$. Hence the nonnegative partial sums (s_n) and (t_k) are either both bounded or both unbounded; apply Theorem 6.1.6.
Structural Commentary: Monotonicity lets dyadic blocks be compared to their endpoint terms.
Key Insight: A sparse dyadic subsequence detects the whole monotone nonnegative series. □

Corollary 6.1.13. *p-Series Test*

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Proof. If $p \leq 0$, the terms do not tend to 0, so Corollary 6.1.5 gives divergence.
If $p > 0$, Theorem 6.1.12 reduces the question to $\sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^p} = \sum_{k=0}^{\infty} 2^{(1-p)k}$.
This geometric series converges exactly when $2^{1-p} < 1$, i.e. $p > 1$, by Theorem 6.1.8.
Structural Commentary: Condensation turns powers of n into powers of 2.
Key Insight: The boundary is $p = 1$ because the condensed ratio is 2^{1-p}. □

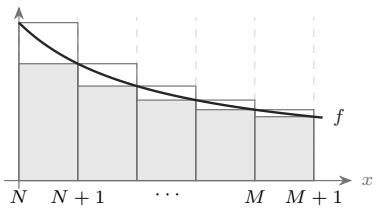


Figure 175: Left and right endpoint estimates for decreasing $f \geq 0$.

Theorem 6.1.14. *Integral Test*

Let $N \in \mathbb{N}$ and let $f : [N, \infty) \to [0, \infty)$ be decreasing and Riemann integrable on compact subintervals.
Then $\sum_{k=N}^{\infty} f(k)$ converges iff $\int_N^{\infty} f(x)\,dx$ converges.
For $M \geq N$, $\int_N^{M+1} f(x)\,dx \leq \sum_{k=N}^{M} f(k) \leq f(N) + \int_N^{M} f(x)\,dx$.

Proof. For a monotone function, compact-subinterval Riemann integrability follows from Theorem 2.1.11 with $\alpha(x) = x$; it is kept in the statement to make the improper-integral dependency explicit.

(⇒) For $x \in [k, k+1]$, monotonicity gives $f(x) \leq f(k)$. Hence $\int_N^{M+1} f(x)\,dx \leq \sum_{k=N}^{M} f(k)$. If the series converges, these truncations are bounded, so $\int_N^{\infty} f$ converges by Theorem 2.3.2.

(⇐) For $k \geq N+1$ and $x \in [k-1, k]$, monotonicity gives $f(k) \leq f(x)$. Hence $\sum_{k=N}^{M} f(k) \leq f(N) + \int_N^{M} f(x)\,dx$. If $\int_N^{\infty} f$ converges, the partial sums are bounded, so the series converges by Theorem 6.1.6.

Structural Commentary: A monotone sequence can be compared to neighboring unit-width areas.
Key Insight: The integral test is comparison of two infinite accumulations with the same tail size. □

Definition 6.1.15. *The Number e*

Define $e := \sum_{n=0}^{\infty} \frac{1}{n!}$, where $0! := 1$; convergence follows by comparison with the geometric series.

Theorem 6.1.16. *A Limit Formula for e*

The limit formula is $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$.

Proof. Let $s_m := \sum_{k=0}^{m} \frac{1}{k!}$ and $t_n := (1 + \frac{1}{n})^n$. The binomial theorem gives $t_n \leq s_n$, hence $\limsup t_n \leq e$.
For fixed $m \leq n$, first m binomial terms give $t_n \geq 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{m!}\left(1 - \frac{1}{n}\right)\cdots\left(1 - \frac{m-1}{n}\right)$.
Letting $n \to \infty$ with m fixed gives $\liminf t_n \geq s_m$, and then $m \to \infty$ gives $e \leq \liminf t_n$.
Structural Commentary: Bound above by the factorial series; bound below by fixed binomial terms.
Key Insight: Limsup and liminf squeeze the binomial limit to the series definition. □

Proposition 6.1.17. *Remainder Bound for the Series of e*
If $n \geq 1$ and $s_n = \sum_{k=0}^{n} 1/k!$, then $0 < e - s_n < \frac{1}{n!\,n}$.

Proof. Tail is $e - s_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \cdots < \frac{1}{(n+1)!}\left(1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots\right) = \frac{1}{n!\,n}$
Structural Commentary: The factorial tail is bounded by a geometric tail.
Key Insight: Rapid convergence of $\sum \frac{1}{n!}$ gives arithmetic control of e. □

Theorem 6.1.18. *e Is Irrational*
The number e is irrational.

Proof. Suppose $e = p/q$ with p, q positive integers. By Proposition 6.1.17, $0 < q!(e - s_q) < \frac{1}{q}$.
But $q!e$ and $q!s_q$ are integers, so $q!(e - s_q)$ is an integer strictly between 0 and 1, impossible.
Structural Commentary: The sharp tail bound turns rationality into an impossible integer.
Key Insight: A positive integer cannot be smaller than 1. □

Theorem 6.1.19. *Root Test*
For a series $\sum a_n$, put $\alpha := \limsup_{n\to\infty} \sqrt[n]{|a_n|}$. Then:

(i) if $\alpha < 1$, then $\sum a_n$ converges absolutely;
(ii) if $\alpha > 1$, then $\sum a_n$ diverges;
(iii) if $\alpha = 1$, the test gives no information.

Proof.

(i) Choose $\alpha < \beta < 1$. Eventually $|a_n| < \beta^n$, so $\sum |a_n|$ converges by comparison with $\sum \beta^n$.
(ii) Since $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$; use Corollary 6.1.5.
(iii) $\sum 1/n$ and $\sum 1/n^2$ both have $\alpha = 1$, but behave differently by Corollary 6.1.13.

Structural Commentary: The root test compares term size directly to geometric decay.
Key Insight: Values below 1 force absolute convergence; values above 1 prevent terms from tending to 0. □

Theorem 6.1.20. *Ratio Test*
Assume $a_n \neq 0$ for all sufficiently large n.

(i) If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum a_n$ converges absolutely.
(ii) If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$, then $\sum a_n$ diverges.
(iii) In the remaining cases, the test gives no information.

Proof.

(i) Choose $\beta < 1$ with $|a_{n+1}/a_n| < \beta$ eventually.
Then $|a_n| = O(\beta^n)$, so $\sum |a_n|$ converges by comparison with a geometric series.
(ii) Choose $\beta > 1$ with $|a_{n+1}/a_n| > \beta$ eventually.
Then the tail $|a_n|$ is eventually increasing and bounded away from 0, so $a_n \not\to 0$.
(iii) The series $\sum 1/n$ and $\sum 1/n^2$ both have ratio limit 1, but behave differently by Corollary 6.1.13.

Structural Commentary: Ratios turn successive terms into a geometric majorant.
Key Insight: Ratios below 1 force geometric decay; ratios above 1 prevent terms from tending to zero. □

Theorem 6.1.21. *Root and Ratio Test Comparison*
For any sequence (c_n) of positive numbers, $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} \sqrt[n]{c_n} \leq \limsup_{n\to\infty} \sqrt[n]{c_n} \leq \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$.

Proof. Let $\alpha = \limsup c_{n+1}/c_n$. If $\alpha < +\infty$ and $\beta > \alpha$, then eventually $c_{n+1} \leq \beta c_n$, hence $c_n \leq c_N \beta^{n-N}$ for all large n. Taking nth roots gives $\limsup \sqrt[n]{c_n} \leq \beta$; let $\beta \downarrow \alpha$. If $\alpha = +\infty$, the rightmost inequality is automatic.
Let $\ell = \liminf c_{n+1}/c_n$. If $\ell = 0$, the leftmost inequality is automatic. If $\gamma < \ell$ with $\gamma > 0$, then eventually $c_{n+1} \geq \gamma c_n$, hence $c_n \geq c_N \gamma^{n-N}$ for all large n. Taking nth roots gives $\liminf \sqrt[n]{c_n} \geq \gamma$; let $\gamma \uparrow \ell$.
The middle inequality is the general fact that $\liminf a_n \leq \limsup a_n$.
Structural Commentary: A tail ratio bound multiplies into a root bound.
Key Insight: Whenever the ratio test proves convergence, the root test does too. □

6.2 Summability Methods

Theorem 6.2.1. *Summation by Parts*
Given two sequences (a_n) *and* (b_n), *put* $A_n := \sum_{k=0}^{n} a_k$ *and* $A_{-1} := 0$.
If $0 \le p \le q$, *then* $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$.

Proof. Since $a_n = A_n - A_{n-1}$, $\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p}^{q} A_{n-1} b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1}$.
Structural Commentary: This is the discrete analogue of integration by parts.
Key Insight: Move the difference from a_n onto b_n by rewriting $a_n = A_n - A_{n-1}$. □

Theorem 6.2.2. *Dirichlet Test*
If the partial sums A_n *of* $\sum a_n$ *are bounded,* $b_n \downarrow 0$, *and* $b_n \ge 0$, *then* $\sum a_n b_n$ *converges.*

Proof. Choose M with $|A_n| \le M$. If $M = 0$, the result is immediate. Otherwise choose N with $b_N \le \varepsilon/(2M)$.
For $N \le p \le q$, $\left|\sum_{n=p}^{q} a_n b_n\right| \le M\left(\sum_{n=p}^{q-1}(b_n - b_{n+1}) + b_q + b_p\right) = 2Mb_p \le 2Mb_N \le \varepsilon$. Theorem 6.1.4 applies.
Structural Commentary: Bounded oscillation in A_n is damped by a monotone scalar tending to 0.
Key Insight: Summation by parts converts cancellation into a total variation estimate for b_n. □

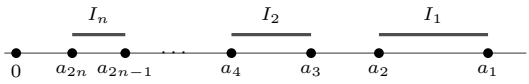


Figure 176: The intervals $I_j = [a_{2j}, a_{2j-1}]$ are disjoint inside $(0, a_1)$.

Corollary 6.2.3. *Alternating Series Test*
Suppose $|c_1| \ge |c_2| \ge |c_3| \ge \cdots$, *the signs alternate, and* $c_n \to 0$. *Then* $\sum c_n$ *converges.*

Proof. After shifting indices to start at 0 if desired, apply Theorem 6.2.2 with $a_n = (-1)^{n+1}$ and $b_n = |c_n|$.
Structural Commentary: Alternation gives bounded sums; decreasing magnitudes damp them.
Key Insight: Alternating convergence is Dirichlet convergence in its simplest form. □

Perspective 6.2.4. *Alternating Partial Sums as Nested Control*
The interval picture explains the boundedness in Corollary 6.2.3: the even partial sums collect disjoint lengths inside a fixed interval, while the odd partial sums differ from the even ones by the next term, which tends to 0.

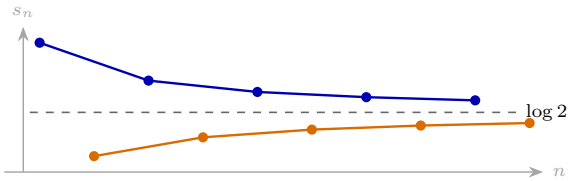


Figure 177: Odd partial sums (blue) decrease and even partial sums (orange) increase toward $\log 2$.

Definition 6.2.5. *Absolute and Nonabsolute Convergence*
The series $\sum a_n$ *converges absolutely* if $\sum |a_n|$ converges.
If $\sum a_n$ converges but $\sum |a_n|$ diverges, then $\sum a_n$ converges *nonabsolutely*.

Theorem 6.2.6. *Absolute Convergence Implies Convergence*
If $\sum a_n$ *converges absolutely, then* $\sum a_n$ *converges.*

Proof. For $m \ge n$, $\left|\sum_{k=n}^{m} a_k\right| \le \sum_{k=n}^{m} |a_k|$. Apply Theorem 6.1.4 to $\sum |a_n|$.
Structural Commentary: Absolute convergence removes cancellation from the convergence question.
Key Insight: Small absolute tails force small signed or complex tails. □

Theorem 6.2.7. *Addition and Scalar Multiplication of Series*
If $\sum a_n = A$ *and* $\sum b_n = B$, *then* $\sum(a_n + b_n) = A + B$ *and* $\sum ca_n = cA$ *for every fixed scalar* c.

Proof. Apply Theorem 2.1.3 to the corresponding partial sums.
Structural Commentary: Linearity of convergent series is just limit algebra for partial sums.
Key Insight: Add series by adding their partial-sum sequences. □

Definition 6.2.8. *Cauchy Product*
Given $\sum a_n$, $\sum b_n$, define $c_n := \sum_{k=0}^{n} a_k b_{n-k}$, $(n = 0, 1, 2, \ldots)$. Series $\sum c_n$ is *Cauchy product* of given series.

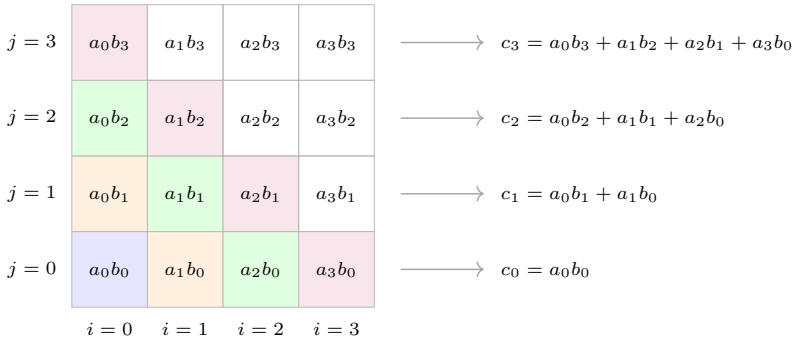


Figure 178: Diagonal sums of $(a_i b_j)$ form the Cauchy product; finite rectangular sums are order-free.

Perspective 6.2.9. *Diagonal Ordering in a Product*
c_n is the diagonal sum of the products $a_i b_j$ with $i + j = n$. Absolute convergence, as in Theorems 6.2.10 and 6.2.13, keeps the value independent of the accumulation order.

Theorem 6.2.10. *Mertens' Theorem for Cauchy Products*
Suppose $\sum a_n$ *converges absolutely,* $\sum a_n = A$, $\sum b_n = B$, *and* $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. *Then* $\sum_{n=0}^{\infty} c_n = AB$.

Proof. Let A_n, B_n, C_n be the partial sums and set $\beta_n := B_n - B$.
Then $C_n = A_n B + \gamma_n$, and $\gamma_n := a_0\beta_n + a_1\beta_{n-1} + \cdots + a_n\beta_0$. Since $A_n B \to AB$, it suffices to show $\gamma_n \to 0$.
Put $\alpha := \sum |a_n|$. If $|\beta_j| \le \varepsilon$ for $j \ge N$, then $|\gamma_n| \le |\beta_0 a_n + \cdots + \beta_N a_{n-N}| + \varepsilon\alpha$.
Let $n \to \infty$ with N fixed. The first term tends to 0, so $\limsup |\gamma_n| \le \varepsilon\alpha$; let $\varepsilon \downarrow 0$.
Structural Commentary: Absolute convergence of one factor controls the convolution error.
Key Insight: One absolute factor makes the Cauchy product converge to AB. □

Definition 6.2.11. *Rearrangement of a Series*
If (k_n) lists each positive integer exactly once and $a'_n := a_{k_n}$, then $\sum a'_n$ is a *rearrangement* of $\sum a_n$.

Theorem 6.2.12. *Riemann Rearrangement Theorem*
If real $\sum a_n$ *converges nonabsolutely and* $-\infty \le \alpha \le \beta \le \infty$,
then some rearrangement has partial sums s'_n *with* $\liminf_{n\to\infty} s'_n = \alpha$ *and* $\limsup_{n\to\infty} s'_n = \beta$.

Proof. Write $a_n = p_n - q_n$, where $p_n = \max(a_n, 0)$ and $q_n = \max(-a_n, 0)$. Since $\sum a_n$ converges but not absolutely, $\sum p_n = \sum q_n = +\infty$, and $p_n, q_n \to 0$. List the positive nonzero p_n as $P_1, P_2, \ldots$ and the positive nonzero q_n as $Q_1, Q_2, \ldots$, preserving their original orders.
Choose finite numbers $\ell_N < u_N$ with $\ell_N \to \alpha$ and $u_N \to \beta$ in the extended sense. Starting from partial sum 0, add unused P terms until the sum is $> u_N$, then add unused terms $-Q$ until the sum is $< \ell_N$; repeat for $N = 1, 2, \ldots$. Divergence of both positive series makes each stopping time finite, and the alternating stages use every nonzero positive and negative term eventually. Insert zero terms anywhere.
After the Nth upward crossing, the partial sum lies between u_N and $u_N + P_j$ for the next positive term used; after the downward crossing, it lies between $\ell_N - Q_j$ and ℓ_N for the next negative term used. Since $P_j, Q_j \to 0$, these overshoots vanish. Hence subsequences of partial sums approach the upper targets and lower targets, while all later partial sums stay within vanishing error of $[\ell_N, u_N]$. Therefore $\liminf s'_n = \alpha$ and $\limsup s'_n = \beta$.
Structural Commentary: Conditional convergence leaves enough mass to steer partial sums.
Key Insight: Vanishing terms make the steering increasingly precise. □

Theorem 6.2.13. *Absolutely Convergent Series Are Rearrangement-Invariant*
Every rearrangement of an absolutely convergent complex series converges to the same sum.

Proof. Let s_n, s'_n be the original and rearranged partial sums.
Given $\varepsilon > 0$, choose N so that $\sum_{i=N}^{m} |a_i| \le \varepsilon$ for $m \ge N$, then choose p so $a_1, \dots, a_N$ occur among $a'_1, \dots, a'_p$.
For $n > p$, $s_n - s'_n$ uses only tail terms, hence $|s_n - s'_n| \le \varepsilon$. Thus s'_n has the same limit as s_n.
Structural Commentary: Absolute tails control any finite reordering of the remaining terms.
Key Insight: Absolute convergence is exactly the protection needed against rearrangement. □

Definition 6.2.14. *Absolute Summation over Countable Sets*
Let X be countable, $f : X \to \mathbb{C}$, and $g : \mathbb{N} \to X$ be a bijection, $g_n := g(n)$.
If $\sum_{n=1}^{\infty} |f(g_n)| < \infty$, then $\sum_{x \in X} f(x)$ is *absolutely convergent*, and $\sum_{x \in X} f(x) := \sum_{n=1}^{\infty} f(g_n)$.

Theorem 6.2.15. *Countable Sums Are Order-Free Under Absolute Convergence*
Let X, Y be countable, $f : X \to \mathbb{C}$ be absolutely summable, and $h : Y \to X$ be a bijection.
Then $f \circ h$ is absolutely summable and $\sum_{y \in Y} f(h(y)) = \sum_{x \in X} f(x)$.
In particular, a countable absolutely convergent sum is independent of the enumeration.

Proof. Let $g : \mathbb{N} \to X$ and $k : \mathbb{N} \to Y$ be enumerations, and set $\sigma := g^{-1} \circ h \circ k : \mathbb{N} \to \mathbb{N}$.
Then σ is a bijection and $h(k_n) = g_{\sigma(n)}$. Hence $\sum_{n=1}^{\infty} |f(h(k_n))| = \sum_{n=1}^{\infty} \left|f(g_{\sigma(n)})\right| < \infty$.
Thus $\sum_{y \in Y} f(h(y)) = \sum_{n=1}^{\infty} f(g_{\sigma(n)}) = \sum_{n=1}^{\infty} f(g_n) = \sum_{x \in X} f(x)$, where the middle equality is Theorem 6.2.13.
Taking $Y = \mathbb{N}$ and h to be any enumeration of X gives enumeration independence.
Structural Commentary: Absolute convergence turns an enumeration into harmless bookkeeping.
Key Insight: Once absolute tails are controlled, the order of a countable listing no longer changes the sum. □

Theorem 6.2.16. *Fubini for Absolutely Summable Arrays*
Let $a_{mn} \in \mathbb{C}$ and suppose $L := \sum_{(m,n) \in \mathbb{N} \times \mathbb{N}} |a_{mn}| < \infty$. Then each row and column is absolutely summable.
With $r_m := \sum_{n=1}^{\infty} a_{mn}$ and $c_n := \sum_{m=1}^{\infty} a_{mn}$, the row-total and column-total series are absolutely convergent,
where $\sum_{m=1}^{\infty} |r_m| < \infty$; $\sum_{n=1}^{\infty} |c_n| < \infty$, and $\sum_{m=1}^{\infty} r_m = \sum_{(m,n) \in \mathbb{N} \times \mathbb{N}} a_{mn} = \sum_{n=1}^{\infty} c_n$.

Proof. For finite $F \subseteq \mathbb{N} \times \mathbb{N}$, $\sum_{(m,n) \in F} |a_{mn}| \le L$. Hence each row and column has bounded absolute partial sums, so rows and columns converge absolutely by Theorem 6.1.6.
Moreover, for $M \ge 1$, $\sum_{m=1}^{M} |r_m| \le \sum_{m=1}^{M} \sum_{n=1}^{\infty} |a_{mn}| = \lim_{N \to \infty} \sum_{m=1}^{M} \sum_{n=1}^{N} |a_{mn}| \le L$, so $\sum_{m} |r_m| < \infty$.
Similarly $\sum_{n} |c_n| < \infty$. Let $\varepsilon > 0$. Choose finite $F \subseteq \mathbb{N} \times \mathbb{N}$ with $\sum_{(m,n) \in (\mathbb{N} \times \mathbb{N}) \setminus F} |a_{mn}| < \varepsilon$, and choose a rectangle $R = \{1, \dots, M\} \times \{1, \dots, N\} \supseteq F$. The full sum, row-iterated sum, and column-iterated sum each differ from $\sum_{(m,n) \in R} a_{mn}$ by at most $\sum_{(m,n) \in (\mathbb{N} \times \mathbb{N}) \setminus R} |a_{mn}| \le \varepsilon$. Let $\varepsilon \downarrow 0$.
Structural Commentary: Finite rectangles plus absolute tails justify Fubini.
Key Insight: Double sums can be rearranged only after the entire array is absolutely summable. □

Method 6.2.17. *Series Proof Patterns*

(i) *Partial sums*: translate every assertion about $\sum a_n$ into an assertion about $s_n = \sum_{k=1}^{n} a_k$.
(ii) *Cauchy tails*: prove convergence by estimating $|\sum_{k=n}^{m} a_k|$ for $m \ge n$.
(iii) *Nonnegative terms*: replace convergence by boundedness of partial sums.
(iv) *Comparison*: dominate only the tail; finitely many early terms are irrelevant.
(v) *Absolute convergence*: use $\sum |a_n|$ when cancellation should not matter.
(vi) *Countable index sets*: choose a bijection $g : \mathbb{N} \to X$; use Theorem 6.2.15 so $\sum_{x \in X} a_x$ is independent of g.
(vii) *Double sums*: prove identities on $R_{M,N} := \{1, \dots, M\} \times \{1, \dots, N\}$; pass to ∞ arrays (Theorem 6.2.16).
(viii) *Conditional convergence*: order matters; use summation by parts or rearrangements.

Key Insight: Series arguments are sequence arguments with special notation for tails.

7 Uniform Convergence

7.1 Uniform Limits

Uniform convergence asks when limits of functions preserve structure. The key distinction is whether N may depend on x, which controls continuity, integration, differentiation, and the supremum metric.

Perspective 7.1.1. *Function Limits Before Function-Sequence Limits*
$\lim_{t\to x} f(t)$ denotes the metric-space function limit of Definition 5.1.4.
Uniform convergence is the hypothesis that lets the limits in n and t commute.

Definition 7.1.2. *Pointwise Limits of Functions*
Let $f_n : E \to (Y, d_Y)$.

(i) $f_n \to f$ *pointwise* on E if $d_Y(f_n(x), f(x)) \to 0$ for every $x \in E$.

(ii) If Y is scalar, $\sum f_n \to f$ *pointwise* on E if $s_N(x) := \sum_{n=1}^{N} f_n(x) \to f(x)$ for every $x \in E$.

No metric on E is required; topology matters only for preserving continuity, limits, or integrals.

Example 7.1.3. *Pointwise Limits Can Lose Structure*

(i) $s_{m,n} := m/(m+n)$ gives $\lim_n \lim_m s_{m,n} = 1$ but $\lim_m \lim_n s_{m,n} = 0$.
(ii) For $f_n(x) = x^2/(1+x^2)^n$, $\sum_{n=0}^{\infty} f_n$ equals 0 at 0 and $1 + x^2$ otherwise, by Theorem 6.1.8.
(iii) $f_n(x) = \sin(nx)/\sqrt{n}$ satisfies $f_n \to 0$ pointwise but $f_n'(0) = \sqrt{n} \to +\infty$.
(iv) $f_n(x) = n^2 x(1-x^2)^n$ on $[0,1]$ satisfies $f_n \to 0$ pointwise, while $\int_0^1 f_n\,dx = n^2/(2n+2) \to +\infty$.

Key Insight: Pointwise control can miss global spikes and oscillations.

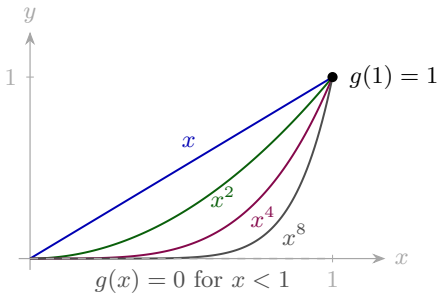


Figure 179: $x^n \to 0$ pointwise on $[0,1)$, but the endpoint prevents uniform convergence on $[0,1]$.

Example 7.1.4. *x^n Converges Pointwise But Not Uniformly*
On $[0,1]$, let $g_n(x) = x^n$. Then $g_n \to g$ pointwise, where $g(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$.
The convergence is not uniform: for each n, some $x < 1$ has $|x^n - g(x)| = x^n > 1/2$.

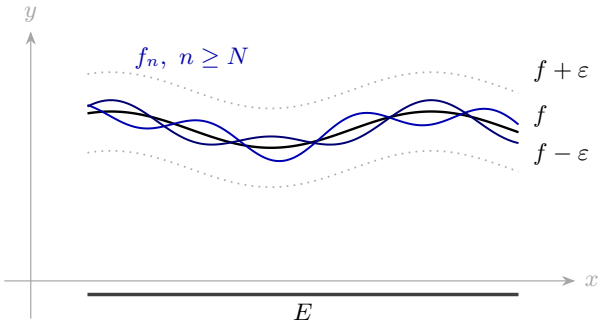


Figure 180: One index N puts all later graphs inside the same ε-strip around f on E.

Definition 7.1.5. *Uniform Convergence*
For $f_n, f : E \to (Y, d_Y)$, $f_n \rightrightarrows f$ on E if $\forall \varepsilon > 0\ \exists\ N \in \mathbb{N}$ such that $n \ge N,\ x \in E \implies d_Y(f_n(x), f(x)) \le \varepsilon$.
In the scalar-valued case, take $d_Y(u,v) = |u - v|$; $\sum f_n$ converges uniformly on E if its partial sums do.

Perspective 7.1.6. *One Tail, All Points*
Pointwise convergence lets N depend on x; uniform convergence uses one N for all x.
Equivalently, late graphs lie in one ε-strip around f.

Theorem 7.1.7. *Cauchy Criterion for Uniform Convergence*
For complex-valued $f_n : E \to \mathbb{C}$, f_n converges uniformly on E if and only if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} : \quad m, n \ge N,\ x \in E \implies |f_n(x) - f_m(x)| \le \varepsilon$$

Proof.

($\Rightarrow$) If $f_n \rightrightarrows f$, choose N so that $|f_k(x) - f(x)| \le \varepsilon/2$ for $k \ge N$ and $x \in E$. Then $|f_n(x) - f_m(x)| \le \varepsilon$.
($\Leftarrow$) For each x, $(f_n(x))$ is Cauchy in $\mathbb{C}$, so define $f(x) := \lim f_n(x)$.
Fix $n \ge N$ and let $m \to \infty$ in the criterion; then $|f_n(x) - f(x)| \le \varepsilon$ for all $x \in E$, hence $f_n \rightrightarrows f$.

Structural Commentary: The same N gives Cauchy control simultaneously for every $x \in E$.
Key Insight: Uniform Cauchy tails are exactly uniform convergence. □

Theorem 7.1.8. *Supremum Test for Uniform Convergence*
Suppose $f_n(x) \to f(x)$ $\forall x \in E$, and $M_n := \sup_{x\in E} |f_n(x) - f(x)|$. Then $f_n \rightrightarrows f$ on E if and only if $M_n \to 0$.

Proof.

($\Rightarrow$) If $f_n \rightrightarrows f$, then $\forall \varepsilon > 0$ $\exists$ N such that $n \ge N$ implies $|f_n(x) - f(x)| \le \varepsilon$ for all $x \in E$. Hence $M_n \le \varepsilon$.
($\Leftarrow$) If $M_n \to 0$, then $\forall \varepsilon > 0$ $\exists$ N such that $n \ge N$ implies $M_n \le \varepsilon$. Hence $|f_n(x) - f(x)| \le \varepsilon$ for all $x \in E$.

Structural Commentary: Uniform convergence is convergence in the supremum error.
Key Insight: The pointwise errors must vanish as one global quantity. □

Theorem 7.1.9. *Weierstrass M-Test*
Let f_n be functions on E and suppose $|f_n(x)| \le M_n$ for every $x \in E$.
If $\sum M_n$ converges, then $\sum f_n$ converges uniformly on E.

Proof. For $m \ge n$ and $x \in E$, $\left|\sum_{k=n}^{m} f_k(x)\right| \le \sum_{k=n}^{m} M_k$.
Since $\sum M_k$ converges, Theorem 6.1.4 makes the right-hand side $< \varepsilon$ for all large m, n.
Thus the partial sums of $\sum f_k$ are uniformly Cauchy, so Theorem 7.1.7 applies.
Structural Commentary: Absolute numerical tails dominate functional tails.
Key Insight: A summable uniform majorant gives uniform convergence of a function series. □

Theorem 7.1.10. *Uniform Convergence Interchanges Limits*
Let E lie in a metric space, let x be a limit point of E, and suppose $f_n \rightrightarrows f$ on E.
If $A_n := \lim_{t\to x} f_n(t)$ exists for every n, then (A_n) converges and $\lim_{t\to x} f(t) = \lim_{n\to\infty} A_n$.

Proof. By Theorem 7.1.7, $\forall \varepsilon > 0$ $\exists$ N such that $m, n \ge N$ and $t \in E$ imply $|f_n(t) - f_m(t)| \le \varepsilon$.
Letting $t \to x$ gives $|A_n - A_m| \le \varepsilon$, hence $A_n \to A$ for some $A \in \mathbb{C}$.
For any n, $|f(t) - A| \le |f(t) - f_n(t)| + |f_n(t) - A_n| + |A_n - A|$. Given $\varepsilon > 0$, choose n so $|f(t) - f_n(t)| < \varepsilon/3$ for all $t \in E$ and $|A_n - A| < \varepsilon/3$; then take t sufficiently close to x so $|f_n(t) - A_n| < \varepsilon/3$. Thus $\lim_{t\to x} f(t) = A$.
Structural Commentary: Uniform convergence controls the outer two terms in the three-term estimate.
Key Insight: Uniformity is precisely what allows the $n \to \infty$ limit to move past the $t \to x$ limit. □

Theorem 7.1.11. *Uniform Limits Preserve Continuity*
If (f_n) is a sequence of continuous functions on E and $f_n \rightrightarrows f$ on E, then f is continuous on E.

Proof. Let $x \in E$. If x is isolated, f is continuous at x.
If x is a limit point, then $\lim_{t\to x} f_n(t) = f_n(x)$ for all n, so Theorem 7.1.10 gives $\lim_{t\to x} f(t) = \lim_{n\to\infty} f_n(x) = f(x)$.
Structural Commentary: Uniform convergence passes continuity to the limit.
Key Insight: Uniform convergence protects continuity from the failure modes in Example 7.1.3. □

Theorem 7.1.12. *Dini's Monotone Uniform Convergence Theorem*
Let K be compact and let $f_n, f : K \to \mathbb{R}$ be continuous. If $f_n(x) \downarrow f(x)$ for every $x \in K$, then $f_n \rightrightarrows f$ on K.

Proof. Set $g_n := f_n - f$. Then g_n is continuous, $g_n \downarrow 0$, and $g_n \ge 0$. Given $\varepsilon > 0$, let $K_n := \{x \in K : g_n(x) \ge \varepsilon\}$.
Then (K_n) is nested compact and $\bigcap_n K_n = \emptyset$, so Corollary 4.1.33 gives $K_N = \emptyset$ for some N.
Thus $0 \le f_n(x) - f(x) < \varepsilon$ for all $x \in K$ and $n \ge N$.
Structural Commentary: Monotone convergence turns the bad sets into a nested compact sequence.
Key Insight: Compactness turns pointwise eventual escape from every bad set into a uniform escape. □

Definition 7.1.13. *The Supremum Metric on Bounded Continuous Functions*
For a metric space X, let $C_b(X)$ be the bounded complex-valued continuous functions and define $\|f\|_\infty := \sup_{x\in X} |f(x)|$; $d(f, g) := \|f - g\|_\infty$.
Then d is a metric on $C_b(X)$, induced by the norm of Definition 1.1.16; d-convergence is uniform convergence by Theorem 7.1.8. If X is compact, then $C(X) = C_b(X)$ by Corollary 5.1.34.

Perspective 7.1.14. *A Metric Space Whose Points Are Functions*
Uniform convergence is sequence convergence in the metric space $C_b(X)$.

Perspective 7.1.15. *Four Limit Notions*
The four limit notions are sequence limits, function limits, pointwise function limits, and uniform function limits. By Theorem 7.1.8, the last is ordinary sequence convergence in the supremum metric.

7.2 Interchange of Limits

Theorem 7.2.1. *Uniform Convergence and Riemann–Stieltjes Integration*
Let α be increasing on $[a,b]$. If $f_n \in \mathcal{R}(\alpha)$ on $[a,b]$ and $f_n \rightrightarrows f$, then $f \in \mathcal{R}(\alpha)$ and $\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f_n\,d\alpha$.

Proof. Put $\varepsilon_n := \sup_{a\le x\le b} |f_n(x) - f(x)|$. Then $\varepsilon_n \to 0$ by Theorem 7.1.8, and $f_n - \varepsilon_n \le f \le f_n + \varepsilon_n$.
Upper and lower integrals of f are therefore squeezed between corresponding integrals of $f_n \pm \varepsilon_n$, and gap is at most $2\varepsilon_n[\alpha(b) - \alpha(a)]$. Thus $f \in \mathcal{R}(\alpha)$. Same estimate gives $\left|\int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha\right| \le \varepsilon_n[\alpha(b) - \alpha(a)] \to 0$.
Structural Commentary: Uniform error becomes an integral error multiplied by the total α-variation.
Key Insight: Uniform convergence is strong enough to pass limits through the Riemann–Stieltjes integral. □

Corollary 7.2.2. *Termwise Integration of Uniformly Convergent Series*
If $f_n \in \mathcal{R}(\alpha)$ on $[a,b]$ and $\sum f_n$ converges uniformly to f, then $\int_a^b f\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$.

Proof. Let $s_N := \sum_{n=1}^{N} f_n$. By linearity, $s_N \in \mathcal{R}(\alpha)$ and $\int_a^b s_N\,d\alpha = \sum_{n=1}^{N} \int_a^b f_n\,d\alpha$.
Since $s_N \rightrightarrows f$, Theorem 7.2.1 gives $\int_a^b f\,d\alpha = \lim_{N\to\infty} \int_a^b s_N\,d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n\,d\alpha$.
Structural Commentary: Reduce the series to its uniformly convergent partial sums.
Key Insight: Uniform convergence lets integration be performed term by term. □

Theorem 7.2.3. *Uniform Convergence and Differentiation*
Let f_n be differentiable on $[a,b]$, suppose $f_n(x_0)$ converges for some $x_0 \in [a,b]$, and suppose $f_n' \rightrightarrows g$ on $[a,b]$. Then $f_n \rightrightarrows f$ for some f, and $f' = g$ on $[a,b]$ (one-sided at endpoints).

Proof. For m, n, MVT 1.1.13 gives $|f_n(x) - f_m(x)| \le |f_n(x_0) - f_m(x_0)| + (b-a) \sup_{t\in[a,b]} |f_n'(t) - f_m'(t)|$.
Thus (f_n) is uniformly Cauchy, so $f_n \rightrightarrows f$. Fix x and set $\phi_n(t) := \frac{f_n(t)-f_n(x)}{t-x}$, $\phi(t) := \frac{f(t)-f(x)}{t-x}$.
For $t \ne x$, the same estimate gives $|\phi_n(t) - \phi_m(t)| \le \|f_n' - f_m'\|_\infty$, hence $\phi_n \rightrightarrows \phi$.
By Theorem 7.1.10, $f'(x) = \lim_{t\to x} \phi(t) = \lim_{n\to\infty} f_n'(x) = g(x)$.
Structural Commentary: Derivative control anchors functions, then quotients.
Key Insight: Uniform derivative convergence plus one anchor passes derivatives to the limit. □

Method 7.2.4. *Uniform Convergence Proof Patterns*

(i) *Pointwise*: fix x, then choose $N = N(x,\varepsilon)$.
(ii) *Uniform*: choose $N = N(\varepsilon)$; equivalently $\sup_E |f_n - f| \to 0$.
(iii) *Uniform Cauchy*: prove $\sup_E |f_n - f_m| \to 0$ as $m, n \to \infty$.
(iv) *Preservation*: use a three-term estimate and choose the uniform index first.
(v) *Series of functions*: apply Weierstrass M-test when numerical absolute tails dominate all functional tails.
(vi) *Termwise operations*: require $f_n \to f$ locally uniformly, i.e. $\sup_{x\in K} |f_n - f| \to 0$ for each compact K, before $\lim_n \int_K f_n = \int_K f$; for derivatives, require $f_n' \to g$ locally uniformly and get $(\lim_n f_n)' = g$.

Key Insight: Uniformity means N is chosen before x.

8 Function-Space Compactness

8.1 Equicontinuity

Function-space compactness treats functions as points. Supremum metrics preview Banach spaces, Baire category describes generic behavior, and equicontinuity leads to Arzela–Ascoli.

Theorem 8.1.1. *Completeness of the Supremum Metric*
The metric space $C_b(X)$ with metric $d(f,g) = \|f - g\|_\infty$ is complete.

Proof. Let (f_n) be Cauchy in $C_b(X)$. Then Theorem 7.1.7 gives a function f with $f_n \rightrightarrows f$. By Theorem 7.1.11, f is continuous. For some n, $\|f - f_n\|_\infty < 1$, hence f_n is bounded, thus $f \in C_b(X)$ and $\|f_n - f\|_\infty \to 0$.
Structural Commentary: Uniform Cauchy convergence builds the limit inside the same function space.
Key Insight: Completeness of $\mathbb{C}$ plus uniform convergence makes $C_b(X)$ complete. □

Definition 8.1.2. *Bounded Target Metric for All Functions*
For metric spaces (X, d_X) and (Y, d_Y), let $\mathcal{F} = Y^X$ and set $\rho(y,z) := \frac{d_Y(y,z)}{1+d_Y(y,z)}$ and $d_\rho(f,g) := \sup_{x\in X} \rho(f(x), g(x))$. Then $\rho < 1$, so d_ρ is finite on all of $\mathcal{F}$; d_ρ-convergence is uniform convergence measured by d_Y.

Theorem 8.1.3. *Completeness and Closed Subspaces for d_ρ*
If the target metric space (Y, d_Y) is complete, then:

(i) $(\mathcal{F}, d_\rho)$ *is complete;*
(ii) $\mathcal{F}_b := \{f \in \mathcal{F} : f(X)$ *is* d_Y*-bounded*$\}$ *is closed in* $\mathcal{F}$*;*
(iii) $C(X,Y)$ *is closed in* $\mathcal{F}$*.*

Proof.

(i) Let (f_n) be d_ρ-Cauchy. For each $x \in X$, $(f_n(x))$ is Cauchy in Y; define $f(x) := \lim_n f_n(x)$.
Given $\varepsilon > 0$, choose N with $d_\rho(f_n, f_m) < \varepsilon$ for $m, n \geq N$. Letting $m \to \infty$ gives $d_\rho(f_n, f) \leq \varepsilon$ for $n \geq N$.

(ii) If $f_n \in \mathcal{F}_b$ and $d_\rho(f_n, f) \to 0$, choose N with $d_\rho(f_N, f) < 1/2$.
Then $d_Y(f_N(x), f(x)) < 1$ for all x. Since $f_N(X)$ is bounded, so is $f(X)$.

(iii) If $f_n \in C(X,Y)$ and $d_\rho(f_n, f) \to 0$, then $f_n \to f$ uniformly in d_Y.
Given $x_0 \in X$ and $\varepsilon > 0$, choose N with $d_Y(f_N(x), f(x)) < \varepsilon/3$ for all x, then use continuity of f_N at x_0 and the triangle inequality. Thus f is continuous.

Structural Commentary: Compress the target metric; then close the useful subspaces.
Key Insight: Completeness and uniform control make boundedness and continuity closed. □

Theorem 8.1.4. *A Continuous Nowhere Differentiable Function Exists*
There exists a real continuous function on $\mathbb{R}$ which is nowhere differentiable.

Proof. Let φ be the 2-periodic extension of $|x|$ on $[-1,1]$. Then $0 \leq \varphi \leq 1$ and $|\varphi(s) - \varphi(t)| \leq |s-t|$.
Define $f(x) := \sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n \varphi(4^n x)$, converges uniformly (Theorem 7.1.9); hence f is continuous (Theorem 7.1.11).
Fix x and $m \in \mathbb{N}$. Choose $\delta_m = \pm\frac{1}{2}4^{-m}$ so that no integer lies between $4^m x$ and $4^m(x+\delta_m)$.
Set $\gamma_{n,m} := \frac{\varphi(4^n(x+\delta_m)) - \varphi(4^n x)}{\delta_m}$. If $n > m$, then $4^n \delta_m$ is an even integer, so $\gamma_{n,m} = 0$.
If $0 \leq n \leq m$, then $|\gamma_{n,m}| \leq 4^n$, and the choice of δ_m gives $|\gamma_{m,m}| = 4^m$.
Hence $\left|\frac{f(x+\delta_m) - f(x)}{\delta_m}\right| = \left|\sum_{n=0}^{m} \left(\frac{3}{4}\right)^n \gamma_{n,m}\right| \geq 3^m - \sum_{n=0}^{m-1} 3^n = \frac{1}{2}(3^m + 1)$
Since $\delta_m \to 0$, the difference quotients are unbounded near x; since x was arbitrary, f is nowhere differentiable.
Structural Commentary: Uniform sums give continuity; scaled cusps force unbounded quotients.
Key Insight: Uniform convergence preserves continuity, not differentiability. □

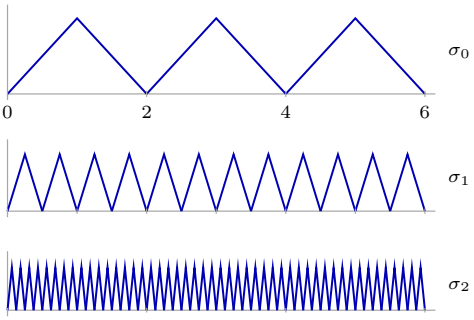


Figure 181: The first compressed sawtooth terms $\sigma_k(x) = (3/4)^k \varphi(4^k x)$: heights shrink, slopes grow.

Perspective 8.1.5. *Explicit Versus Generic Pathology*
The sawtooth series proves existence; Baire category shows nowhere differentiability is residual in $C([a,b], \mathbb{R})$.

Theorem 8.1.6. *Generic Nowhere Differentiability on Compact Intervals*
Let $a < b$, and let $C = C([a,b], \mathbb{R})$ with the supremum metric. There exists a residual set $\mathcal{G} \subseteq C$ such that:

(i) no $f \in \mathcal{G}$ has an available one-sided derivative at any point of $[a,b]$;
(ii) no $f \in \mathcal{G}$ is monotone on any nondegenerate subinterval of $[a,b]$;
(iii) the set $\{f \in C : f$ is differentiable at some point of $[a,b]\}$ is first category.

Proof.

(i) Write $\Delta_h f(x) := f(x+h) - f(x)$, and choose N_0 with $1/N_0 < b - a$. For $n \geq N_0$, define

$$R_n := \left\{ f \in C : \forall x \in [a, b - 1/n] \ \exists h > 0 \text{ with } \left| \frac{\Delta_h f(x)}{h} \right| > n \right\},$$

$$L_n := \left\{ f \in C : \forall x \in [a + 1/n, b] \ \exists h < 0 \text{ with } \left| \frac{\Delta_h f(x)}{h} \right| > n \right\},$$

$$G_n := \{ f \in C : f \text{ is nonmonotone on every subinterval of length } 1/n \}.$$

Each R_n, L_n, G_n is open and dense in C (Weierstrass Approximation Theorem 9.2.1), plus small high-slope sawtooth perturbation; openness follows from compactness and uniform perturbation estimates.

(ii) Set $\mathcal{G} := \bigcap_{n \geq N_0} (R_n \cap L_n \cap G_n)$. By Definition 4.3.3, $\mathcal{G}$ is residual. If $f \in \mathcal{G}$, then the R_n and L_n conditions give one-sided difference quotients of arbitrarily large absolute value; since f is bounded, the corresponding increments tend to 0. Thus no available one-sided derivative exists.

(iii) Let $f \in \mathcal{G}$. Every nondegenerate subinterval contains a subinterval of length $1/n$ for all sufficiently large n, so the G_n conditions make f nonmonotone on every nondegenerate subinterval. If f is differentiable somewhere, then $f \notin \mathcal{G}$; hence the differentiable-somewhere set lies in $C \setminus \mathcal{G}$, which is first category.

Structural Commentary: Baire category turns dense sawtooth perturbability into a generic regularity failure.
Key Insight: The explicit construction shows existence; the category proof shows typicality. □

Definition 8.1.7. *Boundedness and Equicontinuity of Function Families*
Let $\mathcal{F}$ be a family of functions from E into a metric space (Y, d_Y); in scalar cases, take $d_Y(u, v) = |u - v|$.

(i) (f_n) is *pointwise bounded* on E if $(f_n(x))$ is bounded in Y for each $x \in E$.

(ii) A scalar sequence (f_n) is *uniformly bounded* on E if $\exists M$ such that $|f_n(x)| < M$ for all $x \in E, n \in \mathbb{N}$.

(iii) If E lies in a metric space (X, d_X), then $\mathcal{F}$ is *equicontinuous* on E if $\forall \varepsilon > 0 \ \exists \delta > 0$, such that $d_X(x, y) < \delta, \ x, y \in E, \ f \in \mathcal{F} \implies d_Y(f(x), f(y)) < \varepsilon$.

(iv) The family is *pointwise equicontinuous* if for every $p \in E$ and $\varepsilon > 0$ there is $\delta = \delta(p, \varepsilon) > 0$ such that $d_X(p, y) < \delta, \ y \in E, \ f \in \mathcal{F}$ implies $d_Y(f(p), f(y)) < \varepsilon$.

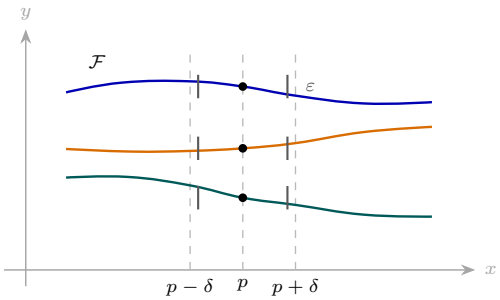


Figure 182: One input scale δ controls every curve in $\mathcal{F}$ within tolerance ε.

Theorem 8.1.8. *Pointwise Equicontinuity Becomes Uniform on Compact Sets*
Every pointwise equicontinuous family $\mathcal{F} \subset C(K, \mathbb{C})$ on compact K is equicontinuous.

Proof. Otherwise, choose $\varepsilon > 0$, $f_n \in \mathcal{F}$, and $x_n, t_n \in K$ with $d(x_n, t_n) < 1/n$ but $|f_n(x_n) - f_n(t_n)| \geq \varepsilon$.
By Theorem 4.1.3, pass to $x_n \to x_0$; then $t_n \to x_0$ too.
Pointwise equicontinuity at x_0 gives $|f_n(x_n) - f_n(t_n)| < \varepsilon$ for large n, a contradiction.
Structural Commentary: Compactness turns moving counterexamples into one base point.
Key Insight: The local $\delta(x, \varepsilon)$ becomes one uniform δ only after compactness prevents escape. □

Example 8.1.9. *Uniform Boundedness Alone Is Not Enough*
On $[0, 1]$, let $f_n(x) := \frac{x^2}{x^2 + (1 - nx)^2}$. Then $|f_n| \leq 1$ and $f_n \to 0$ pointwise, but $f_n(1/n) = 1$; hence no uniformly convergent subsequence. The missing hypothesis is equicontinuity.

Theorem 8.1.10. *Diagonal Selection on a Countable Set*
Every pointwise bounded sequence (f_n) on countable E has a subsequence converging pointwise on E.

Proof. List $E = \{x_1, x_2, \ldots\}$. Inductively choose nested subsequences S_i so that $S_i(x_i)$ converges.
The diagonal subsequence is eventually contained in each S_i, hence converges at every x_i.
Structural Commentary: The diagonal keeps the ith convergence property after the ith step.
Key Insight: Countability permits diagonal convergence. □

Theorem 8.1.11. *Uniform Convergence Implies Equicontinuity on Compact Sets*
If K is compact, $f_n \in C_b(K)$, and $f_n \rightrightarrows f$ on K, then $\{f_n : n \in \mathbb{N}\}$ is equicontinuous.

Proof. Choose N so $\|f_n - f_N\|_\infty < \varepsilon$ for $n > N$.
By Theorem 5.2.3, choose one $\delta > 0$ giving ε-control for $f_1, \dots, f_N$.
If $n > N$, $d(x, y) < \delta$, then $|f_n(x) - f_n(y)| \le |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < 3\varepsilon$.
Together with the finite initial family, $\{f_n\}$ is equicontinuous.
Structural Commentary: A uniformly convergent tail stays close to one uniformly continuous function.
Key Insight: Compactness handles finitely many functions; uniform convergence handles the tail. □

8.2 Arzela-Ascoli-Type Compactness

Theorem 8.2.1. *Arzela–Ascoli Selection Theorem*
Let K be compact. If $(f_n) \subset C_b(K)$ is pointwise bounded and equicontinuous on K, then:

(i) (f_n) is uniformly bounded on K;
(ii) (f_n) contains a uniformly convergent subsequence.

Proof.

(i) Choose $\delta > 0$ from equicontinuity with tolerance 1. Compactness gives $K \subseteq \bigcup_{j=1}^{r} B_\delta(p_j)$.
Pointwise boundedness gives M_j such that $|f_n(p_j)| \le M_j \forall n$.
If $x \in B_\delta(p_j)$, then $|f_n(x)| \le |f_n(x) - f_n(p_j)| + |f_n(p_j)| < 1 + M_j$, seq uniformly bounded by $1 + \max_j M_j$.

(ii) For each m, compactness gives a finite $1/m$-net in K; the union of these finite nets is a countable dense set $E \subset K$. By Theorem 8.1.10, some subsequence (g_i) converges pointwise on E.
Given $\varepsilon > 0$, choose $\delta > 0$ from equicontinuity with tolerance $\varepsilon/3$, and points $x_1, \dots, x_r \in E$ with $K \subseteq \bigcup_s B_\delta(x_s)$. As $(g_i(x_s))$ converges for each of finitely many x_s, $\exists N$ s.t. $i, j \ge N$ implies $|g_i(x_s) - g_j(x_s)| < \varepsilon/3 \ \forall s$. If $d(x, x_s) < \delta$, then $|g_i(x) - g_j(x)| \le |g_i(x) - g_i(x_s)| + |g_i(x_s) - g_j(x_s)| + |g_j(x_s) - g_j(x)| < \varepsilon$.
Thus (g_i) is uniformly Cauchy and converges uniformly by Theorem 7.1.7.

Structural Commentary: Dense convergence plus equicontinuity becomes uniform convergence.
Key Insight: Equicontinuity is exactly the compactness-compatible hypothesis missing from Example 8.1.9. □

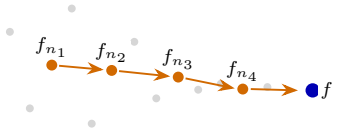


Figure 183: Arzela–Ascoli selects a uniformly convergent subsequence.

Definition 8.2.2. *σ-Compact Metric Space*
A metric space X is *σ-compact* if $X = \bigcup_{j=1}^{\infty} K_j$ for compact subsets $K_j \subseteq X$.

Theorem 8.2.3. *Arzela–Ascoli on σ-Compact Spaces*
Let X be σ-compact, Y a metric space, and (f_n) pointwise equicontinuous maps $X \to Y$. If each orbit $\{f_n(x) : n \in \mathbb{N}\}$ lies in a compact subset of Y, then some subsequence converges uniformly on compact subsets of X.

Proof. Choose an increasing compact exhaustion $X = \bigcup_j K_j$.
By Theorem 8.1.8, each restriction $f_n|_{K_j}$ is uniformly equicontinuous. Applying Theorem 8.2.1 on $K_1, K_2, \dots$ and diagonalizing gives a subsequence (g_m) converging uniformly on every K_j, hence pointwise on X.
If $K \subseteq X$ is compact, then $(g_m|_K)$ is uniformly equicontinuous, and a finite-net argument upgrades pointwise convergence on K to uniform convergence on K.
Structural Commentary: Exhaust X by compact pieces and diagonalize the compact theorem.
Key Insight: Local uniform convergence replaces global uniform convergence. □

Corollary 8.2.4. *Real-Line Local Uniform Subsequence*
If (f_n) is pointwise equicontinuous on $\mathbb{R}$ and bounded at one point, then it has a subsequence converging uniformly on compact subsets of $\mathbb{R}$.

Proof. Let $(f_n(x_0))$ be bounded. On any compact interval containing x_0, Theorem 8.1.8 gives uniform equicontinuity; finitely many δ-steps from x_0 to x bound each orbit $(f_n(x))$. Hence Theorem 8.2.3 applies on $\mathbb{R}$.
Structural Commentary: One bounded anchor propagates across compact intervals.
Key Insight: Pointwise equicontinuity turns local boundedness into compactwise compactness. □

Method 8.2.5. *Function-Space Compactness Proof Patterns*

(i) *Completeness*: prove a function sequence is uniformly Cauchy, build the pointwise limit, and use uniform convergence to keep continuity or boundedness.
(ii) *Baire category*: show a desired property contains a countable intersection of dense open subsets of a complete function space.
(iii) *Compactness/equicontinuity*: pass from finite nets to all of K.
(iv) *Diagonal selection*: get pointwise convergence on a countable dense set, then upgrade using equicontinuity.
(v) *Locally uniform*: exhaust a σ-compact domain by compact sets and diagonalize.

Key Insight: Compactness in function spaces is finite-net control plus equicontinuity.

9 Power Series and Approximation

9.1 Power Series

Power series are controlled infinite polynomials, and approximation is uniform closure. This subsection covers radii, Taylor structure, Abel behavior, Weierstrass, Stone–Weierstrass, and elementary analytic functions.

Motivation 9.1.1. *Power Series as Controlled Infinite Polynomials*
Power-series manipulations rest on uniform convergence: Definition 6.1.1, Theorems 9.1.4, 7.1.9, and 7.2.3.

Definition 9.1.2. *Core Objects: Power Series, Approximation, and Gamma*

(i) A real *power series about a* is $\sum_{n=0}^{\infty} c_n(x-a)^n$; a local representation of f by one makes f *analytic* at a.
(ii) A *polynomial approximant* to f on a compact set K is a polynomial P chosen so that $\|f-P\|_{\infty,K}$ is small.
(iii) A function algebra is a collection closed under addition, multiplication, and scalar multiplication; Stone–Weierstrass asks when such algebras are uniformly dense in $C(K)$.
(iv) For $x>0$, the *gamma function* is $\Gamma(x) := \int_0^\infty t^{x-1}e^{-t}\,dt$ when the improper integral converges.

Key Insight: Power series are local analytic objects; approximation theorems turn algebraic families into dense function families; Γ extends factorials by integration.

Definition 9.1.3. *Power Series and Radius of Convergence*
Given a complex sequence (c_n), the series $\sum_{n=0}^{\infty} c_n z^n$ is a *power series*. If $\alpha := \limsup_{n\to\infty} \sqrt[n]{|c_n|}$, define $R := 1/\alpha$, with $R=+\infty$ when $\alpha=0$ and $R=0$ when $\alpha=+\infty$. The number R is the *radius of convergence*.

Theorem 9.1.4. *Convergence Inside the Circle*
For $\sum c_n z^n$, radius R: converge if $|z|<R$, diverge if $|z|>R$; boundary $|z|=R$ is undecided.

Proof. For $a_n = c_n z^n$, $\limsup_{n\to\infty} \sqrt[n]{|a_n|} = |z| \limsup_{n\to\infty} \sqrt[n]{|c_n|} = \frac{|z|}{R}$, with extended meaning if $R=0$ or $R=+\infty$. If $|z|<R$, this value is <1; if $|z|>R$, it is >1. Apply Theorem 6.1.19.
Structural Commentary: Radius controls the interior and exterior, not the boundary.
Key Insight: The root test is the radius-of-convergence theorem in disguise. □

Theorem 9.1.5. *A Boundary Test for Power Series*
If $\sum c_n z^n$ has radius 1, $c_n \downarrow 0$, and $c_n \geq 0$, then it converges for $|z|=1$, except possibly at $z=1$.

Proof. For $|z|=1$ and $z\neq 1$, put $a_n = z^n$ and $b_n = c_n$.
The partial sums of $\sum z^n$ are bounded because $\left|\sum_{m=0}^{n} z^m\right| = \left|\frac{1-z^{n+1}}{1-z}\right| \leq \frac{2}{|1-z|}$. Theorem 6.2.2 applies.
Structural Commentary: Boundary convergence away from $z=1$ comes from bounded geometric partial sums.
Key Insight: Monotone coefficients can make the unit-circle boundary converge without absolute convergence. □

Perspective 9.1.6. *Centered Series Are Translated Series*
$\sum c_n(x-a)^n$ is $\sum c_n y^n$ with $y = x-a$. Its radius is R from Definition 9.1.3: absolute convergence for $|x-a|<R$, divergence for $|x-a|>R$, and separate endpoint tests for $|x-a|=R$.

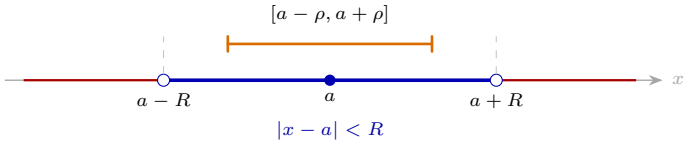


Figure 184: Inside $|x-a|<R$: absolute and locally uniform; outside divergence; endpoints undecided.

Definition 9.1.7. *Real Analytic at a Point*
Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let a be an interior point of E. The function f is *real analytic at a* if there are $r > 0$ and coefficients (c_n) such that $(a-r, a+r) \subseteq E$, the power series $\sum c_n(x-a)^n$ has radius of convergence at least r, and $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ for all $x \in (a-r, a+r)$.
If E is open and f is real analytic at every point of E, then f is *real analytic on E*.

Theorem 9.1.8. *Power Series May Be Differentiated Term by Term*
Let $f(x) = \sum_{n\geq 0} c_n x^n$ have radius $R > 0$, and $I_R = (-R, R)$. For $0 < \rho < R$, let $K_\rho = [-\rho, \rho]$, a closed interval contained in I_R. Then $\sum c_n x^n$ converges uniformly on each K_ρ, and $f \in C^\infty(I_R)$.
For $k \geq 1$, $f^{(k)}(x) = \sum_{n\geq k} \frac{n!}{(n-k)!} c_n x^{n-k}$, $(x \in I_R)$, and $f^{(k)}(0) = k!c_k$.

Proof. For $0 < \rho < R$, Theorem 9.1.4 and the root-test argument give $\sum |c_n|\, \rho^n < \infty$; hence the Weierstrass M-test, Theorem 7.1.9, gives uniform convergence on K_ρ.
Since $\sqrt[n]{n} \to 1$, polynomial factors in n do not change the radius. Thus each differentiated series converges uniformly on K_ρ, and Theorem 7.2.3, iterated, gives the formula and $f^{(k)}(0) = k!c_k$.
Structural Commentary: On each K_ρ, termwise operations are justified.
Key Insight: The derivative coefficients grow only by polynomial factors, which do not change the radius. □

Corollary 9.1.9. *Termwise Calculus for Centered Power Series*
Let $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ have radius $R > 0$. On $(a-R, a+R)$ the series converges locally uniformly and defines a continuous function. For $|x-a| < R$, $f'(x) = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$ with local uniform convergence of the derivative series. If $[y, z] \subset (a-R, a+R)$, then $\int_y^z f(x)\,dx = \sum_{n=0}^{\infty} c_n \frac{(z-a)^{n+1} - (y-a)^{n+1}}{n+1}$.

Proof. Set $g(u) = \sum c_n u^n$ and $u = x - a$.
Theorem 9.1.8 applied to g on $(-R, R)$ gives convergence, continuity, derivative formula after translating.
For $[y, z] \subset (a-R, a+R)$, choose $\rho < R$ with $[y, z] \subset [a-\rho, a+\rho]$; then Corollary 7.2.2 lets the integrals of partial sums pass to the uniform limit.
Structural Commentary: All local termwise operations happen on closed intervals strictly inside the radius.
Key Insight: Translation reduces centered power series to the radius-R theorem about $\sum c_n u^n$. □

Corollary 9.1.10. *Analytic Functions Are Their Taylor Series*
If $f(x) = \sum c_n(x-a)^n$ near a, then f is C^∞, expansion f is unique, and $f(x) = f(a) + \sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$.

Proof. Corollary 9.1.9, applied repeatedly, gives $c_0 = f(a)$ and $c_k = f^{(k)}(a)/k!$.
Hence two expansions about a have the same coefficients.
Structural Commentary: Derivatives determine the local series.
Key Insight: Once a power-series representation exists, the coefficients cannot vary. □

Perspective 9.1.11. *Taylor Coefficients Are Only the First Half*
Power-series coefficients are forced by derivatives (Corollary 9.1.10); for a function not already known to be analytic, proving the Taylor series represents the function needs a remainder estimate (Theorem 1.1.26).

Theorem 9.1.12. *Abel's Endpoint Theorem*
Let $s_n := \sum_{k=0}^{n} c_k$ with $s_n \to s$, and let $f(x) = \sum_{n\geq 0} c_n x^n$ on $(-1, 1)$. Then $f(x) \to s$, for $x \to 1^-$.

Proof. With $s_{-1} = 0$, summation by parts, Theorem 6.2.1, gives for $|x| < 1$, $f(x) = (1-x)\sum_{n=0}^{\infty} s_n x^n$.
Since $(1-x)\sum_{n\geq 0} x^n = 1$, then $f(x) - s = (1-x)\sum_{n\geq 0}(s_n - s)x^n$. Choose N so that $|s_n - s| < \varepsilon/2$ for $n > N$.
The tail is at most $\varepsilon/2$, and the finite part tends to 0 after multiplication by $1-x$.
Structural Commentary: Endpoint convergence becomes one-sided continuity.
Key Insight: The factor $(1-x)$ averages the partial sums s_n with geometric weights. □

Theorem 9.1.13. *Abel Product Theorem for Cauchy Products*
Suppose $\sum a_n = A$, $\sum b_n = B$, $\sum c_n = C$, *and* $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. *Then* $C = AB$.

Proof. For $0 \le x < 1$, set $f(x) := \sum_{n\ge0} a_n x^n$; $g(x) := \sum_{n\ge0} b_n x^n$; and $h(x) := \sum_{n\ge0} c_n x^n$.
Since convergent series have bounded terms, f, g, h converge absolutely for $|x| < 1$. Thus the Cauchy product gives $fg = h$ on $[0, 1)$. By Theorem 9.1.12, $f \to A$, $g \to B$, and $h \to C$ as $x \to 1^-$. Hence $AB = C$.
Structural Commentary: Prove $fg = h$ inside, then pass to $x = 1$.
Key Insight: Inside $|x| < 1$, the Cauchy product is safe; Abel then moves the identity to $x = 1$. □

Theorem 9.1.14. *Re-Expansion and Identity for Power Series*
Let $f(x) = \sum_{n\ge0} c_n x^n$ *have radius* $R > 0$, *and set* $S = (-R, R)$.

(i) If $a \in S$, *then* $f(x) = \sum_{m\ge0} \frac{f^{(m)}(a)}{m!}(x-a)^m$, $(|x-a| < R - |a|)$

(ii) Let $F = \sum_{n\ge0} a_n x^n$ *and* $G = \sum_{n\ge0} b_n x^n$ *on* S.
If $\mathcal{Z}_{F,G} := \{x \in S : F(x) = G(x)\}$ *has a limit point in* S, *then* $a_n = b_n$ *for all* $n \ge 0$, *hence* $F = G$ *on* S.

Proof.

(i) Write $x^n = ((x-a)+a)^n$. If $|x-a| + |a| < R$, then $\sum_{n\ge0} |c_n| \left(|x-a| + |a|\right)^n < \infty$, so the resulting double sum may be rearranged. Theorem 9.1.8 gives the coefficients $f^{(m)}(a)/m!$.

(ii) Subtract the series and call the difference f. Let $Z = \{x \in S : f(x) = 0\}$, and let A be the set of limit points of Z in S. Then $A \neq \emptyset$. For $x_0 \in A$, part (i) gives $f(x) = \sum d_n (x - x_0)^n$.
If d_k were the first nonzero coefficient, then $f(x) = (x - x_0)^k g(x)$ with $g(x_0) \neq 0$, contradicting $x_0 \in A$. Hence $f = 0$ near x_0, so A is open. Also $S \setminus A$ is open; by connectedness, Theorem 5.1.2, $A = S$. Since Z is closed, $f = 0$ on S, and evaluating derivatives at 0 gives equality of coefficients.

Structural Commentary: Power-series equality is local-to-global on an interval.
Key Insight: A limit point of zeros forces all Taylor coefficients to vanish. □

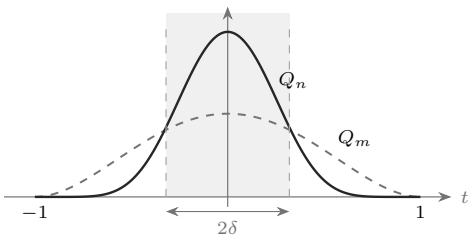


Figure 185: The normalized kernels $Q_n(t) = c_n(1-t^2)^n$ concentrate near 0 while retaining integral 1.

9.2 Approximation Methods

Theorem 9.2.1. *Weierstrass Polynomial Approximation*
If $f : [a, b] \to \mathbb{C}$ *is continuous, then polynomials* P_n *satisfy* $P_n \rightrightarrows f$ *on* $[a, b]$. *If* f *is real-valued, take* P_n *real.*

Proof. By an affine change of variables and subtraction of the endpoint interpolant, assume $[a, b] = [0, 1]$ and $f(0) = f(1) = 0$; extend f by 0 outside $[0, 1]$. For $n \ge 1$, let $Q_n(x) := c_n(1-x^2)^n$ and $\int_{-1}^{1} Q_n(x)\,dx = 1$.
Source estimates give $c_n < \sqrt{n}$ and $Q_n(t) \le \sqrt{n}(1-\delta^2)^n$ for $\delta \le |t| \le 1$.
Define $P_n(x) := \int_{-1}^{1} f(x+t)Q_n(t)\,dt$, $(0 \le x \le 1)$. By change of variables, $P_n(x) = \int_0^1 f(u)Q_n(u-x)\,du$, so P_n is a polynomial in x (real if f is real). Given $\varepsilon > 0$, choose $\delta > 0$ so $|u - v| < \delta$ implies $|f(u) - f(v)| < \varepsilon/2$, and set $M = \sup|f|$. Since $Q_n \ge 0$ and $\int Q_n = 1$, $|P_n(x) - f(x)| \le \int_{-1}^{1} |f(x+t) - f(x)|\, Q_n(t)\,dt \le \frac{\varepsilon}{2} + 4M\sqrt{n}(1-\delta^2)^n$.
The last term tends to 0, so $P_n \rightrightarrows f$ on $[0, 1]$. Undoing the reductions proves the theorem.
Structural Commentary: The kernels Q_n form polynomial approximate identities.
Key Insight: Uniform continuity controls the near part, and concentration of Q_n kills the far part. □

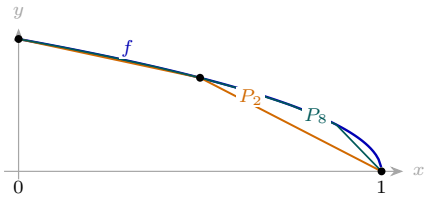


Figure 186: Finer piecewise-linear interpolants approach $f(x) = \sqrt{1-x}$ uniformly on $[0, 1]$.

Corollary 9.2.2. *Polynomial Approximation of $|x|$*
For every interval $[-a, a]$, there are real polynomials P_n such that $P_n(0) = 0$ and $P_n \rightrightarrows |x|$ on $[-a, a]$.

Proof. By Theorem 9.2.1, choose real polynomials P_n^* converging uniformly to $|x|$ on $[-a, a]$.
Since $P_n^*(0) \to 0$, the polynomials $P_n(x) := P_n^*(x) - P_n^*(0)$ have $P_n(0) = 0$ and still converge uniformly to $|x|$.
Structural Commentary: Normalize the approximants at the base point.
Key Insight: Subtracting the value at 0 fixes the normalization without changing the uniform limit. □

Definition 9.2.3. *Algebras and Uniform Closure*
Let $\mathcal{A}$ be a family of complex-valued functions on a set E.

(i) $\mathcal{A}$ is an *algebra* if it is closed under $+$, products, and scalar multiplication (real scalars in the real case).
(ii) $\mathcal{A}$ is *uniformly closed* if $f_n \in \mathcal{A}$ and $f_n \rightrightarrows f$ on E imply $f \in \mathcal{A}$.
(iii) The *uniform closure* of $\mathcal{A}$ is $\{f : f_n \in \mathcal{A},\ f_n \rightrightarrows f \text{ on } E\}$.

Theorem 9.2.4. *Uniform Closures Are Closed Algebras*
The uniform closure of an algebra of bounded functions is a uniformly closed algebra.

Proof. If $f, g \in \mathcal{B}$, choose $f_n, g_n \in \mathcal{A}$ with $f_n \rightrightarrows f$ and $g_n \rightrightarrows g$.
Boundedness and uniform convergence give $f_n + g_n \rightrightarrows f + g$; $f_n g_n \rightrightarrows fg$; and $cf_n \rightrightarrows cf$. Thus $\mathcal{B}$ is an algebra.
If $h_n \in \mathcal{B}$ and $h_n \rightrightarrows h$, approximate each h_n uniformly by some $a_n \in \mathcal{A}$ within $1/n$; then $a_n \rightrightarrows h$, so $h \in \mathcal{B}$.
Structural Commentary: Uniform limits preserve algebraic operations for bounded functions.
Key Insight: The uniform closure of an algebra has already absorbed all further uniform limits. □

Definition 9.2.5. *Separation, Nonvanishing, and Self-Adjointness*
Let $\mathcal{A}$ be a family of functions on E.

(i) $\mathcal{A}$ *separates points* on E if for every distinct $x_1, x_2 \in E$ there is $f \in \mathcal{A}$ with $f(x_1) \neq f(x_2)$.
(ii) $\mathcal{A}$ *vanishes at no point* of E if for every $x \in E$ there is $g \in \mathcal{A}$ with $g(x) \neq 0$.
(iii) A complex algebra $\mathcal{A}$ is *self-adjoint* if $f \in \mathcal{A}$ implies $\overline{f} \in \mathcal{A}$, where $\overline{f}(x) := \overline{f(x)}$.

Theorem 9.2.6. *Two-Point Interpolation in a Separating Algebra*
If an algebra $\mathcal{A}$ on E separates points and vanishes nowhere, then for distinct $x_1, x_2 \in E$ and scalars c_1, c_2, there is $f \in \mathcal{A}$ with $f(x_i) = c_i$, $(i = 1, 2)$.

Proof. Choose $g, h, k \in \mathcal{A}$ with $g(x_1) \neq g(x_2)$, $h(x_1) \neq 0$, and $k(x_2) \neq 0$.
Let $u := gk - g(x_1)k$ and $v := gh - g(x_2)h$. Then $u(x_1) = v(x_2) = 0$ and $u(x_2)v(x_1) \neq 0$.
Hence $f := \frac{c_1 v}{v(x_1)} + \frac{c_2 u}{u(x_2)}$ lies in $\mathcal{A}$ and satisfies $f(x_i) = c_i$.
Structural Commentary: Separation and nonvanishing build two functions that isolate the two points.
Key Insight: Once two point-isolating functions are in the algebra, interpolation is linear algebra. □

Perspective 9.2.7. *Interpolation Is Linear Algebra*
The two-point interpolation step can also be written as a 2×2 linear system. Its invertibility is exactly the finite-dimensional criterion from Theorem 3.2.38, with the determinant test supplied by Theorem 1.1.8. The proof above chooses explicit point-isolating functions instead of reopening the calculation.

Theorem 9.2.8. *Stone–Weierstrass Theorem, Real Form*
$\mathcal{A} \subset C(K, \mathbb{R})$ is algebra on compact K that separates points, vanishes nowhere, then uniform closure is $C(K, \mathbb{R})$.

Proof. Let $\mathcal{B}$ be the uniform closure of $\mathcal{A}$; by Theorem 9.2.4, $\mathcal{B}$ is a uniformly closed algebra.
If $f \in \mathcal{B}$, then $|f| \in \mathcal{B}$: for $a = \|f\|_\infty$, approximate $|y|$ on $[-a, a]$ by polynomials p_n with $p_n(0) = 0$ using Corollary 9.2.2; then $p_n(f) \in \mathcal{B}$ and $p_n(f) \rightrightarrows |f|$.
Hence $\max(f, g), \min(f, g) \in \mathcal{B}$ whenever $f, g \in \mathcal{B}$, as $\max(f, g) = \frac{f+g}{2} + \frac{|f-g|}{2}$; $\min(f, g) = \frac{f+g}{2} - \frac{|f-g|}{2}$.
Fix $f \in C(K, \mathbb{R})$, $x \in K$, $\varepsilon > 0$. $\forall y \in K$, Theorem 9.2.6 gives $h_y \in \mathcal{B}$, $h_y(x) = f(x)$, $h_y(y) = f(y)$.
Compactness gives $y_1, \ldots, y_r$ s.t. $g_x := \max(h_{y_1}, \ldots, h_{y_r})$ satisfies $g_x(x) = f(x)$ and $g_x > f - \varepsilon$ on K.

Compactness gives $x_1, \dots, x_m$ such that $h := \min(g_{x_1}, \dots, g_{x_m})$ satisfies $|h - f| < \varepsilon$ on K.
Thus every $f \in C(K, \mathbb{R})$ is a uniform limit of members of $\mathcal{B}$, so $f \in \mathcal{B}$.
Structural Commentary: Local supports plus finite max/min give global approximation.
Key Insight: Separation plus compactness gives uniform approximation. □

Theorem 9.2.9. *Stone–Weierstrass Theorem, Complex Form*
If $\mathcal{A} \subset C_b(K)$ is self-adjoint, separates points, and vanishes nowhere on compact K, then $\overline{\mathcal{A}} = C_b(K)$.

Proof. Let $\mathcal{A}_{\mathbb{R}} := \{f \in \mathcal{A} : f \text{ real-valued}\}$.
If $f = u + iv \in \mathcal{A}$, self-adjointness gives $u = \frac{f+\overline{f}}{2} \in \mathcal{A}_{\mathbb{R}}$ and $v = \frac{f-\overline{f}}{2i} \in \mathcal{A}_{\mathbb{R}}$.
Thus $\mathcal{A}_{\mathbb{R}}$ separates points; rescaling a nonzero value and taking real parts shows it vanishes nowhere.
By Theorem 9.2.8, $\overline{\mathcal{A}_{\mathbb{R}}} = C(K, \mathbb{R})$. Approximate the real and imaginary parts separately, then combine.
Structural Commentary: Self-adjointness lets the complex algebra expose enough real functions.
Key Insight: The complex theorem is the real theorem applied to real and imaginary parts. □

Theorem 9.2.10. *Exponential and Logarithm from the Power Series*
Define $\mathsf{E}(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}$ for $z \in \mathbb{C}$.
Then, for $z, w \in \mathbb{C}$, $x \in \mathbb{R}$, $\mathsf{E}(z+w) = \mathsf{E}(z)\mathsf{E}(w)$, $\mathsf{E}(x) > 0$, $\mathsf{E}'(x) = \mathsf{E}(x)$, $\mathsf{E}(x) = e^x$, and E is strictly increasing on $\mathbb{R}$. Its inverse $\mathsf{L} : (0, \infty) \to \mathbb{R}$ satisfies $\mathsf{L}'(y) = \frac{1}{y}$, $\mathsf{L}(uv) = \mathsf{L}(u) + \mathsf{L}(v)$ for $u, v > 0$.
For $x > 0$ and $\alpha \in \mathbb{R}$, set $x^\alpha := \mathsf{E}(\alpha \mathsf{L}(x))$; then $(x^\alpha)' = \alpha x^{\alpha-1}$.

Proof. The ratio test (Theorem 6.1.20), gives convergence for all $z \in \mathbb{C}$.
Absolute convergence and Cauchy product give $\mathsf{E}(z)\mathsf{E}(w) = \sum_{n \geq 0} \frac{(z+w)^n}{n!} = \mathsf{E}(z+w)$, hence $\mathsf{E}(z)\mathsf{E}(-z) = 1$.
Theorem 9.1.8 gives $\mathsf{E}' = \mathsf{E}$ on $\mathbb{R}$. Thus $\mathsf{E}(x) > 0$ $\forall$ real x and $\mathsf{E}' > 0$, so E is strictly increasing.
Since $\mathsf{E}(1) = e$, the addition formula gives $\mathsf{E}(p) = e^p$ for $p \in \mathbb{Q}$; continuity extends this to $x \in \mathbb{R}$.
Let $\mathsf{L} = \mathsf{E}^{-1}$. Differentiating $\mathsf{L}(\mathsf{E}(x)) = x$ gives $\mathsf{L}'(y) = 1/y$.
If $u = \mathsf{E}(x)$ and $v = \mathsf{E}(y)$, then $\mathsf{L}(uv) = \mathsf{L}(\mathsf{E}(x+y)) = \mathsf{L}(u) + \mathsf{L}(v)$.
Finally, the chain rule, Theorem 1.1.6, gives $(x^\alpha)' = \alpha x^{\alpha-1}$.
Structural Commentary: Power series define E; inversion gives L and real powers.
Key Insight: The addition formula is the bridge from coefficients to the logarithm law. □

Perspective 9.2.11. *Real Powers by Rational Approximation*
One route defines x^α for $x > 0$ by choosing $q_n \in \mathbb{Q}$ with $q_n \to \alpha$ and setting $x^\alpha = \lim_{n\to\infty} x^{q_n}$ after proving existence and independence of (q_n). The power-series construction sets $x^\alpha = \mathsf{E}(\alpha \mathsf{L}(x))$; both routes must agree with rational powers and be choice-independent.

Perspective 9.2.12. *Logarithmic Integral Route*
With the Riemann integral, set $L(y) := \int_1^y \frac{1}{t}\, dt$ for $y > 0$. Theorem 1.3.1 gives $L'(y) = 1/y$; endpoint limits make L invertible, and Theorem 1.1.6 then recovers $E' = E$ and the product laws.

Perspective 9.2.13. *From Exponentials to Factorials*
Theorems 9.2.10, 1.3.3 supply real-power laws and integration-by-parts pattern used for Γ in Theorem 9.2.16.

Theorem 9.2.14. *Trigonometric Functions from the Exponential*
For real x, define $\mathsf{C}(x) := \frac{\mathsf{E}(ix)+\mathsf{E}(-ix)}{2}$ and $\mathsf{S}(x) := \frac{\mathsf{E}(ix)-\mathsf{E}(-ix)}{2i}$.
Then $\mathsf{E}(ix) = \mathsf{C}(x) + i\mathsf{S}(x)$, $\mathsf{C}^2 + \mathsf{S}^2 = 1$; $\mathsf{C}' = -\mathsf{S}$, and $\mathsf{S}' = \mathsf{C}$.
If $x_0 = \min\{x > 0 : \mathsf{C}(x) = 0\}$ and $\pi := 2x_0$, then $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, $\mathsf{E}(z + 2\pi i) = \mathsf{E}(z)$ for $z \in \mathbb{C}$.
Thus C, S are 2π-periodic, and $\gamma(t) := \mathsf{E}(it)$ traces the unit circle once on $[0, 2\pi]$.

Proof. The definitions give $\mathsf{E}(ix) = \mathsf{C} + i\mathsf{S}$. Since $\mathsf{E}(ix)\mathsf{E}(-ix) = 1$, $|\mathsf{E}(ix)| = 1$, hence $\mathsf{C}^2 + \mathsf{S}^2 = 1$.
Differentiating the definitions and using Theorem 9.1.8 gives $\mathsf{C}' = -\mathsf{S}$ and $\mathsf{S}' = \mathsf{C}$.
If C had no positive zero, then $\mathsf{C} > 0$ and $\mathsf{S}' > 0$ on $(0, \infty)$.
Thus, for $0 < x < y$, $\mathsf{S}(x)(y - x) < \int_x^y \mathsf{S}(t)\, dt = \mathsf{C}(x) - \mathsf{C}(y) \leq 2$, impossible for large y. Hence x_0 exists.
On $(0, \pi/2)$, $\mathsf{C} > 0$, so S increases from 0 to $\mathsf{S}(\pi/2) = 1$; thus $\mathsf{E}(\pi i/2) = i$, and the addition formula gives $\mathsf{E}(\pi i) = -1$, $\mathsf{E}(2\pi i) = 1$, and the period.
Finally, $|\mathsf{E}(it)| = 1$ and the source quadrant argument give the one-time traversal of the unit circle.
Structural Commentary: Trigonometry comes from the path $\mathsf{E}(it)$.
Key Insight: The identity $\mathsf{E}(ix)\mathsf{E}(-ix) = 1$ puts the exponential path on the unit circle. □

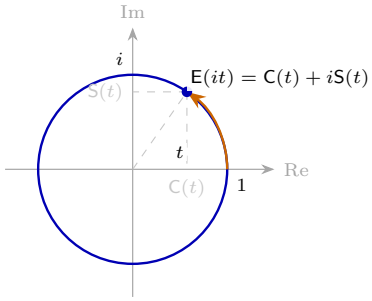


Figure 187: $\mathsf{E}(it) = \mathsf{C}(t) + i\mathsf{S}(t)$ traces the unit circle, defining sine and cosine analytically.

Theorem 9.2.15. *Algebraic Completeness of $\mathbb{C}$*
Every nonconstant complex polynomial has a complex root.

Proof. Let $P(z) = \sum_{j=0}^{n} a_j z^j$, $n \geq 1$, and normalize $a_n = 1$.
Since $|P(z)| \to \infty$, Theorem 5.1.35 gives z_0 with $\mu := |P(z_0)| = \min_{\mathbb{C}} |P|$. If $\mu = 0$, done.
Assume $\mu > 0$ and set $Q(w) := P(z_0 + w)/P(z_0)$. Then $Q(0) = 1$ and $|Q(w)| \geq 1$.
Write $Q(w) = 1 + \sum_{j=k}^{n} b_j w^j$, where $b_k \neq 0$ and k is minimal. By Theorem 9.2.14, choose θ with $b_k e^{ik\theta} = -|b_k|$.
Then $Q(re^{i\theta}) = 1 - |b_k| r^k + r^{k+1} H(r)$, with H bounded near 0, so $\left|Q(re^{i\theta})\right| < 1$ for small $r > 0$, contradiction.
Structural Commentary: Minimize, translate, then perturb.
Key Insight: Choose the phase that lowers the first nonzero term. □

Theorem 9.2.16. *Gamma Function, Beta Function, and Stirling Skeleton*
For $x > 0$, $\Gamma(x+1) = x\Gamma(x)$; $\Gamma(n+1) = n!$ for $n \in \mathbb{N}$; $\log\Gamma$ is convex on $(0,\infty)$.
Moreover, Γ is the unique $f : (0,\infty) \to (0,\infty)$ with $f(1) = 1$, $f(x+1) = xf(x)$, and $\log f$ convex.
For $x, y > 0$, $\mathrm{B}(x,y) := \int_0^1 t^{x-1}(1-t)^{y-1}\,dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.
Consequently, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$; $\int_{-\infty}^{\infty} e^{-s^2}\,ds = \sqrt{\pi}$; $\Gamma(x) = \frac{2^{x-1}}{\sqrt{\pi}}\Gamma\left(\frac{x}{2}\right)\Gamma\left(\frac{x+1}{2}\right)$; and $\lim_{x\to\infty} \frac{\Gamma(x+1)}{(x/e)^x\sqrt{2\pi x}} = 1$.

Proof. Integration by parts gives $\Gamma(x+1) = x\Gamma(x)$; $\Gamma(1) = 1$ gives $\Gamma(n+1) = n!$.
Holder's inequality gives log-convexity. For uniqueness, let $\varphi = \log f$.
If $0 < x < 1$, convexity and $\varphi(x+1) = \varphi(x) + \log x$ give $0 \leq \varphi(x) - \log\frac{n!\,n^x}{x(x+1)\cdots(x+n)} \leq x\log\left(1 + \frac{1}{n}\right) \to 0$.
Thus f is uniquely determined on $(0,1)$, hence on $(0,\infty)$.
For fixed $y > 0$, $\mathrm{B}(1,y) = 1/y$, $\mathrm{B}(x+1,y) = \frac{x}{x+y}\mathrm{B}(x,y)$, and Holder gives log-convexity in x.
Hence $F_y(x) := \frac{\Gamma(x+y)}{\Gamma(y)}\mathrm{B}(x,y)$ satisfies the three uniqueness conditions, so $F_y = \Gamma$.
The consequences follow from $t = \sin^2\theta$, $t = s^2$, the duplication identity, and the substitution $t = x(1+u)$ in the asymptotic proof of Stirling's formula.
Structural Commentary: Functional equation plus convexity gives uniqueness.
Key Insight: Bohr–Mollerup uniqueness turns beta-function identities into gamma identities. □

Method 9.2.17. *Power-Series and Approximation Proof Patterns*

(i) *Power series*: work on $K_\rho \Subset I_R$; use uniform convergence before termwise limits or derivatives.
(ii) *Endpoint theorems*: prove identities inside the radius and pass to endpoints with Abel-type control.
(iii) *Approximate identities*: use kernels $Q_n \geq 0$, $\int Q_n = 1$, with tails $\to 0$ off 0.
(iv) *Uniform closure*: show $a_n \in \mathcal{A}$ and $a_n \rightrightarrows f$.
(v) *Lattice closure*: derive $|f|$, then $\max, \min$ from algebraic formulas.
(vi) *Stone–Weierstrass*: interpolate locally; use compactness and finite max/min to globalize.
(vii) *Gamma function*: use $\Gamma(x+1) = x\Gamma(x)$, $\Gamma(1) = 1$, and log-convexity for uniqueness.

Key Insight: Uniform convergence and compactness are what make infinite algebra behave like finite algebra.

10 Fourier Structure

10.1 Fourier Orthogonality

Fourier structure studies trigonometric series through orthogonality, projection, kernels, and convergence, with inner-product geometry reappearing throughout.

Definition 10.1.1. *Core Fourier Objects*

(i) A *trigonometric polynomial* is a finite sum $a_0 + \sum_{n=1}^{N} (a_n \cos nx + b_n \sin nx)$, equivalently $\sum_{n=-N}^{N} \alpha_n e^{inx}$.

(ii) If f is Riemann integrable on $[-\pi, \pi]$, its *Fourier coefficients* are $\widehat{f}(m) := \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-imx}\, dx$, $m \in \mathbb{Z}$, and $f(x) \sim_{\text{coef}} \sum_{m=-\infty}^{\infty} \widehat{f}(m) e^{imx}$ records coefficients, not convergence.

(iii) A sequence (ϕ_n) on $[a, b]$ is *orthonormal* if $\int_a^b \phi_n(x)\overline{\phi_m(x)}\, dx = 0 \quad (n \neq m)$ and $\int_a^b |\phi_n(x)|^2\, dx = 1$. Its Fourier coefficients are $c_n := \int_a^b f(t)\overline{\phi_n(t)}\, dt$.

Key Insight: Fourier coefficients are orthonormal coordinates; convergence is a separate analytic question.

Definition 10.1.2. *Period-One Fourier Normalization*
Let $C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$ denote the continuous functions $f : \mathbb{R} \to \mathbb{C}$ satisfying $f(x+1) = f(x)$ for all x.
For $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, set $\langle f, g \rangle_{\mathbb{T}} := \int_0^1 f(x)\overline{g(x)}\, dx$; $\|f\|_{L^2(\mathbb{T})} := \sqrt{\langle f, f \rangle_{\mathbb{T}}}$; and $e_n(x) := e^{2\pi i n x}$ for $n \in \mathbb{Z}$.
The *period-one Fourier coefficient* is $\widehat{f}_{\mathbb{T}}(n) := \langle f, e_n \rangle_{\mathbb{T}} = \int_0^1 f(x) e^{-2\pi i n x}\, dx$.
A period-one trigonometric polynomial is a finite sum $\sum_{n=-N}^{N} c_n e_n$.

Perspective 10.1.3. *Two Equivalent Fourier Clocks*
The period-one and 2π-periodic conventions differ by $t = 2\pi x$; only normalizing constants change.
Key Insight: Changing Fourier clocks changes constants, not the projection geometry.

Proposition 10.1.4. *Characters Are an Orthonormal System*
For $m, n \in \mathbb{Z}$, $\langle e_n, e_m \rangle_{\mathbb{T}} = 1$ if $n = m$ and 0 otherwise.
Hence, if $P = \sum_{n=-N}^{N} c_n e_n$, then $c_n = \langle P, e_n \rangle_{\mathbb{T}}$ and $\|P\|^2_{L^2(\mathbb{T})} = \sum_{n=-N}^{N} |c_n|^2$.

Proof. If $n = m$, the integrand is 1. If $n \neq m$, then $\int_0^1 e^{2\pi i (n-m)x}\, dx = 0$.
The coefficient and norm identities are the finite orthonormal-coordinate identities from Theorems 1.3.18 and 1.3.24, applied inside the finite-dimensional span of $e_{-N}, \ldots, e_N$.
Structural Commentary: Fourier coefficients are orthonormal coordinates in a finite trigonometric subspace.
Key Insight: The exponential characters make cross terms vanish. □

10.2 Fourier Approximation

Definition 10.2.1. *Periodic Convolution and Fejer Kernels*
For $f, g \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, define their *periodic convolution* as $(f * g)(x) := \int_0^1 f(y) g(x - y)\, dy$.
For $N \geq 1$, the *Fejer kernel* is $F_N(x) := \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e_n(x) = \frac{1}{N} \left| \sum_{n=0}^{N-1} e_n(x) \right|^2$.

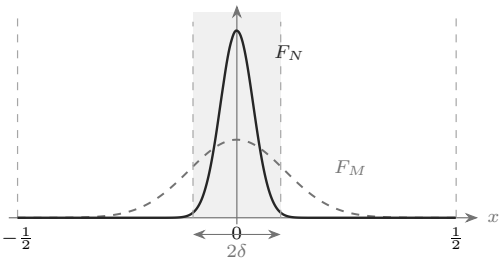


Figure 188: The Fejer kernels are nonnegative trigonometric polynomial approximate identities on $\mathbb{R}/\mathbb{Z}$.

Theorem 10.2.2. *Fejer Approximation on $\mathbb{R}/\mathbb{Z}$*
For $N \geq 1$, F_N is a nonnegative trigonometric polynomial with $\int_0^1 F_N = 1$. For each $0 < \delta < 1/2$, $F_N \to 0$ uniformly on $[\delta, 1 - \delta]$. If $f \in C(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, then $f * F_N$ is a trigonometric polynomial and $f * F_N \rightrightarrows f$.

Proof. From Definition 10.2.1, F_N is a finite linear combination of e_n and $F_N = \frac{1}{N} \left| \sum_{n=0}^{N-1} e_n \right|^2 \geq 0$.
Since $\int_0^1 e_n = 0$ for $n \neq 0$, the e_0 coefficient gives $\int_0^1 F_N = 1$.
For $x \notin \mathbb{Z}$, $F_N(x) = \frac{\sin^2(\pi N x)}{N \sin^2(\pi x)}$. If $m_\delta := \min_{\delta \leq x \leq 1-\delta} \sin^2(\pi x) > 0$, then $0 \leq F_N(x) \leq (N m_\delta)^{-1}$ on $[\delta, 1 - \delta]$.
Also $f * e_n = \widehat{f}_{\mathbb{T}}(n) e_n$, so $f * F_N$ is a trigonometric polynomial. For uniform convergence, write $(f * F_N)(x) =$

$\int_0^1 f(x-t)F_N(t)\,dt$. Let $M = \|f\|_\infty$ and choose δ so $|f(x-t)-f(x)| < \varepsilon$ whenever $t \in [0,\delta]\cup[1-\delta,1]$.
By $F_N \geq 0$ and $\int_0^1 F_N = 1$, uniformly in x, $|(f * F_N)(x) - f(x)| \leq \varepsilon + 2M \sup_{\delta\leq t\leq 1-\delta} F_N(t) \to \varepsilon$.
Structural Commentary: Fejer kernels concentrate mass near the identity element of $\mathbb{R}/\mathbb{Z}$.
Key Insight: Positivity plus concentration turns convolution into uniform approximation. □

Perspective 10.2.3. *Fourier and Plancherel Roadmap*
Path: orthonormal characters, Fejer density, best L^2 projection, then Parseval/Plancherel.
Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

Perspective 10.2.4. *Fourier Coefficients as Projections*
Fourier coefficients are projections onto orthonormal directions, modeled finitely by Theorems 2.1.4, 1.3.18, and 1.3.22; infinite directions make convergence separate.

Theorem 10.2.5. *Fourier Coefficients Give Best Mean-Square Approximation*
Let $\langle u,v\rangle := \int_a^b u\overline{v}$ *and* $\|u\|_2^2 := \langle u,u\rangle$.
If (ϕ_n) *is orthonormal,* $c_m = \langle f,\phi_m\rangle$, $s_n = \sum_{m=1}^{n} c_m\phi_m$, *and* $t_n = \sum_{m=1}^{n} \gamma_m\phi_m$, *then* $\|f-s_n\|_2 \leq \|f-t_n\|_2$, *with equality iff* $\gamma_m = c_m$ *for* $m = 1,\dots,n$. *In particular,* $\sum_{m=1}^{\infty} |c_m|^2 \leq \|f\|_2^2$ *for* $c_m \to 0$.

Proof. For sums over $m = 1,\dots,n$, orthonormality gives
$\|f-t_n\|_2^2 = \|f\|_2^2 - 2\operatorname{Re}\sum c_m\overline{\gamma_m} + \sum|\gamma_m|^2 = \|f\|_2^2 - \sum|c_m|^2 + \sum|\gamma_m - c_m|^2$.
Thus the minimum occurs exactly when $\gamma_m = c_m$.
Taking $t_n = s_n$ gives $\sum_{m=1}^n |c_m|^2 = \|s_n\|_2^2 \leq \|f\|_2^2$; let $n\to\infty$ to get Bessel and $c_m \to 0$.
Structural Commentary: Orthogonality makes the error split into independent coefficient errors.
Key Insight: Fourier coefficients are the coordinates of the orthogonal projection. □

Theorem 10.2.6. *Dirichlet Kernel and Local Fourier Convergence*
Let f *be Riemann integrable and* 2π*-periodic, with* $\widehat{f}(m) := \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-imt}\,dt$.
Let $s_N(f;x) := \sum_{m=-N}^{N} \widehat{f}(m)e^{imx}$ *and* $D_N(t) := \sum_{m=-N}^{N} e^{imt} = \frac{\sin((N+\frac12)t)}{\sin(t/2)}$.
Then $s_N(f;x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x-t)D_N(t)\,dt$.
If, for some x, $|f(x+t)-f(x)| \leq M|t|$ *whenever* $|t|<\delta$, *then* $s_N(f;x)\to f(x)$.
Hence $f = g$ *near* x *implies* $s_N(f;x) - s_N(g;x) \to 0$.

Proof. The displayed form of D_N is the finite geometric sum; substituting $\widehat{f}(m)$ gives the convolution formula.
Also $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N = 1$. Set $q_x(t) := \frac{f(x-t)-f(x)}{\sin(t/2)}$, for $0<|t|\leq\pi$,$q_x(0) := 0$.
Then $s_N(f;x) - f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} q_x(t)\sin\left(N+\frac12\right)t\,dt$.
The hypothesis makes q_x bounded near 0; away from 0 it is Riemann integrable.
Since $\sin((N+\frac12)t) = \sin Nt\cos(t/2) + \cos Nt\sin(t/2)$, Theorem 10.2.5 sends the two Fourier coefficients to 0.
Apply the result to $f-g$ for localization.
Structural Commentary: The Dirichlet kernel rewrites Fourier partial sums as a local oscillatory integral.
Key Insight: Pointwise convergence under a local Lipschitz condition depends only on nearby values of f. □

Corollary 10.2.7. *Uniform Approximation by Trigonometric Polynomials*
If $f \in C(\mathbb{R})$ *is* 2π*-periodic, then for every* $\varepsilon > 0$ *there is a trigonometric polynomial* P *such that* $\|f-P\|_\infty < \varepsilon$.

Proof. Let $\mathbb{T} := \{z\in\mathbb{C} : |z| = 1\}$ and set $F(e^{ix}) := f(x)$.
The algebra $\mathcal{T} := \operatorname{span}\{z^n : n\in\mathbb{Z}\} \subset C(\mathbb{T})$ is self-adjoint, separates points, and vanishes nowhere.
By Theorem 9.2.9, $\overline{\mathcal{T}} = C(\mathbb{T})$. Pulling approximants back by $z = e^{ix}$ gives the desired P.
Structural Commentary: Periodic approximation is Stone–Weierstrass on the unit circle.
Key Insight: The functions e^{inx} are the algebraic coordinates on the circle. □

Theorem 10.2.8. *Parseval's Theorem for Trigonometric Fourier Series*
Let f,g *be Riemann integrable and* 2π*-periodic. Let* $\langle u,v\rangle_2 := \frac{1}{2\pi}\int_{-\pi}^{\pi} u(x)\overline{v(x)}\,dx$ *and* $\|u\|_2^2 := \langle u,u\rangle_2$.
If $f \sim_{\text{coef}} \sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{inx}$, *and* $g \sim_{\text{coef}} \sum_{n\in\mathbb{Z}} \widehat{g}(n)e^{inx}$,
then $\|f-s_N(f)\|_2 \to 0$; $\langle f,g\rangle_2 = \sum_{n\in\mathbb{Z}} \widehat{f}(n)\overline{\widehat{g}(n)}$; *and* $\|f\|_2^2 = \sum_{n\in\mathbb{Z}} \left|\widehat{f}(n)\right|^2$.

Proof. If f is continuous, Corollary 10.2.7 gives trigonometric polynomials P with $\|f-P\|_2$ arbitrarily small.
If P has degree N_0, then Theorem 10.2.5 gives $\|f-s_N(f)\|_2 \leq \|f-P\|_2$ for $N \geq N_0$.
For Riemann integrable f, approximate f in $\|\cdot\|_2$ by a continuous 2π-periodic h.
Then $\|f-s_N(f)\|_2 \leq \|f-h\|_2 + \|h-s_N(h)\|_2 + \|s_N(h-f)\|_2$ and $\|s_N(h-f)\|_2 \leq \|h-f\|_2$ (Theorem 10.2.5).

Hence $s_N(f) \to f$ in $\|\cdot\|_2$. Finally, $\langle s_N(f), g\rangle_2 = \sum_{|n|\le N} \widehat{f}(n)\overline{\widehat{g}(n)}$.
Let $N \to \infty$ and use Schwarz' inequality; the last identity is $g = f$.
Structural Commentary: Density upgrades Bessel to Parseval.
Key Insight: Bessel gives one inequality; density of trigonometric polynomials gives equality. □

Method 10.2.9. *Fourier Proof Patterns*

(i) *Characters*: use orthogonality of e_n to compute coefficients and finite-dimensional projections.
(ii) *Kernels*: rewrite partial sums or averages as convolutions with Dirichlet or Fejer kernels.
(iii) *Approximation*: use Fejer positivity or Stone–Weierstrass to approximate continuous periodic functions by trigonometric polynomials.
(iv) *L^2 control*: project onto finite orthonormal spans, then pass to limits by density.
(v) *Localization*: express the error as an oscillatory integral and use local regularity near the point.

Key Insight: Fourier theory is finite-dimensional orthogonal projection plus an approximation theorem.

Part XI
Measure, Integration, and Functional Analysis

1 Sigma-Algebras, Measures, and Extension

1.1 Sigma-Algebras and Measurable Spaces

Measure theory first builds measurable sets, then measurable functions and integrals. Outer measure covers from outside; measurability splits test sets without losing measure.

Perspective 1.1.1. *Why Abstract Measure Comes First*
Lebesgue integration is built so that, under hypotheses such as domination, $f_n \to f$ implies $\int f_n\,d\mu \to \int f\,d\mu$; see Theorem 2.3.4. The abstract data separate into a measurable space $(X, \mathcal{M})$, a countably additive measure $\mu : \mathcal{M} \to [0, +\infty]$, and an integral defined by approximation from simple functions.
Key Insight: Countable additivity is the set-level structure behind the convergence theorems.

Definition 1.1.2. *σ-Algebras and Measurable Spaces*
Let X be a set. A family $\mathcal{M} \subseteq \mathcal{P}(X)$ is a *σ-algebra* on X if:

(i) $\emptyset, X \in \mathcal{M}$;
(ii) $E \in \mathcal{M}$ implies $X \setminus E \in \mathcal{M}$;
(iii) $E_n \in \mathcal{M}$ for $n \in \mathbb{N}$ implies $\bigcup_{n=1}^{\infty} E_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is a *measurable space*; the members of $\mathcal{M}$ are *measurable sets*.

Proposition 1.1.3. *Closure and Generation for σ-Algebras*
Let $\mathcal{M} \subseteq \mathcal{P}(X)$ be a σ-algebra.

(i) $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{M} \Rightarrow \bigcap_{n=1}^{\infty} E_n \in \mathcal{M}$.
(ii) If $(\mathcal{M}_\alpha)_{\alpha\in A}$ is a nonempty family of σ-algebras on X, then $\bigcap_{\alpha\in A} \mathcal{M}_\alpha$ is a σ-algebra on X.
(iii) For $\mathcal{A} \subseteq \mathcal{P}(X)$, let $\mathscr{S}_{\mathcal{A}}$ be the family of σ-algebras $\mathcal{N}$ on X with $\mathcal{A} \subseteq \mathcal{N}$. Then $\sigma(\mathcal{A}) := \bigcap_{\mathcal{N}\in\mathscr{S}_{\mathcal{A}}} \mathcal{N}$ is the smallest σ-algebra containing $\mathcal{A}$.

Proof.

(i) Since $X \setminus E_n \in \mathcal{M}$, Theorem 1.4.3 gives $\bigcap_n E_n = X \setminus \bigcup_n (X \setminus E_n) \in \mathcal{M}$.
(ii) Set $\mathcal{N} := \bigcap_{\alpha\in A} \mathcal{M}_\alpha$. Then $\emptyset, X \in \mathcal{N}$, and $E \in \mathcal{N}$ implies $X \setminus E \in \mathcal{N}$. Likewise $(E_n) \subseteq \mathcal{N}$ implies $\bigcup_n E_n \in \mathcal{N}$.
(iii) The indexing class in the displayed definition is nonempty since $\mathcal{P}(X)$ belongs to it. By (ii), its intersection is a σ-algebra; it contains $\mathcal{A}$ and is contained in every σ-algebra containing $\mathcal{A}$.

Structural Commentary: Generation is intersection over all admissible σ-algebras.
Key Insight: Intersections preserve universal closure properties. □

Example 1.1.4. *Basic σ-Algebras*
Standard models:

(i) Extremes: $\{\emptyset, X\} \subseteq \mathcal{M} \subseteq \mathcal{P}(X)$ for every σ-algebra $\mathcal{M}$ on X.
(ii) For $E \subseteq X$, $\sigma(\{E\}) = \{\emptyset, E, X \setminus E, X\}$, with repetitions omitted.
(iii) On $\mathbb{N}$, if $O := \{2n-1 : n \in \mathbb{N}\}$, then $\sigma(\{O\}) = \{\emptyset, O, \mathbb{N} \setminus O, \mathbb{N}\}$.
(iv) If X is uncountable, $\mathcal{C}_X := \{E \subseteq X : E \text{ is countable or } X \setminus E \text{ is countable}\}$ is the countable–cocountable σ-algebra.

Key Insight: A σ-algebra records exactly which distinctions the theory is allowed to observe.

Definition 1.1.5. *Borel and Extended Borel σ-Algebras*

(i) For $n \in \mathbb{N}$, the *Borel σ-algebra* on $\mathbb{R}^n$ is $\mathcal{B}(\mathbb{R}^n) := \sigma(\{U \subseteq \mathbb{R}^n : U \text{ is open}\})$. On $\mathbb{R}$, it is also generated by open intervals, closed intervals, half-open intervals, or open half-rays $(a, +\infty)$.
(ii) The *extended Borel σ-algebra* on $\overline{\mathbb{R}}$ is $\overline{\mathcal{B}} := \{E \cup F : E \in \mathcal{B}(\mathbb{R}),\ F \subseteq \{-\infty, +\infty\}\}$.

Definition 1.1.6. *Monotone Classes*
A nonempty $\mathcal{C} \subseteq \mathcal{P}(X)$ is a *monotone class* iff $(E_n) \subseteq \mathcal{C},\ E_n \uparrow E \Rightarrow E \in \mathcal{C}$ and $(F_n) \subseteq \mathcal{C},\ F_n \downarrow F \Rightarrow F \in \mathcal{C}$.
For $\mathcal{A} \subseteq \mathcal{P}(X)$, define $\mathfrak{M}_{\mathcal{A}} := \{\mathcal{C} \subseteq \mathcal{P}(X) : \mathcal{C} \text{ is a monotone class},\ \mathcal{A} \subseteq \mathcal{C}\}$; $\operatorname{mon}(\mathcal{A}) := \bigcap \mathfrak{M}_{\mathcal{A}}$.
Then $\operatorname{mon}(\mathcal{A})$ is the least monotone class containing $\mathcal{A}$, and $\operatorname{mon}(\mathcal{A}) \subseteq \sigma(\mathcal{A})$.
Key Insight: Monotone closure captures limits, not finite unions.

Definition 1.1.7. *Algebras of Sets and Premeasures*
Let X be a set.

(i) An *algebra of sets* (or *field of sets*) on X is $\mathcal{A} \subseteq \mathcal{P}(X)$ such that $\emptyset, X \in \mathcal{A}$; $E \in \mathcal{A} \Rightarrow X \setminus E \in \mathcal{A}$; and $E_1, \dots, E_n \in \mathcal{A} \Rightarrow \bigcup_{j=1}^{n} E_j \in \mathcal{A}$. Equivalently, $\mathcal{A}$ is a ring of sets containing X.
(ii) A *premeasure* (or *measure on an algebra*) on $\mathcal{A}$ is $\mu : \mathcal{A} \to [0, +\infty]$ such that $\mu(\emptyset) = 0$ and, whenever $(E_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ is pairwise disjoint with $\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$, then $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n)$.

Key Insight: Premeasures are countably additive where the algebra permits unions.

Theorem 1.1.8. *Monotone Class Theorem for Algebras*
Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra and set $\mathcal{C} := \mathrm{mon}(\mathcal{A})$. Then $\mathcal{C} = \sigma(\mathcal{A})$.

Proof. Since $\sigma(\mathcal{A})$ is monotone and contains $\mathcal{A}$, $\mathcal{C} \subseteq \sigma(\mathcal{A})$. It suffices to show that $\mathcal{C}$ is a σ-algebra.
For complements, set $\mathcal{D} := \{E \in \mathcal{C} : E^c \in \mathcal{C}\}$. Then $E_n \uparrow E$ in $\mathcal{D}$ implies $E_n^c \downarrow E^c$, and $E_n \downarrow E$ implies $E_n^c \uparrow E^c$; hence $\mathcal{D}$ is monotone. Since $\mathcal{A} \subseteq \mathcal{D}$, minimality gives $\mathcal{C} \subseteq \mathcal{D}$.
For intersections, fix $B \in \mathcal{C}$ and set $\mathcal{D}_B := \{E \in \mathcal{C} : E \cap B \in \mathcal{C}\}$. This is monotone because intersections preserve $\uparrow, \downarrow$ limits. If $B \in \mathcal{A}$, then $\mathcal{A} \subseteq \mathcal{D}_B$, so $\mathcal{C} \subseteq \mathcal{D}_B$; hence $E \cap B \in \mathcal{C}$ for $E \in \mathcal{C}$, $B \in \mathcal{A}$. Fixing $E \in \mathcal{C}$ now gives $\mathcal{A} \subseteq \mathcal{D}_E$, hence $\mathcal{C} \subseteq \mathcal{D}_E$. Thus $\mathcal{C}$ is closed under finite intersections.
Complements and finite intersections give finite unions. If $(E_n) \subseteq \mathcal{C}$, then $F_N := \bigcup_{n=1}^{N} E_n \in \mathcal{C}$ and $F_N \uparrow \bigcup_n E_n$, so $\bigcup_n E_n \in \mathcal{C}$. Hence $\mathcal{C}$ is a σ-algebra containing $\mathcal{A}$, and $\sigma(\mathcal{A}) \subseteq \mathcal{C}$.
Structural Commentary: Finite Boolean closure plus monotone limits gives σ-closure.
Key Insight: Prove closure properties by showing the sets that have the property form a monotone class. □

Corollary 1.1.9. *Monotone-Class Test*
Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be an algebra and let $\mathcal{C} \subseteq \mathcal{P}(X)$ be a monotone class. If $\mathcal{A} \subseteq \mathcal{C}$, then $\sigma(\mathcal{A}) \subseteq \mathcal{C}$.

Proof. Since $\mathcal{A} \subseteq \mathcal{C}$ and $\mathcal{C}$ is monotone, $\mathrm{mon}(\mathcal{A}) \subseteq \mathcal{C}$. By Theorem 1.1.8, $\mathrm{mon}(\mathcal{A}) = \sigma(\mathcal{A})$.
Structural Commentary: Algebra checks plus monotone stability control $\sigma(\mathcal{A})$.
Key Insight: Prove on the algebra; extend by monotone closure. □

1.2 Measures, Charges, and Basic Examples

Definition 1.2.1. *Rings, σ-Rings, and Set Functions*
Let $\mathcal{A} \subseteq \mathcal{P}(X)$.

(i) $\mathcal{A}$ is a *ring of sets* iff $\emptyset \in \mathcal{A}$ and $A, B \in \mathcal{A} \Rightarrow A \cup B,\ A \setminus B \in \mathcal{A}$
(ii) $\mathcal{A}$ is a *σ-ring* iff it is a ring and $(A_n) \subseteq \mathcal{A} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$
(iii) A *set function* on $\mathcal{A}$ is $\phi : \mathcal{A} \to [-\infty, +\infty]$ with $\{+\infty, -\infty\} \nsubseteq \phi(\mathcal{A})$.
(iv) ϕ is *additive* iff $A \cap B = \emptyset \Rightarrow \phi(A \cup B) = \phi(A) + \phi(B)$
(v) ϕ is *countably additive* iff $(A_n) \subseteq \mathcal{A}$ pairwise disjoint, $\bigcup_n A_n \in \mathcal{A} \Rightarrow \phi\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \phi(A_n)$.

Structural Commentary: Rings support finite operations; σ-rings support countable unions.

Definition 1.2.2. *Measures on Measurable Spaces*
Let $(X, \mathcal{M})$ be a measurable space. A *measure* on $(X, \mathcal{M})$ is a map $\mu : \mathcal{M} \to [0, +\infty]$ such that:

(i) $\mu(\emptyset) = 0$;
(ii) if $(E_n) \subseteq \mathcal{M}$ is pairwise disjoint, then $\mu\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n)$

The triple $(X, \mathcal{M}, \mu)$ is a *measure space*.
Key Insight: Measure is countably additive size on the chosen observable sets.

Definition 1.2.3. *Finite and σ-Finite Measures*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) For nonnegative terms, $\sum_{n=1}^{\infty} a_n := \lim_{N\to\infty} \sum_{n=1}^{N} a_n \in [0, +\infty]$.
(ii) μ is *finite* iff $\mu(E) < +\infty$ for every $E \in \mathcal{M}$.
(iii) μ is *σ-finite* iff there exist $E_n \in \mathcal{M}$ such that $X = \bigcup_{n=1}^{\infty} E_n$, $\mu(E_n) < +\infty$ for $n \in \mathbb{N}$

Definition 1.2.4. *Charges*
Let $(X, \mathcal{M})$ be a measurable space. A *charge* is a map $\lambda : \mathcal{M} \to \mathbb{R}$ with $\lambda(\emptyset) = 0$ such that for every pairwise disjoint sequence $(E_n) \subseteq \mathcal{M}$, the series $\sum_{n=1}^{\infty} \lambda(E_n)$ converges in $\mathbb{R}$ and $\lambda(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} \lambda(E_n)$.
Key Insight: Charges are real-valued signed measures with unconditional sums.

Example 1.2.5. *First Abstract Measures*

(i) The zero set function $\mu(E) = 0$ on $\mathcal{M}$ is a finite measure.
(ii) If $X \neq \emptyset$ and $\mathcal{M} = 2^X$, $\mu(\emptyset) = 0$ and $\mu(E) = +\infty$ for $E \neq \emptyset$ give a measure, neither finite nor σ-finite.

(iii) For $p \in X$, the *unit mass* $\delta_p(E) := \mathbf{1}_E(p)$ on 2^X is a finite measure.
(iv) On $2^{\mathbb{N}}$, *counting measure* is $\#(E)$ for finite E and $+\infty$ otherwise; it is σ-finite.
More generally, $(a_n) \subseteq [0,+\infty]$ gives $\mu(E) := \sum_{n\in E} a_n$, and every measure on $2^{\mathbb{N}}$ is uniquely of this form.
(v) If X is uncountable and $\mathcal{M} = 2^X$, $\mu(E) = 0$ for countable E and $\mu(E) = +\infty$ otherwise defines a measure.
(vi) On $2^{\mathbb{N}}$, $\mu(E) = 0$ for finite E and $\mu(E) = +\infty$ otherwise is not a measure, since $\mathbb{N} = \bigcup_{n=1}^{\infty}\{n\}$ gives $+\infty \neq 0$; the constant $+\infty$ rule fails at $\emptyset$.
(vii) Lebesgue measure on $\mathcal{B}(\mathbb{R})$ is σ-finite and gives length $b-a$ to standard intervals; Borel–Stieltjes measures come from increasing right-continuous functions as in Theorem 1.4.4.

Structural Commentary: Measures may be zero, atomic, counting-type, cardinality-based, or geometric.
Key Insight: Countable additivity turns disjoint decompositions into nonnegative series.

Proposition 1.2.6. *Elementary Measure Constructions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) If μ is a measure on $\mathcal{M}$ and $A \in \mathcal{M}$, then $\mu_A(E) := \mu(A \cap E)$ defines a measure μ_A on $\mathcal{M}$.
(ii) If $\mu_1, \dots, \mu_m$ are measures on $\mathcal{M}$ and $a_1, \dots, a_m \geq 0$, then $\lambda := \sum_{j=1}^m a_j\mu_j$ is a measure on $\mathcal{M}$.
(iii) If $(\mu_n)_{n\geq 1}$ are measures on $\mathcal{M}$ with $\mu_n(X) = 1$, then $\lambda := \sum_{n=1}^{\infty} 2^{-n}\mu_n$ is a measure on $\mathcal{M}$ and $\lambda(X) = 1$.

Proof.

(i) $\mu_A(\emptyset) = 0$. If (E_k) is disjoint in $\mathcal{M}$, then $A \cap \bigsqcup_k E_k = \bigsqcup_k (A \cap E_k)$, whence $\mu_A(\bigsqcup_k E_k) = \sum_k \mu_A(E_k)$.
(ii) $\lambda(\emptyset) = 0$. For disjoint (E_k), finite summation gives $\lambda(\bigsqcup_k E_k) = \sum_{j=1}^m a_j \sum_k \mu_j(E_k) = \sum_k \lambda(E_k)$.
(iii) $\lambda(\emptyset) = 0$. For disjoint (E_k), nonnegative reordering gives $\lambda(\bigsqcup_k E_k) = \sum_n 2^{-n} \sum_k \mu_n(E_k) = \sum_k \lambda(E_k)$.
Also $\lambda(X) = \sum_n 2^{-n}\mu_n(X) = \sum_n 2^{-n} = 1$.

Structural Commentary: New measures come from restriction, superposition, and countable convex mixtures.
Key Insight: Nonnegative series make the bookkeeping order-independent. □

Proposition 1.2.7. *Basic Laws for Additive Set Functions*
Let ϕ be additive on a ring $\mathcal{A}$. Then:

(i) $\phi(\emptyset) = 0$;
(ii) $\phi(A_1 \cup \cdots \cup A_n) = \phi(A_1) + \cdots + \phi(A_n)$ for pairwise disjoint $A_i \in \mathcal{A}$;
(iii) $\phi(A \cup B) + \phi(A \cap B) = \phi(A) + \phi(B)$;
(iv) if ϕ is nonnegative and $A \subseteq B$, then $\phi(A) \leq \phi(B)$;
(v) if $B \subseteq A$ and $|\phi(B)| < +\infty$, then $\phi(A \setminus B) = \phi(A) - \phi(B)$.

Proof.

(i) Apply additivity to the empty disjoint union.
(ii) Induct on n, applying additivity to $(A_1 \cup \cdots \cup A_{n-1}) \cup A_n$.
(iii) By disjoint decompositions, $A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$; $A = (A \setminus B) \cup (A \cap B)$; $B = (B \setminus A) \cup (A \cap B)$
(iv) If $A \subseteq B$, then $B = A \cup (B \setminus A)$ disjointly, so nonnegativity gives $\phi(A) \leq \phi(B)$.
(v) If $B \subseteq A$, then $A = B \cup (A \setminus B)$ disjointly; since $|\phi(B)| < +\infty$, subtract $\phi(B)$.

Structural Commentary: Finite additivity plus disjoint decompositions controls all elementary identities.
Key Insight: Rewrite overlaps as disjoint pieces before applying ϕ. □

Theorem 1.2.8. *Continuity from Below*
Suppose ϕ is countably additive on a ring $\mathcal{A}$.
If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$, and $A = \bigcup_{n=1}^{\infty} A_n$ with all $A_n, A \in \mathcal{A}$, then $\phi(A_n) \to \phi(A)$.

Proof. Put $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then B_n are pairwise disjoint, $A_n = B_1 \cup \cdots \cup B_n$, and $A = \bigcup_n B_n$. Countable additivity turns $\phi(A_n)$ and $\phi(A)$ into partial sums and full sum of same series.
Structural Commentary: Increasing unions are disjoint tail decompositions in disguise.
Key Insight: Countable additivity converts set convergence into series convergence. □

Definition 1.2.9. *Variation Candidates of a Charge*
Let λ be a charge on $\mathcal{M}$. For $E \in \mathcal{M}$, define

(i) $\mathfrak{P}_{\mathcal{M}}(E) := \{(A_j)_{j=1}^n : n \in \mathbb{N},\ E = \bigsqcup_{j=1}^n A_j,\ A_j \in \mathcal{M}\}$, the finite measurable partitions of E;
(ii) $\pi_\lambda(E) := \sup\{\lambda(A) : A \subseteq E,\ A \in \mathcal{M}\}$, the *positive variation* candidate;
(iii) $\nu_\lambda(E) := \sup\{\sum_{j=1}^n |\lambda(A_j)| : (A_j)_{j=1}^n \in \mathfrak{P}_{\mathcal{M}}(E)\}$, the *total variation* candidate.

For $E = \emptyset$, the empty partition is allowed and contributes 0.

Proposition 1.2.10. *Charge Operations and Variation Measures*
Let $\lambda, \lambda_1, \dots, \lambda_m$ be charges on $\mathcal{M}$.

(i) If $a_1, \dots, a_m \in \mathbb{R}$, then $\Lambda := \sum_{j=1}^m a_j \lambda_j$ is a charge.
(ii) If $f \in \mathscr{L}^1(\mu)$ on $(X, \mathcal{M}, \mu)$, then $\lambda_f(A) := \int_A f \, d\mu$ is a charge.
(iii) The set functions π_λ and ν_λ are measures on $\mathcal{M}$.

Proof.

(i) If $E = \bigsqcup_n E_n$, then $\Lambda(E) = \sum_{j=1}^m a_j \sum_n \lambda_j(E_n) = \sum_n \Lambda(E_n)$.
(ii) By Theorem 2.2.11, $A \mapsto \int_A f \, d\mu$ is countably additive, and $|\lambda_f(A)| \le \int_X |f| \, d\mu < +\infty$.
(iii) Let $E = \bigsqcup_n E_n$. For $A \subseteq E$, $A = \bigsqcup_n (A \cap E_n)$, so $\lambda(A) \le \sum_n \pi_\lambda(E_n)$; by suprema, $\pi_\lambda(E) \le \sum_n \pi_\lambda(E_n)$. Conversely, for $N \in \mathbb{N}$ choose $A_n \subseteq E_n$ with $\lambda(A_n) > \pi_\lambda(E_n) - \varepsilon 2^{-n}$ for $n \le N$. Then $\pi_\lambda(E) \ge \sum_{n=1}^N \pi_\lambda(E_n) - \varepsilon$. Let $N \to \infty$ and $\varepsilon \downarrow 0$. For ν_λ, every $(A_j)_{j=1}^r \in \mathfrak{P}_{\mathcal{M}}(E)$ satisfies $\sum_{j=1}^r |\lambda(A_j)| \le \sum_n \nu_\lambda(E_n)$ after refining by $A_j \cap E_n$, hence $\nu_\lambda(E) \le \sum_n \nu_\lambda(E_n)$. Conversely, combine near-optimal finite partitions of $E_1, \dots, E_N$, add $\bigcup_{n>N} E_n$ as one part; let $N \to \infty$.

Structural Commentary: Variation turns signed additivity into measure.
Key Insight: Refine partitions against a disjoint decomposition. □

1.3 Outer Measures and Caratheodory Extension

Definition 1.3.1. *Outer Measure Generated by a Premeasure*
Let μ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$. The *outer measure generated by μ* is the set function $\mu^* : \mathcal{P}(X) \to [0, +\infty]$ defined by $\mu^*(B) := \inf\{\sum_{j=1}^\infty \mu(E_j) : E_j \in \mathcal{A},\ B \subseteq \bigcup_{j=1}^\infty E_j\}$ for $B \subseteq X$.

Lemma 1.3.2. *Generated Outer-Measure Properties*
Let μ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, and let μ^* be as in Definition 1.3.1.

(i) $\mu^*(\emptyset) = 0$.
(ii) $C \subseteq D \subseteq X \Rightarrow \mu^*(C) \le \mu^*(D)$.
(iii) $(B_n) \subseteq \mathcal{P}(X) \Rightarrow \mu^*(\bigcup_{n=1}^\infty B_n) \le \sum_{n=1}^\infty \mu^*(B_n)$.
(iv) $E \in \mathcal{A} \Rightarrow \mu^*(E) = \mu(E)$.

Proof.

(i) The cover $(\emptyset)_j$ gives $\mu^*(\emptyset) \le 0$, while $\mu^* \ge 0$ by definition.
(ii) Every algebra cover of D covers C; taking infima gives monotonicity.
(iii) If $\sum_n \mu^*(B_n) = +\infty$, there is nothing to prove. Otherwise, for $\varepsilon > 0$ choose covers $(E_{n,k})_{k \ge 1} \subseteq \mathcal{A}$ of B_n with $\sum_{k=1}^\infty \mu(E_{n,k}) \le \mu^*(B_n) + \varepsilon 2^{-n}$. Then $(E_{n,k})_{n,k}$ covers $\bigcup_n B_n$, hence $\mu^*(\bigcup_n B_n) \le \sum_{n=1}^\infty \sum_{k=1}^\infty \mu(E_{n,k}) \le \sum_n \mu^*(B_n) + \varepsilon$; let $\varepsilon \downarrow 0$.
(iv) For $E \in \mathcal{A}$, the cover $E, \emptyset, \emptyset, \dots$ gives $\mu^*(E) \le \mu(E)$. Conversely, if $E \subseteq \bigcup_n F_n$ with $F_n \in \mathcal{A}$, set $G_n := E \cap F_n \setminus \bigcup_{k<n} F_k$. Then (G_n) is disjoint in $\mathcal{A}$, $\bigcup_n G_n = E$, and $\mu(E) = \sum_n \mu(G_n) \le \sum_n \mu(F_n)$. Taking infima over covers gives $\mu(E) \le \mu^*(E)$.

Structural Commentary: Generated outer measures extend premeasures and are monotone and subadditive.
Key Insight: Intersect an arbitrary cover with E and disjointize inside the algebra. □

Proposition 1.3.3. *Single-Set Envelope Comparison*
Let μ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, and let μ^* be generated by μ. For $B \subseteq X$, set $\mu^\sharp(B) := \inf\{\mu(C) : C \in \mathcal{A},\ B \subseteq C\}$. Then $\mu^\sharp|_{\mathcal{A}} = \mu$ and $\mu^* \le \mu^\sharp$.

Proof. If $E \in \mathcal{A}$, then E is admissible, so $\mu^\sharp(E) \le \mu(E)$.
Conversely, $E \subseteq C \in \mathcal{A}$ implies $\mu(E) \le \mu(C)$; taking infima gives $\mu(E) \le \mu^\sharp(E)$.
If $B \subseteq C \in \mathcal{A}$, then $C, \emptyset, \emptyset, \dots$ covers B, so $\mu^*(B) \le \mu(C)$. Taking infima over C gives $\mu^*(B) \le \mu^\sharp(B)$.
Structural Commentary: One-set envelopes dominate generated outer measures.
Key Insight: A single algebra superset is a countable algebra cover. □

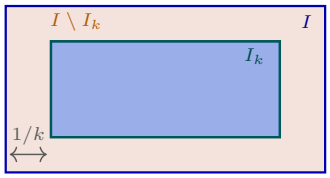


Figure 189: Inner cells I_k exhaust I; the collar $I \setminus I_k$ has vanishing volume.

Example 1.3.4. *Single-Set Envelopes Need Not Be Outer Measures*
Let X be uncountable, let $\mathcal{A}$ be the finite–cofinite algebra on X, and set $\mu(E) := 0$ for finite E, $\mu(E) := +\infty$ for cofinite E. Then μ is a premeasure, $\mu^*(B) = 0$ iff B is countable, and $\mu^\sharp(B) = 0$ iff B is finite.
For any countably infinite $S \subseteq X$, $\mu^\sharp(S) = +\infty$ while $S = \bigcup_{x\in S}\{x\}$ and $\sum_{x\in S}\mu^\sharp(\{x\}) = 0$, QED.
Structural Commentary: Countable covers can be strictly smaller than one-set covers.
Key Insight: Outer measure is built from countable covers precisely to recover countable subadditivity.

Theorem 1.3.5. *Generated Caratheodory Extension*
Let μ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, let μ^* be generated by μ, and set $\mathcal{C}_\mu := \{E \subseteq X : \forall B \subseteq X, \mu^*(B) = \mu^*(B \cap E) + \mu^*(B \setminus E)\}$. Then $\mathcal{A} \subseteq \mathcal{C}_\mu$, $\mathcal{C}_\mu$ is a complete σ-algebra, and $\mu^*|_{\mathcal{C}_\mu}$ is a measure extending μ.

Proof. Lemma 1.3.2 makes μ^* an outer measure; Theorem 1.3.33 makes $\mathcal{C}_\mu$ a σ-algebra, makes $\mu^*|_{\mathcal{C}_\mu}$ a measure, and gives completeness through zero sets. Let $E \in \mathcal{A}$ and $B \subseteq X$.
Subadditivity gives $\mu^*(B) \le \mu^*(B \cap E) + \mu^*(B \setminus E)$.
Conversely, if $B \subseteq \bigcup_n F_n$ with $F_n \in \mathcal{A}$, then $F_n \cap E, F_n \setminus E \in \mathcal{A}$ and $\mu(F_n) = \mu(F_n \cap E) + \mu(F_n \setminus E)$, hence $\mu^*(B \cap E) + \mu^*(B \setminus E) \le \sum_n \mu(F_n)$. Infimizing over covers gives the reverse inequality, so $E \in \mathcal{C}_\mu$.
Thus $\mathcal{A} \subseteq \mathcal{C}_\mu$, and Lemma 1.3.2 gives $\mu^*|_{\mathcal{A}} = \mu$.
Structural Commentary: Caratheodory equality is the additivity domain.
Key Insight: Every algebra set is a clean splitter for the outer measure it generated. □

Proposition 1.3.6. *Caratheodory Splitters for Finitely Subadditive Size*
Let $\alpha : \mathcal{P}(X) \to [0, +\infty)$ satisfy $0 \le \alpha(E) \le \alpha(E \cup F) \le \alpha(E) + \alpha(F)$ for all $E, F \subseteq X$, and set $\mathcal{S} := \{E \subseteq X : (\forall A \subseteq X)\alpha(A) = \alpha(A \cap E) + \alpha(A \setminus E)\}$.

(i) If $\mathcal{S} \neq \emptyset$, then $\mathcal{S}$ is an algebra on X.
(ii) If $E, F \in \mathcal{S}$ and $E \cap F = \emptyset$, then $\alpha(E \cup F) = \alpha(E) + \alpha(F)$.
(iii) Nonemptiness is necessary: $\alpha \equiv 1$ gives $\mathcal{S} = \emptyset$.

Proof.

(i) Choose $E_0 \in \mathcal{S}$. Taking $A = \emptyset$ gives $\alpha(\emptyset) = 0$, hence $\emptyset, X \in \mathcal{S}$. If $E \in \mathcal{S}$, the defining equality is symmetric in E and $X \setminus E$, so $X \setminus E \in \mathcal{S}$. If $E, F \in \mathcal{S}$ and $A \subseteq X$, applying F to A and $A \setminus (E \cap F)$ and applying E to $A \cap F$ gives $\alpha(A) = \alpha(A \cap E \cap F) + \alpha(A \setminus (E \cap F))$; hence $E \cap F \in \mathcal{S}$. Complements give finite unions.
(ii) Apply the defining equality for E to $A = E \cup F$: $\alpha(E \cup F) = \alpha(E) + \alpha(F)$.
(iii) If $\alpha \equiv 1$ and $E \in \mathcal{S}$, taking $A = \emptyset$ gives $1 = 1 + 1$, impossible.

Structural Commentary: The Caratheodory splitting equation is already Boolean at the finite level.
Key Insight: Nonempty splitters force $\alpha(\emptyset) = 0$; without that, the clean-splitting class may vanish. □

Theorem 1.3.7. *σ-Finite Generated Extensions Are Unique*
Let μ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, let μ^* be generated by μ, and let $\mathcal{C}_\mu$ be as in Theorem 1.3.5. If $X = \bigcup_n F_n$ with $F_n \in \mathcal{A}$ and $\mu(F_n) < +\infty$, then $\mu^*|_{\mathcal{C}_\mu}$ is the unique measure ν on $\mathcal{C}_\mu$ satisfying $\nu|_{\mathcal{A}} = \mu$.

Proof. Existence is Theorem 1.3.5. For uniqueness, let ν be a measure on $\mathcal{C}_\mu$ with $\nu|_{\mathcal{A}} = \mu$.
If $\mu(X) < +\infty$, then for $E \in \mathcal{C}_\mu$ and every algebra cover $E \subseteq \bigcup_n A_n$, $\nu(E) \le \sum_n \mu(A_n)$, hence $\nu(E) \le \mu^*(E)$.
Applying this to $X \setminus E$ and using $\nu(E) + \nu(X \setminus E) = \nu(X) = \mu(X) = \mu^*(E) + \mu^*(X \setminus E)$ forces $\nu(E) = \mu^*(E)$.
In general, set $H_n := \bigcup_{j=1}^n F_j$. The finite case on $\mathcal{A}|_{H_n} := \{A \cap H_n : A \in \mathcal{A}\}$ gives $\nu(E \cap H_n) = \mu^*(E \cap H_n)$; since $E \cap H_n \uparrow E$, continuity from below gives $\nu(E) = \mu^*(E)$.
Structural Commentary: Uniqueness reduces to finite-mass exhaustions.
Key Insight: Existence is Caratheodory; uniqueness needs a finite-mass exhaustion. □

Corollary 1.3.8. *Uniqueness on the Generated σ-Algebra*
Let μ be a premeasure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$. If $X = \bigcup_n F_n$ with $F_n \in \mathcal{A}$ and $\mu(F_n) < +\infty$, then any measures ν_1, ν_2 on $\sigma(\mathcal{A})$ with $\nu_1|_{\mathcal{A}} = \nu_2|_{\mathcal{A}} = \mu$ satisfy $\nu_1 = \nu_2$.

Proof. For $E \in \sigma(\mathcal{A})$, repeat the proof of Theorem 1.3.7 with ν_1, ν_2 in place of ν, μ^*; finite-mass comparison on $F_1 \cup \cdots \cup F_n$ and continuity from below give $\nu_1(E) = \nu_2(E)$.
Structural Commentary: σ-finite algebra data determine the generated measure.
Key Insight: Finite algebra exhaustions force uniqueness. □

Example 1.3.9. *Why σ-Finiteness Matters*
Let $X = \mathbb{Q} \cap (0, 1]$, and let $\mathcal{A}$ be the algebra of finite unions of sets $(a, b] \cap X$, where $0 \le a \le b \le 1$ and $a, b \in \mathbb{Q}$. Then every nonempty $A \in \mathcal{A}$ is infinite and $\sigma(\mathcal{A}) = \mathcal{P}(X)$, since each singleton $\{r\}$ is a countable intersection of sets $(a_n, r] \cap X$ with $a_n \uparrow r$. If μ_1 is counting measure and $\mu_2 = 2\mu_1$ on $\mathcal{P}(X)$, then $\mu_1|_{\mathcal{A}} = \mu_2|_{\mathcal{A}}$, but

$\mu_1(\{r\}) \neq \mu_2(\{r\})$. Thus uniqueness may fail without σ-finiteness.
Key Insight: If the starting algebra has no finite nonempty pieces, it cannot detect the later singleton scale.

Definition 1.3.10. *Cells, Intervals, and Volume in $\mathbb{R}^p$*

(i) *Cells in $\mathbb{R}$.* For $a \leq b$, a cell with endpoints a, b is $I \in \{(a,b), [a,b], (a,b], [a,b)\}$, respectively open, closed, right-half-open, left-half-open; if $a = b$, only $[a,a]$ is nonempty.
(ii) *Length.* For a cell I with endpoints $a \leq b$, $\ell(I) := b - a$, independent of endpoint inclusion; $(-\infty, a)$, $(-\infty, a]$, $[b, +\infty)$, $(b, +\infty)$, and $\mathbb{R}$ are intervals, not cells, with length $+\infty$.
(iii) *Cells in $\mathbb{R}^p$.* For $p \geq 1$, a cell is $I = \prod_{i=1}^p I_i$, each I_i a cell in $\mathbb{R}$; its type is open, closed, or half-open, $\prod_i [a_i, b_i)$ is *southwest*, $\prod_i (a_i, b_i]$ is *northeast*, and a *cube* satisfies $\ell(I_1) = \cdots = \ell(I_p)$.
(iv) *Volume.* For $I = \prod_{i=1}^p I_i$, $\mathrm{vol}_p(I) := \prod_{i=1}^p \ell(I_i)$ and $\mathrm{vol}_p(\emptyset) := 0$.
If endpoints are $a_i \leq b_i$, then $\mathrm{vol}_p(I) = \prod_{i=1}^p (b_i - a_i)$.
(v) *Unbounded products.* For cells I, $\mathrm{vol}_p(I^\circ) = \mathrm{vol}_p(\overline{I}) = \mathrm{vol}_p(I)$.
For products $I = \prod_i I_i$ of possibly unbounded intervals, $\exists j : \ell(I_j) = 0$ gives volume 0, while $\ell(I_j) > 0$ for all j and $\exists j : \ell(I_j) = +\infty$ gives volume $+\infty$.

Remark 1.3.11. *Cells versus Intervals*
Cells are precisely bounded intervals with finite endpoints, so every cell is an interval but not conversely. Open cells are open in $\mathbb{R}^p$, closed cells are closed in $\mathbb{R}^p$, and nonempty half-open cells are neither.
In $\mathbb{R}$, $(-\infty, a)$ and $(b, +\infty)$ are open, $(-\infty, a]$ and $[b, +\infty)$ are closed, and $\mathbb{R} = (-\infty, +\infty)$ is clopen.
Key Insight: Topology likes open and closed endpoints; measure bookkeeping often likes half-open endpoints.

Proposition 1.3.12. *Half-Open Cell Calculus*
Fix $\sigma \in \{$southwest, northeast$\}$, and let $\mathcal{H}_\sigma$ be the half-open cells in $\mathbb{R}^p$ with orientation σ.

(i) Any finite Boolean combination of $H_1, \ldots, H_n \in \mathcal{H}_\sigma$ is $\bigsqcup_{r=1}^N K_r$ with $K_r \in \mathcal{H}_\sigma$.
(ii) If $H \in \mathcal{H}_\sigma$, then $\mathbb{R}^p \setminus H = \bigsqcup_{r=1}^\infty K_r$ for some $K_r \in \mathcal{H}_\sigma$.
(iii) For $H \in \mathcal{H}_\sigma$ and $\delta > 0$, $H = \bigsqcup_{r=1}^N K_r$ for some $K_r \in \mathcal{H}_\sigma$ with $\operatorname{diam} K_r < \delta$.

Proof.

(i) For coordinate endpoints $E_j = \{c_{j,0} < \cdots < c_{j,m_j}\}$ of $H_1, \ldots, H_n$, set $[a,b)_\sigma = [a,b)$ if $\sigma =$ *southwest* and $[a,b)_\sigma = (a,b]$ if $\sigma =$ *northeast*, $\mathcal{P}_j = \{[c_{j,q-1}, c_{j,q})_\sigma : 1 \leq q \leq m_j\}$, and $\mathcal{G} = \{\prod_{j=1}^p P_j : P_j \in \mathcal{P}_j\}$; every finite Boolean combination is $\bigsqcup_{G \in \mathcal{G}'} G$, $\mathcal{G}' \subseteq \mathcal{G}$.
(ii) Write $H = \prod_{j=1}^p H_j$ and $D_s = (\prod_{j<s} H_j) \times (\mathbb{R} \setminus H_s) \times \mathbb{R}^{p-s}$; then $\mathbb{R}^p \setminus H = \bigsqcup_{s=1}^p D_s$. Decomposing each one-dimensional ray and $\mathbb{R}$ into countably many bounded σ-half-open intervals and taking products gives the required union; countability follows from Corollary 6.1.11.
(iii) Choose m with $2^{-m}(\sum_{j=1}^p \ell(H_j)^2)^{1/2} < \delta$. After m bisections per coordinate, each subcell has diameter $< \delta$; disjointness and volume preservation follow coordinatewise.

Structural Commentary: Half-open cells behave like grid atoms: finite set operations become finite grid selection.
Key Insight: Half-open endpoints prevent overlap while still partitioning boxes cleanly. □

Definition 1.3.13. *Translation of a Set*
If $A \subseteq \mathbb{R}^p$ and $t \in \mathbb{R}^p$, the translate of A by t is $t + A := \{t + a : a \in A\}$.

Proposition 1.3.14. *Cell Volume Is Translation Invariant*
Let $I = \prod_{i=1}^p I_i$ be a cell in $\mathbb{R}^p$ and $t = (t_i)_{i=1}^p \in \mathbb{R}^p$. Then $t + I = \prod_{i=1}^p (t_i + I_i)$ is a cell with the same endpoint pattern as I, and $\mathrm{vol}_p(t + I) = \mathrm{vol}_p(I)$.

Proof. Let $a_i \leq b_i$ be the endpoints of I_i. Then $t_i + I_i$ has endpoints $t_i + a_i \leq t_i + b_i$, unchanged endpoint inclusion, and $\ell(t_i + I_i) = (t_i + b_i) - (t_i + a_i) = \ell(I_i)$. Hence $\mathrm{vol}_p(t + I) = \prod_{i=1}^p \ell(t_i + I_i) = \prod_{i=1}^p \ell(I_i) = \mathrm{vol}_p(I)$.
Structural Commentary: Translation moves location but not side lengths.
Key Insight: Coordinate volume depends only on side lengths. □

Definition 1.3.15. *Elementary Sets and Regularity in $\mathbb{R}^p$*

(i) *Elementary sets.* $\mathcal{E} := \{\bigcup_{r=1}^n I_r : n \in \mathbb{N}\}$, where each $I_r \subseteq \mathbb{R}^p$ is a bounded cell.
(ii) *Elementary volume.* If $A = \bigsqcup_{r=1}^n I_r$ with cells I_r, set $m(A) := \sum_{r=1}^n \mathrm{vol}_p(I_r)$.
(iii) *Regularity.* A nonnegative additive $\mu : \mathcal{E} \to [0, +\infty]$ is regular iff $\forall A \in \mathcal{E}$, $\forall \varepsilon > 0$, $\exists F, G \in \mathcal{E}$ with $F \subseteq A \subseteq G$, F closed, G open, and $\mu(G) - \varepsilon \leq \mu(A) \leq \mu(F) + \varepsilon$.

$\mathcal{E}$ is a ring; every $A \in \mathcal{E}$ admits a finite disjoint cell decomposition; m is well defined, additive, and regular.

Perspective 1.3.16. *Box Volume and Linear Algebra*
For a box $B = \prod_i (a_i, b_i)$ or $\prod_i [a_i, b_i]$, $m(B) = \prod_i (b_i - a_i)$, the diagonal case of determinant volume scaling. Measure extends this product rule beyond boxes.
Key Insight: Lebesgue measure starts from determinant-style coordinate volume.

Perspective 1.3.17. *Outer Measure as Overestimation*
Elementary $A = \bigsqcup_{r=1}^{n} I_r$ have $m(A) = \sum_{r=1}^{n} \operatorname{vol}_p(I_r)$.
General $E \subseteq \mathbb{R}^p$, e.g. $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$, need not be a finite disjoint cell union.
Replace decompositions by countable cell covers $E \subseteq \bigcup_{k=1}^{\infty} I_k$ and minimize $\sum_{k=1}^{\infty} \operatorname{vol}_p(I_k)$.
Key Insight: Outer covers size every $E \subseteq \mathbb{R}^p$ before measurability.

Definition 1.3.18. *Cell-Cover Outer Measure*
For $E \subseteq \mathbb{R}^p$, let $\mathcal{C}(E)$ be the countable cell covers $(I_k)_{k\geq 1}$ with $E \subseteq \bigcup_{k=1}^{\infty} I_k$. Define $m_p^*(E) := \inf_{(I_k)\in\mathcal{C}(E)} \sum_{k=1}^{\infty} \operatorname{vol}_p(I_k)$.

Here $\mathcal{C}(E) \neq \emptyset$ since $E \subseteq \bigcup_{j=1}^{\infty} [-j, j]^p$, and sum is nonnegative extended-real series, hence order-independent.

Proposition 1.3.19. *Equivalent Covering Families*
For $E \subseteq \mathbb{R}^p$, $\delta > 0$, and fixed half-open orientation σ, let $\mathcal{C}_{\mathsf{cl}}(E)$, $\mathcal{C}_{\mathsf{op}}(E)$, $\mathcal{C}_{\sigma}(E)$, and $\mathcal{C}_{\delta}(E)$ be the subfamilies of $\mathcal{C}(E)$ with cells respectively closed, open, σ-half-open, and satisfying $\operatorname{diam} I_k < \delta$.
If $\alpha(\mathcal{A}) := \inf_{(I_k)\in\mathcal{A}} \sum_{k=1}^{\infty} \operatorname{vol}_p(I_k)$, then $m_p^*(E) = \alpha(\mathcal{C}_{\mathsf{cl}}(E)) = \alpha(\mathcal{C}_{\mathsf{op}}(E)) = \alpha(\mathcal{C}_{\sigma}(E)) = \alpha(\mathcal{C}_{\delta}(E))$.

Proof. Let $\mathcal{C}_*(E)$ be any restricted class above. Since $\mathcal{C}_*(E) \subseteq \mathcal{C}(E)$, $\alpha(\mathcal{C}_*(E)) \geq m_p^*(E)$.
Conversely, for $(I_k) \in \mathcal{C}(E)$ and $\varepsilon > 0$, closures give $\overline{I_k} \supseteq I_k$ and $\operatorname{vol}_p(\overline{I_k}) = \operatorname{vol}_p(I_k)$; open cells $J_k \supseteq I_k$ may be chosen with $\operatorname{vol}_p(J_k) \leq \operatorname{vol}_p(I_k) + \varepsilon 2^{-k}$; each J_k admits a countable disjoint σ-half-open decomposition with total volume $\operatorname{vol}_p(J_k)$; each half-open cell has a finite disjoint subdivision into cells of diameter $< \delta$ with unchanged total volume. Thus $\alpha(\mathcal{C}_*(E)) \leq \sum_k \operatorname{vol}_p(I_k) + \varepsilon$; take infima over (I_k) and let $\varepsilon \downarrow 0$.
Structural Commentary: Near-optimal covers ignore endpoint convention.
Key Insight: Closures, small enlargements, and fine subdivisions preserve the infimum. □

Theorem 1.3.20. *Basic Properties of Cell-Cover Outer Measure*
For $E, F, E_n \subseteq \mathbb{R}^p$,

(i) Null empty set. $0 \leq m_p^*(E) \leq +\infty$ and $m_p^*(\emptyset) = 0$.
(ii) Monotonicity. $E \subseteq F \Longrightarrow m_p^*(E) \leq m_p^*(F)$.
(iii) Subadditivity. $m_p^*(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} m_p^*(E_n)$.

Proof.

(i) Since all cover sums lie in $[0, +\infty]$, $0 \leq m_p^*(E) \leq +\infty$; the cover $(\emptyset, \emptyset, \ldots)$ gives $m_p^*(\emptyset) = 0$.
(ii) If $E \subseteq F$, then $\mathcal{C}(F) \subseteq \mathcal{C}(E)$, hence $\inf_{\mathcal{C}(E)} \leq \inf_{\mathcal{C}(F)}$.
(iii) If some $m_p^*(E_n) = +\infty$, the claim is trivial.
Otherwise, for $\varepsilon > 0$, choose $(I_k^n)_{k\geq 1} \in \mathcal{C}(E_n)$ with $\sum_k \operatorname{vol}_p(I_k^n) \leq m_p^*(E_n) + \varepsilon 2^{-n}$.
Then $(I_k^n)_{(n,k)\in\mathbb{N}^2}$ countable, covers $\bigcup_n E_n$, and $m_p^*(\bigcup_n E_n) \leq \sum_{n,k} \operatorname{vol}_p(I_k^n) \leq \sum_n m_p^*(E_n) + \varepsilon$.

Structural Commentary: Countable subadditivity is proved by merging near-optimal covers.
Key Insight: Nonnegative double sums may be scanned in one sequence. □

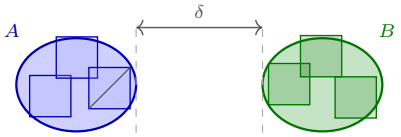


Figure 190: Positive distance separates cells with $\operatorname{diam} I_k < \delta$.

Theorem 1.3.21. *Additivity at Positive Distance*
Let $A, B \subseteq \mathbb{R}^p$ be nonempty and disjoint, and suppose $\operatorname{dist}(A, B) := \inf\{\|a - b\|_2 : a \in A,\ b \in B\} > 0$.
Then $m_p^*(A \cup B) = m_p^*(A) + m_p^*(B)$.

Proof. The inequality $\leq$ follows from Theorem 1.3.20. For $\geq$, set $\delta := \operatorname{dist}(A, B)$. If $m_p^*(A \cup B) = +\infty$, done.
Otherwise, given $\varepsilon > 0$, choose $(I_n) \in \mathcal{C}(A \cup B)$ with $\operatorname{diam} I_n < \delta$ and $\sum_n \operatorname{vol}_p(I_n) \leq m_p^*(A \cup B) + \varepsilon$.
Put $N_A := \{n : I_n \cap A \neq \emptyset\}$ and $N_B := \{n : I_n \cap B \neq \emptyset\}$. Since $N_A \cap N_B = \emptyset$, $(I_n)_{n\in N_A}$ covers A, $(I_n)_{n\in N_B}$ covers B, and $m_p^*(A) + m_p^*(B) \leq \sum_{n\in N_A} \operatorname{vol}_p(I_n) + \sum_{n\in N_B} \operatorname{vol}_p(I_n) \leq \sum_n \operatorname{vol}_p(I_n) \leq m_p^*(A \cup B) + \varepsilon$. Let $\varepsilon \downarrow 0$.
Structural Commentary: Positive separation lets a sufficiently fine cover be sorted unambiguously.
Key Insight: Countable subadditivity becomes equality when no small cell can touch both pieces. □

Theorem 1.3.22. *Outer Measure of a Cell*
If I is a cell in $\mathbb{R}^p$, then $m_p^*(I) = \operatorname{vol}_p(I)$.

Proof. The cover $(I, \emptyset, \ldots)$ gives $m_p^*(I) \le \mathrm{vol}_p(I)$. For $\ge$, the case $\mathrm{vol}_p(I) = 0$ follows by nonnegativity.
Assume $\mathrm{vol}_p(I) > 0$ and fix $\varepsilon > 0$. Choose a closed cell $J \subseteq I$ with $\mathrm{vol}_p(I) - \varepsilon < \mathrm{vol}_p(J)$, taking $J = I$ when I is closed. For any open-cell cover (G_k) of I, compactness gives $J \subseteq \bigcup_{k=1}^m G_k$.
Let $\mathcal{Q}$ be the finite rectangular grid determined by the faces of $J, \overline{G_1}, \ldots, \overline{G_m}$. Up to zero-volume boundary atoms, each $Q \in \mathcal{Q}$ with $Q \subseteq J$ lies in some G_k, hence $\mathrm{vol}_p(J) \le \sum_{k=1}^m \mathrm{vol}_p(G_k) \le \sum_{k=1}^\infty \mathrm{vol}_p(G_k)$.
Infimize over open-cell covers, use Proposition 1.3.19, and let $\varepsilon \downarrow 0$.
Structural Commentary: Compact inner cells reduce covers to finite grids.
Key Insight: Outer measure is calibrated so that cells keep their original volume. □

Theorem 1.3.23. *Translation Invariance of Outer Measure*
For $E \subseteq \mathbb{R}^p$ and $t \in \mathbb{R}^p$, $m_p^*(t + E) = m_p^*(E)$.

Proof. The map $\mathcal{C}(E) \to \mathcal{C}(t + E)$, $(I_k) \mapsto (t + I_k)$, is bijective with inverse $(J_k) \mapsto (-t + J_k)$.
By Proposition 1.3.14, $\sum_k \mathrm{vol}_p(t + I_k) = \sum_k \mathrm{vol}_p(I_k)$, so the two infima are equal.
Structural Commentary: Translate covers forward and backward.
Key Insight: Translation invariance of cell volume lifts directly to translation invariance of outer measure. □

Definition 1.3.24. *Outer Measure and μ-Measurable Sets*
Let μ be additive, regular, nonnegative, and finite on $\mathcal{E}$.
For $E \subseteq \mathbb{R}^p$, define $\mu^*(E) := \inf \sum_{n=1}^\infty \mu(A_n)$, where infimum is over open elementary covers $E \subseteq \bigcup_{n=1}^\infty A_n$.
For $A, B \subseteq \mathbb{R}^p$, set $A \triangle B := (A \setminus B) \cup (B \setminus A)$ and $d_\mu(A, B) := \mu^*(A \triangle B)$.
A set is *finitely μ-measurable* if it is a d_μ-limit of elementary sets.
It is *μ-measurable* if it is a countable union of such sets; denote the classes by $\mathcal{M}_F(\mu)$ and $\mathcal{M}(\mu)$.

Definition 1.3.25. *Box-Cover Outer Measure Model*
An *open box* in $\mathbb{R}^n$ is $B = \prod_{i=1}^n (a_i, b_i) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_i \in (a_i, b_i)\ \forall i\}$, $b_i \ge a_i$, $\mathrm{vol}_n(B) := \prod_{i=1}^n (b_i - a_i)$.
For $\Omega \subseteq \mathbb{R}^n$, the *box-cover outer measure* is $m_n^*(\Omega) := \inf \left\{ \sum_{j \in J} \mathrm{vol}_n(B_j) : (B_j)_{j \in J} \text{ covers } \Omega,\ J \text{ at most countable} \right\}$.

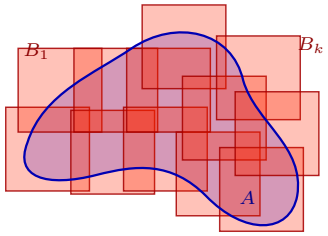


Figure 191: Cover $A \subseteq \bigcup_k B_k$ by squares; minimize $\sum_k |B_k|$.

Perspective 1.3.26. *Outer Measure as Outside Approximation*
Outer measure covers A from outside; $d_\mu(A, B) = \mu^*(A \triangle B)$ measures closeness.
Key Insight: Countable subadditivity collects near-optimal covers.

Method 1.3.27. *Lebesgue Measure Construction*

(i) *Cover from outside*
For $A \subseteq \mathbb{R}^n$, define $m_n^*(A) := \inf \left\{ \sum_{j \in J} \mathrm{vol}_n(B_j) : (B_j)_{j \in J} \text{ covers } A,\ J \text{ at most countable} \right\}$.
This is defined for every set, but it is only countably subadditive in general.

(ii) *Keep the clean splitters*
Call A measurable when every test set $X \subseteq \mathbb{R}^n$ splits additively: $m_n^*(X) = m_n^*(X \cap A) + m_n^*(X \cap A^c)$.
These measurable sets form a σ-algebra.

(iii) *Restrict the outer measure*
Define Lebesgue measure on measurable sets by $m(A) := m_n^*(A)$.
On this restricted class, m is countably additive.

Key Insight: Outer measure sizes all sets; clean splitting restores additivity.

Theorem 1.3.28. *Extension to a Measure*
Let μ be additive, regular, nonnegative, and finite on $\mathcal{E}$. Form μ^* from open elementary covers, set $d_\mu(A, B) = \mu^*(A \triangle B)$, and let $\mathcal{M}(\mu)$ be the countable unions of d_μ-limits of elementary sets. Then:
(i) $\mu^*(A) = \mu(A)$ for $A \in \mathcal{E}$;

(ii) μ^* *is countably subadditive;*
(iii) $\mathcal{M}(\mu)$ *is a* σ*-ring;*
(iv) μ^* *is countably additive on* $\mathcal{M}(\mu)$.

Proof.

(i) Regularity gives $\mu^*(A) \le \mu(A)$; compact closed approximants inside A give the reverse inequality.
(ii) Cover each E_n within $2^{-n}\varepsilon$ of $\mu^*(E_n)$; the combined cover proves subadditivity.
(iii) If $A_k \to A$, $B_k \to B$ in d_μ, $A_k, B_k \in \mathcal{E}$, then $d_\mu(A_k \diamond B_k, A \diamond B) \le d_\mu(A_k, A) + d_\mu(B_k, B) \to 0$, $\diamond \in \{\cup, \cap, \setminus\}$. Thus $\mathcal{M}_F(\mu)$ is a ring. If $A = \bigcup_k A_k$ with $A_k \in \mathcal{M}_F(\mu)$, then $A \setminus B = \bigcup_k (A_k \setminus B)$, so $\mathcal{M}(\mu)$ is a σ-ring.
(iv) For $A = \bigsqcup_n A_n$, $A_n \in \mathcal{M}_F(\mu)$, subadditivity gives $\le$,

$$\text{while finite additivity gives } \mu^*(A) \ge \mu^*\left(\bigsqcup_{n=1}^{N} A_n\right) = \sum_{n=1}^{N} \mu^*(A_n).$$

Let $N \to \infty$, then decompose general disjoint families in $\mathcal{M}(\mu)$ into finite-measurable pieces.

Structural Commentary: Elementary sets are completed under the distance measured by symmetric difference.
Key Insight: Approximate by elementary sets, prove identities there, then pass through $d_\mu(A, B)$. □

Definition 1.3.29. *Extended and Lebesgue Measure*
The *extended measure* is $\mu^*|_{\mathcal{M}(\mu)}$, denoted again by μ.
For the elementary volume m, it is *Lebesgue measure* on $\mathbb{R}^p$.

Definition 1.3.30. *Abstract Outer Measure and Clean Division*
An *abstract outer measure* on M is a monotone, countably subadditive function $\omega : 2^M \to [0, +\infty]$ with $\omega(\emptyset) = 0$.
A set $E \subseteq M$ is *ω-measurable* if it cleanly splits every $X \subseteq M$: $\omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$.
For Lebesgue outer measure, this is Caratheodory measurability.

Lemma 1.3.31. *Finite-Measure Test for Caratheodory Measurability*
Let $\omega : 2^M \to [0, +\infty]$ *be an outer measure. Then* $E \subseteq M$ *is* ω*-measurable iff, for all* $X \subseteq M$,
$\omega(X) < +\infty \implies \omega(X) \ge \omega(X \cap E) + \omega(X \cap E^c)$.

Proof.

($\Rightarrow$) If E is ω-measurable, then $\omega(X) = \omega(X \cap E) + \omega(X \cap E^c)$ $\forall X \subseteq M$, hence finite-test inequality holds.
($\Leftarrow$) For all $X \subseteq M$, $X = (X \cap E) \sqcup (X \cap E^c)$ gives $\omega(X) \le \omega(X \cap E) + \omega(X \cap E^c)$. If $\omega(X) < +\infty$, the hypothesis gives $\ge$; if $\omega(X) = +\infty$, $\ge$ is automatic in $[0, +\infty]$. Thus equality holds for every X.

Structural Commentary: To verify a clean split, only finite outer-measure test sets need real work.
Key Insight: Subadditivity supplies one inequality for free. □

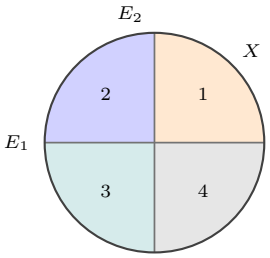


Figure 192: Two cuts split X into four regions; $E_1 \setminus E_2$ is region 2.

Definition 1.3.32. *Zero Sets*
For outer measure $\omega : 2^M \to [0, +\infty]$, *$\omega$-zero set* is $Z \subseteq M$ with $\omega(Z) = 0$; $\mathcal{Z}_\omega := \{Z \subseteq M : \omega(Z) = 0\}$.

Theorem 1.3.33. *Caratheodory Measurable Sets*
Let $\omega : 2^M \to [0, +\infty]$ *be an outer measure, and let* $\mathcal{M}_\omega$ *be the class of* ω*-measurable subsets of* M.
Then $\mathcal{M}_\omega$ *is a* σ*-algebra,* $\omega|_{\mathcal{M}_\omega}$ *is countably additive,* $\mathcal{Z}_\omega \subseteq \mathcal{M}_\omega$, *and* $E \in \mathcal{M}_\omega$, $Z \in \mathcal{Z}_\omega$
imply $E \cup Z, E \setminus Z, E \triangle Z \in \mathcal{M}_\omega$ *and* $\omega(E \cup Z) = \omega(E \setminus Z) = \omega(E \triangle Z) = \omega(E)$.

Proof. If $Z \in \mathcal{Z}_\omega$, then $\omega(X) \le \omega(X \cap Z) + \omega(X \cap Z^c) \le 0 + \omega(X)$, so $Z \in \mathcal{M}_\omega$. Complements are immediate.
If $E, F \in \mathcal{M}_\omega$, iterating the two clean cuts gives $\omega(X) = \sum_{\epsilon,\eta \in \{0,1\}} \omega(X \cap E^\epsilon \cap F^\eta)$, where $E^1 = E$, $E^0 = E^c$; hence all finite Boolean combinations of E, F are measurable.
For disjoint $E_i \in \mathcal{M}_\omega$, set $E = \bigsqcup_i E_i$ and $F_N = (\bigcup_{i=1}^N E_i)^c$.
Finite splitting gives $\omega(X) = \sum_{i=1}^N \omega(X \cap E_i) + \omega(X \cap F_N) \ge \sum_{i=1}^N \omega(X \cap E_i) + \omega(X \cap E^c)$.

Letting $N \to \infty$ and using subadditivity for the reverse inequality yields $\omega(X) = \sum_{i=1}^{\infty} \omega(X \cap E_i) + \omega(X \cap E^c)$; hence $E \in \mathcal{M}_\omega$, and $X = E$ gives countable additivity.
General countable unions follow from $F_1 = E_1$, $F_n = E_n \setminus \bigcup_{i<n} E_i$.
Finally, if $E \in \mathcal{M}_\omega$ and $Z \in \mathcal{Z}_\omega$, then $E = (E \setminus Z) \sqcup (E \cap Z)$ and $E \cup Z = E \sqcup (Z \setminus E)$ give $\omega(E \setminus Z) = \omega(E)$ and $\omega(E \cup Z) = \omega(E)$; also $E \triangle Z = (E \setminus Z) \sqcup (Z \setminus E)$ has measure $\omega(E)$.
Structural Commentary: Measurable sets preserve countable additivity.
Key Insight: Measurable cuts turn subadditivity into equality. □

Definition 1.3.34. *Lebesgue σ-Algebra via Caratheodory*

(i) $\mathcal{L}^n := \{E \subseteq \mathbb{R}^n : E \text{ is } m_n^*\text{-measurable}\}$ is the *Lebesgue σ-algebra*.
(ii) $m := m_n^*|_{\mathcal{L}^n}$ is *Lebesgue measure*; for $E \in \mathcal{L}^n$, write $m(E) = m_n^*(E)$.

Proposition 1.3.35. *Cells Are Lebesgue Measurable*
For every coordinate cell $I \subseteq \mathbb{R}^n$, $I \in \mathcal{L}^n$ and $m(I) = \mathrm{vol}_n(I)$.

Proof. Since $I \in \mathcal{E}$, Theorem 1.3.28 gives $I \in \mathcal{L}^n$ and $m(I) = m_n^*(I) = \mathrm{vol}_n(I)$.
For the open-cell split, fix $A \subseteq \mathbb{R}^n$ with $m_n^*(A) < +\infty$.
Set $I_k := \{x \in I : \mathrm{dist}(x, I^c) > k^{-1}\}$, $\eta_k := m_n^*(I \setminus I_k) \to 0$. Since $\mathrm{dist}(A \cap I_k, A \setminus I) > 0$, Theorem 1.3.21 gives $m_n^*(A) \geq m_n^*(A \cap I_k) + m_n^*(A \setminus I) \geq m_n^*(A \cap I) + m_n^*(A \setminus I) - \eta_k$; let $k \to \infty$ and apply Lemma 1.3.31.
Structural Commentary: Cells are the bridge between coordinate volume and Lebesgue measure.
Key Insight: Shrink, split, let the collar vanish. □

1.4 Borel, Lebesgue, and Stieltjes Measures

Proposition 1.4.1. *Half-Open Interval Algebra on $\mathbb{R}$*
Let $\mathcal{F}$ be the finite-union algebra generated by $(a, b]$, $(-\infty, b]$, $(a, +\infty)$, and $\mathbb{R}$, $a, b \in \mathbb{R}$. Define $\ell((a, b]) := b - a$ and $\ell((-\infty, b]) = \ell((a, +\infty)) = \ell(\mathbb{R}) := +\infty$, extended additively over finite disjoint unions. Then

(i) $\mathcal{F}$ is an algebra on $\mathbb{R}$ and $\sigma(\mathcal{F}) = \mathcal{B}(\mathbb{R})$.
(ii) ℓ is a premeasure on $\mathcal{F}$.

Proof.

(i) Generator complements lie in $\mathcal{F}$, hence $\mathcal{F}$ is an algebra.
Since $\mathcal{F} \subseteq \mathcal{B}(\mathbb{R})$ and $(a, b) = \bigcup_{n > 1/(b-a)} (a, b - 1/n]$, $\sigma(\mathcal{F}) = \mathcal{B}(\mathbb{R})$.
(ii) Normal forms give well-defined finite additivity. For countable additivity, reduce to $(a, b] = \bigsqcup_j (a_j, b_j]$: finite subunions give $\sum_j (b_j - a_j) \leq b - a$, while compactness after enlarging to $(a_j - \varepsilon_j, b_j + \varepsilon_j)$ gives $b - a \leq \sum_j (b_j - a_j) + \varepsilon$. Let $\varepsilon \downarrow 0$. Half-lines follow by truncation, finite disjoint unions componentwise.

Structural Commentary: Lebesgue measure starts as length on a finite interval algebra, not on all subsets.
Key Insight: Compactness supplies the reverse inequality in countable additivity of interval length. □

Remark 1.4.2. *Why the Half-Open Convention Matters*
Finite unions generated by (a, b), $(-\infty, b)$, $(a, +\infty)$, and $\mathbb{R}$ still generate $\mathcal{B}(\mathbb{R})$, but are not an algebra, where $(0, 1)^c = (-\infty, 0] \cup [1, +\infty)$. Half-open endpoints make complements remain in the starting family.

Definition 1.4.3. *Borel–Stieltjes and Lebesgue–Stieltjes Measures*
Let $g : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, set $g(\pm\infty) := \lim_{x \to \pm\infty} g(x)$ in $\overline{\mathbb{R}}$, and let $\mathcal{F}$ be the algebra of finite unions generated by $(a, b]$, $(-\infty, b]$, $(a, +\infty)$, and $\mathbb{R}$. Define μ_g on generators by $\mu_g((a, b]) := g(b) - g(a)$, $\mu_g((-\infty, b]) := g(b) - g(-\infty)$, $\mu_g((a, +\infty)) := g(+\infty) - g(a)$, and $\mu_g(\mathbb{R}) := g(+\infty) - g(-\infty)$; extend to finite disjoint unions by summation. Its unique extension to $\mathcal{B}(\mathbb{R})$ is the *Borel–Stieltjes measure generated by g*; the completion of this Borel measure is the *Lebesgue–Stieltjes measure generated by g*.

Theorem 1.4.4. *Borel–Stieltjes Measure Existence*
Let $g : \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, $g(\pm\infty) := \lim_{x \to \pm\infty} g(x)$,
$\mathcal{I} := \{(a, b] : a < b\} \cup \{(-\infty, b] : b \in \mathbb{R}\} \cup \{(a, +\infty) : a \in \mathbb{R}\} \cup \{\mathbb{R}\}$, and let $\mathcal{F}$ be the finite-union algebra generated by $\mathcal{I}$. Define $G_g((a, b]) := g(b) - g(a)$, $G_g((-\infty, b]) := g(b) - g(-\infty)$, $G_g((a, +\infty)) := g(+\infty) - g(a)$, $G_g(\mathbb{R}) := g(+\infty) - g(-\infty)$, and $\mu_g(\bigsqcup_{j=1}^m I_j) := \sum_{j=1}^m G_g(I_j)$ for disjoint $I_j \in \mathcal{I}$. Then:

(i) μ_g is a σ-finite premeasure on $\mathcal{F}$ and has a unique extension to $\mathcal{B}(\mathbb{R})$, again denoted μ_g.
(ii) If $g = \mathrm{id}_{\mathbb{R}}$, then μ_g is Lebesgue measure.
(iii) If g is continuous, then $\mu_g(J) = g(b) - g(a)$ for $J \in \{(a, b), [a, b), (a, b], [a, b]\}$, $a < b$.

Proof.

(i) Finite normal-form refinement makes μ_g well defined and finitely additive: adjacent g-increments telescope and monotonicity gives positivity.
If $(a,b] = \bigsqcup_{j\geq 1}(a_j,b_j]$, finite additivity gives $\sum_{j=1}^{N} G_g((a_j,b_j]) \leq G_g((a,b])$; conversely, covering $[a+\varepsilon,b]$ by finitely many $(a_j,b_j+\varepsilon_j)$ gives $g(b)-g(a+\varepsilon) \leq \sum_{j\in K}(g(b_j+\varepsilon_j)-g(a_j))$ for finite $K\subseteq\mathbb{N}$, and right-continuity lets $\varepsilon,\varepsilon_j \downarrow 0$. Hence bounded generators are countably additive.
Half-lines follow by truncation and finite unions componentwise, so μ_g is premeasure. As $\mathbb{R} = \bigcup_{n\geq 1}(-n,n]$, $\mu_g((-n,n]) = g(n)-g(-n) < +\infty$, μ_g is σ-finite; Theorem 1.3.7 gives uniqueness.

(ii) If $g = \mathrm{id}_{\mathbb{R}}$, then $G_g((a,b]) = b-a$, so μ_g agrees with half-open length on $\mathcal{F}$.
Uniqueness identifies its Borel extension with Lebesgue measure.

(iii) If g is continuous, then $(x-1/n,x] \downarrow \{x\}$ gives $\mu_g(\{x\}) = \lim_n(g(x)-g(x-1/n)) = 0$.

Structural Commentary: Stieltjes measure replaces ordinary length $b-a$ by the increment $g(b)-g(a)$.
Key Insight: Right-continuity aligns the half-open interval convention with countable covers. □

Proposition 1.4.5. *Finite Borel Measures Give Stieltjes Generators*
Let ν be finite on $\mathcal{B}(\mathbb{R})$ and $g(x) := \nu((-\infty,x])$. Then g is increasing and right-continuous, $g(-\infty)=0$, $g(+\infty)=\nu(\mathbb{R})$, $\nu((a,b]) = g(b)-g(a)$ for $a\leq b$, and $\nu=\mu_g$ on $\mathcal{B}(\mathbb{R})$.

Proof. For $a\leq b$, inclusion gives $g(a)\leq g(b)$ and $(-\infty,b] = (-\infty,a]\sqcup(a,b]$, hence $\nu((a,b]) = g(b)-g(a)$.
If $x_n \downarrow x$, then $(-\infty,x_n]\downarrow(-\infty,x]$, so finite continuity from above gives $g(x_n)\downarrow g(x)$; thus g is right-continuous.
Also $(-\infty,-n]\downarrow\emptyset$ and $(-\infty,n]\uparrow\mathbb{R}$, whence $g(-\infty)=0$ and $g(+\infty)=\nu(\mathbb{R})$.
Thus ν and μ_g agree on the generators of $\mathcal{F}$; Theorem 1.4.4 and uniqueness give $\nu=\mu_g$ on $\mathcal{B}(\mathbb{R})$.
Structural Commentary: A finite real-line measure is encoded by its cumulative mass function.
Key Insight: Right-continuity of distribution functions is continuity from above for measures. □

Example 1.4.6. *Basic Borel–Stieltjes Generators*
The following generators illustrate the interval rule in Theorem 1.4.4.

(i) If $g(x)=2x$, then $\mu_g = 2m$ on $\mathcal{B}(\mathbb{R})$.
(ii) If $g(x)=\arctan x$, then $\mu_g((a,b]) = \arctan b - \arctan a = \int_a^b (1+x^2)^{-1}\,dx$ and $\mu_g(\mathbb{R})=\pi$.
(iii) If $g(x)=0$ for $x<0$ and $g(x)=1$ for $x\geq 0$, then $\mu_g=\delta_0$.
(iv) If $g(x)=0$ for $x<0$ and $g(x)=x$ for $x\geq 0$, then $\mu_g(E) = m(E\cap[0,+\infty))$ for Borel E.

Lemma 1.4.7. *Open Sets Are Countable Unions of Rational Boxes*
For every open $G\subseteq\mathbb{R}^n$, there is a countable $\Sigma\subseteq\{\prod_{i=1}^n(a_i,b_i) : a_i,b_i\in\mathbb{Q},\ a_i<b_i\}$ such that $G=\bigcup_{B\in\Sigma}B$.

Proof. Let $\mathcal{B}_{\mathbb{Q}} := \{\prod_{i=1}^n(a_i,b_i) : a_i,b_i\in\mathbb{Q},\ a_i<b_i\}$; then $\mathcal{B}_{\mathbb{Q}}$ is countable since $\mathbb{Q}^{2n}$ is countable.
Set $\Sigma := \{B\in\mathcal{B}_{\mathbb{Q}} : B\subseteq G\}$. The inclusion $\bigcup_{B\in\Sigma}B\subseteq G$ is immediate.
If $x\in G$, choose $r>0$ with $B_r(x)\subseteq G$, then choose $a_i,b_i\in\mathbb{Q}$ with $x_i - r/n < a_i < x_i < b_i < x_i + r/n$.
Thus $x\in B := \prod_i(a_i,b_i)\subseteq B_r(x)\subseteq G$, so $x\in\bigcup_{B\in\Sigma}B$.
Structural Commentary: Rational boxes give a countable Euclidean base.
Key Insight: Openness supplies a ball; density of $\mathbb{Q}$ supplies rational box endpoints inside it. □

Theorem 1.4.8. *Lebesgue Measure Basic Properties*
Let $m = m_n^*|_{\mathcal{L}^n}$.

(i) $\mathcal{B}(\mathbb{R}^n)\subseteq\mathcal{L}^n$, and $\mathcal{L}^n$ is closed under $E\mapsto E^c$, countable $\cup$, and countable $\cap$.
(ii) For $E,F,E_k\in\mathcal{L}^n$, $0\leq m(E)$, $E\subseteq F\Rightarrow m(E)\leq m(F)$, and $m(\bigcup_k E_k)\leq\sum_k m(E_k)$.
(iii) If $(E_k)\subseteq\mathcal{L}^n$ are pairwise disjoint, then $m(\bigsqcup_k E_k) = \sum_k m(E_k)$.
(iv) $m([0,1]^n)=1$, and $E\in\mathcal{L}^n$, $t\in\mathbb{R}^n$ imply $E+t\in\mathcal{L}^n$ and $m(E+t)=m(E)$.

Proof.

(i) By Theorems 1.3.28, 1.3.33, $\mathcal{L}^n$ is a σ-algebra, hence closed under complements and countable unions/intersections. Open boxes are elementary, hence in $\mathcal{L}^n$; by Lemma 1.4.7, each open $G=\bigcup_j B_j$ with rational open boxes B_j, so $G\in\mathcal{L}^n$. Thus $\mathcal{B}(\mathbb{R}^n)=\sigma(\mathcal{O}_{\mathbb{R}^n})\subseteq\mathcal{L}^n$.

(ii) Since $m = m_n^*|_{\mathcal{L}^n}$, nonnegativity follows from $m_n^*\geq 0$. If $E\subseteq F$, then $m(E)\leq m(F)$ by outer-measure monotonicity. Also $m(\bigcup_k E_k) = m_n^*(\bigcup_k E_k)\leq\sum_k m_n^*(E_k) = \sum_k m(E_k)$ by outer-measure subadditivity.

(iii) If $(E_k)\subseteq\mathcal{L}^n$ pairwise disjoint, by Theorem 1.3.33, m countably additive on $\mathcal{L}^n$, so $m(\bigsqcup_k E_k) = \sum_k m(E_k)$.

(iv) By Theorem 1.3.22, since $[0,1]^n$ is a cell, $m([0,1]^n) = \mathrm{vol}_n([0,1]^n) = 1$.
For translation, let $A\subseteq\mathbb{R}^n$. By Theorem 1.3.23 and $E\in\mathcal{L}^n$, $m_n^*(A) = m_n^*(A-t) = m_n^*((A-t)\cap E) + m_n^*((A-t)\cap E^c) = m_n^*(A\cap(E+t)) + m_n^*(A\cap(E+t)^c)$; hence $E+t\in\mathcal{L}^n$, and $m(E+t) = m_n^*(E+t) = m_n^*(E) = m(E)$.

Structural Commentary: The wish list is recovered only after restricting outer measure to measurable sets.
Key Insight: The box-cover construction gives size; the measurable-set condition restores additivity. □

Theorem 1.4.9. *Uniqueness from Open Cells*
Let ν be a measure on $\mathcal{L}^n$ with $\nu(I) = \mathrm{vol}_n(I)$ for every open cell $I \subseteq \mathbb{R}^n$. Then $\nu = m$ on $\mathcal{L}^n$.

Proof. Fix $N \in \mathbb{N}$, $Q_N = (-N, N)^n$. If $E \in \mathcal{L}^n$ and $E \subseteq Q_N$, then any open-cell cover $E \subseteq \bigcup_j J_j$ gives $\nu(E) \le \sum_j \nu(J_j) = \sum_j \mathrm{vol}_n(J_j)$, hence $\nu(E) \le m(E)$ by infimum.
Applying this to E and $Q_N \setminus E$, and using $\nu(Q_N) = m(Q_N) < +\infty$, yields $\nu(E)+\nu(Q_N \setminus E) = m(E)+m(Q_N \setminus E)$ with each left summand $\le$ its right counterpart; hence $\nu(E) = m(E)$.
For arbitrary $E \in \mathcal{L}^n$, set $A_1 = E \cap Q_1$, $A_N = E \cap (Q_N \setminus Q_{N-1})$ for $N > 1$.
Then $E = \bigsqcup_N A_N$, each $A_N \subseteq Q_N$, and countable additivity gives $\nu(E) = \sum_N \nu(A_N) = \sum_N m(A_N) = m(E)$.
Structural Commentary: Agreement on coordinate cells determines the whole Lebesgue measure.
Key Insight: Covers give $\nu \le m$ locally; finite complements force equality. □

Method 1.4.10. *Translation Invariance by Moving the Test Set*

(i) For $t \in \mathbb{R}^n$ and $A \subseteq \mathbb{R}^n$, translated box covers give $m_n^*(A + t) = m_n^*(A)$.
(ii) If $E \in \mathcal{L}^n$, then for all $A \subseteq \mathbb{R}^n$, $(A \cap (E + t)) - t = (A - t) \cap E$ and $(A \cap (E + t)^c) - t = (A - t) \cap E^c$.
(iii) Apply the Caratheodory split for E to $A - t$ to get $E + t \in \mathcal{L}^n$ and $m(E + t) = m(E)$.

Key Insight: Translation invariance of outer measure transports the clean-splitting identity.

Proposition 1.4.11. *First Borel Hierarchy Examples*
If $G = \bigcap_n U_n$ with U_n open or $F = \bigcup_n C_n$ with C_n closed, then $G, F \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. More generally, countable $\cup, \cap$ operations starting from open or closed sets stay Borel. In $\mathbb{R}$, $\{0\} = \bigcap_{n\ge1}(-1/n, 1/n)$ is G_δ but not open, and $(0, 1) = \bigcup_{n\ge2}[1/n, 1 - 1/n]$ is F_σ but not closed.

Proof. Since $\mathcal{B}(\mathbb{R}^p) = \sigma(\mathcal{O}_{\mathbb{R}^p})$, it contains open sets, closed sets by complements, and countable $\cup, \cap$ by σ-algebra closure; Proposition 1.1.3 gives the stated permanence, and Theorem 1.4.8 gives $\mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. The identities follow from $x = 0 \iff |x| < 1/n$ for all n and $0 < x < 1 \iff \exists n \ge 2 : 1/n \le x \le 1 - 1/n$; $\{0\}$ is not open and $(0, 1)$ is not closed in the usual topology.
Structural Commentary: Borel sets are generated by countable topological operations.
Key Insight: Countable intersections and unions quickly leave the class of open and closed sets. □

Perspective 1.4.12. *Why Not Measure Every Subset?*
Vitali-type sets show m_1^* is not additive on $2^{\mathbb{R}}$: disjoint unions and refinements can break additivity.
Thus measure is m^* restricted to clean splitters E, where $m^*(X) = m^*(X \cap E) + m^*(X \cap E^c)$ for every test set X.
Key Insight: Outer measure is universal; measure lives on clean splitters.

Definition 1.4.13. *Null Sets and Real-Line Measure Zero*
In a measure space, N is *null* if it is measurable and $\mu(N) = 0$.
A set $A \subseteq \mathbb{R}$ has *measure zero* if, for every $\varepsilon > 0$, open intervals O_n cover A and satisfy $\sum_{n=1}^{\infty} |O_n| \le \varepsilon$.
For Lebesgue measure, null sets are also *zero sets*.
A subset of a compact interval has *content zero* if the same condition holds with finitely many intervals.

Proposition 1.4.14. *Outer-Null Sets Are Lebesgue Measurable*
If $Z \subseteq \mathbb{R}^n$ and $m_n^*(Z) = 0$, then $Z \in \mathcal{L}^n$, $m(Z) = 0$, and $\mathcal{P}(Z) \subseteq \mathcal{L}^n$ with $m(W) = 0$ for every $W \subseteq Z$.

Proof. For $A \subseteq \mathbb{R}^n$, monotonicity gives $m_n^*(A \cap Z) = 0$.
Hence $m_n^*(A) \le m_n^*(A \cap Z) + m_n^*(A \cap Z^c) = m_n^*(A \cap Z^c) \le m_n^*(A)$, Z satisfies Caratheodory split and $m(Z) = 0$.
If $W \subseteq Z$, then $m_n^*(W) \le m_n^*(Z) = 0$, and the same argument gives $W \in \mathcal{L}^n$, $m(W) = 0$.
Structural Commentary: Lebesgue measure is complete because outer-null subsets split every test set trivially.
Key Insight: A zero piece cannot disturb additivity. □

Proposition 1.4.15. *Basic Measure-Zero Examples on $\mathbb{R}$*
Every finite $A \subseteq \mathbb{R}$ has content 0; every countable $A \subseteq \mathbb{R}$ has measure 0; Cantor set P has content 0.

Proof. Let $\varepsilon > 0$. If $A = \{a_1, \dots, a_N\}$ with $N \ge 1$, choose open intervals $I_j \ni a_j$ with $|I_j| < \varepsilon/N$; then $A \subseteq \bigcup_{j=1}^N I_j$ and $\sum_{j=1}^N |I_j| < \varepsilon$. The case $A = \emptyset$ is trivial.
If $A = \{a_j : j \in \mathbb{N}\}$, choose open intervals $I_j \ni a_j$ with $|I_j| < \varepsilon 2^{-j}$; then $A \subseteq \bigcup_j I_j$ and $\sum_j |I_j| < \varepsilon$.
For P, choose N with $(2/3)^N < \varepsilon/2$. Write $E_N = \bigcup_{k=1}^{2^N} J_k$, where each J_k is closed and $|J_k| = 3^{-N}$. Choose open $I_k \supseteq J_k$ with $|I_k| < 3^{-N} + \varepsilon/2^{N+1}$. Since $P \subseteq E_N$, the I_k cover P, and $\sum_{k=1}^{2^N} |I_k| < \left(\frac{2}{3}\right)^N + \frac{\varepsilon}{2} < \varepsilon$.
Structural Commentary: Zero size means arbitrarily small interval covers.
Key Insight: Lists split ε by terms; Cantor stages have length $(2/3)^N \to 0$. □

Corollary 1.4.16. *Uncountable Null Sets Exist*
There exists $P \subseteq \mathbb{R}$ with $|P| = 2^{\aleph_0}$ and $m(P) = 0$.

Proof. Let $P := \{\sum_{k=1}^{\infty} \varepsilon_k 3^{-k} : \varepsilon_k \in \{0, 2\}\}$. The map $(\varepsilon_k) \mapsto \sum_k \varepsilon_k 3^{-k}$ identifies P with $\{0, 2\}^{\mathbb{N}}$, so $|P| = 2^{\aleph_0}$. At stage N, P is covered by 2^N closed intervals of length 3^{-N}, hence $m_1^*(P) \leq (2/3)^N$ for all N; letting $N \to \infty$ gives $m_1^*(P) = 0$. By outer-null completeness, $P \in \mathcal{L}^1$ and $m(P) = 0$.
Structural Commentary: Cantor coding separates cardinality from length.
Key Insight: Null need not mean countable. □

Corollary 1.4.17. *Real-Line Lebesgue Measure Checks*
Let m be Lebesgue measure on $\mathbb{R}$.

(i) $a < b \Rightarrow m((a, b)) = m((a, b]) = m([a, b)) = m([a, b]) = b - a$.
(ii) $A \subseteq \mathbb{R}$ *countable* $\Rightarrow A \in \mathcal{B}(\mathbb{R})$ *and* $m(A) = 0$.
(iii) $\emptyset \neq G \subseteq \mathbb{R}$ *open* $\Rightarrow m(G) > 0$.
(iv) $K \subseteq \mathbb{R}$ *compact* $\Rightarrow m(K) < +\infty$.

Proof.

(i) By cell volume, $m([a, b]) = b - a$; endpoint sets finite null; add or delete $\{a\}, \{b\}$ does not change measure.
(ii) If $A = \{a_k : k \in J\}$ with J finite or countable, then $A = \bigcup_{k \in J} \{a_k\} \in \mathcal{B}(\mathbb{R})$; $m(A) \leq \sum_{k \in J} m(\{a_k\}) = 0$.
(iii) Choose $(a, b) \subseteq G$ with $a < b$; by monotonicity and (i), $m(G) \geq m((a, b)) = b - a > 0$.
(iv) Compact K is bounded, so $K \subseteq [-R, R]$ for some $R > 0$; hence $m(K) \leq m([-R, R]) = 2R < +\infty$.

Structural Commentary: On $\mathbb{R}$, length, null endpoints, and compact boundedness give quick measure estimates.
Key Insight: Nonempty open sets contain intervals; compact sets fit inside one. □

Example 1.4.18. *A Positive-Measure Cantor-Type Set with No Interval*
Let $S_0 = [0, 1]$ and, for $n \geq 1$, let R_n be the union of the middle open intervals of length 2^{-2n} removed from the 2^{n-1} components of S_{n-1}; set $S_n := S_{n-1} \setminus R_n$ and $S := \bigcap_{n=0}^{\infty} S_n$.
Then $m(R_n) = 2^{-n-1}$, $\sum_{n \geq 1} m(R_n) = 1/2$, so $S = [0, 1] \setminus \bigcup_n R_n$ is compact with $m(S) = 1/2$.
Since every component of S_n has length $\leq 2^{-n}$, no interval (a, b) can lie in S; hence $\operatorname{int} S = \emptyset$.
Structural Commentary: Summable removals leave measure; repeated splitting kills interior.
Key Insight: Total removed length controls measure; repeated splitting controls interior.

1.5 Continuity and Regularity of Measures

Corollary 1.5.1. *Measure Continuity*
Let μ be a measure. If $E_k \uparrow E$, then $\mu(E_k) \uparrow \mu(E)$. If $F_k \downarrow F$ and $\mu(F_1) < +\infty$, then $\mu(F_k) \downarrow \mu(F)$.

Proof. Disjointize $E_k \uparrow E$ by $E_1' = E_1$ and $E_k' = E_k \setminus E_{k-1}$; countable additivity makes $\mu(E_k)$ the kth partial sum for $\mu(E)$. For $F_k \downarrow F$, apply this to $F_1 \setminus F_k \uparrow F_1 \setminus F$ and subtract from finite $\mu(F_1)$.
Structural Commentary: Monotone set limits are disjoint series or complements of such series.
Key Insight: Downward continuity needs one finite anchor so subtraction is legal. □

Remark 1.5.2. *Borel Sets, Null Sets, and Regular Approximation*
Open, closed, and Borel subsets of $\mathbb{R}^p$ are measurable. Each $A \in \mathcal{M}(\mu)$ admits closed–open approximations $F \subseteq A \subseteq G$ with $\mu(G \setminus A)$ and $\mu(A \setminus F)$ arbitrarily small. In Lebesgue measure, one may take $F_\sigma \subseteq A \subseteq G_\delta$ with $\mu(G_\delta \setminus F_\sigma) = 0$. Null sets form a σ-ring, and countable subsets of $\mathbb{R}^p$ are null.
Structural Commentary: Measurable sets are Borel sets modified by null sets.
Key Insight: Regularity explains why open and closed approximations are enough for integration.

Corollary 1.5.3. *Lebesgue Regularity and Borel Completion on $\mathbb{R}$*
For $A \in \mathcal{L}^1$,

(i) $m(A) = \inf\{m(G) : A \subseteq G,\ G$ *open in* $\mathbb{R}\}$.
(ii) $m(A) = \sup\{m(K) : K \subseteq A,\ K$ *compact in* $\mathbb{R}\}$, *finite-measure case applied to* $A \cap (-n, n]$; $m(A) = +\infty$.
(iii) There exists $B \in \mathcal{B}(\mathbb{R})$ *with* $A \subseteq B$ *and* $m(B \setminus A) = 0$; *equivalently,* $\mathcal{L}^1$ *is the completion of* $\mathcal{B}(\mathbb{R})$.

Proof.

(i) For $\varepsilon > 0$, regular approximation gives open $G \supseteq A$ with $m(G \setminus A) < \varepsilon$, hence $m(G) \leq m(A) + \varepsilon$. Monotonicity gives the reverse inequality.
(ii) If $m(A) < +\infty$, choose closed $F \subseteq A$ with $m(A \setminus F) < \varepsilon$. Since $F \cap [-n, n] \uparrow F$, continuity from below gives compact $K = F \cap [-n, n]$ with $m(F) - m(K) < \varepsilon$, so $m(A) - m(K) < 2\varepsilon$. If $m(A) = +\infty$, apply the finite case to $A_n = A \cap (-n, n]$ and let $n \to \infty$.

(iii) Choose open $G_k \supseteq A$ with $m(G_k \setminus A) < 1/k$, and set $B = \bigcap_k G_k$. Then $B \in \mathcal{B}(\mathbb{R})$, $A \subseteq B$, and $m(B \setminus A) = 0$. Thus every Lebesgue set is Borel plus a null subset.
Conversely, Borel sets and null subsets are Lebesgue measurable.

Structural Commentary: Lebesgue sets are Borel sets plus null-set ambiguity.
Key Insight: Open outer approximation and compact inner approximation are the usable form of regularity. □

Proposition 1.5.4. *Local Measurability on Unit Intervals*
Let $I_n := (n, n+1]$, $n \in \mathbb{Z}$. For $E \subseteq \mathbb{R}$, $E \in \mathcal{L}^1$ iff $E \cap I_n \in \mathcal{L}^1$ for all $n \in \mathbb{Z}$.

Proof.

($\Rightarrow$) Since $I_n \in \mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}^1$, $E \in \mathcal{L}^1$ implies $E \cap I_n \in \mathcal{L}^1$ for all n.
($\Leftarrow$) If $E \cap I_n \in \mathcal{L}^1$ for all n, then $E = \bigcup_{n\in\mathbb{Z}}(E \cap I_n) \in \mathcal{L}^1$.

Structural Commentary: A countable Borel exhaustion localizes measurability.
Key Insight: Local measurable pieces patch by countable unions. □

Proposition 1.5.5. *Finite Measure Meeting Every Unit Interval*
With $I_n := (n, n+1]$, there exists $E \in \mathcal{B}(\mathbb{R})$ such that $0 < m(E \cap I_n)$ for all $n \in \mathbb{Z}$ and $m(E) < +\infty$.

Proof. Set $\varepsilon_n := 2^{-|n|-3}$ and $E := \bigcup_{n\in\mathbb{Z}}(n, n+\varepsilon_n]$. Then $(n, n+\varepsilon_n] \subseteq I_n$ are pairwise disjoint Borel intervals, so $E \in \mathcal{B}(\mathbb{R})$ and $m(E \cap I_n) = \varepsilon_n > 0$. Countable additivity gives $m(E) = \sum_{n\in\mathbb{Z}} \varepsilon_n = \frac{3}{8} < +\infty$.
Structural Commentary: Finite measure can be spread over countably many cells.
Key Insight: Positive local masses can be summable globally. □

Proposition 1.5.6. *Null Sets Form a Sigma-Ideal*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$.
Then $\mathcal{Z}$ is a σ-ring, and $N \in \mathcal{Z}$, $E \in \mathcal{M}$ imply $N \cap E \in \mathcal{Z}$.

Proof. If $(N_k) \subseteq \mathcal{Z}$, then $\mu(\bigcup_k N_k) \le \sum_k \mu(N_k) = 0$, so $\bigcup_k N_k \in \mathcal{Z}$.
If $A, B \in \mathcal{Z}$, then $A \setminus B \subseteq A$, hence $A \setminus B \in \mathcal{Z}$. Finally, $N \cap E \subseteq N$ gives $N \cap E \in \mathcal{Z}$.
Structural Commentary: Null sets are stable under countable unions and measurable cuts.
Key Insight: Monotonicity makes every measurable piece of a null set null. □

Proposition 1.5.7. *Completed Sigma-Algebra*
Let $(X, \mathcal{M}, \mu)$ be measure space, $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$; $\overline{\mathcal{M}}_\mu := \{A \subseteq X : \exists E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ A \triangle E \subseteq N\}$.

(i) $\mathcal{M} \subseteq \overline{\mathcal{M}}_\mu$, and $\overline{\mathcal{M}}_\mu$ is a σ-algebra;
(ii) $A \subseteq N \in \mathcal{Z} \Longrightarrow A \in \overline{\mathcal{M}}_\mu$;
(iii) $\overline{\mathcal{M}}_\mu = \{E \cup Z : E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ Z \subseteq N\}$.

Proof.

(i) For $E \in \mathcal{M}$, $E \triangle E = \emptyset$, so $\mathcal{M} \subseteq \overline{\mathcal{M}}_\mu$. If $A \triangle E \subseteq N$, then $A^c \triangle E^c = A \triangle E \subseteq N$.
If $A_k \triangle E_k \subseteq N_k$, then $(\bigcup_k A_k) \triangle (\bigcup_k E_k) \subseteq \bigcup_k N_k \in \mathcal{Z}$. Hence $\overline{\mathcal{M}}_\mu$ is a σ-algebra.
(ii) If $A \subseteq N \in \mathcal{Z}$, then $A \triangle \emptyset \subseteq N$, so $A \in \overline{\mathcal{M}}_\mu$.
(iii) If $A \triangle E \subseteq N$, then $A = (E \setminus N) \cup (A \cap N)$, with $E \setminus N \in \mathcal{M}$ and $A \cap N \subseteq N$.
Conversely, if $A = E \cup Z$ and $Z \subseteq N \in \mathcal{Z}$, then $A \triangle E \subseteq Z \subseteq N$.

Structural Commentary: Completion adds all null modifications of measurable sets.
Key Insight: Equality modulo null sets is the membership criterion. □

Proposition 1.5.8. *Completed Measure*
Let $(X, \mathcal{M}, \mu)$ be a measure space, $\mathcal{Z} := \{N \in \mathcal{M} : \mu(N) = 0\}$,
and $\overline{\mathcal{M}}_\mu := \{A \subseteq X : \exists E \in \mathcal{M},\ \exists N \in \mathcal{Z},\ A \triangle E \subseteq N\}$. For $A \in \overline{\mathcal{M}}_\mu$, define $\overline{\mu}(A) := \mu(E)$ whenever $E \in \mathcal{M}$ and $A \triangle E \subseteq N \in \mathcal{Z}$. Then $\overline{\mu}$ is a well-defined complete measure on $\overline{\mathcal{M}}_\mu$, and $\overline{\mu}|_{\mathcal{M}} = \mu$.

Proof. If $A \triangle E \subseteq N_1$ and $A \triangle F \subseteq N_2$, with $E, F \in \mathcal{M}$ and $N_1, N_2 \in \mathcal{Z}$, then $E \triangle F \subseteq N_1 \cup N_2$, so $\mu(E) = \mu(F)$.
Hence $\overline{\mu}$ is well defined; since $E \triangle E = \emptyset$, $\overline{\mu}|_{\mathcal{M}} = \mu$.
For disjoint $(A_k) \subseteq \overline{\mathcal{M}}_\mu$, choose $E_k \in \mathcal{M}$ and $N_k \in \mathcal{Z}$ with $A_k \triangle E_k \subseteq N_k$, and set $N := \bigcup_k N_k$.
Then $E_i \cap E_j \subseteq N$ for $i \neq j$. Put $F_1 := E_1$ and $F_k := E_k \setminus \bigcup_{j<k} E_j$.
Then (F_k) disjoint, $A_k \triangle F_k \subseteq N$, $(\bigcup_k A_k) \triangle (\bigcup_k F_k) \subseteq N$, so $\overline{\mu}(\bigcup_k A_k) = \mu(\bigcup_k F_k) = \sum_k \mu(F_k) = \sum_k \overline{\mu}(A_k)$.
If $A \subseteq B \in \overline{\mathcal{M}}_\mu$ and $\overline{\mu}(B) = 0$, choose $E \in \mathcal{M}$ and $N \in \mathcal{Z}$ with $B \triangle E \subseteq N$.
Then $\mu(E) = 0$, so $A \subseteq B \subseteq E \cup N \in \mathcal{Z}$; hence $A \in \overline{\mathcal{M}}_\mu$ and $\overline{\mu}(A) = 0$.
Structural Commentary: The completed measure ignores null errors.
Key Insight: Null perturbations do not change countable additivity. □

Definition 1.5.9. *Limit Inferior and Superior of Sets*
For $(E_n) \subseteq \mathcal{P}(X)$, define $\limsup_n E_n := \bigcap_{m=1}^{\infty} \bigcup_{n \geq m} E_n$ and $\liminf_n E_n := \bigcup_{m=1}^{\infty} \bigcap_{n \geq m} E_n$.

(i) *Infinitely often*: $x \in \limsup_n E_n$ iff $\{n : x \in E_n\}$ is infinite;
(ii) *Eventually always*: $x \in \liminf_n E_n$ iff $\{n : x \notin E_n\}$ is finite;
(iii) *Set limit*: if $\limsup_n E_n = \liminf_n E_n$, write $\lim_n E_n$ for the common set.

Proposition 1.5.10. *Basic Laws for Set Limits*
Let $(E_n) \subseteq \mathcal{P}(X)$.

(i) $\emptyset \subseteq \liminf_n E_n \subseteq \limsup_n E_n \subseteq X$.
(ii) $E_n \uparrow E \Longrightarrow \liminf_n E_n = \limsup_n E_n = E = \bigcup_n E_n$.
(iii) $E_n \downarrow E \Longrightarrow \liminf_n E_n = \limsup_n E_n = E = \bigcap_n E_n$.
(iv) $(E_n) \subseteq \mathcal{M} \Longrightarrow \liminf_n E_n, \limsup_n E_n \in \mathcal{M}$.

Proof.

(i) Since $\bigcap_m \bigcup_{n \geq m} E_n \subseteq \bigcup_{n \geq 1} E_n \subseteq X$ and $\bigcap_{n \geq m} E_n \subseteq \bigcup_{n \geq m} E_n$ for all m, the inclusions follow.
(ii) If $E_n \uparrow E$, then $\{n : x \in E_n\}$ is nonempty iff it is cofinite, hence $\liminf_n E_n = \limsup_n E_n = \bigcup_n E_n = E$.
(iii) If $E_n \downarrow E$, then $\{n : x \in E_n\}$ is infinite iff it is all of $\mathbb{N}$, hence $\liminf_n E_n = \limsup_n E_n = \bigcap_n E_n = E$.
(iv) If $(E_n) \subseteq \mathcal{M}$, then $\bigcup_{n \geq m} E_n, \bigcap_{n \geq m} E_n \in \mathcal{M}$ $\forall m$; countable closure gives $\limsup_n E_n, \liminf_n E_n \in \mathcal{M}$.

Structural Commentary: Set limsup means infinitely often; set liminf means eventually always.
Key Insight: Translate the definitions into pointwise membership statements. □

Proposition 1.5.11. *Measure Bounds for Set Limits*
Let (E_n) be measurable in a measure space $(X, \mathcal{M}, \mu)$.

(i) $\mu(\liminf_n E_n) \leq \liminf_n \mu(E_n)$.
(ii) $\mu(\bigcup_{n=1}^{\infty} E_n) < +\infty \Longrightarrow \limsup_n \mu(E_n) \leq \mu(\limsup_n E_n)$.

Proof.

(i) Set $A_m := \bigcap_{n \geq m} E_n$. Then $A_m \uparrow \liminf_n E_n$ and $A_m \subseteq E_n$ for all $n \geq m$, so $\mu(A_m) \leq \inf_{n \geq m} \mu(E_n)$. By continuity from below, $\mu(\liminf_n E_n) = \lim_m \mu(A_m) \leq \sup_m \inf_{n \geq m} \mu(E_n) = \liminf_n \mu(E_n)$.
(ii) Set $B_m := \bigcup_{n \geq m} E_n$. $B_m \downarrow \limsup_n E_n$, $\mu(B_1) < +\infty$, $E_n \subseteq B_m$ $\forall n \geq m$, so $\sup_{n \geq m} \mu(E_n) \leq \mu(B_m)$. By continuity from above, $\limsup_n \mu(E_n) = \inf_m \sup_{n \geq m} \mu(E_n) \leq \lim_m \mu(B_m) = \mu(\limsup_n E_n)$.

Structural Commentary: Tail intersections increase to liminf; tail unions decrease to limsup.
Key Insight: Compare each tail envelope to the corresponding tail bounds of the measures. □

1.6 Regular Approximation and Caratheodory Tests

The Caratheodory definition is intrinsic to outer measure, but in $\mathbb{R}^p$ it has a topological shadow: measurable sets are exactly those that can be approximated by open, closed, compact, and cell-built sets up to arbitrarily small error. That bridge is also the route to the first nonmeasurable and non-Borel examples.

Theorem 1.6.1. *Lebesgue Measurable Non-Borel Sets Exist*
There exists $E \subseteq \mathbb{R}$ such that $E \in \mathcal{L}^1$ and $E \notin \mathcal{B}(\mathbb{R})$; equivalently, $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}^1$.

Proof. By Lemma 1.4.7, $\mathcal{B}(\mathbb{R})$ is countably generated, hence $|\mathcal{B}(\mathbb{R})| \leq \mathfrak{c}$. Let C be the Cantor set.
Since $|C| = \mathfrak{c}$ and $m(C) = 0$, Proposition 1.4.14 gives $\mathcal{P}(C) \subseteq \mathcal{L}^1$, so $\left|\mathcal{L}^1\right| \geq 2^{\mathfrak{c}}$.
Also $\mathcal{L}^1 \subseteq \mathcal{P}(\mathbb{R})$, so $\left|\mathcal{L}^1\right| \leq 2^{\mathfrak{c}}$; hence $\left|\mathcal{L}^1\right| = 2^{\mathfrak{c}}$. By Cantor's theorem, $\mathfrak{c} < 2^{\mathfrak{c}}$, so $\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}^1$.
Structural Commentary: Countable generation is smaller than completion by null subsets.
Key Insight: Completeness creates more measurable sets than topology can generate. □

Lemma 1.6.2. *Outer Approximation by Open and G_δ Sets*
For $A \subseteq \mathbb{R}^p$:

(i) for every $\varepsilon > 0$, there exists open $G \subseteq \mathbb{R}^p$ with $A \subseteq G$ and $m(G) \leq m_p^*(A) + \varepsilon$;
(ii) $m_p^*(A) = \inf\{m(G) : A \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\}$;
(iii) there exists G_δ set $H \supseteq A$ with $m(H) = m_p^*(A)$.

Proof.

(i) If $m_p^*(A) = +\infty$, take $G = \mathbb{R}^p$.
If $m_p^*(A) < +\infty$, choose an open-cell cover $A \subseteq \bigcup_k I_k$ with $\sum_k \mathrm{vol}_p(I_k) \leq m_p^*(A) + \varepsilon$, and set $G := \bigcup_k I_k$.
Then G is open, $A \subseteq G$, and $m(G) \leq \sum_k m(I_k) = \sum_k \mathrm{vol}_p(I_k) \leq m_p^*(A) + \varepsilon$.

(ii) By (i), $\inf\{m(G) : A \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\} \le m_p^*(A) + \varepsilon$ for all $\varepsilon > 0$; let $\varepsilon \downarrow 0$.
Conversely, $A \subseteq G$ gives $m_p^*(A) \le m_p^*(G) = m(G)$ for open G; take the infimum over G.
(iii) If $m_p^*(A) = +\infty$, take $H = \mathbb{R}^p$.
If $m_p^*(A) < +\infty$, choose open $G_n \supseteq A$ with $m(G_n) \le m_p^*(A) + 1/n$ and set $H := \bigcap_n G_n$.
Then H is G_δ, $A \subseteq H$, and $m_p^*(A) \le m(H) \le m(G_n) \le m_p^*(A) + 1/n$ for all n, so $m(H) = m_p^*(A)$.

Structural Commentary: Outer measure can be realised by a Borel envelope from outside.
Key Insight: Infima over cell covers can be repackaged as open-set approximation. □

Corollary 1.6.3. *Null Sets Sit Inside Borel Null Sets*
If $Z \subseteq \mathbb{R}^p$ and $m_p^*(Z) = 0$, then there exists $B \in \mathcal{B}(\mathbb{R}^p)$ such that $Z \subseteq B$ and $m(B) = 0$.

Proof. By Lemma 1.6.2, choose G_δ set $B \supseteq Z$ with $m(B) = m_p^*(Z) = 0$. Since $G_\delta \subseteq \mathcal{B}(\mathbb{R}^p)$, B is Borel.
Structural Commentary: Null sets admit Borel envelopes of the same measure.
Key Insight: Outer approximation turns arbitrary null sets into Borel null supersets. □

Theorem 1.6.4. *Open and G_δ Characterizations of Measurability*
For $E \subseteq \mathbb{R}^p$, the following are equivalent:

(i) $E \in \mathcal{L}^p$;
(ii) $\forall \varepsilon > 0$, $\exists$ *open* $G \subseteq \mathbb{R}^p$ *such that* $E \subseteq G$ *and* $m_p^*(G \setminus E) < \varepsilon$;
(iii) $\exists$ G_δ *set* $H \subseteq \mathbb{R}^p$ *such that* $E \subseteq H$ *and* $m_p^*(H \setminus E) = 0$;
(iv) $\exists$ G_δ *set* $H \subseteq \mathbb{R}^p$ *and* $Z \in \mathcal{L}^p$ *such that* $Z \subseteq H$, $m(Z) = 0$, *and* $E = H \setminus Z$.

Proof.

(i) $\Rightarrow$ (ii). If $m(E) < +\infty$, choose open $G \supseteq E$ with $m(G) \le m(E) + \varepsilon/2$.
Since $E \in \mathcal{L}^p$, $m_p^*(G \setminus E) = m(G) - m(E) < \varepsilon$. If $m(E) = +\infty$, set $E_1 := E \cap \{x : \|x\|_2 \le 1\}$ and $E_n := E \cap \{x : n-1 < \|x\|_2 \le n\}$ for $n \ge 2$. For each n, choose open $G_n \supseteq E_n$ with $m_p^*(G_n \setminus E_n) < \varepsilon 2^{-n}$; then $G := \bigcup_n G_n$ is open, $E \subseteq G$, and $G \setminus E \subseteq \bigcup_n (G_n \setminus E_n)$, so $m_p^*(G \setminus E) < \varepsilon$.
(ii) $\Rightarrow$ (iii). Choose open $G_n \supseteq E$ with $m_p^*(G_n \setminus E) < 1/n$, and set $H := \bigcap_n G_n$.
Then H is G_δ, $E \subseteq H$, and $H \setminus E \subseteq G_n \setminus E$ for all n; hence $m_p^*(H \setminus E) = 0$.
(iii) $\Rightarrow$ (i). Since H is G_δ, $H \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. Also $m_p^*(H \setminus E) = 0$ implies $H \setminus E \in \mathcal{L}^p$ by Proposition 1.4.14.
Thus $E = H \setminus (H \setminus E) \in \mathcal{L}^p$.
(iii) $\Leftrightarrow$ (iv). If (iii) holds, set $Z := H \setminus E$; then $Z \subseteq H$, $m(Z) = 0$, and $E = H \setminus Z$.
Conversely, if (iv) holds, then $E \subseteq H$ and $H \setminus E = Z$, so $m_p^*(H \setminus E) = 0$.

Structural Commentary: Lebesgue measurable sets are G_δ sets with null material removed.
Key Insight: Small open excess becomes a null G_δ excess after countably many approximations. □

Corollary 1.6.5. *Open Outer Regularity for Measurable Sets*
If $E \in \mathcal{L}^p$, then

(i) $\forall \varepsilon > 0$, $\exists$ *open* $G \supseteq E$ *such that* $m(G) \le m(E) + \varepsilon$;
(ii) $m(E) = \inf\{m(G) : E \subseteq G,\ G \subseteq \mathbb{R}^p \text{ open}\}$.

Proof.

(i) By Theorem 1.6.4, choose open $G \supseteq E$ with $m_p^*(G \setminus E) < \varepsilon$.
If $m(E) < +\infty$, then $m(G) = m(E) + m(G \setminus E) \le m(E) + \varepsilon$; if $m(E) = +\infty$, the inequality is trivial.
(ii) From (i), $\inf_G m(G) \le m(E)$. Conversely, $E \subseteq G$ gives $m(E) \le m(G)$ by monotonicity; take $\inf_G$.

Structural Commentary: Measurable sets are measured from outside by open supersets.
Key Insight: Open excess controls the outer regular infimum. □

Theorem 1.6.6. *Closed and F_σ Characterizations of Measurability*
For $E \subseteq \mathbb{R}^p$, the following are equivalent:

(i) $E \in \mathcal{L}^p$;
(ii) $\forall \varepsilon > 0$, $\exists$ *closed* $F \subseteq E$ *such that* $m_p^*(E \setminus F) < \varepsilon$;
(iii) $\exists$ F_σ *set* $K \subseteq E$ *such that* $m_p^*(E \setminus K) = 0$;
(iv) $\exists$ F_σ *set* $K \subseteq E$ *and* $Z \in \mathcal{L}^p$ *such that* $Z \subseteq E$, $m(Z) = 0$, *and* $E = K \cup Z$.

Proof.

(i) $\Rightarrow$ (ii). Since $E^c \in \mathcal{L}^p$, Theorem 1.6.4 gives open $G \supseteq E^c$ with $m_p^*(G \setminus E^c) < \varepsilon$. Set $F := G^c$.
Then F is closed, $F \subseteq E$, and $E \setminus F = G \setminus E^c$, so $m_p^*(E \setminus F) < \varepsilon$.
(ii) $\Rightarrow$ (iii). Choose closed $F_n \subseteq E$ with $m_p^*(E \setminus F_n) < 1/n$, and set $K := \bigcup_n F_n$.
Then K is F_σ, $K \subseteq E$, and $E \setminus K \subseteq E \setminus F_n$ for all n; hence $m_p^*(E \setminus K) = 0$.

(*iii*) $\Rightarrow$ (*i*). Since K is F_σ, $K \in \mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$. Also $m_p^*(E \setminus K) = 0$ implies $E \setminus K \in \mathcal{L}^p$ by Proposition 1.4.14. Thus $E = K \cup (E \setminus K) \in \mathcal{L}^p$.
(*iii*) $\Leftrightarrow$ (*iv*). If (iii) holds, set $Z := E \setminus K$; then $Z \subseteq E$, $m(Z) = 0$, and $E = K \cup Z$. Conversely, if (iv) holds, then $E \setminus K \subseteq Z$, so $m_p^*(E \setminus K) = 0$.

Structural Commentary: Inside approximation is the complement of outside approximation.
Key Insight: Measurable sets are also F_σ sets with null material added. □

Corollary 1.6.7. *Closed Inner Regularity*
If $E \in \mathcal{L}^p$, then

(i) $\forall \varepsilon > 0$, $\exists$ closed $F \subseteq E$ such that $m(E) \leq m(F) + \varepsilon$;
(ii) $m(E) = \sup\{m(F) : F \subseteq E,\ F \subseteq \mathbb{R}^p \text{ closed}\}$.

Proof.

(i) By Theorem 1.6.6, choose closed $F \subseteq E$ with $m_p^*(E \setminus F) < \varepsilon$. Since $F \in \mathcal{L}^p$ and $E = F \sqcup (E \setminus F)$, $m(E) = m(F) + m(E \setminus F) \leq m(F) + \varepsilon$.
(ii) From (i), $m(E) \leq \sup_F m(F)$. Conversely, $F \subseteq E$ gives $m(F) \leq m(E)$; take $\sup_F$.

Structural Commentary: Measurable sets are measured from inside by closed subsets.
Key Insight: Small discarded measure controls the inner regular supremum. □

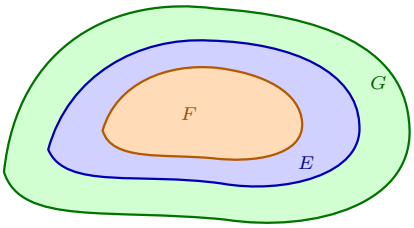


Figure 193: Regular approximation sandwiches E between F and G with small excess.

Theorem 1.6.8. *Compact Approximation at Finite Outer Measure*
Let $E \subseteq \mathbb{R}^p$ and $m_p^*(E) < +\infty$. Then $E \in \mathcal{L}^p \iff \forall \varepsilon > 0,\ \exists C \subseteq$ compact with $m_p^*(E \setminus C) < \varepsilon$.

Proof.

($\Rightarrow$) Set $E_N := E \cap \{x : \|x\|_2 \leq N\}$. Then $E_N \uparrow E$ and $m(E) < +\infty$, so choose N with $m(E \setminus E_N) < \varepsilon/2$. By Theorem 1.6.6, choose closed $C \subseteq E_N$ with $m(E_N \setminus C) < \varepsilon/2$. Then C is closed and bounded, hence compact, and $m_p^*(E \setminus C) \leq m(E \setminus E_N) + m(E_N \setminus C) < \varepsilon$.
($\Leftarrow$) Choose compact $C_n \subseteq E$ with $m_p^*(E \setminus C_n) < 1/n$, and set $K := \bigcup_n C_n$. Then K is F_σ, $K \subseteq E$, and $E \setminus K \subseteq E \setminus C_n$ $\forall n$, so $m_p^*(E \setminus K) = 0$. By Theorem 1.6.6, $E \in \mathcal{L}^p$.

Structural Commentary: Finite measure permits bounded truncation before inner closure.
Key Insight: Compact approximation is closed approximation plus a finite-measure tail estimate. □

Theorem 1.6.9. *Finite Cell Approximation in Symmetric Difference*
If $E \in \mathcal{L}^p$, $m(E) < +\infty$, then $\forall \varepsilon > 0$ $\exists$ bounded open cells $I_1, \ldots, I_N$ s.t. $K_N := \bigcup_{j=1}^N I_j$; $m(E \triangle K_N) < \varepsilon$.

Proof. Choose bounded open cells (J_k) with $E \subseteq G := \bigcup_k J_k$ and $\sum_k \mathrm{vol}_p(J_k) < m(E) + \varepsilon/2$.
Since $G \in \mathcal{L}^p$ and $E \subseteq G$, $m(G \setminus E) = m(G) - m(E) \leq \sum_k \mathrm{vol}_p(J_k) - m(E) < \varepsilon/2$..
By Theorem 1.6.8, choose compact $C \subseteq E$ with $m(E \setminus C) < \varepsilon/2$.
Since $C \subseteq \bigcup_k J_k$, compactness gives $I_1, \ldots, I_N \in \{J_k\}$ with $C \subseteq K_N := \bigcup_{j=1}^N I_j \subseteq G$.
Hence $E \triangle K_N = (E \setminus K_N) \cup (K_N \setminus E) \subseteq (E \setminus C) \cup (G \setminus E)$. Thus $m(E \triangle K_N) \leq m(E \setminus C) + m(G \setminus E) < \varepsilon$.
Structural Commentary: Compactness turns a countable open cover into a finite geometric approximant.
Key Insight: Approximate from outside by cells and from inside by a compact core. □

Corollary 1.6.10. *Cell Approximation Variant Choices*
If $E \in \mathcal{L}^p$, $m(E) < +\infty$, and $\varepsilon > 0$, then there exist bounded cells $I_1, \ldots, I_N$ such that $K := \bigcup_{j=1}^N I_j$ satisfies $m(E \triangle K) < \varepsilon$, with the cells chosen as any one of:

(i) $I_i \cap I_j = \emptyset$ for $i \neq j$, and each I_j is open;
(ii) each I_j is closed;
(iii) each I_j is half-open of a fixed orientation.

Proof.

(i) Choose compact $C \subseteq E$ and open $G \supseteq E$ with $m(E \setminus C) + m(G \setminus E) < \varepsilon$.
Cover C by bounded open cells $J_1, \ldots, J_M$ with $\overline{J_j} \subseteq G$, and set $U := \bigcup_{j=1}^{M} J_j$.
Then $C \subseteq U \subseteq G$, so $m(E \triangle U) \leq m(E \setminus C) + m(G \setminus E) < \varepsilon$.

(ii) Let $\mathcal{G}$ be finite grid from coordinate endpoints of $J_1, \ldots, J_M$. For $\mathcal{G}' \subseteq \mathcal{G}$ selecting U, $\bigsqcup_{Q \in \mathcal{G}'} Q$ gives open or fixed-orientation half-open cells, differs from U only on finitely many faces, hence by a null set.

(iii) Let $K_c := \bigcup_{j=1}^{M} \overline{J_j}$. Then K_c is a finite union of closed cells and $C \subseteq K_c \subseteq G$.
Hence $m(E \triangle K_c) \leq m(E \setminus C) + m(G \setminus E) < \varepsilon$.

Structural Commentary: Cell format changes only boundary-null pieces.
Key Insight: A compact core inside an open envelope permits finite cell variants. □

Theorem 1.6.11. *Outer Measure Additivity with One Measurable Set*
Let $E \in \mathcal{L}^p$ and $F \subseteq \mathbb{R}^p$. Then $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.

Proof. Since $E \in \mathcal{L}^p$, Caratheodory splitting: $m_p^*(E \cup F) = m(E) + m_p^*(F \setminus E)$; $m_p^*(F) = m_p^*(E \cap F) + m_p^*(F \setminus E)$.
Hence $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$
Structural Commentary: One measurable cut splits arbitrary sets exactly.
Key Insight: Split both $E \cup F$ and F along the same measurable cut E. □

Corollary 1.6.12. *Disjoint Additivity and Difference*
Let $E \in \mathcal{L}^p$ and $F \subseteq \mathbb{R}^p$.

(i) $E \cap F = \emptyset \implies m_p^*(E \cup F) = m(E) + m_p^*(F)$.
(ii) $E \subseteq F,\ m(E) < +\infty \implies m_p^*(F \setminus E) = m_p^*(F) - m(E)$.

Proof.

(i) By Theorem 1.6.11, $m_p^*(E \cup F) + m_p^*(E \cap F) = m(E) + m_p^*(F)$.
Since $E \cap F = \emptyset$ and $m_p^*(\emptyset) = 0$, $m_p^*(E \cup F) = m(E) + m_p^*(F)$.

(ii) Since $E \subseteq F$, Caratheodory splitting gives $m_p^*(F) = m(E) + m_p^*(F \setminus E)$; subtract $m(E) < +\infty$.

Structural Commentary: Exact splitting yields disjoint sums and finite differences.
Key Insight: Disjointness removes the overlap; containment isolates the remainder. □

Theorem 1.6.13. *Finite Outer Measure Inner Test*
If $m_p^*(E) < +\infty$, then $E \in \mathcal{L}^p \iff \exists B \in \mathcal{L}^p,\ B \subseteq E,\ m(B) = m_p^*(E)$.

Proof.

($\Rightarrow$) Set $B := E$. Then $B \in \mathcal{L}^p$, $B \subseteq E$, and $m(B) = m(E) = m_p^*(E)$.

($\Leftarrow$) Assume $B \in \mathcal{L}^p$, $B \subseteq E$, and $m(B) = m_p^*(E)$.
Since $m(B) < +\infty$, the difference formula gives $m_p^*(E \setminus B) = m_p^*(E) - m(B) = 0$.
Thus $E \setminus B \in \mathcal{L}^p$, and $E = B \cup (E \setminus B) \in \mathcal{L}^p$.

Structural Commentary: Exact inner mass leaves only a null remainder.
Key Insight: Equality of outer mass converts containment into measurability. □

Theorem 1.6.14. *Single Finite Container Test*
Let $A \in \mathcal{L}^p$, $m(A) < +\infty$, and $E \subseteq A$. Then $E \in \mathcal{L}^p \iff m(A) = m_p^*(E) + m_p^*(A \setminus E)$.

Proof.

($\Rightarrow$) Since $E \in \mathcal{L}^p$ and $E \subseteq A$,
Caratheodory splitting of A by E gives $m(A) = m(E) + m(A \setminus E) = m_p^*(E) + m_p^*(A \setminus E)$.

($\Leftarrow$) Choose G_δ set $H \supseteq A \setminus E$ with $m(H) = m_p^*(A \setminus E)$. Since $A \setminus E \subseteq A \cap H \subseteq H$, $m(A \cap H) = m_p^*(A \setminus E)$.
Set $B := A \setminus H$. $B \in \mathcal{L}^p$, $B \subseteq E$; $m(B) = m(A) - m(A \cap H) = m_p^*(E)$. By Theorem 1.6.13, $E \in \mathcal{L}^p$.

Structural Commentary: Inside a finite measurable container, Caratheodory only needs to be tested once.
Key Insight: Outer-approximate the complement; the remainder has full outer mass. □

Corollary 1.6.15. *Bounded Inner Measure Test*
For bounded cell $Q \subseteq \mathbb{R}^p$ and $E \subseteq Q$, define $m_{*,Q}(E) := m(Q) - m_p^*(Q \setminus E)$. Then:

(i) $E \in \mathcal{L}^p \iff m_{*,Q}(E) = m_p^*(E)$;
(ii) for arbitrary $E \subseteq \mathbb{R}^p$, $E \in \mathcal{L}^p \iff E \cap [-n, n]^p \in \mathcal{L}^p$ *for all* $n \in \mathbb{N}$.

Proof.

(i) ($\Rightarrow$) If $E \in \mathcal{L}^p$, then $m(Q) = m(E) + m_p^*(Q \setminus E)$, hence $m_{*,Q}(E) = m_p^*(E)$.

(i) ($\Leftarrow$) If $m_{*,Q}(E) = m_p^*(E)$, then $m(Q) = m_p^*(E) + m_p^*(Q \setminus E)$, so $E \in \mathcal{L}^p$ by the finite-container test.

(*ii*) ($\Rightarrow$) If $E \in \mathcal{L}^p$, then $E \cap [-n,n]^p \in \mathcal{L}^p$ for all n, since $[-n,n]^p$ is Borel.
(*ii*) ($\Leftarrow$) If $E \cap [-n,n]^p \in \mathcal{L}^p$ for all n, then $E = \bigcup_{n=1}^{\infty}(E \cap [-n,n]^p) \in \mathcal{L}^p$.

Structural Commentary: Bounded inner measure tests one finite container.
Key Insight: Equality of inner and outer measure is a bounded version of Caratheodory. □

1.7 Vitali, Steinhaus, and Nonmeasurable Sets

Definition 1.7.1. *Difference Sets and Vitali Selectors*

(i) *Difference set*: for $A \subseteq \mathbb{R}^p$, $A - A := \{x - y : x, y \in A\}$; $A \subseteq B \Longrightarrow A - A \subseteq B - B$.
(ii) *Rational equivalence*: $x \sim_{\mathbb{Q}^p} y \iff x - y \in \mathbb{Q}^p$; equivalently, $[x]_{\mathbb{Q}^p} = x + \mathbb{Q}^p$.
(iii) *Vitali selector*: choose $\mathcal{V} \subseteq \mathbb{R}^p$ such that $|\mathcal{V} \cap (x + \mathbb{Q}^p)| = 1$ for every $x \in \mathbb{R}^p$.
(iv) *Rational translate*: for $q \in \mathbb{Q}^p$, $\mathcal{V}_q := q + \mathcal{V}$.

Lemma 1.7.2. *Steinhaus Lemma for Compact Sets*
If $K \subseteq \mathbb{R}^p$ is compact and $m(K) > 0$, then $\exists \delta > 0$ such that $B(0,\delta) \subseteq K - K$.

Proof. Choose open $G \supseteq K$ with $m(G) < 2m(K)$. Since K is compact and G^c is closed, set $\delta := \operatorname{dist}(K, G^c) > 0$. If $\|x\|_2 < \delta$, then $x + K \subseteq G$. If $(x + K) \cap K = \emptyset$, translation invariance and additivity give $2m(K) = m(x + K) + m(K) = m((x + K) \cup K) \leq m(G) < 2m(K)$, a contradiction. Hence $(x + K) \cap K \neq \emptyset$, so $x = k_1 - k_2 \in K - K$. Thus $B(0,\delta) \subseteq K - K$.
Structural Commentary: A positive compact set overlaps all sufficiently small translates of itself.
Key Insight: Disjoint small translates cannot fit inside a tight open envelope. □

Theorem 1.7.3. *Steinhaus Theorem*
If $E \in \mathcal{L}^p$ and $m(E) > 0$, then $\exists \delta > 0$ such that $B(0,\delta) \subseteq E - E$.

Proof. Set $E_n := E \cap B(0,n)$. Since $E_n \uparrow E$ and $m(E) > 0$, choose n with $0 < m(E_n) < +\infty$. By compact inner approximation, choose compact $K \subseteq E_n$ with $m(K) > 0$. Lemma 1.7.2 gives $\delta > 0$ with $B(0,\delta) \subseteq K - K \subseteq E - E$.
Structural Commentary: Positive measurable sets contain positive compact cores.
Key Insight: Difference-set thickness follows from compact inner regularity. □

Lemma 1.7.4. *Rational Translates of a Vitali Selector Partition $\mathbb{R}^p$*
Let $(q_i)_{i \geq 1}$ enumerate $\mathbb{Q}^p$.
Then $\mathbb{R}^p = \bigsqcup_{i=1}^{\infty} \mathcal{V}_{q_i}$; equivalently, each $x \in \mathbb{R}^p$ has a unique representation $x = q_i + v$ with $i \in \mathbb{N}$ and $v \in \mathcal{V}$.

Proof. For $x \in \mathbb{R}^p$, choose the unique $v \in \mathcal{V} \cap (x + \mathbb{Q}^p)$; then $x - v = q_i$ for some i, so $x \in \mathcal{V}_{q_i}$. If $q + v = q' + v'$ with $q, q' \in \mathbb{Q}^p$ and $v, v' \in \mathcal{V}$, then $v - v' = q' - q \in \mathbb{Q}^p$, hence $v = v'$ by selector uniqueness and then $q = q'$.
Structural Commentary: Rational translates enumerate the equivalence classes selected by $\mathcal{V}$.
Key Insight: One representative per class makes the rational translates disjoint and exhaustive. □

Theorem 1.7.5. *Vitali Selectors Are Nonmeasurable*
If $\mathcal{V} \subseteq \mathbb{R}^p$ is a Vitali selector, then $\mathcal{V} \notin \mathcal{L}^p$.

Proof. Assume $\mathcal{V} \in \mathcal{L}^p$. If $m(\mathcal{V}) > 0$, Steinhaus gives $\delta > 0$ with $B(0,\delta) \subseteq \mathcal{V} - \mathcal{V}$. Choose $0 \neq q \in \mathbb{Q}^p \cap B(0,\delta)$; then $q = v_1 - v_2$ for $v_1, v_2 \in \mathcal{V}$, so $v_1 \sim_{\mathbb{Q}^p} v_2$, forcing $v_1 = v_2$ and $q = 0$, a contradiction.
If $m(\mathcal{V}) = 0$, then each $\mathcal{V}_{q_i}$ is null by translation invariance, and $\mathbb{R}^p = \bigsqcup_i \mathcal{V}_{q_i}$ would imply every bounded cell has measure 0, contradicting positive cell volume.
Structural Commentary: Positive and zero measure are both impossible.
Key Insight: Steinhaus forbids positive measure; countable translated partition forbids zero measure. □

Theorem 1.7.6. *Nonmeasurable Subsets Are Everywhere*
If $E \subseteq \mathbb{R}^p$ and $m_p^*(E) > 0$, then $\exists N \subseteq E$ with $N \notin \mathcal{L}^p$.

Proof. Let (q_i) enumerate $\mathbb{Q}^p$ and set $E_i := E \cap \mathcal{V}_{q_i}$. If $E_i \in \mathcal{L}^p$ and $m(E_i) > 0$, then $E_i - E_i$ contains a ball about 0, so $0 \neq q \in \mathbb{Q}^p \cap (E_i - E_i)$. But $E_i \subseteq q_i + \mathcal{V}$, so $q = (q_i + v_1) - (q_i + v_2) = v_1 - v_2$ with $v_1, v_2 \in \mathcal{V}$, contradicting selector uniqueness. Hence each measurable E_i is null. If every E_i were measurable, then $E = \bigcup_i E_i$ would satisfy $m_p^*(E) = 0$, contradiction. Thus some $E_i \subseteq E$ is nonmeasurable.
Structural Commentary: Any positive outer-measure set meets the Vitali partition in a nonmeasurable piece.
Key Insight: Vitali pieces cannot be measurable and positive. □

Corollary 1.7.7. *Nonadditive Decompositions*

(i) If $E \in \mathcal{L}^p$ and $0 < m(E) < +\infty$, then $\exists B, C \notin \mathcal{L}^p$ such that $E = B \sqcup C$ and $m(E) < m_p^*(B) + m_p^*(C)$.

(ii) If $B \notin \mathcal{L}^p$, $m_p^*(B) < +\infty$, and H is G_δ with $B \subseteq H$, $m(H) = m_p^*(B)$, then $C := H \setminus B \notin \mathcal{L}^p$ and $m(H) = m_p^*(B \cup C) < m_p^*(B) + m_p^*(C)$.

Proof.

(i) Pick $B \subseteq E$, $B \notin \mathcal{L}^p$, and put $C := E \setminus B$. Then $C \notin \mathcal{L}^p$, else $B = E \setminus C \in \mathcal{L}^p$. Subadditivity gives $\leq$; equality would force $B \in \mathcal{L}^p$ by the finite-container test in E.

(ii) Since $H \in \mathcal{L}^p$, $C \in \mathcal{L}^p$ would give $B = H \setminus C \in \mathcal{L}^p$, contradiction. Also $m_p^*(C) > 0$, else $C \in \mathcal{L}^p$. Thus $B \cup C = H$ and $m(H) = m_p^*(B)$ give $m_p^*(B \cup C) = m(H) < m_p^*(B) + m_p^*(C)$.

Structural Commentary: Nonmeasurability appears exactly as strict failure of additive splitting.
Key Insight: Equality in a finite measurable container is a measurability test. □

Theorem 1.7.8. *A Nonmeasurable Set with Only Null Measurable Pieces*
There exists $\mathcal{U} \subseteq \mathbb{R}^p$ with $\mathcal{U} \notin \mathcal{L}^p$ such that, for every $F \in \mathcal{L}^p$,

$$F \subseteq \mathcal{U} \Longrightarrow m(F) = 0,$$
$$F \subseteq \mathcal{U}^c \Longrightarrow m(F) = 0.$$

Proof. Let $e := (1, \dots, 1) \in \mathbb{R}^p$, $G := \mathbb{Q}^p + \sqrt{2}\,\mathbb{Z}e$, $G_0 := \mathbb{Q}^p + (2\sqrt{2})\,\mathbb{Z}e$, and $G_1 := \sqrt{2}\,e + G_0$. Then $G = G_0 \sqcup G_1$, and G_0, G_1 are dense. Choose $\mathcal{E} \subseteq \mathbb{R}^p$ with one representative from each class of $x - y \in G$, and set $\mathcal{U} := G_0 + \mathcal{E}$.
Unique representatives give $\mathbb{R}^p = (G_0 + \mathcal{E}) \sqcup (G_1 + \mathcal{E})$, hence $\mathcal{U}^c = G_1 + \mathcal{E} = \sqrt{2}\,e + \mathcal{U}$.
Let $F \in \mathcal{L}^p$, $F \subseteq \mathcal{U}$, and $m(F) > 0$. Then $B(0, r) \subseteq F - F$ for some $r > 0$; choose $g_1 \in G_1 \cap B(0, r)$.
Since $F \subseteq G_0 + \mathcal{E}$, $g_1 = (g_0 + s_1) - (g_0' + s_2)$ for some $g_0, g_0' \in G_0$ and $s_1, s_2 \in \mathcal{E}$.
Thus $s_1 - s_2 = g_1 - g_0 + g_0' \in G$, so $s_1 = s_2$ and $g_1 = g_0 - g_0' \in G_0$, impossible.
Hence every measurable $F \subseteq \mathcal{U}$ is null; translation gives the same for $\mathcal{U}^c$.
If $\mathcal{U} \in \mathcal{L}^p$, then $\mathcal{U}$ and $\mathcal{U}^c$ are null, so $m([0,1]^p) = 0$, contradiction.
Structural Commentary: This construction partitions representatives using two dense cosets.
Key Insight: Positive measurable pieces force forbidden coset differences. □

1.8 Cantor–Lebesgue Singular Function

Definition 1.8.1. *Cantor–Lebesgue Singular Function*
Let $P \subseteq [0,1]$ be the Cantor set. For $x = (0.c_1c_2\cdots)_3 \in P$ with $c_k \in \{0, 2\}$, set $\varphi(x) := (0.(c_1/2)(c_2/2)\cdots)_2$.
If (a, b) is a component of $[0,1] \setminus P$, extend by $\varphi(x) := \varphi(a) = \varphi(b)$ for $x \in (a, b)$.
The resulting $\varphi : [0,1] \to [0,1]$ is the *Cantor–Lebesgue singular function*.

Proposition 1.8.2. *Properties of the Singular Function*
Let $P \subseteq [0,1]$ be the Cantor set and let $\varphi : [0,1] \to [0,1]$ be the Cantor–Lebesgue singular function. Then φ is continuous, nondecreasing, onto, nonconstant, and $\varphi' = 0$ on $[0,1] \setminus P$.

Proof. On P, the digit map $\tau((0.c_1c_2\cdots)_3) := (0.(c_1/2)(c_2/2)\cdots)_2$ is nondecreasing and onto $[0,1]$.
The only duplications are endpoint pairs of removed middle-third intervals, where the binary values agree; constant extension across those intervals is therefore nondecreasing with no jumps, hence continuous.
Each component of $[0,1] \setminus P$ is an interval of constancy, so $\varphi' = 0$ there. Since $\varphi([0,1]) = [0,1]$, φ is nonconstant.
Structural Commentary: Variation is confined to P; the range is $[0,1]$.
Key Insight: Ternary $0/2$ digits are re-read as binary digits. □

Theorem 1.8.3. *Lebesgue Measurable Non-Borel Sets Exist*
There exist $A \in \mathcal{L}^1 \setminus \mathcal{B}(\mathbb{R})$ and a homeomorphism $\psi : [0,1] \to [0,2]$ such that $\psi(A) \notin \mathcal{L}^1$.

Proof. Let $\psi(x) := x + \varphi(x)$, where φ is the Cantor–Lebesgue singular function. Since φ is continuous and nondecreasing, ψ is continuous, strictly increasing, maps $[0,1]$ onto $[0,2]$, and hence is a homeomorphism.
Let P be the Cantor set and $U := [0,1] \setminus P$. On each component I of U, φ is constant, so $m(\psi(I)) = m(I)$; the intervals $\psi(I)$ are disjoint. Hence $m(\psi(U)) = m(U) = 1$.
Since $[0,2] = \psi(P) \sqcup \psi(U)$ and $\psi(P)$ is compact, $m(\psi(P)) = 1$. Choose $W \subseteq \psi(P)$ with $W \notin \mathcal{L}^1$, and set $A := \psi^{-1}(W) \subseteq P$. Since $m(P) = 0$, $A \in \mathcal{L}^1$. If $A \in \mathcal{B}(\mathbb{R})$, then $W = \psi(A)$ is Borel in $[0,2]$, hence Lebesgue measurable, contradiction. Thus $A \in \mathcal{L}^1 \setminus \mathcal{B}(\mathbb{R})$ and $\psi(A) = W \notin \mathcal{L}^1$.
Structural Commentary: Completion by null sets creates Lebesgue sets invisible to the Borel hierarchy.
Key Insight: A null Cantor set can be homeomorphically stretched to positive measure. □

2 Measurable Functions and Lebesgue Integration

2.1 Measurable Functions

Definition 2.1.1. *Borel, Lebesgue, and Function Measurability*

(i) *Borel*: $\mathcal{B}(\mathbb{R}^p) := \sigma(\mathcal{O}_{\mathbb{R}^p})$; its elements are *Borel sets*.
(ii) *Lebesgue*: $\mathcal{B}(\mathbb{R}^p) \subseteq \mathcal{L}^p$, and $\mathcal{L}^p$ is obtained by adjoining subsets of Lebesgue null sets.
(iii) *Maps*: measurable spaces $(X,\mathcal{M})$, $(Y,\mathcal{N})$, $f : X \to Y$ is *$(\mathcal{M},\mathcal{N})$-measurable* iff $f^{-1}(E) \in \mathcal{M} \forall E \in \mathcal{N}$.
(iv) *Borel maps*: $f : \mathbb{R}^p \to \mathbb{R}^q$ is *Borel measurable* iff $f^{-1}(B) \in \mathcal{B}(\mathbb{R}^p)$ for all $B \in \mathcal{B}(\mathbb{R}^q)$.
(v) *Thresholds*: write $\{f \bowtie a\} := \{x \in X : f(x) \bowtie a\}$ for $\bowtie \in \{>, \geq, <, \leq\}$; extended-real f is *$\mathcal{M}$-measurable* iff $\{f > a\} \in \mathcal{M}$ for all $a \in \mathbb{R}$.

Structural Commentary: Borel measurability tracks topology; Lebesgue measurability adds null-set completion.
Key Insight: Measurability is inverse-image stability.

Proposition 2.1.2. *Generator Test for Measurable Maps*
Let $f : X \to Y$, $\mathcal{A} \subseteq \mathcal{P}(Y)$, and $\mathcal{N} := \sigma(\mathcal{A})$.

(i) If $f^{-1}(A) \in \mathcal{M}$ for all $A \in \mathcal{A}$, then $f : X \to (Y,\mathcal{N})$ is measurable.
(ii) For real-valued f, $\forall a \in \mathbb{R},\ \{f > a\} \in \mathcal{M} \iff \forall B \in \mathcal{B}(\mathbb{R}),\ f^{-1}(B) \in \mathcal{M}$.

Proof.

(i) Let $\mathcal{G} := \{E \subseteq Y : f^{-1}(E) \in \mathcal{M}\}$. Since $f^{-1}(E^c) = f^{-1}(E)^c$ and $f^{-1}(\bigcup_n E_n) = \bigcup_n f^{-1}(E_n)$, $\mathcal{G}$ is a σ-algebra. Thus $\mathcal{A} \subseteq \mathcal{G} \Longrightarrow \mathcal{N} = \sigma(\mathcal{A}) \subseteq \mathcal{G}$.
(*ii*) ($\Rightarrow$) Put $\mathcal{A} := \{(a,+\infty) : a \in \mathbb{R}\}$. Since $f^{-1}((a,+\infty)) = \{f > a\} \in \mathcal{M}$ for all a and $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$, (i) gives $f^{-1}(B) \in \mathcal{M}$ for all $B \in \mathcal{B}(\mathbb{R})$.
(*ii*) ($\Leftarrow$) Assume $f^{-1}(B) \in \mathcal{M}$ for every $B \in \mathcal{B}(\mathbb{R})$.
For each $a \in \mathbb{R}$, $(a,+\infty) \in \mathcal{B}(\mathbb{R})$, hence $\{f > a\} = f^{-1}((a,+\infty)) \in \mathcal{M}$.

Structural Commentary: Measurability can be checked on any generating family of target sets.
Key Insight: Pullbacks turn generated σ-algebras into generated collections of preimages. □

Proposition 2.1.3. *Open-Set Test for Euclidean Targets*
Let $\Omega \subseteq \mathbb{R}^n$ be Lebesgue measurable and let $f : \Omega \to \mathbb{R}^m$. The following tests are equivalent:

(i) $f^{-1}(V)$ is measurable for every open $V \subseteq \mathbb{R}^m$;
(ii) $f^{-1}(B)$ is measurable for every open box $B \subseteq \mathbb{R}^m$;
(iii) every coordinate function $f_j : \Omega \to \mathbb{R}$ is threshold-measurable.

In particular, every continuous $f : \Omega \to \mathbb{R}^m$ is measurable.

Proof.

(*i*) $\Rightarrow$ (*ii*). Open boxes are open.
(*ii*) $\Rightarrow$ (*i*). By Lemma 1.4.7, write $V = \bigcup_k B_k$ with rational boxes B_k.
Then $f^{-1}(V) = \bigcup_k f^{-1}(B_k)$ is measurable.
(*ii*) $\Rightarrow$ (*iii*). By the previous implication, $f^{-1}(U)$ is measurable for every open $U \subseteq \mathbb{R}^m$.
Hence $\{f_j > a\} = f^{-1}(\{y : y_j > a\})$ is measurable.
(*iii*) $\Rightarrow$ (*ii*). If $B = \prod_{j=1}^m (a_j, b_j)$, then $f^{-1}(B) = \bigcap_{j=1}^m \big(\{f_j > a_j\} \cap \{f_j < b_j\}\big)$,

where $\{f_j < b_j\} = \Omega \setminus \bigcap_{\ell=1}^{\infty} \{f_j > b_j - 1/\ell\}$. Thus each factor is measurable, hence so is $f^{-1}(B)$.

If f is continuous, Theorem 5.1.18 makes $f^{-1}(V)$ open relative to Ω.
By Theorem 3.2.2, it is $W \cap \Omega$ for some open $W \subseteq \mathbb{R}^n$, hence measurable by Theorem 1.4.8.
Structural Commentary: Euclidean-valued measurability is just countable box algebra applied to inverse images.
Key Insight: Open sets reduce to rational boxes; vector functions reduce to coordinate threshold tests. □

Example 2.1.4. *Basic Measurable Functions*
Let $(X,\mathcal{M})$ be a measurable space.

(i) Constant real-valued functions are measurable, since each strict superlevel set is either X or $\emptyset$.
(ii) For $E \subseteq X$, the indicator $\mathbf{1}_E$ is measurable iff $E \in \mathcal{M}$.
(iii) Continuous maps between Euclidean measurable subsets are measurable by Proposition 2.1.3.
(iv) Monotone functions $\mathbb{R} \to \mathbb{R}$ are Borel measurable, since their strict superlevel sets are half-lines, $\mathbb{R}$, or $\emptyset$.

Key Insight: Constants, indicators of measurable sets, continuous functions, and monotone functions form the first reusable stock of measurable functions.

Theorem 2.1.5. *Closure Properties of Measurable Functions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) The four tests $\{f > a\}, \{f \geq a\}, \{f < a\}, \{f \leq a\} \in \mathcal{M}$ for all $a \in \mathbb{R}$ are equivalent.
(ii) f measurable implies $|f|$ measurable.
(iii) If f_n are measurable, then $\sup_n f_n, \inf_n f_n, \limsup_n f_n, \liminf_n f_n$ are measurable.
(iv) If f, g real-valued measurable, $F : \mathbb{R}^2 \to \mathbb{R}$ continuous, then $F(f, g)$ measurable; so are $f + g$ and fg.

Proof.

(i) Use $\{f \geq a\} = \bigcap_{m \geq 1} \{f > a - 1/m\}$; $\{f < a\} = X \setminus \{f \geq a\}$; $\{f \leq a\} = \bigcap_{m \geq 1} \{f < a + 1/m\}$.
(ii) For $a > 0$, $\{|f| < a\} = \{f < a\} \cap \{f > -a\}$; for $a \leq 0$ the set is empty.
(iii) $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$. Also $\limsup_n f_n = \inf_m \sup_{n \geq m} f_n$; $\liminf_n f_n = \sup_m \inf_{n \geq m} f_n$; inf follow from (i).
(iv) For $G_a = \{(u, v) : F(u, v) > a\}$, write $G_a = \bigcup_k (\alpha_k, \beta_k) \times (\gamma_k, \delta_k)$.

Then $\{F(f, g) > a\} = \bigcup_k \big(\{\alpha_k < f < \beta_k\} \cap \{\gamma_k < g < \delta_k\}\big) \in \mathcal{M}$.

Structural Commentary: Measurability is countable set algebra applied to level sets.
Key Insight: Once threshold sets are measurable, ordinary limiting and algebraic operations stay measurable. □

Definition 2.1.6. *Positive and Negative Parts*
For an extended-real-valued function f, define $f^+(x) := \sup\{f(x), 0\}$; $f^-(x) := \sup\{-f(x), 0\}$.
When f is finite-valued, $f = f^+ - f^-$ and $|f| = f^+ + f^-$; equivalently $f^+ = \frac{|f| + f}{2}$; $f^- = \frac{|f| - f}{2}$.

Corollary 2.1.7. *Measurability of Positive and Negative Parts*
If $f : X \to [-\infty, +\infty]$ is measurable, then $f^+, f^-, |f|$ are measurable.
If $f : X \to \mathbb{R}$, then f measurable $\iff$ f^+, f^- measurable.

Proof.

($\Rightarrow$) Since $\{-f > a\} = \{f < -a\}$, $-f$ is measurable.
Thus $f^+ = f \vee 0$, $f^- = (-f) \vee 0$, and $|f| = f \vee (-f)$ are measurable by finite sup-closure.
($\Leftarrow$) If $f : X \to \mathbb{R}$ and f^+, f^- are measurable, then $f = f^+ - f^-$ is measurable.

Structural Commentary: Sign decomposition is compatible with threshold-set measurability.
Key Insight: Reduce signs to suprema and finite algebra. □

Lemma 2.1.8. *Extended-Valued Measurability via the Finite Part*
Let $f : X \to \overline{\mathbb{R}}$ and set $A := f^{-1}(\{+\infty\})$, and $B := f^{-1}(\{-\infty\})$. Define $f_0(x) := \begin{cases} f(x), & x \notin A \cup B, \\ 0, & x \in A \cup B. \end{cases}$.
Then f is measurable iff $A, B \in \mathcal{M}$ and $f_0 : X \to \mathbb{R}$ is measurable.

Proof.

($\Rightarrow$) If f is measurable, then $A = \bigcap_{n=1}^{\infty} \{f > n\}$; and $B = X \setminus \bigcup_{n=1}^{\infty} \{f > -n\}$, so $A, B \in \mathcal{M}$.

Also, $\{f_0 > \alpha\} = \begin{cases} \{f > \alpha\} \setminus A, & \alpha \geq 0, \\ \{f > \alpha\} \cup B, & \alpha < 0, \end{cases}$, so f_0 is measurable.

($\Leftarrow$) If $A, B \in \mathcal{M}$, f_0 is measurable, then $\{f > \alpha\} = \begin{cases} \{f_0 > \alpha\} \cup A, & \alpha \geq 0, \\ \{f_0 > \alpha\} \setminus B, & \alpha < 0, \end{cases}$, so $\{f > \alpha\} \in \mathcal{M}$ $\forall \alpha \in \mathbb{R}$.

Structural Commentary: The infinite-level sets are measurable exactly when the finite core is measurable.
Key Insight: Assign harmless finite values at $\pm\infty$, then put the infinite fibers back by set algebra. □

Corollary 2.1.9. *Extended-Valued Algebra Conventions*
Let $f, g : X \to \overline{\mathbb{R}}$ be measurable.

(i) For $c \in \mathbb{R}$, cf, $|f|$, f^+, and f^- are measurable, with $0(\pm\infty) = 0$.
(ii) Let $E_1 := \{f = -\infty,\ g = +\infty\}$, $E_2 := \{f = +\infty,\ g = -\infty\}$, and $F := E_1 \cup E_2$.
If $h = f + g$ on $X \setminus F$ and $h = 0$ on F, then h is measurable.
(iii) With $0(\pm\infty) = 0$, fg is measurable.

Proof.

(i) By Lemma 2.1.8, $A_\pm(f) := \{f = \pm\infty\} \in \mathcal{M}$ and f_0 is measurable. For $c \in \mathbb{R}$, the infinite fibers of cf are among $A_\pm(f)$ and $(cf)_0 = cf_0$, so cf is measurable. Corollary 2.1.7 gives $|f|, f^+, f^-$.

(ii) $E_1, E_2 \in \mathcal{M}$. Set $f_n := \max\{-n, \min(f, n)\}$ and $g_n := \max\{-n, \min(g, n)\}$.
Then f_n, g_n are real-valued measurable and $(f_n + g_n)\mathbf{1}_{X\setminus F} \to h$ pointwise; hence h is measurable.
(iii) With f_n, g_n as in (ii), $f_n g_n$ measurable, $f_n g_n \to fg$ pointwise under $0(\pm\infty) = 0$; hence fg measurable.

Structural Commentary: Extended algebra is finite algebra plus measurable exceptional fibers.
Key Insight: Truncate first, use real-valued closure, then pass to pointwise limits. □

Definition 2.1.10. *Simple Functions*
Let $(X, \mathcal{M})$ be a measurable space.

(i) *Indicator*: for $E \subseteq X$, $\mathbf{1}_E(x) := 1$ on E and $\mathbf{1}_E(x) := 0$ on $X \setminus E$.
(ii) *Simple*: a real-valued s is simple iff $s(X)$ is finite.
(iii) *Standard representation*: if $s(X) = \{c_1, \dots, c_n\}$ and $E_i := \{s = c_i\} \neq \emptyset$, then $X = \bigsqcup_{i=1}^{n} E_i$; $s = \sum_{i=1}^{n} c_i \mathbf{1}_{E_i}$.
The pairs (c_i, E_i) are unique up to reordering.
(iv) *Measurability*: s measurable iff $E_i \in \mathcal{M}$ $\forall i$; step functions are special case $X \subseteq \mathbb{R}$ with interval level sets.

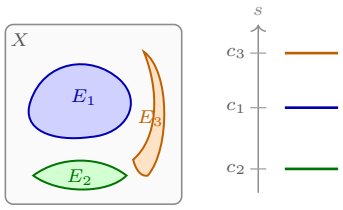


Figure 194: A simple function takes value c_i on each measurable piece E_i.

Proposition 2.1.11. *Simple Functions Form a Lattice Algebra*
Let $s, t : X \to \mathbb{R}$ be simple and $a \in \mathbb{R}$. Then as, $s + t$, st, $s \vee t$, and $s \wedge t$ are simple.
If s, t are measurable, so are these five functions.

Proof. Let $s(X) = \{c_i\}_{i=1}^m$ and $t(X) = \{d_j\}_{j=1}^n$. Each displayed range is contained in the corresponding finite set $\{ac_i\}$, $\{c_i + d_j\}$, $\{c_i d_j\}$, $\{c_i \vee d_j\}$, or $\{c_i \wedge d_j\}$, hence is finite.
If s, t are measurable, the atoms $P_{ij} := \{s = c_i\} \cap \{t = d_j\}$ are measurable and form a finite partition of X. Each displayed function is constant on every P_{ij}, so its level sets are finite unions of P_{ij}.
Structural Commentary: Algebra and lattice operations reduce to finite tables.
Key Insight: Refine two simple functions by the common partition of their level sets. □

Theorem 2.1.12. *Approximation by Simple Functions*
Every real f on X is the pointwise limit of simple functions s_n.
If f is measurable, choose s_n measurable; if $f \geq 0$, choose $0 \leq s_n \uparrow f$.

Proof. For $f \geq 0$, set $E_{ni} = \left\{x : \frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}\right\}$; $F_n = \{f \geq n\}$; and $s_n = \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbf{1}_{E_{ni}} + n\mathbf{1}_{F_n}$.
Then s_n is simple, measurable when f is, and $0 \leq s_n \uparrow f$. For general f, approximate f^+ and f^- and subtract.
Structural Commentary: Simple functions quantize the range of f.
Key Insight: Dyadic range partitions give pointwise approximation while preserving measurability. □

Definition 2.1.13. *Complex-Valued Measurable Functions*
A map $f : X \to \mathbb{C}$ is *measurable* iff $\operatorname{Re} f, \operatorname{Im} f : X \to \mathbb{R}$ are measurable.
Equivalently, $f = \operatorname{Re} f + i \operatorname{Im} f$ with measurable real components.

Proposition 2.1.14. *Complex Measurable Closure*
If $f, g : X \to \mathbb{C}$ are measurable, $a \in \mathbb{C}$, and $h_n : X \to \mathbb{C}$ are measurable with $h_n \to h$ pointwise, then $f + g$, fg, af, and h are measurable.

Proof. Write $f = u + iv$, $g = r + is$, and $a = \alpha + i\beta$. Then $f + g = (u + r) + i(v + s)$, $af = (\alpha u - \beta v) + i(\alpha v + \beta u)$, and $fg = (ur - vs) + i(us + vr)$, so real closure gives measurability.
If $h_n \to h$, then $\operatorname{Re} h_n \to \operatorname{Re} h$ and $\operatorname{Im} h_n \to \operatorname{Im} h$, so pointwise-limit closure gives measurability.
Structural Commentary: Complex measurability is two real measurability statements.
Key Insight: Split into real and imaginary parts. □

Definition 2.1.15. *Almost Everywhere and In Measure*
Let E be measurable.

(i) *A.E. equality*: $f \sim_{\text{a.e.}} g$ on E iff $\mu(\{x \in E : f(x) \neq g(x)\}) = 0$.

(ii) *A.E. property*: a property holds a.e. on E iff its failure set in E is null.
(iii) *Null invariance*: $\sim_{\text{a.e.}}$ is an equivalence relation, and $f \sim_{\text{a.e.}} g$ on E preserves integrals over measurable $A \subseteq E$ whenever they exist.
(iv) *A.E. convergence*: $f_n \to f$ a.e. on E iff $\exists N \subseteq E$ null such that $f_n(x) \to f(x)$ for all $x \in E \setminus N$.
(v) *In measure*: $f_n \to f$ in measure on E iff $\forall \varepsilon > 0$, $\mu(\{x \in E : |f_n(x) - f(x)| > \varepsilon\}) \to 0$.

If $f \in \mathscr{L}^1(\mu)$ on E, then f is finite a.e. on E.

Remark 2.1.16. *Completion and Almost-Everywhere Limits*
If $E \subseteq N \in \mathcal{M}$, $\mu(N) = 0$, and $E \notin \mathcal{M}$, then $f_n \equiv 0$ is measurable and $f_n \to \mathbf{1}_E$ a.e.,
since $\{x : f_n(x) \not\to \mathbf{1}_E(x)\} \subseteq N$. Yet $\mathbf{1}_E$ is not $\mathcal{M}$-measurable.
In the completion, every $E \subseteq N$ is measurable, so null modifications preserve measurability.
Key Insight: Almost-everywhere statements are most stable on complete measure spaces.

Proposition 2.1.17. *Completed Measurable Functions Have Original Representatives*
Let $(X, \mathcal{M}, \mu)$ be a measure space with completion $(X, \overline{\mathcal{M}}_\mu, \overline{\mu})$.
If $f : X \to \overline{\mathbb{R}}$ is $\overline{\mathcal{M}}_\mu$-measurable, then $\exists$ $\mathcal{M}$-measurable $g : X \to \overline{\mathbb{R}}$ with $f = g$ μ-a.e.

Proof. For $q \in \mathbb{Q}$, let $A_q := \{f > q\} \in \overline{\mathcal{M}}_\mu$, let $A_q = E_q \cup Z_q$, with $E_q \in \mathcal{M}$ and $Z_q \subseteq N_q \in \mathcal{M}$, $\mu(N_q) = 0$.
Put $N := \bigcup_{q \in \mathbb{Q}} N_q$, so $N \in \mathcal{M}$ and $\mu(N) = 0$, and define $g(x) := \begin{cases} f(x), & x \notin N, \\ 0, & x \in N. \end{cases}$.
For $q \in \mathbb{Q}$, $\{g > q\} = (E_q \setminus N) \cup \begin{cases} N, & q < 0, \\ \emptyset, & q \geq 0, \end{cases}$ belongs to $\mathcal{M}$. Hence, for $a \in \mathbb{R}$, $\{g > a\} = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} \{g > q\} \in \mathcal{M}$.
Thus g is $\mathcal{M}$-measurable and $f = g$ on $X \setminus N$.
Structural Commentary: Completion only changes measurable functions on null sets.
Key Insight: Replace completed rational levels off one null set. □

Method 2.1.18. *Measure Proof Patterns*

(i) *Disjointize*. If $A_n \uparrow$, set $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$; then $A_n = \bigsqcup_{k \leq n} B_k$ and $\bigcup_n A_n = \bigsqcup_n B_n$.
(ii) *Approximate*. Prove on $\mathcal{E}$; if $d_\mu(A_n, A), d_\mu(B_n, B) \to 0$ with $d_\mu(C, D) := \mu^*(C \triangle D)$, pass limits through $\cup, \cap, \setminus$ via $\triangle$ estimates.
(iii) *Threshold*. For real f, show $\{f > a\} \in \mathcal{M}$ for all $a \in \mathbb{R}$; for $f : X \to \mathbb{R}^m$, use coordinate/open-set tests.
(iv) *Generators*. For $f : X \to (Y, \sigma(\mathcal{A}))$, check $f^{-1}(A) \in \mathcal{M}$ for all $A \in \mathcal{A}$.
(v) *Truncate*. For $f : X \to \overline{\mathbb{R}}$, prove results for $f_n := (-n) \vee (f \wedge n)$, then let $n \to \infty$.
(vi) *Topological bridge*. Continuous maps satisfy $f^{-1}(V)$ open; reduce Borel tests to countable interval or rectangle preimages.

Key Insight: Measure theory alternates between countable set algebra and approximation by elementary sets.

2.2 Lebesgue Integration

Lebesgue integration starts with simple functions, then extends by monotone approximation and positive–negative decomposition.

Definition 2.2.1. *Lebesgue Integral*
Let $(X, \mathcal{M}, \mu)$ be a measure space and $E \in \mathcal{M}$.

(i) *Simple*: if $0 \leq s = \sum_{i=1}^n c_i \mathbf{1}_{E_i}$ with $c_i \geq 0$, $E_i \in \mathcal{M}$ disjoint, set $\mathcal{I}_E(s) := \sum_{i=1}^n c_i \mu(E \cap E_i)$; $0 \cdot (+\infty) := 0$.
(ii) *Nonnegative*: measurable $f \geq 0$, $\mathcal{S}^+(f) := \{s : 0 \leq s \leq f,\ s \text{ simple measurable}\}$; $\int_E f \, d\mu := \sup_{s \in \mathcal{S}^+(f)} \mathcal{I}_E(s)$
(iii) *Signed*: real measurable f, $f^+ := f \vee 0$, $f^- := (-f) \vee 0$.
If at least one of $\int_E f^+ \, d\mu$, $\int_E f^- \, d\mu$ is finite, define $\int_E f \, d\mu := \int_E f^+ \, d\mu - \int_E f^- \, d\mu$.
(iv) *Integrable*: f is Lebesgue integrable on E, written $f \in \mathscr{L}^1(\mu)$ on E, iff $\int_E f^+ \, d\mu, \int_E f^- \, d\mu < +\infty$.
(v) *Whole space*: if $E = X$, this gives the real-valued integrable, or summable, functions on $(X, \mathcal{M}, \mu)$.

Proposition 2.2.2. *Signed Decompositions Give the Same Integral*
Let $E \in \mathcal{M}$ and $f = f_1 - f_2$ on E, with $f_i \geq 0$ measurable and $\int_E f_i \, d\mu < +\infty$ for $i = 1, 2$.
Then $f \in \mathscr{L}^1(\mu)$ on E and $\int_E f \, d\mu = \int_E f_1 \, d\mu - \int_E f_2 \, d\mu$.

Proof. On E, $f^+ \leq f_1$, $f^- \leq f_2$, hence $\int_E f^\pm \, d\mu < +\infty$.
Also $f^+ + f_2 = f_1 + f^-$, so nonnegative additivity gives $\int_E f^+ \, d\mu + \int_E f_2 \, d\mu = \int_E f_1 \, d\mu + \int_E f^- \, d\mu$.
Subtract the finite terms $\int_E f^- \, d\mu$ and $\int_E f_2 \, d\mu$.
Structural Commentary: Any finite nonnegative decomposition gives the same signed value.
Key Insight: Move all positive pieces to one side, use nonnegative additivity, then subtract finite numbers. □

Simple functions are finite linear combinations of indicators of measurable sets: coefficients carry size, while the level sets carry measurability.

Lemma 2.2.3. *Refining a Simple Integral*
Let $s \geq 0$ be simple measurable and suppose $s = \sum_{k=1}^{m} b_k \mathbf{1}_{F_k}$ with $b_k \geq 0$ and $F_k \in \mathcal{M}$.
Then, for every $E \in \mathcal{M}$, $\mathcal{I}_E(s) = \sum_{k=1}^{m} b_k \mu(E \cap F_k)$.

Proof. For $\eta \in \{0,1\}^m$, set $F_k^1 := F_k$, $F_k^0 := F_k^c$, discard empty atoms $G_\eta := \bigcap_{k=1}^{m} F_k^{\eta_k}$. On G_η, $s = c_\eta := \sum_{\eta_k=1} b_k$.
Group equal c_η values to recover the standard representation; finite additivity gives
$\mathcal{I}_E(s) = \sum_\eta c_\eta \mu(E \cap G_\eta) = \sum_{k=1}^{m} b_k \sum_{\eta:\eta_k=1} \mu(E \cap G_\eta) = \sum_{k=1}^{m} b_k \mu(E \cap F_k)$.
Structural Commentary: Refine nonstandard sums to one finite partition.
Key Insight: Refine first, then regroup by equal function values. □

Proposition 2.2.4. *Simple Integral Algebra*
If $s, t \geq 0$ are simple measurable, $c \geq 0$, and $E \in \mathcal{M}$, then $\mathcal{I}_E(cs) = c\mathcal{I}_E(s)$; $\mathcal{I}_E(s+t) = \mathcal{I}_E(s) + \mathcal{I}_E(t)$.
For fixed $s \geq 0$, $\nu_s(A) := \mathcal{I}_A(s)$ is a measure on $\mathcal{M}$.

Proof. Let $s = \sum_i a_i \mathbf{1}_{A_i}$ and $t = \sum_j b_j \mathbf{1}_{B_j}$ be standard.
Homogeneity follows from $\mathcal{I}_E(cs) = \sum_i (ca_i)\mu(E \cap A_i)$ and $0 \cdot (+\infty) = 0$.
On the common refinement, $s + t = \sum_{i,j} (a_i + b_j) \mathbf{1}_{A_i \cap B_j}$.
By Lemma 2.2.3, $\mathcal{I}_E(s+t) = \sum_{i,j} (a_i + b_j)\mu(E \cap A_i \cap B_j) = \sum_i a_i \mu(E \cap A_i) + \sum_j b_j \mu(E \cap B_j)$.
Finally, $\nu_s = \sum_i a_i \, \mu(\cdot \cap A_i)$ is a finite nonnegative linear combination of measures.
Structural Commentary: The integral is additive first on finite measurable partitions.
Key Insight: Add simple functions on the common refinement of their level sets. □

Proposition 2.2.5. *Monotonicity for Nonnegative Integrals*
Let $f, g \geq 0$ be measurable and let $E, F \in \mathcal{M}$.
(i) If $f \leq g$ on E, then $\int_E f \, d\mu \leq \int_E g \, d\mu$.
(ii) If $E \subseteq F$, then $\int_E f \, d\mu \leq \int_F f \, d\mu$.

Proof.
(i) If $s \in \mathcal{S}^+(f)$, then $s\mathbf{1}_E \in \mathcal{S}^+(g)$, $\mathcal{I}_E(s) = \mathcal{I}_E(s\mathbf{1}_E) \leq \int_E g \, d\mu$. Taking $\sup_{s \in \mathcal{S}^+(f)}$ gives $\int_E f \, d\mu \leq \int_E g \, d\mu$.
(ii) If $E \subseteq F$ and $s \in \mathcal{S}^+(f)$, then $\mathcal{I}_E(s) \leq \mathcal{I}_F(s)$. Taking $\sup_{s \in \mathcal{S}^+(f)}$ gives $\int_E f \, d\mu \leq \int_F f \, d\mu$.
Structural Commentary: The nonnegative integral is a supremum over simple functions below the integrand.
Key Insight: More room below the integrand, or over a larger set, can only increase the supremum. □

Method 2.2.6. *Splitting a Simple Minorant Under a Sum*
Let $E \in \mathcal{M}$, $f, g \geq 0$ measurable, and $\omega \in \mathcal{S}^+(f+g)$ simple.
(i) *Grid*: for $n \geq 1$, use $D_n := \{0, \frac{1}{n}, \ldots, 1\}$.
(ii) *Split*: define $\varphi_n(x) := \max\{\theta\omega(x) : \theta \in D_n,\ \theta\omega(x) \leq f(x)\}$; $\psi_n(x) := \left((1 - \frac{1}{n})\,\omega(x) - \varphi_n(x)\right)^+$.
(iii) *Bounds*: φ_n, ψ_n are simple, $\varphi_n \leq f$, $\psi_n \leq g$, and $(1 - \frac{1}{n})\omega \leq \varphi_n + \psi_n$.
(iv) *Use*: simple additivity gives $(1 - \frac{1}{n})\mathcal{I}_E(\omega) \leq \int_E f \, d\mu + \int_E g \, d\mu$; then let $n \to \infty$ and take $\sup_{\omega \in \mathcal{S}^+(f+g)}$.
Key Insight: Grid-split ω between f and g, losing only $1/n$.

Proposition 2.2.7. *Homogeneity and Additivity for Nonnegative Integrals*
Let $f, g \geq 0$ be measurable, let $c \geq 0$, and let $E \in \mathcal{M}$. Then $\int_E cf \, d\mu = c \int_E f \, d\mu$; and $\int_E (f+g) \, d\mu = \int_E f \, d\mu + \int_E g \, d\mu$, where the identities are interpreted in $[0, +\infty]$ with $0 \cdot (+\infty) = 0$.

Proof.
(i) If $c = 0$, both sides are 0. If $c > 0$, $s \mapsto cs$ bijects $\mathcal{S}^+(f)$ with $\mathcal{S}^+(cf)$ and $\mathcal{I}_E(cs) = c\mathcal{I}_E(s)$; take suprema.
(ii) If $s \in \mathcal{S}^+(f)$ and $t \in \mathcal{S}^+(g)$, then $s + t \in \mathcal{S}^+(f+g)$ and $\mathcal{I}_E(s) + \mathcal{I}_E(t) = \mathcal{I}_E(s+t) \leq \int_E (f+g) \, d\mu$. Taking $\sup_{s,t}$ gives $\int_E f \, d\mu + \int_E g \, d\mu \leq \int_E (f+g) \, d\mu$. Conversely, for $\omega \in \mathcal{S}^+(f+g)$, choose simple $\varphi_n \leq f$, $\psi_n \leq g$ with $(1 - \frac{1}{n})\omega \leq \varphi_n + \psi_n$. Then $\left(1 - \frac{1}{n}\right)\mathcal{I}_E(\omega) \leq \mathcal{I}_E(\varphi_n) + \mathcal{I}_E(\psi_n) \leq \int_E f \, d\mu + \int_E g \, d\mu$. Let $n \to \infty$, then take $\sup_{\omega \in \mathcal{S}^+(f+g)}$.
Structural Commentary: The nonnegative integral is linear before signed integrability is introduced.
Key Insight: Scale minorants; split sum-minorants on a finite grid. □

Perspective 2.2.8. *Step Functions versus Simple Functions*
Riemann: $\sup \sum_i m_i \ell(I_i)$ over interval partitions. Lebesgue: $\sup\{\mathcal{I}(s) : 0 \le s \le f,\ s = \sum_i c_i \mathbf{1}_{E_i},\ E_i \in \mathcal{M}\}$.
Structural Commentary: Riemann partitions intervals; Lebesgue partitions level sets.
Key Insight: Same finite constants, measurable pieces.

Perspective 2.2.9. *Undergraph Picture for Nonnegative Functions*
For measurable $f : \mathbb{R} \to [0, +\infty]$, set $\mathcal{U}_f := \{(x, y) \in \mathbb{R} \times [0, +\infty) : 0 \le y < f(x)\}$.
Geometrically, $\int f\, dm = m_2(\mathcal{U}_f)$; analytically, the integral is built as $\sup \mathcal{I}(s)$ over simple $0 \le s \le f$.

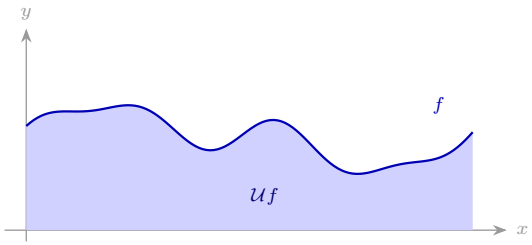


Figure 195: The integral of $f \ge 0$ is modeled by the measure of its undergraph.

Proposition 2.2.10. *Basic Integral Properties*
Let $E \in \mathcal{M}$.

(i) If f is measurable and bounded on E, and $\mu(E) < +\infty$, then $f \in \mathscr{L}^1(\mu)$ on E.
(ii) If $a \le f(x) \le b$ on E and $\mu(E) < +\infty$, then $a\mu(E) \le \int_E f\, d\mu \le b\mu(E)$.
(iii) If $f, g \in \mathscr{L}^1(\mu)$ on E and $f \le g$ on E, then $\int_E f\, d\mu \le \int_E g\, d\mu$.
(iv) If $f \in \mathscr{L}^1(\mu)$ on E, then $cf \in \mathscr{L}^1(\mu)$ on E for finite c, and $\int_E cf\, d\mu = c \int_E f\, d\mu$.
(v) If $\mu(E) = 0$ and f is measurable, then $\int_E f\, d\mu = 0$.
(vi) If $f \in \mathscr{L}^1(\mu)$ on E, $A \in \mathcal{M}$, and $A \subseteq E$, then $f \in \mathscr{L}^1(\mu)$ on A.

Proof.

(i) If $|f| \le M$ on E, then $0 \le f^{\pm} \le M\mathbf{1}_E$ on E, so $\int_E f^{\pm}\, d\mu \le M\mu(E) < +\infty$.
(ii) Since $a\mathbf{1}_E \le f \le b\mathbf{1}_E$ on E, order for the positive integral gives $a\mu(E) \le \int_E f\, d\mu \le b\mu(E)$.
(iii) Since $g - f \ge 0$ on E, $\int_E (g - f)\, d\mu \ge 0$, hence $\int_E f\, d\mu \le \int_E g\, d\mu$.
(iv) For $c \ge 0$, scale the simple approximants. For $c < 0$, use $(cf)^+ = (-c)f^-$ and $(cf)^- = (-c)f^+$.
(v) If $\mu(E) = 0$, then $\mathcal{I}_E(s) = 0$ for every $s \ge 0$ simple, so $\int_E f^{\pm}\, d\mu = 0$.
(vi) Since $A \subseteq E$, $\mathcal{I}_A(s) \le \mathcal{I}_E(s)$ for $s \ge 0$ simple; hence $\int_A f^{\pm}\, d\mu \le \int_E f^{\pm}\, d\mu < +\infty$.

Structural Commentary: Estimate $f^{\pm}$ by simpler functions.
Key Insight: Monotonicity makes bounds, null sets, and restrictions immediate. □

Theorem 2.2.11. *The Integral as a Measure or Charge*
If $f \ge 0$ is measurable, then $\phi_f : \mathcal{M} \to [0, +\infty]$, $\phi_f(A) := \int_A f\, d\mu$, is a measure.
If $f \in \mathscr{L}^1(\mu)$, then $\lambda_f : \mathcal{M} \to \mathbb{R}$, $\lambda_f(A) := \int_A f\, d\mu$, is finite-valued and countably additive.

Proof. For simple $s \ge 0$, $A \mapsto \mathcal{I}_A(s)$ is a measure.
If $A = \dot{\bigcup}_n A_n$ and $s \in \mathcal{S}^+(f)$, then $\mathcal{I}_A(s) = \sum_n \mathcal{I}_{A_n}(s) \le \sum_n \phi_f(A_n)$; taking $\sup_s$ gives $\phi_f(A) \le \sum_n \phi_f(A_n)$.
Conversely, for every N, finite additivity gives $\sum_{n=1}^N \phi_f(A_n) = \phi_f(\bigcup_{n=1}^N A_n) \le \phi_f(A)$; let $N \to \infty$.
Hence ϕ_f is countably additive. If $f \in \mathscr{L}^1(\mu)$, then $\phi_{f^+}(X), \phi_{f^-}(X) < +\infty$ and $\lambda_f = \phi_{f^+} - \phi_{f^-}$.
Thus, for $A = \dot{\bigcup}_n A_n$, $\lambda_f(A) = \sum_n \phi_{f^+}(A_n) - \sum_n \phi_{f^-}(A_n) = \sum_n \lambda_f(A_n)$. *Structural Commentary*: Setwise integration turns densities into measures.
Key Insight: Sup simple measures; subtract finite signs. □

Corollary 2.2.12. *Null and Finiteness Consequences for $f \ge 0$*
Let $f : X \to [0, +\infty]$ be measurable and $\phi_f(A) := \int_A f\, d\mu$.

(i) $\mu(A) = 0 \Longrightarrow \phi_f(A) = 0$ for $A \in \mathcal{M}$.
(ii) $\int_X f\, d\mu = 0 \Longleftrightarrow f = 0$ a.e.
(iii) $\int_X f\, d\mu < +\infty \Longrightarrow \mu\{f = +\infty\} = 0$ and $\{f > 0\}$ is σ-finite.

Proof.

(i) If $\mu(A) = 0$ and $0 \le s \le f$ is simple, then $\mathcal{I}_A(s) = 0$; hence $\phi_f(A) = \sup_s \mathcal{I}_A(s) = 0$.

(*ii*) ($\Rightarrow$) Set $E_n := \{f > 1/n\}$. Since $(1/n)\mathbf{1}_{E_n} \le f$, $0 \le n^{-1}\mu(E_n) \le \int_X f\,d\mu = 0$.
Thus $\mu(E_n) = 0$ for all n, and $\{f > 0\} = \bigcup_n E_n$ is null.
(*ii*) ($\Leftarrow$) If $N := \{f > 0\}$ null, every simple $0 \le s \le f$ vanishes on $X \setminus N$, so $\mathcal{I}_X(s) = \mathcal{I}_N(s) = 0$; take $\sup_s$.
(*iii*) Let $F_n := \{f \ge n\}$. Then $n\mu(F_n) \le \int_X f\,d\mu$, so $\mu\{f = +\infty\} \le \mu(F_n) \le n^{-1}\int_X f\,d\mu \to 0$.
Also $\{f > 0\} = \bigcup_n \{f > 1/n\}$ and $\mu\{f > 1/n\} \le n\int_X f\,d\mu < +\infty$.

Structural Commentary: Finite nonnegative integrals control threshold sets.
Key Insight: Threshold sets convert integral bounds into measure bounds. □

Theorem 2.2.13. *Absolute Integrability and Domination*
If $f \in \mathscr{L}^1(\mu)$ on E, then $|f| \in \mathscr{L}^1(\mu)$ and $\left|\int_E f\,d\mu\right| \le \int_E |f|\,d\mu$.
If f is measurable and $|f| \le g \in \mathscr{L}^1(\mu)$ on E, then $f \in \mathscr{L}^1(\mu)$ on E.

Proof. Use $|f| = f^+ + f^-$, $f \le |f|$, and $-f \le |f|$. If $|f| \le g$, then $0 \le f^\pm \le g$, so $\int_E f^\pm\,d\mu < +\infty$.
Structural Commentary: Lebesgue integrability is absolute integrability.
Key Insight: A single integrable majorant controls both signs. □

Proposition 2.2.14. *Linearity of the Lebesgue Integral*
Let $E \in \mathcal{M}$, $f, g \in \mathscr{L}^1(\mu)$ on E, and $a, b \in \mathbb{R}$.
Then $af + bg \in \mathscr{L}^1(\mu)$ on E and $\int_E (af + bg)\,d\mu = a\int_E f\,d\mu + b\int_E g\,d\mu$.

Proof. For $f, g \ge 0$, choose simple $s_n \uparrow f$ and $t_n \uparrow g$.
Then $s_n + t_n \uparrow f + g$ and $\int_E (s_n + t_n)\,d\mu = \int_E s_n\,d\mu + \int_E t_n\,d\mu \to \int_E f\,d\mu + \int_E g\,d\mu$.
Thus nonnegative additivity holds; homogeneity follows by scaling simple minorants.
For signed f, g, put $a = a^+ - a^-$, $b = b^+ - b^-$ and $P := a^+f^+ + a^-f^- + b^+g^+ + b^-g^-$;
$Q := a^+f^- + a^-f^+ + b^+g^- + b^-g^+$. Then $P, Q \ge 0$, $\int_E P\,d\mu, \int_E Q\,d\mu < +\infty$, and $af + bg = P - Q$.
By Proposition 2.2.2 and nonnegative linearity, the displayed identity follows.
Structural Commentary: Simple additivity plus monotone convergence gives linearity.
Key Insight: Reduce additivity to simple functions before splitting signs. □

Proposition 2.2.15. *L^1 Level Sets, Null Changes, and Indefinite Integrals*
Let $f, g \in \mathscr{L}^1(\mu)$ on X, and write $\lambda_u(A) := \int_A u\,d\mu$.

(i) For $a > 0$, $\mu\{|f| \ge a\} \le a^{-1}\int_X |f|\,d\mu < +\infty$; hence $\{f \ne 0\}$ is σ-finite.
(ii) If measurable $h = 0$ a.e., then $h \in \mathscr{L}^1(\mu)$ and $\lambda_h \equiv 0$.
(iii) $\lambda_f \ge 0$ on $\mathcal{M} \Longrightarrow f \ge 0$ a.e.; $\lambda_f \equiv 0 \Longrightarrow f = 0$ a.e.
(iv) $\lambda_f = \lambda_g$ on $\mathcal{M} \Longleftrightarrow f = g$ a.e.

Proof.

(i) Since $a\mathbf{1}_{\{|f| \ge a\}} \le |f|$, integration gives $a\mu\{|f| \ge a\} \le \int_X |f|\,d\mu$. Also $\{f \ne 0\} = \bigcup_{k \ge 1}\{|f| \ge 1/k\}$.
(ii) Let $N := \{h \ne 0\}$. For $A \in \mathcal{M}$, $h^\pm\mathbf{1}_A$ vanish off $A \cap N$, a null set.
Hence $\int_A h^+\,d\mu = \int_A h^-\,d\mu = 0$, so $\lambda_h(A) = 0$.
(iii) For $k \ge 1$, set $N_k := \{f \le -1/k\}$. By (i), $\mu(N_k) < +\infty$, and $\lambda_f(N_k) \le -k^{-1}\mu(N_k)$.
If $\lambda_f \ge 0$, then $\mu(N_k) = 0$, so $\{f < 0\} = \bigcup_k N_k$ is null. If $\lambda_f \equiv 0$, apply this to f and $-f$.
(iv) If $\lambda_f = \lambda_g$, then $\lambda_{f-g} \equiv 0$, so $f - g = 0$ a.e. by (iii). Conversely, if $f = g$ a.e., apply (ii) to $h = f - g$.

Structural Commentary: An L^1 function is detected by its integrals over all measurable sets, up to null sets.
Key Insight: Test sign on the level sets where the function is bounded away from 0. □

Proposition 2.2.16. *L^1 Approximation and Bounded Multipliers*
Let $f \in \mathscr{L}^1(\mu)$ on X.

(i) $\forall \varepsilon > 0$, $\exists$ measurable simple φ with $\int_X |f - \varphi|\,d\mu < \varepsilon$.
(ii) If g is bounded measurable, then $fg \in \mathscr{L}^1(\mu)$.

Proof.

(i) Choose simple $0 \le s_n \uparrow f^+$ and $0 \le t_n \uparrow f^-$. Then $\int_X (f^+ - s_n)\,d\mu \to 0$ and $\int_X (f^- - t_n)\,d\mu \to 0$.
For large n, $\varphi := s_n - t_n$ satisfies $\int_X |f - \varphi|\,d\mu \le \int_X (f^+ - s_n)\,d\mu + \int_X (f^- - t_n)\,d\mu < \varepsilon$.
(ii) If $|g| \le M$, then fg is measurable and $\int_X |fg|\,d\mu \le M\int_X |f|\,d\mu < +\infty$.

Structural Commentary: L^1 is simple-closed and bounded-multiplier stable.
Key Insight: Approximate signs separately; dominate products by $M|f|$. □

Proposition 2.2.17. *Finite-Measure Indicators by Intervals and Continuous Functions*
Let $A \subseteq \mathbb{R}$ be Lebesgue measurable with $m(A) < +\infty$. For every $\varepsilon > 0$:

(i) $\exists G = \bigcup_{j=1}^N I_j$, I_j open intervals, such that $m(A \triangle G) < \varepsilon$.

(ii) $\exists f \in C(\mathbb{R})$ *such that* $0 \le f \le 1$ *and* $\int_{\mathbb{R}} |\mathbf{1}_A - f|\, dm < \varepsilon$.

Proof.

(i) Choose open $O \supseteq A$ with $m(O \setminus A) < \varepsilon/2$ and $m(O) < +\infty$. Write $O = \bigsqcup_{j\ge1} I_j$ with open intervals I_j, choose N with $m(O \setminus G) < \varepsilon/2$ for $G := \bigcup_{j=1}^{N} I_j$. Then $A \setminus G \subseteq O \setminus G$ and $G \setminus A \subseteq O \setminus A$, so $m(A \triangle G) < \varepsilon$.

(ii) Apply (i) with $\varepsilon/2$ to get finite open G. Choose compact $K \subseteq G$ with $m(G \setminus K) < \varepsilon/2$, and choose piecewise-linear f with $0 \le f \le 1$, $f = 1$ on K, $f = 0$ on $\mathbb{R} \setminus G$. Then $\int_{\mathbb{R}} |\mathbf{1}_G - f|\, dm \le m(G \setminus K) < \varepsilon/2$, hence $\int_{\mathbb{R}} |\mathbf{1}_A - f|\, dm \le m(A \triangle G) + \int_{\mathbb{R}} |\mathbf{1}_G - f|\, dm < \varepsilon$.

Structural Commentary: Finite-measure sets can be made geometrically simple up to small L^1 error.
Key Insight: Approximate by intervals, then smooth the indicator. □

Example 2.2.18. *Counting Measure Gives Absolute Series*
Let μ be counting measure on $(\mathbb{N}, 2^{\mathbb{N}})$. For $f : \mathbb{N} \to \mathbb{R}$,
$f \in \mathscr{L}^1(\mu) \iff \sum_{n=1}^{\infty} |f(n)| < +\infty$; $\int_{\mathbb{N}} f\, d\mu = \sum_{n=1}^{\infty} f(n)$ when these equivalent conditions hold.

Example 2.2.19. *L^1 Is Not Closed Under Squaring*
On $X = \mathbb{N}$ with $\mathcal{M} = 2^{\mathbb{N}}$, define $\mu(E) := \sum_{n\in E} 4^{-n}$ and $f(n) := 2^n/\sqrt{n}$.
Then $\int_X |f|\, d\mu = \sum_{n=1}^{\infty} \frac{1}{2^n\sqrt{n}} < +\infty$; $\int_X f^2\, d\mu = \sum_{n=1}^{\infty} \frac{1}{n} = +\infty$.
The first series is dominated by the geometric series $\sum 2^{-n}$, while the second diverges by Corollary 6.1.13.
Thus $f \in \mathscr{L}^1(\mu)$ need not imply $f^2 \in \mathscr{L}^1(\mu)$.

2.3 Convergence Theorems

This subsection collects the main limit-interchange tools: monotone convergence, Fatou, and dominated convergence.

Theorem 2.3.1. *Monotone Convergence*
Let $E \in \mathcal{M}$, let f_n, f be measurable on E, and assume $0 \le f_n \uparrow f$. Then $\int_E f_n\, d\mu \uparrow \int_E f\, d\mu$.

Proof. By monotonicity, $\int_E f_n\, d\mu \uparrow L \le \int_E f\, d\mu$.
For $s \in \mathcal{S}^+(f)$ and $0 < c < 1$, set $E_n := \{x \in E : f_n(x) \ge c s(x)\}$.
Since $f_n \uparrow f$ and $s \le f$, $E_n \uparrow E$, and $\int_E f_n\, d\mu \ge \int_{E_n} f_n\, d\mu \ge c \int_{E_n} s\, d\mu$.
Continuity from below for $\phi_s(A) := \int_A s\, d\mu$ gives $L \ge c \int_E s\, d\mu$. Let $c \uparrow 1$, then take $\sup_{s\in\mathcal{S}^+(f)}$.
Structural Commentary: Nonnegative integration commutes with monotone limits.
Key Insight: Simple minorants of the limit are eventually captured below the sequence. □

Theorem 2.3.2. *Series Form of Monotone Convergence*
Let $E \in \mathcal{M}$, let $f_n \ge 0$ be measurable, and set $f := \sum_{n=1}^{\infty} f_n$ on E. Then $\int_E f\, d\mu = \sum_{n=1}^{\infty} \int_E f_n\, d\mu$.

Proof. Set $S_N := \sum_{n=1}^{N} f_n$. Then $0 \le S_N \uparrow f$.
By Theorem 2.3.1, $\int_E f\, d\mu = \lim_{N\to\infty} \int_E S_N\, d\mu = \lim_{N\to\infty} \sum_{n=1}^{N} \int_E f_n\, d\mu = \sum_{n=1}^{\infty} \int_E f_n\, d\mu$.
Structural Commentary: Nonnegative series are monotone partial sums.
Key Insight: Integrate partial sums, then pass upward. □

Theorem 2.3.3. *Fatou's Lemma*
Let $E \in \mathcal{M}$ and let $f_n \ge 0$ be measurable. Then $\int_E \liminf_{n\to\infty} f_n\, d\mu \le \liminf_{n\to\infty} \int_E f_n\, d\mu$

Proof. Set $g_n := \inf_{i\ge n} f_i$. Then $0 \le g_n \uparrow \liminf_i f_i$ and, for every $i \ge n$, $g_n \le f_i$.
Hence $\int_E g_n\, d\mu \le \inf_{i\ge n} \int_E f_i\, d\mu$. By Theorem 2.3.1, let $n \to \infty$ to obtain the claim.
Structural Commentary: Fatou compares the liminf to tail lower envelopes.
Key Insight: Apply monotone convergence to $g_n = \inf_{i\ge n} f_i$. □

Theorem 2.3.4. *Dominated Convergence*
Let $E \in \mathcal{M}$, f_n measurable, $f_n \to f$ a.e. on E, and $|f_n| \le g \in \mathscr{L}^1(\mu)$ on E.
Then $f \in \mathscr{L}^1(\mu)$ on E and $\lim_{n\to\infty} \int_E f_n\, d\mu = \int_E f\, d\mu$.

Proof. Choose $N \subseteq E$ null such that $f_n \to f$ on $E \setminus N$; replacing all functions by their restrictions off N does not change integrals. Thus assume $f_n \to f$ everywhere. Then $|f| \le g$, so $f \in \mathscr{L}^1(\mu)$. Put $a_n := \int_E f_n \, d\mu$.
For $g + f_n \ge 0$, by Fatou, $\int_E (g+f)\, d\mu \le \liminf_n \int_E (g+f_n)\, d\mu = \int_E g\, d\mu + \liminf_n a_n$; $\int_E f\, d\mu \le \liminf_n a_n$.
For $g - f_n \ge 0$, by Fautou, $\int_E (g-f)\, d\mu \le \liminf_n \int_E (g-f_n)\, d\mu = \int_E g\, d\mu - \limsup_n a_n$; $\limsup_n a_n \le \int_E f\, d\mu$.
Thus $\liminf_n a_n = \limsup_n a_n = \int_E f\, d\mu$.
Structural Commentary: Domination turns signed convergence into two Fatou inequalities.
Key Insight: Apply Fatou to $g + f_n$ and $g - f_n$ to trap liminf and limsup. □

Corollary 2.3.5. *Monotone Convergence Almost Everywhere*
Let $E \in \mathcal{M}$ and $0 \le f_n \uparrow f$ a.e. on E. Then $\int_E f_n \, d\mu \uparrow \int_E f \, d\mu$.

Proof. Choose $N \subseteq E$ with $\mu(N) = 0$ and $f_n \uparrow f$ on $E \setminus N$. Set $g_n := f_n \mathbf{1}_{E\setminus N}$ and $g := f\mathbf{1}_{E\setminus N}$.
Then $0 \le g_n \uparrow g$, so $\int_E g_n \, d\mu \uparrow \int_E g \, d\mu$. Since $f_n = g_n$ and $f = g$ off N, the integrals over N vanish.
Structural Commentary: Monotone convergence ignores null exceptional sets.
Key Insight: Remove the exceptional set, prove the theorem there, then add back zero. □

Example 2.3.6. *Counting Measure Integrals Are Series*
Let μ be counting measure on $(\mathbb{N}, 2^{\mathbb{N}})$. For $f : \mathbb{N} \to [0, +\infty]$, set $s_N := \sum_{n=1}^{N} f(n)\mathbf{1}_{\{n\}}$; then $0 \le s_N \uparrow f$, with the case $f(k) = +\infty$ giving both sides $+\infty$. Hence $\int_{\mathbb{N}} f \, d\mu = \lim_N \int_{\mathbb{N}} s_N \, d\mu = \lim_N \sum_{n=1}^{N} f(n) = \sum_{n=1}^{\infty} f(n)$.
Structural Commentary: Counting measure turns integration into summation.
Key Insight: Finite-support truncations are the simple-function partial sums.

Example 2.3.7. *Boundary Tests for Monotone Convergence and Fatou*
On $([0, +\infty), \mathcal{B}, m)$:

(i) $f_n := \mathbf{1}_{[0,n]} \uparrow \mathbf{1}_{[0,+\infty)}$ and $\int f_n \, dm = n \uparrow +\infty$; monotone convergence allows $+\infty$.
(ii) $g_n := n^{-1}\mathbf{1}_{[n,+\infty)} \downarrow 0$ uniformly, but $\int g_n \, dm = +\infty \ \forall n$; decreasing convergence needs finite mass.
(iii) $w_n := -n^{-1}\mathbf{1}_{[0,n]} \rightrightarrows 0$, but $\int w_n \, dm = -1$; Fatou can fail without $f_n \ge 0$ or $f_n \ge -h \in L^1$.

Theorem 2.3.8. *Fatou with an Integrable Lower Bound*
Let $h \ge 0$, $h \in \mathscr{L}^1(\mu)$, and let f_n be measurable with $f_n \ge -h$.
Then, whenever the signed integrals are defined in $\overline{\mathbb{R}}$, $\int \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu$.

Proof. Set $F_n := f_n + h \ge 0$. Fatou gives $\int \liminf_n F_n \, d\mu \le \liminf_n \int F_n \, d\mu$.
Since $\liminf_n F_n = h + \liminf_n f_n$ and $\int F_n \, d\mu = \int f_n \, d\mu + \int h \, d\mu$, subtract the finite term $\int h \, d\mu$.
Structural Commentary: Fatou survives controlled negative parts.
Key Insight: Add an integrable floor to make the sequence nonnegative. □

Remark 2.3.9. *Why the Negative Example Has No Floor*
For $w_n := -n^{-1}\mathbf{1}_{[0,n]}$, any $h \in L^1$ with $-h \le w_n \ \forall n$ must satisfy $h \ge n^{-1}$ on $[0, n]$.
Thus $h \ge k^{-1}$ on $[k-1, k]$, and $\int_0^N h \, dm \ge H_N := \sum_{k=1}^{N} \frac{1}{k} \to +\infty$, contradicting $h \in L^1$.

Perspective 2.3.10. *Domination Is the Safeguard*
$f_n \to f$ a.e. need not imply $\int f_n \to \int f$; domination $|f_n| \le g \in L^1$ restores the limit interchange.

Example 2.3.11. *A Nonuniform Limit with Vanishing Integrals*
For $x > 0$, set $f_n(x) := e^{-nx}x^{-1/2}$. Then $f_n \to 0$ pointwise, but $\sup_{0<x<\delta} f_n(x) = +\infty$ for every $\delta > 0$.
For $n \ge 1$, $0 \le f_n \le g := e^{-x}x^{-1/2}$; $g \le x^{-1/2}\mathbf{1}_{(0,1]} + e^{-x}\mathbf{1}_{[1,\infty)} \in \mathscr{L}^1(0, \infty)$.
Thus dominated convergence gives $I_n := \int_0^\infty e^{-nx}x^{-1/2} \, dx \to 0$.
Structural Commentary: Uniform convergence is unnecessary once one integrable envelope controls the sequence.
Key Insight: Domination handles the singular behavior at 0 and the tail at infinity simultaneously.

Proposition 2.3.12. *Uniform Convergence on Finite Measure Spaces*
Assume $\mu(X) < +\infty$.

(i) If $0 \le f_n$ are measurable and $f_n \rightrightarrows f$, then $f \ge 0$ is measurable and $\int_X f \, d\mu = \lim_n \int_X f_n \, d\mu$ in $[0, +\infty]$.
(ii) If $f_n \in \mathscr{L}^1(\mu)$ and $f_n \rightrightarrows f$, then $f \in \mathscr{L}^1(\mu)$ and $\int_X f_n \, d\mu \to \int_X f \, d\mu$ in $\mathbb{R}$.

Proof.

(i) Measurability and $f \ge 0$ pass to pointwise limits. For $\varepsilon > 0$ and $n \gg 1$, $f \le f_n + \varepsilon\mathbf{1}_X$ and $f_n \le f + \varepsilon\mathbf{1}_X$. Thus $\int_X f \, d\mu \le \int_X f_n \, d\mu + \varepsilon\mu(X)$ and $\int_X f_n \, d\mu \le \int_X f \, d\mu + \varepsilon\mu(X)$. Let $\varepsilon \downarrow 0$.

(ii) Choose N with $|f - f_N| \le 1$. Then $f = f_N + (f - f_N) \in \mathscr{L}^1(\mu)$, and $\left|\int_X f_n\,d\mu - \int_X f\,d\mu\right| \le \int_X |f_n - f|\,d\mu \le \mu(X)\sup_X |f_n - f| \to 0$.

Structural Commentary: On finite measure spaces, uniform error controls integral error.
Key Insight: Finiteness of $\mu(X)$ turns a supremum bound into an integral bound. □

Example 2.3.13. *Why the Finite-Measure Hypothesis Matters*
On $(\mathbb{R}, m)$, let $f_n := n^{-1}\mathbf{1}_{[0,n]}$. Then $\|f_n\|_\infty = 1/n \to 0$, but $\int_{\mathbb{R}} f_n\,dm = 1\ \forall n$.
Thus uniform convergence need not commute with integrals when $m(\mathbb{R}) = +\infty$; Fatou only yields $0 \le 1$.

Example 2.3.14. *Why Domination Cannot Be Dropped*
On $(\mathbb{R}, m)$, let $f_n := n\mathbf{1}_{[1/n,2/n]}$. Then $f_n \to 0$ pointwise, but $\int_{\mathbb{R}} f_n\,dm = 1\ \forall n$.
Thus pointwise convergence without domination need not commute with integrals; Fatou only yields $0 \le 1$.

Theorem 2.3.15. *Absolutely Summable L^1 Series*
Let $f_n \in \mathscr{L}^1(\mu)$ and $\sum_{n\ge1} \|f_n\|_1 < +\infty$.
Then $\sum_n f_n$ converges absolutely a.e. to some $f \in \mathscr{L}^1(\mu)$, and $\int_X f\,d\mu = \sum_{n\ge1}\int_X f_n\,d\mu$.

Proof. Set $G := \sum_{n\ge1} |f_n|$. Then $\int_X G\,d\mu = \sum_{n\ge1}\|f_n\|_1 < +\infty$, so $G < +\infty$ a.e. and $G \in \mathscr{L}^1$.
Let $f := \sum_{n\ge1} f_n$ a.e. and $S_N := \sum_{n=1}^N f_n$.
Since $|S_N| \le G$ and $S_N \to f$ a.e., dominated convergence gives $\int_X f\,d\mu = \lim_N \int_X S_N\,d\mu = \sum_{n\ge1}\int_X f_n\,d\mu$.
Structural Commentary: Absolute summability in L^1 gives a.e. absolute convergence and termwise integration.
Key Insight: First integrate the nonnegative majorant $\sum |f_n|$. □

Proposition 2.3.16. *L^1 Convergence Controls Absolute Integrals*
If $f_n, f \in \mathscr{L}^1(\mu)$ and $\|f_n - f\|_1 \to 0$, then $\|f_n\|_1 \to \|f\|_1$; $\int_X |f|\,d\mu = \lim_n \int_X |f_n|\,d\mu$.

Proof. The reverse triangle inequality gives $||f_n| - |f|| \le |f_n - f|$; hence $|\|f_n\|_1 - \|f\|_1| \le \|f_n - f\|_1 \to 0$.
Structural Commentary: L^1 convergence is strong enough to converge the L^1 sizes.
Key Insight: The reverse triangle inequality converts function convergence into norm convergence. □

Proposition 2.3.17. *Finite Integral Localizes to Finite Measure*
If $f : X \to [0, +\infty]$ is measurable, $\int_X f\,d\mu < +\infty$, then $\forall \varepsilon > 0\ \exists\ E \in \mathcal{M}:\quad \mu(E) < +\infty$; $\int_{X\setminus E} f\,d\mu < \varepsilon$.

Proof. Set $E_n := \{1/n < f \le n\}$. Then $\mu(E_n) \le n\int_X f\,d\mu < +\infty$, and $f\mathbf{1}_{E_n} \uparrow f$ a.e.
Hence $\int_{E_n} f\,d\mu \uparrow \int_X f\,d\mu$ by monotone convergence, so $\int_{X\setminus E_n} f\,d\mu \downarrow 0$; choose $E = E_n$.
Structural Commentary: Finite integral mass can be captured on a finite-measure part of the space.
Key Insight: Keep the levels where f is bounded above and away from zero. □

Proposition 2.3.18. *Setwise Convergence from Total Mass Convergence*
Let $f_n, f : X \to [0, +\infty]$ be measurable, $f_n \to f$ pointwise, and $\int_X f\,d\mu = \lim\limits_{n\to\infty}\int_X f_n\,d\mu < +\infty$.
Then $\forall E \in \mathcal{M}$, $\int_E f\,d\mu = \lim\limits_{n\to\infty}\int_E f_n\,d\mu$.

Proof. Fix $E \in \mathcal{M}$ and set $E^c := X \setminus E$. By Fatou, $\int_E f\,d\mu \le \liminf_n \int_E f_n\,d\mu$ and $\int_{E^c} f\,d\mu \le \liminf_n \int_{E^c} f_n\,d\mu$.
Since $\int_X f_n\,d\mu = \int_E f_n\,d\mu + \int_{E^c} f_n\,d\mu \to \int_X f\,d\mu < +\infty$, $\limsup_n \int_E f_n\,d\mu \le \int_X f\,d\mu - \int_{E^c} f\,d\mu = \int_E f\,d\mu$.
Thus $\liminf_n \int_E f_n\,d\mu = \limsup_n \int_E f_n\,d\mu = \int_E f\,d\mu$.
Structural Commentary: Total mass plus Fatou on complements fixes each part.
Key Insight: A lower bound on the complement is an upper bound on the chosen set. □

Definition 2.3.19. *Function Truncates*
For $f : X \to \mathbb{R}$ and $n \in \mathbb{N}$, define the *nth truncate*

$$T_n f(x) := (-n) \vee (f(x) \wedge n) = \begin{cases} -n, & f(x) < -n, \\ f(x), & -n \le f(x) \le n, \\ n, & f(x) > n. \end{cases}$$

Proposition 2.3.20. *Truncates Characterize Integrability*
Let $f : X \to \mathbb{R}$ be measurable and $f_n := T_n f$. Then $f \in \mathscr{L}^1(\mu) \iff \sup_n \|f_n\|_1 < +\infty$.
Moreover, if $f \in \mathscr{L}^1(\mu)$, then $\int_X f_n\,d\mu \to \int_X f\,d\mu$.

Proof. The map $T_n : \mathbb{R} \to [-n, n]$ is continuous, so f_n is measurable; also $f_n \to f$, $|f_n| \le |f|$, and $|f_n| \uparrow |f|$.
If $f \in \mathscr{L}^1(\mu)$, dominated convergence gives $\int_X f_n \, d\mu \to \int_X f \, d\mu$ and $\sup_n \|f_n\|_1 \le \|f\|_1$.
Conversely, monotone convergence gives $\|f\|_1 = \lim_{n\to\infty} \|f_n\|_1 = \sup_n \|f_n\|_1 < +\infty$, so $f \in \mathscr{L}^1(\mu)$.

Structural Commentary: Truncation reduces L^1 questions to bounded approximants.
Key Insight: Dominated convergence gives limits; monotone convergence reads integrability from $|f_n| \uparrow |f|$. □

Example 2.3.21. *Why the Finite Total Mass Assumption Matters*
Let $X = \mathbb{N} \times \{0, 1\}$ with counting measure, set $E := \mathbb{N} \times \{1\}$.
Define $f := \mathbf{1}_{\mathbb{N}\times\{0\}}$; $f_n := \mathbf{1}_{\{1,\dots,n\}\times\{0\}} + n\mathbf{1}_{\{n\}\times\{1\}}$.
Then $f_n \to f$ pointwise and $\int_X f \, d\mu = \lim_n \int_X f_n \, d\mu = +\infty$; $\int_E f \, d\mu = 0$, $\int_E f_n \, d\mu = n$.
Thus Proposition 2.3.18 can fail when the common total mass is infinite.

2.4 Parameter Integrals and Differentiation

Corollary 2.4.1. *Continuity Under a Lebesgue Integral*
Let $f_t(x) := f(x, t)$ be measurable for each $t \in [a, b]$. Fix $t_0 \in [a, b]$.
If $f_t \to f_{t_0}$ a.e. as $t \to t_0$ and $|f_t| \le g \in \mathscr{L}^1(\mu)$ for all t, then $\lim_{t\to t_0} \int_X f_t \, d\mu = \int_X f_{t_0} \, d\mu$.
If $t \mapsto f(x, t)$ is continuous on $[a, b]$ for a.e. x, then $F(t) := \int_X f_t \, d\mu$ is continuous on $[a, b]$.

Proof. For $t_n \to t_0$, set $u_n := f_{t_n}$.
Then $u_n \to f_{t_0}$ a.e. and $|u_n| \le g$, so dominated convergence gives $\int_X u_n \, d\mu \to \int_X f_{t_0} \, d\mu$.
The sequential criterion gives the first claim; applying it for every t_0 gives continuity of F.
Structural Commentary: Parameter continuity reduces to dominated convergence along sequences.
Key Insight: The parameter disappears once t_n is frozen into a sequence of functions. □

Theorem 2.4.2. *Differentiation Under a Lebesgue Integral*
Let $f_t(x) := f(x, t)$ be measurable for each $t \in [a, b]$. Suppose $f_{t_0} \in \mathscr{L}^1(\mu)$ for some $t_0 \in [a, b]$, $\partial_t f$ exists on $X \times (a, b)$,
and $|\partial_t f(x, t)| \le g(x)$ for some $g \in \mathscr{L}^1(\mu)$ and all $(x, t) \in X \times (a, b)$.
Then $f_t \in \mathscr{L}^1(\mu)$ for every $t \in [a, b]$, and, with $F(t) := \int_X f_t \, d\mu$, $F'(t) = \int_X \partial_t f(x, t) \, d\mu$, $t \in (a, b)$.

Proof. For fixed t, $x \mapsto \partial_t f(x, t)$ is measurable as the pointwise limit of measurable difference quotients.
By the mean value theorem, $|f_t - f_{t_0}| \le |t - t_0| \, g$, so $|f_t| \le |f_{t_0}| + |t - t_0| \, g \in \mathscr{L}^1(\mu)$.
If $t_n \to t \in (a, b)$ and $t_n \ne t$, then $\frac{F(t_n)-F(t)}{t_n - t} = \int_X \frac{f_{t_n} - f_t}{t_n - t} \, d\mu$.
Integrand tends pointwise to $\partial_t f(\cdot, t)$ and has modulus $\le g$; dominated convergence gives derivative formula.
Structural Commentary: Bounded parameter derivatives dominate the difference quotients.
Key Insight: Differentiating the integral is just applying dominated convergence to secant slopes. □

Corollary 2.4.3. *A Pointwise Difference-Quotient Test*
Let $f_t(x) := f(x, t)$ be measurable for each $t \in [a, b]$, let $f_{t_0} \in \mathscr{L}^1(\mu)$ for some $t_0 \in [a, b]$, and fix $t_1 \in [a, b]$.
Suppose $\partial_t f(x, t_1)$ exists for every x and $\exists g \in \mathscr{L}^1(\mu)$ such that, for $t \ne t_1$, $Q_t(x) := \frac{f_t(x) - f_{t_1}(x)}{t - t_1}$, $|Q_t| \le g$.
Then $F(t) := \int_X f_t \, d\mu$ is differentiable at t_1 and $F'(t_1) = \int_X \partial_t f(x, t_1) \, d\mu$.

Proof. If $t_0 \ne t_1$, then $|f_{t_1}| \le |f_{t_0}| + |t_0 - t_1| \, g$; otherwise this is given.
Hence $f_{t_1} \in \mathscr{L}^1$, and $|f_t| \le |f_{t_1}| + |t - t_1| \, g$ gives $f_t \in \mathscr{L}^1$. For any $t_n \to t_1$, $t_n \ne t_1$, $Q_{t_n} \to \partial_t f(\cdot, t_1)$ pointwise and $|Q_{t_n}| \le g$, so dominated convergence gives $[F(t_n) - F(t_1)]/(t_n - t_1) \to \int_X \partial_t f(x, t_1) \, d\mu$.
Structural Commentary: One parameter point needs only domination of the secants into that point.
Key Insight: A global derivative bound is sufficient, but not necessary, for a one-point formula. □

Corollary 2.4.4. *One Lebesgue and One Riemann Integral*
Let $f_t(x) := f(x, t)$ be measurable for each $t \in [a, b]$, let $t \mapsto f(x, t)$ be continuous on $[a, b]$ for a.e. x, and suppose $|f_t| \le g \in \mathscr{L}^1(\mu)$ $\forall t$. Then, with Riemann integrals in t, $\int_a^b \left[\int_X f_t \, d\mu\right] dt = \int_X \left[\int_a^b f(x, t) \, dt\right] d\mu(x)$.

Proof. Set $h_t(x) := \int_a^t f(x, s) \, ds$. Then h_t is measurable, $|h_t| \le (b - a)g$, and $\partial_t h_t = f_t$.
By differentiation under the integral, $H(t) := \int_X h_t \, d\mu$ satisfies $H'(t) = \int_X f_t \, d\mu$.
Hence $\int_a^b \left[\int_X f_t \, d\mu\right] dt = H(b) - H(a) = \int_X \left[\int_a^b f(x, t) \, dt\right] d\mu(x)$. *Structural Commentary*: Finite parameter integrals move through by differentiating primitives.
Key Insight: Integrate first in the parameter up to t, then differentiate the resulting Lebesgue integral. □

Corollary 2.4.5. *Improper Parameter Interchange*
Let $f_t(x) := f(x,t)$ be measurable for each $t \in \mathbb{R}$, and let $t \mapsto f(x,t)$ be continuous for a.e. x.
Suppose $\exists g, h \in \mathscr{L}^1(\mu)$ such that $|f_t| \le g$ $\forall t$ and $\int_{\mathbb{R}} |f(x,t)|\,dt \le h(x)$ for a.e. x.
Put $G(t) := \int_X f_t\,d\mu$; $H(x) := \int_{\mathbb{R}} f(x,t)\,dt$. Then $H \in \mathscr{L}^1(\mu)$; $\int_{\mathbb{R}} G(t)\,dt = \int_X H\,d\mu$, $\int_{\mathbb{R}} G := \lim_{R\to+\infty} \int_{-R}^{R} G(t)\,dt$.

Proof. For $R > 0$, finite-interval interchange gives $\int_{-R}^{R} G(t)\,dt = \int_X H_R\,d\mu$ and $H_R(x) := \int_{-R}^{R} f(x,t)\,dt$.
Since $H_R \to H$ a.e. and $|H_R| \le h$, dominated convergence gives $\int_X H_R\,d\mu \to \int_X H\,d\mu$.
Structural Commentary: Finite interchanges pass to improper limits under an L^1 bound.
Key Insight: Truncate in t, interchange on finite intervals, then pass $R \to +\infty$ by domination in x. □

Example 2.4.6. *Differentiating Under the Integral Sign*
For $t > 0$, set $F(t) := \int_0^\infty x^2 e^{-tx}\,dx$. For $t_0 > 0$ and $|h| < t_0/2$, put $Q_h(x) := \frac{x^2 e^{-(t_0+h)x} - x^2 e^{-t_0 x}}{h}$
Then $Q_h \to -x^3 e^{-t_0 x}$ pointwise and $|Q_h| \le x^3 e^{-(t_0/2)x} \in \mathscr{L}^1(0,\infty)$. Hence $F'(t_0) = -\int_0^\infty x^3 e^{-t_0 x}\,dx$.
Structural Commentary: Parameter differentiation is a limit-interchange problem.
Key Insight: Prove a local-in-parameter integrable majorant for the difference quotients.

Example 2.4.7. *Laplace Transform Powers and Factorials*
For $t > 0$, set $I(t) := \int_0^\infty e^{-tx}\,dx$. Then $I(t) = \lim_{b\to+\infty} \frac{1-e^{-tb}}{t} = t^{-1}$.
For $[a,b] \subset (0,+\infty)$ and $k \ge 0$, $0 \le x^k e^{-tx} \le x^k e^{-ax} \in \mathscr{L}^1(0,\infty)$, so repeated differentiation under the integral gives $I^{(n)}(t) = (-1)^n \int_0^\infty x^n e^{-tx}\,dx$. Since $I^{(n)}(t) = (-1)^n n! t^{-n-1}$, for every $n \in \mathbb{N} \cup \{0\}$, $\int_0^\infty x^n e^{-x}\,dx = n!$.
Structural Commentary: A parameter integral turns powers of x into derivatives in t.
Key Insight: The exponential envelope e^{-ax} controls all nearby parameter values $t \ge a$.

2.5 Riemann-Lebesgue Comparison

Theorem 2.5.1. *Comparison with the Riemann Integral*
On $[a,b]$ with Lebesgue measure μ, $f \in \mathcal{R}$ implies $f \in \mathscr{L}^1(\mu)$ and the integrals agree.
If f is bounded with discontinuity set D, then $f \in \mathcal{R} \iff \mu(D) = 0$.

Proof. For refining partitions, let $L_n \uparrow L$ and $U_n \downarrow U$ be lower and upper step envelopes.
Monotone convergence identifies $\int L$ and $\int U$ with the lower and upper Riemann integrals; if $f \in \mathcal{R}$, then $L = U$ a.e., so the common Lebesgue integral equals the Riemann integral.
For bounded f, put $\omega_f(x) := \lim_{r\downarrow 0} \operatorname{diam} f\big([a,b] \cap (x-r, x+r)\big)$, where $D_k := \{x : \omega_f(x) \ge 1/k\}$.
Then f is continuous at x iff $\omega_f(x) = 0$, and its discontinuity set is $D = \bigcup_{k\ge1} D_k$.

($\Rightarrow$) Fix k and $\varepsilon > 0$. Choose P with $U(P,f) - L(P,f) < \varepsilon/k$.
Intervals of P whose interiors meet D_k have total length $< \varepsilon$; adding the finitely many partition points, null by Proposition 1.4.15, covers D_k by arbitrarily short intervals.
Thus $\mu(D_k) = 0$ by Definition 1.4.13, and $\mu(D) = 0$.

($\Leftarrow$) Suppose $\mu(D) = 0$ and $|f| \le B$. If $B = 0$, there is nothing to prove. Given $\varepsilon > 0$, choose k with $(b-a)/k < \varepsilon/2$ and cover D_k by intervals of total length $< \varepsilon/(4B)$.
On the compact complement, $\omega_f < 1/k$, so a finite refinement makes the good intervals contribute $< \varepsilon/2$ to $U - L$; the covered intervals contribute $< 2B \cdot \varepsilon/(4B) = \varepsilon/2$.
Hence $U(P,f) - L(P,f) < \varepsilon$, and Theorem 2.1.7 with $\alpha(x) = x$ gives $f \in \mathcal{R}$.

Structural Commentary: Darboux gap measures oscillation sets.
Key Insight: Lebesgue measure turns "bad discontinuities" into a null-set condition. □

Corollary 2.5.2. *Riemann Integrable Functions as Monotone Step Limits*
Let $f : [a,b] \to \mathbb{R}$ be Riemann integrable. Then $\exists$ step functions $(\varphi_n), (\psi_n)$ such that

(i) $\varphi_n \uparrow$, $\psi_n \downarrow$, and $\varphi_n \le f \le \psi_n$ on $[a,b]$;
(ii) $\lim_n \int_a^b \varphi_n\,dx = \lim_n \int_a^b \psi_n\,dx = \int_a^b f\,dx$;
(iii) $\varphi_n, \psi_n \to f$ a.e.; hence f is Lebesgue measurable and $\int_{[a,b]} f\,dm = \int_a^b f\,dx$.

Proof.

(i) Choose nested partitions P_n with $\max_{I\in P_n} \ell(I) \to 0$ and $U(P_n,f) - L(P_n,f) \to 0$.
Let φ_n, ψ_n be the lower and upper step envelopes on P_n; refinement gives $\varphi_n \uparrow$, $\psi_n \downarrow$, and $\varphi_n \le f \le \psi_n$.

(ii) Since $\int_a^b \varphi_n\,dx = L(P_n,f)$ and $\int_a^b \psi_n\,dx = U(P_n,f)$, by Darboux gap condition, both limits equal $\int_a^b f\,dx$.

(iii) If $x \notin D_f$, then $\max_{I\in P_n} \ell(I) \to 0$ forces $\varphi_n(x), \psi_n(x) \to f(x)$.
Since $m(D_f) = 0$, convergence holds a.e.; hence f is measurable and $\int_{[a,b]} f\,dm = \int_a^b f\,dx$.

Structural Commentary: Riemann integrability is monotone step squeezing.
Key Insight: Refining partitions make Darboux sums into pointwise a.e. step approximations. □

Corollary 2.5.3. *Nonnegative Continuous Functions and Riemann Integrals*

(i) If $f \geq 0$ is continuous on $[a,b]$, then $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.

(ii) If $f \geq 0$ is continuous on $[0,+\infty)$, then $\int_{[0,+\infty)} f\,dm = \lim_{b\to+\infty} \int_0^b f(x)\,dx$ in $[0,+\infty]$.

Proof.

(i) Continuous f on $[a,b]$ is Riemann integrable; by Riemann–Lebesgue comparison, $\int_{[a,b]} f\,dm = \int_a^b f(x)\,dx$.

(ii) Put $f_b := f\mathbf{1}_{[0,b]}$. By (i), $\int_{[0,b]} f\,dm = \int_0^b f(x)\,dx$. As $b \uparrow +\infty$, $f_b \uparrow f$; monotone convergence gives the left limit, and the right limit is the nonnegative improper Riemann integral.

Structural Commentary: Compact comparison plus monotone truncation.
Key Insight: Truncate to compact intervals and pass upward. □

Perspective 2.5.4. *Lebesgue's Criterion as a Roadmap*
For bounded $f : [a,b] \to \mathbb{R}$ with discontinuity set D_f, $f \in \mathcal{R}[a,b] \iff m(D_f) = 0$.
Thus finite, countable, and null Cantor-type D_f are allowed, while $D_f = [a,b]$ is not.

3 Product Measures and Fubini Theory

3.1 Product Measures

Product measure builds measurable rectangles and reads higher-dimensional integrals through slices. The abstract construction comes first; the Euclidean statements below are its geometric specialization.

Definition 3.1.1. *Product Measurable Spaces and Sections*
Let $(X,\mathcal{M})$, $(Y,\mathcal{N})$ be measurable spaces and $Z := X \times Y$.

(i) A *measurable rectangle* is $A \times B$ with $A \in \mathcal{M}$ and $B \in \mathcal{N}$; set $\mathcal{Z}_0 := \{\bigcup_{j=1}^n (A_j \times B_j) : A_j \in \mathcal{M},\ B_j \in \mathcal{N}\}$ and $\mathcal{M} \otimes \mathcal{N} := \sigma\{A \times B : A \in \mathcal{M},\ B \in \mathcal{N}\}$.
(ii) For $E \subseteq Z$, the *sections* are $E_x := \{y \in Y : (x,y) \in E\}$ and $E^y := \{x \in X : (x,y) \in E\}$.
(iii) For $F : Z \to \overline{\mathbb{R}}$, the *function sections* are $F_x(y) := F(x,y)$ and $F^y(x) := F(x,y)$, where defined.

Lemma 3.1.2. *Finite Rectangle Algebra*

(i) If $R_j = A_j \times B_j$, then $R_1 \cap R_2 = (A_1 \cap A_2) \times (B_1 \cap B_2)$ and $R_1 \setminus R_2 = \big[(A_1 \cap A_2) \times (B_1 \setminus B_2)\big] \sqcup \big[(A_1 \setminus A_2) \times B_1\big]$.
(ii) $\mathcal{Z}_0$ is an algebra on Z.
(iii) Every $E \in \mathcal{Z}_0$ admits $E = \bigsqcup_{j=1}^n (A_j \times B_j)$ with $A_j \in \mathcal{M}$ and $B_j \in \mathcal{N}$.

Proof.

(i) For $(x,y) \in Z$, $(x,y) \in R_1 \cap R_2 \iff x \in A_1 \cap A_2$ and $y \in B_1 \cap B_2$. Also, $(x,y) \in R_1 \setminus R_2$ iff $x \in A_1$, $y \in B_1$, and either $x \notin A_2$ or $y \notin B_2$; splitting by $x \in A_2$ gives the disjoint union in (i).

(ii) If $E = \bigcup_j R_j$ and $F = \bigcup_k S_k$, then $E \cap F = \bigcup_{j,k} (R_j \cap S_k) \in \mathcal{Z}_0$ by (i).
Repeatedly applying (i) gives $R_j \setminus F \in \mathcal{Z}_0$, hence $E \setminus F = \bigcup_j (R_j \setminus F) \in \mathcal{Z}_0$. Since $Z = X \times Y \in \mathcal{Z}_0$, complements are differences from Z.

(iii) If $E = \bigcup_{j=1}^n R_j$, set $S_j := R_j \setminus \bigcup_{i<j} R_i$. Then $S_j \in \mathcal{Z}_0$ by (ii), the S_j are disjoint, and $E = \bigsqcup_j S_j$. Writing each S_j as a finite union of rectangles and relabeling gives the required finite disjoint representation.

Structural Commentary: Product rectangles behave like elementary sets for the product space.
Key Insight: Before extending to a σ-algebra, make all finite overlaps into disjoint rectangle pieces. □

Theorem 3.1.3. *Product Measure Theorem*
For measure spaces $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$, there exists a measure π on $\mathcal{M} \otimes \mathcal{N}$ with $\pi(A \times B) = \mu(A)\nu(B)$ for $A \in \mathcal{M}$ and $B \in \mathcal{N}$, where $0 \cdot (+\infty) = 0$. If μ, ν are σ-finite, π is unique; write $\pi = \mu \otimes \nu$.

Proof. For $E = \bigsqcup_{j=1}^n (A_j \times B_j) \in \mathcal{Z}_0$, set $\pi_0(E) := \sum_{j=1}^n \mu(A_j)\nu(B_j)$.
If $A \times B = \bigsqcup_{j\geq 1} (A_j \times B_j)$, then $\mathbf{1}_{A\times B} = \sum_{j\geq 1} \mathbf{1}_{A_j \times B_j}$, so monotone convergence in y, then x, gives $\mu(A)\nu(B) = \sum_{j\geq 1} \mu(A_j)\nu(B_j)$. Refining two finite disjoint rectangle decompositions by intersections gives equal π_0-values; hence π_0 is well defined. If $E = \bigsqcup_k E_k \in \mathcal{Z}_0$ with $E_k \in \mathcal{Z}_0$, write $E = \bigsqcup_i R_i$ and $E_k = \bigsqcup_\ell R_{k\ell}$. Since $R_i = \bigsqcup_{k,\ell} (R_i \cap R_{k\ell})$, the rectangle comparison yields $\pi_0(E) = \sum_k \pi_0(E_k)$; hence π_0 is a premeasure. Extend π_0 to $\sigma(\mathcal{Z}_0) = \mathcal{M} \otimes \mathcal{N}$. If μ, ν are σ-finite, then $\mathcal{Z}_0$ is π_0-σ-finite, so the extension is unique.

Structural Commentary: Rectangles fix the product measure.
Key Insight: Monotone convergence makes rectangle mass respect countable disjoint refinements. □

Lemma 3.1.4. *Measurable Sections*

(i) $E \in \mathcal{M} \otimes \mathcal{N}$ *implies* $E_x \in \mathcal{N}$ *for all* $x \in X$ *and* $E^y \in \mathcal{M}$ *for all* $y \in Y$.
(ii) If $F : Z \to \overline{\mathbb{R}}$ *is* $\mathcal{M} \otimes \mathcal{N}$*-measurable, then* F_x *is* $\mathcal{N}$*-measurable for all* x *and* F^y *is* $\mathcal{M}$*-measurable for all* y.

Proof.

(i) Fix $x \in X$ and set $\mathcal{C}_x := \{E \subseteq Z : E_x \in \mathcal{N}\}$. Since $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$, $\mathcal{C}_x$ is a σ-algebra. Also $(A \times B)_x = B$ if $x \in A$ and $\emptyset$ otherwise; hence $\mathcal{M} \otimes \mathcal{N} \subseteq \mathcal{C}_x$. The E^y proof is identical.
(ii) For $a \in \mathbb{R}$, set $H_a := \{(u, v) \in Z : F(u, v) > a\} \in \mathcal{M} \otimes \mathcal{N}$. By (i), $(H_a)_x \in \mathcal{N}$ and $(H_a)^y \in \mathcal{M}$. Since $(H_a)_x = \{y : F_x(y) > a\}$ and $(H_a)^y = \{x : F^y(x) > a\}$, all threshold sets are measurable.

Structural Commentary: Product measurability is stable under freezing one coordinate.
Key Insight: Section operations preserve the set algebra used to generate the product σ-algebra. □

Proposition 3.1.5. *Section Measures Integrate to Product Measure*
Let $(X, \mathcal{M}, \mu)$ *and* $(Y, \mathcal{N}, \nu)$ *be* σ*-finite,* $\pi := \mu \otimes \nu$*, and* $E \in \mathcal{M} \otimes \mathcal{N}$*. Set* $f_E(x) := \nu(E_x)$ *and* $g_E(y) := \mu(E^y)$*. Then* f_E, g_E *are measurable and* $\int_X f_E \, d\mu = \pi(E) = \int_Y g_E \, d\nu$.

Proof. Assume $\mu(X), \nu(Y) < +\infty$ and let $\mathcal{C}$ be the class of E satisfying the claim.
For $E = A \times B$, $f_E = \nu(B)\mathbf{1}_A$ and $g_E = \mu(A)\mathbf{1}_B$, so $A \times B \in \mathcal{C}$; finite additivity gives $\mathcal{Z}_0 \subseteq \mathcal{C}$.
If $E_n \uparrow E$ in $\mathcal{C}$, then $(E_n)_x \uparrow E_x$ and $(E_n)^y \uparrow E^y$; MCT gives $E \in \mathcal{C}$.
If $E_n \downarrow E$, finite total mass gives downward continuity and DCT, hence $E \in \mathcal{C}$.
Thus $\mathcal{C}$ is a monotone class containing $\mathcal{Z}_0$, so $\mathcal{C} = \mathcal{M} \otimes \mathcal{N}$. For σ-finite μ, ν, choose $X_n \uparrow X$, $Y_n \uparrow Y$ with $\mu(X_n), \nu(Y_n) < +\infty$, apply the finite case to $E_n := E \cap (X_n \times Y_n)$, then let $n \to \infty$ by MCT.
Structural Commentary: Cavalieri's principle is already true for abstract σ-finite products.
Key Insight: Prove rectangles, close under monotone limits, then exhaust by finite-measure rectangles. □

3.2 Fubini and Tonelli Theorems

Theorem 3.2.1. *Tonelli's Theorem for* σ*-Finite Products*
Let $(X, \mathcal{M}, \mu)$ *and* $(Y, \mathcal{N}, \nu)$ *be* σ*-finite,* $\pi := \mu \otimes \nu$*, and* $F : Z \to [0, +\infty]$ *be measurable. Set* $I_F(x) := \int_Y F_x \, d\nu$ *and* $J_F(y) := \int_X F^y \, d\mu$*. Then* I_F, J_F *are measurable and* $\int_X I_F \, d\mu = \int_Z F \, d\pi = \int_Y J_F \, d\nu$.

Proof. For $F = \mathbf{1}_E$, the claim is Proposition 3.1.5. By finite linearity, it holds for simple $\Phi = \sum_{j=1}^n a_j \mathbf{1}_{E_j} \geq 0$.
Choose simple $0 \leq \Phi_n \uparrow F$. Then $(\Phi_n)_x \uparrow F_x$, $(\Phi_n)^y \uparrow F^y$, $I_{\Phi_n} \uparrow I_F$, and $J_{\Phi_n} \uparrow J_F$.
MCT in X, Y, Z passes the identity from Φ_n to F.
Structural Commentary: Tonelli extends sections from indicators to $F \geq 0$.
Key Insight: Nonnegativity lets all three integrals pass through monotone approximation together. □

Theorem 3.2.2. *Fubini's Theorem for* σ*-Finite Products*
Let $(X, \mathcal{M}, \mu)$ *and* $(Y, \mathcal{N}, \nu)$ *be* σ*-finite,* $\pi := \mu \otimes \nu$*, and* $F \in \mathscr{L}^1(\pi)$.

(i) $F_x \in \mathscr{L}^1(\nu)$ *for* μ*-a.e.* x *and* $F^y \in \mathscr{L}^1(\mu)$ *for* ν*-a.e.* y.
(ii) With $f(x) := \int_Y F_x \, d\nu$ *and* $g(y) := \int_X F^y \, d\mu$ *on those full-measure sets, and* 0 *elsewhere,* $f \in \mathscr{L}^1(\mu)$*,* $g \in \mathscr{L}^1(\nu)$*, and* $\int_X f \, d\mu = \int_Z F \, d\pi = \int_Y g \, d\nu$.

Proof.

(i) Tonelli on $|F|$ gives $\int_X (\int_Y |F_x| \, d\nu) d\mu = \int_Z |F| \, d\pi = \int_Y (\int_X |F^y| \, d\mu) d\nu < +\infty$, so the inner integrals are finite a.e.
(ii) Tonelli on F^+ and F^- gives finite identities; subtract them to obtain $\int_X f \, d\mu = \int_Z F \, d\pi = \int_Y g \, d\nu$. Moreover $|f| \leq \int_Y |F_x| \, d\nu$ and $|g| \leq \int_X |F^y| \, d\mu$, so (i) gives $f, g \in \mathscr{L}^1$.

Structural Commentary: Fubini is Tonelli plus subtraction.
Key Insight: $\int |F| < +\infty$ makes the two nonnegative Tonelli identities finite enough to subtract. □

Proposition 3.2.3. *Product Borel and Measurable Building Blocks*
The following hold.

(i) $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) = \mathcal{B}(\mathbb{R}^2)$.
(ii) If $f : X \to \mathbb{R}$ *is* $\mathcal{M}$*-measurable and* $g : Y \to \mathbb{R}$ *is* $\mathcal{N}$*-measurable, then* $(x, y) \mapsto f(x)g(y)$ *is* $\mathcal{M} \otimes \mathcal{N}$*-measurable.*
(iii) If $u : \mathbb{R} \to \mathbb{R}$ *is Borel measurable, then* $(x, y) \mapsto u(x - y)$ *is* $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$*-measurable.*

Proof.

(i) Open rectangles give $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R}) \subseteq \mathcal{B}(\mathbb{R}^2)$.
Conversely, every open $O \subseteq \mathbb{R}^2$ is a countable union of rational rectangles, each in $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$.
(ii) For $\tilde{f}(x,y) := f(x)$ and $\tilde{g}(x,y) := g(y)$, $\{\tilde{f} > a\} = f^{-1}((a,+\infty)) \times Y$ and $\{\tilde{g} > a\} = X \times g^{-1}((a,+\infty))$.
Thus $\tilde{f}, \tilde{g}$ are measurable, so $\tilde{f}\tilde{g}$ is measurable.
(iii) The map $s(x,y) := x - y$ is continuous on $\mathbb{R}^2$, hence measurable by (i); therefore $u \circ s$ is measurable.

Structural Commentary: Product measurability supports two-variable formulas.
Key Insight: Check coordinate pullbacks on rectangles, then use ordinary measurable-function closure. □

Proposition 3.2.4. *Counting Factor Products*
Let $X = \mathbb{N}$, $\mathcal{M} = 2^{\mathbb{N}}$, μ be counting measure, and $(Y, \mathcal{N}, \nu)$ be a measure space.

(i) For $E \subseteq \mathbb{N} \times Y$, $E \in 2^{\mathbb{N}} \otimes \mathcal{N} \iff E_n \in \mathcal{N}$ for all n, and then $E = \bigsqcup_{n\geq 1}(\{n\} \times E_n)$.
(ii) The unique product measure is $\pi = \mu \otimes \nu$ with $\pi(E) = \sum_{n\geq 1} \nu(E_n)$ for $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$.
(iii) For $F : \mathbb{N} \times Y \to \mathbb{R}$, F is product-measurable $\iff F_n$ is $\mathcal{N}$-measurable for all n.
(iv) For product-measurable F, $F \in \mathscr{L}^1(\pi) \iff \sum_{n\geq 1} \int_Y |F_n|\, d\nu < +\infty$;
then $\int_{\mathbb{N}\times Y} F\, d\pi = \sum_{n\geq 1} \int_Y F_n\, d\nu = \int_Y \sum_{n\geq 1} F_n\, d\nu$, with $\sum_n F_n$ absolutely convergent ν-a.e.

Proof.

(i) ($\Rightarrow$) If $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$, Lemma 3.1.4 gives $E_n \in \mathcal{N}$ for all n.
(i) ($\Leftarrow$) If $E_n \in \mathcal{N}$ for all n, then each $\{n\} \times E_n \in 2^{\mathbb{N}} \otimes \mathcal{N}$ and $E = \bigsqcup_{n\geq 1}(\{n\} \times E_n)$, hence $E \in 2^{\mathbb{N}} \otimes \mathcal{N}$.
(ii) Define $\pi(E) := \sum_{n\geq 1} \nu(E_n)$.
If $(E^k)_k$ are disjoint, then $(\bigsqcup_k E^k)_n = \bigsqcup_k E^k_n$, so $\pi(\bigsqcup_k E^k) = \sum_n \sum_k \nu(E^k_n) = \sum_k \pi(E^k)$.
Also $\pi(\{n\} \times A) = \nu(A)$, and (i) forces uniqueness.
(iii) ($\Rightarrow$) If F is product-measurable, Lemma 3.1.4 gives each F_n measurable.
(iii) ($\Leftarrow$) If each F_n is measurable, then for every $a \in \mathbb{R}$, $H_a := \{F > a\}$ satisfies $(H_a)_n = \{F_n > a\} \in \mathcal{N}$,
By (i), $H_a \in 2^{\mathbb{N}} \otimes \mathcal{N}$, so F is product-measurable.
(iv) For $G \geq 0$, (ii) gives the identity for $G = \mathbf{1}_E$; linearity and MCT give $\int_{\mathbb{N}\times Y} G\, d\pi = \sum_{n\geq 1} \int_Y G_n\, d\nu$.
(iv) Apply the preceding identity to $G = |F|$: $F \in \mathscr{L}^1(\pi) \iff \sum_{n\geq 1} \int_Y |F_n|\, d\nu < +\infty$.
If finite, then $\int_Y \sum_n |F_n|\, d\nu < +\infty$, so $\sum_n F_n$ converges absolutely a.e.; applying the nonnegative identity to F^+ and F^- and subtracting gives the integral formulas.

Structural Commentary: A product with counting measure is a measurable sequence of fibers.
Key Insight: Counting atoms turn products into sums. □

Example 3.2.5. *A Diagonal Failure Without σ-Finiteness*
Let $X = Y = [0,1]$, $\mathcal{M} = \mathcal{N} = \mathcal{B}([0,1])$, $\mu = m$, $\nu = \#$, and $D := \{(x,y) \in [0,1]^2 : x = y\}$.
Then $D \in \mathcal{M} \otimes \mathcal{N}$, $D_x = \{x\}$, and $D^y = \{y\}$, so $\nu(D_x) = 1$ and $\mu(D^y) = 0$.
Hence $\int_X \nu(D_x)\, d\mu = 1$ but $\int_Y \mu(D^y)\, d\nu = 0$, so Tonelli/Fubini can fail without σ-finite factors.

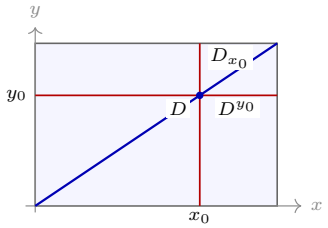


Figure 196: Diagonal sections: $\nu(D_x) = 1$ but $\mu(D^y) = 0$.

Example 3.2.6. *Double Series as Product Integrals*
With counting measure on $\mathbb{N}^2$, Tonelli gives, for $a_{mn} \geq 0$,
$\sum_m \sum_n a_{mn} = \sum_{(m,n)\in\mathbb{N}^2} a_{mn} = \sum_n \sum_m a_{mn} \in [0,+\infty]$.
Fubini needs integrability: set $a_{nn} = 1$, $a_{n,n+1} = -1$, and $a_{mn} = 0$ otherwise.
Then $\sum_m \sum_n a_{mn} = 0$, while $\sum_n \sum_m a_{mn} = 1$, and $\sum_{m,n} |a_{mn}| = +\infty$.

Proposition 3.2.7. *Separated Products*
Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite. If $f \in \mathscr{L}^1(\mu)$, $g \in \mathscr{L}^1(\nu)$, and $H(x,y) := f(x)g(y)$, then $H \in \mathscr{L}^1(\mu\otimes\nu)$ and $\int_{X\times Y} H\, d(\mu \otimes \nu) = (\int_X f\, d\mu)(\int_Y g\, d\nu)$.

Proof. Measurability follows from Proposition 3.2.3.
Tonelli gives $\int_{X\times Y} |H|\, d(\mu\otimes\nu) = \int_X |f(x)|\,(\int_Y |g|\, d\nu)d\mu = (\int_X |f|\, d\mu)(\int_Y |g|\, d\nu) < +\infty$, so $H \in \mathscr{L}^1(\mu\otimes\nu)$.
Fubini applied to H gives the displayed identity.
Structural Commentary: Product-measure integration separates products of one-variable integrable functions.
Key Insight: Absolute integrability factors by Tonelli; signed equality follows by Fubini. □

Corollary 3.2.8. *Equal Section Measures*
Let $(X,\mathcal{M},\mu)$ and $(Y,\mathcal{N},\nu)$ be σ-finite.
If $E,F \in \mathcal{M}\otimes\mathcal{N}$ and $\nu(E_x) = \nu(F_x)$ for all $x \in X$, then $(\mu\otimes\nu)(E) = (\mu\otimes\nu)(F)$.

Proof. Set $f_E(x) := \nu(E_x)$ and $f_F(x) := \nu(F_x)$.
Since $f_E = f_F$, Proposition 3.1.5 gives $(\mu\otimes\nu)(E) = \int_X f_E\, d\mu = \int_X f_F\, d\mu = (\mu\otimes\nu)(F)$.
Structural Commentary: Product mass is recovered from vertical section masses.
Key Insight: Equal fibers give equal product mass. □

Definition 3.2.9. *Convolution*
For Borel $f,g : \mathbb{R}\to\mathbb{R}$, if $y \mapsto f(x-y)g(y)$ is integrable for a.e. x,
define the *convolution* $(f*g)(x) := \int_{\mathbb{R}} f(x-y)g(y)\, dm(y)$ on that full-measure set and 0 elsewhere.

Proposition 3.2.10. *Convolution of L^1 Functions*
If Borel $f,g \in \mathscr{L}^1(m)$, then $f*g$ is a.e. defined, $f*g \in \mathscr{L}^1(m)$, and $\int_{\mathbb{R}} |f*g|\, dm \le (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm)$.

Proof. Set $\Phi(x,y) := f(x-y)g(y)$. By Proposition 3.2.3, Φ is product-measurable. For fixed y, translation invariance gives $\int_{\mathbb{R}} |f(x-y)|\, dm(x) = \int_{\mathbb{R}} |f|\, dm$. Hence Tonelli yields $\int_{\mathbb{R}^2} |\Phi|\, d(m\otimes m) = (\int_{\mathbb{R}} |f|\, dm)(\int_{\mathbb{R}} |g|\, dm) < +\infty$. Fubini gives the a.e. existence of $f*g$, and $\int_{\mathbb{R}} |f*g|\, dm \le \int_{\mathbb{R}^2} |\Phi|\, d(m\otimes m)$ gives the bound.
Structural Commentary: Convolution is Fubini applied to the difference map $(x,y)\mapsto x-y$.
Key Insight: Translation invariance makes every shifted copy of $|f|$ have the same integral. □

Example 3.2.11. *Nonunique Products Without σ-Finiteness*
Let $X=\mathbb{R}$, $\mathcal{M}=2^{\mathbb{R}}$, and $\mu(A)=0$ if A is countable, $\mu(A)=+\infty$ otherwise.
Define $\lambda_i : \mathcal{M}\otimes\mathcal{M}\to\{0,+\infty\}$ by $\lambda_i(E)=0$ exactly on the following null ideals:
$\mathcal{I}_1 = \{E : E = G\cup H,\ \mathrm{pr}_x\, G \text{ countable},\ \mathrm{pr}_y\, H \text{ countable}\}$
and $\mathcal{I}_2 = \{E : E = G\cup H\cup K,\ \mathrm{pr}_x\, G \text{ countable},\ \mathrm{pr}_y\, H \text{ countable},\ \mathrm{pr}_{y=x}\, K \text{ countable}\}$.
Both ideals are closed under countable unions, hence λ_i are measures.
For every rectangle $A\times B$, both give $\mu(A)\mu(B)$: countable sides are null, and uncountable-by-uncountable rectangles are not in either ideal.
For $E=\{(x,y) : x+y=0\}$, $\mathrm{pr}_{y=x}\, E$ is a point, so $\lambda_2(E)=0$ but $\lambda_1(E)=+\infty$.
Structural Commentary: Without σ-finiteness, rectangle values need not determine a product measure.
Key Insight: Extra null lines preserve rectangles but change non-rectangles.

Definition 3.2.12. *Products and Slices*
For $A\subseteq\mathbb{R}^n$, $B\subseteq\mathbb{R}^k$, set $A\times B\subseteq\mathbb{R}^{n+k}$.
For $E\subseteq\mathbb{R}^n\times\mathbb{R}^k$ define $E_x := \{y\in\mathbb{R}^k : (x,y)\in E\}$ as *x-slice*, and $E^y := \{x\in\mathbb{R}^n : (x,y)\in E\}$ as *y-slice*.
For $f : E\to\mathbb{R}$, write $f_x(y) = f^y(x) := f(x,y)$ where defined.

Theorem 3.2.13. *Measurable Products*
If $A\subseteq\mathbb{R}^n$, $B\subseteq\mathbb{R}^k$ Lebesgue measurable, then $A\times B$ measurable, $m(A\times B) = m(A)m(B)$; $0\cdot\infty = \infty\cdot 0 = 0$.

Proof. For boxes I,J, $m(I\times J) = m(I)m(J)$; disjoint box decompositions extend this to open sets.
First assume A,B are bounded with finite measure.
By regularity choose closed $F_A\subseteq A$, $F_B\subseteq B$ and open $G_A\supseteq A$, $G_B\supseteq B$ such that $m(G_A\backslash F_A)$, $m(G_B\backslash F_B) < \varepsilon$.
Then $F_A\times F_B \subseteq A\times B \subseteq G_A\times G_B$, and the open/closed product formula squeezes $m(A\times B)$ to $m(A)m(B)$.
For general A,B, set $A_N = A\cap[-N,N]^n$ and $B_N = B\cap[-N,N]^k$. Since $A_N\times B_N \uparrow A\times B$, continuity from below gives $m(A\times B) = \lim_N m(A_N\times B_N) = \lim_N m(A_N)m(B_N) = m(A)m(B)$ with $0\cdot\infty = \infty\cdot 0 = 0$.
Structural Commentary: Product measure comes from boxes and regularity.
Key Insight: Regularity turns rough measurable products into box products plus zero-set errors. □

Theorem 3.2.14. *Zero Slices*
If $E\subseteq\mathbb{R}^n\times\mathbb{R}^k$ is Lebesgue measurable, then E is null iff E_x is null for a.e. x.

Proof.

($\Rightarrow$) If $m(E)=0$, cover E by an open W with small measure.
For $X_\alpha(W) = \{x : m(W_x) > \alpha\}$, $m(W) \ge \alpha m(X_\alpha(W))$; letting $m(W)\to 0$ gives $m(E_x)=0$ for a.e. x.

($\Leftarrow$) Remove the exceptional null base so every slice is null. For compact $K \subseteq E$, cover each K_x by a small-measure slice neighborhood; compactness gives finitely many product neighborhoods covering K, whose total measure is arbitrarily small by Theorem 3.2.13. Thus $m(K) = 0$, hence $m(E) = 0$ by regularity.

Structural Commentary: Nullity can be checked fiberwise after removing an exceptional base set.
Key Insight: Product measure turns small slice covers into small total measure. □

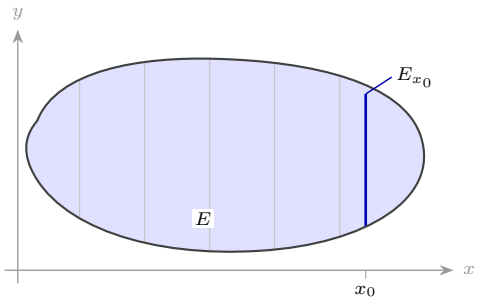


Figure 197: A planar set can be read by vertical slices; Fubini integrates the slice data over the base variable.

Theorem 3.2.15. *Cavalieri's Principle for Lebesgue Measure*
If $E \in \mathcal{L}^{n+k}$, then $E_x \in \mathcal{L}^k$ for a.e. x, and the a.e.-defined function $f_E(x) := m_k(E_x)$ has a measurable representative with $m_{n+k}(E) = \int_{\mathbb{R}^n} f_E \, dm_n$. The analogous statement holds for E^y.

Proof. Let $P(E)$ denote the displayed assertion. For $E = A \times B$, $f_E = m_k(B)\mathbf{1}_A$, so $P(E)$ is Theorem 3.2.13. Fix a bounded box $Q = I \times J$. The class $\{E \subseteq Q : P(E)\}$ contains finite unions of rectangles and is monotone: if $E_j \uparrow E$, then $f_{E_j} \uparrow f_E$ and monotone convergence applies; if $E_j \downarrow E$, then $f_{E_j} \downarrow f_E \le m_k(J)\mathbf{1}_I$ and dominated convergence applies. Hence the monotone class theorem gives $P(E)$ for every Borel $E \subseteq Q$. Exhaustion by $Q_N = [-N, N]^{n+k}$ and monotone convergence give all Borel E.
For Lebesgue E, choose Borel B with $m_{n+k}(E \triangle B) = 0$ (Remark 1.5.2). Then $m_k((E \triangle B)_x) = 0$ for a.e. x (Theorem 3.2.14), so $f_E = f_B$ a.e. and $m_{n+k}(E) = m_{n+k}(B)$. The y-slice statement is identical.
Structural Commentary: Slicing turns a measure in $\mathbb{R}^{n+k}$ into a measurable family of measures in $\mathbb{R}^k$.
Key Insight: Prove boxes, close under monotone limits, then ignore null symmetric differences. □

Perspective 3.2.16. *Fubini Proof Ladder*
Fubini ladder: $\mathbf{1}_E \to$ simple $\Phi \ge 0 \to f \ge 0$ by MCT, then $f = f^+ - f^-$ when $\int |f| < +\infty$.
Theorems 3.2.1 and 3.2.2 are the abstract form.
Key Insight: Fubini is a slice theorem, proof runs through indicators, simple functions, monotone convergence.

Proposition 3.2.17. *Undergraphs and Level-Set Measurability*
Let $f : \mathbb{R}^n \to [0, \infty]$, set $F_a := \{x : f(x) > a\}$, and write $\mathcal{U}f = \{(x, t) \in \mathbb{R}^n \times [0, \infty) : 0 < t < f(x)\}$. Then f is Lebesgue measurable iff $F_q \in \mathcal{L}^n$ $\forall q \in \mathbb{Q}_{>0}$. In that case, $\mathcal{U}f \in \mathcal{L}^{n+1}$ and $(\mathcal{U}f)^t = F_t$, $t > 0$.

Proof. For $a \ge 0$, $F_a = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} F_q$; for $a < 0$, $F_a = \mathbb{R}^n$.
Hence rational positive thresholds are equivalent to measurability.
If they are measurable, then $\mathcal{U}f = \bigcup_{\substack{q,r \in \mathbb{Q} \\ 0 < q < r}} F_r \times (q, r) \in \mathcal{L}^{n+1}$. Finally, $(\mathcal{U}f)^t = \{x : 0 < t < f(x)\} = F_t$ for $t > 0$.
Structural Commentary: A nonnegative function may be studied as the region lying beneath its graph.
Key Insight: Horizontal slices of an undergraph are exactly strict superlevel sets. □

Theorem 3.2.18. *Fubini–Tonelli for Nonnegative Functions*
Let $f : \mathbb{R}^n \times \mathbb{R}^k \to [0, \infty]$ be Lebesgue measurable. Then f_x is measurable for a.e. x; with $F(x) := \int_{\mathbb{R}^k} f_x \, dm_k$ there and arbitrary values elsewhere, F is measurable and $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^n} F \, dm_n$.
The same holds for $G(y) := \int_{\mathbb{R}^n} f^y \, dm_n$, with $\int_{\mathbb{R}^{n+k}} f \, dm_{n+k} = \int_{\mathbb{R}^k} G \, dm_k$.

Proof. Put $U := \mathcal{U}f \subseteq \mathbb{R}^{n+k} \times [0, \infty)$. Proposition 3.2.17 gives $U \in \mathcal{L}^{n+k+1}$. By Cavalieri, Theorem 3.2.15, U_x is measurable for a.e. x, $H(x) := m_{k+1}(U_x)$ is measurable, and $m_{n+k+1}(U) = \int_{\mathbb{R}^n} H \, dm_n$.
For those x, $U_x = \mathcal{U}(f_x)$, so f_x is measurable and $H(x) = \int_{\mathbb{R}^k} f_x \, dm_k$.
Also $m_{n+k+1}(U) = \int_{\mathbb{R}^{n+k}} f \, dm_{n+k}$ by the undergraph formula. The y-formula is identical.
Structural Commentary: Tonelli's theorem is Cavalieri's principle applied one dimension higher.
Key Insight: $(\mathcal{U}f)_x = \mathcal{U}(f_x)$. □

Theorem 3.2.19. *Fubini for L^1 Functions*
If $f \in L^1(\mathbb{R}^2)$, then $f(x,\cdot) \in L^1(\mathbb{R})$ for a.e. x and $f(\cdot,y) \in L^1(\mathbb{R})$ for a.e. y. Defining $F(x) := \int_{\mathbb{R}} f(x,y)\,dy$; $G(y) := \int_{\mathbb{R}} f(x,y)\,dx$ on those sets and as 0 elsewhere gives $F, G \in L^1(\mathbb{R})$ and $\int_{\mathbb{R}} F(x)\,dx = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G(y)\,dy$

Proof. Let $E_x = \{y : (x,y) \in E\}$ and $E^y = \{x : (x,y) \in E\}$. For measurable $E \subseteq \mathbb{R}^2$, the box/regularity argument and Theorems 3.2.13–3.2.14 give $\int_{\mathbb{R}} m(E_x)\,dx = m(E) = \int_{\mathbb{R}} m(E^y)\,dy$.
Hence Fubini holds for $\mathbf{1}_E$, and by linearity for every nonnegative simple function.
Let $0 \le h$ be measurable and choose $0 \le s_n \uparrow h$ by Theorem 2.1.12.
Monotone convergence gives $\int_{\mathbb{R}} \left(\int_{\mathbb{R}} h(x,y)\,dy\right) dx = \lim_n \int_{\mathbb{R}} \left(\int_{\mathbb{R}} s_n(x,y)\,dy\right) dx = \lim_n \int_{\mathbb{R}^2} s_n = \int_{\mathbb{R}^2} h$ and the same argument with x, y interchanged. Now let $f \in L^1(\mathbb{R}^2)$.
Applying the nonnegative case to $|f|$ yields $\int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x,y)|\,dy\right) dx = \int_{\mathbb{R}^2} |f| < \infty$, so $f(x,\cdot) \in L^1(\mathbb{R})$ for a.e. x; similarly $f(\cdot,y) \in L^1(\mathbb{R})$ for a.e. y. With $f = f^+ - f^-$, define, up to null-set choices, $F = F^+ - F^-$, $F^{\pm}(x) = \int_{\mathbb{R}} f^{\pm}(x,y)\,dy$, and analogously $G = G^+ - G^-$. Then $\int_{\mathbb{R}} |F| \le \int_{\mathbb{R}} (F^+ + F^-) = \int_{\mathbb{R}^2} |f| < \infty$, and similarly $G \in L^1(\mathbb{R})$. Subtracting the nonnegative identities for f^+ and f^- gives $\int_{\mathbb{R}} F = \int_{\mathbb{R}^2} f = \int_{\mathbb{R}} G$.
Structural Commentary: Prove identity for $\mathbf{1}_E$, extend by linearity and monotone convergence, apply to $f^{\pm}$.
Key Insight: $\int |f| < \infty$ makes the positive and negative slice integrals finite a.e., so subtraction is legitimate. □

4 Radon-Nikodym and Differentiation

4.1 Signed and Absolutely Continuous Measures

This subsection first decomposes finite signed set functions into honest measures, then compares measures by absolute continuity and singularity. The differentiation results later in the subsection are the local Euclidean counterpart of the Radon–Nikodym viewpoint.

Definition 4.1.1. *Positive, Negative, and Null Sets for a Charge*
Let λ be a charge on $(X, \mathcal{M})$.

(i) $\mathcal{P}_\lambda := \{P \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap P) \ge 0\}$, the *positive* sets.
(ii) $\mathcal{N}_\lambda := \{N \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap N) \le 0\}$, the *negative* sets.
(iii) $\mathcal{Z}_\lambda := \{M \in \mathcal{M} : \forall E \in \mathcal{M}, \lambda(E \cap M) = 0\}$, the *null* sets.

Proposition 4.1.2. *Positive and Negative Set Closure*
For a charge λ:

(i) $A \in \mathcal{M}$ and $A \subseteq P \in \mathcal{P}_\lambda$ imply $A \in \mathcal{P}_\lambda$; $A \subseteq N \in \mathcal{N}_\lambda$ implies $A \in \mathcal{N}_\lambda$.
(ii) $\mathcal{P}_\lambda$ and $\mathcal{N}_\lambda$ are closed under finite unions.

Proof.

(i) If $A \subseteq P \in \mathcal{P}_\lambda$, then $\lambda(E \cap A) = \lambda((E \cap A) \cap P) \ge 0$ for all $E \in \mathcal{M}$, hence $A \in \mathcal{P}_\lambda$.
The negative case reverses the inequality.
(ii) If $P_1, P_2 \in \mathcal{P}_\lambda$, then $\lambda(E \cap (P_1 \cup P_2)) = \lambda(E \cap P_1) + \lambda(E \cap P_1^c \cap P_2) \ge 0$ $\forall E \in \mathcal{M}$, since $E \cap P_1^c \cap P_2 \subseteq P_2$.
Thus $P_1 \cup P_2 \in \mathcal{P}_\lambda$; induction gives finite unions, negative case is identical with reversed inequalities.

Structural Commentary: Sign sets are hereditary and finite-union stable.
Key Insight: Test sets can always be disjointized against the first positive set. □

Theorem 4.1.3. *Hahn Decomposition Theorem*
Let λ be a charge on $(X, \mathcal{M})$. Then there exist $P, N \in \mathcal{M}$ such that

(i) $P \cap N = \emptyset$ and $X = P \sqcup N$.
(ii) $P \in \mathcal{P}_\lambda$.
(iii) $N \in \mathcal{N}_\lambda$.

Such a pair (P, N) is a Hahn decomposition of X for λ.

Proof.

(i) Put $\alpha = \sup\{\lambda(A) : A \in \mathcal{P}_\lambda\}$. If $\alpha = +\infty$, choose $B_m \in \mathcal{P}_\lambda$, $\lambda(B_m) > m$, and set $C_m = \bigcup_{j=1}^m B_j$; then $C_m \in \mathcal{P}_\lambda$, $C_m \uparrow C$, and $\lambda(C_m) \ge \lambda(B_m) > m$, contradicting continuity from below and $\lambda(C) \in \mathbb{R}$.
Choose $A_n \in \mathcal{P}_\lambda$, $\lambda(A_n) \to \alpha$, replace A_n by $\bigcup_{j=1}^n A_j$, and set $P = \bigcup_{n \ge 1} A_n$, $N = X \setminus P$.
Hence $P, N \in \mathcal{M}$, $P \cap N = \emptyset$, and $X = P \sqcup N$.
(ii) For $E \in \mathcal{M}$, $E \cap A_n \uparrow E \cap P$ and $\lambda(E \cap A_n) \ge 0$, so continuity from below gives $\lambda(E \cap P) \ge 0$.
Thus $P \in \mathcal{P}_\lambda$, and $\lambda(P) = \alpha$.

(iii) Suppose $E \in \mathcal{M}$, $E \subseteq N$, and $\lambda(E) > 0$. Then $E \notin \mathcal{P}_\lambda$, else $P \cup E \in \mathcal{P}_\lambda$ and $\lambda(P \cup E) > \alpha$.
Set $R_0 = E$; while $R_{k-1} \notin \mathcal{P}_\lambda$, choose n_k minimal with some $H \in \mathcal{M}$, $H \subseteq R_{k-1}$, $\lambda(H) \leq -1/n_k$, put $E_k = H$, and set $R_k = R_{k-1} \setminus E_k$. If $R_m \in \mathcal{P}_\lambda$, then $\lambda(R_m) = \lambda(E) - \sum_{k=1}^m \lambda(E_k) > 0$, so $P \cup R_m \in \mathcal{P}_\lambda$ and $\lambda(P \cup R_m) > \alpha$; hence the construction is infinite.
For $F_m = \bigcup_{k=1}^m E_k$ and $F = \bigcup_{k \geq 1} E_k$, $\lambda(F_m) = \sum_{k=1}^m \lambda(E_k) \leq -\sum_{k=1}^m 1/n_k$; since $F_m \uparrow F$ and $\lambda(F) \in \mathbb{R}$, $\sum_k 1/n_k < +\infty$ and $n_k \to +\infty$. If $G \in \mathcal{M}$ and $G \subseteq E \setminus F$, then $G \subseteq R_{k-1}$ for every k, so minimality gives $\lambda(G) > -1/(n_k - 1)$ for all large k, hence $\lambda(G) \geq 0$ and $E \setminus F \in \mathcal{P}_\lambda$. But $\lambda(E \setminus F) = \lambda(E) - \lambda(F) > 0$, so $P \cup (E \setminus F) \in \mathcal{P}_\lambda$ and $\lambda(P \cup (E \setminus F)) > \alpha$, contradiction. Thus $N \in \mathcal{N}_\lambda$.

Structural Commentary: A charge splits X into nonnegative-piece and nonpositive-piece regions.
Key Insight: Maximize positive mass; purify any leftover positive piece. □

Lemma 4.1.4. *Hahn Parts Are Unique up to Null Sets*
Let (P_i, N_i), $i = 1, 2$, be Hahn decompositions for a charge λ.
Then $\forall E \in \mathcal{M}$, $\lambda(E \cap P_1) = \lambda(E \cap P_2)$ and $\lambda(E \cap N_1) = \lambda(E \cap N_2)$.

Proof. Fix $E \in \mathcal{M}$. For $\{i, j\} = \{1, 2\}$, $E \cap (P_i \setminus P_j) \subseteq P_i \cap N_j$, hence $\lambda(E \cap (P_i \setminus P_j)) = 0$.
Thus $\lambda(E \cap P_i) = \lambda(E \cap P_1 \cap P_2)$ for $i = 1, 2$.
Similarly, $E \cap (N_i \setminus N_j) \subseteq N_i \cap P_j$, hence $\lambda(E \cap N_i) = \lambda(E \cap N_1 \cap N_2)$ for $i = 1, 2$.
Structural Commentary: Hahn decompositions are not unique as sets, but the measures they define are unique.
Key Insight: Positive-and-negative overlap is necessarily null. □

Definition 4.1.5. *Jordan Variations of a Charge*
Let λ be a charge, (P, N) a Hahn decomposition, and $E \in \mathcal{M}$.

(i) $\lambda^+(E) := \lambda(E \cap P)$, the *positive variation*.
(ii) $\lambda^-(E) := -\lambda(E \cap N)$, the *negative variation*.
(iii) $|\lambda|(E) := \lambda^+(E) + \lambda^-(E)$, the *total variation*.

Lemma 4.1.4 makes these independent of (P, N).

Theorem 4.1.6. *Jordan Decomposition Theorem*
For every charge λ, $\lambda = \lambda^+ - \lambda^-$. If $\lambda = \mu - \nu$ for finite measures μ, ν, then $\lambda^+ \leq \mu$ and $\lambda^- \leq \nu$.

Proof. Let (P, N) be a Hahn decomposition and $E \in \mathcal{M}$. Then $\lambda(E) = \lambda(E \cap P) + \lambda(E \cap N) = \lambda^+(E) - \lambda^-(E)$.
If $\lambda = \mu - \nu$, then $\lambda^+(E) = \mu(E \cap P) - \nu(E \cap P) \leq \mu(E)$ and $\lambda^-(E) = \nu(E \cap N) - \mu(E \cap N) \leq \nu(E)$.
Structural Commentary: Jordan decomposition is the minimal finite-measure difference.
Key Insight: Positive and negative Hahn pieces isolate the irreducible positive masses. □

Corollary 4.1.7. *Variation by Subsets and Null Sets*
For a charge λ and $E, M \in \mathcal{M}$:

(i) $\lambda^+(E) = \sup\{\lambda(F) : F \in \mathcal{M}, F \subseteq E\}$ *and* $\lambda^-(E) = -\inf\{\lambda(F) : F \in \mathcal{M}, F \subseteq E\}$.
(ii) $M \in \mathcal{Z}_\lambda \Leftrightarrow |\lambda|(M) = 0$.
(iii) $\pi_\lambda = \lambda^+$ *and* $\nu_\lambda = |\lambda|$.

Proof.

(i) Let (P, N) be Hahn. If $F \in \mathcal{M}$ and $F \subseteq E$, then $\lambda(F) = \lambda^+(F) - \lambda^-(F) \leq \lambda^+(E)$, with equality at $F = E \cap P$; similarly $\lambda(F) \geq -\lambda^-(E)$, with equality at $F = E \cap N$.
(ii) ($\Rightarrow$) If $M \in \mathcal{Z}_\lambda$, then $\lambda^+(M) = \lambda(M \cap P) = 0$ and $\lambda^-(M) = -\lambda(M \cap N) = 0$, so $|\lambda|(M) = 0$.
(ii) ($\Leftarrow$) If $|\lambda|(M) = 0$, then $\lambda^\pm(G \cap M) = 0$ for every $G \in \mathcal{M}$, hence $\lambda(G \cap M) = 0$ and $M \in \mathcal{Z}_\lambda$.
(iii) The identity $\pi_\lambda = \lambda^+$ is (i). For finite partitions (A_j) of E, $\sum_j |\lambda(A_j)| \leq \sum_j |\lambda|(A_j) = |\lambda|(E)$, and $E = (E \cap P) \sqcup (E \cap N)$ attains equality; hence $\nu_\lambda = |\lambda|$.

Structural Commentary: Variation is intrinsic to λ.
Key Insight: The positive part is the largest charge obtainable inside a set. □

Theorem 4.1.8. *Variations of Integral Charges*
Let $f \in \mathscr{L}^1(\mu)$ and $\lambda(E) := \int_E f \, d\mu$.
Then $\forall E \in \mathcal{M}$, $\lambda^+(E) = \int_E f^+ \, d\mu$, $\lambda^-(E) = \int_E f^- \, d\mu$, and $|\lambda|(E) = \int_E |f| \, d\mu$.

Proof. Set $P = \{f \geq 0\}$ and $N = \{f < 0\}$.
For $A \in \mathcal{M}$, $\lambda(A \cap P) = \int_{A \cap P} f \, d\mu \geq 0$ and $\lambda(A \cap N) = \int_{A \cap N} f \, d\mu \leq 0$, so (P, N) is Hahn.
Thus $\lambda^+(E) = \lambda(E \cap P) = \int_E f^+ \, d\mu$, $\lambda^-(E) = -\lambda(E \cap N) = \int_E f^- \, d\mu$, and $|\lambda|(E) = \lambda^+(E) + \lambda^-(E) = \int_E |f| \, d\mu$.
Structural Commentary: Measure variations match function parts.
Key Insight: Sign sets of the density are Hahn sets. □

Definition 4.1.9. *Absolute Continuity and Singularity*

(i) *Absolute continuity for measures*: $\lambda \ll \mu \Leftrightarrow \forall E \in \mathcal{M}, \mu(E) = 0 \Rightarrow \lambda(E) = 0$.
(ii) *Absolute continuity for charges*: $\lambda \ll \mu \Leftrightarrow |\lambda| \ll \mu$.
(iii) *Mutual singularity*: $\lambda \perp \mu \Leftrightarrow \exists A, B \in \mathcal{M}$ with $A \cap B = \emptyset$, $X = A \cup B$, $\lambda(A) = 0$, and $\mu(B) = 0$.

Proposition 4.1.10. *Absolute Continuity Calculus*
Let λ, μ, ν be measures and η a charge.

(i) $\lambda \ll \lambda$ *and* $\lambda \ll \mu \ll \nu \Rightarrow \lambda \ll \nu$.
(ii) $\ll$ *is not symmetric.*
(iii) $(\forall E \in \mathcal{M}, \mu(E) = 0 \Rightarrow \eta(E) = 0) \Rightarrow \eta^+, \eta^-, |\eta| \ll \mu$.

Proof.

(i) Reflexivity is immediate. If $\lambda \ll \mu \ll \nu$ and $\nu(E) = 0$, then $\mu(E) = 0$, hence $\lambda(E) = 0$.
(ii) On $X = \{0, 1\}$, $\mathcal{M} = 2^X$, $\delta_0 \ll \#$, but $\# \not\ll \delta_0$ since $\delta_0(\{1\}) = 0$ and $\#\{1\} = 1$.
(iii) If $\mu(E) = 0$, then $\mu(F) = 0$ for every $F \in \mathcal{M}$, $F \subseteq E$, so $\eta(F) = 0$.
Hence $E \in \mathcal{Z}_\eta$, and Corollary 4.1.7 gives $|\eta|(E) = 0$; therefore $\eta^{\pm}(E) \leq |\eta|(E) = 0$.

Structural Commentary: Absolute continuity is null-set order; charges reduce to variation.
Key Insight: Null sets pass to measurable subsets, so all signed pieces vanish there. □

Example 4.1.11. *Dominating a Countable Family of Measures*
Let (μ_n) be measures with $\mu_n(X) \leq 1$, and set $\lambda(E) := \sum_{n \geq 1} 2^{-n} \mu_n(E)$.
Then λ is finite, since $\lambda(X) \leq 1$, and countably additive by Tonelli for nonnegative series.
Also $2^{-n}\mu_n(E) \leq \lambda(E)$, hence $\lambda(E) = 0 \Rightarrow \mu_n(E) = 0$ and $\mu_n \ll \lambda$ for all n.
Key Insight: A summable positive mixture packages countably many null-set tests into one reference measure.

Lemma 4.1.12. *Epsilon–Delta Form of Absolute Continuity*
Let λ be finite and μ a measure on $(X, \mathcal{M})$. Then $\lambda \ll \mu \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0, \forall E \in \mathcal{M}, \mu(E) < \delta \Rightarrow \lambda(E) < \varepsilon$.

Proof.

($\Rightarrow$) Suppose the ε-δ condition fails. Then $\exists \varepsilon > 0$ and $E_n \in \mathcal{M}$ with $\mu(E_n) < 2^{-n}$ and $\lambda(E_n) \geq \varepsilon$.
Let $F_n = \bigcup_{k \geq n} E_k$ and $F = \bigcap_n F_n$. Then $\mu(F_n) \leq 2^{-n+1}$, so $\mu(F) = 0$.
Since $F_n \downarrow F$, $\lambda(F_n) \geq \varepsilon$, and λ is finite, continuity from above gives $\lambda(F) \geq \varepsilon$, contradicting $\lambda \ll \mu$.
($\Leftarrow$) If $\mu(E) = 0$, then $\mu(E) < \delta$ for every $\delta > 0$; hence $\lambda(E) < \varepsilon$ for every $\varepsilon > 0$, so $\lambda(E) = 0$.

Structural Commentary: For finite target measure, absolute continuity is uniform smallness on small μ-sets.
Key Insight: A failing sequence is converted into a decreasing null tail with positive λ-mass. □

Example 4.1.13. *Finite Target Is Needed*
On $(\mathbb{N}, 2^{\mathbb{N}})$, let $\lambda = \#$ and $\mu(E) := \sum_{n \in E} 2^{-n}$. Since $\mu(E) = 0 \Leftrightarrow E = \emptyset$, $\lambda \ll \mu$.
But for $E_n = \{n\}$, $\mu(E_n) = 2^{-n} \to 0$ while $\lambda(E_n) = 1$, so the ε-δ condition fails for $\varepsilon = 1$.

Example 4.1.14. *Mutual Absolute Continuity Need Not Preserve Finiteness*
On $(\mathbb{N}, 2^{\mathbb{N}})$, set $\mu(E) := \sum_{n \in E} 2^{-n}$ and $\lambda(\emptyset) = 0$, $\lambda(E) = +\infty$ for $E \neq \emptyset$.
Then $\mu(\mathbb{N}) = 1$, while λ is not σ-finite. Since both measures have null ideal $\{\emptyset\}$, $\lambda \ll \mu$ and $\mu \ll \lambda$.
Key Insight: Absolute continuity compares null sets; it does not compare finiteness, σ-finiteness, or size.

4.2 Radon-Nikodym Theorem

Theorem 4.2.1. *Radon–Nikodym Theorem*
Let λ, μ be σ-finite measures on $(X, \mathcal{M})$ with $\lambda \ll \mu$. Then there is a measurable $f : X \to [0, +\infty]$, unique μ-a.e. and finite μ-a.e., such that $\lambda(E) = \int_E f \, d\mu$ for all $E \in \mathcal{M}$. Write $f = d\lambda/d\mu$.

Proof. Assume first $\lambda(X), \mu(X) < +\infty$.
For $c > 0$, let (P_k, N_k) be Hahn for $\lambda - kc\mu$, put $A_k = N_k \setminus \bigcup_{j<k} A_j$, and $B = X \setminus \bigcup_{k \geq 1} A_k$.
Then $A_k \subseteq N_k$ and, for $k \geq 2$, $A_k \subseteq P_{k-1}$; hence $E \subseteq A_k$ implies $(k-1)c\,\mu(E) \leq \lambda(E) \leq kc\,\mu(E)$.
Also $B \subseteq P_k$ for all k, so $\mu(B) = \lambda(B) = 0$.
Define $f_c := \sum_{k \geq 1} (k-1)c\,\mathbf{1}_{A_k}$. Then $\int_E f_c \, d\mu \leq \lambda(E) \leq \int_E f_c \, d\mu + c\mu(X)$ for all $E \in \mathcal{M}$.
With $c_n = 2^{-n}$ and $f_n = f_{c_n}$, $\|f_n - f_m\|_1 \leq (2^{-n} + 2^{-m})\mu(X) \leq 2^{-n+1}\mu(X)$ for $m \geq n$, so $f_n \to f$ in L^1 for some $f \geq 0$. Letting $n \to \infty$ gives $\lambda(E) = \int_E f \, d\mu$.
For σ-finite λ, μ, choose disjoint $Y_r \in \mathcal{M}$ with $X = \bigsqcup_r Y_r$ and $\lambda(Y_r), \mu(Y_r) < +\infty$.
Apply the finite case to $\lambda|_{Y_r}, \mu|_{Y_r}$, get f_r, and set $f = \sum_r f_r \mathbf{1}_{Y_r}$. Then $\lambda(E) = \sum_r \lambda(E \cap Y_r) = \int_E f \, d\mu$.
Since $\int_{Y_r} f \, d\mu = \lambda(Y_r) < +\infty$, $f < +\infty$ μ-a.e. If f, g both represent λ, then $f = g$ μ-a.e. on each Y_r by Proposition 2.2.15; hence $f = g$ μ-a.e. on X.

Structural Commentary: Hahn decompositions of $\lambda - c\mu$ build lower step approximations to the desired density.
Key Insight: The density is recovered by measuring where λ lies between consecutive multiples of μ. □

Example 4.2.2. *A Non-σ-Finite Failure*
Let $X = [0,1]$, $\mathcal{M} = \mathcal{B}([0,1])$, $\mu = \#$, and $\lambda = m$. Since $\mu(E) = 0 \Leftrightarrow E = \emptyset$, $\lambda \ll \mu$.
If $\lambda(E) = \int_E f\, d\mu$ for some $f \geq 0$, then $0 = \lambda(\{x\}) = \int_{\{x\}} f\, d\mu = f(x)$ for every x, so $\lambda([0,1]) = 0$, contradiction.
Thus RN can fail when μ is not σ-finite.

Example 4.2.3. *Another Non-σ-Finite Failure*
Let X be uncountable and $\mathcal{M} := \{E \subseteq X : E \text{ is countable or } X \setminus E \text{ is countable}\}$.
Set $\mu(E) = \#E$ for finite E and $\mu(E) = +\infty$ otherwise; set $\lambda(E) = 0$ for countable E and $\lambda(E) = +\infty$ otherwise. Since $\mu(E) = 0 \Leftrightarrow E = \emptyset$, $\lambda \ll \mu$. If $\lambda(E) = \int_E f\, d\mu$ with $f \geq 0$, then $0 = \lambda(\{x\}) = \int_{\{x\}} f\, d\mu = f(x)$ for all x, so $\lambda(X) = 0$, contradiction.
Structural Commentary: This is the countable–cocountable version of Example 4.2.2.
Key Insight: Non-σ-finite references can erase all singleton tests.

Remark 4.2.4. *When the Radon–Nikodym Derivative Is Integrable*
In Theorem 4.2.1, $\frac{d\lambda}{d\mu} \in L^1(\mu) \Leftrightarrow \lambda(X) < +\infty$, since $\int_X (d\lambda/d\mu)\, d\mu = \lambda(X)$.

Theorem 4.2.5. *Extended-Valued Radon–Nikodym Theorem*
Let μ be σ-finite, let λ be a measure on $(X, \mathcal{M})$, and assume $\lambda \ll \mu$.
Then $\exists$ measurable $f : X \to [0, +\infty]$ such that $\lambda(E) = \int_E f\, d\mu$ for all $E \in \mathcal{M}$; any two such f agree μ-a.e.

Proof. Assume first $\mu(X) < +\infty$. Let (P_n, N_n) be Hahn for $\lambda - n\mu$, and put $P = \bigcap_{n\geq 1} P_n$, $N = \bigcup_{n\geq 1} N_n$.
If $E \subseteq N_n$, then $\lambda(E) \leq n\mu(E)$; hence $\lambda|_N$ is σ-finite and Theorem 4.2.1 gives $g : N \to [0, +\infty)$ with $\lambda(E \cap N) = \int_{E\cap N} g\, d\mu$. If $E \subseteq P$, then $\lambda(E) \geq n\mu(E)$ for all n, so $\mu(E) > 0 \Rightarrow \lambda(E) = +\infty$, while $\mu(E) = 0 \Rightarrow \lambda(E) = 0$. Thus $f = g\mathbf{1}_N + \infty \cdot \mathbf{1}_P$ represents λ.
For general σ-finite μ, choose $X = \bigsqcup_r Y_r$ with $\mu(Y_r) < +\infty$, apply the finite-μ case to each restriction, and set $f = \sum_r f_r \mathbf{1}_{Y_r}$. Then $\lambda(E) = \sum_r \lambda(E \cap Y_r) = \int_E f\, d\mu$. If f, h represent λ, then on each Y_r, $\int_A f\, d\mu = \int_A h\, d\mu$ for all $A \subseteq Y_r$; applying this to $\{f < h\}$ and $\{h < f\}$ gives $f = h$ μ-a.e.
Structural Commentary: If λ is not σ-finite, the density may have a genuinely infinite part.
Key Insight: Hahn decompositions isolate finite and infinite density regions. □

Theorem 4.2.6. *Lebesgue Decomposition Theorem*
Let λ, μ be σ-finite measures on $(X, \mathcal{M})$. Then $\exists!$ measures λ_s, λ_a with $\lambda = \lambda_s + \lambda_a$, $\lambda_s \perp \mu$, and $\lambda_a \ll \mu$.

Proof. Set $\rho = \lambda + \mu$. Choose $f = d\lambda/d\rho$ and $g = d\mu/d\rho$.
Let $A = \{g = 0\}$, $B = \{g > 0\}$, $\lambda_s(E) := \lambda(E \cap A)$, and $\lambda_a(E) := \lambda(E \cap B)$. Since $\mu(A) = \int_A g\, d\rho = 0$, $\lambda_s \perp \mu$.
If $\mu(E) = 0$, then $g = 0$ ρ-a.e. on E, so $\rho(E \cap B) = 0$ and $\lambda_a(E) = 0$; hence $\lambda_a \ll \mu$.
For uniqueness, let $(\tilde\lambda_s, \tilde\lambda_a)$ be another decomposition.
Since $\lambda_s \perp \mu$ and $\tilde\lambda_a \ll \mu$, also $\lambda_s \perp \tilde\lambda_a$; choose $X = C \cup D$ with $\lambda_s(C) = 0$ and $\tilde\lambda_a(D) = 0$.
Then $\lambda_s(E) = \lambda_s(E \cap D) \leq \lambda(E \cap D) = \tilde\lambda_s(E \cap D) \leq \tilde\lambda_s(E)$. By symmetry, $\lambda_s = \tilde\lambda_s$, hence $\lambda_a = \tilde\lambda_a$.
Structural Commentary: Every measure decomposes into a μ-visible part and a part living on a μ-null set.
Key Insight: Differentiate both measures with respect to their sum, then split where the μ-density vanishes. □

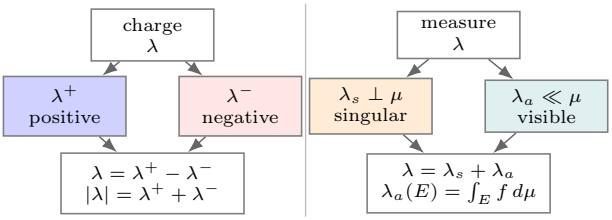


Figure 198: Charges split by sign; measures split into $\lambda_s \perp \mu$ and $\lambda_a \ll \mu$ parts.

Proposition 4.2.7. *Radon–Nikodym Calculus*
Let all measures below be σ-finite.

(i) $\lambda \ll \mu$, $g \geq 0$ *measurable* $\Rightarrow \int_X g\, d\lambda = \int_X g(d\lambda/d\mu)\, d\mu$.
(ii) $\nu \ll \lambda \ll \mu \Rightarrow d\nu/d\mu = (d\nu/d\lambda)(d\lambda/d\mu)$ μ-*a.e.*
(iii) $\lambda_1, \lambda_2 \ll \mu \Rightarrow d(\lambda_1 + \lambda_2)/d\mu = d\lambda_1/d\mu + d\lambda_2/d\mu$ μ-*a.e.*
(iv) $\lambda \ll \mu \ll \lambda \Rightarrow d\lambda/d\mu = 1/(d\mu/d\lambda)$ μ-*a.e.*

Proof.

(i) For $g = \mathbf{1}_E$, $\int_X g\,d\lambda = \lambda(E) = \int_E (d\lambda/d\mu)\,d\mu = \int_X g(d\lambda/d\mu)\,d\mu$.
By linearity the identity holds for nonnegative simple g; for measurable $g \geq 0$, take simple $g_n \uparrow g$ and apply monotone convergence to both sides.

(ii) Let $h := (d\nu/d\lambda)(d\lambda/d\mu)$.
For every $E \in \mathcal{M}$, (i) with $g = \mathbf{1}_E(d\nu/d\lambda)$ gives $\nu(E) = \int_E (d\nu/d\lambda)\,d\lambda = \int_E h\,d\mu$.
Hence h represents ν relative to μ, so $h = d\nu/d\mu$ μ-a.e.

(iii) Let $h := d\lambda_1/d\mu + d\lambda_2/d\mu$. For every $E \in \mathcal{M}$, $\int_E h\,d\mu = \lambda_1(E) + \lambda_2(E) = (\lambda_1 + \lambda_2)(E)$.
Hence h represents $\lambda_1 + \lambda_2$ relative to μ, so uniqueness gives the identity.

(iv) Apply (ii) to $\mu \ll \lambda \ll \mu$. Since $d\mu/d\mu = 1$, $1 = (d\mu/d\lambda)(d\lambda/d\mu)$ μ-a.e.
Also $\mu \ll \lambda$ gives $d\mu/d\lambda > 0$ μ-a.e.; divide to obtain $d\lambda/d\mu = 1/(d\mu/d\lambda)$ μ-a.e.

Structural Commentary: Radon–Nikodym derivatives obey chain and linearity rules.
Key Insight: Prove identities by integrating over arbitrary measurable sets, then invoke uniqueness. □

Corollary 4.2.8. *Absolutely Continuous and Singular Forces Zero*
For measures λ, μ, $\lambda \ll \mu$ and $\lambda \perp \mu$ imply $\lambda = 0$. For a charge η, $|\eta| \perp \mu$ implies $\eta^+ \perp \mu$ and $\eta^- \perp \mu$.

Proof. If $\lambda \perp \mu$, choose $X = A \cup B$ with $\lambda(A) = 0$ and $\mu(B) = 0$. Since $\lambda \ll \mu$, $\lambda(B) = 0$, hence $\lambda(X) = 0$.
If $|\eta| \perp \mu$, choose $X = A \cup B$ with $|\eta|(A) = 0$ and $\mu(B) = 0$. Since $\eta^\pm \leq |\eta|$, $\eta^\pm(A) = 0$, so $\eta^\pm \perp \mu$.
Structural Commentary: Absolute continuity and singular support are incompatible except at zero.
Key Insight: Visible and singular forces zero. □

Corollary 4.2.9. *Radon–Nikodym for Charges*
Let μ be σ-finite and let λ be a charge with $\lambda \ll \mu$. Then $\exists f \in \mathscr{L}^1(\mu)$ such that, for all $E \in \mathcal{M}$, $\lambda(E) = \int_E f\,d\mu$, $\lambda^+(E) = \int_E f^+\,d\mu$, $\lambda^-(E) = \int_E f^-\,d\mu$, and $|\lambda|(E) = \int_E |f|\,d\mu$.

Proof. Since $|\lambda| \ll \mu$, $\lambda^\pm \ll \mu$. By Radon–Nikodym, choose $f_\pm \geq 0$ with $\lambda^\pm(E) = \int_E f_\pm\,d\mu$. Since $\lambda^+ \perp \lambda^-$, $f_+ f_- = 0$ μ-a.e. Set $f = f_+ - f_-$. Then $\int_X |f|\,d\mu = \int_X (f_+ + f_-)\,d\mu = |\lambda|(X) < +\infty$, so $f \in \mathscr{L}^1(\mu)$, $f^+ = f_+$, $f^- = f_-$ μ-a.e., and $\lambda(E) = \lambda^+(E) - \lambda^-(E) = \int_E f\,d\mu$.
Structural Commentary: Signed absolute continuity is reduced to the two positive Jordan pieces.
Key Insight: Differentiate λ^+ and λ^- separately, then subtract. □

Example 4.2.10. *Radon–Nikodym Classification of Basic Stieltjes Generators*
Let m be Lebesgue measure on $\mathcal{B}(\mathbb{R})$, and let λ_i be the Borel–Stieltjes measure generated by g_i.

(i) $g_1(x) = 2x$: $\lambda_1 = 2m$, so $\lambda_1 \ll m$, $d\lambda_1/dm = 2$, $m \ll \lambda_1$, $dm/d\lambda_1 = 1/2$, and $\lambda_1(\mathbb{R}) = +\infty$.

(ii) $g_2(x) = \arctan x$: $\lambda_2(E) = \int_E (1 + x^2)^{-1}\,dm$, so $d\lambda_2/dm = (1 + x^2)^{-1}$, $\lambda_2(\mathbb{R}) = \pi$, $m \ll \lambda_2$, and $dm/d\lambda_2 = 1 + x^2$.

(iii) $g_3 = \mathbf{1}_{[0,+\infty)}$: $\lambda_3 = \delta_0$,
so $\lambda_3 \perp m$, $\lambda_3(\mathbb{R}) = 1$, $\lambda_3 \not\ll m$, and $m \not\ll \lambda_3$.

(iv) $g_4(x) = x\mathbf{1}_{[0,+\infty)}(x)$: $\lambda_4(E) = m(E \cap [0, +\infty))$, so $\lambda_4 \ll m$, $d\lambda_4/dm = \mathbf{1}_{[0,+\infty)}$, $m \not\ll \lambda_4$, and $\lambda_4(\mathbb{R}) = +\infty$.

Thus $\lambda_1, \lambda_2, \lambda_4 \ll m$, $\lambda_3 \perp m$, only λ_2, λ_3 are finite, and $m \ll \lambda_i$ exactly for $i = 1, 2$.
Structural Commentary: Stieltjes generators split naturally into density, atom, and restriction examples.
Key Insight: A positive density makes null sets equivalent; an atom lives on a Lebesgue-null set.

Theorem 4.2.11. *Charges Form a Banach Space*
Let $\mathrm{ca}(X, \mathcal{M}) := \{\lambda : \mathcal{M} \to \mathbb{R} : \lambda \text{ is a charge}\}$ *and* $\|\lambda\|_{\mathrm{var}} := |\lambda|(X)$.
With pointwise operations, $(\mathrm{ca}(X, \mathcal{M}), \|\cdot\|_{\mathrm{var}})$ *is Banach.*

Proof. Linearity follows from charge linearity.
Definiteness follows from $|\lambda|(X) = 0 \Rightarrow |\lambda|(E) = 0\ \forall E$, homogeneity from $|a\lambda| = |a|\,|\lambda|$, and the triangle inequality from $\sum_j |(\lambda + \eta)(A_j)| \leq \sum_j |\lambda(A_j)| + \sum_j |\eta(A_j)|$ over finite measurable partitions (A_j).
Let (λ_n) be variation-Cauchy. Since $|\lambda_n(E) - \lambda_m(E)| \leq \|\lambda_n - \lambda_m\|_{\mathrm{var}}$, define $\lambda(E) := \lim_n \lambda_n(E)$; then λ is finitely additive. For fixed n and finite disjoint (A_j), $\sum_j |(\lambda - \lambda_n)(A_j)| \leq \limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}}$.
For disjoint (E_k), set $E = \bigcup_k E_k$ and $R_K = \bigcup_{k>K} E_k$. Choose n with the preceding limsup $< \varepsilon$.
Since $\lambda_n(R_K) \to 0$, for large K, $|\lambda(R_K)| \leq |\lambda_n(R_K)| + |(\lambda - \lambda_n)(R_K)| < 2\varepsilon$. Hence $\lambda(R_K) \to 0$, so λ is countably additive. Taking suprema over finite partitions gives $\|\lambda - \lambda_n\|_{\mathrm{var}} \leq \limsup_m \|\lambda_m - \lambda_n\|_{\mathrm{var}} \to 0$.
Structural Commentary: Total variation is the signed-measure analogue of an L^1 norm.
Key Insight: Variation-Cauchy controls every finite partition. □

4.3 Differentiation of Measures

Definition 4.3.1. *Averages and Absolute Continuity*

(i) *Average*: for $f \in \mathscr{L}^1(E)$ and $A \subseteq E$, $m(A) > 0$, set $[f : A] := m(A)^{-1} \int_A f \, dm$.
(ii) *Density ratio*: for measurable B and cube or ball Q, $m(Q) > 0$, set $[B : Q] := m(B \cap Q)/m(Q)$.
(iii) *Indefinite integral*: for $f \in \mathscr{L}^1([a,b])$, $F(x) = F(a) + \int_a^x f(t)\, dt$.
(iv) *Absolute continuity*: $F : [a,b] \to \mathbb{R}$ is absolutely continuous if $\forall \varepsilon > 0, \exists \delta > 0$, finite disjoint $(a_j, b_j) \subseteq [a,b]$ satisfy $\sum_j (b_j - a_j) < \delta \Rightarrow \sum_j |F(b_j) - F(a_j)| < \varepsilon$.

Definition 4.3.2. *Bounded Variation*

(i) *Total variation*: for $F : [a,b] \to \mathbb{R}$, $V_a^b(F) := \sup\{\sum_{j=1}^N |F(x_j) - F(x_{j-1})| : a = x_0 < \cdots < x_N = b\}$.
(ii) *Bounded variation*: F has bounded variation on $[a,b]$ if $V_a^b(F) < +\infty$.

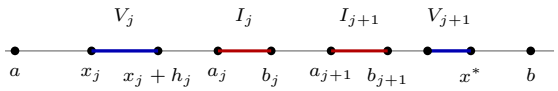


Figure 199: Interval bookkeeping for absolute continuity.

Perspective 4.3.3. *Absolute Continuity as Interval Bookkeeping*
If $F(x) = \int_a^x f(t)\, dt$, by absolute continuity, disjoint intervals with small total length force small F total change.

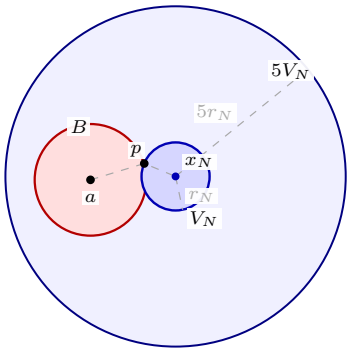


Figure 200: A comparable missed ball lies inside the fivefold dilation of the selected ball.

Perspective 4.3.4. *Vitali Covering Control*
Select balls disjointly by decreasing radius.
If B is rejected, then $B \cap B_j \neq \emptyset$ for some earlier B_j with $r(B) \leq r(B_j)$, hence $B \subseteq 5B_j$.
Key Insight: Disjoint balls add; fixed dilations cover misses.

Definition 4.3.5. *Vitali Coverings and Density Points*

(i) For $A \subseteq M$, $\mathcal{V}$ is a *Vitali covering* of A if $\forall p \in A, \forall r > 0, \exists V \in \mathcal{V}$ with $p \in V$ and $\operatorname{diam}(V) < r$.
(ii) For measurable $E \subseteq \mathbb{R}^n$, its *density-point set* is $\operatorname{dp}(E) := \{p \in \mathbb{R}^n : \lim_{Q \downarrow p} [E : Q] = 1\}$, with Q cubes, equivalently balls, shrinking to p.

Theorem 4.3.6. *Vitali Covering Lemma for Balls and Cubes*
Let $E \subseteq \mathbb{R}^n$, $m^*(E) < +\infty$, and let $\mathcal{V}$ be a Vitali cover of E by closed balls or axis-parallel cubes. Then $\exists$ pairwise disjoint $(V_j) \subseteq \mathcal{V}$ with $m^*(E \setminus \bigcup_{j \geq 1} V_j) = 0$.

Proof. Fix open $O \supseteq E$ with $m(O) < m^*(E) + \varepsilon$, and replace $\mathcal{V}$ by $\{V \in \mathcal{V} : V \subseteq O\}$.
Greedily choose a maximal disjoint (V_j) by dyadic diameter scales.
For a fixed dilation $V \mapsto V^*$, if $W \cap V \neq \emptyset$ and $\operatorname{diam} W \leq 2 \operatorname{diam} V$, then $W \subseteq V^*$ and $m(V^*) \leq C_n m(V)$.
Let $R = E \setminus \bigcup_j V_j$. Fix N and $p \in R$.
Since $V_1, \ldots, V_N$ are closed, choose $W \in \mathcal{V}$ with $p \in W$ and $W \cap \bigcup_{j \leq N} V_j = \emptyset$. Maximality gives some $j > N$ with $W \cap V_j \neq \emptyset$ and comparable diameter, hence $p \in W \subseteq V_j^*$. Thus $R \subseteq \bigcup_{j > N} V_j^*$.
Therefore $m^*(R) \leq C_n \sum_{j > N} m(V_j) \to 0$, because $\sum_j m(V_j) = m(\bigsqcup_j V_j) \leq m(O) < +\infty$. Hence $m^*(R) = 0$.
Structural Commentary: Vitali selection gives a disjoint cover modulo null sets.
Key Insight: $R \subseteq \bigcup_{j > N} V_j^*$ and disjoint tails vanish. □

Theorem 4.3.7. *Lebesgue Density Theorem*
For measurable $E \subseteq \mathbb{R}^n$, $\lim_{Q \downarrow p}[E : Q] = \mathbf{1}_E(p)$ for a.e. p, where Q ranges over cubes, equivalently balls. Equivalently, $m(E \triangle \operatorname{dp}(E)) = 0$.

Proof. Assume $m(E) < +\infty$. For $a \in (0,1)$ set $E_a := \{p \in E : \liminf_{Q \downarrow p}[E : Q] < a\}$.
Cubes $Q \ni p$ with $[E : Q] < a$ form a Vitali cover of E_a. For open $O \supseteq E_a$ with $m(O) < m(E_a) + \varepsilon$, Theorem 4.3.6 gives disjoint $(Q_j) \subseteq O$ covering E_a modulo null sets.
Hence $m(E_a) \leq \sum_j m(E_a \cap Q_j) \leq \sum_j m(E \cap Q_j) < a \sum_j m(Q_j) \leq a\, m(O) < a(m(E_a) + \varepsilon)$.
Let $\varepsilon \downarrow 0$; since $a < 1$, $m(E_a) = 0$. Thus $p \in E \setminus \bigcup_{a \in \mathbb{Q} \cap (0,1)} E_a$ has density 1. Apply same argument to bounded pieces of E^c to get density 0 a.e. on E^c; then exhaust arbitrary E by bounded pieces.
Structural Commentary: Measurable sets are locally 0 or 1 a.e.
Key Insight: Vitali turns bad density into $m(E_a) \leq a\, m(E_a)$. □

Corollary 4.3.8. *Density Points Commute with Slicing Almost Everywhere*
Let $E \subseteq \mathbb{R}^n \times \mathbb{R}^k$ be measurable. For a.e. $x \in \mathbb{R}^n$, $\operatorname{dp}(E_x) = \{y : (x,y) \in \operatorname{dp}(E)\}$ up to a null subset of $\mathbb{R}^k$.

Proof. Set $D = \operatorname{dp}(E)$ and $N = E \triangle D$.
By Theorem 4.3.7, $m_{n+k}(N) = 0$; hence Theorem 3.2.14 gives $m_k(N_x) = 0$ for a.e. x.
Intersecting with the Fubini good set, E_x, D_x are measurable and $m_k(E_x \triangle D_x) = m_k((E \triangle D)_x) = 0$.
Applying Theorem 4.3.7 in $\mathbb{R}^k$ gives $m_k(E_x \triangle \operatorname{dp}(E_x)) = 0$.
Therefore $m_k(\operatorname{dp}(E_x) \triangle D_x) = 0$; and $m_k(\operatorname{dp}(E_x) \triangle D_x) = 0$. This is the claimed equality modulo null sets.
Structural Commentary: Almost-everywhere local fullness is compatible with almost-everywhere slicing.
Key Insight: Null ambient errors have null slices for almost every base point. □

Lemma 4.3.9. *Large Averages Have Small Outer Measure*
For a cube $Q_0 \subseteq \mathbb{R}^n$, $g \geq 0$ in $\mathscr{L}^1(Q_0)$, and $\lambda > 0$,
set $A_\lambda := \{p \in Q_0 : \forall r > 0\ \exists$ cube $Q \subseteq Q_0,\ p \in Q,\ \operatorname{diam} Q < r,\ [g : Q] > \lambda\}$ for $m^*(A_\lambda) \leq \frac{1}{\lambda} \int_{Q_0} g$.

Proof. Let $\mathcal{V}_\lambda = \{Q \subseteq Q_0 : Q$ is a cube and $[g : Q] > \lambda\}$. By definition, $\mathcal{V}_\lambda$ is a Vitali cover of A_λ; choose disjoint $(Q_j) \subseteq \mathcal{V}_\lambda$ with $m^*(A_\lambda \setminus \bigcup_j Q_j) = 0$. Then $\lambda m^*(A_\lambda) \leq \lambda \sum_j m(Q_j) < \sum_j \int_{Q_j} g \leq \int_{Q_0} g$.
Structural Commentary: Weak type for local averages.
Key Insight: Disjoint Vitali cubes turn pointwise large averages into one integral bound. □

Theorem 4.3.10. *Lebesgue Average Value Theorem*
For $f \in \mathscr{L}^1_{\mathrm{loc}}(\mathbb{R}^n)$, $\lim\limits_{Q \downarrow p}[f : Q] = f(p)$ for a.e. $p \in \mathbb{R}^n$ with Q cubes, equivalently balls, shrinking to p.

Proof. Fix a cube Q_0; it suffices to prove the claim for a.e. $p \in Q_0^\circ$.
For $\alpha > 0$, let $B_\alpha = \left\{ p \in Q_0^\circ : \limsup\limits_{Q \downarrow p} |[f : Q] - f(p)| > \alpha \right\}$.
Choose $h \in C(Q_0)$ with $\|f - h\|_{L^1(Q_0)} < \eta$, and set $g = |f - h|$. Since $[h : Q] \to h(p)$, $B_\alpha \subseteq \{g > \alpha/3\} \cup A_{\alpha/3}(g)$.
By Markov and Lemma 4.3.9, $m^*(B_\alpha) \leq \frac{3}{\alpha} \int_{Q_0} g + \frac{3}{\alpha} \int_{Q_0} g < \frac{6\eta}{\alpha}$.
Let $\eta \downarrow 0$; then $m(B_\alpha) = 0$. Count over $\alpha \in \mathbb{Q}_{>0}$ and exhaust $\mathbb{R}^n$ by cube interiors.
Structural Commentary: Local averages recover the original integrable function almost everywhere.
Key Insight: Continuous approximation handles the good part; Vitali's estimate makes the bad part small. □

Theorem 4.3.11. *Lebesgue's Antiderivative Theorem*
For $f \in \mathscr{L}^1([a,b])$, define $F(x) = c + \int_a^x f(t)\, dt$. Then $F \in AC([a,b])$ and $F'(x) = f(x)$ for a.e. $x \in [a,b]$.

Proof. Given $\varepsilon > 0$, choose $\delta > 0$ so that $m(A) < \delta$ implies $\int_A |f| < \varepsilon$.
If (α_j, β_j) are disjoint and $\sum_j (\beta_j - \alpha_j) < \delta$, then $\sum_j |F(\beta_j) - F(\alpha_j)| \leq \sum_j \int_{\alpha_j}^{\beta_j} |f| = \int_{\bigcup_j (\alpha_j, \beta_j)} |f| < \varepsilon$, so $F \in AC([a,b])$. For $x, x+h \in [a,b]$, let $I_h(x) = (\min\{x, x+h\}, \max\{x, x+h\})$.
Then $\frac{F(x+h) - F(x)}{h} = [f : I_h(x)]$. Extending f by 0 outside $[a,b]$, Theorem 4.3.10 in one dimension gives $[f : I_h(x)] \to f(x)$ for a.e. x. Hence $F' = f$ a.e.
Structural Commentary: The indefinite Lebesgue integral is the correct antiderivative for integrable functions.
Key Insight: Differentiating the integral is the same as shrinking the averaging interval. □

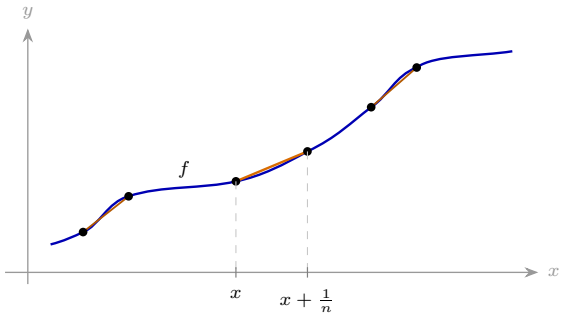


Figure 201: $g_n(x)$ is the slope of the short right secant from x to $x + \frac{1}{n}$.

Perspective 4.3.12. *Right Secants and Differentiation*
Right quotients $g_n(x) = n\big(f(x + 1/n) - f(x)\big)$ encode local slope data.
Key Insight: Secants turn differentiation into slope estimates.

Theorem 4.3.13. *Lebesgue's Last Theorem and Consequences*
On $[a, b]$*:*

(i) F *monotone* $\Rightarrow F'$ *exists a.e.;*
(ii) F *Lipschitz* $\Rightarrow F'$ *exists a.e.;*
(iii) $F \in BV([a, b]) \Rightarrow F'$ *exists a.e.;*
(iv) $F \in AC([a, b]) \Rightarrow F \in BV([a, b])$*, hence* F' *exists a.e.*

Proof.

(i) Assume F nondecreasing. Let $D^+F(x) = \limsup_{h\downarrow 0} \frac{F(x+h)-F(x)}{h}$ and $D_-F(x) = \liminf_{h\downarrow 0} \frac{F(x+h)-F(x)}{h}$.
For $\alpha < \beta$ in $\mathbb{Q}$, set $A_{\alpha,\beta} = \{D_-F < \alpha < \beta < D^+F\}$. The right-secant Vitali selection of Figure 201 gives $m(A_{\alpha,\beta}) = 0$: otherwise selected intervals with slopes $< \alpha$ and $> \beta$ contradict monotonicity of total rise. Also $m^*(\{D^+F > M\}) \le \frac{F(b)-F(a)}{M} \to 0$, so $D^+F = D_-F < \infty$ a.e.; hence F' exists a.e.

(ii) If F is L-Lipschitz, set $G_+(x) = F(x) + Lx$ and $G_-(x) = Lx - F(x)$. Then $G_\pm$ are nondecreasing, by (i) $\exists$ null sets $N_\pm$ s.t. $G'_\pm$ exist off $N_\pm$. For $x \notin N_+ \cup N_-$, $F'(x) = \frac{G'_+(x)-G'_-(x)}{2}$, so F' exists a.e.

(iii) If $F \in BV([a, b])$, write its Jordan decomposition $F = F(a) + V^+ - V^-$ with $V^\pm$ nondecreasing. By (i), $\exists$ null sets $N_\pm$ s.t. $(V^\pm)'$ exist off $N_\pm$. For $x \notin N_+ \cup N_-$, $F'(x) = (V^+)'(x) - (V^-)'(x)$, so F' exists a.e.

(iv) If $F \in AC([a, b])$, choose δ for $\varepsilon = 1$, and partition $a = x_0 < \cdots < x_N = b$ with $x_i - x_{i-1} < \delta$. Grouping any refinement by these subintervals gives $V_a^b(F) \le N$, so $F \in BV([a, b])$. By (iii), $\exists Z \subset [a, b]$ with $m(Z) = 0$ s.t. $F'(x)$ exists for all $x \in [a, b] \setminus Z$.

Structural Commentary: Monotonicity drives all cases.
Key Insight: Vitali controls secants; decompose to monotone. □

5 Lp Spaces and Modes of Convergence

5.1 Lp Spaces

Lebesgue spaces turn integrability into normed linear geometry. The essential point is that an integral norm cannot distinguish functions that differ only on null sets, so the actual vectors are almost-everywhere equivalence classes. Their bounded linear functionals are themselves represented by integration against functions in the conjugate space.

Definition 5.1.1. *Norms and Seminorms*
Let $\mathcal{V}$ be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$.

(i) A *seminorm* is $N : \mathcal{V} \to [0, \infty)$ with $N(\alpha v) = |\alpha|\, N(v)$ and $N(u + v) \le N(u) + N(v)$ $\forall u, v \in \mathcal{V}$, $\alpha \in \mathbb{F}$.
(ii) A *norm* is a seminorm with $N(v) = 0 \iff v = 0$.
(iii) A norm N induces $d_N(u, v) := N(u - v)$.

Example 5.1.2. *Norms and Seminorms Before Quotienting*

(i) On $\mathbb{F}^n$, $\|u\|_p := (\sum_{j=1}^n |u_j|^p)^{1/p}$ for $1 \le p < \infty$ and $\|u\|_\infty := \max_{1\le j\le n} |u_j|$ are norms.
Minkowski gives $\|u + v\|_p \le \|u\|_p + \|v\|_p$.

(ii) On sequences, $\ell^p := \{a : \sum_n |a_n|^p < +\infty\}$ with $\|a\|_p := (\sum_n |a_n|^p)^{1/p}$, and $\ell^\infty := \{a : \sup_n |a_n| < +\infty\}$ with $\|a\|_\infty := \sup_n |a_n|$.

(iii) On $B(X)$, $\|f\|_\infty := \sup_{x\in X} |f(x)|$ is a norm. On $C[0,1]$, $N_0(f) := |f(0)|$ and $N_1(f) := \sup_{0\le x\le 1/2} |f(x)|$ are seminorms. On $C^1[a,b]$, $N_D(f) := \sup_{a\le x\le b} |f'(x)|$ is a seminorm with kernel the constant functions.

Key Insight: Seminorms become norms after quotienting out their zero directions.

Remark 5.1.3. *Integral Norm on Continuous Functions*
On $C[0,1]$, $N(f) := \int_0^1 |f(x)|\,dx$ is a norm: if $f(x_0) \neq 0$, continuity gives $N(f) > 0$. It is not complete, since continuous ramps can be Cauchy in N and converge in L^1 to a discontinuous step function.

Definition 5.1.4. *L^p Spaces and Mean Convergence*
Let $(X, \mathcal{M}, \mu)$ be a measure space.

(i) For $1 \le p < \infty$, the *p-integrable class* is $\mathscr{L}^p(\mu) := \{f : X \to \mathbb{F} : f \text{ measurable}, \int_X |f|^p\,d\mu < +\infty\}$.
(ii) The *L^p space* and *L^p norm* are $L^p(\mu) := \mathscr{L}^p(\mu)/=_{\text{a.e.}}$ and $\|f\|_p := (\int_X |f|^p\,d\mu)^{1/p}$.
(iii) *L^p convergence*, or *mean convergence of order p*, means $f_n \to f \iff \|f_n - f\|_p \to 0$.
L^p-Cauchy means $\|f_m - f_n\|_p \to 0$ as $m, n \to \infty$.
(iv) *Essentially bounded* means $|f| \le M$ a.e. for some $M < +\infty$; $L^\infty(\mu)$ is the a.e.-quotient of such functions, with *essential sup norm* $\|f\|_\infty := \inf\{M \ge 0 : |f| \le M \text{ a.e.}\} = \inf_{\mu(N)=0} \sup_{x\in X\setminus N} |f(x)|$.

Proposition 5.1.5. *L^1 as a Quotient of a Seminorm*
On $\mathscr{L}^1(\mu)$ set $N_\mu(f) := \int_X |f|\,d\mu$. Then N_μ is a seminorm and $N_\mu(f) = 0 \iff f = 0$ a.e.
Hence $L^1(\mu) := \mathscr{L}^1(\mu)/=_{\text{a.e.}}$ with $\|[f]\|_1 := \int_X |f|\,d\mu$ is normed; write f for $[f]$.

Proof. $\mathscr{L}^1(\mu)$ is linear since $|af + bg| \le |a|\,|f| + |b|\,|g|$. Hence $N_\mu(\alpha f) = |\alpha|\,N_\mu(f)$ and $N_\mu(f+g) \le N_\mu(f) + N_\mu(g)$.

($\Rightarrow$) If $N_\mu(f) = 0$, set $E_n := \{|f| \ge 1/n\}$.
Then $(1/n)\mu(E_n) \le \int_{E_n} |f|\,d\mu \le 0$, so $\mu(E_n) = 0$ and $\{f \neq 0\} = \bigcup_n E_n$ is null.
($\Leftarrow$) If $f = 0$ a.e., then $N_\mu(f) = \int_X |f|\,d\mu = 0$.

The quotient kills exactly $\ker N_\mu$, so $\|\cdot\|_1$ is well defined and definite. *Structural Commentary*: The seminorm kernel is exactly the a.e.-zero functions.
Key Insight: Quotient by null differences before calling the integral a norm. □

Definition 5.1.6. *Complex Integrals and L^2*

(i) For $f = u + iv$, *complex measurability* means measurability of u, v.
(ii) On $E \in \mathcal{M}$, *complex integrability* means $u, v \in \mathscr{L}^1(\mu)$ on E; set $\int_E f\,d\mu := \int_E u\,d\mu + i\int_E v\,d\mu$.
(iii) $L^2(\mu)$ consists of complex measurable f with $\int_X |f|^2\,d\mu < +\infty$, modulo a.e. equality; $\|f\|_2 := (\int_X |f|^2\,d\mu)^{1/2}$.
(iv) The *L^2 inner product* is $\langle f, g\rangle_{L^2} := \int_X f\overline{g}\,d\mu$; it induces $\|f\|_2$.

Proposition 5.1.7. *Complex L^1 and Dominated Convergence*

(i) For complex measurable f on E, f is integrable $\iff |f| \in \mathscr{L}^1(\mu)$ on E; then $\left|\int_E f\,d\mu\right| \le \int_E |f|\,d\mu$.
(ii) If $f_n \to f$ a.e. on E and $|f_n| \le g \in \mathscr{L}^1(\mu)$, then f is integrable and $\int_E f\,d\mu = \lim_n \int_E f_n\,d\mu$.

Proof.

(i) ($\Rightarrow$) Write $f = u + iv$. If $u, v \in \mathscr{L}^1$, then $|f| \le |u| + |v|$, so $|f| \in \mathscr{L}^1$.
(i) ($\Leftarrow$) If $|f| \in \mathscr{L}^1$, then $|u|, |v| \le |f|$, so $u, v \in \mathscr{L}^1$ and f is integrable.
If $\int_E f\,d\mu = re^{i\theta}$ and $h = e^{-i\theta}f$, then $\left|\int_E f\,d\mu\right| = r = \operatorname{Re}\int_E h\,d\mu = \int_E \operatorname{Re} h\,d\mu \le \int_E |h|\,d\mu = \int_E |f|\,d\mu$.
(ii) Write $f_n = u_n + iv_n$ and $f = u + iv$. Since $|u_n|, |v_n| \le g$ and $u_n \to u$, $v_n \to v$ a.e., real DCT gives $\int_E u_n\,d\mu \to \int_E u\,d\mu$ and $\int_E v_n\,d\mu \to \int_E v\,d\mu$; hence the complex identity.

Structural Commentary: Complex integration is two real integrals.
Key Insight: Rotate the integral to make its value real and nonnegative before applying the real inequality. □

Theorem 5.1.8. *L^2 Geometry*
For $f, g \in L^2(\mu)$, $\int_X |fg|\,d\mu \le \|f\|_2\,\|g\|_2$, $f + g \in L^2(\mu)$, and $\|f+g\|_2 \le \|f\|_2 + \|g\|_2$.
Modulo equality a.e., $(L^2(\mu), \|\cdot\|_2)$ is complete.

Proof. For $\lambda \in \mathbb{R}$, $0 \le Q(\lambda) := \int_X (|f| + \lambda\,|g|)^2\,d\mu = \|f\|_2^2 + 2\lambda\int_X |fg|\,d\mu + \lambda^2\,\|g\|_2^2$; hence $\Delta(Q) \le 0$ and Schwarz follows. Thus $\|f+g\|_2^2 \le \int_X (|f| + |g|)^2\,d\mu \le (\|f\|_2 + \|g\|_2)^2$.
Let (f_n) be Cauchy. Choose $n_k \uparrow \infty$ so $m, n \ge n_k$ imply $\|f_m - f_n\|_2 \le 2^{-k}$, and set $h_N := \sum_{k<N} |f_{n_{k+1}} - f_{n_k}|$.
Then $\|h_N\|_2 \le \sum_{k<N} 2^{-k}$, so Fatou gives $h := \lim_N h_N \in L^2$; hence $f_{n_k} \to f$ a.e. for some f.
For fixed n, Fatou gives $\|f - f_n\|_2 \le \liminf_k \|f_{n_k} - f_n\|_2$.
The Cauchy property sends the right side to 0 as $n \to \infty$, so $f_n \to f$ in L^2.

Structural Commentary: L^2 is Hilbert-space geometry in measure form.
Key Insight: The same Schwarz mechanism survives because the integral is positive and additive. □

Lemma 5.1.9. *Young's Inequality*
If $A, B \geq 0$, $1 < p < \infty$, and $p^{-1} + q^{-1} = 1$, then $AB \leq A^p/p + B^q/q$, with equality iff $A^p = B^q$.

Proof. If $AB = 0$, the claim is immediate. For $A, B > 0$, let $\alpha = 1/p$.
Since $\varphi(t) = \alpha t + 1 - \alpha - t^\alpha \geq 0$ for $t \geq 0$, with equality iff $t = 1$, apply this to $t = A^p/B^q$ and multiply by B^q.
$AB \leq A^p/p + B^q/q$, with equality iff $A^p = B^q$.
Structural Commentary: Young's inequality is weighted AM–GM in conjugate exponents.
Key Insight: Rescale to A^p and B^q, then use convexity. □

Theorem 5.1.10. *Hölder's Inequality*
Let $1 < p < \infty$ and $p^{-1} + q^{-1} = 1$. Then $f \in L^p(\mu)$ and $g \in L^q(\mu)$ imply $fg \in L^1(\mu)$ and $\|fg\|_1 \leq \|f\|_p \|g\|_q$.

Proof. If $\|f\|_p \|g\|_q = 0$, then $|f|^p = 0$ or $|g|^q = 0$ a.e., so $fg = 0$ a.e.
Otherwise set $A = |f| / \|f\|_p$ and $B = |g| / \|g\|_q$. Young gives $|fg| / (\|f\|_p \|g\|_q) \leq A^p/p + B^q/q$.
The right side lies in L^1, so $fg \in L^1$, and integration yields $\|fg\|_1 / (\|f\|_p \|g\|_q) \leq p^{-1} + q^{-1} = 1$.
Structural Commentary: Normalize both factors to unit L^p and L^q size.
Key Insight: Products are controlled by conjugate powers. □

Definition 5.1.11. *Conjugate Indices*
For $p, q \in (1, \infty)$, p and q are *conjugate indices* if $p^{-1} + q^{-1} = 1$, equivalently $p + q = pq$.
The unique self-conjugate index is $p = q = 2$.

Corollary 5.1.12. *Cauchy–Bunyakovsky–Schwarz in L^2*
If $f, g \in L^2(\mu)$, then $fg \in L^1(\mu)$ and $\left|\int_X fg\, d\mu\right| \leq \int_X |fg|\, d\mu \leq \|f\|_2 \|g\|_2$.

Proof. Hölder with $p = q = 2$ gives $fg \in L^1$ and $\int_X |fg|\, d\mu \leq \|f\|_2 \|g\|_2$.
The first inequality is the complex integral estimate applied to fg.
Structural Commentary: The self-conjugate exponent turns Hölder into Cauchy–Schwarz.
Key Insight: Pair two L^2 factors through the 2–2 conjugacy. □

Theorem 5.1.13. *Minkowski's Inequality*
If $1 \leq p < \infty$, then $f, h \in L^p(\mu)$ imply $f + h \in L^p(\mu)$ and $\|f + h\|_p \leq \|f\|_p + \|h\|_p$.

Proof. For $p = 1$, integrate $|f + h| \leq |f| + |h|$. Let $p > 1$ and $q = p/(p-1)$.
Since $|f + h|^p \leq 2^{p-1}(|f|^p + |h|^p)$, $f + h \in L^p$ and $|f + h|^{p-1} \in L^q$.
From $|f + h|^p \leq (|f| + |h|)\, |f + h|^{p-1}$, Hölder gives $\|f + h\|_p^p \leq (\|f\|_p + \|h\|_p)\, \|f + h\|_p^{p-1}$.
If $\|f + h\|_p > 0$, divide by $\|f + h\|_p^{p-1}$; otherwise the inequality is trivial.
Structural Commentary: Hölder turns the pointwise triangle inequality into a norm triangle inequality.
Key Insight: Minkowski is exactly the closure and triangle estimate for L^p. □

Theorem 5.1.14. *L^p Is Normed*
For $1 \leq p < \infty$, $(L^p(\mu), \|\cdot\|_p)$ is a normed linear space.

Proof. Pointwise operations descend to a.e.-classes, so $L^p(\mu)$ is linear. For $\alpha \in \mathbb{F}$, $\|\alpha f\|_p = |\alpha|\, \|f\|_p$.
If $\|f\|_p = 0$, then $\int_X |f|^p\, d\mu = 0$, hence $|f|^p = 0$ a.e. and $f = 0$ in L^p. The triangle axiom is Minkowski.
Structural Commentary: Quotienting by a.e. equality turns pointwise operations into vector-space operations.
Key Insight: Minkowski supplies the nontrivial norm axiom. □

Definition 5.1.15. *Completeness Language*
A normed space $(V, \|\cdot\|)$ is *complete*, or a *Banach space*, if every Cauchy sequence $(x_n) \subset V$ has $x \in V$ with $\|x_n - x\| \to 0$. Thus $L^p(\mu)$ is complete iff every L^p-Cauchy sequence converges in $\|\cdot\|_p$.

Lemma 5.1.16. *Convergent Implies Cauchy*
If $f_n \to f$ in $L^p(\mu)$, then (f_n) is Cauchy in $L^p(\mu)$.

Proof. Given $\varepsilon > 0$, choose N so $n \geq N$ implies $\|f_n - f\|_p < \varepsilon/2$.
Then $m, n \geq N$ gives $\|f_m - f_n\|_p \leq \|f_m - f\|_p + \|f - f_n\|_p < \varepsilon$ by Minkowski.
Structural Commentary: A convergent sequence has uniformly small tails around its limit.
Key Insight: Use the common limit as the bridge between two tails. □

Theorem 5.1.17. *Riesz–Fischer Completeness Theorem*
For $1 \leq p < \infty$, $(L^p(\mu), \|\cdot\|_p)$ is complete.

Proof. Let (f_n) be Cauchy. Choose $n_k \uparrow \infty$ so $m, n \geq n_k$ imply $\|f_m - f_n\|_p \leq 2^{-k}$, and set $g_k = f_{n_k}$.
For $G_N := |g_1| + \sum_{k<N} |g_{k+1} - g_k|$, Minkowski gives $\|G_N\|_p \leq \|g_1\|_p + \sum_{k<N} 2^{-k} \leq \|g_1\|_p + 1$.
Fatou gives $G := \lim_N G_N \in L^p$. Hence $\sum_k |g_{k+1} - g_k| < +\infty$ a.e., so $f := g_1 + \sum_k (g_{k+1} - g_k)$ exists a.e. and $g_k \to f$ a.e. Since $|g_k - f| \leq G$ a.e., DCT gives $\|g_k - f\|_p \to 0$.
Finally, for $\varepsilon > 0$, choose k with $\|g_k - f\|_p < \varepsilon/2$ and then $n \geq n_k$ with $\|f_n - g_k\|_p < \varepsilon/2$; hence $\|f_n - f\|_p < \varepsilon$.
Structural Commentary: A rapid subsequence converges a.e.; DCT gives norm convergence.
Key Insight: Fatou finds the pointwise limit, dominated convergence brings it back to L^p. □

Proposition 5.1.18. *Essential Supremum Test*
If $f \in L^\infty(\mu)$, then $|f| \leq \|f\|_\infty$ a.e.; if $A < \|f\|_\infty$, then $\mu(\{|f| > A\}) > 0$.

Proof. For each k, $|f| \leq \|f\|_\infty + 1/k$ a.e.; intersect the full-measure sets and let $k \to \infty$.
If $A < \|f\|_\infty$ and $\mu(\{|f| > A\}) = 0$, then A is an essential bound, contradiction.
Structural Commentary: The essential supremum is the least a.e. upper bound.
Key Insight: The sharp bound cannot be lowered on a full-measure set. □

Theorem 5.1.19. *L^∞ Is Banach*
$(L^\infty(\mu), \|\cdot\|_\infty)$ is a Banach space.

Proof. Linearity is pointwise modulo a.e. equality; homogeneity is immediate. The essential bound test gives definiteness, and off the union of two null sets $|f + g| \leq \|f\|_\infty + \|g\|_\infty$, giving the triangle axiom.
Let (f_n) be L^∞-Cauchy. For each m, n, choose null N_{mn} with $|f_m - f_n| \leq \|f_m - f_n\|_\infty$ on $X \setminus N_{mn}$, and put $N = \bigcup_{m,n} N_{mn}$. On $X \setminus N$, (f_n) is uniformly Cauchy, hence uniformly converges to a measurable f.
Thus $\|f_n - f\|_\infty \to 0$; for fixed N_0, $|f| \leq |f_{N_0}| + \sup_{n \geq N_0} \|f_n - f_{N_0}\|_\infty$ a.e., so $f \in L^\infty$.
Structural Commentary: L^∞ completeness is uniform convergence after discarding one null set.
Key Insight: Essential bounds become ordinary uniform bounds off a common null set. □

Proposition 5.1.20. *Simple Approximation, Support, and Tails*
Let $1 \leq p < \infty$ and $f \in L^p(\mu)$.

(i) Simple measurable functions are dense in $L^p(\mu)$; also in $L^\infty(\mu)$ for $\|\cdot\|_\infty$.
(ii) $\{|f| > 0\}$ is σ-finite.
(iii) For $E_n := \{|f| \geq n\}$, $\mu(E_n) \to 0$.
(iv) For every $\varepsilon > 0$, there is $E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset$ implies $\|f \mathbf{1}_F\|_p < \varepsilon$.

Proof.

(i) Choose simple $s_n \to f$ a.e. with $|s_n| \leq |f|$ in the real case; treat real and imaginary parts separately in the complex case. Then $|f - s_n|^p \leq 2^p |f|^p$, so DCT gives $\|f - s_n\|_p \to 0$. For $f \in L^\infty$, discard null N with $|f| \leq \|f\|_\infty + \eta$ on $X \setminus N$, then quantize that bounded range.
(ii) Put $A_k := \{|f| \geq 1/k\}$. Then $\mu(A_k) \leq k^p \|f\|_p^p < +\infty$ and $\{|f| > 0\} = \bigcup_k A_k$.
(iii) $n^p \mu(E_n) \leq \int_{E_n} |f|^p \, d\mu \leq \|f\|_p^p$, hence $\mu(E_n) \leq n^{-p} \|f\|_p^p \to 0$.
(iv) With A_k as in (ii), $A_k \uparrow \{|f| > 0\}$ and $|f|^p \mathbf{1}_{X \setminus A_k} \to 0$ a.e. DCT gives $\int_{X \setminus A_k} |f|^p \, d\mu \to 0$. Choose $E_\varepsilon = A_k$ with this integral $< \varepsilon^p$; if $F \cap E_\varepsilon = \emptyset$, then $\|f \mathbf{1}_F\|_p^p \leq \int_{X \setminus E_\varepsilon} |f|^p \, d\mu < \varepsilon^p$.

Structural Commentary: L^p mass is simple-approximable and finite-piece capturable.
Key Insight: Integrability turns level sets into finite-measure exhaustions. □

Proposition 5.1.21. *Shell Criterion on Finite Measure Spaces*
Let $\mu(X) < +\infty$, f be measurable, and $E_n := \{n - 1 \leq |f| < n\}$ for $n \geq 1$.
For $1 \leq p < \infty$, $f \in L^p(\mu) \iff \sum_{n \geq 1} n^p \mu(E_n) < +\infty$.

Proof. If $\sum_n n^p \mu(E_n) < +\infty$, then $|f|^p \leq \sum_n n^p \mathbf{1}_{E_n}$, so $f \in L^p$.
Conversely, if $f \in L^p$, then $\mu(E_1) \leq \mu(X) < +\infty$ and, for $n \geq 2$, $n^p \mathbf{1}_{E_n} \leq 2^p |f|^p \mathbf{1}_{E_n}$.
Hence $\sum_n n^p \mu(E_n) \leq \mu(X) + 2^p \int_X |f|^p \, d\mu < +\infty$.
Structural Commentary: The finite base shell handles small values; weighted shells control tails.
Key Insight: Shell weights characterize L^p mass. □

Proposition 5.1.22. *Inclusions, Interpolation, and Multipliers*

(i) If $\mu(X) < +\infty$ and $1 \leq r \leq p < \infty$, then $L^p(\mu) \subseteq L^r(\mu)$ and $\|f\|_r \leq \|f\|_p \mu(X)^{1/r - 1/p}$.
(ii) On $(\mathbb{N}, 2^{\mathbb{N}}, \#)$, $1 \leq p \leq s < \infty$ implies $\ell^p \subseteq \ell^s$ and $\|a\|_s \leq \|a\|_p$.
(iii) If $1 \leq p_1 < p_2 < \infty$ and $f \in L^{p_1}(\mu) \cap L^{p_2}(\mu)$, then $f \in L^p(\mu)$ for $p_1 \leq p \leq p_2$.
(iv) If $1 \leq p \leq \infty$, then $L^p(\mu) \cdot L^\infty(\mu) \subseteq L^p(\mu)$ and $\|fg\|_p \leq \|f\|_p \|g\|_\infty$.
(v) $L^\infty(\mu) \subseteq L^1(\mu) \iff \mu(X) < +\infty$. If $\mu(X) = 1$, then $\|f\|_\infty = \lim_{p \to \infty} \|f\|_p$ for $f \in L^\infty(\mu)$.

Proof.

(i) For $r = p$, $L^p \subseteq L^p$ and the estimate reads $\|f\|_p \le \|f\|_p$, i.e. equality.
For $r < p$, Hölder applied to $|f|^r \cdot 1$ gives $\int_X |f|^r \, d\mu \le \|f\|_p^r \, \mu(X)^{1-r/p}$; take rth roots.

(ii) For $a \in \ell^p$, $|a_n| \le \|a\|_p$ for all n, hence $\sum_n |a_n|^s \le \|a\|_p^{s-p} \sum_n |a_n|^p \le \|a\|_p^s$.

(iii) For $p \in [p_1, p_2]$, $|f|^p \le |f|^{p_1} \mathbf{1}_{\{|f|\le 1\}} + |f|^{p_2} \mathbf{1}_{\{|f|>1\}}$.

(iv) For $p < \infty$, $|fg|^p \le \|g\|_\infty^p \, |f|^p$ a.e.; for $p = \infty$, $|fg| \le \|f\|_\infty \, \|g\|_\infty$ a.e.

(v) ($\Rightarrow$) If $L^\infty(\mu) \subseteq L^1(\mu)$, then $\mathbf{1}_X \in L^\infty(\mu)$ lies in $L^1(\mu)$, so $\mu(X) = \int_X \mathbf{1}_X \, d\mu < +\infty$.

(v) ($\Leftarrow$) If $\mu(X) < +\infty$ and $f \in L^\infty(\mu)$, then $\int_X |f| \, d\mu \le \|f\|_\infty \, \mu(X) < +\infty$. If $\mu(X) = 1$, then $\|f\|_p \le \|f\|_\infty$.
For $A < \|f\|_\infty$, $\mu(\{|f| > A\}) > 0$ and $\|f\|_p \ge A\mu(\{|f| > A\})^{1/p} \to A$. Let $A \uparrow \|f\|_\infty$.

Structural Commentary: Inclusions depend on domain size and pointwise bounds.
Key Insight: Check whether size comes from the function values or from the measure of the domain. □

Example 5.1.23. *Membership Models*

(i) On $(\mathbb{N}, 2^{\mathbb{N}}, \#)$, $a_n = 1/n$ satisfies $\|a\|_p^p = \sum_{n\ge 1} n^{-p}$; hence $a \in \ell^p \iff p > 1$, while $a \in \ell^\infty \setminus \ell^1$.

(ii) On $\mathbb{N}$ with $\lambda(\{n\}) = n^{-2}$, $\lambda(\mathbb{N}) < +\infty$ and $f(n) = n^{1/p_0}$ satisfies $\int_{\mathbb{N}} |f|^p \, d\lambda = \sum_{n\ge 1} n^{p/p_0 - 2}$.
Hence $f \in L^p(\lambda) \iff 1 \le p < p_0$. In particular, $f(n) = \sqrt{n}$ gives $p_0 = 2$.

(iii) On $(0, \infty)$, $f(x) = x^{-1/2}(1 + |\log x|)^{-1}$ gives $\int_0^\infty |f|^p \, dx = \int_{-\infty}^\infty e^{(1-p/2)t}(1 + |t|)^{-p} \, dt$ under $x = e^t$.
If $p < 2$ it diverges at $+\infty$; if $p > 2$ at $-\infty$; if $p = 2$ it is $\int (1 + |t|)^{-2} \, dt < +\infty$.

Structural Commentary: L^p membership reduces to series and endpoint tests.
Key Insight: Different ends impose different p-conditions.

Proposition 5.1.24. *Hölder Dual Norm Attainment*
Let $1 < p < \infty$, $p^{-1} + q^{-1} = 1$, and $f \in L^p(\mu)$ be real-valued. Then $\sup_{\|g\|_q \le 1} \left|\int_X fg \, d\mu\right| = \|f\|_p$.
If $f \neq 0$, the supremum is attained by $g_0 = \|f\|_p^{-p/q} \operatorname{sign}(f) \, |f|^{p-1}$.

Proof. Hölder gives $\left|\int_X fg \, d\mu\right| \le \|f\|_p \, \|g\|_q$, hence the supremum is $\le \|f\|_p$.
Since $(p-1)q = p$, $\|g_0\|_q^q = \|f\|_p^{-p} \int_X |f|^p \, d\mu = 1$, and $\int_X fg_0 \, d\mu = \|f\|_p^{-p/q} \int_X |f|^p \, d\mu = \|f\|_p$.
Structural Commentary: Equality in Hölder is obtained by matching the conjugate powers.
Key Insight: The L^q test function points in the power direction of f. □

Definition 5.1.25. *Bounded Linear Functionals on L^p*

(i) A *linear functional* on real $L^p(\mu)$ is a linear map $G : L^p(\mu) \to \mathbb{R}$.

(ii) G is *bounded* if $\exists M < +\infty$ such that $|G(f)| \le M \|f\|_p$ for all $f \in L^p(\mu)$.

(iii) The *operator norm* is $\|G\| := \sup\{|G(f)| : f \in L^p(\mu), \ \|f\|_p \le 1\}$.

(iv) G is *positive* if $f \ge 0$ a.e. implies $G(f) \ge 0$.

Lemma 5.1.26. *Positive Split of a Bounded Functional*
Let $1 \le p < +\infty$ and $G \in (L^p(\mu))^*_{\mathbb{R}}$. Then $G = G^+ - G^-$ for positive bounded linear functionals $G^\pm$.

Proof. For $f \ge 0$, set $G^+(f) := \sup\{G(g) : 0 \le g \le f\}$.
This is finite since $|G(g)| \le \|G\| \, \|g\|_p \le \|G\| \, \|f\|_p$, and positive homogeneity is immediate.
For $f_1, f_2 \ge 0$, $G^+(f_1) + G^+(f_2) \le G^+(f_1 + f_2)$; conversely, if $0 \le h \le f_1 + f_2$, set $h_1 = (h - f_2)^+$ and $h_2 = h - h_1$, so $h = h_1 + h_2$ with $0 \le h_j \le f_j$, hence $G(h) \le G^+(f_1) + G^+(f_2)$.
Thus G^+ is additive on the positive cone. Extend by $G^+(f) = G^+(f^+) - G^+(f^-)$; then G^+ is positive, linear, and bounded. Let $G^- = G^+ - G$. Since $G^+(f) \ge G(f)$ for $f \ge 0$, G^- is positive and bounded.
Structural Commentary: Function-lattice functionals split into positive parts.
Key Insight: Split any competitor $h \le f_1 + f_2$ into the portions assigned to f_1 and f_2. □

Theorem 5.1.27. *Riesz Representation for L^1*
Let $(X, \mathcal{M}, \mu)$ be σ-finite and $G \in (L^1(\mu))^*_{\mathbb{R}}$.

(i) There is $g \in L^\infty(\mu)$ such that $G(f) = \int_X fg \, d\mu$ for all $f \in L^1(\mu)$.

(ii) $\|G\| = \|g\|_\infty$.

(iii) If G is positive, then $g \ge 0$ a.e.

Proof.

(i) First let $\mu(X) < +\infty$ and $G \ge 0$. Set $\lambda(E) := G(\mathbf{1}_E)$. Finite additivity is linearity; if $E_n \uparrow E$, then $\mathbf{1}_{E_n} \to \mathbf{1}_E$ in L^1, hence $\lambda(E_n) \to \lambda(E)$. Thus λ is finite, and $\lambda \ll \mu$ since $\mu(E) = 0$ gives $\mathbf{1}_E = 0$ in L^1.
Radon–Nikodym gives $g \ge 0$ with $\lambda(E) = \int_E g \, d\mu$, so simple functions are represented.
By (ii), $g \in L^\infty$, hence L^1-density of simple functions extends the identity to all $f \in L^1$. For σ-finite μ,

apply the finite case to $f \mapsto G(f\mathbf{1}_{X_n})$ on $X_n \uparrow X$, $\mu(X_n) < +\infty$, and patch densities on overlaps. For general G, use the positive split $G = G^+ - G^-$ and subtract the two densities.

(ii) The pairing estimate gives $\|G\| \le \|g\|_\infty$. Conversely, if $A > \|G\|$ and $E \subseteq \{|g| > A\}$ has $\mu(E) < +\infty$, then with $f = \operatorname{sign}(g)\mathbf{1}_E$, $A\mu(E) < \int_E |g|\, d\mu = G(f) \le \|G\|\,\mu(E)$, so $\mu(E) = 0$.
A σ-finite exhaustion gives $\mu(\{|g| > A\}) = 0$ for every $A > \|G\|$, hence $\|g\|_\infty \le \|G\|$.

(iii) If G is positive and $B = \{g < 0\}$, then for every finite-measure $E \subseteq B$, $0 \le G(\mathbf{1}_E) = \int_E g\, d\mu \le 0$ only if $\int_E (-g)\, d\mu = 0$. Hence $\mu(E) = 0$ on $\{g < -1/k\}$ for all k, and a σ-finite exhaustion gives $\mu(B) = 0$.

Structural Commentary: A bounded functional on L^1 is integration against an essentially bounded density.
Key Insight: Indicators give a measure; Radon–Nikodym gives the density. □

Theorem 5.1.28. *Riesz Representation for L^p, $1 < p < +\infty$*
Let $(X, \mathcal{M}, \mu)$ be any measure space, $1 < p < +\infty$, and $q = p/(p-1)$. For every $G \in (L^p(\mu))^*_{\mathbb{R}}$, there is a unique $g \in L^q(\mu)$ modulo a.e. equality such that $G(f) = \int_X fg\, d\mu$ for all $f \in L^p(\mu)$, and $\|G\| = \|g\|_q$.

Proof. On finite measure spaces, repeat the L^1 proof with $\mathbf{1}_{E_n} \to \mathbf{1}_E$ in L^p and the positive split.
Testing truncations of $\operatorname{sign}(g)\,|g|^{q-1}$ gives $g \in L^q$ and the reverse Hölder inequality, hence $\|G\| = \|g\|_q$; uniqueness follows from testing finite-measure sets.
On σ-finite spaces, patch the finite-measure representatives along $X_n \uparrow X$, $\mu(X_n) < +\infty$.
For arbitrary μ, choose $\|f_n\|_p = 1$ with $G(f_n) \to \|G\|$; the supports of the f_n lie in a σ-finite X_0.
If $E \cap X_0 = \emptyset$ and $\mu(E) < +\infty$, boundedness on $f_n \pm t\mathbf{1}_E$, followed by $n \to \infty$ and $t \downarrow 0$, gives $G(\mathbf{1}_E) = 0$.
Thus G vanishes off X_0; represent it on X_0 and extend the density by 0.
Structural Commentary: For $1 < p < \infty$, every bounded functional comes from the conjugate space L^q.
Key Insight: Localize to σ-finite support, then patch. □

Remark 5.1.29. *L^2 Riesz Gives Another Radon–Nikodym Route*
Let λ, μ be finite and $\lambda \ll \mu$. Put $\rho = \lambda + \mu$ and $G(h) := \int h\, d\lambda$ on $L^2(\rho)$.
If $G(h) = \int hg\, d\rho$, indicator tests give $0 \le g \le 1$ ρ-a.e. and $\mu(\{g = 1\}) = 0$.
Since $\int h(1-g)\, d\lambda = \int hg\, d\mu$, take $E_N = E \cap \{1 - g \ge 1/N\}$ and $h = \mathbf{1}_{E_N}/(1-g) \in L^2(\rho)$ to get $\lambda(E_N) = \int_{E_N} g/(1-g)\, d\mu$. Let $N \to \infty$ by monotone convergence and $\lambda \ll \mu$: $\lambda(E) = \int_E g/(1-g)\, d\mu$.
Key Insight: Riesz representation on $L^2(\lambda + \mu)$ encodes the density in the ratio $g/(1-g)$.

Lemma 5.1.30. *Absolute Continuity of L^p Mass*
For $1 \le p < \infty$ and $f \in L^p(\mu)$, $\forall \varepsilon > 0\, \exists \delta > 0$ such that $\mu(E) < \delta$ implies $\|f\mathbf{1}_E\|_p < \varepsilon$.

Proof. Let $h = |f|^p \in L^1$. Choose $M > 0$ with $\int_{\{h > M\}} h\, d\mu < \varepsilon^p/2$, and set $\delta = \varepsilon^p/(2M)$.
If $\mu(E) < \delta$, then $\|f\mathbf{1}_E\|_p^p = \int_E h\, d\mu \le M\mu(E) + \int_{\{h>M\}} h\, d\mu < \varepsilon^p$.
Structural Commentary: L^p mass is absolutely continuous with respect to measure.
Key Insight: Truncate the tail; small measure controls the bounded part. □

Proposition 5.1.31. *Local Mass Limits for L^p Cauchy Sequences*
For $1 \le p < \infty$ and $f_n \in L^p(\mu)$, set $\beta_n(E) := \|f_n\mathbf{1}_E\|_p$ for $E \in \mathcal{M}$. Then $|\beta_n(E) - \beta_m(E)| \le \|f_n - f_m\|_p$.
Hence, if (f_n) is L^p-Cauchy, then $\beta(E) := \lim_n \beta_n(E)$ exists for every $E \in \mathcal{M}$.

Proof. Minkowski on E gives $\beta_n(E) \le \beta_m(E) + \|(f_n - f_m)\mathbf{1}_E\|_p \le \beta_m(E) + \|f_n - f_m\|_p$; swap m, n.
Thus L^p-Cauchy implies $(\beta_n(E))$ is Cauchy in $\mathbb{R}$.
Structural Commentary: Restricting to E is a 1-Lipschitz operation on L^p mass.
Key Insight: Local mass inherits global Cauchy control. □

Proposition 5.1.32. *Uniform Local Mass Control for L^p Cauchy Sequences*
Let $1 \le p < \infty$, f_n be L^p-Cauchy, and $\beta_n(E) := \|f_n\mathbf{1}_E\|_p$. For every $\varepsilon > 0$:
(i) $\exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset$ implies $\sup_n \beta_n(F) < \varepsilon$;
(ii) $\exists \delta > 0$ such that $\mu(E) < \delta$ implies $\sup_n \beta_n(E) < \varepsilon$.

Proof. Choose N so $n \ge N$ implies $\|f_n - f_N\|_p < \varepsilon/3$.

(i) Choose finite-measure E_j for $1 \le j \le N$ with $F \cap E_j = \emptyset \Rightarrow \beta_j(F) < \varepsilon/3$, and set $E_\varepsilon = \bigcup_{j \le N} E_j$.
For $n \le N$ this works; for $n > N$, $\beta_n(F) \le \|f_n - f_N\|_p + \beta_N(F) < \varepsilon$.

(ii) Choose $\delta_j > 0$ for $1 \le j \le N$ with $\mu(E) < \delta_j \Rightarrow \beta_j(E) < \varepsilon/3$, and set $\delta = \min_{j \le N} \delta_j$.
For $n \le N$ this works; for $n > N$, $\beta_n(E) \le \|f_n - f_N\|_p + \beta_N(E) < \varepsilon$.

Structural Commentary: Cauchy families are uniformly tight and absolutely continuous.
Key Insight: Control finitely many early terms directly and the tail by Cauchy closeness. □

Method 5.1.33. *Lebesgue Proof Patterns*

(i) *Approximate*: for $f \geq 0$, use $s \in \mathcal{S}^+(f)$ and $\int_E f\,d\mu = \sup_{s\in\mathcal{S}^+(f)} \mathcal{I}_E(s)$.
(ii) *Promote*: prove for $\mathbf{1}_A$, extend to $s = \sum_i c_i \mathbf{1}_{A_i}$, then pass through $s_n \uparrow f$.
(iii) *Ignore null sets*: if $f \sim_{\text{a.e.}} g$ on E, replace f by g inside integrals over measurable $A \subseteq E$.
(iv) *Split signs*: use $f = f^+ - f^-$ and $h = u + iv$.
(v) *Exchange limits*: reduce Fatou and dominated convergence to $0 \leq f_n \uparrow f$.
(vi) *Slice*: prove boxes first; use regularity and null slices to pass to products and Fubini.

Key Insight: Reduce to simple functions, monotone convergence, null sets, and product boxes.

5.2 Modes of Convergence

Throughout this subsection $(X, \mathcal{M}, \mu)$ is a fixed measure space and, unless otherwise stated, functions are real-valued and measurable. The point is not to rank convergence modes linearly, but to know which hypotheses convert one mode into another.

Definition 5.2.1. *Pointwise, Uniform, and Almost-Everywhere Convergence*
Let (f_n) and f be functions on X.

(i) $f_n \to f$ *pointwise* if $\forall x \in X$, $\varepsilon > 0$, $\exists N(\varepsilon, x)$ s.t. $n \geq N(\varepsilon, x)$ implies $|f_n(x) - f(x)| < \varepsilon$.
(ii) $f_n \to f$ *uniformly* if $\forall \varepsilon > 0\ \exists N(\varepsilon)$ s.t. $n \geq N(\varepsilon)$ implies $|f_n(x) - f(x)| < \varepsilon\ \forall x \in X$.
(iii) $f_n \to f$ *almost everywhere* if $\exists A \in \mathcal{M}$ with $\mu(A) = 0$ s.t. $\forall x \in X \setminus A$, $\varepsilon > 0$, $\exists N(\varepsilon, x)$ s.t. $n \geq N(\varepsilon, x)$ implies $|f_n(x) - f(x)| < \varepsilon$.

Proposition 5.2.2. *First Implications and First Failures*
Let $f_n, f : X \to \mathbb{R}$.

(i) $f_n \rightrightarrows f \Rightarrow f_n \to f$ pointwise $\Rightarrow f_n \to f$ a.e.; converses fail.
(ii) $|X| < +\infty$ and $f_n \to f$ pointwise $\Rightarrow f_n \rightrightarrows f$.
(iii) $\mu(A) = 0 \Rightarrow A = \emptyset$ and $f_n \to f$ a.e. $\Rightarrow f_n \to f$ pointwise.

Proof.

(i) If $f_n \rightrightarrows f$, then $\forall \varepsilon > 0\ \exists N_\varepsilon$ such that $n \geq N_\varepsilon$ implies $|f_n(x) - f(x)| < \varepsilon$ for all $x \in X$; hence pointwise convergence with $N(\varepsilon, x) = N_\varepsilon$. Pointwise convergence is a.e. convergence with $A = \emptyset$. Conversely, on $([0,1], \lambda)$, $x^n \to \mathbf{1}_{\{1\}}$ pointwise but not uniformly, and $\mathbf{1}_{\{0\}} \to 0$ a.e. but not pointwise.
(ii) If X is finite and $f_n \to f$ pointwise, then for each $\varepsilon > 0$ and $x \in X$ choose $N(\varepsilon, x)$. Set $N_\varepsilon = \max_{x\in X} N(\varepsilon, x)$; then $n \geq N_\varepsilon$ implies $|f_n(x) - f(x)| < \varepsilon$ for all $x \in X$, so $f_n \rightrightarrows f$.
(iii) If $f_n \to f$ a.e., then $\exists A \in \mathcal{M}$ with $\mu(A) = 0$ and $f_n(x) \to f(x)$ for all $x \in X \setminus A$. Since every null set is empty, $A = \emptyset$, so $f_n \to f$ pointwise.

Structural Commentary: Uniformity is bounded x-dependence of $N(\varepsilon, x)$.
Key Insight: Uniformity controls indices; a.e. convergence controls null exceptions. □

Proposition 5.2.3. *Uniform Convergence Gives L^p Convergence on Finite Measure Spaces*
Let $1 \leq p < \infty$, $\mu(X) < +\infty$, and $f_n \in L^p(\mu)$. If $f_n \rightrightarrows f$ on X, then $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.

Proof. For $\varepsilon > 0$, choose N with $n \geq N \Rightarrow |f_n - f| \leq \varepsilon$ on X.
Then $n \geq N \Rightarrow \|f_n - f\|_p \leq (\int_X \varepsilon^p\,d\mu)^{1/p} = \varepsilon\mu(X)^{1/p}$, so $\|f_n - f\|_p \to 0$.
Taking $\varepsilon = 1$, choose n_0 with $|f_{n_0} - f| \leq 1$ on X; hence $f - f_{n_0} \in L^p(\mu)$ and $f = f_{n_0} + (f - f_{n_0}) \in L^p(\mu)$.
Structural Commentary: Uniform error becomes L^p error after multiplication by $\mu(X)^{1/p}$.
Key Insight: Finite measure converts a pointwise sup-bound into an integral bound. □

Theorem 5.2.4. *Dominated Almost-Everywhere Convergence Gives L^p Convergence*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, and $f_n \to f$ a.e.
If $\exists g \in L^p(\mu)$ such that $|f_n| \leq g$ a.e. for all n, then $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.

Proof. Off a null set, $f_n \to f$ and $|f_n| \leq g$ for all n; hence $|f| \leq g$, so $f \in L^p(\mu)$. Also $|f_n - f|^p \to 0$ a.e. and $|f_n - f|^p \leq (2g)^p \in \mathscr{L}^1(\mu)$ a.e.; Theorem 2.3.4 gives $\|f_n - f\|_p^p = \int_X |f_n - f|^p\,d\mu \to 0$.
Structural Commentary: Apply ordinary dominated convergence to the pth-power error.
Key Insight: A single L^p envelope controls both membership of the limit and convergence in mean. □

Corollary 5.2.5. *Uniform Boundedness on Finite Measure Spaces*
Let $\mu(X) < +\infty$, $0 \leq K < +\infty$, and $f_n \to f$ a.e.
If $|f_n| \leq K$ a.e. for all n, then for every $1 \leq p < \infty$, $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.

Proof. Fix $1 \le p < \infty$ and set $g := K\mathbf{1}_X$. Then $\int_X g^p \, d\mu = K^p \mu(X) < +\infty$, so $g \in L^p(\mu)$, and $|f_n| \le g$ a.e. for all n. Since $f_n \to f$ a.e., Theorem 5.2.4 gives $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.
Structural Commentary: Finite measure turns a uniform a.e. bound into an L^p envelope.
Key Insight: Bounded a.e. convergence on finite measure is dominated convergence. □

Example 5.2.6. *Uniform Need Not Imply L^p on Infinite Measure*
On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, fix $1 \le p < \infty$ and set $f_n := n^{-1/p}\mathbf{1}_{[0,n]}$. Then $\|f_n\|_\infty = n^{-1/p} \to 0$, so $f_n \rightrightarrows 0$.
However $\|f_n\|_p^p = \int_0^n n^{-1} \, d\lambda = 1$, hence $\|f_n\|_p \not\to 0$.
Structural Commentary: The support grows exactly enough to preserve L^p mass.
Key Insight: Uniform smallness alone does not control integrals on infinite measure spaces.

Example 5.2.7. *Almost Everywhere and In-Measure Need Not Imply L^p*
On $[0,1]$ with Lebesgue measure, set $f_n := n\mathbf{1}_{[1/n,\,2/n]}$.
For each $x \in [0,1]$, $x \in [1/n, 2/n]$ for only finitely many n, hence $f_n(x) \to 0$.
If $\alpha > 0$ and $n \ge \alpha$, then $\lambda\{|f_n| \ge \alpha\} = 1/n \to 0$, so $f_n \to 0$ in measure.
Yet for $1 \le p < \infty$, $\|f_n\|_p^p = n^p \lambda([1/n, 2/n]) = n^{p-1} \not\to 0$.
Given $\delta > 0$, choose $0 < \eta < \delta$; on $[0,1] \setminus [0,\eta]$, $f_n = 0$ for all $n > 2/\eta$, so $f_n \to 0$ almost uniformly.
Thus a.e., in-measure, and almost-uniform convergence do not imply L^p convergence.

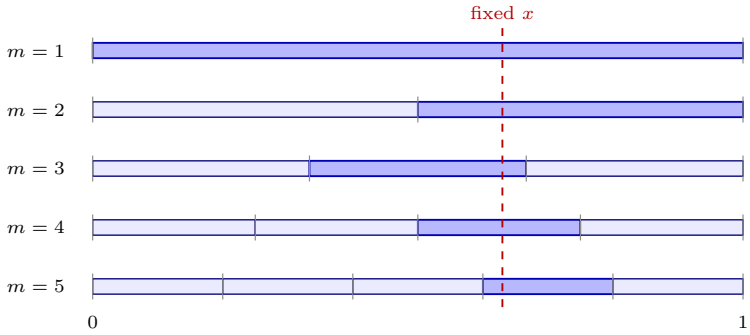


Figure 202: Typewriter intervals shrink, but each fixed x is hit once per partition.

Example 5.2.8. *L^p Convergence Without Pointwise Convergence*
Let $X = [0,1]$ with Lebesgue measure and list $I_{m,k} := [(k-1)/m, k/m]$, $1 \le k \le m$, row by row in m.
Let $f_n := \mathbf{1}_{I_n}$, I_n is nth interval. If I_n in row $m(n)$, then $m(n) \to \infty$, $\|f_n\|_p^p = \lambda(I_n) = 1/m(n) \to 0$; hence $f_n \to 0$ in L^p, and therefore in measure by Proposition 5.2.10.
$\forall x \in [0,1]$, every row contains some $I_{m,k}$ with $x \in I_{m,k}$, giving a subsequence with value 1 at x; for infinitely many rows, some $I_{m,k}$ omits x, giving a subsequence with value 0 at x. Thus $(f_n(x))$ converges nowhere.
The subfamily $I_{m,2} = [1/m, 2/m]$, $m \ge 2$, gives a subsequence converging pointwise to 0 on all of X.

Definition 5.2.9. *Convergence and Cauchy Convergence in Measure*
Let (f_n) be measurable functions on X.

(i) $f_n \to f$ *in measure* if $\forall \alpha > 0$, $\mu\{x \in X : |f_n(x) - f(x)| \ge \alpha\} \to 0$.
(ii) (f_n) is *Cauchy in measure* if $\forall \alpha > 0$, $\mu\{x \in X : |f_m(x) - f_n(x)| \ge \alpha\} \to 0$ as $m, n \to \infty$.

The same definitions result with $>$ in place of $\ge$.

Proposition 5.2.10. *L^p Convergence Implies Convergence in Measure*
Let $1 \le p < \infty$. If $\|f_n - f\|_p \to 0$, then $f_n \to f$ in measure.

Proof. For $\alpha > 0$, set $E_n(\alpha) := \{|f_n - f| \ge \alpha\}$.
Since $\|f_n - f\|_p^p \ge \int_{E_n(\alpha)} |f_n - f|^p \, d\mu \ge \alpha^p \mu(E_n(\alpha))$, $\mu(E_n(\alpha)) \le \alpha^{-p} \|f_n - f\|_p^p \to 0$.
Structural Commentary: Large-error sets are controlled by the integral of the pth-power error.
Key Insight: This is Chebyshev's inequality in deterministic measure form. □

Theorem 5.2.11. *Riesz Subsequence Principle for Convergence in Measure*
If (f_n) is Cauchy in measure, then there exist a subsequence (g_k) and a measurable f such that:

(i) $g_k \to f$ a.e.
(ii) $g_k \to f$ in measure.
(iii) For every $\varepsilon > 0$, there is $E \in \mathcal{M}$ with $\mu(E) < \varepsilon$ such that $g_k \to f$ uniformly on $X \setminus E$.

Proof.

(i) Choose $n_1 < n_2 < \cdots$ so that $m, n \geq n_k \Rightarrow \mu\{|f_m - f_n| \geq 2^{-k}\} < 2^{-k}$, and set $g_k := f_{n_k}$.
For $E_k := \{|g_{k+1} - g_k| \geq 2^{-k}\}$, $\mu(E_k) < 2^{-k}$. Set $F_k := \bigcup_{j=k}^{\infty} E_j$.
Then $F_k \downarrow$, $\mu(F_k) \leq \sum_{j=k}^{\infty} 2^{-j} = 2^{-(k-1)}$, and for $i > j \geq k$, $x \notin F_k$,
$|g_i(x) - g_j(x)| \leq \sum_{r=j}^{i-1} |g_{r+1}(x) - g_r(x)| < 2^{-(j-1)}$. Hence (g_j) is uniformly Cauchy on $X \setminus F_k$ $\forall k$.
Let $F := \bigcap_k F_k$, so $\mu(F) = 0$. Define $f = \lim_j g_j$ on $X \setminus F$, $f = 0$ on F. f is measurable, $g_j \to f$ a.e.

(ii) Letting $i \to \infty$ gives $j \geq k$, $x \notin F_k \Rightarrow |f(x) - g_j(x)| \leq 2^{-(j-1)} \leq 2^{-(k-1)}$. Given $\alpha, \varepsilon > 0$, choose k with $2^{-(k-1)} < \alpha$ and $\mu(F_k) < \varepsilon$; then $j \geq k \Rightarrow \{|f - g_j| \geq \alpha\} \subseteq F_k$, so $g_j \to f$ in measure.

(iii) Given $\varepsilon > 0$, choose k with $\mu(F_k) < \varepsilon$.
The estimate in (ii) gives $\sup_{x \in X \setminus F_k} |f(x) - g_j(x)| \leq 2^{-(j-1)} \to 0$, so $g_j \to f$ uniformly on $X \setminus F_k$.

Structural Commentary: Cauchy in measure lets us select summably small exceptional sets.
Key Insight: Outside the tail union of exceptional sets, the subsequence is uniformly Cauchy. □

Corollary 5.2.12. *Completeness for Convergence in Measure*
(f_n) is Cauchy in measure iff $\exists$ measurable f with $f_n \to f$ in measure.

Proof.

($\Rightarrow$) If (f_n) is Cauchy in measure, Theorem 5.2.11 gives $f_{n_k} \to f$ in measure for some measurable f.
For $\alpha > 0$, $\{|f_n - f| \geq \alpha\} \subseteq \{|f_n - f_{n_k}| \geq \alpha/2\} \cup \{|f_{n_k} - f| \geq \alpha/2\}$.
Choose k large, then n large, to get $f_n \to f$ in measure.

($\Leftarrow$) For $\alpha > 0$, $\{|f_m - f_n| \geq \alpha\} \subseteq \{|f_m - f| \geq \alpha/2\} \cup \{|f_n - f| \geq \alpha/2\}$.
Hence $\mu\{|f_m - f_n| \geq \alpha\} \to 0$ as $m, n \to \infty$.

Structural Commentary: The subsequential in-measure limit attracts the whole Cauchy sequence.
Key Insight: Convergence in measure is complete after measurable limits are admitted. □

Corollary 5.2.13. *Uniqueness for Convergence in Measure*
If $f_n \to f$ and $f_n \to g$ in measure, then $f = g$ a.e.

Proof. For $\alpha > 0$, $\{|f - g| \geq \alpha\} \subseteq \{|f - f_n| \geq \alpha/2\} \cup \{|f_n - g| \geq \alpha/2\}$, so $\mu\{|f - g| \geq \alpha\} = 0$.
Hence $\mu\{|f - g| > 0\} = 0$ and $f = g$ a.e.
Structural Commentary: Two in-measure limits cannot differ on a positive-measure set.
Key Insight: In-measure limits are unique modulo null sets. □

Proposition 5.2.14. *Subsequence and Indicator Consequences*

(i) $f_n \to f$ in measure $\Rightarrow$ every subsequence $f_{n_k} \to f$ in measure; if (f_n) is Cauchy in measure, then every (f_{n_k}) is Cauchy in measure.
(ii) If $\|f_n - f\|_p \to 0$ and $\|f_{n_k} - g\|_p \to 0$, then $f = g$ a.e.
(iii) If $f_n = \mathbf{1}_{E_n}$ and $\|f_n - f\|_p \to 0$, then $f = \mathbf{1}_E$ a.e. for some $E \in \mathcal{M}$.

Proof.

(i) For $\alpha > 0$, $\mu\{|f_{n_k} - f| \geq \alpha\} \to 0$ since $n_k \to \infty$; the Cauchy claim is identical with $m_k, n_\ell \to \infty$.

(ii) Proposition 5.2.10 gives $f_n \to f$ and $f_{n_k} \to g$ in measure; also $f_{n_k} \to f$ in measure by (i).
Uniqueness in Corollary 5.2.13 gives $f = g$ a.e.

(iii) By Proposition 5.2.10 and Theorem 5.2.11, some $f_{n_k} \to f$ a.e.
Since $f_{n_k} \in \{0, 1\}$, $f \in \{0, 1\}$ a.e.; with $E := \{f = 1\}$, $E \in \mathcal{M}$ and $f = \mathbf{1}_E$ a.e.

Structural Commentary: Subsequence permanence and a.e. uniqueness transfer mode limits to representatives.
Key Insight: In-measure subsequences preserve limits; indicator limits remain indicators a.e. □

Theorem 5.2.15. *Dominated Convergence in Measure Gives L^p Convergence*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, $f_n \to f$ in measure, and $|f_n| \leq g$ a.e. for some $g \in L^p(\mu)$.
Then $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.

Proof. Since $f_n \to f$ in measure, (f_n) is Cauchy in measure.
By Theorem 5.2.11, some subsequence (u_j) satisfies $u_j \to h$ a.e. and in measure; uniqueness in Corollary 5.2.13 gives $h = f$ a.e. Since $|u_j| \leq g$ a.e., Theorem 5.2.4 gives $f \in L^p(\mu)$. If $\|f_n - f\|_p \not\to 0$, then $\exists \varepsilon > 0$ and a subsequence (g_k) with $\|g_k - f\|_p \geq \varepsilon$. Since $g_k \to f$ in measure, the same argument gives a subsubsequence (h_j) with $h_j \to f$ a.e.; Theorem 5.2.4 yields $\|h_j - f\|_p \to 0$, contradicting $\|h_j - f\|_p \geq \varepsilon$.
Structural Commentary: Reduce convergence in measure to almost-everywhere convergence along a subsequence.
Key Insight: Domination turns every in-measure subsequence into an L^p-convergent subsubsequence. □

Definition 5.2.16. *Almost Uniform Convergence*
Let (f_n) be measurable functions on X.

(i) $f_n \to f$ *almost uniformly* if $\forall \delta > 0\ \exists E_\delta \in \mathcal{M}$ with $\mu(E_\delta) < \delta$ and $f_n \rightrightarrows f$ on $X \setminus E_\delta$.
(ii) (f_n) is *almost uniformly Cauchy* if $\forall \delta > 0\ \exists E_\delta \in \mathcal{M}$ with $\mu(E_\delta) < \delta$ s.t. (f_n) uniformly Cauchy on $X \setminus E_\delta$.

Lemma 5.2.17. *Almost Uniform Cauchy Sequences Have Almost Uniform Limits*
If (f_n) is almost uniformly Cauchy, then $\exists$ measurable f such that $f_n \to f$ almost uniformly and a.e.

Proof. Choose $E_k \in \mathcal{M}$ with $\mu(E_k) < 2^{-k}$ and (f_n) uniformly Cauchy on $X \setminus E_k$. Set $F_k := \bigcup_{j=k}^{\infty} E_j$ and $F := \bigcap_k F_k$; then $F_k \downarrow$, $\mu(F_k) < 2^{-(k-1)}$, $\mu(F) = 0$, and (f_n) is uniformly Cauchy on $X \setminus F_k$. By Theorem 7.1.7, f_n has a uniform limit on each $X \setminus F_k$; these limits agree on overlaps, so define measurable f on $X \setminus F$, and set $f = 0$ on F. Hence $f_n \to f$ a.e. and uniformly on each $X \setminus F_k$; since $\mu(F_k) \to 0$, $f_n \to f$ almost uniformly.
Structural Commentary: Tail unions convert many small exceptional sets into nested uniform domains.
Key Insight: Almost uniform Cauchy means uniformly Cauchy after discarding a tail-small exceptional set. □

Theorem 5.2.18. *Almost Uniform Convergence and Convergence in Measure*
If $f_n \to f$ almost uniformly, then $f_n \to f$ in measure.
If $f_n \to f$ in measure, then $\exists (f_{n_k})$ with $f_{n_k} \to f$ almost uniformly.

Proof. If $f_n \rightrightarrows f$ on $X \setminus E_\delta$ with $\mu(E_\delta) < \delta$, then for each $\alpha > 0$ and all large n, $\{|f_n - f| \geq \alpha\} \subseteq E_\delta$; hence $\limsup_n \mu\{|f_n - f| \geq \alpha\} \leq \delta$, and $\delta \downarrow 0$ gives convergence in measure. Conversely, $f_n \to f$ in measure implies (f_n) is Cauchy in measure; Theorem 5.2.11 gives $f_{n_k} \to h$ almost uniformly and in measure. By uniqueness, $h = f$ a.e.; absorbing $\{h \neq f\}$ into the exceptional sets gives $f_{n_k} \to f$ almost uniformly.
Structural Commentary: Almost uniform convergence is uniform convergence off arbitrarily small sets.
Key Insight: Convergence in measure always hides an almost-uniform subsequence. □

Theorem 5.2.19. *Egoroff's Theorem*
If $\mu(X) < +\infty$ and $f_n \to f$ a.e., with all functions finite-valued off a null set, then $f_n \to f$ almost uniformly and in measure.

Proof. Absorb the null exceptional set into every later E_δ, and assume $f_n(x) \to f(x)$ for all $x \in X$.
For $m, n \in \mathbb{N}$, set $E_n(m) := \bigcup_{k=n}^{\infty} \{|f_k - f| \geq 1/m\}$. Then $E_n(m) \downarrow \emptyset$, so $\mu(E_n(m)) \to 0$ by Corollary 1.5.1.
Given $\delta > 0$, choose k_m with $\mu(E_{k_m}(m)) < \delta 2^{-m}$ and set $E_\delta := \bigcup_{m \geq 1} E_{k_m}(m)$.
Then $\mu(E_\delta) < \delta$, and $x \notin E_\delta$, $k \geq k_m$ imply $|f_k(x) - f(x)| < 1/m$; hence $f_n \rightrightarrows f$ on $X \setminus E_\delta$.
The in-measure conclusion follows from Theorem 5.2.18.
Structural Commentary: Finite measure turns decreasing bad sets with empty intersection into small bad sets.
Key Insight: A.e. convergence is almost uniform on finite measure spaces. □

Corollary 5.2.20. *Finite-Measure Almost-Everywhere Convergence Implies In-Measure Convergence*
If $\mu(X) < +\infty$ and $f_n \to f$ a.e., with f_n, f finite-valued off a null set, then $f_n \to f$ in measure.

Proof. By Theorem 5.2.19, $f_n \to f$ almost uniformly; Theorem 5.2.18 gives $f_n \to f$ in measure.
Structural Commentary: Finite measure upgrades a.e. convergence through almost uniform convergence.
Key Insight: On finite measure spaces, a.e. convergence implies in-measure convergence. □

Example 5.2.21. *Egoroff Needs a Finite-Valued Limit*
On $([0,1], \lambda)$, set $f_n := n\mathbf{1}_{[0,1]}$. Then $f_n(x) \to +\infty$ for all x, so the pointwise limit is not finite-valued. If $f_n \to f$ almost uniformly for finite-valued measurable f, then for some E with $\lambda(E) < 1/2$, $f_n \rightrightarrows f$ on $[0,1] \setminus E$; this nonempty set has $f_n(x) = n \to +\infty$, contradicting finite-valued pointwise convergence.

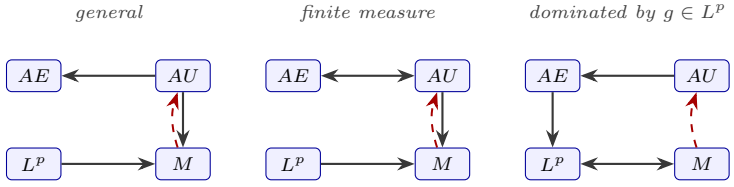


Figure 203: Modes of convergence. Solid arrows are implications; dashed arrows give a convergent subsequence.

Method 5.2.22. *Counterexample Dictionary*

(i) On $(\mathbb{R}, \lambda)$, $f_n = n^{-1/p}\mathbf{1}_{[0,n]}$ satisfies $f_n \rightrightarrows 0$ but $\|f_n\|_p = 1$.
(ii) On $([0,1], \lambda)$, $f_n = n\mathbf{1}_{[1/n, 2/n]}$ satisfies $f_n \to 0$ a.e., in measure, almost uniformly, but $\|f_n\|_p^p = n^{p-1} \nrightarrow 0$.

(iii) Typewriter intervals on $[0,1]$: if $f_n = \mathbf{1}_{I_n}$ and $\lambda(I_n) \to 0$ rowwise, then $\|f_n\|_p \to 0$ and $f_n \to 0$ in measure, but $(f_n(x))$ converges nowhere.
(iv) On $(\mathbb{R}, \lambda)$, $f_n = \mathbf{1}_{[n,n+1]}$ satisfies $f_n \to 0$ pointwise but $\lambda\{|f_n| \geq 1/2\} = 1$ for all n, so $f_n \not\to 0$ in measure.
(v) On $(\mathbb{R}, \lambda)$, $f_n = \mathbf{1}_{[n,n+1]}$ is not almost uniformly convergent to 0: if $\lambda(E) < 1$ and $f_n \rightrightarrows 0$ on $\mathbb{R} \setminus E$, then $[n, n+1] \subseteq E$ for all large n, impossible.

Proposition 5.2.23. *An Integral Metric for Convergence in Measure on Finite Measure Spaces* *Let $\mu(X) < +\infty$ and $r(h) := \int_X |h|\,(1+|h|)^{-1}\,d\mu$. Then $f_n \to f$ in measure iff $r(f_n - f) \to 0$.*

Proof.

(⇒) Set $h_n := f_n - f$. For $\eta > 0$ and $A_n := \{|h_n| \geq \eta\}$, $r(h_n) \leq \mu(A_n) + \eta\mu(X)$.
Let $n \to \infty$, then $\eta \downarrow 0$.
(⇐) For $\alpha > 0$, $\alpha(1+\alpha)^{-1}\mathbf{1}_{\{|h_n| \geq \alpha\}} \leq |h_n|\,(1+|h_n|)^{-1}$, hence $\mu\{|h_n| \geq \alpha\} \leq (1+\alpha)\alpha^{-1} r(h_n) \to 0$.

Structural Commentary: The bounded error transform metrizes convergence in measure on finite measure spaces.
Key Insight: Small integral of the bounded error transform is equivalent to small large-error sets. □

Corollary 5.2.24. *Fatou and Dominated Convergence with In-Measure Hypotheses*

(i) $f_n \geq 0$ and $f_n \to f$ in measure $\Rightarrow \int_X f\,d\mu \leq \liminf_n \int_X f_n\,d\mu$.
(ii) $f_n \to f$ in measure and $|f_n| \leq g \in \mathscr{L}^1(\mu) \Rightarrow \int_X f_n\,d\mu \to \int_X f\,d\mu$.

Proof.

(i) Choose f_{n_k} with $\int f_{n_k}\,d\mu \to \liminf_n \int f_n\,d\mu$. By Theorem 5.2.11 and uniqueness, some $f_{n_{k_j}} \to f$ a.e.; Fatou gives $\int f\,d\mu \leq \liminf_j \int f_{n_{k_j}}\,d\mu = \liminf_n \int f_n\,d\mu$.
(ii) Theorem 5.2.15 with $p = 1$ gives $\|f_n - f\|_1 \to 0$, hence $\left|\int_X (f_n - f)\,d\mu\right| \leq \|f_n - f\|_1 \to 0$.

Structural Commentary: In-measure hypotheses reduce to a.e. subsequences or L^1 convergence.
Key Insight: Fatou and DCT survive convergence in measure through subsequences and domination. □

Theorem 5.2.25. *Vitali Convergence Criterion for L^p*
Let $1 \leq p < \infty$, $f_n \in L^p(\mu)$, and let f be measurable. The following are equivalent:

(i) $f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$.
(ii) The following hold:
  (a) $f_n \to f$ in measure.
  (b) $\forall \varepsilon > 0\ \exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F |f_n|^p\,d\mu < \varepsilon^p$.
  (c) $\forall \varepsilon > 0\ \exists \delta(\varepsilon) > 0$ such that $\mu(E) < \delta(\varepsilon) \Rightarrow \sup_n \int_E |f_n|^p\,d\mu < \varepsilon^p$.

Proof.

(i) Assume (i). Proposition 5.2.10 gives (ii)(a). Also (f_n) is L^p-Cauchy.
By Proposition 5.1.32, for each $\varepsilon > 0$ there are $E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ and $\delta(\varepsilon) > 0$ such that $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \|f_n \mathbf{1}_F\|_p < \varepsilon$ and $\mu(E) < \delta(\varepsilon) \Rightarrow \sup_n \|f_n \mathbf{1}_E\|_p < \varepsilon$; raising to p gives (ii)(b),(ii)(c).
(ii) Assume (ii). Fix $\eta > 0$, choose E_η from (ii)(b) and $\delta(\eta)$ from (ii)(c), and put $F = X \setminus E_\eta$.
Then $\|(f_n - f_m)\mathbf{1}_F\|_p \leq \|f_n \mathbf{1}_F\|_p + \|f_m \mathbf{1}_F\|_p < 2\eta$.
If $\mu(E_\eta) > 0$, set $\alpha := \eta\mu(E_\eta)^{-1/p}$ and $H_{nm} := E_\eta \cap \{|f_n - f_m| \geq \alpha\}$.
By (ii)(a), (f_n) is Cauchy in measure, so $\mu(H_{nm}) < \delta(\eta)$ for large m, n. On $E_\eta \setminus H_{nm}$, $|f_n - f_m| < \alpha$; on H_{nm}, use $|f_n - f_m| \leq |f_n| + |f_m|$. Hence $\left\|(f_n - f_m)\mathbf{1}_{E_\eta}\right\|_p \leq \alpha\mu(E_\eta)^{1/p} + \|f_n \mathbf{1}_{H_{nm}}\|_p + \|f_m \mathbf{1}_{H_{nm}}\|_p < 3\eta$ by (ii)(c). If $\mu(E_\eta) = 0$, the E_η term is 0. Thus $\|f_n - f_m\|_p < 5\eta$ eventually, so (f_n) is L^p-Cauchy.
Riesz–Fischer gives $f_n \to h$ in L^p; Proposition 5.2.10 and Corollary 5.2.13 give $h = f$ a.e., so (i) holds.

Structural Commentary: Vitali = in-measure convergence + tightness + uniform small-set control.
Key Insight: Tightness and small-set control convert in-measure convergence into an L^p Cauchy estimate. □

Corollary 5.2.26. *Domination Gives Tail and Small-Set Control*
Let $1 \leq p < \infty$, $g \in L^p(\mu)$, and $|f_n| \leq g$ a.e. for all n. Then, for every $\varepsilon > 0$:

(i) $\exists E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < +\infty$ such that $F \cap E_\varepsilon = \emptyset \Rightarrow \sup_n \int_F |f_n|^p\,d\mu < \varepsilon^p$.
(ii) $\exists \delta(\varepsilon) > 0$ such that $\mu(E) < \delta(\varepsilon) \Rightarrow \sup_n \int_E |f_n|^p\,d\mu < \varepsilon^p$.

Proof.

(i) Proposition 5.1.20 gives E_ε with $\mu(E_\varepsilon) < +\infty$ and $F \cap E_\varepsilon = \emptyset \Rightarrow \int_F |g|^p\,d\mu < \varepsilon^p$.
Since $|f_n|^p \leq |g|^p$ a.e., $\sup_n \int_F |f_n|^p\,d\mu < \varepsilon^p$.
(ii) Lemma 5.1.30 gives $\delta(\varepsilon) > 0$ with $\mu(E) < \delta(\varepsilon) \Rightarrow \int_E |g|^p\,d\mu < \varepsilon^p$.
Since $|f_n|^p \leq |g|^p$ a.e., $\sup_n \int_E |f_n|^p\,d\mu < \varepsilon^p$.

Structural Commentary: One L^p envelope supplies both tail tightness and small-set control.
Key Insight: Domination transfers both estimates from g to every f_n. □

5.3 Composition and Convergence Modes

Proposition 5.3.1. *Composition and Pointwise Modes*
Let $\varphi : \mathbb{R} \to \mathbb{R}$.

(i) φ continuous, $f_n \to f$ a.e. $\Rightarrow \varphi \circ f_n \to \varphi \circ f$ a.e.
(ii) φ uniformly continuous preserves uniform, a.u., and in-measure convergence.
(iii) If φ is discontinuous, composition need not preserve a.e. convergence.
(iv) Non-uniform continuity can destroy uniform, even in-measure, convergence.

Proof.

(i) Outside a null set, $f_n(x) \to f(x)$; continuity gives $\varphi(f_n(x)) \to \varphi(f(x))$.
(ii) Given $\varepsilon > 0$, choose $\delta > 0$ with $|s-t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$. Uniform and almost-uniform convergence use $\sup |f_n - f| < \delta$ on X or $X \setminus E_\eta$; in measure, $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \delta\}$.
(iii) If φ is discontinuous at a, choose $t_n \to a$ and $\alpha > 0$ with $|\varphi(t_n) - \varphi(a)| \geq \alpha$.
On $X = \{*\}$, $f_n(*) = t_n \to a = f(*)$, but $\varphi(f_n(*)) \nrightarrow \varphi(f(*))$.
(iv) If φ is not uniformly continuous, choose a_n, b_n with $|a_n - b_n| \to 0$ and $|\varphi(a_n) - \varphi(b_n)| \geq \alpha > 0$.
On $(\mathbb{N}, \#)$, set $f(k) = b_k$ and $f_n(k) = a_n$ for $k = n$, $f_n(k) = b_k$ for $k \neq n$.
Then $\|f_n - f\|_\infty \to 0$, but $\#\{|\varphi(f_n) - \varphi(f)| \geq \alpha\} = 1$ for all n.

Structural Commentary: Continuity is pointwise; uniform continuity gives one global δ.
Key Insight: Composition needs the same uniformity as the convergence mode. □

Proposition 5.3.2. *Composition and Convergence in Measure*
Let $f_n, f : X \to \mathbb{R}$ be measurable with $f_n \to f$ in measure, and let $\varphi : \mathbb{R} \to \mathbb{R}$.

(i) If φ is uniformly continuous, then $\mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \to 0$ for every $\varepsilon > 0$.
(ii) If $\mu(X) < +\infty$ and φ is continuous, then $\mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \to 0$ for every $\varepsilon > 0$.

Proof.

(i) Fix $\varepsilon > 0$ and choose $\delta > 0$ with $|s-t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$.
Then $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \delta\}$, so the measures tend to 0.
(ii) Fix $\varepsilon, \eta > 0$. Since $\mu(X) < +\infty$ and f is finite-valued, choose M with $\mu\{|f| > M\} < \eta$. Uniform continuity of φ on $[-M-1, M+1]$ gives $0 < \delta < 1$ such that $s, t \in [-M-1, M+1]$ and $|s-t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \varepsilon$.
Hence $\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \subseteq \{|f| > M\} \cup \{|f_n - f| \geq \delta\}$, so $\limsup_n \mu\{|\varphi(f_n) - \varphi(f)| \geq \varepsilon\} \leq \eta \downarrow 0$.

Structural Commentary: Use global uniformity, or truncate to a bounded range.
Key Insight: On finite measure spaces, measurable finite-valued functions are almost bounded. □

Example 5.3.3. *Continuity Alone Is Not Enough on Infinite Measure*
On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, set $f(x) = x$, $f_n(x) = x + 1/n$, and $\varphi(t) = t^2$.
Then $\|f_n - f\|_\infty = 1/n \to 0$, hence $f_n \rightrightarrows f$ and $f_n \to f$ in measure.
However $\varphi(f_n) - \varphi(f) = 2x/n + 1/n^2$; for $E_n := \{|\varphi(f_n) - \varphi(f)| \geq 1\}$, $[n/2, +\infty) \subseteq E_n$, so $\lambda(E_n) = +\infty$.
Thus $\varphi \circ f_n \nrightarrow \varphi \circ f$ in measure.

Proposition 5.3.4. *Growth Conditions for L^p Composition*
Let $1 \leq p < \infty$, let $\varphi : \mathbb{R} \to \mathbb{R}$ be measurable, and let $f_n, f : X \to \mathbb{R}$ be measurable.

(i) If $|\varphi(t)| \leq A|t| + B$ for all t and some $A, B \geq 0$, $\mu(X) < +\infty$, and $f \in L^p(\mu)$, then $\varphi \circ f \in L^p(\mu)$.
(ii) If $|\varphi(t)| \leq A|t|$ for all t and some $A \geq 0$, and $f \in L^p(\mu)$, then $\varphi \circ f \in L^p(\mu)$.
(iii) If $\mu(X) < +\infty$, φ is uniformly continuous, $f_n, f \in L^p(\mu)$, and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.
(iv) If φ is Lipschitz, $\varphi(0) = 0$, $f_n, f \in L^p(\mu)$, and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.

Proof.

(i) $|\varphi(f)|^p \leq 2^{p-1}(A^p |f|^p + B^p)$, so $\|\varphi \circ f\|_p^p \leq 2^{p-1}(A^p \|f\|_p^p + B^p \mu(X)) < +\infty$.
(ii) $|\varphi(f)|^p \leq A^p |f|^p$, so $\|\varphi \circ f\|_p \leq A \|f\|_p < +\infty$.
(iii) Uniform continuity gives constants $C, D \geq 0$ with $|\varphi(t)| \leq C|t| + D$ and $|\varphi(s) - \varphi(t)| \leq C|s-t| + D$.
Hence all compositions lie in L^p by (i). Let $h_n := |f_n - f|$.
Given $\eta > 0$, choose $\delta > 0$ with $|s-t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \eta$ and set $H_n := \{h_n \geq \delta\}$.
Then $\int_X |\varphi(f_n) - \varphi(f)|^p \, d\mu \leq \eta^p \mu(X) + 2^{p-1} \int_{H_n} (C^p h_n^p + D^p) \, d\mu$.
Since $\int h_n^p \, d\mu \to 0$ and $\mu(H_n) \leq \delta^{-p} \int h_n^p \, d\mu \to 0$, take $\limsup_n$ and then $\eta \downarrow 0$.
(iv) If L is Lipschitz, $|\varphi(f)| \leq L|f|$ and $\|\varphi \circ f_n - \varphi \circ f\|_p \leq L \|f_n - f\|_p \to 0$.

Structural Commentary: Growth gives membership; continuity estimates transfer L^p convergence.
Key Insight: Finite measure makes small uniform errors integrable; Lipschitz bounds give norm control. □

Example 5.3.5. *Uniform Continuity Does Not Suffice on Infinite Measure*
On $(\mathbb{R}, \lambda)$, fix $1 \le p < \infty$, set $f = 0$, $f_n := n^{-1}\mathbf{1}_{[0,n^{p/2}]}$, and $\varphi(t) := \sqrt{|t|}$.
Then φ is uniformly continuous, $\varphi(0) = 0$, $\|f_n - f\|_p^p = n^{-p/2} \to 0$, but $\|\varphi \circ f_n - \varphi \circ f\|_p^p = 1$ for all n.
Thus uniform continuity need not preserve L^p convergence on infinite measure spaces.

Lemma 5.3.6. *L^p Convergence Allows Continuous Composition in Measure*
Let $1 \le p < \infty$, $f_n, f \in L^p(\mu)$, $\|f_n - f\|_p \to 0$, and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous.
Then $\varphi \circ f_n \to \varphi \circ f$ in measure.

Proof. Fix $\alpha, \eta > 0$. Choose M with $\mu\{|f| > M\} < \eta$. By uniform continuity of φ on $[-M-1, M+1]$, choose $0 < \delta < 1$ such that $s, t \in [-M-1, M+1]$ and $|s - t| < \delta \Rightarrow |\varphi(s) - \varphi(t)| < \alpha$.
Then $\{|\varphi(f_n) - \varphi(f)| \ge \alpha\} \subseteq \{|f| > M\} \cup \{|f_n - f| \ge \delta\}$, so $\limsup_n \mu\{|\varphi(f_n) - \varphi(f)| \ge \alpha\} \le \eta$. Let $\eta \downarrow 0$.
Structural Commentary: L^p convergence localizes to a large bounded range where continuity is uniform.
Key Insight: Norm convergence gives in-measure control; bounded-range continuity transfers it. □

Theorem 5.3.7. *Composition Growth Criteria in L^p*
Let $1 \le p < \infty$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous.

(i) If $\mu(X) < +\infty$ and $\exists K > 0$ such that $|\varphi(t)| \le K|t|$ for $|t| \ge K$, then $f \in L^p(\mu) \Rightarrow \varphi \circ f \in L^p(\mu)$.
(ii) Under (i), if $f_n, f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.
(iii) If $\exists K \ge 0$ such that $|\varphi(t)| \le K|t|$ for all t, then $f \in L^p(\mu) \Rightarrow \varphi \circ f \in L^p(\mu)$ on any measure space.
(iv) Under (iii), if $f_n, f \in L^p(\mu)$ and $\|f_n - f\|_p \to 0$, then $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$ on any measure space.

Proof.

(i) Let $M := \sup_{|t| \le K} |\varphi(t)|$ and $C := K + M$.
Then $|\varphi(t)| \le C(1 + |t|)$ for all t. hence $|\varphi(f)|^p \le 2^{p-1}C^p(1 + |f|^p) \in L^1(\mu)$.
(ii) Let C be as in (i). Lemma 5.3.6 gives $\varphi \circ f_n \to \varphi \circ f$ in measure.
Since (f_n) is L^p-Cauchy, Proposition 5.1.32 and $|\varphi(f_n)|^p \le 2^{p-1}C^p(1 + |f_n|^p)$ give the finite-tail estimate with $E_\varepsilon = X$ and the small-set estimate. Theorem 5.2.25 gives $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.
(iii) $|\varphi(f)|^p \le K^p |f|^p \in L^1(\mu)$.
(iv) Lemma 5.3.6 gives $\varphi \circ f_n \to \varphi \circ f$ in measure. Since (f_n) is L^p-Cauchy and $|\varphi(f_n)|^p \le K^p |f_n|^p$, Proposition 5.1.32 gives the finite-tail and small-set estimates for $(\varphi \circ f_n)$. Theorem 5.2.25 gives $\|\varphi \circ f_n - \varphi \circ f\|_p \to 0$.

Structural Commentary: Local growth needs finite measure; global growth does not.
Key Insight: Vitali converts in-measure composition plus inherited mass control into L^p convergence. □

Method 5.3.8. *Atomic Counterexamples for Composition Growth*

(i) If finite-measure tail growth fails, choose $|t_n| \ge n$ with $|\varphi(t_n)| > n^2 |t_n|$. On $\mathbb{N}$, set $\mu(\{n\}) = (n^2 |t_n|^p)^{-1}$ and $f(n) = t_n$. Then $\mu(\mathbb{N}) < +\infty$, $\int |f|^p \, d\mu = \sum_n n^{-2} < +\infty$, but $\int |\varphi(f)|^p \, d\mu \ge \sum_n n^{2p-2} = +\infty$.
(ii) If the same tail growth fails, choose $|t_n| \ge n$ with $|\varphi(t_n) - \varphi(0)| \ge n|t_n|$. With $\mu(\{n\}) = (n^p |t_n|^p)^{-1}$, $f = 0$, and $f_n = t_n \mathbf{1}_{\{n\}}$, one has $\mu(\mathbb{N}) < +\infty$, $\|f_n - f\|_p = 1/n \to 0$, but $\|\varphi \circ f_n - \varphi \circ f\|_p \ge 1$.
(iii) If global linear growth fails and $\varphi(0) = 0$, the constructions in (i)–(ii), without requiring $\mu(\mathbb{N}) < +\infty$, give membership and convergence counterexamples on arbitrary measure spaces.
(iv) If $\varphi(0) \ne 0$, then on $(\mathbb{N}, \#)$, $0 \in L^p$ but $\varphi \circ 0$ is a nonzero constant and hence not in L^p; the constant sequence $f_n = 0$ gives the convergence failure.

6 Banach and Hilbert Spaces

6.1 Normed and Banach Spaces

Normed spaces make vector spaces metric; Banach spaces add completeness. This is the setting for infinite series, fixed points, bounded linear maps, and operator norms.

Definition 6.1.1. *Normed and Banach Spaces*
Let $\mathcal{X}$ be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. A *norm* is a function $\|\cdot\| : \mathcal{X} \to [0, \infty)$ such that, for all $x, y \in \mathcal{X}$ and $\lambda \in \mathbb{F}$, $\|x\| = 0 \iff x = 0$; $\|\lambda x\| = |\lambda| \, \|x\|$; and $\|x + y\| \le \|x\| + \|y\|$.
The metric $d(x, y) := \|x - y\|$ makes $\mathcal{X}$ a *normed space*. If this metric is complete, $\mathcal{X}$ is a *Banach space*.

Definition 6.1.2. *Bounded Linear Maps and Operator Norm*
For normed spaces $\mathcal{X}, \mathcal{Y}$, a linear map $T : \mathcal{X} \to \mathcal{Y}$ is *bounded* if there exists $C \geq 0$ such that $\|Tx\|_{\mathcal{Y}} \leq C \|x\|_{\mathcal{X}}$ for all $x \in \mathcal{X}$. The *operator norm* is $\|T\|_{\text{op}} := \sup_{\|x\|_{\mathcal{X}} \leq 1} \|Tx\|_{\mathcal{Y}} = \sup_{x \neq 0} \frac{\|Tx\|_{\mathcal{Y}}}{\|x\|_{\mathcal{X}}}$.

Theorem 6.1.3. *Bounded Equals Continuous for Linear Maps*
For a linear map $T : \mathcal{X} \to \mathcal{Y}$ between normed spaces, the following are equivalent:

(i) T is bounded.
(ii) T is continuous at 0.
(iii) T is continuous everywhere.

Proof.

(i) $\Rightarrow$ (iii). If T is bounded, then $\|T(x-y)\| \leq \|T\|_{\text{op}} \|x-y\|$, so T is continuous everywhere.
(iii) $\Rightarrow$ (ii). Since $0 \in \mathcal{X}$, continuity everywhere gives continuity at 0.
(ii) $\Rightarrow$ (i). Choose $\delta > 0$ such that $\|x\| < \delta$ implies $\|Tx\| < 1$.
For $x \neq 0$, apply this to $\delta x/(2\|x\|)$ to get $\|Tx\| \leq 2\delta^{-1}\|x\|$; the case $x = 0$ is trivial.

Structural Commentary: Linearity turns local continuity at one point into uniform norm control.
Key Insight: In normed linear spaces, continuity and boundedness are the same operator property. □

Example 6.1.4. *Standard Banach Space Examples*

(i) Every complete supremum space from Theorem 8.1.1 is Banach under $\|\cdot\|_\infty$.
(ii) The sequence spaces ℓ^∞ and ℓ^p, $1 \leq p < \infty$, are Banach by Example 4.1.17.
(iii) The Lebesgue spaces $L^p(\mu)$, $1 \leq p < \infty$, and $L^\infty(\mu)$ are Banach by Theorems 5.1.17 and 5.1.19.
(iv) In particular, L^2 is a Hilbert space by Theorem 5.1.8.

6.2 Hilbert Spaces

Hilbert spaces are complete inner-product spaces. They extend the orthogonal geometry of Definition 1.1.3 to function and sequence spaces, supporting projection, Fourier expansion, and Riesz representation.

Definition 6.2.1. *Hilbert Space*
An inner-product space $\mathcal{H}$ is a *Hilbert space* if the norm $\|x\| := \sqrt{\langle x, x\rangle}$ makes $\mathcal{H}$ complete.
A subset $M \subseteq \mathcal{H}$ is *orthonormal* if $\langle e_i, e_j\rangle = \delta_{ij}$ for all indexed vectors in M.

Proposition 6.2.2. *Continuity of the Inner Product*
Let $\mathcal{V}$ be an inner-product space. For sequences $(x_n), (y_n) \subseteq \mathcal{V}$ and $x, y \in \mathcal{V}$,
$\|x_n - x\| \to 0$, $\|y_n - y\| \to 0 \Longrightarrow \langle x_n, y_n\rangle \to \langle x, y\rangle$, $x_n \to x \Longrightarrow \|x_n\| \to \|x\|$.

Proof.

(i) $|\|x_n\| - \|x\|| \leq \|x_n - x\| \to 0$, hence $\|x_n\| \to \|x\|$.
(ii) Since $y_n \to y$, choose $M < \infty$ with $\|y_n\| \leq M$; Cauchy–Schwarz gives $|\langle x_n, y_n\rangle - \langle x, y\rangle| \leq |\langle x_n - x, y_n\rangle| + |\langle x, y_n - y\rangle| \leq M\|x_n - x\| + \|x\|\|y_n - y\| \to 0$.

Structural Commentary: Cauchy–Schwarz turns norm convergence in either coordinate into scalar convergence.
Key Insight: Inner products are continuous because their errors are controlled by products of norms. □

Theorem 6.2.3. *Hilbert Completion*
For every inner-product space $\mathcal{V}$, $\exists$ Hilbert space $\widehat{\mathcal{V}}$, linear isometry $J : \mathcal{V} \to \widehat{\mathcal{V}}$ such that $\overline{J(\mathcal{V})} = \widehat{\mathcal{V}}$.
If $(\mathcal{K}, I)$ is another such pair, then $\exists!$ surjective linear isometry $U : \widehat{\mathcal{V}} \to \mathcal{K}$ with $U \circ J = I$.

Proof. Let $C(\mathcal{V})$ be Cauchy sequences, $(x_n) \sim (y_n) \iff \|x_n - y_n\| \to 0$, $\widehat{\mathcal{V}} := C(\mathcal{V})/\sim$.
Set $[x_n] + [y_n] := [x_n + y_n]$, $a[x_n] := [ax_n]$, $Jx := [(x)]$, and $\langle [x_n], [y_n]\rangle_{\widehat{\mathcal{V}}} := \lim_{n\to\infty} \langle x_n, y_n\rangle_{\mathcal{V}}$.
As Cauchy sequences bounded, $|\langle x_n, y_n\rangle - \langle x_m, y_m\rangle| \leq M\|x_n - x_m\| + N\|y_n - y_m\|$;
and $|\langle x_n, y_n\rangle - \langle x'_n, y'_n\rangle| \leq M\|x_n - x'_n\| + N\|y_n - y'_n\|$ for suitable $M, N < \infty$, hence limit exists, and is representative-independent. Sesquilinearity and positivity pass to limits, and $\|[x_n] - [y_n]\| = \lim_{n\to\infty} \|x_n - y_n\|$.
Thus induced metric is the completion metric, so $\widehat{\mathcal{V}}$ is Hilbert, J is a linear isometry, and $\overline{J(\mathcal{V})} = \widehat{\mathcal{V}}$.
For another pair $(\mathcal{K}, I)$, metric-completion uniqueness gives a unique surjective isometry $U : \widehat{\mathcal{V}} \to \mathcal{K}$ with $U \circ J = I$; density gives $U(ax + by) = aU(x) + bU(y)$, and Theorem 1.1.23 gives $\langle Ux, Uy\rangle_{\mathcal{K}} = \langle x, y\rangle_{\widehat{\mathcal{V}}}$.
Structural Commentary: Complete the metric first, then recover the inner product by taking limits.
Key Insight: Cauchy–Schwarz makes the lifted inner product well defined. □

Theorem 6.2.4. *Complete, Closed, and Finite-Dimensional Subspaces*
Let $\mathcal{S}$ be a subspace of an inner-product space $\mathcal{V}$.

(i) If $\mathcal{S}$ is complete in the inherited metric, then $\mathcal{S}$ is closed in $\mathcal{V}$.
(ii) If $\mathcal{V}$ is Hilbert, then $\mathcal{S}$ is Hilbert if and only if $\mathcal{S}$ is closed.
(iii) If $\mathcal{S}$ is finite-dimensional, then $\mathcal{S}$ is complete and closed in $\mathcal{V}$.

Proof.

(i) Let $(s_n) \subseteq \mathcal{S}$ and $s_n \to v \in \mathcal{V}$.
Then (s_n) is Cauchy in $\mathcal{S}$, so $\exists s \in \mathcal{S} : s_n \to s$, $v = s \in \mathcal{S}$ by uniqueness, and $\mathcal{S}$ is closed.
(ii) Assume $\mathcal{V}$ is Hilbert; if $\mathcal{S}$ is closed and $(s_n) \subseteq \mathcal{S}$ is Cauchy.
Then $\exists v \in \mathcal{V} : s_n \to v$, closedness gives $v \in \mathcal{S}$, and the converse follows from (i).
(iii) Let $\dim \mathcal{S} = m$ and choose an orthonormal basis $e_1, \dots, e_m$.
If $(s_n) \subseteq \mathcal{S}$ is Cauchy and $s_n = \sum_{j=1}^{m} a_{n,j} e_j$, then $|a_{n,j} - a_{k,j}| = |\langle s_n - s_k, e_j \rangle| \leq \|s_n - s_k\|$.
So $a_{n,j} \to a_j \in \mathbb{F}$ and $s_n \to \sum_{j=1}^{m} a_j e_j \in \mathcal{S}$, hence $\mathcal{S}$ is complete and closed by (i).

Structural Commentary: Closedness is the ambient form of internal completeness.
Key Insight: Finite orthonormal coordinates reduce completeness to completeness of $\mathbb{F}$. □

Definition 6.2.5. *Series in an Inner-Product Space*
Let $(x_k) \subseteq \mathcal{V}$.

(i) The *partial sums* are $s_n := \sum_{k=1}^{n} x_k$.
(ii) The *series* $\sum_{k=1}^{\infty} x_k$ *converges* to $s \in \mathcal{V}$ iff $s_n \to s$.
(iii) The series is *absolutely convergent* iff $\sum_{k=1}^{\infty} \|x_k\| < \infty$.

Theorem 6.2.6. *Completeness via Absolutely Convergent Series*
$\mathcal{V}$ is complete $\iff$ every series $\sum_{k=1}^{\infty} x_k$ in $\mathcal{V}$ with $\sum_{k=1}^{\infty} \|x_k\| < \infty$ converges in $\mathcal{V}$.

Proof.

($\Rightarrow$) Let $s_n = \sum_{k=1}^{n} x_k$ and $\sum_{k=1}^{\infty} \|x_k\| < \infty$.
For $n > m$, $\|s_n - s_m\| \leq \sum_{k=m+1}^{n} \|x_k\| \to 0$, so (s_n) is Cauchy and hence convergent.
($\Leftarrow$) Let (x_n) be Cauchy and choose $N_1 < N_2 < \cdots$ with $i, j \geq N_k \Longrightarrow \|x_i - x_j\| < 2^{-k}$.
Then $\sum_{k=1}^{\infty} \|x_{N_{k+1}} - x_{N_k}\| \leq \sum_{k=1}^{\infty} 2^{-k} < \infty$, so $\sum_{k=1}^{\infty} (x_{N_{k+1}} - x_{N_k})$ converges, whence (x_{N_k}) converges, and Lemma 4.1.11 gives $x_n \to x$.

Structural Commentary: Absolute convergence controls tails; subsequences recover Cauchy limits.
Key Insight: Completeness is exactly the ability to sum every norm-controlled vector tail. □

Example 6.2.7. *ℓ^2 and the Parallelogram Test*
Let $\ell^2 := \{x = (x_n) : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}$. With $\langle x, y \rangle := \sum_{n=1}^{\infty} x_n \overline{y_n}$ and $\|x\|_2 = \langle x, x \rangle^{1/2}$, Cauchy coordinate limits and finite-set estimates give completeness, so ℓ^2 is Hilbert. For $p \neq 2$, $\|\cdot\|_p$ is not inner-product-induced: with $x = (1, 1, 0, \dots)$ and $y = (1, -1, 0, \dots)$, $\|x + y\|_p^2 + \|x - y\|_p^2 = 8$ while $2\|x\|_p^2 + 2\|y\|_p^2 = 4 \cdot 2^{2/p}$, so the parallelogram identity holds only for $p = 2$.
Structural Commentary: Completeness alone does not make a Banach space Hilbert.
Key Insight: The parallelogram identity detects whether a norm really came from an inner product.

Example 6.2.8. *The Supremum Norm Is Not Hilbertian*
For $a < b$, $C([a, b], \mathbb{F})$ with $\|h\|_\infty := \sup_{t \in [a,b]} |h(t)|$ is Banach but not Hilbertian: take $f(t) = 1$ and $g(t) = (t - a)/(b - a)$, so $\|f\|_\infty = \|g\|_\infty = 1$, $\|f + g\|_\infty = 2$, $\|f - g\|_\infty = 1$, and $\|f + g\|_\infty^2 + \|f - g\|_\infty^2 = 5 \neq 4 = 2\|f\|_\infty^2 + 2\|g\|_\infty^2$, violating the parallelogram identity.
Structural Commentary: Banach need not mean Hilbert.
Key Insight: One failed parallelogram identity rules out every possible inner product inducing the norm.

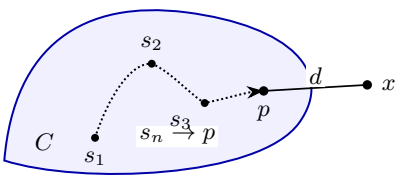


Figure 204: A minimizing sequence in C converges to the nearest point p to x.

Theorem 6.2.9. *Hilbert Projection Theorem*
Let $C \subseteq \mathcal{H}$ be nonempty, closed, and convex. For every $x \in \mathcal{H}$ there is a unique $p \in C$ minimizing $\|x - p\|$. If $M \subseteq \mathcal{H}$ is a closed subspace, then $x = p + (x - p)$ with $p \in M$ and $x - p \in M^\perp$.

Proof. Let $d = \inf_{q \in C} \|x - q\|$ and choose $q_n \in C$ with $\|x - q_n\| \to d$.
Since C is convex, $(q_n + q_m)/2 \in C$. The parallelogram identity gives
$\|q_n - q_m\|^2 = 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4\left\|x - \frac{q_n+q_m}{2}\right\|^2 \le 2\|x - q_n\|^2 + 2\|x - q_m\|^2 - 4d^2$, so (q_n) is Cauchy.
Completeness and closedness give $q_n \to p \in C$, and continuity of the norm gives $\|x - p\| = d$. If p_1, p_2 both minimize, the same identity with $q_n = p_1$, $q_m = p_2$ gives $\|p_1 - p_2\| = 0$. Now let $C = M$ and $r = x - p$.
For $y \in M$ and $t \in \mathbb{R}$, minimality of $p + ty \in M$ gives $0 \le \|r - ty\|^2 - \|r\|^2 = -2t \operatorname{Re}\langle r, y\rangle + t^2\|y\|^2$, hence $\operatorname{Re}\langle r, y\rangle = 0$. Replacing y by iy in the complex case gives $\langle r, y\rangle = 0$. Thus $r \in M^\perp$ and $x = p + r$.
Structural Commentary: Convexity makes minimizing sequences Cauchy; closedness keeps the limit.
Key Insight: The nearest point is characterized by an orthogonal error. □

Definition 6.2.10. *Closed Linear Span and Total Sets*
For $A \subseteq \mathcal{H}$, define the *closed linear span* $\operatorname{cspan}(A) := \overline{\operatorname{span}(A)}$. The set A is *total* in $\mathcal{H}$ if $\operatorname{cspan}(A) = \mathcal{H}$.

Theorem 6.2.11. *Double Orthogonal Complements in Hilbert Spaces*
Let $\mathcal{H}$ be a Hilbert space.

(i) $\forall A \subseteq \mathcal{H}$, $\operatorname{cspan}(A) = A^{\perp\perp}$.
(ii) $\forall \mathcal{S} \le \mathcal{H}$, $\overline{\mathcal{S}} = \mathcal{S}^{\perp\perp}$.
(iii) $\forall \mathcal{K} \le \mathcal{H}$, $\mathcal{K}$ *closed* $\iff \mathcal{K} = \mathcal{K}^{\perp\perp}$.
(iv) $\forall A \subseteq \mathcal{H}$, $\operatorname{span}(A)$ *dense in* $\mathcal{H} \iff A^\perp = \{\mathbf{0}\}$.

Proof.

(i) Let $B = \operatorname{cspan}(A)$. Then B is closed, so Theorem 6.2.9 gives $\mathcal{H} = B \odot B^\perp$ and $B^{\perp\perp} = B$.
Also $B^\perp = A^\perp$, since $A \subseteq B$ and $x \in A^\perp \Rightarrow x \perp \operatorname{span}(A) \Rightarrow x \perp B$ by Proposition 6.2.2.
Hence $A^{\perp\perp} = B^{\perp\perp} = B = \operatorname{cspan}(A)$.
(ii) Apply (i) with $A = \mathcal{S}$: $\mathcal{S}^{\perp\perp} = \operatorname{cspan}(\mathcal{S}) = \overline{\operatorname{span}(\mathcal{S})} = \overline{\mathcal{S}}$.
(iii) By (ii), $\mathcal{K}^{\perp\perp} = \overline{\mathcal{K}}$; hence $\mathcal{K}$ closed $\iff \overline{\mathcal{K}} = \mathcal{K} \iff \mathcal{K}^{\perp\perp} = \mathcal{K}$.
(iv) $\operatorname{span}(A)$ dense $\iff \operatorname{cspan}(A) = \mathcal{H} \iff A^{\perp\perp} = \mathcal{H}$. If $A^\perp = \{\mathbf{0}\}$, then $A^{\perp\perp} = \mathcal{H}$.
Conversely $A^{\perp\perp} = \mathcal{H} \Rightarrow A^\perp \subseteq A^\perp \cap A^{\perp\perp} = \{\mathbf{0}\}$.

Structural Commentary: In a Hilbert space, closure is encoded by taking the orthogonal complement twice.
Key Insight: A missing direction from a closed span is detected by a nonzero orthogonal vector. □

Example 6.2.12. *Completeness Is Needed for Double Complements*
Let $\mathcal{V} := c_{00}(\mathbb{N}; \mathbb{R}) \subseteq \ell^2$ with inherited inner product, $Lx := \sum_{n=1}^\infty x_n/n$, and $\mathcal{K} := \ker L$.
Then $|Lx| \le \|x\|_2 \left(\sum_{n=1}^\infty n^{-2}\right)^{1/2}$, so L is continuous and $\mathcal{K}$ is a closed subspace of $\mathcal{V}$.
If $y \in \mathcal{K}^\perp$, then $e_1 - ne_n \in \mathcal{K}$ for $n \ge 2$, hence $0 = \langle y, e_1 - ne_n\rangle = y_1 - ny_n$.
Since $y \in c_{00}$, choose $n \ge 2$ with $y_n = 0$, so $y_1 = 0$ and then $y_n = 0$ for every n.
Thus $\mathcal{K}^\perp = \{\mathbf{0}\}$ and $\mathcal{K}^{\perp\perp} = \mathcal{V}$. But $e_1 \notin \mathcal{K}$, hence $\mathcal{K} \ne \mathcal{V} = \mathcal{K}^{\perp\perp}$.
Structural Commentary: Closedness need not recover double complements without completeness.
Key Insight: The identity $\mathcal{K} = \mathcal{K}^{\perp\perp}$ is genuinely Hilbert-space geometry.

Theorem 6.2.13. *Hilbert Basis Characterizations*
Let $\mathcal{O} \subseteq \mathcal{H}$ be orthonormal. The following are equivalent:

(i) $\mathcal{O}$ *is maximal orthonormal.*
(ii) $\mathcal{O}^\perp = \{\mathbf{0}\}$.
(iii) $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$.

Proof.

(i) $\Rightarrow$ *(ii)* If $0 \ne v \in \mathcal{O}^\perp$, then $\mathcal{O} \cup \{v/\|v\|\}$ is orthonormal, contradicting maximality; hence $\mathcal{O}^\perp = \{\mathbf{0}\}$.
(ii) $\Rightarrow$ *(iii)* By Theorem 6.2.11, $\operatorname{cspan}(\mathcal{O}) = \mathcal{O}^{\perp\perp} = (\mathcal{O}^\perp)^\perp = \{\mathbf{0}\}^\perp = \mathcal{H}$.
(iii) $\Rightarrow$ *(i)* If $\mathcal{O}$ is not maximal, $\exists$ unit $u \in \mathcal{O}^\perp$; but (iii) and Theorem 6.2.11 give $\mathcal{O}^\perp = \{\mathbf{0}\}$, a contradiction.

Structural Commentary: Maximality means no orthogonal direction remains.
Key Insight: Hilbert bases span by closure, not usually by finite linear combinations. □

Definition 6.2.14. *Arbitrary Sums in a Hilbert Space*
Let $(x_k)_{k \in K} \subseteq \mathcal{H}$.

(i) *Finite partial sums*: $\mathcal{F}(K) := \{J \subseteq K : |J| < \infty\}$, ordered by $\subseteq$, and $s_J := \sum_{k \in J} x_k$.
(ii) *Arbitrary convergence*: $\sum_{k \in K} x_k = x \iff \forall \varepsilon > 0\, \exists S \in \mathcal{F}(K)\, \forall T \in \mathcal{F}(K) : S \subseteq T \Rightarrow \|s_T - x\| \le \varepsilon$.

(iii) *Nonnegative scalar sum*: for $(a_k)_{k\in K} \subseteq [0,\infty)$, $\sum_{k\in K} a_k < \infty \iff \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} a_k < \infty$.

Proposition 6.2.15. *Basic Properties of Arbitrary Sums*
Let $(x_k)_{k\in K}, (y_k)_{k\in K} \subseteq \mathcal{H}$ with $\sum_{k\in K} x_k = x$ and $\sum_{k\in K} y_k = y$. Then $\forall a,b\in\mathbb{F}$, $\sum_{k\in K}(ax_k+by_k) = ax+by$. Moreover, $\forall z\in\mathcal{H}$, $\sum_{k\in K}\langle x_k, z\rangle = \langle x, z\rangle$ and $\sum_{k\in K}\langle z, x_k\rangle = \langle z, x\rangle$.

Proof. For $J\in\mathcal{F}(K)$, set $s_J := \sum_{k\in J} x_k$ and $t_J := \sum_{k\in J} y_k$.
Then $s_J \to x$ and $t_J \to y$, hence $as_J + bt_J \to ax+by$, proving $\sum_{k\in K}(ax_k+by_k) = ax+by$.
For $z\in\mathcal{H}$, Proposition 6.2.2 gives $\langle s_J, z\rangle \to \langle x, z\rangle$ and $\langle z, s_J\rangle \to \langle z, x\rangle$,
i.e. $\sum_{k\in K}\langle x_k, z\rangle = \langle x, z\rangle$ and $\sum_{k\in K}\langle z, x_k\rangle = \langle z, x\rangle$.
Structural Commentary: Arbitrary sums are ordinary finite sums plus a convergence net.
Key Insight: Once the vector sum converges, continuous linear tests may pass through the sum. □

Theorem 6.2.16. *Arbitrary Orthonormal Sums*
Let $(u_k)_{k\in K} \subseteq \mathcal{H}$ be orthonormal.

(i) $\forall (r_k)_{k\in K} \subseteq \mathbb{F}$, $\sum_{k\in K} r_k u_k$ *converges* $\iff \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 < \infty$; *then* $\left\|\sum_{k\in K} r_k u_k\right\|^2 = \sum_{k\in K} |r_k|^2$.
(ii) $\forall (x_k)_{k\in K} \subseteq \mathcal{H}$, $\sum_{k\in K} x_k$ *converges* $\implies |\{k\in K : x_k \neq \mathbf{0}\}| \leq \aleph_0$.

Proof.

(i) ($\Rightarrow$) Let $s_J := \sum_{k\in J} r_k u_k$. If $s_J \to x$, choose $S\in\mathcal{F}(K)$ such that $S\subseteq T \Rightarrow \|s_T - x\| \leq 1$.
For any $J\in\mathcal{F}(K)$, $\sum_{k\in J} |r_k|^2 \leq \sum_{k\in J\cup S} |r_k|^2 = \|s_{J\cup S}\|^2 \leq (\|x\|+1)^2$.

(i) ($\Leftarrow$) Let $M := \sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |r_k|^2 < \infty$. Given $\varepsilon > 0$, choose $S\in\mathcal{F}(K)$ with $M - \sum_{k\in S} |r_k|^2 < \varepsilon^2$.
If $S\subseteq J, L$, then $\|s_J - s_L\|^2 \leq \sum_{k\in(J\cup L)\setminus S} |r_k|^2 < \varepsilon^2$, so (s_J) is Cauchy and converges in $\mathcal{H}$.
Passing $\|s_J\|^2 = \sum_{k\in J} |r_k|^2$ to limits gives $\left\|\sum_{k\in K} r_k u_k\right\|^2 = \sum_{k\in K} |r_k|^2$.

(ii) If $\sum_{k\in K} x_k$ converges, then $\forall n\geq 1$ choose $I_n\in\mathcal{F}(K)$ such that $F\in\mathcal{F}(K\setminus I_n) \Rightarrow \left\|\sum_{k\in F} x_k\right\| \leq 1/n$.
With $I := \bigcup_{n\geq 1} I_n$, I is countable; if $k\notin I$, then $\|x_k\| \leq 1/n$ for all n, hence $x_k = \mathbf{0}$.

Structural Commentary: Orthonormal sums reduce vector convergence to scalar square-summability.
Key Insight: A genuinely convergent arbitrary sum can hide no uncountable collection of nonzero terms. □

Theorem 6.2.17. *Fourier Expansion in a Hilbert Space*
Let $\mathcal{O} = (u_k)_{k\in K} \subseteq \mathcal{H}$ be orthonormal. For $x\in\mathcal{H}$, set $\widehat{x}_{\mathcal{O}} := \sum_{k\in K}\langle x, u_k\rangle u_k$.

(i) $\widehat{x}_{\mathcal{O}}$ *converges in* $\mathcal{H}$ *and* $\|\widehat{x}_{\mathcal{O}}\|^2 = \sum_{k\in K} |\langle x, u_k\rangle|^2 \leq \|x\|^2$.
(ii) $\widehat{x}_{\mathcal{O}}$ *is the unique minimizer of* $\|x-y\|$ *over* $y\in \operatorname{cspan}(\mathcal{O})$.
(iii) $\|\widehat{x}_{\mathcal{O}}\| = \|x\| \iff x\in\operatorname{cspan}(\mathcal{O})$.

Proof.

(i) Finite Bessel gives $\sup_{J\in\mathcal{F}(K)} \sum_{k\in J} |\langle x, u_k\rangle|^2 \leq \|x\|^2$. With $r_k := \langle x, u_k\rangle$, Theorem 6.2.16 applies, so $\widehat{x}_{\mathcal{O}} = \sum_{k\in K} r_k u_k$ converges and $\|\widehat{x}_{\mathcal{O}}\|^2 = \sum_{k\in K} |r_k|^2 = \sum_{k\in K} |\langle x, u_k\rangle|^2 \leq \|x\|^2$.

(ii) Proposition 6.2.15 gives $\langle x - \widehat{x}_{\mathcal{O}}, u_k\rangle = 0$ for all k, hence $r := x - \widehat{x}_{\mathcal{O}} \in \operatorname{cspan}(\mathcal{O})^{\perp}$. Let $M := \operatorname{cspan}(\mathcal{O})$.
For $y\in M$, $x - y = r + (\widehat{x}_{\mathcal{O}} - y)$ orthogonally,
so $\|x-y\|^2 = \|r\|^2 + \|\widehat{x}_{\mathcal{O}} - y\|^2 \geq \|r\|^2 = \|x - \widehat{x}_{\mathcal{O}}\|^2$, with equality iff $y = \widehat{x}_{\mathcal{O}}$.

(iii) ($\Rightarrow$) If $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$, then $x = \widehat{x}_{\mathcal{O}} + (x - \widehat{x}_{\mathcal{O}})$ orthogonally,
so $\|x - \widehat{x}_{\mathcal{O}}\|^2 = \|x\|^2 - \|\widehat{x}_{\mathcal{O}}\|^2 = 0$ and $x = \widehat{x}_{\mathcal{O}} \in \operatorname{cspan}(\mathcal{O})$.

(iii) ($\Leftarrow$) If $x\in\operatorname{cspan}(\mathcal{O})$, then the unique minimizer in (ii) is x, so $\widehat{x}_{\mathcal{O}} = x$ and $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$.

Structural Commentary: Fourier expansion is projection onto the closed span.
Key Insight: Bessel bounds coefficients; completeness sums them. □

Corollary 6.2.18. *Parseval Characterization of Hilbert Bases*
For an orthonormal family $\mathcal{O} = \{u_k : k\in K\}$ in $\mathcal{H}$, the following are equivalent:

(i) $\mathcal{O}$ *is a Hilbert basis.*
(ii) $x = \sum_{k\in K}\langle x, u_k\rangle u_k$ *for every* $x\in\mathcal{H}$.
(iii) Parseval's identity $\langle x, y\rangle = \sum_{k\in K}\langle x, u_k\rangle\overline{\langle y, u_k\rangle}$ *holds for all* $x, y\in\mathcal{H}$.
(iv) $\|x\|^2 = \sum_{k\in K} |\langle x, u_k\rangle|^2$ *for every* $x\in\mathcal{H}$.

Proof.

(i) $\Rightarrow$ (ii) By Theorem 6.2.13, $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$. For $x\in\mathcal{H}$, both x, $\widehat{x}_{\mathcal{O}}$ minimize $\|x-y\|$ over $y\in\operatorname{cspan}(\mathcal{O})$, so Theorem 6.2.17 gives $x = \widehat{x}_{\mathcal{O}} = \sum_{k\in K}\langle x, u_k\rangle u_k$.

(ii) $\Rightarrow$ (iii) For $x, y\in\mathcal{H}$, (ii) and Proposition 6.2.15 give $\langle x, y\rangle = \sum_{k\in K}\langle x, u_k\rangle\langle u_k, y\rangle = \sum_{k\in K}\langle x, u_k\rangle\overline{\langle y, u_k\rangle}$.

(iii) $\Rightarrow$ (iv) Setting $y = x$ in (iii) gives $\|x\|^2 = \langle x, x\rangle = \sum_{k\in K} \langle x, u_k\rangle \overline{\langle x, u_k\rangle} = \sum_{k\in K} |\langle x, u_k\rangle|^2$ for all $x \in \mathcal{H}$.
(iv) $\Rightarrow$ (i) By Theorem 6.2.17, (iv) implies $\|\widehat{x}_{\mathcal{O}}\| = \|x\|$, hence $x \in \operatorname{cspan}(\mathcal{O})$ for all $x \in \mathcal{H}$.
Therefore $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$ and Theorem 6.2.13 gives (i).

Structural Commentary: Completeness is detected by expansion, norms, or inner products.
Key Insight: Parseval says the Hilbert basis loses no length or angle information. □

Definition 6.2.19. *Hilbert Dimension*
For a Hilbert space $\mathcal{H}$ with Hilbert basis $\mathcal{B}$, define $\operatorname{hdim}(\mathcal{H}) := |\mathcal{B}|$.

Theorem 6.2.20. *Hilbert Dimension and $\ell^2(K)$ Models*
Let K be a set.

(i) Every Hilbert space has a Hilbert basis $(\delta_k)_{k\in K}$; if $\mathcal{B}, \mathcal{C}$ are Hilbert bases of $\mathcal{H}$, then $|\mathcal{B}| = |\mathcal{C}|$.
(ii) $\ell^2(K) := \{f : K \to \mathbb{F} : \sum_{k\in K} |f(k)|^2 < \infty\}$ is Hilbert under $\langle f, g\rangle = \sum_{k\in K} f(k)\overline{g(k)}$, with $(\delta_k)_{k\in K}$.
(iii) For Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, $\mathcal{H}_1 \cong_{\text{iso}} \mathcal{H}_2 \iff \operatorname{hdim}(\mathcal{H}_1) = \operatorname{hdim}(\mathcal{H}_2)$.

Proof.

(i) Zorn's lemma, Principle 6.5.2, gives a maximal orthonormal set, hence a Hilbert basis by Theorem 6.2.13. If $\mathcal{B} = \{b_i : i \in I\}$ and $\mathcal{C} = \{c_j : j \in J\}$ are infinite Hilbert bases, set $J_i := \{j \in J : \langle b_i, c_j\rangle \neq 0\}$. Theorem 6.2.16 gives $|J_i| \le \aleph_0$, and totality of $\mathcal{B}$ gives $J = \bigcup_{i\in I} J_i$; hence $|J| \le \aleph_0 |I| = |I|$, and symmetry with Theorem 6.3.1 gives $|I| = |J|$. The finite case is ordinary dimension.
(ii) Finite-subset Cauchy–Schwarz makes $\langle f, g\rangle$ well defined on $\ell^2(K)$. If (f_n) is Cauchy, $f(k) := \lim_n f_n(k)$ exists for each k, and finite-subset estimates give $f \in \ell^2(K)$ and $f_n \to f$; hence $\ell^2(K)$ is Hilbert. The family $(\delta_k)_{k\in K}$ is orthonormal, and $(\delta_k)_{k\in K}^{\perp} = \{\mathbf{0}\}$, so it is a Hilbert basis by Theorem 6.2.13.
(iii) ($\Rightarrow$) If $U : \mathcal{H}_1 \to \mathcal{H}_2$ is a surjective linear isometry and $\mathcal{B}$ is a Hilbert basis of $\mathcal{H}_1$, then $U\mathcal{B}$ is a Hilbert basis of $\mathcal{H}_2$; hence $\operatorname{hdim}(\mathcal{H}_1) = \operatorname{hdim}(\mathcal{H}_2)$.
(iii) ($\Leftarrow$) If $\operatorname{hdim}(\mathcal{H}_1) = \operatorname{hdim}(\mathcal{H}_2)$, choose Hilbert bases $(b_i)_{i\in I}$ and $(c_i)_{i\in I}$. Parseval gives isometric isomorphisms $\Phi_1 x = (i \mapsto \langle x, b_i\rangle)$ and $\Phi_2 y = (i \mapsto \langle y, c_i\rangle)$ onto $\ell^2(I)$. Thus $\Phi_2^{-1}\Phi_1 : \mathcal{H}_1 \to \mathcal{H}_2$ is a surjective linear isometry.

Structural Commentary: Hilbert bases classify Hilbert spaces by cardinality.
Key Insight: Every Hilbert space is a coordinate $\ell^2(K)$ after choosing a Hilbert basis. □

Corollary 6.2.21. *Separable Hilbert Spaces*
$\mathcal{H}$ separable $\iff \operatorname{hdim}(\mathcal{H}) \le \aleph_0$.

Proof.

($\Rightarrow$) Let $D \subseteq \mathcal{H}$ be countable dense and let $\mathcal{O}$ be a Hilbert basis. For each $u \in \mathcal{O}$ choose $d_u \in D$ with $\|u - d_u\| < 1/3$. If $u \neq v$ and $d_u = d_v$, then $\sqrt{2} = \|u - v\| \le \|u - d_u\| + \|d_v - v\| < 2/3$, impossible. Hence $u \mapsto d_u$ injects $\mathcal{O}$ into D, so $\operatorname{hdim}(\mathcal{H}) = |\mathcal{O}| \le \aleph_0$.
($\Leftarrow$) Let $\mathcal{O}$ be a Hilbert basis with $|\mathcal{O}| \le \aleph_0$. The finite $\mathbb{Q}$-span of $\mathcal{O}$ over $\mathbb{R}$, respectively finite $(\mathbb{Q} + i\mathbb{Q})$-span over $\mathbb{C}$, is countable and dense in $\operatorname{cspan}(\mathcal{O}) = \mathcal{H}$.

Structural Commentary: Countable bases and countable dense sets force each other.
Key Insight: Separability of a Hilbert space is exactly countability of its orthogonal coordinate directions. □

Corollary 6.2.22. *Infinite Hilbert Bases Are Not Hamel Bases*
If $\operatorname{hdim}(\mathcal{H})$ is infinite, no Hilbert basis of $\mathcal{H}$ is a Hamel basis.

Proof. Let $\mathcal{O}$ be a Hilbert basis and choose pairwise distinct $(u_n) \subseteq \mathcal{O}$.
Since $\sum_{n=1}^{\infty} 4^{-n} < \infty$, Theorem 6.2.16 gives $x := \sum_{n=1}^{\infty} 2^{-n} u_n \in \mathcal{H}$. Proposition 6.2.15 gives $\langle x, u_n\rangle = 2^{-n} \neq 0$ for all n, so x has infinitely many nonzero $\mathcal{O}$-coordinates and $x \notin \operatorname{span}(\mathcal{O})$.
Structural Commentary: Infinite Hilbert coordinates allow square-summable infinite expansions.
Key Insight: A Hilbert basis spans by limits, while a Hamel basis spans by finite sums. □

Perspective 6.2.23. *Bounded Functionals*
Here $\mathcal{H}^*$ denotes the continuous dual of $\mathcal{H}$. By Definition 6.1.2 and Theorem 6.1.3, $\mathcal{H}^* = \{\varphi : \mathcal{H} \to \mathbb{F} : \varphi \text{ linear}, \ \|\varphi\|_{\text{op}} < \infty\}$.

Proposition 6.2.24. *Operator Norm Algebra*
Let $\mathcal{B}(\mathcal{H})$ be the bounded linear operators $\mathcal{H} \to \mathcal{H}$.
For $S, T \in \mathcal{B}(\mathcal{H})$ and $a \in \mathbb{F}$, $\|aT\|_{op} = |a| \, \|T\|_{op}$, $\|S + T\|_{op} \le \|S\|_{op} + \|T\|_{op}$, and $\|ST\|_{op} \le \|S\|_{op} \, \|T\|_{op}$.
Hence $\mathcal{B}(\mathcal{H})$ is a normed algebra.

Proof. Taking $\sup_{\|x\|\le 1}$ in $\|aTx\| = |a|\,\|Tx\|$ and $\|(S+T)x\| \le \|Sx\| + \|Tx\| \le (\|S\|_{\mathrm{op}} + \|T\|_{\mathrm{op}})\,\|x\|$ gives homogeneity and subadditivity. Also $\|STx\| \le \|S\|_{\mathrm{op}}\,\|Tx\| \le \|S\|_{\mathrm{op}}\,\|T\|_{\mathrm{op}}\,\|x\|$, so $\|ST\|_{\mathrm{op}} \le \|S\|_{\mathrm{op}}\,\|T\|_{\mathrm{op}}$.
Structural Commentary: Composition multiplies stretch bounds.
Key Insight: Bounded operators behave like matrices with a compatible norm. □

Theorem 6.2.25. *Riesz Representation for Hilbert Spaces*
For every bounded linear functional $\varphi : \mathcal{H} \to \mathbb{F}$, $\exists! y \in \mathcal{H}$ such that $\forall x \in \mathcal{H}$, $\varphi(x) = \langle x, y\rangle$; moreover $\|\varphi\|_{op} = \|y\|$.

Proof. If $\varphi = 0$, take $y = 0$. If $\varphi \neq 0$, let $M := \ker\varphi$ and choose $a \notin M$. Since M is closed, Theorem 6.2.9 gives $a = p + u$ with $p \in M$ and $0 \neq u \in M^{\perp}$; hence $\varphi(u) = \varphi(a) \neq 0$. For $x \in \mathcal{H}$, $x - \varphi(x)\varphi(u)^{-1}u \in M$, so $\langle x, u\rangle = \varphi(x)\varphi(u)^{-1}\,\|u\|^2$. Put $y := \overline{\varphi(u)}\,\|u\|^{-2}\,u$; conjugate linearity in the second slot gives $\langle x, y\rangle = \varphi(x)$.
If z also represents φ, then $\langle x, y - z\rangle = 0$ for all x, so $x = y - z$ gives $y = z$.
Finally, $|\varphi(x)| \le \|x\|\,\|y\|$ gives $\|\varphi\|_{\mathrm{op}} \le \|y\|$, while $y \neq 0$ gives $\|\varphi\|_{\mathrm{op}} \ge |\varphi(y)|\,/\,\|y\| = \|y\|$; $y = 0$ is immediate.
Structural Commentary: The kernel is a hyperplane; its orthogonal direction represents the functional.
Key Insight: Bounded functionals are exactly inner products with fixed vectors. □

Corollary 6.2.26. *Riesz Identifies the Continuous Dual*
The Riesz map $R : \mathcal{H} \to \mathcal{H}^*$, $Ry := (x \mapsto \langle x, y\rangle)$, is an isometric bijection.
It is conjugate-linear over $\mathbb{C}$ and linear over $\mathbb{R}$.

Proof. Theorem 6.2.25 gives $\forall \varphi \in \mathcal{H}^*\ \exists! y \in \mathcal{H} : \varphi = Ry$ and $\|Ry\|_{\mathrm{op}} = \|y\|$, so R is bijective and isometric.
Also $R(\alpha y + \beta z) = \overline{\alpha}Ry + \overline{\beta}Rz$ over $\mathbb{C}$, while $R(\alpha y + \beta z) = \alpha Ry + \beta Rz$ over $\mathbb{R}$.
Structural Commentary: Continuous functionals are represented by vectors.
Key Insight: Riesz identifies $\mathcal{H}^*$ with $\mathcal{H}$, up to complex conjugation. □

Theorem 6.2.27. *Orthonormal Systems in L^2*
Let $(\phi_n)_{n\ge 1} \subseteq L^2(\mu)$ be orthonormal, $\langle \phi_n, \phi_m\rangle = \delta_{nm}$, and $c_n(f) := \langle f, \phi_n\rangle = \int_X f\overline{\phi_n}\,d\mu$.

(i) $\forall f \in L^2(\mu)$, $\sum_{n=1}^{\infty} |c_n(f)|^2 \le \|f\|_2^2$ and $f \sim_{\mathrm{coef}} \sum c_n(f)\phi_n$.
(ii) If $(c_n) \in \ell^2$, then $s_N := \sum_{k=1}^{N} c_k\phi_k$ converges in $L^2(\mu)$.
(iii) If (ϕ_n) is complete, then $\forall f \in L^2(\mu)$, $\|f\|_2^2 = \sum_{n=1}^{\infty} |c_n(f)|^2$.

For the trigonometric system on $[-\pi, \pi]$, this is L^2 Parseval and a.e. uniqueness.

Proof.

(i) Finite orthonormality gives $\left\| f - \sum_{k=1}^{N} c_k(f)\phi_k \right\|_2^2 = \|f\|_2^2 - \sum_{k=1}^{N} |c_k(f)|^2 \ge 0$.
Hence $\sum_{n=1}^{\infty} |c_n(f)|^2 \le \|f\|_2^2$ and $f \sim_{\mathrm{coef}} \sum c_n(f)\phi_n$.
(ii) For $m < N$, $\|s_N - s_m\|_2^2 = \sum_{k=m+1}^{N} |c_k|^2 \to 0$ since $(c_n) \in \ell^2$.
Completeness of $L^2(\mu)$ gives $s_N \to g \in L^2(\mu)$.
(iii) If (ϕ_n) is complete, then (i) gives $(c_n(f)) \in \ell^2$, so (ii) gives $s_N \to g$. For every j, $\langle f - g, \phi_j\rangle = c_j(f) - \lim_N \langle s_N, \phi_j\rangle = 0$. Completeness gives $f = g$, hence $\|f\|_2^2 = \lim_N \|s_N\|_2^2 = \sum_{n=1}^{\infty} |c_n(f)|^2$.

Structural Commentary: Complete orthonormal systems coordinatize L^2 by square-summable sequences.
Key Insight: Parseval is Pythagoras after passing from finite coordinates to an L^2 limit. □

Method 6.2.28. *Hilbert Space Proof Patterns*

(i) *Complete*: Cauchy tails, completions, and convergent subsequences force limits.
(ii) *Parallelogram*: test Hilbertian norms and make distance-minimizing sequences Cauchy.
(iii) *Project*: for closed M, write $x = p + r$, $p \in M$, $r \in M^{\perp}$.
(iv) *Separate*: density/equality via $M^{\perp} = \{\mathbf{0}\}$ and $M^{\perp\perp} = \overline{M}$.
(v) *Sum*: arbitrary convergence is net convergence over $\mathcal{F}(K)$; control finite tails.
(vi) *Expand*: orthonormal finite identities + Bessel bounds + completeness ⇒ Fourier expansions.
(vii) *Classify*: Hilbert bases give hdim, separability, and $\mathcal{H} \cong \ell^2(K)$.
(viii) *Dualize*: $\mathcal{B}(\mathcal{H})$ norm algebra and Riesz $\mathcal{H}^* \cong \mathcal{H}$.

Key Insight: Hilbert-space arguments are finite orthogonal geometry plus completeness.

6.3 Core Functional Analytic Theorems

The core theorems explain why complete normed spaces behave rigidly under linear maps. Baire category turns pointwise control, surjectivity, and closed graphs into operator statements.

Theorem 6.3.1. *Hahn–Banach Extension Theorem*
Let $\mathcal{X}$ be a real normed space, let $M \subseteq \mathcal{X}$ be a linear subspace, and let $p : \mathcal{X} \to \mathbb{R}$ be sublinear. If $\varphi : M \to \mathbb{R}$ is linear and $\varphi(x) \le p(x)$ for all $x \in M$, then φ extends to a linear functional $\Phi : \mathcal{X} \to \mathbb{R}$ with $\Phi(x) \le p(x)$ for all $x \in \mathcal{X}$. In the normed complex form, every bounded functional on a subspace extends with the same norm.

Theorem 6.3.2. *Positive Riesz Representation on $C([a,b])$*
Let $J = [a,b]$ and let $G \in C(J)^*$ be positive, i.e. $f \ge 0$ on J implies $G(f) \ge 0$.
Then $\exists$ a finite positive Borel measure γ on $\mathbb{R}$ with $\operatorname{supp}\gamma \subseteq J$ s.t. $G(f) = \int_J f\,d\gamma$ $\forall f \in C(J)$.
Moreover, $\|G\|_{op} = G(\mathbf{1}_J) = \gamma(J)$.

Proof. For $a \le t < b$ and n large, set

$$\varphi_{t,n}(s) := \begin{cases} 1, & a \le s \le t, \\ n(t + 1/n - s), & t < s < t + 1/n, \\ 0, & t + 1/n \le s \le b. \end{cases}$$

Then $\varphi_{t,n} \downarrow \mathbf{1}_{[a,t]}$ pointwise, hence $G(\varphi_{t,n})$ decreases.
Define $g(t) := \lim_{n\to\infty} G(\varphi_{t,n})$; $g(t) = 0$ for $t < a$; and $g(t) = G(\mathbf{1}_J)$ for $t \ge b$.
Ramp comparisons give g increasing and right-continuous.
By Theorem 1.4.4, g induces a finite Borel–Stieltjes measure γ with $\operatorname{supp}\gamma \subseteq J$ and $\gamma(J) = g(b) = G(\mathbf{1}_J)$.
Fix $f \in C(J)$ and $\varepsilon > 0$. Choose $P : a = t_0 < \cdots < t_m = b$ with oscillation $\omega_f(P) \le \varepsilon$.
Put $\Delta g_k := g(t_k) - g(t_{k-1})$. For n large, ramp increments $\varphi_{t_k,n} - \varphi_{t_{k-1},n}$ give continuous approximant $f_{P,n}$ whose G-increments approximate $\sum_k f(t_k)\Delta g_k$ within ε, while step approximant s_P satisfies $\|f - s_P\|_{L^1(\gamma)} \le \varepsilon\gamma(J)$.
Hence $\left|G(f) - \int_J f\,d\gamma\right| \le \varepsilon(2\|G\|_{\mathrm{op}} + 1)$ after absorbing constants into ε. Letting $\varepsilon \downarrow 0$ gives $G(f) = \int_J f\,d\gamma$.
Finally, positivity gives $|G(f)| \le G(\mathbf{1}_J)\,\|f\|_\infty = \gamma(J)\,\|f\|_\infty$, and $f = \mathbf{1}_J$ gives $\|G\|_{\mathrm{op}} \ge \gamma(J)$.
Structural Commentary: Positivity turns ramp limits into interval masses, hence a Stieltjes measure.
Key Insight: Ramps turn order on $C(J)$ into interval masses. □

Corollary 6.3.3. *Signed Riesz Representation by a Charge*
Let $J = [a,b]$ and let $G \in C(J)^*$ admit $G = G^+ - G^-$, $G^\pm \in C(J)^*$; for $G^\pm \ge 0$.
Then $\exists$ a charge $\lambda : \mathcal{B}(\mathbb{R}) \to \mathbb{R}$ with $\operatorname{supp}\lambda \subseteq J$ such that $G(f) = \int_J f\,d\lambda$ $\forall f \in C(J)$.

Proof. By Theorem 6.3.2, $\exists$ finite positive Borel measures $\gamma^\pm$ with $\operatorname{supp}\gamma^\pm \subseteq J$; $G^\pm(f) = \int_J f\,d\gamma^\pm$ $\forall f \in C(J)$.
Set $\lambda := \gamma^+ - \gamma^-$, i.e. $\lambda(E) := \gamma^+(E) - \gamma^-(E)$ for $E \in \mathcal{B}(\mathbb{R})$. Then λ is a charge by Proposition 1.2.10, $\operatorname{supp}\lambda \subseteq J$, and $G(f) = G^+(f) - G^-(f) = \int_J f\,d\gamma^+ - \int_J f\,d\gamma^- = \int_J f\,d\lambda$.
Structural Commentary: Positive representations subtract to a charge representation.
Key Insight: Signed representation is positive representation plus subtraction. □

Theorem 6.3.4. *Uniform Boundedness Principle*
Let $\mathcal{X}$ be Banach and let $\mathcal{Y}$ be normed. If $\mathcal{F} \subseteq \mathcal{L}(\mathcal{X},\mathcal{Y})$ is pointwise bounded, meaning $\sup_{T\in\mathcal{F}} \|Tx\| < \infty$ for every $x \in \mathcal{X}$, then $\sup_{T\in\mathcal{F}} \|T\|_{op} < \infty$.

Theorem 6.3.5. *Open Mapping Theorem*
A surjective bounded linear map $T : \mathcal{X} \to \mathcal{Y}$ between Banach spaces is open. Equivalently, T maps neighborhoods of 0 onto neighborhoods of 0.

Theorem 6.3.6. *Closed Graph Theorem*
Let $\mathcal{X}$ and $\mathcal{Y}$ be Banach spaces. A linear map $T : \mathcal{X} \to \mathcal{Y}$ is bounded if and only if its graph

$$\Gamma(T) := \{(x, Tx) : x \in \mathcal{X}\}$$

is closed in $\mathcal{X} \times \mathcal{Y}$.

7 Weak Topologies, Compact Operators, and Spectral Theory

7.1 Weak Topologies and Duality

Weak topologies replace norm control with scalar tests by functionals, exposing compactness and duality that strong topology can hide.

Definition 7.1.1. *Dual Space and Weak Convergence*
For a normed space $\mathcal{X}$, its continuous dual is $\mathcal{X}^* := \mathcal{L}(\mathcal{X},\mathbb{F})$. A sequence (x_n) *converges weakly* to x, written $x_n \rightharpoonup x$, if

$$\varphi(x_n) \to \varphi(x) \qquad \text{for every } \varphi \in \mathcal{X}^*.$$

Definition 7.1.2. *Weak-* Convergence*
If $(\varphi_n) \subseteq \mathcal{X}^*$ and $\varphi \in \mathcal{X}^*$, then φ_n *converges weak-** to φ if

$$\varphi_n(x) \to \varphi(x) \qquad \text{for every } x \in \mathcal{X}.$$

This is weak convergence in the dual tested only against evaluation at points of the original space.

Definition 7.1.3. *Reflexivity*
The canonical map $J : \mathcal{X} \to \mathcal{X}^{**}$ is defined by $Jx(\varphi) = \varphi(x)$. A Banach space is *reflexive* if J is onto.

Theorem 7.1.4. *Banach–Alaoglu Theorem*
The closed unit ball of $\mathcal{X}^*$ is compact in the weak- topology. Consequently, bounded nets of functionals have weak-* cluster points.*

7.2 Compact Operators

Compact operators carry bounded sets to relatively compact sets. Their spectral and solvability theory therefore resembles matrix theory inside infinite-dimensional spaces.

Definition 7.2.1. *Compact and Finite-Rank Operators*
Let $\mathcal{X}, \mathcal{Y}$ be normed spaces. A linear map $T : \mathcal{X} \to \mathcal{Y}$ is *compact* if $T(B)$ has compact closure in $\mathcal{Y}$ whenever $B \subseteq \mathcal{X}$ is bounded.
The map has *finite rank* if $\dim \operatorname{range}(T) < \infty$.

Proposition 7.2.2. *Finite Rank Operators Are Compact*
If $\mathcal{X}$ is normed, $\mathcal{Y}$ is Banach, and $T : \mathcal{X} \to \mathcal{Y}$ is bounded with finite-dimensional range, then T is compact.

Theorem 7.2.3. *Norm Limits of Compact Operators*
If $T_n : \mathcal{X} \to \mathcal{Y}$ are compact operators and $T_n \to T$ in operator norm, with $\mathcal{Y}$ Banach, then T is compact.

Theorem 7.2.4. *Fredholm Alternative*
Let $K : \mathcal{X} \to \mathcal{X}$ be compact on a Banach space. For $\lambda \neq 0$, the equation

$$(\lambda I - K)x = y$$

has a solution for every y exactly when the homogeneous equation $(\lambda I - K)x = 0$ has only the zero solution. In this case $(\lambda I - K)^{-1}$ is bounded.

7.3 Spectral Theory of Operators

For bounded operators, spectrum replaces the finite-dimensional eigenvalue list. The resolvent tracks invertibility, while the spectrum records every obstruction to it.

Definition 7.3.1. *Spectrum and Resolvent*
Let $T \in \mathcal{L}(\mathcal{X})$ be a bounded operator on a Banach space. The *resolvent set* is

$$\rho(T) := \{\lambda \in \mathbb{C} : \lambda I - T \text{ is bijective and } (\lambda I - T)^{-1} \in \mathcal{L}(\mathcal{X})\}.$$

The *spectrum* is $\sigma(T) := \mathbb{C} \setminus \rho(T)$.

Definition 7.3.2. *Spectral Radius*
The *spectral radius* of $T \in \mathcal{L}(\mathcal{X})$ is

$$r(T) := \sup\{|\lambda| : \lambda \in \sigma(T)\}.$$

Theorem 7.3.3. *Spectrum of a Complex Bounded Operator*
If $\mathcal{X}$ is a nonzero complex Banach space and $T \in \mathcal{L}(\mathcal{X})$, then $\sigma(T)$ is a nonempty compact subset of $\mathbb{C}$ and $\sigma(T) \subseteq \{\lambda : |\lambda| \leq \|T\|_{op}\}$.

Theorem 7.3.4. *Spectral Radius Formula*
For $T \in \mathcal{L}(\mathcal{X})$ on a complex Banach space,

$$r(T) = \lim_{n \to \infty} \|T^n\|_{op}^{1/n}.$$

Perspective 7.3.5. *Hilbert Operator Spectral Theory*
For bounded self-adjoint and normal operators on Hilbert spaces, spectral theory recovers a functional-calculus form of diagonalization. Compact self-adjoint operators have an orthonormal eigenbasis away from the kernel, giving the closest infinite-dimensional analogue of the finite spectral theorem.

7.4 Distributions and Sobolev Preview

Distributions test against smooth compactly supported functions; Sobolev spaces package weak derivatives in normed or Hilbert spaces as a bridge to PDE.

Definition 7.4.1. *Test Functions and Distributions*
Let $\Omega \subseteq \mathbb{R}^n$ be open. A *test function* is a smooth function $\varphi \in C_c^\infty(\Omega)$. A *distribution* on Ω is a continuous linear functional on the test-function space, written $u \in \mathcal{D}'(\Omega)$ and paired as $\langle u, \varphi \rangle$.

Part XII
Probability and Stochastic Systems

1 Probability Spaces, Random Variables, and Expectation

1.1 Probability Spaces and Random Variables

Probability assigns numbers to experimental outcomes; measure theory specifies which subsets are events.

Definition 1.1.1. *Sample Spaces, Events, and Probability Spaces*

(i) A *sample space* is a set Ω of possible outcomes.
(ii) An *outcome* or *realization* is a point $\omega \in \Omega$.
(iii) An *event* is a subset $A \subseteq \Omega$; the admissible events form a *σ-algebra* $\mathcal{F} \subseteq \mathcal{P}(\Omega)$.
(iv) A *probability measure* on $(\Omega, \mathcal{F})$ is a measure $\mathbb{P} : \mathcal{F} \to [0,1]$ with $\mathbb{P}(\Omega) = 1$.
(v) The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Key Insight: Probability is countably additive size normalized so that the whole sample space has mass 1.

Method 1.1.2. *Sample Space Construction Method*
Choose Ω as the set of complete outcomes and encode the target condition as an event $A \subseteq \Omega$:

(i) *Finite product method*: for m discrete trials with state set S, use $\Omega = S^m$ and coordinate constraints. For two coin tosses, $S = \{H, T\}$, $\Omega = S^2$, and first toss heads gives $A = \{\omega \in \Omega : \omega_1 = H\} = \{HH, HT\}$.
(ii) *Interval method*: for real-valued measurements, use $\Omega = \mathbb{R}$ and translate inequalities into intervals. $10 < \omega \leq 23$ gives $A = (10, 23]$.
(iii) *Cylinder method*: for infinite sequences with state set S, use $\Omega = S^{\mathbb{N}}$ and constrain finitely many coordinates; for coin tosses, $S = \{H, T\}$ and first head on toss 3 gives $A = \{\omega \in \Omega : \omega_1 = T, \omega_2 = T, \omega_3 = H\}$.

Example 1.1.3. *Introductory Sample Spaces and Events*

(i) If a commute passes three traffic lights, each with outcome stop s or continue c, then $\Omega = \{c, s\}^3$. The event that the first light is a stop is $\{sss, ssc, scc, scs\}$.
(ii) If a print queue length is modeled as random, one possible sample space is $\Omega = \{0, 1, 2, \ldots\}$; with a known capacity N, one may instead use $\Omega = \{0, 1, \ldots, N\}$.
(iii) If the measured waiting time between events is nonnegative, a natural sample space is $\Omega = [0, \infty)$.

Key Insight: Choose Ω at the event resolution needed.

Definition 1.1.4. *Event Algebra, Indicators, and Monotone Events*

(i) For $A \in \mathcal{F}$, the *complement* is $A^c := \Omega \setminus A$.
(ii) For $A, B \in \mathcal{F}$, the *union* and *intersection* are $A \cup B$ and $A \cap B$.
(iii) For $A, B \in \mathcal{F}$, the *difference* is $A \setminus B := A \cap B^c$.
(iv) *Commutativity*: $A \cup B = B \cup A$ and $A \cap B = B \cap A$.
(v) *Associativity*: $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.
(vi) *Distributivity*: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
(vii) *De Morgan laws*: $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.
(viii) Events $(A_i)_{i \in I}$ are *disjoint* or *mutually exclusive* if $A_i \cap A_j = \emptyset$ for $i \neq j$.
(ix) Events $(A_i)_{i \in I}$ form a *partition* of Ω if $A_i \cap A_j = \emptyset$ for $i \neq j$ and $\bigcup_{i \in I} A_i = \Omega$.
(x) The *indicator* of $A \in \mathcal{F}$ is $\mathbf{1}_A : \Omega \to \{0, 1\}$, where $\mathbf{1}_A(\omega) = 1$ for $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ for $\omega \notin A$.
(xi) A sequence (A_n) is *increasing* to A, written $A_n \uparrow A$, if $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_{n \geq 1} A_n$.
(xii) A sequence (A_n) is *decreasing* to A, written $A_n \downarrow A$, if $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_{n \geq 1} A_n$.

Proposition 1.1.5. *Indicator Algebra for Unions and Intersections*
For events $A, B \in \mathcal{F}$: $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B = \min\{\mathbf{1}_A, \mathbf{1}_B\}$, and $\mathbf{1}_{A \cup B} = \max\{\mathbf{1}_A, \mathbf{1}_B\}$.

Proof. Fix $\omega \in \Omega$ and set $a := \mathbf{1}_A(\omega)$, $b := \mathbf{1}_B(\omega)$. Then $a, b \in \{0, 1\}$, $\mathbf{1}_{A \cap B}(\omega) = ab = \min\{a, b\}$, and $\mathbf{1}_{A \cup B}(\omega) = \max\{a, b\}$. Since ω was arbitrary, the identities hold pointwise.
Structural Commentary: Boolean operations on events become arithmetic operations on 0-1 variables.
Key Insight: Indicators let event algebra be manipulated inside sums and expectations. □

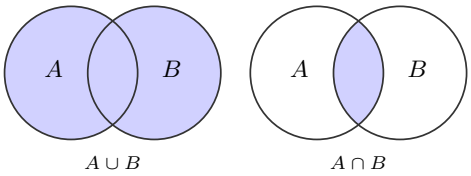


Figure 205: Set operations become event operations in a probability space.

Example 1.1.6. *Endpoint Sensitivity in Monotone Limits*
Let $\Omega = \mathbb{R}$.

(i) If $A_i = [0, \frac{1}{i}]$, then $\bigcup_{i=1}^{\infty} A_i = [0,1]$ and $\bigcap_{i=1}^{\infty} A_i = \{0\}$.

(ii) If $A_i = (0, \frac{1}{i})$, then $\bigcup_{i=1}^{\infty} A_i = (0,1)$ and $\bigcap_{i=1}^{\infty} A_i = \emptyset$.

Key Insight: Monotone set limits remember whether boundary points are included.

Proposition 1.1.7. *First Consequences of the Probability Axioms*
For events $A, B \in \mathcal{F}$:

(i) Null event: $\mathbb{P}(\emptyset) = 0$.
(ii) Monotonicity: $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$.
(iii) Bounds: $0 \leq \mathbb{P}(A) \leq 1$.
(iv) Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
(v) Disjoint additivity: $A \cap B = \emptyset \Rightarrow \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
(vi) Two-set inclusion-exclusion: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof.

(i) Countable additivity on $\Omega = \Omega \sqcup \emptyset \sqcup \emptyset \sqcup \cdots$ gives $1 = 1 + \sum_{n \geq 1} \mathbb{P}(\emptyset)$, hence $\mathbb{P}(\emptyset) = 0$.
(ii) If $A \subseteq B$, then $B = A \sqcup (B \setminus A)$, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.
(iii) Since $\emptyset \subseteq A \subseteq \Omega$, (i) and (ii) give $0 = \mathbb{P}(\emptyset) \leq \mathbb{P}(A) \leq \mathbb{P}(\Omega) = 1$.
(iv) Since $\Omega = A \sqcup A^c$, $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$, so $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
(v) If $A \cap B = \emptyset$, then $A \cup B = A \sqcup B$, so $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.
(vi) Decompose $A \cup B = (A \setminus B) \sqcup (A \cap B) \sqcup (B \setminus A)$, $A = (A \setminus B) \sqcup (A \cap B)$, and $B = (A \cap B) \sqcup (B \setminus A)$. Hence $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Structural Commentary: The algebraic identities are set partitions in disguise.
Key Insight: Split overlapping sets into disjoint pieces before adding probabilities. □

Example 1.1.8. *Two Coin Tosses and Inclusion-Exclusion*
Let $\Omega = \{HH, HT, TH, TT\}$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$ for all $\omega \in \Omega$. Set $H_1 = \{HH, HT\}$ and $H_2 = \{HH, TH\}$. Since $H_1 \cap H_2 = \{HH\}$, $\mathbb{P}(H_1 \cup H_2) = \mathbb{P}(H_1) + \mathbb{P}(H_2) - \mathbb{P}(H_1 \cap H_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$.

Proposition 1.1.9. *Three-Set Inclusion-Exclusion and Bonferroni*
For events $A, B, C \in \mathcal{F}$:

(i) $\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$.
(ii) $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$.

Proof.

(i) Let $U = A \cup B$. Then $A \cup B \cup C = U \cup C$ and $U \cap C = (A \cap C) \cup (B \cap C)$.
By two-set inclusion-exclusion, $\mathbb{P}(U \cup C) = \mathbb{P}(U) + \mathbb{P}(C) - \mathbb{P}(U \cap C)$, $\mathbb{P}(U) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $\mathbb{P}(U \cap C) = \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) - \mathbb{P}(A \cap B \cap C)$. Substitution gives (i).
(ii) Since $\mathbb{P}(A \cup B) \leq 1$ and $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, rearrange.

Structural Commentary: Inclusion-exclusion alternates between overcounting and correcting overlaps.
Key Insight: Bonferroni inequalities are one-sided forms of inclusion-exclusion. □

Theorem 1.1.10. *Continuity of Probabilities*

(i) Continuity from below: if $A_n \uparrow A$, then $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.
(ii) Continuity from above: if $A_n \downarrow A$, then $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.

Proof.

(i) Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then (B_n) is disjoint, $A_n = \bigcup_{j=1}^{n} B_j$, and $A = \bigcup_{j=1}^{\infty} B_j$, so countable additivity gives $\mathbb{P}(A_n) = \sum_{j=1}^{n} \mathbb{P}(B_j) \to \sum_{j=1}^{\infty} \mathbb{P}(B_j) = \mathbb{P}(A)$.

(ii) Set $C_n = A_1 \setminus A_n$. Then $C_n \uparrow A_1 \setminus A$, so (i) gives $\mathbb{P}(C_n) \to \mathbb{P}(A_1 \setminus A)$; since $A_1 = A_n \sqcup C_n$ and $A_1 = A \sqcup (A_1 \setminus A)$, finite additivity gives $\mathbb{P}(A_n) = \mathbb{P}(A_1) - \mathbb{P}(C_n) \to \mathbb{P}(A_1) - \mathbb{P}(A_1 \setminus A) = \mathbb{P}(A)$.

Structural Commentary: Monotone limits convert countable additivity into limit rules.
Key Insight: Increasing uses increments; decreasing uses complements. □

Definition 1.1.11. *Events Occurring Infinitely Often and Eventually Always*
For events $(A_n)_{n\geq 1}$:

(i) *Tail union*: $B_n := \bigcup_{i=n}^{\infty} A_i$.

(ii) *Tail intersection*: $C_n := \bigcap_{i=n}^{\infty} A_i$.

(iii) *Limsup*: $\limsup_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} B_n = \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} A_i$.

(iv) *Liminf*: $\liminf_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} A_i$.

(v) *Infinitely often*: $\omega \in \limsup_n A_n$ iff $\omega \in A_n$ for infinitely many n.

(vi) *Eventually always*: $\omega \in \liminf_n A_n$ iff $\omega \in A_n$ for all but finitely many n.

Proposition 1.1.12. *Tail Monotonicity*
For events $(A_n)_{n\geq 1}$ with tail union B_n and tail intersection C_n:

(i) Tail unions: $B_{n+1} \subseteq B_n$ for all n, hence $B_n \downarrow \bigcap_{n\geq 1} B_n$.

(ii) Tail intersections: $C_n \subseteq C_{n+1}$ for all n, hence $C_n \uparrow \bigcup_{n\geq 1} C_n$.

Proof.

(i) Since $\{i : i \geq n+1\} \subseteq \{i : i \geq n\}$, $B_{n+1} = \bigcup_{i=n+1}^{\infty} A_i \subseteq \bigcup_{i=n}^{\infty} A_i = B_n$.

(ii) Since $\{i : i \geq n+1\} \subseteq \{i : i \geq n\}$, $C_n = \bigcap_{i=n}^{\infty} A_i \subseteq \bigcap_{i=n+1}^{\infty} A_i = C_{n+1}$.

Structural Commentary: Tail shifts turn recurrence and permanence into monotone limits.
Key Insight: Tail unions track recurrence; tail intersections track permanence. □

Proposition 1.1.13. *Union Bound and Countable Intersections of Almost-Sure Events*
For events $(A_n)_{n\geq 1}$:

(i) Union bound: $\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

(ii) Countable almost-sure intersection: if $\mathbb{P}(A_n) = 1$ for all n, then $\mathbb{P}(\bigcap_{n=1}^{\infty} A_n) = 1$.

Proof.

(i) Set $B_n := A_n \setminus \bigcup_{j<n} A_j$. Then (B_n) is disjoint, $\bigcup_n B_n = \bigcup_n A_n$, and $B_n \subseteq A_n$.
Hence $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(B_n) \leq \sum_n \mathbb{P}(A_n)$.

(ii) If $\mathbb{P}(A_n) = 1$ for all n, then $\mathbb{P}(A_n^c) = 0$ for all n.
By (i), $\mathbb{P}((\bigcap_n A_n)^c) = \mathbb{P}(\bigcup_n A_n^c) \leq \sum_n \mathbb{P}(A_n^c) = 0$, so $\mathbb{P}(\bigcap_n A_n) = 1$.

Structural Commentary: Countable subadditivity is countable additivity after disjointization.
Key Insight: Almost sure statements survive countable intersections. □

Definition 1.1.14. *Uniform Probability on a Countable Space*
Let Ω be countable.

(i) A probability measure $\mathbb{P}$ on $(\Omega, \mathcal{P}(\Omega))$ is *uniform* if $\exists c \in [0,1]$ such that $\mathbb{P}(\{\omega\}) = c$ for all $\omega \in \Omega$.

(ii) The constant c is the *common atom mass*.

Proposition 1.1.15. *No Uniform Probability on a Countably Infinite Set*
Let $\Omega = \{0, 1, 2, \ldots\}$. No uniform probability measure exists on $(\Omega, \mathcal{P}(\Omega))$.

Proof. Assume uniform $\mathbb{P}$ exists, and let c be its common atom mass.
Since $\Omega = \bigsqcup_{n=0}^{\infty} \{n\}$, countable additivity gives $1 = \mathbb{P}(\Omega) = \sum_{n=0}^{\infty} \mathbb{P}(\{n\}) = \sum_{n=0}^{\infty} c$.
If $c = 0$, the sum is 0; if $c > 0$, the sum diverges. Both contradict $1 = \mathbb{P}(\Omega)$.
Structural Commentary: Countable additivity forces total mass to equal the sum of the atom masses.
Key Insight: Equal mass on countably many atoms is incompatible with total mass 1. □

Method 1.1.16. *Finite Uniform Probability by Counting*
Let $0 < |\Omega| < \infty$ and assume the *uniform model* $\mathbb{P}(\{\omega\}) = |\Omega|^{-1}$ for all $\omega \in \Omega$.
For $A \subseteq \Omega$, $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) = \frac{|A|}{|\Omega|}$, so probability reduces to counting.

Method 1.1.17. *Counting Workflow for Uniform Finite Probability*
For a finite uniform sample space Ω and an event $A \subseteq \Omega$:

(i) choose complete outcomes so that each $\omega \in \Omega$ is equally likely;
(ii) count $|\Omega|$; and count $|A|$, usually by products, complements, or disjoint cases;
(iii) conclude $\mathbb{P}(A) = |A| \,/\, |\Omega|$.

Key Insight: The hard step is often choosing equally likely outcomes, not applying the ratio.

Example 1.1.18. *Simpson's Paradox in Two Urn Games*
Let $p(U)$ be the red probability in urn U, and let B_i, W_i be the black and white urns in game i.
Within games, $p(B_1) = 5/11 > 3/7 = p(W_1)$ and $p(B_2) = 6/9 > 9/14 = p(W_2)$.
After pooling by color into B and W, $p(B) = 11/20 < 12/21 = p(W)$, so the aggregate winner is white.
Key Insight: A within-group advantage can reverse after aggregation when the group sizes change.

Example 1.1.19. *Complement Counting in Uniform Models*

(i) For plates $\Omega = L^3 \times D^3$ with $|L| = 26$ and $|D| = 10$, $|\Omega| = 26^3 \cdot 10^3$. If E is the event of no repeated letters and no repeated digits, then $|E| = (26)_3 (10)_3$, so $\mathbb{P}(E) = \frac{(26)_3 (10)_3}{26^3 \cdot 10^3}$.
(ii) For $1 \le n \le 365$ iid uniform birthdays, take $\Omega = \{1, \ldots, 365\}^n$. If C is the event of a shared birthday, then $C^c = D_n(\{1, \ldots, 365\})$, so $\mathbb{P}(C) = 1 - \frac{(365)_n}{365^n}$; for $n = 23$, $\mathbb{P}(C) > 1/2$.

Key Insight: "At least one collision" is often easier to count by subtracting the no-collision event.

Definition 1.1.20. *Independent Events*

(i) Events $A, B \in \mathcal{F}$ are *independent*, written $A \perp\!\!\!\perp B$, if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
(ii) A family $(A_i)_{i \in I} \subseteq \mathcal{F}$ is *pairwise independent* if A_i and A_j are independent whenever $i \neq j$.
(iii) A family $(A_i)_{i \in I} \subseteq \mathcal{F}$ is *mutually independent* if $\mathbb{P}(\bigcap_{i \in J} A_i) = \prod_{i \in J} \mathbb{P}(A_i)$ for every finite $J \subseteq I$.

Key Insight: Mutual independence means multiplicativity for every finite subfamily, not just pairs.

Example 1.1.21. *Pairwise Independence Need Not Be Mutual*
Let $\Omega = \{HH, HT, TH, TT\}$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$ $\forall \omega \in \Omega$.
Set $A = \{HH, HT\}$, $B = \{HH, TH\}$, $C = \{HT, TH\}$.
Then $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$, $\mathbb{P}(A \cap B) = \mathbb{P}(A \cap C) = \mathbb{P}(B \cap C) = \frac{1}{4}$, so A, B, C pairwise independent.
But $A \cap B \cap C = \emptyset$, so $\mathbb{P}(A \cap B \cap C) = 0 \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$.
Key Insight: Pairwise checks do not control higher-order intersections.

Proposition 1.1.22. *Basic Independence Tests*
Let $A, B \in \mathcal{F}$.

(i) If $A \cap B = \emptyset$ and $\mathbb{P}(A)\mathbb{P}(B) > 0$, then $A \not\perp\!\!\!\perp B$.
(ii) If $A \perp\!\!\!\perp B$, then $A^c \perp\!\!\!\perp B$, $A \perp\!\!\!\perp B^c$, and $A^c \perp\!\!\!\perp B^c$.
(iii) If $\mathbb{P}(A) \in \{0, 1\}$, then $A \perp\!\!\!\perp C$ for every $C \in \mathcal{F}$.
(iv) If $A \perp\!\!\!\perp A$, then $\mathbb{P}(A) \in \{0, 1\}$.

Proof.

(i) Since $A \cap B = \emptyset$, $\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B)$.
(ii) If $A \perp\!\!\!\perp B$, then $\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = (1 - \mathbb{P}(A))\mathbb{P}(B) = \mathbb{P}(A^c)\mathbb{P}(B)$
and $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A)\mathbb{P}(B^c)$.
Applying the first identity to $A \perp\!\!\!\perp B^c$ gives $A^c \perp\!\!\!\perp B^c$.
(iii) Let $C \in \mathcal{F}$. If $\mathbb{P}(A) = 0$, then $0 \le \mathbb{P}(A \cap C) \le \mathbb{P}(A) = 0$, so $\mathbb{P}(A \cap C) = 0 = \mathbb{P}(A)\mathbb{P}(C)$.
If $\mathbb{P}(A) = 1$, then $\mathbb{P}(A^c) = 0$, hence $A^c \perp\!\!\!\perp C$ by the previous case and $A \perp\!\!\!\perp C$ by (ii).

(iv) Let $p = \mathbb{P}(A)$. From $A \perp\!\!\!\perp A$, $p = \mathbb{P}(A \cap A) = p^2$, hence $p \in \{0, 1\}$.

Structural Commentary: Independence is probabilistic, not visual; disjointness usually means strong dependence.
Key Insight: Complements preserve independence because B splits into $A \cap B$ and $A^c \cap B$. □

Example 1.1.23. *Independence, Complements, and First Success*

(i) *Independence by counting*: for a fair die, $\Omega = \{1, \dots, 6\}$, $A = \{2, 4, 6\}$, and $B = \{1, 2, 3, 4\}$.
Then $\mathbb{P}(A \cap B) = \frac{2}{6} = (\frac{3}{6})(\frac{4}{6}) = \mathbb{P}(A)\mathbb{P}(B)$, hence $A \perp\!\!\!\perp B$.

(ii) *Complement method*: for independent fair tosses $X_1, \dots, X_{10}$, let $A = \{\exists j \le 10 : X_j = H\}$ and $T_j = \{X_j = T\}$. Then $A^c = \bigcap_{j=1}^{10} T_j$ and $\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \prod_{j=1}^{10} \mathbb{P}(T_j) = 1 - (\frac{1}{2})^{10}$.

(iii) *First-success decomposition*: let $p_1 = \frac{1}{3}$, $p_2 = \frac{1}{4}$.
Let A_j be player 1's first success on turn j before player 2 succeeds.
Then $E = \bigsqcup_{j \ge 1} A_j$, $\mathbb{P}(A_j) = ((1 - p_1)(1 - p_2))^{j-1} p_1 = (\frac{1}{2})^{j-1}/3$, and $\mathbb{P}(E) = \sum_{j \ge 1} \mathbb{P}(A_j) = \frac{2}{3}$.

Method 1.1.24. *Independence, Complement, and Decomposition Methods*

(i) *Verify independence*: compute $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(A \cap B)$; conclude $A \perp\!\!\!\perp B$ iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

(ii) *Use complements*: if A^c is simpler, compute $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$.
If $A^c = \bigcap_{j=1}^{n} C_j$ with independent C_j, then $\mathbb{P}(A) = 1 - \prod_{j=1}^{n} \mathbb{P}(C_j)$.

(iii) *Disjoint decomposition*: write $E = \bigsqcup_j A_j$ with disjoint A_j, then $\mathbb{P}(E) = \sum_j \mathbb{P}(A_j)$.

(iv) *Geometric summation*: if $\mathbb{P}(A_j) = ar^{j-1}$ for $j \ge 1$ and $|r| < 1$, then $\mathbb{P}(E) = \sum_{j \ge 1} ar^{j-1} = \frac{a}{1-r}$.

Example 1.1.25. *Reliability and Match Probabilities*
Let A_i be the event that component i works, and assume all relevant events are mutually independent.

(i) Series: if $\mathbb{P}(A_i^c) = p$ for $1 \le i \le n$ and $W_s = \bigcap_{i=1}^{n} A_i$, then $\mathbb{P}(W_s) = (1 - p)^n$ and $\mathbb{P}(W_s^c) = 1 - (1 - p)^n$.

(ii) Parallel: if $W_p = \bigcup_{i=1}^{n} A_i$, then $\mathbb{P}(W_p^c) = \mathbb{P}(\bigcap_{i=1}^{n} A_i^c) = p^n$.

(iii) DNA match: let $N_j \in \{A, C, G, T\}$ with $\mathbb{P}(N_j = x) = p_{xj}$ for $j = 1, 2$.
If $N_1 \perp\!\!\!\perp N_2$, then $\mathbb{P}(N_1 = N_2) = \sum_{x \in \{A,C,G,T\}} p_{x1} p_{x2}$.

Key Insight: Independence multiplies intersections; complements handle failures; matches sum disjoint cases.

Definition 1.1.26. *Random Variables, Laws, and Distribution Functions*

(i) A *real-valued random variable* on $(\Omega, \mathcal{F}, \mathbb{P})$ is map $X : \Omega \to \mathbb{R}$ s.t. $\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}\ \forall B \in \mathcal{B}(\mathbb{R})$.

(ii) Its *law* or *distribution* is $\mathcal{L}(X) : \mathcal{B}(\mathbb{R}) \to [0, 1]$ given by $\mathcal{L}(X)(B) := \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B)$.

(iii) Its *cumulative distribution function* is $F_X : \mathbb{R} \to [0, 1]$ given by $F_X(x) := \mathbb{P}(X \le x) = \mathcal{L}(X)((-\infty, x])$.

Definition 1.1.27. *Distribution Functions and Densities*

(i) A *distribution function* is $F : \mathbb{R} \to [0, 1]$ such that $x \le y \Rightarrow F(x) \le F(y)$, $F(x) = \lim_{t \downarrow x} F(t)$ for all $x \in \mathbb{R}$, $\lim_{x \to -\infty} F(x) = 0$, and $\lim_{x \to \infty} F(x) = 1$.

(ii) A *density* for F is an integrable $g : \mathbb{R} \to [0, \infty)$ with $\int_{-\infty}^{\infty} g(x)\,dx = 1$ and $F(t) = \int_{-\infty}^{t} g(x)\,dx\ \forall t \in \mathbb{R}$.
If g is continuous, then $F'(t) = g(t)$.

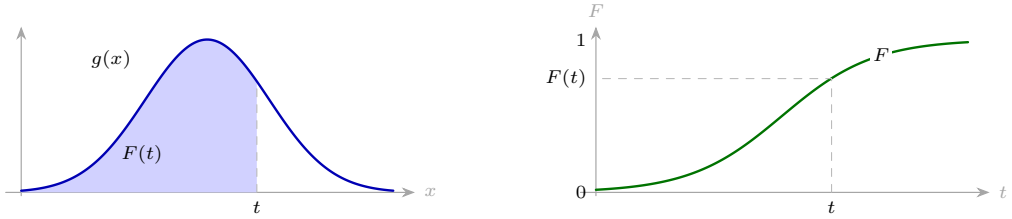


Figure 206: A normal density accumulates area into the distribution function $\Phi(t) = \int_{-\infty}^{t} g(x)\,dx$.

Definition 1.1.28. *Discrete and Continuous Random Variables*

(i) X is *discrete* if $X(\Omega) \subseteq S$ for some finite or countable $S = \{x_1, x_2, \dots\} \subseteq \mathbb{R}$.

(ii) The *probability mass function* of discrete X is $f_X : S \to [0,1]$, $f_X(x) := \mathbb{P}(X = x)$, $\sum_{x \in S} f_X(x) = 1$.

(iii) The *discrete CDF* is $F_X(x) = \mathbb{P}(X \le x) = \sum_{x_i \le x} f_X(x_i)$.

(iv) X is *continuous* if its law has a density $f_X : \mathbb{R} \to [0, \infty)$.

(v) The *continuous probabilities* satisfy $\mathbb{P}(a < X < b) = \int_a^b f_X(t)\,dt$ and $F_X(x) = \int_{-\infty}^x f_X(t)\,dt$.

(vi) The *point-mass distinction*: for continuous X, $\mathbb{P}(X = x) = 0$ for every fixed x, so $f_X(x) \neq \mathbb{P}(X = x)$.

A law may be mixed. It can have atoms and a continuous density on the remaining part.

Remark 1.1.29. *Mixed Laws Are Natural*
For a component lifetime X, one may have $\mathbb{P}(X = 0) = p > 0$ and, conditional on $X > 0$, a density g on $(0, \infty)$. Then $\mathcal{L}(X)$ has an atom plus a continuous part, so it is mixed.

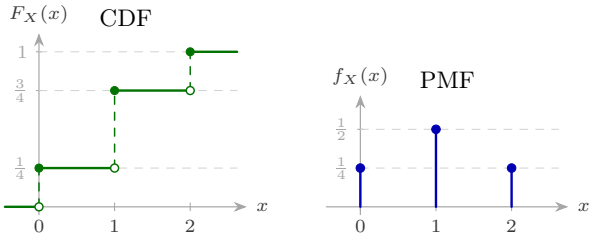


Figure 207: CDF and PMF for the number of heads in two independent fair coin tosses.

Example 1.1.30. *Counting Heads in Two Tosses*
Let $\Omega = \{H, T\}^2$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$, and set $X(\omega) = |\{j \in \{1,2\} : \omega_j = H\}|$.
Then $f_X(k) = \mathbb{P}(X = k) = \binom{2}{k} 2^{-2}$ for $k \in \{0,1,2\}$, so $(f_X(0), f_X(1), f_X(2)) = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$.
Hence $F_X(x) = 0$ for $x < 0$, $F_X(x) = \frac{1}{4}$ for $0 \le x < 1$, $F_X(x) = \frac{3}{4}$ for $1 \le x < 2$, $F_X(x) = 1$ for $x \ge 2$.
Key Insight: A CDF is defined for every real x, even when a discrete random variable takes only a few values.

Proposition 1.1.31. *CDF Probability Calculus*
Let X have CDF F and define $F(x^-) := \lim_{y \uparrow x} F(y)$.

(i) Point mass: $\mathbb{P}(X = x) = F(x) - F(x^-)$.
(ii) Half-open interval: if $x < y$, then $\mathbb{P}(x < X \le y) = F(y) - F(x)$.
(iii) Right tail: $\mathbb{P}(X > x) = 1 - F(x)$.
(iv) Continuous endpoints: if X is continuous and $a < b$, then $\mathbb{P}(a < X < b) = \mathbb{P}(a \le X < b) = \mathbb{P}(a < X \le b) = \mathbb{P}(a \le X \le b)$.

Proof.

(i) Let $E_n = \{x - \frac{1}{n} < X \le x\}$. Then $E_n \downarrow \{X = x\}$ and $\mathbb{P}(E_n) = F(x) - F(x - \frac{1}{n})$. Hence $\mathbb{P}(X = x) = \lim_n \mathbb{P}(E_n) = F(x) - F(x^-)$.

(ii) Since $\{X \le y\} = \{X \le x\} \sqcup \{x < X \le y\}$, $F(y) = F(x) + \mathbb{P}(x < X \le y)$.

(iii) Since $\{X > x\} = \{X \le x\}^c$, $\mathbb{P}(X > x) = 1 - \mathbb{P}(X \le x) = 1 - F(x)$.

(iv) If X is continuous, then $\mathbb{P}(X = t) = 0$ for all $t \in \mathbb{R}$. For $I_1 = (a,b)$, $I_2 = [a,b)$, $I_3 = (a,b]$, and $I_4 = [a,b]$, each $X^{-1}(I_j) \setminus X^{-1}(I_1) \subseteq \{X = a\} \cup \{X = b\}$, hence all four probabilities are equal.

Structural Commentary: CDF calculus subtracts nested lower-level events.
Key Insight: CDF jumps are point masses; continuous laws have none. □

Definition 1.1.32. *Quantiles and Equality in Distribution*

(i) For a CDF F, the *quantile function* is $F^{-1}(q) := \inf\{x \in \mathbb{R} : F(x) \ge q\}$, $q \in (0,1)$.

(ii) If F is continuous and strictly increasing, then F^{-1} is the ordinary inverse: $F(F^{-1}(q)) = q$ for $q \in (0,1)$ and $F^{-1}(F(x)) = x$ for $x \in \mathbb{R}$.

(iii) The *pth quantile* is $x_p = F^{-1}(p)$; under (ii), $F(x_p) = p$ uniquely. Here $x_{\frac{1}{2}}$ is the median and $x_{\frac{1}{4}}, x_{\frac{3}{4}}$ are the lower and upper quartiles.

(iv) Random variables X, Y are *equal in distribution*, written $X \overset{\text{dist}}{=} Y$, iff $F_X = F_Y$ on $\mathbb{R}$.

Example 1.1.33. *Value at Risk as a Quantile*
Let $V_0 > 0$, $R = \frac{V_1 - V_0}{V_0}$, and $F_R(x) = \mathbb{P}(R \le x)$.
The loss is $L = V_0 - V_1 = -V_0 R$, so $\{L \ge v\} = \{R \le -\frac{v}{V_0}\}$ and $\mathbb{P}(L \ge v) = F_R(-\frac{v}{V_0})$.

If F_R is invertible and $\mathbb{P}(L \geq v_\alpha) = \alpha$, then $v_\alpha = -V_0 F_R^{-1}(\alpha)$.
Key Insight: VaR is a lower-tail return quantile converted back into a dollar loss.

Definition 1.1.34. *Core Discrete Distributions*
Common discrete laws include:

(i) *Point mass*: $X \sim \delta_a$ for $a \in \mathbb{R}$ if $\mathbb{P}(X = a) = 1$.
(ii) *Discrete uniform*: $X \sim \text{Unif}\{1, \dots, k\}$ for $k \in \mathbb{N}$ if $\mathbb{P}(X = x) = \frac{1}{k}$ for $x = 1, \dots, k$.
(iii) *Bernoulli*: $X \sim \text{Bernoulli}(p)$ for $p \in [0,1]$ if $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$.
Equivalently $\mathbb{P}(X = x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$.
(iv) *Binomial*: $X \sim \text{Binomial}(n, p)$ for $n \in \mathbb{N}$ and $p \in [0,1]$ if $\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$ for $x = 0, \dots, n$.
(v) *Geometric*: $X \sim \text{Geom}(p)$, $p \in (0,1]$, if $\mathbb{P}(X = k) = p(1-p)^{k-1}$ for $k \geq 1$; X counts trials to first success.
(vi) *Negative binomial*: $X \sim \text{NegBin}(r, p)$, $r \in \mathbb{N}$, $p \in (0,1]$, if $\mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$ for $k \geq r$;
X counts trials to the rth success.
(vii) *Hypergeometric*: $X \sim \text{Hypergeom}(n, r, m)$ if $\mathbb{P}(X = k) = \frac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}$
for $\max\{0, m - n + r\} \leq k \leq \min\{r, m\}$.
(viii) *Poisson*: $X \sim \text{Poisson}(\lambda)$ for $\lambda \geq 0$ if $\mathbb{P}(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ for $x = 0, 1, 2, \dots$.

Definition 1.1.35. *Core Continuous Distributions*
Common continuous laws include:

(i) *Uniform*: $X \sim \text{Uniform}(a, b)$ for $a < b$ if $f_X(x) = (b-a)^{-1}\mathbf{1}_{[a,b]}(x)$.
(ii) *Normal*: $X \sim N(\mu, \sigma^2)$ for $\mu \in \mathbb{R}$, $\sigma > 0$, if $f_X(x) = (\sigma\sqrt{2\pi})^{-1} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$ for $x \in \mathbb{R}$.
$N(0,1)$ has density ϕ and CDF Φ.
(iii) *Exponential*: $X \sim \text{Exp}(\lambda)$ for $\lambda > 0$ if $f_X(x) = \lambda e^{-\lambda x}\mathbf{1}_{[0,\infty)}(x)$.
(iv) *Gamma*: $X \sim \text{Gamma}(\alpha, \lambda)$ for $\alpha, \lambda > 0$ if $f_X(x) = \lambda^\alpha \Gamma(\alpha)^{-1} x^{\alpha-1} e^{-\lambda x}\mathbf{1}_{[0,\infty)}(x)$; $\text{Exp}(\lambda) = \text{Gamma}(1, \lambda)$.
(v) *Beta*: $X \sim \text{Beta}(\alpha, \beta)$ for $\alpha, \beta > 0$ if $f_X(x) = \Gamma(\alpha+\beta)(\Gamma(\alpha)\Gamma(\beta))^{-1} x^{\alpha-1}(1-x)^{\beta-1}\mathbf{1}_{(0,1)}(x)$.
(vi) *Cauchy*: $X \sim \text{Cauchy}(0,1)$ if $f_X(x) = (\pi(1+x^2))^{-1}$ for $x \in \mathbb{R}$.
(vii) χ^2: $X \sim \chi^2_p$ for $p > 0$ if $f_X(x) = (\Gamma(\frac{p}{2})2^{\frac{p}{2}})^{-1} x^{\frac{p}{2}-1} e^{-\frac{x}{2}}\mathbf{1}_{(0,\infty)}(x)$.

Method 1.1.36. *Discrete Distribution Model Recipes*

(i) *Indicators*: for $A \in \mathcal{F}$, $\mathbf{1}_A \sim \text{Bernoulli}(\mathbb{P}(A))$.
(ii) *Binomial counts*: if $B_1, \dots, B_n$ are independent $\text{Bernoulli}(p)$ variables, then $\sum_{i=1}^n B_i \sim \text{Binomial}(n, p)$.
(iii) *Geometric waiting*: the probability of first success on trial k is $(1-p)^{k-1}p$.
(iv) *Negative-binomial waiting*: for iid $B_i \sim \text{Bernoulli}(p)$ and $T_r = \min\{k : \sum_{i=1}^k B_i = r\}$,
$\{T_r = k\} = \{B_k = 1, \sum_{i=1}^{k-1} B_i = r-1\}$, so $\mathbb{P}(T_r = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$.
(v) *Hypergeometric sampling*: if $M \subseteq [n]$, $|M| = r$, and S is a uniform m-subset,
then $X = |S \cap M|$ and $\mathbb{P}(X = k) = \frac{\binom{r}{k}\binom{n-r}{m-k}}{\binom{n}{m}}$ for $\max\{0, m-n+r\} \leq k \leq \min\{r, m\}$.

Key Insight: Name the event; PMFs come from independence or subset counts.

Proposition 1.1.37. *Poisson Limit of Rare Binomial Counts*
Let $\lambda > 0$, $k \in \mathbb{N}_0$, and $X_n \sim \text{Binomial}(n, \frac{\lambda}{n})$ for all sufficiently large n. Then $\lim_{n\to\infty} \mathbb{P}(X_n = k) = \frac{e^{-\lambda}\lambda^k}{k!}$.

Proof. For large n, $\mathbb{P}(X_n = k) = \binom{n}{k}(\frac{\lambda}{n})^k(1-\frac{\lambda}{n})^{n-k} = \frac{\lambda^k}{k!}\prod_{j=0}^{k-1}(1-\frac{j}{n})(1-\frac{\lambda}{n})^n(1-\frac{\lambda}{n})^{-k}$.
Since $\prod_{j=0}^{k-1}(1-\frac{j}{n}) \to 1$, $(1-\frac{\lambda}{n})^n \to e^{-\lambda}$, and $(1-\frac{\lambda}{n})^{-k} \to 1$, the limit is $\frac{e^{-\lambda}\lambda^k}{k!}$.
Structural Commentary: Rare binomial masses split into fixed-k choices and an exponential no-event factor.
Key Insight: Large n, small p, fixed np turns binomial masses into Poisson masses. □

Method 1.1.38. *Recursive Binomial and Poisson Mass Computation*
Let $q := 1 - p$.

(i) If $X \sim \text{Binomial}(n, p)$, $p_k := \mathbb{P}(X = k)$, and $q > 0$, then $p_0 = q^n$ and $\frac{p_k}{p_{k-1}} = \frac{(n-k+1)p}{kq}$ for $1 \leq k \leq n$.
(ii) If $X \sim \text{Poisson}(\lambda)$ and $p_k := \mathbb{P}(X = k)$, then $p_0 = e^{-\lambda}$ and $\frac{p_k}{p_{k-1}} = \frac{\lambda}{k}$ for $k \geq 1$.

Key Insight: Successive mass ratios avoid repeatedly evaluating large factorials.

Example 1.1.39. *Majority Decoding as a Binomial Tail*
Let $n = 2m+1$ and let $C_i \sim \text{Bernoulli}(p)$ be independent corruption indicators.
Then $X = \sum_{i=1}^{n} C_i \sim \text{Binomial}(n,p)$ and the majority decoder is correct iff $X \le m$.
For $n=5$, $p=0.1$, $\mathbb{P}(X \le 2) = 0.9^5 + 5(0.1)(0.9)^4 + 10(0.1)^2(0.9)^3 \approx 0.9914$.
Key Insight: Repetition coding converts reliability into a lower binomial tail.

Example 1.1.40. *Poisson Approximation for Rare Repeated Events*
Let $I_j = \mathbf{1}_{\{\text{double six on roll } j\}}$ for $1 \le j \le 100$.
Then $I_j \sim \text{Bernoulli}(\frac{1}{36})$ iid, $X = \sum_{j=1}^{100} I_j \sim \text{Binomial}(100, \frac{1}{36})$, and with $\lambda = \frac{100}{36} = \frac{25}{9}$, $\mathbb{P}(X=k) \approx \frac{e^{-\frac{25}{9}}(\frac{25}{9})^k}{k!}$.
Key Insight: Rare repeated event counts are often well approximated by a Poisson law with $\lambda = np$.

Proposition 1.1.41. *Exponential Memorylessness*
If $T \sim \text{Exp}(\lambda)$ with $\lambda > 0$, then $\mathbb{P}(T > s+t \mid T > s) = \mathbb{P}(T > t)$ for all $s, t \ge 0$.

Proof. For $u \ge 0$, $\mathbb{P}(T > u) = e^{-\lambda u}$. Hence $\mathbb{P}(T > s+t \mid T > s) = \frac{\mathbb{P}(T>s+t)}{\mathbb{P}(T>s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t)$.
Structural Commentary: Conditioning on survival to time s leaves the residual tail unchanged.
Key Insight: Exponential tails convert elapsed time into a cancellable factor. □

Remark 1.1.42. *Poisson Counts and Exponential Gaps*
In homogeneous Poisson process, counts satisfy $N(t) \sim \text{Poisson}(\lambda t)$; interarrival gaps satisfy $S_i \sim \text{Exp}(\lambda)$ iid.

Proposition 1.1.43. *Standardizing Normal Variables*
For $\sigma, \sigma_i > 0$:

(i) Standardization: if $X \sim N(\mu, \sigma^2)$, then $\frac{X-\mu}{\sigma} \sim N(0,1)$.
(ii) Affine reconstruction: if $Z \sim N(0,1)$, then $\mu + \sigma Z \sim N(\mu, \sigma^2)$.
(iii) Independent sums: if $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then $\sum_{i=1}^{n} X_i \sim N(\sum_{i=1}^{n} \mu_i, \sum_{i=1}^{n} \sigma_i^2)$.
(iv) Interval probabilities: if $X \sim N(\mu, \sigma^2)$ and $a < b$, then $\mathbb{P}(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$.

Proof.

(i) Let $Y = \frac{X-\mu}{\sigma}$. By change of variables, $f_Y(z) = \sigma f_X(\mu + \sigma z) = (2\pi)^{-\frac{1}{2}} e^{-\frac{z^2}{2}}$, so $Y \sim N(0,1)$.
(ii) Let $Y = \mu + \sigma Z$. Then $f_Y(y) = \sigma^{-1}\phi(\frac{y-\mu}{\sigma}) = (\sigma\sqrt{2\pi})^{-1}\exp(-\frac{(y-\mu)^2}{2\sigma^2})$, hence $Y \sim N(\mu, \sigma^2)$.
(iii) It suffices by induction to treat $n = 2$.
Let $X_i \sim N(\mu_i, v_i)$ be independent, $S = X_1 + X_2$, $v = v_1 + v_2$, $w = \frac{v_1 v_2}{v}$, and $a_s = \frac{v_2\mu_1 + v_1(s-\mu_2)}{v}$.
Then $f_S(s) = \int_{\mathbb{R}} f_1(u) f_2(s-u)\, du$, and completing the square gives
$f_1(u)f_2(s-u) = (2\pi v)^{-\frac{1}{2}} e^{-\frac{(s-\mu_1-\mu_2)^2}{2v}} (2\pi w)^{-\frac{1}{2}} e^{-\frac{(u-a_s)^2}{2w}}$.
The second factor integrates to 1, so $S \sim N(\mu_1+\mu_2, v_1+v_2)$; induction gives (iii).
(iv) Let $Z = \frac{X-\mu}{\sigma}$. Since $\sigma > 0$, $\{a < X < b\} = \{\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\}$.
Hence $\mathbb{P}(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$.

Structural Commentary: Normal laws are stable under affine maps and independent addition.
Key Insight: Standardize to $N(0,1)$, compute with Φ, then transform back. □

Example 1.1.44. *Normal Tail and Quantile Standardization*
Let $X \sim N(3,5)$ and $Z = \frac{X-3}{\sqrt{5}} \sim N(0,1)$. Then $\mathbb{P}(X > 1) = \mathbb{P}(Z > -\frac{2}{\sqrt{5}}) = 1 - \Phi(-\frac{2}{\sqrt{5}}) \approx 0.81$.
If $\mathbb{P}(X < q) = 0.2$, then $\Phi(\frac{q-3}{\sqrt{5}}) = 0.2$, so $q = 3 + \sqrt{5}\,\Phi^{-1}(0.2) \approx 3 - 0.8416\sqrt{5} \approx 1.1181$.

1.2 Joint Distributions, Multivariate Laws, and Transformations

Definition 1.2.1. *Joint and Marginal Distributions*

(i) The *joint mass function* of discrete X, Y is $f_{X,Y}(x,y) := \mathbb{P}(X = x, Y = y)$.
(ii) The *discrete marginals* are $f_X(x) = \sum_y f_{X,Y}(x,y)$ and $f_Y(y) = \sum_x f_{X,Y}(x,y)$.
(iii) The *continuous marginals* are $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy$, and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx$.
(X,Y) has joint density $f_{X,Y}$.
(iv) The *joint CDF* is $F_{X,Y}(x,y) := \mathbb{P}(X \le x, Y \le y)$.

Proposition 1.2.2. *Rectangular Increments of a Joint CDF*
Let $F(x,y) := \mathbb{P}(X \le x, Y \le y)$.
For $x_1 < x_2$ *and* $y_1 < y_2$, $\mathbb{P}(x_1 < X \le x_2,\ y_1 < Y \le y_2) = F(x_2,y_2) - F(x_1,y_2) - F(x_2,y_1) + F(x_1,y_1)$.
If (X,Y) *has density* $f_{X,Y}$ *and* $F \in C^2$, *then* $f_{X,Y}(x,y) = \partial_x \partial_y F(x,y)$ *and* $F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$.

Proof. Let $A_{ij} = \{X \le x_i, Y \le y_j\}$ and $R = \{x_1 < X \le x_2,\ y_1 < Y \le y_2\}$.
Pointwise, $\mathbf{1}_R = \mathbf{1}_{A_{22}} - \mathbf{1}_{A_{12}} - \mathbf{1}_{A_{21}} + \mathbf{1}_{A_{11}}$; taking expectations gives the four-corner identity.
If $F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{X,Y}(u,v)\,dv\,du$ and $F \in C^2$, the two-variable FTC gives $\partial_x\partial_y F = f_{X,Y}$.
Structural Commentary: Joint CDFs encode southwest rectangles; finite rectangles are four-corner increments.
Key Insight: Mixed derivatives are infinitesimal rectangular increments. □

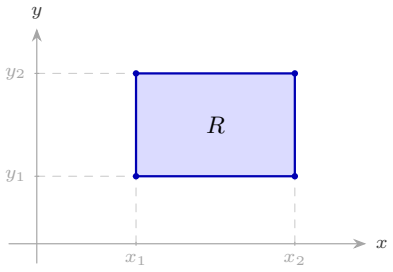


Figure 208: The rectangular probability is a four-term increment of the joint CDF.

Example 1.2.3. *A Three-Toss Joint Mass Table*
Let $\Omega = \{H,T\}^3$ with $\mathbb{P}(\{\omega\}) = 2^{-3}$, set $X = \mathbf{1}_{\{\omega_1 = H\}}$, $Y = \sum_{i=1}^{3} \mathbf{1}_{\{\omega_i = H\}}$, and write $p_{xy} := \mathbb{P}(X = x, Y = y)$.

	$Y=0$	$Y=1$	$Y=2$	$Y=3$
$X=0$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0
$X=1$	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$

Thus $p_{0,2} = \frac{1}{8}$, $\sum_y p_{0y} = \sum_y p_{1y} = \frac{1}{2}$, and $\sum_x p_{xj} = \binom{3}{j}2^{-3}$ for $j = 0,\dots,3$.

Definition 1.2.4. *Independent Random Variables and IID Samples*
(i) Random variables $X_1,\dots,X_n$ are *independent* if $\mathbb{P}(X_1 \in B_1,\dots,X_n \in B_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i)$ for all $B_i \in \mathcal{B}(\mathbb{R})$.
(ii) They are *identically distributed* if $\mathcal{L}(X_1) = \cdots = \mathcal{L}(X_n)$.
(iii) They are *iid* if they are independent and identically distributed.
(iv) The *random vector* $X = (X_1,\dots,X_n)$ is a *random sample* when $X_1,\dots,X_n$ are iid from a common law.

Proposition 1.2.5. *Factorization Tests for Independence*
(i) Discrete factorization: discrete X, Y *are independent iff* $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ *for all* x, y.
(ii) Density factorization: if (X,Y) *has joint density* $f_{X,Y}$ *and marginals* f_X, f_Y, *then* X, Y *are independent iff* $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ *a.e.*
(iii) Rectangular support test: if $f_{X,Y}(x,y) = g(x)h(y)\mathbf{1}_{A\times B}(x,y)$ *for measurable* A, B, *then* X, Y *are independent after normalization.*

Proof.
(*i*) (⇒) If $X \perp\!\!\!\perp Y$, then $f_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) = f_X(x)f_Y(y)$.
(*i*) (⇐) If $f_{X,Y} = f_X f_Y$, then for A, B, $\mathbb{P}(X \in A, Y \in B) = \sum_{x\in A}\sum_{y\in B} f_X(x)f_Y(y) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$.
(*ii*) (⇒) If $X \perp\!\!\!\perp Y$, the product law has density $f_X f_Y$, so uniqueness of densities gives $f_{X,Y} = f_X f_Y$ a.e.
(*ii*) (⇐) If $f_{X,Y} = f_X f_Y$ a.e., then for Borel A, B,
$\mathbb{P}(X \in A, Y \in B) = \int_A \int_B f_X(x)f_Y(y)\,dy\,dx = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$.
(*iii*) Let $G = \int_A g$ and $H = \int_B h$. Normalization gives $GH = 1$, while $f_X(x) = g(x)H\mathbf{1}_A(x)$ and $f_Y(y) = h(y)G\mathbf{1}_B(y)$, hence $f_X(x)f_Y(y) = f_{X,Y}(x,y)$ a.e.; apply (ii).
Structural Commentary: Independence is product structure of the joint law.
Key Insight: Independence means the joint law factors. □

Example 1.2.6. *A Non-Rectangular Bivariate Density*
Let $S = \{(x,y) : x^2 \le y \le 1\}$ and $f_{X,Y}(x,y) = cx^2y\mathbf{1}_S(x,y)$.
Since $S = \{(x,y) : -1 \le x \le 1,\ x^2 \le y \le 1\}$, $1 = c\int_{-1}^{1}\int_{x^2}^{1} x^2y\,dy\,dx = \frac{4c}{21}$, so $c = \frac{21}{4}$.
For $E = \{X \ge Y\}$, $S \cap E = \{(x,y) : 0 \le x \le 1,\ x^2 \le y \le x\}$, hence $\mathbb{P}(E) = (\frac{21}{4})\int_0^1\int_{x^2}^{x} x^2y\,dy\,dx = \frac{3}{20}$.
Key Insight: In bivariate integration, pick one variable first and let the other range over the corresponding slice.

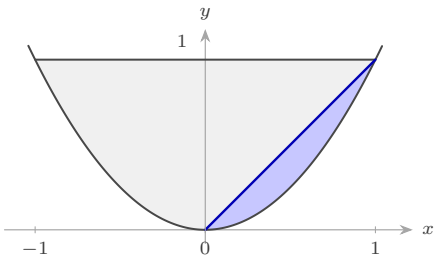


Figure 209: Gray: support $x^2 \le y \le 1$; blue: event $X \ge Y$ $(x^2 \le y \le x)$.

Definition 1.2.7. *Two Multivariate Families*

(i) *Multinomial*: for $n \in \mathbb{N}$ and $p \in [0,1]^k$ with $\sum_{j=1}^{k} p_j = 1$, $X \sim \text{Multinomial}(n,p)$
if $\mathbb{P}(X = x) = n!(x_1!\cdots x_k!)^{-1}\prod_{j=1}^{k} p_j^{x_j}$ for $x \in \mathbb{N}_0^k$ with $\sum_{j=1}^{k} x_j = n$; then $X_j \sim \text{Binomial}(n,p_j)$.

(ii) *Multivariate normal*: for $\mu \in \mathbb{R}^k$ and symmetric positive definite $\Sigma \in \mathbb{R}^{k\times k}$, $X \sim N(\mu,\Sigma)$
if $f_X(x) = ((2\pi)^k \det\Sigma)^{-\frac{1}{2}}\exp(-\frac{(x-\mu)^t\Sigma^{-1}(x-\mu)}{2})$ for $x \in \mathbb{R}^k$.
$N(0,I_k)$ has independent $N(0,1)$ coordinates.

Proposition 1.2.8. *Multivariate Normal Affine Standardization*
Let $k \in \mathbb{N}$, $\mu \in \mathbb{R}^k$, $\Sigma \in \mathbb{R}^{k\times k}$ with $\Sigma \succ 0$, and $A = \Sigma^{\frac{1}{2}}$. For $X \in \mathbb{R}^k$, $X \sim N(\mu,\Sigma)$ iff $A^{-1}(X-\mu) \sim N(0,I_k)$. Equivalently, if $Z \sim N(0,I_k)$, then $\mu + AZ \sim N(\mu,\Sigma)$.

Proof.
($\Rightarrow$) Set $Z = \Sigma^{-\frac{1}{2}}(X-\mu)$, so $x = \mu + \Sigma^{\frac{1}{2}}z$ and $|\det\Sigma^{\frac{1}{2}}| = (\det\Sigma)^{\frac{1}{2}}$.
By change of variables, $f_Z(z) = f_X(\mu+\Sigma^{\frac{1}{2}}z)(\det\Sigma)^{\frac{1}{2}} = (2\pi)^{-\frac{k}{2}}\exp(-\frac{z^tz}{2})$, hence $Z \sim N(0,I_k)$.
($\Leftarrow$) Let $X = \mu + \Sigma^{\frac{1}{2}}Z$ with $Z \sim N(0,I_k)$. Since $z = \Sigma^{-\frac{1}{2}}(x-\mu)$ and $|\det\Sigma^{-\frac{1}{2}}| = (\det\Sigma)^{-\frac{1}{2}}$,
$f_X(x) = f_Z(\Sigma^{-\frac{1}{2}}(x-\mu))(\det\Sigma)^{-\frac{1}{2}} = ((2\pi)^k\det\Sigma)^{-\frac{1}{2}}\exp(-\frac{(x-\mu)^t\Sigma^{-1}(x-\mu)}{2})$, so $X \sim N(\mu,\Sigma)$.
Structural Commentary: Positive definite covariance is an affine deformation of the standard normal.
Key Insight: Standardize by $\Sigma^{-\frac{1}{2}}$; reconstruct by $\Sigma^{\frac{1}{2}}$. □

Proposition 1.2.9. *Basic Multivariate Normal Calculus*
Let $X = (X_a, X_b) \sim N(\mu,\Sigma)$ with $X_a \in \mathbb{R}^r$, $X_b \in \mathbb{R}^s$, $\mu = (\mu_a,\mu_b)$, and $\Sigma = \begin{pmatrix}\Sigma_{aa} & \Sigma_{ab}\\ \Sigma_{ba} & \Sigma_{bb}\end{pmatrix} \succ 0$.

(i) Marginal: $X_a \sim N(\mu_a,\Sigma_{aa})$.
(ii) Conditional: $X_b \mid X_a = x_a \sim N(\mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a-\mu_a), \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab})$.
(iii) Linear form: for any vector a, $a^tX \sim N(a^t\mu, a^t\Sigma a)$.

Proof.
(i) Let $P_aX = X_a$. By Proposition 1.2.8, $X = \mu + \Sigma^{\frac{1}{2}}Z$ with $Z \sim N(0,I)$.
Hence $X_a = P_a\mu + P_a\Sigma^{\frac{1}{2}}Z \sim N(\mu_a, P_a\Sigma P_a^t) = N(\mu_a,\Sigma_{aa})$.
(ii) Set $S := \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}$ and $m(x_a) := \mu_b + \Sigma_{ba}\Sigma_{aa}^{-1}(x_a-\mu_a)$.
The block determinant and inverse identities give $\det\Sigma = \det\Sigma_{aa}\det S$ and
$(x-\mu)^t\Sigma^{-1}(x-\mu) = (x_a-\mu_a)^t\Sigma_{aa}^{-1}(x_a-\mu_a) + (x_b-m(x_a))^tS^{-1}(x_b-m(x_a))$.
Thus $f_{b|a}(x_b \mid x_a) = \frac{f_{a,b}(x_a,x_b)}{f_a(x_a)}$ is the $N(m(x_a),S)$ density.
(iii) With $X = \mu + \Sigma^{\frac{1}{2}}Z$, $a^tX = a^t\mu + (\Sigma^{\frac{1}{2}}a)^tZ$.
Since Z has independent $N(0,1)$ coordinates, $(\Sigma^{\frac{1}{2}}a)^tZ \sim N(0,a^t\Sigma a)$, hence $a^tX \sim N(a^t\mu, a^t\Sigma a)$.

Structural Commentary: Marginals, conditionals, and linear projections of Gaussian vectors remain Gaussian.
Key Insight: Gaussian calculus is affine calculus plus Schur complements. □

Definition 1.2.10. *Copulas*
A *copula* is a joint CDF $C : [0,1]^2 \to [0,1]$ with uniform marginals: $C(u,1) = u$ and $C(1,v) = v$ for $u, v \in [0,1]$. If $C \in C^2$, its *copula density* is $c(u,v) := \partial_u \partial_v C(u,v)$.

Proposition 1.2.11. *Constructing a Joint Law from Marginals and a Copula*
Let F_X, F_Y be continuous CDFs with densities f_X, f_Y, and let C be a copula.
Define $F_{X,Y}(x,y) := C(F_X(x), F_Y(y))$. Then $F_{X,Y}$ has marginals F_X, F_Y.
If $C \in C^2$ with density $c = \partial_u \partial_v C$, then $f_{X,Y}(x,y) = c(F_X(x), F_Y(y)) f_X(x) f_Y(y)$.

Proof. $F_{X,Y}(x,\infty) = C(F_X(x), 1) = F_X(x)$ and $F_{X,Y}(\infty, y) = C(1, F_Y(y)) = F_Y(y)$.
If densities exist, $\partial_x \partial_y C(F_X(x), F_Y(y)) = (\partial_u \partial_v C)(F_X(x), F_Y(y)) f_X(x) f_Y(y)$ by the chain rule.
Structural Commentary: Marginals and dependence are separated into F_X, F_Y and C.
Key Insight: Marginals alone do not determine a joint law; the copula supplies the dependence. □

Example 1.2.12. *Farlie–Morgenstern Copulas*
For $|\alpha| \le 1$, $C_\alpha(u,v) = uv\{1 + \alpha(1-u)(1-v)\}$ is a copula with density $c_\alpha(u,v) = 1 + \alpha(1-2u)(1-2v)$. Here $\alpha = 0$ gives the independence copula uv, while $\alpha = -1$ and $\alpha = 1$ give densities $2u + 2v - 4uv$ and $2 - 2u - 2v + 4uv$ on $[0,1]^2$, respectively; all have uniform marginals.

Method 1.2.13. *One-Variable Transformations*
Let $Y = r(X)$.

(i) *Discrete pushforward*: if X is discrete, then $f_Y(y) = \mathbb{P}(Y = y) = \mathbb{P}(X \in r^{-1}(\{y\})) = \sum_{x : r(x) = y} f_X(x)$.

(ii) *CDF method*: for $Y = r(X)$, compute $F_Y(y) = \int_{\{x : r(x) \le y\}} f_X(x)\, dx$; then $f_Y(y) = F_Y'(y)$ where differentiable.

(iii) *Monotone change*: if r is strictly monotone on $\operatorname{supp} X$ with inverse s, then $f_Y(y) = f_X(s(y))\, |s'(y)|\, \mathbf{1}_{r(\operatorname{supp} X)}(y)$.

Key Insight: The CDF method works because it translates the transformed event back into an event about X.

Example 1.2.14. *Log Transform of an Exponential Variable*
Let $X \sim \text{Exp}(1)$ and $Y = \log X$. Since $r(x) = \log x$ has inverse $s(y) = e^y$ with $s'(y) = e^y$, the change-of-variables formula gives $f_Y(y) = f_X(e^y) e^y = e^y e^{-e^y}$ for $y \in \mathbb{R}$.

Example 1.2.15. *A Nonmonotone Square Transform*
Let $X \sim \text{Uniform}(-1,3)$ and $Y = X^2$. Since $f_X = \frac{1}{4}$ on $[-1,3]$, $r^{-1}(\{y\}) \cap [-1,3] = \{-\sqrt{y}, \sqrt{y}\}$ for $0 < y < 1$ and $\{\sqrt{y}\}$ for $1 < y < 9$. Hence $f_Y(y) = \frac{1}{4\sqrt{y}} \mathbf{1}_{(0,1)}(y) + \frac{1}{8\sqrt{y}} \mathbf{1}_{(1,9)}(y)$.
Key Insight: Non-one-to-one transformations require splitting the range into cases.

Example 1.2.16. *Squaring a Standard Normal*
Let $Z \sim N(0,1)$ and $X = Z^2$. For $x > 0$, $F_X(x) = \mathbb{P}(|Z| \le \sqrt{x}) = \Phi(\sqrt{x}) - \Phi(-\sqrt{x})$,
so $f_X(x) = x^{-\frac{1}{2}} \phi(\sqrt{x}) = (2\pi)^{-\frac{1}{2}} x^{-\frac{1}{2}} e^{-\frac{x}{2}}$. Hence $X \sim \chi_1^2 = \text{Gamma}(\frac{1}{2}, \frac{1}{2})$ in the rate convention.

Example 1.2.17. *Reciprocal of a Uniform Variable*
Let $U \sim \text{Uniform}(0,1)$ and $V = \frac{1}{U}$. Then $F_V(v) = 0$ for $v < 1$, while for $v \ge 1$, $F_V(v) = \mathbb{P}(U \ge \frac{1}{v}) = 1 - \frac{1}{v}$.
Hence $f_V(v) = v^{-2} \mathbf{1}_{[1,\infty)}(v)$.

Proposition 1.2.18. *Probability Integral Transform and Inverse Transform*
Let $F : I \to (0,1)$ be a continuous strictly increasing CDF with inverse $F^{-1} : (0,1) \to I$.

(i) If $X \sim F$, then $F(X) \sim \text{Uniform}(0,1)$.
(ii) If $U \sim \text{Uniform}(0,1)$ and $Y = F^{-1}(U)$, then $F_Y = F$.

Proof.

(i) For $0 < u < 1$, $\mathbb{P}(F(X) \le u) = \mathbb{P}(X \le F^{-1}(u)) = F(F^{-1}(u)) = u$, with endpoint values $0, 1$.
(ii) For $x \in I$, $\mathbb{P}(Y \le x) = \mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$.

Structural Commentary: F maps its law to uniform scale; F^{-1} maps uniform scale back.
Key Insight: Uniform random numbers become target laws by applying inverse CDFs. □

Method 1.2.19. *Inverse-Transform Generation*

(i) *Exponential*: if $U \sim \text{Uniform}(0,1)$ and $\lambda > 0$, then $T = -\lambda^{-1} \log(1-U) \sim \text{Exp}(\lambda)$. (or $T = -\lambda^{-1} \log U$)

(ii) *Weibull*: if $F(x) = 1 - \exp(-(\frac{x}{\alpha})^\beta)$ for $x \ge 0$ and $\alpha, \beta > 0$,
then $F^{-1}(u) = \alpha[-\log(1-u)]^{\frac{1}{\beta}}$ and $W = \alpha[-\log(1-U)]^{\frac{1}{\beta}} \sim F$.

(iii) *Integer-valued inverse CDF*: if F is integer-valued and $U \sim \text{Uniform}(0,1)$, set $Y = \inf\{k \in \mathbb{Z} : F(k) \ge U\}$.
Equivalently $F(k-1) < U \le F(k)$. Then $F_Y = F$.

Key Insight: The simulation problem is reduced to solving $u = F(x)$ for x.

Method 1.2.20. *Several-Variable Transformations*
For $Z = r(X, Y)$ with joint density $f_{X,Y}$:

(i) *Sublevel set*: define $A_z := \{(x, y) : r(x, y) \le z\}$.
(ii) *CDF integral*: compute $F_Z(z) = \mathbb{P}(Z \le z) = \iint_{A_z} f_{X,Y}(x, y)\, dx\, dy$.
(iii) *Density*: differentiate where possible, $f_Z(z) = F_Z'(z)$.

Proposition 1.2.21. *Two-Dimensional Density Change of Variables*
Let (X, Y) have density $f_{X,Y}$ on $D \subseteq \mathbb{R}^2$, and let $T : D \to E$ be one-to-one C^1 with inverse $h : E \to D$ and $\det DT \neq 0$ on D. If $(U, V) = T(X, Y)$, then $f_{U,V}(u, v) = f_{X,Y}(h(u, v))\, |\det Dh(u, v)|\, \mathbf{1}_E(u, v)$.

Proof. For Borel $B \subseteq E$, $\mathbb{P}((U, V) \in B) = \mathbb{P}((X, Y) \in h(B))$
$= \int_{h(B)} f_{X,Y}(x, y)\, dx\, dy = \int_B f_{X,Y}(h(u, v))\, |\det Dh(u, v)|\, du\, dv$ (by Theorem 5.1.13)
Structural Commentary: Transforming a density means preserving probability mass while rescaling area.
Key Insight: The absolute Jacobian accounts for local volume distortion. □

Example 1.2.22. *Sums and Quotients*
Let (X, Y) have joint density $f_{X,Y}$.

(i) Sum: for $S = X + Y$, $f_S(s) = \int_{\mathbb{R}} f_{X,Y}(x, s - x)\, dx$; if $X \perp\!\!\!\perp Y$, then $f_S(s) = \int_{\mathbb{R}} f_X(x) f_Y(s - x)\, dx$.
(ii) Quotient: for $Z = \frac{Y}{X}$, $f_Z(z) = \int_{\mathbb{R}} |x|\, f_{X,Y}(x, xz)\, dx$; if $X \perp\!\!\!\perp Y$, then $f_Z(z) = \int_{\mathbb{R}} |x|\, f_X(x) f_Y(xz)\, dx$.

Key Insight: Add one auxiliary coordinate, change variables, integrate it out.

Example 1.2.23. *Sum of Two Independent Uniforms*
Let $X_1, X_2 \sim \text{Uniform}(0, 1)$ be independent and $Y = X_1 + X_2$.
Since $f_{X_1,X_2} = \mathbf{1}_{[0,1]^2}$, $F_Y(y)$ is the area of $A_y = \{(x_1, x_2) \in [0, 1]^2 : x_1 + x_2 \le y\}$,
so $F_Y(y) = \frac{y^2}{2}\mathbf{1}_{[0,1]}(y) + (1 - \frac{(2-y)^2}{2})\mathbf{1}_{(1,2]}(y) + \mathbf{1}_{(2,\infty)}(y)$. Hence $f_Y(y) = y\mathbf{1}_{(0,1)}(y) + (2 - y)\mathbf{1}_{(1,2)}(y)$.

Example 1.2.24. *Polar Transform of Two Standard Normals*
Let $X, Y \overset{\text{iid}}{\sim} N(0, 1)$, $R = (X^2 + Y^2)^{\frac{1}{2}}$, and $\Theta = \text{atan2}(Y, X)$.
Polar change of variables gives $f_{R,\Theta}(r, \theta) = (2\pi)^{-1} r e^{-\frac{r^2}{2}} \mathbf{1}_{[0,\infty)}(r)\mathbf{1}_{[0,2\pi)}(\theta)$,
so $R \perp\!\!\!\perp \Theta$, $\Theta \sim \text{Uniform}(0, 2\pi)$, and $R^2 \sim \text{Exp}(\frac{1}{2})$.
Hence for iid $U_1, U_2 \sim \text{Uniform}(0, 1)$, $(\sqrt{-2\log U_1}\cos(2\pi U_2), \sqrt{-2\log U_1}\sin(2\pi U_2)) \sim N(0, I_2)$.

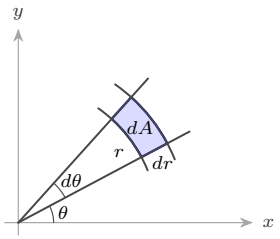


Figure 210: In polar coordinates, a small patch has area approximately $r\, dr\, d\theta$.

1.3 Expectation and Integration in Probability

Expectation is the averaging operation attached to a probability law. In the notation of Section 1.1.1, if X has law F_X, then integration against F_X is the common notation that covers both sums for discrete variables and ordinary integrals for variables with densities.

Definition 1.3.1. *Expectation Through the Law*
Let X be real-valued with distribution function F_X.

(i) *Expectation*: if $\int_{\mathbb{R}} |x|\, dF_X(x) < \infty$, then $\mathbb{E}(X) := \int_{\mathbb{R}} x\, dF_X(x)$.
(ii) *Discrete form*: if X has support S, mass function f_X, $\mathbb{E}(X) = \sum_{x \in S} x f_X(x)$ when $\sum_{x \in S} |x| f_X(x) < \infty$.
(iii) *Density form*: if X has density f_X, then $\mathbb{E}(X) = \int_{\mathbb{R}} x f_X(x)\, dx$ when $\int_{\mathbb{R}} |x| f_X(x)\, dx < \infty$.
(iv) *Function form*: for measurable $r : \mathbb{R} \to \mathbb{R}$, if $\int_{\mathbb{R}} |r(x)|\, dF_X(x) < \infty$, then $\mathbb{E}(r(X)) := \int_{\mathbb{R}} r(x)\, dF_X(x)$.

Example 1.3.2. *Expectation of the Two-Toss Head Count*
Let $\Omega = \{H,T\}^2$ with $\mathbb{P}(\{\omega\}) = \frac{1}{4}$, and let $X(\omega)$ be the number of heads in ω.
Then $f_X(k) = \binom{2}{k}2^{-2}$ for $k = 0,1,2$, so $\mathbb{E}(X) = \sum_{k=0}^{2} k f_X(k) = 0\cdot\frac{1}{4} + 1\cdot\frac{1}{2} + 2\cdot\frac{1}{4} = 1$.

Definition 1.3.3. *Variance and Covariance*
(i) The *variance* of X is $\operatorname{Var}(X) := \mathbb{E}[(X - \mathbb{E}X)^2]$, assuming $\mathbb{E}(X^2) < \infty$.
(ii) The *standard deviation* of X is $\operatorname{sd}(X) := \sqrt{\operatorname{Var}(X)}$.
(iii) The *covariance* of X, Y is $\operatorname{Cov}(X,Y) := \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, assuming $\mathbb{E}(X^2), \mathbb{E}(Y^2) < \infty$.

Proposition 1.3.4. *Indicators Turn Probabilities into Expectations*
For $A \in \mathcal{F}$, let $\mathbf{1}_A : \Omega \to \{0,1\}$ be $\mathbf{1}_A(\omega) = 1$ for $\omega \in A$ and $\mathbf{1}_A(\omega) = 0$ for $\omega \notin A$. Then $\mathbb{E}(\mathbf{1}_A) = \mathbb{P}(A)$.

Proof. Since $\Omega = A \sqcup A^c$ and $\mathbf{1}_A = 1$ on A, 0 on A^c, $\mathbb{E}(\mathbf{1}_A) = 1\cdot\mathbb{P}(A) + 0\cdot\mathbb{P}(A^c) = \mathbb{P}(A)$.
Structural Commentary: Indicators encode events as 0-1 random variables.
Key Insight: Probabilities are expectations of indicators. □

Remark 1.3.5. *Law-Based Integration*
For measurable r, $\mathbb{E}(r(X))$ depends only on the law of X: $\mathbb{E}(r(X)) = \int_{\mathbb{R}} r(x)\, dF_X(x)$ whenever integrable.
Discrete laws give $\sum_x r(x) f_X(x)$; density laws give $\int_{\mathbb{R}} r(x) f_X(x)\, dx$.

Example 1.3.6. *First Expectation Computations*
(i) If $X \sim \text{Bernoulli}(p)$, then $\mathbb{E}X = 0(1-p) + 1p = p$.
(ii) If $X \sim \text{Uniform}(-1,3)$, then $\mathbb{E}X = (\frac{1}{4})\int_{-1}^{3} x\, dx = 1$.
(iii) If $X \sim \text{Cauchy}(0,1)$, then $\mathbb{E}|X| = (\frac{2}{\pi})\int_0^\infty x(1+x^2)^{-1}\, dx = \infty$, so $\mathbb{E}X$ does not exist.

Theorem 1.3.7. *Expectation of a Function*
Let r be measurable and assume the displayed expectations exist.
(i) One variable: for real-valued X, $\mathbb{E}(r(X)) = \int_{\mathbb{R}} r(x)\, dF_X(x)$.
(ii) Several variables: for (X,Y) with joint law $F_{X,Y}$, $\mathbb{E}(r(X,Y)) = \int_{\mathbb{R}^2} r(x,y)\, dF_{X,Y}(x,y)$.

Proof.
(i) Since $\mathcal{L}(X) = X_*\mathbb{P}$, $\mathbb{E}(r(X)) = \int_\Omega r(X(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} r(x)\, d\mathcal{L}(X)(x) = \int_{\mathbb{R}} r(x)\, dF_X(x)$.
(ii) Let $W = (X,Y)$. Since $\mathcal{L}(W) = W_*\mathbb{P} = F_{X,Y}$,
$\mathbb{E}(r(X,Y)) = \int_\Omega r(W(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}^2} r(x,y)\, d\mathcal{L}(W)(x,y) = \int_{\mathbb{R}^2} r(x,y)\, dF_{X,Y}(x,y)$.
Structural Commentary: Expectations of transformations integrate over the transformed law.
Key Insight: Pushforward laws let functions move from random variables to their values. □

Example 1.3.8. *Using the Function Rule*
(i) If $X \sim \text{Uniform}(0,1)$ and $Y = e^X$, then $\mathbb{E}Y = \mathbb{E}(e^X) = \int_0^1 e^x\, dx = e - 1$.
(ii) If $X \sim \text{Uniform}(0,1)$ and $Y = \max\{X, 1-X\}$, then $\mathbb{E}Y = \int_0^{\frac{1}{2}}(1-x)\, dx + \int_{\frac{1}{2}}^1 x\, dx = \frac{3}{4}$.
(iii) If $(X,Y) \sim \text{Uniform}([0,1]^2)$ and $Z = X^2 + Y^2$, then $\mathbb{E}Z = \int_0^1\int_0^1 (x^2+y^2)\, dx\, dy = \frac{2}{3}$.

Definition 1.3.9. *Moments, Correlation, and Sample Statistics*
(i) The *kth moment* of X is $\mathbb{E}(X^k)$, assuming $\mathbb{E}|X|^k < \infty$.
(ii) The *correlation* of X,Y is $\operatorname{Corr}(X,Y) := \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)}\sqrt{\operatorname{Var}(Y)}}$, assuming $\operatorname{Var}(X), \operatorname{Var}(Y) > 0$.
(iii) The *sample mean* of $X_1,\dots,X_n$ is $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$.
(iv) The *sample variance* of $X_1,\dots,X_n$ is $S_n^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X}_n)^2$.

Theorem 1.3.10. *Expectation, Variance, and Covariance Algebra*
Assume all displayed quantities exist and $a_i, a, b \in \mathbb{R}$.
(i) Linearity: $\mathbb{E}(\sum_i a_i X_i) = \sum_i a_i \mathbb{E}X_i$.
(ii) Independent product: if $X_1,\dots,X_n$ are independent, then $\mathbb{E}(\prod_i X_i) = \prod_i \mathbb{E}X_i$.
(iii) Variance identities: $\operatorname{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$ and $\operatorname{Var}(aX+b) = a^2\operatorname{Var}(X)$.
(iv) Covariance identity: $\operatorname{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y$.

(v) Variance of a linear combination: $\operatorname{Var}(\sum_i a_i X_i) = \sum_i a_i^2 \operatorname{Var}(X_i) + 2\sum_{i<j} a_i a_j \operatorname{Cov}(X_i, X_j)$.

Proof.

(i) $\mathbb{E}(\sum_i a_i X_i) = \int_\Omega \sum_i a_i X_i \, d\mathbb{P} = \sum_i a_i \int_\Omega X_i \, d\mathbb{P} = \sum_i a_i \mathbb{E}X_i$.

(ii) By independence, $\mathcal{L}(X_1, \dots, X_n) = \prod_i F_{X_i}$; $\mathbb{E}(\prod_i X_i) = \int_{\mathbb{R}^n} \prod_i x_i \, d\prod_i F_{X_i}(x_i) = \prod_i \int_{\mathbb{R}} x_i \, dF_{X_i}(x_i) = \prod_i \mathbb{E}X_i$.

(iii) $\operatorname{Var}(X) = \mathbb{E}(X^2 - 2X\mathbb{E}X + (\mathbb{E}X)^2) = \mathbb{E}(X^2) - (\mathbb{E}X)^2$.
Also $\operatorname{Var}(aX + b) = \mathbb{E}[(aX + b - a\mathbb{E}X - b)^2] = a^2 \operatorname{Var}(X)$.

(iv) $\operatorname{Cov}(X, Y) = \mathbb{E}(XY - X\mathbb{E}Y - Y\mathbb{E}X + \mathbb{E}X\mathbb{E}Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y$.

(v) With $S = \sum_i a_i X_i$, (iv) gives
$\operatorname{Var}(S) = \operatorname{Cov}(S, S) = \sum_{i,j} a_i a_j \operatorname{Cov}(X_i, X_j) = \sum_i a_i^2 \operatorname{Var}(X_i) + 2\sum_{i<j} a_i a_j \operatorname{Cov}(X_i, X_j)$.

Structural Commentary: Expectation is linear; variance and covariance are quadratic and bilinear consequences.
Key Insight: Expand products, then use linearity. □

Example 1.3.11. *Indicator Sums as a Method*

(i) If $X \sim \text{Binomial}(n, p)$, write $X = \sum_{i=1}^n I_i$ with independent $I_i \sim \text{Bernoulli}(p)$.
Then $\mathbb{E}X = np$ and $\operatorname{Var}(X) = np(1-p)$.

(ii) Let $\xi_i \in \{-1, 1\}$ with $\mathbb{P}(\xi_i = -1) = p$ and $\mathbb{P}(\xi_i = 1) = 1 - p$, independently, and $X_n = \sum_{i=1}^n \xi_i$.
Then $\mathbb{E}X_n = n(1-2p)$ and $\operatorname{Var}(X_n) = 4np(1-p)$.

(iii) Let $F_n = c\prod_{i=1}^n M_i$ with independent $M_i \in \{2, \frac{1}{2}\}$ and $\mathbb{P}(M_i = 2) = \mathbb{P}(M_i = \frac{1}{2}) = \frac{1}{2}$.
Then $\mathbb{E}F_n = c\prod_{i=1}^n \mathbb{E}M_i = c(\frac{5}{4})^n$.

Proposition 1.3.12. *Zero Variance Means Almost Surely Constant*
For $X \in L^2$ and $m := \mathbb{E}X$, $\operatorname{Var}(X) = 0$ iff $X = m$ a.s.

Proof.
($\Rightarrow$) If $\operatorname{Var}(X) = 0$, then $\mathbb{E}[(X-m)^2] = 0$. Since $(X-m)^2 \geq 0$, $\mathbb{P}((X-m)^2 > 0) = 0$, hence $X = m$ a.s.
($\Leftarrow$) If $X = m$ a.s., then $(X-m)^2 = 0$ a.s., so $\operatorname{Var}(X) = \mathbb{E}[(X-m)^2] = 0$.

Structural Commentary: Variance is the mean squared distance from the mean.
Key Insight: Zero nonnegative expectation forces zero almost surely. □

Proposition 1.3.13. *Tail Integral Formula*
If $X \geq 0$ has density f_X, distribution function F_X, and $\mathbb{E}X < \infty$,
then $\mathbb{E}X = \int_0^\infty \mathbb{P}(X > x)\, dx = \int_0^\infty (1 - F_X(x))\, dx$.

Proof. For $b > 0$, integration by parts gives $\int_0^b x f_X(x)\, dx = -b(1 - F_X(b)) + \int_0^b (1 - F_X(x))\, dx$.
Since $\mathbb{E}X < \infty$, $b(1 - F_X(b)) \leq \int_b^\infty x f_X(x)\, dx \to 0$.
Letting $b \to \infty$ yields $\mathbb{E}X = \int_0^\infty (1 - F_X(x))\, dx = \int_0^\infty \mathbb{P}(X > x)\, dx$.
Structural Commentary: A nonnegative variable is integrated by summing its survival probabilities.
Key Insight: Expectation equals area under the tail. □

Example 1.3.14. *Maximum of Uniform Samples*
Let $X_1, \dots, X_n$ be independent $\text{Uniform}(0, 1)$ variables and $Y_n = \max_i X_i$.
Then $F_{Y_n}(y) = \mathbb{P}(X_1 \leq y, \dots, X_n \leq y) = y^n$ for $0 \leq y \leq 1$.
So $\mathbb{E}Y_n = \int_0^1 (1 - F_{Y_n}(y))\, dy = \int_0^1 (1 - y^n)\, dy = \frac{n}{n+1}$.

Theorem 1.3.15. *Mean and Variance of the Basic Sample Statistics*
Let $X_1, \dots, X_n$ be iid with $\mathbb{E}X_i = \mu$ and $\operatorname{Var}(X_i) = \sigma^2 < \infty$.

(i) Sample-mean expectation: $\mathbb{E}\overline{X}_n = \mu$.

(ii) Sample-mean variance: $\operatorname{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$.

(iii) Unbiased sample variance: $\mathbb{E}S_n^2 = \sigma^2$.

Proof.

(i) $\mathbb{E}\overline{X}_n = n^{-1}\sum_{i=1}^{n}\mathbb{E}X_i = n^{-1}n\mu = \mu$.

(ii) Independence gives $\mathrm{Var}(\overline{X}_n) = n^{-2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = n^{-2}n\sigma^2 = \frac{\sigma^2}{n}$.

(iii) Since $\sum_{i=1}^{n}(X_i-\overline{X}_n)^2 = \sum_{i=1}^{n}(X_i-\mu)^2 - n(\overline{X}_n-\mu)^2$. Thus $\mathbb{E}[(n-1)S_n^2] = n\sigma^2 - n(\frac{\sigma^2}{n}) = (n-1)\sigma^2$.

Structural Commentary: Averaging preserves mean and scales variance.
Key Insight: Center at μ, then subtract the fluctuation of $\overline{X}_n$. □

Definition 1.3.16. *Mean Vector and Covariance Matrix*
For a random vector $X = (X_1,\dots,X_k)^t$:

(i) *Mean vector*: $\mathbb{E}X := (\mathbb{E}X_1,\dots,\mathbb{E}X_k)^t$.
(ii) *Covariance matrix*: $\mathrm{Var}(X) := [\mathrm{Cov}(X_i,X_j)]_{i,j=1}^{k}$.

Proposition 1.3.17. *Linear Transformations of Random Vectors*
Let $X \in \mathbb{R}^k$ satisfy $\mathbb{E}X = \mu$ and $\mathrm{Var}(X) = \Sigma$.

i. Linear form: for $a \in \mathbb{R}^k$, $\mathbb{E}(a^tX) = a^t\mu$ and $\mathrm{Var}(a^tX) = a^t\Sigma a$.
ii. Matrix transform: for $A \in \mathbb{R}^{m\times k}$, $\mathbb{E}(AX) = A\mu$ and $\mathrm{Var}(AX) = A\Sigma A^t$.

Proof.

i. $\mathbb{E}(a^tX) = a^t\mathbb{E}X = a^t\mu$, and $\mathrm{Var}(a^tX) = \mathbb{E}[(a^t(X-\mu))^2] = a^t\mathbb{E}[(X-\mu)(X-\mu)^t]a = a^t\Sigma a$.
ii. $\mathbb{E}(AX) = A\mathbb{E}X = A\mu$, and $\mathrm{Var}(AX) = \mathbb{E}[(AX-A\mu)(AX-A\mu)^t] = A\mathbb{E}[(X-\mu)(X-\mu)^t]A^t = A\Sigma A^t$.

Structural Commentary: Linear maps push means forward and conjugate covariance matrices.
Key Insight: Center first; then move constants outside expectations. □

Example 1.3.18. *Multinomial Covariance Structure*
If $X \sim \mathrm{Multinomial}(n,p)$ with $p = (p_1,\dots,p_k)$, then $\mathbb{E}X = np$ and $\mathrm{Var}(X_i) = np_i(1-p_i)$.
For $i \neq j$, $X_i + X_j \sim \mathrm{Binomial}(n, p_i+p_j)$, so $n(p_i+p_j)(1-p_i-p_j) = np_i(1-p_i) + np_j(1-p_j) + 2\,\mathrm{Cov}(X_i,X_j)$.
Hence $\mathrm{Cov}(X_i,X_j) = -np_ip_j$.

Theorem 1.3.19. *Markov and Chebyshev Inequalities*

i. Markov: if $X \geq 0$ and $\mathbb{E}X < \infty$, then for $t > 0$, $\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}X}{t}$.
ii. Chebyshev: if $\mu := \mathbb{E}X$ and $\sigma^2 := \mathrm{Var}(X) < \infty$, then for $t > 0$, $\mathbb{P}(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}$.

Proof.

i. Since $X \geq t\mathbf{1}_{\{X\geq t\}}$, taking expectations gives $\mathbb{E}X \geq t\mathbb{P}(X \geq t)$.
ii. Apply (i) to $Y := (X-\mu)^2 \geq 0$ with threshold t^2, then $\mathbb{P}(|X-\mu| \geq t) = \mathbb{P}(Y \geq t^2) \leq \frac{\mathbb{E}Y}{t^2} = \frac{\sigma^2}{t^2}$.

Structural Commentary: Tail bounds convert moment control into probability control.
Key Insight: Dominate the tail indicator by a scaled nonnegative random variable. □

Example 1.3.20. *A Distribution-Free Error-Rate Bound*
Let $X_1,\dots,X_n$ be independent Bernoulli(p) error indicators and $\overline{X}_n := n^{-1}\sum_{i=1}^{n}X_i$. Since $\mathbb{E}\overline{X}_n = p$ and $\mathrm{Var}(\overline{X}_n) = \frac{p(1-p)}{n}$, Chebyshev gives, for $\varepsilon > 0$, $\mathbb{P}(|\overline{X}_n - p| > \varepsilon) \leq \frac{p(1-p)}{n\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}$.

Theorem 1.3.21. *Hoeffding's Bounded-Sum Bound*
Let $Y_1,\dots,Y_n$ be independent, $\mathbb{E}Y_i = 0$, $a_i \leq Y_i \leq b_i$ a.s., and $V := \sum_{i=1}^{n}(b_i-a_i)^2$. For $S_n := \sum_{i=1}^{n}Y_i$ and $t,\varepsilon > 0$, $\mathbb{P}(S_n \geq \varepsilon) \leq \exp(-t\varepsilon + \frac{t^2V}{8})$; if $V > 0$, then $\mathbb{P}(S_n \geq \varepsilon) \leq \exp(-\frac{2\varepsilon^2}{V})$.

Proof. For $t > 0$, Markov gives $\mathbb{P}(S_n \geq \varepsilon) = \mathbb{P}(e^{tS_n} \geq e^{t\varepsilon}) \leq e^{-t\varepsilon}\mathbb{E}e^{tS_n}$.
Independence gives $\mathbb{E}e^{tS_n} = \prod_{i=1}^{n}\mathbb{E}e^{tY_i}$, and Hoeffding's lemma gives $\mathbb{E}e^{tY_i} \leq \exp(\frac{t^2(b_i-a_i)^2}{8})$.
Hence $\mathbb{P}(S_n \geq \varepsilon) \leq \exp(-t\varepsilon + \frac{t^2V}{8})$. If $V > 0$, minimize the exponent at $t = \frac{4\varepsilon}{V}$ to obtain $\exp(-\frac{2\varepsilon^2}{V})$.
Structural Commentary: Hoeffding turns bounded independent increments into exponential tail decay.
Key Insight: Exponentiate the sum, factor the MGF, then optimize t. □

Corollary 1.3.22. *Hoeffding for Bernoulli Averages*
If $X_1,\dots,X_n$ independent Bernoulli(p) *variables, $\overline{X}_n := n^{-1}\sum_{i=1}^{n}X_i$, then $\mathbb{P}(|\overline{X}_n - p| > \varepsilon) \leq 2e^{-2n\varepsilon^2}$.*

Proof. Set $Y_i^+ := \frac{X_i - p}{n}$.
Then $\mathbb{E}Y_i^+ = 0$, $Y_i^+ \in [-\frac{p}{n}, \frac{1-p}{n}]$, and $V = \sum_i (\frac{1}{n})^2 = \frac{1}{n}$, by Theorem 1.3.21, $\mathbb{P}(\overline{X}_n - p \geq \varepsilon) \leq e^{-2n\varepsilon^2}$.

Apply same argument to $Y_i^- := \frac{p - X_i}{n}$, $\mathbb{P}(p - \overline{X}_n \geq \varepsilon) \leq e^{-2n\varepsilon^2}$; the union bound yields the claim.
Structural Commentary: Bernoulli averages are sums of bounded centered increments.
Key Insight: Control both tails separately, then add them. □

Corollary 1.3.23. *Hoeffding Confidence Interval for a Bernoulli Parameter*
Let $X_1, \ldots, X_n$ be independent Bernoulli(p) *variables, $\widehat{p}_n := \overline{X}_n$, and $\alpha \in (0,1)$.*
Set $\varepsilon_n := \sqrt{(2n)^{-1} \log(\frac{2}{\alpha})}$ and $C_n := [\widehat{p}_n - \varepsilon_n, \widehat{p}_n + \varepsilon_n]$.

(i) Coverage: $\mathbb{P}(p \in C_n) \geq 1 - \alpha$.
(ii) Length: $|C_n| = 2\varepsilon_n \leq 0.05$ whenever $n \geq 800 \log(\frac{2}{\alpha})$.

Proof.

(i) Since $\{p \notin C_n\} = \{|\widehat{p}_n - p| > \varepsilon_n\}$, Corollary 1.3.22 gives $\mathbb{P}(p \notin C_n) \leq 2e^{-2n\varepsilon_n^2} = \alpha$.
(ii) If $n \geq 800 \log(\frac{2}{\alpha})$, then $\varepsilon_n^2 \leq \frac{1}{1600}$, hence $|C_n| = 2\varepsilon_n \leq 0.05$.

Structural Commentary: Confidence intervals invert a concentration inequality.
Key Insight: Choose ε_n so the Hoeffding failure bound equals α. □

Theorem 1.3.24. *Cauchy–Schwarz and Jensen*
If $X, Y \in L^2$, then $\mathbb{E}|XY| \leq \sqrt{\mathbb{E}(X^2)\mathbb{E}(Y^2)}$.
If $g : I \to \mathbb{R}$ convex, $X \in I$ a.s., $X, g(X) \in L^1$, then $\mathbb{E}g(X) \geq g(\mathbb{E}X)$; for concave g, inequality reverses.

Proof. For Cauchy–Schwarz, apply the L^2 inner-product inequality to $|X|$ and $|Y|$.
For Jensen, set $m := \mathbb{E}X$ and take an affine support $\ell(x) := g(m) + s(x - m)$ with $\ell \leq g$ on I.
Then $\mathbb{E}g(X) \geq \mathbb{E}\ell(X) = g(m) + s(\mathbb{E}X - m) = g(\mathbb{E}X)$. For concave g, apply the convex case to $-g$.
Structural Commentary: Cauchy–Schwarz controls products; Jensen moves convexity through expectation.
Key Insight: Use Hilbert-space geometry for products and supporting lines for convex functions. □

Definition 1.3.25. *Moment Generating Function*
The *moment generating function* of X is $\psi_X(t) := \mathbb{E}(e^{tX})$ on $D_X := \{t \in \mathbb{R} : \mathbb{E}e^{tX} < \infty\}$.
If $(-\delta, \delta) \subseteq D_X$ for some $\delta > 0$, then $\psi_X^{(k)}(0) = \mathbb{E}(X^k)$ for all $k \geq 1$.

Proposition 1.3.26. *Basic MGF Properties*
Assume the MGFs below are finite on a common open interval about 0.

(i) Affine transform: if $a, b \in \mathbb{R}$ and $Y = aX + b$, then $\psi_Y(t) = e^{bt}\psi_X(at)$.
(ii) Independent sum: if $X_1, \ldots, X_n$ are independent and $Y := \sum_{i=1}^n X_i$, then $\psi_Y(t) = \prod_{i=1}^n \psi_{X_i}(t)$.
(iii) Uniqueness: if $\psi_X(t) = \psi_Y(t)$ for all t in an open interval containing 0, then $\mathcal{L}(X) = \mathcal{L}(Y)$.

Proof.

(i) $\psi_Y(t) = \mathbb{E}e^{t(aX+b)} = e^{bt}\mathbb{E}e^{(at)X} = e^{bt}\psi_X(at)$.
(ii) For admissible t, $e^{tX_1}, \ldots, e^{tX_n}$ are independent and integrable.
Hence $\psi_Y(t) = \mathbb{E}e^{t\sum_{i=1}^n X_i} = \mathbb{E}\prod_{i=1}^n e^{tX_i} = \prod_{i=1}^n \mathbb{E}e^{tX_i} = \prod_{i=1}^n \psi_{X_i}(t)$.
(iii) Let $\mu := \mathcal{L}(X)$ and $\nu := \mathcal{L}(Y)$. If $\psi_X = \psi_Y$ on $(-\delta, \delta)$, then $\int_{\mathbb{R}} e^{tx}\, d\mu(x) = \int_{\mathbb{R}} e^{tx}\, d\nu(x)$ for all $|t| < \delta$.
Uniqueness of the two-sided Laplace transform gives $\mu = \nu$.

Structural Commentary: MGFs turn affine maps and independent sums into algebraic operations.
Key Insight: Exponentials convert sums into products, and the MGF near 0 determines the law. □

Example 1.3.27. *MGF Computations*

(i) If $X \sim \text{Exp}(1)$, then $\psi_X(t) = \mathbb{E}e^{tX} = \int_0^\infty e^{(t-1)x}\, dx = (1-t)^{-1}$ for $t < 1$.
Hence $\mathbb{E}X = \psi_X'(0) = 1$ and $\text{Var}(X) = \psi_X''(0) - (\psi_X'(0))^2 = 1$.
(ii) If $X \sim \text{Binomial}(n, p)$, then $X = \sum_{i=1}^n B_i$ with independent $B_i \sim \text{Bernoulli}(p)$.
Hence $\psi_X(t) = \prod_{i=1}^n \psi_{B_i}(t) = (pe^t + 1 - p)^n$.

Method 1.3.28. *Common Means, Variances, and MGFs*
MGFs are stated on their natural neighborhood of 0; values at $t = 0$ are by continuity.
The geometric and negative-binomial rows use the total-trials convention.

Distribution	$\mathbb{E}X$	$\mathrm{Var}(X)$	$\psi_X(t)$
point mass at a	a	0	e^{at}
Bernoulli(p)	p	$p(1-p)$	$1-p+pe^t$
Binomial(n,p)	np	$np(1-p)$	$(1-p+pe^t)^n$
Geometric(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
NegBin(r,p)	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$
Poisson(λ)	λ	λ	$e^{\lambda(e^t-1)}$
Uniform(a,b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{(b-a)t}$
$N(\mu,\sigma^2)$	μ	σ^2	$\exp(\mu t+\frac{\sigma^2t^2}{2})$
Exp(λ)	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Gamma(α,λ)	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha$
Beta(α,β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	not useful
χ^2_p	p	$2p$	$(1-2t)^{-\frac{p}{2}}$

1.4 Order Statistics, Extrema, and Moment Methods

Order statistics describe the sorted values of an independent sample. They turn joint density calculations into formulas for minima, maxima, medians, ranges, and distribution-free tolerance intervals.

Definition 1.4.1. *Order Statistics*
For real-valued $X_1,\dots,X_n$,

(i) The *order statistics* are the pointwise rearrangement $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ of $X_1,\dots,X_n$.
(ii) $X_{(1)} = \min_{1\le i\le n} X_i$, $X_{(n)} = \max_{1\le i\le n} X_i$, and if $n = 2m+1$, then $X_{(m+1)}$ is the sample median.

Proposition 1.4.2. *Minimum and Maximum of an IID Continuous Sample*
Let $n \ge 1$ and $X_1,\dots,X_n \overset{\text{iid}}{\sim} F$, where F is differentiable with density $f = F'$.
Set $V = X_{(1)} = \min_i X_i$ and $U = X_{(n)} = \max_i X_i$; then for $u, v \in \mathbb{R}$, $F_U(u) = F(u)^n$, $f_U(u) = nf(u)F(u)^{n-1}$, $F_V(v) = 1-(1-F(v))^n$, and $f_V(v) = nf(v)(1-F(v))^{n-1}$.

Proof. Since $\{U \le u\} = \bigcap_{i=1}^n \{X_i \le u\}$, independence gives $F_U(u) = \prod_{i=1}^n F(u) = F(u)^n$.
Differentiation gives $f_U(u) = nf(u)F(u)^{n-1}$.
Since $\{V > v\} = \bigcap_{i=1}^n \{X_i > v\}$, independence gives $1 - F_V(v) = \prod_{i=1}^n (1-F(v)) = (1-F(v))^n$.
Differentiation of $F_V(v) = 1-(1-F(v))^n$ gives $f_V(v) = nf(v)(1-F(v))^{n-1}$.
Structural Commentary: Extremes convert order events into products over independent coordinates.
Key Insight: Maxima use all observations to the left; minima use all observations to the right. □

Example 1.4.3. *Series and Parallel Exponential Lifetimes*
Let $T_1,\dots,T_n \overset{\text{ind}}{\sim} \text{Exp}(\lambda)$ be component lifetimes.

(i) For a series system, all components must survive, so system failure occurs at the first component failure: $V = \min_i T_i = T_{(1)}$. Thus $F_V(v) = 1-e^{-n\lambda v}$ and $f_V(v) = n\lambda e^{-n\lambda v}$ for $v \ge 0$; hence $V \sim \text{Exp}(n\lambda)$.
(ii) For a parallel system, one surviving component suffices, so system failure occurs only after all components fail: $U = \max_i T_i = T_{(n)}$. Thus $F_U(u) = (1-e^{-\lambda u})^n$ and $f_U(u) = n\lambda e^{-\lambda u}(1-e^{-\lambda u})^{n-1}$ for $u \ge 0$.

Theorem 1.4.4. *Density of the kth Order Statistic*
Let $X_1,\dots,X_n$ be iid real-valued random variables with differentiable CDF F and density $f = F'$.
For $1 \le k \le n$ and $x \in \mathbb{R}$, $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} f(x)F(x)^{k-1}(1-F(x))^{n-k}$.

Proof. $N_x = \sum_{i=1}^{n} \mathbf{1}_{\{X_i \le x\}}$; $N_x \sim \mathrm{Bin}(n, F(x))$, $\{X_{(k)} \le x\} = \{N_x \ge k\}$; $F_{X_{(k)}}(x) = \sum_{j=k}^{n} \binom{n}{j} F(x)^j (1-F(x))^{n-j}$.

With $p = F(x)$, $\frac{d}{dp} \sum_{j=k}^{n} \binom{n}{j} p^j (1-p)^{n-j} = \frac{n!}{(k-1)!(n-k)!} p^{k-1}(1-p)^{n-k}$, and the chain rule gives stated density.

Structural Commentary: The multinomial count of left, middle, and right observations supplies the coefficient.
Key Insight: An order statistic is located by counting how many sample points fall on each side. □

Corollary 1.4.5. *Uniform Order Statistics Are Beta*
If $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathrm{Uniform}(0,1)$, *then for* $1 \le k \le n$, $X_{(k)} \sim \mathrm{Beta}(k, n-k+1)$
with density $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k} \mathbf{1}_{(0,1)}(x)$.

Proof. On $(0,1)$, $F(x) = x$ and $f(x) = 1$, so Theorem 1.4.4 gives $f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k}$.
Also $B(k, n-k+1) = \frac{(k-1)!(n-k)!}{n!}$, hence this density is the $\mathrm{Beta}(k, n-k+1)$ density.
Structural Commentary: Uniform order statistics reduce the general density to the beta normalizing constant.
Key Insight: Counting positions in a uniform sample produces beta powers of left and right mass. □

Proposition 1.4.6. *Joint Density of Minimum and Maximum*
Let $n \ge 2$, *let* $X_1, \dots, X_n \overset{\text{iid}}{\sim} F$ *with differentiable CDF* F, *density* $f = F'$; $V = X_{(1)}$, $U = X_{(n)}$, $R = U - V$.

(i) For $v < u$, $f_{V,U}(v,u) = n(n-1)f(v)f(u)(F(u)-F(v))^{n-2}$, *and* $f_{V,U}(v,u) = 0$ *for* $v \ge u$.
(ii) If $F(t) = t$ *on* $[0,1]$, *then* $f_{V,U}(v,u) = n(n-1)(u-v)^{n-2}\mathbf{1}_{\{0<v<u<1\}}$.
(iii) In the uniform case, $f_R(r) = n(n-1)r^{n-2}(1-r)\mathbf{1}_{(0,1)}(r)$, *with* $F_R(r) = 0$ *for* $r < 0$,
$F_R(r) = nr^{n-1} - (n-1)r^n$ *for* $0 \le r \le 1$, *and* $F_R(r) = 1$ *for* $r \ge 1$.

Proof.

(i) For $v < u$, $G(v,u) = \mathbb{P}(V \ge v, U \le u) = \mathbb{P}(v \le X_i \le u,\ 1 \le i \le n) = (F(u) - F(v))^n$,
so $f_{V,U}(v,u) = -\partial_v \partial_u G(v,u) = n(n-1)f(v)f(u)(F(u)-F(v))^{n-2}$.
(ii) If $F(t) = t$ and $f(t) = 1$ on $(0,1)$, by (i), $f_{V,U}(v,u) = n(n-1)(u-v)^{n-2}$ on $0 < v < u < 1$, else 0.
(iii) For $0 < r < 1$, $f_R(r) = \int_0^{1-r} f_{V,U}(v, v+r)\, dv = n(n-1)r^{n-2}(1-r)$.
Integration gives $F_R(r) = \int_0^r f_R(s)\, ds = nr^{n-1} - (n-1)r^n$; outside $[0,1]$, the support gives $F_R(r) \in \{0,1\}$.

Structural Commentary: Mixed derivatives select the two boundary observations.
Key Insight: Fixed extremes leave only interior mass. □

Example 1.4.7. *Distribution-Free Tolerance Interval*
Let $X_1, \dots, X_n$ be iid with continuous CDF F, and set $Q = F(X_{(n)}) - F(X_{(1)})$.
By the probability integral transform, $U_i = F(X_i) \overset{\text{iid}}{\sim} \mathrm{Uniform}(0,1)$, so $Q \overset{d}{=} U_{(n)} - U_{(1)}$.
Therefore, for $0 \le \alpha \le 1$, $\mathbb{P}(Q > \alpha) = 1 - F_R(\alpha) = 1 - n\alpha^{n-1} + (n-1)\alpha^n$. For $n = 100$ and $\alpha = .95$, $\mathbb{P}(Q > .95) \approx .96$: $[X_{(1)}, X_{(n)}]$ covers at least .95 of the F-mass with probability about .96.

The expectation calculus developed above also supports a useful collection of moment methods: identifying whether the mean exists, replacing difficult distributions of functions by direct expectation, controlling spread by covariance, and using generating functions or Taylor expansion when direct integration is cumbersome.

Example 1.4.8. *Existence and First Distributional Means*

(i) If $X \sim \mathrm{Geom}(p)$ on $\{1, 2, \dots\}$ and $q = 1 - p$, then $\mathbb{E}X = p \sum_{k \ge 1} k q^{k-1} = p \frac{d}{dq} \frac{q}{1-q} = \frac{1}{p}$.
(ii) If $X \sim \mathrm{Poisson}(\lambda)$, then $\mathbb{E}X = e^{-\lambda} \sum_{k \ge 1} \frac{k\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{j \ge 0} \frac{\lambda^j}{j!} = \lambda$.
(iii) If $X \sim \mathrm{Cauchy}(0,1)$, then $\mathbb{E}|X| = \frac{2}{\pi} \int_0^\infty x(1+x^2)^{-1}\, dx = \infty$, so X is not integrable and $\mathbb{E}X$ is undefined.
(iv) If $\mathbb{P}(Y = 2^k) = 2^{-(k+1)}$ for $k \ge 0$, then $\mathbb{E}Y = \sum_{k \ge 0} 2^{-1} = \infty$; truncation is essential.

Key Insight: Expectation is a finite absolute average, not merely a formal symmetry or a plausible center.

Method 1.4.9. *Indicator Sums Without Independence*

(i) For a finite index set I and events $(A_i)_{i \in I}$, if $W = \sum_{i \in I} \mathbf{1}_{A_i}$, then $\mathbb{E}W = \sum_{i \in I} \mathbb{P}(A_i)$.
(ii) The identity uses only linearity of expectation; the events A_i need not be independent.

Example 1.4.10. *Counting Word Occurrences*
Let $S_1, \dots, S_N \overset{\text{iid}}{\sim} \mathrm{Uniform}\{A, C, G, T\}$ and fix a word $w \in \{A, C, G, T\}^q$.
For $1 \le m \le N - q + 1$, set $I_m = \mathbf{1}_{\{(S_m, \dots, S_{m+q-1}) = w\}}$ and $W = \sum_{m=1}^{N-q+1} I_m$.

Then $\mathbb{P}(I_m = 1) = 4^{-q}$ and $\mathbb{E}W = (N - q + 1)4^{-q}$.
Overlaps may create dependence; linearity does not require independence.

Example 1.4.11. *Coupon Collection by Waiting Times*
Draw uniformly from n coupon types, and let X_r be the waiting time for the rth new type. Then $X_r \sim \text{Geom}(\frac{n-r+1}{n})$ on $\{1, 2, \ldots\}$, so $\mathbb{E}X_r = \frac{n}{n-r+1}$.
For $T = \sum_{r=1}^{n} X_r$, linearity gives $\mathbb{E}T = \sum_{r=1}^{n} \frac{n}{n-r+1} = nH_n$, where $H_n = \sum_{j=1}^{n} \frac{1}{j} \sim \log n + \gamma$.

Theorem 1.4.12. *Correlation Bound and Equality Case*
Let $X, Y \in L^2$ have means μ_X, μ_Y and standard deviations $0 < \sigma_X, \sigma_Y < \infty$, set $\rho = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$, and define $Z_X = \frac{X-\mu_X}{\sigma_X}$, $Z_Y = \frac{Y-\mu_Y}{\sigma_Y}$.
(i) $\mathbb{E}Z_X = \mathbb{E}Z_Y = 0$, $\text{Var}(Z_X) = \text{Var}(Z_Y) = 1$, and $\text{Cov}(Z_X, Z_Y) = \rho$.
(ii) $-1 \le \rho \le 1$, equivalently $|\rho| \le 1$.
(iii) $\rho = \pm 1$ iff $Z_Y = \pm Z_X$ a.s., equivalently $Y = \mu_Y \pm (\frac{\sigma_Y}{\sigma_X})(X - \mu_X)$ a.s., with the same sign.

Proof.
(i) $\mathbb{E}Z_X = 0$, $\text{Var}(Z_X) = \sigma_X^{-2} \text{Var}(X) = 1$, and similarly for Z_Y.
Moreover, $\text{Cov}(Z_X, Z_Y) = \sigma_X^{-1}\sigma_Y^{-1} \text{Cov}(X, Y) = \rho$.
(ii) Since $0 \le \text{Var}(Z_X \pm Z_Y) = 2 \pm 2\rho$, both $1 + \rho \ge 0$ and $1 - \rho \ge 0$, hence $|\rho| \le 1$.
(iii) ($\Rightarrow$) If $\rho = 1$, then $\mathbb{E}(Z_X - Z_Y)^2 = 0$, so $Z_X = Z_Y$ a.s..
If $\rho = -1$, then $\mathbb{E}(Z_X + Z_Y)^2 = 0$, so $Z_X = -Z_Y$ a.s.
(iii) ($\Leftarrow$) If $Z_Y = \pm Z_X$, then $\rho = \text{Cov}(Z_X, \pm Z_X) = \pm 1$, and $\frac{Y-\mu_Y}{\sigma_Y} = \pm\frac{X-\mu_X}{\sigma_X}$
iff $Y = \mu_Y \pm (\frac{\sigma_Y}{\sigma_X})(X - \mu_X)$ a.s.
Structural Commentary: Correlation is centred covariance after unit-variance normalization.
Key Insight: Perfect correlation means no residual scatter around an affine line. □

Example 1.4.13. *Bivariate Uniform Association*
For (X, Y) with uniform marginals on $[0, 1]$, $\mathbb{E}X = \mathbb{E}Y = \frac{1}{2}$ and $\text{Var}(X) = \text{Var}(Y) = \frac{1}{12}$.
Hence $\text{Cov}(X, Y) = \mathbb{E}(XY) - \frac{1}{4}$ and $\rho = 12\,\text{Cov}(X, Y)$.
If $\mathbb{E}(XY) = \frac{2}{9}$, then $\text{Cov}(X, Y) = -\frac{1}{36}$, $\rho = -\frac{1}{3}$; if $\mathbb{E}(XY) = \frac{5}{18}$, then $\text{Cov}(X, Y) = \frac{1}{36}$, $\rho = \frac{1}{3}$.
The sign of ρ records negative or positive association.

Method 1.4.14. *MGF Calculus for Classical Laws*
If $\psi_X(t) = \mathbb{E}e^{tX}$ exists near 0, then $\mathbb{E}X = \psi_X'(0)$ and $\text{Var}(X) = \psi_X''(0) - \psi_X'(0)^2$.
(i) If $X \sim \text{Poisson}(\lambda)$, then $\psi_X(t) = \exp(\lambda(e^t - 1))$, $\mathbb{E}X = \lambda$, and $\text{Var}(X) = \lambda$.
(ii) If $X \sim \text{Gamma}(\alpha, \lambda)$ with rate λ, then $\psi_X(t) = (\frac{\lambda}{\lambda - t})^\alpha$ for $t < \lambda$, $\mathbb{E}X = \frac{\alpha}{\lambda}$, and $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.
(iii) If $Z \sim N(0, 1)$, then $\psi_Z(t) = e^{\frac{t^2}{2}}$; for $Y = \mu + \sigma Z$, $\psi_Y(t) = \exp(\mu t + \frac{\sigma^2 t^2}{2})$.

Corollary 1.4.15. *Independent Sums from MGFs*
Let $X \perp\!\!\!\perp Y$, and interpret all MGFs on a common neighborhood of 0.
(i) If $\lambda, \mu > 0$, $X \sim \text{Poisson}(\lambda)$, and $Y \sim \text{Poisson}(\mu)$, then $X + Y \sim \text{Poisson}(\lambda + \mu)$.
(ii) If $X \sim \text{Gamma}(\alpha_1, \lambda)$ and $Y \sim \text{Gamma}(\alpha_2, \lambda)$ with $\alpha_1, \alpha_2, \lambda > 0$, then $X + Y \sim \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$.
(iii) If $\mu, \nu \in \mathbb{R}$, $\sigma, \tau > 0$, $X \sim N(\mu, \sigma^2)$, and $Y \sim N(\nu, \tau^2)$, then $X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2)$.

Proof.
(i) $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = \exp(\lambda(e^t - 1)) \exp(\mu(e^t - 1)) = \exp((\lambda + \mu)(e^t - 1))$, QED.
(ii) For $t < \lambda$, $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = (\frac{\lambda}{\lambda - t})^{\alpha_1 + \alpha_2}$, QED.
(iii) $\psi_{X+Y}(t) = \psi_X(t)\psi_Y(t) = \exp(\mu t + \frac{\sigma^2 t^2}{2}) \exp(\nu t + \frac{\tau^2 t^2}{2}) = \exp((\mu + \nu)t + \frac{(\sigma^2 + \tau^2)t^2}{2})$, QED.
Structural Commentary: Independence turns convolution into multiplication of MGFs.
Key Insight: Closed distribution families are stable when their MGF parameters add. □

Method 1.4.16. *Moment Propagation by Taylor Expansion*
Let the relevant variables have finite second moments, and let g be C^2 near the relevant mean point.
(i) If $Y = g(X)$, $\mu = \mathbb{E}X$, and $\sigma^2 = \text{Var}(X)$, then $\mathbb{E}Y \approx g(\mu)$ and $\text{Var}(Y) \approx [g'(\mu)]^2\sigma^2$.
(ii) The second-order mean correction is $\mathbb{E}g(X) \approx g(\mu) + \frac{1}{2}g''(\mu)\sigma^2$.
(iii) For $Z = g(\mathbf{X})$, $\mu = \mathbb{E}\mathbf{X}$, and $\Sigma = \text{Cov}(\mathbf{X})$, $\text{Var}(Z) \approx \nabla g(\mu)^\top \Sigma \nabla g(\mu)$ and $\mathbb{E}Z \approx g(\mu) + \frac{1}{2}\,\text{tr}(\Sigma H_g(\mu))$.
Structural Commentary: Taylor expansion propagates moments through local derivatives.
Key Insight: Smooth nonlinear functions are locally linear for variance and locally quadratic for mean correction.

Example 1.4.17. *Approximate Moments of a Ratio*
Let $Z = \frac{Y}{X}$, $\mu_X \neq 0$, $\mu = (\mu_X, \mu_Y)$, and write $\sigma_X^2 = \mathrm{Var}(X)$, $\sigma_Y^2 = \mathrm{Var}(Y)$, $\sigma_{XY} = \mathrm{Cov}(X,Y)$.
For $g(x,y) = \frac{y}{x}$, Taylor propagation gives $\mathbb{E}Z \approx \frac{\mu_Y}{\mu_X} + \frac{\sigma_X^2 \mu_Y}{\mu_X^3} - \frac{\sigma_{XY}}{\mu_X^2}$
and $\mathrm{Var}(Z) \approx \frac{\sigma_X^2 \mu_Y^2}{\mu_X^4} + \frac{\sigma_Y^2}{\mu_X^2} - \frac{2\sigma_{XY}\mu_Y}{\mu_X^3}$. The approximation is ill-conditioned as $|\mu_X| \downarrow 0$.

1.5 Normal-Derived Distributions

Independent $N(0,1)$ variables generate the normal-derived inference laws used below: χ^2 sums, Student t ratios, and F ratios.

Proposition 1.5.1. *Chi-Square Sums*
Let $n \geq 1$ and $Z_1, \ldots, Z_n \overset{\text{ind}}{\sim} N(0,1)$.

(i) For $V = \sum_{i=1}^{n} Z_i^2$, $V \sim \chi_n^2 = \mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$ in the rate convention,
$f_V(v) = 2^{-\frac{n}{2}}\Gamma(\frac{n}{2})^{-1} v^{\frac{n}{2}-1} e^{-\frac{v}{2}} \mathbf{1}_{(0,\infty)}(v)$, $\psi_V(t) = (1-2t)^{-\frac{n}{2}}$ for $t < \frac{1}{2}$, $\mathbb{E}V = n$, and $\mathrm{Var}(V) = 2n$.
(ii) If $U \sim \chi_m^2$, $W \sim \chi_n^2$, and $U \perp\!\!\!\perp W$, then $U + W \sim \chi_{m+n}^2$.

Proof.

(i) For $Z \sim N(0,1)$ and $t < \frac{1}{2}$, $\mathbb{E}e^{tZ^2} = (2\pi)^{-\frac{1}{2}} \int_{\mathbb{R}} e^{-\frac{(1-2t)z^2}{2}}\, dz = (1-2t)^{-\frac{1}{2}}$.
By independence, $\psi_V(t) = \prod_{i=1}^{n} \mathbb{E}e^{tZ_i^2} = (1-2t)^{-\frac{n}{2}}$. By MGF uniqueness, $V \sim \mathrm{Gamma}(\frac{n}{2}, \frac{1}{2})$.
Differentiating ψ_V at 0 gives $\mathbb{E}V = n$ and $\mathrm{Var}(V) = 2n$.
(ii) If $U \perp\!\!\!\perp W$, then $\psi_{U+W}(t) = (1-2t)^{-\frac{m}{2}}(1-2t)^{-\frac{n}{2}} = (1-2t)^{-\frac{m+n}{2}}$, hence $U + W \sim \chi_{m+n}^2$.

Structural Commentary: Chi-square degrees of freedom add by MGF exponents.
Key Insight: Squared independent standard normals form gamma sums. □

Definition 1.5.2. *Student t Distribution*
If $Z \sim N(0,1)$, $U \sim \chi_n^2$, $Z \perp\!\!\!\perp U$, then $T = Z/\sqrt{\frac{U}{n}}$ has *Student t distribution* with $df = n \geq 1$, $T \sim t_n$.

Proposition 1.5.3. *Student t Density*
Let $n \geq 1$ and $T = Z/\sqrt{\frac{U}{n}}$ with $Z \sim N(0,1)$, $U \sim \chi_n^2$, and $Z \perp\!\!\!\perp U$.

(i) For $t \in \mathbb{R}$, $f_T(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$.
(ii) For all $t \in \mathbb{R}$, $f_T(-t) = f_T(t)$.
(iii) For each fixed $t \in \mathbb{R}$, $\lim_{n\to\infty} f_T(t) = \phi(t) = (2\pi)^{-\frac{1}{2}} e^{-\frac{t^2}{2}}$.

Proof.

(i) For fixed $u > 0$, $z = t\sqrt{\frac{u}{n}}$ has Jacobian $\sqrt{\frac{u}{n}}$, so $f_T(t) = \int_0^\infty \phi(t\sqrt{\frac{u}{n}})\sqrt{\frac{u}{n}} f_U(u)\, du$.
Substituting $\phi(z) = (2\pi)^{-\frac{1}{2}} e^{-\frac{z^2}{2}}$ and $f_U(u) = 2^{-\frac{n}{2}}\Gamma(\frac{n}{2})^{-1} u^{\frac{n}{2}-1} e^{-\frac{u}{2}} \mathbf{1}_{(0,\infty)}(u)$ gives
$f_T(t) = \frac{1}{\sqrt{n\pi}\, 2^{\frac{n+1}{2}}\Gamma(\frac{n}{2})} \int_0^\infty u^{\frac{n+1}{2}-1} e^{-\frac{(1+\frac{t^2}{n})u}{2}}\, du$. Then $F_T(t) = \Gamma(\frac{n+1}{2}) 2^{\frac{n+1}{2}} (1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$, yielding density.
(ii) By (i), $f_T(t) = c_n h_n(t^2)$ for $c_n > 0$, so $f_T(-t) = f_T(t)$ for all t.
(iii) For fixed t, $f_T(t) = (2\pi)^{-\frac{1}{2}} \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\frac{n}{2}}\Gamma(\frac{n}{2})}(1 + \frac{t^2}{n})^{-\frac{n+1}{2}}$, gamma ratio tends to 1, last factor tends to $e^{-\frac{t^2}{2}}$.

Structural Commentary: Student t is a normal scale mixture with random chi-square scale.
Key Insight: Random denominator uncertainty thickens the normal tails. □

Example 1.5.4. *t_1 is Cauchy*
If $T \sim t_1$, then $f_T(t) = \Gamma(1)(\sqrt{\pi}\Gamma(\frac{1}{2}))^{-1}(1+t^2)^{-1} = \frac{1}{\pi(1+t^2)}$, so $T \sim \mathrm{Cauchy}(0,1)$.

Definition 1.5.5. *F Distribution*
If $U \sim \chi_m^2$, $V \sim \chi_n^2$, $U \perp\!\!\!\perp V$, then $W = (\frac{U}{m})/(\frac{V}{n})$ has *F distribution*, $df = (m,n)$, $W \sim F_{m,n}$.

Proposition 1.5.6. *F Density*
Let $m, n \geq 1$ and $W = (\frac{U}{m})/(\frac{V}{n})$, where $U \sim \chi_m^2$, $V \sim \chi_n^2$, and $U \perp\!\!\!\perp V$.

(i) $f_W(w) = \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})}(\frac{m}{n})^{\frac{m}{2}} w^{\frac{m}{2}-1}(1 + \frac{mw}{n})^{-\frac{m+n}{2}} \mathbf{1}_{(0,\infty)}(w)$.
(ii) If $n > 2$, then $\mathbb{E}W = \frac{n}{n-2}$.

Proof.

(i) For $w > 0$, set $u = \frac{mwv}{n}$; the Jacobian is $\frac{mv}{n}$, so $f_W(w) = \int_0^\infty f_U(\frac{mwv}{n}) f_V(v)(\frac{mv}{n})\, dv$. By substitution, $f_W(w) = Cw^{\frac{m}{2}-1} \int_0^\infty v^{\frac{m+n}{2}-1} e^{-\frac{(1+\frac{mw}{n})v}{2}}\, dv$; $C = \frac{(\frac{m}{n})^{\frac{m}{2}}}{2^{\frac{m+n}{2}}\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})}$, by gamma integral, QED.

(ii) Since $U \perp\!\!\!\perp V$, $\mathbb{E}W = (\frac{n}{m})\mathbb{E}U\, \mathbb{E}(V^{-1}) = n\, \mathbb{E}(V^{-1}) = \frac{n}{n-2}$ for $n > 2$.

Structural Commentary: An F variable is a ratio of independent chi-square averages.
Key Insight: Denominator variability controls the right tail and the mean condition $n > 2$. □

Corollary 1.5.7. *Reciprocal F and Squared t*

(i) For $m, n \geq 1$, if $W \sim F_{m,n}$, then $W^{-1} \sim F_{n,m}$.
(ii) If $T \sim t_n$, then $T^2 \sim F_{1,n}$.

Proof.

(i) Write $W = (\frac{U}{m})/(\frac{V}{n})$ with $U \sim \chi^2_m$, $V \sim \chi^2_n$, and $U \perp\!\!\!\perp V$; then $W^{-1} = (\frac{V}{n})/(\frac{U}{m}) \sim F_{n,m}$.

(ii) Write $T = Z/\sqrt{\frac{V}{n}}$ with $Z \sim N(0,1)$, $V \sim \chi^2_n$, and $Z \perp\!\!\!\perp V$; since $Z^2 \sim \chi^2_1$, $T^2 = (\frac{Z^2}{1})/(\frac{V}{n}) \sim F_{1,n}$.

Structural Commentary: Reciprocals swap F degrees of freedom, and squaring t removes the normal sign.
Key Insight: Both identities are immediate from the same chi-square-ratio representation. □

2 Conditional Probability and Conditional Expectation

2.1 Conditional Probability

Definition 2.1.1. *Conditional Probability*
For $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the *conditional probability* of A given B is $\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, i.e. the probability of A under the normalised restriction of $\mathbb{P}$ to B.

Example 2.1.2. *Diagnostic Table*
Let T be the event of a positive test and D the event of disease, with observed counts

	D	D^c	total
T	25	14	39
T^c	18	78	96
total	43	92	135

Let $N(\cdot)$ be counts, $\widehat{\mathbb{P}}(A \mid B) = \frac{N(A \cap B)}{N(B)}$.
Thus $\widehat{\mathbb{P}}(D \mid T) = \frac{25}{39}$, $\widehat{\mathbb{P}}(D^c \mid T^c) = \frac{78}{96}$, $\widehat{\mathbb{P}}(D^c \mid T) = \frac{14}{39}$, and $\widehat{\mathbb{P}}(D \mid T^c) = \frac{18}{96}$.

Proposition 2.1.3. *Conditioning on a Fixed Event Gives a Probability Measure*
For $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, define $\mathbb{P}_B : \mathcal{F} \to [0,1]$ by $\mathbb{P}_B(A) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$.
Then $\mathbb{P}_B$ is a probability measure on $(\Omega, \mathcal{F})$.

Proof. For $A \in \mathcal{F}$, $\mathbb{P}_B(A) \geq 0$, and $\mathbb{P}_B(\Omega) = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1$.
If $(A_i)_{i \geq 1}$ are pairwise disjoint, then $(A_i \cap B)_{i \geq 1}$ are pairwise disjoint and $(\bigcup_i A_i) \cap B = \bigcup_i (A_i \cap B)$.
Thus $\mathbb{P}_B(\bigcup_i A_i) = \frac{\mathbb{P}(\bigcup_i (A_i \cap B))}{\mathbb{P}(B)} = \sum_i \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_i \mathbb{P}_B(A_i)$.
Structural Commentary: Conditioning normalizes the restriction of $\mathbb{P}$ to B.
Key Insight: Once B is fixed, conditional probability is ordinary probability on the restricted world. □

Remark 2.1.4. *Conditional Probability Is Not Symmetric*
In general, $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$; conditioning fixes the reference population. For disease event D and positive-test event T, large $\mathbb{P}(T \mid D)$ need not imply large $\mathbb{P}(D \mid T)$ when $\mathbb{P}(D)$ is small.

Proposition 2.1.5. *Multiplication and Chain Rules*
Let $A, B, C \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, $\mathbb{P}(B) > 0$, and $\mathbb{P}(B \cap C) > 0$.

(i) $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$.
(ii) $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)\mathbb{P}(C)$.

Proof.

(i) For $\mathbb{P}(F) > 0$, $\mathbb{P}(E \mid F)\mathbb{P}(F) = \mathbb{P}(E \cap F)$; apply this with $(E, F) = (A, B)$ and (B, A).

(ii) Since $\mathbb{P}(B \cap C) > 0$, $\mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C)\mathbb{P}(C) = \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(B \cap C)} \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(C)} \mathbb{P}(C) = \mathbb{P}(A \cap B \cap C)$.

Structural Commentary: The chain rule orders information as C, then B within C, then A within $B \cap C$.
Key Insight: A joint probability factors into conditional increments along a chosen order. □

Example 2.1.6. *Cards Without Replacement*
Draw two cards from a standard deck without replacement. Let A be the event that the first card is the ace of clubs and B the event that the second card is the queen of diamonds. Since $\mathbb{P}(A) = \frac{1}{52}$ and $\mathbb{P}(B \mid A) = \frac{1}{51}$, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) = \frac{1}{52 \cdot 51}$. The factor $\mathbb{P}(B \mid A)$ uses the 51-card deck remaining after A occurs.

Proposition 2.1.7. *Independence Through Conditioning*
Let $A, B \in \mathcal{F}$ with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. The following are equivalent:

(i) *A and B are independent.*
(ii) $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.
(iii) $\mathbb{P}(B \mid A) = \mathbb{P}(B)$.

Proof.

(*i*) $\Rightarrow$ (*ii*) If A and B are independent, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, so $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A)$.
(*ii*) $\Rightarrow$ (*iii*) If $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, so $\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \mathbb{P}(B)$.
(*iii*) $\Rightarrow$ (*i*) If $\mathbb{P}(B \mid A) = \mathbb{P}(B)$, then $\mathbb{P}(A \cap B) = \mathbb{P}(B \mid A)\mathbb{P}(A) = \mathbb{P}(A)\mathbb{P}(B)$, hence A and B are independent.

Structural Commentary: Each condition is $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ normalized by a positive marginal.
Key Insight: Independence means conditioning on either event preserves the probability of the other. □

Theorem 2.1.8. *Law of Total Probability*

(i) *Let $A_1, \ldots, A_k \in \mathcal{F}$ form a finite partition of Ω with $\mathbb{P}(A_i) > 0$ for all i.*
(ii) *For every $B \in \mathcal{F}$, $\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$.*

Proof.

(i) The partition assumptions give $B = \bigsqcup_{i=1}^{k} (B \cap A_i)$, hence $\mathbb{P}(B) = \sum_{i=1}^{k} \mathbb{P}(B \cap A_i)$.
(ii) For each i, $\mathbb{P}(B \cap A_i) = \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$; substituting in (i) gives the formula.

Structural Commentary: The partition $\{A_i\}_{i=1}^{k}$ decomposes B into disjoint cases $B \cap A_i$.
Key Insight: Total probability is weighted averaging over cases. □

Example 2.1.9. *Second Draw by Total Probability*
An urn has 3 red balls, 1 blue ball; draw twice without replacement, let R_i be the event that draw i is red. By total probability, $\mathbb{P}(R_2) = \mathbb{P}(R_2 \mid R_1)\mathbb{P}(R_1) + \mathbb{P}(R_2 \mid R_1^c)\mathbb{P}(R_1^c) = \frac{2}{3} \cdot \frac{3}{4} + 1 \cdot \frac{1}{4} = \frac{3}{4}$.
Key Insight: The second draw has the same marginal red probability as the first.

Definition 2.1.10. *Prior and Posterior Probabilities*
Let $A_1, \ldots, A_k \in \mathcal{F}$ be a partition with $\mathbb{P}(A_i) > 0$ for all i, and let $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$.

(i) $\mathbb{P}(A_i)$ is the *prior probability* of A_i.
(ii) $\mathbb{P}(A_i \mid B)$ is the *posterior probability* of A_i after conditioning on B.

Theorem 2.1.11. *Bayes' Theorem*

Let $A_1, \ldots, A_k \in \mathcal{F}$ satisfy $A_i \cap A_j = \varnothing$ for $i \neq j$, $\bigcup_{i=1}^{k} A_i = \Omega$, and $\mathbb{P}(A_i) > 0$ for all i.
If $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, then for each i, $\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}$.

Proof. For fixed i, $\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\mathbb{P}(B)}$.
Since $B = \bigsqcup_{j=1}^{k} (B \cap A_j)$, $\mathbb{P}(B) = \sum_{j=1}^{k} \mathbb{P}(B \cap A_j) = \sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)$. Substitution gives the formula.
Structural Commentary: Bayes reweights priors by likelihoods, then normalises.
Key Insight: Posterior equals likelihood times prior divided by evidence. □

Corollary 2.1.12. *Posterior Mass Must Balance*
Let $A_1, \ldots, A_k \in \mathcal{F}$ be a partition of Ω with $\mathbb{P}(A_i) > 0$ for all i, and let $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$.
If $\mathbb{P}(A_1 \mid B) < \mathbb{P}(A_1)$, then there exists $i \in \{2, \ldots, k\}$ such that $\mathbb{P}(A_i \mid B) > \mathbb{P}(A_i)$.

Proof. Suppose $\mathbb{P}(A_i \mid B) \le \mathbb{P}(A_i)$ for all $i \ge 2$
As $\sum_{i=1}^{k} \mathbb{P}(A_i \mid B) = \sum_{i=1}^{k} \mathbb{P}(A_i) = 1$, hence $1 = \sum_{i=1}^{k} \mathbb{P}(A_i \mid B) < \mathbb{P}(A_1) + \sum_{i=2}^{k} \mathbb{P}(A_i) = 1$, a contradiction.
Structural Commentary: Posterior mass is conserved across the partition.
Key Insight: A decrease in one cell must be offset by an increase in another. □

Example 2.1.13. *Transition Probabilities*
Let states U, M, L denote upper, middle, lower groups, and let $P = (p_{ij})$ with $p_{ij} = \mathbb{P}(j_2 \mid i_1)$:

	U_2	M_2	L_2
U_1	.45	.48	.07
M_1	.05	.70	.25
L_1	.01	.50	.49

For $\pi_1 = (.10, .40, .50)$, total probability gives $\pi_2 = \pi_1 P$ and $\mathbb{P}(U_2) = .45(.10) + .05(.40) + .01(.50) = .07$.
Bayes gives the inverse query $\mathbb{P}(U_1 \mid U_2) = \frac{\mathbb{P}(U_2|U_1)\mathbb{P}(U_1)}{\mathbb{P}(U_2)} = \frac{.45(.10)}{.07} \approx .64$.
Key Insight: Use πP forward and Bayes backward.

Example 2.1.14. *Base Rates in a Medical Test*
Let D be disease and T a positive test, with joint law

	D	D^c
T	0.0081	0.0900
T^c	0.0009	0.9010

Then $\mathbb{P}(T \mid D) = \frac{0.0081}{0.0081+0.0009} = 0.9$, $\mathbb{P}(T^c \mid D^c) = \frac{0.9010}{0.9010+0.0900} \approx 0.91$, $\mathbb{P}(D \mid T) = \frac{0.0081}{0.0081+0.0900} \approx 0.083$.
Key Insight: A rare condition can have small posterior probability after a positive test.

Example 2.1.15. *Rare Classes and Screening Errors*
Let L denote lying and $T = L^c$ truth, with $\mathbb{P}(+ \mid L) = .88$, $\mathbb{P}(+ \mid T) = .14$, $\mathbb{P}(L) = .01$, and $\mathbb{P}(T) = .99$.
Bayes gives $\mathbb{P}(T \mid +) = \frac{\mathbb{P}(+|T)\mathbb{P}(T)}{\mathbb{P}(+|T)\mathbb{P}(T)+\mathbb{P}(+|L)\mathbb{P}(L)} = \frac{(.14)(.99)}{(.14)(.99)+(.88)(.01)} \approx .94$.
Key Insight: Rare-class positives can mostly come from the common class.

Example 2.1.16. *Monty Hall*
The player chooses door 1. Let $\omega = (\omega_1, \omega_2)$, where ω_1 is the prize door and ω_2 is Monty's opened door.
Monty opens neither door 1 nor the prize door, and chooses uniformly from $\{2, 3\}$ when $\omega_1 = 1$.
The induced probabilities are $\mathbb{P}(1, 2) = \mathbb{P}(1, 3) = \frac{1}{6}$ and $\mathbb{P}(2, 3) = \mathbb{P}(3, 2) = \frac{1}{3}$.
Let $S = \{\omega_1 = 1\}$ be the event that staying wins.
Given $\omega_2 = 2$, the staying-win probability is $\mathbb{P}(S \mid \omega_2 = 2) = \frac{\frac{1}{6}}{\frac{1}{6}+\frac{1}{3}} = \frac{1}{3}$.
Thus switching wins with probability $\frac{2}{3}$, and the case $\omega_2 = 3$ is symmetric.
Key Insight: Once Monty's rule is part of the sample space, switching has conditional probability $\frac{2}{3}$.

Example 2.1.17. *Three Cards and Conditional Sampling*
Cards are GG, RR, and GR. Choose a card uniformly and then a side uniformly.
Let V be the event that the visible side is green and H the event that the hidden side is green.
Among the three equally likely green visible sides, two have green hidden side, so $\mathbb{P}(H \mid V) = \frac{2}{3}$.

Example 2.1.18. *Conditioning Changes the Reference Class*
Let $X \sim \text{Binomial}(5, \frac{1}{4})$ be number of blue-eyed children, Y be event that youngest child has blue eyes.

(i) Given $X \ge 1$, $\mathbb{P}(X \ge 3 \mid X \ge 1) = \frac{\binom{5}{3}\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^2+\binom{5}{4}\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)+\left(\frac{1}{4}\right)^5}{1-\left(\frac{3}{4}\right)^5} = \frac{106}{781}$.

(ii) Given Y, $\mathbb{P}(X \ge 3 \mid Y) = 1 - \left(\frac{3}{4}\right)^4 - 4\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^3 = \frac{67}{256}$.

Key Insight: Similar information can determine different conditioning events.

Example 2.1.19. *Bayes with Several Categories*
Let $O \in \{M, W, L\}$ denote operating system with probabilities $(0.30, 0.50, 0.20)$.
Let I be infection, with $\mathbb{P}(I \mid O = M) = 0.65$, $\mathbb{P}(I \mid O = W) = 0.82$, and $\mathbb{P}(I \mid O = L) = 0.50$.
Bayes' theorem gives $\mathbb{P}(W \mid I) = \frac{0.82 \cdot 0.50}{0.65 \cdot 0.30 + 0.82 \cdot 0.50 + 0.50 \cdot 0.20} = \frac{82}{141}$.

Example 2.1.20. *Posterior and Predictive Probabilities*
Let C be uniform on $\{C_1, \ldots, C_5\}$, and suppose coin C_i has head probability p_i.
The head-probability vector is $(p_1, p_2, p_3, p_4, p_5) = \left(0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\right)$. Let H_n be the event that toss n is heads.
The posterior after one head is $\mathbb{P}(C_i \mid H_1) = \frac{p_i}{p_1+p_2+p_3+p_4+p_5} = \left(0, \frac{1}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{5}\right)_i$.
The predictive probability is $\mathbb{P}(H_2 \mid H_1) = \sum_{i=1}^{5} p_i \mathbb{P}(C_i \mid H_1) = \frac{3}{4}$.

If B_4 is the event that the first head occurs on toss 4,
the posterior is $\mathbb{P}(C_i \mid B_4) = \frac{(1-p_i)^3 p_i}{\sum_{j=1}^{5}(1-p_j)^3 p_j}$, and $\big(\mathbb{P}(C_i \mid B_4)\big)_{i=1}^{5} = \big(0, \frac{27}{46}, \frac{16}{46}, \frac{3}{46}, 0\big)$.

Key Insight: Bayes' theorem updates hidden parameters; prediction averages over the posterior.

Definition 2.1.21. *Conditional Mass and Density Functions*

(i) *Discrete*: if X, Y are discrete and $f_Y(y) > 0$, the conditional mass function is $f_{X|Y}(x \mid y) := \mathbb{P}(X = x \mid Y = y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$.

(ii) *Continuous*: if X, Y have a joint density and $f_Y(y) > 0$, the conditional density is $f_{X|Y}(x \mid y) := \frac{f_{X,Y}(x,y)}{f_Y(y)}$, so $\mathbb{P}(X \in A \mid Y = y) = \int_A f_{X|Y}(x \mid y)\,dx$ for measurable A.

Proposition 2.1.22. *Joint Factorizations and Total Probability for Conditional Laws*
Assume the displayed sums or integrals are well defined.

(i) If (X, Y) is discrete, $p_Y(y) > 0$, then $p_{X,Y}(x,y) = p_{X|Y}(x \mid y)p_Y(y)$, $p_X(x) = \sum_y p_{X|Y}(x \mid y)p_Y(y)$.

(ii) If (X, Y) has density, $f_X(x) > 0$, then $f_{X,Y}(x,y) = f_{Y|X}(y \mid x)f_X(x)$, $f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y \mid x)f_X(x)\,dx$.

Proof.

(i) For $p_Y(y) > 0$, $p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$.
Hence $p_{X,Y}(x,y) = p_{X|Y}(x \mid y)p_Y(y)$ and $p_X(x) = \sum_y p_{X,Y}(x,y) = \sum_y p_{X|Y}(x \mid y)p_Y(y)$.

(ii) For $f_X(x) > 0$, $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$.
Hence $f_{X,Y}(x,y) = f_{Y|X}(y \mid x)f_X(x)$ and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx = \int_{-\infty}^{\infty} f_{Y|X}(y \mid x)f_X(x)\,dx$.

Structural Commentary: Conditional laws disintegrate a joint law into a marginal law and a fiber law.
Key Insight: Total probability is marginalization after conditional factorization. □

Example 2.1.23. *Conditional Masses from the Three-Toss Table*
Let Y be the number of heads in three fair tosses, and let $X = 1$ iff the first toss is heads.
Since $p_Y(1) = \frac{3}{8}$, $p_{X,Y}(0,1) = \frac{2}{8}$, and $p_{X,Y}(1,1) = \frac{1}{8}$, one has $p_{X|Y}(0 \mid 1) = \frac{2}{3}$ and $p_{X|Y}(1 \mid 1) = \frac{1}{3}$.
Thus, given $Y = 1$, the first toss is more likely tails.

Example 2.1.24. *Uniform Square*
If $(X, Y) \sim \text{Uniform}([0,1]^2)$, then for $0 \le y \le 1$, $0 \le x \le 1$, $f_{X|Y}(x \mid y) = 1$. Hence $X \mid Y = y \sim \text{Uniform}(0,1)$.

Example 2.1.25. *Conditioning in the Parabolic Region*
Let (X, Y) have density $f(x,y) = \frac{21}{4}x^2 y$, $x^2 \le y \le 1$. Then $f_X(x) = \frac{21}{8}x^2(1-x^4)$, $-1 \le x \le 1$.
For $x^2 \le y \le 1$, $f_{Y|X}(y \mid x) = \frac{\frac{21}{4}x^2 y}{\frac{21}{8}x^2(1-x^4)} = \frac{2y}{1-x^4}$.
At $x = \frac{1}{2}$, $f_{Y|X}(y \mid \frac{1}{2}) = \frac{32y}{15}$, so $\mathbb{P}(Y \ge \frac{3}{4} \mid X = \frac{1}{2}) = \int_{\frac{3}{4}}^{1} \frac{32y}{15}\,dy = \frac{7}{15}$.

Example 2.1.26. *Poisson Thinning*
Let $\lambda > 0$, $0 \le p \le 1$, $N \sim \text{Poisson}(\lambda)$, and $X \mid N = n \sim \text{Binomial}(n,p)$.
For $k \in \mathbb{N}_0$, by total probability, $\mathbb{P}(X = k) = \sum_{n=k}^{\infty} e^{-\lambda}\frac{\lambda^n}{n!}\binom{n}{k}p^k(1-p)^{n-k} = e^{-\lambda p}\frac{(\lambda p)^k}{k!}$, thus $X \sim \text{Poisson}(\lambda p)$.

Example 2.1.27. *A Triangular Exponential Joint Density*
Let $\lambda > 0$ and $f_{X,Y}(x,y) = \lambda^2 e^{-\lambda y}\mathbf{1}_{\{0 \le x \le y\}}$.
For $x \ge 0$, $f_X(x) = \int_x^{\infty} \lambda^2 e^{-\lambda y}\,dy = \lambda e^{-\lambda x}$; for $y \ge 0$, $f_Y(y) = \int_0^y \lambda^2 e^{-\lambda y}\,dx = \lambda^2 y e^{-\lambda y}$.
Hence $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \lambda e^{-\lambda(y-x)}\mathbf{1}_{[x,\infty)}(y)$ and $f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = y^{-1}\mathbf{1}_{[0,y]}(x)$.
Thus $Y \mid X = x \sim x + \text{Exp}(\lambda)$ and $X \mid Y = y \sim \text{Uniform}(0,y)$.

Example 2.1.28. *Bivariate Normal Conditional Density*
Let (X, Y) be bivariate normal with means μ_X, μ_Y, standard deviations $\sigma_X, \sigma_Y > 0$, and correlation $\rho \in (-1,1)$.
For fixed x, $f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \{\sigma_Y\sqrt{2\pi(1-\rho^2)}\}^{-1}\exp\left[-\frac{(y-\mu_Y-\rho\sigma_Y\sigma_X^{-1}(x-\mu_X))^2}{2\sigma_Y^2(1-\rho^2)}\right]$.
Thus $Y \mid X = x \sim N(\mu_Y + \rho\sigma_Y\sigma_X^{-1}(x-\mu_X), \sigma_Y^2(1-\rho^2))$; if $\rho = 0$, this law is independent of x.

2.2 Conditional Expectation

Definition 2.2.1. *Conditional Expectation Given a Random Variable*
Let X, Y be random variables, and assume the relevant sums or integrals exist.

(i) *Discrete case*: If X, Y are discrete, then $\mathbb{E}(X \mid Y = y) := \sum_x x\,p_{X|Y}(x \mid y)$.

(ii) *Density case*: If X, Y have conditional density $f_{X|Y}$, then $\mathbb{E}(X \mid Y = y) := \int_{\mathbb{R}} x f_{X|Y}(x \mid y)\, dx$
(iii) *Measurable transforms*: For measurable r,

$$\mathbb{E}(r(X,Y) \mid Y = y) = \begin{cases} \sum_x r(x,y) p_{X|Y}(x \mid y), & \text{discrete case,} \\ \int_{\mathbb{R}} r(x,y) f_{X|Y}(x \mid y)\, dx, & \text{density case.} \end{cases}$$

(iv) *Random-variable form*: $\mathbb{E}(X \mid Y)$ is the function $g(Y)$ with $g(y) = \mathbb{E}(X \mid Y = y)$.

Example 2.2.2. *A Conditional Uniform Model*
Let $X \sim \text{Uniform}(0,1)$ and, conditionally on $X = x$, let $Y \sim \text{Uniform}(x,1)$.
For $0 \le x < 1$ and $x < y < 1$, $f_{Y|X}(y \mid x) = \frac{1}{1-x}$, so $\mathbb{E}(Y \mid X = x) = \frac{1}{1-x}\int_x^1 y\, dy = \frac{1+x}{2}$; $\mathbb{E}(Y \mid X) = \frac{1+X}{2}$.

Theorem 2.2.3. *Tower Property*
Let X, Y be real-valued random variables whose joint law is either discrete or admits a density, and let $r : \mathbb{R}^2 \to \mathbb{R}$ be measurable. If $\mathbb{E}|Y| < \infty$ and $\mathbb{E}|r(X,Y)| < \infty$, then $\mathbb{E}[\mathbb{E}(Y \mid X)] = \mathbb{E}Y$ and $\mathbb{E}[\mathbb{E}(r(X,Y) \mid X)] = \mathbb{E}(r(X,Y))$.

Proof. Let $h(x,y)$ denote either y or $r(x,y)$, and set $g_h(x) := \mathbb{E}(h(X,Y) \mid X = x)$.
In the density case, $g_h(x) = \int h(x,y) f_{Y|X}(y \mid x)\, dy$ and $f(x,y) = f_X(x) f_{Y|X}(y \mid x)$,
so Fubini gives $\mathbb{E}[g_h(X)] = \int\int h(x,y) f(x,y)\, dy\, dx = \mathbb{E}h(X,Y)$.
In the discrete case, $\mathbb{E}[g_h(X)] = \sum_x \sum_y h(x,y) p_{Y|X}(y \mid x) p_X(x) = \sum_x \sum_y h(x,y) p_{X,Y}(x,y) = \mathbb{E}h(X,Y)$.
Taking $h(x,y) = y$ and $h(x,y) = r(x,y)$ gives the two identities.
Structural Commentary: Conditional expectation first averages over fibers $\{X = x\}$, and the outer expectation averages those fiber means over the law of X.
Key Insight: Iterated expectation removes the conditioning variable without changing the total mean. □

Example 2.2.4. *Using the Tower Property*
Let $X \sim \text{Uniform}(0,1)$ and, conditionally on $X = x$, let $Y \sim \text{Uniform}(x,1)$.
Since $\mathbb{E}(Y \mid X) = \frac{1+X}{2}$ and $\mathbb{E}X = \frac{1}{2}$, the tower property gives $\mathbb{E}Y = \mathbb{E}[\mathbb{E}(Y \mid X)] = \mathbb{E}\left[\frac{1+X}{2}\right] = \frac{3}{4}$.

Proposition 2.2.5. *Pulling Out What Is Known*
Let X, Y be real-valued random variables whose joint law is either discrete or admits a density, and let $r, s : \mathbb{R} \to \mathbb{R}$ be measurable with $\mathbb{E}|s(Y)| < \infty$, $\mathbb{E}|r(X)s(Y)| < \infty$, and $\mathbb{E}|r(X)| < \infty$. Then $\mathbb{E}(r(X)s(Y) \mid X) = r(X)\mathbb{E}(s(Y) \mid X)$ and $\mathbb{E}(r(X) \mid X) = r(X)$.

Proof. Let K_x be the conditional law of Y given $X = x$, so $K_x(\{y\}) = p_{Y|X}(y \mid x)$ in the discrete case and $K_x(dy) = f_{Y|X}(y \mid x)\, dy$ in the density case. Set $m_s(x) := \int s(y) K_x(dy) = \mathbb{E}(s(Y) \mid X = x)$, with the integral read as a sum in the discrete case. For admissible x, $\mathbb{E}(r(X)s(Y) \mid X = x) = \int r(x)s(y) K_x(dy) = r(x) m_s(x)$, and $\mathbb{E}(r(X) \mid X = x) = \int r(x) K_x(dy) = r(x)$. Evaluating at $x = X$ gives the identities.
Structural Commentary: Conditioning on X makes every function of X fixed inside remaining average over Y.
Key Insight: Known factors can be treated as constants under conditional expectation. □

Definition 2.2.6. *Conditional Variance*
Let X, Y be real-valued random variables with finite second conditional moment for each x under consideration.

(i) The conditional mean is $\mu(x) := \mathbb{E}(Y \mid X = x)$.
(ii) The conditional variance at x is $v(x) := \text{Var}(Y \mid X = x) := \mathbb{E}((Y - \mu(x))^2 \mid X = x)$.
(iii) The conditional-variance random variable is $\text{Var}(Y \mid X) := v(X)$.

Theorem 2.2.7. *Law of Total Variance*
Let X, Y be real-valued random variables with $\mathbb{E}Y^2 < \infty$. Set $b(x) := \mathbb{E}(Y \mid X = x)$ and $v(x) := \text{Var}(Y \mid X = x)$, and write $b(X)$ and $v(X)$ for the induced random variables. Then $\text{Var}(Y) = \mathbb{E}[v(X)] + \text{Var}(b(X))$.

Proof. Let $m := \mathbb{E}Y$ and $e := Y - b(X)$.
The tower property gives $\mathbb{E}b(X) = m$, and $\mathbb{E}(e \mid X) = \mathbb{E}(Y \mid X) - b(X) = 0$.
Since $Y - m = e + (b(X) - m)$, expansion gives $\text{Var}(Y) = \mathbb{E}e^2 + 2\mathbb{E}[e(b(X) - m)] + \mathbb{E}(b(X) - m)^2$.
The cross term is zero because $\mathbb{E}[e(b(X) - m)] = \mathbb{E}[(b(X) - m)\mathbb{E}(e \mid X)] = 0$.
Also $\mathbb{E}e^2 = \mathbb{E}[\mathbb{E}(e^2 \mid X)] = \mathbb{E}[v(X)]$, $\mathbb{E}(b(X) - m)^2 = \text{Var}(b(X))$. Thus $\text{Var}(Y) = \mathbb{E}[v(X)] + \text{Var}(b(X))$.
Structural Commentary: Variance splits into within-fiber variance and between-fiber mean variation.
Key Insight: Total variance equals average conditional variance plus variance of conditional expectation. □

Example 2.2.8. *Random Binomial Proportion*
Let $Q \sim \text{Uniform}(0,1)$ and, conditionally on $Q = q$, let $X \sim \text{Binomial}(n,q)$.
Then $\mathbb{E}(X \mid Q) = nQ$, so $\mathbb{E}X = \mathbb{E}(nQ) = \frac{n}{2}$. Also $\text{Var}(X \mid Q) = nQ(1-Q)$.
Hence $\mathbb{E}[\text{Var}(X \mid Q)] = n\int_0^1 q(1-q)\, dq = \frac{n}{6}$, $\text{Var}(\mathbb{E}(X \mid Q)) = \text{Var}(nQ) = \frac{n^2}{12}$. Thus $\text{Var}(X) = \frac{n}{6} + \frac{n^2}{12}$.

Corollary 2.2.9. *Constant Conditional Mean Implies Uncorrelated*
Let $X, Y \in L^2$ and suppose $\mathbb{E}(X \mid Y) = c$ a.s. for some constant $c \in \mathbb{R}$. Then $c = \mathbb{E}X$ and $\operatorname{Cov}(X, Y) = 0$.

Proof. The tower property gives $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X \mid Y)] = c$.
Since Y is known when conditioning on Y, $\mathbb{E}(XY \mid Y) = Y\mathbb{E}(X \mid Y) = cY$.
Taking expectations gives $\mathbb{E}(XY) = c\mathbb{E}Y = \mathbb{E}X\,\mathbb{E}Y$, hence $\operatorname{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}X\,\mathbb{E}Y = 0$.
Structural Commentary: A constant conditional mean kills covariance with all functions of Y, hence with Y.
Key Insight: Mean independence implies uncorrelatedness, but not conversely. □

2.3 Conditional Moment Methods and Prediction

Conditional expectation is not only a way to compute means. It is also the geometric projection rule behind random-sum formulas and least-squares prediction.

Proposition 2.3.1. *Random Sums*
Let $N \in \mathbb{N}_0$ a.s. with $\mathbb{E}N^2 < \infty$, and let $(X_i)_{i\geq 1}$ be iid, independent of N, with $\mathbb{E}X_1 = m$ and $\operatorname{Var}(X_1) = v < \infty$. Set $T = \sum_{i=1}^{N} X_i$, with $T = 0$ on $\{N = 0\}$.

(i) $\mathbb{E}T = m\mathbb{E}N$.
(ii) $\operatorname{Var}(T) = v\mathbb{E}N + m^2 \operatorname{Var}(N)$.

Proof.

(i) On $\{N = n\}$, $T = \sum_{i=1}^{n} X_i$, so $\mathbb{E}(T \mid N = n) = nm$ and $\mathbb{E}(T \mid N) = mN$; hence $\mathbb{E}T = \mathbb{E}\mathbb{E}(T \mid N) = m\mathbb{E}N$.

(ii) On $\{N = n\}$, independence gives $\operatorname{Var}(T \mid N = n) = nv$, so $\operatorname{Var}(T \mid N) = vN$.
Total variance gives $\operatorname{Var}(T) = \mathbb{E}(vN) + \operatorname{Var}(mN) = v\mathbb{E}N + m^2 \operatorname{Var}(N)$.

Structural Commentary: The randomness of the number of summands contributes the second variance term.
Key Insight: First condition on how many summands exist; then average over that count. □

Proposition 2.3.2. *MGF of a Random Sum*
Let $T = \sum_{i=1}^{N} X_i$, where $N \in \mathbb{N}_0$ is independent of iid $(X_i)_{i\geq 1}$.
If ψ_N and ψ_X are finite at the relevant arguments, then $\psi_T(t) = \psi_N(\log \psi_X(t))$.

Proof. On $\{N = n\}$, independence gives $\mathbb{E}(e^{tT} \mid N = n) = \mathbb{E}e^{t\sum_{i=1}^{n} X_i} = \prod_{i=1}^{n} \psi_X(t) = [\psi_X(t)]^n$.
Hence $\psi_T(t) = \mathbb{E}[\psi_X(t)^N] = \mathbb{E}e^{N \log \psi_X(t)} = \psi_N(\log \psi_X(t))$.
Structural Commentary: Conditioning on N turns the random sum into a fixed product of MGFs.
Key Insight: Random counts compose MGFs by substituting $\log \psi_X(t)$ into ψ_N. □

Example 2.3.3. *Compound Poisson Count*
Let $N \sim \text{Poisson}(\mu)$ and, given N, let $T = \sum_{i=1}^{N} X_i$ with iid $X_i \sim \text{Poisson}(\lambda)$ independent of N.
Since $\psi_N(s) = \exp(\mu(e^s - 1))$ and $\psi_X(t) = \exp(\lambda(e^t - 1))$, one has $\psi_T(t) = \psi_N(\log \psi_X(t)) = \exp(\mu(e^{\lambda(e^t-1)} - 1))$.

Proposition 2.3.4. *Best Constant Predictor*
For $Y \in L^2$, $\arg\min_{c\in\mathbb{R}} \mathbb{E}(Y - c)^2 = \mathbb{E}Y$, and $\min_{c\in\mathbb{R}} \mathbb{E}(Y - c)^2 = \operatorname{Var}(Y)$.

Proof. Let $\mu = \mathbb{E}Y$. Since $\mathbb{E}(Y - \mu) = 0$, $\mathbb{E}(Y - c)^2 = \mathbb{E}[(Y - \mu) + (\mu - c)]^2 = \operatorname{Var}(Y) + (\mu - c)^2$.
Since $(\mu - c)^2 \geq 0$ with equality iff $c = \mu$, the unique minimizer is $c = \mu$.
Structural Commentary: The mean is the orthogonal projection of Y onto the constants.
Key Insight: Best constant prediction equals the average. □

Theorem 2.3.5. *Conditional Expectation as Best Predictor*
Let $X, Y \in L^2$ and $\mathcal{G} := \{h(X) : h(X) \in L^2\}$. Let $M := \mathbb{E}(Y \mid X)$, then $M \in \mathcal{G}$, $\mathbb{E}(Y - M)^2 \leq \mathbb{E}(Y - Z)^2 \forall Z \in \mathcal{G}$.

Proof. For $Z \in \mathcal{G}$, both M and Z are $\sigma(X)$-measurable.
Since $\mathbb{E}(Y - M \mid X) = 0$, $\mathbb{E}[(Y - M)(M - Z)] = \mathbb{E}[(M - Z)\mathbb{E}(Y - M \mid X)] = 0$.
Hence $\mathbb{E}(Y - Z)^2 = \mathbb{E}(Y - M)^2 + \mathbb{E}(M - Z)^2 \geq \mathbb{E}(Y - M)^2$.
Structural Commentary: Prediction from X is performed fiber by fiber over the conditional laws of Y.
Key Insight: The optimal nonlinear least-squares predictor is the conditional mean. □

Definition 2.3.6. *Best Linear Predictor*
For $X, Y \in L^2$, set $\mathcal{A}_X := \{\alpha + \beta X : \alpha, \beta \in \mathbb{R}\}$.
A *best linear predictor* of Y from X is any $\widehat{Y}^* \in \arg\min_{Z \in \mathcal{A}_X} \mathbb{E}(Y - Z)^2$.

Theorem 2.3.7. *Best Linear Predictor*
Let $X, Y \in L^2$, $\sigma_X^2 := \mathrm{Var}(X) > 0$, $\mu_X := \mathbb{E}X$, and $\mu_Y := \mathbb{E}Y$.

(i) $\arg\min_{Z \in \mathcal{A}_X} \mathbb{E}(Y - Z)^2 = \{\alpha^* + \beta^* X\}$, *where* $\beta^* = \frac{\mathrm{Cov}(X,Y)}{\sigma_X^2}$ *and* $\alpha^* = \mu_Y - \beta^* \mu_X$.

(ii) $\min_{Z \in \mathcal{A}_X} \mathbb{E}(Y - Z)^2 = \mathrm{Var}(Y) - \frac{\mathrm{Cov}(X,Y)^2}{\sigma_X^2}$.

(iii) If $\mathrm{Var}(Y) > 0$ *and* $\rho = \mathrm{Corr}(X, Y)$, *then the minimum error is* $\mathrm{Var}(Y)(1 - \rho^2)$.

Proof.

(i) Let $q(\alpha, \beta) := \mathbb{E}(Y - \alpha - \beta X)^2$. For fixed β, $q(\alpha, \beta) = \mathrm{Var}(Y - \beta X) + (\mu_Y - \alpha - \beta \mu_X)^2$, so $\alpha(\beta) = \mu_Y - \beta \mu_X$ uniquely minimizes over α.

(ii) Then $q(\alpha(\beta), \beta) = \mathrm{Var}(Y - \beta X) = \mathrm{Var}(Y) + \beta^2 \sigma_X^2 - 2\beta \,\mathrm{Cov}(X, Y)$. Since $\sigma_X^2 > 0$, this quadratic has unique minimizer $\beta^* = \frac{\mathrm{Cov}(X,Y)}{\sigma_X^2}$.

(iii) Substituting β^* gives $\mathrm{Var}(Y) - \frac{\mathrm{Cov}(X,Y)^2}{\sigma_X^2}$. If $\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$, then this equals $\mathrm{Var}(Y)(1 - \rho^2)$.

Structural Commentary: Linear prediction projects Y onto $\mathcal{A}_X$.
Key Insight: The optimal slope is normalized covariance. □

Example 2.3.8. *Bivariate Normal Prediction*
If (X, Y) is bivariate normal with means μ_X, μ_Y, standard deviations $\sigma_X, \sigma_Y > 0$, and correlation $\rho \in (-1, 1)$, then $Y \mid X \sim N(\mu_Y + \rho \sigma_Y \sigma_X^{-1}(X - \mu_X), \sigma_Y^2(1 - \rho^2))$.
Thus $\mathbb{E}(Y \mid X) = \mu_Y + \rho \sigma_Y \sigma_X^{-1}(X - \mu_X)$ is linear, with conditional variance $\sigma_Y^2(1 - \rho^2)$.

Example 2.3.9. *Regression to the Mean*
Let $\mathbb{E}X = \mathbb{E}Y = \mu$, $\mathrm{Var}(X) = \mathrm{Var}(Y) = \sigma^2 > 0$, and $\mathrm{Corr}(X, Y) = \rho$. Best linear predictor is $\widehat{Y} = \mu + \rho(X - \mu)$.
For $0 < \rho < 1$, $|\widehat{Y} - \mu| = \rho |X - \mu| < |X - \mu|$, so predicted scores regress toward μ.

3 Probabilistic Convergence

3.1 Modes of Probabilistic Convergence

Pointwise convergence is too strict for most random sequences. The useful replacement is to ask whether probabilities, laws, or moments of the errors become small.

Definition 3.1.1. *Modes of Probabilistic Convergence*
Let $(X_n)_{n \geq 1}$ and X be real-valued random variables with distribution functions F_n and F.

(i) *Convergence in probability*: $X_n \xrightarrow{\mathbb{P}} X$ iff $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$.
(ii) *Convergence in distribution*: $X_n \rightsquigarrow X$ iff $F_n(t) \to F(t)$ at every continuity point t of F.
(iii) *Convergence in L^p*: for $p \geq 1$, $X_n \xrightarrow{L^p} X$ iff $\mathbb{E}\,|X_n - X|^p \to 0$.
(iv) *Convergence in quadratic mean*: $X_n \xrightarrow{qm} X$ iff $X_n \xrightarrow{L^2} X$.
(v) *Almost-sure convergence*: $X_n \xrightarrow{\text{a.s.}} X$ iff $\mathbb{P}(\{\omega \in \Omega : X_n(\omega) \to X(\omega)\}) = 1$.
(vi) *Constant limits*: if $X = c$ almost surely for a constant c, then $X_n \xrightarrow{\mathbb{P}} c$, $X_n \rightsquigarrow c$, $X_n \xrightarrow{L^p} c$, $X_n \xrightarrow{qm} c$, and $X_n \xrightarrow{\text{a.s.}} c$ denote convergence to that constant random variable in the corresponding modes.

Example 3.1.2. *A Shrinking Normal Law*
Let $X_n \sim N(0, \frac{1}{n})$, and let F be the distribution function of δ_0.
Since $\sqrt{n} X_n \sim N(0, 1)$, $F_n(t) = \mathbb{P}(X_n \leq t) = \Phi(\sqrt{n} t)$, so $F_n(t) \to 0$ for $t < 0$ and $F_n(t) \to 1$ for $t > 0$.
Hence $F_n(t) \to F(t)$ at every continuity point $t \neq 0$ of F, and therefore $X_n \rightsquigarrow 0$.

Theorem 3.1.3. *Basic Implications Between Convergence Modes*
Let X_n and X be real-valued random variables with distribution functions F_n and F, and let $c \in \mathbb{R}$.

(i) $X_n \xrightarrow{qm} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$.
(ii) $X_n \xrightarrow{\mathbb{P}} X \Rightarrow X_n \rightsquigarrow X$.
(iii) $X_n \rightsquigarrow c \Rightarrow X_n \xrightarrow{\mathbb{P}} c$.
(iv) $X_n \xrightarrow{\text{a.s.}} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$.

(v) $X_n \xrightarrow{L^1} X \Rightarrow X_n \xrightarrow{\mathbb{P}} X$, *and* $X_n \xrightarrow{qm} X \Rightarrow X_n \xrightarrow{L^1} X$.

Proof.

(i) If $X_n \xrightarrow{qm} X$, then for each $\varepsilon > 0$, Markov gives $\mathbb{P}(|X_n - X| > \varepsilon) \le \varepsilon^{-2}\mathbb{E}\,|X_n - X|^2 \to 0$.

(ii) Let x be a continuity point of F.
For each $\varepsilon > 0$, $F(x-\varepsilon) - \mathbb{P}(|X_n - X| > \varepsilon) \le F_n(x) \le F(x+\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon)$,
so $F(x-\varepsilon) \le \liminf_n F_n(x) \le \limsup_n F_n(x) \le F(x+\varepsilon)$; letting $\varepsilon \downarrow 0$ gives $F_n(x) \to F(x)$.

(iii) For $\varepsilon > 0$, the distribution function of c is continuous at $c-\varepsilon$ and $c+\varepsilon$.
Hence $\mathbb{P}(|X_n - c| > \varepsilon) \le F_n(c-\varepsilon) + 1 - F_n(c+\varepsilon) \to 0$.

(iv) Fix $\varepsilon > 0$ and set $Y_n = \mathbf{1}_{\{|X_n - X| > \varepsilon\}}$. If $X_n \to X$ a.s., then $Y_n \to 0$ a.s. and $0 \le Y_n \le 1$.
Hence dominated convergence gives $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{E}Y_n \to 0$.

(v) If $X_n \xrightarrow{L^1} X$, then $\mathbb{P}(|X_n - X| > \varepsilon) \le \varepsilon^{-1}\mathbb{E}\,|X_n - X| \to 0$ by Markov.
If $X_n \xrightarrow{qm} X$, then $\mathbb{E}\,|X_n - X| \le (\mathbb{E}\,|X_n - X|^2)^{\frac{1}{2}} \to 0$ by Cauchy–Schwarz.

Structural Commentary: Strong modes give tail bounds; distribution convergence gives c.d.f. control.
Key Insight: Convert to tail bounds; use c.d.f. squeezes when laws enter. □

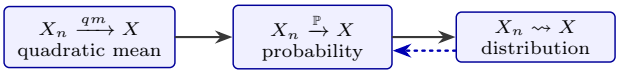


Figure 211: Solid: always; dotted: point-mass limit.

Example 3.1.4. *Why the Reverse Arrows Fail*

(i) Let $U \sim \text{Uniform}(0,1)$ and $X_n = \sqrt{n}\mathbf{1}_{(0,\frac{1}{n})}(U)$. Then for each $\varepsilon > 0$, $\mathbb{P}(|X_n| > \varepsilon) \le \frac{1}{n} \to 0$,
so $X_n \xrightarrow{\mathbb{P}} 0$, but $\mathbb{E}X_n^2 = 1$ for all n; hence $X_n \not\to 0$ in quadratic mean.

(ii) Let $X \sim N(0,1)$ and set $X_n = -X$. Then $X_n \sim X$ for all n, so $X_n \rightsquigarrow X$,
but for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(2\,|X| > \varepsilon) > 0$ for all n; hence $X_n \not\to X$ in probability.

Remark 3.1.5. *A Warning About Expectations*
Convergence in probability does not imply convergence of expectations.
If $\mathbb{P}(X_n = n^2) = \frac{1}{n}$, $\mathbb{P}(X_n = 0) = 1 - \frac{1}{n}$, then $\forall \varepsilon > 0$, $\mathbb{P}(|X_n| > \varepsilon) = \frac{1}{n} \to 0$, so $X_n \xrightarrow{\mathbb{P}} 0$, $\mathbb{E}X_n = n \to +\infty$.

Definition 3.1.6. *Asymptotic Uniform Integrability*
A sequence (X_n) is *asymptotically uniformly integrable* if $\lim_{M\to\infty} \limsup_{n\to\infty} \mathbb{E}(|X_n|\,\mathbf{1}_{\{|X_n|>M\}}) = 0$.

Proposition 3.1.7. *Recovering Expectations from Probability Convergence*
Let $b \in \mathbb{R}$. *If* $X_n \xrightarrow{\mathbb{P}} b$ *and* (X_n) *is asymptotically uniformly integrable, then* $\mathbb{E}X_n \to b$.

Proof. For $M > |b|$, set $\tau_M(x) = (-M) \vee (x \wedge M)$. As τ_M bounded, continuous; $\tau_M(b) = b$, $\tau_M(X_n) \xrightarrow{\mathbb{P}} b$.
For $\eta > 0$, $\mathbb{E}\,|\tau_M(X_n) - b| \le \eta + 2M\mathbb{P}(|\tau_M(X_n) - b| > \eta)$, so $\mathbb{E}\tau_M(X_n) \to b$ after $n \to \infty$ and $\eta \downarrow 0$.
Moreover, $|X_n - \tau_M(X_n)| \le |X_n|\,\mathbf{1}_{\{|X_n|>M\}}$, hence $\limsup_n |\mathbb{E}X_n - b| \le \limsup_n \mathbb{E}(|X_n|\,\mathbf{1}_{\{|X_n|>M\}})$.
Letting $M \to \infty$ and using asymptotic uniform integrability gives $\mathbb{E}X_n \to b$.
Structural Commentary: Truncation makes convergence bounded; asymptotic uniform integrability kills tails.
Key Insight: Probability convergence handles the truncations; tail control restores expectations. □

Theorem 3.1.8. *Continuous Mapping and Slutsky Rules*
Let $g : \mathbb{R} \to \mathbb{R}$ *be continuous, and let* $c \in \mathbb{R}$.

(i) $X_n \xrightarrow{\mathbb{P}} X \Rightarrow g(X_n) \xrightarrow{\mathbb{P}} g(X)$.
(ii) $X_n \rightsquigarrow X \Rightarrow g(X_n) \rightsquigarrow g(X)$.
(iii) $X_n \rightsquigarrow X$ *and* $Y_n \xrightarrow{\mathbb{P}} c \Rightarrow X_n + Y_n \rightsquigarrow X + c$ *and* $X_nY_n \rightsquigarrow cX$.
(iv) $X_n \xrightarrow{\mathbb{P}} X$ *and* $Y_n \xrightarrow{\mathbb{P}} Y \Rightarrow X_n + Y_n \xrightarrow{\mathbb{P}} X + Y$ *and* $X_nY_n \xrightarrow{\mathbb{P}} XY$.

Proof.

(i) Fix $\varepsilon, \eta > 0$. Choose K, $\mathbb{P}(|X| > K) < \eta$, $\delta \le 1$ s.t. $|x-y| < \delta$, $|x|, |y| \le K+1$; $|g(x) - g(y)| < \varepsilon$.
Then $\mathbb{P}(|g(X_n) - g(X)| > \varepsilon) \le \mathbb{P}(|X| > K) + \mathbb{P}(|X_n - X| > \delta)$, so lim sup at most η.

(ii) Let t be a continuity point of the law of $g(X)$ and set $A_t = \{x : g(x) \le t\}$.
Since $\partial A_t \subseteq \{x : g(x) = t\}$ and $\mathbb{P}(g(X) = t) = 0$, the continuity-set form of weak convergence gives
$\mathbb{P}(g(X_n) \le t) = \mathbb{P}(X_n \in A_t) \to \mathbb{P}(X \in A_t) = \mathbb{P}(g(X) \le t)$.

(iii) If $Z_n \rightsquigarrow Z$ and $R_n \xrightarrow{\mathbb{P}} 0$, then $Z_n + R_n \rightsquigarrow Z$ by the c.d.f. squeeze
$F_{Z_n}(t-\eta) - \mathbb{P}(|R_n| > \eta) \le F_{Z_n+R_n}(t) \le F_{Z_n}(t+\eta) + \mathbb{P}(|R_n| > \eta)$,
followed by $n \to \infty$ and $\eta \downarrow 0$ through continuity points of F_Z.
Since $X_n + c \rightsquigarrow X + c$ by (ii), applying this with $Z_n = X_n + c$ and $R_n = Y_n - c$ gives $X_n + Y_n \rightsquigarrow X + c$.
Also $X_n \rightsquigarrow X$ implies (X_n) is tight, so $\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \le \mathbb{P}(|X_n| > K) + \mathbb{P}(|Y_n - c| > \frac{\varepsilon}{K})$.
Letting $n \to \infty$, then $K \uparrow \infty$ gives $X_n(Y_n - c) \xrightarrow{\mathbb{P}} 0$.
Since $cX_n \rightsquigarrow cX$ by (ii), the additive perturbation just proved gives $X_nY_n = cX_n + X_n(Y_n - c) \rightsquigarrow cX$.

(iv) For sums, $\mathbb{P}(|(X_n + Y_n) - (X + Y)| > \varepsilon) \le \mathbb{P}(|X_n - X| > \frac{\varepsilon}{2}) + \mathbb{P}(|Y_n - Y| > \frac{\varepsilon}{2}) \to 0$.
For products, (X_n) and Y are tight, $\mathbb{P}(|X_n(Y_n - Y)| > \frac{\varepsilon}{2}) \le \mathbb{P}(|X_n| > K) + \mathbb{P}(|Y_n - Y| > \frac{\varepsilon}{2K})$,
while $\mathbb{P}(|Y(X_n - X)| > \frac{\varepsilon}{2}) \le \mathbb{P}(|Y| > K) + \mathbb{P}(|X_n - X| > \frac{\varepsilon}{2K})$; let $n \to \infty$ and then $K \uparrow \infty$.

Structural Commentary: Continuity transports convergence; Slutsky absorbs probability-small errors.
Key Insight: Control the bounded high-probability part; let the tails vanish. □

3.2 Probability on Metric Spaces

4 Limit Theorems

4.1 Laws of Large Numbers

The law of large numbers formalizes the idea that a large sample average is close to the mean of the distribution.

Perspective 4.1.1. *Law of the Sample Mean*

If $X_1, \dots, X_n$ are IID, mean μ, variance σ^2, then $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$ is a random variable with law $\mathcal{L}(\overline{X}_n)$.

By Theorem 1.3.15, $\mathbb{E}\overline{X}_n = \mu$ and $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$, so $\mathcal{L}(\overline{X}_n)$ concentrates around μ as $n \to \infty$.

Theorem 4.1.2. *Weak Law of Large Numbers*

Let $(X_i)_{i\ge1}$ be IID real-valued random variables with $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 < +\infty$. Set $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$.

Then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$, equivalently $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \to 0$ for every $\varepsilon > 0$.

Proof. For $n \ge 1$, $\mathbb{E}\overline{X}_n = n^{-1} \sum_{i=1}^{n} \mathbb{E}X_i = \mu$ and, by independence, $\mathrm{Var}(\overline{X}_n) = n^{-2} \sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$.
For every $\varepsilon > 0$, Chebyshev gives $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \le \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0$, hence $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.
Structural Commentary: Averaging keeps the mean fixed and divides variance by n.
Key Insight: Chebyshev turns vanishing variance into convergence in probability. □

Proposition 4.1.3. *Weak LLN with Vanishing Average Variance*

Let $(X_i)_{i\ge1}$ be independent, $X_i \in L^2$, with $\mu_i := \mathbb{E}X_i$ and $\sigma_i^2 := \mathrm{Var}(X_i)$.
Set $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$ and $m_n := n^{-1} \sum_{i=1}^{n} \mu_i$. If $m_n \to \mu$ and $n^{-2} \sum_{i=1}^{n} \sigma_i^2 \to 0$, then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. $\mathbb{E}\overline{X}_n = m_n$. By independence, $\mathrm{Var}(\overline{X}_n) = n^{-2} \sum_{i=1}^{n} \sigma_i^2 \to 0$. Fix $\varepsilon > 0$. $\forall$ large n, $|m_n - \mu| < \frac{\varepsilon}{2}$.
Hence $\{|\overline{X}_n - \mu| > \varepsilon\} \subseteq \{|\overline{X}_n - m_n| > \frac{\varepsilon}{2}\}$. By Chebyshev, $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \le \frac{4\,\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} \to 0$.
Structural Commentary: Identical distribution is not essential for a weak law.
Key Insight: It is enough that the mean target stabilises and the average variance vanishes. □

Example 4.1.4. *Coin Proportions and Chebyshev Scale*

Let $X_i \sim \mathrm{Bernoulli}(p)$ be IID head indicators, $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$. Then $\mathbb{E}\overline{X}_n = p$, $\mathrm{Var}(\overline{X}_n) = \frac{p(1-p)}{n}$, $\overline{X}_n \xrightarrow{\mathbb{P}} p$.
For $p = \frac{1}{2}$, Chebyshev gives $\mathbb{P}(0.4 \le \overline{X}_n \le 0.6) \ge 1 - \frac{1}{4n(0.1)^2} = 1 - \frac{25}{n}$, at least 0.7 for $n \ge 84$.
If $n = 10000$ fair tosses produce 5067 heads, then $\overline{X}_n = 0.5067$, while $\mathrm{Var}(\overline{X}_n) = \frac{1}{4n} = 2.5 \cdot 10^{-5}$ and $\mathrm{sd}(\overline{X}_n) = 0.005$. The observed deviation is about 1.34 standard deviations from $\frac{1}{2}$.

Theorem 4.1.5. *Strong Law of Large Numbers*

Let $(X_i)_{i\ge1}$ be IID real-valued random variables with $X_1 \in L^1$, $\mu := \mathbb{E}X_1$, and $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$.

Then $\overline{X}_n \xrightarrow{\text{a.s.}} \mu$, equivalently $\mathbb{P}(\lim_{n\to\infty} \overline{X}_n = \mu) = 1$.

Proof. After replacing X_i by $X_i - \mu$, assume $\mu = 0$. Set $Y_i := X_i \mathbf{1}_{\{|X_i| \le i\}}$.
Since $\sum_{i=1}^{\infty} \mathbb{P}(X_i \neq Y_i) = \sum_{i=1}^{\infty} \mathbb{P}(|X_1| > i) < +\infty$, Borel–Cantelli gives $X_i = Y_i$ eventually a.s.
Hence $n^{-1} \sum_{i=1}^{n} (X_i - Y_i) \to 0$ a.s. For $Z_i := Y_i - \mathbb{E}Y_i$, independence gives
$\sum_{i=1}^{\infty} \frac{\mathrm{Var}(Z_i)}{i^2} \le \sum_{i=1}^{\infty} \frac{\mathbb{E}(X_1^2 \mathbf{1}_{\{|X_1| \le i\}})}{i^2} < +\infty$, by Kolmogorov's variance criterion, $n^{-1} \sum_{i=1}^{n} Z_i \to 0$ a.s.
By dominated convergence, $\mathbb{E}Y_i \to 0$, hence Cesaro averaging gives $n^{-1} \sum_{i=1}^{n} \mathbb{E}Y_i \to 0$.
Combining $X_i - Y_i$, Z_i, and $\mathbb{E}Y_i$ gives $\overline{X}_n \to 0$ a.s.; adding back μ completes the proof.
Structural Commentary: Truncate tails; centered truncations have summable variance.
Key Insight: Replace by an eventually equal truncated sequence, then average the centered part. □

Corollary 4.1.6. *Strong Implies Weak*
Let $(X_i)_{i \ge 1}$ be IID real-valued random variables, $X_1 \in L^1$, $\mu := \mathbb{E}X_1$, $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$. Then $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. By Theorem 4.1.5, $\overline{X}_n \to \mu$ a.s. For $\varepsilon > 0$, set $Y_n := \mathbf{1}_{\{|\overline{X}_n - \mu| > \varepsilon\}}$.
Then $Y_n \to 0$ a.s. and $0 \le Y_n \le 1$, so dominated convergence gives $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) = \mathbb{E}Y_n \to 0$.
Structural Commentary: Almost-sure convergence gives pointwise tail indicators tending to zero.
Key Insight: Dominate the indicators by 1 and integrate. □

Proposition 4.1.7. *Consistency of Empirical Moments and Variance*
Let $(X_i)_{i \ge 1}$ be iid, $X_1 \in L^2$, $\mu := \mathbb{E}X_1$, and $\sigma^2 := \mathrm{Var}(X_1)$.
For $n \ge 2$, set $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$ and $S_n^2 := (n-1)^{-1} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$. Then $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2$.
More generally, if $X_1 \in L^r$ for an integer $r \ge 1$, then $n^{-1} \sum_{i=1}^{n} X_i^r \xrightarrow{\mathbb{P}} \mathbb{E}X_1^r$.

Proof. Since $(X_i^r)_{i \ge 1}$ is iid and integrable, the strong law gives $n^{-1} \sum_{i=1}^{n} X_i^r \to \mathbb{E}X_1^r$ a.s., hence in probability.
Let $m_2 := \mathbb{E}X_1^2 = \sigma^2 + \mu^2$. The cases $r = 1, 2$ give $\overline{X}_n \to \mu$ and $n^{-1} \sum_{i=1}^{n} X_i^2 \to m_2$ a.s.
Since $S_n^2 = \frac{n}{n-1}(n^{-1} \sum_{i=1}^{n} X_i^2 - \overline{X}_n^2)$, continuous mapping gives $S_n^2 \to m_2 - \mu^2 = \sigma^2$ a.s., hence in probability.
Structural Commentary: Sample moments are averages of transformed observations.
Key Insight: Apply the strong law to X_i^r, then pass through the continuous variance formula. □

Remark 4.1.8. *Why Integrability Matters*
If $X_1 \notin L^1$, the mean $\mathbb{E}X_1$ need not exist and sample averages need not converge to any finite deterministic limit; integrability is the condition that makes a law of large numbers meaningful.

4.2 Central Limit Theory

The law of large numbers says that $\overline{X}_n$ is close to μ for large n. The central limit theorem describes the scale and shape of the remaining fluctuation.

Theorem 4.2.1. *Central Limit Theorem*
Let $(X_i)_{i \ge 1}$ be IID real-valued random variables, $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 \in (0, +\infty)$.
Let $\overline{X}_n := n^{-1} \sum_{i=1}^{n} X_i$, $Z_n := \sigma^{-1} \sqrt{n} (\overline{X}_n - \mu)$.
Then $Z_n \rightsquigarrow N(0,1)$, equivalently $\mathbb{P}(Z_n \le z) \to \Phi(z) := \int_{-\infty}^{z} (2\pi)^{-\frac{1}{2}} e^{-\frac{x^2}{2}} \, dx \ \forall z \in \mathbb{R}$.

Proof. Set $Y_i := \frac{X_i - \mu}{\sigma}$, so $\mathbb{E}Y_i = 0$, $\mathrm{Var}(Y_i) = 1$, and $Z_n = n^{-\frac{1}{2}} \sum_{i=1}^{n} Y_i$.
For $\varphi(t) := \mathbb{E}e^{itY_1}$, the finite second moment expansion gives $\varphi(u) = 1 - \frac{u^2}{2} + o(u^2)$ as $u \to 0$.
By independence, $\varphi_{Z_n}(t) = [\varphi(\frac{t}{\sqrt{n}})]^n = [1 - \frac{t^2}{2n} + o(\frac{1}{n})]^n \to e^{-\frac{t^2}{2}}$ for every $t \in \mathbb{R}$.
Since $e^{-\frac{t^2}{2}}$ is the characteristic function of $N(0,1)$, Levy's continuity theorem gives $Z_n \rightsquigarrow N(0,1)$; continuity of Φ gives the displayed c.d.f. convergence.
Structural Commentary: Centering and scaling turn the sum into unit-variance IID increments.
Key Insight: Characteristic functions turn normalized sums into powers, whose limit is Gaussian. □

Method 4.2.2. *Equivalent CLT Approximation Forms*
Let $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$ with mean μ and variance $\sigma^2 \in (0,+\infty)$, and set $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$.

(i) Standardized form: $Z_n \approx N(0,1)$.
(ii) Centered form: $\sqrt{n}(\overline{X}_n - \mu) \approx N(0,\sigma^2)$.
(iii) Sample-mean form: $\overline{X}_n \approx N(\mu, \frac{\sigma^2}{n})$.

Use any equivalent form to approximate probabilities involving $\overline{X}_n$.

Lemma 4.2.3. *MGF Continuity Theorem*
Let Z_n, Z be real-valued with MGFs $\psi_n(t) := \mathbb{E}e^{tZ_n}$ and $\psi(t) := \mathbb{E}e^{tZ}$ finite on $(-a,a)$ for some $a > 0$. If $\psi_n(t) \to \psi(t)$ for all $t \in (-a,a)$, then $Z_n \rightsquigarrow Z$.

Proof. For each $s \in \mathbb{R}$, the characteristic functions satisfy $\varphi_n(s) = \mathbb{E}e^{\mathrm{i}sZ_n} = \psi_n(\mathrm{i}s)$ by analytic continuation on the strip $\{z : |\mathrm{Re}\, z| < a\}$. Pointwise MGF convergence near 0 gives $\varphi_n(s) \to \varphi(s)$, the characteristic function of Z; Levy's continuity theorem yields $Z_n \rightsquigarrow Z$.
Structural Commentary: Local MGF convergence determines local characteristic-function convergence.
Key Insight: MGFs near 0 identify the weak limit. □

Proposition 4.2.4. *MGF Proof of the IID CLT*
Let $(X_i)_{i\geq 1}$ be iid with $\mathbb{E}X_1 = 0$, $\mathrm{Var}(X_1) = \sigma^2 \in (0,+\infty)$, and MGF $M(t) := \mathbb{E}e^{tX_1}$ finite near 0. Set $S_n := \sum_{i=1}^{n} X_i$ and $Z_n := \frac{S_n}{\sigma\sqrt{n}}$. Then $Z_n \rightsquigarrow N(0,1)$, equivalently $\mathbb{P}(Z_n \leq x) \to \Phi(x)$ for all $x \in \mathbb{R}$.

Proof. For fixed t near 0, independence gives $M_{Z_n}(t) = \mathbb{E}e^{\frac{tS_n}{\sigma\sqrt{n}}} = \prod_{i=1}^{n} \mathbb{E}e^{\frac{tX_i}{\sigma\sqrt{n}}} = [M(\frac{t}{\sigma\sqrt{n}})]^n$.
Since $M(0) = 1$, $M'(0) = 0$, and $M''(0) = \sigma^2$, Taylor expansion gives $M(s) = 1 + \frac{\sigma^2 s^2}{2} + o(s^2)$ as $s \to 0$.
Hence $M_{Z_n}(t) = (1 + \frac{t^2}{2n} + o(\frac{1}{n}))^n \to e^{\frac{t^2}{2}}$, the MGF of $N(0,1)$.
Lemma 4.2.3 gives $Z_n \rightsquigarrow N(0,1)$, and continuity of Φ gives the c.d.f. convergence.
Structural Commentary: The MGF proof needs the stronger near-zero MGF assumption.
Key Insight: Standardization gives $\frac{t^2}{2n}$; independence exponentiates. □

Example 4.2.5. *A Poisson Error-Count Approximation*
Let $X_1, \ldots, X_{125} \sim \mathrm{Poisson}(5)$ be IID and set $\overline{X}_{125} = 125^{-1}\sum_{i=1}^{125} X_i$.
Since $\mu = 5$ and $\sigma^2 = 5$, the CLT gives $\mathbb{P}(\overline{X}_{125} < 5.5) \approx \Phi(\frac{\sqrt{125}(5.5-5)}{\sqrt{5}}) = \Phi(2.5) \approx 0.9938$.

Example 4.2.6. *Normal Approximation to a Poisson Count*
Let $X_\lambda \sim \mathrm{Poisson}(\lambda)$, $Z_\lambda := \frac{X_\lambda - \lambda}{\sqrt{\lambda}}$. Then $M_{Z_\lambda}(t) = \exp(-t\sqrt{\lambda} + \lambda(e^{\frac{t}{\sqrt{\lambda}}} - 1))$ and $\log M_{Z_\lambda}(t) = \frac{t^2}{2} + o(1)$, so $Z_\lambda \rightsquigarrow N(0,1)$. If $X \sim \mathrm{Poisson}(900)$, then $\mathbb{P}(X > 950) = \mathbb{P}(\frac{X-900}{30} > \frac{5}{3}) \approx 1 - \Phi(\frac{5}{3}) \approx 0.04779$, close to 0.04712.

Example 4.2.7. *Normal Approximation to the Binomial*
If $X \sim \mathrm{Binomial}(n,p)$, $np > 5$, $n(1-p) > 5$, then $X \approx N(np, np(1-p))$; $\frac{X-np}{\sqrt{np(1-p)}} \approx N(0,1)$.
For $X \sim \mathrm{Binomial}(100, \frac{1}{2})$, $\mathbb{P}(X \geq 60) \approx 1 - \Phi(\frac{60-50}{5}) = 1 - \Phi(2) \approx 0.0228$.

Example 4.2.8. *Skewed Summands Need More Averaging*
Let $E_1, \ldots, E_n$ be iid Exp(1) and $G_n := \sum_{i=1}^{n} E_i \sim \mathrm{Gamma}(n,1)$. Then $\frac{G_n - n}{\sqrt{n}} \rightsquigarrow N(0,1)$.
Skewness makes the normal approximation slower for small n.

Example 4.2.9. *Products and Lognormal Shape*
Let $Y_n = y_0 \prod_{i=1}^{n} X_i$ with iid $X_i > 0$, and suppose $\mu_\ell := \mathbb{E}\log X_1$ and $\sigma_\ell^2 := \mathrm{Var}(\log X_1) \in (0,+\infty)$.
Since $\log Y_n = \log y_0 + \sum_{i=1}^{n} \log X_i$, the CLT gives $\log Y_n \approx N(\log y_0 + n\mu_\ell, n\sigma_\ell^2)$, so Y_n is approximately lognormal.
Likewise $V_n = v_0 \prod_{i=1}^{n} R_i$ when iid log-returns $\log R_i$ have finite positive variance.

Theorem 4.2.10. *Studentized Central Limit Theorem*
Let $(X_i)_{i\geq 1}$ be IID real-valued random variables with $\mathbb{E}X_1 = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 \in (0,+\infty)$. Set $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$, $S_n^2 := (n-1)^{-1}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$, and $T_n := S_n^{-1}\sqrt{n}(\overline{X}_n - \mu)$. Then $T_n \rightsquigarrow N(0,1)$.

Proof. The CLT gives $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N(0,1)$, and Proposition 4.1.7 gives $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2$, hence $S_n \xrightarrow{\mathbb{P}} \sigma$. Thus $\frac{\sigma}{S_n} \xrightarrow{\mathbb{P}} 1$, and Slutsky's rule from Theorem 3.1.8 gives $T_n = Z_n(\frac{\sigma}{S_n}) \rightsquigarrow N(0,1)$.
Structural Commentary: Studentization replaces the unknown σ by the consistent estimate S_n.
Key Insight: Slutsky permits multiplying the CLT statistic by $\frac{\sigma}{S_n} \to 1$. □

Theorem 4.2.11. *Berry–Esseen Rate*
Let $(X_i)_{i\geq 1}$ be IID real-valued random variables, $\mathbb{E}X_1 = \mu$, $\mathrm{Var}(X_1) = \sigma^2 \in (0,+\infty)$, $\rho_3 := \mathbb{E}\,|X_1-\mu|^3 < +\infty$. Let $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$, $Z_n := \sigma^{-1}\sqrt{n}(\overline{X}_n - \mu)$, $\Phi(z) := \int_{-\infty}^z (2\pi)^{-\frac{1}{2}} e^{-\frac{x^2}{2}}\,dx$; $\sup_{z\in\mathbb{R}} |\mathbb{P}(Z_n \leq z) - \Phi(z)| \leq \frac{33}{4}\frac{\rho_3}{\sigma^3\sqrt{n}}$.

Proof. Set $Y_i := \frac{X_i-\mu}{\sigma}$, $\rho := \mathbb{E}\,|Y_1|^3 = \frac{\rho_3}{\sigma^3}$, and $W_n := n^{-\frac{1}{2}}\sum_{i=1}^n Y_i = Z_n$.
For $\varphi(t) := \mathbb{E}e^{itY_1}$, Taylor's formula with $\mathbb{E}Y_1 = 0$, $\mathbb{E}Y_1^2 = 1$, $\mathbb{E}\,|Y_1|^3 = \rho$ gives $\varphi(t) = 1 - \frac{t^2}{2} + O(\rho\,|t|^3)$ near 0.
Thus $\varphi_{W_n}(t) = [\varphi(\frac{t}{\sqrt{n}})]^n$ differs from $e^{-\frac{t^2}{2}}$ by $O(\rho n^{-\frac{1}{2}})$ in Esseen's smoothing integral.
Standard optimised smoothing bound gives $\sup_z |\mathbb{P}(W_n \leq z) - \Phi(z)| \leq \frac{33}{4}\frac{\rho}{\sqrt{n}}$. Substitute $\rho = \frac{\rho_3}{\sigma^3}$, QED.
Structural Commentary: Berry–Esseen bounds the CLT error by standardised third absolute moment.
Key Insight: Normalise, compare characteristic functions, then apply Esseen smoothing. □

Theorem 4.2.12. *Multivariate Central Limit Theorem*
Let $(X_i)_{i\geq 1}$ be IID random vectors in $\mathbb{R}^k$ with $\mathbb{E}X_1 = \mu \in \mathbb{R}^k$ and $\mathrm{Cov}(X_1) = \Sigma$, and set $\overline{X}_n := n^{-1}\sum_{i=1}^n X_i$.
Then $\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N_k(0,\Sigma)$.

Proof. For $a \in \mathbb{R}^k$, set $Y_{i,a} := a^t(X_i - \mu)$ and $s_a^2 := a^t\Sigma a$.
Then $(Y_{i,a})_{i\geq 1}$ are IID, $\mathbb{E}Y_{1,a} = 0$, $\mathrm{Var}(Y_{1,a}) = s_a^2$, and $a^t\sqrt{n}(\overline{X}_n - \mu) = n^{-\frac{1}{2}}\sum_{i=1}^n Y_{i,a}$.
If $s_a^2 > 0$, the scalar CLT gives $a^t\sqrt{n}(\overline{X}_n - \mu) \rightsquigarrow N(0,s_a^2)$; if $s_a^2 = 0$, then $\mathbb{E}Y_{1,a}^2 = 0$, so the same convergence holds with degenerate limit $N(0,0)$. Since $a^tZ \sim N(0,s_a^2)$ for $Z \sim N_k(0,\Sigma)$, by Cramer–Wold, QED.
Structural Commentary: Multivariate convergence is checked through one-dimensional projections.
Key Insight: Project first, apply the scalar CLT, then use Cramer–Wold. □

Theorem 4.2.13. *Delta Method*
Let $(Y_n)_{n\geq 1}$ be real-valued random variables, let $\mu \in \mathbb{R}$ and $\sigma > 0$, and assume $\sigma^{-1}\sqrt{n}(Y_n - \mu) \rightsquigarrow N(0,1)$. If $g : \mathbb{R} \to \mathbb{R}$ is differentiable at μ and $g'(\mu) \neq 0$, then $(|g'(\mu)|\,\sigma)^{-1}\sqrt{n}(g(Y_n) - g(\mu)) \rightsquigarrow N(0,1)$. Equivalently $g(Y_n) \approx N(g(\mu), \frac{(g'(\mu))^2\sigma^2}{n})$.

Proof. Set $A_n := \sigma^{-1}\sqrt{n}(Y_n - \mu)$ and define $\rho(\mu) := 0$, $\rho(y) := \frac{g(y)-g(\mu)}{y-\mu} - g'(\mu)$ for $y \neq \mu$.
Then $A_n \rightsquigarrow Z \sim N(0,1)$, $A_n = O_{\mathbb{P}}(1)$, and $Y_n - \mu = \frac{\sigma A_n}{\sqrt{n}} \xrightarrow{\mathbb{P}} 0$.
Differentiability gives $\rho(y) \to 0$ as $y \to \mu$, hence $\rho(Y_n) \xrightarrow{\mathbb{P}} 0$.
For all n, $g(Y_n) - g(\mu) = (g'(\mu) + \rho(Y_n))(Y_n - \mu)$. Thus, with $T_n := (|g'(\mu)|\,\sigma)^{-1}\sqrt{n}(g(Y_n) - g(\mu))$,
$T_n = \mathrm{sgn}(g'(\mu))A_n(1 + \frac{\rho(Y_n)}{g'(\mu)}) \rightsquigarrow \mathrm{sgn}(g'(\mu))Z \sim N(0,1)$ by Theorem 3.1.8.
Structural Commentary: The delta method is first-order Taylor expansion at the limit point.
Key Insight: The derivative rescales the fluctuation; the remainder is negligible in probability. □

Example 4.2.14. *Exponentiating a Sample Mean*
Let $(X_i)_{i=1}^n$ be IID with mean μ and variance $\sigma^2 < +\infty$, and set $W_n := e^{\overline{X}_n}$.
For $g(s) = e^s$, $g'(\mu) = e^\mu$, so the delta method gives $W_n \approx N(e^\mu, \frac{e^{2\mu}\sigma^2}{n})$.

Theorem 4.2.15. *Multivariate Delta Method*
Let $(Y_n)_{n\geq 1}$ be random vectors in $\mathbb{R}^k$ with $\sqrt{n}(Y_n - \mu) \rightsquigarrow N_k(0,\Sigma)$, and let $g : \mathbb{R}^k \to \mathbb{R}$ be differentiable at μ with gradient $\nabla g(\mu)$. Then $\sqrt{n}(g(Y_n) - g(\mu)) \rightsquigarrow N(0, \nabla g(\mu)^t\Sigma\nabla g(\mu))$.

Proof. Set $A_n := \sqrt{n}(Y_n - \mu)$, so $A_n \rightsquigarrow N_k(0,\Sigma)$ and $Y_n \xrightarrow{\mathbb{P}} \mu$.
Differentiability gives $g(Y_n) - g(\mu) = \nabla g(\mu)^t(Y_n - \mu) + r(Y_n)\,\|Y_n - \mu\|$ with $r(Y_n) \xrightarrow{\mathbb{P}} 0$.
Since $A_n = O_{\mathbb{P}}(1)$, $\sqrt{n}r(Y_n)\,\|Y_n - \mu\| = r(Y_n)\,\|A_n\| \xrightarrow{\mathbb{P}} 0$.
Thus $\sqrt{n}(g(Y_n) - g(\mu)) = \nabla g(\mu)^t A_n + o_{\mathbb{P}}(1) \rightsquigarrow N(0, \nabla g(\mu)^t\Sigma\nabla g(\mu))$ by Theorem 3.1.8.
Structural Commentary: Multivariate delta method linearizes g by its gradient.
Key Insight: The gradient projection carries the Gaussian limit; the differentiability remainder vanishes. □

Example 4.2.16. *Product and Ratio of Sample Means*
Let $X_i = (X_{1i}, X_{2i})^t$ be IID with mean $\mu = (\mu_1, \mu_2)^t$ and covariance $\Sigma = (\sigma_{ij})$, and set $\overline{X}_{jn} := n^{-1} \sum_{i=1}^{n} X_{ji}$.
For $g(x_1, x_2) = x_1 x_2$, $\nabla g(\mu) = (\mu_2, \mu_1)^t$, so $\sqrt{n}(\overline{X}_{1n}\overline{X}_{2n} - \mu_1\mu_2) \rightsquigarrow N(0, \mu_2^2\sigma_{11} + 2\mu_1\mu_2\sigma_{12} + \mu_1^2\sigma_{22})$.
If $\mu_2 \neq 0$, then for $h(x_1, x_2) = \frac{x_1}{x_2}$ and $a := \nabla h(\mu) = (\frac{1}{\mu_2}, -\frac{\mu_1}{\mu_2^2})^t$, $\sqrt{n}(\frac{\overline{X}_{1n}}{\overline{X}_{2n}} - \frac{\mu_1}{\mu_2}) \rightsquigarrow N(0, a^t\Sigma a)$.

5 Stochastic Processes

5.1 Process-Level Probability

Definition 5.1.1. *Stochastic Process, State Space, and Index Set*

(i) A *stochastic process* is a family $(X_t)_{t\in T}$ of random variables.
(ii) The set T is the *index set*, often representing time.
(iii) Each X_t takes values in a *state space* $\mathcal{X}$.

Typical choices are discrete time $T = \{0, 1, 2, \ldots\}$ and continuous time $T = [0, \infty)$.

Example 5.1.2. *IID Sequences as Processes*
An IID sequence $(X_n)_{n\geq 1}$ is a stochastic process indexed by $T = \{1, 2, \ldots\}$.

Example 5.1.3. *Discrete States and Discrete Time*
If $\mathcal{X} = \{\text{sunny}, \text{cloudy}\}$ and X_n is the weather on day n, then $(X_n)_{n\geq 1}$ is a stochastic process with discrete state space and discrete index set.

Example 5.1.4. *The Empirical CDF as a Process*
Let $X_1, \ldots, X_n$ be IID with CDF F on $[0, 1]$, and define $\widehat{F}_n(t) = n^{-1} \sum_{i=1}^{n} \mathbf{1}\{X_i \leq t\}$.
Then $(\widehat{F}_n(t))_{t\in[0,1]}$ is a stochastic process with continuous index set $[0, 1]$ and state space $[0, 1]$.

Remark 5.1.5. *Sample Paths*
A *sample path* is one realized function $t \mapsto X_t(\omega)$ of a stochastic process.
The state space and index set may each be discrete or continuous.

Proposition 5.1.6. *Finite-Dimensional Chain Rule*
Let $X_{1:n} = (X_1, \ldots, X_n)$ have joint mass function or density $f_{1:n}$.
At points where the needed marginals are positive, $f_{1:n}(x_{1:n}) = f_1(x_1) \prod_{i=2}^{n} f_{i|1:i-1}(x_i \mid x_{1:i-1})$.

Proof. For $i \geq 2$, the definition of conditional density gives $f_{1:i}(x_{1:i}) = f_{i|1:i-1}(x_i \mid x_{1:i-1}) f_{1:i-1}(x_{1:i-1})$.
Iterating this identity from $i = n$ down to 2 yields the product formula.
Structural Commentary: The joint law factors by revealing one coordinate at a time.
Key Insight: Conditional densities convert successive joint marginals into a telescoping product. □

5.2 Filtrations and Adapted Processes

5.3 Martingales in Discrete and Continuous Time

5.4 Markov Chains and Markov Processes

Definition 5.4.1. *Probability Vectors and Markov Matrices*
A vector $\mathbf{x} \in \mathbb{R}^n$ is a *probability vector* if its entries are nonnegative and sum to 1. Let $\mathbf{e} := (1, \ldots, 1)^t$. A matrix $A = [a_{ij}] \in \mathrm{Mat}(n, n, \mathbb{R})$ is a *Markov matrix* if $a_{ij} \geq 0$ for all i, j and $A\mathbf{e} = \mathbf{e}$; equivalently, every row sum is 1.

Lemma 5.4.2. *Basic Properties of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $1 \in \mathrm{spec}(A)$, and each A^p is Markov with entries in $[0, 1]$.

Proof. Since $A\mathbf{e} = \mathbf{e}$, 1 is an eigenvalue. Inductively, A^p has nonnegative entries and $A^p\mathbf{e} = \mathbf{e}$, so A^p is Markov.
Each nonnegative row sums to 1, so all entries lie in $[0, 1]$.
Structural Commentary: Markov structure is closed under powers.
Key Insight: Nonnegative row sums of 1 bound every entry. □

Theorem 5.4.3. *Unit-Circle Spectrum of Markov Matrices*
If $A \in \mathrm{Mat}(n, n, \mathbb{R})$ is Markov, then $\rho(A) = 1$, and $\dim E_\lambda(A) = m_A(\lambda)$ for every $\lambda \in \mathrm{spec}(A)$ with $|\lambda| = 1$.

Proof. Lemma 5.4.2 makes A power bounded. Theorem 3.3.4 gives $\rho(A) \leq 1$ and $\dim E_\lambda(A) = m_A(\lambda)$ on $|\lambda| = 1$. Since $A\mathbf{e} = \mathbf{e}$, $1 \in \operatorname{spec}(A)$, so $\rho(A) \geq 1$. Thus $\rho(A) = 1$.
Structural Commentary: Bounded Markov powers rule out unit-circle Jordan growth.
Key Insight: $A\mathbf{e} = \mathbf{e}$ forces the eigenvalue 1. □

Theorem 5.4.4. *Positive Markov Matrices Converge to a Stationary Distribution*
Let $n \geq 2$, $\mathbf{e} := (1, \ldots, 1)^t$, and $A = (a_{ij}) \in \operatorname{Mat}(n, n, \mathbb{R})$ satisfy $a_{ij} > 0$ and $A\mathbf{e} = \mathbf{e}$.
Let $\Delta_{n-1} := \{\mathbf{u} \in \mathbb{R}^n : \mathbf{u} \geq 0,\ \mathbf{e}^t\mathbf{u} = 1\}$. Then $\exists!\, \mathbf{x} \in \Delta_{n-1}$ s.t. $A^t\mathbf{x} = \mathbf{x}$, $\mathbf{x} > 0$, $\lim_{p\to\infty} A^p = \mathbf{e}\mathbf{x}^t$.

Proof. Let $A\mathbf{y} = \lambda\mathbf{y}$ with $\mathbf{y} \neq 0$ and $|\lambda| = 1$, and choose k with $|y_k| = \|\mathbf{y}\|_\infty$.
Since $\sum_j a_{kj} = 1$, then $\|\mathbf{y}\|_\infty = |\lambda y_k| = \left|\sum_{j=1}^n a_{kj} y_j\right| \leq \sum_{j=1}^n a_{kj} |y_j| \leq \|\mathbf{y}\|_\infty$. Thus all inequalities are equalities.
Since $a_{kj} > 0$, $|y_j| = \|\mathbf{y}\|_\infty\ \forall j$; equality in the triangle inequality gives $\mathbf{y} = c\mathbf{e}$, $c \neq 0$.
Hence $\lambda = 1$, $E_1(A) = \operatorname{span}\{\mathbf{e}\}$, and $\operatorname{spec}(A) \cap \{z \in \mathbb{C} : |z| = 1\} = \{1\}$.
Theorem 5.4.3 makes 1 semisimple, so $A = S([1] \oplus J)S^{-1}$ with $S = [\mathbf{e}\ Y]$ and $\rho(J) < 1$.
First row of S^{-1} is $\mathbf{x}^t$. Then $\mathbf{x}^t A = \mathbf{x}^t$, $\mathbf{x}^t\mathbf{e} = 1$, and $A^p = S([1] \oplus J^p)S^{-1} \longrightarrow S([1] \oplus 0)S^{-1} = \mathbf{e}\mathbf{x}^t$.
Since $A^p \geq 0$, the limit gives $\mathbf{x} \geq 0$; since $A^t\mathbf{x} = \mathbf{x}$, $\mathbf{x}^t\mathbf{e} = 1$, and $A^t > 0$, we get $\mathbf{x} > 0$.
Finally, $\dim E_1(A^t) = n - \operatorname{rank}(A^t - I_n) = n - \operatorname{rank}(A - I_n) = \dim E_1(A) = 1$, so any $\mathbf{u} \in \Delta_{n-1}$ with $A^t\mathbf{u} = \mathbf{u}$ satisfies $\mathbf{u} = c\mathbf{x}$ and $\mathbf{e}^t\mathbf{u} = 1 = \mathbf{e}^t\mathbf{x}$, hence $\mathbf{u} = \mathbf{x}$.
Structural Commentary: Positivity leaves one simple unit-circle block; all other blocks decay.
Key Insight: The limit is the rank-one projection $\mathbf{e}\mathbf{x}^t$. □

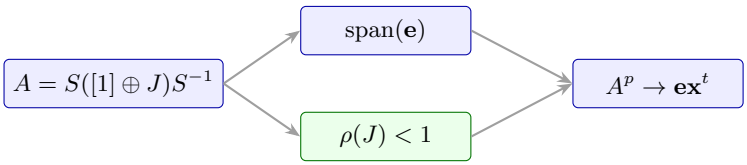


Figure 212: The line span($\mathbf{e}$) remains; all other Jordan directions decay, so $A^p \to \mathbf{e}\mathbf{x}^t$.

The preceding results are matrix facts. In probability language, a row-stochastic matrix becomes a transition matrix for a discrete-time chain.

Definition 5.4.5. *Markov Chain and Transition Matrix*
(i) Let $\mathcal{X}$ be countable and let $(X_n)_{n\geq 0}$ take values in $\mathcal{X}$.
(ii) Process is *Markov chain* if $\mathbb{P}(X_n = x \mid X_0, \ldots, X_{n-1}) = \mathbb{P}(X_n = x \mid X_{n-1})$ $\forall$ admissible n and $x \in \mathcal{X}$.
(iii) It is *homogeneous* if $p_{ij} := \mathbb{P}(X_{n+1} = j \mid X_n = i)$ is independent of n.
(iv) The matrix $P = [p_{ij}]$ is the *transition matrix*; $p_{ij} \geq 0$ and $\sum_{j\in\mathcal{X}} p_{ij} = 1$ for each i.

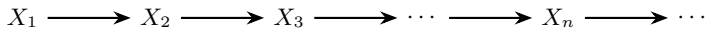


Figure 213: A Markov chain remembers the current state, not the full past.

Example 5.4.6. *Random Walk with Absorbing Barriers*
Let $\mathcal{X} = \{1, \ldots, N\}$ and $q = 1 - p$. The absorbing random walk has $p_{11} = p_{NN} = 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q$ for $2 \leq i \leq N - 1$, and all other transition probabilities equal to 0.

Definition 5.4.7. *n-Step Transition Probabilities*
For a homogeneous Markov chain, the *n-step transition probability* from i to j is $p_{ij}(n) := \mathbb{P}(X_{m+n} = j \mid X_m = i)$; set $P_n = [p_{ij}(n)]$.

Theorem 5.4.8. *Chapman-Kolmogorov Equations*
Let $(X_r)_{r\geq 0}$ be homogeneous Markov chain on countable $\mathcal{X}$. Let $p_{ij}(r) := \mathbb{P}_i(X_r = j)$, $P_r = (p_{ij}(r))_{i,j\in\mathcal{X}}$.
$\forall i, j \in \mathcal{X}$ and $m, n \in \mathbb{N}_0$, $p_{ij}(m+n) = \sum_{k\in\mathcal{X}} p_{ik}(m)p_{kj}(n)$; equivalently, $P_{m+n} = P_m P_n$ and $P_n = P^n$ for $P = P_1$.

Proof. Fix $i, j \in \mathcal{X}$ and $m, n \in \mathbb{N}_0$. By total probability, the Markov property, and homogeneity,
$p_{ij}(m+n) = \mathbb{P}_i(X_{m+n} = j) = \sum_{k\in\mathcal{X}} \mathbb{P}_i(X_{m+n} = j \mid X_m = k)\mathbb{P}_i(X_m = k) = \sum_{k\in\mathcal{X}} p_{kj}(n)p_{ik}(m)$.

Thus $(P_m P_n)_{ij} = \sum_{k \in \mathcal{X}} p_{ik}(m) p_{kj}(n) = p_{ij}(m+n)$ for all i, j, so $P_{m+n} = P_m P_n$.
With $P_0 = I$ and $P = P_1$, induction gives $P_n = P^n$.
Structural Commentary: Transition over $m+n$ steps factors through the intermediate state after m steps.
Key Insight: Conditioning over intermediate states is matrix multiplication. □

Lemma 5.4.9. *Marginal Evolution*
Let $(X_n)_{n \geq 0}$ be a homogeneous Markov chain on countable $\mathcal{X}$ with $P = P_1 = (p_{ij})$, where $p_{ij}(n) := \mathbb{P}_i(X_n = j)$ and $P_n = (p_{ij}(n))$. Define row laws $\mu_n(j) := \mathbb{P}(X_n = j)$. Then $\mu_n = \mu_0 P^n$ for all $n \in \mathbb{N}_0$.

Proof. $\forall j \in \mathcal{X}$, by total probability $\mu_n(j) = \sum_{i \in \mathcal{X}} \mathbb{P}(X_n = j \mid X_0 = i)\mathbb{P}(X_0 = i) = \sum_{i \in \mathcal{X}} \mu_0(i) p_{ij}(n) = (\mu_0 P_n)_j$.
By Theorem 5.4.8, $P_n = P^n$, so $\mu_n(j) = (\mu_0 P^n)_j$ for every $j \in \mathcal{X}$; hence $\mu_n = \mu_0 P^n$.
Structural Commentary: Marginals evolve by multiplying the initial law by the transition semigroup.
Key Insight: Conditioning on X_0 turns distribution evolution into row-vector matrix multiplication. □

Definition 5.4.10. *Communication Classes*

(i) State i *reaches* j, written $i \to j$, if $p_{ij}(n) > 0$ for some $n \geq 0$.
(ii) States i, j *communicate*, written $i \leftrightarrow j$, if $i \to j$ and $j \to i$.
(iii) A *communication class* is an equivalence class under $\leftrightarrow$.
(iv) The chain is *irreducible* if all states lie in one communication class.
(v) A set C is *closed* if $p_{ij} = 0$ for all $i \in C$ and $j \notin C$; a singleton closed class is an *absorbing state*.

Proposition 5.4.11. *Communication Classes Partition the State Space*
Let $(X_n)_{n \geq 0}$ be a homogeneous Markov chain on countable $\mathcal{X}$ with n-step probabilities $p_{ij}(n)$. Define $i \to j$ iff $\exists n \in \mathbb{N}_0$ such that $p_{ij}(n) > 0$, and $i \leftrightarrow j$ iff $i \to j$ and $j \to i$. Then $\leftrightarrow$ is equivalence relation, and $\mathcal{X} = \bigsqcup_{C \in \mathcal{C}} C$ where $\mathcal{C} := \{[i]_{\leftrightarrow} : i \in \mathcal{X}\}$ is set of communication classes.

Proof. Since $p_{ii}(0) = 1$, $i \leftrightarrow i$ for all $i \in \mathcal{X}$; symmetry is immediate from the definition.
If $i \leftrightarrow j$ and $j \leftrightarrow k$, choose $m, n, r, s \in \mathbb{N}_0$ with $p_{ij}(m) > 0$, $p_{jk}(n) > 0$, $p_{kj}(r) > 0$, and $p_{ji}(s) > 0$.
By Theorem 5.4.8, $p_{ik}(m+n) \geq p_{ij}(m) p_{jk}(n) > 0$ and $p_{ki}(r+s) \geq p_{kj}(r) p_{ji}(s) > 0$, hence $i \leftrightarrow k$.
Thus $\leftrightarrow$ is reflexive, symmetric, and transitive.
Its equivalence classes are pairwise disjoint, cover $\mathcal{X}$, and are exactly the communication classes.
Structural Commentary: Communication classes are the equivalence classes generated by mutual reachability.
Key Insight: Positive transition paths concatenate by Chapman-Kolmogorov. □

Definition 5.4.12. *Recurrent and Transient States*

(i) Let $\tau_i^+ := \inf\{n \geq 1 : X_n = i\}$ be the first return time to i.
(ii) State i is *recurrent* or *persistent* if $\mathbb{P}_i(\tau_i^+ < \infty) = 1$.
(iii) State i is *transient* if $\mathbb{P}_i(\tau_i^+ < \infty) < 1$.

Theorem 5.4.13. *Recurrence Criterion*

(i) State i is recurrent iff $S_i := \sum_{n=0}^{\infty} p_{ii}(n) = \infty$.
(ii) State i is transient iff $S_i < \infty$.

Proof. Let $N_i := \sum_{n=0}^{\infty} \mathbf{1}\{X_n = i\}$ and $r_i := \mathbb{P}_i(\tau_i^+ < \infty)$. By Tonelli, $\mathbb{E}_i N_i = \sum_{n=0}^{\infty} p_{ii}(n) = S_i$.
By the Markov property after returns to i, $\mathbb{P}_i(N_i \geq k) = r_i^{k-1}$ for $k \geq 1$, hence $\mathbb{E}_i N_i = \sum_{k=1}^{\infty} r_i^{k-1}$.

(i) ($\Rightarrow$) If i is recurrent, then $r_i = 1$, so $S_i = \mathbb{E}_i N_i = \infty$.
(i) ($\Leftarrow$) If $S_i = \infty$, then $r_i < 1$ would give $\mathbb{E}_i N_i = (1 - r_i)^{-1} < \infty$, a contradiction; hence $r_i = 1$.
(ii) ($\Rightarrow$) If i is transient, then $r_i < 1$, so $S_i = \mathbb{E}_i N_i = (1 - r_i)^{-1} < \infty$.
(ii) ($\Leftarrow$) If $S_i < \infty$, then $r_i = 1$ would give $\mathbb{E}_i N_i = \infty$, a contradiction; hence $r_i < 1$.

Structural Commentary: Recurrence is detected by the expected total number of returns.
Key Insight: Visits to i repeat with geometric return probability r_i. □

Definition 5.4.14. *Period and Ergodicity*

(i) For recurrent i, it is *positive recurrent* if $\mathbb{E}_i \tau_i^+ < \infty$ and *null recurrent* otherwise.
(ii) The *period* of i is $d(i) := \gcd\{n \geq 1 : p_{ii}(n) > 0\}$.
(iii) State i is *aperiodic* if $d(i) = 1$.
(iv) State i is *ergodic* if it is recurrent, positive recurrent, and aperiodic.

(v) The chain is *ergodic* if every state is ergodic.

Definition 5.4.15. *Stationary and Limiting Distributions*

(i) A probability row vector π is *stationary* or *invariant* if $\pi = \pi P$.
(ii) A finite chain has *limiting distribution* π if $p_{ij}(n) \to \pi_j$ for all states i, j.

Equivalently, every row of P^n converges to π.

Theorem 5.4.16. *Finite Ergodic Chain Convergence*
Let P be the transition matrix of a finite irreducible aperiodic Markov chain on $\mathcal{X}$.
There is a unique $\pi \in \Delta(\mathcal{X})$ with $\pi P = \pi$, and $P^n \to \mathbf{e}\pi$.
For every bounded $g : \mathcal{X} \to \mathbb{R}$, $N^{-1} \sum_{n=1}^{N} g(X_n) \to \pi(g) := \sum_{j \in \mathcal{X}} g(j)\pi_j$ almost surely.

Proof. Finite irreducibility and aperiodicity imply P is primitive: $P^m > 0$ for some $m \geq 1$.
Row-vector form of Theorem 5.4.4 on $A = P^m$, gives unique $\pi \in \Delta(\mathcal{X})$ with $\pi P^m = \pi$, $(P^m)^k \to \mathbf{e}\pi$.
Since $(\pi P)P^m = \pi P$ and $\pi P \in \Delta(\mathcal{X})$, uniqueness gives $\pi P = \pi$.
For $n = mk + r$ with $0 \leq r < m$, $P^n = (P^m)^k P^r \to \mathbf{e}\pi P^r = \mathbf{e}\pi$.
The finite-state Markov-chain strong law yields $N^{-1} \sum_{n=1}^{N} g(X_n) \to \pi(g)$ almost surely for bounded g.
Structural Commentary: Irreducibility gives one communicating class; aperiodicity removes cyclic obstruction.
Key Insight: Primitivity makes P^n converge to the rank-one projection $\mathbf{e}\pi$. □

Definition 5.4.17. *Detailed Balance*
Let $P = [p_{ij}]$ be a transition matrix on $\mathcal{X}$.
A probability row vector π satisfies *detailed balance* for P if $\pi_i p_{ij} = \pi_j p_{ji}$ for all $i, j \in \mathcal{X}$.

Theorem 5.4.18. *Detailed Balance Implies Stationarity*
Let $P = [p_{ij}]$ be row-stochastic on countable $\mathcal{X}$, and let $\pi \in \Delta(\mathcal{X})$. If $\pi_i p_{ij} = \pi_j p_{ji}$ $\forall i, j \in \mathcal{X}$, then $\pi P = \pi$.

Proof. $\forall j \in \mathcal{X}$, $(\pi P)_j = \sum_{i \in \mathcal{X}} \pi_i p_{ij} = \sum_{i \in \mathcal{X}} \pi_j p_{ji} = \pi_j \sum_{i \in \mathcal{X}} p_{ji} = \pi_j$, since row j of P sums to 1. QED.
Structural Commentary: Detailed balance gives pairwise cancellation of probability fluxes.
Key Insight: Reversibility is stronger than stationarity. □

Remark 5.4.19. *Stationary Need Not Mean Convergent*
For the deterministic three-cycle on $\{1, 2, 3\}$, set $p_{12} = p_{23} = p_{31} = 1$ and all other $p_{ij} = 0$.
Then $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ satisfies $\pi P = \pi$, but $P^{n+3} = P^n$, so P^n does not converge.

Proposition 5.4.20. *Two-State Chain Limit*
Let $0 < a, b < 1$, $P = \begin{bmatrix} 1-a & a \\ b & 1-b \end{bmatrix}$. Set $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$. Then $\pi P = \pi$, π unique, $P^n \to \mathbf{e}\pi = \begin{bmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{bmatrix}$.

Proof. Solving $\pi P = \pi$ with $\pi \mathbf{e} = 1$ gives $\pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$.
The characteristic polynomial is $(\lambda - 1)(\lambda - (1 - a - b))$, so $\operatorname{spec}(P) = \{1, 1 - a - b\}$ and $|1 - a - b| < 1$.
Thus stationary component is projected onto $\mathbf{e}\pi$, transverse component decays geometrically, giving $P^n \to \mathbf{e}\pi$.
Structural Commentary: The dynamics split into the stationary line and one transverse eigendirection.
Key Insight: The non-unit eigenvalue $1 - a - b$ controls convergence. □

5.5 Stationary Processes and Ergodicity

5.6 Gaussian Processes

5.7 Point Processes and Renewal Structure

Counting processes model random event arrivals over time. The Poisson process is the basic model in which disjoint time intervals contribute independent counts and short intervals usually contain either zero or one event.

Definition 5.7.1. *Counting Process*
A *counting process* is a stochastic process $\{X(t) : t \geq 0\}$ with state space $\{0, 1, 2, \ldots\}$ such that $X(t)$ records the number of events observed up to and including time t.

Definition 5.7.2. *Poisson Process with Intensity Function*
Let $\lambda : [0, \infty) \to [0, \infty)$ be locally integrable.
A counting process $X = \{X(t) : t \geq 0\}$ is a *Poisson process* with *intensity function* λ if:

(i) $X(0) = 0$.

(ii) For $0 = t_0 < t_1 < \cdots < t_n$, the increments $\Delta_k X := X(t_k) - X(t_{k-1})$, $1 \le k \le n$, are independent.
(iii) For each $t \ge 0$, as $h \downarrow 0$, $\mathbb{P}(X(t+h) - X(t) = 1) = \lambda(t)h + o(h)$ and $\mathbb{P}(X(t+h) - X(t) \ge 2) = o(h)$.

Theorem 5.7.3. *Increment Distribution*
Let X be a Poisson process with intensity λ, and set $\Lambda(s,t) := \int_s^t \lambda(u)\,du$ for $0 \le s \le t$.
Then $X(t) - X(s) \sim \text{Poisson}(\Lambda(s,t))$. In particular, $X(t) \sim \text{Poisson}(\Lambda(0,t))$, so $\mathbb{E}X(t) = \text{Var}(X(t)) = \Lambda(0,t)$.

Proof. Fix $s \ge 0$ and let $p_k(u) := \mathbb{P}(X(s+u) - X(s) = k)$ with $p_{-1} := 0$. By small-time conditions and independent increments, $p_k(u+h) = p_k(u)(1 - \lambda(s+u)h + o(h)) + p_{k-1}(u)\lambda(s+u)h + o(h)$.
Hence $p_k'(u) = -\lambda(s+u)p_k(u) + \lambda(s+u)p_{k-1}(u)$, with $p_0(0) = 1$ and $p_k(0) = 0$ for $k \ge 1$.
Writing $M(u) := \Lambda(s, s+u)$, the unique solution is $p_k(u) = e^{-M(u)}\frac{M(u)^k}{k!}$ by induction on k.
Taking $u = t - s$ yields $X(t) - X(s) \sim \text{Poisson}(\Lambda(s,t))$. Mean-variance identities follow from Poisson law.
Structural Commentary: The infinitesimal jump rules form a triangular system for the increment probabilities.
Key Insight: Integrating the local rate λ gives the Poisson mean over the interval. □

Definition 5.7.4. *Homogeneous Poisson Process*
If $\lambda(t) \equiv \lambda > 0$, then $X(t)$ is a *homogeneous Poisson process* with *rate* λ. In this case, $X(t) \sim \text{Poisson}(\lambda t)$.

Definition 5.7.5. *Waiting and Interarrival Times*
For a homogeneous Poisson process:

(i) Set $W_0 = 0$ and $W_n := \inf\{t \ge 0 : X(t) \ge n\}$ for $n \ge 1$; $W_1, W_2, \ldots$ are *waiting times*.
(ii) Set $S_n := W_{n+1} - W_n$ for $n \ge 0$; $S_0, S_1, \ldots$ are *sojourn* or *interarrival times*.

Theorem 5.7.6. *Poisson Interarrival Times*
Let X be a homogeneous Poisson process with rate $\lambda > 0$.
Then $S_0, S_1, \ldots \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, with density $f(s) = \lambda e^{-\lambda s}$ for $s \ge 0$.
For $n \ge 1$, $W_n = \sum_{r=0}^{n-1} S_r \sim \text{Gamma}(n, \lambda)$, so $\mathbb{E}W_n = \frac{n}{\lambda}$ and $\text{Var}(W_n) = \frac{n}{\lambda^2}$.

Proof. For $t \ge 0$, $\mathbb{P}(S_0 > t) = \mathbb{P}(X(t) = 0) = e^{-\lambda t}$, so $S_0 \sim \text{Exp}(\lambda)$.
By independent stationary increments, $\mathbb{P}(S_n > t \mid \mathcal{F}_{W_n}) = \mathbb{P}(X(W_n + t) - X(W_n) = 0 \mid \mathcal{F}_{W_n}) = e^{-\lambda t}$.
Thus each $S_n \sim \text{Exp}(\lambda)$ and is independent of the past, so the S_n are IID.
Since $W_n = \sum_{r=0}^{n-1} S_r$, the Gamma law and its mean and variance follow.
Structural Commentary: Arrival times are cumulative sums of IID interarrival gaps.
Key Insight: No events in a length-t interval has probability $e^{-\lambda t}$. □

Remark 5.7.7. *Uniform Event-Time Diagnostic*
For a homogeneous Poisson process on $[0,T]$, conditional on $X(T) = n$, the event times $(W_1, \ldots, W_n)$ have the law of the order statistics of n IID Uniform$(0,T)$ variables.
Marked nonuniformity in the observed event times is evidence against homogeneity.

Proposition 5.7.8. *Conditional Binomial Count*
Let X be a homogeneous Poisson process with rate $\lambda > 0$.
For $s, t > 0$ and $n \in \mathbb{N}_0$, $X(t) \mid X(t+s) = n \sim \text{Binomial}\left(n, \frac{t}{t+s}\right)$.

Proof. Let $Y = X(t)$ and $Z = X(t+s) - X(t)$. Then $Y \sim \text{Poisson}(\lambda t)$, $Z \sim \text{Poisson}(\lambda s)$, and $Y \perp Z$.
For $0 \le k \le n$, $\mathbb{P}(Y = k \mid Y + Z = n) = \frac{\frac{(\lambda t)^k}{k!}\frac{(\lambda s)^{n-k}}{(n-k)!}}{\frac{(\lambda(t+s))^n}{n!}} = \binom{n}{k}\left(\frac{t}{t+s}\right)^k\left(\frac{s}{t+s}\right)^{n-k}$; QED.
Structural Commentary: Conditioning on the total allocates independent Poisson counts.
Key Insight: Independent Poisson counts become multinomial after conditioning on their sum. □

Proposition 5.7.9. *Odd Poisson Count*
Let X be a homogeneous Poisson process with rate $\lambda > 0$. For $t \ge 0$, $\mathbb{P}(X(t) \in 2\mathbb{N}_0 + 1) = \frac{1 - e^{-2\lambda t}}{2}$.

Proof. Let $\mu = \lambda t$. Since $X(t) \sim \text{Poisson}(\mu)$, $\mathbb{P}(X(t) \in 2\mathbb{N}_0 + 1) = e^{-\mu}\sum_{k=0}^{\infty}\frac{\mu^{2k+1}}{(2k+1)!} = e^{-\mu}\sinh\mu = \frac{1 - e^{-2\mu}}{2}$.
Structural Commentary: Split the Poisson mass by parity.
Key Insight: The odd terms of e^{μ} sum to $\sinh\mu$. □

Proposition 5.7.10. *Nearest Neighbor in a Planar Poisson Process*
Let Φ be a planar Poisson point process with intensity $\lambda > 0$, fix $x_0 \in \mathbb{R}^2$, and set $R := \inf_{Y \in \Phi}\|Y - x_0\|$.
Then for $t \ge 0$, $\mathbb{P}(R > t) = e^{-\lambda\pi t^2}$, and $\mathbb{E}R = \frac{1}{2\sqrt{\lambda}}$.

Proof. Let $B_t := B(x_0, t)$, $N_t := \#(\Phi \cap B_t)$. Then $N_t \sim \text{Poisson}(\lambda \pi t^2)$, $\{R > t\} = \{N_t = 0\}$, so $\mathbb{P}(R > t) = e^{-\lambda \pi t^2}$. Since $R \geq 0$, $\mathbb{E}R = \int_0^\infty \mathbb{P}(R > t)\, dt = \int_0^\infty e^{-\lambda \pi t^2}\, dt = \frac{1}{2\sqrt{\lambda}}$.
Structural Commentary: Nearest-neighbor distance is encoded by empty disks around x_0.
Key Insight: The tail of R is the zero-count probability of a Poisson disk count. □

6 Stochastic Calculus and SDEs

6.1 Brownian Motion

6.2 Stochastic Integration

6.3 Itô Calculus

6.4 Stochastic Differential Equations

6.5 Change of Measure and Girsanov Theory

6.6 Diffusions and PDE Connections

6.7 Stochastic Flows and Advanced Dynamics

Part XIII
Dynamical and PDE Systems

1 Dynamical Systems and ODEs

1.1 ODE Foundations and Flows

1.2 Linear Systems and Stability

Method 1.2.1. *Solving $\mathbf{x}'(t) = A\mathbf{x}(t)$ by Jordan Form*
To solve $\mathbf{x}'(t) = A\mathbf{x}(t)$ with $\mathbf{x}(0) = \mathbf{x}_0$:

(i) Put $A = SJS^{-1}$, where $J = \bigoplus_{i=1}^{r} J_{k_i}(\lambda_i)$.
(ii) Set $\mathbf{y}(t) = S^{-1}\mathbf{x}(t)$ and $\mathbf{y}_0 = S^{-1}\mathbf{x}_0$. Then $\mathbf{y}'(t) = J\mathbf{y}(t)$ and $\mathbf{x}(t) = S\mathbf{y}(t)$.
(iii) Split $\mathbf{y} = (\mathbf{y}_1, \ldots, \mathbf{y}_r)$ according to the Jordan blocks. Then $\mathbf{y}_i'(t) = J_{k_i}(\lambda_i)\mathbf{y}_i(t)$.
(iv) For each block, use Proposition 7.3.23: $\mathbf{y}_i(t) = e^{tJ_{k_i}(\lambda_i)}\mathbf{y}_i(0) = e^{\lambda_i t}\sum_{q=0}^{k_i-1} \frac{t^q}{q!} J_{k_i}(0)^q \mathbf{y}_i(0)$.
(v) Reassemble $\mathbf{y}(t)$ and transform back: $\mathbf{x}(t) = S\mathbf{y}(t)$.

Thus a size-k block with eigenvalue λ contributes entries of the form $e^{\lambda t}p(t)$, where $\deg p < k$.

Perspective 1.2.2. *Jordan Functions and Polynomial Factors*

For a Jordan block $J_k(\lambda) = \lambda I_k + N$ with $N^k = 0$, any polynomial f satisfies $f(J_k(\lambda)) = \sum_{r=0}^{k-1} \frac{f^{(r)}(\lambda)}{r!} N^r$.

Nilpotent part adds derivative terms; exponentials gives polynomial factors (Proposition 7.3.23, Method 1.2.1).

Example 1.2.3. *Nilpotent Jordan Block Evolution*
$N = J_3(0) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Since $N^3 = 0$, $e^{tN} = I_3 + tN + \frac{t^2}{2}N^2 = \begin{bmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$.
Nilpotent Jordan blocks create polynomial, not exponential, growth along chains.

1.3 Nonlinear Stability Theory

1.4 Bifurcation Theory

1.5 Dynamical Systems Structure

1.6 Semigroups and Evolution Viewpoint

2 Partial Differential Equations

2.1 First-Order PDE and Characteristics

2.2 Elliptic Equations

2.3 Parabolic Equations

2.4 Hyperbolic Equations

3 Weak Solutions and Evolution Equations

3.1 Sobolev Spaces for PDE

Definition 3.1.1. *Weak Derivative*
Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and let α be a multi-index. A function $v \in L^1_{\mathrm{loc}}(\Omega)$ is the *weak derivative* $D^\alpha u$ if

$$\int_\Omega u\, D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_\Omega v\, \varphi \, dx \qquad \text{for every } \varphi \in C_c^\infty(\Omega).$$

Definition 3.1.2. *Sobolev Spaces*
For $1 \le p \le \infty$ and an integer $k \ge 0$, the Sobolev space $W^{k,p}(\Omega)$ consists of functions $u \in L^p(\Omega)$ whose weak derivatives $D^\alpha u$ lie in $L^p(\Omega)$ for every $|\alpha| \le k$. The norm is

$$\|u\|_{W^{k,p}} := \sum_{|\alpha| \le k} \|D^\alpha u\|_{L^p}$$

with the usual supremum modification when $p = \infty$.

3.2 Weak Solutions and Variational Methods

3.3 Evolution Equations

Part XIV
Numerical and Stochastic Computation

1 Numerical Analysis

1.1 Floating Point and Error Analysis

1.2 Nonlinear Equations and Optimization Basics

1.3 Interpolation and Approximation

1.4 Numerical Differentiation and Quadrature

1.5 Numerical Stability and Convergence

2 Numerical Differential Systems

2.1 Numerical ODEs

2.2 Geometric Numerical Integration

2.3 Finite Difference PDE Methods

2.4 Finite Element Methods

2.5 Numerical SDEs

2.6 Computational Stability Across ODE/PDE/SDE

3 Monte Carlo and Stochastic Computation

3.1 Monte Carlo Foundations

Monte Carlo methods turn integrals into averages of simulated random variables. This is especially useful when the integral is high-dimensional or when the target density is known only up to a normalizing constant.

Remark 3.1.1. *Bayesian Integrals as a Motivation*
For prior density $\pi(\theta)$ and likelihood $\mathcal{L}(\theta)$, write $Z = \int \mathcal{L}(u)\pi(u)\,du$.
Then $\pi(\theta \mid X^n) = Z^{-1}\mathcal{L}(\theta)\pi(\theta)$ and $\mathbb{E}[\theta \mid X^n] = Z^{-1}\int \theta\,\mathcal{L}(\theta)\pi(\theta)\,d\theta$.
For $\theta = (\theta_1, \dots, \theta_k)$, high-dimensional marginalisation is often analytically impractical.

Method 3.1.2. *Basic Monte Carlo Integration*

(i) Let $I = \int h(x)f(x)\,dx$, where f is a probability density.

(ii) For IID draws $X_1, \dots, X_N \sim f$, define $\widehat{I}_N = N^{-1}\sum_{i=1}^{N} h(X_i)$; then $\widehat{I}_N \xrightarrow{p} I$ by the law of large numbers.

(iii) With $Y_i = h(X_i)$, set $s^2 = (N-1)^{-1}\sum_{i=1}^{N}(Y_i - \widehat{I}_N)^2$ and $\widehat{\mathrm{se}} = \frac{s}{\sqrt{N}}$.

(iv) An approximate $1-\alpha$ Monte Carlo interval is $\widehat{I}_N \pm z_{\frac{\alpha}{2}}\widehat{\mathrm{se}}$.

Example 3.1.3. *Monte Carlo Integration from the LLN*
Let $X_1, \dots, X_n$ be iid Uniform$(0,1)$ and $f \in L^1([0,1])$.
The estimator $\widehat{I}_n(f) := n^{-1}\sum_{i=1}^{n} f(X_i)$ satisfies $\widehat{I}_n(f) \xrightarrow{\mathbb{P}} \mathbb{E}f(X_1) = \int_0^1 f(x)\,dx$.
For $f(x) = (2\pi)^{-\frac{1}{2}}e^{-\frac{x^2}{2}}$ on $[0,1]$, a 1000-point estimate near 0.3417 is consistent with $\int_0^1 f(x)\,dx \approx 0.3413$.

Method 3.1.4. *Rejection Sampling*
Let f be a target density on an interval $[a,b]$, possibly with infinite endpoints.
Choose an envelope M such that $M(x) \geq f(x)$ on $[a,b]$ and $A := \int_a^b M(t)\,dt < \infty$. Let $m(x) := \frac{M(x)}{A}$.

(i) Generate T with density m.

(ii) Generate $U \sim$ Uniform$(0,1)$ independently of T.

(iii) Accept $X = T$ if $U \leq \frac{f(T)}{M(T)}$; otherwise reject T and repeat.

Key Insight: Sample under an easy envelope and keep only points that fall below the target curve.

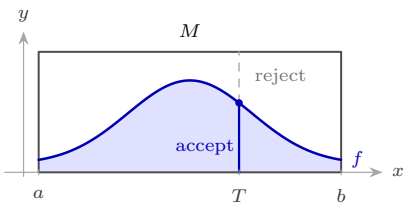


Figure 214: Rejection sample accepts when the vertical draw lies below $f(T)$.

Proposition 3.1.5. *Rejection Sampling Correctness*
Let f be a density, let $M \geq f$ with $A := \int M(x)\,dx < \infty$, and set $m := \frac{M}{A}$. If $T \sim m$, $U \sim \text{Uniform}(0,1)$ independently, and $\mathcal{A} := \{U \leq \frac{f(T)}{M(T)}\}$, then $\mathbb{P}(\mathcal{A}) = \frac{1}{A}$ and $T \mid \mathcal{A}$ has density f.

Proof. For measurable B, $\mathbb{P}(T \in B, \mathcal{A}) = \int_B \frac{m(x)f(x)}{M(x)}\,dx = A^{-1}\int_B f(x)\,dx$.
Taking $B = \mathbb{R}$ gives $\mathbb{P}(\mathcal{A}) = A^{-1}$, and hence $\mathbb{P}(T \in B \mid \mathcal{A}) = \int_B f(x)\,dx$.
Structural Commentary: Acceptance reweights the envelope m back to f.
Key Insight: Conditional on acceptance, the normalized accepted density is exactly the target. □

Example 3.1.6. *Integrating on an Interval*
For $X \sim \text{Uniform}(a,b)$, $I = \int_a^b h(x)\,dx = (b-a)\mathbb{E}[h(X)]$.
Given IID draws $X_1, \ldots, X_N \sim \text{Uniform}(a,b)$, estimate I by $\widehat{I}_N = (b-a)N^{-1}\sum_{i=1}^{N} h(X_i)$.

Example 3.1.7. *Estimating a Normal CDF*
Let $Z \sim N(0,1)$ and $\Phi(x) = \mathbb{P}(Z \leq x)$. For IID draws $Z_1, \ldots, Z_N \sim N(0,1)$, use $h(s) = \mathbf{1}\{s \leq x\}$ and $\widehat{\Phi}(x) = N^{-1}\sum_{i=1}^{N} \mathbf{1}\{Z_i \leq x\}$. Thus $\widehat{\Phi}(x)$ is the empirical fraction of draws not exceeding x.

Example 3.1.8. *Two Binomial Proportions*
Suppose $X \sim \text{Binomial}(n, p_1)$ and $Y \sim \text{Binomial}(m, p_2)$ are independent, with independent flat priors on p_1, p_2.
Then $p_1 \mid X \sim \text{Beta}(X+1, n-X+1)$ and $p_2 \mid Y \sim \text{Beta}(Y+1, m-Y+1)$.
For $\delta = p_2 - p_1$, draw $P_1^{(i)} \sim \text{Beta}(X+1, n-X+1)$ and $P_2^{(i)} \sim \text{Beta}(Y+1, m-Y+1)$, set $\delta^{(i)} = P_2^{(i)} - P_1^{(i)}$, and summarize the empirical distribution of $\delta^{(i)}$.

Method 3.1.9. *Simulating a Posterior Function*
(i) Let $\delta = g(p_1, \ldots, p_k)$ be the posterior target.
(ii) Draw posterior samples $(P_1^{(i)}, \ldots, P_k^{(i)})$ for $i = 1, \ldots, N$.
(iii) Set $\delta^{(i)} = g(P_1^{(i)}, \ldots, P_k^{(i)})$, use empirical distribution of $\delta^{(i)}$ to estimate posterior distribution of δ.

3.2 Variance Reduction

Method 3.2.1. *Importance Sampling*
(i) For $I = \int h(x)f(x)\,dx$, choose a tractable proposal density g with $g(x) > 0$ wherever $h(x)f(x) \neq 0$.
(ii) Then $I = \int h(x)f(x)g(x)^{-1}g(x)\,dx = \mathbb{E}_g\left[\frac{h(X)f(X)}{g(X)}\right]$.
(iii) For IID draws $X_1, \ldots, X_N \sim g$, use $\widehat{I}_N = N^{-1}\sum_{i=1}^{N} \frac{h(X_i)f(X_i)}{g(X_i)}$.

Remark 3.2.2. *Choosing the Proposal*
The importance weights have large, possibly infinite, variance when g is too small on regions where $|h|\,f$ is large. Choose g to approximate the shape of $|h|\,f$ while giving it tails at least as thick as the target integrand.

Theorem 3.2.3. *Optimal Importance Density*
Let $I = \int h(x)f(x)\,dx$ and $A = \int |h(x)|\,f(x)\,dx \in (0,\infty)$.
For proposal densities g with $g(x) > 0$ wherever $|h(x)|\,f(x) > 0$, define $W_g = \frac{h(X)f(X)}{g(X)}$ for $X \sim g$.
The variance $\text{Var}_g(W_g)$ is minimised by $g^*(x) = \frac{|h(x)|f(x)}{A}$, and the minimum variance is $A^2 - I^2$.

Proof. For each admissible g, $\mathbb{E}_g W_g = I$, so $\text{Var}_g(W_g) = M(g) - I^2$, where $M(g) = \mathbb{E}_g(W_g^2) = \int \frac{h(x)^2 f(x)^2}{g(x)}\,dx$.
For any admissible g, Cauchy-Schwarz gives $A^2 = (\int |h(x)|\,f(x)g(x)^{-\frac{1}{2}}g(x)^{\frac{1}{2}}\,dx)^2 \leq M(g)\int g(x)\,dx = M(g)$.

Equality holds precisely when $|h(x)|\, f(x) g(x)^{-\frac{1}{2}} = c\, g(x)^{\frac{1}{2}}$ a.e., equivalently $g(x) \propto |h(x)|\, f(x)$.
Normalization gives $g^*(x) = \frac{|h(x)| f(x)}{A}$, hence $M(g) \geq M(g^*) = A^2$ and $\mathrm{Var}_g(W_g) \geq A^2 - I^2 = \mathrm{Var}_{g^*}(W_{g^*})$.
Structural Commentary: Fixed mean reduces variance to second-moment minimization.
Key Insight: Cauchy-Schwarz is sharp exactly when the proposal is proportional to $|h|\, f$. □

Example 3.2.4. *Rare Tail Probability*
For $Z \sim N(0,1)$, write $I = \mathbb{P}(Z > 3) = \int \mathbf{1}\{x > 3\} f(x)\, dx$.
If g is centered near the right tail and $X_1, \ldots, X_N \sim g$, estimate I by $\widehat{I}_N = N^{-1} \sum_{i=1}^{N} \frac{\mathbf{1}\{X_i > 3\} f(X_i)}{g(X_i)}$.

3.3 Markov Chain Monte Carlo

Monte Carlo integration in Method 3.1.2 uses independent draws from the target density. Markov chain Monte Carlo instead constructs a Markov chain $X_0, X_1, \ldots$ whose stationary density is the target f. When the chain mixes well, sample averages from the chain are used as approximations to expectations under f.

Method 3.3.1. *Metropolis-Hastings*
Let f be a target density available up to proportionality, and let $q(\cdot \mid x)$ be a tractable proposal density. Choose X_0 arbitrarily; given $X_i = x$, define X_{i+1} as follows.

(i) Draw $Y \sim q(\cdot \mid x)$.
(ii) Set $r(x, Y) = \min\left\{\frac{f(Y) q(x \mid Y)}{f(x) q(Y \mid x)}, 1\right\}$.
(iii) Set $X_{i+1} = Y$ with probability $r(x, Y)$, and set $X_{i+1} = x$ otherwise.

Key Insight: Only ratios of f enter r, so unknown normalising constants cancel.

Proposition 3.3.2. *Detailed Balance for Metropolis-Hastings*
Let $\Phi(x, y) = f(x) q(y \mid x)$ and assume $\Phi(x, y) > 0$ iff $\Phi(y, x) > 0$.
For $x \neq y$, define $\alpha(x, y) = \min\left\{\frac{\Phi(y,x)}{\Phi(x,y)}, 1\right\}$ when $\Phi(x, y) > 0$ and $\alpha(x, y) = 0$ otherwise.
Set $p(x, y) = q(y \mid x)\alpha(x, y)$. Then $f(x)p(x, y) = f(y)p(y, x)$ for $x \neq y$.

Proof. If $\Phi(x, y) = 0$, the support condition gives $\Phi(y, x) = 0$, hence $p(x, y) = p(y, x) = 0$.
If $\Phi(x, y) > 0$, then $f(x)p(x, y) = \Phi(x, y) \min\left\{\frac{\Phi(y,x)}{\Phi(x,y)}, 1\right\} = \min\{\Phi(x, y), \Phi(y, x)\}$.
Interchanging x, y gives the same expression for $f(y)p(y, x)$, QED.
Structural Commentary: The acceptance rule symmetrizes the weighted proposal flow.
Key Insight: Both directions reduce to the same minimum of the two proposal fluxes. □

Proposition 3.3.3. *Detailed Balance Implies Stationarity*
Let $\pi(dx) = f(x)\, dx$ and let P be a Markov kernel on state space X.
If $\pi(dx)P(x, dy) = \pi(dy)P(y, dx)$ as measures on $\mathsf{X} \times \mathsf{X}$, then $\pi P = \pi$.

Proof. For measurable $A \subseteq \mathsf{X}$, $\pi P(A) = \int P(x, A)\, \pi(dx) = \int \mathbf{1}_A(y)\, \pi(dx) P(x, dy)$.
By detailed balance, this equals $\int \mathbf{1}_A(y)\, \pi(dy) P(y, dx) = \int_A P(y, \mathsf{X})\, \pi(dy) = \pi(A)$, proving $\pi P = \pi$.
Structural Commentary: Detailed balance makes the joint law of consecutive states symmetric.
Key Insight: Marginalising the symmetric joint measure gives the same one-step density. □

Method 3.3.4. *Random-Walk Metropolis-Hastings*

(i) Use the proposal $Y = X_i + \varepsilon_i$, where ε_i has a symmetric density.
(ii) Then $q(y \mid x) = q(x \mid y)$, so the acceptance probability is $r(x, y) = \min\left\{\frac{f(y)}{f(x)}, 1\right\}$.
(iii) For $\varepsilon_i \sim N(0, b^2)$, small b gives slow local moves, large b gives frequent rejection. Tune b for adequate mixing.

Example 3.3.5. *Cauchy Target*
For Cauchy target $f(x) = \frac{1}{\pi(1+x^2)}$ and symmetric random-walk proposal, $r(x, y) = \min\left\{\frac{1+x^2}{1+y^2}, 1\right\}$.
The proposal scale changes mixing efficiency, not the stationary target.

Method 3.3.6. *Independence Metropolis-Hastings*

(i) Use a state-independent proposal $Y \sim g$, so $q(y \mid x) = g(y)$.
(ii) The acceptance probability is $r(x, y) = \min\{\frac{f(y) g(x)}{f(x) g(y)}, 1\}$.
(iii) Choose g as a tractable approximation to the target with sufficient mass on important regions.

Method 3.3.7. *Gibbs Sampling*

(i) Let (X, Y) have joint density $f_{X,Y}$ with tractable conditionals $f_{X|Y}$ and $f_{Y|X}$.
(ii) Starting from (X_0, Y_0), iterate $X_{n+1} \sim f_{X|Y}(\cdot \mid Y_n)$ and $Y_{n+1} \sim f_{Y|X}(\cdot \mid X_{n+1})$.
(iii) In higher dimension, update one coordinate or block at a time from its full conditional.

Method 3.3.8. *Metropolis within Gibbs*

(i) If $f_{X_j|X_{-j}}$ is not directly simulable, update X_j by Metropolis-Hastings conditional on X_{-j}.
(ii) In the bivariate case, propose $Z \sim q(\cdot \mid X_n)$ and accept $X_{n+1} = Z$ with probability $\min\{\frac{f(Z,Y_n)q(X_n|Z)}{f(X_n,Y_n)q(Z|X_n)}, 1\}$.
(iii) Then update Y analogously, conditional on the newly updated X.

3.4 Sequential Monte Carlo

3.5 Rare-Event Simulation

3.6 Stochastic Optimization

Part XV
Statistical Inference and Time Series

Statistical inference turns probabilistic models into procedures for learning from data. The probability, measure-theoretic, stochastic-process, optimization, and Monte Carlo material developed in Parts 11–14 is treated here as prerequisite structure.

1 Statistical Models and Experiments

1.1 Statistical Experiments and Likelihoods

A statistical problem starts with observable data and a family of probability laws that could have generated it. The model is not the data itself; it is the mathematical language in which uncertainty about the data-generating mechanism is expressed.

Definition 1.1.1. *Statistical Experiment and Model*

(i) A *statistical experiment* is $(\mathcal{X}, \mathcal{A}, \mathcal{P})$, where $(\mathcal{X}, \mathcal{A})$ is measurable and $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is a family of probability measures on $(\mathcal{X}, \mathcal{A})$.
(ii) Θ is the *parameter space*, $X \in \mathcal{X}$ is the *sample*, and $\mathcal{P}$ is the *statistical model*.
(iii) The model is *well specified* if the true law $P_0 \in \mathcal{P}$.
(iv) The model is *misspecified* if $P_0 \notin \mathcal{P}$.

Definition 1.1.2. *Dominated Models, Likelihood, Score, and Information*

(i) A model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is *dominated* by μ if $P_\theta \ll \mu$ for all $\theta \in \Theta$; write $p_\theta := \frac{dP_\theta}{d\mu}$.
(ii) Given x, the *likelihood* and *log-likelihood* are $L_x(\theta) := p_\theta(x)$ and $\ell_x(\theta) := \log p_\theta(x)$ when $p_\theta(x) > 0$.
(iii) If $\Theta \subseteq \mathbb{R}^d$ and differentiation is valid, the *score* is $S_\theta(X) := \nabla_\theta \ell_X(\theta)$.
(iv) The *Fisher information* is $I(\theta) := \mathbb{E}_\theta[S_\theta(X) S_\theta(X)^\top]$.

Example 1.1.3. *Bernoulli and Normal Likelihoods*
Let $X_1, \dots, X_n$ be independent observations.

(i) If $X_i \sim \text{Bernoulli}(p)$, then $L_x(p) = p^s (1-p)^{n-s}$, where $s := \sum_{i=1}^{n} x_i$.

(ii) If $X_i \sim N(\mu, \sigma^2)$ and σ^2 is known, then $\ell_x(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$, maximized at $\widehat{\mu} = \overline{x}$.

Example 1.1.4. *Multinomial Cell Probabilities*
Let $X = (X_1, \dots, X_m)$ have total count n.

(i) If $X \sim \text{Multinomial}(n; p_1, \dots, p_m)$ and $\sum_{j=1}^{m} p_j = 1$, then $L_x(p) \propto \prod_{j=1}^{m} p_j^{x_j}$ and $\widehat{p}_j = \frac{x_j}{n}$.

(ii) If $p_j = p_j(\theta)$, then $L_x(\theta) \propto \prod_{j=1}^{m} p_j(\theta)^{x_j}$ and $\widehat{\theta} \in \arg\max_\theta \sum_{j=1}^{m} x_j \log p_j(\theta)$.

Method 1.1.5. *Estimating a Finite Transition Matrix*

(i) Observe finite homogeneous chain $X_0, \dots, X_n$, let $n_{ij} := \sum_{t=0}^{n-1} \mathbf{1}\{X_t = i, X_{t+1} = j\}$, $n_i := \sum_j n_{ij}$.
(ii) The row likelihood is proportional to $\prod_j p_{ij}^{n_{ij}}$ subject to $\sum_j p_{ij} = 1$.
(iii) If $n_i > 0$, the rowwise MLE is $\widehat{p}_{ij} = \frac{n_{ij}}{n_i}$.
(iv) If $n_i = 0$, row i is not identifiable without a convention or prior.

Example 1.1.6. *Capture–Recapture Likelihood*
Suppose t population units are tagged and a later sample of size m contains $r > 0$ tagged units. For population size n, likelihood is $L_n = \frac{\binom{t}{r}\binom{n-t}{m-r}}{\binom{n}{m}}$. Since $\frac{L_n}{L_{n-1}} = \frac{(n-t)(n-m)}{n(n-t-m+r)}$, L_n increases for $n < \frac{mt}{r}$ and decreases for $n > \frac{mt}{r}$. Hence $\widehat{n} = \lfloor \frac{mt}{r} \rfloor$; for $(t, m, r) = (10, 20, 4)$, $\widehat{n} = 50$.

Definition 1.1.7. *Identifiability and Functionals*

(i) The parameterization is *identifiable* if $P_\theta = P_{\theta'}$ implies $\theta = \theta'$.
(ii) A *statistical functional* is a map $T : \mathcal{P} \to \mathcal{T}$, such as $T(P) = \int x \, dP(x)$ or $T(P) = \text{median}(P)$.

(iii) In a parametric model, a *target parameter* may be a function $\psi(\theta)$ rather than the full parameter θ.

Key Insight: Estimation should target a law or functional that the experiment can actually distinguish.

Remark 1.1.8. *Likelihood as Relative Evidence*
For fixed data x, the likelihood is a function of θ, not a probability distribution on Θ.
Ratios $\frac{L_x(\theta_1)}{L_x(\theta_0)}$ compare how well two model laws explain the same observation.

1.2 Sufficiency, Exponential Families, and Local Asymptotics

The shape of a model determines which summaries of the data preserve information and which approximations become valid in large samples.

Definition 1.2.1. *Statistics, Sufficiency, and Ancillarity*

(i) A *statistic* is a measurable map $T : \mathcal{X} \to \mathcal{T}$.
(ii) T is *sufficient* if $\mathcal{L}_\theta(X \mid T)$ is independent of θ.
(iii) T is *ancillary* if $\mathcal{L}_\theta(T)$ is independent of θ.

Theorem 1.2.2. *Factorization Criterion*
Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be dominated by μ, with $p_\theta := \frac{dP_\theta}{d\mu}$. A statistic $T : \mathcal{X} \to \mathcal{T}$ is sufficient iff there exist measurable $g_\theta : \mathcal{T} \to [0, \infty)$ and $h : \mathcal{X} \to [0, \infty)$ such that $p_\theta(x) = g_\theta(T(x))h(x)$ μ-a.e. for all θ.

Proof.
($\Rightarrow$) If T is sufficient, write $\nu_\theta := \mathcal{L}_\theta(T)$ and choose a θ-free kernel $K_t = \mathcal{L}_\theta(X \mid T = t)$,
so $P_\theta(A) = \int K_t(A)\nu_\theta(dt)$. Choosing densities for ν_θ and for K_t on each fiber gives $p_\theta(x) = g_\theta(T(x))h(x)$.
($\Leftarrow$) If $p_\theta = g_\theta \circ T\, h$, then on the fiber $T = t$ the factor $g_\theta(t)$ cancels:
for $c(t) := \int_{T^{-1}\{t\}} h\, d\mu_t > 0$, $\mathbb{P}_\theta(X \in A \mid T = t) = c(t)^{-1} \int_{A \cap T^{-1}\{t\}} h\, d\mu_t$, independent of θ.

Structural Commentary: Factorisation separates parameter information $g_\theta(T)$ from residual shape h.
Key Insight: Sufficiency keeps exactly the likelihood's θ-dependent part. □

Example 1.2.3. *Bernoulli and Normal Sufficient Statistics*
Let $X_1, \dots, X_n$ be independent and let $x = (x_1, \dots, x_n)$.

(i) If $X_i \sim \text{Bernoulli}(\theta)$, then $p_\theta(x) = \theta^s(1-\theta)^{n-s}$ with $s := \sum_i x_i$.
Hence $T = \sum_i X_i$ is sufficient for θ, and $\mathcal{L}_\theta(X \mid T = t)$ is uniform on $\{x : \sum_i x_i = t\}$.

(ii) If $X_i \sim N(\mu, \sigma^2)$ with (μ, σ^2) unknown, then $p_{\mu,\sigma^2}(x) \propto (\sigma^2)^{-\frac{n}{2}} \exp\{-\frac{\sum_i x_i^2 - 2\mu \sum_i x_i + n\mu^2}{2\sigma^2}\}$.
Hence $T = (\sum_i X_i, \sum_i X_i^2)$ is sufficient for (μ, σ^2).

Corollary 1.2.4. *Likelihood Estimators Use Sufficient Statistics*
Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be dominated with densities p_θ, and let T be sufficient with $p_\theta(x) = g_\theta(T(x))h(x)$.
If $a(t) \in \arg\max_{\theta \in \Theta} g_\theta(t)$ exists, then $\widehat{\theta}(X) := a(T(X))$ is an MLE.

Proof. For fixed x, $L_x(\theta) = p_\theta(x) = h(x)g_\theta(T(x))$. If $h(x) > 0$, then $\arg\max_\theta L_x(\theta) = \arg\max_\theta g_\theta(T(x))$.
If $h(x) = 0$, then $L_x(\theta) = 0$ for all θ. Thus an MLE may be chosen as $a(T(x))$.
Structural Commentary: The θ-free factor $h(x)$ is irrelevant to maximization.
Key Insight: MLEs can depend on data only through a sufficient statistic. □

Theorem 1.2.5. *Rao–Blackwell Improvement*
Let T be sufficient and let $\widehat{\psi} \in L^2(P_\theta)$ estimate a real target $\psi(\theta)$. Choose the θ-free version $\widetilde{\psi}(T) := \mathbb{E}_\theta(\widehat{\psi} \mid T)$.
Then $\mathbb{E}_\theta\widetilde{\psi} = \mathbb{E}_\theta\widehat{\psi}$ and $\mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2 \le \mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2$ for all θ, with equality iff $\widehat{\psi} = \widetilde{\psi}$ P_θ-a.s.

Proof. Sufficiency makes $\mathcal{L}_\theta(X \mid T)$ independent of θ, so $\widetilde{\psi}$ is a T-statistic. Also $\mathbb{E}_\theta\widetilde{\psi} = \mathbb{E}_\theta\widehat{\psi}$.
Since $\mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi} \mid T) = 0$ and $\widetilde{\psi} - \psi(\theta)$ is T-measurable, $\mathbb{E}_\theta[(\widehat{\psi} - \widetilde{\psi})(\widetilde{\psi} - \psi(\theta))] = 0$.
Hence $\mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2 = \mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi})^2 + \mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2 \ge \mathbb{E}_\theta(\widetilde{\psi} - \psi(\theta))^2$, with equality iff $\mathbb{E}_\theta(\widehat{\psi} - \widetilde{\psi})^2 = 0$.
Structural Commentary: Rao–Blackwellization projects an estimator onto the $\sigma(T)$-measurable statistics.
Key Insight: Conditioning on a sufficient statistic preserves bias and removes residual noise. □

Definition 1.2.6. *Exponential Family*

(i) A dominated model $\{P_\eta : \eta \in \mathcal{N}\}$ is a *k-parameter exponential family*
if $\mathcal{N} \subseteq \mathbb{R}^k$ and $p_\eta(x) = \exp\{\eta^\top T(x) - A(\eta)\}h(x)$.

(ii) η is the *natural parameter*, $T(X)$ is the *natural statistic*, and h is the carrier density.
(iii) $A(\eta)$ is the *log-partition function*: $A(\eta) = \log \int \exp\{\eta^\top T(x)\} h(x)\, d\mu(x)$ when finite.

Proposition 1.2.7. *Moment Identities in Full Exponential Families*
Let $p_\eta(x) = \exp\{\eta^\top T(x) - A(\eta)\}h(x)$ be a full exponential family with open natural parameter space, and assume differentiation under the integral is valid. Then $\nabla A(\eta) = \mathbb{E}_\eta T(X)$ and $\nabla^2 A(\eta) = \mathrm{Cov}_\eta(T(X))$.

Proof. From $1 = \int p_\eta\, d\mu$, differentiation gives $0 = \int (T - \nabla A(\eta)) p_\eta\, d\mu$, hence $\nabla A(\eta) = \mathbb{E}_\eta T(X)$.
For coordinates i, j, differentiating $\mathbb{E}_\eta T_i = \int T_i p_\eta\, d\mu$ gives $\partial_j \mathbb{E}_\eta T_i = \mathbb{E}_\eta[T_i(T_j - \partial_j A(\eta))] = \mathrm{Cov}_\eta(T_i, T_j)$, QED.
Structural Commentary: The log-partition function cumulatively encodes the natural statistic.
Key Insight: First and second derivatives of A give mean and covariance. □

Definition 1.2.8. *Local Asymptotic Normality*
For experiments $\{P_\theta^{(n)} : \theta \in \Theta \subseteq \mathbb{R}^d\}$, set $\theta_{n,h} := \theta_0 + \frac{h}{\sqrt{n}}$.

(i) The sequence is *locally asymptotically normal* at θ_0 with information $I(\theta_0)$ if,
for each fixed $h \in \mathbb{R}^d$, $\log(\frac{dP^{(n)}_{\theta_{n,h}}}{dP^{(n)}_{\theta_0}}) = h^\top \Delta_n - \frac{1}{2} h^\top I(\theta_0) h + o_{P^{(n)}_{\theta_0}}(1)$.
(ii) The *central sequence* satisfies $\Delta_n \Rightarrow N(0, I(\theta_0))$ under $P^{(n)}_{\theta_0}$.

Remark 1.2.9. *Why Local Asymptotics Matter*
LAN replaces local likelihood ratios by the Gaussian quadratic experiment $h^\top \Delta - \frac{1}{2} h^\top I(\theta_0) h$, giving normal approximations for estimators, tests, and confidence regions.

1.3 Design, Randomization, and Observational Comparisons

Comparative inference is only as meaningful as the experiment or study that produced the groups. The same two-sample statistic can estimate a causal treatment effect in a randomised experiment, describe an association in an observational study, or amplify bias if the groups were formed by a hidden mechanism.

Definition 1.3.1. *Controls, Randomisation, and Blinding*

(i) A *controlled experiment* compares $Y \mid T = 1$ and $Y \mid T = 0$ under common conditions.
(ii) *Randomisation* assigns $T \in \{0, 1\}$ so that T is independent of pre-treatment factors C by design.
(iii) *Blinding* conceals T when knowledge of treatment could affect Y or its measurement.

Example 1.3.2. *Mammary Artery Ligation*
A blinded trial randomised patients after preliminary incisions to ligation ($T = 1$) or sham surgery ($T = 0$). Similar outcomes show that controls remove background improvement; blinding limits placebo, observer effects.

Definition 1.3.3. *Placebo Effect and Confounding*

(i) A *placebo effect* is variation in Y caused by belief in treatment rather than the active mechanism.
(ii) Placebos and blinding separate expectation effects from the causal effect of T.
(iii) A pre-treatment variable C is *confounded* with treatment T for response Y if $\mathcal{L}(C \mid T = 1) \neq \mathcal{L}(C \mid T = 0)$ and C affects Y.
(iv) An *observational study* compares groups without assigning T. Group differences may reflect confounding rather than the factor of interest.

Principle 1.3.4. *Confounding, Blocking, and Pairing*
Randomisation balances unmeasured covariates C in probability; blocking, matching, and pairing condition on known influential C before randomisation. Pairing is one-factor blocking; blocked and factorial designs extend this control to multiple factors.

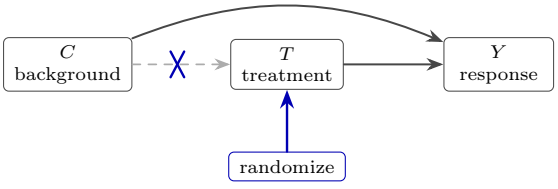


Figure 215: Randomization breaks the confounding path from C to T.

Example 1.3.5. *Uncontrolled Adjustments and Position Effects*
In the Lanarkshire milk experiment, post-randomisation movement between groups made T depend on perceived nourishment C, reintroducing selection bias; weighing children in clothes also confounded measurement with socioeconomic status. In the FD&C Red No. 40 study, cage location was systematic in sex and dose, so position effects confounded treatment comparisons; randomising or blocking cage location would protect inference.

Example 1.3.6. *Graduate Admissions and Aggregation*
In the Berkeley admissions example, the aggregate admission rate for men was higher than for women.
Within the six listed majors, however, women were not consistently admitted at lower rates.
The apparent aggregate disadvantage was driven by application patterns: women applied more often to majors with lower admission rates. Major was therefore confounded with sex in the observational comparison.

Remark 1.3.7. *Comparative Study Checklist*
Before interpreting group comparisons, record sampling, treatment assignment T, controls, blinding, blocking or pairing factors C, dependence among repeated measurements, and the number of tested comparisons.
These determine whether the analysis supports causality, association, or hypothesis generation.

2 Estimation Theory

2.1 Estimators, Bias, Variance, and Risk

An estimator is a statistic interpreted as an answer to a target question. Its quality depends on the target, the loss function, and the model law under which performance is measured.

Definition 2.1.1. *Estimator, Loss, and Risk*

(i) A target parameter is a map $\psi : \Theta \to \Psi$.
(ii) An *estimator* of $\psi(\theta)$ is a statistic $\delta(X) \in \mathcal{D}$.
(iii) A *loss function* is $L : \Theta \times \mathcal{D} \to [0, \infty]$.
(iv) The *risk* of δ is $R(\theta, \delta) := \mathbb{E}_\theta L(\theta, \delta(X))$.
(v) For squared-error loss $L(\theta, d) = \|d - \psi(\theta)\|^2$, $R(\theta, \delta)$ is mean squared error.

Definition 2.1.2. *Bias, Variance, and Consistency*
For real-valued $\widehat{\psi}$, $\mathrm{Bias}_\theta(\widehat{\psi}) := \mathbb{E}_\theta\widehat{\psi} - \psi(\theta)$, and $\mathrm{Var}_\theta(\widehat{\psi}) := \mathbb{E}_\theta(\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi})^2$.

(i) $\widehat{\psi}$ is *unbiased* if $\mathrm{Bias}_\theta(\widehat{\psi}) = 0$ for all θ.
(ii) $\widehat{\psi}_n$ is *consistent* at θ if $\widehat{\psi}_n \to \psi(\theta)$ in P_θ-probability.

Proposition 2.1.3. *Bias-Variance Decomposition*
Let $\widehat{\psi} \in L^2(P_\theta)$ be real-valued. Under squared-error loss, $\mathrm{MSE}_\theta(\widehat{\psi}) := \mathbb{E}_\theta(\widehat{\psi} - \psi(\theta))^2 = \mathrm{Var}_\theta(\widehat{\psi}) + \mathrm{Bias}_\theta(\widehat{\psi})^2$.

Proof. Write $\widehat{\psi} - \psi(\theta) = (\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi}) + (\mathbb{E}_\theta\widehat{\psi} - \psi(\theta))$. Square, take $\mathbb{E}_\theta$ gives the identity since $\mathbb{E}_\theta(\widehat{\psi} - \mathbb{E}_\theta\widehat{\psi}) = 0$.
Structural Commentary: MSE splits into random fluctuation plus squared systematic error.
Key Insight: Accuracy requires controlling both variance and bias. □

Example 2.1.4. *Sample Mean*
If $X_1, \ldots, X_n$ are independent with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$,
then $\overline{X}_n := n^{-1}\sum_{i=1}^{n} X_i$ satisfies $\mathbb{E}\overline{X}_n = \mu$, $\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n}$, and $\overline{X}_n \xrightarrow{P} \mu$.

Definition 2.1.5. *Plug-In and M-Estimators*

(i) A *plug-in estimator* estimates $T(P)$ by $T(\widehat{P}_n)$, where $\widehat{P}_n$ is empirical or fitted.
(ii) An *M-estimator* satisfies $\widehat{\theta}_n \in \arg\min_{\theta\in\Theta} M_n(\theta)$ for an empirical criterion M_n.
(iii) Common choices are $M_n =$ negative log-likelihood, squared loss, or absolute loss.

2.2 Information Bounds and Asymptotic Efficiency

Information inequalities describe the best possible local accuracy under regular models. Efficient procedures are those whose asymptotic error reaches these bounds.

Lemma 2.2.1. *Two Forms of Fisher Information*
For a regular one-parameter dominated model, let $\ell_\theta(x) := \log p_\theta(x)$ and $S_\theta(X) := \partial_\theta \ell_\theta(X)$.
Then $I(\theta) := \mathbb{E}_\theta S_\theta(X)^2 = -\mathbb{E}_\theta \partial_\theta^2 \ell_\theta(X)$.

Proof. Differentiating $\int p_\theta \, d\mu = 1$ gives $\mathbb{E}_\theta S_\theta = 0$.
Differentiating, $0 = \partial_\theta \mathbb{E}_\theta S_\theta = \mathbb{E}_\theta[\partial_\theta S_\theta + S_\theta^2] = \mathbb{E}_\theta[\partial_\theta^2 \ell_\theta + S_\theta^2]$, since $\partial_\theta p_\theta = S_\theta p_\theta$. Thus $I(\theta) = \mathbb{E}_\theta S_\theta^2 = -\mathbb{E}_\theta \partial_\theta^2 \ell_\theta$.
Structural Commentary: Fisher information is both score variance and expected log-likelihood curvature.
Key Insight: Normalisation forces the mean score to vanish, linking slope variance to curvature. □

Theorem 2.2.2. *Cramer–Rao Bound*
Let $X_1, \dots, X_n \sim P_\theta$ be iid from a regular one-parameter dominated model with one-observation information $I(\theta) > 0$. If $\widehat{\psi} \in L^2(P_\theta)$ is unbiased for differentiable $\psi(\theta)$, then $\mathrm{Var}_\theta(\widehat{\psi}) \geq \frac{\psi'(\theta)^2}{nI(\theta)}$.

Proof. Let $S_n(\theta) := \partial_\theta \log L_n(\theta) = \sum_{i=1}^n S_\theta(X_i)$. Regularity gives $\mathbb{E}_\theta S_n = 0$ and $\mathrm{Var}_\theta(S_n) = \mathbb{E}_\theta S_n^2 = nI(\theta)$.
Differentiating $\mathbb{E}_\theta \widehat{\psi} = \psi(\theta)$ gives $\psi'(\theta) = \mathbb{E}_\theta[\widehat{\psi} S_n] = \mathrm{Cov}_\theta(\widehat{\psi}, S_n)$.
Thus $\psi'(\theta)^2 \leq \mathrm{Var}_\theta(\widehat{\psi}) \, \mathrm{Var}_\theta(S_n) = \mathrm{Var}_\theta(\widehat{\psi}) nI(\theta)$ by Cauchy–Schwarz.
Structural Commentary: The score is the direction in which unbiased estimators must covary to track $\psi(\theta)$.
Key Insight: Information lower-bounds variance through Cauchy–Schwarz. □

Example 2.2.3. *Poisson Mean Attains the Bound*
For $X \sim \mathrm{Poisson}(\lambda)$, $\ell_\lambda(X) = X \log \lambda - \lambda - \log(X!)$ and $\partial_\lambda^2 \ell_\lambda(X) = -\frac{X}{\lambda^2}$, so $I(\lambda) = -\mathbb{E}_\lambda \partial_\lambda^2 \ell_\lambda(X) = \frac{1}{\lambda}$.
For iid $X_1, \dots, X_n$, any unbiased estimator T of λ satisfies $\mathrm{Var}_\lambda(T) \geq \frac{\lambda}{n}$.
$\overline{X}$ attains equality since $\mathbb{E}\overline{X} = \lambda$ and $\mathrm{Var}(\overline{X}) = \frac{\lambda}{n}$.

Definition 2.2.4. *Maximum Likelihood Estimator*
For likelihood $L_X(\theta)$ and log-likelihood $\ell_X(\theta)$, a *maximum likelihood estimator* is any measurable $\widehat{\theta}_n(X) \in \arg\max_{\theta \in \Theta} L_X(\theta) = \arg\max_{\theta \in \Theta} \ell_X(\theta)$; equivalently, $\widehat{\theta}_n \in \arg\min_{\theta \in \Theta}[-\ell_X(\theta)]$.

Theorem 2.2.5. *Regular MLE Asymptotic Normality*
Let $X_1, \dots, X_n \sim P_{\theta_0}$ iid in a regular one-parameter model, with θ_0 interior, $I(\theta_0) > 0$, and a consistent interior MLE $\widehat{\theta}_n$. Then $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$.

Proof. Let $S_n(\theta) := \partial_\theta \ell_n(\theta)$ and $J_n(\theta) := -n^{-1} \partial_\theta S_n(\theta)$.
Since $S_n(\widehat{\theta}_n) = 0$, Taylor expansion gives $0 = S_n(\theta_0) - nJ_n(\widetilde{\theta}_n)(\widehat{\theta}_n - \theta_0)$ for some $\widetilde{\theta}_n$ between $\widehat{\theta}_n$ and θ_0.
Hence $\sqrt{n}(\widehat{\theta}_n - \theta_0) = J_n(\widetilde{\theta}_n)^{-1} n^{-\frac{1}{2}} S_n(\theta_0)$.
Regularity gives $J_n(\widetilde{\theta}_n) \xrightarrow{P} I(\theta_0)$ and $n^{-\frac{1}{2}} S_n(\theta_0) \Rightarrow N(0, I(\theta_0))$, so Slutsky gives the result.
Structural Commentary: Local likelihood curvature converts score noise into estimator error.
Key Insight: The MLE is asymptotically score divided by information. □

Proposition 2.2.6. *Delta Method*
If $\sqrt{n}(T_n - \theta) \Rightarrow N(0, \Sigma)$ and $g : \mathbb{R}^d \to \mathbb{R}^k$ is differentiable at θ with derivative $Dg(\theta)$, then $\sqrt{n}(g(T_n) - g(\theta)) \Rightarrow N(0, Dg(\theta) \Sigma Dg(\theta)^\top)$.

Proof. Differentiability gives $g(T_n) - g(\theta) = Dg(\theta)(T_n - \theta) + r_n$ with $\frac{\|r_n\|}{\|T_n - \theta\|} \to 0$ in probability.
Since $\sqrt{n}(T_n - \theta) = O_P(1)$, $\sqrt{n} r_n = o_P(1)$; $\sqrt{n}(g(T_n) - g(\theta)) = Dg(\theta)\sqrt{n}(T_n - \theta) + o_P(1)$, by Slutsky QED.
Structural Commentary: The derivative transports the limiting normal fluctuation through g.
Key Insight: First-order linearisation determines asymptotic variance. □

Definition 2.2.7. *Asymptotic Efficiency*
For a scalar regular parameter with information $I(\theta_0) > 0$, an estimator sequence $\widehat{\theta}_n$ is *asymptotically efficient* at θ_0 if $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$, i.e. it attains the inverse Fisher-information variance.

Remark 2.2.8. *Efficiency and Approximation*
Finite-sample efficiency means equality in the Cramer–Rao bound.
Asymptotic efficiency means $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1})$; finite-sample bias or boundary effects still matter.

2.3 Method of Moments and Parametric Likelihood Examples

Definition 2.3.1. *Sampling Distribution and Standard Error*

(i) For a statistic $T_n = T_n(X_1, \dots, X_n)$ under P_θ, its *sampling distribution* is $\mathcal{L}_\theta(T_n)$.
(ii) If $T_n \in L^2(P_\theta)$, its *standard error* is $\mathrm{SE}_\theta(T_n) := \sqrt{\mathrm{Var}_\theta(T_n)}$.
(iii) An *estimated standard error* estimates $\mathrm{SE}_\theta(T_n)$, typically by plug-in, asymptotics, or resampling.

Method 2.3.2. *Method of Moments*

(i) Let $X_1, \dots, X_n$ be independent from P_θ, and set $\mu_k(\theta) := \mathbb{E}_\theta X^k$ for $k = 1, \dots, r$.

(ii) If $\mu(\theta) = (\mu_1(\theta), \dots, \mu_r(\theta))$ identifies $\theta \in \Theta \subseteq \mathbb{R}^r$, define $\widehat{\mu}_k := n^{-1} \sum_{i=1}^{n} X_i^k$.

(iii) The *method of moments* estimator solves $\mu_k(\widehat{\theta}) = \widehat{\mu}_k$ for $k = 1, \dots, r$.

Key Insight: Match population moments to sample moments.

Proposition 2.3.3. *Consistency of Continuous Moment Estimates*
Let $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu_0$ and let g be continuous at μ_0 with $g(\mu_0) = \theta_0$. Then $\widehat{\theta}_n := g(\widehat{\mu}_n) \xrightarrow{P_{\theta_0}} \theta_0$. Thus method-of-moments estimators are consistent whenever $\widehat{\mu}_n \xrightarrow{P} \mu(\theta_0)$, inversion map $g = \mu^{-1}$ is continuous at $\mu(\theta_0)$.

Proof. Since $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu_0$ and g is continuous at μ_0, by continuous mapping theorem, $g(\widehat{\mu}_n) \xrightarrow{P_{\theta_0}} g(\mu_0) = \theta_0$.
For MoM, if the required moments are finite, the weak law gives $\widehat{\mu}_n \xrightarrow{P_{\theta_0}} \mu(\theta_0)$.
Structural Commentary: Moment consistency is convergence plus continuous inversion.
Key Insight: Estimate moments first, then pass through g. □

Example 2.3.4. *Moment Estimation for Small Open Probabilities*
Let $N \sim \text{Binomial}(m, p)$ be the number of open channels and let $S = cN$ be total current.
Then $\mathbb{E}S = cmp$, $\text{Var}(S) = c^2 mp(1-p)$, and $\frac{\text{Var}(S)}{\mathbb{E}S} = c(1-p) \approx c$ for small p.
Thus independent measurements estimate c by the empirical ratio $\frac{\widehat{\text{Var}}(S)}{\widehat{\mathbb{E}}S}$, without knowing m.

Example 2.3.5. *Poisson Mean*
Let $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$ be independent. Since $\mathbb{E}_\lambda X = \lambda$, $\widehat{\lambda}_{\text{mom}} = \overline{X}$.
Also $\ell(\lambda) = \sum_{i=1}^{n} \{X_i \log \lambda - \lambda - \log(X_i!)\}$ and $\ell'(\lambda) = \frac{S}{\lambda} - n$, where $S = \sum_i X_i$, so $\widehat{\lambda}_{\text{mle}} = \frac{S}{n} = \overline{X}$.
If true value is λ_0, then $S \sim \text{Poisson}(n\lambda_0)$. Hence $\mathbb{E}_{\lambda_0}\widehat{\lambda} = \lambda_0$, $\text{Var}_{\lambda_0}(\widehat{\lambda}) = \frac{\lambda_0}{n}$, $\text{SE}_{\lambda_0}(\widehat{\lambda}) = \sqrt{\frac{\lambda_0}{n}}$, $\widehat{\text{SE}}(\widehat{\lambda}) = \sqrt{\frac{\widehat{\lambda}}{n}}$.

Remark 2.3.6. *Grouped Counts and Expected Frequencies*
Let $Y_1, \dots, Y_N$ be observations, let $A_1, \dots, A_m$ partition sample space, with observed counts $O_j := \sum_{i=1}^{N} \mathbf{1}_{\{Y_i \in A_j\}}$.
Given a fitted parametric law $P_{\widehat{\theta}}$, set $p_j(\widehat{\theta}) := P_{\widehat{\theta}}(Y \in A_j)$; the fitted expected count is $E_j := N p_j(\widehat{\theta})$.
For grouped Poisson tails, $p_j(\widehat{\theta})$ is a sum of Poisson masses, $(O_1, \dots, O_m)$ is modelled as multinomial under $P_{\widehat{\theta}}$.

Example 2.3.7. *Normal Mean and Variance*
For iid $X_i \sim N(\mu, \sigma^2)$, $\mu_1 = \mu$, $\mu_2 = \mu^2 + \sigma^2$. Thus by MoM $\widehat{\mu} = \overline{X}$; $\widehat{\sigma}^2 = n^{-1} \sum_{i=1}^{n} X_i^2 - \overline{X}^2 = n^{-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$.
The normal MLEs are the same; $\widehat{\sigma}^2$ uses denominator n, not the unbiased denominator $n-1$.

Example 2.3.8. *Gamma Shape and Rate*
Let X_i be iid gamma with shape α and rate λ, so $\mathbb{E}X = \frac{\alpha}{\lambda}$ and $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.
With $\widehat{\sigma}^2 := n^{-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$, MoM gives $\widehat{\alpha}_{\text{mom}} = \frac{\overline{X}^2}{\widehat{\sigma}^2}$ and $\widehat{\lambda}_{\text{mom}} = \frac{\overline{X}}{\widehat{\sigma}^2}$.
The log-likelihood is $\ell(\alpha, \lambda) = n\alpha \log \lambda + (\alpha - 1) \sum_i \log X_i - \lambda \sum_i X_i - n \log \Gamma(\alpha)$.
The MLE satisfies $\widehat{\lambda} = \frac{\widehat{\alpha}}{\overline{X}}$ and $\log \widehat{\alpha} - \log \overline{X} + n^{-1} \sum_i \log X_i - \psi(\widehat{\alpha}) = 0$, where $\psi = \frac{\Gamma'}{\Gamma}$.
The MoM estimate is a natural starting value.

Example 2.3.9. *Angular Density*
Let $f(x \mid \alpha) = \frac{1+\alpha x}{2}$ on $[-1, 1]$, with $\alpha \in [-1, 1]$. Since $\mathbb{E}_\alpha X = \int_{-1}^{1} \frac{x(1+\alpha x)}{2}\, dx = \frac{\alpha}{3}$, MoM gives $\widehat{\alpha} = 3\overline{X}$.
Also $\text{Var}_\alpha(X) = \frac{1}{3} - \frac{\alpha^2}{9}$, so $\text{Var}_\alpha(\widehat{\alpha}) = 9 \text{Var}_\alpha(\overline{X}) = \frac{3 - \alpha^2}{n}$.

Definition 2.3.10. *Overdispersed Counts*
Count data are *overdispersed* relative to a Poisson model if $\text{Var}(X) > \mathbb{E}X$; empirically, this is $\widehat{\sigma}^2 > \overline{X}$.

Example 2.3.11. *Negative Binomial Overdispersion*
For overdispersed counts, negative-binomial mass $f(x \mid m, k) = (1 + \frac{m}{k})^{-k} \Gamma(k + x) \{x! \Gamma(k)\}^{-1} (\frac{m}{m+k})^x$, $x \in \mathbb{N}_0$, has $\mathbb{E}X = m$ and $\text{Var}(X) = m + \frac{m^2}{k}$. By MoM $\widehat{m} = \overline{X}$, $\widehat{k} = \frac{\overline{X}^2}{\widehat{\sigma}^2 - \overline{X}}$ when $\widehat{\sigma}^2 > \overline{X}$.

3 Hypothesis Testing and Confidence

3.1 Tests, Likelihood Ratios, and Error Control

Testing formalises decisions between hypotheses. The central quantities are the probability of false rejection under the null and the probability of rejection under alternatives.

Definition 3.1.1. *Hypotheses, Tests, Size, and Power*

(i) A *null hypothesis* is a subset $\Theta_0 \subseteq \Theta$ of the parameter space.
(ii) An *alternative* is a subset $\Theta_1 \subseteq \Theta$ disjoint from Θ_0, so $\Theta_0 \cap \Theta_1 = \emptyset$.
(iii) Randomised *test* is a measurable map $\varphi : \mathcal{X} \to [0,1]$, where $\varphi(x)$ is rejection probability at observation x.
(iv) The *size* of φ is $\alpha(\varphi) = \sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\varphi(X)]$.
(v) The *power function* of φ is $\beta_\varphi(\theta) = \mathbb{E}_\theta[\varphi(X)]$.

Definition 3.1.2. *Simple and Composite Hypotheses; Type Errors*

(i) A *simple* hypothesis specifies one law; a *composite* hypothesis specifies several.
(ii) A *type I error* rejects a true H_0; its probability is bounded by $\alpha(\varphi)$.
(iii) For a simple alternative P_1, a *type II error* is failure to reject H_0 under P_1, with probability $\mathbb{E}_1[1-\varphi(X)]$.
(iv) The *power* against P_1 is $1 - \mathbb{E}_1[1-\varphi(X)]$.

Proposition 3.1.3. *Posterior Odds and Likelihood Ratios*
Let H_0, H_1 be simple hypotheses with $\mathbb{P}(H_i) = \pi_i > 0$ and likelihood $p_i(x) = p(x \mid H_i)$ for $i = 0, 1$. For $m(x) := \pi_0 p_0(x) + \pi_1 p_1(x) > 0$, $\mathbb{P}(H_i \mid x) = \frac{\pi_i p_i(x)}{m(x)}$ and $\frac{\mathbb{P}(H_0|x)}{\mathbb{P}(H_1|x)} = (\frac{\pi_0}{\pi_1})(\frac{p_0(x)}{p_1(x)})$.

Proof. Bayes' rule gives $\mathbb{P}(H_i \mid x) = \frac{\pi_i p_i(x)}{m(x)}$ for $i = 0, 1$. Therefore $\frac{\mathbb{P}(H_0|x)}{\mathbb{P}(H_1|x)} = \frac{\frac{\pi_0 p_0(x)}{m(x)}}{\frac{\pi_1 p_1(x)}{m(x)}} = (\frac{\pi_0}{\pi_1})(\frac{p_0(x)}{p_1(x)})$.
Structural Commentary: Posterior odds factor as prior odds times the data likelihood ratio.
Key Insight: Bayes updating multiplies odds by $\frac{p_0(x)}{p_1(x)}$. □

Theorem 3.1.4. *Neyman–Pearson Lemma*
Let $P_1 \ll P_0$, $\Lambda := \frac{dP_1}{dP_0}$, $\alpha \in [0,1]$. If $c \geq 0$ and $\gamma \in [0,1]$ make $\varphi^* := \mathbf{1}_{\{\Lambda > c\}} + \gamma \mathbf{1}_{\{\Lambda = c\}}$ satisfy $\mathbb{E}_0\varphi^* = \alpha$, then φ^* is most powerful among level-α tests: for every $\psi : \mathcal{X} \to [0,1]$ with $\mathbb{E}_0\psi \leq \alpha$, $\mathbb{E}_1\psi \leq \mathbb{E}_1\varphi^*$.

Proof. For any level-α competitor ψ, $(\varphi^* - \psi)(\Lambda - c) \geq 0$ pointwise.
Since $dP_1 = \Lambda\, dP_0$, $\mathbb{E}_1(\varphi^* - \psi) = \mathbb{E}_0[\Lambda(\varphi^* - \psi)] \geq c\mathbb{E}_0(\varphi^* - \psi) = c(\alpha - \mathbb{E}_0\psi) \geq 0$.
Structural Commentary: Null controls size constraint, while alternative supplies power objective.
Key Insight: Use the rejection budget where $\Lambda = \frac{dP_1}{dP_0}$ is largest. □

Remark 3.1.5. *Choosing the Null*
At level a, for simple P_0 versus P_1, the *Neyman–Pearson problem* is $\sup\{\mathbb{E}_1\varphi : 0 \leq \varphi \leq 1,\ \mathbb{E}_0\varphi \leq a\}$.
Thus H_0 is the protected claim: choose it so the false-rejection probability $\mathbb{E}_0\varphi$ is the error to be bounded by a.
The choice of H_0 is substantive, not determined by $\{P_0, P_1\}$.

Definition 3.1.6. *Uniformly Most Powerful Test*
For a composite alternative Θ_1, a level-α test φ is *uniformly most powerful* if, for every $\theta \in \Theta_1$, it has power at least as large as every other level-α test at that same θ.

Example 3.1.7. *Normal Mean, Known Variance*
Let $X_1, \ldots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ^2. For $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1 > \mu_0$, the likelihood ratio is increasing in $\overline{X}$, so the level-α test is $\varphi^* = \mathbf{1}_{\{\overline{X} > \mu_0 + \frac{z_{1-\alpha}\sigma}{\sqrt{n}}\}}$, where $z_{1-\alpha}$ is the $N(0,1)$ quantile. Since the cutoff is independent of μ_1, φ^* is uniformly most powerful for $H_1 : \mu > \mu_0$; no single likelihood-ratio direction is uniformly most powerful for $H_1 : \mu \neq \mu_0$.

Definition 3.1.8. *Generalized Likelihood-Ratio Test*
For $H_0 : \theta \in \Theta_0 \subseteq \Theta$, the *generalized likelihood ratio* is $\lambda(x) := \frac{\sup_{\theta \in \Theta_0} L_x(\theta)}{\sup_{\theta \in \Theta} L_x(\theta)} \in [0,1]$.
A GLR test rejects H_0 for small $\lambda(x)$, typically $\lambda(x) \leq c_\alpha$ with c_α chosen to give level α.

Theorem 3.1.9. *Large-Sample Likelihood-Ratio Calibration*

(i) Let $H_0 : \theta \in \Theta_0 \subseteq \Theta$, $\dim \Theta = d$, $\dim \Theta_0 = d_0$, $r := d - d_0$, and $X^{(n)} \sim P_{\theta_0}$ with $\theta_0 \in \Theta_0$.
(ii) Let $L_n(\theta) := e^{\ell_n(\theta)}$ and $\lambda_n := \frac{\sup_{\theta \in \Theta_0} L_n(\theta)}{\sup_{\theta \in \Theta} L_n(\theta)}$.
(iii) Assume local coordinates with $\theta_0 = 0$, $\Theta_0 = \{(u,0) : u \in \mathbb{R}^{d_0}\}$, $h = (u,v) \in \mathbb{R}^{d_0} \times \mathbb{R}^r$, and $\ell_n(\frac{h}{\sqrt{n}}) - \ell_n(0) = h^\top Z_n - \frac{\|h\|^2}{2} + o_{P_{\theta_0}}(1)$, where $Z_n = (U_n, V_n) \Rightarrow N_d(0, I_d)$, $U_n \in \mathbb{R}^{d_0}$, $V_n \in \mathbb{R}^r$.
(iv) Then under H_0, $-2\log\lambda_n \Rightarrow \chi^2_r$.

Proof.

(i) Since $L_n = e^{\ell_n}$, $\log\lambda_n = \sup_{\theta \in \Theta_0} \ell_n(\theta) - \sup_{\theta \in \Theta} \ell_n(\theta)$, hence $-2\log\lambda_n = 2\{\sup_{\theta \in \Theta} \ell_n(\theta) - \sup_{\theta \in \Theta_0} \ell_n(\theta)\}$.

(ii) In the local coordinates, write $\theta = \frac{h}{\sqrt{n}}$ for the full model and $\theta = \frac{(u,0)}{\sqrt{n}}$ for the null model.
With $Q_n(h) := h^\top Z_n - \frac{\|h\|^2}{2}$ and $Q_{0,n}(u) := Q_n(u,0) = u^\top U_n - \frac{\|u\|^2}{2}$, (iii) and cancellation of $\ell_n(0)$ give $-2\log\lambda_n = 2\{\sup_{h\in\mathbb{R}^d} Q_n(h) - \sup_{u\in\mathbb{R}^{d_0}} Q_{0,n}(u)\} + o_{P_{\theta_0}}(1)$.

(iii) For $a \in \mathbb{R}^m$, $x^\top a - \frac{\|x\|^2}{2} = \frac{\|a\|^2}{2} - \frac{\|x-a\|^2}{2}$, so supremum over $x \in \mathbb{R}^m$ is $\frac{\|a\|^2}{2}$ at $x = a$.
Applying this with $a = Z_n$ and $a = U_n$ gives $\sup_{h\in\mathbb{R}^d} Q_n(h) = \frac{\|Z_n\|^2}{2}$ and $\sup_{u\in\mathbb{R}^{d_0}} Q_{0,n}(u) = \frac{\|U_n\|^2}{2}$,
hence $-2\log\lambda_n = \|Z_n\|^2 - \|U_n\|^2 + o_{P_{\theta_0}}(1) = \|V_n\|^2 + o_{P_{\theta_0}}(1)$.

(iv) Since $Z_n = (U_n, V_n) \Rightarrow N_d(0, I_d)$, marginal convergence gives $V_n \Rightarrow Y$ with $Y \sim N_r(0, I_r)$.
Therefore $\|V_n\|^2 \Rightarrow \|Y\|^2 \sim \chi^2_r$ by continuous mapping, and Slutsky gives $-2\log\lambda_n \Rightarrow \chi^2_r$.

Structural Commentary: The likelihood ratio compares unconstrained fit with null-constrained fit.
Key Insight: Wilks' statistic is the squared Gaussian fluctuation in the $r = d - d_0$ constrained directions. □

Example 3.1.10. *Two-Sided Normal Mean GLRT*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ^2, and test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.
Since $\widehat{\mu} = \overline{X}$, $\lambda(X) = \frac{L_n(\mu_0)}{L_n(\overline{X})} = \exp\{-\frac{n(\overline{X}-\mu_0)^2}{2\sigma^2}\}$, so $-2\log\lambda(X) = \frac{n(\overline{X}-\mu_0)^2}{\sigma^2}$.
Under H_0, $\frac{\sqrt{n}(\overline{X}-\mu_0)}{\sigma} \sim N(0,1)$; hence $-2\log\lambda(X) \sim \chi^2_1$,
and the level-α GLR rejects iff $\frac{n(\overline{X}-\mu_0)^2}{\sigma^2} > \chi^2_{1,1-\alpha}$, equivalently $\left|\overline{X} - \mu_0\right| > \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}$.

Example 3.1.11. *Sign Test as a Binomial GLRT*
Let $X \sim \text{Binomial}(n, p)$ and test $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$.
For observed x, $\widehat{p} = \frac{x}{n}$ and $\lambda(x) = \frac{L_x(\frac{1}{2})}{L_x(\widehat{p})} = \frac{(\frac{1}{2})^n}{[(\frac{x}{n})^x(1-\frac{x}{n})^{n-x}]}$, $0^0 = 1$ at $x = 0, n$.
Since $\lambda(x)$ decreases as $\left|x - \frac{n}{2}\right|$ increases, a level-α GLR rejects on the largest values of $\left|X - \frac{n}{2}\right|$, with cutoff chosen under $X \sim \text{Binomial}(n, \frac{1}{2})$; this is the two-sided sign test.

Definition 3.1.12. *p-Value*
A *p-value* for $H_0 : \theta \in \Theta_0$ is a statistic $p : \mathcal{X} \to [0,1]$ such that $\sup_{\theta\in\Theta_0} \mathbb{P}_\theta\{p(X) \leq u\} \leq u$ for every $u \in [0,1]$.
Then $\varphi_\alpha(X) := \mathbf{1}_{\{p(X)\leq\alpha\}}$ has size $\alpha(\varphi_\alpha) \leq \alpha$.

Proposition 3.1.13. *Continuous Null p-Values Are Uniform*
Let $H_0 : \theta \in \Theta_0$ and let $T : \mathcal{X} \to \mathbb{R}$ satisfy $\mathbb{P}_\theta(T \leq t) = F_0(t)$ for all $\theta \in \Theta_0$, where F_0 is continuous.
For the right-tail convention, set $p(X) := 1 - F_0(T(X))$.
Then $p(X) \sim \text{Uniform}(0,1)$ under every $\theta \in \Theta_0$, and $\varphi_\alpha(X) := \mathbf{1}_{\{p(X)\leq\alpha\}}$ has size α for $\alpha \in [0,1]$.

Proof. Fix $\theta \in \Theta_0$ and set $U := F_0(T(X))$.
By probability integral transform, Proposition 1.2.18, $U \sim \text{Uniform}(0,1)$ under $\mathbb{P}_\theta$,
hence $p(X) = 1 - U \sim \text{Uniform}(0,1)$. Thus $\mathbb{E}_\theta\varphi_\alpha = \mathbb{P}_\theta(p(X) \leq \alpha) = \alpha$ $\forall\theta \in \Theta_0$, so $\alpha(\varphi_\alpha) = \sup_{\theta\in\Theta_0} \mathbb{E}_\theta\varphi_\alpha = \alpha$.
Structural Commentary: A continuous null CDF converts the statistic to a canonical uniform scale.
Key Insight: Exact calibration is the probability integral transform applied under H_0. □

Remark 3.1.14. *p-Values Are Tail Probabilities*
A p-value is a null tail probability, e.g. $p(x) = \mathbb{P}_{H_0}\{T(X) \geq T(x)\}$ for a right-tail statistic T.
It is not $\mathbb{P}(H_0 \mid x)$; posterior probabilities require priors and an alternative likelihood, as in Proposition 3.1.3.

Remark 3.1.15. *Error Rates*
For one test φ, size is $\alpha(\varphi) = \sup_{\theta\in\Theta_0} \mathbb{E}_\theta\varphi$. For a family of tests, let R be the number of rejections and V the number of false rejections; common targets are FWER $= \mathbb{P}\{V \geq 1\}$ and FDR $= \mathbb{E}[\frac{V}{R\vee 1}]$, while directional or selective rates replace V by the corresponding error count.

Method 3.1.16. *Fishing Expeditions and Bonferroni Control*
(i) For N null hypotheses, let R_i be the event that null i is rejected.
(ii) By the union bound, $\mathbb{P}(\bigcup_{i=1}^N R_i) \leq \sum_{i=1}^N \mathbb{P}(R_i)$.
(iii) To control family-wise error at level α, test each null at level $\frac{\alpha}{N}$; then $\mathbb{P}$(at least one false rejection) $\leq \alpha$.
(iv) Without such correction, many tests can produce small p-values even when all nulls are true.

3.2 Confidence Sets, Pivots, and Asymptotic Intervals

Confidence procedures attach uncertainty to estimators by controlling coverage probability under repeated sampling.

Definition 3.2.1. *Confidence Sets and Pivots*

(i) A random set $C(X) \subseteq \Psi$ is a $1-\alpha$ *confidence set* for $\psi : \Theta \to \Psi$ if $\inf_{\theta \in \Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \geq 1-\alpha$. The left side is the *coverage guarantee*.

(ii) A *pivot* is a statistic $Q(X, \theta)$ whose distribution under $\mathbb{P}_\theta$ is free of θ.

(iii) If constants a_α, b_α satisfy $\mathbb{P}_\theta\{a_\alpha \leq Q(X,\theta) \leq b_\alpha\} = 1-\alpha$ for all $\theta \in \Theta$, then $C(X) := \{\theta \in \Theta : a_\alpha \leq Q(X,\theta) \leq b_\alpha\}$ is an exact $1-\alpha$ confidence set for θ.

Example 3.2.2. *Normal Mean with Known Variance*

Let $X_1, \ldots, X_n$ be independent $N(\mu, \sigma^2)$ variables with known σ.

The pivot $Z_\mu := \frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim N(0,1)$ gives $\mathbb{P}_\mu\{|Z_\mu| \leq z_{1-\frac{\alpha}{2}}\} = 1-\alpha$.

Solving for μ gives the exact interval $C_\alpha(X) = [\overline{X} - \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}, \overline{X} + \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}]$.

Test Connection: The two-sided level-α test of $H_0 : \mu = \mu_0$ rejects iff $\mu_0 \notin C_\alpha(X)$.

Example 3.2.3. *Measurement Error and Sample Size*

Let $X_1, \ldots, X_n$ be independent measurements with $\mathbb{E}X_i = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$.

For $c > 0$, the CLT gives $\mathbb{P}(|\overline{X}_n - \mu| < c) \approx \Phi(\frac{c\sqrt{n}}{\sigma}) - \Phi(\frac{-c\sqrt{n}}{\sigma})$.

For $(n, \sigma, c) = (16, 1, 0.5)$, this is $\Phi(2) - \Phi(-2) \approx 0.954$.

Example 3.2.4. *Normal Mean and Variance with Unknown Variance*

Let $X_1, \ldots, X_n$ be independent $N(\mu, \sigma^2)$ variables, and set $S^2 := (n-1)^{-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$.

The exact pivots are $T_\mu := \frac{\sqrt{n}(\overline{X}-\mu)}{S} \sim t_{n-1}$ and $W_{\sigma^2} := \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$. Writing $t_{\nu,p}$ and $\chi^2_{\nu,p}$ for p-quantiles, pivot inversion gives $C_\mu(X) = [\overline{X} - \frac{t_{n-1,1-\frac{\alpha}{2}} S}{\sqrt{n}}, \overline{X} + \frac{t_{n-1,1-\frac{\alpha}{2}} S}{\sqrt{n}}]$ and $C_{\sigma^2}(X) = [\frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}]$.

Both intervals have exact marginal coverage $1-\alpha$; $C_{\sigma^2}(X)$ is asymmetric because the χ^2 pivot is not symmetric.

Proposition 3.2.5. *Test–Confidence Duality*

For each $\psi_0 \in \Psi$, let $\varphi_{\psi_0} : \mathcal{X} \to \{0,1\}$ satisfy $\sup_{\theta : \psi(\theta) = \psi_0} \mathbb{E}_\theta \varphi_{\psi_0} \leq \alpha$.

Then $C(X) := \{\psi_0 \in \Psi : \varphi_{\psi_0}(X) = 0\}$ is a $1-\alpha$ confidence set for $\psi(\theta)$.

Proof. Fix $\theta \in \Theta$ and set $\psi_\theta := \psi(\theta)$.

By definition, $\{\psi(\theta) \notin C(X)\} = \{\varphi_{\psi_\theta}(X) = 1\}$, so $\mathbb{P}_\theta\{\psi(\theta) \notin C(X)\} = \mathbb{E}_\theta \varphi_{\psi_\theta} \leq \sup_{\eta : \psi(\eta) = \psi_\theta} \mathbb{E}_\eta \varphi_{\psi_\theta} \leq \alpha$.

Hence $\mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \geq 1-\alpha$ for every θ, and $\inf_{\theta \in \Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \geq 1-\alpha$.

Structural Commentary: Inverting all accepted point-null tests turns non-rejection regions into a confidence set.

Key Insight: Coverage failure equals rejection of the true null value. □

Proposition 3.2.6. *Confidence–Test Converse*

Let $C(X) \subseteq \Psi$ satisfy $\inf_{\theta \in \Theta} \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \geq 1-\alpha$.

For $\psi_0 \in \Psi$, let $\varphi_{\psi_0}(X) := \mathbf{1}_{\{\psi_0 \notin C(X)\}}$ for testing $H_0 : \psi(\theta) = \psi_0$. Then φ_{ψ_0} is level α: $\sup_{\theta : \psi(\theta) = \psi_0} \mathbb{E}_\theta \varphi_{\psi_0} \leq \alpha$.

Proof. Fix θ with $\psi(\theta) = \psi_0$. Then $\mathbb{E}_\theta \varphi_{\psi_0} = \mathbb{P}_\theta\{\psi_0 \notin C(X)\} = \mathbb{P}_\theta\{\psi(\theta) \notin C(X)\} = 1 - \mathbb{P}_\theta\{\psi(\theta) \in C(X)\} \leq \alpha$.

Taking $\sup_{\theta : \psi(\theta) = \psi_0}$ gives the level bound.

Structural Commentary: A confidence set tests a proposed value by checking membership.

Key Insight: Rejection of ψ_0 is exactly non-coverage when ψ_0 is true. □

Remark 3.2.7. *Acceptance Regions and Confidence Regions*

For level-α point-null tests with acceptance regions $A_{\psi_0} \subseteq \mathcal{X}$, the inversion $C(X) := \{\psi_0 \in \Psi : X \in A_{\psi_0}\}$ is the set of null values not rejected at X. Conversely, any confidence set $C(X)$ induces point-null tests $\varphi_{\psi_0}(X) = \mathbf{1}_{\{\psi_0 \notin C(X)\}}$.

Definition 3.2.8. *Wald, Score, and Likelihood Intervals*

For scalar regular θ with estimator $\widehat{\theta}$, standard error $\widehat{\mathrm{se}}(\widehat{\theta})$, score U_n, and likelihood ratio $\lambda_n(\theta_0)$:

(i) *Wald*: use $\widehat{\theta} \pm z_{1-\frac{\alpha}{2}} \widehat{\mathrm{se}}(\widehat{\theta})$.

(ii) *Score*: invert $\frac{|U_n(\theta_0)|}{\sqrt{I_n(\theta_0)}} \leq z_{1-\frac{\alpha}{2}}$ over θ_0.

(iii) *Likelihood*: invert $-2\log\lambda_n(\theta_0) \le \chi^2_{1,1-\alpha}$ over θ_0.

Key Insight: Wald, score, and likelihood intervals are first-order equivalent in regular models.

Example 3.2.9. *Poisson Mean Wald Interval*
Let $X_1, \dots, X_n$ be independent Poisson(λ) variables.
Then $\widehat{\lambda} = \overline{X}$ and $\frac{\sqrt{n}(\overline{X}-\lambda)}{\sqrt{\lambda}} \Rightarrow N(0,1)$, so the plug-in standard error is $\widehat{\mathrm{se}}(\widehat{\lambda}) = \sqrt{\frac{\overline{X}}{n}}$.
The Wald interval is $C_\alpha(X) = [\overline{X} - z_{1-\frac{\alpha}{2}}\sqrt{\frac{\overline{X}}{n}}, \overline{X} + z_{1-\frac{\alpha}{2}}\sqrt{\frac{\overline{X}}{n}}]$, with asymptotic coverage $1-\alpha$.

3.3 Goodness-of-Fit and Diagnostic Plots

Goodness-of-fit methods compare an observed empirical pattern with the pattern predicted by a probability model. Formal tests give calibrated evidence against a specified model, while diagnostic plots show where the model fits or fails.

Definition 3.3.1. *Multinomial Goodness-of-Fit Model*

(i) *Observed counts*: $O = (O_1, \dots, O_m)$ records cell frequencies from n trials, with $O_i \in \mathbb{N}_0$ and $\sum_{i=1}^{m} O_i = n$.
(ii) *Null model*: $\mathcal{M}_0 = \{p(\theta) = (p_1(\theta), \dots, p_m(\theta)) : \theta \in \Theta_0\} \subseteq \Delta_{m-1}$ is parametric.
(iii) *Saturated model*: $\Delta_{m-1} = \{p \in [0,1]^m : \sum_i p_i = 1\}$ allows every multinomial probability vector.
(iv) *Fitted counts*: if $\widehat{\theta}$ is the null MLE, then $E_i := np_i(\widehat{\theta})$ for $i = 1, \dots, m$.
(v) *Degrees of freedom*: if $\dim\Theta_0 = k$, then $\mathrm{df} = (m-1) - k = m - k - 1$.

Definition 3.3.2. *Pearson and Likelihood-Ratio Goodness-of-Fit Statistics*

(i) *Inputs*: $O = (O_1, \dots, O_m)$ are observed counts, $E = (E_1, \dots, E_m)$ are fitted null counts with $E_i > 0$, and the null model has k free parameters.
(ii) *Pearson statistic*: $X^2 := \sum_{i=1}^{m} \frac{(O_i - E_i)^2}{E_i}$.
(iii) *Likelihood-ratio statistic*: $G^2 := 2\sum_{i=1}^{m} O_i \log(\frac{O_i}{E_i})$, with $0\log(\frac{0}{E_i}) := 0$.
(iv) *Null calibration*: under the regular null model, $X^2 \Rightarrow \chi^2_{m-k-1}$ and $G^2 = -2\log\lambda(X) \Rightarrow \chi^2_{m-k-1}$. Moreover $G^2 - X^2 = o_P(1)$ under H_0.

Example 3.3.3. *Hardy–Weinberg Fit*
For genotype cells (M, MN, N), the Hardy–Weinberg model is $p(\theta) = ((1-\theta)^2, 2\theta(1-\theta), \theta^2)$.
With $n = 1029$ and $\widehat{\theta} = 0.4247$, the observed and fitted counts are

	M	MN	N
O_i	342	500	187
E_i	340.6	502.8	185.6

Since $m = 3$ and $k = 1$, $\mathrm{df} = m - k - 1 = 1$. Pearson statistic is $X^2 = 0.0319$, so $p = \mathbb{P}\{\chi^2_1 \ge 0.0319\}$ is large. The data give no evidence against Hardy–Weinberg equilibrium.

Method 3.3.4. *Reading Cell Contributions*

(i) Compute signed residuals $r_i := O_i - E_i$ and Pearson contributions $c_i := \frac{(O_i - E_i)^2}{E_i}$.
(ii) Large c_i identify cells driving the rejection; the sign of r_i indicates overfit or underfit in that cell.
(iii) If some E_i are small, group adjacent or tail cells before testing so the chi-square approximation is reliable.

Example 3.3.5. *Poisson Clumps*
For grouped bacterial clump frequencies from $n = 400$ grid squares, fit $H_0 : X \sim \mathrm{Poisson}(\lambda)$ with $\widehat{\lambda} = 2.44$ and grouped tails so E_i are adequate. The Pearson statistic is $X^2 = 75.4$ with $\mathrm{df} = 6$, so $p = \mathbb{P}\{\chi^2_6 \ge 75.4\} < 0.005$; the largest $c_i = \frac{(O_i - E_i)^2}{E_i}$ are in the first and last cells, indicating excess small and large counts.

Definition 3.3.6. *Poisson Dispersion Test*

The data is $x_1, \dots, x_n$ are independent counts and $\overline{x} := n^{-1}\sum_{i=1}^{n} x_i$.

(i) *Models*: $H_0 : x_i \sim \mathrm{Poisson}(\lambda)$ with common λ; the saturated alternative uses separate rates λ_i.
(ii) *Likelihood-ratio statistic*: $G^2_D := 2\sum_{i=1}^{n} x_i \log(\frac{x_i}{\overline{x}})$, with $0\log(\frac{0}{\overline{x}}) := 0$.
(iii) *Pearson approximation*: $T := \overline{x}^{-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$.

(iv) *Calibration*: under H_0, G_D^2 and T are calibrated by χ^2_{n-1}
Large values indicate overdispersion relative to $\mathrm{Var}(X) = \mathbb{E}X$.

Example 3.3.7. *Dispersion in Bacterial Clumps*
For bacterial clump counts with $n = 400$, $\overline{x} = 2.44$ and $s^2 = 4.59$, so $\frac{s^2}{\overline{x}} = 1.88 > 1$.
Dispersion statistic is $T \approx 752.7$ with $\mathrm{df} = n - 1 = 399$, so $p = \mathbb{P}\{\chi^2_{399} \geq 752.7\} \approx 0$.
The data show strong overdispersion relative to a Poisson model.

Definition 3.3.8. *Hanging Rootograms and Chi-Grams*
Let n_j be observed bin counts and $\widehat{n}_j$ fitted counts in ordered bins.

(i) A *hanging histogram* plots raw residuals $r_j := n_j - \widehat{n}_j$, whose variance depends on $\widehat{n}_j$.
(ii) A *hanging rootogram* plots $q_j := \sqrt{n_j} - \sqrt{\widehat{n}_j}$, using $\mathrm{Var}(\sqrt{n_j}) \approx \frac{1}{4}$ for count data.
(iii) A *hanging chi-gram* plots standardized residuals $z_j := \frac{n_j - \widehat{n}_j}{\sqrt{\widehat{n}_j}}$.

Runs of same-sign residuals, tail spikes, or systematic curvature indicate model departure.

Method 3.3.9. *Probability Plots*
Let $X_{(1)} \leq \cdots \leq X_{(n)}$ be order statistics and let F be the candidate continuous strictly increasing CDF.

(i) A *P-P plot* compares $F(X_{(k)})$ with $u_k := \frac{k}{n+1}$ for $k = 1, \ldots, n$.
(ii) A *Q-Q plot* compares $X_{(k)}$ with $F^{-1}(u_k)$ for $k = 1, \ldots, n$.

The model is plausible when the plotted points are approximately linear.
For $F(x) = G(\frac{x-\mu}{\sigma})$, Q-Q slope and intercept estimate σ and μ.

Method 3.3.10. *Probability Plots for Grouped Data*
Let bins boundaries $x_1 < \cdots < x_{m+1}$, counts $n_1, \ldots, n_m$, total $n := \sum_{j=1}^{m} n_j$, cumulative counts $N_j := \sum_{i=1}^{j} n_i$.
Let G be the standardised CDF in the proposed location-scale family $F_{\mu,\sigma}(x) = G(\frac{x-\mu}{\sigma})$.

(i) Plot empirical boundary quantiles x_{j+1} against theoretical quantiles $G^{-1}(\frac{N_j}{n+1})$ for $j = 1, \ldots, m$.
(ii) Use this grouped Q-Q plot when individual order statistics are unavailable.

Definition 3.3.11. *Normality Diagnostics*
Let $X_{(1)} \leq \cdots \leq X_{(n)}$, $\overline{X} := n^{-1} \sum_i X_i$, and $s^2 := n^{-1} \sum_i (X_i - \overline{X})^2$.

(i) *Normal Q-Q plot*: plot $X_{(k)}$ against $\Phi^{-1}(\frac{k}{n+1})$; normality predicts approximate linearity.
(ii) *Skewness*: $b_1 := n^{-1} \sum_{i=1}^{n} \frac{(X_i - \overline{X})^3}{s^3}$ measures asymmetry; large $|b_1|$ indicates nonnormality.
(iii) *Kurtosis*: $b_2 := n^{-1} \sum_{i=1}^{n} \frac{(X_i - \overline{X})^4}{s^4}$ measures tail weight and peakedness; normality gives $b_2 \approx 3$.
(iv) *Plot correlation*: Q-Q point correlation; small values indicate curvature.

Remark 3.3.12. *Grouped Normality Tests*
For binned normal-fit testing, use $X^2 = \sum_j \frac{(O_j - E_j)^2}{E_j}$ with $E_j = n\{F_{\widehat{\mu},\widehat{\sigma}}(b_{j+1}) - F_{\widehat{\mu},\widehat{\sigma}}(b_j)\}$.
The usual χ^2_{m-k-1} calibration requires $(\widehat{\mu}, \widehat{\sigma})$ from the grouped counts; ungrouped estimates change the limit.

3.4 Normal Sampling Pivots

For normal samples, the sample mean and sample variance separate exactly. This is the finite-sample source of the classical z, χ^2, and Student t pivots used in normal-theory confidence intervals and tests.

Definition 3.4.1. *Normal Sample Statistics*

(i) *Sample*: $X_1, \ldots, X_n$ are independent with $X_i \sim N(\mu, \sigma^2)$, $n \geq 2$, and $\sigma^2 > 0$.
(ii) *Sample mean*: $\overline{X} := n^{-1} \sum_{i=1}^{n} X_i$ estimates the location μ.
(iii) *Sample variance*: $S^2 := (n-1)^{-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$ estimates the scale σ^2.

Proposition 3.4.2. *Normal Sample Mean*
Let $X_1, \ldots, X_n$ be independent with $X_i \sim N(\mu, \sigma^2)$, and set $\overline{X} := n^{-1} \sum_{i=1}^{n} X_i$. Then $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.

Proof. Since independent normal sums are normal, $\overline{X}$ is normal.
Also $\mathbb{E}\overline{X} = n^{-1}\sum_i \mathbb{E}X_i = \mu$ and $\mathrm{Var}(\overline{X}) = n^{-2}\sum_i \mathrm{Var}(X_i) = \frac{\sigma^2}{n}$. Thus $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$.
Structural Commentary: Averaging preserves normality and divides variance by n.
Key Insight: The sample mean is an exact normal pivot center, not an asymptotic approximation. □

Theorem 3.4.3. *Mean-Residual Independence*
Let $X_1, \dots, X_n$ be independent with $X_i \sim N(\mu, \sigma^2)$, set $\overline{X} := n^{-1}\sum_i X_i$, $R := (X_1 - \overline{X}, \dots, X_n - \overline{X})$, and $S^2 := (n-1)^{-1}\sum_i R_i^2$. Then $\overline{X} \perp\!\!\!\perp R$, hence $\overline{X} \perp\!\!\!\perp S^2$.

Proof. Let $Z_i := \frac{X_i - \mu}{\sigma}$, so $Z \sim N_n(0, I_n)$. Choose orthogonal Q with first row $n^{-\frac{1}{2}}\mathbf{1}^\top$, set $Y := QZ$.
Then $Y \sim N_n(0, I_n)$, $Y_1 \perp\!\!\!\perp (Y_2, \dots, Y_n)$. Since $\overline{X} = \mu + \frac{\sigma Y_1}{\sqrt{n}}$ and $R = \sigma(I - n^{-1}\mathbf{1}\mathbf{1}^\top)Z = \sigma Q^\top(0, Y_2, \dots, Y_n)^\top$, $\overline{X}$ is function of Y_1, R is function of $(Y_2, \dots, Y_n)$. Therefore $\overline{X} \perp\!\!\!\perp R$, by $S^2 = (n-1)^{-1}\sum_i R_i^2$, $\overline{X} \perp\!\!\!\perp S^2$.
Structural Commentary: Mean and residuals are orthogonal projections.
Key Insight: For Gaussian vectors, orthogonal components are independent. □

Theorem 3.4.4. *Sample Variance Chi-Square Law*
Given independent $X_1, \dots, X_n$; $X_i \sim N(\mu, \sigma^2)$;$S^2 := (n-1)^{-1}\sum_{i=1}^n (X_i - \overline{X})^2$. Then $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$.

Proof. Let $Z_i := \frac{X_i - \mu}{\sigma}$, so $Z = (Z_1, \dots, Z_n) \sim N_n(0, I_n)$.
Choose orthogonal matrix Q with first row $n^{-\frac{1}{2}}(1, \dots, 1)$, set $Y := QZ$.
Orthogonal invariance gives $Y \sim N_n(0, I_n)$ with independent coordinates.
Then $Y_1 = \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma}$ and $\sum_{j=2}^n Y_j^2 = \sum_{i=1}^n (Z_i - \overline{Z})^2 = \frac{(n-1)S^2}{\sigma^2}$.
Thus $\frac{(n-1)S^2}{\sigma^2}$ is the sum of squares of $n-1$ independent $N(0,1)$ variables, hence χ^2_{n-1}.
Structural Commentary: Cantering leaves $n-1$ residual directions.
Key Insight: The sample variance is the squared Gaussian residual length scaled by σ^2. □

Corollary 3.4.5. *Student Pivot for a Normal Mean*
Let $X_1, \dots, X_n$ be independent; $X_i \sim N(\mu, \sigma^2)$, $\overline{X} := n^{-1}\sum_i X_i$, $S^2 := (n-1)^{-1}\sum_i (X_i - \overline{X})^2$; $\frac{\overline{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$.

Proof. Let $Z := \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma}$ and $V := \frac{(n-1)S^2}{\sigma^2}$. By normal mean law, $Z \sim N(0,1)$. By sample-variance chi-square law, $V \sim \chi^2_{n-1}$. By mean-residual independence, $Z \perp\!\!\!\perp V$. Therefore $\frac{\overline{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{Z}{\sqrt{\frac{V}{n-1}}} \sim t_{n-1}$.
Structural Commentary: Replacing σ by S yields a Student pivot.
Key Insight: Student's t is a normal variable divided by an independent chi-square scale estimate. □

Method 3.4.6. *Exact Normal Mean Intervals*
Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$ variables, with $\overline{X} := n^{-1}\sum_i X_i$ and $S^2 := (n-1)^{-1}\sum_i (X_i - \overline{X})^2$.
Below cases have exact coverage $1 - \alpha$.

(i) *Known variance*: if σ is known, $C_\mu(X) = [\overline{X} - \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}, \overline{X} + \frac{z_{1-\frac{\alpha}{2}}\sigma}{\sqrt{n}}]$
(ii) *Unknown variance*: if σ is unknown, $C_\mu(X) = [\overline{X} - \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}, \overline{X} + \frac{t_{n-1,1-\frac{\alpha}{2}}S}{\sqrt{n}}]$
(iii) *Variance*: $C_{\sigma^2}(X) = [\frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}]$

3.5 Two-Sample Normal Inference

Two-sample inference compares the location of two response distributions. In a controlled experiment, one sample may receive a treatment and the other a control; in an observational comparison, the same formulas may be computed but their causal interpretation depends on the design.

Definition 3.5.1. *Two Independent Samples and the Location Contrast*
$X_1, \dots, X_n \sim F$ are independent, $Y_1, \dots, Y_m \sim G$ are independent, and the two samples are independent.

(i) *Means*: $\mu_X := \mathbb{E}X_i$ and $\mu_Y := \mathbb{E}Y_j$ summarize the two average levels, assuming both exist.
(ii) *Location contrast*: $\Delta := \mu_X - \mu_Y$ measures the shift from group Y to group X.
(iii) *Estimator*: $\widehat{\Delta} := \overline{X} - \overline{Y}$ estimates Δ, where $\overline{X} := n^{-1}\sum_i X_i$ and $\overline{Y} := m^{-1}\sum_j Y_j$.

Proposition 3.5.2. *Known Common Variance Pivot*
Let $X_1,\dots,X_n$ and $Y_1,\dots,Y_m$ be independent samples, $X_i \sim N(\mu_X,\sigma^2)$, $Y_j \sim N(\mu_Y,\sigma^2)$, known $\sigma>0$. Set $\Delta := \mu_X-\mu_Y$, $\widehat{\Delta} := \overline{X}-\overline{Y}$, and $\mathrm{se}_0 := \sigma\sqrt{n^{-1}+m^{-1}}$.

(i) $\widehat{\Delta} \sim N(\Delta,\mathrm{se}_0^2)$ *and* $Z := \frac{\widehat{\Delta}-\Delta}{\mathrm{se}_0} \sim N(0,1)$.

(ii) $C_\alpha(X,Y) := [\widehat{\Delta} - z_{1-\frac{\alpha}{2}}\,\mathrm{se}_0, \widehat{\Delta} + z_{1-\frac{\alpha}{2}}\,\mathrm{se}_0]$ *satisfies* $\mathbb{P}_\Delta\{\Delta \in C_\alpha(X,Y)\} = 1-\alpha$.

Proof.

(i) Since $\overline{X} \sim N(\mu_X,\frac{\sigma^2}{n})$, $\overline{Y} \sim N(\mu_Y,\frac{\sigma^2}{m})$, and $\overline{X} \perp\!\!\!\perp \overline{Y}$, $\widehat{\Delta} = \overline{X}-\overline{Y} \sim N(\Delta,\frac{\sigma^2}{n}+\frac{\sigma^2}{m}) = N(\Delta,\mathrm{se}_0^2)$. Standardization gives $Z \sim N(0,1)$.

(ii) By (i), $\mathbb{P}_\Delta\{|Z| \le z_{1-\frac{\alpha}{2}}\} = 1-\alpha$, equivalently $\mathbb{P}_\Delta\{\Delta \in C_\alpha(X,Y)\} = 1-\alpha$.

Structural Commentary: The contrast is a linear combination of independent sample means.
Key Insight: Known common variance gives an exact normal pivot for Δ. □

Definition 3.5.3. *Pooled Sample Variance*
Let $X_1,\dots,X_n \sim N(\mu_X,\sigma^2)$ and $Y_1,\dots,Y_m \sim N(\mu_Y,\sigma^2)$ be independent samples with common unknown σ^2.

(i) *Sample variances*: $S_X^2 := (n-1)^{-1}\sum_{i=1}^{n}(X_i-\overline{X})^2$ and $S_Y^2 := (m-1)^{-1}\sum_{j=1}^{m}(Y_j-\overline{Y})^2$.

(ii) *Pooled variance*: $S_p^2 := \frac{(n-1)S_X^2+(m-1)S_Y^2}{n+m-2}$.

(iii) *Degrees-of-freedom weighting*: S_p^2 weights S_X^2 and S_Y^2 by $n-1$ and $m-1$, respectively.

Theorem 3.5.4. *Pooled Two-Sample Student Pivot*
Let $X_i \sim N(\mu_X,\sigma^2)$, $i=1,\dots,n$, and $Y_j \sim N(\mu_Y,\sigma^2)$, $j=1,\dots,m$, be mutually independent, with unknown common $\sigma^2>0$. Set $\Delta := \mu_X-\mu_Y$ and $\nu := n+m-2$. Then $T := \frac{(\overline{X}-\overline{Y})-\Delta}{S_p\sqrt{n^{-1}+m^{-1}}} \sim t_\nu$.

Proof. Let $Z := \frac{(\overline{X}-\overline{Y})-\Delta}{\sigma\sqrt{n^{-1}+m^{-1}}}$ and $W := \frac{\nu S_p^2}{\sigma^2} = \frac{(n-1)S_X^2+(m-1)S_Y^2}{\sigma^2}$.
Since $\overline{X}-\overline{Y} \sim N(\Delta,\sigma^2(n^{-1}+m^{-1}))$, $Z \sim N(0,1)$. Let $U_X := \frac{(n-1)S_X^2}{\sigma^2}$ and $U_Y := \frac{(m-1)S_Y^2}{\sigma^2}$.
By normal mean-residual independence and sample independence, Z, U_X, U_Y mutually independent, with $U_X \sim \chi^2_{n-1}$, $U_Y \sim \chi^2_{m-1}$. Hence $W = U_X+U_Y \sim \chi^2_\nu$ and $W \perp\!\!\!\perp Z$. Since $S_p^2 = \frac{\sigma^2 W}{\nu}$, $T = \frac{Z}{\sqrt{\frac{W}{\nu}}} \sim t_\nu$.
Structural Commentary: The pooled variance combines the two residual sums of squares.
Key Insight: A common variance permits replacing σ by an independent pooled estimate. □

Method 3.5.5. *Pooled Two-Sample t Procedures*
Under the equal-variance normal model, set $\Delta := \mu_X-\mu_Y$, $\nu := n+m-2$, and $\mathrm{se}_p := S_p\sqrt{n^{-1}+m^{-1}}$.

(i) *Interval*: $C_\alpha := [(\overline{X}-\overline{Y}) - t_{\nu,1-\frac{\alpha}{2}}\,\mathrm{se}_p, (\overline{X}-\overline{Y}) + t_{\nu,1-\frac{\alpha}{2}}\,\mathrm{se}_p]$ is exact $1-\alpha$ for Δ.

(ii) *Statistic*: for $H_0:\Delta=0$, use $t_0 := \frac{\overline{X}-\overline{Y}}{\mathrm{se}_p}$, with $t_0 \sim t_\nu$ under H_0.

(iii) *Rejection*: reject H_0 for $|t_0| > t_{\nu,1-\frac{\alpha}{2}}$ under $H_1:\Delta\neq 0$
reject H_0 for $t_0 > t_{\nu,1-\alpha}$ under $H_1:\Delta>0$
reject H_0 for $t_0 < -t_{\nu,1-\alpha}$ under $H_1:\Delta<0$.

(iv) *Duality*: the two-sided test rejects iff $0 \notin C_\alpha$.

Example 3.5.6. *Latent Heat of Fusion: Two Methods*
For methods A and B, let $n_A=13$, $n_B=8$, $\overline{x}_A=80.02$, $\overline{x}_B=79.98$, $s_A=0.024$, $s_B=0.031$, and $S_p=0.027$. Then $\widehat{\Delta} = \overline{x}_A-\overline{x}_B = 0.04$, $\widehat{\mathrm{se}} = S_p\sqrt{13^{-1}+8^{-1}} \approx 0.012$, pooled 95% interval for $\Delta=\mu_A-\mu_B$ is $\approx (0.015, 0.065)$. For $H_0:\Delta=0$, $t_0 = \frac{\widehat{\Delta}}{\widehat{\mathrm{se}}} \approx 3.33$ with $\nu=19$, strong evidence that method means differ.

Proposition 3.5.7. *Likelihood-Ratio View of the Pooled Test*
Let $X_i \sim N(\mu_X,\sigma^2)$, $i=1,\dots,n$, and $Y_j \sim N(\mu_Y,\sigma^2)$, $j=1,\dots,m$, be mutually independent, with unknown common $\sigma^2>0$. Set $\Delta := \mu_X-\mu_Y$, $\nu := n+m-2$, and $T := \frac{\overline{X}-\overline{Y}}{S_p\sqrt{n^{-1}+m^{-1}}}$.
For $H_0:\Delta=0$ against $H_1:\Delta\neq 0$, $\lambda(X,Y) = (1+\frac{T^2}{\nu})^{-\frac{n+m}{2}}$.
Hence the GLR rejects for large $|T|$; at level α, reject iff $|T| > t_{\nu,1-\frac{\alpha}{2}}$.

Proof. Set $N := n+m$, $\nu := N-2$, $d := \overline{X}-\overline{Y}$, $R_1 := \sum_i (X_i-\overline{X})^2 + \sum_j (Y_j-\overline{Y})^2$, $\widehat{\mu}_0 := \frac{n\overline{X}+m\overline{Y}}{N}$, and $R_0 := \sum_i (X_i-\widehat{\mu}_0)^2 + \sum_j (Y_j-\widehat{\mu}_0)^2$. Under H_1, $(\widehat{\mu}_X,\widehat{\mu}_Y,\widehat{\sigma}_1^2) = (\overline{X},\overline{Y},\frac{R_1}{N})$; under H_0, $(\widehat{\mu},\widehat{\sigma}_0^2) = (\widehat{\mu}_0,\frac{R_0}{N})$.
The maximised Gaussian likelihood is proportional to $\widehat{\sigma}^{-N}$, hence $\lambda = (\frac{R_1}{R_0})^{\frac{N}{2}}$.
ANOVA decomposition gives $R_0 = R_1 + \{\frac{nm}{N}\}d^2$, while $T^2 = \frac{d^2}{(\frac{R_1}{\nu})(n^{-1}+m^{-1})} = \frac{\nu\{\frac{nm}{N}\}d^2}{R_1}$.

Therefore $\frac{R_0}{R_1} = 1 + \frac{T^2}{\nu}$ and $\lambda = (1 + \frac{T^2}{\nu})^{-\frac{N}{2}}$, decreasing in T^2; small λ iff large $|T|$.
Structural Commentary: The null fit adds only the between-sample mean discrepancy to the unrestricted fit.
Key Insight: The GLR depends on the data through the pooled t statistic. □

Method 3.5.8. *Unequal-Variance Two-Sample Procedure*
For independent samples without the equal-variance assumption, set $\widehat{v} := \frac{S_X^2}{n} + \frac{S_Y^2}{m}$.

(i) *Standard error*: estimate $\mathrm{Var}(\overline{X} - \overline{Y})$ by $\widehat{v}$.
(ii) *Welch statistic*: for $H_0 : \Delta = \Delta_0$, use $T_W := \frac{(\overline{X}-\overline{Y})-\Delta_0}{\sqrt{\widehat{v}}}$.
(iii) *Degrees of freedom*: calibrate T_W by t_ν, where $\nu := \frac{\widehat{v}^2}{\frac{(S_X^2/n)^2}{n-1}+\frac{(S_Y^2/m)^2}{m-1}}$.
(iv) *Use*: the procedure avoids pooling variances, at the cost of approximate calibration.

Corollary 3.5.9. *Variance Ratio Test Under Normality*
Let $X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$ be independent samples, with $n, m \geq 2$. Set $\rho := \frac{\sigma_X^2}{\sigma_Y^2}$ and $R := \frac{S_X^2}{S_Y^2}$. Then $\frac{R}{\rho} \sim F_{n-1,m-1}$.

Proof. Let $U := \frac{(n-1)S_X^2}{\sigma_X^2}$ and $V := \frac{(m-1)S_Y^2}{\sigma_Y^2}$.
Normal residual chi-square laws give $U \sim \chi^2_{n-1}$ and $V \sim \chi^2_{m-1}$, and sample independence gives $U \perp\!\!\!\perp V$.
Thus $\frac{\frac{U}{n-1}}{\frac{V}{m-1}} = \frac{R}{\rho} \sim F_{n-1,m-1}$. If $H_0 : \rho = 1$, then $R \sim F_{n-1,m-1}$, yielding exact normal-theory tests and confidence intervals for ρ; the calibration is normality-sensitive.
Structural Commentary: The comparison standardises two independent residual variance estimates.
Key Insight: Normality turns each sample variance into an independent chi-square pivot. □

Method 3.5.10. *Approximate Power for Balanced Two-Sample z Tests*
For planning, assume balanced samples of size n, known common σ, and target effect $\Delta := \mu_X - \mu_Y > 0$.

(i) *Statistic*: use $Z := \frac{\overline{X}-\overline{Y}}{\sigma\sqrt{\frac{2}{n}}}$ and reject two-sided at level α when $|Z| > z_{1-\frac{\alpha}{2}}$.
(ii) *Shift*: under $\Delta > 0$, $Z \sim N(\delta, 1)$ with $\delta := (\frac{\Delta}{\sigma})\sqrt{\frac{n}{2}}$.
(iii) *Power*: $\pi(\Delta) := 1 - \Phi(z_{1-\frac{\alpha}{2}} - \delta) + \Phi(-z_{1-\frac{\alpha}{2}} - \delta)$.
(iv) *Sample size*: if the lower-tail term is negligible, target power $1 - \beta$ gives $n \approx \frac{2\sigma^2(z_{1-\frac{\alpha}{2}}+z_{1-\beta})^2}{\Delta^2}$.

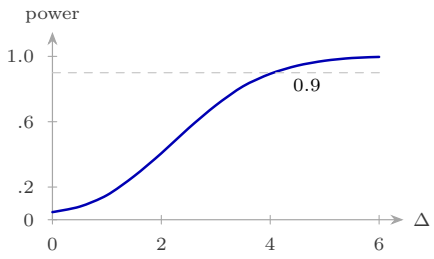


Figure 216: Power rises as the detectable mean difference Δ increases.

Remark 3.5.11. *Transformations and Normal Diagnostics*
Normal-theory two-sample procedures are often CLT-robust for large samples, but small skewed or outlier-prone samples require diagnostics. Transformations may stabilize spread and improve symmetry, but they change the estimand, e.g. a log-scale mean difference corresponds to a ratio-scale comparison.

3.6 Paired-Sample Inference

Paired comparisons arise when observations are naturally matched: before and after measurements on the same unit, littermates, matched subjects, or two measurement methods applied to the same specimen. Pairing changes the stochastic structure because the two samples are no longer independent.

Definition 3.6.1. *Paired Differences*

(i) *Pairs*: (X_i, Y_i), $i = 1, \dots, n$, are independent across i; dependence within each pair is allowed.
(ii) *Differences*: $D_i := X_i - Y_i$ and $\overline{D} := n^{-1} \sum_{i=1}^{n} D_i$.
(iii) *Mean*: if $\mathbb{E}X_i = \mu_X$ and $\mathbb{E}Y_i = \mu_Y$, then $\mathbb{E}D_i = \mu_X - \mu_Y$.

(iv) *Variance*: if $\mathrm{Var}(X_i) = \sigma_X^2$, $\mathrm{Var}(Y_i) = \sigma_Y^2$, and $\mathrm{Cov}(X_i, Y_i) = \sigma_{XY}$, then $\mathrm{Var}(D_i) = \sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}$.

Proposition 3.6.2. *Variance Effect of Pairing*
Let (X_i, Y_i), $i = 1, \ldots, n$, *be independent pairs with* $\mathrm{Var}(X_i) = \sigma_X^2$, $\mathrm{Var}(Y_i) = \sigma_Y^2$, *and* $\mathrm{Corr}(X_i, Y_i) = \rho$. *Set* $D_i := X_i - Y_i$ *and* $\overline{D} := n^{-1} \sum_i D_i$. *Then* $\mathrm{Var}(\overline{D}) = n^{-1}(\sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y)$.

Proof. Since $\mathrm{Cov}(X_i, Y_i) = \rho\sigma_X\sigma_Y$, $\mathrm{Var}(D_i) = \mathrm{Var}(X_i - Y_i) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$.
Independence across pairs gives $\mathrm{Var}(\overline{D}) = n^{-2} \sum_i \mathrm{Var}(D_i) = \frac{\mathrm{Var}(D_i)}{n}$.

If $\sigma_X = \sigma_Y = \sigma$, substitution gives $\mathrm{Var}(\overline{D}) = \frac{2\sigma^2(1-\rho)}{n}$; independent unpaired samples of size n per group have $\mathrm{Var}(\overline{X} - \overline{Y}) = \frac{2\sigma^2}{n}$, so the paired-to-unpaired variance ratio is $1 - \rho$.
Structural Commentary: Pairing replaces two marginal measurements by one within-pair contrast.
Key Insight: Positive within-pair correlation cancels shared variation and lowers variance. □

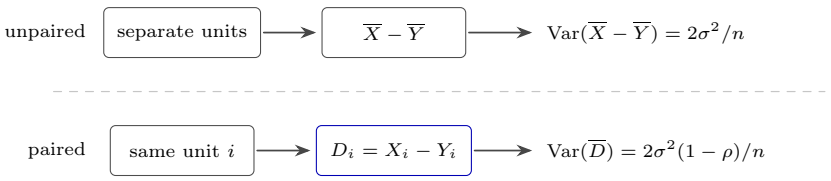


Figure 217: Pairing cancels shared variation; positive within-pair correlation lowers variance.

Method 3.6.3. *Paired t Procedure*
Let $D_1, \ldots, D_n \sim N(\mu_D, \sigma_D^2)$, where $\mu_D := \mu_X - \mu_Y$.

(i) *Variance*: $S_D^2 := (n-1)^{-1} \sum_{i=1}^{n} (D_i - \overline{D})^2$.
(ii) *Pivot*: $T := \frac{\overline{D} - \mu_D}{S_D/\sqrt{n}} \sim t_{n-1}$.
(iii) *Interval*: $C_\alpha := [\overline{D} - \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}, \overline{D} + \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}]$ is exact $1 - \alpha$ for μ_D.
(iv) *Test*: for $H_0 : \mu_D = 0$, reject two-sided at level α iff $|\overline{D}| > \frac{t_{n-1,1-\frac{\alpha}{2}} S_D}{\sqrt{n}}$.

For large n, use same rule by CLT; for small nonnormal differences, prefer rank or sign methods.

Example 3.6.4. *Platelet Aggregation Before and After Smoking*
For $n = 11$ before-after platelet measurements, let $D_i = \text{after}_i - \text{before}_i$ with $\overline{D} = 10.27$ and $\frac{S_D}{\sqrt{11}} = 2.40$.
Then $t_0 = \frac{\overline{D}}{\frac{S_D}{\sqrt{11}}} = 4.28$ with $\nu = 10$, so $p < 0.01$ two-sided; a 90% confidence interval is approximately $(5.9, 14.6)$, indicating a positive mean increase after smoking.

Example 3.6.5. *Mercury Measurement Methods*
For $n = 25$ paired mercury measurements, let $D_i = M_{1i} - M_{2i}$ with $\overline{D} = 0.04$ and $S_D = 0.116$.
Then $t_0 = \frac{\overline{D}}{\frac{S_D}{\sqrt{25}}} = 1.724$ with $\nu = 24$, giving $p \approx 0.094$ two-sided.
Plot of D_i against measurement level should be checked: level-dependent discrepancies not captured by $\overline{D}$ alone.

Remark 3.6.6. *Pairing Is Not Extra Independent Sampling*
If unit i contributes repeated measurements $Z_{i1}, \ldots, Z_{ik}$, independence is across units, not within unit.
Treating all nk observations as iid ignores within-unit dependence and overstates the effective sample size.

4 Decision Theory

4.1 Loss, Risk, Bayes, and Minimax Rules

Decision theory compares statistical procedures by their consequences, not only by their formulas. The same data model can lead to different optimal rules under different loss functions.

Definition 4.1.1. *Decision Problem and Optimality Criteria*

(i) *Problem*: a statistical decision problem is $(\{P_\theta : \theta \in \Theta\}, \mathcal{A}, L)$ with $L : \Theta \times \mathcal{A} \to [0, \infty]$.
(ii) *Rule*: a decision rule is a measurable $\delta : \mathcal{X} \to \mathcal{A}$.
(iii) *Risk*: $R(\theta, \delta) := \mathbb{E}_\theta[L(\theta, \delta(X))]$.
(iv) *Bayes risk*: for prior Π on Θ, $r(\Pi, \delta) := \int R(\theta, \delta)\, d\Pi(\theta)$.

(v) *Bayes rule*: δ_Π is Bayes if $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.
(vi) *Maximum risk*: $M(\delta) := \sup_{\theta \in \Theta} R(\theta, \delta)$; $\delta^\star$ is minimax if $M(\delta^\star) = \inf_\delta M(\delta)$.
(vii) *Admissibility*: δ inadmissible if $\exists \delta'$ with $R(\theta, \delta') \le R(\theta, \delta)\ \forall \theta$ and $<$ for some θ; otherwise δ admissible.

Proposition 4.1.2. *Posterior Risk Characterization*
Let Π be a prior, let $\Pi(\cdot \mid x)$ be a posterior, and set $\rho_x(a) := \int L(\theta, a)\, d\Pi(\theta \mid x)$.
If a measurable selector $\delta_\Pi(x) \in \arg\min_{a \in \mathcal{A}} \rho_x(a)$ exists, then δ_Π is Bayes: $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.

Proof. By iterated expectation, $r(\Pi, \delta) = \mathbb{E}_\Pi[\rho_X(\delta(X))]$, where X has the prior predictive law.
Since $\rho_x(\delta_\Pi(x)) \le \rho_x(\delta(x))\ \forall$ rule δ and almost every x, $r(\Pi, \delta_\Pi) \le r(\Pi, \delta)\ \forall \delta$. Thus $r(\Pi, \delta_\Pi) = \inf_\delta r(\Pi, \delta)$.
Structural Commentary: Bayes risk is the prior-predictive average of posterior risks.
Key Insight: Global Bayes optimization reduces to pointwise posterior-risk minimization. □

Example 4.1.3. *Bayes Rules Under Common Losses*
For a real parameter and posterior distribution $\Pi(\cdot \mid x)$:

(i) Squared-error loss is minimized by the posterior mean.
(ii) Absolute-error loss is minimized by a posterior median.
(iii) Zero-one loss on a finite action set is minimized by a posterior mode.

Remark 4.1.4. *Least Favorable Priors*
If a prior $\Pi^\star$ has Bayes risk equal to the minimax value, then any Bayes rule for $\Pi^\star$ with constant risk on the support of $\Pi^\star$ is minimax. Such priors are called least favorable.

4.2 Shrinkage, Empirical Bayes, and Regularization

Risk can often be reduced by intentionally pulling noisy estimates toward a structured target. The price is bias; the benefit is variance reduction.

Definition 4.2.1. *Shrinkage Estimator*

(i) *Input*: let $\widehat{\theta} = \widehat{\theta}(X)$ be a data-based estimator.
(ii) *Target*: choose θ_{tar} as a reference value or lower-complexity model.
(iii) *Weight*: let $a(X) \in [0, 1]$ control the amount of shrinkage.
(iv) *Estimator*: $\delta(X) := a(X)\widehat{\theta} + \{1 - a(X)\}\theta_{\text{tar}}$.

Example 4.2.2. *Ridge Regression as Shrinkage*
For design matrix X, response y, and $\lambda \ge 0$, ridge regression sets $\widehat{\beta}_\lambda \in \arg\min_\beta \{\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2\}$.

If $X^\top X$ is invertible, $\widehat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$, shrinking $\widehat{\beta}_{\text{LS}}$ toward 0 most in weak eigendirections.

Theorem 4.2.3. *James–Stein Phenomenon*
Let $X \sim \mathcal{N}_d(\theta, \sigma^2 I_d)$ with $d \ge 3$, known σ^2, and squared-error risk $R(\theta, \delta) := \mathbb{E}_\theta \|\delta(X) - \theta\|^2$.
Define $\delta_{\text{JS}}(X) := (1 - \frac{(d-2)\sigma^2}{\|X\|^2})X$ for $X \ne 0$, with arbitrary value at $X = 0$.
Then $R(\theta, \delta_{\text{JS}}) < R(\theta, X) = d\sigma^2$ for every $\theta \in \mathbb{R}^d$.

Proof. Let $a := (d-2)\sigma^2$ and $g(x) := -\frac{ax}{\|x\|^2}$, so $\delta_{\text{JS}}(X) = X + g(X)$.
Stein's identity gives $R(\theta, X + g) = d\sigma^2 + \mathbb{E}_\theta[\|g(X)\|^2 + 2\sigma^2 \nabla \cdot g(X)]$.
For $x \ne 0$, $\|g(x)\|^2 = \frac{a^2}{\|x\|^2}$ and $\nabla \cdot g(x) = -\frac{a(d-2)}{\|x\|^2}$.
Thus $R(\theta, \delta_{\text{JS}}) = d\sigma^2 - (d-2)^2\sigma^4 \mathbb{E}_\theta[\|X\|^{-2}] < d\sigma^2$, with $\mathbb{E}_\theta[\|X\|^{-2}] < \infty$ since $d \ge 3$.
Structural Commentary: Shrinkage trades bias toward 0 for enough variance reduction to lower total risk.
Key Insight: In dimension $d \ge 3$, the usual estimator X is inadmissible under squared-error loss. □

Definition 4.2.4. *Empirical Bayes*

(i) *Hierarchy*: $\theta_i \mid \eta \sim \Pi_\eta$ and $X_i \mid \theta_i \sim P_{\theta_i}$ for $i = 1, \dots, m$.
(ii) *Hyperparameter*: estimate η by $\widehat{\eta} = \widehat{\eta}(X_1, \dots, X_m)$ from the marginal law of the data.
(iii) *Plug-in Bayes rule*: use the Bayes rule computed under $\Pi_{\widehat{\eta}}$.

Remark 4.2.5. *Regularization as Decision-Theoretic Structure*
Penalization, shrinkage, and empirical Bayes impose structure to reduce risk, typically by trading increased bias for reduced variance. The criterion is $R(\theta, \delta)$, not unbiasedness.

5 Bayesian Inference

5.1 Priors, Posteriors, and Conjugacy

Bayesian inference treats unknown parameters as uncertain quantities and updates a prior distribution by the likelihood supplied by the data.

Definition 5.1.1. *Prior, Posterior, Predictive, and Conjugacy*
Let Π be a prior on Θ and $p_\theta(x)$ a dominated likelihood.

(i) *Prior*: Π encodes uncertainty about $\theta \in \Theta$ before observing x.
(ii) *Evidence*: $m(x) := \int_\Theta p_\vartheta(x)\, d\Pi(\vartheta)$, assumed finite and positive.
(iii) *Posterior*: $d\Pi(\theta \mid x) := \frac{p_\theta(x)\, d\Pi(\theta)}{m(x)}$.
(iv) *Predictive*: for future X_{new}, $\mathbb{P}(X_{\text{new}} \in A \mid x) = \int_\Theta P_\theta(A)\, d\Pi(\theta \mid x)$.
(v) *Conjugacy*: $\{\Pi_\eta : \eta \in H\}$ is conjugate if $\Pi_\eta(\cdot \mid x) = \Pi_{\eta(x)}$ for some $\eta(x) \in H$.

Example 5.1.2. *Beta–Binomial Updating*
If $X \mid p \sim \text{Binomial}(n, p)$ and $p \sim \text{Beta}(a, b)$, then $p \mid X = x \sim \text{Beta}(a + x, b + n - x)$.
Thus $\mathbb{E}[p \mid x] = \frac{a+x}{a+b+n}$, a weighted average of $\frac{a}{a+b}$ and $\frac{x}{n}$.

Example 5.1.3. *Gamma–Poisson Updating*
Let $X_1, \dots, X_n \mid \lambda$ be iid Poisson(λ) and $\lambda \sim \text{Gamma}(\alpha, \nu)$ in rate form.
Writing $s := \sum_{i=1}^{n} x_i$, the posterior kernel is $\pi(\lambda \mid x) \propto \lambda^{\alpha+s-1} e^{-(\nu+n)\lambda}$, hence $\lambda \mid x \sim \text{Gamma}(\alpha + s, \nu + n)$.
Thus $\mathbb{E}[\lambda \mid x] = \frac{\alpha+s}{\nu+n}$, $\text{Var}(\lambda \mid x) = \frac{\alpha+s}{(\nu+n)^2}$, $\text{mode}(\lambda \mid x) = \frac{\alpha+s-1}{\nu+n}$ if $\alpha + s > 1$.

Example 5.1.4. *Normal Mean with Normal Prior*
Let $X_1, \dots, X_n \mid \mu$ be iid $N(\mu, \sigma^2)$ with known σ^2, and let $\mu \sim N(m_0, \tau_0^2)$.
Then $\mu \mid X \sim N(m_n, \tau_n^2)$, where $\tau_n^{-2} = \tau_0^{-2} + n\sigma^{-2}$ and $m_n = \tau_n^2(\tau_0^{-2} m_0 + n\sigma^{-2}\overline{X})$.

Example 5.1.5. *Known Mean and Unknown Precision*
Let $X_1, \dots, X_n \mid \xi$ be iid $N(\theta_0, \xi^{-1})$ with known θ_0, and let $\xi \sim \text{Gamma}(\alpha, \lambda)$ in rate form.
Writing $q := \sum_{i=1}^{n} (x_i - \theta_0)^2$, the posterior is $\xi \mid x \sim \text{Gamma}(\alpha + \frac{n}{2}, \lambda + \frac{q}{2})$.
With a flat gamma prior, $\sigma^2 = \xi^{-1}$ centers near the MLE $\frac{q}{n}$.

Definition 5.1.6. *Improper Prior*
An *improper prior* is a σ-finite prior measure Π on Θ with $\Pi(\Theta) = \infty$. For likelihood $p_\theta(x)$, it is usable only when $0 < m_\Pi(x) := \int_\Theta p_\theta(x)\, d\Pi(\theta) < \infty$, in which case $d\Pi(\theta \mid x) := \frac{p_\theta(x)\, d\Pi(\theta)}{m_\Pi(x)}$ is a proper posterior.

Remark 5.1.7. *Conditioning Viewpoint*
Bayesian updating is conditioning under the joint law $dP(\theta, x) = p_\theta(x)\, d\Pi(\theta)\, dx$; the posterior $\Pi(\cdot \mid x)$ is a regular conditional law of θ given $X = x$.

5.2 Computation, Prediction, and Asymptotic Bayes

Except in conjugate models, posterior summaries are integrals that usually require approximation. Monte Carlo and optimization methods from numerical computation become statistical tools here.

Definition 5.2.1. *Credible Sets and Posterior Monte Carlo*
Let $\Pi(\cdot \mid x)$ be the posterior on Θ.

(i) *Credible set*: $C(x) \subseteq \Theta$ has credibility $1 - \alpha$ if $\Pi(C(x) \mid x) \geq 1 - \alpha$.
(ii) *Conditioning*: credibility is a posterior probability conditional on the observed data x.
(iii) *Monte Carlo sample*: let $\theta^{(1)}, \dots, \theta^{(M)}$ be draws, or approximate draws, from $\Pi(\cdot \mid x)$.
(iv) *Posterior mean*: for integrable g, $\int g(\theta)\, d\Pi(\theta \mid x) \approx M^{-1} \sum_{m=1}^{M} g(\theta^{(m)})$.
(v) *Dependent draws*: MCMC uses dependent samples but targets the same posterior integrals.

Method 5.2.2. *Laplace Approximation*

(i) *Setup*: let $\pi_n(\theta \mid x) \propto \exp\{-n q_n(\theta)\}$, with interior minimiser $\widehat{\theta}$.
(ii) *Curvature*: set $H_n(\widehat{\theta}) := \nabla^2 q_n(\widehat{\theta})$, assumed positive definite.
(iii) *Quadratic approximation*: $q_n(\theta) \approx q_n(\widehat{\theta}) + \frac{1}{2}(\theta - \widehat{\theta})^\top H_n(\widehat{\theta})(\theta - \widehat{\theta})$.
(iv) *Normal approximation*: $\Pi(\cdot \mid x) \approx N(\widehat{\theta}, \{n H_n(\widehat{\theta})\}^{-1})$.

Method 5.2.3. *Gibbs Sampling from Full Conditionals*

(i) *Kernel*: unknown mean θ, precision ξ, flat-prior posterior satisfies $\pi(\theta,\xi \mid x) \propto \xi^{\frac{n}{2}-1}\exp\{-\frac{\xi}{2}\sum_i (x_i-\theta)^2\}$.

(ii) *Decomposition*: $\sum_i (x_i-\theta)^2 = \sum_i (x_i-\overline{x})^2 + n(\theta-\overline{x})^2$.

(iii) *Mean conditional*: $\theta \mid \xi, x \sim N(\overline{x}, (n\xi)^{-1})$.

(iv) *Precision conditional*: $\xi \mid \theta, x \sim \text{Gamma}(\frac{n}{2}, \frac{1}{2}\sum_i (x_i-\theta)^2)$ in rate form.

(v) *Gibbs sampler*: alternate conditionals; post-burn-in averages estimate posterior summaries.

Theorem 5.2.4. *Bernstein–von Mises Principle*
Let $X_1,\dots,X_n \sim P_{\theta_0}$ in a regular d-dimensional model with nonsingular information $I(\theta_0)$.
Assume the prior density π is positive and continuous near θ_0, and let $\widehat{\theta}_n$ be an efficient center.
Then $\Pi(\sqrt{n}(\theta-\widehat{\theta}_n) \in \cdot \mid X_1,\dots,X_n) \Rightarrow N_d(0, I(\theta_0)^{-1})$ in P_{θ_0}-probability.

Proof. Set $h := \sqrt{n}(\theta-\widehat{\theta}_n)$. Local asymptotic normality gives $\ell_n(\widehat{\theta}_n + \frac{h}{\sqrt{n}}) - \ell_n(\widehat{\theta}_n) = -\frac{1}{2}h^\top I(\theta_0)h + o_{P_{\theta_0}}(1)$ uniformly on bounded h. Prior continuity gives $\frac{\pi(\widehat{\theta}_n+\frac{h}{\sqrt{n}})}{\pi(\theta_0)} \to 1$ in P_{θ_0}-probability on bounded h.
After normalization, the posterior density of h converges to the $N_d(0, I(\theta_0)^{-1})$ density, with tail mass outside bounded sets negligible. Thus the displayed posterior weak convergence holds.
Structural Commentary: Locally, the likelihood is quadratic and the prior is flat to first order.
Key Insight: In regular finite-dimensional models, posterior uncertainty is asymptotically frequentist normal. □

Remark 5.2.5. *Alignment and Limits*
In regular fixed-dimensional models, Bayesian credible sets based on the Bernstein–von Mises normal approximation have asymptotic frequentist coverage. This alignment can fail in high-dimensional, nonparametric, weakly identified, or boundary problems, where the prior may remain first-order relevant.

5.3 Bayesian Normal Comparisons

The normal two-sample model also has a Bayesian interpretation. With diffuse priors, the resulting posterior formulas resemble frequentist t intervals, but the conditioning and interpretation are different.

Definition 5.3.1. *Common-Precision Two-Sample Normal Model*

(i) *Common-Precision Model*: $\theta = (\mu_X, \mu_Y, \xi) \in \mathbb{R}^2 \times (0,\infty)$ with shared precision $\xi = \sigma^{-2}$ and variance ξ^{-1}.

(ii) *Conditional model*: Given (μ_X, μ_Y, ξ), the variables $X_1,\dots,X_n, Y_1,\dots,Y_m$ are independent, with $X_i \mid (\mu_X,\xi) \sim N(\mu_X, \xi^{-1})$ and $Y_j \mid (\mu_Y,\xi) \sim N(\mu_Y, \xi^{-1})$.

(iii) *Improper prior*: $\pi(\mu_X,\mu_Y,\xi) \propto \xi^{-1}$ on $\mu_X, \mu_Y \in \mathbb{R}$ and $\xi > 0$.

(iv) *Observed data*: Write $x = (x_1,\dots,x_n)$ and $y = (y_1,\dots,y_m)$.

(v) *Sample means*: $\overline{x} := n^{-1}\sum_{i=1}^n x_i$ and $\overline{y} := m^{-1}\sum_{j=1}^m y_j$.

(vi) *Sample variances*: $s_X^2 := (n-1)^{-1}\sum_{i=1}^n (x_i-\overline{x})^2$ and $s_Y^2 := (m-1)^{-1}\sum_{j=1}^m (y_j-\overline{y})^2$.

(vii) *Pooled variance*: $s_p^2 := \frac{(n-1)s_X^2+(m-1)s_Y^2}{m+n-2}$.

(viii) *Posterior kernel*: $\pi(\mu_X,\mu_Y,\xi \mid x,y) \propto \xi^{\frac{m+n}{2}-1}\exp\left\{-\frac{\xi}{2}\left[\sum_{i=1}^n (x_i-\mu_X)^2 + \sum_{j=1}^m (y_j-\mu_Y)^2\right]\right\}$.

Proposition 5.3.2. *Conditional Posterior Factors*
For $n,m \geq 2$, put $\nu := m+n-2$ and take Gamma *shape-rate.*
In the common-precision model, $\pi(\mu_X,\mu_Y,\xi \mid x,y) = \pi(\mu_X \mid \xi,x,y)\pi(\mu_Y \mid \xi,x,y)\pi(\xi \mid x,y)$,
where $(\mu_X,\mu_Y) \mid \xi,x,y \sim N(\overline{x},(n\xi)^{-1}) \otimes N(\overline{y},(m\xi)^{-1})$ and $\xi \mid x,y \sim \text{Gamma}(\frac{\nu}{2}, \frac{\nu s_p^2}{2})$. $\mu_X \perp\!\!\!\perp \mu_Y \mid (\xi,x,y)$.

Proof. Let $S_X(\mu_X) := \sum_{i=1}^n (x_i-\mu_X)^2$, $S_Y(\mu_Y) := \sum_{j=1}^m (y_j-\mu_Y)^2$, and $A := (n-1)s_X^2 + (m-1)s_Y^2 = \nu s_p^2$.
Completing squares gives $S_X(\mu_X) = (n-1)s_X^2 + n(\mu_X-\overline{x})^2$ and $S_Y(\mu_Y) = (m-1)s_Y^2 + m(\mu_Y-\overline{y})^2$.
Thus $\pi(\mu_X,\mu_Y,\xi \mid x,y) \propto \xi^{\frac{m+n}{2}-1}\exp\{-\frac{\xi A}{2}\}\exp\{-\frac{n\xi(\mu_X-\overline{x})^2}{2}\}\exp\{-\frac{m\xi(\mu_Y-\overline{y})^2}{2}\}$.
For fixed ξ, last two factors are independent normal kernels for μ_X, μ_Y with variances $(n\xi)^{-1}$, $(m\xi)^{-1}$.
Also $\int \exp\{-\frac{n\xi(\mu_X-\overline{x})^2}{2}\}\,d\mu_X \propto (n\xi)^{-\frac{1}{2}}$ and $\int \exp\{-\frac{m\xi(\mu_Y-\overline{y})^2}{2}\}\,d\mu_Y \propto (m\xi)^{-\frac{1}{2}}$.
Hence marginalising μ_X,μ_Y leaves $\pi(\xi \mid x,y) \propto \xi^{\frac{\nu}{2}-1}\exp\{-\frac{\xi\nu s_p^2}{2}\}$.
Structural Commentary: Completing squares isolates location kernels and pooled residual $A = \nu s_p^2$.
Key Insight: Marginalising μ_X,μ_Y leaves the pooled gamma law for ξ. □

Proposition 5.3.3. *Posterior t Law for a Mean Difference*
In the common-precision model with $n, m \geq 2$, set $\Delta := \mu_X - \mu_Y$, $d := \overline{x} - \overline{y}$, $c := n^{-1} + m^{-1}$, $\nu := m + n - 2$, and $T := \frac{\Delta - d}{s_p \sqrt{c}}$. Then $T \mid x, y \sim t_\nu$.

Proof. By Proposition 5.3.2, $\Delta \mid \xi, x, y \sim N(d, \frac{c}{\xi})$ and $\xi \mid x, y \sim \text{Gamma}(\frac{\nu}{2}, \frac{\nu s_p^2}{2})$.
Equivalently, take independent $E \sim N(0,1)$, $V \sim \chi^2_\nu$, $\xi = \frac{V}{\nu s_p^2}$, $\Delta = d + \sqrt{\frac{c}{\xi}}\, E$. Thus $T = \frac{\Delta - d}{s_p \sqrt{c}} = \frac{E}{\sqrt{\frac{V}{\nu}}}$, QED.
Structural Commentary: Normal given ξ; Student after mixing over ξ.
Key Insight: The t law is exactly a standard normal divided by an independent estimated-scale factor $\sqrt{\frac{V}{\nu}}$. □

Remark 5.3.4. *Same Algebra, Different Conditioning*
The statistic has the same algebra as the pooled two-sample t pivot in Theorem 3.5.4: frequentist inference treats Δ as fixed and X, Y as random, whereas the posterior treats x, y as fixed and $\Delta \mid x, y$ as random.

Example 5.3.5. *Latent Heat Posterior Comparison*
For method contrast $\Delta = \mu_X - \mu_Y$, the common-precision posterior gives $\mathbb{P}(\Delta > 0 \mid X, Y) \approx 0.998$ and posterior interval $\mathbb{P}(0.015 \leq \Delta \leq 0.065 \mid X, Y) \approx 0.95$, numerically matching the pooled normal-theory interval.

6 Nonparametric and High-Dimensional Statistics

6.1 Empirical Processes, Density, and Regression Smoothing

Nonparametric statistics replaces finite-dimensional parameter assumptions by smoothness, shape, sparsity, or complexity restrictions.

Definition 6.1.1. *Empirical Measure and Empirical Process*
Let $X_1, \ldots, X_n$ be observations in $(\mathcal{X}, \mathcal{A})$ with reference law P.

(i) *Empirical measure*: $\widehat{P}_n := n^{-1} \sum_{i=1}^n \delta_{X_i}$, so $\widehat{P}_n f = n^{-1} \sum_{i=1}^n f(X_i)$.
(ii) *Empirical process*: For measurable $\mathcal{F}$, $\mathbb{G}_n(f) := \sqrt{n}(\widehat{P}_n f - P f)$ for $f \in \mathcal{F}$.

Theorem 6.1.2. *Glivenko–Cantelli Theorem*
Let $X_1, X_2, \ldots$ be i.i.d. real variables with CDF F, and set $\widehat{F}_n(t) := n^{-1} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq t\}}$.
Then $\sup_{t \in \mathbb{R}} \left| \widehat{F}_n(t) - F(t) \right| \to 0$ almost surely.

Proof. Add independent $R_i \sim \text{Unif}[0,1]$ and set $U_i := F(X_i-) + R_i\{F(X_i) - F(X_i-)\}$.
Then U_i are i.i.d. Unif$[0,1]$, and with $H_n(u) := n^{-1} \sum_i \mathbf{1}_{\{U_i \leq u\}}$ one has $\widehat{F}_n(t) = H_n(F(t))$ a.s.
Hence $\sup_t \left| \widehat{F}_n(t) - F(t) \right| \leq \sup_{0 \leq u \leq 1} |H_n(u) - u|$.
Fix $k \geq 1$, $u_j := \frac{j}{k}$, let $M_{n,k} := \max_{0 \leq j \leq k} |H_n(u_j) - u_j|$.
By the SLLN, $M_{n,k} \to 0$ a.s. If $u_{j-1} \leq u \leq u_j$, monotonicity gives $H_n(u_{j-1}) - u_j \leq H_n(u) - u \leq H_n(u_j) - u_{j-1}$,
Hence $\sup_{0 \leq u \leq 1} |H_n(u) - u| \leq M_{n,k} + k^{-1}$, and $\limsup_{n \to \infty} \sup_{0 \leq u \leq 1} |H_n(u) - u| \leq k^{-1}$ a.s. $\forall k$. Let $k \to \infty$.
Structural Commentary: Finite-grid SLLN plus monotonicity gives uniform CDF control.
Key Insight: Monotonicity bounds between-grid drift by the mesh. □

Definition 6.1.3. *Kernel Density Estimator*

(i) *Kernel and bandwidth*: Let $\int K(u)\, du = 1$ and $h > 0$.
(ii) *Estimator*: For data $X_1, \ldots, X_n$, $\widehat{f}_h(x) := (nh)^{-1} \sum_{i=1}^n K(\frac{x - X_i}{h})$.

Larger h lowers variance and raises bias; smaller h does the reverse.

Definition 6.1.4. *Nonparametric Regression*

(i) *Model*: $Y = m(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X] = 0$, so $m(x) = \mathbb{E}[Y \mid X = x]$.
(ii) *Weights*: For kernel K and bandwidth $h > 0$, set $w_i(x) := K(\frac{x - X_i}{h})$.
(iii) *Estimator*: If $\sum_i w_i(x) > 0$, define $\widehat{m}_h(x) := \frac{\sum_{i=1}^n w_i(x) Y_i}{\sum_{i=1}^n w_i(x)}$.

Definition 6.1.5. *Kernel-Weighted Local Average*
Let $(X_i, Y_i)_{i=1}^n$ be regression data and fix a query point x_0.

(i) *Local kernel*: For a nonnegative profile D and width $h_\lambda(x_0) > 0$, set $K_\lambda(x_0, x) := D(\|x - x_0\| / h_\lambda(x_0))$.

(ii) *Estimator*: If $\sum_{i=1}^n K_\lambda(x_0, X_i) > 0$, local-constant (Nadaraya–Watson) estimate is $\widehat{m}(x_0) := \frac{\sum_{i=1}^n K_\lambda(x_0, X_i) Y_i}{\sum_{i=1}^n K_\lambda(x_0, X_i)}$.

Metric windows set $h_\lambda(x_0) = \lambda$; nearest-neighbor windows set $h_\lambda(x_0) = \mathrm{dist}(x_0, X_{(k)}(x_0))$.

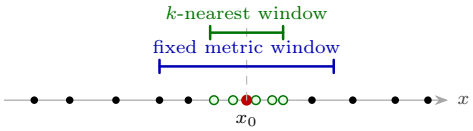


Figure 218: Fixed metric and k-nearest windows around x_0.

Example 6.1.6. *Common Local-Smoothing Kernels*
In one dimension, common compact profiles are $D_E(t) := \frac{3}{4}(1 - t^2)\mathbf{1}_{\{|t| \le 1\}}$ for Epanechnikov smoothing and $D_T(t) := (1 - |t|^3)^3 \mathbf{1}_{\{|t| \le 1\}}$ for tri-cube smoothing. Both have support $[-1, 1]$; Gaussian profiles have full support, and normalization constants cancel in regression-weight averages but matter for density estimation.

Method 6.1.7. *Local Linear Regression Smoothing*
Fix x_0 and bandwidth $h > 0$.

(i) *Weighted fit*: Set $(\widehat{\beta}_0(x_0), \widehat{\beta}_1(x_0)) \in \operatorname{argmin}_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1(X_i - x_0))^2 K(\frac{X_i - x_0}{h})$.

(ii) *Estimate*: Define $\widehat{m}_h(x_0) := \widehat{\beta}_0(x_0)$.

Smaller h gives a more variable fit; larger h gives more bias.

Method 6.1.8. *Local Polynomial Regression*
Fix a query point $x_0 \in \mathbb{R}^p$ and degree $d \in \mathbb{N}_0$.

(i) *Basis*: Let $\mathcal{A}_d := \{\alpha \in \mathbb{N}_0^p : |\alpha| \le d\}$ and $b_d(u) := (u^\alpha)_{\alpha \in \mathcal{A}_d}$, where $u^\alpha := \prod_{j=1}^p u_j^{\alpha_j}$.

(ii) *Weights*: Set $K_i(x_0) := K_\lambda(x_0, X_i)$.

(iii) *Weighted fit*: For $\beta = (\beta_\alpha)_{\alpha \in \mathcal{A}_d}$, choose $\widehat{\beta}(x_0) \in \operatorname{argmin}_{\beta \in \mathbb{R}^{|\mathcal{A}_d|}} \sum_{i=1}^n K_i(x_0)(Y_i - b_d(X_i - x_0)^\top \beta)^2$.

(iv) *Estimator*: With $\mathbf{0} \in \mathcal{A}_d$ the constant multi-index, define $\widehat{m}_d(x_0) := \widehat{\beta}_{\mathbf{0}}(x_0)$.

$d = 0$ is local constant; $d = 1$ is local linear with boundary correction.
Larger d trades lower curvature bias for higher variance.

Proposition 6.1.9. *Equivalent Kernel Moments*
Fix $x_0 \in \mathbb{R}$ and set $u_i := X_i - x_0$, $K_i := K_\lambda(x_0, X_i)$, $z_i := (1, u_i)^\top$, $Z := (z_1^\top, \dots, z_n^\top)^\top$, $W := \mathrm{diag}(K_1, \dots, K_n)$, $Y := (Y_1, \dots, Y_n)^\top$, and $G := Z^\top W Z$.
Assume G is nonsingular, let $e_1 := (1, 0)^\top$, $\widehat{\beta}(x_0) := G^{-1} Z^\top W Y$, and $\widehat{m}(x_0) := e_1^\top \widehat{\beta}(x_0)$.

(i) Equivalent weights: $\widehat{m}(x_0) = \sum_{i=1}^n \ell_i(x_0) Y_i$ with $\ell_i(x_0) := K_i z_i^\top G^{-1} e_1$.

(ii) Moment constraints: $\sum_{i=1}^n \ell_i(x_0) = 1$ and $\sum_{i=1}^n u_i \ell_i(x_0) = 0$.

(iii) Affine reproduction: If $Y_i = a + b X_i$, then $\widehat{m}(x_0) = a + b x_0$.

Proof.

(i) *Equivalent weights*: The weighted criterion $(Y - Z\beta)^\top W (Y - Z\beta)$ has normal equations $G\widehat{\beta} = Z^\top W Y$, so $\widehat{\beta} = G^{-1} Z^\top W Y$. Since G^{-1} is symmetric, $\widehat{m}(x_0) = e_1^\top G^{-1} Z^\top W Y = \sum_{i=1}^n K_i e_1^\top G^{-1} z_i Y_i = \sum_{i=1}^n \ell_i(x_0) Y_i$.

(ii) *Moment constraints*: With $\ell(x_0) := (\ell_1(x_0), \dots, \ell_n(x_0))^\top = W Z G^{-1} e_1$, $Z^\top \ell(x_0) = Z^\top W Z G^{-1} e_1 = e_1$, hence $\sum_i \ell_i(x_0) = 1$ and $\sum_i u_i \ell_i(x_0) = 0$.

(iii) *Affine reproduction*: If $Y_i = a + b X_i = (a + b x_0) + b u_i$,
then $\widehat{m}(x_0) = (a + b x_0) \sum_i \ell_i(x_0) + b \sum_i u_i \ell_i(x_0) = a + b x_0$.

Structural Commentary: Weighted projection onto span$\{1, x - x_0\}$ imposes two weight moments.
Key Insight: $Z^\top \ell = e_1$ encodes constant reproduction and zero first moment. □

Definition 6.1.10. *Structured Kernel Metric*
Fix $x_0, x \in \mathbb{R}^p$, bandwidth $\lambda > 0$, profile $D \geq 0$, and a positive semidefinite matrix $A \succeq 0$.

(i) *Quadratic scale*: Define $\rho_A(x, x_0) := (x - x_0)^\top A(x - x_0)$.
(ii) *Kernel*: Set $K_{\lambda,A}(x_0, x) := D(\rho_A(x, x_0)/\lambda)$.

A sets coordinate weights, correlations, and active directions.
Standardise predictors first since ρ_A is unit-dependent.

Method 6.1.11. *Local Likelihood and Varying Coefficients*
Let $\ell_i(\theta)$ be the log-likelihood contribution of observation i under parameter θ.

(i) *Local likelihood*: At target x_0, set $K_i(x_0) := K_\lambda(x_0, X_i)$ and choose $\widehat{\theta}(x_0) \in \operatorname{argmax}_\theta \sum_{i=1}^n K_i(x_0)\ell_i(\theta)$.
(ii) *Local parameterisation*: In generalised models, replace a constant natural parameter by $\eta_i(\gamma; x_0) := b_d(X_i - x_0)^\top \gamma$ and maximize $\sum_i K_i(x_0)\ell_i(\eta_i(\gamma; x_0))$ over γ.
(iii) *Varying coefficients*: For predictors (U, Z) with $U \in \mathbb{R}^q$, model $m(U, Z) := \alpha(Z) + \sum_{r=1}^q \beta_r(Z)U_r$.
(iv) *Local fit*: At z_0, choose $(\widehat{\alpha}, \widehat{\beta}) \in \operatorname{argmin}_{a,b} \sum_i K_\lambda(z_0, Z_i)(Y_i - a - b^\top U_i)^2$ and set $(\widehat{\alpha}(z_0), \widehat{\beta}(z_0)) := (\widehat{\alpha}, \widehat{\beta})$.

The coefficients are ordinary regression coefficients locally in U, but functions of the index variable Z.

Remark 6.1.12. *Bandwidth, Boundaries, and Dimension*
Let $N_\lambda(x_0) := \{i : K_\lambda(x_0, X_i) > 0\}$ and $n_\lambda(x_0) := |N_\lambda(x_0)|$.

(i) *Bias–variance*: Larger λ usually increases $n_\lambda(x_0)$, lowering variance, raising bias; smaller λ does reverse.
(ii) *Windows*: Metric windows fix radius; k-nearest-neighbor windows choose $\lambda(x_0)$ so that $n_{\lambda(x_0)}(x_0) \approx k$.
(iii) *Dimension*: In $\mathbb{R}^p$, neighborhoods must expand with p to keep $n_\lambda(x_0)$ stable, reducing local resolution.
(iv) *Cost*: Without indexing, one query evaluates $K_\lambda(x_0, X_i)$ for all i, costing $O(n)$.

Method 6.1.13. *Bandwidth Choice by Leave-One-Out Cross-Validation*

(i) *Candidates*: Fix bandwidths $h_1, \ldots, h_M > 0$.
(ii) *Leave-one-out score*: Let $\widehat{m}_{h_k,-i}$ be local linear fit excluding (X_i, Y_i). Set $\mathrm{CV}(h_k) := \sum_{i=1}^n (Y_i - \widehat{m}_{h_k,-i}(X_i))^2$.
(iii) *Selection*: Choose $\widehat{h} \in \operatorname{argmin}_{1 \leq k \leq M} \mathrm{CV}(h_k)$, with a visual smoothness check.

Definition 6.1.14. *Kernel Density Classification*
Let $G \in \{1, \ldots, J\}$ be the class label.
Estimate $\pi_j := \mathbb{P}(G = j)$ by $\widehat{\pi}_j$ and $f_j(x) := f_{X|G=j}(x)$ by a kernel density estimate $\widehat{f}_j(x)$.

(i) *Posterior*: Define $\widehat{p}_j(x) := \widehat{\mathbb{P}}(G = j \mid X = x) := \frac{\widehat{\pi}_j \widehat{f}_j(x)}{\sum_{k=1}^J \widehat{\pi}_k \widehat{f}_k(x)}$ whenever the denominator is positive.
(ii) *Classifier*: The plug-in Bayes rule is $\widehat{g}(x) \in \operatorname{argmax}_{1 \leq j \leq J} \widehat{p}_j(x)$.

Full density estimation is unnecessary if only the boundary is needed.

Remark 6.1.15. *Complexity Control*
For a function class $\mathcal{F}$, estimation is stable when $\sup_{f \in \mathcal{F}} \left| \widehat{P}_n f - Pf \right|$ is small with high probability or a.s.
Smoothness and entropy control this supremum in place of finite dimension.

6.2 Sparse High-Dimensional Inference

High-dimensional statistics studies regimes where the number of candidate parameters is comparable to, or much larger than, the sample size. Estimation is possible only after adding structure.

Definition 6.2.1. *Sparsity, Lasso, and Restricted Eigenvalues*

(i) *High dimension*: A sequence has dimension $p = p_n$, often with $p_n \gg n$.
(ii) *Sparsity*: For $\theta \in \mathbb{R}^p$, $\|\theta\|_0 := |\{j : \theta_j \neq 0\}|$; θ is s-sparse if $\|\theta\|_0 \leq s$.
(iii) *Lasso*: For $y = X\beta_0 + \varepsilon$ and $\lambda \geq 0$, $\widehat{\beta}_\lambda \in \operatorname{arg\,min}_{\beta \in \mathbb{R}^p} \{(2n)^{-1} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1\}$.

The convex ℓ^1 penalty promotes sparsity.

(iv) *Restricted eigenvalue*: For $S \subseteq [p] := \{1, \dots, p\}$, $C(S) := \{v : \|v_{S^c}\|_1 \le 3\|v_S\|_1\}$, and $\kappa > 0$, X satisfies $\mathrm{RE}(S, \kappa)$ if $n^{-\frac{1}{2}}\|Xv\|_2 \ge \kappa\|v_S\|_2$ for all $v \in C(S)$.

Theorem 6.2.2. *Basic Lasso Rate Form*
Let $y = X\beta_0 + \varepsilon$, $S := \mathrm{supp}(\beta_0)$, $|S| \le s$, and let $\widehat{\beta}_\lambda$ be any lasso solution.
If X satisfies $\mathrm{RE}(S, \kappa)$ and $\lambda > 0$ with $\lambda \ge 2\left\|\frac{X^\top \varepsilon}{n}\right\|_\infty$, then $n^{-1}\left\|X(\widehat{\beta}_\lambda - \beta_0)\right\|_2^2 \le \frac{9s\lambda^2}{\kappa^2}$ and $\left\|\widehat{\beta}_\lambda - \beta_0\right\|_1 \le \frac{12s\lambda}{\kappa^2}$.

Proof. Set $\delta := \widehat{\beta}_\lambda - \beta_0$. The lasso optimality inequality gives $(2n)^{-1}\|X\delta\|_2^2 + \lambda\|\beta_0 + \delta\|_1 \le n^{-1}\varepsilon^\top X\delta + \lambda\|\beta_0\|_1$.
Since $\lambda \ge 2\left\|\frac{X^\top\varepsilon}{n}\right\|_\infty$, $n^{-1}\varepsilon^\top X\delta \le \frac{\lambda}{2}\|\delta\|_1$, and since $\beta_{0,S^c} = 0$, $\|\beta_0\|_1 - \|\beta_0 + \delta\|_1 \le \|\delta_S\|_1 - \|\delta_{S^c}\|_1$.
Thus $(2n)^{-1}\|X\delta\|_2^2 + \frac{\lambda}{2}\|\delta_{S^c}\|_1 \le \frac{3\lambda}{2}\|\delta_S\|_1$, so $\delta \in C(S)$. By $\mathrm{RE}(S, \kappa)$, $\|\delta_S\|_1 \le \sqrt{s}\|\delta_S\|_2 \le \frac{\sqrt{s}\|X\delta\|_2}{\kappa\sqrt{n}}$.
Hence $n^{-1}\|X\delta\|_2^2 \le \frac{3\lambda\sqrt{s}\|X\delta\|_2}{\kappa\sqrt{n}}$, giving $n^{-1}\|X\delta\|_2^2 \le \frac{9s\lambda^2}{\kappa^2}$; also $\|\delta\|_1 \le 4\|\delta_S\|_1 \le \frac{12s\lambda}{\kappa^2}$.
Structural Commentary: Basic inequality plus RE converts score control into prediction and ℓ^1 error bounds.
Key Insight: The ℓ^1 penalty forces the error into the RE cone. □

Definition 6.2.3. *Multiple-Testing Error Counts and FDR*
For a multiple-testing rule over M hypotheses.
Let M_0 be the number of true nulls and $M_1 = M - M_0$ the number of false nulls.
Let V be false rejections, S true rejections, U true nonrejections, and T false nonrejections.
The total number of discoveries is $R = V + S$.

(i) *Family-wise error rate*: $\mathrm{FWER} := \mathbb{P}(V \ge 1)$.
(ii) *False discovery proportion*: $\mathrm{FDP} := \frac{V}{\max(R,1)}$.
(iii) *False discovery rate*: $\mathrm{FDR} := \mathbb{E}[\mathrm{FDP}]$.
(iv) *Power*: One aggregate power measure is $\mathbb{E}[S]/M_1$ when $M_1 > 0$.

Thus FWER controls the chance of any false discovery.
FDR controls the expected false fraction among the reported discoveries.

Method 6.2.4. *Featurewise Two-Sample Screening*
Let x_{ij} be feature $j \in [M]$ for observation i, with groups C_1, C_2 and $n_\ell := |C_\ell|$.

(i) *Group means*: For $\ell \in \{1, 2\}$, set $\bar{x}_{\ell j} := n_\ell^{-1} \sum_{i \in C_\ell} x_{ij}$.
(ii) *Pooled scale*: Set $\widehat{\sigma}_j^2 := (n_1 + n_2 - 2)^{-1}\{\sum_{i \in C_1}(x_{ij} - \bar{x}_{1j})^2 + \sum_{i \in C_2}(x_{ij} - \bar{x}_{2j})^2\}$ and $\mathrm{se}_j := \widehat{\sigma}_j\sqrt{n_1^{-1} + n_2^{-1}}$.
(iii) *Statistic*: Define $t_j := (\bar{x}_{2j} - \bar{x}_{1j})/\mathrm{se}_j$; large $|t_j|$ flags a featurewise mean shift.
(iv) *Family*: Screening acts on $\{t_j : 1 \le j \le M\}$, so error control targets the family, not a single statistic.

Method 6.2.5. *Benjamini–Hochberg Step-Up Rule*
For M hypotheses with p-values $p_1, \dots, p_M$ and target level $\alpha \in (0, 1)$:

(i) *Order*: Let $p_{(1)} \le \cdots \le p_{(M)}$ be the order statistics.
(ii) *Step-up index*: Set $L := \max\{j : p_{(j)} \le \alpha j/M\}$, with $L := 0$ if the set is empty.
(iii) *Rejections*: Reject $\mathcal{R} := \{m : p_m \le p_{(L)}\}$ if $L > 0$, and reject none if $L = 0$.
(iv) *Guarantee*: If true-null p-values are independent and uniform, then $\mathrm{FDR} \le (M_0/M)\alpha \le \alpha$, where M_0 is the number of true nulls.

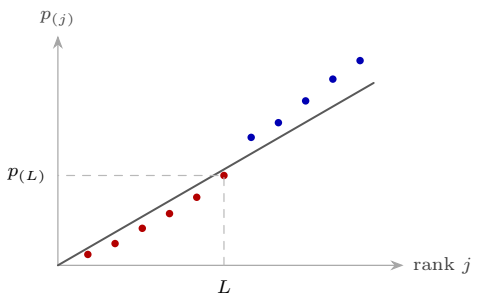


Figure 219: Benjamini–Hochberg threshold: red points reject, blue points do not; line is $\alpha j/M$.

Method 6.2.6. *Permutation Plug-In FDR Estimate*
Let t_j be observed statistics for $j \in [M]$, and let $t_j^{(k)}$ be the statistic after label permutation $k \in [K]$.

(i) *Observed discoveries*: For cutpoint $C > 0$, set $R_{\text{obs}}(C) := \sum_{j=1}^{M} \mathbf{1}_{\{|t_j|>C\}}$.

(ii) *Null discoveries*: Estimate the expected false discoveries by $\widehat{\mathbb{E}}(V;C) := K^{-1} \sum_{k=1}^{K} \sum_{j=1}^{M} \mathbf{1}_{\{|t_j^{(k)}|>C\}}$.

(iii) *Plug-in FDR*: If $R_{\text{obs}}(C) > 0$, set $\widehat{\text{FDR}}(C) := \widehat{\mathbb{E}}(V;C)/R_{\text{obs}}(C)$.

(iv) *Permutation p-values*: Use $\{t_j^{(k)}\}$ to compare $|t_j|$ with featurewise or pooled permutation null statistics.

Remark 6.2.7. *Asymmetric Cutpoints and SAM*
Let $t_{(1)} \le \cdots \le t_{(M)}$ and $\bar{t}^0_{(j)} := K^{-1} \sum_{k=1}^{K} t_{(j)}^{(k)}$, where $t_{(j)}^{(k)}$ is the jth ordered permutation statistic.
For $d_j := t_{(j)} - \bar{t}^0_{(j)}$, $a > 0$, set $c_+(a) := \min\{t_{(j)} : d_j > a\}$, $c_-(a) := \max\{t_{(j)} : d_j < -a\}$ when nonempty.
Reject $t_j \in A(c_-, c_+) := (-\infty, c_-] \cup [c_+, \infty)$, estimate FDR by Method 6.2.6, $A(c_-, c_+)$ replace $\{|t| > C\}$.

Definition 6.2.8. *Positive FDR, Local FDR, and q-Values*
Let $Z_j = 0$ indicate a true null, $Z_j = 1$ a false null, t_j a test statistic, and Γ a rejection region.

(i) *pFDR*: For false discoveries V and discoveries R, $\text{pFDR} := \mathbb{E}[V/R \mid R > 0]$.

(ii) *Mixture model*: If $\mathbb{P}(Z_j = 0) = \pi_0$, $\mathcal{L}(t_j \mid Z_j = 0) = F_0$, and $\mathcal{L}(t_j \mid Z_j = 1) = F_1$, then $\mathcal{L}(t_j) = \pi_0 F_0 + (1 - \pi_0) F_1$.

(iii) *Bayes form*: For a rejection region Γ, $\text{pFDR}(\Gamma) = \mathbb{P}(Z_j = 0 \mid t_j \in \Gamma)$.

(iv) *Local FDR*: At value t_0, $\text{lfdr}(t_0) := \mathbb{P}(Z_j = 0 \mid t_j = t_0)$.

(v) *q-value*: For feature j, $q_j := \inf\{\alpha : j \in \mathcal{R}_\alpha\}$, where $\mathcal{R}_\alpha$ is the discovery set at FDR level α.

Remark 6.2.9. *FDR Versus Family-Wise Error*
With false discoveries V and total discoveries R, $\text{FWER} := \mathbb{P}(V \ge 1)$ controls any false rejection, while $\text{FDR} := \mathbb{E}[V/\max(R,1)]$ controls the expected false fraction among discoveries. Thus FDR is less conservative for large screening lists and measures list reliability directly.

Remark 6.2.10. *Statistical and Computational Structure*
High-dimensional inference couples concentration bounds, convex optimization, and selection error control; computation and statistics are part of the same guarantee.

6.3 Rank-Based Comparative Tests

Rank methods replace numerical values by their relative order. They are invariant under strictly increasing transformations and reduce the influence of extreme observations, which makes them natural companions to the two-sample and paired t procedures.

Definition 6.3.1. *Mann–Whitney and Rank-Sum Statistics*
Let $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$ be independent continuous samples.
Rank the pooled observations increasingly.

(i) *Pooled size*: $N := m + n$.

(ii) *Y-ranks*: $R_j^Y := \text{rank}(Y_j)$ in the pooled sample.

(iii) *Rank sum*: $T_Y := \sum_{j=1}^{m} R_j^Y$.

(iv) *Mann–Whitney statistic*: $U_Y := T_Y - \frac{m(m+1)}{2}$.

Under continuity, $U_Y = \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{1}_{\{X_i < Y_j\}}$ a.s., so U_Y counts cross-sample pairs with $Y_j > X_i$.

Theorem 6.3.2. *Null Mean and Variance of the Rank Sum*
Let $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$ be independent continuous samples.
Let rank-sum T_Y and Mann–Whitney statistic U_Y. Assume $F = G$ and set $N := m + n$.

(i) Rank law: $T_Y = \sum_{r \in A} r$, where A is uniform over all m-subsets of $\{1, \dots, N\}$.

(ii) Rank-sum moments: $\mathbb{E}T_Y = \frac{m(N+1)}{2}$ and $\text{Var}(T_Y) = \frac{mn(N+1)}{12}$.

(iii) Mann–Whitney moments: $\mathbb{E}U_Y = \frac{mn}{2}$ and $\text{Var}(U_Y) = \frac{mn(N+1)}{12}$.

Proof.

(i) Continuity gives no ties and $F = G$ makes labels exchangeable, so the Y-rank set $A \subseteq \{1, \dots, N\}$ is uniform over all m-subsets and $T_Y = \sum_{r \in A} r$.

(ii) For ranks $1, \dots, N$, $\overline{r} = \frac{N+1}{2}$ and $S_r^2 = \frac{N^2-1}{12}$; sampling m ranks without replacement gives $\mathbb{E}T_Y = m\overline{r} = \frac{m(N+1)}{2}$ and $\mathrm{Var}(T_Y) = \frac{mS_r^2(N-m)}{N-1} = \frac{mn(N+1)}{12}$.

(iii) Since $U_Y = T_Y - \frac{m(m+1)}{2}$, shifting by a constant gives $\mathbb{E}U_Y = \frac{m(N+1)}{2} - \frac{m(m+1)}{2} = \frac{mn}{2}$ and $\mathrm{Var}(U_Y) = \mathrm{Var}(T_Y) = \frac{mn(N+1)}{12}$.

Structural Commentary: Under $F = G$, ranks reduce to simple random sampling without replacement.
Key Insight: U_Y is just a centered rank sum, so only the mean shifts. □

Method 6.3.3. *Mann–Whitney Test*
Given independent samples $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$, test $H_0 : F = G$:

(i) *Compute*: Pool and rank $m + n$ observations, use midranks for ties. Compute T_Y, $U_Y = T_Y - \frac{m(m+1)}{2}$.
(ii) *Calibrate*: For small m, n, enumerate m-subsets of $\{1, \dots, m+n\}$ to get the exact null law. For large m, n, use $Z := \frac{U_Y - \frac{mn}{2}}{\sqrt{\frac{mn(m+n+1)}{12}}} \approx N(0,1)$, with tie correction when needed.
(iii) *Decide*: Compute the relevant one- or two-sided tail probability under the chosen null law. Reject H_0 at level α when $p \le \alpha$.

Example 6.3.4. *Latent Heat Data by Ranks*
For the latent-heat data, take $m = 8$, $n = 13$, $N = 21$, and midranks for ties.
The smaller-sample rank sum is $R = 51$, its complement is $R' = m(N+1) - R = 125$, and $R^* = \min(R, R') = 51$.
Under equality of the two method distributions, $\mu_T = \frac{m(N+1)}{2} = 88$ and $\sigma_T = \sqrt{\frac{mn(N+1)}{12}} \approx 13.8$,
hence $z = \frac{R - \mu_T}{\sigma_T} \approx -2.68$ and the two-sided normal p-value is ≈ 0.007.
The exact rank-sum test therefore rejects equality at level 0.01.

Definition 6.3.5. *Mann–Whitney Effect Probability*
Let $X \sim F$ and $Y \sim G$ be independent, with samples $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$.

(i) *Effect probability*: $\pi := \mathbb{P}(X < Y)$.
(ii) *Estimator*: $\widehat{\pi} := (mn)^{-1} \sum_{i=1}^{n} \sum_{j=1}^{m} \mathbf{1}_{\{X_i < Y_j\}} = \frac{U_Y}{mn}$.

Thus π is the probability that a random Y exceeds a random X.

Method 6.3.6. *Shift Intervals from Pairwise Differences*
Assume the shift model $G(x) = F(x - \Delta)$.

(i) *Compute*: Set $D_{ij} := Y_j - X_i$ and order $D_{(1)} \le \cdots \le D_{(mn)}$.
(ii) *Calibrate*: Choose k so the two-sided rank test accepts a candidate shift when $k \le U_Y(\Delta) \le mn - k$.
(iii) *Invert*: Report $[D_{(k)}, D_{(mn-k+1)}]$ as the inverted shift interval.

The Hodges–Lehmann point estimate is $\widehat{\Delta}_{HL} := \mathrm{median}\{D_{ij} : 1 \le i \le n, 1 \le j \le m\}$.

Definition 6.3.7. *Signed-Rank Statistic*
Let $D_1, \dots, D_n$ be paired differences.

(i) *Nonzero index set*: $I := \{i : D_i \neq 0\}$.
(ii) *Absolute ranks*: $R_i := \mathrm{rank}(|D_i|)$ among $\{|D_\ell| : \ell \in I\}$, using midranks for ties.
(iii) *Signed-rank statistic*: $W_+ := \sum_{i \in I} R_i \mathbf{1}_{\{D_i > 0\}}$.

Method 6.3.8. *Signed-Rank Test*
For paired differences D_i, test $H_0 : D_i \overset{d}{=} -D_i$.

(i) *Compute*: Discard zero differences, rank nonzero $|D_i|$ using midranks for ties, and compute W_+.
(ii) *Calibrate*: Conditional on the nonzero absolute ranks, enumerate the 2^r sign assignments, where $r := |I|$, to get the null law of W_+.
(iii) *Decide*: Compute the relevant tail probability and reject H_0 at level α when $p \le \alpha$.

The sign test is the coarser paired alternative using only signs, not ranks.

Theorem 6.3.9. *Signed-Rank Mean and Variance*
Under the signed-rank null, assume no zero differences and no ties in $|D_i|$.

(i) Mean: $\mathbb{E}W_+ = \frac{n(n+1)}{4}$.

(ii) Variance: $\mathrm{Var}(W_+) = \frac{n(n+1)(2n+1)}{24}$.

Proof. Let $\mathcal{R}$ be the absolute-rank information and let I_k indicate positivity at absolute-rank k. Then $W_+ = \sum_{k=1}^{n} kI_k$ and, under symmetry, $I_k \mid \mathcal{R}$ are independent Bernoulli($\frac{1}{2}$).

(i) *Mean*: $\mathbb{E}(W_+ \mid \mathcal{R}) = \frac{1}{2}\sum_{k=1}^{n} k = \frac{n(n+1)}{4}$; the right side is deterministic, hence $\mathbb{E}W_+ = \frac{n(n+1)}{4}$.

(ii) *Variance*: $\mathrm{Var}(W_+ \mid \mathcal{R}) = \frac{1}{4}\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{24}$, while $\mathbb{E}(W_+ \mid \mathcal{R})$ is deterministic.

Total variance gives $\mathrm{Var}(W_+) = \frac{n(n+1)(2n+1)}{24}$.

Structural Commentary: Condition on absolute ranks; only independent fair signs remain.
Key Insight: W_+ is a weighted sum of Bernoulli signs with weights $1, \ldots, n$. □

Example 6.3.10. *Paired Rank Analyses*
Two paired-rank calculations are:

(i) *Platelet data*: With one negative signed rank, use W_-; exact signed-rank test rejects symmetry at 0.01.
(ii) *Mercury data*: Dropping one zero difference and using midranks gives $n = 24$ and $W_+ = 194.5$. Under H_0, $\mathbb{E}_0 W_+ = \frac{24\cdot 25}{4} = 150$ and $\mathrm{Var}_0(W_+) = \frac{24\cdot 25\cdot 49}{24} = 1225$, so $Z = \frac{194.5-150}{35} \approx 1.27$ and two-sided $p \approx 0.20$; there is no strong evidence of a systematic shift.

Remark 6.3.11. *Rank Tests and Normal Theory*
Rank tests use order information, require no normal model, and are robust to outliers; under normal models their efficiency relative to t tests is often close to 1.

7 Time Series Models

7.1 Stationarity, Autocovariance, and Linear Processes

Time-series statistics studies data ordered by time, where dependence is part of the signal rather than a nuisance to be removed.

Definition 7.1.1. *Strict and Weak Stationarity*
Let $(X_t)_{t\in\mathbb{Z}}$ be a real-valued stochastic process.

(i) *Strict stationarity*: $\forall k \in \mathbb{N},\ t_1, \ldots, t_k, h \in \mathbb{Z}$, $(X_{t_1}, \ldots, X_{t_k}) \overset{\text{dist}}{=} (X_{t_1+h}, \ldots, X_{t_k+h})$.
(ii) *Weak stationarity*: $\exists \mu \in \mathbb{R}$ and $\gamma : \mathbb{Z} \to \mathbb{R}$ with $\forall t, h \in \mathbb{Z}$, $X_t \in L^2$, $\mathbb{E}X_t = \mu$, and $\mathrm{Cov}(X_t, X_{t-h}) = \gamma(h)$.

Key Insight: (i) plus L^2 implies (ii); (ii) fixes only μ and γ, not finite-dimensional laws.

Definition 7.1.2. *Autocorrelation, Partial Autocorrelation, and Spectral Density*
Let $(X_t)_{t\in\mathbb{Z}}$ be weakly stationary with autocovariance $\gamma(h) := \mathrm{Cov}(X_t, X_{t-h})$ and $\gamma(0) > 0$.
For $h \geq 2$ and $t \in \mathbb{Z}$, set $\mathcal{H}_{t,h} := \mathrm{span}\{1, X_{t-1}, \ldots, X_{t-h+1}\}$ let $P_{t,h}$ be L^2-orthogonal projection onto $\mathcal{H}_{t,h}$.
Assume $\mathrm{Var}(X_t - P_{t,h}X_t) > 0$ and $\mathrm{Var}(X_{t-h} - P_{t,h}X_{t-h}) > 0$ $\forall h \geq 2,\ t \in \mathbb{Z}$.

(i) *Autocorrelation function*: $\rho : \mathbb{Z} \to [-1, 1]$, $\rho(h) := \frac{\gamma(h)}{\gamma(0)}$.
(ii) *Partial autocorrelation function*: $\alpha : \mathbb{N}_0 \to [-1, 1]$, $\alpha(0) := 1$, $\alpha(1) := \frac{\gamma(1)}{\gamma(0)}$, and $\alpha(h) := \mathrm{Corr}(X_t - P_{t,h}X_t, X_{t-h} - P_{t,h}X_{t-h})$ $\forall h \geq 2,\ t \in \mathbb{Z}$.
(iii) *Spectral density*: If $\sum_{h\in\mathbb{Z}} |\gamma(h)| < \infty$, $f_X : [-\pi, \pi] \to \mathbb{C}$, $f_X(\lambda) := (2\pi)^{-1} \sum_{h\in\mathbb{Z}} \gamma(h)e^{-ih\lambda}$ $\forall \lambda \in [-\pi, \pi]$.

Definition 7.1.3. *White Noise and Linear Process*
Let $\sigma^2 \geq 0$, $\mu \in \mathbb{R}$, and $(\psi_j)_{j\geq 0} \subset \mathbb{R}$ satisfy $\sum_{j=0}^{\infty} |\psi_j| < \infty$; all processes below are real-valued, indexed by $\mathbb{Z}$.

(i) *White noise*: $(\varepsilon_t)_{t\in\mathbb{Z}} \subset L^2$, $\mathbb{E}\varepsilon_t = 0$ and $\mathrm{Var}(\varepsilon_t) = \sigma^2$ $\forall t \in \mathbb{Z}$, and $\mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0$ $\forall s, t \in \mathbb{Z}$ with $s \neq t$.
(ii) *Linear process*: $(X_t)_{t\in\mathbb{Z}}$ satisfies $X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ $\forall t \in \mathbb{Z}$ for some $(\varepsilon_t)_{t\in\mathbb{Z}}$ satisfying (i).

Structural Commentary: $\sum_j |\psi_j| < \infty$ gives an L^2-stable one-sided innovation filter.

Example 7.1.4. *Autoregression of Order One*
Let $(\varepsilon_t)_{t\in\mathbb{Z}}$ be white noise, variance σ^2, $|\phi| < 1$. For $X_t = \phi X_{t-1} + \varepsilon_t$, stationary solution is $X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$.
Thus $\gamma(h) = \mathrm{Cov}(X_t, X_{t-h}) = \frac{\sigma^2\phi^{|h|}}{1-\phi^2}$, $\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^{|h|}$. Autocorrelation decays geometrically at rate $|\phi|$.

Remark 7.1.5. *Process-Level Prerequisites*
Time-series inference uses filtrations $(\mathcal{F}_t)$, adapted X_t, predictors $\mathbb{E}(X_{t+h} \mid \mathcal{F}_t)$, martingale differences ε_t, and stationarity or ergodicity assumptions.

7.2 ARIMA, State-Space, and Volatility Models

Classical time-series models combine deterministic filters, stochastic innovations, and likelihood-based estimation.

Definition 7.2.1. *ARMA Model*
Let $p, q \in \mathbb{N}_0$, $\phi_i \in \mathbb{R}$ $(1 \le i \le p)$, $\theta_j \in \mathbb{R}$ $(1 \le j \le q)$, and let B denote the backshift, $(BX)_t = X_{t-1}$.
Set $\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$ and $\theta(z) := 1 + \sum_{j=1}^{q} \theta_j z^j$; empty sums are zero.
$(X_t)_{t\in\mathbb{Z}}$ is ARMA(p, q) with innovations $(\varepsilon_t)_{t\in\mathbb{Z}}$ iff (ε_t) is white noise and $\phi(B)X_t = \theta(B)\varepsilon_t$ $\forall t \in \mathbb{Z}$.
Structural Commentary: Causal iff $\phi(z) \neq 0$ for $|z| \le 1$; invertible iff $\theta(z) \neq 0$ for $|z| \le 1$.
Key Insight: AR roots control recursion; MA roots control innovation recovery.

Definition 7.2.2. *ARIMA Model*
Let $p, q, d \in \mathbb{N}_0$, $\phi_i \in \mathbb{R}$ $(1 \le i \le p)$, $\theta_j \in \mathbb{R}$ $(1 \le j \le q)$, and let B denote the backshift.
Set $\Delta^d := (1 - B)^d$, $\phi(z) := 1 - \sum_{i=1}^{p} \phi_i z^i$, and $\theta(z) := 1 + \sum_{j=1}^{q} \theta_j z^j$.
$(X_t)_{t\in\mathbb{Z}}$ is ARIMA(p, d, q) with innovations $(\varepsilon_t)_{t\in\mathbb{Z}}$ iff (ε_t) is white noise and $\phi(B)\Delta^d X_t = \theta(B)\varepsilon_t$ $\forall t \in \mathbb{Z}$.
Structural Commentary: $Y_t := \Delta^d X_t$ satisfies ARMA(p, q) with the same innovations.
Key Insight: $d > 0$ shifts stationarity from X_t to $\Delta^d X_t$.

Definition 7.2.3. *Linear Gaussian State-Space Model*
Let $m, r, n \in \mathbb{N}$, $F \in \mathbb{R}^{m\times m}$, $G \in \mathbb{R}^{m\times r}$, $H \in \mathbb{R}^{n\times m}$, and $Q \in \mathbb{R}^{r\times r}$, $R \in \mathbb{R}^{n\times n}$ with $Q, R \succeq 0$.
Let (η_t) and (ε_t) be mutually independent Gaussian noises with $\eta_t \sim N(0, Q)$ and $\varepsilon_t \sim N(0, R)$.
Let α_0 be independent Gaussian. $(\alpha_t, Y_t)_{t\in\mathbb{N}_0}$ is *linear Gaussian state-space* iff
$\alpha_t \in \mathbb{R}^m$, $Y_t \in \mathbb{R}^n$, $\alpha_{t+1} = F\alpha_t + G\eta_t$, $Y_t = H\alpha_t + \varepsilon_t$ $\forall t \in \mathbb{N}_0$.
Key Insight: With $\mathcal{F}_t^Y := \sigma(Y_0, \dots, Y_t)$, Kalman filtering recurses $(a_t, P_t) := (\mathbb{E}(\alpha_t \mid \mathcal{F}_t^Y), \mathrm{Var}(\alpha_t \mid \mathcal{F}_t^Y))$.

Definition 7.2.4. *ARCH and GARCH Volatility*
Let $p, q \in \mathbb{N}_0$, $\omega > 0$, $\alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_p \ge 0$, and let $(z_t)_{t\in\mathbb{Z}}$ be standardized white noise.
All processes below are real-valued and indexed by $\mathbb{Z}$; set $A := \sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j$.

(i) *ARCH(q)*: $(X_t, \sigma_t)_{t\in\mathbb{Z}}$, $\sigma_t \ge 0$, satisfies $X_t = \sigma_t z_t$ and $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i X_{t-i}^2$ $\forall t \in \mathbb{Z}$.

(ii) *GARCH(p, q)*: $(X_t, \sigma_t)_{t\in\mathbb{Z}}$, $\sigma_t \ge 0$, satisfies $X_t = \sigma_t z_t$ and $\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i X_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2$ $\forall t \in \mathbb{Z}$.

Structural Commentary: $\omega > 0$, $\alpha_i, \beta_j \ge 0$ force $\sigma_t^2 \ge 0$; by stationary GARCH, $A < 1$ has $\mathbb{E}X_t^2 = \frac{\omega}{1-A}$.

Remark 7.2.5. *Likelihood by Filtering*
If one-step densities exist, likelihood filtering has $S_t = \mathcal{T}_\vartheta(S_{t-1}, W_t)$ and $\ell_T(\vartheta) = \sum_{t=1}^{T} \log p_\vartheta(W_t \mid S_{t-1})$.
Inference depends on stability of S_t.

8 Sequential Inference and Resampling

8.1 Resampling, Bootstrap, and Permutation Methods

Resampling methods approximate sampling distributions by recomputing statistics on data-generated pseudo-samples.

Definition 8.1.1. *Nonparametric Bootstrap*
For $X_{1:n} \sim P^{\otimes n}$, set $\widehat{P}_n := n^{-1} \sum_{i=1}^{n} \delta_{X_i}$. Conditional on $X_{1:n}$, let $X_{1:n}^* \sim \widehat{P}_n^{\otimes n}$ and $\widehat{P}_n^* := n^{-1} \sum_{i=1}^{n} \delta_{X_i^*}$.
For any law functional T, $\mathcal{L}(T(\widehat{P}_n^*) - T(\widehat{P}_n) \mid X_{1:n})$ approximates $\mathcal{L}(T(\widehat{P}_n) - T(P))$.
Key Insight: P is replaced by $\widehat{P}_n$ before pseudo-sampling.

Method 8.1.2. *Bootstrap Standard Error*
Let $\widehat{\psi} = s(X_{1:n})$ be observed statistic, $\mathcal{R}_*(X_{1:n})$ be the chosen bootstrap law on pseudo-samples, $B \ge 2$.
(i) *Resample*: $X_{1:n}^{*b} \sim \mathcal{R}_*(X_{1:n})$ for $b = 1, \dots, B$.

(ii) *Recompute*: $\widehat{\psi}^{*b} := s(X_{1:n}^{*b})$ for $b = 1, \dots, B$.

(iii) *Estimate*: $\widehat{\mathrm{se}}_*(\widehat{\psi}) := \{(B-1)^{-1} \sum_{b=1}^{B} (\widehat{\psi}^{*b} - \overline{\psi}^*)^2\}^{\frac{1}{2}}$, where $\overline{\psi}^* := B^{-1} \sum_{b=1}^{B} \widehat{\psi}^{*b}$.

Definition 8.1.3. *Parametric Bootstrap*
Let $\{P_\theta : \theta \in \Theta\}$ be parametric model for $X_{1:n} := (X_1, \dots, X_n)$, let $\widehat{\theta} := \widehat{\theta}(X_{1:n})$, and let S be a statistic. $X_1^*, \dots, X_n^* \mid X_{1:n} \sim P_{\widehat{\theta}}^{\otimes n}$, and $\mathcal{L}(S(X_{1:n}^*) \mid X_{1:n})$ is the fitted-model resampling law for $S(X_{1:n})$.
Key Insight: The model class $\{P_\theta\}$ is retained; only θ is replaced by $\widehat{\theta}$.

Method 8.1.4. *Parametric Bootstrap Standard Errors and Intervals*
Let $\{P_\theta : \theta \in \Theta\}$ be parametric model, $\widehat{\psi} := s(X_{1:n})$ estimate $\psi(\theta)$, fix $B \geq 2$, fix confidence level $1 - \alpha$.

(i) *Fit*: $\widehat{\theta} := \widehat{\theta}(X_{1:n})$.

(ii) *Simulate*: $X_{1:n}^{*b} \sim P_{\widehat{\theta}}^{\otimes n}$ and $\widehat{\psi}^{*b} := s(X_{1:n}^{*b})$ for $b = 1, \dots, B$.

(iii) *Standard error*: $\widehat{\mathrm{se}}_*(\widehat{\psi}) := \{(B-1)^{-1} \sum_{b=1}^{B} (\widehat{\psi}^{*b} - \overline{\psi}^*)^2\}^{\frac{1}{2}}$, where $\overline{\psi}^* := B^{-1} \sum_{b=1}^{B} \widehat{\psi}^{*b}$.

(iv) *Pivotal interval*: for empirical quantiles $q_{\frac{\alpha}{2}}^*$ and $q_{1-\frac{\alpha}{2}}^*$ of $\{\widehat{\psi}^{*b} - \widehat{\psi}\}_{b=1}^{B}$, use $[\widehat{\psi} - q_{1-\frac{\alpha}{2}}^*, \widehat{\psi} - q_{\frac{\alpha}{2}}^*]$.

The percentile interval uses empirical quantiles of $\{\widehat{\psi}^{*b}\}_{b=1}^{B}$ directly.
The pivotal and percentile intervals coincide when the bootstrap error law is symmetric about 0.

Remark 8.1.5. *Plug-In Monte Carlo*
The parametric bootstrap estimates model-dependent quantities by replacing the unknown parameter with its estimate before simulation. If $h(X, \theta)$ is a loss, error indicator, or other accuracy summary, then Monte Carlo from $P_{\widehat{\theta}}$ approximates $\mathbb{E}_\theta h(X, \theta)$ by the empirical average of $h(X^{*b}, \widehat{\theta})$.

Definition 8.1.6. *Permutation Test*
Let a finite group G act on the sample space $\mathcal{X}$, let P_0 satisfy $gX \overset{\text{dist}}{=} X$ under P_0 $\forall g \in G$, and let $T : \mathcal{X} \to \mathbb{R}$ be a test statistic. With U_G uniform on G and independent of X, $\mathcal{L}(T(U_G X) \mid X)$, the empirical law on $\{T(gX) : g \in G\}$, is the reference law for tail probabilities.
Key Insight: Calibration follows from the invariance $gX \overset{\text{dist}}{=} X$ under P_0.

Remark 8.1.7. *Validity Depends on the Resampling Law*
Bootstrap validity is asymptotic; permutation validity is finite-sample exact under invariance; both require the resampling law to match the target sampling variation.

8.2 Sequential Experiments, Filtering, and Online Inference

Sequential methods update decisions as observations arrive. They must account for optional stopping, adaptive sampling, and computational constraints.

Definition 8.2.1. *Sequential Test and Stopping Time*
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $I \subseteq \mathbb{N}_0$ be a time index, let $(\mathcal{F}_t)_{t \in I}$ be a filtration, and let $(D, \mathcal{D})$ be a decision space. For any $\tau : \Omega \to I \cup \{\infty\}$, set $\mathcal{F}_\tau := \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \ \forall t \in I\}$.

(i) *Stopping time*: $\tau : \Omega \to I \cup \{\infty\}$ satisfies $\{\tau \leq t\} \in \mathcal{F}_t$ $\forall t \in I$.

(ii) *Sequential test*: (τ, d) with τ a stopping time and $d : (\Omega, \mathcal{F}_\tau) \to (D, \mathcal{D})$ measurable.

Key Insight: Stop and decide using only pre-stopping information.

Definition 8.2.2. *Sequential Probability Ratio Test*
Let P_0 and P_1 be simple iid models with one-observation densities p_0 and p_1 for $(X_t)_{t \geq 1}$,
let $\Lambda_0 := 1$, $\Lambda_t := \prod_{i=1}^{t} \frac{p_1(X_i)}{p_0(X_i)}$, and choose $0 < A < B < \infty$.
$\tau := \inf\{t \geq 1 : \Lambda_t \in (0, A] \cup [B, \infty)\}$, with terminal decision $d_\tau = P_0$ on $\{\Lambda_\tau \leq A\}$ and $d_\tau = P_1$ on $\{\Lambda_\tau \geq B\}$.
Key Insight: First exit of $\log \Lambda_t$ from $(\log A, \log B)$ converts cumulative likelihood evidence into a stopping rule.

Definition 8.2.3. *Online Estimator*
Let $(X_t)_{t \geq 1}$ be observations in $\mathcal{X}$, let $\widehat{\theta}_0 \in \Theta \subseteq \mathbb{R}^d$, let $(a_t)_{t \geq 0}$ be positive step sizes, and let $H : \Theta \times \mathcal{X} \to \mathbb{R}^d$.
$\widehat{\theta}_{t+1} := \widehat{\theta}_t + a_t H(\widehat{\theta}_t, X_{t+1})$ for $t \geq 0$.
Structural Commentary: $(\widehat{\theta}_t)$ is adapted to $\mathcal{F}_t := \sigma(\widehat{\theta}_0, X_1, \dots, X_t)$.
Key Insight: Under stationarity, analyze the recursion through $h(\theta) := \mathbb{E} H(\theta, X)$.

Definition 8.2.4. *Filtering Problem*
Let $(Z_t, Y_t)_{t \geq 1}$ be a hidden-state model with state space $(E, \mathcal{E})$, and set $\mathcal{F}_t^Y := \sigma(Y_1, \dots, Y_t)$.

For $A \in \mathcal{E}$, $\pi_t(A) := \mathbb{P}(Z_t \in A \mid \mathcal{F}_t^Y)$; the task is recursive propagation or approximation of $(\pi_t)_{t\geq 1}$.
Structural Commentary: (π_t) is an $\mathcal{F}_t^Y$-adapted probability-kernel-valued process.
Key Insight: Conditioning converts hidden dynamics into an observable belief-state recursion.

Remark 8.2.5. *Sequential Cross-References*
Sequential inference uses filtrations and martingales from stochastic processes, stability of recursions from dynamical systems, and simulation methods from Monte Carlo computation.

8.3 Two-Sample Resampling Methods

Two-sample resampling procedures must preserve the source of randomness relevant to the question. Bootstrap procedures usually resample within groups to estimate sampling variability under the fitted empirical distributions; permutation procedures pool labels when the null asserts exchangeability.

Method 8.3.1. *Two-Sample Nonparametric Bootstrap*

Let $x_{1:n}$ and $y_{1:m}$ be independent samples from F and G, let $\widehat{F}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i}$ and $\widehat{G}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j}$.

Let $T(F,G)$ be a two-sample functional, and fix $B \geq 2$.

(i) *Resample*: $x_{1:n}^{*b} \sim \widehat{F}_n^{\otimes n}$ and $y_{1:m}^{*b} \sim \widehat{G}_m^{\otimes m}$ independently for $b = 1,\dots,B$.

(ii) *Recompute*: with $\widehat{F}_n^{*b} := n^{-1}\sum_{i=1}^{n}\delta_{x_i^{*b}}$ and $\widehat{G}_m^{*b} := m^{-1}\sum_{j=1}^{m}\delta_{y_j^{*b}}$, set $\widehat{T}^{*b} := T(\widehat{F}_n^{*b}, \widehat{G}_m^{*b})$.

(iii) *Summarize*: $s_T^* := \mathrm{sd}\{\widehat{T}^{*b} : 1 \leq b \leq B\}$; $I_{1-\alpha}^* := [q_{\frac{\alpha}{2}}^*, q_{1-\frac{\alpha}{2}}^*]$, where q_γ^* is the empirical γ-quantile.

This is the two-sample version of Method 8.1.2.

Example 8.3.2. *Bootstrap for $\pi = \mathbb{P}(X < Y)$*
Let $x_{1:n} \sim F^{\otimes n}$ and $y_{1:m} \sim G^{\otimes m}$ independently.
Set $\widehat{F}_n := n^{-1}\sum_{i=1}^{n}\delta_{x_i}$, $\widehat{G}_m := m^{-1}\sum_{j=1}^{m}\delta_{y_j}$, and $\widehat{\pi} := (mn)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbf{1}_{\{x_i<y_j\}}$.
For $b = 1,\dots,B$, draw $x_{1:n}^{*b} \sim \widehat{F}_n^{\otimes n}$ and $y_{1:m}^{*b} \sim \widehat{G}_m^{\otimes m}$ independently.
Set $\widehat{\pi}^{*b} := (mn)^{-1}\sum_{i=1}^{n}\sum_{j=1}^{m}\mathbf{1}_{\{x_i^{*b}<y_j^{*b}\}}$, and use $s_\pi^* := \mathrm{sd}\{\widehat{\pi}^{*b} : 1 \leq b \leq B\}$.
Key Insight: Separate resampling targets $F \otimes G$ variation; rank-sum calibration imposes $F = G$.

Method 8.3.3. *Permutation Test for a Difference of Means*

Let $x_{1:n}$ be control responses, let $y_{1:m}$ be treatment responses, set $\overline{x} := n^{-1}\sum_{i=1}^{n}x_i$ and $\overline{y} := m^{-1}\sum_{j=1}^{m}y_j$.
Let $z_{1:N}$ with $N = n+m$ be the pooled responses, and set $T_{\mathrm{obs}} := \overline{y} - \overline{x}$.
Let H_0 denote no treatment effect, set $\mathcal{S}_{m,N} := \{S \subseteq \{1,\dots,N\} : |S| = m\}$, and use $S \mid H_0 \sim \mathrm{Unif}(\mathcal{S}_{m,N})$.

(i) *Relabel*: $S \in \mathcal{S}_{m,N}$, by enumeration or Monte Carlo draws from $\mathrm{Unif}(\mathcal{S}_{m,N})$.

(ii) *Recompute*: $T_S^* := m^{-1}\sum_{i\in S}z_i - n^{-1}\sum_{i\notin S}z_i$.

(iii) *Calibrate*: $p_{\mathrm{exact}} := \binom{N}{m}^{-1}\#\{S \in \mathcal{S}_{m,N} : |T_S^*| \geq |T_{\mathrm{obs}}|\}$, Monte Carlo approximation if $\binom{N}{m}$ large.

Remark 8.3.4. *Permutation and Rank Tests*
The permutation test for means and the Mann–Whitney test both use the randomization principle that, under the null, labels are exchangeable. They differ in the statistic: the permutation mean test uses the numerical values and is sensitive to outliers, while the Mann–Whitney test first replaces values by ranks.

Method 8.3.5. *Bootstrap for a Hodges–Lehmann Shift*
For independent samples $x_{1:n}$ and $y_{1:m}$ and $B \geq 2$, set $\widehat{\Delta}_{\mathrm{HL}} := \mathrm{median}\{y_j - x_i : 1 \leq i \leq n,\ 1 \leq j \leq m\}$, resample the two groups separately for $b = 1,\dots,B$, and use the empirical spread or quantiles of the recomputed medians $\widehat{\Delta}_{\mathrm{HL}}^{*b}$ for a standard error or interval.

Part XVI
Statistical Data Analysis and Applied Models

Applied statistical analysis organizes data summaries, sampling designs, contingency tables, designed experiments, and regression models into reusable workflows. The inferential tools of Part 15 supply the testing, estimation, and uncertainty language used here.

1 Survey Sampling

1.1 Finite Population Parameters and Simple Random Sampling

Survey sampling treats a finite population as fixed and the sample-selection mechanism as random. The inferential target is often a population mean, total, or proportion, while the random variables come from the sampling design.

Definition 1.1.1. *Finite Population Parameters*
Let $U := \{1, \dots, N\}$ and fix $x_i \in \mathbb{R}$, $i \in U$.

(i) *Population mean*: $\mu := \frac{1}{N} \sum_{i=1}^{N} x_i$.

(ii) *Population total*: $\tau := \sum_{i=1}^{N} x_i = N\mu$.

(iii) *Population variance*: $\sigma^2 := \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^{N} x_i^2 - \mu^2$.

(iv) *Population proportion*: $\{x_i : i \in U\} \subseteq \{0, 1\} \implies p := \mu = \frac{1}{N} \sum_{i=1}^{N} x_i$ and $\sigma^2 = p(1-p)$.

Definition 1.1.2. *Simple Random Sampling*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, and fix $x_i \in \mathbb{R}$, $i \in U$.
Let S be a random n-element subset of U; for a random ordering $J_1, \dots, J_n$ of S, set $X_k := x_{J_k}$.

(i) *Simple random sample*: $\mathbb{P}(S = s) = \frac{1}{\binom{N}{n}}$, $\forall s \subseteq U$ with $|s| = n$.

(ii) *Sample mean*: $\overline{X} := \frac{1}{n} \sum_{i=1}^{n} X_i$.

(iii) *Total estimator*: $T := N\overline{X}$.

(iv) *Sample proportion*: $\{x_i : i \in U\} \subseteq \{0, 1\} \implies \widehat{p} := \overline{X}$.

Lemma 1.1.3. *Law of One Sample Position*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Let $\mu := \frac{1}{N} \sum_{a=1}^{N} x_a$, $\sigma^2 := \frac{1}{N} \sum_{a=1}^{N} (x_a - \mu)^2$, $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
If $(J_1, \dots, J_n)$ is a random ordering of S, set $X_k := x_{J_k}$.
Let $\zeta_1, \dots, \zeta_m$ be distinct values in $\{x_a : a \in U\}$, let $n_j := \#\{a \in U : x_a = \zeta_j\}$. For each $k \in \{1, \dots, n\}$:

(i) Distribution: the selected value X_k has the finite-population law $\mathbb{P}(X_k = \zeta_j) = \frac{n_j}{N}$ for $j = 1, \dots, m$.
(ii) Moments: the finite-population law has mean $\mathbb{E}X_k = \mu$ and variance $\mathrm{Var}(X_k) = \sigma^2$.

Proof.

(i) For every $a \in U$, $\mathbb{P}(J_k = a) = \frac{1}{N}$, so $\mathbb{P}(X_k = \zeta_j) = \sum_{a : x_a = \zeta_j} \mathbb{P}(J_k = a) = \frac{n_j}{N}$.

(ii) From (i), $\mathbb{E}X_k = \frac{1}{N} \sum_{a=1}^{N} x_a = \mu$. By finite-population law, $\mathrm{Var}(X_k) = \frac{1}{N} \sum_{a=1}^{N} (x_a - \mu)^2 = \sigma^2$.

Structural Commentary: Each sample position is marginally uniform on U.
Key Insight: Without replacement changes joint dependence, not the one-position marginal distribution. □

Theorem 1.1.4. *Unbiased SRS Estimators*
Let $U := \{1, \dots, N\}$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N} \sum_{a=1}^{N} x_a$, $\tau := \sum_{a=1}^{N} x_a = N\mu$. Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
Given random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n} \sum_{k=1}^{n} X_k$, and $T := N\overline{X}$. Then:

(i) Mean estimator: the sample mean is design-unbiased for the finite-population mean, $\mathbb{E}\overline{X} = \mu$.

(ii) Total estimator: the expansion estimator is design-unbiased for the finite-population total, $\mathbb{E}T = \tau$.

Proof.

(i) For each k, $\mathbb{P}(J_k = a) = \frac{1}{N}\ \forall a \in U$, so $\mathbb{E}X_k = \frac{1}{N}\sum_{a=1}^{N} x_a = \mu$. By linearity $\mathbb{E}\overline{X} = \frac{1}{n}\sum_{k=1}^{n} \mathbb{E}X_k = \mu$.

(ii) Since $T = N\overline{X}$, $\mathbb{E}T = N\mathbb{E}\overline{X} = N\mu = \tau$.

Structural Commentary: $T = N\overline{X}$ rescales the mean estimate.
Key Insight: Equal position marginals give unbiasedness; independence is unnecessary. □

Lemma 1.1.5. *Negative Dependence Without Replacement*
Let $U := \{1, \dots, N\}$, $2 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\tau := \sum_{a=1}^{N} x_a$, $\sigma^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu)^2$. Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
For random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$. If $i, j \in \{1, \dots, n\}$ and $i \ne j$, then $\mathrm{Cov}(X_i, X_j) = -\frac{\sigma^2}{N-1}$.

Proof. Let $(Y_1, \dots, Y_N)$ be a uniformly random ordering of $(x_1, \dots, x_N)$.
Each Y_a has variance σ^2, and symmetry gives $\mathrm{Cov}(Y_a, Y_b) = \gamma$ for $a \ne b$.
Since $\sum_{a=1}^{N} Y_a = \tau$ is fixed, $0 = \mathrm{Var}(\sum_{a=1}^{N} Y_a) = N\sigma^2 + N(N-1)\gamma$.
Thus $\gamma = -\frac{\sigma^2}{N-1}$, and (X_i, X_j) has same law as any two distinct positions in full random ordering.
Structural Commentary: A fixed total forces negative pairwise covariance.
Key Insight: Without replacement, one large draw leaves less for the rest. □

Theorem 1.1.6. *Variance and Finite Population Correction*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$.
Set $\mu := \frac{1}{N}\sum_{a=1}^{N} x_a$ and $\sigma^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu)^2$. Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$.
For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, and $T := N\overline{X}$. Then:

(i) Mean variance: the sample mean has design variance $\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right)$.
(ii) Total variance: the total estimator has design variance $\mathrm{Var}(T) = N^2\,\mathrm{Var}(\overline{X})$.

Proof.

(i) Each X_k has variance σ^2. Let $(Y_1, \dots, Y_N)$ be a uniformly random ordering of $(x_1, \dots, x_N)$, set $\tau := \sum_{a=1}^{N} x_a$, and let γ be the common covariance of Y_a, Y_b for $a \ne b$.
Since $\sum_{a=1}^{N} Y_a = \tau$, $0 = \mathrm{Var}(\sum_{a=1}^{N} Y_a) = N\sigma^2 + N(N-1)\gamma$, so $\gamma = -\frac{\sigma^2}{N-1}$.
Therefore $\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\left(n\sigma^2 + n(n-1)\left[-\frac{\sigma^2}{N-1}\right]\right) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right)$.

(ii) The total estimator satisfies $T = N\overline{X}$, so $\mathrm{Var}(T) = \mathrm{Var}(N\overline{X}) = N^2\,\mathrm{Var}(\overline{X})$.

Structural Commentary: The factor $\frac{N-n}{N-1}$ is the *finite population correction*.
Key Insight: Averaging gives $\frac{1}{n}$ scaling; without replacement subtracts negative covariance. □

Corollary 1.1.7. *Estimated Standard Errors*
Let $U := \{1, \dots, N\}$, $1 < n \le N$, fix $x_a \in \mathbb{R}$ for $a \in U$, and let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$. For a random ordering $(J_1, \dots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $T := N\overline{X}$, and $s^2 := \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \overline{X})^2$.
Then practical plug-in estimated variances are:

(i) Mean: the plug-in estimated variance of the sample mean is $s^2_{\overline{X}} = \frac{s^2}{n}\left(1 - \frac{n}{N}\right)$.
(ii) Total: the plug-in estimated variance of the total estimator is $s^2_T = N^2 s^2_{\overline{X}}$.
(iii) Proportion: if $x_a \in \{0, 1\}\ \forall a \in U$ and $\widehat{p} := \overline{X}$, then plug-in estimated variance of $\widehat{p}$ is $s^2_{\widehat{p}} = \frac{\widehat{p}(1-\widehat{p})}{n-1}\left(1 - \frac{n}{N}\right)$.

Proof. Set $c_N := 1 - \frac{n}{N}$.

(i) Replace unknown finite-population variance by s^2, correction by c_N gives $s^2_{\overline{X}} = (\frac{s^2}{n})c_N = \frac{s^2}{n}(1 - \frac{n}{N})$.
(ii) Since $T = N\overline{X}$, variance scales quadratically, so $s^2_T = N^2 s^2_{\overline{X}}$.

(iii) If $X_k \in \{0,1\}$ and $\widehat{p} = \overline{X}$, then $s^2 = \frac{1}{n-1}\sum_{k=1}^{n}(X_k - \widehat{p})^2 = \frac{1}{n-1}(n\widehat{p} - n\widehat{p}^2) = \frac{n\widehat{p}(1-\widehat{p})}{n-1}$.

Substituting this into (i) gives $s_{\widehat{p}}^2 = \frac{\widehat{p}(1-\widehat{p})c_N}{n-1} = \frac{\widehat{p}(1-\widehat{p})}{n-1}(1-\frac{n}{N})$.

Structural Commentary: Plug-in SEs use s^2 and the $1-\frac{n}{N}$ correction.
Key Insight: Binary-data variance depends only on $\widehat{p}$. □

Example 1.1.8. *Finite Population Correction in a Hospital Survey*
For a finite population of $N = 393$ hospitals, let x_i denote discharges and suppose the finite-population standard deviation is $\sigma \approx 589.7$. Under simple random sampling without replacement with $n = 32$, the sample-mean standard deviation is $\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}\sqrt{1-\frac{n-1}{N-1}} \approx \frac{589.7}{\sqrt{32}}\sqrt{1-\frac{31}{392}} \approx 100.0$. Since $\frac{n}{N} = \frac{32}{393}$ is small, the finite-population correction is near 1, and the standard error is mainly governed by the $\frac{1}{\sqrt{n}}$ averaging scale.

Method 1.1.9. *Sequential Simple Random Sampling*
Fix target size $n \geq 1$, assume terminal stream length is $N \geq n$, N unknown while streaming.
Population units arrive sequentially as $1, \ldots, N$. Let $S_t \subseteq \{1, \ldots, t\}$ denote current reservoir after t arrivals.

(i) *Initialize*: for $1 \leq t \leq n$, set $S_t = \{1, \ldots, t\}$.
(ii) *Admit*: for $t > n$, draw $B_t \sim \text{Bernoulli}(\frac{n}{t})$.
(iii) *Update*: if $B_t = 0$, set $S_t = S_{t-1}$; if $B_t = 1$, draw $J_t \mid S_{t-1} \sim \text{Unif}(S_{t-1})$ and set $S_t = (S_{t-1} \setminus \{J_t\}) \cup \{t\}$.
(iv) *Output*: after the stream ends, return S_N.

Then for every $t \geq n$, $S_t \sim \text{Unif}\{s \subseteq \{1, \ldots, t\} : |s| = n\}$.
Hence S_N is a simple random sample without replacement.

1.2 Normal Approximation and Confidence Intervals

The exact design formulas in Theorem 1.1.6 give the center and scale of the sampling distribution. For large samples, a normal approximation turns those formulas into error probabilities and confidence intervals.

Method 1.2.1. *Normal Approximation for SRS*

(i) Let $U := \{1, \ldots, N\}$, $N > 1$, $1 \leq n \leq N$, $x_i \in \mathbb{R}$ for $i \in U$, and $S \sim \text{Unif}\{s \subseteq U : |s| = n\}$.
(ii) For a random ordering $(J_1, \ldots, J_n)$ of S, set $X_k := x_{J_k}$, $\overline{X} := n^{-1}\sum_{k=1}^{n} X_k$, $T := N\overline{X}$, $\mu := N^{-1}\sum_{i=1}^{N} x_i$, $\sigma^2 := N^{-1}\sum_{i=1}^{N}(x_i - \mu)^2$, $\tau := N\mu$, and $\sigma_{\overline{X}}^2 := \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$.
(iii) Let Φ be the standard normal cdf. For large n, use $\overline{X} \approx N(\mu, \sigma_{\overline{X}}^2)$.
(iv) If $\sigma_{\overline{X}} > 0$, then $\mathbb{P}(|\overline{X} - \mu| \leq \delta) \approx 2\Phi(\frac{\delta}{\sigma_{\overline{X}}}) - 1$ for $\delta > 0$, and $T - \tau = N(\overline{X} - \mu) \approx N(0, N^2\sigma_{\overline{X}}^2)$.

Definition 1.2.2. *Normal Right-Tail Quantile*
For $0 < \alpha < 1$ and $Z \sim N(0,1)$, $z(\alpha) := \Phi^{-1}(1-\alpha)$, so $\mathbb{P}(Z > z(\alpha)) = \alpha$.

Method 1.2.3. *Approximate SRS Confidence Intervals*

(i) Let S be an SRS without replacement, with targets μ, τ and estimators $\overline{X}, T$.
(ii) Fix $0 < \alpha < 1$, let Φ be the standard normal cdf, set $z(\frac{\alpha}{2}) := \Phi^{-1}(1-\frac{\alpha}{2})$, and use plug-in SEs $s_{\overline{X}}, s_T$, and, when binary, $s_{\widehat{p}}$.
(iii) $\text{CI}_{1-\alpha}(\mu) = \overline{X} \pm z(\frac{\alpha}{2})s_{\overline{X}}$ and $\text{CI}_{1-\alpha}(\tau) = T \pm z(\frac{\alpha}{2})s_T$.
(iv) In the dichotomous case, with population proportion p and sample proportion $\widehat{p}$, use $\text{CI}_{1-\alpha}(p) = \widehat{p} \pm z(\frac{\alpha}{2})s_{\widehat{p}}$.
(v) Intervals are random before sampling; after observation, parameters are fixed and coverage is procedural.

Example 1.2.4. *Generic SRS Intervals*
Suppose finite population has $N = 8000$ units. Simple random sample of $n = 100$ gives $\overline{X} = 1.6$ for variable with sample standard deviation $s = 0.8$. Use $z(0.025) \approx 1.96$, suppose sample gives $\widehat{p} = 0.12$ for binary attribute.

(i) *Mean*: $s_{\overline{X}} = \frac{s}{\sqrt{n}}\sqrt{1-\frac{n}{N}} = \frac{0.8}{10}\sqrt{1-\frac{100}{8000}} \approx 0.0795$, then 95% CI for μ is $1.6 \pm 1.96(0.0795) = (1.44, 1.76)$.
(ii) *Total*: $T = N\overline{X} = 8000(1.6) = 12800$ and $s_T = Ns_{\overline{X}} \approx 636$, then 95% CI for τ is $(11553, 14047)$.
(iii) *Proportion*: $s_{\widehat{p}} = \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n-1}\left(1-\frac{n}{N}\right)} = \sqrt{\frac{0.12(0.88)}{99}\left(1-\frac{100}{8000}\right)} \approx 0.0325$, then 95% CI for p is $(0.056, 0.184)$. The estimated attribute total is $N\widehat{p} = 960$ with $Ns_{\widehat{p}} \approx 260$ and interval $(451, 1469)$.

These intervals quantify SRS uncertainty for fixed finite-population quantities.

Method 1.2.5. *Sample Size from Desired Half-Width*

(i) Fix $0 < \alpha < 1$, half-width $h > 0$, planning SD $\sigma > 0$, and planned size n.
Let Φ be the standard normal cdf and set $z(\frac{\alpha}{2}) := \Phi^{-1}(1-\frac{\alpha}{2})$.

(ii) Ignoring the finite population correction, $h \approx z(\frac{\alpha}{2})\frac{\sigma}{\sqrt{n}}$ gives $n \approx \left(\frac{z(\frac{\alpha}{2})\sigma}{h}\right)^2$.

(iii) For total based on population proportion over N units, use planning value $p_0 \in [0,1]$, SE $N\sqrt{\frac{p_0(1-p_0)}{n}}$.

(iv) With negligible finite population correction, halving interval width requires roughly $4n$ observations.

1.3 Ratio Estimation

Ratio estimation uses an auxiliary variable whose population mean or total is known. It is especially useful when the auxiliary variable is strongly correlated with the target variable.

Definition 1.3.1. *Population and Sample Ratios*
Let $U := \{1, \dots, N\}$ and fixed pairs $(x_i, y_i) \in \mathbb{R}^2$, $i \in U$.
Set $\mu_x := \frac{1}{N}\sum_{i=1}^{N} x_i$, $\mu_y := \frac{1}{N}\sum_{i=1}^{N} y_i$, $\tau_x := \sum_{i=1}^{N} x_i$, and $\tau_y := \sum_{i=1}^{N} y_i$, with $\mu_x \neq 0$.
For SRS pairs $(X_i, Y_i)_{i=1}^n$, set $\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$ and $\overline{Y} := \frac{1}{n}\sum_{i=1}^{n} Y_i$, with $\overline{X} \neq 0$.

(i) *Population ratio*: $r := \sum_{i=1}^{N} y_i / \sum_{i=1}^{N} x_i = \frac{\mu_y}{\mu_x}$.

(ii) *Sample ratio*: $R := \frac{\overline{Y}}{\overline{X}}$.

(iii) *Ratio mean estimator*: $\overline{Y}_R := \mu_x R$ when μ_x is known.

(iv) *Ratio total estimator*: $T_R := \tau_x R$ when τ_x is known.

Key Insight: $\{i \in U : x_i = 0\} = \emptyset$ does not imply $r = \frac{1}{N}\sum_{i=1}^{N} \frac{y_i}{x_i}$.

Definition 1.3.2. *Population Covariance and Correlation*
Let $U := \{1, \dots, N\}$ and fixed pairs $(x_i, y_i) \in \mathbb{R}^2$, $i \in U$.
Set $\mu_x := \frac{1}{N}\sum_{i=1}^{N} x_i$, $\mu_y := \frac{1}{N}\sum_{i=1}^{N} y_i$, $\sigma_x^2 := \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_x)^2$, and $\sigma_y^2 := \frac{1}{N}\sum_{i=1}^{N}(y_i - \mu_y)^2$.
For sample pairs $(X_i, Y_i)_{i=1}^n$, set $\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$ and $\overline{Y} := \frac{1}{n}\sum_{i=1}^{n} Y_i$.

(i) *Population covariance*: $\sigma_{xy} := \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)$.

(ii) *Population correlation*: $\rho := \frac{\sigma_{xy}}{\sigma_x \sigma_y}$ when $\sigma_x, \sigma_y > 0$.

(iii) *Sample covariance*: $s_{xy} := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$.

Theorem 1.3.3. *Approximate Moments of the Ratio Estimator*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 \le n \le N$, and let $(x_a, y_a) \in \mathbb{R}^2$ be fixed for $a \in U$.
Set $\lambda_n := \frac{1}{n}(1 - \frac{n-1}{N-1})$, $\mu_x := \frac{1}{N}\sum_{a=1}^{N} x_a$, $\mu_y := \frac{1}{N}\sum_{a=1}^{N} y_a$, $r := \frac{\mu_y}{\mu_x}$
with $\mu_x \neq 0$, $\sigma_x^2 := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu_x)^2$, $\sigma_y^2 := \frac{1}{N}\sum_{a=1}^{N}(y_a - \mu_y)^2$, and $\sigma_{xy} := \frac{1}{N}\sum_{a=1}^{N}(x_a - \mu_x)(y_a - \mu_y)$.
Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$, let $(J_1, \dots, J_n)$ be a random ordering of S, set $(X_k, Y_k) := (x_{J_k}, y_{J_k})$, $\overline{X} := \frac{1}{n}\sum_{k=1}^{n} X_k$, $\overline{Y} := \frac{1}{n}\sum_{k=1}^{n} Y_k$, and $R := \frac{\overline{Y}}{\overline{X}}$ on $\{\overline{X} \neq 0\}$. The subscript d denotes design moments.

(i) Variance: first-order Taylor gives $\mathrm{Var}_d(R) \approx \frac{\lambda_n}{\mu_x^2}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$.

(ii) Mean: second-order Taylor gives $\mathbb{E}_d R \approx r + \frac{\lambda_n}{\mu_x^2}(r\sigma_x^2 - \sigma_{xy})$.

Proof. Let $(A_q, B_q)_{q=1}^N$ be a uniformly random ordering of the population pairs; the ordered sample has the law of the first n entries. For $q \neq r$, write $\gamma_x := \mathrm{Cov}_d(A_q, A_r)$, $\gamma_y := \mathrm{Cov}_d(B_q, B_r)$, and $\gamma_{xy} := \mathrm{Cov}_d(A_q, B_r)$.
The fixed sums give $0 = \mathrm{Var}_d(\sum_{q=1}^{N} A_q) = N\sigma_x^2 + N(N-1)\gamma_x$, $0 = \mathrm{Var}_d(\sum_{q=1}^{N} B_q) = N\sigma_y^2 + N(N-1)\gamma_y$,
and $0 = \mathrm{Cov}_d(\sum_{q=1}^{N} A_q, \sum_{q=1}^{N} B_q) = N\sigma_{xy} + N(N-1)\gamma_{xy}$.
Hence $\gamma_x = -\frac{\sigma_x^2}{N-1}$, $\gamma_y = -\frac{\sigma_y^2}{N-1}$, $\gamma_{xy} = -\frac{\sigma_{xy}}{N-1}$, $V_x := \mathrm{Var}_d(\overline{X}) = \lambda_n \sigma_x^2$, $V_y := \mathrm{Var}_d(\overline{Y}) = \lambda_n \sigma_y^2$,

and $C_{xy} := \mathrm{Cov}_d(\overline{X}, \overline{Y}) = \lambda_n \sigma_{xy}$. Let $\Sigma := \begin{pmatrix} V_x & C_{xy} \\ C_{xy} & V_y \end{pmatrix}$.

For $g(x,y) = \frac{y}{x}$ and $\mu := (\mu_x, \mu_y)$, $g(\mu) = r$, $\nabla g(\mu) = (-\frac{r}{\mu_x}, \frac{1}{\mu_x})^\top$, and $H_g(\mu) = \begin{pmatrix} 2r/\mu_x^2 & -1/\mu_x^2 \\ -1/\mu_x^2 & 0 \end{pmatrix}$.

(i) *Variance*: $\mathrm{Var}_d(R) \approx \nabla g(\mu)^\top \Sigma \nabla g(\mu) = \frac{\lambda_n}{\mu_x^2}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$.

(ii) *Mean*: $\mathbb{E}_d R \approx g(\mu) + \frac{1}{2}\,\mathrm{tr}(H_g(\mu)\Sigma) = r + \frac{\lambda_n}{\mu_x^2}(r\sigma_x^2 - \sigma_{xy})$.

Structural Commentary: The ratio estimator is locally linear for variance and locally quadratic for bias.
Key Insight: Ratio variance is scaled residual variation in $Y - rX$. □

Corollary 1.3.4. *Estimated Standard Error of a Ratio*
Let $U := \{1, \dots, N\}$, $N > 1$, $1 < n \le N$, and let $(x_a, y_a) \in \mathbb{R}^2$ be fixed for $a \in U$.
Set $\lambda_n := \frac{1}{n}(1 - \frac{n-1}{N-1})$, $\mu_x := \frac{1}{N}\sum_{a=1}^N x_a$, $\tau_x := \sum_{a=1}^N x_a$, $\tau_y := \sum_{a=1}^N y_a$, and $r := \frac{\tau_y}{\tau_x}$ with $\tau_x \neq 0$.
Let $S \sim \mathrm{Unif}\{s \subseteq U : |s| = n\}$. For a random ordering $(J_1, \dots, J_n)$ of S,
set $(X_i, Y_i) := (x_{J_i}, y_{J_i})$, $\overline{X} := \frac{1}{n}\sum_{i=1}^n X_i$, and $\overline{Y} := \frac{1}{n}\sum_{i=1}^n Y_i$.
Assume $\overline{X} \neq 0$ and set $R := \frac{\overline{Y}}{\overline{X}}$, $s_x^2 := \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})^2$, $s_y^2 := \frac{1}{n-1}\sum_{i=1}^n (Y_i - \overline{Y})^2$, $s_{xy} := \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})(Y_i - \overline{Y})$, and $q_R := R^2 s_x^2 + s_y^2 - 2R s_{xy}$. If $0 < \alpha < 1$ and $z(\frac{\alpha}{2}) := \Phi^{-1}(1 - \frac{\alpha}{2})$ for the standard normal cdf Φ, then:

(i) Ratio variance: $s_R^2 := \frac{\lambda_n}{\overline{X}^2} q_R$ estimates $\mathrm{Var}_d(R)$.
(ii) Ratio interval: $\mathrm{CI}_{1-\alpha}(r) := R \pm z(\frac{\alpha}{2}) s_R$.
(iii) Mean and total: for $\overline{Y}_R := \mu_x R$ and $T_R := \tau_x R$, $\mathrm{SE}(\overline{Y}_R) = |\mu_x|\, s_R$ and $\mathrm{SE}(T_R) = |\tau_x|\, s_R$ when the corresponding auxiliary constant is known.

Proof.

(i) The plug-in map $(\mu_x, r, \sigma_x^2, \sigma_y^2, \sigma_{xy}) \mapsto (\overline{X}, R, s_x^2, s_y^2, s_{xy})$ in Theorem 1.3.3 yields $\widehat{\mathrm{Var}}_d(R) = \frac{\lambda_n}{\overline{X}^2}(R^2 s_x^2 + s_y^2 - 2R s_{xy}) = s_R^2$.

(ii) If $Z_R := (R - r)/s_R \approx \mathcal{N}(0,1)$, then $\mathbb{P}_d(|Z_R| \le z(\frac{\alpha}{2})) \approx 1 - \alpha$, equivalently $r \in R \pm z(\frac{\alpha}{2}) s_R$.

(iii) For known c, $\widehat{\mathrm{Var}}_d(cR) = c^2 s_R^2$, so $\mathrm{SE}(cR) = |c|\, s_R$; set $c = \mu_x$ and $c = \tau_x$.

Structural Commentary: Plug in sample moments to estimate the ratio variance.
Key Insight: Ratio uncertainty is residual uncertainty in $Y - RX$. □

Method 1.3.5. *When the Ratio Estimate Helps*
Let $U = \{1, \dots, N\}$ contain fixed pairs (x_i, y_i) with $\mu_x, \mu_y > 0$, $\sigma_x, \sigma_y > 0$, σ_{xy}, $\rho := \frac{\sigma_{xy}}{\sigma_x \sigma_y}$, and $r := \frac{\mu_y}{\mu_x}$.

For an SRS $(X_k, Y_k)_{k=1}^n$, set $\overline{X} := \frac{1}{n}\sum_{k=1}^n X_k$, $\overline{Y} := \frac{1}{n}\sum_{k=1}^n Y_k$, $R := \frac{\overline{Y}}{\overline{X}}$, and $\overline{Y}_R := \mu_x R$.

Ignoring finite-population corrections:

(i) *Variances*: $V_0 := \mathrm{Var}(\overline{Y}) = \frac{\sigma_y^2}{n}$ and $V_R^* := \frac{1}{n}(r^2\sigma_x^2 + \sigma_y^2 - 2r\sigma_{xy})$ approximate $\mathrm{Var}(\overline{Y}_R)$.
(ii) *Improvement*: $V_R^* < V_0 \iff 2\sigma_{xy} > r\sigma_x^2 \iff 2\rho\sigma_y > r\sigma_x$.
(iii) *Coefficient form*: with $C_x := \frac{\sigma_x}{\mu_x}$ and $C_y := \frac{\sigma_y}{\mu_y}$, $V_R^* < V_0 \iff \rho > \frac{1}{2}\frac{C_x}{C_y}$.
(iv) *Favorable case*: large positive ρ and small C_x/C_y.

Example 1.3.6. *Mortgage-Payment Ratio*
For an SRS of $n = 100$ recent house buyers, let Y_i be monthly mortgage payment and X_i gross income. Suppose $\overline{X} = 3100$, $\overline{Y} = 868$, $s_x = 1200$, $s_y = 250$, and $\widehat{\rho} := s_{xy}/(s_x s_y) = 0.85$.

(i) *Ratio*: $R = \frac{\overline{Y}}{\overline{X}} = \frac{868}{3100} = 0.28$.
(ii) *Variance input*: $q := R^2 s_x^2 + s_y^2 - 2R\widehat{\rho} s_x s_y \approx 32596$.
(iii) *Standard error*: ignoring the finite-population correction, $s_R \approx \frac{\sqrt{q}}{\sqrt{100}\,\overline{X}} \approx 0.00582$.
(iv) *Interval*: $\mathrm{CI}_{0.95}(r) = R \pm 1.96 s_R \approx 0.28 \pm 1.96(0.00582) = (0.269, 0.291)$.

Thus the population mortgage-payment-to-income ratio is about 28%.

Remark 1.3.7. *Bias and Resampling Corrections*
For SRS from fixed pairs (x_i, y_i), set $R := \overline{Y}/\overline{X}$, $\overline{Y}_R := \mu_x R$, and $T_R := \tau_x R$.

(i) Since $g(x,y) = y/x$ is nonlinear, generally $\mathbb{E}_d R \neq r$, hence $\mathbb{E}_d \overline{Y}_R \neq \mu_y$ and $\mathbb{E}_d T_R \neq \tau_y$.
(ii) Bias corrections target $B_d(R) := \mathbb{E}_d R - r$; use resampling when Taylor bias matters.

1.4 Stratified Random Sampling

Stratified sampling partitions the population into subpopulations and samples within each one. It can guarantee representation of important groups and can reduce variance when strata are internally homogeneous or have different means.

Definition 1.4.1. *Strata and the Stratified Mean*

Let U be a finite population, $L \geq 1$, nonempty subsets $U_1, \ldots, U_L \subseteq U$, $N_l := |U_l|$, $N := \sum_{l=1}^{L} N_l$, and fixed values $x_{il} \in \mathbb{R}$ for $i = 1, \ldots, N_l$, $l = 1, \ldots, L$.
For stratum samples of sizes n_l, write sampled values as X_{il}, $i = 1, \ldots, n_l$.

(i) *Strata*: $U = \bigsqcup_{l=1}^{L} U_l$.

(ii) *Stratum weight*: $W_l := \frac{N_l}{N}$.

(iii) *Stratum mean*: $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$.

(iv) *Stratum variance*: $\sigma_l^2 := \frac{1}{N_l} \sum_{i=1}^{N_l} (x_{il} - \mu_l)^2$.

(v) *Stratum sample mean*: $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$.

(vi) *Stratified mean*: $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$.

Structural Commentary: With $\mu := \frac{1}{N} \sum_{l=1}^{L} \sum_{i=1}^{N_l} x_{il}$, $\mu = \sum_{l=1}^{L} W_l \mu_l$.

Theorem 1.4.2. *Mean and Variance of the Stratified Estimate*

Let $L \geq 1$, $N_l > 1$, $1 \leq n_l \leq N_l$, $N := \sum_{l=1}^{L} N_l$, $W_l := \frac{N_l}{N}$, and $\lambda_l := \frac{1}{n_l}(1 - \frac{n_l - 1}{N_l - 1})$.

For fixed $x_{il} \in \mathbb{R}$, $i = 1, \ldots, N_l$, $l = 1, \ldots, L$, set $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$, $\sigma_l^2 := \frac{1}{N_l} \sum_{i=1}^{N_l} (x_{il} - \mu_l)^2$, and $\mu := \sum_{l=1}^{L} W_l \mu_l$.
Independently sample $S_l \sim \mathrm{Unif}\{s \subseteq \{1, \ldots, N_l\} : |s| = n_l\}$; for random orderings $(J_{il})_{i=1}^{n_l}$ of S_l, set $X_{il} := x_{J_{il},l}$, $\overline{X}_l := \frac{1}{n_l} \sum_{i=1}^{n_l} X_{il}$, and $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$. Let $\mathbb{E}_d, \mathrm{Var}_d$ denote design moments.

(i) Mean: $\mathbb{E}_d \overline{X}_s = \mu$.

(ii) Variance: $\mathrm{Var}_d(\overline{X}_s) = \sum_{l=1}^{L} W_l^2 \lambda_l \sigma_l^2$.

(iii) Small fractions: if $n_l / N_l \ll 1$ for all l, then $\mathrm{Var}_d(\overline{X}_s) \approx \sum_{l=1}^{L} W_l^2 \frac{\sigma_l^2}{n_l}$.

Proof.

(i) For $a \in \{1, \ldots, N_l\}$, $\mathbb{P}_d(J_{il} = a) = 1/N_l$, hence $\mathbb{E}_d X_{il} = \mu_l$, $\mathbb{E}_d \overline{X}_l = \mu_l$, and $\mathbb{E}_d \overline{X}_s = \sum_{l=1}^{L} W_l \mu_l = \mu$.

(ii) Independence across strata gives $\mathrm{Var}_d(\overline{X}_s) = \sum_{l=1}^{L} W_l^2 \mathrm{Var}_d(\overline{X}_l)$.
For stratum l, let $(Y_{1,l}, \ldots, Y_{N_l,l})$ uniformly order $\{x_{il} : 1 \leq i \leq N_l\}$, put $\gamma_l := \mathrm{Cov}_d(Y_{a,l}, Y_{b,l})$ for $a \neq b$.
As $\sum_{a=1}^{N_l} Y_{a,l}$ is fixed, $0 = N_l \sigma_l^2 + N_l(N_l - 1)\gamma_l$, so $\gamma_l = -\frac{\sigma_l^2}{N_l - 1}$, $\mathrm{Var}_d(\overline{X}_l) = \frac{1}{n_l^2}(n_l \sigma_l^2 + n_l(n_l - 1)\gamma_l) = \lambda_l \sigma_l^2$.

(iii) If $n_l / N_l \ll 1$, then $1 - \frac{n_l - 1}{N_l - 1} \approx 1$, so $\lambda_l \sigma_l^2 \approx \sigma_l^2 / n_l$.

Structural Commentary: Stratified estimation is a weighted sum of independent within-stratum estimators.
Key Insight: Independent strata make variance add as $\sum_l W_l^2 \mathrm{Var}_d(\overline{X}_l)$. □

Corollary 1.4.3. *Totals and Estimated Variances*

Let $L \geq 1$, $N_l > 1$, $1 < n_l \leq N_l$, $N := \sum_{l=1}^{L} N_l$, $W_l := \frac{N_l}{N}$, and $f_l := n_l / N_l$.

Fix $x_{il} \in \mathbb{R}$ for $i = 1, \ldots, N_l$, $l = 1, \ldots, L$, set $\mu_l := \frac{1}{N_l} \sum_{i=1}^{N_l} x_{il}$, $\mu := \sum_{l=1}^{L} W_l \mu_l$, and $\tau := N\mu$.
Assume independently $S_l \sim \mathrm{Unif}\{s \subseteq \{1, \ldots, N_l\} : |s| = n_l\}$; for random orderings $(J_{il})_{i=1}^{n_l}$ of S_l.

Set $X_{il} := x_{J_{il},l}$, $\overline{X}_l := \frac{1}{n_l}\sum_{i=1}^{n_l} X_{il}$, $\overline{X}_s := \sum_{l=1}^{L} W_l \overline{X}_l$, $T_s := N\overline{X}_s$, *and* $s_l^2 := \frac{1}{n_l-1}\sum_{i=1}^{n_l}(X_{il}-\overline{X}_l)^2$.
For $0<\alpha<1$, *let* $z(\frac{\alpha}{2}) := \Phi^{-1}(1-\frac{\alpha}{2})$.

(i) Total moments: $\mathbb{E}_d T_s = \tau$ *and* $\mathrm{Var}_d(T_s) = N^2 \mathrm{Var}_d(\overline{X}_s)$.

(ii) Estimated mean variance: $s^2_{\overline{X}_s} := \sum_{l=1}^{L} W_l^2(1-f_l)\frac{s_l^2}{n_l}$.

(iii) Intervals: $\mathrm{CI}_{1-\alpha}(\mu) = \overline{X}_s \pm z(\frac{\alpha}{2}) s_{\overline{X}_s}$ *and* $\mathrm{CI}_{1-\alpha}(\tau) = T_s \pm z(\frac{\alpha}{2}) N s_{\overline{X}_s}$.

Proof.

(i) Since $T_s = N\overline{X}_s$ and $\tau = N\mu$, Theorem 1.4.2 gives $\mathbb{E}_d T_s = N\mu = \tau$ and $\mathrm{Var}_d(T_s) = N^2 \mathrm{Var}_d(\overline{X}_s)$.

(ii) Let $S_l^2 := \frac{1}{N_l-1}\sum_{i=1}^{N_l}(x_{il}-\mu_l)^2$. Then $\mathrm{Var}_d(\overline{X}_l) = \frac{(1-f_l)S_l^2}{n_l}$, $\mathbb{E}_d s_l^2 = S_l^2$, so $S_l^2 \mapsto s_l^2$; $s^2_{\overline{X}_s} = \sum_{l=1}^{L} W_l^2(1-f_l)\frac{s_l^2}{n_l}$.

(iii) With $\frac{\overline{X}_s-\mu}{s_{\overline{X}_s}} \approx \mathcal{N}(0,1)$, inversion gives $\mathrm{CI}_{1-\alpha}(\mu)$; multiplying by N gives $\mathrm{CI}_{1-\alpha}(\tau)$.

Structural Commentary: Totals and intervals scale the stratified mean.
Key Insight: Estimated variance adds weighted stratum variances. □

Theorem 1.4.4. *Neyman Allocation*
Let $L \geq 1$, $W_l > 0$, $\sum_{l=1}^{L} W_l = 1$, $\sigma_l > 0$, $S_\sigma := \sum_{k=1}^{L} W_k\sigma_k$, *and* $n > 0$.
On $\mathcal{A} := \{(n_1,\ldots,n_L) \in (0,\infty)^L : \sum_{l=1}^{L} n_l = n\}$, *set* $V(n_1,\ldots,n_L) := \sum_{l=1}^{L} \frac{W_l^2\sigma_l^2}{n_l}$.
Then V *has the unique minimiser* $n_l^* := n\frac{W_l\sigma_l}{S_\sigma}$, $l = 1,\ldots,L$.

Proof. V is strictly convex on $(0,\infty)^L$ since $\partial_{ll}V = \frac{2W_l^2\sigma_l^2}{n_l^3} > 0$ and $\partial_{lm}V = 0$ for $l \neq m$.

For $\mathcal{L}(n,\lambda) := V(n) + \lambda(\sum_{l=1}^{L} n_l - n)$, the first-order equations are $-\frac{W_l^2\sigma_l^2}{n_l^2} + \lambda = 0$, hence $n_l = \frac{W_l\sigma_l}{\sqrt{\lambda}}$.
The constraint gives $\frac{1}{\sqrt{\lambda}} = \frac{n}{S_\sigma}$, so $n_l = n_l^*$; strict convexity gives uniqueness.
Structural Commentary: The constrained optimum balances marginal variance reduction across strata.
Key Insight: A stratum receives more sample not only when it is large, but also when it is internally variable. □

Corollary 1.4.5. *Allocation Comparisons*
Let $L \geq 1$, $W_l > 0$, $\sum_{l=1}^{L} W_l = 1$, $\mu_l \in \mathbb{R}$, $\sigma_l > 0$, $\mu := \sum_{l=1}^{L} W_l\mu_l$, $\overline{\sigma} := \sum_{l=1}^{L} W_l\sigma_l$, *and* $n > 0$.
Ignoring finite-population corrections, let $\overline{X}_{so}$ *be the optimal-allocation stratified mean,* $\overline{X}_{sp}$ *the proportional-allocation stratified mean with* $n_l = nW_l$, *and* $\overline{X}$ *the ordinary SRS mean.*

(i) Allocation variances: $\mathrm{Var}(\overline{X}_{so}) \approx \frac{\overline{\sigma}^2}{n}$ *and* $\mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l\sigma_l^2$.

(ii) Optimal-allocation gain: $\mathrm{Var}(\overline{X}_{sp}) - \mathrm{Var}(\overline{X}_{so}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l(\sigma_l-\overline{\sigma})^2$.

(iii) Proportional-allocation gain: $\mathrm{Var}(\overline{X}) - \mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l(\mu_l-\mu)^2$.

Proof.

(i) With finite-population corrections suppressed, $\mathrm{Var}(\overline{X}_s) \approx \sum_{l=1}^{L} \frac{W_l^2\sigma_l^2}{n_l}$. Neyman allocation gives $n_l = \frac{nW_l\sigma_l}{\overline{\sigma}}$, hence $\mathrm{Var}(\overline{X}_{so}) \approx \frac{\overline{\sigma}^2}{n}$; proportional allocation $n_l = nW_l$ gives $\mathrm{Var}(\overline{X}_{sp}) \approx \frac{1}{n}\sum_{l=1}^{L} W_l\sigma_l^2$.

(ii) Since $\sum_{l=1}^{L} W_l = 1$, $\sum_{l=1}^{L} W_l\sigma_l^2 - \overline{\sigma}^2 = \sum_{l=1}^{L} W_l(\sigma_l-\overline{\sigma})^2$.

(iii) For ordinary SRS, $\mathrm{Var}(\overline{X}) \approx \frac{1}{n}\{\sum_{l=1}^{L} W_l\sigma_l^2 + \sum_{l=1}^{L} W_l(\mu_l-\mu)^2\}$; subtract the proportional-allocation variance.

Structural Commentary: Proportional handles means; optimal also handles spreads.
Key Insight: Mean gaps help stratification; spread gaps help Neyman allocation. □

Example 1.4.6. *Four-Stratum Stratification Summary*
Let $N = 393$ and strata $l \in \{A,B,C,D\}$ have rounded summaries $(N_l) = (98,98,98,99)$, $(W_l) = (.249,.249,.249,.251)$,

$(\mu_l) = (182.9, 526.5, 956.3, 1591.2)$, and $(\sigma_l) = (103.4, 204.8, 243.5, 419.2)$. Let n be the total sample size, ignore finite-population corrections, and write $\overline{X}_s$ for the stratified mean.

(i) *Equal allocation*: if $n_l = n/4$, then $\mathrm{Var}(\overline{X}_s) \approx \sum_{l=1}^{4} W_l^2 \sigma_l^2/(n/4) \approx 72042.6/n$, so $\mathrm{SE}(\overline{X}_s) \approx 268.4/\sqrt{n}$.
(ii) *SRS benchmark*: with overall standard deviation $\sigma \approx 589.7$, $\mathrm{SE}(\overline{X}) \approx 589.7/\sqrt{n}$.
(iii) *Neyman allocation*: $a_l := \frac{W_l \sigma_l}{\sum_{k=1}^{4} W_k \sigma_k} \approx (.106, .210, .250, .434)$ for (A, B, C, D).

Stratification reduces variance, and Neyman allocation assigns the largest share to the stratum with largest σ_l.

Remark 1.4.7. *Model Limits and Other Designs*
For a finite-population survey:

(i) *Systematic*: fixed-interval sampling after a random start is SRS-like only when list order is noninformative; periodic order can bias selection.
(ii) *Cluster*: sampling groups lowers cost but can inflate variance when within-cluster correlation is positive.
(iii) *Scope*: design-based SEs measure random selection variation, not nonresponse, coverage error, selection bias, interviewer effects, or wording effects.

2 Exploratory Data Analysis and Data Summaries

2.1 Empirical Distribution and Survival Summaries

Exploratory summaries first treat the observed values as a batch of numbers. A stochastic model may later explain their variability, but the displays themselves are useful before a model has been chosen.

Definition 2.1.1. *Empirical CDF and Empirical Survival Function*
Let $x_1, \ldots, x_n \in \mathbb{R}$ be data; for survival summaries assume $x_i \geq 0$ for $1 \leq i \leq n$.

(i) *Empirical CDF*: $F_n : \mathbb{R} \to [0,1]$, $F_n(x) := n^{-1}\#\{i \in \{1, \ldots, n\} : x_i \leq x\}$.
(ii) *Empirical Survival Function*: $S_n : [0, \infty) \to [0,1]$, $S_n(t) := 1 - F_n(t) = n^{-1}\#\{i \in \{1, \ldots, n\} : x_i > t\}$.

Key Insight: F_n and S_n encode complementary empirical tails, with $S_n(t) = 1 - F_n(t)$ for $t \geq 0$.

Remark 2.1.2. *Step-Function Convention*
Let $v_1 < \cdots < v_m$ be the distinct observed values, with multiplicities $r_1, \ldots, r_m$. Then F_n is right-continuous, constant on each component of $\mathbb{R} \setminus \{v_1, \ldots, v_m\}$, and has jump $\Delta F_n(v_j) = \frac{r_j}{n}$.

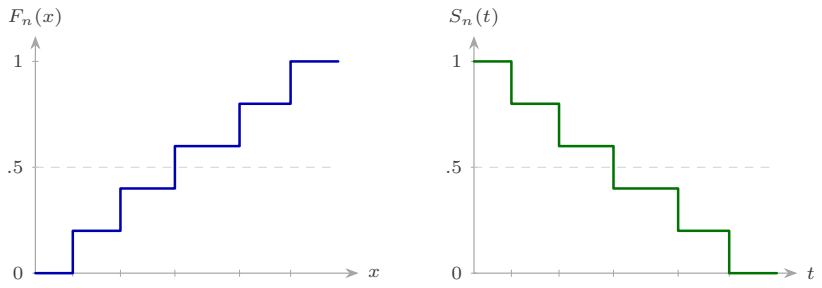


Figure 220: Empirical CDF (blue) rises as empirical survival (green) falls.

Proposition 2.1.3. *Fixed-Point Behavior of the Empirical CDF*
Let $X_1, \ldots, X_n$ be independent with common CDF F.
For $x \in \mathbb{R}$, set $I_i(x) := \mathbf{1}_{(-\infty, x]}(X_i)$ and $F_n(x) := n^{-1} \sum_{i=1}^{n} I_i(x)$. Then, for each fixed $x \in \mathbb{R}$:

(i) Law: $nF_n(x) \sim \mathrm{Binomial}(n, F(x))$.
(ii) Mean: $\mathbb{E}F_n(x) = F(x)$.
(iii) Variance: $\mathrm{Var}(F_n(x)) = n^{-1}F(x)(1 - F(x))$.

Proof. Fix $x \in \mathbb{R}$ and put $p := F(x)$ and $I_i := I_i(x)$.

(i) Since $I_i \sim \mathrm{Bernoulli}(p)$ independently, $nF_n(x) = \sum_{i=1}^{n} I_i \sim \mathrm{Binomial}(n, p)$.

(ii) Linearity gives $\mathbb{E}F_n(x) = n^{-1} \sum_{i=1}^{n} \mathbb{E}I_i = n^{-1}np = p = F(x)$.

(iii) Independence gives $\operatorname{Var}(F_n(x)) = n^{-2}\sum_{i=1}^{n}\operatorname{Var}(I_i) = n^{-2}np(1-p) = n^{-1}F(x)(1-F(x))$.

Structural Commentary: Pointwise, $F_n(x) = n^{-1}\sum_{i=1}^{n} I_i(x)$ with $I_i(x) \sim \text{Bernoulli}(F(x))$ independently.
Key Insight: The empirical CDF at one point is a normalised binomial count. □

Proposition 2.1.4. *Two-Point Covariance of the Empirical CDF*
Let $X_1, \dots, X_n$ be independent with common CDF F.
For $w \in \mathbb{R}$, set $I_i(w) := \mathbf{1}_{(-\infty,w]}(X_i)$, $F_n(w) := n^{-1}\sum_{i=1}^{n} I_i(w)$. For $u, v \in \mathbb{R}$, put $m := \min\{u, v\}$. Then:

(i) Covariance: $\operatorname{Cov}\big(F_n(u), F_n(v)\big) = n^{-1}\big(F(m) - F(u)F(v)\big)$.
(ii) Sign: $\operatorname{Cov}\big(F_n(u), F_n(v)\big) \geq 0$.

Proof. Fix $u, v \in \mathbb{R}$ and set $m := \min\{u, v\}$.

(i) *Covariance*: $\operatorname{Cov}(F_n(u), F_n(v)) = n^{-2}\sum_{i,j}\operatorname{Cov}(I_i(u), I_j(v))$.
For $i \neq j$ the term is 0 by independence, while for $i = j$,
$\operatorname{Cov}(I_i(u), I_i(v)) = \mathbb{E}[I_i(u)I_i(v)] - \mathbb{E}I_i(u)\mathbb{E}I_i(v) = F(m) - F(u)F(v)$.
Hence $\operatorname{Cov}(F_n(u), F_n(v)) = n^{-1}(F(m) - F(u)F(v))$.
(ii) *Sign*: If $u \leq v$, then $F(m) - F(u)F(v) = F(u)(1 - F(v)) \geq 0$.
If $v \leq u$, then $F(m) - F(u)F(v) = F(v)(1 - F(u)) \geq 0$.

Structural Commentary: $F_n(u)$ and $F_n(v)$ average iid Bernoulli pairs, so covariance scales by $1/n$.
Key Insight: Cross-index independence leaves only the same-observation overlap $\{X_i \leq m\}$. □

Definition 2.1.5. *Survival and Hazard Functions*
Let $T \geq 0$ have CDF F and density f, and set $D := \{t \geq 0 : 1 - F(t) > 0\}$.

(i) *Survival Function*: $S : [0, \infty) \to [0, 1]$, $S(t) := \mathbb{P}(T > t) = 1 - F(t)$.
(ii) *Hazard Function*: $h : D \to [0, \infty)$, $h(t) := \frac{f(t)}{1-F(t)} = \frac{f(t)}{S(t)}$.

Structural Commentary: On intervals where S is differentiable and positive, $h(t) = -\frac{d}{dt}\log S(t)$.

Proposition 2.1.6. *Hazard Determines the Density*
Let $T \geq 0$ have survival $S(t) := \mathbb{P}(T > t)$ with $S \in C^1([0, \infty))$, $S(t) > 0$, $S(0) = 1$, density $f(t) := -S'(t)$, and hazard $h(t) := \frac{f(t)}{S(t)}$. Put $H(t) := \int_0^t h(s)\,ds$. Then, for $t \geq 0$:

(i) Survival: $S(t) = e^{-H(t)}$.
(ii) Density: $f(t) = h(t)e^{-H(t)}$.

Proof. Fix $t \geq 0$.

(i) *Survival*: Since $(\log S)' = S'/S = -h$, $\log S(t) - \log S(0) = -H(t)$. Since $S(0) = 1$, $S(t) = e^{-H(t)}$.
(ii) *Density*: Since $h = f/S$, $f(t) = h(t)S(t) = h(t)e^{-H(t)}$.

Structural Commentary: $h = -S'/S = -(\log S)'$, so H is the cumulative log-decay of survival.
Key Insight: The hazard determines the law through $S = e^{-H}$ and $f = hS$. □

Example 2.1.7. *Constant and Power Hazards*
For $t \geq 0$:

(i) *Exponential*: If $F(t) = 1 - e^{-\lambda t}$ with $\lambda > 0$, then $S(t) = e^{-\lambda t}$, $f(t) = \lambda e^{-\lambda t}$, and $h(t) = \lambda$.
(ii) *Weibull-Type*: If $F(t) = 1 - e^{-\alpha t^\beta}$ with $\alpha, \beta > 0$, then $S(t) = e^{-\alpha t^\beta}$, $f(t) = \alpha\beta t^{\beta-1}e^{-\alpha t^\beta}$, $h(t) = \alpha\beta t^{\beta-1}$.

Key Insight: Power hazards increase for $\beta > 1$, decrease for $0 < \beta < 1$, and are constant for $\beta = 1$.

Method 2.1.8. *Empirical Log-Survival Plot*
Suppose $0 \leq T_{(1)} < \cdots < T_{(n)}$ and there are no ties. Set $p_i := \frac{i}{n+1}$ and $y_i := \log(1 - p_i)$.

(i) *Plot*: Draw $P^{(n)}_{\log S} := \{(T_{(i)}, y_i) : 1 \leq i \leq n\}$; using $n + 1$ keeps y_n finite.
(ii) *Hazard Reading*: Since $(\log S)' = -h$, more negative local slopes indicate larger hazards.
(iii) *Tail Variability*: $\operatorname{Var}(\log[1 - F_n(t)]) \approx n^{-1}\frac{F(t)}{1-F(t)}$, which grows as $F(t) \uparrow 1$.

Remark 2.1.9. *Pointwise and Simultaneous CDF Uncertainty*
Proposition 2.1.3 is pointwise: x is fixed and $F_n(x)$ is a normalised binomial count.
Simultaneous uncertainty controls $\|F_n - F\|_\infty := \sup_x |F_n(x) - F(x)|$, yielding bands and diagnostics.

2.2 Quantile and Shape Displays

Distributional shape can be read from cumulative displays, density displays, or quantile comparisons. These views emphasize different questions: cumulative proportion, local concentration, and relative change between two distributions.

Definition 2.2.1. *Quantile–Quantile Plot*
Let F and G be continuous strictly increasing CDFs with quantile inverses $F^{-1}, G^{-1} : (0,1) \to \mathbb{R}$. For equal-sized batches $x_1, \dots, x_n$ and $y_1, \dots, y_n$, let $x_{(1)} \le \cdots \le x_{(n)}$ and $y_{(1)} \le \cdots \le y_{(n)}$.

(i) *Q–Q Plot*: $Q_{F,G} := \{(F^{-1}(p), G^{-1}(p)) : 0 < p < 1\} \subseteq \mathbb{R}^2$.
(ii) *Empirical Q–Q Plot*: $Q^{(n)}_{x,y} := \{(x_{(i)}, y_{(i)}) : 1 \le i \le n\} \subseteq \mathbb{R}^2$.

Key Insight: $(u, v) \in Q_{F,G}$ implies $F(u) = G(v)$, so plotted pairs match common cumulative probability levels.

Proposition 2.2.2. *Location and Scale Effects in Q–Q Plots*
Let F, G be continuous strictly increasing CDFs.
Set $x_p := F^{-1}(p)$, $y_p := G^{-1}(p)$ for $p \in (0,1)$, and $Q_{F,G} := \{(x_p, y_p) : 0 < p < 1\}$. Then:

(i) Location: If $y_p = x_p + h$ $\forall p \in (0,1)$ with $h \in \mathbb{R}$, then $G(y) = F(y-h)$ $\forall y \in \mathbb{R}$, $Q_{F,G} \subseteq \{(x,y) : y = x + h\}$.
(ii) Scale: If $y_p = cx_p$ $\forall p \in (0,1)$ with $c > 0$, then $G(y) = F(\frac{y}{c})$ $\forall y \in \mathbb{R}$, $Q_{F,G} \subseteq \{(x,y) : y = cx\}$.

Proof.

(i) *Location*: Fix $y \in \mathbb{R}$ and set $p := G(y)$. Then $p \in (0,1)$, $y = y_p = x_p + h$, and $x_p = y - h$, so $G(y) = p = F(x_p) = F(y-h)$. Also $(x_p, y_p) \in Q_{F,G}$ implies $y_p = x_p + h$.
(ii) *Scale*: Fix $y \in \mathbb{R}$ and set $p := G(y)$. Then $p \in (0,1)$, $y = y_p = cx_p$, and $x_p = \frac{y}{c}$, so $G(y) = p = F(x_p) = F(\frac{y}{c})$. Also $(x_p, y_p) \in Q_{F,G}$ implies $y_p = cx_p$.

Structural Commentary: A Q–Q plot is the graph of the quantile transport $F^{-1}(p) \mapsto G^{-1}(p)$.
Key Insight: Straight Q–Q lines encode location shifts and scale changes. □

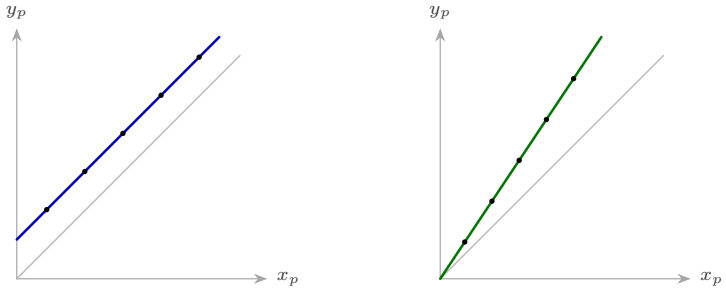


Figure 221: Q–Q lines: blue $y = x + h$ for location shift, green $y = cx$ for scale change.

Example 2.2.3. *Normal and Exponential Q–Q Lines*
For $p \in (0,1)$, set $x_p := F^{-1}(p)$ and $y_p := G^{-1}(p)$.

(i) *Normal Location*: If $F = N(0,1)$ and $G = N(1,1)$, then $y_p = x_p + 1$, so the Q–Q line is $y = x + 1$.
(ii) *Normal Location–Scale*: If $F = N(0,1)$ and $G = N(1,4)$, then $y_p = 1 + 2x_p$, so the Q–Q line is $y = 1 + 2x$.
(iii) *Exponential Rates*: If $F = \mathrm{Exp}(\lambda_1)$ and $G = \mathrm{Exp}(\lambda_2)$, then $x_p = -\frac{\log(1-p)}{\lambda_1}$ and $y_p = -\frac{\log(1-p)}{\lambda_2} = \frac{\lambda_1}{\lambda_2} x_p$, so the Q–Q line is $y = \frac{\lambda_1}{\lambda_2} x$.

Method 2.2.4. *Interpolated Empirical Quantiles*
Let $X_{(1)} \le \cdots \le X_{(n)}$ and set $p_k := \frac{k}{n+1}$ for $1 \le k \le n$.

(i) *Grid*: Assign $X_{(k)}$ to probability p_k.
(ii) *Interpolation*: If $p_k \le p \le p_{k+1}$ for $1 \le k < n$, set $\lambda := \frac{p - p_k}{p_{k+1} - p_k} = (n+1)p - k$, $\widehat{x}_p := (1-\lambda)X_{(k)} + \lambda X_{(k+1)}$.
(iii) *Q–Q Use*: For unequal samples, evaluate both interpolated quantile functions on a common grid $p \in (0,1)$.

Proposition 2.2.5. *Approximate Variance of a Sample Quantile*
Let $Y_1, \dots, Y_n$ be iid with continuous CDF F and density f. Fix $p \in (0,1)$, set $x_p := F^{-1}(p)$ with $f(x_p) > 0$, choose k with $\frac{k}{n+1} \approx p$, and put $\widehat{x}_p := Y_{(k)}$. Then:

(i) Probability Scale: $U_{(k)} := F(Y_{(k)})$ has kth uniform order-statistic law. $\mathrm{Var}(U_{(k)}) = \frac{k(n+1-k)}{(n+1)^2(n+2)} \approx \frac{p(1-p)}{n}$.
(ii) Quantile Scale: $\mathrm{Var}(\widehat{x}_p) \approx \frac{p(1-p)}{n f(x_p)^2}$.

Proof.

(i) By probability integral transform, $U_i := F(Y_i)$ are iid Uniform$(0,1)$, $U_{(k)} = F(Y_{(k)})$. Theorem 1.4.4 gives the stated exact variance, hence $\operatorname{Var}(U_{(k)}) \approx \frac{p(1-p)}{n}$ when $\frac{k}{n+1} \approx p$.

(ii) Let $g := F^{-1}$. Since $g'(p) = \frac{1}{f(x_p)}$ and $\widehat{x}_p = g(U_{(k)})$, delta approximation $g(U_{(k)}) \approx g(p) + g'(p)(U_{(k)} - p)$ gives $\operatorname{Var}(\widehat{x}_p) \approx g'(p)^2 \operatorname{Var}(U_{(k)}) \approx \frac{p(1-p)}{n f(x_p)^2}$.

Structural Commentary: Quantile variance is probability-scale variance multiplied by $(F^{-1})'(p)^2$.
Key Insight: Low density near x_p inflates sample-quantile variability by $1/f(x_p)^2$. □

Definition 2.2.6. *Histogram, Density Display, and Stem-and-Leaf Plot*
Let $x_1, \ldots, x_n \in \mathbb{R}$ be data. For disjoint intervals $B_1, \ldots, B_m$ with widths $w_j > 0$.
Set $N_j := \#\{i \in \{1, \ldots, n\} : x_i \in B_j\}$. Let K be a kernel, $h > 0$, and $b > 0$ be a chosen scale.

(i) *Histogram*: $H_j^c := N_j$, $H_j^p := \frac{N_j}{n}$, or $H_j^d := \frac{N_j}{n w_j}$ on B_j.

(ii) *Density Display*: $\widehat{f}_h : \mathbb{R} \to [0, \infty)$, $\widehat{f}_h(x) := (nh)^{-1} \sum_{i=1}^{n} K(\frac{x - x_i}{h})$.

(iii) *Stem-and-Leaf Plot*: (s_i, ℓ_i) satisfies $x_i = b s_i + \ell_i$, $s_i \in \mathbb{Z}$, $\ell_i \in [0, b)$. Entries with common s_i are printed together in increasing s_i.

Structural Commentary: For density-scaled heights $H_j^d = \frac{N_j}{n w_j}$, $\sum_j H_j^d w_j = 1$ when the bins cover the sample.
Key Insight: Smaller w_j or h retains more local variation; larger w_j or h increases smoothing.

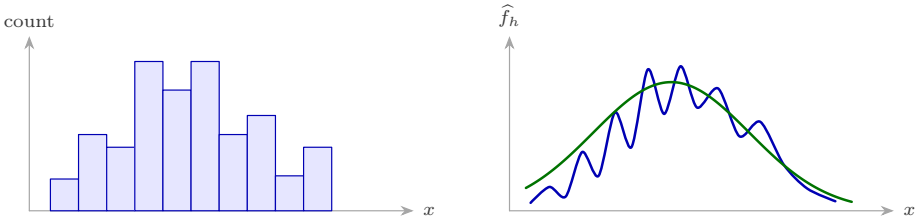


Figure 222: Shape displays: narrow bins at left; small h (blue) is rougher than large h (green).

Remark 2.2.7. *What Shape Displays Preserve*

(i) *ECDF*: Preserves ranks and supports direct quantile reading.
(ii) *Histogram*: Preserves bin counts (N_j), not exact values.
(iii) *Kernel Estimate*: Preserves smoothed local mass; bandwidth h controls resolution.
(iv) *Stem–Leaf*: Preserves coarse shape and recoverable numerical values.

2.3 Location, Scale, and Robust Summaries

Numerical summaries compress a batch more aggressively than a plot. They should therefore be chosen with attention to the feature being summarized and to the influence of extreme observations.

Definition 2.3.1. *Location Summaries for a Batch*
Let $x_1, \ldots, x_n \in \mathbb{R}$ have order statistics $x_{(1)} \le \cdots \le x_{(n)}$.
When a trim fraction is used, let $0 \le \alpha < \frac{1}{2}$ and $m_\alpha := \lfloor n\alpha \rfloor$.

(i) *Arithmetic Mean*: $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$.

(ii) *Median*: $\widetilde{x} := x_{(\frac{n+1}{2})}$ if n is odd, and $\widetilde{x} := \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2}$ if n is even.

(iii) $100\alpha\%$ *Trimmed Mean*: $\overline{x}_\alpha := (n - 2m_\alpha)^{-1} \sum_{i=m_\alpha+1}^{n-m_\alpha} x_{(i)}$.

Structural Commentary: Under $x_i \mapsto x_i + \Delta$, $\overline{x} \mapsto \overline{x} + \frac{\Delta}{n}$; $\widetilde{x}$ and $\overline{x}_\alpha$ depend only on central order statistics.
Key Insight: Extreme magnitudes affect $\overline{x}$ directly, but affect $\widetilde{x}$ and $\overline{x}_\alpha$ mainly through rank changes.

Remark 2.3.2. *Outliers and Center*
Replacing one x_i by M sends $\overline{x}$ to $\overline{x} + \frac{M - x_i}{n}$, hence unbounded as $M \to \pm\infty$.
Once an observation is extreme, moving it farther outward leaves $\widetilde{x}$ unchanged and leaves $\overline{x}_\alpha$ unchanged if it remains trimmed. This is descriptive robustness, not an independence or model.

Proposition 2.3.3. *Distribution-Free Median Interval*
Let $X_1, \dots, X_n$ be independent with continuous CDF F and median η, so $F(\eta) = \frac{1}{2}$.
Let $X_{(1)} \le \cdots \le X_{(n)}$ and $1 \le k \le \lfloor \frac{n}{2} \rfloor$. Then:

(i) Interval: $C_k := [X_{(k)}, X_{(n-k+1)}]$.

(ii) Coverage: $\mathbb{P}(\eta \in C_k) = 1 - \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$.

Proof.

(i) Let $B_- := \sum_{i=1}^{n} \mathbf{1}_{\{X_i < \eta\}}$, $B_+ := \sum_{i=1}^{n} \mathbf{1}_{\{X_i > \eta\}}$. By continuity, $B_- + B_+ = n$ a.s. and $B_- \sim \text{Binomial}(n, \frac{1}{2})$.

(ii) $\eta \notin C_k$ iff $B_- < k$ or $B_+ < k$. Since $k \le \lfloor \frac{n}{2} \rfloor$, these two events are disjoint.

(iii) By symmetry, $\mathbb{P}(\eta \notin C_k) = 2\mathbb{P}(B_- < k) = 2 \sum_{j=0}^{k-1} \binom{n}{j} 2^{-n} = \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$, so $\mathbb{P}(\eta \in C_k) = 1 - \frac{1}{2^{n-1}} \sum_{j=0}^{k-1} \binom{n}{j}$.

Structural Commentary: The coverage depends only on the binomial count of observations below η, not on F.
Key Insight: Continuity turns median coverage into a symmetric binomial tail. □

Definition 2.3.4. *Location M-Estimates*
For data $x_1, \dots, x_n \in \mathbb{R}$, scale $\sigma > 0$, and loss $\Psi : \mathbb{R} \to \mathbb{R}$, $\widehat{\mu} \in \arg\min_{\mu \in \mathbb{R}} \sum_{i=1}^{n} \Psi(\frac{x_i - \mu}{\sigma})$.

Key Insight: Quadratic loss gives $\overline{x}$; absolute or subquadratic loss improves outlier resistance.

Definition 2.3.5. *Dispersion Summaries*
Let $x_1, \dots, x_n \in \mathbb{R}$ have mean $\overline{x}$, median $\widetilde{x}$, and selected sample quartiles Q_1, Q_3.

(i) *Sample Variance*: $s^2 := (n-1)^{-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$.

(ii) *Sample Standard Deviation*: $s := \sqrt{s^2}$.

(iii) *Interquartile Range*: $\text{IQR} := Q_3 - Q_1$.

(iv) *Median Absolute Deviation*: $\text{MAD} := \text{median}\{|x_i - \widetilde{x}| : 1 \le i \le n\}$.

Structural Commentary: s^2 is residual-based; IQR and MAD are order-based.
Key Insight: A single extreme value can dominate s, but not IQR or MAD unless it changes central ranks.

Remark 2.3.6. *Normal Scale Calibrations*
For approximately normal data, $\frac{\text{IQR}}{1.35}$ and $\frac{\text{MAD}}{0.675}$ are common robust scale estimates for the normal standard deviation. The constants calibrate the population normal IQR and median absolute deviation to σ.

Method 2.3.7. *Bootstrap Variability for Location Summaries*
Let $T = T(x_1, \dots, x_n)$ be a location summary and let $B \ge 1$.

(i) *Resample*: For $1 \le b \le B$, draw $x_1^{*(b)}, \dots, x_n^{*(b)}$ iid from $\widehat{F}_n := n^{-1} \sum_{i=1}^{n} \delta_{x_i}$.

(ii) *Recompute*: Set $T_b^* := T(x_1^{*(b)}, \dots, x_n^{*(b)})$ and $\overline{T}^* := B^{-1} \sum_{b=1}^{B} T_b^*$.

(iii) *Spread*: Estimate variability by $\widehat{\text{se}}_*(T) := \left((B-1)^{-1} \sum_{b=1}^{B} (T_b^* - \overline{T}^*)^2 \right)^{1/2}$.

Key Insight: This is useful for medians and trimmed means, where analytic standard errors can be tail-sensitive.

Remark 2.3.8. *Compare More Than One Center*
Let $\mathcal{C} := \{\overline{x}, \widetilde{x}, \overline{x}_\alpha, \widehat{\mu}_\Psi\}$. For symmetric light-tailed data, range($\mathcal{C}$) is typically small; a large range signals skewness, heavy tails, outliers, or mixture structure.

2.4 Boxplots, Scatterplots, and Relationship Summaries

Boxplots and scatterplots are compact exploratory displays. A boxplot summarizes one batch or compares several batches; a scatterplot displays relationships between paired variables.

Method 2.4.1. *Boxplot Construction*
A standard boxplot is built from the median, lower quartile Q_1, upper quartile Q_3, and $\text{IQR} = Q_3 - Q_1$:

(i) Draw a box from Q_1 to Q_3 and mark the median inside the box.

(ii) Draw whiskers to the most extreme observations lying within 1.5 IQR of the nearer quartile.

(iii) Mark observations beyond the whiskers individually as possible outliers.

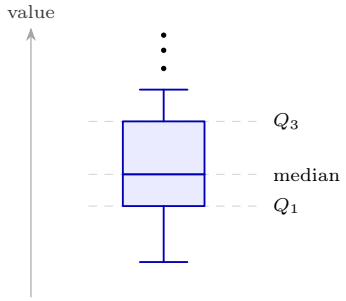


Figure 223: Boxplot: box Q_1 to Q_3, line median, dots possible outliers.

Remark 2.4.2. *Comparing Batches with Boxplots*

(i) *Location*: Differences in medians M_g indicate group shifts.
(ii) *Spread*: Differences in $\mathrm{IQR}_g = Q_{3,g} - Q_{1,g}$ indicate central-spread changes.
(iii) *Tails*: Unequal whiskers or one-sided outliers indicate skewness or tail imbalance.

Definition 2.4.3. *Scatterplot, Pearson Correlation, and Rank Correlation*

Let $(x_i, y_i) \in \mathbb{R}^2$ for $1 \le i \le n$, set $\overline{x} := n^{-1} \sum_{i=1}^{n} x_i$ and $\overline{y} := n^{-1} \sum_{i=1}^{n} y_i$.

Let R_i, S_i be chosen ranks of x_i, y_i, using average ranks for ties.

Set $\overline{R} := n^{-1} \sum_{i=1}^{n} R_i$ and $\overline{S} := n^{-1} \sum_{i=1}^{n} S_i$. Set $a_i := x_i - \overline{x}$, $b_i := y_i - \overline{y}$, $u_i := R_i - \overline{R}$, and $v_i := S_i - \overline{S}$.

Let $S_{xx} := \sum_i a_i^2$, $S_{yy} := \sum_i b_i^2$, $S_{xy} := \sum_i a_i b_i$, $S_{RR} := \sum_i u_i^2$, $S_{SS} := \sum_i v_i^2$, and $S_{RS} := \sum_i u_i v_i$.

(i) *Scatterplot*: $P := \{(x_i, y_i) : 1 \le i \le n\} \subset \mathbb{R}^2$, displayed in the xy-plane.
(ii) *Pearson Correlation Coefficient*: if $S_{xx} S_{yy} > 0$, $r := \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$.
(iii) *Rank Correlation Coefficient*: if $S_{RR} S_{SS} > 0$, $r_{\mathrm{rank}} := \frac{S_{RS}}{\sqrt{S_{RR} S_{SS}}}$.

Structural Commentary: r is affine-scale invariant; r_{rank} is monotone-transform invariant.
Key Insight: r measures linear association; r_{rank} measures rank association.

Method 2.4.4. *Scatterplot Diagnostics*

For paired data $P = \{(x_i, y_i) : 1 \le i \le n\}$:

(i) *Raw Plot*: Inspect P for outliers, curvature, clusters, and nonconstant spread.
(ii) *Transform*: If scale obscures structure, compare $(g(x_i), h(y_i))$ for domain-meaningful g, h such as log.
(iii) *Correlation*: Use r only when the displayed relation is approximately linear.
(iv) *Multiple Variables*: For $z_i \in \mathbb{R}^d$ with $d > 2$, inspect all pairwise plots before relying on pairwise summaries.

Example 2.4.5. *Changing Scale Can Reveal Structure*

If $x_i > 0$ and $\max_i x_i / \min_i x_i \gg 1$, the raw plot $P = \{(x_i, y_i)\}$ may compress variation. The transformed plot $P_{\log} := \{(\log x_i, y_i)\}$ can reveal monotone or linear structure, and generally $r(P_{\log}) \neq r(P)$ because Pearson correlation depends on the chosen coordinates.

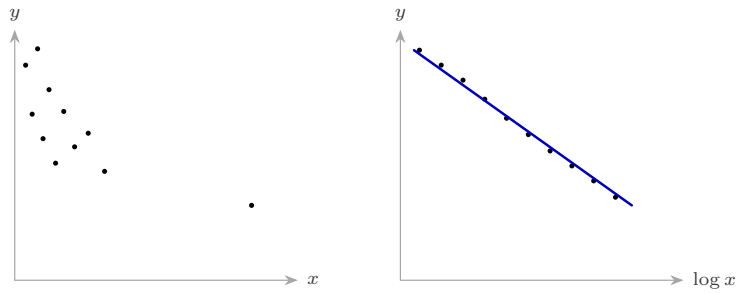


Figure 224: Scatterplots: raw x compresses points; $\log x$ reveals a linear trend.

Remark 2.4.6. *Exploration Before Formal Inference*

Exploratory displays choose the inferential target: transformations $g(X)$, influential subsets, comparable spreads IQR_g, and data-quality checks should be settled before intervals, tests, or models are trusted.

3 Categorical Data Analysis

3.1 Fisher's Exact Test and Fixed Margins

Categorical data record counts in classes rather than measurements on a continuous scale. In a two-way table, the rows and columns encode two classifications, and the cells record how many units fall in each cross-classification. The probability model is therefore built from multinomial and hypergeometric count laws rather than from normal sampling pivots.

Definition 3.1.1. *Two-Way Contingency Table*

Let $I, J \in \mathbb{N}$ and $n_{ij} \in \mathbb{N}_0$ for $1 \le i \le I$, $1 \le j \le J$. Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, and $n_{\cdot\cdot} := \sum_{i=1}^{I}\sum_{j=1}^{J} n_{ij}$.

For corresponding random counts, write N_{ij} and use capitalized totals.

(i) *Two-way contingency table*: $(n_{ij})_{1\le i\le I,\ 1\le j\le J}$, indexed by row and column classifications.
(ii) *Margins*: $\{n_{i\cdot} : 1 \le i \le I\}$, $\{n_{\cdot j} : 1 \le j \le J\}$, and $n_{\cdot\cdot}$.

Structural Commentary: Fixed-margin exact tests condition on the margins of random table $N = (N_{ij})$.

Method 3.1.2. *Fisher's Exact Test for a 2×2 Table*

Let a 2×2 table have fixed margins

	1	2	Total
1	N_{11}	N_{12}	$n_{1\cdot}$
2	N_{21}	N_{22}	$n_{2\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 2}$	$n_{\cdot\cdot}$

Let R_1, R_2 be fixed disjoint row sets with $|R_i| = n_{i\cdot}$.
Let C be the random column-1 set with $|C| = n_{\cdot 1}$, and set $\Omega_0 := \{A \subseteq R_1 \cup R_2 : |A| = n_{\cdot 1}\}$.

(i) $H_0 : \mathbb{P}_0(C = A) = \binom{n_{\cdot\cdot}}{n_{\cdot 1}}^{-1}$ $\forall A \in \Omega_0$.
(ii) $N_{11} = |C \cap R_1|$, $N_{12} = n_{1\cdot} - N_{11}$, $N_{21} = n_{\cdot 1} - N_{11}$, and $N_{22} = n_{2\cdot} - n_{\cdot 1} + N_{11}$.
(iii) $\mathcal{S} := \{x \in \mathbb{Z} : 0 \le x \le n_{1\cdot},\ 0 \le n_{\cdot 1} - x \le n_{2\cdot}\}$ and $p_0(x) := \mathbb{P}_0(N_{11} = x) = \frac{\binom{n_{1\cdot}}{x}\binom{n_{2\cdot}}{n_{\cdot 1}-x}}{\binom{n_{\cdot\cdot}}{n_{\cdot 1}}}$, $x \in \mathcal{S}$.
(iv) $p_+(x_{\text{obs}}) := \sum_{x\in\mathcal{S}:x\ge x_{\text{obs}}} p_0(x)$, $p_-(x_{\text{obs}}) := \sum_{x\in\mathcal{S}:x\le x_{\text{obs}}} p_0(x)$, and $p_2(x_{\text{obs}}) := \sum_{x\in\mathcal{S}:p_0(x)\le p_0(x_{\text{obs}})} p_0(x)$.
Reject when the selected value is $\le \alpha$.

Example 3.1.3. *Promotion Decisions and Fixed Margins*

In randomised label experiment with 24 male-labeled,24 female-labeled files, the observed table is

	Male	Female	Total
Promote	21	14	35
Hold file	3	10	13
Total	24	24	48

Under H_0 and fixed margins $(35, 13)$, $(24, 24)$, N_{11} satisfies $\mathbb{P}_0(N_{11} = x) = \frac{\binom{35}{x}\binom{13}{24-x}}{\binom{48}{24}}$, $x \in \mathcal{S} := \{11, \dots, 24\}$.

At level 0.05, $R^{(2)}_{0.05} := \{11, 12, 13, 14, 21, 22, 23, 24\}$; since $x_{\text{obs}} = 21 \in R^{(2)}_{0.05}$, reject H_0.

Remark 3.1.4. *What Is Exact Here*

For $N = (N_{ij})$ and $\mathcal{M} := (\{N_{i\cdot}\}_i, \{N_{\cdot j}\}_j)$, exactness means evaluating $\mathcal{L}_0(N \mid \mathcal{M})$ without asymptotics. Conditioning is design-justified when $\mathcal{M}$ is ancillary to H_0; rejection interpretation is design-dependent.

3.2 Chi-Square Tests for Contingency Tables

For an $I \times J$ table with observed counts n_{ij} and fitted null counts E_{ij}, use $X^2 := \sum_{i=1}^{I}\sum_{j=1}^{J}(n_{ij} - E_{ij})^2/E_{ij}$; in contingency tables, E_{ij} is determined by the row-column null model.

Definition 3.2.1. *Homogeneity of Multinomial Distributions*

Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and $\Delta_{I-1} := \{\pi \in [0,1]^I : \sum_{i=1}^{I} \pi_i = 1\}$.

For independent columns $j \in [J]$, let $(N_{1j}, \dots, N_{Ij}) \sim \text{Multinomial}(n_{\cdot j}; \pi_{1j}, \dots, \pi_{Ij})$ with observed counts n_{ij}. The column multinomial laws are *homogeneous* iff $H_0 : \exists \pi \in \Delta_{I-1}$ with $\pi_{ij} = \pi_i$ $\forall (i,j) \in [I] \times [J]$; under H_0, π_i is the common category probability.

Proposition 3.2.2. *Common Probability MLE Under Homogeneity*

Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, $\Delta_{I-1} := \{\pi \in [0,1]^I : \sum_{i=1}^{I} \pi_i = 1\}$, and $N_{\cdot j} := (N_{1j}, \dots, N_{Ij})$, independently

for $j \in [J]$. Under H_0, $N_{\cdot j} \sim \text{Multinomial}(n_{\cdot j}; \pi)$ for a common $\pi \in \Delta_{I-1}$.
With observed counts n_{ij}, set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$ and $n_{\cdot\cdot} := \sum_{i=1}^{I} n_{i\cdot} > 0$.
Then $\widehat{\pi}_i = \frac{n_{i\cdot}}{n_{\cdot\cdot}}$ and $E_{ij} = n_{\cdot j}\widehat{\pi}_i = \frac{n_{i\cdot}n_{\cdot j}}{n_{\cdot\cdot}}$ for $(i,j) \in [I] \times [J]$.

Proof. Up to factors independent of π, $L(\pi) \propto \prod_{j=1}^{J}\prod_{i=1}^{I} \pi_i^{n_{ij}} = \prod_{i=1}^{I} \pi_i^{n_{i\cdot}}$, hence $\ell(\pi) = \sum_{i=1}^{I} n_{i\cdot} \log \pi_i$ on Δ_{I-1}.

Stationarity of $\ell(\pi) + \lambda(1 - \sum_{i=1}^{I} \pi_i)$ gives $\frac{n_{i\cdot}}{\pi_i} - \lambda = 0$ when $n_{i\cdot} > 0$ and $\widehat{\pi}_i = 0$ when $n_{i\cdot} = 0$,
Normalisation gives $\lambda = n_{\cdot\cdot}$ and $\widehat{\pi}_i = \frac{n_{i\cdot}}{n_{\cdot\cdot}}$. Therefore $E_{ij} = \mathbb{E}_{\widehat{\pi}} N_{ij} = n_{\cdot j}\widehat{\pi}_i$.
Structural Commentary: H_0 reduces IJ cell probabilities to one shared $\pi \in \Delta_{I-1}$.
Key Insight: $\widehat{\pi}$ is the pooled row proportion, and E_{ij} redistributes each column total by $\widehat{\pi}_i$. □

Definition 3.2.3. *Homogeneity Fitted Counts*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, $n_{\cdot\cdot} := \sum_{i=1}^{I}\sum_{j=1}^{J} n_{ij} > 0$, and $\widehat{\pi}_i := \frac{n_{i\cdot}}{n_{\cdot\cdot}}$.
The homogeneity *fitted counts* are $E_{ij} := n_{\cdot j}\widehat{\pi}_i = \frac{n_{i\cdot}n_{\cdot j}}{n_{\cdot\cdot}}$ for $(i,j) \in [I] \times [J]$.

Method 3.2.4. *Pearson Test of Homogeneity*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, $n_{\cdot\cdot} := \sum_{i=1}^{I}\sum_{j=1}^{J} n_{ij} > 0$, and $E_{ij} := \frac{n_{i\cdot}n_{\cdot j}}{n_{\cdot\cdot}}$. Assume $E_{ij} > 0$ for cells used.

(i) *Statistic*: $X^2 := \sum_{i=1}^{I}\sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$.

(ii) *Calibration*: If $n_{\cdot\cdot}$ large, $\min_{i,j} E_{ij}$ not small, then $X^2 \overset{H_0}{\approx} \chi^2_\nu$; $\nu := J(I-1) - (I-1) = (I-1)(J-1)$.
(iii) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_{(I-1)(J-1)} \geq X^2_{\text{obs}}\} \leq \alpha$.

Example 3.2.5. *Word Counts by Text Group*
Let $W = \{a, an, this, that, with, without\}$, let $G_1, \dots, G_4$ be text groups. Observed word-count table n_{wj} is

Word	G_1	G_2	G_3	G_4
a	147	186	101	83
an	25	26	11	29
this	32	39	15	15
that	94	105	37	22
with	59	74	28	43
without	18	10	10	4
Total	375	440	202	196

For $C_A := \{1, 2, 3\}$, by homogeneity test: $X^2_A = 12.27$, $\nu_A = (6-1)(3-1) = 10$, $p_A = \mathbb{P}\{\chi^2_{10} \geq 12.27\} \approx 0.27$.
Pooling C_A and comparing it with $C_B := \{4\}$ gives $X^2_{A,B} = 32.81$, $\nu_{A,B} = 6 - 1 = 5$, and $p_{A,B} < 0.001$.
$w \in \{an, that\}$ gives the largest Pearson contributions, localising nonhomogeneity by row.

Definition 3.2.6. *Independence in a Contingency Table*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and $N = (N_{ij}) \sim \text{Multinomial}(n; (\pi_{ij})_{[I]\times[J]})$ with $n > 0$, observed counts n_{ij}.
Set $\pi_{i\cdot} := \sum_{j=1}^{J} \pi_{ij}$, $\pi_{\cdot j} := \sum_{i=1}^{I} \pi_{ij}$, $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, and $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$.
Rows and columns are *independent* iff $H_0 : \pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$ $\forall (i,j) \in [I] \times [J]$; then $\widehat{\pi}_{ij} := \frac{n_{i\cdot}n_{\cdot j}}{n^2}$ and $E_{ij} := \frac{n_{i\cdot}n_{\cdot j}}{n}$.

Method 3.2.7. *Pearson Test of Independence*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ for $(i,j) \in [I] \times [J]$.
Set $n := \sum_{i=1}^{I}\sum_{j=1}^{J} n_{ij} > 0$, $n_{i\cdot} := \sum_{j=1}^{J} n_{ij}$, $n_{\cdot j} := \sum_{i=1}^{I} n_{ij}$, and $E_{ij} := \frac{n_{i\cdot}n_{\cdot j}}{n}$. Assume $E_{ij} > 0$ for cells used.

(i) *Statistic*: $X^2 := \sum_{i=1}^{I}\sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$.

(ii) *Calibration*: If n is large and $\min_{i,j} E_{ij}$ is not small, then $X^2 \overset{H_0}{\approx} \chi^2_\nu$ with $\nu := (I-1)(J-1)$.
(iii) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_\nu \geq X^2_{\text{obs}}\} \leq \alpha$.
(iv) *Degrees of Freedom*: $\nu = (IJ - 1) - ((I-1) + (J-1)) = (I-1)(J-1)$.

Example 3.2.8. *Binary Classifications*
Let $n = 1436$, $[n] := \{1, \dots, n\}$, and $(A_k, B_k) \in \{a_1, a_2\} \times \{b_1, b_2\}$ for $k \in [n]$.
Set $n_{ij} := \#\{k \in [n] : A_k = a_i,\ B_k = b_j\}$ and $E_{ij} := \frac{n_{i\cdot} n_{\cdot j}}{n}$. The observed and independence-fitted tables are

n_{ij}

$a_i \backslash b_j$	b_1	b_2	Total
a_1	550	61	611
a_2	681	144	825
Total	1231	205	1436

E_{ij}

$a_i \backslash b_j$	b_1	b_2
a_1	523.8	87.2
a_2	707.2	117.8

Under $H_0 : A \perp B$, $X^2 = \sum_{i=1}^{2} \sum_{j=1}^{2} \frac{(n_{ij} - E_{ij})^2}{E_{ij}} = 16.01$, $\nu = 1$, and $p < 0.001$; reject H_0.
Thus $A \not\perp B$; absent randomised A, inference is associative, not causal.

Remark 3.2.9. *Homogeneity Versus Independence*
Both tests use $E_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n_{\cdot\cdot}}$, $X^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$, and $\nu = (I-1)(J-1)$.

(i) *Homogeneity*: $N_{\cdot j} \sim \text{Multinomial}(n_{\cdot j}; \pi_{\cdot j})$ independently for $j \in [J]$, with fixed $n_{\cdot j}$, $H_0 : \pi_{\cdot 1} = \cdots = \pi_{\cdot J}$.
(ii) *Independence*: $N \sim \text{Multinomial}(n_{\cdot\cdot}; (\pi_{ij})_{[I]\times[J]})$ with $H_0 : \pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}\ \forall (i,j) \in [I] \times [J]$.

If $\pi_{\cdot j} > 0$, then independence iff $\pi_{i|j} := \frac{\pi_{ij}}{\pi_{\cdot j}} = \pi_{i\cdot}\ \forall (i,j) \in [I] \times [J]$.

Method 3.2.10. *Likelihood-Ratio Version*
Let $I, J \in \mathbb{N}$, $[K] := \{1, \dots, K\}$, and let $n_{ij} \in \mathbb{N}_0$ with null fitted counts E_{ij} for $(i,j) \in [I] \times [J]$.
Set $n_{\cdot\cdot} := \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} > 0$ and $\nu := (I-1)(J-1)$.
Let H_0 be the homogeneity or independence null producing E_{ij}, and assume $E_{ij} > 0$ whenever $n_{ij} > 0$.

(i) *Statistic*: $G^2 := 2 \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij} \log\left(\frac{n_{ij}}{E_{ij}}\right)$.
(ii) *Zero Counts*: $n_{ij} = 0$ contributes 0 to G^2.
(iii) *Calibration*: Under the large-sample likelihood-ratio conditions of Theorem 3.1.9, $G^2 \overset{H_0}{\approx} \chi^2_\nu$.
(iv) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_\nu \geq G^2_{\text{obs}}\} \leq \alpha$.

3.3 Matched Pairs and McNemar's Test

Matched categorical data require a different table from unmatched samples. The pair, not the individual measurement, is the independent sampling unit.

Definition 3.3.1. *Matched Binary Pair Table*
Let $(Y_k^{(1)}, Y_k^{(2)}) \in \{0,1\}^2$ for $k \in [n]$, and set $R_k := Y_k^{(1)} + 1$, $S_k := Y_k^{(2)} + 1$.
For $(r,s) \in \{1,2\}^2$, set $n_{rs} := \#\{k \in [n] : (R_k, S_k) = (r,s)\}$, so $n = \sum_{r=1}^{2} \sum_{s=1}^{2} n_{rs}$.

(i) *Matched Binary Pair Table*: $N := (n_{rs})_{(r,s)\in\{1,2\}^2}$.
(ii) *Concordant Cells*: $C := \{(r,s) \in \{1,2\}^2 : r = s\} = \{(1,1),(2,2)\}$; counts $(n_{rs})_{(r,s)\in C} = (n_{11}, n_{22})$.
(iii) *Discordant Cells*: $D := \{(r,s) \in \{1,2\}^2 : r \neq s\} = \{(1,2),(2,1)\}$; counts $(n_{rs})_{(r,s)\in D} = (n_{12}, n_{21})$.

Key Insight: (n_{12}, n_{21}) determines marginal change; (n_{11}, n_{22}) records agreement.

Definition 3.3.2. *Marginal Homogeneity for Matched Pairs*
Let $\pi = (\pi_{rs})_{(r,s)\in\{1,2\}^2} \in [0,1]^4$ satisfy $\sum_{r=1}^{2} \sum_{s=1}^{2} \pi_{rs} = 1$. Set $\pi_{r\cdot} := \sum_{s=1}^{2} \pi_{rs}$ and $\pi_{\cdot s} := \sum_{r=1}^{2} \pi_{rs}$.
The matched-pair law is *marginally homogeneous* iff $H_0 : (\pi_{1\cdot}, \pi_{2\cdot}) = (\pi_{\cdot 1}, \pi_{\cdot 2})$, equivalently $\pi_{12} = \pi_{21}$.
Key Insight: H_0 constrains only the discordant probabilities π_{12}, π_{21}.

Method 3.3.3. *McNemar's Test*
Let $n_{rs} \in \mathbb{N}_0$ and cell probabilities π_{rs} for $(r,s) \in \{1,2\}^2$, with $n := \sum_{r=1}^{2} \sum_{s=1}^{2} n_{rs} > 0$. Set $d := n_{12} + n_{21}$.

(i) *Null*: $H_0 : \pi_{12} = \pi_{21}$.
(ii) *MLE Under H_0*: $\widehat{\pi}_{11} = \frac{n_{11}}{n}$, $\widehat{\pi}_{22} = \frac{n_{22}}{n}$, and $\widehat{\pi}_{12} = \widehat{\pi}_{21} = \frac{d}{2n}$.
(iii) *Statistic*: If $d > 0$, $X^2 := \frac{(n_{12} - n_{21})^2}{d}$.
(iv) *Calibration*: If d is large, then $X^2 \overset{H_0}{\approx} \chi^2_1$.
(v) *Decision*: Reject H_0 at level α when $\mathbb{P}\{\chi^2_1 \geq X^2_{\text{obs}}\} \leq \alpha$.

Example 3.3.4. *Matched Case-Control Pairs*
In a matched case-control study, 85 cases were each paired with one control, and a binary exposure was recorded for both members of each pair.

	Control	
Case	Unexposed	Exposed
Unexposed	37	7
Exposed	15	26

An unmatched chi-square test treats cases and controls as independent samples.
McNemar's test uses only discordant pairs; here $X^2 = (7-15)^2/(7+15) = 2.91$ and $p \approx 0.09$, so accounting for the matching weakens the evidence of an exposure difference.

Example 3.3.5. *Matched Case-Crossover Exposure*
In a matched case-crossover study, each subject is observed during an event window and a comparable control window, with a binary exposure recorded in both windows.

Event window	Control window		
	Exposed	Unexposed	Total
Exposed	13	157	170
Unexposed	24	505	529
Total	37	662	699

The discordant counts are 157 and 24, so $X^2 = (157-24)^2/(157+24) = 97.7$, giving strong evidence that exposure prevalence differs between the two windows. This does not by itself establish causation, since time-varying covariates may affect both exposure and event risk.

Remark 3.3.6. *Matched Does Not Mean Independent*
McNemar's null hypothesis is marginal homogeneity, not independence of paired responses. Thus the analysis must preserve the matching structure rather than treat the two margins as independent samples.

3.4 Odds Ratios and Case-Control Sampling

Odds ratios summarize association in a binary exposure–binary outcome table. They are especially useful because the same cross-product ratio can be estimated under several sampling designs, including retrospective case-control designs where risks themselves cannot be estimated.

Definition 3.4.1. *Odds and Odds Ratio*
Let A satisfy $0 < \mathbb{P}(A) < 1$.
Let X and D be binary exposure and disease events with $0 < \mathbb{P}(X) < 1$ and $0 < \mathbb{P}(D \mid X), \mathbb{P}(D \mid \overline{X}) < 1$.

(i) *Odds*: $\operatorname{odds}(A) := \frac{\mathbb{P}(A)}{1-\mathbb{P}(A)}$.
(ii) *Odds ratio*: $\Delta := \frac{\operatorname{odds}(D|X)}{\operatorname{odds}(D|\overline{X})}$.

Structural Commentary: If $o = \operatorname{odds}(A)$, then $\mathbb{P}(A) = \frac{o}{1+o}$; $p \mapsto p/(1-p)$ is a bijection $(0,1) \to (0,\infty)$.

Proposition 3.4.2. *Cross-Product Formula*
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $X, D \in \mathcal{F}$, and set $X_0 := \overline{X}$, $X_1 := X$, $D_0 := \overline{D}$, $D_1 := D$.
Let $\pi_{ij} := \mathbb{P}(X_i \cap D_j) > 0$ for $i, j \in \{0,1\}$, with margins $\pi_{i\cdot} := \pi_{i0} + \pi_{i1}$ and $\pi_{\cdot j} := \pi_{0j} + \pi_{1j}$.

	$\overline{D}$	D	
$\overline{X}$	π_{00}	π_{01}	$\pi_{0\cdot}$
X	π_{10}	π_{11}	$\pi_{1\cdot}$
	$\pi_{\cdot 0}$	$\pi_{\cdot 1}$	1

With $\Delta := \frac{\operatorname{odds}(D|X)}{\operatorname{odds}(D|\overline{X})}$, the cross-product identity is $\Delta = \frac{\pi_{11}\pi_{00}}{\pi_{01}\pi_{10}}$.

Proof. Since $\pi_{1\cdot} = \pi_{10} + \pi_{11}$ and $\pi_{0\cdot} = \pi_{00} + \pi_{01}$, $\mathbb{P}(D \mid X) = \frac{\pi_{11}}{\pi_{1\cdot}}$ and $1 - \mathbb{P}(D \mid X) = \frac{\pi_{10}}{\pi_{1\cdot}}$, so $\operatorname{odds}(D \mid X) = \frac{\pi_{11}}{\pi_{10}}$.
Similarly, $\operatorname{odds}(D \mid \overline{X}) = \frac{\pi_{01}}{\pi_{00}}$. Hence $\Delta = \frac{\frac{\pi_{11}}{\pi_{10}}}{\frac{\pi_{01}}{\pi_{00}}} = \frac{\pi_{11}\pi_{00}}{\pi_{01}\pi_{10}}$.
Structural Commentary: $\pi_{1\cdot}, \pi_{0\cdot}$ enter conditional probabilities but cancel after passage to odds.
Key Insight: Δ equals diagonal product $\pi_{11}\pi_{00}$ divided by off-diagonal product $\pi_{01}\pi_{10}$. □

Method 3.4.3. *Estimating an Odds Ratio by Design*
Let $N = (n_{ij})_{i,j\in\{0,1\}}$ be a 2×2 table for (X_i, D_j), where $X_1 := X$, $X_0 := \overline{X}$, $D_1 := D$, and $D_0 := \overline{D}$.
Assume $n_{ij} > 0$, set $n_{i\cdot} := n_{i0} + n_{i1}$, $n_{\cdot j} := n_{0j} + n_{1j}$, $n := \sum_{i=0}^{1}\sum_{j=0}^{1} n_{ij}$.

(i) *Simple Random Sampling*: $\widehat{\pi}_{ij} := \frac{n_{ij}}{n}$ estimates π_{ij}; hence $\widehat{\Delta} := \frac{n_{00}n_{11}}{n_{01}n_{10}}$ estimates Δ.

(ii) *Prospective Sampling*: $n_{0\cdot}, n_{1\cdot}$ fixed; $\frac{n_{01}}{n_{0\cdot}}$, $\frac{n_{11}}{n_{1\cdot}}$ estimate $\mathbb{P}(D \mid \overline{X})$, $\mathbb{P}(D \mid X)$; $\mathbb{P}(X)$ not identified.
(iii) *Retrospective Case-Control Sampling*: $n_{\cdot 0}, n_{\cdot 1}$ are fixed; disease risks are not identified, but $\widehat{\Delta} = \frac{n_{11}}{n_{01}} / \frac{n_{10}}{n_{00}}$ is estimable from exposure odds.

Key Insight: Design-fixed margins cancel in $\widehat{\Delta}$, so odds ratio is estimable even when risks or margins are not.

Example 3.4.4. *Unmatched Case-Control Odds Ratio*
Consider an unmatched case-control table with binary exposure:

	Exposed	Unexposed
Case	67	34
Control	43	64

The estimated odds ratio is $\widehat{\Delta} = \frac{67 \cdot 64}{43 \cdot 34)} = 2.93$, indicating a positive exposure–disease association.

Method 3.4.5. *Parametric Bootstrap for an Odds-Ratio Standard Error*
Let $a, b, c, d \in \mathbb{N}$ denote the observed counts in rows $D, \overline{D}$ and columns $X, \overline{X}$: $a := n_{DX}$, $b := n_{D\overline{X}}$, $c := n_{\overline{D}X}$, and $d := n_{\overline{D}\overline{X}}$. Set $m_D := a + b$, $m_{\overline{D}} := c + d$, $\widehat{p}_D := \frac{a}{m_D}$, $\widehat{p}_{\overline{D}} := \frac{c}{m_{\overline{D}}}$, and $\widehat{\Delta} := \frac{ad}{bc}$.

(i) *Bootstrap Law*: For $r \in [R]$, draw independently $A_r^* \sim \text{Binomial}(m_D, \widehat{p}_D)$ and $C_r^* \sim \text{Binomial}(m_{\overline{D}}, \widehat{p}_{\overline{D}})$.
(ii) *Replicate*: If $0 < A_r^* < m_D$ and $0 < C_r^* < m_{\overline{D}}$, set $\widehat{\Delta}_r^* := \frac{A_r^*(m_{\overline{D}} - C_r^*)}{(m_D - A_r^*)C_r^*}$.
(iii) *Standard Error*: With $\overline{\Delta}^* := \frac{1}{R}\sum_{r=1}^{R} \widehat{\Delta}_r^*$, set $\widehat{\text{se}}_*(\widehat{\Delta}) := \left(\frac{1}{R-1}\sum_{r=1}^{R}(\widehat{\Delta}_r^* - \overline{\Delta}^*)^2\right)^{1/2}$.

Key Insight: Resample exposure counts conditional on fixed row totals; $\text{sd}\{(\widehat{\Delta}_r^*)_{r\in[R]}\}$ estimates $\text{se}(\widehat{\Delta})$.

Remark 3.4.6. *Odds Ratios Are Not Automatically Risks*
The odds ratio $\Delta = \text{odds}(D \mid X)/\text{odds}(D \mid \overline{X})$ differs from risk contrasts such as $\mathbb{P}(D \mid X) - \mathbb{P}(D \mid \overline{X})$.
In retrospective sampling, $n_{\cdot 0}, n_{\cdot 1}$ are design-fixed, absolute risks are not identified without population margins; Δ remains estimable because those margins cancel in the cross-product ratio.

3.5 Higher-Order Tables and Sampling Design

Two-way tables are the simplest categorical-data structure. Higher-order tables introduce several possible notions of dependence: full independence, conditional independence given a third factor, marginal homogeneity, and interactions among factors.

Definition 3.5.1. *Two-Way Log-Linear Form*
Let $I, J \in \mathbb{N}$ and let $\pi_{ij} > 0$ be two-way cell probabilities for $1 \le i \le I$, $1 \le j \le J$,
with $\pi_{i\cdot} := \sum_{j=1}^{J} \pi_{ij}$, $\pi_{\cdot j} := \sum_{i=1}^{I} \pi_{ij}$, $\alpha_i := \log \pi_{i\cdot}$, and $\beta_j := \log \pi_{\cdot j}$.

(i) *Independence form*: $\pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}\ \forall 1 \le i \le I, 1 \le j \le J \iff \log \pi_{ij} = \alpha_i + \beta_j\ \forall 1 \le i \le I, 1 \le j \le J$.
(ii) *Interaction terms*: $\gamma_{ij} := \log \pi_{ij} - \alpha_i - \beta_j$, giving $\log \pi_{ij} = \alpha_i + \beta_j + \gamma_{ij}$.

Key Insight: $\gamma_{ij} = 0\ \forall 1 \le i \le I, 1 \le j \le J \iff \pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}\ \forall 1 \le i \le I, 1 \le j \le J$.

Remark 3.5.2. *Three-Way Interactions*
Let $\pi_{ijk} > 0$ for $(i, j, k) \in [I] \times [J] \times [K]$. A log-linear model with all two-factor interactions and no three-factor interaction is $\log \pi_{ijk} = \mu + \alpha_i + \beta_j + \eta_k + \delta_{ij} + \varepsilon_{ik} + \gamma_{jk}$.
The absent term λ_{ijk} encodes the constraint of no third-order interaction; nested models are compared by deleting interaction terms and using the corresponding loss of degrees of freedom.

Principle 3.5.3. *Choose the Test from the Sampling Mechanism*
For a contingency table $N = (n_{ij})$, let $n_{i\cdot}$ and $n_{\cdot j}$ denote row and column margins.
Select the test from the design-fixed quantities:

(i) *Fixed Margins*: If $(n_{i\cdot})$ and $(n_{\cdot j})$ are fixed in a 2×2 table, use Fisher's exact test.
(ii) *Independent Samples*: If rows are independent multinomial samples with fixed $(n_{i\cdot})$, test homogeneity of column probabilities.
(iii) *Single Multinomial Sample*: If $N \sim \text{Multinomial}(n, (\pi_{ij}))$, test independence, i.e. $\pi_{ij} = \pi_{i\cdot}\pi_{\cdot j}$.
(iv) *Matched Pairs*: If observations are paired counts n_{rs}, test marginal homogeneity using discordant cells.
(v) *Case-Control Sampling*: If $(n_{\cdot 0}, n_{\cdot 1})$ are fixed by disease status, estimate odds ratios. Absolute risks require population margins.

Key Insight: The same numeric table can encode different null hypotheses under different sampling laws.

Remark 3.5.4. *Use Counts, Not Percentages*
Let $Q(O,E) := \sum_i \frac{(O_i-E_i)^2}{E_i}$. Pearson's chi-square calibration applies to counts: $Q(O/n, E/n) = \frac{1}{n}Q(O,E)$ and $Q(100O/n, 100E/n) = \frac{100}{n}Q(O,E)$. Thus relative frequencies or percentages give a rescaled statistic unless the original count scale is restored.

4 Analysis of Variance and Designed Experiments

4.1 One-Way Layouts and the F Test

Analysis of variance compares several mean levels by splitting total variation into variation within treatment groups and variation between treatment groups. The method is a multisample continuation of the equal-variance two-sample procedure, with design and randomisation assumptions kept explicit.

Definition 4.1.1. *Balanced One-Way Layout*
For $I, J \in \mathbb{N}$, one factor has levels $i = 1, \dots, I$, each with exactly J independent replicates Y_{ij}, $j = 1, \dots, J$.
The normal fixed-effects model satisfies $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $\sum_{i=1}^{I} \alpha_i = 0$, and independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Structural Commentary: $\mathbb{E}Y_{ij} = \mu + \alpha_i$, and $\sum_{i=1}^{I} \alpha_i = 0$ identifies μ as the average treatment mean.
Key Insight: $H_0 : \alpha_i = 0 \ \forall i$ iff all treatment means are equal.

Definition 4.1.2. *One-Way Sums of Squares*
For a balanced one-way layout, set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_{j=1}^{J} Y_{ij}$ and $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J} Y_{ij}$.

(i) *Total*: $SS_{\text{tot}} := \sum_{i=1}^{I}\sum_{j=1}^{J}(Y_{ij} - \overline{Y}_{\cdot\cdot})^2$; grand-mean dispersion.

(ii) *Within-treatment*: $SS_W := \sum_{i=1}^{I}\sum_{j=1}^{J}(Y_{ij} - \overline{Y}_{i\cdot})^2$; residual dispersion within treatments.

(iii) *Between-treatment*: $SS_B := J\sum_{i=1}^{I}(\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$; treatment-mean dispersion scaled by J.

Key Insight: $SS_{\text{tot}} = SS_W + SS_B$ separates within-treatment noise from between-treatment mean differences.

Proposition 4.1.3. *One-Way Sum-of-Squares Decomposition*
For $I, J \in \mathbb{N}$ and observations Y_{ij}, set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$,
$SS_{\text{tot}} := \sum_i\sum_j (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$, $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, *and* $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$. *Then* $SS_{\text{tot}} = SS_W + SS_B$.

Proof. Set $r_{ij} := Y_{ij} - \overline{Y}_{i\cdot}$ and $b_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$. Then $Y_{ij} - \overline{Y}_{\cdot\cdot} = r_{ij} + b_i$ and $\sum_j r_{ij} = 0 \ \forall i$.
Thus $SS_{\text{tot}} = \sum_i\sum_j (r_{ij} + b_i)^2 = \sum_i\sum_j r_{ij}^2 + 2\sum_i b_i \sum_j r_{ij} + J\sum_i b_i^2 = SS_W + SS_B$.
Structural Commentary: $Y_{ij} - \overline{Y}_{\cdot\cdot} = r_{ij} + b_i$, where $r_{ij} := Y_{ij} - \overline{Y}_{i\cdot}$ and $b_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$.
Key Insight: $\sum_j r_{ij} = 0 \ \forall i$ annihilates the cross term $2\sum_i b_i \sum_j r_{ij}$. □

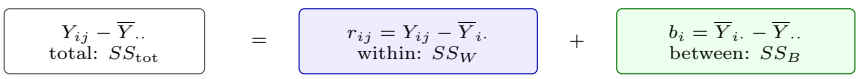


Figure 225: Squaring and summing gives $SS_{\text{tot}} = SS_W + SS_B$; large MS_B/MS_W suggests treatment effects.

Lemma 4.1.4. *Expected Squared Deviation from a Sample Mean*
Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = \mu_i$ and $\mathrm{Var}(X_i) = \sigma^2$ for $1 \le i \le n$.
Set $\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$ and $\overline{\mu} := \frac{1}{n}\sum_{i=1}^{n}\mu_i$. Then $\mathbb{E}(X_i - \overline{X})^2 = (\mu_i - \overline{\mu})^2 + \frac{n-1}{n}\sigma^2$ for $1 \le i \le n$.

Proof. Set $U_i := X_i - \overline{X}$. Then $\mathbb{E}U_i = \mu_i - \overline{\mu}$, $\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$, and $\mathrm{Cov}(X_i, \overline{X}) = \frac{\sigma^2}{n}$ by independence.
Hence $\mathrm{Var}(U_i) = \mathrm{Var}(X_i) + \mathrm{Var}(\overline{X}) - 2\,\mathrm{Cov}(X_i, \overline{X}) = \sigma^2 + \frac{\sigma^2}{n} - 2\frac{\sigma^2}{n} = \frac{n-1}{n}\sigma^2$.
Thus $\mathbb{E}U_i^2 = (\mathbb{E}U_i)^2 + \mathrm{Var}(U_i) = (\mu_i - \overline{\mu})^2 + \frac{n-1}{n}\sigma^2$.
Structural Commentary: $X_i - \overline{X}$ has mean $\mu_i - \overline{\mu}$ and variance $\mathrm{Var}(X_i - \overline{X}) = \frac{n-1}{n}\sigma^2$.
Key Insight: $\mathrm{Cov}(X_i, \overline{X}) = \frac{\sigma^2}{n}$ removes exactly one $\frac{\sigma^2}{n}$ from the raw variance σ^2. □

Theorem 4.1.5. *Expected One-Way Sums of Squares*
For $I, J \in \mathbb{N}$, let $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = 0$, independent errors satisfying $\mathbb{E}\varepsilon_{ij} = 0$, $\mathrm{Var}(\varepsilon_{ij}) = \sigma^2$. Set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$.
Then $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$; $\mathbb{E}(SS_W) = I(J-1)\sigma^2$ and $\mathbb{E}(SS_B) = (I-1)\sigma^2 + J\sum_i \alpha_i^2$. Hence $S_p^2 := \frac{SS_W}{I(J-1)}$ satisfies $\mathbb{E}S_p^2 = \sigma^2$.

Proof. For fixed i, Lemma 4.1.4 with $\mu_{ij} = \mu + \alpha_i$ gives $\mathbb{E}\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2 = (J-1)\sigma^2$, hence $\mathbb{E}SS_W = I(J-1)\sigma^2$.
For $\overline{Y}_{i\cdot}$, $\mathbb{E}\overline{Y}_{i\cdot} = \mu + \alpha_i$, $\mathrm{Var}(\overline{Y}_{i\cdot}) = \frac{\sigma^2}{J}$, and $\frac{1}{I}\sum_i (\mu + \alpha_i) = \mu$. By Lemma 4.1.4, $(\overline{Y}_{i\cdot})_{i=1}^I$ gives
$\mathbb{E}\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2 = \sum_i \alpha_i^2 + \frac{I-1}{J}\sigma^2$, so $\mathbb{E}SS_B = J\sum_i \alpha_i^2 + (I-1)\sigma^2$. Thus $\mathbb{E}S_p^2 = \frac{\mathbb{E}SS_W}{I(J-1)} = \sigma^2$.
Structural Commentary: $\mathbb{E}SS_W$ depends only on σ^2; $\mathbb{E}SS_B$ adds $J\sum_i \alpha_i^2$.
Key Insight: Under $\alpha_i = 0\ \forall i$, $\mathbb{E}\{\frac{SS_B}{I-1}\} = \mathbb{E}\{\frac{SS_W}{I(J-1)}\} = \sigma^2$. □

Theorem 4.1.6. *Normal-Theory One-Way F Test*
For $I, J \in \mathbb{N}$, let $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = 0$ and independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$.
Set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_j Y_{ij}$, $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_i\sum_j Y_{ij}$, $SS_W := \sum_i\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := J\sum_i (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$.
$MS_W := \frac{SS_W}{I(J-1)}$, $MS_B := \frac{SS_B}{I-1}$; $\frac{SS_W}{\sigma^2} \sim \chi^2_{I(J-1)}$.
If $H_0 : \alpha_i = 0\ \forall i$, then $\frac{SS_B}{\sigma^2} \sim \chi^2_{I-1}$, $SS_B \perp\!\!\!\perp SS_W$, $F := \frac{MS_B}{MS_W} \sim F_{I-1, I(J-1)}$.

Proof. Let $C_m := \mathrm{Id}_m - m^{-1}\mathbf{1}_m\mathbf{1}_m^\top$, so $C_m^2 = C_m = C_m^\top$ and $\mathrm{rank}\, C_m = m - 1$.
For each i, $Z_i := \frac{1}{\sigma}(Y_{i1} - \mu - \alpha_i, \ldots, Y_{iJ} - \mu - \alpha_i)^\top \sim N_J(0, \mathrm{Id}_J)$ independently,
$\frac{1}{\sigma^2}\sum_j (Y_{ij} - \overline{Y}_{i\cdot})^2 = Z_i^\top C_J Z_i \sim \chi^2_{J-1}$. Summing over independent treatments gives $\frac{SS_W}{\sigma^2} \sim \chi^2_{I(J-1)}$.
Under H_0, $\widetilde{Z} := \sqrt{J}\sigma^{-1}(\overline{Y}_{1\cdot} - \mu, \ldots, \overline{Y}_{I\cdot} - \mu)^\top \sim N_I(0, \mathrm{Id}_I)$, and $\frac{SS_B}{\sigma^2} = \widetilde{Z}^\top C_I \widetilde{Z} \sim \chi^2_{I-1}$.
For each i, $C_J Z_i \perp\!\!\!\perp \mathbf{1}_J^\top Z_i$ by orthogonal normal components.
Hence $SS_W \perp\!\!\!\perp (\overline{Y}_{1\cdot}, \ldots, \overline{Y}_{I\cdot})$ and $SS_W \perp\!\!\!\perp SS_B$. Thus $F = \frac{SS_B}{I-1} / \frac{SS_W}{I(J-1)} = \frac{MS_B}{MS_W} \sim F_{I-1, I(J-1)}$.
Structural Commentary: $IJ = 1 + I(J-1) + (I-1)$ partitions df into mean, residual, and contrast subspaces.
Key Insight: Under H_0, MS_B and MS_W are independent σ^2 estimators, so $\frac{MS_B}{MS_W}$ has an F law. □

Method 4.1.7. *Unequal Sample Sizes*

For $I \in \mathbb{N}$, treatment i has observations Y_{ij}, $1 \le j \le J_i$, with $J_i \in \mathbb{N}$ and $N := \sum_{i=1}^I J_i$.

(i) $\overline{Y}_{i\cdot} := \frac{1}{J_i}\sum_{j=1}^{J_i} Y_{ij}$ and $\overline{Y}_{\cdot\cdot} := \frac{1}{N}\sum_{i=1}^I\sum_{j=1}^{J_i} Y_{ij}$.

(ii) $SS_W := \sum_{i=1}^I\sum_{j=1}^{J_i} (Y_{ij} - \overline{Y}_{i\cdot})^2$, $SS_B := \sum_{i=1}^I J_i(\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$, $SS_{\mathrm{tot}} := \sum_{i=1}^I\sum_{j=1}^{J_i} (Y_{ij} - \overline{Y}_{\cdot\cdot})^2$ satisfy $SS_{\mathrm{tot}} = SS_W + SS_B$.

(iii) $\mathrm{df}_W = N - I$ and $\mathrm{df}_B = I - 1$.

(iv) If $Y_{ij} = \mu_i + \varepsilon_{ij}$ with independent $\varepsilon_{ij} \sim N(0, \sigma^2)$ and $H_0 : \mu_1 = \cdots = \mu_I$, then $F := \frac{\frac{SS_B}{I-1}}{\frac{SS_W}{N-I}} \sim F_{I-1, N-I}$.

Corollary 4.1.8. *Two Treatments Give $F = t^2$*
Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ be independent normal samples with common variance, means $\overline{X}, \overline{Y}$, pooled variance $S_p^2 := \frac{SS_W}{n+m-2}$, and $t := \frac{\overline{X} - \overline{Y}}{S_p\sqrt{n^{-1} + m^{-1}}}$. The one-way ANOVA statistic for the two groups satisfies $F = t^2$.

Proof. Let $N := n + m$ and $\overline{Z} := \frac{n\overline{X} + m\overline{Y}}{N}$. Then $SS_B = n(\overline{X} - \overline{Z})^2 + m(\overline{Y} - \overline{Z})^2 = \frac{nm}{N}(\overline{X} - \overline{Y})^2$.
Since $df_B = 1$ and $MS_W = \frac{SS_W}{N-2} = S_p^2$, $F = \frac{SS_B}{MS_W} = \frac{(\overline{X} - \overline{Y})^2}{S_p^2(n^{-1} + m^{-1})} = t^2$.
Structural Commentary: Two groups leave one between-group contrast.
Key Insight: ANOVA squares the pooled t contrast. □

Example 4.1.9. *Generic Balanced One-Way ANOVA*
Let $I = 7$, $J = 10$, and $Y_{ij} = \mu_i + \varepsilon_{ij}$ with independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$. Suppose the ANOVA table is

Component	df	SS	MS	F
Treatment	6	.125	.021	5.66
Error	63	.231	.0037	
Total	69	.356		

Then $F = 5.66$ has null law $F_{6,63}$, $p = \mathbb{P}(F_{6,63} \geq 5.66) < .01$, and $H_0 : \mu_1 = \cdots = \mu_7$ is rejected at level .01.
If diagnostics are adequate, between-treatment variation exceeds within-treatment noise.

The same one-way decomposition also supports a random-effects interpretation, where groups are sampled from a population of possible group levels rather than fixed in advance.

Definition 4.1.10. *Balanced One-Way Random Effects Model*
A *balanced one-way random-effects model* has I groups and common replication J, with $Y_{ij} = \mu + A_i + \varepsilon_{ij}$ for $1 \leq i \leq I$, $1 \leq j \leq J$, independent A_i satisfying $\mathbb{E}A_i = 0$ and $\mathrm{Var}(A_i) = \sigma_A^2$, and independent errors ε_{ij} independent of (A_i) with $\mathbb{E}\varepsilon_{ij} = 0$ and $\mathrm{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$.
Structural Commentary: $\mathrm{Var}(Y_{ij}) = \sigma_A^2 + \sigma_\varepsilon^2$, and $\mathrm{Cov}(Y_{ij}, Y_{ij'}) = \sigma_A^2$ for $j \neq j'$ within the same group.

Proposition 4.1.11. *Random-Effects Expected Mean Squares*
Assume $Y_{ij} = \mu + A_i + \varepsilon_{ij}$ for $1 \leq i \leq I$, $1 \leq j \leq J$, with independent A_i and ε_{ij}, $\mathbb{E}A_i = \mathbb{E}\varepsilon_{ij} = 0$, $\mathrm{Var}(A_i) = \sigma_A^2$, and $\mathrm{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$. Then $\mathbb{E}(MS_W) = \sigma_\varepsilon^2$ and $\mathbb{E}(MS_B) = \sigma_\varepsilon^2 + J\sigma_A^2$.
The method-of-moments estimators are $\widehat{\sigma}_\varepsilon^2 = MS_W$ and $\widehat{\sigma}_A^2 = \frac{MS_B - MS_W}{J}$, truncated at 0 if required.

Proof. Condition on (A_i). Since $Y_{ij} - \overline{Y}_{i\cdot} = \varepsilon_{ij} - \overline{\varepsilon}_{i\cdot}$, $\mathbb{E}(SS_W \mid A) = I(J-1)\sigma_\varepsilon^2$, hence $\mathbb{E}(MS_W) = \sigma_\varepsilon^2$.
Also $\overline{Y}_{i\cdot} = \mu + A_i + \overline{\varepsilon}_{i\cdot}$ has variance $\sigma_A^2 + \frac{\sigma_\varepsilon^2}{J}$.
Hence the one-way between mean square satisfies $\mathbb{E}(MS_B) = J\,\mathrm{Var}(\overline{Y}_{i\cdot}) = \sigma_\varepsilon^2 + J\sigma_A^2$.
Solving these two moment equations gives the estimators.
Structural Commentary: Within estimates error; between adds group heterogeneity.
Key Insight: The excess $MS_B - MS_W$ estimates $J\sigma_A^2$. □

Example 4.1.12. *Dye Strength Batch Variation*
Suppose $I = 6$ batches each have $J = 6$ subsamples, with $\overline{y}_{\cdot\cdot} = 93.61$
and batch means $(92.00, 90.50, 91.25, 98.25, 91.17, 98.50)$.
Given $SS_B = 415.47$ and $SS_W = 134.08$, the mean squares are $MS_B = 83.09$ and $MS_W = 4.47$.
The moment estimates are $\widehat{\sigma}_\varepsilon^2 = MS_W = 4.47$ and $\widehat{\sigma}_A^2 = \frac{MS_B - MS_W}{6} = 13.10$.
Batch variation dominates measurement-error variation.

Remark 4.1.13. *Assumptions and Robustness*
For $Y_{ij} = \mu_i + \varepsilon_{ij}$, the one-way F calibration assumes independent $\varepsilon_{ij} \sim N(0, \sigma^2)$. Balance softens variance heterogeneity; clustered, paired, blocked, or repeated data require dependence modeling.

4.2 Multiple Comparisons After ANOVA

The one-way F test assesses $H_0 : \mu_1 = \cdots = \mu_I$; after rejection, inference targets contrasts $\mu_a - \mu_b$ with error control over the chosen comparison family.

Definition 4.2.1. *Studentized Range Quantile*
Let $Q_{I,\nu}$ denote a random variable with the studentized range distribution for I treatment means and ν error degrees of freedom. The *studentized range quantile* is $q_{I,\nu}(\alpha) := \inf\{q \in \mathbb{R} : \mathbb{P}(Q_{I,\nu} > q) \leq \alpha\}$.
Structural Commentary: Balanced Tukey comparisons use pooled scale S_p.

Method 4.2.2. *Tukey Pairwise Comparisons*
In a balanced normal one-way layout, let $\mathcal{P} := \{(a, b) : 1 \leq a < b \leq I\}$, $\nu := I(J-1)$, and $h_\alpha := q_{I,\nu}(\alpha) S_p / \sqrt{J}$.

(i) For $(a, b) \in \mathcal{P}$, the simultaneous $100(1-\alpha)\%$ confidence interval for $\mu_a - \mu_b$ is $(\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}) \pm h_\alpha$.
(ii) Reject $H_{0,ab} : \mu_a = \mu_b$ iff $\left|\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}\right| > h_\alpha$.

Simultaneity over $\mathcal{P}$ uses the studentized range cutoff.

Example 4.2.3. *Generic Tukey Pairwise Calculation*
Consider a balanced one-way layout with $I = 7$, $J = 10$, $S_p = .06$, and ordered sample means
$(\overline{Y}_{1\cdot}, \overline{Y}_{3\cdot}, \overline{Y}_{7\cdot}, \overline{Y}_{2\cdot}, \overline{Y}_{5\cdot}, \overline{Y}_{6\cdot}, \overline{Y}_{4\cdot}) = (4.062, 4.003, 3.998, 3.997, 3.957, 3.955, 3.920)$.
With $q_{7,60}(.05) = 4.31$, the Tukey half-width is $h_{.05} = 4.31(.06)/\sqrt{10} \approx .082$.
Rejected pairs at simultaneous level .05 are $(1,4), (1,5), (1,6), (3,4)$; remaining pairwise nulls not rejected.

Method 4.2.4. *Bonferroni Pairwise Comparisons*
Let $\mathcal{C} \subseteq \{(a,b) : 1 \le a < b \le I\}$ be the planned pairwise family with $|\mathcal{C}| = k$.
Set $\nu := N - I$ and $h_{ab} := t_{\nu,1-\alpha/(2k)} S_p \sqrt{1/J_a + 1/J_b}$.

(i) For $(a,b) \in \mathcal{C}$, the Bonferroni confidence interval for $\mu_a - \mu_b$ is $(\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}) \pm h_{ab}$.
(ii) Reject $H_{0,ab} : \mu_a = \mu_b$ iff $\left|\overline{Y}_{a\cdot} - \overline{Y}_{b\cdot}\right| > h_{ab}$.

The family satisfies FWER $\le \alpha$ and allows unequal treatment sizes through J_a, J_b.

Example 4.2.5. *Generic Bonferroni Pairwise Calculation*
For $I = 7$ treatments and $k = \binom{7}{2} = 21$ planned pairwise comparisons at $\alpha = .05$, take $t_{60,1-.05/(2\cdot 21)} \approx 3.16$.
The resulting simultaneous 95% Bonferroni half-width is $h \approx .085$.
Rejected pairwise nulls are $(1,4), (1,5), (1,6)$; all other planned pairwise nulls are not rejected.

Remark 4.2.6. *The Global Test Does Not Name the Difference*
Rejecting $H_0 : \mu_1 = \cdots = \mu_I$ establishes only $\exists\, a < b$ with $\mu_a \ne \mu_b$.
It does not identify such pairs or provide simultaneous sign guarantees.
Select Tukey, Bonferroni, or planned-contrast procedures according to the preassigned comparison family.

4.3 Rank-Based One-Way ANOVA

When normality or outlier sensitivity is a concern, a one-way comparison can be made after replacing observations by pooled ranks.

Definition 4.3.1. *Kruskal–Wallis Ranks*
Let $\mathcal{Y} := \{Y_{ij} : 1 \le i \le I, 1 \le j \le J_i\}$ be an unequal-size one-way sample with $N := \sum_{i=1}^{I} J_i$.
Assign increasing pooled ranks to $\mathcal{Y}$, averaging ranks over ties.

(i) *Pooled rank*: $R_{ij} := \mathrm{rank}_{\mathrm{pool}}(Y_{ij})$.
(ii) *Treatment mean rank*: $\overline{R}_{i\cdot} := \frac{1}{J_i} \sum_{j=1}^{J_i} R_{ij}$.
(iii) *Grand mean rank*: $\overline{R}_{\cdot\cdot} := \frac{N+1}{2}$.

Method 4.3.2. *Kruskal–Wallis Test*
Let $SS_B^{(R)} := \sum_{i=1}^{I} J_i(\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2$ and $K := \frac{12}{N(N+1)} SS_B^{(R)} = \frac{12}{N(N+1)} \sum_{i=1}^{I} J_i \overline{R}_{i\cdot}^2 - 3(N+1)$.

(i) The exact null law enumerates all allocations of pooled ranks to group sizes $J_1, \ldots, J_I$.
(ii) For moderate samples, $K \overset{H_0}{\approx} \chi^2_{I-1}$.
(iii) Reject $H_0 : F_1 = \cdots = F_I$ at level α when $\mathbb{P}\{\chi^2_{I-1} \ge K_{\mathrm{obs}}\} \le \alpha$.

Example 4.3.3. *Generic Kruskal–Wallis Rank Calculation*
For $I = 7$ groups, suppose the pooled ranks give $K_{\mathrm{obs}} = 29.51$.
Using $K \overset{H_0}{\approx} \chi^2_6$, the tail probability satisfies $\mathbb{P}\{\chi^2_6 \ge 29.51\} < .005$.
Reject $H_0 : F_1 = \cdots = F_7$; the group response distributions are not all equal.

Corollary 4.3.4. *Two Groups Recover Mann–Whitney*
For $I = 2$, let $T_i := \sum_{j=1}^{J_i} R_{ij}$ and $N = J_1 + J_2$. Then $K = \frac{12}{J_1 J_2 (N+1)} \left(T_1 - \frac{J_1(N+1)}{2}\right)^2$, so the Kruskal–Wallis rejection order is the two-sided Mann–Whitney rank-sum order.

Proof. Let $G := \frac{N+1}{2}$. Since $T_1 + T_2 = NG$, $\overline{R}_{1\cdot} - G = \frac{T_1 - J_1 G}{J_1}$ and $\overline{R}_{2\cdot} - G = -\frac{T_1 - J_1 G}{J_2}$.
Hence $SS_B^{(R)} = (T_1 - J_1 G)^2 (\frac{1}{J_1} + \frac{1}{J_2}) = \frac{N(T_1 - J_1 G)^2}{J_1 J_2}$, and the formula for K follows.
Since $U_1 = T_1 - \frac{J_1(J_1+1)}{2}$ is affine in T_1, both tests order evidence by $|T_1 - J_1 G|$.
Structural Commentary: Two groups leave one rank contrast.
Key Insight: K is a squared centered rank sum. □

Method 4.3.5. *Rank-Based Follow-Up Comparisons*
After rejecting global rank null, let $\mathcal{C} \subseteq \{(a,b) : 1 \le a < b \le I\}$ be planned follow-up family with $|\mathcal{C}| = k$.

(i) For $(a,b) \in \mathcal{C}$, test $H_{0,ab} : F_a = F_b$ by a two-sample rank-sum statistic and let p_{ab} be its two-sided p-value.
(ii) Reject $H_{0,ab}$ iff $p_{ab} \le \frac{\alpha}{k}$, equivalently $k p_{ab} \le \alpha$.
(iii) The selected family satisfies FWER $\le \alpha$ and is most efficient when $\mathcal{C}$ is fixed before inspection.

Remark 4.3.6. *What Rank ANOVA Tests*
The null is $H_0 : F_1 = \cdots = F_I$, not merely $\mu_1 = \cdots = \mu_I$. Under comparable distributional shapes, rejection is read as a location difference; the rank form avoids normality and reduces sensitivity to extremes.

4.4 Two-Way Layouts and Interaction

A two-way layout studies two factors at once. It estimates main effects for each factor and checks whether the effect of one factor changes across levels of the other.

Definition 4.4.1. *Two-Way Additive Parametrization*
For one observation Y_{ij} at factor levels $i = 1, \dots, I$ and $j = 1, \dots, J$,
set $\overline{Y}_{i\cdot} := J^{-1} \sum_{j=1}^{J} Y_{ij}$, $\overline{Y}_{\cdot j} := I^{-1} \sum_{i=1}^{I} Y_{ij}$, and $\overline{Y}_{\cdot\cdot} := (IJ)^{-1} \sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij}$.

(i) *Intercept*: $\widehat{\mu} := \overline{Y}_{\cdot\cdot}$.
(ii) *Factor A, B coefficient*: $\widehat{\alpha}_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$, $1 \le i \le I$ and $\widehat{\beta}_j := \overline{Y}_{\cdot j} - \overline{Y}_{\cdot\cdot}$, $1 \le j \le J$.
(iii) *Additive fitted value*: $\widehat{Y}_{ij} := \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j$.
(iv) *Interaction residual*: $\widehat{\delta}_{ij} := Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}_{\cdot\cdot}$, so $Y_{ij} = \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j + \widehat{\delta}_{ij}$.

Structural Commentary: Exact fits satisfy zero-sum effects and margins: $\sum_i \widehat{\alpha}_i = \sum_j \widehat{\beta}_j = \sum_i \widehat{\delta}_{ij} = \sum_j \widehat{\delta}_{ij} = 0$.

Remark 4.4.2. *Interaction and Parallel Profiles*
Under additivity, $\widehat{Y}_{ij} - \widehat{Y}_{i'j} = \widehat{\alpha}_i - \widehat{\alpha}_{i'}$ is independent of j, so factor-A profiles are parallel across B. Nonparallel profiles indicate $\delta_{ij} \neq 0$ structure: the contrast between factor levels depends on the conditioning level.

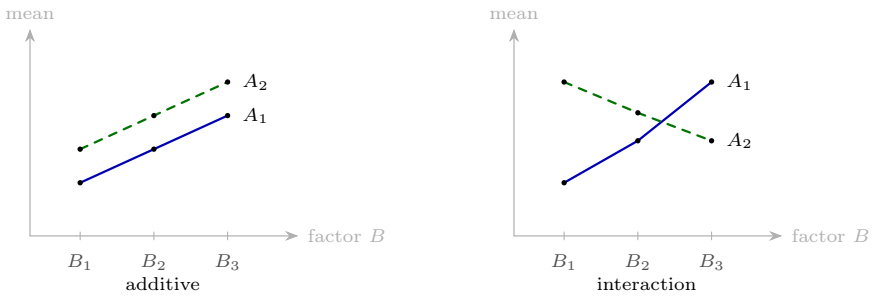


Figure 226: Parallel profiles indicate additivity; changing gaps indicate interaction.

Definition 4.4.3. *Balanced Two-Way Normal Model*
A *balanced two-way normal fixed-effects model* observes Y_{ijk} for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$, with common cell size $K > 1$, and assumes $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$ for mutually independent $\varepsilon_{ijk} \sim N(0, \sigma^2)$, under $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$, and $\sum_j \delta_{ij} = 0\ \forall i$.
Key Insight: $\delta_{ij} = 0\ \forall (i,j)$ iff the cell means are additive in the two factors.

Definition 4.4.4. *Two-Way Sums of Squares*
Let Y_{ijk} be balanced observations for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$.

(i) *Means*: $\overline{Y}_{ij\cdot} := \frac{1}{K} \sum_k Y_{ijk}$, $\overline{Y}_{i\cdot\cdot} := \frac{1}{JK} \sum_j \sum_k Y_{ijk}$, $\overline{Y}_{\cdot j\cdot} := \frac{1}{IK} \sum_i \sum_k Y_{ijk}$, $\overline{Y}_{\cdots} := \frac{1}{IJK} \sum_i \sum_j \sum_k Y_{ijk}$.
(ii) *Between-cell components*: $SS_A := JK \sum_i (\overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdots})^2$, $SS_B := IK \sum_j (\overline{Y}_{\cdot j\cdot} - \overline{Y}_{\cdots})^2$,
and $SS_{AB} := K \sum_i \sum_j (\overline{Y}_{ij\cdot} - \overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdot j\cdot} + \overline{Y}_{\cdots})^2$.
(iii) *Within-cell and total*: $SS_E := \sum_i \sum_j \sum_k (Y_{ijk} - \overline{Y}_{ij\cdot})^2$, $SS_{\text{tot}} := \sum_i \sum_j \sum_k (Y_{ijk} - \overline{Y}_{\cdots})^2$,
and $SS_{\text{tot}} = SS_A + SS_B + SS_{AB} + SS_E$.

Structural Commentary: Orthogonal projections split total variation.

Theorem 4.4.5. *Expected Two-Way Sums of Squares*
Assume $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$ *with independent* $\varepsilon_{ijk} \sim N(0, \sigma^2)$,
and $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$; $\sum_j \delta_{ij} = 0\ \forall i$.

(i) $\mathbb{E}(SS_A) = (I-1)\sigma^2 + JK \sum_i \alpha_i^2$.
(ii) $\mathbb{E}(SS_B) = (J-1)\sigma^2 + IK \sum_j \beta_j^2$.
(iii) $\mathbb{E}(SS_{AB}) = (I-1)(J-1)\sigma^2 + K \sum_i \sum_j \delta_{ij}^2$.
(iv) $\mathbb{E}(SS_E) = IJ(K-1)\sigma^2$.

Proof. Let $Y = m + \varepsilon \in \mathbb{R}^{IJK}$ with $m_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij}$ and $\varepsilon \sim N(0, \sigma^2 I_{IJK})$.
For an orthogonal projection P, $\mathbb{E}\|PY\|^2 = \|Pm\|^2 + \sigma^2 \operatorname{rank} P$.

(i) $P_A m$ has value α_i repeated over j, k, so $\|P_A m\|^2 = JK \sum_i \alpha_i^2$ and $\operatorname{rank} P_A = I - 1$.

(ii) $P_B m$ has value β_j repeated over i, k, so $\|P_B m\|^2 = IK \sum_j \beta_j^2$ and $\operatorname{rank} P_B = J - 1$.

(iii) $P_{AB} m$ has value δ_{ij} repeated over k, so $\|P_{AB} m\|^2 = K \sum_i \sum_j \delta_{ij}^2$ and $\operatorname{rank} P_{AB} = (I-1)(J-1)$.

(iv) P_E centers within cells; since m_{ijk} is constant in k for fixed (i, j), $P_E m = 0$ and $\operatorname{rank} P_E = IJ(K-1)$.

Applying the projection identity to P_A, P_B, P_{AB}, P_E gives the four expectations.
Structural Commentary: Balanced orthogonality separates noise degrees of freedom from effect energy.
Key Insight: Each expectation is σ^2 times rank plus squared signal norm. □

Theorem 4.4.6. *Two-Way Normal-Theory F Tests*
Assume observations Y_{ijk} for $1 \le i \le I$, $1 \le j \le J$, $1 \le k \le K$ satisfy $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$ with independent $\varepsilon_{ijk} \sim N(0, \sigma^2)$, $\sum_i \alpha_i = \sum_j \beta_j = 0$, $\sum_i \delta_{ij} = 0\ \forall j$, and $\sum_j \delta_{ij} = 0\ \forall i$.

Let $d_E := IJ(K-1)$ and $MS_E := \frac{SS_E}{d_E}$.

(i) $SS_A, SS_B, SS_{AB}, SS_E$ are independent, and $\frac{SS_E}{\sigma^2} \sim \chi^2_{d_E}$.

(ii) Under $H_A : \alpha_i = 0\ \forall i$, $\frac{SS_A}{\sigma^2} \sim \chi^2_{I-1}$; under $H_B : \beta_j = 0\ \forall j$, $\frac{SS_B}{\sigma^2} \sim \chi^2_{J-1}$; under $H_{AB} : \delta_{ij} = 0\ \forall (i,j)$, $\frac{SS_{AB}}{\sigma^2} \sim \chi^2_{(I-1)(J-1)}$.

(iii) $F_A := \frac{\frac{SS_A}{I-1}}{MS_E} \sim F_{I-1, d_E}$ under H_A, $F_B := \frac{\frac{SS_B}{J-1}}{MS_E} \sim F_{J-1, d_E}$ under H_B, and $F_{AB} := \frac{\frac{SS_{AB}}{(I-1)(J-1)}}{MS_E} \sim F_{(I-1)(J-1), d_E}$ under H_{AB}.

Proof.

(i) Vectorise $Y = m + \varepsilon \in \mathbb{R}^{IJK}$ with $\varepsilon \sim N(0, \sigma^2 I_{IJK})$.
Let P_A, P_B, P_{AB}, P_E be the mutually orthogonal projections onto the factor-A, factor-B, interaction, and within-cell subspaces, so $SS_R = \|P_R Y\|^2$ for $R \in \{A, B, AB, E\}$.
Orthogonal projections of a spherical normal vector are independent, and $\frac{\|P_R Y\|^2}{\sigma^2} \sim \chi^2_{d_R}(\lambda_R)$ with $d_R = \operatorname{rank} P_R$ and $\lambda_R = \frac{\|P_R m\|^2}{\sigma^2}$. Since $d_E = IJ(K-1)$ and $\lambda_E = 0$, claim (i) follows.

(ii) Here $d_A = I - 1$, $d_B = J - 1$, and $d_{AB} = (I-1)(J-1)$. Under H_A, H_B, H_{AB} respectively, $\lambda_A, \lambda_B, \lambda_{AB}$ vanish, so the central chi-square laws in claim (ii) follow.

(iii) In each null case the numerator chi-square in claim (ii) is independent of SS_E by claim (i); dividing by its degrees of freedom and by $\frac{SS_E}{d_E} = MS_E$ gives the stated F ratios in claim (iii).

Structural Commentary: Balanced normal ANOVA is an orthogonal Gaussian projection decomposition.
Key Insight: Null effects remove noncentrality, leaving independent chi-square ratios. □

Example 4.4.7. *Generic Transformed Two-Way ANOVA*
Consider balanced two-way study with factor A having 2 levels, factor B having 3 levels, $K = 18$ observations per cell. After log transformation stabilises cell spread, ANOVA table on transformed scale is

Source	*df*	*SS*	*MS*	*F*
Factor A	1	2.074	2.074	5.99
Factor B	2	15.588	7.794	22.53
Interaction	2	.810	.405	1.17
Error	102	35.296	.346	
Total	107	53.768		

For factor A, $F_A = 5.99$ gives $p < .025$, and an estimated log contrast .28 corresponds to the original-scale multiplier $e^{.28} \approx 1.32$ with approximate interval $(1.06, 1.65)$. For interaction, $F_{AB} = 1.17$ gives $p = .31$, so there is no strong evidence that the A contrast depends on the level of B.

Remark 4.4.8. *Transformations and Diagnostics*
Two-way ANOVA inherits the normal-theory assumptions of the one-way model. Residual plots, normal probability plots, profile plots, and plots of cell spread against cell mean help decide whether a transformation is needed. A transformation can stabilize variance and improve symmetry, but it also changes additive effects on the transformed scale into multiplicative or otherwise nonlinear effects on the original scale.

Remark 4.4.9. *Factorial Economy and Incomplete Designs*
Factorial designs use each observation to estimate multiple factor contrasts and interactions.
As the factor set grows, full factorials may be replaced by fractional factorials that omit cells while preserving

chosen estimable effects; when blocks cannot contain all treatments, incomplete block designs retain partial blocking without all block-treatment pairs.

4.5 Randomised Blocks and Friedman's Test

Randomised block designs compare treatments within relatively homogeneous blocks. Blocking removes major background variation from the treatment comparison, much as pairing removes shared variation.

Definition 4.5.1. *Randomised Block Design*
A randomised block design has J blocks, each containing one experimental unit for each of the I treatments; within every block, treatments are assigned by an independent random permutation.
The additive block model is $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$ for $1 \le i \le I$, $1 \le j \le J$, with $\sum_i \alpha_i = \sum_j \beta_j = 0$.

Key Insight: Blocks absorb heterogeneity; $\alpha_i - \alpha_{i'}$ is a within-block treatment contrast.

Method 4.5.2. *Randomized Block F Test*
Treat $Y_{ij} = \mu+\alpha_i+\beta_j+\varepsilon_{ij}$ as a two-way layout with one observation per treatment-block cell and no interaction.

(i) Define $MS_A := \frac{SS_A}{I-1}$, $MS_B := \frac{SS_B}{J-1}$, and $MS_{AB} := \frac{SS_{AB}}{(I-1)(J-1)}$.
(ii) Under additivity, $\mathbb{E}(MS_A) = \sigma^2 + \frac{J}{I-1}\sum_i \alpha_i^2$, $\mathbb{E}(MS_B) = \sigma^2 + \frac{I}{J-1}\sum_j \beta_j^2$, and $\mathbb{E}(MS_{AB}) = \sigma^2$.
(iii) Test $H_0 : \alpha_1 = \cdots = \alpha_I = 0$ by $F := \frac{MS_A}{MS_{AB}}$.
(iv) Under the normal additive null, $F \sim F_{I-1,(I-1)(J-1)}$.

Remark 4.5.3. *Interaction in a Block Design*
If $\delta_{ij} \neq 0$, then MS_{AB} estimates σ^2 plus treatment-block interaction energy.
The denominator of $F = \frac{MS_A}{MS_{AB}}$ is inflated, so the additive-model treatment test is conservative.

Example 4.5.4. *Generic Randomized Block ANOVA*
Suppose $I = 7$ treatments are compared across $J = 10$ blocks, with each block receiving all treatments in randomized order. The randomized-block ANOVA table is

Source	*df*	*SS*	*MS*	*F*
Treatments	6	53013	8835	2.85
Blocks	9	103280	11476	3.71
Interaction	54	167130	3095	
Total	69	323422		

$F = 2.85$ on $(6, 54)$ degrees of freedom gives $p < .025$, rejecting equal treatment effects.
Blocking isolates nuisance block heterogeneity.

Method 4.5.5. *Friedman's Rank Test for Blocks*
For observations Y_{ij} from treatment i in block j, rank $Y_{1j}, \ldots, Y_{Ij}$ within each block.

(i) Let R_{ij} be within-block rank, use average ranks for ties, set $\overline{R}_{i\cdot} := \frac{1}{J}\sum_j R_{ij}$, $\overline{R}_{\cdot\cdot} := \frac{I+1}{2}$.
(ii) Define $Q := \frac{12J}{I(I+1)}\sum_i (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2$.
(iii) Under $H_0 : F_1 = \cdots = F_I$, use $Q \overset{H_0}{\approx} \chi^2_{I-1}$ for moderate J.
(iv) Reject at level α when $\mathbb{P}\{\chi^2_{I-1} \ge Q_{\text{obs}}\} \le \alpha$.

Example 4.5.6. *Generic Friedman Rank Calculation*
For $I = 7$ treatments, suppose the average within-block ranks are

Treatment	T_1	T_2	T_3	T_4	T_5	T_6	T_7
Average rank	5.10	4.90	2.30	3.50	3.05	4.90	4.25

With $\overline{R}_{\cdot\cdot} = 4$, $\sum_i (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2 = 6.935$ and $Q = 14.86$.
Using $Q \overset{H_0}{\approx} \chi^2_6$, the tail probability is below .025, so the treatment response distributions are not all equal.

Corollary 4.5.7. *Two Treatments Recover the Sign Test*
For $I = 2$, set $D_j := Y_{1j} - Y_{2j}$ and $B := \#\{j : D_j > 0\}$. Friedman's statistic is a function of B alone, so the two-treatment Friedman test is equivalent to the paired sign test.

Proof. Ignoring ties, $R_{1j} = \mathbf{2}_{\{D_j>0\}} + \mathbf{1}_{\{D_j<0\}} = 1 + \mathbf{1}_{\{D_j>0\}}$.
Hence $\sum_j R_{1j} = J + B$, so $\overline{R}_{1\cdot}$, $\overline{R}_{2\cdot}$, and Q are determined by B.
Under H_0, signs are exchangeable and $B \sim \text{Bin}(J, 1/2)$, which is the paired sign-test calibration.
Structural Commentary: Two within-block ranks encode only the sign of D_j.
Key Insight: The rank-sum statistic collapses to a binomial sign count. □

5 Linear Models and Regression

5.1 Least Squares Curve Fitting and Linearization

Least squares begins as a geometric fitting rule and becomes a statistical model only after assumptions are made about the errors. The optimization step is asymmetric in the two plotted variables: ordinary regression measures vertical prediction errors.

Definition 5.1.1. *Least-Squares Fit to a Parametric Family*
Let $[n] := \{1, \dots, n\}$, let $(x_i, y_i)_{i\in[n]} \subseteq \mathcal{X} \times \mathbb{R}$, let $\Theta \subseteq \mathbb{R}^p$, and let $m : \mathcal{X} \times \Theta \to \mathbb{R}$.

(i) *Criterion*: $S : \Theta \to [0, \infty)$, $S(\theta) := \sum_{i=1}^{n} \big(y_i - m(x_i; \theta)\big)^2$.

(ii) *Estimate*: $\widehat{\theta} \in \arg\min_{\theta\in\Theta} S(\theta)$.

(iii) *Residual*: $\widehat{e}_i := y_i - m(x_i; \widehat{\theta})$, $i \in [n]$.

Key Insight: $y_i = m(x_i; \widehat{\theta}) + \widehat{e}_i$, so $\widehat{e}_i > 0$ iff $y_i > m(x_i; \widehat{\theta})$.

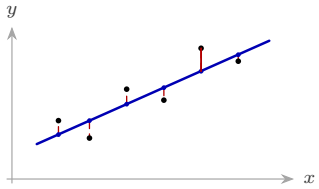


Figure 227: Least squares minimizes squared vertical residuals.

Proposition 5.1.2. *Least-Squares Line*
Let $[n] := \{1, \dots, n\}$, $(x_i, y_i)_{i\in[n]} \subseteq \mathbb{R}^2$, $\overline{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$, $\overline{y} := \frac{1}{n}\sum_{i=1}^{n} y_i$, *assume* $D := \sum_{i=1}^{n}(x_i - \overline{x})^2 > 0$.

For $b = (b_0, b_1) \in \mathbb{R}^2$, *set* $S(b) := \sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2$.

Then $\exists!\widehat{\beta} = (\widehat{\beta}_0, \widehat{\beta}_1) \in \arg\min_{b\in\mathbb{R}^2} S(b)$; $\widehat{y}(x) := \widehat{\beta}_0 + \widehat{\beta}_1 x$ *satisfies* $\widehat{\beta}_1 = \frac{1}{D}\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})$, $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1\overline{x}$.

Proof. Set $u_i := x_i - \overline{x}$, $v_i := y_i - \overline{y}$, and $c(b) := b_0 + b_1\overline{x} - \overline{y}$.
Since $\sum_{i=1}^{n} u_i = \sum_{i=1}^{n} v_i = 0$, $S(b) = \sum_{i=1}^{n}(v_i - b_1 u_i)^2 + nc(b)^2$.
Thus $c(\widehat{\beta}) = 0$, so $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1\overline{x}$, and $\widehat{\beta}_1$ uniquely minimizes $Q(t) := \sum_{i=1}^{n}(v_i - tu_i)^2$.
Since $Q'(t) = -2\sum_{i=1}^{n} u_i(v_i - tu_i)$ and $\sum_{i=1}^{n} u_i^2 = D > 0$, the unique critical point is $t = \frac{1}{D}\sum_{i=1}^{n} u_i v_i$.
Structural Commentary: The fitted line passes through $(\overline{x}, \overline{y})$.
Key Insight: Centering turns the slope into covariance divided by variance. □

Definition 5.1.3. *Linear in the Parameters*
Let $\mathcal{X}$ be an input set, let $\Theta \subseteq \mathbb{R}^p$, and let $m : \mathcal{X} \times \Theta \to \mathbb{R}$. m is linear in β if and only if
$\exists$ fixed maps $\phi_0, \dots, \phi_{p-1} : \mathcal{X} \to \mathbb{R}$ with $m(x; \beta) = \sum_{j=0}^{p-1} \beta_j \phi_j(x)$ $\forall x \in \mathcal{X}$ and $\beta \in \Theta$.
Structural Commentary: Polynomial regression uses $\phi_j(x) = x^j$ and estimates only the coefficients β_j.

Example 5.1.4. *Linear and Nonlinear Parameters*
For fixed $\omega = (\omega_1, \omega_2)$, $f_{\theta,\omega}(t) := A\cos(\omega_1 t) + B\sin(\omega_1 t) + C\cos(\omega_2 t) + D\sin(\omega_2 t) = \theta^t \phi_\omega(t)$
with $\theta = (A, B, C, D)^t$ and $\phi_\omega(t) = (\cos(\omega_1 t), \sin(\omega_1 t), \cos(\omega_2 t), \sin(\omega_2 t))^t$.
Estimating ω makes $f_{\theta,\omega}$ nonlinear in (θ, ω).
For fixed $\lambda = (\alpha, \beta)$, $g_{\eta,\lambda}(t) := Ae^{-\alpha t} + Be^{-\beta t} = \eta^t \psi_\lambda(t)$ with $\eta = (A, B)^t$ and $\psi_\lambda(t) = (e^{-\alpha t}, e^{-\beta t})^t$.
Estimating λ makes $g_{\eta,\lambda}$ nonlinear in (η, λ).

Method 5.1.5. *Linearizing Transformations*
Let $[n] := \{1, \dots, n\}$, $K > 0$, $C > 0$, $e_A \in \mathbb{R}$, $T > 0$, and $\alpha(T) := C\exp\left(-\frac{e_A}{KT}\right)$.
For observations $(T_i, \alpha_i)_{i\in[n]} \subset (0, \infty)^2$, set $x := \frac{1}{T}$, $z := \log\alpha(T)$, $x_i := \frac{1}{T_i}$, and $z_i := \log\alpha_i$.

(i) $z = a + bx$ with $a := \log C$ and $b := -\frac{e_A}{K}$.

(ii) $(\widehat{a},\widehat{b}) \in \arg\min_{(a,b)\in\mathbb{R}^2} \sum_{i=1}^{n}(z_i - a - bx_i)^2$, $\widehat{C} := e^{\widehat{a}}$, and $\widehat{e}_A := -K\widehat{b}$.

(iii) $\widehat{\varepsilon}_i := z_i - \widehat{a} - \widehat{b}x_i$ is diagnosed against x_i and $\widehat{z}_i := \widehat{a} + \widehat{b}x_i$.
Systematic structure indicates misspecification on the transformed scale.

5.2 Simple Linear Regression Inference and Diagnostics

Simple linear regression adds a noise model to the least-squares line. The x_i values are treated as fixed design points, and randomness enters through the observed responses.

Definition 5.2.1. *Standard Simple Linear Regression Model*
Let $[n] := \{1,\dots,n\}$, let $x_i \in \mathbb{R}$ be fixed for $i \in [n]$, let $\beta_0,\beta_1 \in \mathbb{R}$, and let $\sigma^2 > 0$.

(i) *Standard model*: $Y_i = \beta_0 + \beta_1 x_i + e_i$, $\mathbb{E}e_i = 0$, $\mathrm{Var}(e_i) = \sigma^2$, and $\mathrm{Cov}(e_i,e_j) = 0$ for $i \neq j$, $i,j \in [n]$.
(ii) *Normal model*: $Y_i = \beta_0 + \beta_1 x_i + e_i$ with $e_1,\dots,e_n$ independent and $e_i \sim N(0,\sigma^2)$, $i \in [n]$.

Structural Commentary: $(x_i)_{i\in[n]}$ is fixed; randomness enters through $(e_i)_{i\in[n]}$.

Theorem 5.2.2. *Mean and Variance of the Simple Regression Coefficients*
Let $[n] := \{1,\dots,n\}$, let $x_i \in \mathbb{R}$, and set $\overline{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$ and $D := \sum_{i=1}^{n}(x_i - \overline{x})^2 > 0$.
Let $\beta := (\beta_0,\beta_1)$ and assume $Y_i = \beta_0 + \beta_1 x_i + e_i$, $\mathbb{E}e_i = 0$, $\mathrm{Var}(e_i) = \sigma^2$, and $\mathrm{Cov}(e_i,e_j) = 0$ for $i \neq j$.
If $\widehat{\beta} = (\widehat{\beta}_0,\widehat{\beta}_1)$ minimizes $\sum_{i=1}^{n}(Y_i - b_0 - b_1x_i)^2$ over $b = (b_0,b_1) \in \mathbb{R}^2$, then $\mathbb{E}\widehat{\beta} = \beta$; $\mathrm{Var}(\widehat{\beta}) = \sigma^2 \begin{pmatrix} \frac{1}{n} + \frac{\overline{x}^2}{D} & -\frac{\overline{x}}{D} \\ -\frac{\overline{x}}{D} & \frac{1}{D} \end{pmatrix}$.

Proof. Set $u_i := x_i - \overline{x}$ and $\overline{e} := \frac{1}{n}\sum_{i=1}^{n} e_i$.

Proposition 5.1.2 gives $\widehat{\beta}_1 - \beta_1 = D^{-1}\sum_{i=1}^{n} u_i e_i$ and $\widehat{\beta}_0 - \beta_0 = \overline{e} - \overline{x}(\widehat{\beta}_1 - \beta_1)$.

Because $\sum_{i=1}^{n} u_i = 0$, $\mathbb{E}(\widehat{\beta} - \beta) = 0$, $\mathrm{Var}(\widehat{\beta}_1) = D^{-2}\sum_{i=1}^{n} u_i^2\sigma^2 = \frac{\sigma^2}{D}$, $\mathrm{Cov}(\overline{e},\widehat{\beta}_1) = \frac{1}{n}D^{-1}\sum_{i=1}^{n} u_i\sigma^2 = 0$, and $\mathrm{Var}(\overline{e}) = \frac{\sigma^2}{n}$. Therefore $\mathrm{Var}(\widehat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\overline{x}^2\sigma^2}{D}$ and $\mathrm{Cov}(\widehat{\beta}_0,\widehat{\beta}_1) = -\frac{\overline{x}\sigma^2}{D}$.
Structural Commentary: The slope is a fixed linear combination of the responses.
Key Insight: Centering the design isolates the uncertainty in the slope. □

Definition 5.2.3. *Residual Sum of Squares and Error Scale*
Let $[n] := \{1,\dots,n\}$ with $n > 2$, and let $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ for $i \in [n]$.

(i) *Residual Sum of Squares*: $\mathrm{RSS} := \sum_{i=1}^{n}(Y_i - \widehat{y}_i)^2$.
(ii) *Error Scale*: $s^2 := \frac{\mathrm{RSS}}{n-2}$ and $s := \sqrt{s^2}$.

Key Insight: $n-2$ subtracts the fitted coefficients $\widehat{\beta}_0,\widehat{\beta}_1$.

Corollary 5.2.4. *Simple Regression t Intervals and Tests*
Let $n > 2$, $[n] := \{1,\dots,n\}$, $\alpha \in (0,1)$, $\beta \in \mathbb{R}^2$, $\sigma^2 > 0$, $x = (x_1,\dots,x_n)^t \in \mathbb{R}^n$, $X := (\mathbf{1},x) \in \mathbb{R}^{n\times 2}$ with $\mathrm{rank}(X) = 2$, $Y = X\beta + e$, and $e \sim N(0,\sigma^2 I_n)$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$, $s^2 := \frac{\|Y - X\widehat{\beta}\|_2^2}{n-2}$, $\overline{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$, $D := \sum_{i=1}^{n}(x_i - \overline{x})^2$, $c_0 := \frac{1}{n} + \frac{\overline{x}^2}{D}$, and $c_1 := D^{-1}$.
For $j \in \{0,1\}$ and $\beta_{j0} \in \mathbb{R}$, set $s_{\widehat{\beta}_j} := s\sqrt{c_j}$.
Then $T_j := \frac{\widehat{\beta}_j - \beta_j}{s_{\widehat{\beta}_j}} \sim t_{n-2}$, $C_j(Y) := [\widehat{\beta}_j - t_{n-2,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}, \widehat{\beta}_j + t_{n-2,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}]$ satisfies $\mathbb{P}_\beta(\beta_j \in C_j(Y)) = 1-\alpha$,
and the two-sided level-α test of $H_0 : \beta_j = \beta_{j0}$ rejects iff $\left|\frac{\widehat{\beta}_j - \beta_{j0}}{s_{\widehat{\beta}_j}}\right| > t_{n-2,1-\frac{\alpha}{2}}$.

Proof. Let $P := X(X^tX)^{-1}X^t$, $M := I - P$, and $G := (X^tX)^{-1} = (g_{jk})_{j,k=0}^{1}$.
Then $\widehat{\beta} - \beta = GX^te$, $g_{00} = c_0$, $g_{11} = c_1$, $\mathrm{Var}(\widehat{\beta}_j) = \sigma^2 c_j$, and $Z_j := \frac{\widehat{\beta}_j - \beta_j}{\sigma\sqrt{c_j}} \sim N(0,1)$.
Also $\frac{(n-2)s^2}{\sigma^2} = \frac{e^tMe}{\sigma^2}$, $M = M^t = M^2$, $\mathrm{rank}(M) = n-2$, and $MX = 0$, hence $U := \frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{n-2}$ and $\mathrm{Cov}(GX^te, Me) = GX^t\sigma^2M = 0$. Joint normality gives $Z_j \perp U$, so $T_j = \frac{Z_j}{\sqrt{\frac{U}{n-2}}} \sim t_{n-2}$.
The interval and rejection rule follow from $\mathbb{P}(|T_j| \le t_{n-2,1-\frac{\alpha}{2}}) = 1-\alpha$.
Structural Commentary: $T_j = \frac{Z_j}{\sqrt{\frac{U}{n-2}}}$ with $Z_j \sim N(0,1)$, $U \sim \chi^2_{n-2}$, and $Z_j \perp U$.
Key Insight: Normality and orthogonal residual projection make replacing σ by s exact. □

Example 5.2.5. *Calibration by Peak Area*
Let $[n] := \{1, \dots, n\}$, with known standard concentrations x_i and measured peak areas A_i, $i \in [n]$.
Fit $A_i = \beta_0 + \beta_1 x_i + e_i$ by least squares, giving $\widehat{A}(x) = \widehat{\beta}_0 + \widehat{\beta}_1 x$.
For a new area A_*, the inverse calibration estimate is $\widehat{x}_* := \frac{A_* - \widehat{\beta}_0}{\widehat{\beta}_1}$ when $\widehat{\beta}_1 \neq 0$.
In the reported fit, $\widehat{\beta}_0 = .0729$, $\widehat{\beta}_1 = 10.77$, $s = .068$, $\mathrm{se}(\widehat{\beta}_0) = .0297$, and $\mathrm{se}(\widehat{\beta}_1) = .27$, hence $t_0 = 2.45$, $t_1 = 40.20$, and the reported 95% intervals are $C_0 = (.011, .135)$ and $C_1 = (10.21, 11.33)$.
Thus $\widehat{\beta}_1 > 0$ makes peak area a monotone calibration signal.

Method 5.2.6. *Residual Diagnostics*
After fitting a regression, examine residuals against the fitted values and against important predictors.

(i) A random horizontal band supports the chosen mean form.
(ii) Curvature suggests a transformation or added nonlinear term.
(iii) Changing spread suggests heteroscedasticity.
(iv) A normal probability plot checks large departures from normality and possible outliers.
(v) Points with extreme x values may have high leverage and can strongly affect the fitted line.

These checks refine the modeling assumptions; they do not replace them.

Definition 5.2.7. *Homoscedastic and Heteroscedastic Errors*
Let $[n] := \{1, \dots, n\}$, let $e = (e_1, \dots, e_n)^t$ be errors at design points x_i, and set $v_i := \mathrm{Var}(e_i)$.

(i) *Homoscedastic*: $\exists \sigma^2 \geq 0$ with $v_i = \sigma^2$, $i \in [n]$.
(ii) *Heteroscedastic*: $\exists i, j \in [n]$ with $v_i \neq v_j$.

Key Insight: $\mathrm{Var}(e) \neq \sigma^2 I_n$ invalidates formulas requiring $\mathrm{Var}(e) = \sigma^2 I_n$.

Example 5.2.8. *Diagnostic Patterns*
Let $[n] := \{1, \dots, n\}$, $r_i := Y_i - \widehat{Y}_i$, $f_i := \widehat{Y}_i$, $\ell_i \in \mathcal{G}$,
$m_r(x) := \mathbb{E}(r_i \mid x_i = x)$, $v_r(f) := \mathrm{Var}(r_i \mid f_i = f)$, and $m_a := \mathbb{E}(r_i \mid \ell_i = a)$.

(i) $m_r \not\equiv 0$ or m_r non-affine; transform variables or add nonlinear terms.
(ii) v_r nonconstant; transform Y_i or use heteroscedastic errors.
(iii) $\exists a, b \in \mathcal{G}$ with $m_a \neq m_b$ or $\exists i \neq j$ with $\mathrm{Cov}(r_i, r_j \mid \ell_i = \ell_j) \neq 0$; add group effects or correlated errors.

Proposition 5.2.9. *Correlation and the Regression Line*
Let $[n] := \{1, \dots, n\}$, *let* $(x_i, y_i) \in \mathbb{R}^2$ *for* $i \in [n]$.
Set $\overline{x} := \frac{1}{n}\sum_{i=1}^n x_i$, $\overline{y} := \frac{1}{n}\sum_{i=1}^n y_i$, $s_{xx} := \frac{1}{n}\sum_{i=1}^n (x_i - \overline{x})^2$, $s_{yy} := \frac{1}{n}\sum_{i=1}^n (y_i - \overline{y})^2$, $s_{xy} := \frac{1}{n}\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})$.
If $s_{xx}, s_{yy} > 0$, $r := \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}}$, $(\widehat{\beta}_0, \widehat{\beta}_1)$ *minimizes* $\sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$ *over* $(b_0, b_1) \in \mathbb{R}^2$,
and $\widehat{y}(x) := \widehat{\beta}_0 + \widehat{\beta}_1 x$, *then* $\widehat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$, $r = \widehat{\beta}_1 \sqrt{\frac{s_{xx}}{s_{yy}}}$, *and* $\frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}} = r \frac{x - \overline{x}}{\sqrt{s_{xx}}}$.

Proof. Set $D := \sum_{i=1}^n (x_i - \overline{x})^2 = n s_{xx}$ and $N := \sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y}) = n s_{xy}$.
Proposition 5.1.2 gives $\widehat{\beta}_1 = \frac{N}{D} = \frac{s_{xy}}{s_{xx}}$ and $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$.
Hence $r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}} = \widehat{\beta}_1 \sqrt{\frac{s_{xx}}{s_{yy}}}$ and $\widehat{y}(x) - \overline{y} = \widehat{\beta}_1 (x - \overline{x})$, so $\frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}} = r \frac{x - \overline{x}}{\sqrt{s_{xx}}}$.
Structural Commentary: $z_x := \frac{x - \overline{x}}{\sqrt{s_{xx}}}$ and $z_{\widehat{y}} := \frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}}$ give $z_{\widehat{y}} = r z_x$.
Key Insight: $\widehat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$ becomes $r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}}$ after standardising both coordinates. □

Remark 5.2.10. *Regression to the Mean*
With $s_{xx}, s_{yy} > 0$ and $0 < r < 1$, set $z_x := \frac{x - \overline{x}}{\sqrt{s_{xx}}}$ and $z_{\widehat{y}} := \frac{\widehat{y}(x) - \overline{y}}{\sqrt{s_{yy}}}$.
Proposition 5.2.9 gives $z_{\widehat{y}} = r z_x$, hence $|z_{\widehat{y}}| < |z_x|$ for $z_x \neq 0$.

5.3 Matrix Linear Models and Normal Equations

Matrix notation separates the geometry of least squares from the statistical assumptions on the errors. It also makes multiple regression a direct extension of simple regression.

Definition 5.3.1. *Fixed-Design Linear Model*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta, b \in \mathbb{R}^p$, and $e \in \mathbb{R}^n$.

(i) *Model*: $Y = X\beta + e$.
(ii) *Least-Squares Estimator*: $\widehat{\beta} \in \arg\min_{b \in \mathbb{R}^p} \|Y - Xb\|_2^2$.

Structural Commentary: An intercept is encoded by $X_{\cdot 1} = \mathbf{1}$.

Theorem 5.3.2. *Normal Equations for the Linear Model*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, and $Q(b) := \|Y - Xb\|_2^2$ for $b \in \mathbb{R}^p$.

(i) $\widehat{\beta} \in \arg\min_{b\in\mathbb{R}^p} Q(b)$ iff $X^tX\widehat{\beta} = X^tY$.

(ii) If $\operatorname{rank}(X) = p$, then X^tX is invertible and $\arg\min_{b\in\mathbb{R}^p} Q(b) = \{(X^tX)^{-1}X^tY\}$.

Proof.

(*i*) ($\Rightarrow$) If $\widehat{\beta}$ minimizes Q, then $\forall (t,h) \in \mathbb{R}\times\mathbb{R}^p$, $0 \le Q(\widehat{\beta}+th) - Q(\widehat{\beta}) = 2th^tX^t(X\widehat{\beta} - Y) + t^2\|Xh\|_2^2$.
Hence $\forall h \in \mathbb{R}^p$, $h^tX^t(X\widehat{\beta} - Y) = 0$, so $X^tX\widehat{\beta} = X^tY$.

(*i*) ($\Leftarrow$) If $X^tX\widehat{\beta} = X^tY$, then $\forall h \in \mathbb{R}^p$, $Q(\widehat{\beta}+h) - Q(\widehat{\beta}) = \|Xh\|_2^2 \ge 0$, so $\widehat{\beta}$ minimizes Q.

(*ii*) If $\operatorname{rank}(X) = p$, then $\forall u \in \mathbb{R}^p \setminus \{0\}$, $u^tX^tXu = \|Xu\|_2^2 > 0$, so X^tX is invertible.
Applying (i) gives $\arg\min_{b\in\mathbb{R}^p} Q(b) = \{(X^tX)^{-1}X^tY\}$.

Structural Commentary: $\widehat{Y} = X\widehat{\beta}$ and $\widehat{e} = Y - \widehat{Y}$ satisfy $X^t\widehat{e} = 0$, hence $\widehat{e} \perp \operatorname{col}(X)$.
Key Insight: The projection $\widehat{Y} \in \operatorname{col}(X)$ is unique; $\operatorname{rank}(X) = p$ makes $b \mapsto Xb$ injective, so $\widehat{\beta}$ is unique. □

Lemma 5.3.3. *Rank Criterion for the Normal Matrix*
Let $X \in \mathbb{R}^{n\times p}$. Then X^tX is nonsingular iff $\operatorname{rank}(X) = p$.

Proof.

($\Rightarrow$) If X^tX is nonsingular and $\operatorname{rank}(X) < p$, then $\exists u \in \mathbb{R}^p \setminus \{0\}$ with $Xu = 0$, hence $X^tXu = 0$, contradiction.

($\Leftarrow$) If $\operatorname{rank}(X) = p$ and $X^tXu = 0$, then $0 = u^tX^tXu = \|Xu\|_2^2$, so $Xu = 0$ and hence $u = 0$.
Therefore $\ker(X^tX) = \{0\}$ and X^tX is nonsingular.

Structural Commentary: $\ker(X^tX) = \ker(X)$.
Key Insight: X^tX is invertible exactly when $Xu = Xv \Rightarrow u = v$. □

Definition 5.3.4. *Hat Matrix, Fitted Values, and Residuals*
Assume $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\operatorname{rank}(X) = p$, $\widehat{\beta} = (X^tX)^{-1}X^tY$, and $I = I_n$.

(i) *Hat Matrix*: $P := X(X^tX)^{-1}X^t$.

(ii) *Fitted Values*: $\widehat{Y} := X\widehat{\beta} = PY$.

(iii) *Residuals*: $\widehat{e} := Y - \widehat{Y} = (I - P)Y$.

Key Insight: $\operatorname{range}(P) = \operatorname{col}(X)$ and $PY = \widehat{Y}$.

Proposition 5.3.5. *Projection Properties of the Hat Matrix*
Let $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank}(X) = p$, $I = I_n$, $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, and $\widehat{e} := (I - P)Y$.
Then $P = P^t = P^2$, $I - P = (I - P)^t = (I - P)^2$, $\operatorname{range}(P) = \operatorname{col}(X)$, $\widehat{Y} = P_{\operatorname{col}(X)}Y$, and $X^t\widehat{e} = 0$.

Proof. Since $(X^tX)^t = X^tX$, $P^t = X((X^tX)^{-1})^tX^t = P$, and $P^2 = X(X^tX)^{-1}X^tX(X^tX)^{-1}X^t = P$.
Hence $(I - P)^t = I - P$ and $(I - P)^2 = I - 2P + P^2 = I - P$.
If $z = Xb \in \operatorname{col}(X)$, then $Pz = X(X^tX)^{-1}X^tXb = z$; also $\operatorname{range}(P) \subseteq \operatorname{col}(X)$, so $\operatorname{range}(P) = \operatorname{col}(X)$.
Finally, $X^t\widehat{e} = X^t(I - P)Y = X^tY - X^tX(X^tX)^{-1}X^tY = 0$, so $\widehat{e} \perp \operatorname{col}(X)$ and $\widehat{Y} = P_{\operatorname{col}(X)}Y$.
Structural Commentary: $P = P_{\operatorname{col}(X)}$ and $I - P = P_{\operatorname{col}(X)^\perp}$.
Key Insight: Fitting decomposes $Y = PY + (I - P)Y = \widehat{Y} + \widehat{e}$ with $\widehat{Y} \perp \widehat{e}$. □

Theorem 5.3.6. *Unbiased Estimate of the Error Variance*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, $\mathbb{E}e = 0$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, $\widehat{e} := Y - \widehat{Y}$, and $s^2 := \frac{\|\widehat{e}\|_2^2}{n-p}$. Then $\mathbb{E}\|\widehat{e}\|_2^2 = (n-p)\sigma^2$ and $\mathbb{E}s^2 = \sigma^2$.

Proof. Let $M := I_n - P$. Since $PX = X$, $\widehat{e} = MY = M(X\beta + e) = Me$.
Since $M = M^t = M^2$, $\|\widehat{e}\|_2^2 = e^tM^tMe = e^tMe$. For fixed $A \in \mathbb{R}^{n\times n}$ and centered e with $\operatorname{Var}(e) = \sigma^2 I_n$,
$\mathbb{E}(e^tAe) = \operatorname{tr}(A\sigma^2 I_n) = \sigma^2 \operatorname{tr}(A)$. Thus $\mathbb{E}\|\widehat{e}\|_2^2 = \sigma^2 \operatorname{tr}(M)$,
and $\operatorname{tr}(M) = n - \operatorname{tr}(P) = n - \operatorname{tr}((X^tX)^{-1}X^tX) = n - p$. Therefore $\mathbb{E}s^2 = \frac{1}{n-p}\mathbb{E}\|\widehat{e}\|_2^2 = \sigma^2$.
Structural Commentary: $M = I_n - P = P_{\operatorname{col}(X)^\perp}$ and $\operatorname{tr}(M) = n - p$.
Key Insight: $\widehat{e} = Me$, so residual variation is noise projected onto the $n - p$ directions orthogonal to $\operatorname{col}(X)$. □

Definition 5.3.7. *Standardized Residual*
Let $[n] := \{1, \dots, n\}$, $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$ with $\operatorname{rank}(X) = p$, $P = (p_{ij}) := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, $\widehat{e} := Y - \widehat{Y}$, $s^2 := \frac{\|\widehat{e}\|_2^2}{n-p}$, and $s > 0$. The *standardized residual* is $r_i := \frac{\widehat{e}_i}{s\sqrt{1-p_{ii}}}$ for $i \in [n]$.
Key Insight: Under $\operatorname{Var}(e) = \sigma^2 I_n$, $\operatorname{Var}(\widehat{e}_i) = \sigma^2(1 - p_{ii})$.

Proposition 5.3.8. *Residuals Are Uncorrelated with Fitted Values*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $P := X(X^tX)^{-1}X^t$, $\widehat{Y} := PY$, and $\widehat{e} := (I_n - P)Y$. Then $\operatorname{Cov}(\widehat{e}, \widehat{Y}) = 0$.

Proof. Since $\operatorname{Cov}(Y) = \operatorname{Cov}(e) = \sigma^2 I_n$ and $P = P^t = P^2$,
$\operatorname{Cov}(\widehat{e}, \widehat{Y}) = (I_n - P)\operatorname{Cov}(Y)P^t = \sigma^2(I_n - P)P = \sigma^2(P - P^2) = 0$.
Structural Commentary: $\widehat{Y} = PY$ and $\widehat{e} = (I_n - P)Y$ are images of Y under complementary projections.
Key Insight: Spherical error covariance makes orthogonal projection components uncorrelated. □

Perspective 5.3.9. *Solve Least Squares with Factors*
For $X \in \mathbb{R}^{n \times p}$ with $\operatorname{rank}(X) = p$, normal equations use $X^tX\widehat{\beta} = X^tY$ but $\kappa_2(X^tX) = \kappa_2(X)^2$.
Prefer $X = QR$ with $Q^tQ = I_p$ and R nonsingular, giving $\widehat{\beta} = R^{-1}Q^tY$.
Use Cholesky on X^tX only when X^tX is well conditioned.

5.4 Inference, Prediction, and Multiple Regression

Once the linear model is written in matrix form, standard errors for coefficients and predictions are consequences of linear transformations of random vectors.

Theorem 5.4.1. *Mean and Covariance of Least-Squares Estimates*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, $\mathbb{E}e = 0$, and $\operatorname{Var}(e) = \sigma^2 I_n$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$. Then $\mathbb{E}\widehat{\beta} = \beta$ and $\operatorname{Var}(\widehat{\beta}) = \sigma^2(X^tX)^{-1}$.

Proof. Let $A := (X^tX)^{-1}X^t$. Since $AX = I_p$, $\widehat{\beta} = A(X\beta + e) = \beta + Ae$.
Hence $\mathbb{E}\widehat{\beta} = \beta + A\mathbb{E}e = \beta$, and $\operatorname{Var}(\widehat{\beta}) = A\operatorname{Var}(e)A^t = \sigma^2(X^tX)^{-1}X^tX(X^tX)^{-1} = \sigma^2(X^tX)^{-1}$.
Structural Commentary: $\widehat{\beta} - \beta = (X^tX)^{-1}X^te$ is a fixed linear transformation of e.
Key Insight: X^tX controls precision: $\operatorname{Var}(\widehat{\beta}) = \sigma^2(X^tX)^{-1}$. □

Method 5.4.2. *Coefficient Inference in a Normal Linear Model*
Let $Y = X\beta + e$, $e \sim N_n(0, \sigma^2 I_n)$, $X \in \mathbb{R}^{n \times p}$ with $\operatorname{rank}(X) = p$, $\widehat{\beta} := (X^tX)^{-1}X^tY$,
$s^2 := \frac{1}{n-p}\left\|Y - X\widehat{\beta}\right\|_2^2$, $C = (c_{jk}) := (X^tX)^{-1}$, $j \in \{1, \dots, p\}$, and $\alpha \in (0, 1)$.

(i) *Standard Error*: $s_{\widehat{\beta}_j} := s\sqrt{c_{jj}}$.

(ii) *Pivot*: $T_j := \frac{\widehat{\beta}_j - \beta_j}{s_{\widehat{\beta}_j}} \sim t_{n-p}$.

(iii) *Confidence Interval*: $C_{j,1-\alpha} := \left[\widehat{\beta}_j - t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}, \widehat{\beta}_j + t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\beta}_j}\right]$.

(iv) *Test Statistic*: For $H_0 : \beta_j = \beta_{j0}$, $t_j := \frac{\widehat{\beta}_j - \beta_{j0}}{s_{\widehat{\beta}_j}}$.

If a coefficient CLT holds, $T_j \approx N(0, 1)$.

Proposition 5.4.3. *Mean Response at a New Design Point*
Let $Y = X\beta + e$ with $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, and $e \sim N_n(0, \sigma^2 I_n)$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$, $s^2 := \frac{1}{n-p}\left\|Y - X\widehat{\beta}\right\|_2^2$, $x_0 \in \mathbb{R}^p$, $c_0 := x_0^t(X^tX)^{-1}x_0 > 0$,
$\mu_0 := x_0^t\beta$, $\widehat{\mu}_0 := x_0^t\widehat{\beta}$, $s_{\widehat{\mu}_0} := s\sqrt{c_0}$, and $\alpha \in (0, 1)$. Then $\mathbb{E}\widehat{\mu}_0 = \mu_0$, $\operatorname{Var}(\widehat{\mu}_0) = \sigma^2 c_0$, $T_0 := \frac{\widehat{\mu}_0 - \mu_0}{s_{\widehat{\mu}_0}} \sim t_{n-p}$,
and $C_{0,1-\alpha} := \left[\widehat{\mu}_0 - t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\mu}_0}, \widehat{\mu}_0 + t_{n-p,1-\frac{\alpha}{2}}s_{\widehat{\mu}_0}\right]$ has coverage $1 - \alpha$ for μ_0.

Proof. Let $A := (X^tX)^{-1}X^t$, $P := XA$, and $M := I_n - P$.
Since $AX = I_p$, $\widehat{\mu}_0 - \mu_0 = x_0^tA(X\beta + e) - x_0^t\beta = x_0^tAe$, so $\mathbb{E}\widehat{\mu}_0 = \mu_0$ and $\operatorname{Var}(\widehat{\mu}_0) = x_0^tA(\sigma^2 I_n)A^tx_0 = \sigma^2 c_0$.
Normality gives $Z := \frac{\widehat{\mu}_0 - \mu_0}{\sigma\sqrt{c_0}} \sim N(0, 1)$.

Since $M = M^t = M^2$, $\operatorname{rank}(M) = n - p$, and $MA^tx_0 = 0$, $Z \perp U := \frac{e^tMe}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p}$.
Thus $T_0 = \frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}$, and $\mathbb{P}(\mu_0 \in C_{0,1-\alpha}) = \mathbb{P}(|T_0| \le t_{n-p,1-\frac{\alpha}{2}}) = 1 - \alpha$.
Structural Commentary: $\widehat{\mu}_0 - \mu_0 = x_0^t(X^tX)^{-1}X^te$ and $s^2 = \frac{e^t(I_n - P)e}{n-p}$ use orthogonal components of e.
Key Insight: Estimating a mean response adds uncertainty only through $c_0 = x_0^t(X^tX)^{-1}x_0$. □

Corollary 5.4.4. *Prediction Interval for a New Observation*
Let $Y = X\beta + e$ *with* $Y, e \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$, $\beta \in \mathbb{R}^p$, $\operatorname{rank}(X) = p$, *and* $e \sim N_n(0, \sigma^2 I_n)$.
Let $x_0 \in \mathbb{R}^p$, $e_0 \sim N(0,\sigma^2)$ *with* $e_0 \perp e$, *and* $Y_0 := x_0^t\beta + e_0$.
Set $\widehat{\beta} := (X^tX)^{-1}X^tY$, $\widehat{\mu}_0 := x_0^t\widehat{\beta}$, $s^2 := \frac{\|Y - X\widehat{\beta}\|_2^2}{n-p}$, $c_0 := x_0^t(X^tX)^{-1}x_0$, $s_{\text{pred}} := s\sqrt{1+c_0}$, *and* $\alpha \in (0,1)$.
Then $\operatorname{Var}(Y_0 - \widehat{\mu}_0) = \sigma^2(1+c_0)$, $T_{\text{pred}} := \frac{Y_0 - \widehat{\mu}_0}{s_{\text{pred}}} \sim t_{n-p}$,
and $I_{0,1-\alpha} := \left[\widehat{\mu}_0 - t_{n-p,1-\frac{\alpha}{2}} s_{\text{pred}}, \widehat{\mu}_0 + t_{n-p,1-\frac{\alpha}{2}} s_{\text{pred}}\right]$ *has coverage* $1-\alpha$ *for* Y_0.

Proof. Let $A := (X^tX)^{-1}X^t$, $P := XA$, and $M := I_n - P$.
Since $AX = I_p$, $Y_0 - \widehat{\mu}_0 = e_0 - x_0^tAe$, so $\operatorname{Var}(Y_0 - \widehat{\mu}_0) = \sigma^2 + \sigma^2 x_0^tAA^tx_0 = \sigma^2(1+c_0)$.
Normality gives $Z := \frac{Y_0 - \widehat{\mu}_0}{\sigma\sqrt{1+c_0}} \sim N(0,1)$. Since $AM = 0$, $Z \perp U := \frac{e^tMe}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p}$.
Thus $T_{\text{pred}} = \frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}$, and $\mathbb{P}(Y_0 \in I_{0,1-\alpha}) = \mathbb{P}(|T_{\text{pred}}| \le t_{n-p,1-\frac{\alpha}{2}}) = 1-\alpha$.
Structural Commentary: $Y_0 - \widehat{\mu}_0 = e_0 - x_0^t(X^tX)^{-1}X^te$ combines future noise and estimation error.
Key Insight: Prediction variance is mean-response variance plus irreducible noise, $\sigma^2c_0 + \sigma^2$. □

Definition 5.4.5. *Multiple Regression Coefficients*
Let $[n] := \{1,\dots,n\}$, $z_i = (x_{i1},\dots,x_{i,p-1}) \in \mathbb{R}^{p-1}$ for $i \in [n]$, $u_1,\dots,u_{p-1}$ be the standard basis of $\mathbb{R}^{p-1}$, and $m(z) := \beta_0 + \sum_{k=1}^{p-1} \beta_k z_k$ In $Y_i = m(z_i) + e_i$, the *multiple regression coefficient* β_k satisfies $\forall z \in \mathbb{R}^{p-1}$ and $\forall \delta \in \mathbb{R}$, $m(z + \delta u_k) - m(z) = \delta\beta_k$ for $k \in \{1,\dots,p-1\}$.
Structural Commentary: Polynomial regression replaces z_i by transformed design columns such as z_{ik}^2.

Definition 5.4.6. *Coefficient of Determination*
Let $[n] := \{1,\dots,n\}$, $Y, \widehat{Y} \in \mathbb{R}^n$, $\overline{Y} := n^{-1}\sum_{i=1}^n Y_i$, $\mathbf{1} \in \operatorname{col}(X)$, and $\sum_{i=1}^n (Y_i - \overline{Y})^2 > 0$.

(i) *Residual Sum of Squares*: $\text{RSS} := \sum_{i=1}^n (Y_i - \widehat{Y}_i)^2$.

(ii) *Total Sum of Squares*: $\text{TSS} := \sum_{i=1}^n (Y_i - \overline{Y})^2$.

(iii) *Coefficient of Determination*: $R^2 := 1 - \frac{\text{RSS}}{\text{TSS}}$.

Structural Commentary: $R^2 = \operatorname{corr}(Y, \widehat{Y})^2$ for an intercept model.
Key Insight: $R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}$ is the observed response variation removed by the fitted model.

Remark 5.4.7. *Collinearity and Interpretation*
Let X_{-j} be X without column $X_{\cdot j}$, set $d_j := \operatorname{dist}(X_{\cdot j}, \operatorname{col}(X_{-j}))$, and let $C = (c_{kl}) := (X^tX)^{-1}$.
If $0 < d_j \ll \|X_{\cdot j}\|_2$, then X^tX is nearly singular and $s_{\widehat{\beta}_j} = s\sqrt{c_{jj}}$ can be large even when $\widehat{Y} = X\widehat{\beta}$ is stable.
If $d_j = 0$, then $\operatorname{rank}(X) < p$ and $\widehat{\beta}$ is not unique.

Example 5.4.8. *Quadratic Terms as a Lack-of-Fit Check*
Compare $M_1 : Y_i = \beta_0 + \beta_1 x_i + e_i$ with $M_2 : Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + e_i$.
If $t_2 = \widehat{\beta}_2 / s_{\widehat{\beta}_2}$ is small and residual curvature is absent, retain M_1.
If $|t_2|$ is large and residual curvature decreases under M_2, use the quadratic term.
Add x^2 only when residual structure or scale supports $\beta_2 \neq 0$.

5.5 Random Design, Bootstrap, and Extensions

The fixed-design model conditions on the observed predictor values. Many observational studies are more naturally viewed as sampling random predictor-response pairs, with the fixed-design analysis recovered after conditioning on the realised design.

Definition 5.5.1. *Random-Design Linear Regression*
Let $[n] := \{1,\dots,n\}$, $(Y_i, \xi_i)_{i\in[n]}$ be independent copies of (Y, ξ), $\xi \in \mathbb{R}^p$, $\beta \in \mathbb{R}^p$, and $\sigma^2 > 0$.
A *Random-Design Linear Regression* is an iid sample $(Y_i, \xi_i)_{i\in[n]}$ satisfying $\mathbb{E}(Y \mid \xi = x) = x^t\beta$ and $\operatorname{Var}(Y \mid \xi = x) = \sigma^2$ for $x \in \operatorname{supp}(\xi)$.
Structural Commentary: Conditioning on $X_{i\cdot} = \xi_i^t$ gives $Y_{1:n} = X\beta + e$ with $e_i = Y_i - \xi_i^t\beta$.

Proposition 5.5.2. *Conditional Coverage Implies Unconditional Coverage*
Let $C(X,Y)$ *be a confidence set for* β_j, $\alpha \in (0,1)$, *and* $F := \{\operatorname{rank}(X) = p\}$.
If $\mathbb{P}(F) = 1$ *and* $\mathbb{P}(\beta_j \in C(X,Y) \mid X) = 1-\alpha$ *on* F, *then* $\mathbb{P}(\beta_j \in C(X,Y)) = 1-\alpha$.

Proof. Set $A := \{\beta_j \in C(X, Y)\}$. Since $\mathbb{P}(F) = 1$, $\mathbb{P}(A) = \mathbb{E}[\mathbb{P}(A \mid X)] = \mathbb{E}[(1-\alpha)\mathbf{1}_F + \mathbb{P}(A \mid X)\mathbf{1}_{F^c}] = 1 - \alpha$.
Structural Commentary: $\mathbb{P}(F) = 1$ and $\mathbb{P}(A \mid X) = 1 - \alpha$ on F.
Key Insight: $\mathbb{P}(A \mid X) = 1 - \alpha$ a.s. implies $\mathbb{P}(A) = 1 - \alpha$. □

Remark 5.5.3. *Unconditional Variability Can Differ*
Under random design, if $\mathbb{E}(\widehat{\beta} \mid X) = \beta$, then $\mathbb{E}\widehat{\beta} = \beta$.
Moreover, $\mathrm{Var}(\widehat{\beta}) = \mathbb{E}[\mathrm{Var}(\widehat{\beta} \mid X)] + \mathrm{Var}(\mathbb{E}[\widehat{\beta} \mid X]) = \sigma^2\mathbb{E}[(X^tX)^{-1}]$, not generally $\sigma^2(X^tX)^{-1}$ for frozen design.

Method 5.5.4. *Pairs Bootstrap for Random-Design Regression*
Let $D := \{(Y_i, x_i) : i \in [n]\}$, $x_i \in \mathbb{R}^p$, and let $T(D)$ be a fitted coefficient, prediction, or smooth statistic.

(i) Draw $D_b^* := \{(Y_{i_m}, x_{i_m}) : m \in [n]\}$ with $i_1, \ldots, i_n$ sampled iid from $[n]$.
(ii) Fit least squares on D_b^* and compute $T_b^* := T(D_b^*)$.
(iii) Repeat for $b \in [B]$ and estimate $\mathrm{se}(T(D))$ by $\widehat{\mathrm{se}}_*(T) := \left(\frac{1}{B-1}\sum_{b=1}^{B}(T_b^* - \overline{T}^*)^2\right)^{1/2}$, where $\overline{T}^* := B^{-1}\sum_{b=1}^{B} T_b^*$.

Perspective 5.5.5. *Extensions Beyond Ordinary Least Squares*

(i) *Robust regression*: Replace $\sum_i \widehat{e}_i^2$ by $\sum_i \Psi(\widehat{e}_i)$; $\Psi(u) = u^2$ gives least squares.
(ii) *Variable selection*: For selected predictors $S \subseteq \{1, \ldots, p\}$, $\widehat{\beta}_S$ and $\mathrm{se}(\widehat{\beta}_S)$ are conditional on S.
(iii) *Influence*: Large leverage $h_{ii} = x_i^t(X^tX)^{-1}x_i$ can make observation i dominate $\widehat{\beta}$.
(iv) *Inverse calibration*: Given a fitted response curve $\widehat{m}$ and measured response y_0, estimating x_0 from $\widehat{m}(x_0) = y_0$ is an inverse problem with uncertainty beyond $\mathrm{se}(\widehat{\beta})$.

Part XVII
Statistical Learning and Deep Learning

Statistical learning studies prediction rules, representation maps, and trained systems as mathematical objects. The statistical foundations of Parts 15 and 16 remain the language for experiments, inference, and model assessment; here the emphasis is on risk, generalization, optimization, architectures, and evaluation of learned predictors.

1 Learning Problems and Risk

1.1 Prediction, Loss Functions, and Risk

A learning problem turns observations into an action rule. Unlike inferential problems, the primary object is often the future loss of a predictor rather than a confidence statement about a parameter.

Definition 1.1.1. *Prediction Problem and Risk*
Inputs lie in $(\mathcal{X}, \mathcal{A})$, outputs in $(\mathcal{Y}, \mathcal{B})$, and $(X, Y) \sim P$ on $\mathcal{X} \times \mathcal{Y}$. Let $(\mathcal{A}_0, \mathcal{C})$ be an action space.

(i) *Predictor*: A predictor is a measurable map $f : \mathcal{X} \to \mathcal{A}_0$.
(ii) *Loss and risk*: For loss $\ell : \mathcal{A}_0 \times \mathcal{Y} \to [0, \infty]$, define $R(f) := \mathbb{E}_P \ell(f(X), Y)$.
(iii) *Bayes risk*: $R^* := \inf_f R(f)$, over measurable $f : \mathcal{X} \to \mathcal{A}_0$.

Definition 1.1.2. *Supervised Learning Terminology*
Let $S := \{(x_i, y_i) : 1 \le i \le n\}$ with $x_i = (x_{i1}, \dots, x_{ip})^\top \in \mathcal{X}$ and $y_i \in \mathcal{Y}$.

(i) *Features*: Coordinates x_{ij} are variables, inputs, predictors, independent variables, or features.
(ii) *Response*: Values y_i are outputs, responses, targets, or dependent variables.
(iii) *Regression*: If $\mathcal{Y} \subseteq \mathbb{R}^q$, the task is quantitative prediction.
(iv) *Classification*: If $\mathcal{Y} = \mathcal{G}$ is a finite unordered label set, the task is classification.
(v) *Ordered labels*: If $\mathcal{Y}$ is ordered categorical, order matters but arithmetic on labels usually does not.
(vi) *One-hot coding*: For $|\mathcal{G}| = K$, encode class g by $t_g \in \{0, 1\}^K$ with $(t_g)_g = 1$.
Classify vector scores $s(x) \in \mathbb{R}^K$ by $\operatorname{argmax}_k s_k(x)$.

If y_i are unobserved, the problem is unsupervised and targets structure in $\{x_i\}_{i=1}^n$.

Example 1.1.3. *Representative Learning Problems*

(i) *Email filtering*: x contains word and punctuation frequencies, $y \in \{\text{spam}, \text{ham}\}$, and asymmetric loss can penalize false deletion more than missed spam.
(ii) *Clinical regression*: $y = \log(\text{PSA})$ and x contains log cancer volume, prostate weight, age, and related measurements; correlated predictors are common.
(iii) *Digit recognition*: $x \in [0, 1]^{16 \times 16}$ or $[0, 1]^{256}$ and $y \in \{0, \dots, 9\}$; action set may add rejection option $\perp$.
(iv) *Microarrays*: An expression matrix $X \in \mathbb{R}^{G \times n}$ has genes as rows and samples as columns.
Typical first tasks are unsupervised clustering of genes or samples.

Example 1.1.4. *Common Prediction Losses*

(i) Regression commonly uses squared loss $\ell(a, y) = \|a - y\|_2^2$ or absolute loss $\ell(a, y) = |a - y|$.
(ii) Binary classification with labels in $\{-1, 1\}$ uses zero-one loss $\mathbf{1}_{\{ay \le 0\}}$ for a signed score a.
(iii) Probabilistic classification uses log loss $-\log q_y$ when q is a predicted distribution on labels.

Definition 1.1.5. *Bayes Predictors*

(i) *Conditional risk*: For action a, set $r_x(a) := \mathbb{E}[\ell(a, Y) \mid X = x]$; any $f^*(x) \in \operatorname{argmin}_a r_x(a)$ is Bayes.
(ii) *Squared loss*: If $\ell(a, y) = (a - y)^2$ and $\mathbb{E}[|Y| \mid X = x] < \infty$, then $f^*(x) = \mathbb{E}[Y \mid X = x]$.
(iii) *Absolute loss*: If $\ell(a, y) = |a - y|$, any conditional median $m(x)$
with $\mathbb{P}(Y \le m(x) \mid X = x) \ge \frac{1}{2}$ and $\mathbb{P}(Y \ge m(x) \mid X = x) \ge \frac{1}{2}$ is Bayes.
(iv) *Zero-one loss*: For labels $\mathcal{G}$ and $\ell(g, y) = \mathbf{1}_{\{g \ne y\}}$, a Bayes classifier is $g^*(x) \in \operatorname{argmax}_{g \in \mathcal{G}} \mathbb{P}(Y = g \mid X = x)$.

Remark 1.1.6. *Risk as a Prediction Target*
Here risk means predictive loss $R(f) = \mathbb{E}_P \ell(f(X), Y)$ under a data law P.
Decision-theoretic risk instead studies $R(\theta, \delta) = \mathbb{E}_\theta L(\theta, \delta(X))$ over a model family.

1.2 Empirical Risk Minimization

The population risk is the target, but only a sample is observed. Empirical risk minimisation replaces the unknown expectation by an average over the training data.

Definition 1.2.1. *Empirical Risk Minimization*
Let $S := ((x_1, y_1), \dots, (x_n, y_n))$.

(i) *Empirical risk*: For loss ℓ, define $\widehat{R}_S(f) := n^{-1} \sum_{i=1}^{n} \ell(f(x_i), y_i)$.
(ii) *Hypothesis class*: Let $\mathcal{F}$ be the admissible predictor class.
(iii) *ERM*: Any $\widehat{f} \in \operatorname{argmin}_{f \in \mathcal{F}} \widehat{R}_S(f)$ is an empirical risk minimiser.

Definition 1.2.2. *Regularised Empirical Risk*
Let $\Omega : \mathcal{F} \to [0, \infty]$ be a complexity penalty and $\lambda \geq 0$.
Define $Q_\lambda(f) := \widehat{R}_S(f) + \lambda \Omega(f)$ and choose $\widehat{f}_\lambda \in \operatorname{argmin}_{f \in \mathcal{F}} Q_\lambda(f)$.
The penalty Ω may encode smoothness, sparsity, norm control, margin size, or architecture.

Theorem 1.2.3. *Finite-Class ERM Bound*
Let $S = ((X_1, Y_1), \dots, (X_n, Y_n))$ be i.i.d. from P, let $\emptyset \neq \mathcal{F}$ be finite.
Assume $0 \leq \ell(f(X), Y) \leq 1$ for all $f \in \mathcal{F}$, and fix $\delta \in (0, 1)$. Set $\varepsilon_{n,\delta} := \sqrt{\log(2 |\mathcal{F}| / \delta)/(2n)}$.

(i) Uniform deviation: With probability at least $1 - \delta$, $\sup_{f \in \mathcal{F}} \left| R(f) - \widehat{R}_S(f) \right| \leq \varepsilon_{n,\delta}$.

(ii) ERM excess risk: If $\widehat{f} \in \operatorname{argmin}_{f \in \mathcal{F}} \widehat{R}_S(f)$, then $R(\widehat{f}) \leq \inf_{f \in \mathcal{F}} R(f) + 2\varepsilon_{n,\delta}$.

Proof.

(i) For fixed f, set $Z_i(f) := \ell(f(X_i), Y_i) \in [0, 1]$, so $R(f) = \mathbb{E} Z_i(f)$ and $\widehat{R}_S(f) = n^{-1} \sum_{i=1}^{n} Z_i(f)$.
Hoeffding and a union bound give $\mathbb{P}(\sup_{f \in \mathcal{F}} \left| \widehat{R}_S(f) - R(f) \right| > \varepsilon) \leq 2 |\mathcal{F}| e^{-2n\varepsilon^2}$; set $\varepsilon = \varepsilon_{n,\delta}$.

(ii) On the uniform-deviation event, choose $f^* \in \operatorname{argmin}_{f \in \mathcal{F}} R(f)$.
Since $\widehat{R}_S(\widehat{f}) \leq \widehat{R}_S(f^*)$, $R(\widehat{f}) \leq \widehat{R}_S(\widehat{f}) + \varepsilon_{n,\delta} \leq \widehat{R}_S(f^*) + \varepsilon_{n,\delta} \leq R(f^*) + 2\varepsilon_{n,\delta}$.

Structural Commentary: Union-bounded concentration gives uniform control; ERM turns it into risk control.
Key Insight: Control every gap $\left| R - \widehat{R}_S \right|$ by $\varepsilon_{n,\delta}$; the ERM comparison pays this gap twice. □

Remark 1.2.4. *Statistical and Computational Error*
In practice the selected predictor may be only an approximate minimizer of the empirical criterion.
Its excess risk is then decomposed into approximation error, estimation error, and optimisation error.

1.3 Surrogate Losses and Calibration

Many prediction losses are discontinuous or combinatorial. Learning algorithms often optimise a smoother surrogate and rely on calibration to connect that surrogate to the target loss.

Definition 1.3.1. *Surrogate Loss*
For binary labels $y \in \{-1, 1\}$, let $g : \mathcal{X} \to \mathbb{R}$ be score function. Define $m_g(x, y) := yg(x)$ and $\widehat{y}_g(x) := \operatorname{sign}(g(x))$.

(i) *Zero-one loss*: $\ell_{01}(g(x), y) := \mathbf{1}_{\{m_g(x,y) \leq 0\}}$.
(ii) *Surrogate*: $\ell_\phi(g(x), y) := \phi(m_g(x, y))$ for tractable $\phi : \mathbb{R} \to [0, \infty)$.

Example 1.3.2. *Margin Surrogates*

(i) Hinge loss: $\phi(t) = \max\{0, 1 - t\}$.
(ii) Logistic loss: $\phi(t) = \log(1 + \exp(-t))$.
(iii) Exponential loss: $\phi(t) = \exp(-t)$.
(iv) Squared margin loss: $\phi(t) = (1 - t)^2$.

Definition 1.3.3. *Classification Calibration*
Let $R_\phi(g) := \mathbb{E}\phi(Yg(X))$, $R_{01}(g) := \mathbb{P}(Yg(X) \leq 0)$, $R_\phi^* := \inf_g R_\phi(g)$, and $R_{01}^* := \inf_g R_{01}(g)$.
Margin surrogate ϕ is classification calibrated if $R_\phi(g_m) \to R_\phi^*$ implies $R_{01}(g_m) \to R_{01}^*$ $\forall$ score sequence (g_m).
Equivalently, $R_\phi(g) - R_\phi^* \to 0$ forces $R_{01}(g) - R_{01}^* \to 0$.

Remark 1.3.4. *Proper Scoring Rules*
A scoring rule $S(q, y)$ is proper if $\operatorname{argmin}_q \mathbb{E}[S(q, Y) \mid X = x] = \mathcal{L}(Y \mid X = x)$.
Thus log and Brier losses learn calibrated conditional probabilities.

1.4 Bias-Variance and Approximation-Estimation Tradeoffs

Learning quality depends both on the richness of the model class and on the variability introduced by fitting that class from a finite sample.

Proposition 1.4.1. *Bias-Variance Decomposition*
Let $Y = f_0(X) + \varepsilon$ *with* $\mathbb{E}[\varepsilon \mid X = x] = 0$ *and* $\mathrm{Var}(\varepsilon \mid X = x) = \sigma^2(x)$.
Let random S *produce* $\widehat{f}_S$, *set* $\bar{f}(x) := \mathbb{E}_S \widehat{f}_S(x)$, *and let* $Y_x \perp S$ *be a fresh response at* x.
Then $\mathbb{E}_{S,Y_x}[(Y_x - \widehat{f}_S(x))^2] = \sigma^2(x) + (f_0(x) - \bar{f}(x))^2 + \mathrm{Var}_S(\widehat{f}_S(x))$.

Proof. Write $Y_x - \widehat{f}_S(x) = \varepsilon_x + f_0(x) - \widehat{f}_S(x)$, where $\varepsilon_x := Y_x - f_0(x)$.
Since $\mathbb{E}[\varepsilon_x] = 0$ and Y_x is conditionally independent of S, $\mathbb{E}_{S,Y_x}[\varepsilon_x\{f_0(x) - \widehat{f}_S(x)\}] = 0$.
Thus $\mathbb{E}_{S,Y_x}[(Y_x - \widehat{f}_S(x))^2] = \sigma^2(x) + \mathbb{E}_S[(f_0(x) - \widehat{f}_S(x))^2]$.
Finally, $\mathbb{E}_S[(f_0(x) - \widehat{f}_S(x))^2] = (f_0(x) - \bar{f}(x))^2 + \mathbb{E}_S[(\widehat{f}_S(x) - \bar{f}(x))^2]$.
Structural Commentary: Squared prediction error equals noise plus fitted-function error.
Key Insight: Center $\widehat{f}_S(x)$ at $\bar{f}(x)$; the remaining cross term has mean zero. □

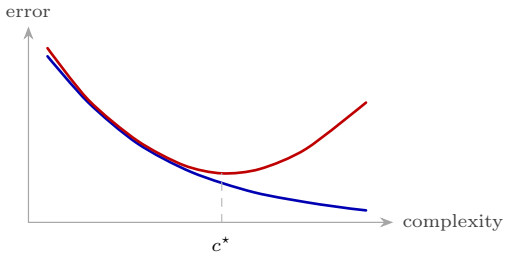


Figure 228: Bias–variance tradeoff: blue: training error, red: test error; dashed line marks test minimum.

Method 1.4.2. *Nearest-Neighbor Local Averaging*
Let $\mathcal{T} := \{(x_i, y_i) : 1 \le i \le n\}$ be training data in a metric space $(\mathcal{X}, d)$.
Let $N_k(x) \subseteq \{1, \dots, n\}$ be the indices of the k smallest distances $d(x_i, x)$.
For numerical responses, define the local average $\widehat{f}_k(x) := \frac{1}{k} \sum_{i \in N_k(x)} y_i$.
For class labels $g_i \in \mathcal{G}$, define $\widehat{p}_g(x) := \frac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}_{\{g_i = g\}}$ and classify by $\widehat{g}_k(x) \in \operatorname{argmax}_{g \in \mathcal{G}} \widehat{p}_g(x)$.
Thus k-nearest neighbours replace conditioning at $X = x$ by averaging over $N_k(x)$.

Remark 1.4.3. *Locality in High Dimension*
The parameter k trades local bias against variance: small k reacts strongly to S, while large k averages more but may wash out local structure. In $[0,1]^p$, a subcube with volume fraction r has side length $e_p(r) = r^{1/p}$, so $e_{10}(0.01) \approx 0.63$ and $e_{10}(0.10) \approx 0.80$. For N uniform points in the unit ball, the median nearest-neighbour radius is $d(p, N) = (1 - 2^{-1/N})^{1/p}$, about 0.52 for $(p, N) = (10, 500)$. Thus high-dimensional neighbourhoods large enough to contain data may not be local; useful local methods need structure or an adapted metric.

Definition 1.4.4. *Approximation and Estimation Errors*
Let $R^* := \inf_f R(f)$ and $R^*_{\mathcal{F}} := \inf_{f \in \mathcal{F}} R(f)$.

(i) *Approximation error*: $\mathcal{A}(\mathcal{F}) := R^*_{\mathcal{F}} - R^*$.
(ii) *Estimation error*: For a data-dependent $\widehat{f} \in \mathcal{F}$, $\mathcal{E}(\widehat{f}; \mathcal{F}) := R(\widehat{f}) - R^*_{\mathcal{F}}$.
(iii) *Excess risk*: $R(\widehat{f}) - R^* = \mathcal{A}(\mathcal{F}) + \mathcal{E}(\widehat{f}; \mathcal{F})$.

Larger $\mathcal{F}$ usually lowers $\mathcal{A}(\mathcal{F})$ and raises estimation difficulty.

Remark 1.4.5. *Model Complexity*
As $c(\mathcal{F})$ grows, $\widehat{R}_S(\widehat{f}_{\mathcal{F}})$ usually decreases, while $R(\widehat{f}_{\mathcal{F}}) - R^* = \mathcal{A}(\mathcal{F}) + \mathcal{E}(\widehat{f}_{\mathcal{F}}; \mathcal{F})$ trades smaller approximation error for larger estimation error. For zero-one loss, the relevant failure is crossing the wrong side of the decision boundary, not an additive squared-error bias–variance split.

2 Generalization Theory

2.1 Uniform Convergence and VC Theory

Generalisation asks when small training loss implies small population loss. Uniform convergence controls the gap for all predictors in a class at once.

Definition 2.1.1. *Uniform Convergence and VC Dimension*

(i) *Uniform convergence*: For samples S_n of size n and bounded loss, $\mathcal{F}$ has uniform convergence if $\sup_{f\in\mathcal{F}}\left|R(f)-\widehat{R}_{S_n}(f)\right|\to 0$ in probability, or almost surely, as $n\to\infty$.

(ii) *Shattering and VC dimension*: For $\mathcal{C}\subseteq 2^{\mathcal{X}}$, a finite $A\subseteq\mathcal{X}$ is shattered if $\{A\cap C : C\in\mathcal{C}\}=2^A$, and $\operatorname{VCdim}(\mathcal{C})$ is the supremum of $|A|$ over finite shattered $A\subseteq\mathcal{X}$.

Example 2.1.2. *Shattering Examples*
For $\mathcal{H}_p:=\{\{x\in\mathbb{R}^p : a^\top x+b\ge 0\} : a\in\mathbb{R}^p, b\in\mathbb{R}\}$, $\operatorname{VCdim}(\mathcal{H}_p)=p+1$; in particular, $\mathcal{H}_2$ shatters any three noncollinear points and shatters no four-point set.
Parameter count not decisive: one-parameter class $\mathcal{C}:=\{\{x\in\mathbb{R}:\sin(\alpha x)>0\}:\alpha\in\mathbb{R}\}$ has infinite VCdim.

Theorem 2.1.3. *VC Generalization Bound Form*
Let $\mathcal{F}\subseteq\{0,1\}^{\mathcal{X}}$ have $1\le d:=\operatorname{VCdim}(\mathcal{F})<\infty$, let $S_n=((X_i,Y_i))_{i=1}^n$ be i.i.d., and use $\ell_f(x,y):=\mathbf{1}_{\{f(x)\neq y\}}$, $R(f):=\mathbb{E}\ell_f(X,Y)$, and $\widehat{R}_{S_n}(f):=\frac{1}{n}\sum_{i=1}^n\ell_f(X_i,Y_i)$.
Define $\Pi_{\mathcal{F}}(m):=\max|\{(f(x_1),\dots,f(x_m)) : f\in\mathcal{F}\}|$, where the maximum is over $x_1,\dots,x_m\in\mathcal{X}$.

(i) Growth bound: If $m\ge d\ge 1$, then $\Pi_{\mathcal{F}}(m)\le\sum_{j=0}^d\binom{m}{j}\le\left(\frac{em}{d}\right)^d$.

(ii) Uniform deviation: For every $\delta\in(0,1)$ and $n\ge d$, with probability at least $1-\delta$, $\sup_{f\in\mathcal{F}}\left|R(f)-\widehat{R}_{S_n}(f)\right|\le C\sqrt{\frac{d\log\left(\frac{2en}{d}\right)+\log\left(\frac{1}{\delta}\right)}{n}}$ for a universal constant C.

Proof.

(i) *Growth bound*: Since no set of size $d+1$ is shattered, Sauer's lemma gives $\Pi_{\mathcal{F}}(m)\le\sum_{j=0}^d\binom{m}{j}$, and the binomial-sum bound gives $\sum_{j=0}^d\binom{m}{j}\le\left(\frac{em}{d}\right)^d$ for $m\ge d$.

(ii) *Uniform deviation*: Symmetrization with an independent ghost sample and Hoeffding's inequality on the at most $\Pi_{\mathcal{F}}(2n)$ realized labelings give $\mathbb{P}(\sup_{f\in\mathcal{F}}\left|R(f)-\widehat{R}_{S_n}(f)\right|>\varepsilon)\le 8\Pi_{\mathcal{F}}(2n)\exp(-cn\varepsilon^2)$ for a universal $c>0$; applying (i) with $m=2n$ and solving the right side at level δ yields (ii).

Structural Commentary: VC dimension polynomially bounds sample labelings.
Key Insight: In finite-class concentration, use $|\mathcal{F}|\rightsquigarrow\Pi_{\mathcal{F}}(2n)\le\left(\frac{2en}{d}\right)^d$. □

Remark 2.1.4. *Growth Functions*
For $d=\operatorname{VCdim}(\mathcal{F})$, $\Pi_{\mathcal{F}}(m)$ is the effective number of labelings on m sample points. Sauer's lemma gives $\log\Pi_{\mathcal{F}}(m)\le d\log\left(\frac{em}{d}\right)$, so VC bounds replace $\log|\mathcal{F}|$ by the sample-level complexity $\log\Pi_{\mathcal{F}}(m)$.

Method 2.1.5. *Structural Risk Minimization*

(i) *Model ladder*: Choose $\mathcal{F}_1\subseteq\mathcal{F}_2\subseteq\cdots$ with complexities c_m, such as $c_m=\operatorname{VCdim}(\mathcal{F}_m)$.

(ii) *Uniform penalties*: Choose $\delta_m>0$ with $\sum_m\delta_m\le\delta$ and penalties $\psi_m(n,\delta_m)$ such that $\mathbb{P}(\sup_{f\in\mathcal{F}_m}\left|R(f)-\widehat{R}_{S_n}(f)\right|\le\psi_m(n,\delta_m))\ge 1-\delta_m$.

(iii) *SRM rule*: Choose $(\widehat{m},\widehat{f})\in\operatorname*{argmin}_{m\in\mathbb{N},f\in\mathcal{F}_m}\{\widehat{R}_{S_n}(f)+\psi_m(n,\delta_m)\}$.

Larger m improves fit but raises ψ_m; SRM balances flexibility against search cost.

2.2 Rademacher Complexity and Covering Numbers

For real-valued predictors, complexity is often measured by how well the class can fit random noise.

Definition 2.2.1. *Empirical Rademacher Complexity*
Given sample points $x_1,\dots,x_n$ and independent Rademacher signs $\sigma_i\in\{-1,1\}$,

$$\widehat{\mathfrak{R}}_S(\mathcal{F})=\mathbb{E}_\sigma\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\sigma_i f(x_i)\right].$$

The expectation over the sample gives the population Rademacher complexity.

Theorem 2.2.2. *Rademacher Generalization Bound*
For a loss-composed class $\ell \circ \mathcal{F}$ bounded in $[0,1]$, with probability at least $1-\delta$,

$$R(f) \le \widehat{R}_S(f) + 2\widehat{\mathfrak{R}}_S(\ell \circ \mathcal{F}) + 3\sqrt{\frac{\log(2/\delta)}{2n}}$$

for every $f \in \mathcal{F}$.

Definition 2.2.3. *Covering Number*
For a metric space (T,d), the covering number $N(\varepsilon, T, d)$ is the smallest number of closed d-balls of radius ε needed to cover T. Covering numbers measure metric size rather than parameter count.

Remark 2.2.4. *Entropy Methods*
Bounds on $\log N(\varepsilon, \mathcal{F}, d)$ can be integrated over scales to control empirical processes. This is the route from functional-analytic compactness to statistical-learning rates.

2.3 Concentration and Stability

Generalization can also follow from concentration of the training criterion or from stability of the learning algorithm itself.

Definition 2.3.1. *Subgaussian Random Variable*
A centered random variable Z is *subgaussian* with variance proxy σ^2 if

$$\mathbb{E}\exp(\lambda Z) \le \exp(\lambda^2\sigma^2/2)$$

for every $\lambda \in \mathbb{R}$. Subgaussian variables have Gaussian-type tail bounds.

Theorem 2.3.2. *Bounded Differences Inequality*
Let $F(X_1,\dots,X_n)$ change by at most c_i when only X_i is replaced. Then

$$P\{F - \mathbb{E}F \ge t\} \le \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Definition 2.3.3. *Uniform Stability*
A learning algorithm A has *uniform stability* β_n for loss ℓ if replacing one training example changes the loss on any test point by at most β_n:

$$\sup_z |\ell(A(S), z) - \ell(A(S'), z)| \le \beta_n.$$

Remark 2.3.4. *Algorithm-Dependent Generalization*
Stability bounds depend on the training procedure, not just the hypothesis class. This makes them useful for regularized empirical risk minimization and for understanding why some optimization algorithms generalize better than the raw size of their model class suggests.

2.4 PAC-Bayes and Information-Theoretic Bounds

PAC-Bayes bounds analyze randomized predictors by comparing a data-dependent posterior distribution on hypotheses to a fixed prior.

Definition 2.4.1. *Randomized Predictor*
Let $\mathcal{H}$ be a hypothesis space. A distribution Q on $\mathcal{H}$ defines a randomized predictor by sampling $h \sim Q$ and predicting with h. Its risk is

$$R(Q) = \mathbb{E}_{h\sim Q} R(h).$$

Theorem 2.4.2. *PAC-Bayes Bound Form*
Let Π be a prior on $\mathcal{H}$ independent of the sample. For bounded losses, with probability at least $1-\delta$, every posterior Q satisfies a bound of the form

$$R(Q) \le \widehat{R}_S(Q) + O\left(\sqrt{\frac{\mathrm{KL}(Q\|\Pi) + \log(1/\delta)}{n}}\right).$$

Remark 2.4.3. *Prior and Posterior*
The prior in PAC-Bayes is not required to be a subjective Bayesian prior. Its role is to anchor a complexity penalty: posteriors close to the prior are simpler in the sense of relative entropy.

Definition 2.4.4. *Information-Theoretic Generalization*
If $W = A(S)$ is the learned output of an algorithm, information-theoretic bounds control the generalization gap by quantities such as the mutual information $I(W;S)$. Such bounds measure how much the trained model reveals about the particular sample.

3 Classical Supervised Learning

3.1 Linear Prediction and Generalized Linear Models

Linear prediction is the local model against which many nonlinear learning methods are compared.

Definition 3.1.1. *Linear Predictor*
For features $x \in \mathbb{R}^p$, a linear score has the form

$$f_{w,b}(x) = w^\top x + b.$$

Regression interprets this score directly, while classification often passes it through a sign, sigmoid, or softmax map.

Definition 3.1.2. *Least-Squares Linear Regression*
Given a design matrix $X \in \mathbb{R}^{n\times d}$ and response vector $y \in \mathbb{R}^n$, ordinary least squares solves

$$\widehat{\beta} \in \arg\min_{\beta\in\mathbb{R}^d} \|y - X\beta\|_2^2 .$$

When X^TX is nonsingular,

$$\widehat{\beta} = (X^TX)^{-1}X^Ty, \qquad \widehat{y} = X\widehat{\beta}.$$

The columns of X may include raw quantitative variables, transformations such as logarithms or powers, dummy variables for categorical inputs, interaction products, or basis functions. The model is linear in the coefficients even when the fitted function is nonlinear in the original variables.

Method 3.1.3. *Projection View of Least Squares*
If X has full column rank, the fitted vector is

$$\widehat{y} = Hy, \qquad H = X(X^TX)^{-1}X^T.$$

The matrix H is the orthogonal projection onto the column space of X, and the residual $y - \widehat{y}$ is orthogonal to every fitted column. If the columns are linearly dependent, the coefficient vector need not be unique, but the fitted vector is still the projection of y onto $\mathrm{col}(X)$. Thus rank deficiency is a representation problem for coefficients, not an ambiguity in the least-squares fitted values. See also Definition 5.3.4 and Proposition 5.3.5.

Remark 3.1.4. *Adjusted Coefficients*
The coefficient of a predictor in a multiple regression is its contribution after adjustment for the other columns. If z_j is the residual from regressing x_j on the remaining columns, then the coefficient of x_j can be read as the simple regression coefficient of y on z_j:

$$\widehat{\beta}_j = \frac{z_j^T y}{z_j^T z_j}.$$

When x_j is nearly explained by the other predictors, $\|z_j\|_2$ is small and the coefficient estimate is unstable. This is the linear-model form of the collinearity problem that motivates shrinkage and selection in Definition 3.2.2.

Algorithm 1: Least Squares by Successive Orthogonalization

Require: Full-column-rank design columns $x_0, \dots, x_p$ and response vector y
Ensure: Least-squares coefficients $\widehat{\beta}$
  for $j = 0, \dots, p$ **do**
    $z_j \leftarrow x_j$
    for $\ell = 0, \dots, j-1$ **do**
      $r_{\ell j} \leftarrow q_\ell^T x_j$
      $z_j \leftarrow z_j - r_{\ell j} q_\ell$
    end for
    $r_{jj} \leftarrow \|z_j\|_2$
    $q_j \leftarrow z_j / r_{jj}$
  end for
  Form $Q = [q_0, \dots, q_p]$ and the upper-triangular matrix $R = (r_{\ell j})$
  Solve $R\widehat{\beta} = Q^T y$ by back-substitution

Remark 3.1.5. *Linear-Model Inference After Selection*
Under a fixed-design Gaussian model $y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$,

$$\widehat{\beta} \sim N\big(\beta, \sigma^2(X^TX)^{-1}\big), \qquad \widehat{\sigma}^2 = \frac{\left\|y - X\widehat{\beta}\right\|_2^2}{n-d}.$$

This gives the usual coefficient standard errors, t- or large-sample z-scores, and nested-model F tests. In a learning workflow these summaries are descriptive unless the model was fixed before looking at the data; after subset search, shrinkage tuning, or repeated validation, the displayed standard errors do not account for the selection process.

Method 3.1.6. *Multiple-Output Least Squares*
For K quantitative responses collected in $Y \in \mathbb{R}^{n\times K}$, the linear model

$$Y = XB + E$$

has least-squares solution

$$\widehat{B} = (X^T X)^{-1} X^T Y$$

when X has full column rank. The columns decouple: fitting all responses together gives the same coefficient matrix as fitting K separate least-squares regressions. If the responses share a common low-dimensional structure, one can instead impose a rank constraint on B or shrink common response directions, borrowing strength across outputs rather than treating them independently.

Method 3.1.7. *Canonical Correlation and Reduced-Rank Regression*
Multiple-output problems can borrow strength across responses by first finding paired directions in input and output space. Canonical correlation analysis chooses vectors v_m and u_m so that the derived variables Xv_m and Yu_m have large correlation, with later pairs constrained to be uncorrelated with earlier pairs. At most $\min(p, K)$ such paired directions are available.

Reduced-rank regression uses the same idea inside a prediction model. For a rank budget m, one constrains the coefficient matrix in

$$Y \approx XB$$

to satisfy $\operatorname{rank}(B) \le m$, equivalently forcing the fitted responses to live in an m-dimensional response subspace. In the common covariance-weighted form, the target is

$$\widehat{B}^{\mathrm{rr}}_m \in \operatorname*{arg\,min}_{\operatorname{rank}(B)\le m} \sum_{i=1}^{n} (y_i - B^T x_i)^T \widehat{\Sigma}^{-1} (y_i - B^T x_i).$$

With an estimated common response covariance, the solution can be read as ordinary least squares on the leading canonical response directions, followed by mapping the fitted values back to the original K responses. The rank m is a complexity parameter: $m = K$ recovers the separate least-squares fits, while smaller m pools information through shared response structure.

Definition 3.1.8. *Discriminant Functions and Linear Boundaries*
For classes $\mathcal{G} = \{1, \dots, K\}$, a discriminant classifier assigns

$$\widehat{G}(x) = \operatorname*{arg\,max}_{k\in\mathcal{G}} \delta_k(x).$$

The boundary between classes k and ℓ is contained in the comparison set

$$\{x : \delta_k(x) = \delta_\ell(x)\}.$$

Thus a classifier has linear decision boundaries whenever all pairwise differences $\delta_k - \delta_\ell$ are affine functions of x. The same conclusion holds if an order-preserving transform of posterior odds is affine; for example, in the binary case

$$\log \frac{\mathbb{P}(G = 1 \mid X = x)}{\mathbb{P}(G = 2 \mid X = x)} = \beta_0 + \beta^\top x$$

has boundary $\{x : \beta_0 + \beta^\top x = 0\}$.

Method 3.1.9. *Indicator-Matrix Regression Classifier*
Code the K classes by target vectors $t_k = e_k \in \mathbb{R}^K$. The ith response row is $Y_i = t_{g_i}$, and with intercept-augmented rows $\widetilde{x}_i^\top = (1, x_i^\top)$ the least-squares target fit solves

$$\widehat{B} \in \operatorname*{arg\,min}_{B} \sum_{i=1}^{n} \left\| Y_i - \widehat{B}^\top \widetilde{x}_i \right\|_2^2, \qquad \widehat{B} = (\widetilde{X}^\top \widetilde{X})^{-1} \widetilde{X}^\top Y$$

when the normal matrix is invertible. A new point is scored by $\widehat{f}(x) = \widehat{B}^\top \widetilde{x}$ and classified by

$$\widehat{G}(x) = \operatorname*{arg\,max}_{k} \widehat{f}_k(x),$$

equivalently by choosing the closest target vector t_k in squared Euclidean distance.

With an intercept, the fitted components sum to one, reflecting the conditional-expectation identity $\mathbb{E}[Y_k \mid X = x] = \mathbb{P}(G = k \mid X = x)$. They need not individually lie in $[0, 1]$, however, and for $K \ge 3$ the rigid linear fit can *mask* a middle class: the fitted component for that class may never be largest even when linear separating boundaries exist.

Definition 3.1.10. *Generalized Linear Model*
A generalized linear model specifies a conditional distribution from an exponential family and links its mean to a linear score:

$$g(\mathbb{E}[Y \mid X = x]) = w^\top x + b.$$

Logistic regression is the Bernoulli case with logit link.

Definition 3.1.11. *Multiclass Logistic Regression*
Using class K as a reference, multinomial logistic regression models the posterior odds by

$$\log \frac{\mathbb{P}(G = k \mid X = x)}{\mathbb{P}(G = K \mid X = x)} = \beta_{k0} + \beta_k^\top x, \qquad k = 1, \dots, K-1.$$

Equivalently, for $k < K$,

$$\mathbb{P}(G = k \mid X = x) = \frac{\exp(\beta_{k0} + \beta_k^\top x)}{1 + \sum_{\ell=1}^{K-1} \exp(\beta_{\ell 0} + \beta_\ell^\top x)}$$

while the reference-class probability has numerator 1. The parameters are fit by maximizing the conditional multinomial log-likelihood

$$\ell(\theta) = \sum_{i=1}^{n} \log p_{g_i}(x_i; \theta),$$

so the fitted probabilities sum to one and remain in $[0, 1]$ by construction. The choice of reference class changes the parametrization but not the fitted probabilities.

Method 3.1.12. *Binary Logistic Regression by IRLS*
For $y_i \in \{0, 1\}$ and design rows $x_i^\top$ including the intercept, write $p_i(\beta) = \{1 + \exp(-x_i^\top \beta)\}^{-1}$. The Bernoulli log-likelihood has score and Hessian

$$\nabla \ell(\beta) = X^\top (y - p), \qquad \nabla^2 \ell(\beta) = -X^\top W X,$$

where $W = \operatorname{diag}(p_i(1 - p_i))$. Newton's method can therefore be written as a weighted least-squares update with adjusted response

$$z = X\beta^{\text{old}} + W^{-1}(y - p), \qquad \beta^{\text{new}} = (X^\top W X)^{-1} X^\top W z.$$

This is the iteratively reweighted least-squares form. When the maximum-likelihood estimate is finite, the concavity of the log-likelihood makes Newton updates natural, and step-halving is a common safeguard against overshooting.

Method 3.1.13. *L_1-Regularized Logistic Regression*
For binary logistic regression with unpenalized intercept, the lasso version maximizes

$$\sum_{i=1}^{n} \{y_i(\beta_0 + x_i^\top \beta) - \log(1 + \exp(\beta_0 + x_i^\top \beta))\} - \lambda \sum_{j=1}^{p} |\beta_j|.$$

For active variables the score equations take the generalized correlation form

$$x_j^\top (y - p) = \lambda \operatorname{sign}(\beta_j),$$

while inactive variables have correlations with the residual vector controlled by the same threshold. Coordinate-descent and quadratic-approximation methods exploit this structure.

Method 3.1.14. *Gaussian Discriminant Analysis*
Suppose $X \mid G = k$ has density f_k and prior probability π_k. Bayes' rule gives

$$\mathbb{P}(G = k \mid X = x) = \frac{f_k(x)\pi_k}{\sum_{\ell=1}^{K} f_\ell(x)\pi_\ell}.$$

For a Gaussian class model with mean μ_k and covariance Σ_k, one may compare the scores

$$\delta_k(x) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2}(x - \mu_k)^\top \Sigma_k^{-1}(x - \mu_k) + \log \pi_k.$$

If all classes share a common covariance Σ, terms common to all classes cancel and the equivalent linear discriminant scores are

$$\delta_k^{\text{LDA}}(x) = x^\top \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^\top \Sigma^{-1} \mu_k + \log \pi_k.$$

Separate covariances Σ_k give quadratic discriminant analysis. In practice, π_k, μ_k, and the covariance matrices are estimated from the training data; a pooled covariance gives LDA, while classwise covariances give QDA.

Method 3.1.15. *Regularized Discriminant Analysis*
Regularized discriminant analysis interpolates between LDA and QDA by shrinking classwise covariance esti-

mates toward the pooled covariance:

$$\widehat{\Sigma}_k(\alpha) = \alpha\widehat{\Sigma}_k + (1-\alpha)\widehat{\Sigma}, \qquad 0 \le \alpha \le 1.$$

The endpoint $\alpha = 0$ is LDA, and $\alpha = 1$ is QDA. A further shrinkage toward a scalar covariance,

$$\widehat{\Sigma}(\gamma) = \gamma\widehat{\Sigma} + (1-\gamma)\widehat{\sigma}^2 I,$$

reduces variance when covariance estimation is unstable.

Definition 3.1.16. *Naive Bayes Classifier*
Naive Bayes assumes conditional independence of features within each class:

$$f_j(x) = \prod_{k=1}^{p} f_{jk}(x_k), \qquad j = 1, \dots, J.$$

With class priors π_j, the posterior is proportional to $\pi_j \prod_k f_{jk}(x_k)$. Relative to a reference class J, the log odds have the additive form

$$\log \frac{\mathbb{P}(G=\ell \mid X=x)}{\mathbb{P}(G=J \mid X=x)} = \log \frac{\pi_\ell}{\pi_J} + \sum_{k=1}^{p} \log \frac{f_{\ell k}(x_k)}{f_{Jk}(x_k)}.$$

The independence assumption can be false while still reducing variance enough to give useful classification probabilities near decision regions.

Remark 3.1.17. *Least Squares Versus LDA*
For two classes, least-squares regression on any distinct numeric coding of the labels produces a coefficient direction proportional to the LDA direction $\widehat{\Sigma}^{-1}(\widehat{\mu}_2 - \widehat{\mu}_1)$. The resulting cutpoint need not match the LDA rule unless the class sizes are balanced or the cutpoint is chosen separately. For more than two classes, indicator-matrix regression and LDA differ substantially; LDA avoids the masking pathology by using distances to class centroids in the pooled covariance geometry.

Remark 3.1.18. *Logistic Regression Versus LDA*
LDA and multinomial logistic regression can lead to the same logit-linear form, but they estimate it differently. Logistic regression maximizes the conditional likelihood of G given X and leaves the marginal distribution of X unrestricted. LDA maximizes a joint Gaussian mixture likelihood for (X, G), using information in the marginal feature distribution. This can reduce variance when the Gaussian assumptions are reasonable, but it also makes LDA more sensitive to model misspecification and outliers.

Remark 3.1.19. *Whitening and Reduced-Rank LDA*
With common within-class covariance W, LDA can be viewed by first whitening the features with respect to W and then comparing distances to class centroids, adjusted by $\log \pi_k$. The K whitened centroids lie in an affine subspace of dimension at most $K-1$, so all LDA distance comparisons can be made after projecting to this centroid-spanning subspace. Fisher's discriminant directions arise from maximizing

$$\frac{a^\top B a}{a^\top W a},$$

where B measures between-class centroid variation and W measures pooled within-class variation; equivalently, they solve a generalized eigenvalue problem.

Remark 3.1.20. *Feature Maps*
Applying a linear model to transformed features $\phi(x)$ gives nonlinear prediction in the original input space. Kernel methods and neural networks can both be viewed as systematic ways to choose or learn such feature maps.

3.2 Regularization and Model Selection

Regularization controls model complexity by penalizing unstable parameter directions or by constraining the feasible set.

Definition 3.2.1. *Penalized Linear Estimation*
For empirical loss $L_n(\beta)$, a penalized estimator solves

$$\widehat{\beta}_\lambda \in \arg\min_{\beta} \{L_n(\beta) + \lambda\Omega(\beta)\}.$$

The choices $\Omega(\beta) = \|\beta\|_2^2$ and $\Omega(\beta) = \|\beta\|_1$ give ridge and lasso penalties.

Definition 3.2.2. *Subset Selection, Ridge, and Lasso*
Let $y \in \mathbb{R}^n$ be centered and let $X \in \mathbb{R}^{n \times p}$ have centered, standardized columns. Three standard ways to restrict least squares are:

(i) *Best subset*: for a size m, solve

$$\widehat{\beta}_m^{\text{sub}} \in \operatorname*{arg\,min}_{|\operatorname{supp}(\beta)| \le m} \|y - X\beta\|_2^2 .$$

(ii) *Ridge*: for $\lambda \ge 0$, solve

$$\widehat{\beta}_\lambda^{\text{ridge}} \in \arg\min_\beta \{\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2\}.$$

(iii) *Lasso*: for $\lambda \ge 0$, solve

$$\widehat{\beta}_\lambda^{\text{lasso}} \in \arg\min_\beta \left\{ \frac{1}{2} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}.$$

Centering separates the intercept: for raw data the intercept is estimated by the sample mean after centering the columns, and it is not penalized.

Remark 3.2.3. *Penalty Geometry*
The ℓ^2 penalty shrinks coefficients smoothly along directions of weak design information. The ℓ^1 penalty has corners and can set coefficients exactly to zero, linking prediction to variable selection. With orthonormal columns and least-squares coordinate $z_j = x_j^T y$, best subset keeps the largest coordinates, ridge sends z_j to $z_j/(1+\lambda)$, and lasso applies the soft-threshold map

$$z_j \longmapsto \operatorname{sign}(z_j)(|z_j| - \lambda)_+.$$

Figure 229 shows the same contrast in the two-parameter constrained picture.

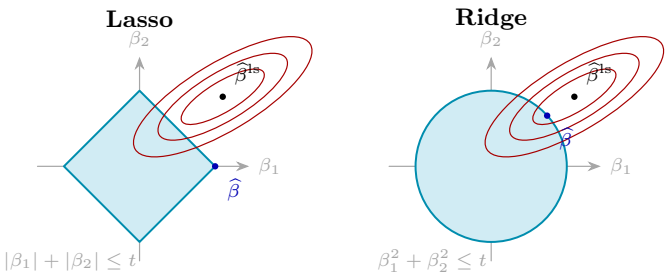


Figure 229: Constraint geometry for lasso and ridge. The lasso diamond has corners where an elliptical least-squares contour can first touch with one coefficient equal to zero; the ridge disk shrinks smoothly without creating such corners.

Proposition 3.2.4. *Ridge Shrinkage and Effective Degrees*
Assume X is centered and has thin singular value decomposition $X = UDV^T$, with singular values $d_1, \dots, d_r > 0$. The ridge fitted vector satisfies

$$X\widehat{\beta}_\lambda^{\text{ridge}} = X(X^TX + \lambda I)^{-1}X^Ty = \sum_{j=1}^{r} \frac{d_j^2}{d_j^2 + \lambda}\, u_j u_j^T y.$$

Thus ridge keeps the principal directions of X but shrinks directions with smaller singular values more strongly. Its linear smoother matrix has effective degrees of freedom

$$\operatorname{df}(\lambda) = \operatorname{tr}\big(X(X^TX + \lambda I)^{-1}X^T\big) = \sum_{j=1}^{r} \frac{d_j^2}{d_j^2 + \lambda}.$$

Definition 3.2.5. *Model Selection by Validation*
A validation criterion estimates predictive risk for candidate models or hyperparameters. In K-fold cross-validation, the data are split into K folds; each fold is held out once while the model is trained on the remaining folds, and the held-out losses are averaged.

Method 3.2.6. *One-Standard-Error Rule*
Suppose cross-validation gives an estimated error curve and standard errors over an ordered complexity parameter. Let $\widehat{E}_{\min}$ be the smallest estimated error and let $\widehat{\text{se}}_{\min}$ be its standard error. The one-standard-error rule chooses the simplest candidate with estimated error at most $\widehat{E}_{\min} + \widehat{\text{se}}_{\min}$. A final test set, if available, is reserved for judging the selected model rather than for choosing the complexity parameter.

Example 3.2.7. *A Selection and Shrinkage Workflow*
In the prostate-cancer regression example from the source pages, the response is the log prostate-specific antigen level, and the candidate predictors include log cancer volume, log prostate weight, age, log benign-prostatic-hyperplasia amount, seminal vesicle invasion, log capsular penetration, Gleason score, and the percentage of high Gleason scores. A typical comparison first standardizes the predictors, splits the data into training and test parts, and then uses cross-validation on the training part to choose the complexity parameter for each method.

The methods compared there illustrate the main families in this subsection: best subset selection keeps a small selected set, ridge keeps all variables with smooth shrinkage, the lasso both shrinks and sets some coefficients to zero, and PCR/PLS replace the predictors by a smaller number of derived directions. The final test set is used only after the tuning step, so the reported test error evaluates the selected model rather than helping choose it.

Remark 3.2.8. *Selection Bias*
The same validation data should not be used repeatedly as if it were untouched test data. Nested validation or a final test set is needed when model selection itself is part of the learning procedure.

Method 3.2.9. *Derived Input Directions*
When predictors are numerous or highly correlated, a regression may first replace the original columns $x_1, \dots, x_p$ by derived directions $z_m = Xv_m$ and then regress y on $z_1, \dots, z_M$.

(i) *Principal components regression*: if v_m are right singular vectors of X, set $z_m = Xv_m$ and fit

$$\widehat{y}_M^{\mathrm{pcr}} = \sum_{m=1}^{M} z_m \frac{z_m^T y}{z_m^T z_m}.$$

This keeps the M high-variance directions and discards the rest.

(ii) *Partial least squares*: construct orthogonal directions using both X and y. The first direction weights predictors by their univariate inner products with y; later directions repeat the same idea after orthogonalizing the remaining predictor columns. Equivalently, the directions favor high variance in X together with high correlation with the response.

The number M is a complexity parameter and is usually selected by validation.

Algorithm 2: Partial Least Squares Directions

Require: Centered response y, standardized columns $x_1, \dots, x_p$, number of directions $M \le p$
Ensure: Fitted vectors $\widehat{y}^{(1)}, \dots, \widehat{y}^{(M)}$
  Set $\widehat{y}^{(0)} \leftarrow 0$ and $x_j^{(0)} \leftarrow x_j$ for $j = 1, \dots, p$
  for $m = 1, \dots, M$ **do**
    $z_m \leftarrow \sum_{j=1}^{p} (x_j^{(m-1)})^T y \, x_j^{(m-1)}$
    $\widehat{\theta}_m \leftarrow z_m^T y / (z_m^T z_m)$
    $\widehat{y}^{(m)} \leftarrow \widehat{y}^{(m-1)} + \widehat{\theta}_m z_m$
    for $j = 1, \dots, p$ **do**
      $x_j^{(m)} \leftarrow x_j^{(m-1)} - \left(z_m^T x_j^{(m-1)} / z_m^T z_m \right) z_m$
    end for
  end for

Remark 3.2.10. *PLS Versus PCR*
PCR chooses directions that explain variance in the predictors, whether or not those directions are predictive. PLS lets the response enter the direction construction through the inner products $(x_j^{(m-1)})^T y$, and then removes the part of each predictor column already represented by z_m. Thus PLS tends to behave like a compromise between high-variance predictor directions and high-response-correlation directions.

Method 3.2.11. *Piecewise-Linear Path Algorithms*
For objectives of the form

$$\widehat{\beta}(\lambda) \in \arg\min_{\beta} \{ R(\beta) + \lambda J(\beta) \},$$

quadratic or piecewise-quadratic losses paired with piecewise-linear penalties often produce piecewise-linear coefficient paths. Least angle regression moves in directions that keep active predictors tied in absolute correlation with the current residual; the lasso modification drops an active predictor if its coefficient crosses zero. Coordinate descent takes a complementary view: fix a penalty value, solve by cycling through one-coordinate lasso subproblems, then warm-start the next penalty value.

Algorithm 3: Least Angle Regression Path

Require: Centered response y and standardized columns $x_1, \dots, x_p$

Ensure: A piecewise-linear sequence of coefficient vectors
Initialize $\beta \leftarrow 0$, $r \leftarrow y$, and active set $\mathcal{A} \leftarrow \varnothing$
repeat
Choose an inactive predictor x_j maximizing $|x_j^T r|$ and add j to $\mathcal{A}$
Find the least-squares direction in the span of $X_{\mathcal{A}}$ whose signed correlations with the active predictors are equal in magnitude
Move $\beta_{\mathcal{A}}$ in this direction and update r until either an inactive predictor reaches the same absolute correlation or the full least-squares fit is reached
until all predictors have entered or the residual is orthogonal to the active span

Method 3.2.12. *LAR, Lasso, and Correlation Ties*
For the lasso objective

$$\frac{1}{2}\|y - X\beta\|_2^2 + \lambda \|\beta\|_1 ,$$

the optimality conditions can be written in terms of correlations with the residual $r = y - X\beta$:

$$x_j^T r = \lambda \operatorname{sign}(\beta_j) \quad \text{if } \beta_j \neq 0, \qquad |x_j^T r| \leq \lambda \quad \text{if } \beta_j = 0.$$

Thus active variables have tied absolute residual correlations, and inactive variables are no larger. LAR follows the same tied-correlation geometry. It agrees with the lasso path until an active coefficient would cross zero; the lasso modification then removes that variable from the active set and recomputes the joint direction. This is why lasso paths are piecewise linear even though the active set can both gain and lose variables.

Algorithm 4: Incremental Forward Stagewise Regression

Require: Centered response y, standardized columns $x_1, \ldots, x_p$, small step size $\varepsilon > 0$
Ensure: A stagewise coefficient path
Initialize $\beta \leftarrow 0$ and $r \leftarrow y$
repeat
Choose j maximizing $|x_j^T r|$
$\delta \leftarrow \varepsilon \operatorname{sign}(x_j^T r)$
$\beta_j \leftarrow \beta_j + \delta$
$r \leftarrow r - \delta x_j$
until the residual has negligible correlation with every predictor

Remark 3.2.13. *Stagewise and Lasso Paths*
Small-step forward stagewise regression is more cautious than forward stepwise regression: it changes only the coefficient of the currently most correlated predictor by a small amount, then checks correlations again. In many linear examples its limiting path is close to the lasso/LAR path; when signs would reverse, the stagewise path is more constrained and avoids changing direction too abruptly. This connection is useful because the same stagewise idea reappears in boosting as additive model fitting.

Algorithm 5: Pathwise Coordinate Descent for the Lasso

Require: Centered response y, standardized columns x_j with $\|x_j\|_2 = 1$, decreasing penalties $\lambda_1 > \cdots > \lambda_L \geq 0$
Ensure: Lasso coefficient vectors $\widehat{\beta}(\lambda_\ell)$ on the penalty grid
Define $S(z, \lambda) = \operatorname{sign}(z)(|z| - \lambda)_+$
Initialize $\beta \leftarrow 0$ and $r \leftarrow y - X\beta$
for $\ell = 1, \ldots, L$ **do**
repeat
for $j = 1, \ldots, p$ **do**
$r \leftarrow r + x_j \beta_j$
$z_j \leftarrow x_j^T r$
$\beta_j \leftarrow S(z_j, \lambda_\ell)$
$r \leftarrow r - x_j \beta_j$
end for
until the coefficient vector has converged
$\widehat{\beta}(\lambda_\ell) \leftarrow \beta$
end for

Definition 3.2.14. *Effective Degrees of Freedom*
For a linear smoother $\widehat{y} = S_\lambda y$, where S_λ depends on the inputs and tuning parameter but not on the response values, the effective number of parameters is

$$\operatorname{df}(S_\lambda) = \operatorname{trace}(S_\lambda).$$

If S_λ is the orthogonal projection onto an m-dimensional feature space, this equals m; for shrinkage smoothers it is usually noninteger.

Remark 3.2.15. *Optimism of Training Error*
Training error is typically optimistic because each response helps produce its own fitted value. For squared-error linear smoothers under an additive noise model, the expected in-sample optimism has the form

$$\frac{2}{n}\sum_{i=1}^{n}\operatorname{Cov}(\widehat{y}_i, y_i) = \frac{2\sigma^2}{n}\operatorname{trace}(S_\lambda).$$

This is the covariance reason that parameter counts or effective degrees of freedom appear in model-selection penalties.

Method 3.2.16. *AIC, C_p, and BIC Selection*
For squared-error fitting with training error $\overline{\mathrm{err}}(\lambda) = n^{-1}\sum_i (y_i - \widehat{y}_{\lambda,i})^2$ and noise estimate $\widehat{\sigma}^2$, a C_p- or AIC-style criterion is

$$\mathrm{AIC}(\lambda) = \overline{\mathrm{err}}(\lambda) + 2\,\frac{\mathrm{df}(S_\lambda)}{n}\widehat{\sigma}^2.$$

For likelihood fits, the corresponding scaled form replaces the squared-error term by $-\frac{2}{n}\,\mathrm{loglik}(\lambda)$ and adds $2d(\lambda)/n$. BIC uses the same likelihood term with the heavier penalty $(\log n)d(\lambda)/n$, favoring simpler models as n grows.

Remark 3.2.17. *Bayesian and MDL Readings of BIC*
For training data Z and candidate likelihood models $\mathcal{M}_1, \ldots, \mathcal{M}_M$ with approximately uniform prior model probabilities, the BIC approximation gives posterior model weights of the form

$$\widehat{\mathbb{P}}(\mathcal{M}_m \mid Z) = \frac{\exp\{-\,\mathrm{BIC}_m/2\}}{\sum_{\ell=1}^{M}\exp\{-\,\mathrm{BIC}_\ell/2\}}.$$

Thus minimizing BIC is an approximate maximum-posterior model choice, not merely a stronger AIC penalty. The minimum-description-length interpretation is parallel: a model is preferred when the code length for the fitted discrepancies plus the code length for the model parameters is shortest. Both views reward fit but charge for the information needed to specify a more flexible model.

Method 3.2.18. *Generalized Cross-Validation*
For a linear smoother $\widehat{y} = S_\lambda y$, leave-one-out residuals often have the form

$$y_i - \widehat{f}^{-i}(x_i) = \frac{y_i - \widehat{f}(x_i)}{1 - S_{\lambda,ii}}.$$

Generalized cross-validation replaces the individual leverages $S_{\lambda,ii}$ by their average $\operatorname{trace}(S_\lambda)/n$:

$$\mathrm{GCV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \widehat{f}_\lambda(x_i)}{1 - \operatorname{trace}(S_\lambda)/n}\right)^2.$$

This gives a cheap approximation to leave-one-out cross-validation when the diagonal entries of the smoother matrix are inconvenient to compute one by one.

3.3 Basis Expansions, Splines, and Wavelets

Basis expansions keep the fitting step linear while moving nonlinearity into the representation of the inputs.

Definition 3.3.1. *Linear Basis Expansion*
A basis expansion replaces the raw input x by derived features $h_1(x), \ldots, h_M(x)$ and fits

$$f(x) = \sum_{m=1}^{M}\beta_m h_m(x).$$

The functions h_m may include original coordinates, polynomial and interaction terms, transformations of individual coordinates, or indicators of regions. A linear rule in the expanded variables can produce nonlinear decision boundaries in the original input space.

Remark 3.3.2. *Dictionaries Need Complexity Control*
Large dictionaries make the representation flexible but unstable unless some restriction, selection rule, or penalty controls the fitted coefficients. Additive models restrict interactions by writing $f(x) = \sum_j f_j(x_j)$; selection methods search the dictionary; regularization keeps the dictionary but shrinks the coefficients.

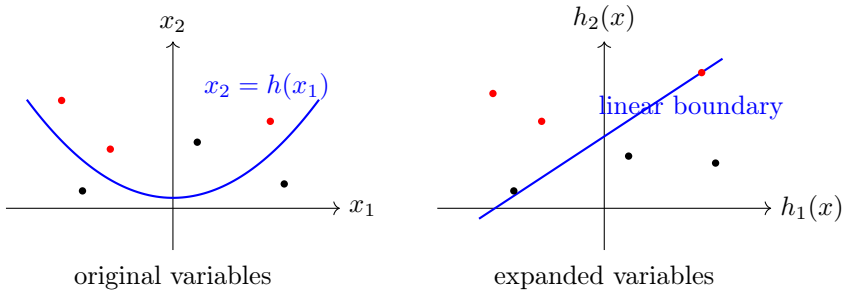


Figure 230: A linear boundary after a basis expansion can pull back to a curved boundary in the original coordinates.

Definition 3.3.3. *Regression Splines*
For one-dimensional x, an order-M spline with knots $\xi_1, \ldots, \xi_K$ is a piecewise polynomial of degree $M-1$ whose derivatives agree at the knots up to order $M-2$. A cubic regression spline can be represented by a truncated-power basis

$$1, \quad x, \quad x^2, \quad x^3, \quad (x-\xi_1)_+^3, \ldots, (x-\xi_K)_+^3.$$

A natural cubic spline adds boundary constraints that make the function linear outside the boundary knots, reducing erratic extrapolation near the edges. B-spline bases span the same spline spaces with local support, which is numerically preferable when many knots are used.

Method 3.3.4. *Smoothing Splines as Penalized Fits*
For one-dimensional regression data (x_i, y_i), a smoothing spline balances fidelity and curvature by minimizing

$$\sum_{i=1}^{n} (y_i - f(x_i))^2 + \lambda \int \{f''(t)\}^2 \, dt.$$

The solution is a natural cubic spline with knots at the observed x_i. Its fitted values can be written $\widehat{f} = S_\lambda y$ for a smoother matrix S_λ, and $\operatorname{trace}(S_\lambda)$ serves as an effective degrees of freedom. Leave-one-out cross-validation can be computed from the diagonal entries:

$$\mathrm{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \widehat{f}(x_i)}{1 - S_\lambda(i,i)} \right)^2.$$

Method 3.3.5. *Spline-Filtered Logistic Features*
For functional or highly correlated inputs $x \in \mathbb{R}^p$, choose a smooth basis matrix $H \in \mathbb{R}^{p \times M}$ and write a smooth coefficient vector as $\beta = H\theta$. Then

$$x^\top \beta = x^\top H \theta = (H^\top x)^\top \theta,$$

so the original features are replaced by filtered features $x^* = H^\top x$, and an ordinary logistic regression can be fit in the lower-dimensional coefficient θ. This trades raw flexibility for smooth, interpretable contrasts.

Definition 3.3.6. *Tensor-Product and Thin-Plate Splines*
If h_{1j} is a basis for functions of x_1 and h_{2k} is a basis for functions of x_2, then

$$g_{jk}(x_1, x_2) = h_{1j}(x_1) h_{2k}(x_2)$$

forms a tensor-product basis for two-dimensional functions. This generalizes directly but grows exponentially with dimension. A thin-plate spline instead controls a multivariate function by a roughness penalty; in two dimensions a canonical penalty is

$$J[f] = \iint_{\mathbb{R}^2} \left\{ \left(\frac{\partial^2 f}{\partial x_1^2} \right)^2 + 2 \left(\frac{\partial^2 f}{\partial x_1 \partial x_2} \right)^2 + \left(\frac{\partial^2 f}{\partial x_2^2} \right)^2 \right\} dx_1 dx_2.$$

Its finite-dimensional solution has radial basis terms centered at data or knot locations.

Method 3.3.7. *Wavelet Shrinkage*
For data observed on a regular lattice, an orthonormal wavelet matrix W transforms the response into coefficients $y^* = W^\top y$. A sparse wavelet smoother solves the orthonormal lasso problem

$$\min_\theta \|y - W\theta\|_2^2 + 2\lambda \|\theta\|_1,$$

so each coefficient is soft-thresholded:

$$\widehat{\theta}_j = \operatorname{sign}(y_j^*)(|y_j^*| - \lambda)_+, \qquad \widehat{f} = W\widehat{\theta}.$$

The basis is localized in position and scale, making shrinkage selective for isolated features while discarding small noisy coefficients.

3.4 Kernel Methods and RKHS Theory

Kernel methods keep the algorithm linear in an implicit feature space while computing only inner products.

Definition 3.4.1. *Positive Definite Kernel*
A symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is a *positive definite kernel* if for every finite set $x_1, \dots, x_n$ the Gram matrix $K_{ij} = k(x_i, x_j)$ is positive semidefinite.

Definition 3.4.2. *Reproducing Kernel Hilbert Space*
A Hilbert space $\mathcal{H}$ of functions on $\mathcal{X}$ is a *reproducing kernel Hilbert space* if point evaluation is continuous. Its kernel k satisfies

$$f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}}$$

for every $f \in \mathcal{H}$.

Theorem 3.4.3. *Representer Theorem Form*
For many objectives of the form

$$\sum_{i=1}^{n} \ell(f(x_i), y_i) + \lambda \Phi(\|f\|_{\mathcal{H}}),$$

where Φ is increasing, a minimizer in an RKHS has the finite expansion

$$f(\cdot) = \sum_{i=1}^{n} \alpha_i k(x_i, \cdot).$$

Thus an infinite-dimensional regularized problem reduces to coefficients on the observed data and the Gram matrix $K_{ij} = k(x_i, x_j)$.

Example 3.4.4. *Common Kernels* Polynomial kernels encode interaction order:

$$k(x, y) = (\langle x, y \rangle + 1)^d.$$

For example, when $p = 2$ and $d = 2$ this corresponds to the six-dimensional feature vector

$$h(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2).$$

Gaussian radial basis kernels

$$k_\nu(x, y) = \exp(-\nu \|x - y\|_2^2)$$

instead encode locality: nearby inputs have large inner product, and distant inputs have small inner product.

Remark 3.4.5. *Localization Kernels and RKHS Kernels*
The word "kernel" is used in two related but distinct ways. In local smoothing, weights $K_\lambda(x_0, x_i)$ localize a fit around the target point x_0; see Definition 6.1.5. In RKHS methods, a positive definite kernel computes inner products in an implicit feature space and usually appears inside a regularized global optimization problem.

Example 3.4.6. *Radial Basis Expansions*
A radial basis model uses localized basis functions centered at prototype points ξ_j:

$$f(x) = \sum_{j=1}^{M} \beta_j D\left(\frac{\|x - \xi_j\|}{\lambda_j} \right).$$

With a common scale λ, normalized radial bases

$$h_j(x) = \frac{D(\|x - \xi_j\| / \lambda)}{\sum_{k=1}^{M} D(\|x - \xi_k\| / \lambda)}$$

avoid regions where all unnormalized basis functions are small. The Nadaraya–Watson smoother is the special normalized expansion with one center at each training point and coefficients Y_i.

Method 3.4.7. *Kernel Ridge Regression*
With squared-error loss and RKHS norm penalty, the representer form gives

$$\widehat{f}(x) = \sum_{i=1}^{n} \widehat{\alpha}_i k(x, x_i).$$

Writing $K_{ij} = k(x_i, x_j)$, the finite criterion is

$$\min_{\alpha \in \mathbb{R}^n} (y - K\alpha)^\top (y - K\alpha) + \lambda \alpha^\top K \alpha,$$

and the fitted coefficients satisfy

$$\widehat{\alpha} = (K + \lambda I)^{-1} y$$

when the displayed system is nonsingular.

Remark 3.4.8. *Spline and Kernel Viewpoints*
Smoothing splines, thin-plate splines, radial basis expansions, kernel ridge regression, and support-vector classifiers all fit the representer pattern with different losses, kernels, and unpenalized low-dimensional components; compare Method 3.3.4 and Definition 3.3.6.

3.5 Support Vector Machines and Margin Methods

Margin methods prefer classifiers that separate classes with a buffer, not merely with zero training error. The support-vector construction makes this preference explicit by trading margin width against margin violations.

Definition 3.5.1. *Affine Hyperplane and Signed Distance*
For $w \neq 0$ and $b \in \mathbb{R}$, the affine hyperplane

$$H = \{x : w^\top x + b = 0\}$$

has normal direction w. The signed distance from x to H is $(w^\top x + b)/\|w\|_2$. Thus multiplying both w and b by the same positive constant changes the score scale but not the separating hyperplane.

Definition 3.5.2. *Margin*
For labels $y_i \in \{-1, 1\}$ and score $g(x) = w^\top x + b$, the signed margin of (x_i, y_i) is $y_i g(x_i)$. Large positive margins indicate confident correct classification.

Definition 3.5.3. *Hard-Margin Support Vector Machine*
For linearly separable data, the maximum-margin separator fixes the arbitrary score scale by requiring $y_i(w^\top x_i + b) \geq 1$ and solves

$$\min_{w,b} \frac{1}{2}\|w\|_2^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1, \qquad i = 1, \ldots, n.$$

The margin slab has width proportional to $1/\|w\|_2$, so minimizing $\|w\|_2$ maximizes the geometric margin.

Definition 3.5.4. *Soft-Margin Support Vector Machine*
For overlapping classes, the standard soft-margin SVM introduces slack variables and solves

$$\min_{w,b,\xi} \left\{ \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \xi_i \right\}$$

subject to

$$y_i(w^\top x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0, \qquad i = 1, \ldots, n.$$

The parameter $C > 0$ prices violations of the unit margin. A larger C discourages positive slack and tends to fit a narrower, less tolerant margin; a smaller C allows more observations to lie inside the margin.

Remark 3.5.5. *Why the Standard Slack Scaling*
Before fixing the score scale, one might try to maximize a geometric margin M while allowing either additive violations $y_i(w^\top x_i + b) \geq M - \xi_i$ or relative violations $y_i(w^\top x_i + b) \geq M(1 - \xi_i)$. The relative version is the one that becomes the convex formulation in Definition 3.5.4 after setting the functional margin scale to one. The slack ξ_i then measures proportional margin failure: $\xi_i = 0$ is outside or on the correct margin, $0 < \xi_i \leq 1$ is inside the margin but still correctly classified, and $\xi_i > 1$ is misclassified.

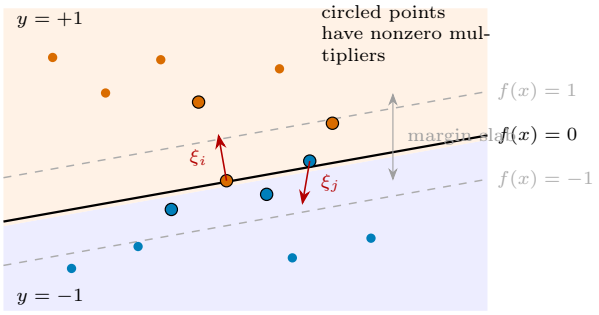


Figure 231: Soft-margin support-vector geometry. The solid line is the decision boundary, dashed lines are the unit-margin boundaries, circled observations have nonzero multipliers, and arrows show slack for points inside or across their margin.

Proposition 3.5.6. *Hinge-Loss Penalty Form*
Writing $f(x) = w^\top x + b$, Definition 3.5.4 is equivalent, up to multiplying the objective by a positive constant, to

$$\min_{w,b} \left\{ \sum_{i=1}^{n} [1 - y_i f(x_i)]_+ + \frac{\lambda}{2} \|w\|_2^2 \right\}, \qquad \lambda = \frac{1}{C}.$$

For fixed w, b, the smallest feasible slack is exactly $\xi_i = [1 - y_i f(x_i)]_+$. Thus the SVM is a regularized empirical-risk problem with hinge loss and an unpenalized intercept.

Proposition 3.5.7. *Hard-Margin Dual and Support Vectors*
The hard-margin problem has Lagrange multipliers $\alpha_i \geq 0$ satisfying $\sum_i \alpha_i y_i = 0$ and dual objective

$$\max_{\alpha} \left\{ \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j x_i^\top x_j \right\}.$$

At the solution, $w = \sum_i \alpha_i y_i x_i$. Points with $\alpha_i > 0$ lie on the margin boundary and are the support vectors.

Proposition 3.5.8. *Soft-Margin Dual and Support Types*
The soft-margin dual has the same quadratic objective as Proposition 3.5.7, but with box constraints

$$0 \leq \alpha_i \leq C, \qquad \sum_{i=1}^{n} \alpha_i y_i = 0.$$

At the solution,

$$\widehat{w} = \sum_{i=1}^{n} \widehat{\alpha}_i y_i x_i.$$

The Karush–Kuhn–Tucker conditions separate the training points into geometric roles:

$$\widehat{\alpha}_i = 0 \Rightarrow y_i \widehat{f}(x_i) \geq 1, \qquad 0 < \widehat{\alpha}_i < C \Rightarrow y_i \widehat{f}(x_i) = 1.$$

Points with $\widehat{\alpha}_i = C$ may lie on the margin or inside it, including on the wrong side of the decision boundary. The nonzero coefficients are precisely the support vectors.

Method 3.5.9. *Kernelized Support Vector Classifier*
Let $h(x)$ be a feature map and let $k(x, x') = \langle h(x), h(x') \rangle$ be its positive definite kernel. The SVM dual depends on the data only through inner products, so the kernelized dual replaces $x_i^\top x_j$ by $k(x_i, x_j)$:

$$\max_{\alpha} \left\{ \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j k(x_i, x_j) \right\}$$

subject to $0 \leq \alpha_i \leq C$ and $\sum_i \alpha_i y_i = 0$. The fitted score has the finite form

$$\widehat{f}(x) = \sum_{i=1}^{n} \widehat{\alpha}_i y_i k(x, x_i) + \widehat{b}, \qquad \widehat{G}(x) = \operatorname{sign}(\widehat{f}(x)).$$

The intercept can be obtained from any margin support vector with $0 < \widehat{\alpha}_i < C$, or averaged over such points for numerical stability. Common polynomial and radial choices are listed in Example 3.4.4; the finite expansion is a concrete instance of the representer pattern in Theorem 3.4.3.

Remark 3.5.10. *Hinge Loss and Probability-Oriented Losses*
The hinge loss estimates a decision rule directly: its population target is the sign of the class-posterior imbalance. Logistic deviance instead estimates log posterior odds, while squared-error losses estimate linear transformations of posterior probabilities. Geometrically, hinge loss ignores correctly classified observations beyond the margin, logistic deviance keeps a smooth tail, and squared error can continue to give leverage to points well inside their own class region.

Method 3.5.11. *SVM Regularization Path View*
In the hinge-loss form, decreasing $\lambda = 1/C$ narrows the margin and makes the fit less tolerant of margin violations. Along a linear SVM path, training points move among three states: outside the margin with $\alpha_i = 0$, on the margin with $0 \leq \alpha_i \leq 1$ in a normalized path parametrization, and inside the margin with saturated multiplier. Between changes of state, only the margin points need varying coefficients, so the coefficient paths are piecewise linear. The same path idea applies after kernelization, with $w^\top x_i$ replaced by kernel evaluations.

Method 3.5.12. *Epsilon-Insensitive Support Vector Regression*
For a quantitative response, support-vector regression fits $f(x) = w^\top x + b$ by replacing classification hinge loss with the ϵ-insensitive error

$$V_\epsilon(r) = \begin{cases} 0, & |r| < \epsilon, \\ |r| - \epsilon, & |r| \geq \epsilon. \end{cases}$$

The penalized criterion is

$$\min_{w,b}\left\{\sum_{i=1}^{n} V_\epsilon\big(y_i - f(x_i)\big) + \frac{\lambda}{2}\|w\|_2^2\right\}.$$

Residuals inside the ϵ-tube make no contribution, while residuals outside the tube receive linear loss. In kernel form, the fitted function has coefficients of opposite signs for upper- and lower-tube violations,

$$\widehat{f}(x) = \sum_{i=1}^{n}(\widehat{\alpha}_i^* - \widehat{\alpha}_i)k(x, x_i) + \widehat{b}.$$

Method 3.5.13. *Perceptron Update as a Contrast*
The perceptron searches for some separating hyperplane by updating only on misclassified points. If $y_i(w^\top x_i + b) \le 0$, then with learning rate $\rho > 0$,

$$(w, b) \leftarrow (w, b) + \rho\, y_i(x_i, 1).$$

On linearly separable data this can find a separator, but it does not choose the maximum-margin separator and depends on initialization and data order. The SVM criterion replaces this order-dependent search by a convex margin optimization problem.

Remark 3.5.14. *Support Vectors*
Only examples on or inside the margin determine the fitted separating hyperplane. In kernel form, these examples are the nonzero coefficients in the representer expansion, so prediction cost is controlled by the number of support vectors rather than the full size of the feature expansion.

Remark 3.5.15. *Margins and Generalization*
Margin bounds refine raw VC counting by using the geometry of the fitted classifier. The effective complexity depends on both norm control and the achieved margin.

3.6 Trees and Ensemble Averaging

Tree-based methods partition the input space into simple regions and fit a local rule in each region. Ensemble averaging methods keep this local modeling idea but stabilize it by resampling, voting, and decorrelating the individual trees. Sequential boosting of trees is developed separately in Subsection 3.8.

Definition 3.6.1. *Decision Tree*
A decision tree recursively splits the input space by simple tests, typically of the form $x_j \le t$. Each terminal node stores a local prediction, such as an average response for regression or a class-probability vector for classification.

Definition 3.6.2. *Bagging and Random Forests*
Bootstrap aggregating, or *bagging*, trains predictors on bootstrap resamples of the training set and averages their outputs. A *random forest* adds random feature selection at tree splits, reducing correlation among the trees before averaging.

Algorithm 6: Random Forest

Require: Training data $(x_i, y_i)_{i=1}^n$, number of trees B, candidate split count $m \le p$, tree stopping rule
Ensure: Trees $T_1, \dots, T_B$ and an aggregated predictor
  for $b = 1, \dots, B$ **do**
    Draw a bootstrap resample S^{*b} of size n
    Grow a tree T_b on S^{*b}
    for each nonterminal node while growing T_b **do**
      Select m of the p predictors at random
      Choose the best split using only those m predictors
      Split the node and continue until the stopping rule is met
    end for
  end for
  For regression, return $\widehat{f}_{\mathrm{rf}}(x) = B^{-1}\sum_{b=1}^{B} T_b(x)$
  For classification, return a majority vote or averaged class-probability rule from $T_1(x), \dots, T_B(x)$

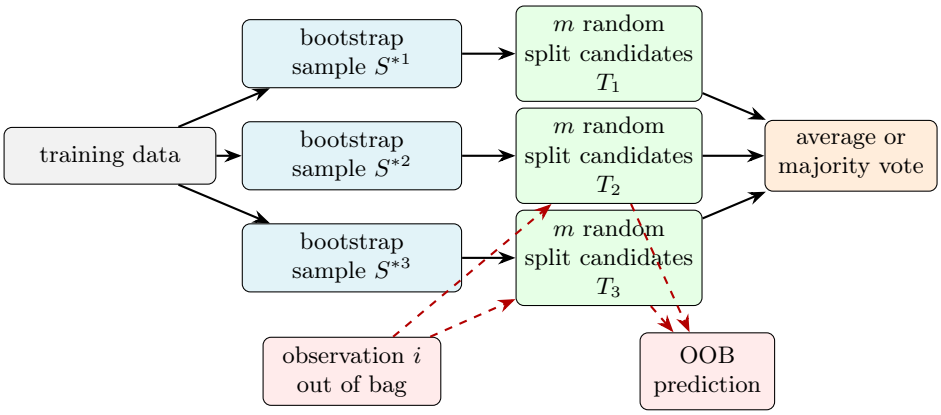


Figure 232: A random forest combines bootstrap resampling with random split-variable selection. Out-of-bag predictions use only trees whose bootstrap samples omitted the target observation.

Method 3.6.3. *CART-Style Regression Trees*
For a regression tree with terminal regions $R_1, \dots, R_M$, use the constant-leaf model

$$\widehat{f}(x) = \sum_{m=1}^{M} \widehat{c}_m \mathbf{1}_{\{x \in R_m\}}, \qquad \widehat{c}_m = \frac{1}{N_m} \sum_{x_i \in R_m} y_i.$$

A greedy binary split searches over coordinates j and thresholds s, compares the two child regions

$$R_L(j,s) = \{x : x_j \le s\}, \qquad R_R(j,s) = \{x : x_j > s\},$$

and chooses the split with the smallest within-child squared error after refitting the two child means. The usual strategy grows a large tree T_0 and then prunes. If $T \subseteq T_0$ is a subtree, $|T|$ is its number of terminal nodes, and

$$Q_m(T) = \frac{1}{N_m} \sum_{x_i \in R_m} (y_i - \widehat{c}_m)^2,$$

then cost-complexity pruning uses

$$C_\alpha(T) = \sum_{m=1}^{|T|} N_m Q_m(T) + \alpha |T|.$$

The parameter $\alpha \ge 0$ is selected by validation or cross-validation, often with the one-standard-error rule from Method 3.2.6. This separates tree growth, which is deliberately aggressive, from tree selection, which controls variance.

Definition 3.6.4. *Classification Tree Impurity*
In a classification tree, let $\widehat{p}_{mk}$ be the empirical class-k proportion in node m, and let $k(m) = \arg\max_k \widehat{p}_{mk}$. Common node impurity scores are

$$\text{misclassification error} = 1 - \widehat{p}_{m,k(m)},$$

$$\text{Gini index} = \sum_{k=1}^{K} \widehat{p}_{mk}(1 - \widehat{p}_{mk}), \qquad \text{cross-entropy} = -\sum_{k=1}^{K} \widehat{p}_{mk} \log \widehat{p}_{mk}.$$

For growing a tree, Gini and cross-entropy are often more sensitive to changes in class proportions than misclassification error; for pruning, misclassification error is also commonly used. Unequal losses can be handled by changing the terminal-node decision rule or by weighting observations during tree growth. If $L_{kk} = 0$ and $L_{k\ell}$ is the loss for predicting ℓ when the true class is k, the terminal-node empirical Bayes rule predicts a class minimizing $\sum_k L_{k\ell} \widehat{p}_{mk}$.

Remark 3.6.5. *Tree Practicalities*
For unordered categorical predictors, binary partitions of the levels can be searched directly when the number of levels is small; for binary responses, ordering levels by the class-one proportion reduces the search to ordered splits. Missing predictor values may be handled by a missing category for categorical inputs or by surrogate splits that mimic the selected primary split. These conveniences are part of the practical appeal of trees, but high-cardinality categorical inputs and repeated split searches can also encourage overfitting.

Method 3.6.6. *Bagged Prediction and Votes*
Let S^{*b}, $b = 1, \dots, B$, be bootstrap resamples of the training set, and let $\widehat{f}^{*b}$ be the predictor trained on S^{*b}.

The bagged regression estimate is

$$\widehat{f}_{\mathrm{bag}}(x) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}^{*b}(x).$$

For classification, one may average indicator votes or average estimated class probabilities; averaging probabilities usually preserves more information, especially when the base learner already estimates node probabilities. Bagging is most useful for unstable learners such as large trees, because averaging reduces variance. Random forests add extra randomization at split selection to reduce correlation among the bootstrap trees before averaging.

Proposition 3.6.7. *Squared-Error Benefit of Averaging*
Fix an input value x and view a resampled predictor $\widehat{f}^*(x)$ as random through the resampling or training perturbation, independently of a future response Y at the same x. Let

$$f_{\mathrm{ag}}(x) = \mathbb{E}_*\{\widehat{f}^*(x)\},$$

where $\mathbb{E}_*$ averages over the perturbation. Then

$$\mathbb{E}_{*,Y}\left[(Y - \widehat{f}^*(x))^2\right] = \mathbb{E}_Y\left[(Y - f_{\mathrm{ag}}(x))^2\right] + \mathbb{E}_*\left[(\widehat{f}^*(x) - f_{\mathrm{ag}}(x))^2\right].$$

Thus ideal aggregation removes the perturbation variance at the fixed input. Finite bagging approximates f_{ag} by Monte Carlo averaging over bootstrap fits, so it is most useful when the base learner is unstable.

Proposition 3.6.8. *Correlated Ensemble Variance*
Let $Z_1, \dots, Z_B$ have common variance σ^2 and common pairwise correlation ρ. Then

$$\mathrm{Var}\left(\frac{1}{B}\sum_{b=1}^{B} Z_b\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Thus increasing B removes only the independent part of the variation. The split-level feature restriction in Algorithm 6 is useful because it tries to reduce ρ among tree predictions without making each individual tree too weak.

Method 3.6.9. *Out-of-Bag Error and Forest Importance*
For an observation (x_i, y_i), its *out-of-bag* prediction averages only those trees whose bootstrap samples did not contain that observation. The resulting OOB error estimates prediction error while the forest is being grown, so it can be used to monitor when adding more trees has stopped materially changing the fitted forest.

Two common importance summaries are local to the fitted forest. Split-improvement importance accumulates the improvement in the tree splitting criterion over all splits using a variable. Permutation importance sends OOB observations through a tree, records predictive accuracy, randomly permutes one variable in those OOB observations, and records the resulting loss in accuracy. The latter measures how much the fitted forest relied on that variable in its present context; it is not the same as refitting the model without that variable.

Remark 3.6.10. *Proximities, Tuning, and Overfitting*
A random forest can define a proximity matrix by counting how often two training observations land in the same terminal node, often using OOB trees. Low-dimensional displays of this matrix can be suggestive, but their geometry reflects the forest's terminal-node partitioning rather than an intrinsic data metric.

The split count m, terminal-node size, and number of trees B play different roles. Smaller m tends to decorrelate trees but can increase bias if relevant variables are rarely proposed at splits. Increasing B does not make the finite average chase the training data in the same way as repeatedly enlarging a single model, but the limiting fully grown forest can still be too variable or too rich for a problem. Defaults such as $m \approx \sqrt{p}$ for classification and $m \approx p/3$ for regression are starting points rather than guarantees.

Method 3.6.11. *Error-Correcting Output Codes*
For a K-class problem, choose a binary coding matrix $C \in \{0,1\}^{K\times L}$. Column ℓ defines a two-class problem by merging the original classes according to the entries of $C_{\cdot\ell}$; fit one probabilistic classifier $\widehat{p}_\ell(x)$ for each column. A test point is decoded by comparing the predicted code to each row, for example by

$$\delta_k(x) = \sum_{\ell=1}^{L} |C_{k\ell} - \widehat{p}_\ell(x)|, \qquad \widehat{k}(x) \in \arg\min_k \delta_k(x).$$

Redundant code columns can reduce variance and tolerate some column-level classification errors, much like other ensemble averages.

Method 3.6.12. *Stacking and Bumping*
Given fitted candidates $\widehat{f}_1, \dots, \widehat{f}_M$, stacking estimates weights from out-of-sample predictions rather than from

the same fitted values used to train the candidates. With leave-one-out predictions $\widehat{f}_m^{-i}(x_i)$, a squared-error stacking fit solves

$$\widehat{w}^{\mathrm{st}} \in \arg\min_{w} \sum_{i=1}^{n} \left(y_i - \sum_{m=1}^{M} w_m \widehat{f}_m^{-i}(x_i) \right)^2,$$

often with constraints $w_m \geq 0$ and $\sum_m w_m = 1$. The final predictor is $\sum_m \widehat{w}_m^{\mathrm{st}} \widehat{f}_m(x)$. Restricting w to the coordinate vectors recovers model choice by cross-validated error; allowing general weights gives model averaging. Bumping uses bootstrap resamples differently: train several bootstrap predictors, evaluate each on the original training set, and keep the single bootstrap-trained model with the best original-sample criterion. Thus stacking averages candidates, while bumping uses resampling as a stochastic search over fitted models.

3.7 Additive Models and Adaptive Basis Methods

Additive and adaptive basis methods sit between rigid linear models and fully local learners. They keep prediction interpretable by building the fitted function from one-dimensional effects, boxes, hinges, or soft routing weights.

Definition 3.7.1. *Generalized Additive Model*
For predictors $X = (X_1, \ldots, X_p)$ and conditional mean $\mu(x) = \mathbb{E}[Y \mid X = x]$, an additive regression model has

$$\mu(x) = \alpha + \sum_{j=1}^{p} f_j(x_j).$$

More generally, a generalized additive model uses a link function g and writes

$$g(\mu(x)) = \alpha + \sum_{j=1}^{p} f_j(x_j).$$

The identity, logit, probit, and log links give Gaussian, binomial, and count-data variants analogous to generalized linear models. To identify the intercept separately from the component functions, one usually imposes centering constraints such as $\sum_i f_j(x_{ij}) = 0$ for each j.

Algorithm 7: Backfitting for an Additive Regression Model

Require: Training data $(x_i, y_i)_{i=1}^n$, univariate smoothers S_j for predictors $j = 1, \ldots, p$
Ensure: Centered component fits $\widehat{f}_1, \ldots, \widehat{f}_p$ and intercept $\widehat{\alpha}$
  Set $\widehat{\alpha} \leftarrow n^{-1} \sum_{i=1}^n y_i$ and initialize $\widehat{f}_j(x_{ij}) \leftarrow 0$
  repeat
    for $j = 1, \ldots, p$ **do**
      $r_{ij} \leftarrow y_i - \widehat{\alpha} - \sum_{k \neq j} \widehat{f}_k(x_{ik})$ for $i = 1, \ldots, n$
      $\widehat{f}_j \leftarrow S_j(r_{1j}, \ldots, r_{nj})$
      Center $\widehat{f}_j$ by subtracting $n^{-1} \sum_i \widehat{f}_j(x_{ij})$
    end for
  until the component functions change by less than the chosen tolerance

Method 3.7.2. *Local Scoring for Additive Logistic Regression*
For binary responses, write

$$\eta_i = \alpha + \sum_{j=1}^{p} f_j(x_{ij}), \qquad p_i = \frac{1}{1 + \exp(-\eta_i)}.$$

An IRLS step constructs the working response and weights

$$z_i = \eta_i + \frac{y_i - p_i}{p_i(1 - p_i)}, \qquad w_i = p_i(1 - p_i),$$

then fits an additive model to z_i using weighted backfitting with weights w_i. Iterating these weighted additive fits gives the generalized additive analogue of logistic IRLS from Method 3.1.12.

Method 3.7.3. *MARS: Adaptive Piecewise-Linear Basis Search*
Multivariate adaptive regression splines use reflected hinge pairs

$$(x_j - t)_+, \qquad (t - x_j)_+,$$

with knots t drawn from observed predictor values. The fitted model is linear in selected basis functions,

$$\widehat{f}(x) = \widehat{\beta}_0 + \sum_{m=1}^{M} \widehat{\beta}_m h_m(x),$$

where each h_m is a hinge function or a product of previously selected hinge functions. A forward pass adds the candidate pair or product that most reduces residual sum of squares; a backward pass deletes terms to control overfitting. A common deletion score is generalized cross-validation,

$$\mathrm{GCV}(\lambda) = \frac{\sum_{i=1}^{n}(y_i - \widehat{f}_\lambda(x_i))^2}{\{1 - M(\lambda)/n\}^2},$$

where $M(\lambda)$ is an effective model size along the backward sequence. Limiting the maximum product order gives an interaction cap; order one gives an additive piecewise-linear model.

Method 3.7.4. *PRIM Box Peeling*
The patient rule induction method searches for rectangular boxes whose response average is high. Starting with a box containing all training points, it repeatedly peels away a small fraction of observations from one face, such as those with the largest or smallest values of a chosen predictor, and keeps the peel that maximizes the remaining box mean. After a minimum box size is reached, a pasting step may expand faces when doing so improves the box mean. Cross-validation and subject-matter judgment choose a box along the sequence; the selected observations can then be removed and the procedure repeated to find additional boxes.

Algorithm 8: Patient Rule Induction Method

Require: Training data, response values, peeling fraction α, minimum box size
Ensure: A sequence of high-response boxes
Start with all training observations in a maximal box
repeat
Consider each predictor and each face of the current box
Peel away an α fraction of observations from one face
Keep the peel whose remaining box has the largest response mean
until the box reaches the minimum allowed size
Paste back along any face when the expansion increases the box mean
Use validation or subject-matter judgment to choose one box from the sequence
Remove the selected box and repeat the peeling-pasting search if more boxes are desired

Remark 3.7.5. *Adaptive Structure Versus Tree Structure*
CART represents a partition by a binary tree, which is interpretable but can make additive structure hard to discover. PRIM keeps box rules but drops the requirement that the boxes arise from a binary tree. MARS drops box indicators in favor of hinge functions and products, gaining smoother local fits while retaining a linear-in-coefficients final model.

Remark 3.7.6. *Soft Trees and Mixtures of Experts*
Hierarchical mixtures of experts replace hard tree branches by gating probabilities. A node may route an input by a softmax rule

$$g_j(x;\gamma) = \frac{\exp(\gamma_j^\top x)}{\sum_{\ell=1}^{K}\exp(\gamma_\ell^\top x)}.$$

Terminal experts supply local regression or classification models, and the final prediction is a mixture of expert predictions with input-dependent weights. This keeps the tree-like decomposition but turns split fitting into a smooth likelihood problem, commonly handled with an EM-type procedure; see Method 4.1.18.

Remark 3.7.7. *Missing Predictors*
Structured learners need an explicit missing-data convention. Trees can use surrogate splits or, for categorical predictors, a missing category. Additive and basis methods often require imputation before fitting, and model-based imputation can be treated as a separate prediction problem for each missing feature. When imputation uncertainty matters, fitting the response model across multiple imputed data sets gives a more honest assessment than treating imputed values as observed.

3.8 Boosting and Additive Trees

Boosting is a stagewise way to build a strong predictor from simple base learners. The key object is an additive score: later terms are trained against the current fit rather than fit independently.

Definition 3.8.1. *Boosting*
Boosting builds an additive predictor

$$F_M(x) = \sum_{m=1}^{M} \alpha_m h_m(x)$$

by fitting each weak learner h_m to the residuals, gradients, or weighted examples emphasized by the current ensemble. In classification the final prediction is usually obtained from the sign or largest component of the additive score.

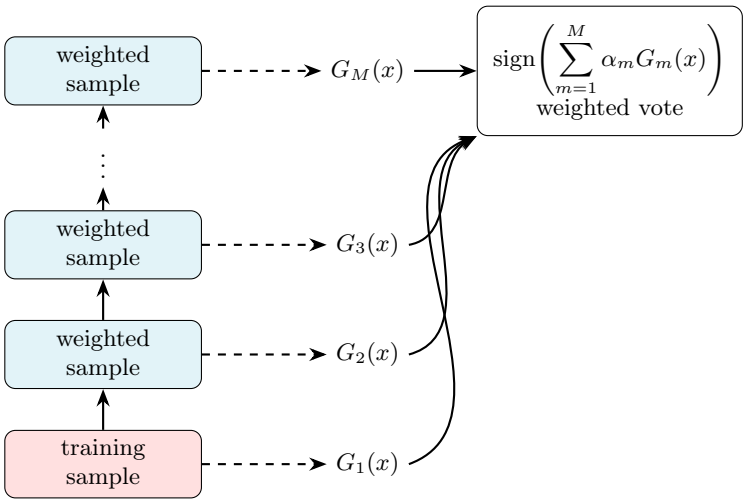


Figure 233: AdaBoost repeatedly trains a base classifier on reweighted observations, then combines the classifiers by a weighted vote.

Method 3.8.2. *Forward Stagewise Additive Modeling*
Let $b(x;\gamma)$ range over a class of simple basis functions, and let $L(y, f(x))$ be a pointwise loss. A forward stagewise additive fit starts with f_0 and, at step m, solves the one-term problem

$$(\beta_m, \gamma_m) \in \arg\min_{\beta,\gamma} \sum_{i=1}^{n} L\{y_i, f_{m-1}(x_i) + \beta b(x_i;\gamma)\}.$$

It then updates

$$f_m(x) = f_{m-1}(x) + \beta_m b(x;\gamma_m).$$

Previously added terms are not refit. For squared-error loss, the new term is fit to the current residuals.

Remark 3.8.3. *Stagewise Paths and Lasso Geometry*
If $\mathcal{T} = \{T_1, \dots, T_K\}$ is a large dictionary of possible tree basis functions, a formal additive-tree model has the linear expansion

$$f(x) = \sum_{k=1}^{K} \alpha_k T_k(x).$$

Direct least squares over such a dictionary is ill posed when K is large, so one can view boosting with small steps as a forward-stagewise approximation to a penalized fit

$$\min_{\alpha} \left\{ \sum_{i=1}^{n} \left(y_i - \sum_{k=1}^{K} \alpha_k T_k(x_i) \right)^2 + \lambda J(\alpha) \right\}.$$

The lasso choice $J(\alpha) = \sum_k |\alpha_k|$ favors sparse active dictionaries and connects this view to the path algorithms in Method 3.2.11. Shrinkage in tree boosting plays the role of taking smaller stagewise moves along this path.

Algorithm 9: AdaBoost.M1

Require: Binary training data (x_i, y_i) with $y_i \in \{-1, 1\}$, weighted classification learner, iterations M
Ensure: Classifier $G(x)$
Initialize $w_i \leftarrow 1/n$ for $i = 1, \dots, n$
for $m = 1, \dots, M$ **do**
Fit a classifier G_m using observation weights w_i
$\mathrm{err}_m \leftarrow \dfrac{\sum_{i=1}^{n} w_i \mathbf{1}_{\{y_i \neq G_m(x_i)\}}}{\sum_{i=1}^{n} w_i}$
$\alpha_m \leftarrow \log\left(\dfrac{1 - \mathrm{err}_m}{\mathrm{err}_m}\right)$
for $i = 1, \dots, n$ **do**
$w_i \leftarrow w_i \exp\{\alpha_m \mathbf{1}_{\{y_i \neq G_m(x_i)\}}\}$

end for
end for
return $G(x) = \operatorname{sign}\left(\sum_{m=1}^{M} \alpha_m G_m(x)\right)$

Proposition 3.8.4. *AdaBoost as Exponential-Loss Stagewise Fitting*
For binary labels $y_i \in \{-1, 1\}$, consider the exponential loss

$$L(y, f(x)) = \exp\{-y f(x)\}.$$

At stage m, set $w_i^{(m)} = \exp\{-y_i f_{m-1}(x_i)\}$. If the next basis function is a classifier $G(x) \in \{-1, 1\}$, then for any fixed $\beta > 0$ the stagewise objective is minimized over G by minimizing the weighted misclassification error

$$\sum_{i=1}^{n} w_i^{(m)} \mathbf{1}_{\{y_i \neq G(x_i)\}}.$$

For the selected classifier G_m, the optimal coefficient is

$$\beta_m = \frac{1}{2} \log\left(\frac{1 - \mathrm{err}_m}{\mathrm{err}_m}\right), \qquad \mathrm{err}_m = \frac{\sum_{i=1}^{n} w_i^{(m)} \mathbf{1}_{\{y_i \neq G_m(x_i)\}}}{\sum_{i=1}^{n} w_i^{(m)}}.$$

The induced update

$$w_i^{(m+1)} = w_i^{(m)} \exp\{-\beta_m y_i G_m(x_i)\}$$

is equivalent, up to a common multiplicative factor, to Algorithm 9 with $\alpha_m = 2\beta_m$.

Remark 3.8.5. *Exponential Loss and Class Probabilities*
The population minimizer of $\mathbb{E}\{\exp(-Y f(X)) \mid X = x\}$ is

$$f^*(x) = \frac{1}{2} \log \frac{\mathbb{P}(Y = 1 \mid X = x)}{\mathbb{P}(Y = -1 \mid X = x)}.$$

Thus the sign of the fitted additive score estimates the Bayes classification rule. The binomial deviance

$$\log\{1 + \exp(-2Y f(X))\}$$

has the same population minimizer, but on finite samples it penalizes very negative margins less aggressively than the exponential loss. This connects the boosting view here with the margin surrogates in Example 1.3.2.

Definition 3.8.6. *Additive Tree Expansion*
A tree with terminal regions $R_1, \ldots, R_J$ can be written as

$$T(x; \Theta) = \sum_{j=1}^{J} \gamma_j \mathbf{1}_{\{x \in R_j\}}, \qquad \Theta = \{(R_j, \gamma_j) : 1 \leq j \leq J\}.$$

A boosted tree model is a stagewise sum

$$f_M(x) = \sum_{m=1}^{M} T(x; \Theta_m).$$

Given the regions of a new tree, the terminal constants are one-dimensional location fits,

$$\widehat{\gamma}_{jm} \in \arg\min_{\gamma} \sum_{x_i \in R_{jm}} L\{y_i, f_{m-1}(x_i) + \gamma\}.$$

Algorithm 10: Gradient Tree Boosting

Require: Training data (x_i, y_i), differentiable loss $L(y, f)$, tree sizes J_m, learning rate $0 < \nu \leq 1$, iterations M
Ensure: Additive predictor f_M

$f_0(x) \leftarrow \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma)$
for $m = 1, \ldots, M$ **do**
  for $i = 1, \ldots, n$ **do**
    $r_{im} \leftarrow -\left[\dfrac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f = f_{m-1}}$
  end for
  Fit a regression tree to targets r_{im}, giving terminal regions R_{jm}, $j = 1, \ldots, J_m$
  for $j = 1, \ldots, J_m$ **do**
    $\gamma_{jm} \leftarrow \arg\min_{\gamma} \sum_{x_i \in R_{jm}} L\{y_i, f_{m-1}(x_i) + \gamma\}$

end for

$$f_m(x) \leftarrow f_{m-1}(x) + \nu \sum_{j=1}^{J_m} \gamma_{jm} \mathbf{1}_{\{x \in R_{jm}\}}$$

end for
return f_M

Remark 3.8.7. *Pseudo-Residuals*
Gradient boosting replaces the difficult direct stagewise search by fitting each tree to negative gradient values at the training points. For squared-error loss, these pseudo-residuals are ordinary residuals. For absolute-error loss they are signs of residuals, and for multinomial deviance they have class components of the form $\mathbf{1}_{\{y_i = \mathcal{G}_k\}} - p_k(x_i)$, where p_k is the current softmax probability.

Algorithm 11: Gradient Boosting for K-Class Classification

Require: Encoded labels $y_{ik} \in \{0, 1\}$, classes $k = 1, \dots, K$, tree sizes J_m, iterations M
Ensure: Class-score functions $\widehat{f}_1, \dots, \widehat{f}_K$
Initialize $f_{k0}(x) \leftarrow 0$ for $k = 1, \dots, K$
for $m = 1, \dots, M$ **do**
$$p_k(x) \leftarrow \frac{\exp(f_{k,m-1}(x))}{\sum_{\ell=1}^{K} \exp(f_{\ell,m-1}(x))} \text{ for } k = 1, \dots, K$$
for $k = 1, \dots, K$ **do**
Compute $r_{ikm} \leftarrow y_{ik} - p_k(x_i)$ for $i = 1, \dots, n$
Fit a regression tree to targets r_{ikm}, giving terminal regions R_{jkm}, $j = 1, \dots, J_m$
for $j = 1, \dots, J_m$ **do**
$$\gamma_{jkm} \leftarrow \frac{K-1}{K} \frac{\sum_{x_i \in R_{jkm}} r_{ikm}}{\sum_{x_i \in R_{jkm}} |r_{ikm}|(1 - |r_{ikm}|)}$$
end for
$$f_{km}(x) \leftarrow f_{k,m-1}(x) + \sum_{j=1}^{J_m} \gamma_{jkm} \mathbf{1}_{\{x \in R_{jkm}\}}$$
end for
end for
return $\widehat{f}_k(x) = f_{kM}(x)$ for $k = 1, \dots, K$

Remark 3.8.8. *Tree Size, Shrinkage, and Subsampling*
The number of boosting iterations M, the terminal-node count J_m, and the shrinkage factor ν all control model complexity. Restricting all trees to a common small size J makes J a global tuning parameter: stumps ($J = 2$) give additive main effects, while larger trees allow interactions of order up to $J - 1$. Shrinkage replaces a full step by the smaller update in Algorithm 10, usually requiring more iterations but often improving test risk. Stochastic gradient boosting additionally fits each new tree on a subsample of the training observations, reducing computation and adding averaging-like randomness.

Method 3.8.9. *Relative Importance and Partial Dependence*
For a single tree, a predictor's squared relative importance is the sum of the split-improvement scores over internal nodes where that predictor is used. For an additive tree expansion, average the same quantity over the fitted trees and then rescale if a relative display is desired.

For a fitted predictor f and a subset of coordinates S with complement C, the partial dependence function is the marginal average

$$f_S(x_S) = \mathbb{E}_{X_C}\{f(x_S, X_C)\},$$

estimated empirically by

$$\widehat{f}_S(x_S) = \frac{1}{n} \sum_{i=1}^{n} f(x_S, x_{iC}).$$

This describes the effect of X_S after averaging over the observed distribution of the other coordinates; it is not the same as the conditional expectation $\mathbb{E}\{f(X) \mid X_S = x_S\}$ unless the dependence structure makes those averages coincide.

Principle 3.8.10. *Bet on Sparsity*
In a very large dictionary, an ℓ^1 penalty is attractive when a good predictor uses relatively few nonzero basis coefficients. Dense problems remain difficult because there may be too little data to estimate many small effects accurately; sparse problems at least give the lasso-style search a chance to isolate a small active set. The useful notion of sparsity is relative to the sample size, noise level, and dictionary being searched.

Method 3.8.11. *Lasso Post-Processing of Ensembles*
An ensemble can be used first as a dictionary generator and only second as a predictor. Suppose a fitting procedure produces basis functions

$$\mathcal{T}_L = \{T_1, \dots, T_M\}.$$

Post-processing fits a regularized path in this fixed dictionary, for instance

$$\widehat{\alpha}(\lambda) \in \arg\min_{\alpha} \left\{ \sum_{i=1}^{n} L\left(y_i, \alpha_0 + \sum_{m=1}^{M} \alpha_m T_m(x_i) \right) + \lambda \sum_{m=1}^{M} |\alpha_m| \right\}.$$

The resulting model can use far fewer trees than the original ensemble. A good generated dictionary should contain basis functions that are relevant but not nearly identical; subsampling observations while growing trees is one way to broaden the dictionary.

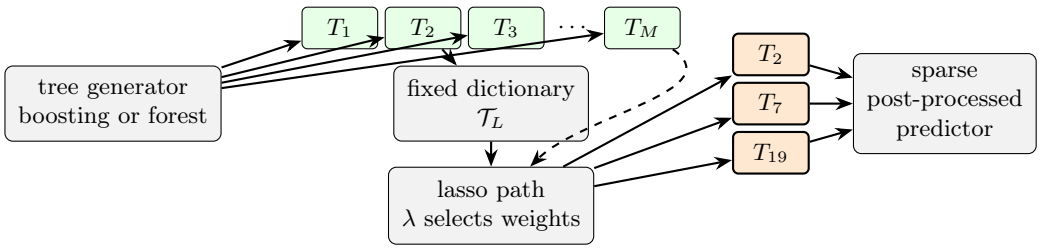


Figure 234: Lasso post-processing treats a large generated ensemble as a fixed dictionary, then selects a smaller weighted subset of basis functions.

Method 3.8.12. *Rule Ensembles*
Every path from the root of a tree to one of its nodes defines a rule, namely an indicator of the conjunction of split conditions along that path. Replacing each tree by the collection of its induced rules gives an enlarged dictionary

$$\mathcal{T}_{\text{rule}} = \bigcup_{m=1}^{M} \mathcal{T}_{\text{rule}}^{m},$$

which can then be fit by lasso or another regularized linear method. Adding the original variables as linear terms lets the rule dictionary capture simple linear effects as well as interactions represented by tree paths.

3.9 High-Dimensional Supervised Learning

High-dimensional supervised learning studies prediction problems in which the number of candidate features is comparable to, or much larger than, the number of observations. The central difficulty is not merely computational size: many rules can fit the training data too well, while the scientific question often asks which features or feature groups are driving the prediction.

Principle 3.9.1. *Less Fitting in the $p \gg n$ Regime*
When $p \gg n$, many flexible procedures can interpolate or nearly interpolate the training data. Full least squares is nonunique if the design rank is below p, and full-covariance discriminant rules rely on singular or unstable covariance estimates. The useful procedures therefore fit less than the data allow: they use strong shrinkage, diagonal covariance structure, low-dimensional row-space reductions, sparse penalties, or screening before fitting. Effective degrees of freedom, as in Proposition 3.2.4, are often more interpretable than the raw value of a tuning parameter. The main tradeoff is between prediction error and interpretability: ridge-like shrinkage stabilizes all coordinates, while sparse or thresholded rules identify a smaller set of features.

Method 3.9.2. *Diagonal LDA and Nearest Shrunken Centroids*
For class k, let $\bar{x}_{kj}$ be the sample mean of feature j, let s_j be the pooled within-class standard deviation, and let π_k be the class prior. Diagonal-covariance LDA scores a test point x by

$$\delta_k(x) = -\sum_{j=1}^{p} \frac{(x_j - \bar{x}_{kj})^2}{s_j^2} + 2\log \pi_k, \qquad \widehat{G}(x) = \arg\max_k \delta_k(x).$$

This is nearest-centroid classification after coordinatewise standardization. Nearest shrunken centroids further shrink class means toward the overall mean. Let $\bar{x}_{\cdot j}$ be the overall mean, $m_k^2 = 1/n_k - 1/n$, and let $s_0 > 0$ stabilize small standard deviations. Define

$$d_{kj} = \frac{\bar{x}_{kj} - \bar{x}_{\cdot j}}{m_k(s_j + s_0)}.$$

For a threshold $\Delta \geq 0$, apply soft thresholding

$$d'_{kj} = \operatorname{sign}(d_{kj})(|d_{kj}| - \Delta)_+, \qquad \bar{x}'_{kj} = \bar{x}_{\cdot j} + m_k(s_j + s_0)d'_{kj}.$$

The diagonal LDA rule is then evaluated with $\bar{x}'_{kj}$ in place of $\bar{x}_{kj}$. A feature disappears from the classifier if $d'_{kj} = 0$ for every class.

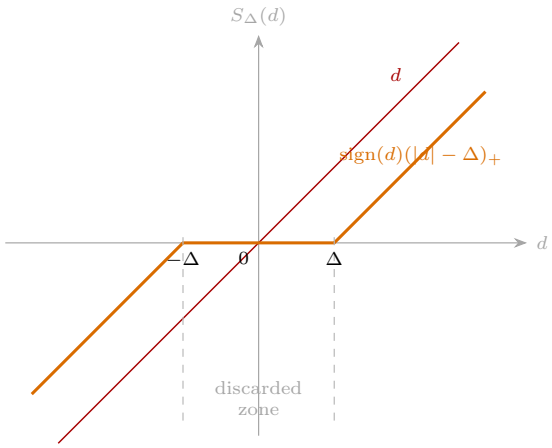


Figure 235: Soft thresholding for nearest shrunken centroids and lasso-type estimators. Contrasts with magnitude at most Δ are set to zero, while larger contrasts are pulled toward zero by Δ.

Method 3.9.3. *Quadratic Regularization for Linear Rules*
Quadratic penalties stabilize high-dimensional linear classifiers without producing sparsity. For the symmetric multinomial model

$$\mathbb{P}(G = k \mid X = x) = \frac{\exp(\beta_{k0} + x^T\beta_k)}{\sum_{\ell=1}^{K} \exp(\beta_{\ell 0} + x^T\beta_\ell)},$$

an L_2-regularized fit maximizes

$$\sum_{i=1}^{n} \log \mathbb{P}(g_i \mid x_i) - \frac{\lambda}{2} \sum_{k=1}^{K} \|\beta_k\|_2^2,$$

with intercepts left unpenalized. For discriminant analysis, an analogous regularization shrinks a pooled covariance estimate toward its diagonal,

$$\widehat{\Sigma}(\gamma) = \gamma\widehat{\Sigma} + (1-\gamma)\operatorname{diag}(\widehat{\Sigma}), \qquad 0 \leq \gamma \leq 1.$$

The endpoint $\gamma = 0$ gives diagonal LDA; larger γ reintroduces covariance information when enough data support it.

Proposition 3.9.4. *Row-Space Reduction for Quadratic Penalties*
Let $X \in \mathbb{R}^{n\times p}$ have thin singular value decomposition

$$X = UDV^T = RV^T, \qquad R = UD \in \mathbb{R}^{n\times r},$$

where $r = \operatorname{rank}(X)$ and $V^TV = I_r$. For ridge regression,

$$\widehat{\beta}_\lambda^{\text{ridge}} = V(R^TR + \lambda I_r)^{-1}R^Ty.$$

More generally, for objectives

$$\sum_{i=1}^{n} L(y_i, \beta_0 + x_i^T\beta) + \lambda \|\beta\|_2^2,$$

with $\lambda > 0$, there is an optimal coefficient vector in the row space of X, so fitting can be done in the r coordinates r_i^T and mapped back by V.

Method 3.9.5. *Sparse and Structured Penalties*
Replacing the quadratic penalty by an L_1 penalty gives automatic feature selection. For multinomial logistic regression, one uses

$$\sum_{i=1}^{n} \log \mathbb{P}(g_i \mid x_i) - \lambda \sum_{k=1}^{K} \sum_{j=1}^{p} |\beta_{kj}|.$$

The elastic-net version replaces $|\beta_{kj}|$ by

$$\alpha|\beta_{kj}| + (1-\alpha)\beta_{kj}^2, \qquad 0 \leq \alpha \leq 1,$$

combining sparsity with shrinkage of correlated features toward shared participation. When the features have a natural order, a fused lasso penalty adds adjacent differences:

$$\lambda_1 \sum_{j=1}^{p} |\beta_j| + \lambda_2 \sum_{j=1}^{p-1} |\beta_{j+1} - \beta_j|.$$

The first term selects coordinates; the second encourages neighboring coefficients to be equal or piecewise constant.

Method 3.9.6. *Dantzig Selector*

The Dantzig selector is another sparse linear rule based on residual correlations. In centered, standardized least-squares notation it solves

$$\widehat{\beta}_s^{\mathrm{DS}} \in \arg\min_{\beta} \|\beta\|_1 \quad \text{subject to} \quad \left\|X^T(y - X\beta)\right\|_\infty \le s.$$

Equivalently, for a coefficient-size budget t, one can minimize the maximum absolute residual correlation subject to $\|\beta\|_1 \le t$. This makes the connection with the lasso transparent: both procedures use an ℓ^1 size control and both watch the correlations $x_j^T(y - X\beta)$, but the lasso decreases the residual sum of squares along an active set whereas the Dantzig selector directly controls the largest residual correlation.

Method 3.9.7. *Grouped Lasso*

When predictors arrive in prespecified groups, such as dummy variables for one categorical predictor or features from the same scientific pathway, coordinatewise sparsity may be the wrong unit of selection. Partition the predictors into L groups and write $X_\ell\beta_\ell$ for the contribution of group ℓ, with group size p_ℓ. The grouped lasso solves

$$\arg\min_{\beta_0,\beta_1,\ldots,\beta_L} \left\{ \left\| y - \beta_0 \mathbf{1} - \sum_{\ell=1}^{L} X_\ell\beta_\ell \right\|_2^2 + \lambda \sum_{\ell=1}^{L} \sqrt{p_\ell}\, \|\beta_\ell\|_2 \right\}.$$

The Euclidean norm is zero only when the whole group vector is zero, so the penalty can drop an entire group at once. The factor $\sqrt{p_\ell}$ keeps larger groups from being favored simply because they contain more coordinates.

Method 3.9.8. *Relaxed, Adaptive, and Nonconvex Lasso Variants*

The ordinary lasso shrinks nonzero coefficients toward zero, which is helpful for variance but can leave a noticeable bias. A simple two-stage correction is the *relaxed lasso*: use the lasso to select an active set, and then refit, or refit with a smaller penalty, using only that selected set.

The *adaptive lasso* keeps the convex ℓ^1 form but gives different coordinates different weights,

$$\frac{1}{2}\|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} w_j |\beta_j|, \qquad w_j = \frac{1}{|\widetilde{\beta}_j|^\nu},$$

where $\widetilde{\beta}$ is an initial estimate and $\nu > 0$. Large initial coefficients receive smaller weights and are shrunk less strongly.

Nonconvex penalties pursue the same goal more directly by penalizing large coefficients less than the lasso does. One example is the smoothly clipped absolute deviation penalty, whose derivative can be written

$$\frac{d}{dt} J_{a,\lambda}(t) = \lambda \operatorname{sign}(t) \left[\mathbf{1}_{\{|t| \le \lambda\}} + \frac{(a\lambda - |t|)_+}{(a-1)\lambda} \mathbf{1}_{\{|t| > \lambda\}} \right], \qquad a \ge 2.$$

These penalties can reduce shrinkage bias for large signals, but the resulting optimization is generally less straightforward than the convex lasso or elastic net.

Remark 3.9.9. *Three Roles for Feature Selection*

Feature selection can play three different roles in high-dimensional learning. It can reduce variance before prediction, as in nearest shrunken centroids; it can produce a sparse predictive model, as in lasso or elastic-net logistic regression; or it can produce a scientific list of features to be assessed separately, where multiple-testing control becomes the primary concern. The last role belongs to high-dimensional inference rather than prediction; see Definition 6.2.3 and Method 6.2.5.

Algorithm 12: Supervised Principal Components

Require: Standardized feature matrix X, outcome y, association scores s_j, thresholds $0 \le \theta_1 < \cdots < \theta_K$, candidate component counts $\mathcal{M}$

Ensure: A selected principal-component predictor

for $k = 1, \ldots, K$ **do**

$A_k \leftarrow \{j : |s_j| > \theta_k\}$

for $m \in \mathcal{M}$ **do**

Compute the first m principal components of the submatrix X_{A_k}

Fit the outcome model using those components as predictors
Estimate held-out prediction error by cross-validation
end for
end for
Choose (θ_k, m) by the cross-validation criterion and refit using the selected columns

Remark 3.9.10. *Screening Before Components*
Ordinary principal components search for directions of large variance in X, whether or not those directions predict the outcome. Supervised principal components first screen for features with sizable marginal association to y, then compute principal components only on the screened set. For ordinary regression, the score s_j can be a standardized univariate regression coefficient; for censored survival outcomes, it can be a one-feature Cox score. The resulting component can be read as a denoised latent factor assembled from many weakly informative features.

Method 3.9.11. *Preconditioned Lasso*
Preconditioning combines the variance reduction of supervised principal components with the sparsity of the lasso. First compute a supervised principal-component predictor $\widehat{y}_i$ for each training point. Then fit a lasso regression using $\widehat{y}$ as the response and the original feature matrix as predictors:

$$\widehat{\beta}^{\mathrm{pre}} \in \arg\min_{\beta} \left\{ \frac{1}{2} \left\| \widehat{y} - X\beta \right\|_2^2 + \lambda \left\| \beta \right\|_1 \right\}.$$

The first stage denoises the target; the second stage seeks a sparse set of original features that reproduces that denoised signal.

Method 3.9.12. *Inner-Product and Distance-Only Classifiers*
Some high-dimensional or structured objects are easier to compare than to featurize directly. If a positive semidefinite Gram matrix $K_{ii'} = \langle x_i, x_{i'} \rangle$ is available, then many linear procedures can be kernelized. Nearest-neighbor distances follow from

$$\left\| x_i - x_{i'} \right\|^2 = K_{ii} + K_{i'i'} - 2K_{ii'}.$$

Nearest-centroid comparisons can also be written using only inner products:

$$\left\| x_0 - \bar{x}_k \right\|^2 = \langle x_0, x_0 \rangle - \frac{2}{n_k} \sum_{g_i = k} \langle x_0, x_i \rangle + \frac{1}{n_k^2} \sum_{g_i = k} \sum_{g_{i'} = k} \langle x_i, x_{i'} \rangle.$$

If only pairwise squared distances $\Delta_{ii'}^2$ are available, define $B_{ii'} = -\Delta_{ii'}^2/2$ and double-center it:

$$\widetilde{K} = (I - M)B(I - M), \qquad M = \frac{1}{n}\mathbf{1}\mathbf{1}^T.$$

Then $\widetilde{K}$ is the centered inner-product matrix used by kernel PCA and related classifiers.

Example 3.9.13. *String Kernels*
For strings over a finite alphabet, let $\mathcal{A}_m$ be the set of all substrings of length m. The explicit feature map

$$\Phi_m(x) = \{\phi_a(x)\}_{a \in \mathcal{A}_m}$$

records how many times each substring a appears in x, and the string kernel is

$$K_m(x, x') = \langle \Phi_m(x), \Phi_m(x') \rangle.$$

The dimension $|\mathcal{A}_m|$ grows exponentially in m, but the kernel value can often be computed without materializing the full feature vector. This lets support-vector or centroid-style classifiers operate on sequence objects through their pairwise similarities.

3.10 Prototype and Nearest-Neighbor Methods

Prototype and nearest-neighbor classifiers are model-free rules built from distances in feature space. They can fit irregular decision boundaries and are often strong predictors, but their behavior is controlled by geometry: scaling, metric choice, neighborhood size, and storage of representative points. The local-averaging view appears in Method 1.4.2; this subsection records the classification and metric-design side.

Definition 3.10.1. *Labeled Prototype Rule*
Let the training data be $(x_i, g_i)_{i=1}^n$, where $x_i \in \mathbb{R}^p$ and $g_i \in \{1, \ldots, K\}$. A prototype rule replaces the training set by labeled representatives

$$m_j(k) \in \mathbb{R}^p, \qquad j = 1, \ldots, R_k, \quad k = 1, \ldots, K.$$

Given a distance d, it classifies a query point by

$$(\widehat{j}, \widehat{k}) \in \operatorname*{arg\,min}_{1 \le k \le K,\ 1 \le j \le R_k} d\big(x, m_j(k)\big), \qquad \widehat{G}(x) = \widehat{k}.$$

For quantitative features, Euclidean distance is meaningful only after putting coordinates on comparable scales. When every training observation is retained as a prototype with its own label, this reduces to one-nearest-neighbor classification.

Method 3.10.2. *k-Means Prototypes*
Classwise k-means gives a simple prototype classifier. For each class k, run Algorithm 15 on the inputs with label k, using R centers. Label the resulting centers by k, and classify a new point by the nearest center among all KR prototypes.

This compresses the training set and produces piecewise-linear decision boundaries in Euclidean geometry. Its weakness is also clear: the prototypes for one class are placed without looking at the other classes, so a center may sit near a competing boundary even when moving it would improve classification.

Algorithm 13: Learning Vector Quantization

Require: Labeled data $(x_i, g_i)_{i=1}^n$, prototypes per class R, learning rates $\epsilon_t \downarrow 0$
Ensure: Labeled prototypes $m_j(k)$
Initialize $m_1(k), \dots, m_R(k)$ from training observations in class k, for each k
for $t = 1, 2, \dots$ **do**
Sample a training index i
Choose the closest prototype (j, k) to x_i over all classes and prototype indices
if $g_i = k$ **then**
$m_j(k) \leftarrow m_j(k) + \epsilon_t\big(x_i - m_j(k)\big)$
else
$m_j(k) \leftarrow m_j(k) - \epsilon_t\big(x_i - m_j(k)\big)$
end if
end for

Remark 3.10.3. *Boundary-Aware Prototype Motion*
Learning vector quantization modifies prototypes online. A sampled point attracts the closest prototype when their labels match and repels it when their labels differ. Thus all classes influence the placement of all prototypes, unlike classwise k-means. The tradeoff is that LVQ is primarily algorithmic: the update rule is easy to state, but it is not the exact minimization of a fixed objective like the k-means criterion.

Method 3.10.4. *Gaussian Mixtures as Soft Prototypes*
A Gaussian mixture component has a center and covariance, so mixtures can be read as soft prototype models. If each class is represented by a fitted mixture density, Bayes' rule gives posterior class probabilities, as in Method 3.1.14. The responsibilities in Definition 4.1.15 replace hard nearest-center membership by fractional membership.

With common spherical component covariance, EM for mixtures is especially close to k-means: the E-step assigns responsibilities according to relative likelihoods, while the M-step updates weighted centers and covariances; compare Method 4.1.18. As the common variance becomes small, the responsibilities concentrate on nearest centers, matching the hard-assignment limit in Remark 4.1.17.

Method 3.10.5. *k-Nearest-Neighbor Classification*
For a query point x_0, order the training inputs by

$$d_i = d(x_i, x_0), \qquad i = 1, \dots, n,$$

and let $N_k(x_0)$ be the indices of the k smallest distances. The k-nearest-neighbor classifier predicts by majority vote:

$$\widehat{G}(x_0) \in \operatorname*{arg\,max}_{g \in \{1,\dots,K\}} \sum_{i \in N_k(x_0)} \mathbf{1}_{\{g_i = g\}}.$$

Small k gives low-bias but high-variance decision boundaries; larger k smooths the boundary by averaging over more labels. The tuning parameter is therefore usually chosen by validation or cross-validation, as in Method 8.1.4.

Proposition 3.10.6. *Asymptotic One-Nearest-Neighbor Error*
At a fixed feature value x, write

$$p_k(x) = \mathbb{P}(G = k \mid X = x), \qquad k^\star \in \operatorname*{arg\,max}_{1 \le k \le K} p_k(x).$$

The Bayes error at x is

$$L^\star(x) = 1 - p_{k^\star}(x).$$

In the asymptotic one-nearest-neighbor idealization, the nearest training feature coincides with x but its label is an independent draw from the same conditional distribution. The resulting local error is

$$L_{1\mathrm{NN}}(x) = \sum_{k=1}^{K} p_k(x)\big(1 - p_k(x)\big) = 1 - \sum_{k=1}^{K} p_k(x)^2.$$

It satisfies

$$L^\star(x) \le L_{1\mathrm{NN}}(x) \le 2L^\star(x) - \frac{K}{K-1}\big(L^\star(x)\big)^2 \le 2L^\star(x).$$

Proof idea. The first inequality follows from $\sum_k p_k(x)^2 \le p_{k^\star}(x)$. For the upper bound, fix the mass $q = 1 - p_{k^\star}(x)$ outside the dominant class. The quantity $1 - \sum_k p_k(x)^2$ is largest when the remaining $K-1$ probabilities share this mass equally, giving $2q - Kq^2/(K-1)$. □

Example 3.10.7. *Tuning Depends on the Problem*
The source comparison used ten independent uniform features and two binary classification problems. In the easy problem, the classes are separated by the hyperplane $X_1 = 1/2$:

$$Y = \mathbf{1}_{\{X_1 > 1/2\}}.$$

In the difficult problem, the first three coordinates define a checkerboard-like pattern:

$$Y = \mathbf{1}_{\left\{\mathrm{sign}\left(\prod_{j=1}^{3}(X_j - 1/2)\right) > 0\right\}}.$$

Across repeated simulations, the best neighborhood size or number of prototypes was problem-dependent. This is the operational message: for memory-based and prototype methods, the tuning parameter is not a universal complexity constant; it has to be estimated for the geometry at hand.

Example 3.10.8. *Image Scene Classification by Local Pixel Features*
In the image-scene example, each pixel has four spectral-band measurements and a land-use label from seven classes. A five-nearest-neighbor rule was applied after replacing each pixel by the pixel itself and its eight immediate spatial neighbors in each of the four bands, giving $(1+8)4 = 36$ input features; see Figure 236. The reported test error was about 9.5%, illustrating that nearest-neighbor rules can be competitive when the class boundaries in the constructed feature space are irregular.

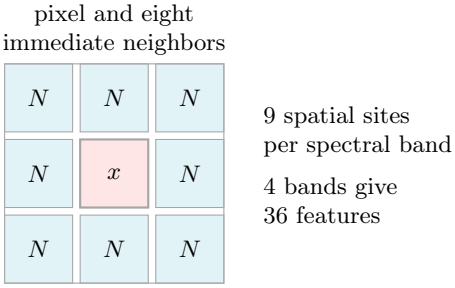


Figure 236: A local pixel feature map uses the target pixel and its eight immediate spatial neighbors. With four spectral bands this gives $9 \cdot 4 = 36$ input features per pixel.

Method 3.10.9. *Invariant Metrics and Tangent Distance*
For structured inputs such as images, ordinary Euclidean distance can treat harmless transformations as large changes. If $\mathcal{A}$ is a class of transformations and $T_\alpha x$ is the transformed input, the invariant distance is

$$d_{\mathrm{inv}}(x, x') = \inf_{\alpha,\beta \in \mathcal{A}} \|T_\alpha x - T_\beta x'\|_2 .$$

This compares transformation orbits rather than only the original vectors.

Tangent distance replaces each orbit near x by a tangent affine space. If V_x spans the small-transformation directions at x, use

$$d_{\tan}(x, x') = \inf_{a,b} \|x + V_x a - x' - V_{x'} b\|_2 .$$

This keeps the invariance local: small rotations, translations, scalings, shears, or thickness changes can be discounted without declaring large transformations harmless. In the handwritten ZIP-code example, the reported test-error rates were

neural network	0.049
one-nearest-neighbor, Euclidean distance	0.055
one-nearest-neighbor, tangent distance	0.026.

Adding transformed versions of each training image is another way to encode invariance, but it becomes infeasible when many transformation directions must be represented.

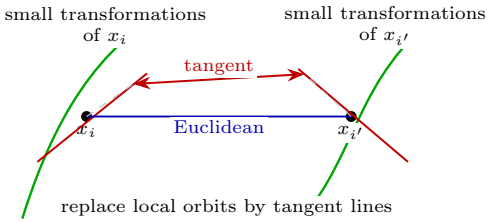


Figure 237: Tangent distance approximates each transformation orbit by its local tangent affine space, then measures distance between these tangent approximations rather than between the original points alone.

Method 3.10.10. *Discriminant Adaptive Nearest Neighbors*

High-dimensional nearest-neighbor rules can have large bias because the nearest points are still far away. In the cube model from the source pages, if N points are uniformly distributed in $[-1/2, 1/2]^p$ and R is the radius of the one-nearest-neighbor ball centered at the origin, then

$$\operatorname{median}(R) = v_p^{-1/p}\left(1 - \frac{1}{2^{1/N}}\right)^{1/p},$$

where $v_p r^p$ is the volume of a radius-r ball in $\mathbb{R}^p$. This is the same geometric pressure described in Remark 1.4.3.

Discriminant adaptive nearest neighbors choose a local metric at each query point. Around x_0, use a pilot neighborhood to estimate local class proportions π_k, class means $\bar{x}_k$, the local mean $\bar{x}$, pooled within-class covariance $\mathbf{W}$, and between-class covariance

$$\mathbf{B} = \sum_{k=1}^{K} \pi_k (\bar{x}_k - \bar{x})(\bar{x}_k - \bar{x})^T.$$

Define the sphered between-class matrix and metric

$$\mathbf{B}^\star = \mathbf{W}^{-1/2}\mathbf{B}\mathbf{W}^{-1/2}, \qquad \mathbf{\Sigma}_{x_0} = \mathbf{W}^{-1/2}\big(\mathbf{B}^\star + \epsilon I\big)\mathbf{W}^{-1/2},$$

and use

$$D_{x_0}(x, x_0) = (x - x_0)^T \mathbf{\Sigma}_{x_0} (x - x_0)$$

inside the nearest-neighbor vote at x_0. Whitening by $\mathbf{W}$ removes local within-class scale, and the factor $\mathbf{B}^\star + \epsilon I$ stretches neighborhoods along directions with little local class separation. The parameter $\epsilon > 0$ keeps those neighborhoods from degenerating into infinite strips.

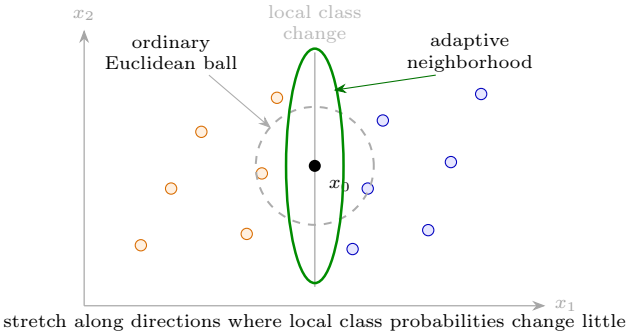


Figure 238: DANN replaces an ordinary Euclidean neighborhood by a query-dependent ellipsoid that stretches along directions where the local class probabilities change slowly.

Method 3.10.11. *Global Dimension Reduction for Nearest Neighbors*

The DANN idea can also be globalized. Compute a local between-class matrix $\mathbf{B}_i$ at each training point and average

$$\overline{\mathbf{B}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{B}_i.$$

Let $e_1, \dots, e_p$ be eigenvectors of $\overline{\mathbf{B}}$, ordered from largest to smallest eigenvalue θ_ℓ. Projecting onto the leading L directions gives a nearest-neighbor rule in a supervised reduced subspace. Equivalently, the rank-L approximation

$$\overline{\mathbf{B}}_{[L]} = \sum_{\ell=1}^{L} \theta_\ell e_\ell e_\ell^T$$

solves

$$\min_{\operatorname{rank}(M)=L} \sum_{i=1}^{N} \operatorname{trace}\big[(\mathbf{B}_i - M)^2\big].$$

Thus the reduced space is selected from local class-separation information, not from unsupervised variance alone.

Remark 3.10.12. *Computation and Storage*
A direct nearest-neighbor query with N stored observations and p features requires about Np coordinate operations and stores the whole training set. Fast nearest-neighbor search can reduce the query cost in favorable geometries, while prototype rules reduce storage by replacing many observations with representatives.

Editing and condensing procedures reduce storage in a different way. Editing removes points whose neighborhoods look inconsistent, trying to keep homogeneous regions. Condensing starts from a small retained set and adds a training point only when the current retained set misclassifies it. Both ideas aim to keep points that determine the decision boundary and discard points deep inside stable regions.

3.11 Flexible and Mixture Discriminant Analysis

Linear discriminant analysis is useful partly because it is simple: each class is represented by a centroid, distances are measured in a common covariance metric, and the decision boundaries are linear. Its limitations are equally structural. A single centroid may not represent an inhomogeneous class, linear boundaries may be too rigid, and high-dimensional predictors may require smoothness or shrinkage beyond ordinary covariance pooling. Flexible, penalized, and mixture discriminant methods keep the LDA comparison geometry while changing the regression fit, the metric, or the number of class prototypes.

Method 3.11.1. *Flexible and Penalized Discriminant Analysis*
Optimal scoring recasts LDA as regression on transformed class labels. Let $g_i \in \{1, \dots, K\}$ and choose up to $L \le K-1$ score functions θ_ℓ on the class labels together with fitted score maps $\eta_\ell(x)$. Flexible discriminant analysis minimizes an average scored residual of the form

$$\frac{1}{n} \sum_{\ell=1}^{L} \sum_{i=1}^{n} \{\theta_\ell(g_i) - \eta_\ell(x_i)\}^2 + \lambda \sum_{\ell=1}^{L} J(\eta_\ell),$$

with centering and orthogonality normalizations on the label scores to remove trivial constant solutions. If $\eta_\ell(x) = x^\top \beta_\ell$ and no extra roughness penalty is used, the resulting directions agree with the LDA discriminant coordinates up to scale. Replacing the linear regression fit by splines, additive models, MARS, kernels, or another smoother gives nonlinear decision boundaries while keeping the LDA idea of class separation in a low-dimensional fitted-score space.

Penalized discriminant analysis is the same construction when the score maps are linear in an expanded representation, $\eta_\ell(x) = h(x)^\top \beta_\ell$, and a quadratic roughness penalty $\beta_\ell^\top \Omega \beta_\ell$ is imposed. Equivalently, it compares class centroids in the expanded variables using the penalized Mahalanobis metric

$$D_\lambda(x, k) = \big(h(x) - \bar{h}_k\big)^\top (\widehat{\Sigma}_{W,h} + \lambda\Omega)^{-1} \big(h(x) - \bar{h}_k\big),$$

where $\bar{h}_k$ is the class-k centroid in the derived variables and $\widehat{\Sigma}_{W,h}$ is their pooled within-class covariance. The penalty gives less weight to rough or unstable directions in the derived feature space.

Algorithm 14: FDA Coordinates by Optimal Scoring

Require: Class labels $g_i \in \{1, \dots, K\}$, indicator response matrix $Y \in \mathbb{R}^{n \times K}$, linear smoother S_λ for the chosen regression method
Ensure: Nonconstant discriminant coordinate functions $\eta_1, \dots, \eta_{K-1}$
Fit the multivariate regression of Y on the features: $\widehat{Y} \leftarrow S_\lambda Y$
Let $\eta^*(x)$ denote the K fitted regression functions evaluated at a new input x
Set $D_\pi \leftarrow Y^T Y / n$
Solve $Y^T \widehat{Y} \theta = \gamma D_\pi \theta$ and normalize eigenvectors so that $\Theta^T D_\pi \Theta = I$
Discard the constant scoring direction
for each remaining eigenvector θ_ℓ **do**
  Set $\eta_\ell(x) \leftarrow \theta_\ell^T \eta^*(x)$

end for

Remark 3.11.2. *FDA as LDA After a Regression Map*
When the fitted maps η_ℓ come from polynomial regression, spline regression, additive regression, or a kernel smoother, FDA can be read as LDA after passing the original inputs through learned fitted-score coordinates. Thus quadratic regression scores lead to quadratic decision surfaces in the original variables, while additive or spline scores can express more irregular boundaries. This is the same linear-in-expanded-space paradigm used by basis expansions in Definition 3.3.1 and by kernel SVMs in Method 3.5.9.

Method 3.11.3. *Mixture Discriminant Analysis*
Mixture discriminant analysis replaces the single Gaussian prototype in LDA by several prototypes inside each class, while keeping a common covariance metric. With R_k components in class k, the class-conditional density is modeled as

$$f_k(x) = \sum_{r=1}^{R_k} \pi_{kr} \phi(x; \mu_{kr}, \Sigma), \qquad \pi_{kr} \geq 0, \quad \sum_{r=1}^{R_k} \pi_{kr} = 1,$$

and, with class prior ρ_k, Bayes' rule gives

$$\mathbb{P}(G = k \mid X = x) = \frac{\rho_k \sum_{r=1}^{R_k} \pi_{kr} \phi(x; \mu_{kr}, \Sigma)}{\sum_{\ell=1}^{K} \rho_\ell \sum_{r=1}^{R_\ell} \pi_{\ell r} \phi(x; \mu_{\ell r}, \Sigma)}.$$

The case $R_k = 1$ for every class recovers the common-covariance Gaussian discriminant model in Method 3.1.14.

The natural fitting scheme is EM with subclass responsibilities computed only within the observed class. If $g_i = k$, set

$$r_{ikr} = \frac{\pi_{kr} \phi(x_i; \mu_{kr}, \Sigma)}{\sum_{s=1}^{R_k} \pi_{ks} \phi(x_i; \mu_{ks}, \Sigma)}, \qquad r = 1, \dots, R_k.$$

The M-step updates mixture weights, subclass centers, and the shared covariance by weighted Gaussian maximum likelihood. Equivalently, one can form a "blurred" response matrix whose columns are subclasses c_{kr}: a class-k row has entries $r_{ik1}, \dots, r_{ikR_k}$ in the class-k subclass columns and zeros elsewhere. The M-step is then a weighted discriminant-analysis problem on these blurred subclasses, so rank restrictions, FDA, or PDA can be inserted exactly as in the ordinary LDA step. This is the supervised analogue of the mixture responsibilities in Definition 4.1.15.

Remark 3.11.4. *MDA and Prototype Rules*
MDA is a probabilistic prototype classifier: each subclass center is a prototype, but classification uses Gaussian posterior weights rather than a hard nearest-center rule. Compared with classwise k-means prototypes in Method 3.10.2, MDA also estimates component weights and a shared metric. Compared with ordinary Gaussian mixtures in Method 4.1.18, the component labels are constrained by the observed class labels.

Remark 3.11.5. *Computational Scale*
For N training cases and p predictors, LDA and PDA require covariance-scale linear algebra, roughly of order $Np^2 + p^3$ before any low-rank reductions. FDA inherits the cost of its regression smoother: additive and adaptive basis fits can be close to linear in N for fixed model size, whereas general kernel or spline smoothers may require solving dense $N \times N$ systems. MDA adds EM iterations and is sensitive to initialization because mixture likelihoods can have multiple local optima.

4 Unsupervised and Latent-Variable Learning

4.1 Clustering and Mixture Models

Unsupervised learning searches for structure in inputs without direct labels or a direct prediction loss. Clustering summarizes the input distribution by grouping observations, while mixture models replace hard groups by latent component probabilities.

Definition 4.1.1. *k-Means Clustering*
Given observations $x_1, \dots, x_n \in \mathbb{R}^d$, k-means chooses assignments $C(i) \in \{1, \dots, k\}$ and centers $m_1, \dots, m_k \in \mathbb{R}^d$ to minimize the within-cluster sum of squares

$$\Phi(C, m_1, \dots, m_k) = \sum_{r=1}^{k} \sum_{C(i)=r} \|x_i - m_r\|_2^2 .$$

For fixed assignments, the minimizing center m_r is the empirical mean of the points assigned to cluster r; for fixed centers, each point is assigned to a nearest center.

Definition 4.1.2. *Object Dissimilarities*
A clustering procedure may start from pairwise dissimilarities $d_{ii'} = d(x_i, x_{i'})$ rather than from Euclidean coordinates. A usual dissimilarity matrix has nonnegative entries, zero diagonal, and is symmetrized if the original proximity measurements are directional. When observations have attributes x_{ij}, an overall dissimilarity is often assembled as

$$D(x_i, x_{i'}) = \sum_{j=1}^{p} w_j \, d_j(x_{ij}, x_{i'j}), \qquad w_j \geq 0, \quad \sum_{j=1}^{p} w_j = 1.$$

Quantitative attributes commonly use absolute or squared differences, ordinal attributes can first be mapped to ordered scores such as $(r-1/2)/M$, and unordered categorical attributes require an explicit loss matrix between levels.

Remark 4.1.3. *Weights and Missing Values*
Equal numerical weights do not necessarily give equal practical influence: the contribution of an attribute also depends on its average dissimilarity over the data. Missing values can be handled by omitting missing coordinates from a pairwise comparison, imputing them before clustering, or treating "missing" as a categorical level when that convention matches the problem.

Definition 4.1.4. *Within- and Between-Cluster Scatter*
For a fixed number k of clusters and dissimilarities $d_{ii'}$, an assignment C has within-cluster scatter

$$W(C) = \frac{1}{2} \sum_{r=1}^{k} \sum_{C(i)=r} \sum_{C(i')=r} d_{ii'}.$$

The total scatter

$$T = \frac{1}{2} \sum_{i=1}^{n} \sum_{i'=1}^{n} d_{ii'}$$

does not depend on C. Writing $T = W(C) + B(C)$ defines the between-cluster scatter $B(C)$, so minimizing $W(C)$ is equivalent to maximizing $B(C)$. Exhaustive search is infeasible except for very small n, which motivates iterative descent heuristics.

Algorithm 15: k-Means Alternating Minimization

Require: Data $x_1, \dots, x_n \in \mathbb{R}^d$, number of clusters k, initial centers or assignments
Ensure: Cluster assignments $C(i)$ and centers $m_1, \dots, m_k$
repeat
Given assignments, set each m_r to the mean of the observations with $C(i) = r$
Given centers, assign each observation by $C(i) \leftarrow \arg\min_{1 \leq r \leq k} \|x_i - m_r\|_2^2$
until the assignments no longer change

Remark 4.1.5. *k-Means Local Minima*
Each step in Algorithm 15 does not increase the objective in Definition 4.1.1. The final assignment can still be only a local optimum, so multiple initializations are commonly compared by their final objective values.

Algorithm 16: k-Medoids Alternating Representatives

Require: Pairwise dissimilarities $D(x_i, x_{i'})$, number of clusters k, initial assignments
Ensure: Cluster assignments $C(i)$ and representative indices $i_1^\star, \dots, i_k^\star$
repeat
for $r = 1, \dots, k$ **do**
Set $i_r^\star \leftarrow \arg\min_{i:\, C(i)=r} \sum_{C(i')=r} D(x_i, x_{i'})$
end for
Assign each observation by $C(i) \leftarrow \arg\min_{1 \leq r \leq k} D(x_i, x_{i_r^\star})$
until the assignments no longer change

Remark 4.1.6. *k-Medoids Versus k-Means*
k-medoids replaces each Euclidean mean by an actual observation:

$$i_r^\star \in \arg\min_{i:\, C(i)=r} \sum_{C(i')=r} D(x_i, x_{i'}).$$

It only needs pairwise dissimilarities and is less tied to squared Euclidean geometry, but the representative update is more computationally expensive than taking a cluster mean.

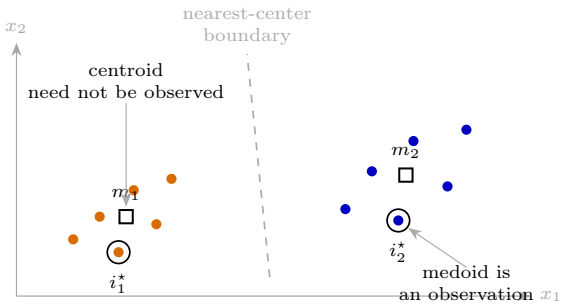


Figure 239: Centroid representatives used by k-means can lie between observations, while medoids are actual observed points. The dashed line indicates the nearest-center assignment boundary for two fitted centers.

Method 4.1.7. *Hierarchical Clustering and Linkage*
Hierarchical clustering produces a nested sequence of partitions. Agglomerative methods begin with singleton clusters and repeatedly merge the closest pair of groups; divisive methods begin with all observations in one group and recursively split one group into two. For two groups G and H, common agglomerative linkages are

$$d_{\mathrm{SL}}(G,H) = \min_{i\in G,\ i'\in H} d_{ii'}, \qquad d_{\mathrm{CL}}(G,H) = \max_{i\in G,\ i'\in H} d_{ii'},$$

$$d_{\mathrm{GA}}(G,H) = \frac{1}{N_G N_H} \sum_{i\in G} \sum_{i'\in H} d_{ii'}.$$

The resulting binary tree is displayed as a dendrogram. Single linkage uses the closest bridge between groups and can chain through intermediate points; complete linkage controls the largest within-merge diameter; group average is a compromise based on average pairwise separation.

Remark 4.1.8. *Choosing the Number of Clusters*
For k-means or k-medoids one must choose both k and an initialization. A basic diagnostic plots within-cluster scatter W_k as a function of k and looks for a sharp bend. The gap-statistic idea compares the observed curve with a reference distribution spread over the data range:

$$G(k) = \mathbb{E}_0[\log W_k] - \log W_k.$$

With simulation standard error s'_{k+1}, one formal selection rule is

$$\widehat{k} = \min\{k : G(k) \geq G(k+1) - s'_{k+1}\}.$$

This is still a heuristic, but it makes the "bend" comparison explicit.

Remark 4.1.9. *Dendrograms and Cophenetic Distances*
Cutting a dendrogram at a chosen height gives a partition into clusters. The tree also defines a cophenetic dissimilarity $C_{ii'}$: the height at which observations i and i' first join the same cluster. These tree-induced dissimilarities obey the ultrametric inequality

$$C_{ii'} \leq \max\{C_{ih}, C_{i'h}\},$$

so a dendrogram should be read as an algorithmic summary of hierarchical structure, not as an exact reproduction of arbitrary original dissimilarities.

Method 4.1.10. *Self-Organizing Map*
A self-organizing map is a constrained prototype method. Choose prototypes $m_j \in \mathbb{R}^p$ indexed by grid locations $\ell_j \in \mathcal{Q}_1 \times \mathcal{Q}_2$. For an observation x_i, find the closest prototype m_j in feature space, but update prototypes according to distance on the index grid:

$$m_k \leftarrow m_k + \alpha h(\|\ell_j - \ell_k\|_2)(x_i - m_k).$$

The simplest neighborhood rule takes h to be an indicator of $\|\ell_j - \ell_k\|_2 \leq r$, always including the winning prototype itself. The learning rate α and neighborhood radius r are typically decreased during training, so early updates preserve a smooth topological arrangement while later updates refine local fit. If each neighborhood contains only the winner, the spatial constraint disappears and the method becomes an online version of k-means.

Remark 4.1.11. *Checking the Topological Constraint*
Since a self-organizing map restricts prototypes to respect a low-dimensional lattice, its reconstruction error

$$\sum_i \|x_i - m_{C(i)}\|_2^2$$

should be compared with ordinary k-means using the same number of prototypes. The k-means error is necessarily no larger; a much smaller k-means error is evidence that the lattice constraint is too restrictive for the data.

Method 4.1.12. *Spectral Clustering*
Spectral clustering replaces Euclidean cluster geometry by graph geometry. From similarities $s_{ii'} \geq 0$, form a weighted graph with adjacency matrix $W = (w_{ii'})$, degrees $g_i = \sum_{i'} w_{ii'}$, diagonal degree matrix G, and unnormalized Laplacian

$$L = G - W.$$

For example, one may use a radial similarity $s_{ii'} = \exp(-\|x_i - x_{i'}\|_2^2 / c)$ and either keep all edges or retain only mutual nearest-neighbor edges. The identity

$$f^T L f = \frac{1}{2} \sum_{i=1}^{n} \sum_{i'=1}^{n} w_{ii'} (f_i - f_{i'})^2$$

shows why eigenvectors with small nonzero eigenvalues vary slowly across highly weighted edges. A standard procedure computes the eigenvectors corresponding to the first few nontrivial small eigenvalues and clusters the rows of the resulting eigenvector matrix, often by k-means.

Remark 4.1.13. *Connected Components and Random Walks*
If the similarity graph has m connected components, the zero-eigenvalue eigenspace of L is spanned by the component indicator vectors; in practice, nearly separated clusters give small rather than exactly zero eigenvalues. With the normalized view $P = G^{-1}W$, spectral clusters are also groups of nodes for which the random walk with transition matrix P tends to remain inside the group and rarely crosses to another group.

Definition 4.1.14. *Finite Mixture Model*
A finite mixture has density

$$p(x) = \sum_{z=1}^{k} \pi_z p_z(x; \theta_z), \qquad \pi_z \geq 0, \quad \sum_{z=1}^{k} \pi_z = 1,$$

where the component label Z is latent.

Definition 4.1.15. *Mixture Responsibilities*
After fitting a mixture model, the posterior probability that observation x_i belongs to component m is

$$\widehat{r}_{im} = \frac{\widehat{\pi}_m p_m(x_i; \widehat{\theta}_m)}{\sum_{\ell=1}^{k} \widehat{\pi}_\ell p_\ell(x_i; \widehat{\theta}_\ell)}.$$

These responsibilities can summarize soft clusters, or be thresholded when a hard partition is needed.

Remark 4.1.16. *Mixtures as Density Estimates*
Gaussian mixtures can be used as flexible density estimates. If separate mixtures are fit within each class, Bayes' rule gives a density-based classifier, as in Definition 6.1.14. If the mixture is fit without labels, any agreement between components and labels is discovered structure rather than supervised training.

Remark 4.1.17. *Gaussian Mixtures as Soft k-Means*
For a Gaussian mixture with common spherical covariance $\sigma^2 I$, the responsibility in Definition 4.1.15 is a soft assignment whose relative size is a monotone function of Euclidean distance to the component center, after accounting for the mixing weights. As σ^2 becomes small, responsibilities concentrate on the nearest center, giving the hard-assignment behavior of k-means.

Method 4.1.18. *Expectation-Maximization*
For the finite mixture in Definition 4.1.14, the observed log-likelihood is

$$\ell(\pi, \theta) = \sum_{i=1}^{n} \log \left(\sum_{m=1}^{k} \pi_m p_m(x_i; \theta_m) \right).$$

EM introduces latent component labels and alternates:

(i) *E-step*: with current parameters, compute responsibilities

$$r_{im} = \frac{\pi_m p_m(x_i; \theta_m)}{\sum_{\ell=1}^{k} \pi_\ell p_\ell(x_i; \theta_\ell)}.$$

(ii) *M-step*: update

$$\pi_m^{\text{new}} = \frac{1}{n} \sum_{i=1}^{n} r_{im}, \qquad \theta_m^{\text{new}} \in \arg\max_{\theta_m} \sum_{i=1}^{n} r_{im} \log p_m(x_i; \theta_m).$$

For a Gaussian mixture, the component parameters in the M-step are weighted means and variances or covariances. With an exact M-step, the observed log-likelihood is nondecreasing, though the procedure may converge to a local maximum depending on initialization.

Remark 4.1.19. *Identifiability*
Mixture components are identifiable only up to label permutations. This is a benign symmetry for density estimation but important when interpreting clusters.

4.2 Dimension Reduction and Manifold Learning

Dimension reduction replaces high-dimensional observations, similarities, or dissimilarities by lower-dimensional coordinates that preserve selected structure. Linear methods preserve variance or inner products; manifold methods preserve neighborhoods, graph distances, or local affine geometry.

Definition 4.2.1. *Principal Component Analysis*
For centered observations $x_1, \dots, x_n \in \mathbb{R}^p$, principal component analysis chooses an affine rank-r linear approximation by minimizing squared reconstruction error over orthonormal directions $v_1, \dots, v_r$. Equivalently, the first direction maximizes projected variance, the second maximizes projected variance among directions orthogonal to the first, and so on.

Method 4.2.2. *PCA by SVD*
Let the centered observations be the rows of $X \in \mathbb{R}^{n \times p}$ and write

$$X = UDV^T.$$

The rank-r PCA directions are the first r columns of V, the principal-component scores are the rows of $U_r D_r$, and the rank-r fitted values are

$$\widehat{X}_r = U_r D_r V_r^T = X V_r V_r^T.$$

Thus PCA is the statistical form of the SVD and truncated-SVD approximation in Theorem 4.2.10 and Theorem 5.1.14: it gives the orthogonal rank-r linear manifold with smallest squared reconstruction error.

Definition 4.2.3. *Multidimensional Scaling*
Given pairwise dissimilarities $d_{ii'}$ among n observations, multidimensional scaling seeks coordinates $z_1, \dots, z_n \in \mathbb{R}^r$ whose Euclidean distances reproduce those dissimilarities. Least-squares, or Kruskal–Shepard, scaling minimizes the stress

$$S_{\mathrm{MDS}}(z_1, \dots, z_n) = \sum_{i \neq i'} \left(d_{ii'} - \|z_i - z_{i'}\|_2\right)^2.$$

A Sammon-weighted variant, for positive dissimilarities, uses

$$S_{\mathrm{Sammon}}(z_1, \dots, z_n) = \sum_{i \neq i'} \frac{\left(d_{ii'} - \|z_i - z_{i'}\|_2\right)^2}{d_{ii'}},$$

which puts relatively more weight on preserving small pairwise dissimilarities.

Method 4.2.4. *Classical Scaling*
Classical scaling starts from centered similarities $s_{ii'}$, often centered inner products, and chooses coordinates $z_i \in \mathbb{R}^r$ to approximate them by inner products:

$$S_{\mathrm{classical}}(z_1, \dots, z_n) = \sum_{i,i'} \left(s_{ii'} - \langle z_i - \bar{z}, z_{i'} - \bar{z} \rangle\right)^2.$$

If $S = (s_{ii'})$ is centered and has leading positive eigenpairs (λ_j, e_j), then the row coordinates of

$$E_r D_r, \qquad E_r = (e_1, \dots, e_r), \qquad D_r = \mathrm{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_r}).$$

solve the rank-r approximation problem in the same spectral form as PCA.

If only Euclidean distances $\Delta_{ii'} = \|x_i - x_{i'}\|_2$ are supplied, set $\Delta^{(2)} = (\Delta_{ii'}^2)$ and

$$H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T, \qquad B = -\frac{1}{2} H \Delta^{(2)} H.$$

Then B is the centered Gram matrix for the data, so applying the preceding eigenvector construction to B recovers classical-scaling coordinates. This is the same double-centering operation used for distance-only kernel calculations in Method 3.9.12.

Method 4.2.5. *Nonmetric Scaling*
Nonmetric MDS preserves the rank order of dissimilarities rather than their numerical values. It fits an increasing function θ and coordinates z_i by minimizing a normalized stress of the form

$$S_{\mathrm{NM}}(z_1, \dots, z_n) = \frac{\sum_{i \neq i'} \left(\|z_i - z_{i'}\|_2 - \theta(d_{ii'})\right)^2}{\sum_{i \neq i'} \|z_i - z_{i'}\|_2^2}.$$

With θ fixed, the coordinates can be updated by stress minimization. With the coordinates fixed, isotonic regression gives the best monotone approximation $\theta(d_{ii'})$ to the fitted distances. Thus nonmetric scaling uses the order of the dissimilarities rather than their original units.

Definition 4.2.6. *Encoder and Decoder*
An *encoder* maps data x to a representation $h = e(x)$, and a *decoder* maps h back to a reconstruction $d(h)$. Linear PCA is the special case where both maps are linear and the representation subspace is orthogonal.

Method 4.2.7. *Procrustes Shape Alignment*
For two landmark configurations $X_1, X_2 \in \mathbb{R}^{n\times p}$ with corresponding rows, an orthogonal Procrustes alignment chooses $R^T R = I$ and translation μ to minimize

$$\left\| X_2 - (X_1 R + \mathbf{1}\mu^T) \right\|_F .$$

Let $\bar{x}_1, \bar{x}_2$ be the column means and let $\widetilde{X}_1, \widetilde{X}_2$ be the centered matrices. If

$$\widetilde{X}_1^T \widetilde{X}_2 = U D V^T,$$

then $\widehat{R} = UV^T$ and $\widehat{\mu} = \bar{x}_2 - \widehat{R}\bar{x}_1$. A scaled version of the centered problem minimizes $\|X_2 - \beta X_1 R\|_F$, with $\widehat{\beta} = \operatorname{tr}(D)/\|X_1\|_F^2$ after centering.

Method 4.2.8. *Procrustes Shape Average*
For shapes $X_1, \dots, X_L$ with corresponding landmarks, a Procrustes average seeks a template M that is close, after rotations, to the whole collection:

$$\min_{\{R_\ell\}, M} \sum_{\ell=1}^{L} \|X_\ell R_\ell - M\|_F^2 .$$

A simple alternating scheme initializes M to one observed shape, aligns each X_ℓ to the current M, replaces M by the average of the aligned shapes, and repeats until the criterion stabilizes. The template is determined only up to a common rotation unless an additional convention is imposed.

Remark 4.2.9. *Manifold Hypothesis*
Many representation methods assume that high-dimensional observations concentrate near lower-dimensional sets. Global Euclidean distances can be misleading on a curved set: points near opposite ends of a curve may be close in ambient space but far along the curve. Manifold learning tries to preserve local neighborhoods, geodesic structure, or tangent information rather than only global Euclidean variance.

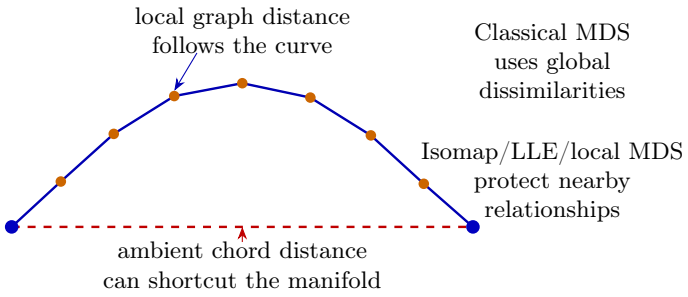


Figure 240: On a curved data set, an ambient chord can be short even when the path along nearby samples is long. Local manifold methods replace or modify the global dissimilarities before extracting coordinates.

Example 4.2.10. *Manifold Methods* Isomap, locally linear embedding, Laplacian eigenmaps, and diffusion maps construct low-dimensional coordinates from a neighborhood graph or kernel on the sample.

Method 4.2.11. *Isomap*
Isomap replaces ambient Euclidean distance by an estimated geodesic distance along the data cloud. Construct a neighborhood graph by connecting nearby observations, compute shortest-path distances in that graph, and then apply classical scaling from Method 4.2.4 to the graph-distance matrix. The graph step is the nonlinear part; the final coordinate extraction is still spectral.

Method 4.2.12. *Locally Linear Embedding*
Locally linear embedding first records how each observation is reconstructed from its K nearest neighbors. For each i, choose weights with $w_{ik} = 0$ outside the neighbor set $\mathcal{N}(i)$ and $\sum_k w_{ik} = 1$ by minimizing

$$\left\| x_i - \sum_{k\in\mathcal{N}(i)} w_{ik} x_k \right\|_2^2 .$$

The affine constraint makes the local reconstruction insensitive to translations of the neighborhood. Then keep the weights fixed and choose low-dimensional coordinates $y_i \in \mathbb{R}^r$ minimizing

$$\sum_{i=1}^{n} \left\| y_i - \sum_{k=1}^{n} w_{ik} y_k \right\|_2^2 .$$

Writing $W = (w_{ik})$ and Y for the coordinate matrix, this objective is

$$\operatorname{tr}\{Y^T (I - W)^T (I - W) Y\}.$$

The constant vector gives a trivial zero-eigenvalue direction, so the embedding discards it and uses the next low-eigenvalue directions. The resulting coordinates are mean-centered.

Method 4.2.13. *Local MDS*

Local MDS keeps the stress local while pushing non-neighboring points apart. Let $\mathcal{N}$ be the symmetric set of nearby pairs, for example pairs where either point is among the other's K nearest neighbors. A limiting form of the local stress is

$$S_{\text{local}}(z_1, \dots, z_n) = \sum_{(i,i') \in \mathcal{N}} \left(d_{ii'} - \|z_i - z_{i'}\|_2 \right)^2 - \tau \sum_{(i,i') \notin \mathcal{N}} \|z_i - z_{i'}\|_2 ,$$

with tuning parameter $\tau > 0$. The first term preserves local distances, while the second discourages unrelated parts of a curved manifold from collapsing together in the low-dimensional representation.

Definition 4.2.14. *Principal Curves and Surfaces*

A principal curve is a smooth one-dimensional curve $f(\lambda) \in \mathbb{R}^p$ that represents data by nearest points on the curve and satisfies the self-consistency condition

$$f(\lambda) = \mathbb{E}\{X \mid \lambda_f(X) = \lambda\},$$

where $\lambda_f(x)$ indexes the closest point of the curve to x. Principal surfaces use two or more coordinates, for example

$$f(\lambda_1, \lambda_2) = \big(f_1(\lambda_1, \lambda_2), \dots, f_p(\lambda_1, \lambda_2)\big).$$

For finite data, the usual alternating idea is to smooth each coordinate of X as a function of current curve or surface coordinates, then reproject the observations to the fitted manifold.

Remark 4.2.15. *SOMs Versus Principal Surfaces*

Self-organizing maps in Method 4.1.10 and principal surfaces both approximate data by a low-dimensional manifold-like object. Principal surfaces keep a smooth parametrization and can preserve local distances in the fitted coordinates, while SOMs use a finite lattice of shared prototypes. With many prototypes and kernel-type neighborhood weights, the batch SOM update resembles coordinate-wise surface smoothing, but the fitted objects remain conceptually different.

Method 4.2.16. *Kernel PCA*

Kernel PCA applies the PCA construction to an implicit feature matrix through its Gram matrix. Given a positive definite kernel k as in Definition 3.4.1, let $K_{ii'} = k(x_i, x_{i'})$ and double-center it:

$$\widetilde{K} = (I - M) K (I - M), \qquad M = \frac{1}{n} \mathbf{1}\mathbf{1}^T .$$

If $\widetilde{K} = U D^2 U^T$, then UD gives the principal-component coordinates for the training observations in the implicit feature space. Radial kernels of the form $\exp(-\|x - x'\|_2^2 / c)$ use the same locality signal as the similarity matrices in spectral clustering; kernel PCA extracts large-variance directions of a centered kernel matrix, while spectral clustering uses small-eigenvalue directions of a graph Laplacian.

Method 4.2.17. *Sparse Principal Components*

Sparse PCA modifies the loading directions so that only a subset of variables has substantial weight. A variance-based version chooses

$$\max_{v} v^T X^T X v \qquad \text{subject to} \qquad \sum_{j=1}^{p} |v_j| \le t, \quad v^T v = 1.$$

A reconstruction-based version for one component solves

$$\min_{\theta, v} \sum_{i=1}^{n} \left\| x_i - \theta v^T x_i \right\|_2^2 + \lambda \|v\|_2^2 + \lambda_1 \|v\|_1 \qquad \text{subject to} \qquad \|\theta\|_2 = 1.$$

The L_1 term encourages sparse loadings, while the reconstruction term keeps the component tied to the PCA geometry. For multiple components, one alternates between sparse regression-type updates for the loadings and an orthogonal Procrustes-type update for the score directions.

Remark 4.2.18. *Local versus Global Scaling*
Classical MDS is faithful when the important geometry is already captured by global Euclidean distances or centered inner products. Isomap, LLE, and local MDS instead use neighborhoods to protect the ordering or local shape of points along a curved manifold. Their quality therefore depends strongly on the neighborhood scale.

4.3 Matrix Factorization and Topic Models

Many unsupervised models express data as combinations of latent factors.

Definition 4.3.1. *Low-Rank Matrix Factorization*
For a data matrix $X \in \mathbb{R}^{m\times n}$, a rank-$r$ factorization approximates

$$X \approx UV^\top, \qquad U \in \mathbb{R}^{m\times r}, \quad V \in \mathbb{R}^{n\times r}.$$

The factors may represent latent coordinates, user-item preferences, or basis dictionaries.

Definition 4.3.2. *Nonnegative Matrix Factorization*
For nonnegative data $X \in \mathbb{R}_+^{n\times p}$, nonnegative matrix factorization approximates

$$X \approx WH, \qquad W \in \mathbb{R}_+^{n\times r}, \quad H \in \mathbb{R}_+^{r\times p}.$$

The constraint encourages additive parts-based representations when the data are counts, intensities, or images whose negative components would be hard to interpret.

Method 4.3.3. *Poisson NMF Multiplicative Updates*
When x_{ij} is modeled with mean $(WH)_{ij}$, the Poisson log-likelihood, up to constants independent of W, H, is

$$L(W,H) = \sum_{i=1}^{n}\sum_{j=1}^{p}\{x_{ij}\log(WH)_{ij} - (WH)_{ij}\}.$$

An alternating multiplicative scheme updates

$$w_{ik} \leftarrow w_{ik}\frac{\sum_{j=1}^{p} h_{kj}x_{ij}/(WH)_{ij}}{\sum_{j=1}^{p} h_{kj}}, \qquad h_{kj} \leftarrow h_{kj}\frac{\sum_{i=1}^{n} w_{ik}x_{ij}/(WH)_{ij}}{\sum_{i=1}^{n} w_{ik}}.$$

The iteration is a local factorization procedure; different starting values can lead to different decompositions.

Remark 4.3.4. *NMF Non-Uniqueness*
Nonnegativity alone does not make a factorization identifiable. Even when $X = WH$ holds exactly, a low-dimensional data cloud may leave open cones of possible nonnegative basis vectors, so many bases reconstruct the same observations. This is why the interpretability of NMF depends on both the data geometry and the optimization path.

Definition 4.3.5. *Archetypal Analysis*
Archetypal analysis also writes $X \approx WH$, but the rows of W are convex-combination weights over archetypes and the archetypes themselves are constrained to be convex combinations of observed rows:

$$H = BX, \qquad w_{ik} \geq 0, \quad \sum_{k=1}^{r} w_{ik} = 1, \qquad b_{ki} \geq 0, \quad \sum_{i=1}^{n} b_{ki} = 1.$$

Equivalently, it minimizes

$$J(W,B) = \|X - WBX\|_F^2$$

over the two sets of simplex-constrained weights. The archetypes are forced toward the convex hull of the data, so they represent extreme profiles rather than central prototypes.

Remark 4.3.6. *Archetypes, k-Means, and NMF*
If each row of W were restricted to have a single nonzero entry, the approximation would resemble a prototype assignment as in k-means. Unlike k-means, archetypal analysis represents each observation by a convex combination of several prototypes; unlike ordinary NMF, the prototype rows of H must themselves come from the convex hull of the observed data.

Definition 4.3.7. *Topic Model*
A topic model represents a document as a mixture of latent topics, with each topic a distribution over words. Latent Dirichlet allocation places Dirichlet priors on document-topic and topic-word distributions.

Remark 4.3.8. *Regularized Factors*
Low-rank, sparse, nonnegative, and probabilistic constraints turn an underdetermined factorization into a learning problem with interpretable inductive bias.

4.4 Representation Learning Objectives

Representation learning trains maps that make downstream prediction, reconstruction, comparison, or generation easier.

Definition 4.4.1. *Autoencoder*
An autoencoder is an encoder-decoder pair (e, d) trained to minimize a reconstruction criterion such as

$$\frac{1}{n}\sum_{i=1}^{n}\|x_i - d(e(x_i))\|_2^2 .$$

Bottlenecks, noise, sparsity, or probabilistic latent variables prevent the identity map from being the only useful solution.

Definition 4.4.2. *Sparse Representation Objective*
Given a dictionary W, sparse coding seeks representations h_i with small reconstruction error and many zero coordinates:

$$\min_{W,h_1,\dots,h_n}\sum_{i=1}^{n}\|x_i - Wh_i\|_2^2 + \lambda\|h_i\|_1 .$$

Definition 4.4.3. *Contrastive Objective*
Contrastive learning trains representations so related examples have high similarity while unrelated examples have low similarity. The negative examples and augmentation distribution are part of the statistical model of the task.

Remark 4.4.4. *From Regularization Source to Representation Learning*
The old sparse-representation notes appeared inside deep-learning regularization. They belong here as representation objectives, while their neural-network penalties reappear in the neural-network regularization section.

4.5 Association Rules and Density-Ratio Views

Association-rule analysis searches for simple descriptions of regions where the input distribution has relatively large probability. The classical market-basket setting is the binary special case, but the same idea extends to conjunctions of coordinate conditions.

Definition 4.5.1. *Conjunctive Region*
Let $X = (X_1, \dots, X_p)$, and let $\mathcal{S}_j$ be the set of possible values of X_j. For an index set $\mathcal{J} \subseteq \{1, \dots, p\}$ and subsets $s_j \subseteq \mathcal{S}_j$, define

$$R(\mathcal{J}, s) = \bigcap_{j\in\mathcal{J}}\{X_j \in s_j\}.$$

If $s_j = \mathcal{S}_j$, then the jth variable imposes no restriction and can be omitted from the rule. For quantitative variables the sets s_j are often intervals; for categorical variables they may be explicit subsets of levels.

Definition 4.5.2. *Market-Basket Item Sets*
In the binary market-basket formulation, each observation has item indicators $z_i = (z_{i1}, \dots, z_{iK}) \in \{0,1\}^K$. An item set is a subset $\mathcal{K} \subseteq \{1, \dots, K\}$, and its empirical support or prevalence is

$$T(\mathcal{K}) = \frac{1}{n}\sum_{i=1}^{n}\prod_{k\in\mathcal{K}} z_{ik}.$$

Observation i contains $\mathcal{K}$ exactly when every item in $\mathcal{K}$ is present. Finite categorical variables can be placed in this form by using one dummy indicator for each attainable level, so a data set with supports $\mathcal{S}_1, \dots, \mathcal{S}_p$ yields $K = \sum_{j=1}^{p}|\mathcal{S}_j|$ possible item indicators.

Definition 4.5.3. *Association Rule, Confidence, and Lift*
An association rule partitions an item set into disjoint nonempty antecedent and consequent sets,

$$A \Rightarrow B, \qquad A \cap B = \emptyset.$$

Its support is $T(A \cup B)$, its confidence is

$$C(A \Rightarrow B) = \frac{T(A \cup B)}{T(A)},$$

and its lift is

$$L(A \Rightarrow B) = \frac{C(A \Rightarrow B)}{T(B)}.$$

Thus confidence estimates the conditional prevalence of B among observations containing A, while lift compares that confidence with the marginal prevalence of B.

Example 4.5.4. *A Market-Basket Rule*
For the rule

$$\{\text{peanut butter}, \text{jelly}\} \Rightarrow \{\text{bread}\},$$

support 0.03 means that all three items appear together in about 3% of baskets. Confidence 0.82 means that among baskets containing peanut butter and jelly, about 82% also contain bread. If bread has marginal support about 0.43, the lift is about 0.82/0.43, so the rule predicts bread substantially more often than its baseline prevalence.

Algorithm 17: Apriori Association-Rule Search

Require: Binary item data $z_1, \dots, z_n$, support threshold t, confidence threshold c
Ensure: Rules with support above t and confidence above c
Compute supports for all one-item sets and retain $\mathcal{F}_1 = \{\mathcal{K} : |\mathcal{K}| = 1,\ T(\mathcal{K}) > t\}$
Set $m \leftarrow 2$
while $\mathcal{F}_{m-1} \neq \emptyset$ **do**
Form candidate item sets of size m whose every size-$(m-1)$ subset lies in $\mathcal{F}_{m-1}$
Scan the data to compute $T(\mathcal{K})$ for each candidate
Retain $\mathcal{F}_m = \{\mathcal{K} : |\mathcal{K}| = m,\ T(\mathcal{K}) > t\}$
Set $m \leftarrow m + 1$
end while
For each retained item set $\mathcal{K}$, test nonempty proper antecedents $A \subset \mathcal{K}$
Report $A \Rightarrow \mathcal{K} \setminus A$ when $C(A \Rightarrow \mathcal{K} \setminus A) > c$

Remark 4.5.5. *Apriori Monotonicity and Limits*
Algorithm 17 relies on the monotonicity

$$\mathcal{L} \subseteq \mathcal{K} \quad \Longrightarrow \quad T(\mathcal{L}) \geq T(\mathcal{K}).$$

An infrequent subset rules out all of its supersets, making the search feasible when the support threshold leaves only a small survivor set. The same threshold can also hide rare-consequent rules that have high confidence or high lift but low joint support.

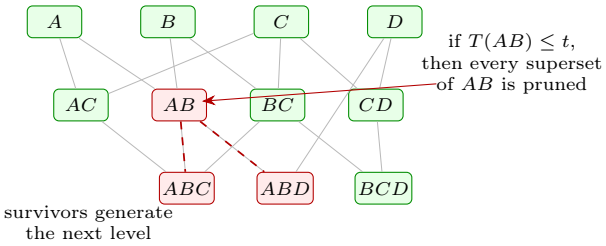


Figure 241: Apriori pruning uses the monotonicity of support: once an item set is below the support threshold, every item set containing it is also below threshold and need not be scanned as a candidate.

Method 4.5.6. *Generalized Association Rules*
For a conjunctive region $R(\mathcal{J}, s)$ from Definition 4.5.1, the empirical support is

$$\widehat{\Pr}\{R(\mathcal{J}, s)\} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x_{ij} \in s_j \text{ for every } j \in \mathcal{J}\}.$$

Generalized rules allow intervals for quantitative variables and multi-level subsets for categorical variables. Because an exhaustive search over all such regions is typically too ambitious, one uses heuristic rule learners and then checks both the fitted score and the observed data support.

Method 4.5.7. *Unsupervised Learning as Supervised Density-Ratio Estimation*
Let g be the data density and let g_0 be a chosen reference density. Draw a reference sample from g_0, pool it with the observed data, and weight the two samples so that the two labels have equal total mass. If $Y = 1$ marks a data observation and $Y = 0$ marks a reference observation, then

$$\mu(x) = \mathbb{E}[Y \mid X = x] = \frac{g(x)}{g(x) + g_0(x)} = \frac{g(x)/g_0(x)}{1 + g(x)/g_0(x)}.$$

A probabilistic classifier fit to the pooled sample gives an estimate $\widehat{\mu}(x)$, hence the density-ratio estimate

$$\frac{\widehat{g}(x)}{g_0(x)} = \frac{\widehat{\mu}(x)}{1-\widehat{\mu}(x)}.$$

A uniform g_0 emphasizes nonuniform high-density regions; a moment-matched Gaussian g_0 emphasizes departures from joint normality; a product of marginal data densities emphasizes departures from independence.

Remark 4.5.8. *Rule Learners for Density Ratios*
After the density-ratio construction in Method 4.5.7, supervised learners that return conjunctive regions can be used to search for generalized item sets. Terminal nodes of a decision tree from Definition 3.6.1 and boxes from PRIM in Method 3.7.4 both have the form of intersections of coordinate conditions. This cross-reference is conceptual: the unsupervised target is a high density ratio together with adequate data support, not ordinary predictive risk.

4.6 Graph Ranking and PageRank

Graph ranking turns link structure into a numerical importance score. PageRank is a spectral example: a page receives weight from incoming links, and a link from an already important page contributes more than a link from an unimportant one.

Definition 4.6.1. *PageRank Equations*
For N pages, let $L_{ij} = 1$ if page j links to page i, and let $L_{ij} = 0$ otherwise. Let

$$c_j = \sum_{i=1}^{N} L_{ij}$$

be the number of outgoing links from page j, and assume $c_j > 0$ for the displayed normalization. For damping parameter $0 < d < 1$, the PageRank scores satisfy

$$p_i = (1-d) + d\sum_{j=1}^{N} \frac{L_{ij}}{c_j} p_j.$$

Each page distributes its current score evenly over its outgoing links, while the term $1-d$ gives every page a baseline score.

Proposition 4.6.2. *PageRank Normalization*
Any solution of the PageRank equations in Definition 4.6.1 has total score $\sum_{i=1}^{N} p_i = N$.

Proof. Summing the defining equations over i gives

$$\sum_i p_i = N(1-d) + d\sum_j \frac{p_j}{c_j}\sum_i L_{ij} = N(1-d) + d\sum_j p_j.$$

Since $0 < d < 1$, this forces $\sum_i p_i = N$. □

Method 4.6.3. *Eigenvector Form*
Let $\mathbf{e}$ be the vector of ones and $D_c = \operatorname{diag}(c_1, \ldots, c_N)$. With the normalization $\mathbf{e}^T\mathbf{p} = N$, the PageRank equations are

$$\mathbf{p} = \left((1-d)\frac{\mathbf{e}\mathbf{e}^T}{N} + dLD_c^{-1}\right)\mathbf{p} = A\mathbf{p}.$$

The matrix A is column-stochastic, and its teleportation term has positive entries. Thus PageRank is the positive eigenvector for eigenvalue 1, normalized to have average entry 1. In Markov-chain language this is a stationary distribution after converting between column-vector and row-vector conventions; compare Definition 5.4.15 and Theorem 5.4.16.

Algorithm 18: PageRank by Power Iteration

Require: Link matrix L, positive outlink counts c_j, damping parameter $0 < d < 1$, initial vector $\mathbf{p}_0 > 0$
Ensure: PageRank vector normalized by $\mathbf{e}^T\mathbf{p} = N$
  Form $A = (1-d)\mathbf{e}\mathbf{e}^T/N + dLD_c^{-1}$
  repeat
    $\mathbf{p}_k \leftarrow A\mathbf{p}_{k-1}$
    $\mathbf{p}_k \leftarrow N\mathbf{p}_k/(\mathbf{e}^T\mathbf{p}_k)$
  until the iterates stabilize

Remark 4.6.4. *Random-Surfer Normalization*
Some presentations put $(1-d)/N$ in the scalar recurrence instead of $1-d$ and normalize scores to sum to 1. That version has the same random-surfer interpretation at probability scale: with probability d the surfer follows an outgoing link, and with probability $1-d$ the surfer jumps to a uniformly chosen page. The equations above use the equivalent average-score normalization $\mathbf{e}^T\mathbf{p} = N$.

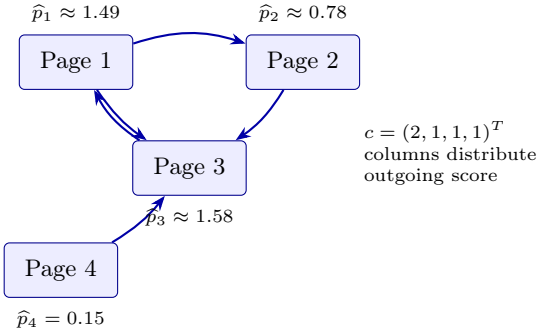


Figure 242: The four-page PageRank example. The third page receives score from three incoming links, while the fourth page has no incoming links and keeps only the baseline score.

Example 4.6.5. *Four-Page Network*
For the network in Figure 242, the link matrix is

$$L = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathbf{c} = (2,1,1,1)^T,$$

with damping $d = 0.85$, the PageRank solution is approximately

$$\widehat{\mathbf{p}} = (1.49, 0.78, 1.58, 0.15)^T.$$

The fourth page has no incoming links in this small network, so it receives the baseline score $1-d = 0.15$.

4.7 Latent Factor and Component Analysis

Latent factor methods model observed coordinates as mixtures of unobserved sources. The main question is which structure makes the sources interpretable: covariance structure, statistical independence, or non-Gaussian projections.

Definition 4.7.1. *Factor Analysis*
A latent-factor model represents observed variables as noisy linear mixtures of fewer hidden factors:

$$X = AS + \varepsilon, \qquad \mathrm{Cov}(S) = I, \quad \mathrm{Cov}(\varepsilon) = D_\varepsilon.$$

At the covariance level this gives

$$\Sigma = AA^T + D_\varepsilon,$$

where the columns of A are factor loadings. The model is rotationally non-identifiable: replacing A by AR^T and S by RS for an orthogonal R preserves the same covariance representation. Factor rotations can therefore change interpretability without changing the fitted covariance structure.

Definition 4.7.2. *Independent Component Analysis*
Independent component analysis keeps the same linear mixture form $X = AS$, but assumes the latent components S_j are statistically independent rather than merely uncorrelated. After whitening the data so that $\mathrm{Cov}(X) = I$, one may take A orthogonal and search for a rotation such that

$$Y = A^T X$$

has components as independent and as non-Gaussian as possible. The non-Gaussian condition is essential: Gaussian independent components are identifiable only up to rotations.

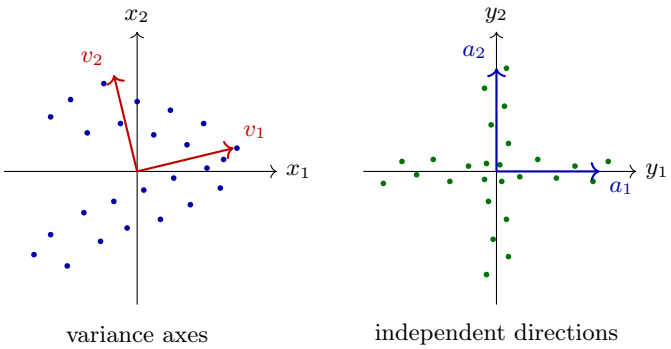


Figure 243: After whitening, PCA has no preferred orientation inside equal-variance directions. ICA searches among rotations for coordinates whose marginal distributions reveal non-Gaussian source structure.

Method 4.7.3. *Entropy View of ICA*
For a continuous random variable with density g, the differential entropy is

$$H(Y) = -\int g(y) \log g(y)\, dy.$$

For a random vector $Y = (Y_1, \ldots, Y_p)$, the mutual information

$$I(Y) = \sum_{j=1}^{p} H(Y_j) - H(Y)$$

measures departure from a product density. When X is whitened and $Y = A^T X$ with A orthogonal, $H(Y) = H(X)$, so minimizing $I(Y)$ is equivalent to minimizing the sum of marginal entropies. Since Gaussian variables maximize entropy among equal-variance variables, ICA criteria often search for orthogonal projections with large departures from Gaussianity, for example through negentropy approximations.

Remark 4.7.4. *Projection Pursuit and ICA*
Exploratory projection pursuit searches for one- or two-dimensional projections that look least Gaussian, revealing clusters, long tails, or other structure hidden in high dimensions. In the representation here, projection indices have the same form as ICA non-Gaussianity criteria applied to $Y_j = a_j^T X$, though projection-pursuit directions need not be constrained to be mutually orthogonal. For supervised ridge-function models using learned projected coordinates, see Definition 6.1.3.

Method 4.7.5. *Product-Density ICA*
A direct ICA approach models each independent component density as a tilted Gaussian

$$f_j(s_j) = \phi(s_j) \exp(g_j(s_j)),$$

where ϕ is the standard Gaussian density and g_j is a smooth contrast function chosen so that f_j is normalized. For whitened data, the regularized log-likelihood separates into one-dimensional smooth density fits plus an orthogonality-constrained search over A. The roughness penalty on each g_j controls how far the fitted marginal density can move away from the Gaussian baseline.

Algorithm 19: Product-Density ICA Alternation

Require: Whitened observations $x_1, \ldots, x_n$, initial orthogonal matrix A
Ensure: Orthogonal unmixing matrix A and components $s_{ij} = a_j^T x_i$
  repeat
    for each component j **do**
      Fit a smooth tilted-Gaussian density $f_j(s) = \phi(s) \exp(g_j(s))$ to the current scores $a_j^T x_i$
      Update $a_j \leftarrow \mathbb{E}\{X g_j'(a_j^T X)\} - \mathbb{E}\{g_j''(a_j^T X)\} a_j$
    end for
    Orthogonalize A: if $A = UDV^T$ is its SVD, set $A \leftarrow UV^T$
  until the directions stabilize

5 Probabilistic Graphical and Causal Learning

5.1 Directed and Undirected Graphical Models

Graphical models use graph structure to encode conditional independence and to organize high-dimensional probability laws.

Definition 5.1.1. *Directed Graphical Model*
For a directed acyclic graph G with vertices $1, \dots, d$, a distribution factorizes according to G if

$$p(x_1, \dots, x_d) = \prod_{j=1}^{d} p(x_j \mid x_{\mathrm{pa}(j)}),$$

where $\mathrm{pa}(j)$ denotes the parents of vertex j.

Definition 5.1.2. *Undirected Graphical Model*
For an undirected graph $G = (V, E)$, a positive distribution is Markov with respect to G when missing edges encode conditional independence. Equivalently, for positive distributions, the joint law factorizes over maximal cliques:

$$p(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}} \psi_C(x_C),$$

where $\mathcal{C}$ is the set of maximal cliques, each ψ_C is a nonnegative clique potential, and Z is the normalizing constant. The potentials score compatible local configurations; they need not themselves be marginal densities.

Definition 5.1.3. *Markov Graph Vocabulary*
Vertices j and k are adjacent when $(j, k) \in E$. A path is a sequence of adjacent vertices, and a set C separates sets A and B when every path from A to B meets C. In a Markov graph,

$$C \text{ separates } A \text{ and } B \quad \Longrightarrow \quad X_A \perp\!\!\!\perp X_B \mid X_C.$$

A clique is a complete subgraph, and a maximal clique is one that is not contained in a larger clique.

Proposition 5.1.4. *Gaussian Graphical Zeros*
Let $X \sim N(\mu, \Sigma)$ and let $\Theta = \Sigma^{-1}$. Partition $X = (Z, Y)$, where Y is one coordinate and Z contains the rest. Then

$$\mathbb{E}[Y \mid Z = z] = \mu_Y + (z - \mu_Z)^T \Sigma_{ZZ}^{-1} \sigma_{ZY}, \qquad \Sigma_{ZZ}^{-1} \sigma_{ZY} = -\theta_{ZY} / \theta_{YY}.$$

Consequently zero entries of the precision matrix encode missing conditional linear dependences. In particular, for a Gaussian graphical model, $\theta_{jk} = 0$ is the edge-absence condition corresponding to

$$X_j \perp\!\!\!\perp X_k \mid X_{-(j,k)}.$$

Definition 5.1.5. *Ising Model*
For binary variables $X \in \{0, 1\}^p$ on an edge set E, a pairwise Ising model has probabilities of the form

$$p_\Theta(x) = \exp\left\{ \sum_{(j,k) \in E} \theta_{jk} x_j x_k - \Phi(\Theta) \right\},$$

with a constant node included when intercept or main-effect terms are needed. Its nodewise conditional distributions are logistic:

$$\mathbb{P}_\Theta(X_j = 1 \mid X_{-j} = x_{-j}) = \frac{1}{1 + \exp\{-\theta_{j0} - \sum_{k:(j,k) \in E} \theta_{jk} x_k\}}.$$

Thus an edge parameter measures conditional dependence between two binary nodes after the remaining nodes are held fixed.

Definition 5.1.6. *Restricted Boltzmann Machine*
A restricted Boltzmann machine is a binary Markov network with visible and hidden layers and no within-layer edges. The restriction makes units in one layer conditionally independent given the other layer, so Gibbs updates can sample an entire visible or hidden layer in parallel using the logistic conditional form from Definition 5.1.5. In supervised uses, the visible layer may be split into feature units and label units.

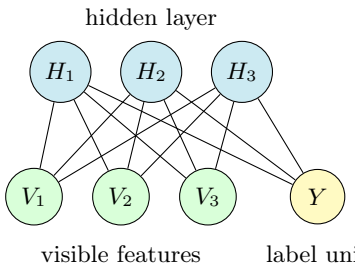


Figure 244: A restricted Boltzmann machine has bipartite visible-hidden connectivity; conditional on one layer, the units in the other layer decouple.

Remark 5.1.7. *Separation*
Graph separation rules translate paths in the graph into conditional independence statements. Directed models use d-separation; undirected models use ordinary graph separation. The absence of an edge is only a pairwise statement, while separation can imply larger conditional independences by cutting all paths between two groups of vertices.

5.2 Inference and Learning in Graphical Models

Inference in a graphical model computes marginal, conditional, or most likely latent quantities from the joint law encoded by the graph.

Definition 5.2.1. *Inference Tasks*
Common tasks include computing marginals $p(x_A)$, conditionals $p(x_A \mid x_B)$, maximum a posteriori assignments, and posterior distributions over latent variables.

Method 5.2.2. *Message Passing*
On trees, belief propagation sends local messages along edges and computes exact marginals. On loopy graphs, analogous iterations often serve as approximations.

Definition 5.2.3. *Variational Inference*
Variational inference approximates an intractable posterior $p(z \mid x)$ by a tractable distribution $q(z)$, commonly by minimizing $\mathrm{KL}(q(z)\|p(z \mid x))$ or maximizing an evidence lower bound.

Remark 5.2.4. *Learning Parameters and Structure*
Parameter learning uses likelihood or regularized likelihood when the graph is fixed. Structure learning searches over graphs and therefore combines statistical scoring with combinatorial optimization.

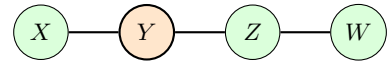
Method 5.2.5. *Gaussian Graphical Model Learning*
For centered Gaussian data with empirical covariance S, the precision matrix $\Theta = \Sigma^{-1}$ has log-likelihood, up to constants,

$$\ell(\Theta) = \log\det\Theta - \operatorname{trace}(S\Theta), \qquad \Theta \succ 0.$$

If the graph is known, missing edges impose equality constraints $\theta_{jk} = 0$. If the graph is unknown, the graphical lasso estimates a sparse precision matrix by maximizing the penalized criterion

$$\log\det\Theta - \operatorname{trace}(S\Theta) - \lambda \|\Theta\|_1 ,$$

usually leaving the diagonal unpenalized in the coordinate updates. This turns graph selection into shrinkage of inverse-covariance entries.



Y separates X from $\{Z, W\}$

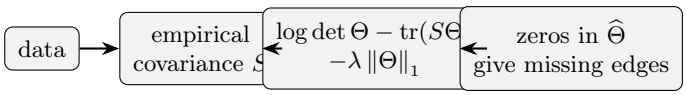


Figure 245: Undirected graph separation gives conditional independence statements, while graphical-lasso learning estimates a sparse precision matrix whose zero entries remove edges.

Algorithm 20: Graphical Lasso by Block Coordinate Descent

Require: Empirical covariance $S \in \mathbb{R}^{p\times p}$, penalty $\lambda \geq 0$
Ensure: Covariance estimate W and sparse precision estimate $\widehat{\Theta} = W^{-1}$
Initialize $W \leftarrow S + \lambda I$ and keep the diagonal fixed
repeat
  for $j = 1, \ldots, p$ **do**
    Partition W into W_{11}, w_{12}, and w_{22} by isolating row and column j; partition S similarly to get s_{12}
    Define $\mathcal{S}_\lambda(z) = \operatorname{sign}(z)(|z| - \lambda)_+$
    repeat
      for $k = 1, \ldots, p-1$ **do**

$$\beta_k \leftarrow \frac{\mathcal{S}_\lambda\left(s_{12,k} - \sum_{\ell\neq k}(W_{11})_{k\ell}\beta_\ell\right)}{(W_{11})_{kk}}$$

end for
until β has converged for this block
Update $w_{12} \leftarrow W_{11}\beta$ and symmetrically update row and column j of W
end for
until W has converged
In the final cycle recover $\widehat{\theta}_{12} = -\widehat{\beta}\widehat{\theta}_{22}$ with $1/\widehat{\theta}_{22} = w_{22} - w_{12}^T\widehat{\beta}$

Method 5.2.6. *Discrete Markov Network Learning*
For the Ising model in Definition 5.1.5, the likelihood equations match empirical edge products to their model expectations:

$$\widehat{\mathbb{E}}(X_jX_k) - \mathbb{E}_\Theta(X_jX_k) = 0.$$

Exact evaluation of the model expectation requires summing over all binary configurations, so it becomes expensive as p grows. For smaller graphs one can use log-linear Poisson modeling, gradient or Newton methods, or iterative proportional fitting. For larger or hidden-node models, mean-field approximations and Gibbs sampling replace exact expectations.

With hidden nodes, the empirical term is itself conditional on the observed variables. Restricted Boltzmann machines simplify this step because hidden units are conditionally independent given visible units and visible units are conditionally independent given hidden units. Contrastive-divergence training uses this structure by starting the Markov chain at observed data and running only a small number of alternating Gibbs steps to obtain an approximate learning direction.

5.3 Causal Graphs, Interventions, and Identifiability

Causal learning asks which predictions remain valid under interventions, not merely under passive observation.

Definition 5.3.1. *Structural Causal Model*
A structural causal model specifies variables by assignments

$$X_j = f_j(X_{\mathrm{pa}(j)}, U_j),$$

where the exogenous variables U_j represent disturbances. The associated directed graph records direct causal inputs.

Definition 5.3.2. *Intervention*
An intervention $\mathrm{do}(X_j = x)$ replaces the structural assignment for X_j by the constant x and leaves the remaining assignments unchanged. The resulting law is an interventional distribution.

Definition 5.3.3. *Identifiability*
A causal quantity is *identifiable* if it is uniquely determined by the observational distribution together with the assumed causal graph or model class.

Remark 5.3.4. *Causal and Predictive Targets*
A predictor can be accurate under the training distribution while failing after an intervention. Causal assumptions specify which conditional relationships are stable enough to transport across regimes.

5.4 Counterfactuals and Policy Evaluation

Policy learning and evaluation combine prediction with causal questions about actions not actually taken.

Definition 5.4.1. *Potential Outcomes*
For treatment $A \in \{0, 1\}$, the potential outcomes $Y(1)$ and $Y(0)$ denote the outcomes that would occur under each treatment level. The individual causal effect is $Y(1) - Y(0)$, while average effects take expectations over a population.

Definition 5.4.2. *Ignorability and Propensity*
Treatment assignment is ignorable given covariates X if $(Y(0), Y(1))$ is conditionally independent of A given X. The propensity score is $e(X) = P(A = 1 \mid X)$.

Definition 5.4.3. *Off-Policy Evaluation*
Off-policy evaluation estimates the value of a target policy using data collected under another policy. Importance weighting corrects the action distribution when the logging probabilities are known and overlap holds.

Remark 5.4.4. *Feedback*
When predictions influence actions, the data-generating process changes. This connects causal learning, reinforcement learning, and deployment monitoring.

6 Neural Network Foundations

6.1 Network Architectures and Expressivity

The migrated deep-feedforward-network notes describe neural networks as learned feature maps followed by prediction heads.

Definition 6.1.1. *Deep Feedforward Network*
A deep feedforward network represents a map

$$f(x;\theta) = f_L \circ f_{L-1} \circ \cdots \circ f_1(x),$$

where typical layers have the form

$$h^{(k)} = g^{(k)}(W^{(k)} h^{(k-1)} + b^{(k)}).$$

Training chooses parameters θ to make this map small-risk for a task.

Remark 6.1.2. *Learned Feature Maps*
Classical nonlinear prediction fixes a feature map $\phi(x)$ and learns a linear head. Deep learning learns both the representation $\phi(x;\theta)$ and the head, usually losing convexity but gaining adaptable features.

Definition 6.1.3. *Projection Pursuit and Ridge Functions*
A projection-pursuit regression model represents a response surface by

$$f(x) = \sum_{m=1}^{M} g_m(\omega_m^T x), \qquad \|\omega_m\|_2 = 1.$$

Each term is a *ridge function*: it varies only along the one-dimensional derived coordinate $\omega_m^T x$. This is an additive model in learned directions rather than in the original input coordinates.

Definition 6.1.4. *Output and Hidden Units*
Linear output units model conditional means, sigmoid units parameterize Bernoulli probabilities, and softmax units parameterize distributions over finitely many classes. Hidden units commonly use rectified linear, leaky rectified, maxout, hyperbolic tangent, or other nonlinear activations.

Definition 6.1.5. *Single Hidden Layer Network*
A classical single-hidden-layer feedforward network forms

$$Z_m = \sigma(\alpha_{0m} + \alpha_m^T x), \qquad m = 1, \ldots, M,$$

then computes

$$T_k = \beta_{0k} + \beta_k^T Z, \qquad f_k(x) = g_k(T), \qquad k = 1, \ldots, K.$$

For regression, $g_k(T) = T_k$ is common. For K-class classification, the softmax output

$$g_k(T) = \frac{\exp(T_k)}{\sum_{\ell=1}^{K} \exp(T_\ell)}$$

gives positive class-probability estimates summing to one.

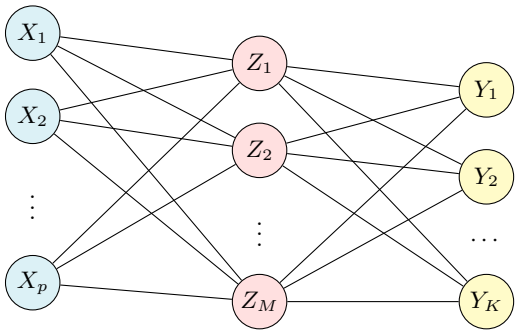


Figure 246: A single-hidden-layer feedforward network learns hidden features Z_m and then combines them in output scores T_k.

Remark 6.1.6. *Neural Networks as Ridge Expansions*
With one hidden layer and scalar output, the network is a projection-pursuit-type expansion with simpler

learned ridge functions. If $\omega_m = \alpha_m / \|\alpha_m\|_2$, then a hidden unit contributes a term of the form

$$\beta_m \sigma(\alpha_{0m} + \alpha_m^T x) = \beta_m \sigma(\alpha_{0m} + \|\alpha_m\|_2 \, \omega_m^T x).$$

Thus the direction, threshold, scale, and output coefficient are learned together.

Theorem 6.1.7. *Universal Approximation Theorem Form*
A feedforward network with a linear output layer and at least one sufficiently wide hidden layer with a suitable non-polynomial activation can approximate every continuous function on a compact subset of $\mathbb{R}^d$ arbitrarily well in the uniform norm.

Remark 6.1.8. *Depth and Expressivity*
Universal approximation is an existence result, not a training guarantee. Depth can represent some compositional and piecewise-linear functions with far fewer units than a shallow network.

6.2 Backpropagation and Automatic Differentiation

Backpropagation is reverse-mode differentiation applied to the computational graph of the network and loss.

Definition 6.2.1. *Computational Graph*
A computational graph is a directed acyclic graph whose nodes are intermediate variables and whose edges record functional dependence. Neural-network training differentiates a scalar loss through this graph with respect to many parameters.

Method 6.2.2. *Backpropagation*
For a multilayer network, the forward pass computes activations layer by layer. The backward pass propagates adjoints from the loss to the parameters using the chain rule. For layer preactivation $z^{(k)}$ and activation $h^{(k)}$, the local Jacobians determine how the error signal at layer $k+1$ becomes the error signal at layer k.

Method 6.2.3. *Single-Hidden-Layer Backpropagation Equations*
For the single-hidden-layer model in Definition 6.1.5, write

$$a_{mi} = \alpha_{0m} + \alpha_m^T x_i, \qquad z_{mi} = \sigma(a_{mi}),$$
$$t_{ki} = \beta_{0k} + \beta_k^T z_i, \qquad f_k(x_i) = g_k(t_{ki}).$$

For a componentwise output map, use the squared-error contribution

$$R_i = \sum_{k=1}^{K} \{y_{ik} - f_k(x_i)\}^2.$$

Define output-layer errors

$$\delta_{ki} = -2\{y_{ik} - f_k(x_i)\} g_k'(t_{ki}).$$

Then the output-layer derivatives are

$$\frac{\partial R_i}{\partial \beta_{km}} = \delta_{ki} z_{mi}, \qquad \frac{\partial R_i}{\partial \beta_{0k}} = \delta_{ki}.$$

The hidden-layer error is the output error back-propagated through the weights and local activation derivative:

$$s_{mi} = \sigma'(a_{mi}) \sum_{k=1}^{K} \beta_{km} \delta_{ki}.$$

Thus the input-to-hidden derivatives are

$$\frac{\partial R_i}{\partial \alpha_{m\ell}} = s_{mi} x_{i\ell}, \qquad \frac{\partial R_i}{\partial \alpha_{0m}} = s_{mi}.$$

Summing these derivatives over the training cases gives the batch gradient. A forward pass computes a, z, t, and f; a backward pass computes δ and s, then accumulates the displayed products.

Algorithm 21: Minibatch Backpropagation Step

Require: Minibatch B, parameters θ, learning rate η
Compute predictions $f(x;\theta)$ for all $(x,y) \in B$
Compute minibatch loss $\widehat{L}_B(\theta)$
Run reverse-mode automatic differentiation to compute $g = \nabla_\theta \widehat{L}_B(\theta)$
Update parameters $\theta \leftarrow \theta - \eta g$

Remark 6.2.4. *Automatic Differentiation*
Automatic differentiation is neither numerical finite differencing nor purely symbolic simplification. It evaluates exact chain-rule derivatives of the implemented computation up to floating-point arithmetic.

6.3 Optimization for Deep Learning

The old optimization notes are migrated here only where the algorithms are used as training methods for neural networks. General numerical optimization belongs to Part 14.

Definition 6.3.1. *Neural Empirical Objective*
For network $f(x;\theta)$ and per-example loss L, training commonly minimizes

$$J(\theta) = \frac{1}{n}\sum_{i=1}^{n} L(f(x_i;\theta), y_i).$$

The population target replaces the empirical average by expectation under the data-generating distribution.

Method 6.3.2. *Minibatch Stochastic Gradient Descent*
At iteration k, sample a minibatch B_k and use

$$g_k = \frac{1}{|B_k|}\sum_{i\in B_k} \nabla_\theta L(f(x_i;\theta_k), y_i), \qquad \theta_{k+1} = \theta_k - \eta_k g_k.$$

The minibatch gradient is noisy but cheaper than a full-batch gradient.

Definition 6.3.3. *Momentum and Adaptive Methods*
Momentum keeps an exponentially decaying velocity of past gradients. AdaGrad, RMSProp, and Adam rescale updates coordinatewise using accumulated first or second moment estimates of gradients.

Remark 6.3.4. *Deep Optimization Challenges*
Ill-conditioning, saddle regions, plateaus, exploding gradients, vanishing gradients, and noisy objectives make neural-network optimization different from convex empirical risk minimization. Initialization, normalization, clipping, skip connections, and learning-rate schedules are therefore part of the training method.

Remark 6.3.5. *Second-Order Methods*
Newton, conjugate-gradient, BFGS, and L-BFGS ideas can be adapted to learning problems, but full Hessian methods are usually too costly for large networks. Their practical role is often indirect: preconditioning, curvature approximations, or analysis.

6.4 Regularization, Normalization, and Generalization

Deep-learning regularization combines penalties, stochastic training effects, architecture, and validation-based stopping.

Definition 6.4.1. *Parameter-Norm Regularization*
Given objective $J(\theta)$, a regularized neural-network objective is

$$\widetilde{J}(\theta) = J(\theta) + \alpha\Omega(\theta), \qquad \alpha \geq 0.$$

Weight decay uses an ℓ^2 penalty on weights, while ℓ^1 penalties encourage sparsity. For the single-hidden-layer notation of Definition 6.1.5, a typical weight-decay penalty is

$$\Omega(\theta) = \sum_{k,m} \beta_{km}^2 + \sum_{m,\ell} \alpha_{m\ell}^2,$$

usually leaving intercept weights unpenalized.

Definition 6.4.2. *Dropout*
Dropout trains subnetworks obtained by multiplying input or hidden units by random Bernoulli masks. It can be interpreted as optimizing an average loss over a large tied-parameter ensemble.

Method 6.4.3. *Early Stopping*
During training, track validation loss and return the parameter value with the best validation performance. Early stopping limits the reachable region of parameter space and can act like an implicit regularizer.

Remark 6.4.4. *Starting Values, Scaling, and Multiple Fits*
Neural-network objectives are generally nonconvex. Exact zero initialization can freeze hidden units by symmetry, while very large initial weights can put sigmoid units into nearly saturated regions. Small random starting weights and standardized inputs are therefore part of the model specification, not merely software details. Because different starts can reach different local minima, it is common to try several starts and keep the fitted model with the best validation or penalized training criterion.

Remark 6.4.5. *Hidden Units and Weight Decay*
With too few hidden units, the network may not represent the relevant nonlinearities; with too many, the extra weights can overfit unless controlled. Weight decay and early stopping both shrink the fitted model toward a simpler, more nearly linear regime. In practice the number of hidden units and the penalty or stopping time are tuning parameters, often chosen by validation.

Definition 6.4.6. *Batch Normalization*
Batch normalization replaces a minibatch of activations H by standardized activations

$$H' = \frac{H - \mu}{\sigma},$$

then often applies learned scale and shift parameters. At evaluation time, running averages of μ and σ replace minibatch statistics.

Remark 6.4.7. *Augmentation and Noise*
Data augmentation, label smoothing, noise injection, adversarial training, and multitask learning regularize by changing the effective training distribution or by constraining learned representations.

7 Modern Deep Models

7.1 Convolutional and Geometric Architectures

The convolutional-network source is migrated here because convolution is an architectural prior for data with grid, group, or geometric structure.

Definition 7.1.1. *Convolution*
For functions on the real line, convolution is

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau)\, d\tau.$$

For sequences on $\mathbb{Z}$, the discrete convolution is

$$(f * g)[n] = \sum_{m=-\infty}^{\infty} f[m] g[n - m].$$

Definition 7.1.2. *Convolutional Neural Network*
A convolutional neural network uses convolution or cross-correlation in at least one layer. Neural-network libraries often implement cross-correlation while referring to it as convolution.

Remark 7.1.3. *Sparse Interactions, Sharing, and Equivariance*
Convolutional layers use small kernels, share parameters across spatial locations, and produce representations that are equivariant to translations. Pooling then creates approximate invariance over local neighborhoods.

Definition 7.1.4. *Pooling, Padding, and Stride*
Pooling replaces nearby activations by summary statistics such as maxima or averages. Padding controls boundary behavior, and stride controls the spacing between evaluated output locations.

Remark 7.1.5. *Geometric Extensions*
Graph neural networks, equivariant networks, and manifold-based architectures generalize the convolutional idea: encode the symmetries and locality of the domain directly into the model class.

7.2 Sequence Models and Attention

Sequence models process ordered data where context and position matter.

Definition 7.2.1. *Recurrent Neural Network*
A recurrent neural network updates a hidden state by

$$h_t = \phi(W_x x_t + W_h h_{t-1} + b),$$

then predicts from h_t. Sharing W_h across time lets the model process variable-length sequences.

Remark 7.2.2. *Long-Term Dependencies*
Repeated multiplication by transition Jacobians can cause gradients to vanish or explode. Gated recurrent units, long short-term memory units, clipping, and skip-like paths mitigate this problem.

Definition 7.2.3. *Attention*
Given queries Q, keys K, and values V, scaled dot-product attention has the form

$$\operatorname{Attn}(Q, K, V) = \operatorname{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V.$$

It lets each position form a data-dependent weighted average of value vectors.

Remark 7.2.4. *Attention as Routing*
Attention replaces a fixed local recurrence by content-dependent communication among tokens or locations. Masks and positional encodings specify which interactions are allowed and how order is represented.

7.3 Transformers and Large-Scale Training

Transformers combine attention, feedforward blocks, normalization, and residual connections into architectures that scale well on sequence and multimodal data.

Definition 7.3.1. *Transformer Block*
A transformer block applies multi-head self-attention, residual connections, normalization layers, and a positionwise feedforward network. Stacking these blocks yields deep contextual representations.

Definition 7.3.2. *Self-Attention*
Self-attention uses the same sequence to form queries, keys, and values. Multi-head attention runs several attention maps in parallel and recombines their outputs.

Remark 7.3.3. *Large-Scale Training*
Large-scale deep learning is constrained by optimization, data quality, memory, communication, and evaluation. Batch size, learning-rate schedules, mixed precision, parallelism, and checkpointing are part of the mathematical and systems design of training.

Remark 7.3.4. *Inductive Bias*
Transformers have weaker built-in locality bias than convolutional networks but stronger capacity for long-range interaction. Domain-specific variants add locality, recurrence, sparsity, or equivariance when the task demands it.

7.4 Generative Models

Generative modeling learns distributions, simulators, or conditional samplers rather than only point predictions.

Definition 7.4.1. *Latent-Variable Generative Model*
A latent-variable model samples $Z \sim p(z)$ and then $X \sim p_\theta(x \mid z)$. Learning fits θ so the marginal $p_\theta(x) = \int p_\theta(x \mid z)p(z)\,dz$ matches the data distribution.

Definition 7.4.2. *Variational Autoencoder*
A variational autoencoder combines a decoder $p_\theta(x \mid z)$ with an encoder $q_\phi(z \mid x)$ and maximizes an evidence lower bound consisting of reconstruction and latent regularization terms.

Definition 7.4.3. *Generative Adversarial Network*
A generative adversarial network trains a generator against a discriminator that tries to distinguish generated samples from data. The game defines a learned discrepancy between distributions.

Definition 7.4.4. *Diffusion Model*
A diffusion model corrupts data through a noising process and learns a reverse denoising process. Sampling starts from noise and iteratively applies learned denoising transitions.

7.5 Reinforcement Learning and Decision Systems

Reinforcement learning studies agents whose actions affect future observations and rewards.

Definition 7.5.1. *Markov Decision Process*
A Markov decision process consists of states $\mathcal{S}$, actions $\mathcal{A}$, transition law $P(s' \mid s, a)$, reward function $r(s, a)$, and discount factor $\gamma \in [0, 1)$. A policy $\pi(a \mid s)$ chooses actions from states.

Definition 7.5.2. *Value Functions*
The value of a policy is

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \,\middle|\, S_0 = s\right].$$

The action-value function $Q^\pi(s, a)$ conditions additionally on the first action.

Definition 7.5.3. *Bellman Equation*
For a fixed policy,

$$V^{\pi}(s) = \sum_{a} \pi(a \mid s) \left(r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V^{\pi}(s') \right).$$

Optimality equations replace the policy average by a maximum over actions.

Remark 7.5.4. *RL as a Learning System*
Reinforcement learning combines prediction, causal feedback, optimization, exploration, and evaluation under distribution shift. Its validation must account for sequential dependence and policy-induced data.

8 Learning Systems and Evaluation

8.1 Data Splitting, Validation, and Benchmarking

Evaluation estimates how the trained system behaves on data not used to fit it.

Definition 8.1.1. *Training, Validation, and Test Splits*
The training set fits parameters, the validation set selects models and hyperparameters, and the test set estimates final performance after selection. A typical data-rich workflow reserves disjoint portions of the data for these roles; when data are scarce, cross-validation or bootstrap methods approximate the validation step. Mixing roles produces optimistic error estimates.

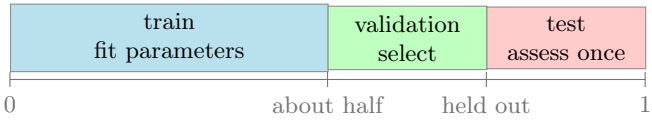


Figure 247: Training data fit the candidate rules, validation data choose among them, and a test set is saved for one final assessment of the selected rule.

Definition 8.1.2. *Model Selection and Model Assessment*
Model selection estimates performance across candidate models or tuning parameters in order to choose one. Model assessment estimates the prediction error of the final chosen rule on new data. A test set used repeatedly for selection no longer gives an honest assessment of the selected model, because the choice has adapted to that set.

Definition 8.1.3. *Conditional and Expected Test Error*
Given a training set $\mathcal{T}$ and fitted predictor $\widehat{f}_{\mathcal{T}}$, the conditional test error is

$$\mathrm{Err}_{\mathcal{T}} = \mathbb{E}_{X^0,Y^0}\left[\ell(\widehat{f}_{\mathcal{T}}(X^0), Y^0) \mid \mathcal{T} \right],$$

where (X^0, Y^0) is an independent test point from the same population. The expected test error averages over training sets:

$$\mathrm{Err} = \mathbb{E}_{\mathcal{T}} \, \mathrm{Err}_{\mathcal{T}}.$$

Most reusable error estimators target Err more reliably than the particular conditional error $\mathrm{Err}_{\mathcal{T}}$.

Method 8.1.4. *K-Fold Cross-Validation*
Let $\kappa : \{1, \dots, n\} \to \{1, \dots, K\}$ assign observations to folds, and let $\widehat{f}^{-k}$ be the predictor trained with fold k removed. For loss ℓ, the cross-validation estimate is

$$\mathrm{CV}(\widehat{f}) = \frac{1}{n} \sum_{i=1}^{n} \ell\Big(\widehat{f}^{-\kappa(i)}(x_i), y_i \Big).$$

For a tuning parameter α, compute the same quantity for every candidate α, choose a minimizer or a one-standard-error simplification, and then refit the chosen model on all training data. Common choices such as $K = 5$ or $K = 10$ trade the low bias of leave-one-out cross-validation against the lower variance and lower computation of using larger validation folds.

Remark 8.1.5. *What Cross-Validation Estimates*
Cross-validation is usually best viewed as estimating expected prediction error for the learning procedure, not the exact conditional error of the particular fitted model at hand. Leave-one-out cross-validation nearly uses the full training set in each refit, but its fold estimates are highly dependent. Five- or tenfold cross-validation fits on smaller samples, so it can be biased upward when the learning curve is steep, but it often has lower variance.

Definition 8.1.6. *Benchmark*
A benchmark specifies data, preprocessing rules, train-test splits, metrics, and allowed external resources. Without these conventions, reported performance numbers need not be comparable.

Remark 8.1.7. *Data Leakage*
Leakage occurs when information from validation or test data enters training, feature construction, preprocessing, hyperparameter selection, or manual iteration. Leakage is a systems failure as much as a statistical one.

Method 8.1.8. *Cross-Validation with Screening or Preprocessing*
When a modeling pipeline screens variables, chooses transformations, or performs any supervised feature selection, the entire pipeline must be repeated inside each training fold:

(i) split the observations into folds;
(ii) for fold k, choose features and tuning choices using only observations outside fold k;
(iii) fit the model on those same outside-fold observations;
(iv) predict the held-out fold k.

Unsupervised preprocessing that does not use the response is less dangerous, but its allowed scope should still be stated in the benchmark protocol. Selecting variables on the full data set before cross-validation lets the held-out labels influence the predictor and can make error estimates wildly optimistic.

Remark 8.1.9. *Metric Choice*
Accuracy, log loss, calibration error, ranking metrics, cost-sensitive utility, and latency measure different aspects of a system. A benchmark metric should match the decision problem it stands in for.

Method 8.1.10. *Bootstrap Estimates of Prediction Error*
For bootstrap samples S^{*b}, train $\widehat{f}^{*b}$ on S^{*b}. The naive bootstrap error reuses the original sample as the evaluation set,

$$\widehat{\mathrm{Err}}_{\mathrm{boot}} = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{n}\sum_{i=1}^{n}\ell\left(\widehat{f}^{*b}(x_i), y_i\right),$$

and can be too optimistic because many evaluation observations also appear in the bootstrap training sample. Let C^{-i} be the set of bootstrap indices b for which observation i is absent from S^{*b}. The out-of-bag, or leave-one-out bootstrap, estimate is

$$\widehat{\mathrm{Err}}^{(1)} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|C^{-i}|}\sum_{b\in C^{-i}}\ell\left(\widehat{f}^{*b}(x_i), y_i\right),$$

omitting terms with $|C^{-i}| = 0$. The .632 estimate combines the training error $\overline{\mathrm{err}}$ and this out-of-bag estimate:

$$\widehat{\mathrm{Err}}^{(.632)} = .368\,\overline{\mathrm{err}} + .632\,\widehat{\mathrm{Err}}^{(1)}.$$

The .632+ variant increases the out-of-bag weight when the fitted rule appears heavily overfit, interpolating between the .632 estimate and the out-of-bag estimate.

Remark 8.1.11. *Limits of Reuse-Based Assessment*
Cross-validation and bootstrap estimates can be excellent for comparing nearby tuning choices, because an additive bias common to all candidates may not change the selected model. Estimating the absolute conditional error of the final fitted rule is harder. Highly adaptive procedures can still overfit the validation process itself, in which case only a genuinely untouched test set gives an unbiased final assessment.

8.2 Robustness, Shift, and Distributional Uncertainty

Learning systems are deployed under distributions that may differ from the training law.

Definition 8.2.1. *Distribution Shift*
Common forms of distribution shift include:

(i) Covariate shift changes P_X while preserving $P_{Y|X}$.
(ii) Label shift changes P_Y while preserving $P_{X|Y}$.
(iii) Concept shift changes the relationship between inputs and outputs.

Definition 8.2.2. *Adversarial Example*
An adversarial example is a perturbed input x' near x for which the model output changes in a harmful way. Adversarial training adds such perturbed inputs to the training objective to improve local robustness.

Definition 8.2.3. *Distributionally Robust Optimization*
Distributionally robust optimization minimizes worst-case risk over an uncertainty set $\mathcal{U}$ of plausible distributions:

$$\inf_f \sup_{Q \in \mathcal{U}} \mathbb{E}_Q \ell(f(X), Y).$$

Remark 8.2.4. *Uncertainty*
Reliable systems distinguish aleatoric noise, epistemic uncertainty, and out-of-distribution uncertainty. Calibration and abstention policies are part of robustness, not decorative outputs.

8.3 Interpretability, Fairness, and Safety

Evaluation also asks whether a model can be understood, audited, and constrained in the setting where it is used.

Definition 8.3.1. *Interpretability*
Interpretability methods assign human-usable explanations to model behavior. Examples include feature attribution, counterfactual explanations, surrogate models, and mechanistic analysis of internal representations.

Definition 8.3.2. *Fairness Criteria*
Group fairness criteria compare error rates, acceptance rates, calibration, or other performance quantities across protected groups. Different fairness criteria can conflict, so the mathematical condition must be tied to a policy goal.

Definition 8.3.3. *Safety Constraint*
A safety constraint restricts model behavior outside ordinary average-case loss, for example by enforcing abstention, monotonicity, bounded action sets, human review, or worst-case risk limits.

Remark 8.3.4. *Beyond Accuracy*
Interpretability, fairness, and safety are not single metrics. They are collections of mathematical requirements, measurement protocols, and governance choices around a deployed learning system.

8.4 Deployment, Monitoring, and Feedback Loops

Deployment turns a fitted model into part of a larger stochastic system.

Definition 8.4.1. *Monitoring*
Monitoring tracks input distributions, prediction distributions, losses when labels arrive, calibration, latency, missingness, and operational failures. It detects when the assumptions behind validation no longer hold.

Definition 8.4.2. *Feedback Loop*
A feedback loop occurs when model predictions influence the future data observed by the model. Recommendation, credit, policing, and medical triage systems can all change their own training distribution.

Method 8.4.3. *Model Update Policy*
An update policy specifies when to retrain, what data are eligible, how to validate a replacement, and how to roll back. It is part of the statistical procedure because it determines the data seen by future models.

Remark 8.4.4. *Human-System Boundary*
When a learning model supports human decisions, the deployed predictor is only one component. The effective policy includes interfaces, incentives, overrides, monitoring, and accountability.

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